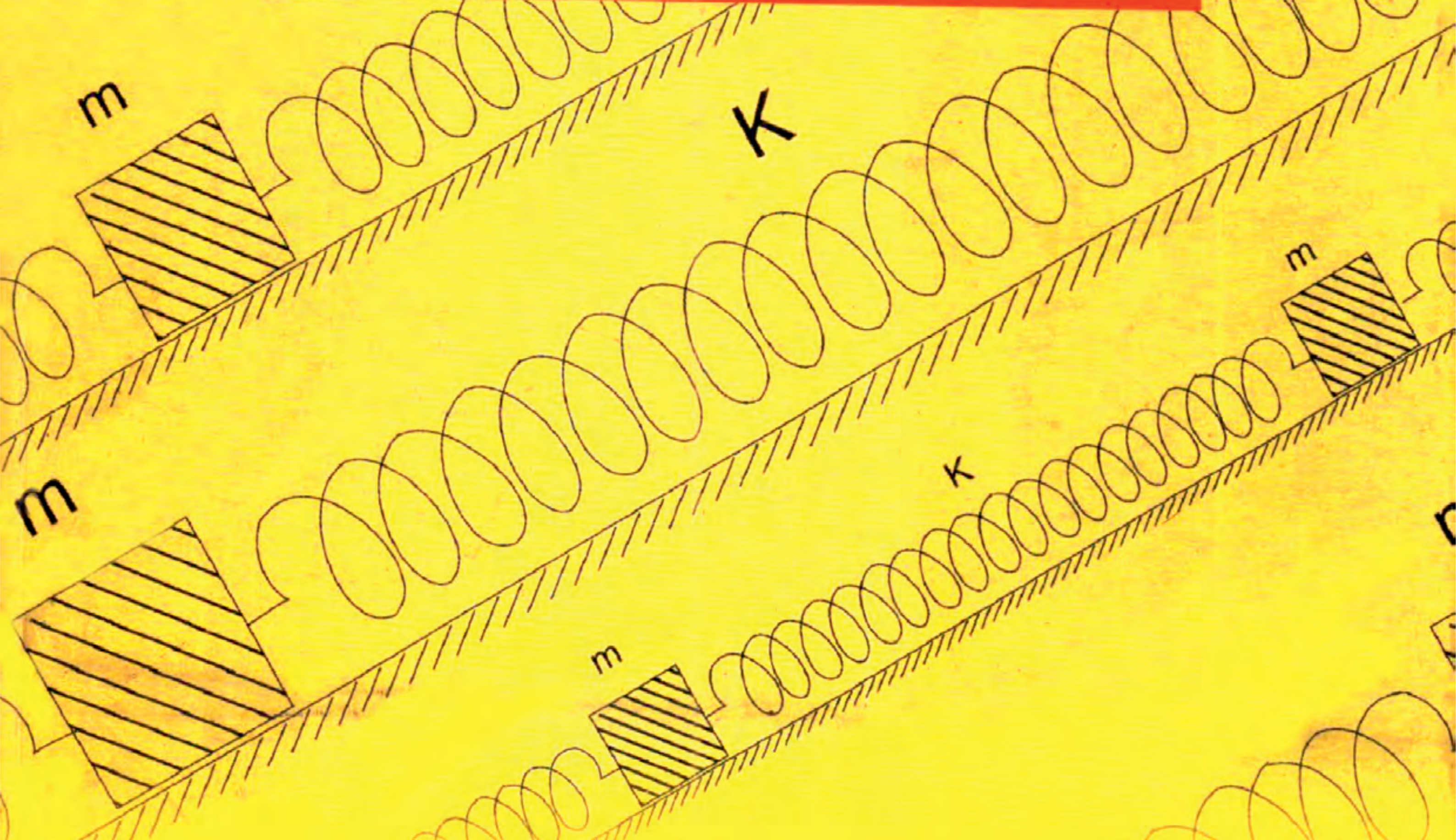
University Physics-I

J.C. Upadhyaya



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UNIVERSITY PHYSICS - I

Part - I: MECHANICS, OSCILLATIONS AND PROPERTIES OF MATTER
Part - II: ELECTRICITY, MAGNETISM AND ELECTROMAGNETIC

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PREFACE

Present book has been written according to the Unified syllabus prescibed for B.Sc. (I) Physics students for all Indian universities by University Grants Commission (U.G.C.), New Delhi. The presecribed course for B.Sc. (I) has been divided into two parts—(1) Mechanics, Oscillations and Properties of Matter and (2) Electricity, Magnetism and Electromagnetic theory. First part of the book basically deals with mechanics which in fact forms the foundation of entire physics. This is why this course is taught first of all to the students in any physics syllabus. Second part is related to electromagnetism, having sufficient formulations similar to mechanics. The learned experts of U.G.C. syllabus have placed, therefore, the two papers together related to mechanics and electromagnetism in the B.Sc. (Physics) First Year course.

First part of the book has been divided into fifteen chapters. First six chapters deal with the fundamental principles of mechanics. These chapters begin with the well known Newton's laws of motion and conservation of energy. Middle chapters discuss the Kepler's laws of planetary motion, gravitational field and potential and consevation laws of linear and angular momentum. The sixth chapter deals with the rigid body motion where conceptually the climax of mechanics subject is reached and with this chapter, the basic discussion on the principles of mechanics comes to an end. Next five chapters are related to the oscillations of physical systems in which starting the subject matter by simple harmonic motion with suitable examples, superposition of harmonic motions and coupled oscillators are discussed in the middle chapters and finally damped and driven harmonic oscillators are given in two chapters. Then the book deals with a chapter on 'Motion of Charged Particles in Electric and Magnetic Fields', related to electromagnetism and modern physics. Basically the subject matter of this chapter discusses the applications of principles of mechanics. Finally the last three chapters, namely Elasticity, Kinematics of Moving Fluids (Viscosity) and Surface Tension, are devoted to the properties of matter whose importance is well known is physics and every day life.

In the second part of the book, we deal with the topics, related to mathematical methods, electricity, magnetism and electromagnetic theory into twelve chapters. First three chapters on 'Mathematical Background'discuss the vectors and their calculus, differentiation and integration of functions of two and three variables. Next three chapters deal with the matter of electrostatics and further two chapters that of steady and alternating currents. Then a chapter is devoted to magnetostatics, where we discuss the effect of magnetic field on current carrying conductors, Biot-Savart's law, Ampere's law and magnetic field in matter. Last three chapters of this part are based on electromagnetic theory, dealing with the time varying fields, Maxwell's equations, electromagnetic waves and their reflection and refraction.

In order to grasp the basic concepts of physics and to understand the complicacy of subject matter, it is necessary that the students have sufficient practice to solve the related problems. To stress this point, several Indian Universities have made compulsory to ask numerical problems in the examination paper. This is why we have given a sufficient number of selected informative and modern solved problems in few groups at differet places inside a chapter and each group of problems has been followed by an exercise (set of unsolved problems). The unsolved problems in an exercise have been systematically and methodically so arranged that after solving some simpler problems in the begining an average student is encouraged to tackle relatively difficult problems and a good student feels pleasure and intellectually satisfied after solving some difficult problems given in the last. Regarding the solved problems, we advise the students : "*Always try to solve a problem in writing before looking to its solution. This will help you in locating your difficulties and consequently in removing them.*"

In view of the latest pattern of university examination papers, theoretical questions have been systematically arranged at the end of each chapter of the book in the following three heads :

(1) Long Answer Quetions, (2) Short Answer Questions and (3) Objective Type Questions.

In writing this book, we have tried to present the subject matter in a simple language so that an average student feels interesting in following the text. We have also kept in mind that after studying this book, the base of knowledge of the students becomes so strong that they may not feel difficulty in understanding the subject matter in higher classes.

Though best effort has been made not to have any mistake inside the book, even then some mistakes may have inadvertently crept in. The author shall feel highly grateful to the readers who will point out and inform regarding any mistake. Further any suggestion towards the improvement of the subject matter will be thankfully received.

Finally the author shall feel highly satisfied and rewarded, if the student community is benefilted to any substantial extent.

This book is dedicated to my family members, Smt. Raj Kumari (wife), Dr. Sharad, Ram (sons) and Dr. Tanuja Upadhyaya (daughter), for their constant support and assistance in so many ways during the preparation of the book.

Date: 7th August, 2007.

—J.C. Upadhyaya

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PART-I

Mechanics, Oscillations and Properties of Matter

1

Laws of Motion and Frame of Reference

1·1. Mechanics

When we look around, we observe the objects in *motion* and at *rest*. *Mechanics is a branch of physics which deals with physical objects in motion and at rest under the influence of external and internal interactions.* The motion of a body is influenced by the bodies, surrounding it. The cause of this influence is the *interaction* between the bodies. In order to analyze and predict the motion of bodies due to different kinds of interaction, some important concepts have been discovered such as momentum, force and energy. We may say that *mechanics is the study of the principles, related to momentum, force and energy which apply to the analysis of all kinds of motion.*

Mechanics was developed since ancient times on the basis of observations on the motion of material objects. Although efforts were made earlier to propose theoretical hypotheses regarding the relationship of force and motion, but it was not until Newton announced his famous laws of motion in 1687. Later the subject of mechanics had been greatly developed by a large number of scientists, specifically by Lagrange and Hamilton, and more recently by Eienstein, Schrodinger, Heisenberg, Dirac etc.

1·2. Motion and Rest

When the position of a body changes with time, we ordinarily say that the body is in the state of motion and if its position does not change with time it is said to be at rest. There may be two kinds of rest and motion—**absolute** and **relative**. A body is said to be in absolute motion when its motion is observed with respect to a point which is stationary in space. But in the universe we do not know any point which remains always stationary. Therefore, absolute motion can only be imagined and practically it has no importance. In the universe every body is in the state of motion; earth, sun and other stars are moving all the time and thus any body does not remain stationary. Hence absolute rest is also imaginary.

In fact **motion and rest both are relative**. A body will be said to be in the state of motion or rest with respect to another, accordingly if its position changes or does not change relative to the other. For example, a tree on the ground is at rest relative to earth; but the same tree is in the state of motion relative to sun because firstly, the earth with this tree is constantly rotating round its axis and secondly, it is moving in its orbit round the sun. Relative to a passenger sitting inside the train other passengers are at rest, but the outside trees, buildings etc. are in the state of motion.

1·3. Frame of Reference—Velocity and Acceleration

From the above description we see that in order to describe the motion of a body we must know that with respect to what this motion has been measured. *If we imagine a coordinate system attached to a rigid body and we describe the position of any particle in space relative to it, then such a coordinate system is called* **frame of reference.** For the location of the objects, the position vectors are drawn from the **origin** of the coordinate system. An observer uses a frame of reference which is stationary relative to him and generally the position of the observer coincides with the origin.

The simplest frame of reference is a *cartesian coordinate system* X, Y, Z. To study the motion of a particle P relative to this system, we draw position vector $\vec{r}$ from the origin O at an instant. In terms of (x,y,z) coordinates, the position vector $\vec{r}$ is represented as

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \qquad ...(1)$$

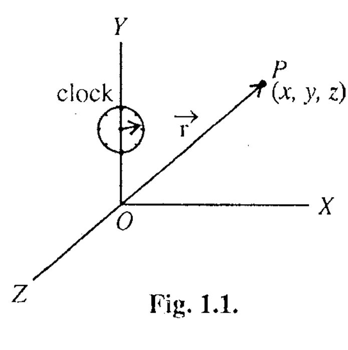


Fig. 1.1.

To describe fully the motion of the particle, we must know its position at different instants of time, *i.e.*, the position coordinate $\vec{r}$

should be expressed as a function of time. Then the **velocity** and **acceleration** of the particle will be given by

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{\mathbf{i}} + \frac{dy}{dt}\hat{\mathbf{j}} + \frac{dz}{dt}\hat{\mathbf{k}} \quad ...(2)$$

and

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = \frac{d^2x}{dt^2}\hat{\mathbf{i}} + \frac{d^2y}{dt^2}\hat{\mathbf{j}} + \frac{d^2z}{dt^2}\hat{\mathbf{k}} \quad ...(3)$$

If a physical phenomenon (*e.g.*, a particle passing through some point P) occurs in space, its position is known as the *point*; the *time* of occurrence and the *point* taken together are called **an event.** In general, for the location of an event in a frame of reference we require its position and time of occurrence and therefore for this purpose four coordinates (x, y, z, t) are required. The reference system, employed for this purpose, is called *space-time frame of reference*.

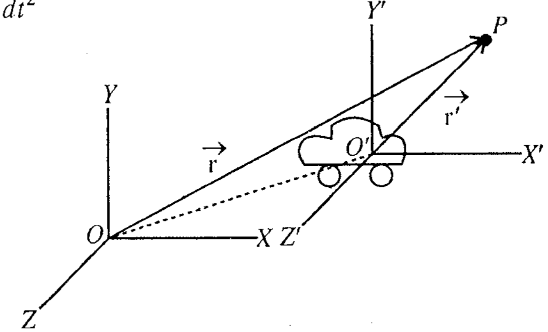
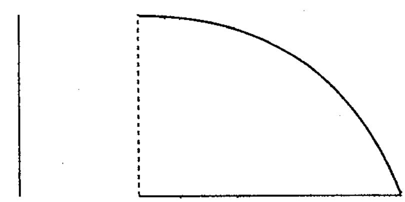


Fig. 1.2.

Frames of reference in relative motion

If two frames are in relative motion, the motion of a particle will be described differently. Consider an observer O, standing on the ground with XYZ frame of reference. Another observer O', sitting in a moving car, uses $X'Y'Z'$ frame. If O and O' are at rest relative to one another, they will observe the same motion of a particle. But if O and O' are in relative motion, their observations of the motion of the particle will be different. But for convenience that frame of reference is used in which the motion is easy to describe.



Straight line (a) Parabolic path (b)

Fig. 1.3.

Examples : (1) Suppose that an observer is sitting inside a train, moving with constant velocity. Now, if he drops a stone, he will see that the stone moves along a straight line [**fig. 1.3(a)**]. But an observer, standing stationary on the ground, sees the stone, moving with constant horizontal velocity (*i.e.*, the velocity of the train) and thus the stone will move on a parabolic path relative to this observer [**fig. 1.3 (b)**].

(2) The motion of the planets will appear very complicated relative to a frame attached with the earth. But the planetary motions are much simpler to describe when we assume that the sun is the centre of the solar system and the frame is situated on the sun (*heliocentric frame*). Thus to describe the motion in a simpler way we always try to select a proper origin of frame of reference; however sometimes we cannot reach physically there.

(3) Suppose we stand on the earth and look at the motion of a point on the edge of a slowly moving wheel of a car. The point will be seen to move through a complicated curve, known as *cycloid* [**fig. 1.4**]. Thus a position vector from us as origin to a point on the wheel performs an extremely complicated motion. But a point on the earth is not always the most convenient origin for position vectors. If we reach mentally at the centre of the wheel and taking it as the origin if we study the motion, the point will appear to move on a circular path and thus the motion will be far simpler.

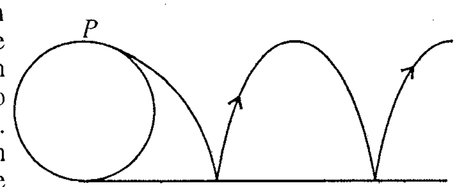


Fig. 1.4.

1·4. Newton's Laws of Motion—Laws of Mechanics

Sir Isaac Newton expressed his ideas of motion in the form of three laws which are known as the **laws of mechanics.** In fact, mechanics is a study of certain general relations that describe the interactions of material bodies. One general property of a material body is its **inertial mass**. Another new concept useful in describing interactions is **force.** These two concepts, inertial mass and force, were defined in a quantitative manner by Isaac Newton. The definitions of mass and force are contained in his three laws of motion.

(1) Law of inertia : *A body remains at rest or moves with constant velocity, unless not disturbed by some external influence.* The property of a body that it cannot change its state of rest or uniform motion is called **inertia** and the influence under which the velocity of a particle changes is called **force.** The quantitative definitions of *force* and measure of inertia of a body, which we call **mass,** are contained in second and third laws of motion.

(2) Law of force. *The time-rate of change of momentum is proportional to the impressed force, i.e.,*

$$\vec{F} = \frac{d\vec{p}}{dt} \qquad ...(1)$$

Every body possesses the property of inertia or resistance to motion. This inertia is different for different bodies. The measure to this inertia for translation is called the **mass** of a body and is denoted by m. If v be the velocity of a body of mass m, then its momentum is defined by

$$\vec{p} = m\vec{v}, \quad \therefore \quad \vec{F} = \frac{d}{dt}(m\vec{v})^*$$

Newton considered that mass of a body remains constant during its motion. Therefore,

$$\vec{F} = m\frac{d\vec{v}}{dt} = m\vec{a} \qquad ...(2)$$

i.e., *Force = mass × acceleration*

This is the fundamental law of classical mechanics. Quantitatively, first law is the special case of second law, because if force is not acting on a body, *i.e.,* $\vec{F} = 0$, then

$$\frac{d\vec{v}}{dt} = 0, \quad \therefore \quad \vec{v} = \text{constant} \qquad ...(3)$$

This constant may be zero, which represents the state of rest.

Evidently, Newton's second law uses two new quantities (mass and force) neither of which have been rigorously defined. However, his third law in combination with the second law defines uniquely mass and force.

Newton's first and second laws of motion are valid only in inertial frames, defined later.

(3) Law of action and reaction. *To every action there is always equal and opposite reaction.* This means that if 1 and 2 bodies are interacting mutually, then

$$\vec{F}_{12} = -\vec{F}_{21} \qquad ...(4)$$

i.e., force on 1st body due to 2nd = – force on 2nd body due to 1st.

But $\vec{F}_{12}$ = Rate of change of momentum of the 1st body $= \frac{d}{dt}(m_1\vec{v}_1)$ ...(5)

and similarly $\vec{F}_{21} = \frac{d}{dt}(m_2\vec{v}_2)$ ...(6)

Now, from (4), (5) and (6), we have

$$m_1\frac{d\vec{v}_1}{dt} = -m_2\frac{d\vec{v}_2}{dt}$$

If $a_1 = \left|\frac{d\vec{v}_1}{dt}\right|$ and $a_2 = \left|\frac{d\vec{v}_2}{dt}\right|$ denote for accelerations, then in magnitude

$$m_1a_1 = m_2a_2 \text{ or } m_2 = m_1a_1/a_2 \qquad ...(7)$$

* Common experience tells us that a greater pull or force is required to change a definite amount of velocity in a certain time for a massive body than a lighter body. So Newton considered that mass also be included in the definition of force by defining the momentum as mass times velocity.

This means that if an isolated system of two bodies is interacting among themselves, then by measuring their accelerations the ratio of their masses can be determined. If one is the standard body of mass 1 kg (say m_1), then the other (m_2) can be determined. Thus, *Newton's third law defines mass uniquely and hence equation* (2) *can be used to define the force in a unique way.*

Limitations of the Newton's laws

Newton's first and second laws appear to be violated if the observer himself is accelerated. If a particle is experiencing no force and its acceleration is zero relative to an observer, then another observer, moving with an acceleration relative to the first one, will see that particle is in acceleration, *i.e.*, a fictitious force is acting on the particle, relative to the second observer. Newton realised the difficulty and he specified that his laws of motion are valid only when the observer is stationed in an inertial system. Newton defined an **inertial system** as any system which is not accelerating relative to the fixed stars.

According to modern views, Newton's third law of motion is not correct when a force is acting at a distance because the forces or actions cannot move faster than the speed of light. In other words, if the first particle produces any change in the second particle, changed force (or reaction) reaches on the first particle after a finite interval of time. This means simultaneously action is not equal to reaction.* Remember that Newton's third law is still correct for the bodies at rest and for contact forces.

1·5. Motion Under Constant Force *(Motion in a Uniform Field)*

If a particle moves in a uniform field, then it is acted by a constant force and consequently the particle moves with constant acceleration. Ordinarily we consider two types of field ($\vec{E}$) : (i) Gravitational field (*e.g.*, near the surface of the earth), where the force on a particle of mass m is given by $\vec{F} = m\vec{E} = m\vec{g}$, $\vec{E} = \vec{g}$ being the field or acceleration due to gravity and (ii) Electric field (*e.g.*, in between the plates of parallel plate condenser), where the force on a particle of charge q and mass m is given by $\vec{F} = q\vec{E}$. For uniform or constant field, the force $\vec{F}$, acting on the particle, is constant.

According to Newton's second law of motion,

$$\vec{F} = m\vec{a} \qquad ...(1)$$

where $\vec{a} = d\vec{v}/dt = d^2\vec{r}/dt^2$ is the acceleration of the particle. In the Newtonian mechanics, the mass of a moving particle remains constant and hence when a particle is moving under a constant force, its acceleration $\vec{a}$ will remain constant. Thus, for constant force

$$\vec{a} = \frac{\vec{F}}{m} = \frac{d\vec{v}}{dt} = \text{constant} \qquad ...(2)$$

Eq. (2) can be used to deduce the three equations of motion.

Equations of motion

From eq. (2), the integration of the relation $d\vec{v} = \vec{a}\,dt$ gives

$$\int d\vec{v} = \int \vec{a}\,dt \quad \text{or} \quad \vec{v} = \vec{a}t + \vec{C} \qquad ...(3)$$

where $\vec{C}$ is the constant of integration. Let at $t = 0$, $\vec{v} = \vec{u}$, the initial velocity. Then

$$\vec{u} = 0 + \vec{C} \quad \text{or} \quad \vec{C} = \vec{u}$$

Therefore, eq., (3) assumes the form

$$\vec{v} = \vec{u} + \vec{a}t \qquad ...(4)$$

---

* A simple example for the gross violation of third law may be quoted here. Let a charged particle move away from a wire carrying a current. The wire exerts a magnetic force on the charged particle, while at the same time the net force on the wire due to the particle is exactly zero. (See Chap. 8, Fundamental Physics by *J. Orear,* John Wiley & Sons, Inc., New York, 1965).

which is the **first equation of motion** in vector form.

Now, $$\vec{v} = \frac{d\vec{r}}{dt} = \vec{u} + \vec{a}\,t$$

Integration of it yields

$$\int d\vec{r} = \int \vec{u}\,dt + \int \vec{a}\,t\,dt \quad \text{or} \quad \vec{r} = \vec{u}\,t + \frac{1}{2}\vec{a}\,t^2 + \vec{C_1}$$

where $\vec{C_1}$ is the constant of integration

If at $t = 0$, $\vec{r} = \vec{r_0}$ is the initial position vector, then

$$\vec{r_0} = 0 + \vec{C_1} \quad \text{or} \quad \vec{C_1} = \vec{r_0}$$

Therefore, $$\vec{r} = \vec{r_0} + \vec{u}\,t + \frac{1}{2}\vec{a}\,t^2 \qquad ...(5)$$

If we write $\vec{r} - \vec{r_0} = \vec{s}$, the displacement of the particle in time t, then

$$\vec{s} = \vec{u}\,t + \frac{1}{2}\vec{a}\,t^2 \qquad ...(6)$$

This is the **second equation of motion.**
From eq. (2)

$$\vec{a} = \frac{d\vec{v}}{dt}$$

Scalar product of $\vec{v}$ on both sides of this equation gives

$$\vec{v}\cdot\vec{a} = \vec{v}\cdot\frac{d\vec{v}}{dt} \quad \text{or} \quad \frac{d\vec{s}}{dt}\cdot\vec{a} = \vec{v}\cdot\frac{d\vec{v}}{dt} \qquad \left(\because \frac{d\vec{r}}{dt} = \frac{d\vec{s}}{dt} = \vec{v}\right)$$

This can be integrated as

$$\int \vec{a}\cdot d\vec{s} = \int \vec{v}\cdot d\vec{v} \quad \text{or} \quad \vec{a}\cdot\vec{s} = \frac{1}{2}\vec{v}\cdot\vec{v} + C_2 \qquad ...(7)$$

where C_2 is the constant of integration. Remember that $\vec{a}$ is constant and $d\left(\frac{1}{2}\vec{v}\cdot\vec{v}\right) = \frac{1}{2}\left(d\,\vec{v}\cdot\vec{v} + \vec{v}\cdot d\,\vec{v}\right)$ $= \vec{v}.d\vec{v}$.

When $\vec{s} = 0$, $\vec{v} = \vec{u}$ and therefore from eq. (7)

$$0 = \frac{1}{2}\vec{u}\cdot\vec{u} + C_2 \quad \text{or} \quad C_2 = -u^2/2.$$

Hence, eq. (7) becomes

$$\vec{a}\cdot\vec{s} = \frac{v^2}{2} - \frac{u^2}{2} \qquad \left(\because \vec{v}\cdot\vec{v} = v^2\right)$$

or $$v^2 = u^2 + 2\,\vec{a}\cdot\vec{s} \qquad ...(8)$$

which is the **third equation of motion**. Thus the **three equations of motion in vector form** are :

$$\vec{v} = \vec{u} + \vec{a}\,t\,;\ \vec{s} = \vec{u}\,t + \frac{1}{2}\vec{a}\,t^2;\ v^2 = u^2 + 2\,\vec{a}\cdot\vec{s} \qquad ...(9)$$

In general, these equations of motion in vector form represent the **motion of a particle in three dimensions. If the motion is along a straight or in one dimension,** then

$$v = u + at,\ s = ut + \frac{1}{2}at^2 \text{ and } v^2 = u^2 + 2as \qquad ...(10)$$

Writing $\vec{u}$, $\vec{v}$ and $\vec{a}$ in the component form,

$$\vec{u} = u_x\hat{\mathbf{i}} + u_y\hat{\mathbf{j}} + u_z\hat{\mathbf{k}}, \quad \vec{v} = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} + v_z\hat{\mathbf{k}} \text{ and } \vec{a} = a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}} + a_z\hat{\mathbf{k}}$$

and if at $t = 0$, the particle is at the origin, *i.e.*, $\vec{r}_0 = 0$, then

$$\vec{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

Thus, when a particle moves with constant acceleration in three dimensions, the form of the equations of motion will be

$$v_x = u_x + a_x t; \qquad v_y = u_y + a_y t; \qquad v_z = u_z + a_z t \qquad \text{...(11a)}$$

$$x = u_x + \frac{1}{2}a_x t^2; \qquad y = u_y + \frac{1}{2}a_y t^2; \qquad z = u_z + \frac{1}{2}a_z t^2 \qquad \text{...(11b)}$$

and $$v_x^2 = u_x^2 + 2a_x x; \quad v_y^2 = u_y^2 + 2a_y y; \quad v_z^2 = u_z^2 + 2a_z z \qquad \text{...(11c)}$$

In case of motion in two dimensions or in a plane, eqs. (11a) and (11b) are sufficient.

Motion in a uniform gravitational field

We can assume a uniform gravitational field near the surface of the earth. When a particle is thrown close to the surface of the earth, a constant force $m\vec{g}$ acts on it, where $\vec{g}$ is the acceleration due to gravity acting downwards.

(a) Vertical motion—Freely falling body : This is an example of motion in a straight line with constant acceleration. When a body is thrown vertically or falls freely near the earth's surface, by taking Y-axis vertically upwards from the earth, we can write

$$\vec{g} = -g\hat{\mathbf{j}} \qquad \text{...(12)}$$

where g is the magnitude of the acceleration due to gravity.

In this case, the equations of motion are obtained as

$$v_y = u_y - gt; \quad y = u_y t - \frac{1}{2}gt^2; \quad v_y^2 = u_y^2 - 2gy$$

or $$v = u - gt; \quad h = ut - \frac{1}{2}gt^2; \quad v^2 = u^2 - 2gh \qquad \text{...(13)}$$

where we have written the equations of motion in the commonly used form ($y = h$) in the gravitational field near the surface of the earth.

(b) Motion in a Plane—Projectile motion : Projectile motion is an example of motion in a plane with constant acceleration. When a body is thrown obliquely near the surface of the earth, it moves on a curved path. Such a body is called **projectile** and its motion is called **projectile motion.**

Let a particle be projected from the point O at $t = 0$ with an initial velocity u at an angle θ with the horizontal. The particle P goes through the highest point A and falls at B on the horizontal surface through O. Here, the angle θ is called the **angle of projection** and the distance OB is called **horizontal range.** We take XY as the plane of motion. Horizontal line OX is the X-axis and vertical line OY is the Y-axis. By taking the acceleration, acting on the particle, $\vec{a} = \vec{g} = -g\hat{\mathbf{j}}$, equations of motion will be

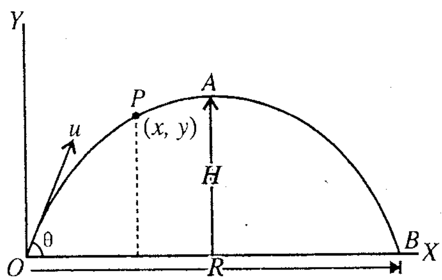


Fig. 1.5.

$$v_x = u_x = u\cos\theta \qquad (\because a_x = 0)$$

$$x = u_x t = u\cos\theta\ t \qquad \text{...(14)}$$

$$v_y = u_y - gt = u\sin\theta - gt \qquad (\because a_y = -g) \text{ ...(15)}$$

$$y = u_y t - \frac{1}{2}gt^2 = u \sin\theta\, t - \frac{1}{2}gt^2 \qquad ...(16)$$

and
$$v_y^2 = u_y^2 - 2gy = u^2 \sin^2\theta - 2gy^2 \qquad ...(17)$$

where (x, y) are the coordinates of the particle at time t.

From eq. (14), if we put $t = x/u \cos\theta$ in eq. (16), we get

$$y = \tan\theta\, x - \frac{1}{2}\frac{g}{u^2 \cos^2\theta}x^2 \qquad ...(18)$$

This is the equation of a parabola. Thus, *the projectile moves on a* **parabolic path** *in the gravitational field near the surface of the earth.*

Time of flight : The time in which the projected particle covers the path OAB is called **time of flight**. At the point B, $y = 0$ and then from eq. (16), we have

$$0 = u \sin\theta\, t - \frac{1}{2}gt^2 \quad \text{or} \quad t\left(u \sin - \frac{1}{2}gt\right) = 0$$

Therefore, either $t = 0$ or $t = 2u \sin\theta/g$

Since $t = 0$ corresponds to the initial position O of the particle, the time $t = T$ at which the particle reaches the position B is thus,

$$\text{T} = \frac{2u \sin\theta}{g} \qquad ...(19)$$

This is the formula for the time of flight.

Maximum height attained : At the maximum height $y = H$, let t be the time required. At $y = H$, $v_y = 0$ and hence from eq. (15), we get

$$0 = u \sin\theta - gt, \text{ when } t = \frac{u \sin\theta}{g}$$

Using eq. (16) with $y = H$, $H = u \sin\theta\, t - \frac{1}{2}gt^2 = u \sin\theta\left(\frac{u \sin\theta}{g}\right) - \frac{1}{2}g\frac{u^2 \sin^2\theta}{g^2}$

or
$$H = \frac{u^2 \sin^2\theta}{2g} \qquad ...(20)$$

Horizontal range : Horizontal range $R = OB$ is the distance OB covered by the projected particle in the time of flight $T = 2u \sin\theta/g$, *i.e.*, using eq. (14) for $x = R$ and $t = T$,

$$R = u \cos\theta\; T = (u \cos\theta)\left(\frac{2u \sin\theta}{g}\right) = \frac{2u^2 \sin\theta \cos\theta}{g} = \frac{u^2 \sin 2\theta}{g} \qquad ...(21)$$

Ex. 1. *A force* $\vec{P} = 25\hat{i} + 6\hat{j}$ *newtons acts on a particle of mass* 2.5 *kg for* 5 *sec. If the initial position of the particle is* $\vec{r}_0 = 6\hat{j} + 8\hat{k}$ *metre and the initial velocity* $\vec{u} = 2.5\hat{i} + 3\hat{k}$ *metre/sec, caculate*

(i) *the final velocity of the particle,* (ii) *the final position of the particle and* (iii) *the work done by the force on the particle.* **(Agra 1970)**

Sol. (i) Final velocity, $\vec{v} = \vec{u} + \vec{a}\,t$

Here, $\vec{a} = \frac{\vec{P}}{m} = \frac{25\hat{i} + 6\hat{j}}{2.5}$ m/sec^2, $\vec{u} = 2.5\hat{i} + 3\hat{k}$ m/sec and $t = 5$ sec.

$\vec{v} = 2.5\hat{i} + 3\hat{k} + (25\hat{i} + 6\hat{j})\,5/2.5 = 2.5\hat{i} + 3\hat{k} + 50\hat{i} + 12\hat{j} = \mathbf{52.5\hat{i} + 12\hat{j} + 3\hat{k}}$ **m/sec.**

(ii) Final position, $\vec{r} = \vec{r}_0 + \vec{u}\,t + \frac{1}{2}\vec{a}\,t^2 = 6\hat{j} + 8\hat{k} + (2.5\hat{i} + 3\hat{k})\,5 + \frac{1}{2}\frac{(25\hat{i} + 6\hat{j})}{2.5}\cdot 5^2$

$= 6\hat{j} + 8\hat{k} + 12.5\hat{i} + 15\hat{k} + 125\hat{i} + 30\hat{j} = \mathbf{137.5\hat{i} + 36\hat{j} + 23\hat{k}}$

(iii) Work done on the particle $= \vec{P}\cdot(\vec{r} - \vec{r}_0) = (25\hat{i} + 6\hat{j})\,.\,(137.5\hat{i} + 30\hat{j} + 15\hat{k})$

$= 3437.5 + 180 =$ **3617.5 joules.**

Ex. 2. *A force of* 10 *N is applied on an object of mass* 5 *kg for* 5 *seconds. Calculate :* (i) *impulse of the force,* (ii) *change in momentum of the object,* (iii) *change in velocity of object,* (iv) *acceleration of object.* **(Bilaspur 2003)**

Sol. Here, $m = 5$ kg, $F = 10$ N and $\Delta t = 5$ sec.

(i) Impulse of force = $F\Delta t = 10 \times 5 =$ **50 newton-sec.**

(ii) Now, change in momentum = Impulse of the force = **50 kg-ms^{-1}**

(iii) Change in velocity of object $\Delta v = \dfrac{\text{Change in momentum}}{\text{Mass of object}} = \dfrac{50}{5} =$ **10 m/sec.**

(iv) Acceleration $a = \dfrac{\Delta v}{\Delta t} = \dfrac{10}{5} =$ **2 m/sec.**

[**Impulse of a force :** If a constant force $\vec{F}$ is acting on a particle for a short interval of time Δt, then the impluse of this force is given by

$$\text{Impulse} = \vec{F}\,\Delta t \qquad ...(i)$$

If the change in the velocity of the body in the interval Δt is $\Delta\vec{v}$, then

$$\text{Impulse} = \vec{F}\,\Delta t = m\,\Delta\vec{v} = \Delta\vec{p} \quad \left(\because \vec{F} = \frac{m\Delta\vec{v}}{\Delta t} \text{ or } \vec{F}\,\Delta t = m\Delta\vec{v}\right) \qquad ...(ii)$$

where $\Delta\vec{p} = m\,\Delta\vec{v}$ = change in momentum of the body. If the force $\vec{F}$ varies in the time interal from $t = t_1$ to $t = t_2$, then the impluse transferred will be

$$\int_{t_1}^{t_2}\vec{F}\,dt = m(\vec{v}_2 - \vec{v}_1) = \vec{p}_2 - \vec{p}_1 \text{ or } \int_{t_1}^{t_2}\vec{F}dt = m\Delta\vec{v} = \Delta\vec{p} \qquad ...(iii)$$

I.E., *Impulse transferred = change in momentum.*]

Ex. 3. *Two blocks of masses $m_1 = 1$ kg and $m_2 = 3$ kg are connected by a light string and placed on a frictionless table. When m_1 is pulled by a force F, both blocks move with an acceleration a = 5 m/s^2* [**fig. 1.6**]. *Determine the forces F and tension in the string.*

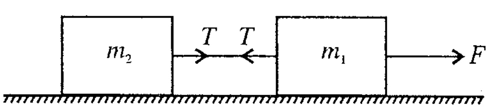


Fig. 1.6.

Sol. Let T be the tension in the string. Obviously, the force T is acting on the block m_2 and $F - T$ on the block m_1. Hence according to Newton's second law of motion,

$$F - T = m_1 a = 1 \times 5 = 5 \text{ newton and } T = m_2 a = 3 \times 5 = \mathbf{15\ newton}$$

Therefore $\quad F =$ **20 newton.**

Ex. 4. *A block of metal weighing* 4 *kg resting on a frictionless plane. It is struck by a jet releasing water at a rate of* 1 *kg/s and at a speed of* 5 *m/s* [**fig. 1.7**]. *Calculate the resulting acceleration of the block.*

Sol. According to Newton's second law,

$$F = \frac{dp}{dt} = \frac{d}{dt}(mv),$$

where m is the mass and v its velocity at any instant.

If v remains constant and m changes with time, then

$$F = v\,dm/dt \qquad ...(i)$$

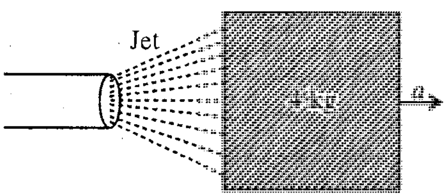


Fig. 1.7.

This is the situation in the present problem. Here a jet of water strikes a block [**fig. 1.7**] at a rate of $dm/dt = 1$ kg/s with speed $v = 5$ m/s. Hence, the force acting on the block due to the jet of water is

$$F = 5 \times 1 = 5 \text{ N}$$

If this force produces an acceleration a in the block of mass $M = 4$ kg, then

$$a = F/M = 5/4 = \mathbf{1.25 \ m/s^2}.$$

Ex. 5. *An aeroplane is flying horizontally at a height of 1960 m with a velocity of 360 km/hour above the ground towards a point directly over a person struggling in flood water. A survival kit is dropped so as to reach the person. How far horizontally from the person should the kit be released ?*

Sol. As the aeroplane is flying horizontally, the initial downward velocity of the kit is zero. If t be the time taken by it to reach the ground, then

$$h = ut + \frac{1}{2} gt^2$$

Here, $u = 0$, $h = 1960$ m and therefore

$$1960 = 0 + \frac{1}{2} \times 9.8 \times t^2 \quad \text{or} \quad t = \sqrt{\frac{2 \times 1960}{9.8}} = 20 \text{ s}$$

Velocity in horizontal direction, $v = 360 \text{ km/hr} = \dfrac{360 \times 1000}{60 \times 60} = 100$ m/s

The horizontal distance from the person where the kit is to be released

$$x = vt = 100 \times 20 = \mathbf{2000 \ m.}$$

Ex. 6. *Consider a massless string going over a frictionless pulley. Blocks of masses m_1 and m_2 are attached at the ends of the string. Such a system is called* **Atwood machine [fig. 1.8]**. *Find the acceleration of the blocks and tension in the string.*

Sol. The pulley is frictionless and hence it will not rotate. If m_2 is greater than m_1 and T is the tension in the string, then according to Newton's second law, equations of motion of the two masses are

$$T - m_1 g = m_1 a \qquad \text{...(i)}$$

and

$$m_2 g - T = m_2 a \qquad \text{...(ii)}$$

where a is the acceleration which will be the same for the two masses because the string is continuous.

Adding eqs. (i) and (ii), we obtain

$$(m_2 - m_1)\, g = (m_1 + m_2)\, a \text{ or } a = \frac{m_2 - m_1}{m_1 + m_2} g \qquad \text{...(iii)}$$

Therefore,

$$T = m_1 \left(g + \frac{m_2 - m_1}{m_1 + m_2} g \right) = \frac{2m_1 m_2}{m_1 + m_2} g \qquad \text{...(iv)}$$

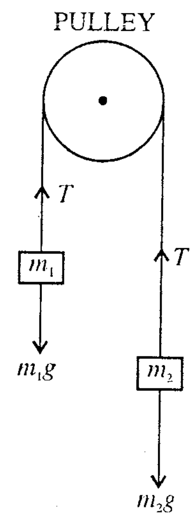


Fig. 1.8 : Atwood machine

Ex. 7. *An empty plastic box, having mass m, is found to move up with an acceleration g/6 inside deep water. How much sand should be put inside the box so that it may move down with an acceleration g/6.*

Sol. Let the force of buoyancy on the box be F. When the box is moving down,

$$F - mg = mg/6 \qquad \text{...(i)}$$

Suppose some sand of mass m' is placed inside the box so that it moves down with an acceleration $g/6$. In this situation

$$(m + m')\, g - F = (m + m')\, g/6 \qquad \text{...(ii)}$$

Adding eqs. (i) and (ii), we get

$$m'g = \frac{mg}{6} + (m + m')\frac{g}{6}$$

or

$$m'g - m'\frac{g}{6} = \frac{mg}{6} + \frac{mg}{6}$$

or

$$\frac{5m'g}{6} = \frac{mg}{3} \quad \text{or} \quad m' = \mathbf{\frac{2m}{5}}.$$

EXERCISE 1.1

1. A force produces an acceleration of 8 m/s^2 in body of mass 1 kg and an acceleration of 4 m/s^2 in another body. If both bodies are tied together, find the acceleration produced in the system by this force.
   [**Ans.** 2.67 m/s^2.]
2. A uniform rope of length l, resting on a frictionless surface, is pulled at one end by a force F. Determine the tension in the rope at a distance x from this end.
   [**Ans.** $F(l-x)/l$.]
3. A rod of length l and mass m is acted by two unequal forces F_1 and F_2 ($< F_1$) [**fig. 1.9**]. What is the tension in the rod at a distance x from the end of the rod, where F_1 is acting.
   [**Ans.** $[F_1(l-x) + F_2 x]/l$.]

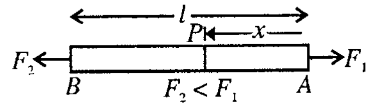


Fig. 1.9.

4. **fig. 1.10** shows a uniform rod of length 60 cm and mass 3 kg. The strings shown in the figure are pulled by constant forces of 10 N and 16 N. Find the force exerted by the 40 cm part of the rod on the 20 cm part. All the surfaces are smooth and the strings and pulleys are light.
   [**Ans.** 12 N.]

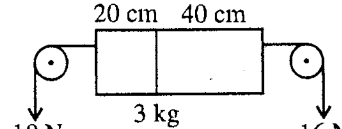
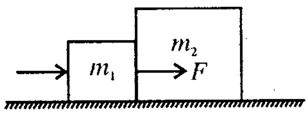


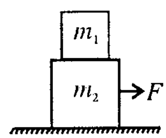
Fig. 1.10.

5. Two blocks of masses m_1 and m_2 are kept in contact on a frictionless table [**fig. 1.11**]. When the block m_1 is pushed from behind, the two blocks accelerate. If m_1 block exerts a force F on the block m_2, find the pushing force on m_1
   [**Ans.** $F[1 + (m_1/m_2)]$]



Frictionless table
Fig. 1.11.

6. The coefficient of friction between the two blocks of masses m_1 and m_2 is μ [**fig. 1.12**] and the floor is smooth. Find the maximum horizontal force F, applied on m_2, so that the two blocks move together.
   [**Ans.** $F = \mu(m_1 + m_2)g$.]



Smooth
Fig. 1.12.

7. Two blocks are connected by a string as shown in **fig. 1.13**. The upper block is suspended by another string. (i) Find the tension in the upper string due to the two blocks suspended in the rest position. (ii) What force F will have to be applied on the upper string to produce an acceleration of 2 m/s^2 in the upward direction in both the blocks ? (iii) In this situation what is the tension in the string between the two blocks ?
   [**Ans.** (i) 58.8 N, (ii) 70.8 N, (iii) 47.2 N]

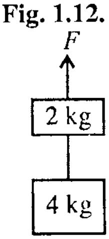


Fig. 1.13.

8. **Fig. 1.14.** shows a system of two masses m_1 and m_2 connected by a string. If a constant force $F = m_1g/2$ is applied, find the acceleration of m_1. Assume the string and pulley to be light and the surface of the table to be smooth.
   [**Ans.** $\dfrac{m_1 g}{2(m_1+m_2)}$ towards right.]

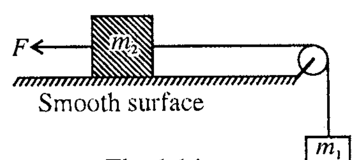


Fig. 1.14.

9. A body is thrown horizontally from a point 100 m above the ground with a speed of 20 m/s. Find (i) the time it takes to reach the ground, (ii) the horizontal distance it travels before reaching the ground, (iii) the velocity (direction and magnitude) with which it strikes the ground.
   [**Ans.** (i) 4.5 s; (ii) 90 m; (iii) 49 m/s, θ = 66° with horizontal.]
10. A gun, kept on a straight horizontal road, used to hit a car travelling along the same road away from the gun with a uniform speed of 72 km/hr. The car is at a distance of 500 m from the gun, when the gun is fired at an angle of 45° with the horizontal. Find (a) the distance of the car from the gun when

the shell hits it. (b) The speed of projection of the shell from the gun. ($g = 9.8\ m/s^2$)
[**Ans.** (a) 746.9 m; (b) 85.56 m/s.]

11. A stone is thrown from the ground towards a wall 6 m high at a distance of 4 m such that it just clears the top of the wall. Find the speed of projection of the ball.
[**Ans.** 11.43 m/s]

1·6. Inertial Frames of Reference

Those unaccelerated frames of reference, in which Newton's first and second laws hold, are called **inertial frames.** In such frames, if a body is not acted by external force, it continues in its state of rest or uniform translatory motion (*law of inertia*) and hence they are called **inertial frames.** Thus, in an inertial frame, if a body is not experiencing any external force its acceleration $\vec{a}$ is given by

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = 0 \quad (\because\ \vec{F} = m\vec{a} = 0) \qquad ...(1)$$

The law of inertia was first stated by Galileo, therefore the inertial frames are also named as *Galilean frames of reference.*

The position $\vec{r}$, velocity $\vec{v}$ and the acceleration of a particle depend on the frame chosen. Let us see how can we connect the position, velocity and acceleration of a particle measured in two different frames in constant relative motion. Consider an inertial frame S and another frame S', which is moving with constant velocity $\vec{v}_0$ relative to S. Initially (at t = 0) if the positions of the origins of the two frames coincide, then in the two frames the position vectors of any particle P at any instant t can be related by the following expressions :

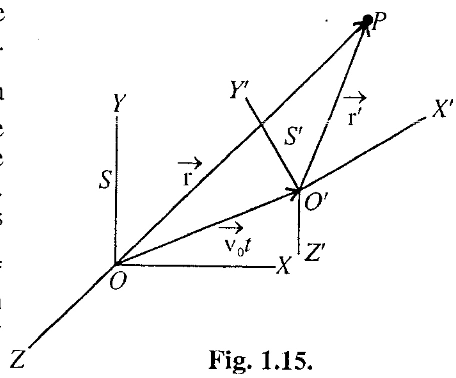


Fig. 1.15.

$$\vec{r} = \vec{OO'} + \vec{r'} \quad \text{or} \quad \vec{r} = \vec{v}_0 t + \vec{r'}$$

or

$$\vec{r'} = \vec{r} - \vec{v}t \qquad ...(2)$$

This relation connects the position vector $\vec{r'}$ of the particle in frame S' to the position vector $\vec{r}$ of the particle in S.

In Newtonian mechanics, two observers O and O' in constant relative motion will measure the same time, *i.e.*, $t = t'$, when the particle is at P. Differentiating equation (2) with respect to time,

$$\frac{d\vec{r'}}{dt} = \frac{d\vec{r}}{dt} - \vec{v}_0 \quad \text{or} \quad \frac{d\vec{r'}}{dt'} = \frac{d\vec{r}}{dt} - \vec{v}_0 \qquad (\because\ \vec{v}_0 = \text{constant})$$

$$\vec{v'} = \vec{v} - \vec{v}_0 \qquad ...(3)$$

where $\vec{v}$ is the velocity of the particle with respect to S, $\vec{v'}$ is the velocity of the particle with respect to S' and $\vec{v}_0$ the velocity of S' with respect to S. Eq. (3) connects the velocities of the particle P in S' and S frames.

Differentiating eq. (3) with respect to time, we obtain

$$\frac{d\vec{v'}}{dt} = \frac{d\vec{v}}{dt} \quad \text{or} \quad \vec{a'} = \vec{a} \qquad ...(4)$$

Thus, the acceleration of the particle as observed in frame S' is the same as observed in frame S, which is moving with constant velocity relative to S.

According to Newton's second law, the force on the particle P in the frame S is

$$\vec{F} = m\vec{a}$$

and the force on the particle P in the frame S' is

$$\vec{F}' = m\vec{a}'$$

But $\vec{a}' = \vec{a}$, hence

$$\vec{F}' = m\vec{a}' = m\vec{a} = \vec{F} \quad ...(5)$$

where m is assumed to be the same (independent of velocity) in two frames.

Thus the force on the particle, as measured in S', using Newton's second law, is the same as measured in S, using the same law. This means that if S is an inertial frame, then S' is also an inertial frame, because Newton's second law holds good in both the frames.

Further if in absence of external force ($\vec{F} = 0$) in frame S, the acceleration of the particle $\vec{a}$ is zero, then from eq. (5) in frame S' the acceleration $\vec{a}'$ is also zero, *i.e.*, no force ($\vec{F}' = m\vec{a}' = m\vec{a} = 0$) is acting on the particle in the frame S'. This means that if law of inertia (*i.e.*, first law) is valid in frame S, then it is also valid in frame S'.

Thus, we conclude that *if a frame is an inertial frame, then all those frames, which are moving with constant velocity relative to the first frame, are also inertial.*

1·7. Galilean Transformations

At any instant, the coordinates of a point or particle in space will be different in different coordinate systems. The equations which provide the relationship between the coordinates of two reference systems are called **transformation equations.**

We have shown above that a frame S' which is moving with constant velocity $\vec{v}_0$ relative to an inertial frame S, is itself inertial.

For convenience, if we assume (i) that the origins of the two frames coincide at $t = 0$, (ii) that the coordinate axes of the second frame are parallel to that of the first and (iii) that the velocity of the second frame relative to the first is $\vec{v}_0$ along X-axis, then the position vectors of a particle at any instant t in the two frames are related by the equation.

$$\vec{r}' = \vec{r} - \vec{v}\,t \quad ...(1)$$

In the component form, the coordinates are related by the equations

$$x' = x - v_0 t \ ; \ y' = y \ ; \ y' = z \quad ...(2)$$

Eq. (1) or (2) expresses the transformation of coordinates from one inertial frame to another and they are referred as **Galilean transformations.**

The form of eq. (1) or (2) depends, of course, on the relative motion of two frames of reference, but it also depends upon certain assumptions regarding the nature of *time* and *space*. It is assumed that the time is independent of any particular frame of reference, *i.e.*, if t and t' be the times recorded by the observers O and O' of an event occurring at P, then $t' = t$. If we add the equation $t' = t$, then the **Galilean transformation equations** are expressed as

$$x' = x - v_0 t \ ; \ y' = y \ ; \ z' = z \ ; \ t' = t \quad ...(3)$$

We can also consider that frame S is moving with velocity $- v_0$ along the negative X-axis with respect to S' frame. Then the transformation equations from frame s' to S are

$$x = x' + v_0 t' \ ; \ y = y' \ ; \ z = z' \ ; \ t = t' \quad ...(3')$$

These are known as **inverse Galilean transformations.**

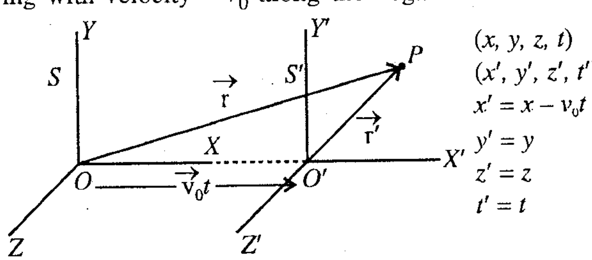


Fig. 1.16 : Representation of Galilean transformations

The other assumption, regarding the nature of the space, is that the distance between two points (or two particles) is independent of any particular frame of reference. Evidently, this assumption is expressed by the form of the transformation eq. (1) or (3). If a rod has length L in the frame S with the end coordinates x_1 and x_2 then $L = x_2 - x_1$.

If at the same time the end coordinates of the rod in S' are x_1' and x_2', then $L' = x_2' - x_1'$.

But for any time t, from eq. (3), we have

$$x_2' - x_1' = x_2 - x_1$$

Therefore,

$$L' = L \qquad \text{...(4)}$$

Thus, *the length or distance between two points is invariant under Galilean transformations.*

Differentiating eq. (1) with respect to time, we get

$$\frac{d\vec{r}}{dt} = \vec{v}_0 + \frac{d\vec{r}'}{dt} = \vec{v}_0 + \frac{d\vec{r}'}{dt'} \qquad [\because t = t']$$

or

$$\vec{v} = \vec{v}_0 + \vec{v}' \qquad \text{...(5)}$$

where $\vec{v}$ and $\vec{v}'$ are the observed velocities in S and S' frames respectively.

Eq. (5) transforms the velocity of a particle from one frame to another and is known as **Galilean law of addition of velocities.**

Again differentiating eq. (5) with respect to time t, we have

$$\frac{d\vec{v}}{dt} = 0 + \frac{d\vec{v}'}{dt'} = \frac{d\vec{v}'}{dt'} \qquad [\because t = t']$$

or

$$\vec{a} = \vec{a}' \qquad \text{...(6)}$$

Hence, the accelerations of a particle relative to S and S' frames are equal.

Hence **according to Galilean transformations,** *the accelerations of a particle relative to S and S' frames are equal.*

It is to be mentioned that the Galilean transformations are based basically on two assumptions :

(1) ***Threre exists a universal time t which is the same in all reference systems.***

(2) ***The distance between two points in various inertial systems is the same.***

Thus if any two events occur simultaneously for any observer, then they must occur simultaneously for all observers. In other words, *the time interval between two given events must be identical for all systems of reference.*

Relative motion and Law of Addition of Velocities

Now, suppose there are two bodies ; one body is particle P, having velocity $\vec{v}_1$ $(= \vec{v})$ with respect to frame S, and second body is attached at O' in frame S', having velocity $\vec{v}_2$ $(= \vec{v}_0)$ relative to frame S. Hence, if the velocities of the two bodies are known with respect to a common frame S, the velocity of one body with respect to the second body $(\vec{v}_1 - \vec{v}_2)$ $(\vec{v}' = \vec{v} - \vec{v}_0)$ can be obtained by subtracting the velocity $(\vec{v}_2)$ of second body from the velocity $(\vec{v}_1)$ of the first body. From eq. (5),

$$\vec{v}' = \vec{v} - \vec{v}_0 \quad \text{or} \quad \vec{v} = \vec{v}' + \vec{v}_0 \qquad \text{...(7)}$$

This means that *the velocity of the particle P with respect to S is equal to the velocity of the particle with respect to S' plus the velocity of S' with respect to S.* This is the Galilean law of addition of velocities. (This law is not correct when the velocities are comparable with the speed of light 3×10^8 m/s and in fact, the law is modified at relativistic speeds.)

For example, when we say that a swimmer can swim at a speed of 3 km/h, we understand 3 km/h is the velocity of swimmer with respect to water. Now, if the water is also flowing at 2 km/h speed with respect to ground, then the swimmer moves in the direction of flow at the speed of 5 km/h with respect to the ground.

Notes : (1) Necessarily an inertial frame is not accelerated because if a frame is accelerated, a force-free particle, whose velocity is constant, will be seen accelerated. In the absence of an external force on a particle, if its path is not observed in straight line, then the employed frame will not be inertial. For example, the path of a force-free particle, moving in a straight line in an inertial frame, will be seen curved in a rotating frame. Consequently, this rotating frame is not inertial.

(2) For many purposes, the earth is fairly good approximation to an inertial frame. However the earth is rotating and the frame attached to it will be non-inertial. At the equator, the change in the value of acceleration due to gravity (g) will be 3.4×10^{-2} m/s^2 and for several practical purposes, this variation may be neglected and the earth may be assumed as inertial frame.

Ex. 1. *Show that the motion of one projectile as seeen from another projectile will always be a straight line motion.* **(Agra 1994, 80; Ajmer 88; Raj. 84; Rohil. 80)**

Sol. Suppose that two projectiles are projected at the same time in space with initial velocities u_1 and u_2 and with angles of projection α_1 and α_2 respectively. Let us consider for our convenience the motion of the projectiles only in the X-Y plane. Taking the point of projection as the origin, if at any time t the position coordinates of the projectiles are (x_1, y_1) and (x_2, y_2) [**fig. 1.17**], then

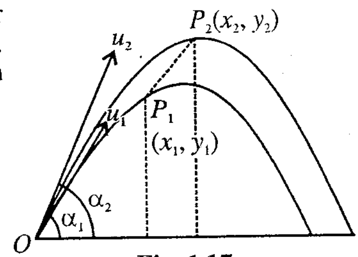


Fig. 1.17.

$$x_1 = u_1 \cos \alpha_1 t; \qquad x_2 = u_2 \cos \alpha_2 t$$

$$y_1 = u_1 \sin \alpha_1 t - \frac{1}{2} gt^2; \quad y_2 = u_2 \sin \alpha_2 t - \frac{1}{2} gt^2$$

$$\therefore \quad x_2 - x_1 = (u_2 \cos \alpha_2 - u_1 \cos \alpha_1)\, t$$

and $$y_2 - y_1 = (u_2 \sin \alpha_2 - u_1 \sin \alpha_1)\, t$$

or $$\frac{y_2 - y_1}{x_2 - x_1} = \frac{u_2 \cos\alpha_2 - u_1 \cos\alpha_1}{u_2 \sin\alpha_2 - u_1 \sin\alpha_1} = m, \text{ say.}$$

But $(y_2 - y_1) = Y$ and $(x_2 - x_1) = X$ represent the coordinates of the second projectile relative to the first. Therefore,

$$\frac{Y}{X} = m \quad \text{or} \quad Y = mX$$

Hence, *the motion of one projectile as seen from another will always be a straight line motion.*

Alternatively, both of the projectiles are acted by the same acceleration $- g$, hence the first projectile has zero acceleration relative to the second. This means that the path of the first projectile with respect to the second projectile is a **straight line.**

Ex. 2. *Water in a river moves east at 6 km/hour and a boat heads north at 8 km/hour with respect to water. Find the velocity of the boat with respect to ground.*

Sol. Velocity of the water relative to the ground $\vec{v}_0 = 6\hat{i}$ km/sec.

Velocity of the boat relative to water $\vec{v} = 8\hat{j}$ km/sec.

where $\hat{i}$ and $\hat{j}$ are unit vectors along east and north directions respectively.

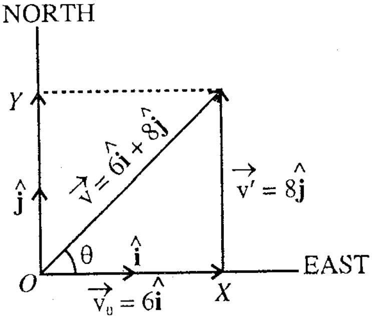


Fig. 1.18.

Velocity of the boat with respect to ground is given by

$$\vec{v} = \vec{v}' + \vec{v}_0 = 8\hat{j} + 6\hat{i}$$

Its magnitude, $$v = \sqrt{8^2 + 6^2} = \mathbf{10\ km/hour.}$$

Direction of the boat relative to the ground,

$$\tan\theta = 8/6 \text{ or } \theta = \mathbf{\tan^{-1} (4/3) \text{ from east.}}$$

Ex. 3. *A person is travelling in the east direction with a speed of 2 km per hour with respect to ground. He finds that the wind seems to blow directly from the North. When he doubles his speed, the wind appears to come from North-East. Determine the velocity of the wind.*

Sol. Suppose $\hat{i}$ and $\hat{j}$ represent unit vectors along east and the north directions respectively. Let the velocity of the wind with respect to the ground be $\vec{v}$ and expressed as :

$$\vec{v} = (x\hat{i} + y\hat{j}) \text{ km/hour.} \quad ...(i)$$

Velocity of the man relative to the ground, $\vec{v_1} = 2\hat{i}$ km/hour.

Wind velocity relative to person, $\vec{v} - \vec{v_1} = (x\hat{i} + y\hat{j}) - 2\hat{i} = (x-2)\hat{i} + y\hat{j}$ **[fig. 1.19(a)]** ...(ii)

The wind appears to come from North. This means it has no X-component. For which $x - 2 = 0$ *i.e.*, $x = 2$.

Next, when he doubles his speed relative to ground, wind velocity relative to the person is

$$\vec{v} - \vec{v_2} = (x\hat{i} + y\hat{j}) - 4\hat{i} = (x-4)\hat{i} + y\hat{j} = -2\hat{i} + y\hat{j} \quad ...(iii)$$

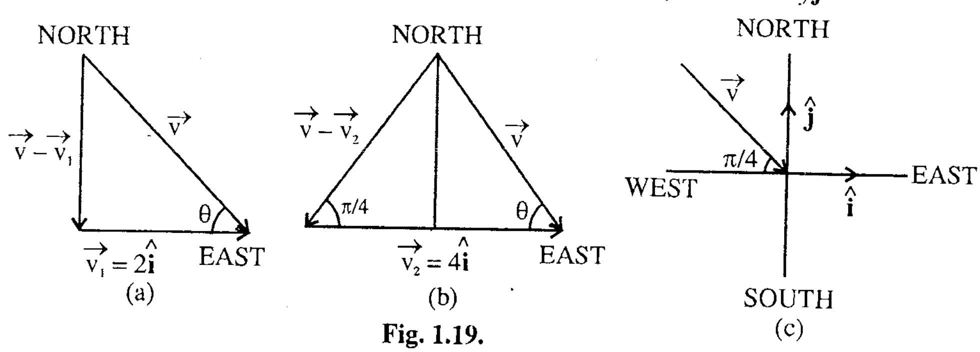


Fig. 1.19.

However the wind appears to come from North-East direction **[fig. 1.19(b)]**. But the unit vector along the North-East direction is

$$\hat{u} = \cos\frac{\pi}{4}\hat{i} + \cos\frac{\pi}{4}\hat{j} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} \quad ...(iv)$$

Therefore, the direction of the wind is along $-\hat{u} = -\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$. Hence, comparing it with unit vector $\hat{u}' = -\frac{2}{\sqrt{2^2 + y^2}}\hat{i} + \frac{y}{\sqrt{2^2 + y^2}}$ represented by eq. (iii), we obtain

$$-\frac{2}{\sqrt{2^2 + y^2}} = -\frac{1}{\sqrt{2}} \quad ...(v)$$

$$\frac{y}{\sqrt{2^2 + y^2}} = -\frac{1}{\sqrt{2}} \quad ...(vi)$$

From (v) $y = \pm 2$ and eq. (vi) is satisfied for $y = -2$. Hence the velocity of the wind relative to the ground is **[fig. 1.19(c)]**

$$\vec{v} = \mathbf{2\hat{i} - 2\hat{j}} \textbf{ km/hour} \quad ...(vii)$$

Its magnitude, $v =$ $\mathbf{2\sqrt{2}}$ **km/hour.**

Unit vector along it $= \frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} = \cos\frac{\pi}{4}\hat{i} - \cos\frac{\pi}{4}\hat{j}$.

Thus the direction of the wind from **North-West** (or along South-East).

Alternatively : Let $\vec{v}$ be the velocity of the wind with respect to ground, making an angle θ with the east direction. When the person travels with speed 2 km/h due east, he finds that the wind seems to blow directly from the North. Obviously the east component of the wind is equal to the velocity of the person, *i.e.*, $v \cos\theta = 2$ km/h **[fig. 1.19 (a)]**. When he doubles his speed, *i.e.*, 4 km/h along east, then the wind appears to come from North-East, *i.e.*, making $\pi/4$ angle from East **[fig. 1.19 (b)]**. Applying Lami's theorem,

$$\frac{v}{\sin\frac{\pi}{4}} = \frac{4}{\sin\left[\pi - \left(\frac{\pi}{4}+\theta\right)\right]} = \frac{4}{\sin\left(\frac{\pi}{4}+\theta\right)}$$

or $$v = \frac{4}{\sqrt{2}\sin\left(\frac{\pi}{4}+\theta\right)} \text{ or } \frac{4}{\sqrt{2}\sin\left(\frac{\pi}{4}+\theta\right)} = \frac{2}{\cos\theta} \quad (\because v\cos\theta = 2)$$

or $$\sqrt{2}\cos\theta = \sin\left(\frac{\pi}{4}+\theta\right) = \frac{1}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta$$

or $$2\cos\theta = \cos\theta + \sin\theta \quad \text{or} \quad \cos\theta = \sin\theta \quad \text{or} \quad \sin\theta = \sin\left(\frac{\pi}{2}-\theta\right)$$

or $$\theta = \frac{\pi}{2} - \theta \quad \text{or} \quad \theta = \frac{\pi}{4}$$

Thus, $$v = \frac{2}{\cos\theta} = \frac{2}{\cos\pi/4} = 2\sqrt{2} \text{ km/h}$$

Hence the wind is blowing with **$2\sqrt{2}$ km/h** speed from **North-West** direction. **[fig. 1.9 (c)]**.

Ex. 4. ***Coordinate transformation, when the axes of two cartesian systems are inclined to each other :*** *If the axes of a frame S′ are inclined to the axes of another frame S and the origins coincide, find the transformation equations. Show that if S is in inertial frame, then S′ is also inertial.*

Sol. Let S and S' be the two frames, whose axes are inclined with each other and origins coincide **[fig. 1.20]**. If the coordinates of a point P in the frame S are (x, y, z) and in the frome S' are (x', y', z'), then we have to find a relationship between (x, y, z) and (x', y', z').

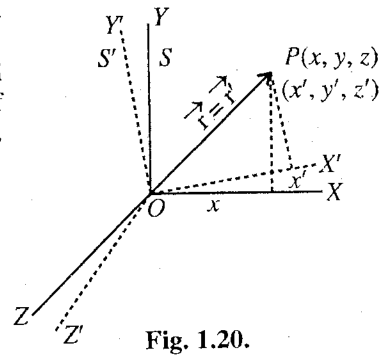


Fig. 1.20.

Now, x' = sum of the components of x, y, z along OX

i.e., $$x' = x\cos(X, X') + y\cos(Y, X') + z\cos(Z, X')$$

or $$x' = xC_{xx'} + yC_{yx'} + zC_{zx'} \quad ...(i)$$

where $C_{xx'} = \cos(X, X')$, $C_{yx'} = \cos(Y, X')$ etc.

Similary, $$y' = xC_{xy'} + yC_{yy'} + zC_{zy'} \quad ...(ii)$$

$$z' = xC_{xz'} + yC_{yz'} + zC_{zz'} \quad ...(iii)$$

These equations give the transformation equations from the system S to S'. If we want to find the transformation equations from S' to S, then

$$x = x'C_{x'x} + y'C_{y'x} + z'C_{z'x} \text{ etc.} \quad ...(iv)$$

These are the **inverse transformation equations.**

If S is the inertial frame and no force is acting on the particle at P, then

$$\frac{d^2x}{dt^2} = 0 = \frac{d^2y}{dt^2} = \frac{d^2z}{dt^2} \quad ...(v)$$

If we differentiate eqs. (i), (ii) and (iii) two times t, then

$$\left.\begin{aligned} \frac{d^2x'}{dt^2} &= C_{xx'}\frac{d^2x}{dt^2} + C_{yx'}\frac{d^2y}{dt^2} + C_{zx'}\frac{d^2z}{dt^2} = 0; \\ \frac{d^2y'}{dt^2} &= C_{xy'}\frac{d^2x}{dt^2} + C_{yy'}\frac{d^2y}{dt^2} + C_{zy'}\frac{d^2z}{dt^2} = 0; \\ \frac{d^2z'}{dt^2} &= C_{xz'}\frac{d^2x}{dt^2} + C_{yz'}\frac{d^2y}{dt^2} + C_{zz'}\frac{d^2z}{dt^2} = 0 \end{aligned}\right\} \quad ...(vi)$$

Thus, the frame S' will also be inertial.

Ex. 5. *If the position vector of a particle is* $\vec{r} = 3\hat{i} + 4\hat{j} - \hat{k}$ *with a stationary frame S, then find its transformation* $\vec{r}'$ *with respect to a frame S' which has rotated through an angle* 30° *clockwise about Z, Z' axis with origins coinciding. Show that* $|\vec{r}'| = |\vec{r}|$. **(Ajmer 1989)**

Sol. $x' = x \cos(X, X') + y \cos(Y, X') + z \cos(Z, X')$ etc.

Here, [see **fig. 1.20**] $x = 3$, $y = 4$, $z = -1$, $\cos(X, X') = \cos 30° = \frac{\sqrt{3}}{2}$,

$$\cos(Y, X') = \cos\left(\frac{\pi}{2} - 30°\right) = \sin 30° = \frac{1}{2} \text{ and } \cos(Z, X') = \cos\frac{\pi}{2} = 0$$

$$\therefore \quad x' = 3 \times \frac{\sqrt{3}}{2} + 4 \times \frac{1}{2} + 0 = \frac{3\sqrt{3}}{2} + 2$$

$$y' = x\cos(X, Y') + y\cos(Y, Y') = \cos\left(\frac{\pi}{2} + 30°\right) + 4\cos 30°$$

$$= -3\sin 30° + 4 \times \frac{\sqrt{3}}{2} = -\frac{3}{2} + 2\sqrt{3}$$

$$z' = z = -1$$

Thus the transformation vector $\vec{r}'$ is

$$\vec{r}' = x'\hat{i}' + y'\hat{j}' + z'\hat{k}' = \left(\frac{3\sqrt{3}}{2} + 2\right)\hat{i}' + \left(-\frac{3}{2} + 2\sqrt{3}\right)\hat{j}' - \hat{k}'$$

where $\hat{i}'$ and $\hat{j}'$ are unit vectors along X' and Y' axes of frame S'.

Now, $|\vec{r}| = \sqrt{3^2 + 4^2 + 1^2} = \sqrt{26}$

and $|\vec{r}'| = \sqrt{\left(\frac{3\sqrt{3}}{2} + 2\right)^2 + \left(-\frac{3}{2} + 2\sqrt{3}\right)^2 + 1^2} = \sqrt{\frac{27}{4} + 4 + 6\sqrt{3} + \frac{9}{4} + 12 - 6\sqrt{3} + 1} = \sqrt{26}$

Hence $|\vec{r}'| = |\vec{r}|$.

Proved.

Ex. 6. Coordinate Transformation under rotation of cartesian coordinates : *Find the transformation equations, connecting two frames S and S' such that their origins and Z-axes are coincident and the frame S' rotates about the common Z-axis with uniform angular velocity.*

Show that if S is an inertial frame, then S' is a non-inertial frame. **(Ajmer 1988)**

Sol. Consider two cartesian frames S and S', whose origins and Z-axes are coinciding Z-axis with a constant angular velocity ω. If we assume that initially at $t = 0$, the axes of the frame S' coincides with the axes of the frame S, then the axes (X' and Y') of the frame S' will rotate through an angle ωt in time t. We have to find the relationship between the coordinates of P, (x, y, z) and (x', y', z') in two frames [**fig 1.21**].

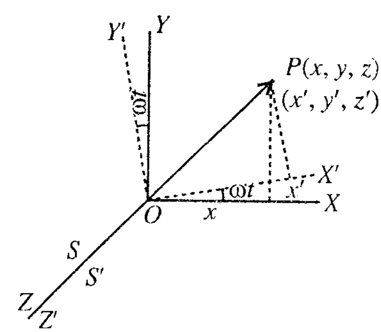


Fig. 1.21 : Inertial and rotating frames

Now, x' = sum of components of x, y, z along OX'

$$= x\cos(X, X') + y\cos(Y, X') + z\cos(Z, X')$$

$$= x\cos\omega t + y\cos\left(\frac{\pi}{2} - \omega t\right) + z\cos\left(\frac{\pi}{2}\right)$$

[Here, the axes X, X'; Y, Y' are in the plane of the paper and Z (or Z') axis is perpendicular to it.]

$$= x\cos\omega t + y\sin\omega t$$

y' = sum of components of x, y, z along OY'

$$= x\cos\left(\frac{\pi}{2} + \omega t\right) + y\cos\omega t + z\cos\frac{\pi}{2}$$

$= -x \sin \omega t + y \cos \omega t$

$z' = z$

Thus, the transformation equations are :

$$x' = x \sin \omega t + y \cos \omega t;$$
$$y' = -x \sin \omega t + y \cos \omega t;$$
$$z' = z \qquad \text{...(i)}$$

The inverse transformation equations are :

$$x = x' \cos \omega t - y' \sin \omega t;$$
$$y = x' \sin \omega t + y' \cos \omega t; \; z = z' \qquad \text{...(ii)}$$

Equations (i) or (ii) connect two frames which have a uniform relative rotatory motion.

Now, if S is an inertial frame and no force is acting on the particle at any point P, then

$$\frac{d^2x}{dt^2} = 0 = \frac{d^2y}{dt^2} = \frac{d^2z}{dt^2} \qquad \text{...(iii)}$$

Differentiating eq. (i) with respect to t, we get

$$\frac{dx'}{dt} = -\omega x \sin \omega t + \omega y \cos \omega t + \frac{dx}{dt}\cos \omega t + \frac{dy}{dt}\sin \omega t = \omega y' + \frac{dx}{dt}\cos \omega t + \frac{dy}{dt}\sin \omega t \text{ ...(iva)}$$

$$\frac{dy'}{dt} = -\omega x \cos \omega t - \omega y \sin \omega t - \frac{dx}{dt}\sin \omega t + \frac{dy}{dt}\cos \omega t = -\omega x' - \frac{dx}{dt}\sin \omega t + \frac{dy}{dt}\cos \omega t \text{ ...(ivb)}$$

and $$\frac{dz'}{dt} = \frac{dz}{dt} \qquad \text{...(ivc)}$$

Again differentiating eq. (iv) relative to t, we have (using $\frac{d^2x}{dt^2} = \frac{d^2y}{dt^2} = 0$ for a force-free particle in S)

$$\frac{d^2x'}{dt^2} = \omega\frac{dy'}{dt} - \omega\frac{dx}{dt}\sin \omega t + \omega\frac{dy}{dt}\cos \omega t$$

$$= \omega\frac{dy'}{dt} + \omega\left(\frac{dy'}{dt} + \omega x'\right) \text{ [from eq. (ivb)]} = 2\omega\frac{dy'}{dt} + \omega^2 x' \qquad \text{...(va)}$$

$$\frac{d^2y'}{dt^2} = -\omega\frac{dx'}{dt} - \omega\frac{dx}{dt}\cos \omega t - \omega\frac{dy}{dt}\sin \omega t$$

$$= -\omega\frac{dx'}{dt} - \omega\left(\frac{dx'}{dt} - \omega y'\right) \text{ [from eq. (ivb)]} = -2\omega\frac{dx'}{dt} + \omega^2 y' \qquad \text{...(vb)}$$

and $$\frac{d^2z'}{dt^2} = \frac{d^2z}{dt^2} = 0 \qquad \text{...(vc)}$$

Eq. (va) and (vb) show that a force-free particle in the inertial frame S has an acceleration in the frame S'. Thus the frame S' is **non-inertial.**

EXERCISE 1.2

1. The position of a particle P in a coordinate system S is measured as $\vec{r} = (6t^2 - 4t)\hat{\mathbf{i}} + (-3t^3)\hat{\mathbf{j}} + 3\hat{\mathbf{k}}m.$
   (a) Determine the constant relative velocity of system S' with respect to S if the position of P is measured as $\vec{r}' = (6t^2 - 3t)\hat{\mathbf{i}} + (-3t^3)\hat{\mathbf{j}} + 3\hat{\mathbf{k}}\, m.$

   [**Ans.** $-7\hat{\mathbf{i}}$ m/sec.]

   (b) Show that the acceleration of the particle is the same in both systems.
2. A swimmer can swim in still water at the speed of 2 km/hr. If he swims in a river flowing at 1.5 km/h and

keeps his direction perpendicular to the flow direction, find his velocity relative to the ground.
[**Ans.** 2.5 km/hr at an angle $\theta = 53°$ with the direction of flow.]

3. A cyclist is running at a speed of 12 km/hr along East direction. Raindrops fall vertically with a speed of 16 km/hr. Determine the velocity of the raindrops with respect to the man.
[**Ans.** 20 km/hr at an angle $\theta = 37°$ from East of the vertical.]

4. The pilot of an aeroplane wishes to reach a point 200 miles East of his present position. A wind blows 30 miles per hour from the North-West. Calculate his vector velocity with respect to the moving air mass if his schedule requires him to arrive at his destination in 40 minutes (take $\sqrt{2} = 1.4$). **(Agra 1972)**
[**Ans.** $300\,\hat{\mathrm{i}} - \frac{30}{\sqrt{2}}(\hat{\mathrm{i}} - \hat{\mathrm{j}})$ m/h, where $\hat{\mathrm{i}}$ and $\hat{\mathrm{j}}$ are the unit vectors along East and North directions respectively.]

5. A pilot of an aeroplane wishes to reach a place 100 km East to its present position. The velocity of the wind is 15 km/hour from the North-East. Find his velocity relative to wind, if he reaches in 20 minutes at the required place.
$\left[\textbf{Ans. } \left(300 + \frac{15}{\sqrt{2}}\right)\hat{\mathrm{i}} + \frac{15}{\sqrt{2}}\hat{\mathrm{j}}.\right]$

1·8. Non-inertial Frames and Fictitious Forces

Those frames of reference in which Newton's law of inertia does not hold, are called **non-inertial frames**. All the accelerated and rotating frames are the non-inertial frames of reference. According to Newton's second law, the applied force $\vec{F}_i$ on a body of mass m is given by

$$\vec{F}_i = m\vec{a}_i \qquad ...(1)$$

where $\vec{a}_i$ is the acceleration, observed in an inertial frame. This equation, as it stands, is not valid for accelerated frames because due to its own acceleration, the accelerated observer will observe an acceleration $\vec{a}_n$ different to $\vec{a}_i$ and hence

$$\vec{F}_i \neq m\vec{a}_n \qquad ...(2)$$

If no external force is acting on a particle, even then in the accelerated frame it will appear that a force is acting on it. Suppose that S is an *inertial frame* and another frame S' is moving with an acceleration $\vec{a}_0$ relative to S. The acceleration of a particle P, on which no external force is acting, will be zero in the frame S; but in frame S' the observer will find that an acceleration $-\vec{a}_0$ is acting on it. Thus, in frame S', the observed force on the particle is $-m\vec{a}_0$, where m is the mass of the particle.

Such a force, which does not really act on the particle but appears due to the acceleration of the frame, is called a **fictitious** *or* **pseudo force.** Hence the fictitious force on the particle P is

$$\vec{F}_0 = -m\vec{a}_0 \qquad ...(3)$$

Here, accelerated frame S' is non-inertial, because in this frame Newton's law of inertia is not valid, *i.e.*, in absence of any external force, the observer of S' frame is observing a force $(-m\vec{a}_0)$ on the particle.

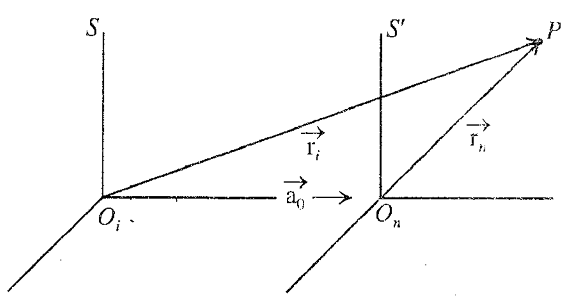


Fig. 1.22.

Suppose frame S' coincides at $t = 0$ with the inertial frame S. Then at any time t, the position vectors of a particle $\vec{r}_i$ and $\vec{r}_n$ in the inertial and non-inertial frames respectively are connected as **[fig. 1.22]**

$$\vec{r}_i = \vec{r}_n + \frac{1}{2}\vec{a}_0 t^2 \qquad ...(4)$$

where $\vec{a}_0$ is the acceleration of the frame S' with respect to S.

Double differentiation with respect to time gives

$$\frac{d^2 \vec{r}_i}{dt^2} = \frac{d^2 \vec{r}_n}{dt^2} + \vec{a}_0$$

or
$$\vec{a}_i = \vec{a}_n + \vec{a}_0 \qquad ...(5)$$

or
$$\vec{a}_i - \vec{a}_0 = \vec{a}_n$$

Thus
$$m\vec{a}_i - m\vec{a}_0 = m\vec{a}_n$$

or
$$\vec{F}_0 + \vec{F}_i = \vec{F}_n \qquad \text{[from eq. (1) and (3)]} \qquad ...(6)$$

This formula gives the observed force $\vec{F}_n$ in the accelerated system.

For example, suppose that a box is falling in the gravitational field of the earth with an acceleration g. Now, if we consider about a particle, falling freely inside the box, the fictitious force on the particle is mg upward, because the acceleration of the frame (box) is g downward or $-g$ (we have taken +ve in the upward direction). As the real force on the particle due to the attraction of the earth $-mg$, the force observed by the observer inside the box is

$$F_n = F_i + F_0 = -mg + mg = 0$$

where we have assumed earth as an inertial frame (stationary).

If the particle has no initial velocity relative to the box, it will seem to remain suspended in mid-air at the same place inside the falling box.

Now, suppose that the box is moved with an acceleration $a_0 = g$ in the upward direction relative to the ground. In such a case, the real force (F_i) and fictitious force (F_0) on the particle are given by

$$F_i = -mg, \quad F_0 = -ma_0 = -mg$$

Hence the total force in the accelerated frame (box) is

$$F_n = F_i + F_0 = -mg - mg = -2mg$$

This means that the observer, stationed in the box having an acceleration g upward, will measure a force $2mg$ downward on the particle.

Weight of a man in a moving lift : We also consider the example of a lift which is moving relative to the ground :

(a) with constant velocity upward or downward,

(b) with an acceleration $\vec{a}_0 = (= a_0\hat{n})$ upward,

(c) with an acceleration $-\vec{a}_0 (= -a_0\hat{n})$ downward.

where $\hat{n}$ is the unit vector in the upward direction.

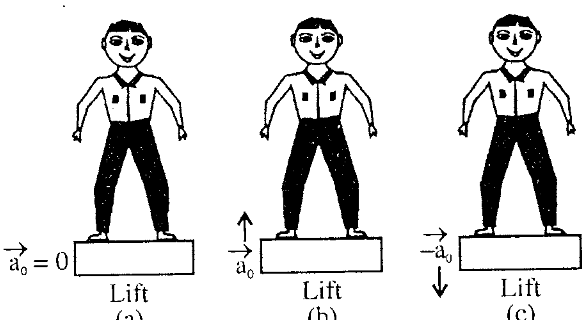


Fig. 1.23 : Lift moving—(a) up or down with uniform velocity ($\vec{a}_0 = 0$), (b) with an acceleration $\vec{a}_0$ (= $a_0\hat{n}$) upward, (c) with an acceleration $\vec{a}_0$ (= $-a_0\hat{n}$) downward.

Let us determine the weight of a

man of mass m, standing on the lift. We assume here the earth to be stationary and a frame fixed with it to be inertial one. In all the cases, the force in the inertial frame (real force) is

$$\vec{F}_i = -mg\hat{n} \qquad ...(7)$$

The observer in case (a) is moving up or down with constant velocity so that the acceleration of the frame (attached with the lift) relative to the ground is zero, *i.e.*, $\vec{a}_0 = 0$. Thus the frame of the lift is inertial and in both frames the force on the man is same, *i.e.*,

$$\vec{F}(\text{Lift}) = \vec{F}_i = -mg\hat{n}$$

Thus in the first case, the weight of the man is mg acting downward.

In case (b), $\vec{F}_0 = -m\vec{a}_0 = -ma_0\hat{n}$ and the observed force on the mass in the frame of the lift (accelerated upward) is

$$\vec{F}_n = \vec{F}_i + \vec{F}_0 = -mg\hat{n} - ma_0\hat{n} = -m(g + a_0)\hat{n} \qquad ...(8)$$

In case (c), $\vec{F}_0 = -ma_0 = -ma_0\hat{n}$ and the force in the accelerated (lift) frame downward is

$$\vec{F}_n = \vec{F}_i + \vec{F}_0 = -mg\hat{n} + ma_0\hat{n} = -m(g - a_0)\hat{n} \qquad ...(9)$$

Thus *when the lift is accelerated upward, the weight of the man is $m(g + a_0)$ and hence he becomes heavier, while for the lift accelerating downward, the man loses his weight $[m(g - a_0)]$. If the lift falls freely ($a_0 = g$), then the weight of the man becomes zero, i.e., he feels weightlessness.*

1·9. Centripetal Acceleration and Centrifugal Force *(Circular Motion)*

Let us consider a particle P of mass m, moving on the circumference of a circle of constant radius r in X-Y plane in an inertial frame S (XYZ). The position vector $\vec{r}$, which is the function of the scalar variable t, is rotating with a constant angular velocity ω. At any time t, the position vector $\vec{r}$ can be expressed in the components form as

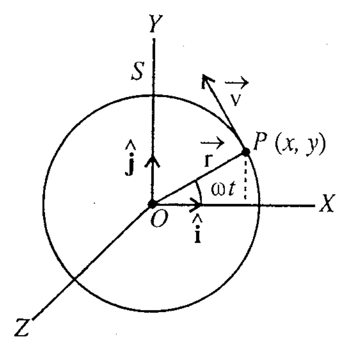


Fig. 1.24.

$$\vec{r} = x\hat{i} + y\hat{j}$$

or
$$\vec{r} = r\cos\omega t\,\hat{i} + r\sin\omega t\,\hat{j}$$

or
$$\vec{r} = r(\cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}) \qquad ...(1)$$

Velocity of the particle P is obtained as

$$\vec{v} = \frac{d\vec{r}}{dt} = r(-\omega\cos\omega t\,\hat{i} + \omega\sin\omega t\,\hat{j})$$

[$\because$ $dr/dt = 0$, because radius r is constant for a circle.]

or
$$\vec{v} = \omega r(-\sin\omega t\,\hat{i} + \cos\omega t\,\hat{j}) \qquad ...(2)$$

Its magnitude, $\quad v = \sqrt{\omega^2 r^2(\sin^2\omega t + \cos^2\omega t)} = \omega r \qquad ...(3)$

Differentiating eq. (2), we have the acceleration of the particle in circular motion, *i.e.*,

$$\vec{a} = \frac{d\vec{v}}{dt} = \omega^2 r(-\cos\omega t\,\hat{i} - \sin\omega t\,\hat{j}) = -\omega^2 r(\cos\omega t\,\hat{i} + \sin\omega t\,\hat{j})$$

But $\quad \vec{r} = r(\cos\omega t\,\hat{i} + \sin\omega t\,\hat{j})$.

Therefore, $\quad \vec{a} = -\omega^2\vec{r} \qquad ...(4)$

The direction of this acceleration is directed towards the centre of the circle and hence this acceleration is called **centripetal** (*centre seeking*) **acceleration.** If m is the mass of the particle, then according to Newton's second law, the force acting on the particle in the inertial frame (XYZ) is

$$\vec{F}_i = -m\omega^2\vec{r} \qquad ...(5)$$

This is called **centripetal** (*centre seeking*) **force**, which is provided by some actual force, needed to rotate the particle on a circular path. Its magnitude is

$$F_i = m\omega^2 r = mv^2/r \qquad (\because\ v = r\omega) \qquad ...(6)$$

For example, if a mass tied to a string, is rotating in a circle, then the centripetal force, necessary for rotation on the circular path, is provided by the tension of the string, acting on the mass towards the centre of the circle.

Now, let us consider a frame S', which is rotating with an angular velocity ω with respect to the frame S so that the particle is at rest in the rotating frame [**fig. 1.25**]. Therefore in the rotating (non inertial) frame S', the acceleration or force on the particle is zero, *i.e.*, $\vec{a}_n = 0$ or $\vec{F}_n = 0$.

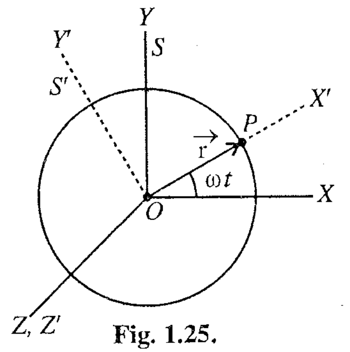


Fig. 1.25.

Using the relation $\vec{F}_i + \vec{F}_0 = \vec{F}_n$ and putting $\vec{F}_i = -m\omega^2\vec{r}$ and $\vec{F}_n = 0$, we obtain

$$-m\omega^2\vec{r} + \vec{F}_0 = 0 \quad \text{or} \quad \vec{F}_0 = m\omega^2\vec{r} \qquad ...(7)$$

This fictitious froce $\vec{F}_0$ is called **centrifugal force** and is obviously *directed away from the centre* along the radius vector $(\vec{r})$. In the rotating (non inertial) frame, in order to produce zero acceleration of the particle, the centrifugal force is balanced by some actual force (*e.g.*, inward tension in a rotating string with a mass).

1·10. Satellite in Circular Orbit

A body, which revolves constantly round a comparatively much larger body, is said to be a **satellite**. We know that the earth rotates round the sun and moon round the earth in specified orbits. The earth is the satellite of the sun and the moon that of the earth. These are called **natural satellites**. Each one of these satellites is attracted by its primary by the gravitational force. Scientists have also been able to place man-made satellites. They are called **artificial satellites** and are usually sent in space with the help of high velocity rockets.

In the circular orbit of a satellite around the earth, the gravitational attraction is balanced by centrifugal force, [**fig. 1.26**] *i.e.*,

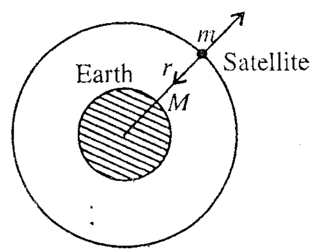


Fig. 1.26.

$$GMm/r^2 = m\omega^2 r \qquad ...(1)$$

where M = mass of the earth, m = mass of the satellite, r = distance between the centres of earth and satellite and ω = angular speed of satellite.

From (1) $\qquad \omega^2 = GM/r^3 \qquad ...(2)$

If T is the *time of revolution, then*

$$\left(\frac{2\pi}{T}\right)^2 = \frac{GM}{r^3} \quad \text{or} \quad T^2 = \frac{4\pi^2}{GM}r^3$$

But $g = \dfrac{GM}{R^2}$, then

$$T^2 = \frac{4\pi^2}{gR^2}\cdot r^3 \quad \text{or} \quad T = \frac{2\pi r}{R}\sqrt{\frac{r}{g}} \qquad ...(3)$$

If the satellite revolves very close to earth, then $r = R$

$$T = 2\pi\sqrt{\frac{R}{g}} \qquad ...(4)$$

Taking $R = 6.4 \times 10^6$ m and $g = 9.8$ m/sec^2, the period of revolution of the satellite near the surface of the earth is obtained to be

$$T = 2 \times 3.14\sqrt{6.4\times 10^6/9.8}\ \text{sec} = 84\ \text{minutes (approx.)}$$

Velocity of satellite, $v = r\omega = r\sqrt{GM/r^3} = \sqrt{GM/r} = \sqrt{gR^2/r}$

or $$v = R\sqrt{\frac{g}{r}} \quad ...(5)$$

When $r = R$, $v = \sqrt{gR}$. ...(6)

Taking $R = 6.4 \times 10^6$ m and $g = 9.8$ m/sec^2,

$$v = \sqrt{9.8 \times 6.4 \times 10^6} \text{ m/sec} = 7.92 \text{ or } 8 \text{ km/sec approx.} \quad ...(7)$$

Hence *if we want to have an artificial satellite close to the surface of the earth, we have to give the orbital velocity about* 8 *km/sec.*

Ex. 1. *Calculate the fictitious force and the total force acting on a freely falling body of mass* 20 *kg with reference to a frame moving with a downward acceleration of* 6 *m/sec*^{2}. **(Ajmer 1989)**

Sol. Assuming earth as inertial frame and positive direction upward

$$F_i = -mg = -20 \times 9.8 = -196$$

Here $a_0 = -6$ m/sec^2 and $g = -9.8$ m/sec^2.

Fictitious force, $F_0 = -ma_0 = -20 \times -6 = 120$ n

i.e., fictitious force of amount **120 newtons** is acting in the upward direction.

Total force in the accelerated frame

$$F_n = ma_n = F_i + F_0 = -196 + 120 = -76 \text{ n}.$$

i.e., total force of amount **76 newtons** is acting downward in the accelerated frame.

Ex. 2. *Calculate the fictitious and total force on a body of mass* 2.5 *kg relative to a frame moving vertically upwards on earth with an acceleration of* 10 *m/sec*^{2}. **(Mysore 1981)**

Sol. If we take positive direction, then the true force on the body is

$$F_i = mg = -2.5 \times 9.8 \text{ n}$$

Here $a_0 = 10$ m/sec^2.

$\therefore$ Fictitious force, $F_0 = -ma_0 = -2.5 \times 10 = -25$ n = **25 newtons downward.**

Total force, $F_N = F_i + F_0 = -2.5\ (9.8 + 10) = -2.5 \times 19.8 =$ **49.50 newtons downward.**

Ex. 3. *Calculate the effective weight of an astronaut ordinarily weighing* 60 *kg when his rocket moves vertically* (i) *upward and* (ii) *downward with* 6 *g acceleration.* **(Kanpur 1990)**

Sol. True force, $F_i = -mg$

Force in accelerated frame or effective weight

$$F_n = F_i + F_0 = -mg - ma_0 = -m\ (g + a_0)$$

In case (i) $m = 60$ kg and $a_0 = 6\ g$

$\therefore$ effective weight $= -60\ (g + 6g) = -420\ g =$ **420 kg-weight downward.**

In case (ii) $a_0 = -6g$ and hence

effective weight $= -60\ (g - 6g) = 300\ g =$ **300 kg-weight upward.**

Ex. 4. *With what acceleration 'a' should the box of* **fig. 1.27** *move downward or upward so that the mass m exerts a force mg/*3 *on the floor of the box ?*

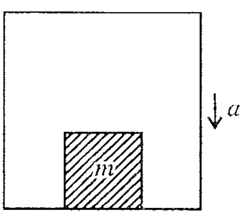


Fig. 1.27.

Sol. $F_n = F_i + F_0$

Let positive direction be upward.

Here, $F_n = -mg/3$, $F_i = -mg$, $F_0 = ma$

$\therefore$ $-mg/3 = -mg + ma$ or $a = -2g/3$

The box should move with an acceleration **2*g*/3 downward.**

Ex. 5. *A* 1 *kg stone at the end of a* 2 *metres long string makes* 5 *revolutions per second. Calculate the force on the stone as measured in an inertial frame and in a frame which is rotating with the string.*

Sol. In an inertial frame, the necessary force to rotate the stone in the circular path is centripetal force $-m\omega^2 r$, *i.e.*,

Force on the stone in inertial frame $= - m\,\omega^2 r = - 1\ (2 \times 3.14 \times 5)^2 \times 2 = - 1974$ newtons
$=$ **1974 newtons towards the centre (centripetal)**

This force is supplied by the tension in the string.

In case of the non-inertial frame, rotating with the stone, the acceleration of the stone is **zero**. Thus, the total force in this frame is **zero**. In order to have equilibrium of the stone in the rotating frame, the inward force of tension on the stone is balanced by the centrifugal force (fictitious force), amounting $m\omega^2 r =$ **1974 newtons.**

Ex. 6. *An artificial satellite is revolving round the earth at a distance of 620 km. Calculate the minimum velocity and the period of revolution. Radius of earth is 6380 km and acceleration due to earth's gravity at the surface of the earth is 9.8 metres/sec^2.*

Sol. Radius of earth's satellite orbit, r = Radius of earth + Distance of satellite from earth's surface.

$= 6380 + 620 = 7000$ km $= 7 \times 10^6$ m/sec.

Radius of earth, $R = 6380 \times 10^3$ m; $g = 9.8$ m/sec^2.

$$\therefore \quad \text{Period of revolution, } T = \frac{2\pi r}{R}\sqrt{\frac{r}{g}} = \frac{2\pi \times 7 \times 10^6}{6380 \times 10^3}\sqrt{\frac{7 \times 10^6}{9.8}} = \mathbf{5775\ sec.}$$

$$\text{and} \quad \text{orbital velocity, } v = R\sqrt{\frac{g}{r}} = 6380 \times 10^3 \sqrt{\frac{9.8}{7 \times 10^6}} = \mathbf{7.55 \times 10^6\ m/sec.}$$

Ex. 7. *Calculate the height of an equitorial satellite which is always seen to be over the same point of earth's surface. ($G = 6.66 \times 10^{-11}$ S.I. Units; $M = 5.98 \times 10^{24}$ kg.)*

Sol. Let the height of the equatorial satellilte be h. So that the equatorial satellite seems over the same point of earth's surface, *i.e.*, the angular velocity of satellite is the same as that of the earth itself.

$$\text{Here,} \quad \text{angular velocity of satellite} = \frac{2\pi}{24 \times 60 \times 60} = 7.3 \times 10^{-5}\ \text{sec}^{-1}.$$

$$\text{Also,} \quad \frac{GMm}{r^2} = m\omega^2 r \quad \text{or} \quad r^3 = \frac{GM}{\omega^2} = \frac{6.66 \times 10^{-11} \times 5.98 \times 10^{24}}{(7.3 \times 10^{-5})^2} = 75 \times 10^{27} \text{ or } r = 4.2 \times 10^7 \text{ m}$$

$$\therefore \quad h = r - R = 4.2 \times 10^7 - 6.4 \times 10^6 = \mathbf{3.56 \times 10^7\ m.}$$

EXERCISE 1.3

1. What is the fictitious force and total force acting on a freely falling body of 3 kg mass with reference to a frame moving with a downward acceleration of 4 m/sec^2 ?
   [**Ans.** 12 newtons upward; 17.4 newtons downward.]
2. Calculate fictitious and total force acting on a body of mass 5 kg relative to a frame moving vertically upwards on earth with an acceleration of 5 m/sec^2.
   [**Ans.** 25 newtons downward; 74 newtons downward.]
3. A rocket is moving upward with acceleration of $3g$. Calculate the effective weight of a man sitting in it, if his actual weight is 75 kg. **(Agra 1995, 92)**
   [**Ans.** 300 kg-wt.]
4. A 2 kg stone at the end of 0.5 metre long string makes 4 revolutions in 3 seconds. Calculate the forces on the stone as measured in an inertial frame and in a frame which is rotating with the string.
   [**Ans.** $F_i = - 70.1\ n$, Total force $F_n = 0$, $F_0 = 70.1\ n$.]
5. A body is attached to the lower end of a spring suspended from the roof of a lift. When the lift is at rest, the body produces an elongation of 2 cm in the spring. When the lift moves down with an acceleration, the elongation is changed to 1.95 cm. Find the acceleration of the lift. **(Mysore 1981)**
   [**Ans.** 0.245 ms^{-2}.]
6. A massless string pulls a mass of 50 kg upwards against gravity. The string would break if subjected to a tension greater than 600 newtons. What is the maximum acceleration with which mass can be moved upwards ? **(Lucknow 1996, 80)**
   [**Ans.** 2.2 ms^{-2}.]

7. An elevator is coming down with uniform acceleration. To measure the acceleration, a person inside the elevator drops a coin at the instant the elevator starts to move. The coin is 2 m above the floor of the elevator at the time it is dropped. The coin strikes the floor in 1 second. Determine the acceleration of the elevator.
[**Ans.** 5.8 m/sec^2.]
8. The first artificial satellite was circling round the earth at a distance of 900 km. Taking the radius of the earth equal to 6371 km and mass 5.983×10^{24} kg and G equal to 6.61×10^{-11} S.I. units, find the velocity of satellite and its period of revolution.
[**Ans.** Velocity = 7.40 km/sec.; Time period = 1 hr. 42 min. 52 sec.]
9. A satellite moves in a circular orbit round the earth at height $R_e/2$ from earth's surface where R_e is the radius of the earth. Calculate its period of revolution. ($R_e = 6.38 \times 10^6$ m.) **(Agra 1982)**
[**Ans.** 9.3×10^3 sec.]
10. A small satellite revolves round a planet in an orbit just above the planet's surface. Taking mean density of the planet as 8000 kg/m^3 and G as 6.66×10^{-11} S.I. units, find the time period of satellite.
[**Ans.** 4206 sec.]
11. An astronaut, in an artificial satellite orbitting the earth, is in a state of weightlessness. Explain why ? If the satellite is orbitting the earth at an altitude of 100 km, find its period of revolution. Assume the orbit to be circular. (Mass of the earth = 5.98×10^{24} kg; mean radius of the earth = 6371 km; G = 6.668×10^{-24} S.I. units). **(I.A.S. 1977)**
[**Ans.** 5,177 sec.]
12. An object fixed with respect to the surface of a planet identical in mass and radius of the earth experiences zero gravitational attraction at the equator. What is the length of a day on the planet ? (g = 9.8 m/sec^2, $R = 6.4 \times 10^6$ m.)
[**Ans.** 84.5 minutes.]

1·11. Rotating Frame of Reference and Coriolis Force

The earth itself rotates about its axis in 24 hours. Therefore, the frame fixed on the earth will also rotate with it and so it will be a **non-inertial frame.** Suppose that a frame S' (X_r, Y_r, Z_r) is rotating with an angular velocity $\vec{\omega}$ relative to an inertial frame S (X_i, Y_i, Z_i). For simplicity, we assume that both of the frames have common origin O and common Z-axis **[fig. 1.28]**. In case of earth, the common origin O may be thought as the centre of the earth, Z-axis as coinciding with its rotational axis and the frame S' as rotating with earth relative to the non-rotating frame S.*

The position vector of a particle P in both frames will be the same**, *i.e.*, $\vec{R}_i = \vec{R}_r = \vec{R}$, because the origins are coincident. Now, if the particle P is stationary in the frame S, the observer in the rotating frame S' will see that the particle is moving oppositely with linear velocity $-\vec{\omega} \times \vec{R}$. Thus, if the velocity of the particle in the frame S is $(d\vec{R}/dt)_i$, then its velocity $(d\vec{R}/dt)_r$ in the rotating frame will be given by

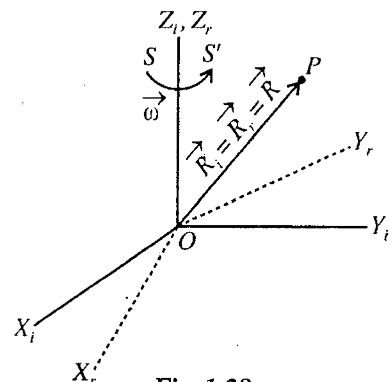


Fig. 1.28.

$$\left(\frac{d\vec{R}}{dt}\right)_r = \left(\frac{d\vec{R}}{dt}\right)_i - \vec{\omega} \times \vec{R} \quad \text{or} \quad \left(\frac{d\vec{R}}{dt}\right)_i = \left(\frac{d\vec{R}}{dt}\right)_r + \vec{\omega} \times \vec{R} \qquad ...(1)$$

Actually this equation holds for all vectors and relates the time derivatives of a vector in the frames S and S'. Therefore, relation (1) may be written in the form of operator equation :

* Truly speaking, this frame is rotating about sun. But the assumption of S to be an inertial frame will not add any perceptible error in our measurements on earth.

** The components may have different values, but the direction and magnitude of the position vector will be the same in two frames.

$$\left(\frac{d}{dt}\right)_i = \left(\frac{d}{dt}\right)_r + \vec{\omega} \qquad ...(2)$$

Writing $d\vec{R}/dt = \vec{v}$ for the velocity of the particle, we have

$$\vec{v}_i = \vec{v}_r + \vec{\omega} \times \vec{R} \qquad ...(3)$$

This relation connects the **velocity** in inertial frame and rotating frame.

Now, if we operate eq. (2) on velocity vector $\vec{v}_i$, we have

$$\left(\frac{d\vec{v}_i}{dt}\right)_i = \left(\frac{d\vec{v}_i}{dt}\right)_r + \vec{\omega} \times \vec{v}_i$$

Substituting the value of $\vec{v}_i$ in the right hand side of this relation, from eq. (3), we get

$$\left(\frac{d\vec{v}_i}{dt}\right)_i = \left[\frac{d}{dt}(\vec{v}_r + \vec{\omega} \times \vec{R})\right]_r + \vec{\omega} \times (\vec{v}_r + \vec{\omega} \times \vec{R})$$

$$= \left(\frac{d\vec{v}_r}{dt}\right)_r + \frac{d\vec{\omega}}{dt} \times \vec{R} + \vec{\omega} \times \left(\frac{d\vec{R}}{dt}\right)_r + \vec{\omega} \times \vec{v}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R})$$

Writing the acceleration $d\vec{v}/dt = \vec{a}$, we have

$$\vec{a}_i = \vec{a}_r + 2\vec{\omega} \times \vec{v}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R}) + \frac{d\vec{\omega}}{dt} \times \vec{R} \qquad ...(4)$$

This equation tells us the relation between the accelerations in the inertial and relation frames.

If the angular velocity $d\vec{\omega}/dt = 0$. In such a case, $\vec{\omega}$ is constant as in the case of earth, then

$$\vec{a}_i = \vec{a}_r + 2\vec{\omega} \times \vec{v}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R}) \qquad ...(5)$$

If m is the mass of the particle, then force ($\vec{F}_r$) in the rotating frame is

$$m\vec{a}_r = m\vec{a}_i - 2m\,\vec{\omega} \times \vec{v}_r - m\,\vec{\omega} \times (\vec{\omega} \times \vec{R}) \qquad ...(6)$$

or

$$\vec{F}_r = m\vec{a}_i - 2m\,\vec{\omega} \times \vec{v}_r - m\,\vec{\omega} \times (\vec{\omega} \times \vec{R}) \qquad ...(7)$$

But the force in the rotating frame,

$$\vec{F}_r = \vec{F}_i + \vec{F}_0 \qquad ...(8)$$

Comparing eqs. (7) and (8), we get

$$\therefore \quad \text{Fictitious force } \vec{F}_0 = -2m\,\vec{\omega} \times \vec{v}_r - m\,\vec{\omega} \times (\vec{\omega} \times \vec{R}) \qquad ...(9)$$

where $-2m\vec{\omega} \times \vec{v}_r$ is called **Coriolis force*** and $-m\,\vec{\omega} \times (\vec{\omega} \times \vec{R})$, the **centrifugal force.** Both the Coriolis force and centrifugal force are the fictitious forces, resulting from the rotational motion of the frame S' relative to the inertial frame S. These forces are not due to any specific action applied to the particle. If the particle is at rest (*i.e.*, $\vec{v}_r = 0$) in the rotating frame, then centrifugal force is only the fictitious force acting on the particle. On the other hand, if the particle is moving with velocity $\vec{v}_r \neq 0$ in the rotating frame, then in addition to centrifugal force, Coriolis force acts on the particle.

* The name Coriolis force comes from the name of the scientist G.G. de Coriolis, who described this effect in the rotational motion.

1·12. Motion Relative to the Earth

In the rotating frame with the earth, the acceleration of a particle is given by

$$\vec{a}_r = \vec{a}_i - \vec{\omega} \times (\vec{\omega} \times \vec{R}) - 2\vec{\omega} \times \vec{v}_r \quad ...(1)$$

where $\vec{a}_i$ is is the acceleration in the non-rotating or inertial frame, $-\vec{\omega} \times (\vec{\omega} \times \vec{R})$ is the **centrifugal acceleration** and $-2\vec{\omega} \times \vec{v}_r$ is the **Coriolis acceleration.**

An interesting application of eq. (1) is the study of the motion of a body relative to a frame (S') rotating with the earth. The angular velocity of the earth is $\vec{\omega}$ along its axis of rotation. Now consider a particle P close to the earth's surface [**fig. 1.29**]. If we call $\vec{g}_0$ as the acceleration due to gravity with respect to an inertial or non-rotating frame, then $\vec{a}_i = \vec{g}_0$ and hence the acceleration as measured by an observer rotating with the earth is [from eq. (1)]

$$\vec{a}_r = \vec{g}_0 - \vec{\omega} \times (\vec{\omega} \times \vec{R}) - 2\vec{\omega} \times \vec{v}_r \quad ...(2)$$

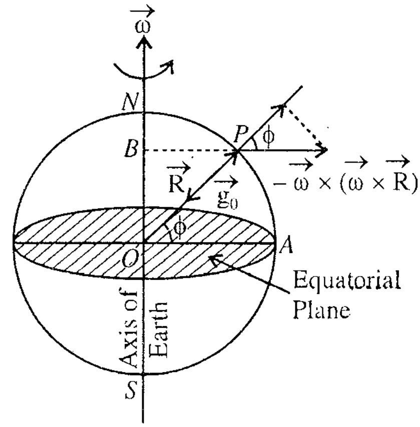


Fig. 1.29.

Hence the acceleration of a body relative to the rotating earth depends on its velocity relative to the earth and on the position vector $\vec{R}$ of the body.

Now we shall discuss the effect of centrifugal and Coriolis accelerations separately.

(a) **Effect of centrifugal force :** If the particle P is at rest on the earth's surface, then Coriolis term is zero and the acceleration $\vec{a}_r$ measured relative to the earth is called **effective acceleration due to gravity** $\vec{g}$. Thus

$$\vec{g} = \vec{g}_0 - \vec{\omega} \times (\vec{\omega} \times \vec{R}) \quad ...(3)$$

Obviously the centrifugual acceleration $-\vec{\omega} \times (\vec{\omega} \times \vec{R})$ acts in the outward direction along P and its magnitude is $\omega^2 R \cos\phi$ *, where ϕ is the latitude at P.

Assuming earth to be spherical, we may consider $\vec{g}_0$ to be pointing towards the centre of the earth along the radial direction. Due to the second term in (3), the direction of $\vec{g}$, called the vertical, deviates slightly from the radial direction and is determined by a plumb line. For practical purposes, the vertical may be assumed to coincide with the radial direction. The magnitude of $\vec{g}$ is slightly less than the value of $\vec{g}_0$ and can be expressed as

$$g = g_0 - \omega^2 R \cos^2 \phi \quad ...(4)$$

The second term in eq. (4) is very small (about 0.3 %) compared to g_0 and accounts for most of the observed variations of acceleration due to gravity with latitude at earth's surface.

(b) **Effect of Coriolis force :** *Coriolis force* $(-2m\vec{\omega} \times \vec{v}_r)$ *is a fictitious force which acts on a particle only if it is in motion with respect to the*

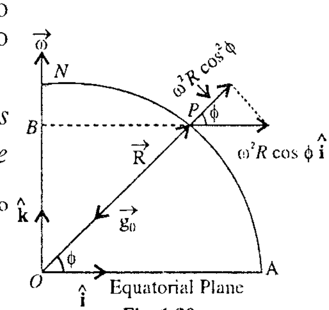


Fig. 1.30.

* Consider the plane $ONPA$ and take unit vectors $\hat{i}$ and $\hat{k}$ along OA and ON [**fig. 1.30**]. So that $\vec{\omega} = \omega\hat{k}$ and $\vec{R} = R\cos\phi\hat{i} + R\sin\phi\hat{k}$

Now, $\vec{\omega} \times (\vec{\omega} \times \vec{R}) = (\vec{\omega}\cdot\vec{R})\,\vec{\omega} - (\vec{\omega}\cdot\vec{\omega})\,\vec{R} = \omega^2 R\sin\phi\,\hat{k} - \omega^2 R[\hat{i}\cos\phi + \hat{k}\sin\phi]$

$= -\omega^2 R\cos\phi\,\hat{i}$

rotating frame. Hence, in the rotating frame if a particle moves with velocity $\vec{v}_r$, then it always experiences a force $(-2m\vec{\omega}\times\vec{v}_r)$ perpendicular to its path opposite to the direction of vector product $\vec{\omega}\times\vec{v}_r$. The effect of Coriolis force is not considerable at small (particle) speeds. Let us discuss the effect of Coriolis force, when the particle is moving in a horizontal plane or falling freely on the earth.

(i) Particle moving in a horizontal plane : The Coriolis force causes a moving particle in a horizontal plane in the northern hemisphere to deflect towards the right of its path. In the southern hemisphere, the deflection is towards the left of the path. Let a particle of mass m be projected with velocity $\vec{v}$ in a horizontal plane at a point P on earth's surface with latitude ϕ. Consider a frame XYZ fixed on the earth at P so that X-axis is vertical and YZ is horizontal plane [**fig. 1.31**]. Now,

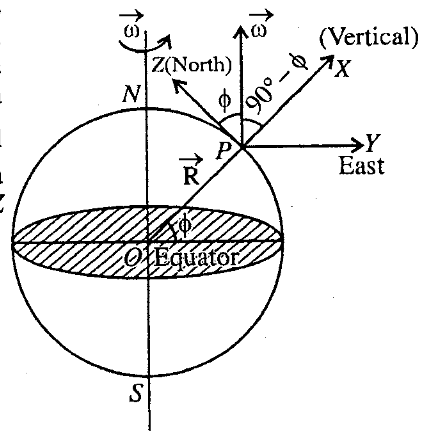


Fig. 1.31.

$$\vec{v} = v_y\hat{\mathbf{j}} + v_z\hat{\mathbf{k}} \quad \text{and} \quad \vec{\omega} = \omega\sin\phi\,\hat{\mathbf{i}} + \omega\cos\phi\,\hat{\mathbf{k}}$$

The Coriolis force acting on the particle is

$$\vec{F} = -2m(\vec{\omega}\times\vec{v})$$
$$= -2m(\omega\sin\phi\,\hat{\mathbf{i}} + \omega\cos\phi\,\hat{\mathbf{k}}) \times (v_y\hat{\mathbf{j}} + v_z\hat{\mathbf{k}})$$
$$= 2m\,\omega\,v_y\cos\phi\,\hat{\mathbf{i}} - 2m\,\omega\,v_z\sin\phi\,\hat{\mathbf{j}} + 2m\,\omega\,v_y\sin\phi\,\hat{\mathbf{k}} \quad ...(5)$$

If $\vec{F}_H$ and $\vec{F}_V$ are the horizontal and vertical components of the Coriolis force, then

$$\vec{F}_H = 2m\,\omega\,v_z\sin\phi\,\hat{\mathbf{j}} - 2m\,\omega\,v_y\sin\phi\,\hat{\mathbf{k}} \quad ...(6)$$

Its magnitude $\quad F_H = 2m\,\omega\,v\sin\phi \quad (\because \sqrt{v_y^2 + v_z^2} = v) \quad ...(7)$

Also $\quad \vec{F}_V = 2m\,\omega\,v_y\cos\phi\,\hat{\mathbf{i}} \quad ...(8)$

Its magnitude $\quad F_V = 2m\,\omega\,v_y\cos\phi \quad ...(9)$

Obviously the horizontal component of the Coriolis force tends to deviate the path of the particle toward the right in the northern hemisphere. For example, if the particle is projected towards north, due to the Coriolis force it is deflected toward east.

The magnitude of horizontal Coriolis force is $2m\omega v\sin\phi$ and is zero at $\phi = 0$, *i.e.*, at equator. The effect of Coriolis force is appreciable due to its horizontal component because in the vertical direction, its effect is masked by the large gravitational force.

(ii) Freely falling body at earth's surface : One of the important effect of Coriolis force is that a freely falling body deviates from its true vertical path. This deviation is always towards east direction in either of the hemisphere of the earth. Due to the horizontal component of earth's angular velocity, Coriolis force starts to act on freely falling body in the horizontal direction and deviates it from the true vertical direction. We deduce below an expression for this deviation.

Let us consider the free fall of a body from a height h on the surface of the earth at a latitude ϕ. The earth is rotating about its axis with an angular velocity $\vec{\omega}$. At the point P of the earth, take X-axis vertically, Y-axis along east and Z-axis along north. [**fig. 1.31**]. If $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$, are unit vectors along these axes, then the angular velocity $\vec{\omega}$ can be represented as

$$\vec{\omega} = \omega\cos\left[\frac{\pi}{2} - \phi\right]\hat{\mathbf{i}} + \omega\cos\phi\,\hat{\mathbf{k}} \text{ or } \vec{\omega} = \omega(\sin\phi\,\hat{\mathbf{i}} + \cos\phi\,\hat{\mathbf{k}}) \quad ...(10)$$

As the effective value of the acceleration due to gravity g is the combined effect of the centrifugal acceleration and accelerlation in the inertial frame, then substituting $\vec{a}_i - \vec{\omega} \times (\vec{\omega} \times \vec{R}) = -\hat{\mathbf{i}}g$ in the equation

we get
$$\vec{a}_i = \vec{a}_r + 2\vec{\omega} \times \vec{v}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R}),$$

$$-\hat{\mathbf{i}}g = \vec{a}_r + 2\vec{\omega} \times \vec{v}_r \qquad ...(11)$$

Here the velocity of the body $\vec{v}_r$ is almost along X-axis with negligible Y and Z components and hence we can have

$$\vec{v}_r = \hat{\mathbf{i}}\frac{dx}{dt} \qquad ...(12)$$

Writting $\vec{a}_r$ in component form, we get from eq. (11)

$$-\hat{\mathbf{i}}g = \hat{\mathbf{i}}\frac{d^2x}{dt^2} + \hat{\mathbf{j}}\frac{d^2y}{dt^2} + \hat{\mathbf{k}}\frac{d^2z}{dt^2} + 2\omega(\sin\phi\,\hat{\mathbf{i}} + \cos\phi\,\hat{\mathbf{k}}) \times \hat{\mathbf{i}}\frac{dx}{dt}$$

or
$$-\hat{\mathbf{i}}g = \hat{\mathbf{i}}\frac{d^2x}{dt^2} + \hat{\mathbf{j}}\left[\frac{d^2y}{dt^2} + 2\omega\frac{dx}{dt}\cos\phi\right] + \hat{\mathbf{k}}\frac{d^2z}{dt^2}$$

Now, the three component equations are

$$\frac{d^2x}{dt^2} = -g \quad ...(13a) \qquad \frac{d^2y}{dt^2} = -2\omega\frac{dx}{dt}\cos\phi \quad ...(13b) \qquad \frac{d^2z}{dt^2} = 0 \quad ....(13c)$$

Integrating eq. (13a), we get

$$\frac{dx}{dt} = -gt + C$$

But initially at $t = 0$, $dx/dt = 0$, therefore $C = 0$.

Thus
$$\frac{dx}{dt} = -gt. \qquad ...(14)$$

Integrating it further, we get

$$x = -\frac{1}{2}gt^2 + C'$$

Initially at $t = 0$, the distance of the stone from the earth $x = h$. This gives $h = C'$, and hence

$$x = -\frac{1}{2}gt^2 + h$$

Finally, when at $t = T$, the stone touches the ground $x = 0$. Therefore

$$h - \frac{1}{2}gT^2 = 0 \quad \text{or} \quad T = \sqrt{2h/g} \qquad ...(15)$$

Now,
$$\frac{d^2y}{dt^2} = -2\omega\frac{dx}{dt}\cos\phi \quad \text{or} \quad \frac{d^2y}{dt^2} = 2\omega gt\cos\phi \qquad \text{[Using eq. (14)]}$$

Integrating it, we get

$$\frac{dy}{dt} = \omega gt^2\cos\phi \quad \text{and hence } y = \frac{\omega gt^3}{3}\cos\phi \qquad ...(16)$$

The constants of integration in (16) have been taken zero, because initially the stone has no displacement and velocity in the Y-direction is zero.

Finally, at $t = T$, the maximum horizontal displacement will be given by

$$Y = \frac{\omega gT^3}{3}\cos\phi \qquad ...(17)$$

Substituting the value of T from eq. (15), we get

$$Y = \frac{\omega g}{3}\left[\frac{2h}{g}\right]^{3/2}\cos\phi \quad \text{or} \quad Y = \left[\frac{8}{9g}\right]^{1/2} h^{3/2}\omega\cos\phi \qquad ...(18)$$

Thus a freely falling body at latitude ϕ is displaced horizontally due east by Coriolis force by an amount, given by eq. (18). **At the equator** ($\phi = 0$), the easterly deflection is given by

$$Y = \left[\frac{8}{9g}\right]^{1/2} h^{3/2}\omega \qquad ...(19)$$

Some other effects of Coriolis force

(1) **Formation of cyclones :** When a low pressure zone is created at a place on earth, the wind will flow radially from high pressure regions to the central low pressure region. However the Coriolis force deviates the air molecules toward the right of their path in the northern hemisphere, resulting in anticlockwise motion of the air particles and formation of the cyclone [**fig. 1.32(a)**]. The cyclone also moves along with the centre of low pressure region. In the southern hemisphere, the formation of cyclone is clockwise [**fig. 1.32(b)**].

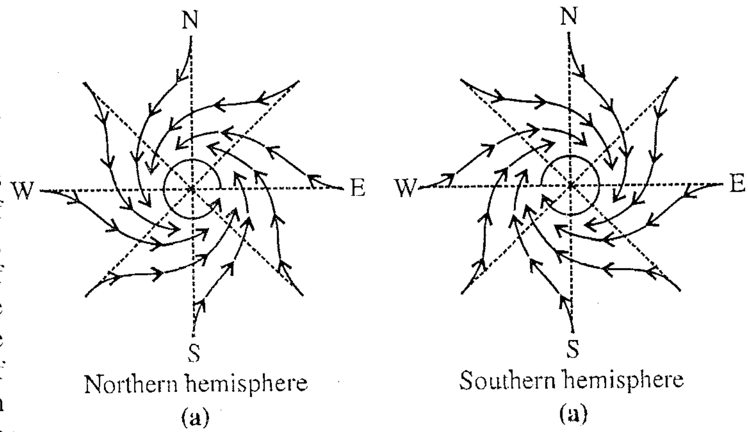


Fig. 1.32 : Formation of cyclones.

(2) In the northern hemisphere, flowing water in a river is acted by Coriolis force towards right of its path so that the right bank of the river is eroded more in comparison to its left bank. In the southern hemisphere, the effect is more on the left bank.

(3) **Rotation of plane of oscillation of Foucault's pendulum :** Foucault's pendulum is similar to a simple pendulum, with a heavy bob and long strong suspension wire. If the pendulum is oscillated, Coriolis force acts toward the right of its path which results in the rotation of the plane of oscillation slowly in the clockwise direction in the northern hemisphere. This experimental fact demonstrates that the earth rotates about its axis. (For details see next article).

1·13. Foucault's Pendulum

Experimentally this fact, that earth rotates about its axis and hence the frame attached to it is not an inertial frame, is demonstrated by Foucault's pendulum. The experiment was first performed by Foucault in 1851. Foucault's pendulum is similar to a simple pendulum in which a very heavy bob is suspended by means of a long strong wire [**fig. 1.33**]. Thus if it is once vibrated, it vibrates for considerably large time with a large period ($\approx$17 sec.). Foucault took in his experiment a bob of 28 kg mass and a wire of 70 metres length. The upper end of the wire is attached to a rigid support in such a way that the pendulum may vibrate with equal freedom in any direction.

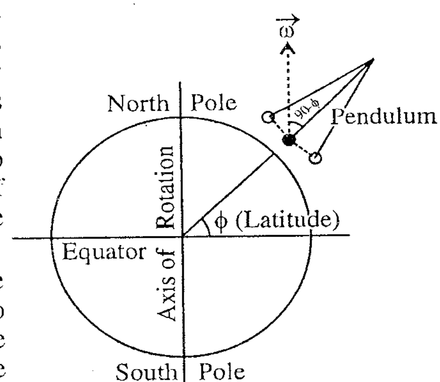


Fig. 1.33 : Foucault's Pendulum.

When Foucault's pendulum is allowed to oscillate in the northern hemisphere, its plane of oscillation rotates from east to west (or clockwise as seen from above). It is not possible that the point of support might have turned the plane of vibration. The only possibility is this that the floor, *i.e.*, the earth under the pendulum may be rotating. If such a pendulum is imagined to vibrate at the north pole of the earth, *the plane of oscillation will remain fixed in an inertial frame or solar reference frame. But the earth under the pendulum is rotating once every 24 hours, therefore an observer on the earth will see that the plane of oscillation is turning from east to west, opposite to earth's rotation.*

At any other latitude ϕ, the angular velocity $\vec{\omega}$ can be resolved into two components : vertical component $\omega \sin \phi$ and horizontal component $\omega \cos \phi$ in the north direction. Evidently the later component will not have any perceptible effect on the pendulum. The vertical component will make the plane of its oscillation rotate with an angular velocity $\omega \sin \phi$. Hence the *time of* **one rotation of the plane of oscillation** will be given by

$$T = \frac{2\pi}{\omega \sin \phi} \qquad ...(1)$$

As the earth rotates once in 24 hours with an angular velocity ω, *i.e.*, $2\pi/\omega$ = 24 hours, hence

$$T = \frac{24 \text{ hours}}{\sin \phi} \qquad ...(2)$$

This rotation will be clockwise in the northern hemisphere and opposite to it in the southern hemisphere. In fact, this rotation, as seen by the observer on the earth, is due to Coriolis force. When the pendulum moves along a horizontal line, Coriolis force acts in the perpendicular direction to its velocity and in consequence the plane of oscillation turns. Thus, the *rotation of the plane of oscillation of Foucault's pendulum demonstrates that the earth rotates about its axis.*

Ex. 1. *A bullet is fired horizontally in the north direction with a velocity of* 500 *m/sec at* 30°*N latitude. Calculate the horizontal component of Coriolis acceleration and the consequent deflection of the bullet as it hits a target* 250 *metres away. Also determine the vertical displacement of the bullet due to gravity. If the mass of the bullet is* 100 *gm, find the Coriolis force.* **(Raipur 2003; Ajmer 1988)**

Sol. If we take X-axis vertically, Z-axis towards north and Y-axis along east, then the velocity of the bullet is

$$\vec{v} = 500\ \hat{k} \text{ m/sec.}$$

and angular velocity $\qquad \vec{\omega} = \omega\,(\hat{k} \cos 30° + \hat{i} \sin 30°)$

since the angular velocity vector $\vec{\omega}$ of the earth is directed parallel to its axis and is inclined at 30° to the horizontal **[fig. 1.34]**.

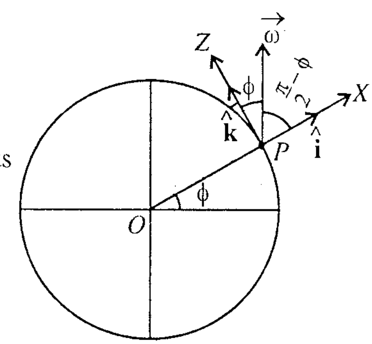


Fig. 1.34.

Here, $\qquad \omega = \dfrac{2\pi}{24 \times 60 \times 60} = 7.2 \times 10^{-5}$ rad./sec.

$\therefore$ Coriolis acceleration $= -2\,\vec{\omega} \times \vec{v}$

$$= -2\omega\,(\hat{k} \cos 30° + \hat{i} \sin 30°) \times 500\,\hat{k}$$

$$= 2 \times 7.2 \times 10^{-5} \times 10^{2} \times \frac{1}{2}\,\hat{j} = \mathbf{0.036\ m/sec^2\ towards\ east.}$$

Time of journey $= \dfrac{250}{500} = \dfrac{1}{2}$ sec.

$\therefore$ Deflection of the bullet due to Coriolis acceleration

$$= \frac{1}{2}at^2 = \frac{1}{2} \times 0.036 \times \left(\frac{1}{2}\right)^2 = \mathbf{4.5 \times 10^{-3}\ m\ towards\ east.}$$

Vertical displacement of the bullet due to gravity $= \dfrac{1}{2}gt^2 = \dfrac{1}{2} \times 9.8 \times \left(\dfrac{1}{2}\right)^2 = \mathbf{1.23\ m}$

Coriolis force $= -2m\,\vec{\omega} \times \vec{v} = 0.1 \times 0.036\,\hat{i} = \mathbf{3.6 \times 10^{-3}}$ **newton towards east.**

Ex. 2. *Prove that the observed acceleration due to gravity* g_ϕ *at the latitude* ϕ *is related to its real value* g *by the relation*

$$g_\phi^{\,2} = (g \cos \phi - \omega^2 R \cos \phi)^2 + (g \sin \phi)^2. \qquad \textbf{(Ajmer 1988; Raj. 83)}$$

Sol. If a particle is at rest at ϕ latitude, then it is not acted by Coriolis force.

Therefore, $\vec{a}_i = \vec{a}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R})$

If we take Z-axis along the axis of the earth and X-axis perpendicular to it, then

$$\vec{a}_r = \vec{a}_i - \vec{\omega} \times (\vec{\omega} \times \vec{R})$$

or

$$g\phi = -g\,(\hat{\mathbf{i}} \cos\phi + \hat{\mathbf{k}} \sin\phi) - \omega\,\hat{\mathbf{k}} \times [\omega \hat{\mathbf{k}} \times R\,(\hat{\mathbf{i}} \cos\phi + \hat{\mathbf{k}} \sin\phi)]$$
$$= -g\,(\hat{\mathbf{i}} \cos\phi + \hat{\mathbf{k}} \sin\phi) + \omega^2 R \cos\phi\, \hat{\mathbf{i}}$$
$$= -\hat{\mathbf{i}}\,(g \cos\phi - \omega^2 R \cos\phi) - \hat{\mathbf{k}}\, g \sin\phi$$

$\therefore$

$$g_\phi^{\,2} = g_\phi \cdot g_\phi = (g \cos\phi - \omega^2 R \cos\phi)^2 + (g \sin\phi)^2.$$

Ex. 3. *Prove that the plane of oscillation of Foucault's pendulum rotates* 15° sin ϕ *per hour, where* ϕ *is the latitude of the place.*

Sol. Period of one rotation of plane of oscillation $T = \dfrac{24 \text{ hours}}{\sin\phi}$

If the plane of oscillation be rotating with an angular velocity α per hour, then

$$T = \frac{2\pi}{\alpha} = \frac{360°}{\alpha} \text{ hour}, \therefore \frac{360°}{\alpha} = \frac{24}{\sin\phi}$$

whence $\alpha = 15° \sin\phi$ per hour.

Ex. 4. *In how much time will the plane of oscillation of a Foucault's pendulum rotate through* 90° *at* 30° *latitude ?*

Sol. Period of one rotation, (*i.e.*, 360°) of the plane of oscillation of Foucault's pendulum is

$$T = \frac{24 \text{ hours}}{\sin\phi}$$

Hence the plane of oscillation of the pendulum will rotate through 90° in time

$$= \frac{24 \text{ hours}}{\sin\phi} \times \frac{90°}{360°} = \frac{6 \text{ hours}}{\sin 30°} = \textbf{12 hours.}$$

EXERCISE 1.4

1. Find the horizontal component of Coriolis force acting on a body of mass 0.1 kg moving northward with a horizontal velocity of 100 m/sec at 30° N latitude on the earth.
   [**Ans.** 7.2×10^{-4} newtons towards east.]'
2. A person in a jet plane is flying along the equator due east with a speed of 450 m/sec. What is his Coriolis acceleration ?
   [**Ans.** 6.56×10^{-3} m/sec^2.]
3. A stone is dropped with zero initial velocity from the top of a 100 m high tower at the equator. Calculate horizontal displacement of the stone due to earth's rotation.
   (Take g = 10 m/sec^2 and $\omega = 7.2 \times 10^{-5}$ rad./sec.)
   [**Ans.** 2.15 cm.]
4. Calculate the eastward deviation of a stone falling freely from a height 50 metres above ground at latitudes (i) 60° N, (ii) 45° N and (iii) at equator. ($\omega = 7.3 \times 10^{-5}$ rad./sec, g = 9.8 m/sec^2.)
   [**Ans.** (i) 0.388 cm; (ii) 0.549 cm; (iii) 0.776 cm.]
5. 'A' suggests that if a stone is thrown vertically upwards (from a plane in the northern hemisphere) so as to attain a maximum height 'h' on its return to the earth, it will have no horizontal displacement. He argues that during the return journey the stone experiences exactly equal and opposite Coriolis force as it does during the onwards journey and, therefore, the net displacement on the ground will be zero. 'B' disagrees to this view. What is your answer to this riddle ? Support your answer with proper derivation.
   [**Ans.** The net displacement will be double of the displacement at maximum height towards west.]
6. A body is thrown vertically upward with a velocity u. Prove that it will fall back on a point displaced

to the west by a distance equal to

$$\frac{4}{3}\left(\frac{8h^3}{g}\right)^{1/2} \omega \cos\phi,$$

where ϕ is the latitude and $h = u^2/2g$. **(Ajmer 1989; Rajasthan 84)**

7. A body of mass 10 gm is at rest in an inertial frame. Describe its motion in a frame rotating with an angular speed of 10 rad./sec. The particle is situated at a distance of 5 cm from the axis of rotation. Determine the Coriolis and centrifugal forces.

   [**Ans.** Centrifugal force = 0.05 n directed outwardly; Coriolis force = 0.1 n directed inwardly. The addition of the two fictitious forces is 0.05 n inwardly, *i.e.*, this explains the observed circular motion and it seems that a force (centripetal) 0.05 n is acting towards the axis of rotation.]

8. What is the fictitious acceleration of the sun relative to a frame rotating with the earth. (Distance between the sun and the earth = 1.5×10^{11} m) **(Agra 1981)**

   [**Ans.** 7.8×10^2 m/sec^2 towards earth.]

9. The acceleration due to gravity measured in an earth bound coordinate system is represented by g. Due to earth's rotation, g differs from the true acceleration due to gravity g_0. Assuming that the earth is a sphere of radius R and it is rotating with angular speed ω, find g as a function of latitude ϕ.

   [**Ans.** $g = g_0\,[1 - (2x - x^2)\cos^2\lambda]^{1/2}$, where $x = \omega^2 R/g_0$]

   [**Hint :** In the rotating frame with the earth the total fictitious acceleration on the sun

   $$= -2\vec{\omega}\times\vec{v}_r - \vec{\omega}\times(\vec{\omega}\times\vec{R}) = 2\vec{\omega}\times(\vec{\omega}\times\vec{R}) - \vec{\omega}\times(\vec{\omega}\times\vec{R}) = \vec{\omega}\times(\vec{\omega}\times\vec{R}) = -\vec{\omega}^2\vec{R}.]$$

10. In how much time will the plane of oscillation of a Foucault's pendulum turn through 180° at 30° latitude ?

    [**Ans.** 24 hours.]

11. A Foucault's pendulum is oscillating at any time in the east-west direction. Calculate the time in which its plane of oscillation will start pointing 18° north of east, if the latitude of the place be 45°.

    [**Ans.** 32.2 hours.]

1·14. Coordinate Systems

In order to describe the motion of a particle (or a system of particles), we need to know its position and how it is changing with time. The position of a particle is represented by choosing a coordinate system. Some of the commonly used coordinate systems are cartesian coordinates, plane polar coordinates, cylindrical coordinates and spherical coordinates.

(1) Cartesian coordinates : In this coordinate system, X, Y, Z axes are perpendicular to each other and the position of a particle P in this system is represented by the coordinates (x, y, z). **[fig. 1.35]**. The position vector $\vec{r}$ of the particle is represented as

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad \text{...(1)}$$

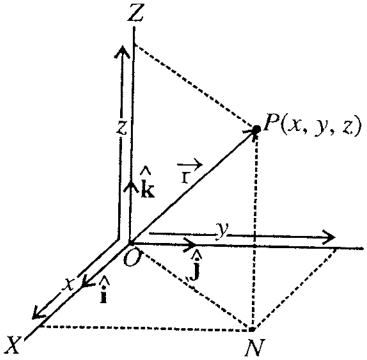


Fig. 1.35 : Cartesian coordinates.

Remember that the three mutually orthogonal axes form a right handed coordinate system.

(2) Plane polar coordinates : In a two dimensional motion, the position of a particle can be represented by rectangular coordinates (x, y), or plane polar coordinates (r, θ). The position of the particle P is at a distance r from the origin O and the line OP makes an angle θ with the X-axis **[fig. 1.36]**. The cartesian coordinates (x, y) and polar coordinates (r, θ) of the particle are related as

$$x = r\cos\theta, \quad y = r\sin\theta \quad \text{...(2)}$$

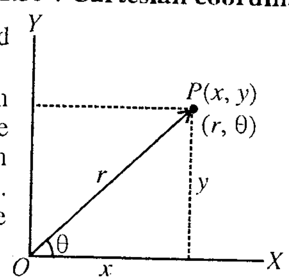


Fig. 1.36 : Polar coordinates

Also $r^2 \cos^2 \theta + r^2 \sin^2 \theta = x^2 + y^2$

or $$r^2 = x^2 + y^2$$

Thus $$r = \sqrt{x^2 + y^2} \quad ...(3a)$$

Further $$\tan \theta = \frac{y}{x} \text{ or } \theta = \tan^{-1} \frac{y}{x} \quad ...(3b)$$

(3) Cylindrical coordinates : Consider the position of a particle P, located at a distance r from the origin O. This position P can be expressed in rectangular (x, y, z) or cylindrical (ρ, ϕ, z) coordinates as shown in [**fig. 1.37**]. The two systems of coordinates are related as

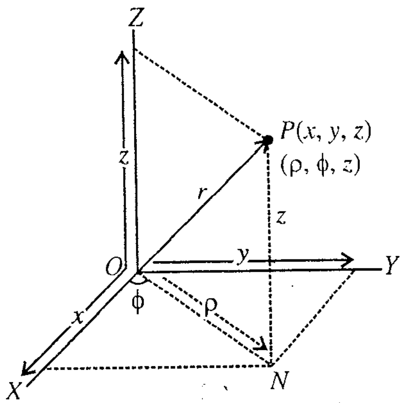


Fig. 1.37 : Cylindrical coordinates.

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z \quad ...(4)$$

$$\text{Also, } \rho = \sqrt{x^2 + y^2}, \quad \tan \phi = \frac{y}{x} \text{ or } \phi = \tan^{-1} \frac{y}{x}, \quad z = z \quad ...(5)$$

Observe that in cylindrical coordinates ρ has replaced r and ϕ has replaced θ of plane polar coordinates.

(4) Spherical coordinates : Let us consider the position of a particle P, located at a distance r from the origin O. The spherical coordinates (r, θ, ϕ) representing the position P are related to the cartesian coordinates as follows [**fig. 1.38**] :

$$x = OA = ON \cos \phi = r \sin \theta \cos \phi$$
$$y = OB = ON \sin \phi = r \sin \theta \sin \phi$$
$$z = OC = OP \cos \theta = r \cos \theta$$

i.e., $x = r \sin \theta \cos \phi, \ y = r \sin \theta \sin \phi, \ z = r \cos \theta \quad ...(6)$

Also, $x^2 + y^2 + z^2 = r^2 \sin^2 \theta (\sin^2 \phi + \cos^2 \phi) + r^2 \cos^2 \theta$
$$= r^2 \sin^2 \theta + r^2 \cos^2 \theta = r^2$$

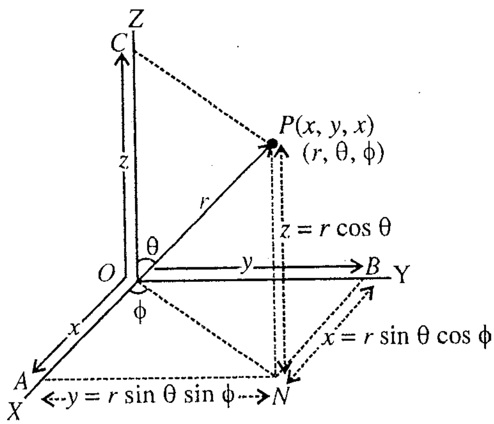


Fig. 1.38 : Spherical coordinates.

or $$r = \sqrt{x^2 + y^2 + z^2}$$

and $$\tan \phi = \frac{AN}{OA} = \frac{y}{x}. \qquad \text{Further } \tan\theta = \frac{PC}{OC} = \frac{ON}{z} = \frac{\sqrt{x^2+y^2}}{z}$$

i.e., $$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1} \frac{\sqrt{x^2+y^2}}{z} \text{ and } \phi = \tan^{-1} \frac{y}{x} \quad ...(7)$$

The relations (7) express r, θ, ϕ in terms of x, y, z while relations (6) represent x, y, z in terms of r, θ, ϕ.

1·15. Velocity and Acceleration in Different Coordinate Systems

(1) Cartersian coordinates : Let $\vec{r}$ be the position vector of a particle at an instant t. Then

$$\vec{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}} \quad ...(1)$$

Its velocity $\vec{v}$ and acceleration $\vec{a}$ are given by

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{\mathbf{i}} + \frac{dy}{dt}\hat{\mathbf{j}} + \frac{dz}{dt}\hat{\mathbf{k}} \text{ or } \vec{v} = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} + v_z\hat{\mathbf{k}} \quad ...(2)$$

and $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = \frac{dv_x}{dt}\hat{\mathbf{i}} + \frac{dv_y}{dt}\hat{\mathbf{j}} + \frac{dv_z}{dt}\hat{\mathbf{k}}$ or $\vec{a} = a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}} + a_z\hat{\mathbf{k}}$ ...(3)

where v_x, v_y, v_z are the cartesian components of $\vec{v}$ and a_x, a_y, a_z are those of $\vec{a}$.

(2) Plane polar coordinates : When a particle is moving in a plane, many times (*e.g.*, circular motion) it is convenient to describe the motion in plane polar coordinates (r, θ). The position vector $\vec{r}$ of the particle at any instant t can be expressed as

$$\vec{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} \quad \text{or} \quad \vec{r} = r\cos\theta\,\hat{\mathbf{i}} + r\sin\theta\,\hat{\mathbf{j}} \qquad ...(4)$$

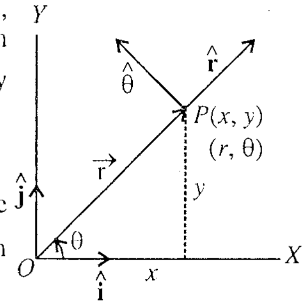


Fig. 1.39.

We define two unit vectors $\hat{\mathbf{r}}$ and $\hat{\theta}$ in polar coordinates which are prependicular to each other. $\hat{\mathbf{r}}$ is the unit vector along $\vec{r}$ and $\hat{\theta}$ in a direction perpendicular to $\vec{r}$ with θ increasing [**fig. 1.39**].

Dividing $\vec{r}$ by r in (4), we get

$$\frac{\vec{r}}{r} = \hat{\mathbf{r}} = \cos\theta\,\hat{\mathbf{i}} + \sin\theta\,\hat{\mathbf{j}} \qquad ...(5)$$

which has the direction in increasing r.

Also obviously from **fig. 1.40**

$$\hat{\theta} = -\cos\left(\frac{\pi}{2} - \theta\right)\hat{\mathbf{i}} + \sin\left(\frac{\pi}{2} - \theta\right)\hat{\mathbf{j}}$$

or
$$\hat{\theta} = -\sin\theta\,\hat{\mathbf{i}} + \cos\theta\,\hat{\mathbf{j}} \qquad ...(6)$$

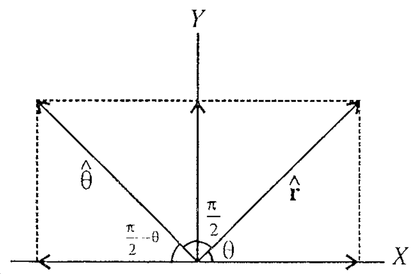


Fig. 1.40.

Observe that both unit vectors $\hat{\mathbf{r}}$ and $\hat{\theta}$ are the functions of θ.

The velocity $\vec{v}$ of the particle P is given by

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\cos\theta\,\hat{\mathbf{i}} + r\sin\theta\,\hat{\mathbf{j}})$$

$$= \left(\frac{dr}{dt}\cos\theta - r\sin\theta\frac{d\theta}{dt}\right)\hat{\mathbf{i}} + \left(\frac{dr}{dt}\sin\theta + r\cos\theta\frac{d\theta}{dt}\right)\hat{\mathbf{j}}$$

$$= \frac{dr}{dt}(\cos\theta\,\hat{\mathbf{i}} + \sin\theta\,\hat{\mathbf{j}}) + r\frac{d\theta}{dt}(-\sin\theta\,\hat{\mathbf{i}} + \cos\theta\,\hat{\mathbf{j}})$$

Using (5) and (6), we get

$$\vec{v} = \frac{dr}{dt}\hat{\mathbf{r}} + r\frac{d\theta}{dt}\hat{\theta} \qquad ...(7)$$

$\therefore$ Radial component of velocity, $v_r = \frac{dr}{dt} = \dot{r}$

and its transverse component, $v_\theta = \frac{rd\theta}{dt} = r\dot{\theta}$ ...(8)

Again from (5) $\quad \frac{d\hat{\mathbf{r}}}{dt} = (-\sin\theta\,\hat{\mathbf{i}} + \cos\theta\,\hat{\mathbf{j}})\frac{d\theta}{dt} = \hat{\theta}\frac{d\theta}{dt}$ ...(9a)

and from (6) $\quad \frac{d\hat{\theta}}{dt} = (-\cos\theta\,\hat{\mathbf{i}} - \sin\theta\,\hat{\mathbf{j}})\frac{d\theta}{dt} = -\hat{\mathbf{r}}\frac{d\theta}{dt}$ ...(9b)

Now, $\quad$ acceleration, $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}\left(\frac{dr}{dt}\hat{\mathbf{r}} + r\frac{d\theta}{dt}\hat{\theta}\right)$

$$= \frac{d^2r}{dt^2}\hat{\mathbf{r}} + \frac{dr}{dt}\frac{d\hat{\mathbf{r}}}{dt} + \frac{dr}{dt}\frac{d\theta}{dt}\hat{\theta} + r\frac{d^2\theta}{dt^2}\hat{\theta} + r\frac{d\theta}{dt}\frac{d\hat{\theta}}{dt}$$

$$= \frac{d^2r}{dt^2}\hat{\mathbf{r}} + 2\frac{dr}{dt}\frac{d\theta}{dt}\hat{\theta} + r\frac{d^2\theta}{dt^2}\hat{\theta} - r\left(\frac{d\theta}{dt}\right)^2\hat{\mathbf{r}} \qquad \text{[from (9a) and (9b)]}$$

Thus
$$\vec{a} = \left\{\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right\}\hat{\mathbf{r}} + \left(2\frac{dr}{dt}\frac{d\theta}{dt} + r\frac{d^2\theta}{dt^2}\right)\hat{\theta} \qquad ...(10)$$

Therefore, *the radial acceleration* $a_r = \frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2 = \ddot{r} - r\dot{\theta}^2$ ...(11a)

and *the transverse acceleration* $a_\theta = 2\frac{dr}{dt}\frac{d\theta}{dt} + r\frac{d^2\theta}{dt^2} = \frac{1}{r}\frac{d}{dt}\left(r^2\frac{d\theta}{dt}\right) = 2\dot{r}\dot{\theta} + r\ddot{\theta}$ ...(11b)

Thus the acceleration, $\vec{a} = a_r\hat{\mathbf{r}} + a_\theta\hat{\theta}$ ...(12)

which has the magnitude $a = \sqrt{a_r^2 + a_\theta^2}$.

and direction with the radial vector r as $\tan\alpha = \frac{a_\theta}{a_r} = \frac{2\dot{r}\dot{\theta} + r\ddot{\theta}}{\ddot{r} - r\dot{\theta}^2}$ **[fig. 1.41]** ...(13)

This is to be noted that $r\dot{\theta}^2$ in (11a) is given by

$$r\dot{\theta}^2 = r\left(\frac{v_\theta}{r}\right)^2 = \frac{v_\theta^2}{r} \qquad ...(14)$$

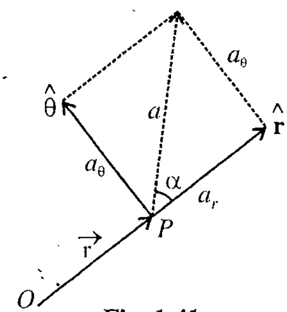


Fig. 1.41.

which represents the centripetal acceleration arising from the motion in the $\hat{\theta}$ direction.

If r is held constant in time, $\dot{r} = \ddot{r} = 0$, the particle moves on a **circular path.** There may be two kinds of circular motion : (i) Non-uniform and (ii) Uniform.

(i) Non-uniform circular motion : When r is held constant, the particle in general describes *non-uniform circular motion.* In such a case $\dot{r} = \ddot{r} = 0$ and eqs. (11a) and (11b) give

radial acceleration, $a_r = -\, r\dot{\theta}^2 = -\frac{v_\theta^2}{r}$ and transverse acceleration, $a_\theta = r\ddot{\theta} = \frac{dv_\theta}{dt}$ ...(15)

because $v_\theta = r\ddot{\theta}$ and $\frac{dv_\theta}{dt} = r\ddot{\theta}$.

Also $\vec{v} = r\dot{\theta}\hat{\theta} = v_\theta\hat{\theta}$ [from (7)]; $\therefore\ v = v_\theta = r\dot{\theta}$ or $v = r\omega$ ...(16)

where $\omega = d\theta/dt$ is the angular velocity.

Thus $a_r = -\frac{v^2}{r}$, $a_\theta = \frac{dv}{dt}$ *i.e.,* $\vec{a} = -\frac{v^2}{r}\hat{\mathbf{r}} + \frac{dv}{dt}\hat{\theta}$ ...(17)

Here the speed v of the particle, moving in a circle, is not constant, and the acceleration has radial (v^2/r) as well as transverse (dv/dt) components. The transverse component in the circular motion is along the tangent and hence it is called **tangential acceleration** for circular motion. The radial component is the **centripetal acceleration.** The magnitude of the acceleration in non-uniform circular motion is given by

$$a = \sqrt{\left(\frac{v^2}{r}\right)^2 + \left(\frac{dv}{dt}\right)^2} \qquad ...(18a)$$

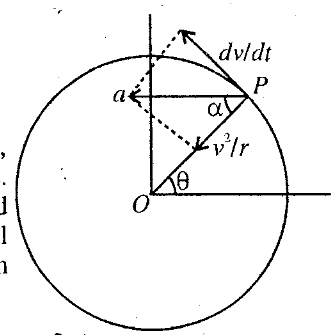


Fig. 1.42 : Non-uniform circular motion.

whose direction makes an angle α with the radius PO [**fig. 1.42**] so that

$$\tan \alpha = \frac{dv/dt}{v^2/r} \quad \text{...(18b)}$$

(ii) Uniform circular motion : If the particle moves in the circle with constant speed v, it is said to perform **uniform circular motion**. In such a case $dv/dt = 0$ and the resultant acceleration [from (17)] on the particle is

$$\vec{a} = -\frac{v^2}{r}\hat{\mathbf{r}} \quad \text{[fig. 1.43]} \quad \text{...(19)}$$

which is the centripetal acceleration with magnitude

$$a = \frac{v^2}{r} = r\dot{\theta}^2 = \omega^2 r \quad \text{...(20)}$$

Particle velocity [from (16)]

$$\vec{v} = r\dot{\theta}\hat{\theta} = v\hat{\theta} \quad \text{...(21)}$$

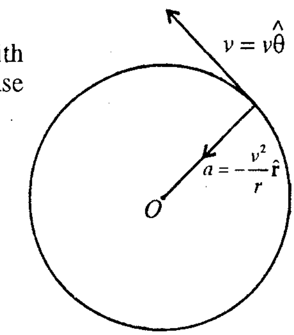


Fig. 1.43 : Uniform circular motion.

whose magnitude, *i.e.*, speed $v = r\dot{\theta} = r\omega$ remains constant, but direction continuously changes with time. Thus in uniform circular motion, velocity v changes with time and hence there is acceleration in the circular motion.

(3) Cylindrical coordinates : If we add a Z component to the plane polar coordinates, we get cylindrical coordinates to describe the motion in three dimensions. The three unit vectors ρ, $\hat{\phi}$ and $\hat{\mathbf{z}}$ in the direction of increasing $\hat{\rho}$, ϕ and z respectively are shown in **fig. 1.44**. They are mutually perpendicular to each other. The relations between the rectangular coordinates (x, y, z) and cylindrical coordinates (ρ, ϕ, z) are

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z$$

Replacing (r, θ) of plane polar coordinates by (ρ, ϕ) of cylindrical coordinates, we may write

$$\hat{\rho} = \cos \phi\, \hat{\mathbf{i}} + \sin \phi\, \hat{\mathbf{j}} \text{ [from eq. (5)]} \quad \text{...(22)}$$

$$\hat{\phi} = -\sin \phi\, \hat{\mathbf{i}} + \cos \phi\, \hat{\mathbf{j}} \quad \text{[from eq. (6)]} \text{...(23)}$$

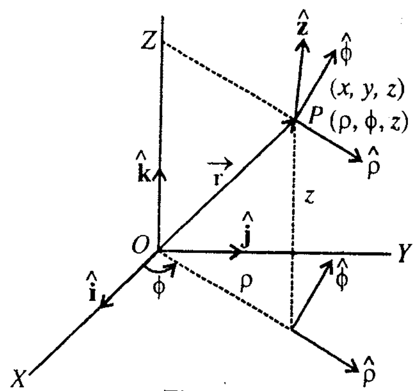


Fig. 1.44.

Obviously unit vectors $\hat{\rho}$ and $\hat{\phi}$ are functions of ϕ and change, but the unit vector $\hat{\mathbf{z}}$ remains constant, *i.e.*, $\frac{d\hat{\mathbf{z}}}{dt} = 0$.

The position vector $\vec{r}$, describing the location of a particle P, is given by

$$\vec{r} = \rho\hat{\rho} + z\hat{\mathbf{z}} \quad \text{...(24)}$$

Therefore, the velocity $\vec{v}$ of the particle is given by

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(\rho\hat{\rho} + z\hat{\mathbf{z}}) = \frac{d\rho}{dt}\hat{\rho} + \rho\frac{d\hat{\rho}}{dt} + \frac{dz}{dt}\hat{\mathbf{z}}$$

From (22) and (23), we have

$$\frac{d\hat{\rho}}{dt} = -\sin \phi\dot{\phi}\hat{\mathbf{i}} + \cos \phi\dot{\phi}\hat{\mathbf{j}} = (-\sin \phi\hat{\mathbf{i}} + \cos \phi\hat{\mathbf{j}})\dot{\phi} = \dot{\phi}\hat{\phi}$$

$$\therefore \quad \vec{v} = \dot{\rho}\hat{\rho} + \rho\dot{\phi}\hat{\phi} + \dot{z}\hat{\mathbf{z}} \quad \text{...(25)}$$

Hence, acceleration, $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{\rho}\hat{\rho} + \rho\dot{\phi}\hat{\phi} + \dot{z}\hat{\mathbf{z}}) = \frac{d}{dt}(\dot{\rho}\hat{\rho} + \rho\dot{\phi}\hat{\phi}) + \frac{d}{dt}(\dot{z}\hat{\mathbf{z}})$

As shown in plane polar coordinates, we have

$$\vec{a} = (\ddot{\rho} - \rho\dot{\phi}^2)\hat{\rho} + (2\dot{\rho}\dot{\phi} + \rho\ddot{\phi})\hat{\phi} + \ddot{z}\hat{\mathbf{z}} \quad ...(26)$$

Eqs. (25) and (26) express the **velocity** and **acceleration** in cylindrical coordinates.

(4) Spherical coordinates : The position of a particle P in spherical coordinates is represented (r, θ, ϕ) where r is called radial distance, ϕ the azimuthal angle and θ the polar angle. The polar angle θ is measured down from the Z-axis. Cartesian coordinates (x, y, z) are related to spherical coordinates (r, θ, ϕ) as **[fig. 1.45]**

$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta \quad ...(27)$$

The three mutually perpendicular unit vetors in spherical coordinates are $\hat{r}, \hat{\theta}$ and $\hat{\phi}$. The position vector $\vec{r}$ can be represented as

$$\vec{r} = r\sin\theta\cos\phi\,\hat{\mathbf{i}} + r\sin\theta\sin\phi\,\hat{\mathbf{j}} + r\cos\theta\,\hat{\mathbf{k}}$$

Unit vector $\hat{\mathbf{r}}$ along increasing r is given by

$$\hat{\mathbf{r}} = \frac{\vec{r}}{r} = \sin\theta\cos\phi\,\hat{\mathbf{i}} + \sin\theta\sin\phi\,\hat{\mathbf{j}} + \cos\theta\,\hat{\mathbf{k}} \quad ...(28a)$$

If $\hat{\rho}$ is the unit vector along ZP (or ON), then **[fig. 1.46(a)]**

$$\hat{\rho} = \cos\phi\,\hat{\mathbf{i}} + \sin\phi\,\hat{\mathbf{j}}$$

Unit vector $\hat{\theta}$ can be expressed as **[fig. 1.46(b)]**

$$\hat{\theta} = \cos\theta\,\hat{\rho} + \cos\left(\frac{\pi}{2} - \theta\right)(-\hat{\mathbf{k}})$$

where $\hat{\theta}$ is the unit vector perpendicular to $\vec{OP} = \vec{r}$ in the plane $ZPNO$ and it makes angle θ with $\hat{\rho}$ and $\left(\frac{\pi}{2} - \theta\right)$ with $\vec{PN}$; $\vec{PN}$ is along negative Z-axis with $-\hat{\mathbf{k}}$ unit vector in this direction.

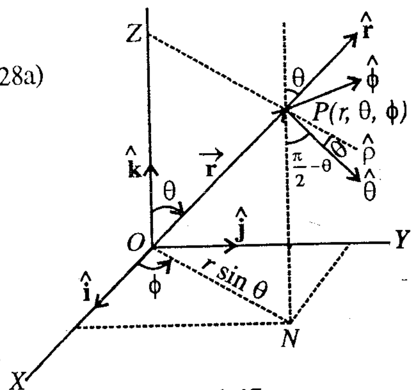
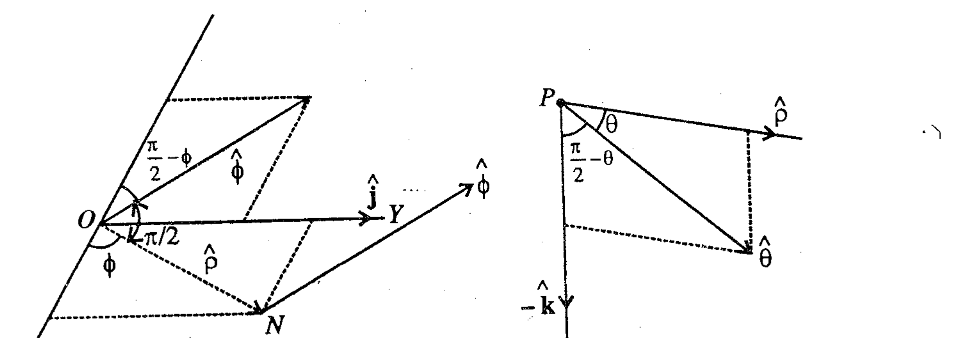


Fig. 1.45.

Hence $$\hat{\theta} = \cos\theta\cos\phi\,\hat{\mathbf{i}} + \cos\theta\sin\phi\,\hat{\mathbf{j}} - \sin\theta\,\hat{\mathbf{k}} \quad ...(28b)$$

and $$\hat{\phi} = -\sin\phi\,\hat{\mathbf{i}} + \cos\phi\,\hat{\mathbf{j}} \quad \textbf{[fig. 1.46(a)]} \quad ...(28c)$$



$$\hat{\rho} = \cos\phi\hat{\mathbf{i}} + \sin\phi\hat{\mathbf{j}}$$
$$\hat{\phi} = -\cos\left(\frac{\pi}{2} - \phi\right)\hat{\mathbf{i}} + \sin\left(\frac{\pi}{2} - \phi\right)\hat{\mathbf{j}}$$
or $$\hat{\phi} = -\sin\phi\hat{\mathbf{i}} + \cos\phi\hat{\mathbf{j}}$$

(a)

$$\hat{\theta} = \cos\theta\hat{\rho} - \sin\theta\hat{\mathbf{k}}$$
$$= \cos\theta\cos\phi\hat{\mathbf{i}} + \cos\theta\sin\phi\hat{\mathbf{j}} - \sin\theta\hat{\mathbf{k}}$$

(b)

Fig. 1.46.

where $\hat{\phi}$ is unit vector perpendicular to ON, making $\left(\frac{\pi}{2}-\phi\right)$ angle with –ve X-axis

We observe that $\hat{\mathbf{r}}, \hat{\theta}$ and $\hat{\phi}$ are orthogonal to each other and they are functions of θ and ϕ. Differentiating eq. (28a), we obtain

$$\frac{\partial \hat{\mathbf{r}}}{\partial \theta} = \cos\theta\cos\phi\,\hat{\mathbf{i}} + \cos\theta\sin\phi\,\hat{\mathbf{j}} - \sin\theta\,\hat{\mathbf{k}} = \hat{\theta}$$

Similarly, $\dfrac{\partial\hat{\theta}}{\partial\theta} = -\hat{\mathbf{r}}$, $\dfrac{\partial\hat{\phi}}{\partial\theta} = 0$, $\dfrac{\partial\hat{\mathbf{r}}}{\partial\phi} = \hat{\phi}\sin\theta$, $\dfrac{\partial\hat{\theta}}{\partial\phi} = \hat{\phi}\cos\theta$, and $\dfrac{\partial\hat{\phi}}{\partial\theta}$

$$= -\cos\phi\,\hat{\mathbf{i}} - \sin\phi\,\hat{\mathbf{j}} = -\hat{\mathbf{r}}\sin\theta - \hat{\theta}\cos\theta \qquad ...(30)$$

In spherical coordinates, the position of the point P is given by $\vec{\mathrm{r}} = r\hat{\mathbf{r}} = r\hat{\mathbf{r}}(\theta, \phi)$

The velocity $\vec{\mathrm{v}}$ of the particle is given by $\vec{\mathrm{v}} = \dfrac{d\vec{\mathrm{r}}}{dt} = r\dfrac{d\hat{\mathbf{r}}}{dt} + \dot{r}\hat{\mathbf{r}}$

But $$\frac{d\hat{\mathbf{r}}}{dt} = \frac{\partial\hat{\mathbf{r}}}{\partial\theta}\frac{d\theta}{dt} + \frac{\partial\hat{\mathbf{r}}}{\partial\phi}\frac{d\phi}{dt} = \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\hat{\phi}$$

Thus $$\vec{\mathrm{v}} = \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\hat{\phi}$$

Hence the acceleration $\vec{\mathrm{a}}$ is given by

$$\vec{\mathrm{a}} = \frac{d\vec{\mathrm{v}}}{dt} = \frac{d}{dt}[\dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\hat{\phi}] = \ddot{r}\hat{\mathbf{r}} + \dot{r}\frac{d\hat{\mathbf{r}}}{dt} + \frac{d}{dt}(r\dot{\theta})\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt} + \frac{d}{dt}(r\sin\theta\,\dot{\phi})\hat{\phi} + r\sin\theta\,\dot{\phi}\frac{d\hat{\phi}}{dt}$$

which gives on simplification

$$a = (\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\,\dot{\phi}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\cos\theta\,\dot{\phi}^2)\hat{\theta} + (r\sin\theta\,\ddot{\phi} + 2\dot{r}\dot{\phi}\sin\theta + 2r\dot{\theta}\dot{\phi}\cos\theta)\hat{\phi} \quad ...(31)$$

Eqs. (30) and (31) represent the velocity ($\vec{\mathrm{v}}$) and acceleration ($\vec{\mathrm{a}}$) of a particle in spherical coordinates.

Ex. 1. *Consider the motion of a satellite in equatorial orbit about earth with a constant radius b and angular velocity ω. Find its velocity and acceleration in rectangular and spherical coordinates.*

Sol. Let O be the centre (origin) of the earth and $\vec{\mathrm{r}}$ $(r = b)$ be the position vector of the equatorial satellite. If X and Y axes are the equatorial plane and Z-axis along the axis of the earth, then in rectangular coordinates,

$$x = b\cos\omega t, \quad y = b\sin\omega t, \quad z = 0$$

So that $$\vec{\mathrm{r}} = b\cos\omega t\,\hat{\mathbf{i}} + b\sin\omega t\,\hat{\mathbf{j}} \qquad ...(i)$$

$$\vec{\mathrm{v}} = \frac{d\vec{\mathrm{r}}}{dt} = -b\omega\sin\omega t\,\hat{\mathbf{i}} + b\omega\cos\omega t\,\hat{\mathbf{j}} \qquad ...(ii)$$

and $$\vec{\mathrm{a}} = \frac{d\vec{\mathrm{v}}}{dt} = -b\omega^2\cos\omega t\,\hat{\mathbf{i}} - b\omega^2\sin\omega t\,\hat{\mathbf{j}} \qquad ...(iii)$$

In spherical coordinates,

$$r = b,\ \theta = \frac{\pi}{2},\ \phi = \omega t;\ \dot{r} = 0,\ \dot{\theta} = 0,\ \dot{\phi} = \omega;\ \ddot{r} = 0,\ \ddot{\theta} = 0,\ \ddot{\phi} = 0 \qquad ...(iv)$$

Using the exprerssions of velocity and acceleration in spherical coordinates, we have

$$\vec{\mathrm{v}} = b\omega\hat{\phi} \qquad ...(v)$$

$$\vec{\mathrm{a}} = -b\omega^2\hat{\mathbf{r}} \qquad ...(vi)$$

Note : Of course, the expressions for velocity and acceleration in spherical coordinates are somewhat long and little tedius in their derivation, but look that their use is in the simplicity in certain problems.

Ex. 2. *An object moves on the surface of a sphere along a path given by*

$$r = R_0, \quad \theta = \omega t, \quad \phi = bt$$

in spherical coordinates. Find the speed as a function of time.

Sol. In spherical coordinates,

$$r = R_0, \quad \theta = \omega t, \quad \phi = bt; \qquad \dot{r} = 0, \quad \dot{\theta} = \omega, \quad \dot{\phi} = b$$

Hence velocity, $\vec{v} = R_0\,\omega\,\hat{\theta} + R_0\,b\,\sin\omega t\,\hat{\phi}$ and speed, $v = R_0\sqrt{\omega^2 + b^2\sin^2\omega t}$.

EXERCISE 1.5

1. An object is located at $x = 3$ m, $y = 4$ m and $z = 5$ m in rectangular coordinates. Find the cylindrical and spherical coordinates of the object in frames whose origins coincide with the origin of the cartesian frame.

   [**Ans.** $\rho = 5$ m, $\phi = \tan^{-1}(4/3)$, $z = 5$ m; $r = 5\sqrt{2}$ m, $\theta = \tan^{-1}(5/3)$, $\phi = \tan^{-1}(4/3)$.]

2. An object moves on the surface of a sphere along a path given by $r = r_0$, $\theta = \omega t$, $\phi = bt$ in spherical coordinates. Find the acceleration as a function of time.

   [**Ans.** $\vec{a} = -r_0(\omega^2 + b^2\sin^2\omega t)\hat{r} - r_0 b^2\sin^2\omega t\cos\omega t\,\hat{\theta} + 2r_0 b\,\omega\cos bt\,\hat{\phi}$.]

3. A particle moves in a straight path along the line $x = 2$ m with a constant velocity $\vec{v} = 3\hat{j}$ m/s. Find its position and velocity in polar coordinates.

   [**Ans.** $r = \sqrt{4+9t^2}$, $\theta = \tan^{-1}(3t/2)$; $\vec{v} = \dfrac{3}{\sqrt{4+9t^2}}(3t\hat{r} + 2\hat{\theta})$.]

4. The location of a body in plane polar coordinates is given by $r = r_0 e^{kt}$, $\theta = \omega t$. Show that the angle between the velocity and acceleration vectors remains constant with time.

5. A particle moves along a helical path so that its position is given by $\rho = r_0$, $\phi = \omega t$, $z = bt$ in cylindrical polar coordinates. Find the speed and acceleration.

   [**Ans.** $v = \sqrt{r_0^2\omega^2 + b^2}$; $a = -r_0\omega^2$.]

QUESTIONS

Long Answer Questions

1. State Newton's laws of motion. How do these laws define uniquely the mass and force. Discuss the limitations of Newton's laws.
2. What are inertial frames ? Show that a frame of reference having a uniform rectilinear motion relative to an inertial frame is also inertial. **(Kanpur 1984)**
3. What do you mean by frame of reference ? Differentiate the inertial and non-inertial of reference.
4. Calculate the total force and fictitious force acting on a body in a non-inertial frame. **(Bhopal 2004)**
5. What are non-inertial frames and fictitious forces ? Is the centrifugal force fictitious one ? **(Agra 1984; Kanpur 84)**
6. What are fictitious forces. Illustrate with examples. Find out the fictitious acceleration of sun in a frame fixed with the earth and rotating about its axis. **(Agra 1977)**
7. What are fictitious forces ? How are these related to non-inertial frames ? What are the effects of Coriolis force due to earth's rotation ? Explain. **(Ajmer 1989)**
8. What is Coriolis force ? Under what conditions does it come into play. Discuss in general terms the effect of Coriolis force produced as a result of earth's motion. **(Kumaun 1996)**
9. Find out horizontal and vertical components of Coriolis force due to the rotation of the earth on a particle of mass m, projected horizontally with velocity $\vec{v}$.
10. What is Coriolis force ? Show that the total Coriolis force acting on a body of mass m in a rotating frame is $-2m\vec{\omega}\times\vec{v}_r$, where $\vec{\omega}$ is the angular velocity of rotating frame and $\vec{v}_r$ is the velocity of the body in rotating frame. **(Ajmer 1988; Gulberga 95)**

11. A reference frame a ratates with respect to another reference frame b with uniform angular velocity $\vec{\omega}$. If the position, velocity and acceleration of a particle in frame a are represented by $\vec{R}$, $\vec{v}_a$ and $\vec{f}_a$ respectively, show that the acceleration of that particle in frame b is given by $\vec{f}_b$, where $\vec{f}_b = \vec{f}_a + 2\vec{\omega} \times \vec{v}_a + \vec{\omega} \times (\vec{\omega} \times \vec{R})$. Interpret this equation with reference to the motion of bodies on earth's surface. **(Agra 1970 S)**
12. Explain inertial frames of reference. Frame of reference R rotates about its origin fixed in an inertial frames of reference I. Find how velocities and accelerations in the two reference frames are related to each other. What are pseudo forces ? Explain. **(Agra 1992)**
13. What is Foucault's pendulum ? From this how is it proved that the earth rotates ? **(Ajmer 1989)**
14. Prove that for the motion of a particle in a plane, the acceleration is given by

    $\vec{a} = (\ddot{r} - r\dot{\theta}^2)\,\hat{\mathbf{r}} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$, where symbols have their normal meanings.
15. Derive the expressions for the velocity and acceleration of a particle in plane polar coordianates. **(Jabalpur 2004; Ujjain 2003; Indore 2004)**
16. Derive the expressions for the velocity and acceleration of a moving particle in cylindrical coodinates. **(Ujjain 2004; Raipur 2003)**
17. Write the expressions for the velocity and acceleration of a moving particle in spherical coordinates and derive them. **(Gwalior 2004; Bilaspur 2003; Indore 2004)**

Short Answer Questions

1. State and explain Newton's laws of motion.
2. Give an example of (i) position dependent force; (ii) velocity dependent force.
   [**Ans.** (i) Gravitational force; (ii) Viscous force.]
3. When a horse pulls a cart, which force helps the horse to move forward : the ground on the horse or the horse on the ground ?
4. A book is placed on a table. What are the forces acting on the book ? How are the action and reaction forces acting ?
5. A uniform string, having mass, is suspended from ceiling with a load at the lower end. Will the tension be uniform in the string ? Explain. Where will the tension be maximum ?
6. According to Newton's third law every force is accompanied by an equal and opposite force. How does the motion take place ?
7. What are the limitations of Newton's laws of motion ?
8. What do you understand by frame of reference ? **(Lucknow 2005)**
9. A coin is left free to fall on the ground from a moving train with constant velocity. Explain the path as seen by an observer on the ground and on the train.
10. Explain the meaning of fictitious force with reference to the non-inertail frame. **(Sagar 2005)**
11. What are fictitious forces ? Discuss centrifugal force.
12. Differentiate between real and fictitious forces.
13. What is Coriolis force ?
14. Prove that a frame of reference fixed with the earth is non-inertial frame.
15. Discuss the effect of centrifugal and Coriolis forces due to earth's rotation.
16. What is Coriolis force ? Obtain the effect of Coriolis force over a mass freely falling from a height H. **(Kanpur 2005)**
17. How is it demonstated that the earth rotates about its own axis by Foucault's pendulum ? **(Kanpur 2004)**
18. Discuss some geographical effects of Coriolis force due to rotation of the earth.
19. Express the kinetic energy of a moving particle in spherical coordinates. **(Agra 2005)**

    [**Ans.** $\frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2)$.]
20. Express spherical and cylindrical coordinates in terms of cartesian coordinates. **(Agra 2004)**
21. Prove that in plane polar coordinates

$$\frac{d\hat{\mathbf{r}}}{dt} = \hat{\theta}\frac{d\theta}{dt}, \quad \frac{d\hat{\theta}}{dt} = -\hat{\mathbf{r}}\frac{d\theta}{dt}.$$

Objective Type Questions

1. A stone is dropped from an aeroplane moving with a constant horizontal velocity, the path of the stone observed by the pilot will be : **(Agra 2003)**
   (a) parabola (b) straight line (c) hyperbola (d) arc of circle
2. The path of a particle situated on the circumference of the wheel of a moving car will be seen by an observer on earth as : **(Kanpur 2004; Rewa 1992)**
   (a) cycloid (b) circular (c) parabolic (d) straight line
3. If S is an inertial frame of reference, then another frame accelerated with constant acceleration with respect to it will be :
   (a) inertial (b) non-inertial
   (c) inertial or non-inertial (d) neither inertial nor non-inertial
4. Inertial frames are those frames of reference in which a free particle moves : **(Bhopal 1989)**
   (a) along a straight line with a constant speed (b) with a variable speed along a straight line
   (c) with variable speed on a curved path (c) with constant speed on a curved path
5. A block of mass m is placed on a smooth inclined plane of inclination angle θ. What will be the magnitude of the force exerted on the block by the plane ?
   (a) mg (b) $mg/\cos\theta$ (c) $mg\tan\theta$ (d) $mg\cos\theta$
6. A block of mass m is placed on an inclined plane of inclination θ and the whole system is accelerated in such a way that the block does not slide along the inclined plane. The force exerted by the plane on the block will be :
   (a) mg (b) $mg/\cos\theta$ (c) $mg\tan\theta$ (d) $mg\cos\theta$

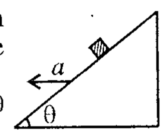


Fig. 1.47.

7. A frame is rotating with an angular velocity $\vec{\omega}$. If $\vec{v}$ is the velocity observed in this frame, then the Coriolis force acts :
   (a) along $\vec{v}$ (b) along $\vec{\omega}$ (c) perpendicular to $\vec{v}$ (d) independent of $\vec{v}$ and $\vec{\omega}$
8. A stone is allowed to fall under gravity from the top of a high tower at the equator. The horizontal displacement of the stone due to the rotation of the earth will be along :
   (a) east (b) west (c) north (d) south
9. The frame attached to any point on earth's surface is : **(Bundelkhand 2001)**
   (a) non-inertial (b) inertial
   (c) neither inertial nor non-inertial (d) Newtonian frame of reference.
10. The motion of one projectile as seen from another projectile will always be : **(Kanpur 2001)**
    (a) straight line (b) parabola (c) ellipse (d) circle
11. The fictitious force acting on a freely falling body of mass 20 kg with reference to a frame moving with a downward acceleration of 6 m/s^2 would be : **(Kanpur 2001)**
    (a) 120 N in the upward direction (b) 120 N in the downward direction
    (c) 76 N in the upward direction (d) 76 N in the downward direction
12. If a body is projected on the earth along south, the particle due to Coriolis force is deflected toward :
    (a) east (b) west (c) vertically up (d) vertically down
13. A freely falling body in southern hemisphere deviates from its true vertical path toward :
    (a) east (b) west (c) north (d) south
14. A Foucault's pendulum is located at the equator. Its plane of oscillation will make a complete revolution in time equal to :
    (a) 24 hrs (b) 0 (c) ∞ (d) $24\sqrt{2}$ hrs
15. What is the direction of the Coriolis force acting on a train, moving from south to north at the latitude 30° N ? **(Agra 2006)**
    (a) Eastward (b) Westward (c) Northward (d) Southward
16. The value of coriolis force on a mass 20g placed at a distance of 10 cm from the axis of rotating frame will be (angular speed of rotation of the frame is 10 rad/sec) : **(Agra 2006)**
    (a) 2×10^4 dyne (b) 4×10^4 dyne (c) 1×10^4 dyne (d) 8×10^4 dyne

ANSWERS

1. (b), **2.** (a) **3.** (b), **4.** (a), **5.** (d), **6.** (b), **7.** (c), **8.** (a), **9.** (a), **10.** (a),
11. (a), **12.** (b), **13.** (a), **14.** (c), **15.** (a), **16.** (b).

2

Conservation of Energy

2·1. Conservation of Energy

A number of conservation laws are known to us; familiar examples are the laws of conservation of energy, linear momentum, angular momentum, charge etc. These laws are usually the consequences of some underlying symmetry in the universe. Conservation laws are something very fundamental in nature. A violation of any one of these laws has never been observed. These laws are very powerful tools for solving the problems in physics and predicting new phenomena.

Energy is a quantity which can be converted from one form to another but it cannot be created or destroyed. In any process, the total energy, *i.e.*, the sum of all energy present in all different forms remains the same. This is known as **principle of conservation of energy**. In this chapter we plan to discuss the conservation of energy related to mechanics, *i.e.*, mechanical energy. This involves the ideas of work, kinetic energy and potential energy.

2·2. Work

We consider a particle P, moving along a curve AB under the action of a variable force $\vec{F}$. Let at any instant the position vector of the particle be $\vec{r}$. If this force $\vec{F}$ displaces the particle through a distance $d\vec{r}$, then the infinitesimal small amount of work dW done by this foce on the particle is defined as

$$dW = \vec{F} \cdot d\vec{r}$$

If the particle is moved from a point A, when position vector is $\vec{r_1}$ to the point B, whose position vector is $\vec{r_2}$, then the total work done by the force on the particle is given by

$$W = \int_A^B \vec{F} \cdot d\vec{r} = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r} \qquad ...(1)$$

If the force makes an angle θ with the tangent to the path at any point, then

$$W = \int_A^B F \cos\theta \, dr \qquad ...(2)$$

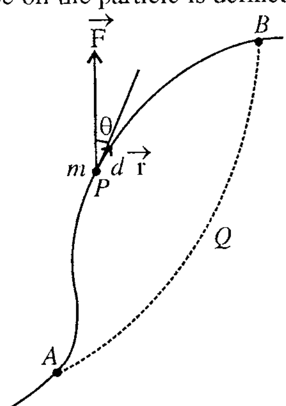


Fig. 2.1.

Remember that the values of $\vec{F}$ and θ may change from point to point of the path of the particle. In general for any vector, which is function of position, an integral of the form $\int_A^B \vec{F} \cdot d\vec{r} = \int_A^B \vec{F} \cos\theta \, dr$ along some path (line) AB is called **line integral.** Since $F \cos\theta$ is the component of force along the path, so *the work may be defined as the line integral of the tangential component of the force, taken over the actual line of motion.*

If the force $\vec{F}$ remains constant and the displacement $\vec{r}$ is along a straight line, then

$$W = \int_A^B \vec{F} \cdot d\vec{r} = \vec{F} \cdot \vec{r} = F \cos\theta \, r \qquad ...(3)$$

This expression for the amount of work done is in accordance with the elementary definition of work, *i.e.*, work is the product of the component of force along the displacement and the distance moved by the particle.

If F_x, F_y and F_z are the rectangular components of the force $\vec{F}$ and $d\vec{r} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$, then from eq. (1), we get

$$W = \int_A^B (F_x\hat{i} + F_y\hat{j} + F_z\hat{k}) \cdot (dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}) = \int_A^B (F_x dx + F_y dy + F_z dz) \qquad ...(4)$$

The rate of doing work (dW/dt) is defined as **power**, *i.e.*,

$$P = \frac{dW}{dt} \qquad ...(5)$$

If $\vec{F}$ is the instantaneous force acting on the particle, then

$$P = \frac{dW}{dt} = \vec{F} \cdot \frac{d\vec{r}}{dt} = \vec{F} \cdot \vec{v} \qquad ...(6)$$

which is the expression for the instantaneous power. The average power during a time interval t is obtained by dividing the total work W by time t so that $P_{av} = W/t$.

2·3. Kinetic Energy : Work-energy Theorem

The energy of a particle is defined as the capacity of doing work and it is measured by the amount of work which it can do in the position in which it is. *The energy due to the speed of a particle is called its kinetic energy.* It is generally represented by the letter K.

According to Newton's second law of motion,

$$\vec{F} = m\frac{d\vec{v}}{dt} \qquad ...(1)$$

where $d\vec{v}/dt$ is the acceleration, produced in a particle of mass m by the force $\vec{F}$.

The amount of work done on the particle, when it is taken from the point A to the point B, is given by

$$W = \int_A^B \vec{F} \cdot d\vec{r} = m\int_A^B \frac{d\vec{v}}{dt} \cdot d\vec{r} \qquad ...(2)$$

Now, $d\vec{r} = \frac{d\vec{r}}{dt}dt = \vec{v}\,dt$ and so $\frac{d\vec{v}}{dt} \cdot d\vec{r} = \left(\frac{d\vec{v}}{dt} \cdot \vec{v}\right)dt = \frac{d}{dt}\left(\frac{1}{2}\vec{v} \cdot \vec{v}\right)dt = d\left(\frac{v^2}{2}\right)$.

$$\therefore \qquad W = \frac{m}{2}\int_A^B d(v^2) = \frac{m}{2}\left[v^2\right]_A^B = \frac{m}{2}(v_B{}^2 - v_A{}^2)$$

where v_A and v_B are the speeds of the particle at the points A and B respectively.

Thus
$$W = \int_A^B \vec{F} \cdot d\vec{r} = \frac{1}{2}mv_B{}^2 - \frac{1}{2}mv_A{}^2 \qquad ...(3)$$

This equation shows that the work done on a free particle depends only upon initial and final speeds of the particle and is *independent of the nature of the force* and the path followed by the particle.

The energy $\frac{1}{2}mv^2$ is called the kinetic energy* of the particle, *i.e.*, $K = \frac{1}{2}mv^2$.

Thus, W = Kinetic energy (K_B) at B – Kinetic energy (K_A) at A = Change in kinetic energy.

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* In case, if a particle has initially speed v and finally it acquires zero speed against a force $\vec{F}$, then the work done by the particle against the forces is $W = \int_A^B \vec{F} \cdot d\vec{r} = -m\int_A^B \frac{d\vec{v}}{dt} \cdot d\vec{r} = -\frac{m}{2}\int_v^0 d(v^2) = \frac{1}{2}mv^2$.

Thus, a particle of speed v has the capacity to do $\frac{1}{2}mv^2$ work. This means that $\frac{1}{2}mv^2$ is the kinetic energy of the particle.

Thus, *the work done by a force on a particle is equal to the change in its kinetic energy.* This is known as **work-energy theorem** and has been represented in equation form by eq. (3).

When the work done on the particle by the external force is positive, its kinetic energy increases. In case, if the work done on the particle by the force is negative, there is a decrease in its kinetic energy. Actually, in such a case the particle does work in moving itself against the force and loses its kinetic energy.

2·4. Conservative Force and Conservative Force Field

A force is said to be conservative if the work done by it in moving a particle from one point to another point depends only on these points and not on the path followed. This means that if a particle is taken from the point A to the point B through APB or AQB paths and if each time the amount of work done $\left(\int_A^B \vec{F} \cdot d\vec{r}\right)$ by the force is the same, then this force is conservative. Thus, for a conservative force

$$\underset{(\text{Path } APB)}{\int_A^B \vec{F} \cdot d\vec{r}} = \underset{(\text{Path } AQB)}{\int_A^B \vec{F} \cdot d\vec{r}} \qquad ...(1)$$

Paths APB and AQB are any two arbitrary paths, joining the points A and B.

The region in which a particle experiences a conservative force is called a **conservative force field.**

Central force is an example of a conservative force : If a force acts on a particle in such a way that it is always directed towards or away from a point and its magnitude depends only upon the distance r from the point, then the force is called a **central force.** Hence if $\vec{r}$ is the position vector of a particle relative to origin (point) O and in case of a central force acting on it, then

$$\vec{F} = F(r)\hat{\mathbf{r}} \qquad ...(2)$$

where $F(r)$ is some function of r. Forces of the type $F(r) = -k/r^n$, (specific case) s-inverse square law force, $F(r) = -k/r^2$ and linear restoring force $F(r) = -kr$, are the examples of central force. Gravitational, electrostatic and elastic forces are such type of forces, *i.e.*, all of these forces are the examples of conservative forces. Experimentally we also know that for these forces, the work done does not depend on the path. In case of earth, the gravitational force, acting on a particle at a distace from its centre around the earth, is a conservative force and the region around the earth in which a particle experiences the gravitational force is a conservative force field.

In **fig. 2.2**, a central force has been shown to act away from (or toward) the centre. In the central force field* any two arbitrary paths APB and AQB have been drawn between two points A and B. Taking centre O, two dashed sectors of circles have been drawn with radii r and $r + dr$ respectively. Now take two points P and Q on APB and AQB paths respectively, lying between the two sectors [**fig. 2.2**]. Let $\vec{F}_1$ and $\vec{F}_2$ be the central forces at P and Q and let $d\vec{r}_1$ and $d\vec{r}_2$ be the displacements of the particle between two sectors along two paths respectively. If θ_1 and θ_2 are the angles between $\vec{F}_1$, $d\vec{r}_1$ and $\vec{F}_2$, $d\vec{r}_2$ respectively, then

$$\vec{F}_1 \cdot d\vec{r}_1 = F_1\, dr_1 \cos\theta_1 \text{ and } \vec{F}_2 \cdot d\vec{r}_2 = F_2\, dr_2 \cos\theta_2$$

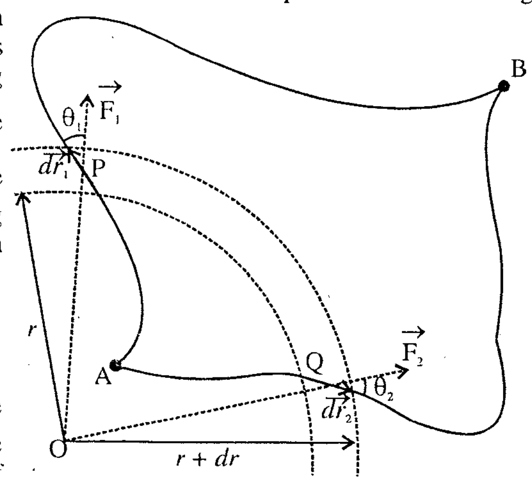

Fig. 2.2

As $\vec{F}_1$ and $\vec{F}_2$ lie at equal distances from the centre O, their magnitudes F_1 and F_2 are **equal.** Also, the separation of the circles measured along the direction of $\vec{F}_1$ is equal to the separation measured along $\vec{F}_2$, hence

* The region in which the force acting on a particle is directed away or towards a centre is called central force field.

the projections of $d\vec{r}_1$ and $d\vec{r}_2$ on $\vec{F}_1$ and $\vec{F}_2$ respectively are equal, *i.e.*, $dr_1 \cos\theta_1 = dr_2 \cos\theta_2$. Thus

$$F_1\, dr_1 \cos\theta_1 = F_2\, dr_2 \cos\theta_2 \quad \text{or} \quad \vec{F}_1 \cdot d\vec{r}_1 = \vec{F}_2 \cdot d\vec{r}_2 .$$

Similarly, the same argument can be given for every pair of segments of the paths APB and AQB so that

$$\underset{(\text{Path } APB)}{\int_A^B \vec{F}_1 \cdot d\vec{r}_1} = \underset{(\text{Path } AQB)}{\int_A^B \vec{F}_2 \cdot d\vec{r}_2} \qquad ...(3)$$

This means that the work done $W = \int_A^B \vec{F} \cdot d\vec{r}$ by a central force is independent of the path. Hence *the central force is conservative.*

For conservative forces, the work done around a closed path is always zero : The amount of work done, when a particle reaches from the point A to the point B in a conservative force field, is

$$W_{(A\to B)} = \int_A^B \vec{F} . d\vec{r} = -\int_B^A \vec{F} \cdot d\vec{r} = -W_{(B\to A)} \qquad ...(4)$$

Hence, if the particle moves in the same conservative force field from the point B to the point A, then the amount of work done is

$$W_{(B\to A)} = -W_{(A\to B)}$$

Thus, the total amount of work done on the particle by a conservative force, when it moves from A to B and then from B to A, *i.e.*, along a closed path, is

$$\oint F \cdot d\vec{r} = W_{(A\to B)} + W_{(B\to A)} = W_{(A\to B)} - W_{(A\to B)} = 0 \qquad ...(5)$$

where $\oint$ sign represents the integral over the closed path.

The points A and B are any two arbitrary points in the conservative force field. Hence we see that *for conservative forces the work done around any closed path is zero.* In fact, this is a very important characteristic of conservative forces. Hence we can define a conservative force as follows :

A force is said to be conservative, if the work done on a particle is zero, when the particle moves under such a force around a closed path.

Theorem 1. *Show that a constant force is an example of conservative force.*

Proof : In a constant and uniform force field, *i.e.*, for $\vec{F}$ = constant, the work done around a closed path is given by

$$\oint \vec{F} \cdot d\vec{r} = \vec{F} \cdot \oint d\vec{r} = 0$$

because for a closed curve $\oint d\vec{r} = 0$. Thus a uniform constant force field is always conservative.

Theorem 2. *Lorentz force* $\vec{F} = q(\vec{v} \times \vec{B})$ *is a conservative force.*

Proof : If a charge q is moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$, then the force acting on this particle is called the Lorentz force and it is given by

$$\vec{F} = q(\vec{v} \times \vec{B})$$

The direction of this force is perpendicular to both $\vec{v}$ and $\vec{B}$. For such a force, the work done on the particle is

$$W = \int \vec{F} \cdot d\vec{r} = \int q(\vec{v} \times \vec{B}) . \frac{d\vec{r}}{dt} dt = q\int (\vec{v} \times \vec{B}) \cdot \vec{v}\, dt = q\int \vec{v} \cdot (\vec{v} \times \vec{B}) dt = q\int (\vec{v} \times \vec{v}) \cdot \vec{B}\, dt = 0$$

because in a scalar triple product the positions of dot and cross can be interchanged and $\vec{v} \times \vec{v} = 0$. Thus the work done on a charged particle due to Lorentz force along any path is always zero and does not depend on the path followed. Thus Lorentz force is a conservative force.

2·5. Potential Energy

The potential energy of a particle is defined as the energy stored in it due to its position. It is measured by that amount of work which this particle can do, when it moves from its present position to some standard position. It is generally represented by the letter U.

Let us consider a particle in a conservative force field (*e.g.*, gravitational, electrostatic etc.). The conservative force ($\vec{F}$), which is acting on the particle can be just counterbalanced by a suitable applied force $\vec{F}_a$ *i.e.*,

$$\vec{F} + \vec{F}_a = 0 \quad \text{or} \quad \vec{F}_a = -\vec{F} \qquad \text{...(1)}$$

Now, if the particle is moved very slowly against the conservative force from one position to another without developing the kinetic energy, the work done in moving the particle will appear as its potential energy. *We define the difference in potential energy between two positions of the particle as equal to the work done by such an applied force on the particle in moving it from the* 1*st position to* 2nd *position, i.e.*,

$$U_2 - U_1 = \int_1^2 \vec{F}_a \cdot dr \qquad \text{...(2a)}$$

But $\qquad \vec{F}_a = -\vec{F}$,

$$\therefore \qquad U_2 - U_1 = -\int_1^2 \vec{F} \cdot dr \qquad \text{...(2b)}$$

Remember that the definition of difference of potential energy $U_2 - U_1$ in eq. (2b) is true only for conservative forces (not for non-conservative forces, described later).

Now, if the work done on the particle by conservative force $\vec{F}$ is represented by W, then

$$U_2 - U_1 = -W = -\int_1^2 \vec{F} \cdot d\vec{r} \qquad \text{...(3)}$$

Hence the difference of potential energy at two points of the field is the negative of the work done by the conservative force. When positive work is done by the conservative force, the potential energy is reduced and for negative work, the potential energy increases.

The potential energy at a point is obtained by assigning zero potential energy to any convenient point and then calculating the difference of potential energies. Now, if *we assume the* 1*st position at* ∞ *and the potential energy to be* **zero** *there,* then the potential energy of the particle at a point distant $\vec{r}$ is given by

$$U(\vec{r}) = -\int_\infty^{\vec{r}} \vec{F} \cdot d\vec{r} \qquad \text{...(4)}$$

or

$$U(\vec{r}) = \int_\infty^{\vec{r}} \vec{F}_a \cdot d\vec{r} \qquad \text{...(5)}$$

Thus, *the potential energy of a particle at a point* $\vec{r}$ *is defined as the amount of work done by an applied force in moving the particle from infinity to that point.*

It is often convenient to choose the reference or standard position to be that at which the force acting on the particle is zero. Thus the force exerted by a spring is zero when the spring has its normal unstretched length; so we generally say that in this position, the potential energy is zero. In case of the gravitational and electrostatic forces, the particle experiences no force of interaction, when it is at infinite distance. Hence in such cases we assign **zero potential energy at** ∞. In case of the earth, the gravitational force (mg) is constant near its surface and hence we find it convenient to take the zero of potential energy at the earth's surface, not at infinity.

Force as the negative gradient of potential energy : From eq. (4), we have

$$U(\vec{r}) = -\int_{\infty}^{r} \vec{F}\cdot d\vec{r}$$

Expressing $\vec{F}$ and $d\vec{r}$ in rectangular components, we have

$$U(\vec{r}) = \int (F_x\hat{\mathbf{i}} + F_y\hat{\mathbf{j}} + F_z\hat{\mathbf{k}})\cdot(dx\,\hat{\mathbf{i}} + dy\,\hat{\mathbf{j}} + dz\,\hat{\mathbf{k}})$$

or
$$U(\vec{r}) = -\int F_x dx - \int F_y dy - \int F_z dz \qquad ...(6)$$

Differentiating this equation partially with respect to x, y, z, we get

$$F_x = -\frac{\partial U}{\partial x},\ F_y = -\frac{\partial U}{\partial y},\ F_z = -\frac{\partial U}{\partial z}$$

Hence the force
$$\vec{F} = \hat{\mathbf{i}}F_x + \hat{\mathbf{j}}F_y + \hat{\mathbf{k}}F_z = -\hat{\mathbf{i}}\frac{\partial U}{\partial x} - \hat{\mathbf{j}}\frac{\partial U}{\partial y} - \hat{\mathbf{k}}\frac{\partial U}{\partial z}$$

or
$$\vec{F} = -\left(\hat{\mathbf{i}}\frac{\partial U}{\partial x} + \hat{\mathbf{j}}\frac{\partial U}{\partial y} + \hat{\mathbf{k}}\frac{\partial U}{\partial z}\right) = -\text{ grad } U \quad \text{or} \quad -\nabla U \qquad ...(7)$$

Thus, a *conservative force is the negative gradient of potential energy U.*

Only for one dimension, $F = -\dfrac{\partial U}{\partial x}$ or $U = -\int F\,dx$ ...(8)

For a conservative force curl $\vec{F} = 0$: A conservative force can be represented as $\vec{F} = -\vec{\nabla}U$.

$$\therefore \qquad \text{curl } \vec{F} = \vec{\nabla}\times\vec{F} = -\vec{\nabla}\times\vec{\nabla}U = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = -\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial U}{\partial x} & \frac{\partial U}{\partial y} & \frac{\partial U}{\partial z} \end{vmatrix}$$

$$= -\left[\hat{\mathbf{i}}\left(\frac{\partial^2 U}{\partial y\partial z} - \frac{\partial^2 U}{\partial z\partial y}\right) + \hat{\mathbf{j}}\left(\frac{\partial^2 U}{\partial z\partial x} - \frac{\partial^2 U}{\partial x\partial z}\right) + \hat{\mathbf{k}}\left(\frac{\partial^2 U}{\partial x\partial y} - \frac{\partial^2 U}{\partial y\partial x}\right)\right]$$

But U is taken as a perfect differential, *i.e.*, $\dfrac{\partial^2 U}{\partial y\partial z} = \dfrac{\partial^2 U}{\partial z\partial y}$ etc.

Hence, $\qquad$ curl $\vec{F} = 0$ ...(9)

Thus, *the curl of a conservative force is zero.*

Theorem 1. *Prove that a central force is a conservative force.*

Proof : For a central force,

$$\vec{F} = F(r)\hat{\mathbf{r}}$$

where $F(r)$ is a scalar function of distance r and $\hat{\mathbf{r}} = \vec{r}/r$.

$$\text{Now,curl } \vec{F} = \vec{\nabla}\times\hat{\mathbf{r}}F(r)^* = \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{\mathbf{r}} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ F(r) & 0 & 0 \end{vmatrix} = \frac{1}{r^2\sin\theta}\left[r\hat{\theta}\frac{\partial F(r)}{\partial\phi} - r\sin\theta\hat{\phi}\frac{\partial F(r)}{\partial\theta}\right]$$

Since $F(r)$ is the function of distance r only and is independent of θ and ϕ,

$$\text{curl } \vec{F} = 0.$$

Thus a central force is always conservative.

As special cases, $F(r) = -k/r^n$, $F(r) = -k/r^2$, $F(r) = -kr$, all are the examples of central force and they are conservative forces.

Theorem 2. *Prove that a linear restoring force is a conservative force.*

Proof : *A force is said to be a linear restoring force, if it is proportional to the displacement of the particle from a fixed point and is directed opposite to the displacement.* This means that such a force is the linear function of the displacement, having direction opposite to it.

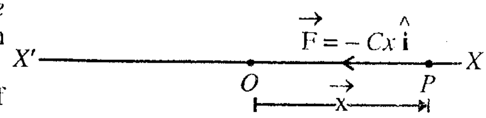

Fig. 2.3.

Let a particle move along X-axis in the influence of linear restoring force $\vec{F}$ [**fig. 2.3**]. If the fixed point is the origin O and at any time t the displacement is $\vec{x} = x\hat{i}$ (where $\hat{i}$ is the unit vector along X-axis), then according to the definition,

$$\vec{F} = -Cx\hat{i} \quad \text{or} \quad F = -Cx \qquad ...(10)$$

where C is some positive constant, called the **force constant** or **spring factor.** The negative sign indicates that the restoring force acts opposite to the displacement from the origin at $x = 0$.

For sufficiently small displacements, a linear restoring force may be produced by a stretched or compressed spring.

If $\vec{F} = -Cx\hat{i}$ is a conservaive force, then for it curl $\vec{F} = 0$, *i.e.*,

$$\text{curl } \vec{F} = -\vec{\nabla} \times Cx\hat{i} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ Cx & 0 & 0 \end{vmatrix} = 0$$

Thus *a linear restoring force is a conservative force.*

2·6. Conservation of Mechanical Energy for a Particle

Let a particle of mass m move from one point to another under the influence of a conservative force. Then according to eq. (3) (Art. 2·3) and eq. (2b) (Art. 2·5) the amount of work done by the force on the particle will be given by

$$W = \int_1^2 \vec{F} \cdot d\vec{r} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = U_1 - U_2 \qquad ...(1)$$

where v_1 and v_2 are the speeds of the particle at the first and second points respectively.

Eq. (1) expresses that in a conservative force field, if the kinetic energy of the particle increases in moving from one point to another, then its potential energy should decrease.

From eq. (1), we get

$$\frac{1}{2}m_1v_1^2 + U_1 = \frac{1}{2}m_2v_2^2 + U_2 \qquad ...(2)$$

Writing the kinetic energy $\frac{1}{2}mv^2 = K$, we have

$$K_1 + U_1 = K_2 + U_2. \qquad ...(3)$$

This means that *the sum of kinetic and potential energies of a particle remains constant at any point of a conservative force field.* This is known as the **law of conservation of mechanical energy.** Thus *if a particle moves under the action of such a force so that its mechanical energy (potential energy and kinetic energy) remains constant, this force is said to be a conservative force.*

* In spherical coordinates, curl of any vector $\vec{A}$ is given by

$$\text{curl } \vec{A} = \vec{\nabla} \times \vec{A} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \partial/\partial r & \partial/\partial\theta & \partial/\partial\phi \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

The total energy (E) of a particle is the sum of its kinetic energy (K) and potential energy (U), *i.e.*, $E = K + U$ and hence the total energy of a particle remains constant in a conservative force field. Sometimes it is convenient to call the quantity $E = K + U$ as the **energy function**. This energy function is invariant with respect to change in time.

An important note on potential energy : If a particle interacts with another particle or a system of particles and experiences similar to gravitational or electrostatic forces, then such interaction forces are conservative forces. If this one particle system moves in a conservative force field due to an external source, then

$$U_2 - U_1 = -\int_1^2 \vec{F} \cdot d\vec{r} = -W = -(K_2 - K_1) = K_1 - K_2$$

where U_1 and K_1 are the potential and kinetic energies in the first position and U_2 and K_2 in the second position. Hence in a conservative force field,

$$K_1 + U_1 = K_2 + U_2 = E \text{ (total mechanical energy)} \qquad ...(4)$$

i.e., total mechanical energy of the particle remains constant.

In fact, potential energy is not the property of a particle (or body) of its own, but it is the property of the system; this system consists of the particle (or body) and another body due to which a force is acting on the particle. The potential energy of a body above the surface of the earth is shared by the body and earth both. When the body is allowed to fall, the body and earth move towards each other. However, due to the large mass of the earth, its speed is negligible and practically all the kinetic energy developed is in the share of the body. This is why all the potential energy or kinetic energy is assumed to be associated with the body. In case, if there is a system of two masses whose masses are comparable with each other, the potential energy will be said to be associated with the system of two bodies, not with any one of them.

Hence *potential energy is that energy which is stored in a body or particle due to its position in a force-field or stored in the system due to its configuration.* When the configuration of the system changes, its potential energy changes. Remember that when we say the potential energy of a system, the conservative forces are internal.

Ex. 1. *If a force $F = Ax + Bx^2$ acts parallel to X-axis on an object and moves it from $x = 1$ to $x = 2$, calculate the work done.*

Sol. Work done $= \int_1^2 F_x\, dx = \int_1^2 (Ax + Bx^2)\, dx = \left[\frac{Ax^2}{2} + \frac{Bx^3}{3}\right]_1^2$

$$= \frac{A}{2}[2^2 - 1^2] + \frac{B}{3}[2^3 - 1^3] = \frac{3}{2}A + \frac{7}{3}B.$$

Ex. 2. *A particle moves along half the circumference of a circle of 1 metre radius. Calculate the work done if the force at any point is inclined at 60° to the tangent at that point and has 5 newtons magnitude.*

Sol. Work done $= \int_A^B \vec{F} \cdot d\vec{s} = \int_A^B F \cos\theta\, ds$

Here, $F = 5$ newtons and $\cos\theta = \cos 60° = \frac{1}{2}$

$\therefore$ Work $= \int_A^B 5 \cdot \frac{1}{2}\, ds = \frac{5}{2}$ (length s of the semicircle) $= \frac{5}{2}(\pi r) = \frac{5}{2} \times 3.14 \times 1$ = **7.85 joules.**

Ex. 3. *A body of mass m is revolving in a circle of radius r under the influence of a centripetal force $F = -k/r^2$, where k is a constant. Determine the total energy of the body.* **(Ujjain 1999, 95)**

Sol. Centripetal force, $\frac{mv^2}{r} = \frac{k}{r^2}$, $\therefore$ Kinetic energy $\frac{1}{2}mv^2 = \frac{k}{2r}$

Potential energy, $U = -\int_\infty^r \vec{F} \cdot d\vec{r} = \int_\infty^r \left(\frac{k\hat{r}}{r^2}\right) . d\vec{r} = \int_\infty^r \frac{k}{r^2} dr = -\frac{k}{r}$

Hence Total energy, E = Kinetic energy + Potential energy

$$= \frac{k}{2r} - \frac{k}{r} = -\frac{k}{2r}.$$

Ex. 4. *The potential energy of a particle is represented as*

$$U = \frac{1}{2}k(x^2 + y^2)$$

Determine the force acting on the particle. **(Rewa 1993)**

Sol. Force $\vec{F} = -\vec{\nabla}U = -\hat{i}\frac{\partial U}{\partial x} - \hat{j}\frac{\partial U}{\partial y} - \hat{k}\frac{\partial U}{\partial z}$

Here, $U = \frac{1}{2}k(x^2 + y^2)$

$\therefore$ $\frac{\partial U}{\partial x} = kx, \quad \frac{\partial U}{\partial y} = ky \quad \text{and} \quad \frac{\partial U}{\partial z} = 0$

Hence $\vec{F} = -kx\hat{i} - ky\hat{j} \quad \text{or} \quad \vec{F} = -k(x\hat{i} + y\hat{j})$

Ex. 5. (a). *If* $\vec{F} = (2xy + x^2)\hat{i} + x^2\hat{j} + 2xz\hat{k}$, *then show that it is conservative.* **(Indore 1990)**

(b) *Calculate the amount of work done by the above force in moving a particle form* (0, 1, 2) *to* (5, 2, 7). **(Agra 1998)**

Sol. (a) For a conservative force curl $\vec{F} = 0$

$$\text{Here, curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^2 & x^2 & 2xz \end{vmatrix}$$

$$= \hat{i}\left[\frac{\partial}{\partial x}(2xz) - \frac{\partial}{\partial z}(x^2)\right] + \hat{j}\left[\frac{\partial}{\partial z}(2xy + z^2) - \frac{\partial}{\partial x}(2xz)\right] + \hat{k}\left[\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(2xy + z^2)\right]$$

$$= 0 + \hat{j}(2z - 2z) + \hat{k}(2x - 2x) = 0.$$

Thus, $\vec{F} = (2xy + x^2)\hat{i} + x^2\hat{j} + 2xz\hat{k}$ is a conservative force.

(b) Work done $= \int_A^B \vec{F}\cdot d\vec{r} = \int_A^B (F_x dx + F_y dy + F_z dz) = \int_A^B [(2xy + z^2)\,dx + x^2 dy + 2xz\,dz]$

$$= \int_A^B [(2xy\,dx + x^2\,dy) + (z^2\,dx + 2xz\,dz)] = \int_A^B [d(x^2y) + d(z^2x)]$$

$$= \int_A^B d(x^2y + z^2x) = \left[x^2y + z^2x\right]_{A(0,1,2)}^{B(5,2,7)} = [5\times5\times2 + 7\times7\times5 - 0 - 0]$$

$$= 50 + 245 = \mathbf{295\ units.}$$

Ex. 6. *At an instant, the position vector of a particle, moving with constant angular velocity, is given by*

$$\vec{r} = r\cos\theta\,\hat{i} + r\sin\theta\,\hat{j} \text{ with } \theta = \omega t$$

Prove that the force acting on the particle is conservative.

Sol. $\vec{r} = r\cos\theta\,\hat{i} + r\sin\theta\,\hat{j}$

Unit vector along $\vec{r}$ is $\hat{r} = \frac{\vec{r}}{r} = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$

Now, velocity $\vec{v} = \frac{d\vec{r}}{dt} = -r\sin\theta\frac{d\theta}{dt}\hat{i} + r\cos\theta\frac{d\theta}{dt}\hat{j} = -r\omega\sin\theta\,\hat{i} + r\omega\cos\theta\,\hat{j}$

where $\frac{d\theta}{dt} = \omega$, because $\theta = \omega t$.

Hence, acceleration, $\vec{a} = \frac{d\vec{v}}{dt} = -r\,\omega\cos\theta\frac{d\theta}{dt}\hat{i} - r\,\omega\sin\theta\frac{d\theta}{dt}\hat{j}$

or $\vec{a} = -\omega^2 r\,(\cos\theta\,\hat{i} + \sin\theta\,\hat{j}) = -\omega^2\vec{r} = -\omega^2 r\hat{r}$

Thus the force, $\vec{F} = m\vec{a} = -m\omega^2 r\hat{r}$

This force is a central force and hence conservative.

EXERCISE 2·1

1. Find the work done by a force $F = F_0 + kx$ acting parallel to X-axis on an object which moves along X-axis from x_1 to x_2.
$\left[\textbf{Ans.}\ (x_2 - x_1)\left\{F_0 + k\frac{x_2 + x_1}{2}\right\}\right]$
2. Calculate the amount of work done by a force $F = kx^2$ acting on a particle at an angle 60° to displace it from x_1 to x_2 along X-axis.
$\left[\textbf{Ans.}\ \frac{k}{6}(x_2^{\,2} - x_1^{\,3})\right]$
3. If the force acting on a particle at any point (x, y, z) is $(5x\,\hat{i} + xy\,\hat{j} + z\,\hat{k})$, how much work is done, when the particle moves from the point (5, 2, 1) to the point (5, 3, 2) ?
[**Ans.** 14 units.]
4. A constant force of 5 newton acts on a body of mass 0.5 kg for 2 seconds. If the initial velocity of the body is 4 m/sec in the direction of the force, find (i) the work done by the force, (ii) the final kinetic energy, (iii) the power developed, and (iv) the increase in kinetic energy.
[**Ans.** (i) 140 joule, (ii) 144 joule, (iii) 70 joule/sec, (iv) 140 joule.]
5. If the magnitude of the force of attraction between two particles of masses m_1 and m_2 is given by
$$F = -K\frac{m_1 m_2}{r^2}$$
where K is a constant and r is the distance between the particles, determine (a) the potential energy function, (b) the work required to increase the separation of masses from $r = r_0$ to $r = r_0 + R$.
[**Ans.** (a) $U(r) = -\frac{Km_1m_2}{r}$, if $U_{(\infty)} = 0$; (b) $\frac{Km_1m_2R}{r_0(r_0+R)}$.]
6. The potential energy of a body is given by
$$U = 40 + 6x^2 - 7xy + 32z \text{ joule}$$
Find the formula for the force acting on the particle. What is the force at the point (– 2, 0, 5)
[**Ans.** $\vec{F} = (-12x+7y)\,\hat{i} + (7x-16y)\,\hat{j} - 32\hat{k}\,n;\ 24\,\hat{i} - 14\,\hat{j} - 32\hat{k}\,n$]
7. (i) Show that the force
$$\vec{F} = (2xy + yz^2)\,\hat{i} + (x^2 + xz^2)\,\hat{j} + xyz\,\hat{k} \text{ is conservative.}$$
(ii) Show that the force $\vec{F} = yz\,\hat{i} + zx\,\hat{j} + xy\,\hat{k}$ is conservative. **(Allahabad 1978)**
(iii) The position of a moving particle at any instant is given by $\vec{r} = A\cos\theta\,\hat{i} + A\sin\theta\,\hat{j}$.
Show that the force acting on it is a conservative one. Also calculate the total energy of the particle. **(Allahabad 1990)**
(iv) If $\vec{F} = \cos y\,\hat{i} - x\sin y\,\hat{j} - \cos z\,\hat{k}$, prove that the force-field is conservative.

8. You are given a force $\vec{F} = (y^2 - x^2)\,\hat{i} + 3xy\,\hat{j}$. Compute the line integral from the point (0, 0) to the point (x_0, y_0) along the path made up of two straight sections (0, 0) to $(x_0, 0)$ and $(x_0, 0)$ to (x_0, y_0). Compare with the result you get by taking other two sides of the rectangle as the path of integration. Is the force conservative ?
[**Ans.** Non-conservative.]

9. Yukawa potential energy is given by $U = -\frac{r_0}{r} U_0\, e^{-r/r_0}$.
What is the force corresponding to this energy ? Is it a conservative force ?
[**Ans.** $\vec{F} = -\frac{r_0 U_0}{r^3}\left(1 + \frac{r}{r_0}\right) e^{-r/r_0}\,\hat{\mathbf{r}}$; Yes.]

10. Show that the force $\vec{F} = \frac{k}{r^n}\hat{\mathbf{r}}$ is a conservative force, where k and n are constants.

[**Hint :** $\quad \text{curl } \vec{F} = \vec{\nabla} \times \frac{k}{r^n}\hat{\mathbf{r}} = \vec{\nabla} \times \frac{k\,\vec{\mathbf{r}}}{r^{n+1}} \qquad \left(\because \ \hat{\mathbf{r}} = \frac{\vec{\mathbf{r}}}{r}\right)$

But $\quad \vec{\mathbf{r}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$; $r = (x^2 + y^2 + z^2)^{1/2}$ and hence, $r^{n+1} = (x^2 + y^2 + z^2)^{\frac{n+1}{2}}$

Now, $\quad \text{curl } \vec{F} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{kx}{(x^2+y^2+z^2)^{\frac{n+1}{2}}} & \frac{ky}{(x^2+y^2+z^2)^{\frac{n+1}{2}}} & \frac{kz}{(x^2+y^2+z^2)^{\frac{n+1}{2}}} \end{vmatrix} = 0$]

2·7. Non-Conservative Forces : General Law of Conservation of Energy

A force is said to be non-conservative, if the work done by that force on a particle that moves between two points depends on the path taken between those points.

Frictional and viscous forces are the examples of non-conservative forces. In case of such forces, if a particle moves from A to B and then from B to A, the work done around the closed path is not zero, but the particle itself loses its kinetic energy along both paths. Thus for non-conservative force,

$$\oint \vec{F} \cdot d\vec{r} = W_{(A \to B)} + W_{(B \to A)} \neq 0. \qquad \text{...(1)}$$

For example, if we push a block over a rough table between any two points by various paths, the distance traversed varies and accordingly the work done by the frictional force and hence the work done depends on the path. Hence we cannot define a potential energy like function for non-conservative force.

Difference between conservative and non-conservative forces :

(1) When a particle moves under the influence of a conservative force, the work done by it in moving the particle from one point to another depends only on these points and is independent of the path followed, while in case of a non-conservative force, the work done on the particle depends on the length of the path. For example, for gravitational (conservative) force, the work done is independent of the paths between two points but when frictional (non-conservative) force is present the work done has different values for different paths between two points.

(2) The work done under the influence of conservative force is equal to the difference in potential energies $(U_1 - U_2)$ in initial and final positions, while for non-conservative forces we can not define potential energy or difference of potential energies $(U_1 - U_2)$.

(3) The work done on a particle is zero, when it moves along a closed path under a conservative force. However for a non-conservative force, the work done on the particle along a closed path is non zero.

(4) The mechanical energy (sum of kinetic and potential energies) of a particle, moving under the

influence of a conservative force, remains constant, while it does not remain constant in case of a non-conservative force.

Work done by conservative, non-conservative and external forces and work-energy theorem : For conservative forces, *the sum of potential and kinetic energies, i.e., mechanical energy is constant, i.e.,*

$$K + U = E, \text{ a constant (mechanical energy)}$$

or
$$\Delta K + \Delta U = 0 = \Delta E \quad \text{(for conservative forces)} \qquad ...(2)$$

But the work done by conservative forces is equal to negative of the increase in potential energy, *i.e.*, $W_c = -\Delta U = -(U_2 - U_1)$.

Now let us suppose that in addition to the conservative forces, a non-conservative force due to friction acts on the particle. If W_{nc} is the work done by the frictional force and W_c by conservative force, then according to work-energy theorem, the work done $(W_{nc} + W_c)$ on the particle should be equal to the change (ΔK) in the kinetic energy of the particle. Thus,

$$W_{nc} + W_c = \Delta K = K_2 - K_1 \qquad ...(3)$$

If in addition to frictional and conservative forces, an external force is also acting on the particle, then according to work-energy theorem,

$$W_{ex} + W_{nc} + W_c = \Delta K = K_2 - K_1 \qquad ...(4)$$

Remember that the work done due to all the forces, acting on the particle, is equal to the change in its kinetic energy.

But
$$W_c = -\Delta U = U_1 - U_2,$$

then
$$W_{ex} + W_{nc} = \Delta K + \Delta U = \Delta E$$

or
$$W_{ex} + W_{nc} = E_2 - E_1 \qquad ...(5)$$

Hence *if external and frictional forces are acting on the particle, then the work done by them is equal to the change in the mechanical energy.*

When external force is not acting, the work done by the non-conservative or frictional force is given by
$$W_{nc} = E_2 - E_1 \qquad ...(6)$$

This equation shows that if a frictional force acts on a particle, the total mechanical energy is not constant, but changes by the amount of work done by the frictional force.

But W_{nc}, which is the work done by friction on the particle, is always negative, hence we see from eq. (6) that the final mechanical energy E_2 $(= K_2 + U_2)$ is less than the initial mechanical energy E_1 $(= K_1 + U_1)$. The lost mechanical energy appears in the form of heat. The heat energy developed is exactly equal to the mechanical energy dissipated. This heat energy, so produced, is equal to the work done by the particle. If Q is the heat energy developed, then evidently $W_{nc} = -Q$ and hence we get

$$\Delta E + Q = 0 \qquad ...(7)$$

This shows that there is no change in the sum of the mechanical energy and heat energy of the system so far conservative and frictional forces are acting on the system. Therefore, the total energy may again be said to be **conserved.**

Similarly, if other non-conservative forces are acting, then the mechanical energy lost is converted into other forms of energy, therefore, *the total energy in general is conserved, i.e.,*

$$\Delta E + Q + \text{change in other forms of energy} = 0. \qquad ...(8)$$

This is the *general law of conservation of energy.*

Thus, the total energy is always conserved while kinetic and potential energies are conserved only when conservative forces act.

In case, non-conservative forces, *e.g.*, friction are not acting on the particle and in addition to the conservative force, an external force is acting then from eq. (5)

$$W_{ex} = E_2 - E_1 \qquad ...(9)$$

Hence *the work done by external force is equal to the change in mechanical energy of the particle.*

2·8. One Dimensional Conservative Systems

Here we will discuss two important one dimensional conservative systems :

(1) Spring-mass syatem and (2) Body-earth system.

In case of the first system spring force (linear restoring force) between the spring and mass and in case of second system, the gravitational force between a body and earth both are the conservative forces.

(1) Spring-mass system : For sufficiently small displacements, a linear restoring force may be produced by a stretched or compressed spring.

Let us consider a massless spring whose one end is fixed to a rigid wall and other end is attached to a particle of mass m. Let the system be placed horizontally on a frictionless table [**fig. 2.4**]. Let us take the equilibrium position of the particle to be the origin $x = 0$ and the line along the length of the spring as X-axis. Now, if the spring is extended or compressed slowly by the application of an external force, work will be done on the system (spring + particle) against the restoring force $\vec{F}$ and this work will become the potential energy of the system. If we assume the potential energy of the system in the undisplaced position to be zero, then the potential energy at the displacement x of the particle will be given by

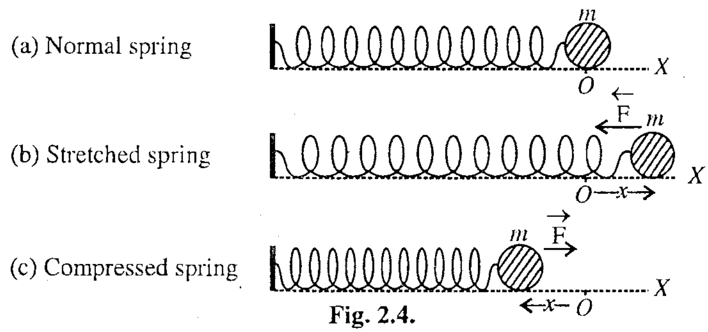

Fig. 2.4.

$$U = -\int_0^x \vec{F} \cdot d\vec{r} = -\int_0^x F\,dx \qquad \text{[from eq. (2b), Art (2·5)]}$$

But $\qquad F = -Cx.$

$$\therefore \qquad U = \int_0^x Cx\,dx = \frac{1}{2}Cx^2 \qquad ...(1)$$

If a graph between U and x is plotted, it will be **parabola [fig. 2.5]**.

If the maximum displacement of the particle is a, then the potential energy developed in the system is

$$U = \frac{1}{2}Ca^2$$

Now, if the particle of mass m be released from the **rest** position at $x = a$, let us calculate the kinetic energy of the particle at any displacement x.

On releasing the particle, it will move under restoring force, which is **conservative** in nature, towards the origin. So the total energy E of the system will remain constant. Initially, total energy E is given by

$$E = K + U = 0 + \frac{1}{2}Ca^2 = \frac{1}{2}Ca^2.$$

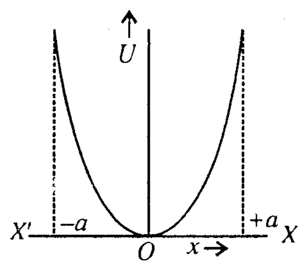

Fig. 2.5.

If v be the speed of the particle at the displacement x, then

kinetic energy of the system $K = \frac{1}{2}mv^2$; potential energy of the system $U = \frac{1}{2}Cx^2$

Hence, $\qquad$ total energy $E = \frac{1}{2}mv^2 + \frac{1}{2}Cx^2$

Using the law of conservation of energy, we have

$$\frac{1}{2}Ca^2 = \frac{1}{2}mv^2 + \frac{1}{2}Cx^2 \qquad ...(2)$$

or

$$\frac{1}{2}mv^2 = \frac{1}{2}C(a^2 - x^2), \text{ whence } v = \pm\sqrt{\frac{C}{m}(a^2 - x^2)} \qquad ...(3)$$

This equation gives the kinetic energy and the velocity of the particle at any displacement x.

When the particle reaches the origin $x = 0$, its kinetic energy and hence the velocity (v_0) will be maximum. Thus,

$$\frac{1}{2}mv_0^2 = \frac{1}{2}Ca^2 \text{ and } v_0 = \pm\sqrt{\frac{C}{m}}\,a \quad ...(4)$$

This means that the maximum potential energy $\left(\frac{1}{2}Ca^2\right)$ of the particle with respect to the origin converts entirely in the form of kinetic energy $\left(\frac{1}{2}mv_0^2\right)$ at $x = 0$.

(2) Body-earth system–Force of gravity : We know that the radius of the earth is nearly 6,400 km. Hence we can have practically the value of acceleration due to gravity (g) to be the same for a height of few kilometers from the earth's surface.

For body-earth system, we take Z-axis of the coordinate system vertically up with origin at the earth's surface. We assume the surface of the earth as standard position so that at $z = 0$, $U_{(0)} = 0$. The position of a body under consideration will be taken close to the surface of the earth so that the acceleration due to gravity acting on the body may be taken as constant. If the height of the body above the earth's surface is z, then its potential energy will be

$$U_{(z)} = -\int_0^z \vec{F}\cdot d\vec{r} = -\int_0^z (-mg\hat{k})\cdot dz\hat{k} \quad i.e., \ U_{(z)} = \int_0^z mg\,dz = mgz \quad ...(5)$$

where $\vec{F} = m\vec{g} = -mg\hat{k}$ is the force of gravity on the body and $\vec{g} = -g\hat{k}$ is the acceleration due to gravity near the surface of the earth.

Therefore the work done in carrying out the body upto a height z will be mgz. In this position the capacity of doing the work is mgz and hence its potential energy is mgz. This satisfies the relation $F = -\frac{\partial U}{\partial z} = -mg$ for a conservative force.

Let the body be thrown up with initial velocity v_0. As the force of gravity is a conservative force, hence from the law of conservation of mechanical energy,

Initial mechanical energy (at $z = 0$) = mechanical energy at z height

i.e.,
$$\frac{1}{2}mv_0^2 + 0 = \frac{1}{2}mv^2 + mgz$$

where v is the speed of the body at a height z.

Hence
$$\frac{1}{2}mv_0^2 - \frac{1}{2}mv^2 = mgz \quad ...(6)$$

Thus the loss in the kinetic energy of the body is equal to the work done by the body on the gravitational field and will exist in the field. This is the potential energy of the body or body-earth system. Here the work done on the body by gravitational field is $-mgz$. At the maximum height h, the potential energy will be mgh and kinetic energy zero, *i.e.,*

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + mgz = 0 + mgh \quad \text{or} \quad \frac{1}{2}mv_0^2 = mgh \quad ...(7)$$

When the body comes back, gravitational field will do the work on the body and its kinetic energy will increase and on reaching the earth's surface, the kinetic energy of the body will be equal to its initial kinetic energy $\frac{1}{2}mv_0^2$.

Ex. 1. *A body of mass 4 kg slides on a horizontal frictionless table with a speed of 2 m/sec. It is brought to rest in compressing a spring in its path. By how much is the spring compressed if the force constant of the spring is 64 kg/m.*

Sol. Kinetic energy of the body $= \frac{1}{2}mv^2 = \frac{1}{2} \times 4 \times 2^2 = 8$ joule.

Let the spring be compressed through a distance x so that the body loses its all kinetic energy and comes to rest. Obviously, the lost kinetic energy will be stored in the form of potential energy $\frac{1}{2}Cx^2$

of the system. Thus,

$$\frac{1}{2}Cx^2 = \frac{1}{2}mv^2 = 8 \text{ joule} \quad \text{or } x^2 = \frac{16}{C} = \frac{16}{64} = \frac{1}{4}, \text{ whence } x = \mathbf{0.5 \text{ metre.}}$$

Ex. 2. *A body of mass* 2 *kg is attached to a horizontal spring of force constant* 8 *n/m and then a constant force of* 6 *newtons is applied on the body along the length of the spring. Find the speed of the body when it has been displaced through* 0.5 *metre. Now, if the force is removed, how much farther the body will move before coming to rest ?*

Sol. Work done by the constant force 6 newtons in displacing the mass through 0.5 metre,

$$W = 6 \times 0.5 = 3 \text{ joule.}$$

The potential energy of the system at this displacement

$$U = \frac{1}{2}Cx^2 = \frac{1}{2} \times 8 \times (0.5)^2 = 1 \text{ joule}$$

The work done W on the system increases the potential energy and kinetic energy of the system. If the speed acquired by the body is v m/sec, then

$$W = \frac{1}{2}Cx^2 + \frac{1}{2}mv^2 \quad \text{or} \quad 3 = 1 + \frac{1}{2} \times 2 \times v^2, \text{ whence } v = \sqrt{2} = \mathbf{1.4 \text{ m/sec.}}$$

At the displacement 0.5 metre, the total energy of the system is 3 joule. Now, if the constant force is removed, only 3 joule energy will remain with the system. Now, if the body is further displaced through a distance α against the elastic force of he spring so that it comes to rest, the total energy will become in the form of potential energy, *i.e.*,

$$\frac{1}{2} C (0.5 + \alpha)^2 = 3 \quad \text{or} \quad \frac{8}{2} (0.5 + \alpha)^2 = 3$$

or $$0.5 + \alpha = \sqrt{3}/2 = 1.732/2 = 0.866, \text{ whence } \alpha = \mathbf{0.366 \text{ metre.}}$$

Ex. 3. *A body of mass m is suspended from a spring. It comes to rest after a downward displacement* x_0. *If a linear force is acting, then prove that*

(i) *the spring (force constant) is* mg/x_0,

(ii) *the gravitational energy lost is* mgx_0.

(iii) *the elastic potential energy gained is* $\frac{1}{2} mgx_0$.

What happened to the remaining energy ?

Sol. In the equilibrium position, the downward force mg on the spring will be balanced by the restoring force in the spring. So that

$$mg = -\text{ Restoring force} = -(-Cx_0) = Cx_0$$

whence $$C = mg/x_0 \qquad \text{...(i)}$$

Loss in gravitational potential energy $= mgx_0$ ...(ii)

$$\text{Elastic potential energy} = \int_0^{x_0} Cx\,dx = \frac{1}{2}Cx_0^2 = \frac{1}{2}mgx_0 \qquad \text{...(iii)}$$

$$\text{Remaining energy} = mgx_0 - \frac{1}{2}mgx_0 = \frac{1}{2}mgx_0 \qquad \text{...(iv)}$$

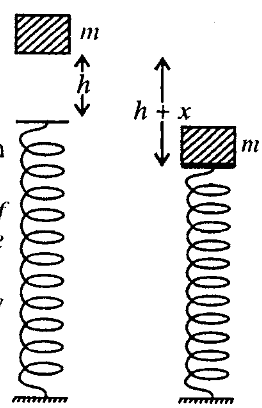

Fig. 2.6.

This energy is first changed to kinetic energy and finally to heat, when intial oscillations stop.

Ex. 4. *A body of mass* 1 *kg, initially at rest is dropped from a height of* 2 *metres onto a vertical spring having force constant* 490 *N/m. Calculate the maximum distance x through which the spring will be compressed.*

Sol. Loss in gravitational potential energy = Gain in potential energy by the spring

i.e., $$mg(h + x) = \frac{1}{2}Cx^2$$

Here, $m = 1$ kg, $g = 9.8$ m/sec^2, $h = 2$ m and $C = 490$ N/m

$\therefore \quad 1 \times 9.8\,(2 + x) = \dfrac{490}{2}x^2$ or $100x^2 = 4x + 8$

or $\quad 25x^2 - x - 2 = 0$

whence $\quad x = \dfrac{1 \pm \sqrt{1+200}}{50} = \dfrac{15.177}{50} = \mathbf{0.3035\ m}$ (Negative value is inadmissible)

Ex. 5. *A body is allowed to slide on a frictionless track from rest position under gravity. The track ends in a circular loop of diameter D. What is the minimum height of the body in terms of D so that it may complete successfully the loop.* **(Delhi 1989 (II))**

Sol. When the body is at A, the kinetic energy gained by the body is

$$\frac{1}{2}mv^2 = mgh \text{ or } v^2 = 2gh \qquad ...(i)$$

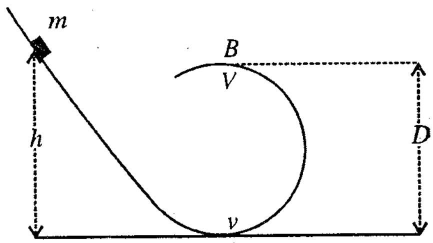

Fig. 2.7.

For the least value of h, the value of v should be least in such a way that the body just reaches up to B. In this position, the necessary centripetal force $2mV^2/D$ is provided by the weight of the body [**fig. 2.7**], *i.e.*,

$$\frac{2mV^2}{D} = mg$$

or $$V^2 = \frac{Dg}{2} \qquad ...(ii)$$

Also $\quad \frac{1}{2}mv^2 - \frac{1}{2}mV^2 = mgD \quad$ or $\quad v^2 - V^2 = 2gD \qquad ...(iii)$

Putting the values from (i) and (ii), we get

$$2gh - \frac{1}{2}Dg = 2gD \qquad \text{or} \qquad h = \frac{5}{4}D.$$

Ex. 6. *A mass m is revolving in a vertical circle at the end of a string of r meter length. Calculate the difference in kinetic energies at the top and bottom of the circle. Prove that the tension of the string at the lowest point exceeds the tension at the topmost point by 6 mg.*

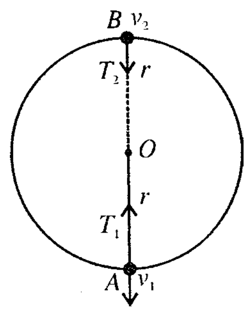

Fig. 2.8.

Sol. As the mass m reaches from position A to the position B, it will lose kinetic energy, but its potential energy will increase by the same amount. Thus,

$$\frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2 = mg \times 2r \qquad ...(i)$$

At the lowest point $\quad T_1 - mg = mv_1^2/r$

At the top point $\quad T_2 + mg = mv_2^2/r$

where T_1 and T_2 are the tensions of the string at A and B respectively.

$$\therefore \quad T_1 - T_2 - 2mg = \frac{mv_1^2}{r} - \frac{mv_2^2}{r} = 4\,mg \text{ [From eq (i)]}$$

or $$T_1 - T_2 = \mathbf{6\ mg}.$$

Ex. 7. *A particle slides along a track with elevated ends and a flat central part 3 metres long* [**fig. 2.9**]. *The curved portions of the track are frictionless while for flat part the coefficient of friction is 0.2. If the particle is released from A (height 1.5 metres above the flat part), find the position where the particle will come to rest.*

Sol. Potential energy of the particle at A relative to B $= mgh$.

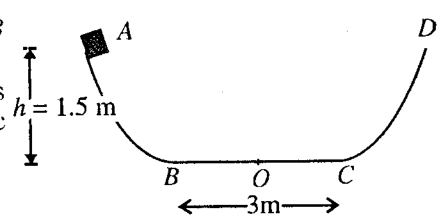

Fig. 2.9.

When the particle will reach at B along the frictionless track, its potential energy will convert in the form of its kinetic energy $\frac{1}{2}mv^2$. So that

$$\frac{1}{2}mv^2 = mgh \qquad ...(i)$$

Now, the particle, moves a distance, say d, along the flat part BC (for which $\mu = 0.2$) and then comes to rest. Hence the kinetic energy $\frac{1}{2}mv^2$ of the particle will be used against the friction. So that

$$\frac{1}{2}mv^2 = \mu Rd = \mu mgd \qquad [\because \text{ Reaction } R = mg] \qquad ...(ii)$$

From eq. (i) and (ii), we have

$$mgh = \mu mgd \quad \text{or} \quad d = h/\mu = 1.5/0.2 = 7.5 \text{ m}.$$

This distance 7.5 m is greater than the distance $BC = 3$ m. So first, the particle will move frictional part $BC = 3$ m and then it will rise up towards D and will come back to C with the same speed, which it was having previously at C (since elevated part is frictionless). Then the particle will travel the frictional distance $CB = 3$ metres and will rise at B towards A and again on returning it will move a distance **1.5 m** upto O.

EXERCISE 2·2

1. Calculate the amount of work done in stretching a spring by 0.1 m, if it is known that the increase in the length of the spring caused by a force of 1 kg-wt is 0.98 metre.
[**Ans.** 0.5 joule.]
2. The scale of a spring balance reads from 0 to 250 kg and is 25 cm long. What is the potential energy of the spring when a 100 kg weight hangs from it ? **(Lucknow 1996)**
[**Ans.** 49 J.]
3. A body of mass 16 kg slides on a horizontal frictionless table with a speed of 1 metre per second. It is brought to rest in compressing a spring in its path. By how much is the spring compressed if the force constant of the spring is 0 25 n/m.
[**Ans.** 8 metres.]
4. A block of mass m, initially at rest, is dropped from a height h onto a spring of force constant C. Find the maximum distance y through which the spring will be compressed ?
$$\left[\textbf{Ans. } y = \frac{1}{2}\left\{\frac{2mg}{C} \pm \sqrt{\left(\frac{2mg}{C}\right)^2 + \frac{8mgh}{C}}\right\}\right]$$
5. A body slides down a curved track which is one quadrant of a circle of radius 1 m. If it starts from rest and there is no friction, find its speed at the bottom of track. ($g = 9.8$ m/sec^2)
[**Ans.** 4.43 m/sec.]
6. A pendulum bob of mass M is held out in the horizontal position and then released. If the string is of length l, what is the bob velocity and what is the force on the string when the bob reaches the lowest position ?
[**Ans.** $v = \sqrt{2gl}$; $T = 3\ Mg$.]

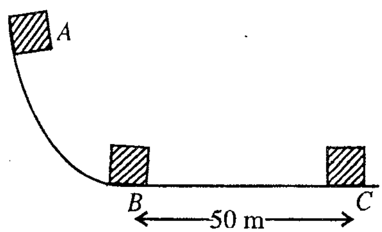

Fig. 2.10.

7. A block weighing 1 kg is released from rest at a point A on a track which is one quadrant of a circle of radius 2 metres [**fig. 2.10**]. It slides down the track and reaches point B with a speed of 5 m/sec. Now, it slides on a level surface a distance 50 metres upto the point C, where it comes to rest. Calculate the coefficient of friction and the amount of work done against friction as the body slides down the circular arc from A to B.
[**Ans.** 0.25; 7.1 joule]
[**Hint :** Work done against friction $= mgh - \frac{1}{2}mv_B{}^2$, where h is the vertical height of A from B.
and $\mu Rd = \frac{1}{2}mv_B{}^2$ or $\mu \times 1 \times 50 = \frac{1}{2} \times 1 \times 5^2$.]

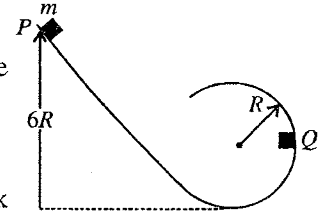

Fig. 2.11.

8. A small block of mass m slides along the frictionless loop-the-loop track shown in **fig. 2.11**. If it starts from P (having height $6R$ from the bottom),

find the tangential and normal components of the force at Q.
[Ans. mg; 10 mg]

QUESTIONS

Long Answer Questions

1. What is the meaning of the line integral of a force ? Show that the work done by an external force on a particle increases its kinetic energy.
2. What do you mean by conservative force ? Write down the essential conditions for it. Show that a central force is conservative. **(Rewa 2001, 1999; Gwalior 2001, 96; Bhopal 98, 96; Raipur 99, 97)**
3. What do you mean by the conservative and non-conservative forces ? Write down the essential conditions for it. Show that a central force is a conservative force. **(Jabalpur 2003; Bhopal 2003)**
4. (a) Prove work-energy theorem. **(Agra 1995, 93)**
   (b) Define conservative force. Show that a central force is conservative. Hence prove that the work done by a conservative force round a closed path is zero. **(Agra 1991; Lucknow 96)**
5. What do you understand by conservative forces ? Show that such a force is expressible as the gradient of a scalar and explain the physical meaning of this scalar. **(Allahabad 1990)**
6. Show that a conservative force is expressed as $\vec{F} = -\text{grad } U = -\vec{\nabla} U$, where U is the potential energy. Hence show that the value of curl $\vec{F}$ is zero. **(Bundelkhand 2004; Gwalior 2001, 1994; Bhopal 2003, 01, 97; Agra 2004; Sagar 99, 98; Bilaspur 97; Ujjain 2004, 03)**
7. What do you mean by potential energy ? Obtain an expression for potential energy of a particle in a conservative force field. Why this expression is negative ?**(Ujjain 1997; Sagar 97; Gwalior 97)**
8. Prove that the total mechanical energy (= kinetic energy + potential energy) of a system acted upon by conservative force is always conserved. **(Bhopal 2003; Gwalior 1999; Bhopal 97)**
9. What are conservative and non-conservative forces ? Show that conservative force is given by the negative gradient of potential energy. **(Garwal 1995; Meerut 95)**
10. (a) What are central forces ? Give two examples of them. Show that a central force is conservative. **(Garwal 1995)**
    (b) Show that Lorentz force is conservative. **(Agra 1978)**
11. What is central force ? Show that the central forces are always conservative.
    A particle is under the influence of a central force $\vec{F}$. Show that the force $\vec{F}$ is the negative gradient of potential energy and curl of it is always zero. **(Allahabad 1979)**

Short Answer Questions

1. State the work-energy theorem. **(Kanpur 2005)**
2. What are conservative and non-conservative forces ? Give two examples of each. **(Kanpur 2005; Ujjain 2004; Bhopal 1998, 96; Gwalior 94, 91; Sagar 98, 96; Raipur 98)**
3. Prove that in case of a conservative force, the work done along a closed path is zero. **(Sagar 1998; Bhopal 98; Raipur 98)**
4. Prove that the negative gradient of potential energy is equal to the conservative force. **(Agra 2004; Osmania 2005)**
5. Show that the central forces is conservative. **(Lucknow 2005; Osmania 2005)**
6. Show that for a conservative force F.
   $$\oint \vec{F} \cdot d\vec{r} = 0$$ **(Lucknow 2005)**
7. Prove that the centripetal force acting on a particle in uniform circular motion is conservative. **(Rewa 1994)**

[**Ans.** Centripetal force $\vec{F} = -m\omega^2 \vec{r} = m\omega^2 r\hat{\mathbf{r}}$ is a central force of the form $\vec{F} = F(r)\hat{\mathbf{r}}$, hence it is a conservative force. **Alternatively** the kinetic energy of a particle, moving under central force remains constant. Hence the work done on the particle along a closed path (circle) is $\oint \vec{F} \cdot d\vec{r} = K_2 - K_1 = 0$. Thus the centripetal force is conservative.]

8. Gravitational field is conservative or not ? **(Rewa 1995)**

[**Ans.** Since gravitational force is a central force, this force and the associated field are conservative in nature.

9. Prove that the Lorentz force acting on a charged particle in a uniform magnetic field is conservative.
10. Prove that curl of a conservative force is zero. **(Ujjain 1999)**
11. You lift a suitcase from the floor to keep on a table. The work done by you on the suitcase depends or does not depend on the path followed by suitcase.
12. A body of mass m fastened at one end of rope is whirled in a vertical circle of radius r. Calculate the tension in the rope when the body happens to be at the lowest point of the circle. **(Lucknow 2005)**
13. What is a non-conservative force ? Distinguish between conservative and non-conservative forces. **(Lucknow 2005)**
14. Prove that $\vec{F} = (x^2 + y^2)\hat{\mathbf{i}} + 3xy\hat{\mathbf{j}}$ is a non-conservative force. **(Kanpur 2005)**

Objective Type Questions

1. Negative value of work done by a conservative force on a particle is equal to :
   (a) total energy (b) kinetic energy
   (c) potential energy (d) none of these
2. The work done by all the forces acting on a system is equal to :
   (a) total energy (b) kinetic energy
   (c) potential energy (d) none of these
3. Change in potential energy of a particle moving in a conservative force field depends on : **(Bhopal 1989)**
   (a) the particle's initial or final position (b) the particle's intial position and velocity
   (c) the particle's initial position and its path (d) path
4. Correct relation for conservative force is : **(Rewa 2000)**
   (a) $\vec{\nabla} \cdot \vec{F} = 0$ (b) $\vec{\nabla} \times \vec{F} = 0$
   (c) $\vec{\nabla}\vec{F} = 0$ (d) $\nabla^2 \vec{F} = 0$.
5. A force acts on a body that always remains normal to the direction of the velocity of a particle and the particle moves in a plane. It is inferred that its :
   (a) velocity is constant (b) acceleration is constant
   (c) momentum is constant (d) kinetic energy is constant
6. Central force is :
   (a) conservative (b) non-conservative
   (c) conservative or non-conservative (d) none of them
7. The work done around any closed path in a conservative force field is :
   (a) zero (b) positive
   (c) negative (d) directly proportional to the length of path
8. A body of mass m is moved by a force from rest and it travels d distance. The kinetic energy of the particle is directly proportional to :
   (a) $\sqrt{m}$ (b) m
   (c) $1/m$ (d) d

9. A conservative force $\vec{F}$ is related to the potential energy U by the relation :

(a) $\vec{F} \neq \vec{\nabla} U$ (b) $\vec{F} = \vec{\nabla} U$ (c) $\vec{F} = \vec{\nabla} \times U$ (d) $\vec{F} = -\vec{\nabla} U$.

10. If the work done by a force in moving a particle from pont A to point B does not depend on the path taken by it, then the force is : **(Bundelshand 2001)**

(a) central (b) non-conservative

(c) conservative (d) inverse square law force.

11. The gravitational field is : **(Bhopal 1990)**

(a) a conservative field (b) a non-conservative field

(c) a fictitious field (d) a magnetic field

12. Due to conservative force, the potential energy is : **(Agra 2003; Rewa 1998)**

(a) $\frac{1}{2} kx$, (b) $\frac{1}{2} k^2x$, (c) $\frac{1}{2} kx^2$, (d) $\frac{1}{2} k^2x^2$.

ANSWERS

1. (c), **2.** (b) **3.** (a), **4.** (b), **5.** (d), **6.** (a), **7.** (a), **8.** (d), **9.** (d), **10.** (c), **11.** (a), **12.** (c).

3

Motion Under Central Force and Kepler's Laws

3·1. Central Force

When a force acts on a particle in such a way that it is always directed towards or away from a fixed point and its magnitude depends only upon the distance (r) from the point, then this force is called a **central force.** Thus, a central force is represented by

$$\vec{F} = F_{(r)}\hat{r} = F_{(r)}\vec{r}/r \qquad ...(1)$$

where $F_{(r)}$ is any function of distance r and $\hat{r} = \vec{r}/r$ is a unit vector along $\vec{r}$ from the fixed centre. The force $\vec{F}$ is attractive or repulsive, if $F_{(r)} < 0$ or $F_{(r)} > 0$ respectively.

Now, if a particle is moving under the influence of a central force $\vec{F} = F_{(r)}\vec{r}/r$ so that the torque acting on it is given by

$$\vec{T} = \frac{d\vec{J}}{dt} = \vec{r} \times \vec{F} = \vec{r} \times F_{(r)}\vec{r}/r = 0 \qquad [\because \vec{r} \times \vec{r} = 0] \qquad ...(2)$$

where $\vec{J}$ is the angular momentum about the origin.

Integration of (2) gives $\vec{J}$ = constant (vector) ...(3)

Thus, *when a particle moves under the action of a central force, its angular momentum* ($\vec{J}$) *is conserved, i.e.,* $\vec{J}$ *remains the same in the magnitude and direction.*

But $\vec{J} = \vec{r} \times \vec{p}$, where $\vec{p}$ is the linear momentum.

Taking dot product with $\vec{r}$ of both sides of this equation, we have

$$\vec{r} \cdot \vec{J} = \vec{r} \cdot (\vec{r} \times \vec{p}) = (\vec{r} \times \vec{r}) \cdot \vec{p} = 0$$

becuase in a scalar triple product the position of dot and cross may be interchanged and $\vec{r} \times \vec{r} = 0$.

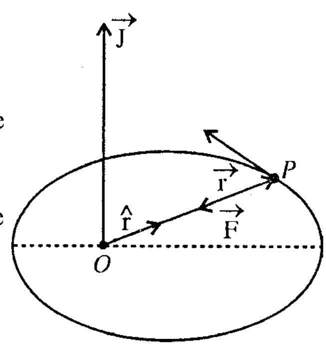

Fig. 3.1.

Therefore, $\vec{r}$ is always perpendicular to the constant vector $\vec{J}$ **[fig. 3.1]**, *i.e., the motion takes place in a plane for a central force.*

A central force, frequently occurring in practice, is represented by

$$F_{(r)} = -\frac{K}{r^n} \qquad ...(4)$$

where K and n are constants. This is known as the n**th power law of force.**

Now, if $K > 0$, the force is one of the attraction between the two particles, and if $K < 0$, the force is repulsive.

We have seen already in Chapter 2 that a central force is essentially a conservative force. Thus if

$U_{(r)}$ is the potential function, then taking integration constant to be zero,

$$F_{(r)} = -\frac{dU}{dr} = -\frac{K}{r^n}, \quad \therefore \quad U_{(r)} = -\frac{K}{n-1}\cdot\frac{1}{r^{n-1}} \qquad ...(5)$$

Here, we will discuss only the two important cases, namely :

(a) When $n = -1$, $K > 0$,

$$F_{(r)} = -Kr \qquad \text{and} \qquad U_{(r)} = \frac{K}{2}r^2 \qquad ...(6)$$

This represents the case of **harmonic oscillator.**

(b) When $n = 2$, K may be +ve or –ve,

$$F_{(r)} = -\frac{K}{r^2} \qquad \text{and} \qquad U_{(r)} = -\frac{K}{r} \qquad ...(7)$$

This is the **inverse square law force**, being the most important example of a central force. We will deal in this chapter this law and its consequences in some detail.

3·2. Inverse Square Law Force

This is one of the basic laws of Classical Mechanics. According to this law, the force, acting between two particles at a distance r apart, is given by

$$F_{(r)} = -\frac{K}{r^2} \qquad ...(1)$$

where K is a constant. If one particle is at the origin and the other at the position $\vec{r}$, then the force acting on the second particle is given by

$$\vec{F} = -\frac{K}{r^2}\hat{r} \qquad ...(2)$$

Examples of such forces are gravitational and electrostatic forces. For gravitational force between two point masses m_1 and m_2; $K = Gm_1m_2$ and for the electrostatic force between point charges q_1 and q_2;

$K = -\frac{1}{4\pi\varepsilon_0}q_1q_2$ (in S.I. system), when they are situated in air or vacuum, *i.e.*,

$$\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{r} \quad ; \quad \vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}\hat{r} \qquad ...(3)$$

Newton's law of Gravitation ; Coulomb's law

The gravitational force is always attractive. The electrostatic (coulomb) force is attractive if the charges q_1, q_2 have opposite signs and repulsive if q_1, q_2 have the same signs.

Inverse square law of force (central force) is a conservative force. So that

$$F_{(r)} = -\frac{dU}{dr} = -\frac{K}{r^2} \qquad ...(4)$$

Integrating it, we get

$$U_{(r)} = -\frac{K}{r} + \text{a constant.}$$

If we choose $U_{(r)}$ to be zero, when the particles are at infinite separation (*i.e.*, $r = \infty$), then the constant of integration is zero and we have

$$U_{(r)} = -\frac{K}{r} \qquad ...(5)$$

Thus, the gravitational potential energy of a system of two particles of masses m_1 and m_2 is given by

$$U_{(r)} = -\frac{Gm_1m_2}{r} \qquad ...(6)$$

and the electrostatic potential energy of charges q_1 and q_2 is given by

$$U_{(r)} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r} \qquad ...(7)$$

3·3. Motion of a System of Two Particles Under Central Force

We plan to study the motion of two particle system, interacting mutually through central force or inverse square law force. We will see that when central force is acting between two bodies, this two-body problem can always be reduced to one-body problem. Let us consider two particles of masses m_1 and m_2, whose instantaneous position vectors in an inertial frame with origin O_i are $\vec{r}_1$ and $\vec{r}_2$ **[fig. 3.2]**.

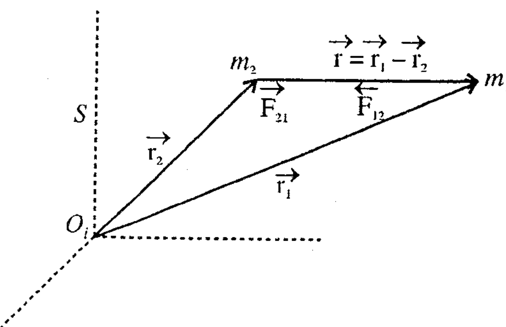

Fig. 3.2.

Hence the vector distance of m_1 from m_2 is $\vec{r} = \vec{r}_1 - \vec{r}_2$. If m_1 and m_2 interact via central force* collinear with vector $\vec{r}$, then the forces acting on the two particles are given by

$$\vec{F}_{12} = m_1 \frac{d^2 \vec{r}_1}{dt^2} \quad \text{and} \quad \vec{F}_{21} = m_2 \frac{d^2 \vec{r}_2}{dt^2} \qquad ...(1)$$

But according to Newton's third law of motion

$$\vec{F}_{12} = -\vec{F}_{21} = \vec{F} \text{ (say)}, \therefore \frac{d^2 \vec{r}_1}{dt^2} = \frac{\vec{F}}{m_1} \quad \text{and} \quad \frac{d^2 \vec{r}_2}{dt^2} = -\frac{\vec{F}}{m_2}$$

Hence $$\frac{d^2}{dt^2}(\vec{r}_1 - \vec{r}_2) = \vec{F}\left(\frac{1}{m_1} + \frac{1}{m_2}\right)$$

If $\hat{\mathbf{r}}$ is the unit vector along $\vec{r} = \vec{r}_1 - \vec{r}_2$ then $\vec{F} = F\hat{\mathbf{r}}$, where F is negative for attractive force and positive for repulsive force. For central force, F is any function of distance r. Now,

$$\frac{d^2 \vec{r}}{dt^2} = \left(\frac{1}{m_1} + \frac{1}{m_2}\right) F\hat{\mathbf{r}}. \qquad ...(2)$$

If we put $\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$ in this equation, then

$$\frac{d^2 \vec{r}}{dt^2} = \frac{\vec{F}}{\mu} = \frac{F\hat{\mathbf{r}}}{\mu} \qquad ...(3)$$

or $$\vec{F} = \mu \frac{d^2 \vec{r}}{dt^2} \qquad ...(4)$$

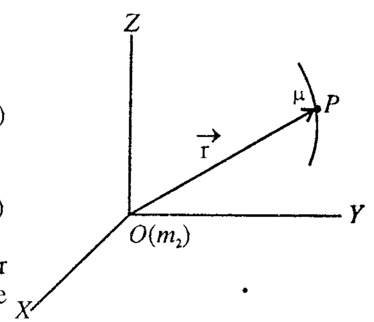

Fig. 3.3.

This equation represents a one-body problem, because it is similar to the equation of motion of a single particle of mass μ at vector distance $\vec{r}$ from one of the two particles, considered as centre of force O**. Thus, the original two-body problem, involving the calculation of $\vec{r}_1$ and $\vec{r}_2$ has been reduced to the calculation of a single vector $\vec{r}$ as function of time. The quantity $\mu = \dfrac{m_1 m_2}{m_1 + m_2}$ is therefore called the **reduced mass** of the two particles, having masses m_1 and m_2. The reduced mass μ has a value less than m_1 or m_2.

If $m_2 >> m_1$, then

---

* This central force may be electrostatic or gravitational.

** Remember that the distance of m_1 from m_2 is $\vec{r}$ and also the distance (reduced mass) from the origin O is $\vec{r}$ *i.e.*, the path of m_1 with respect to m_2 will be same as that of μ mass relative to O.

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m_1}{1 + m_1 / m_2} \simeq m_1 \text{ (say } m) \qquad ...(5)$$

In such a case, we have to consider only a one-body problem. When we do not need much accuracy, this approximation will be sufficient.

In **hydrogen atom,** an electron of mass m revolves round a proton of much heavier mass M. The **reduced mass of the hydrogen** atom will be given by

$$\mu = \frac{mM}{m + M} = m\left(1 + \frac{m}{M}\right)^{-1} = m\left(1 - \frac{m}{M}\right) \qquad ...(6)$$

But $\dfrac{\text{mass of the electron}}{\text{mass of the proton}} = \dfrac{m}{M} = \dfrac{1}{1836}$

$$\therefore \qquad \mu = m\left(1 - \frac{1}{1836}\right) \qquad ...(7)$$

Hence in case of hydrogen atom, if the energy, period etc. are calculated by assuming that an electron moves round a fixed proton, there will occur some error. To get the correct results, m should be replaced by the reduced mass μ in the expressions of energy, period etc. Actually, in case of hydrogen atom, this error will be very small, because

$$m\left(1 - \frac{1}{1836}\right) \cong m \text{ nearly} \qquad ...(8)$$

Positronium is a temporary combination of a positron and an electron similar to a hydrogen atom. A positron is a particle which has mass equal to the electron mass but has positive charge. The reduced mass of the positronium is given by

$$\mu' = \frac{mm}{m + m} = \frac{m}{2} \qquad ...(9)$$

($\because$ m = mass of the electron = mass of the positron)

Thus the reduced mass of the positronium particle is one-half the mass of the electron.

3·4. Centre of Mass and Relative Coordinate System

When reducing two-body problem to one-body problem, we have taken the position vectors $\vec{r}_1$ and $\vec{r}_2$ of masses m_1 and m_2 respectively in an inertial frame. The relative position vector of m_1 relative to m_2 is $\vec{r} = \vec{r}_1 - \vec{r}_2$. If a frame of reference is attached to m_2, shown in **[fig. 3.3]** by frame of reference XYZ with origin O, then this frame of reference (XYZ) is a relative coordinate system (with respect to the frames attached to O_i).

As a force is acting on m_2, this m_2 or O is in the state of acceleration in the inertial frame S, hence $OXYZ$ frame will be a non-inertial frame. Looking at the equation,

$$\vec{F} = \mu \frac{d^2 \vec{r}}{dt^2}, \qquad ...(1)$$

one may forget this fact. *Actual paths of both particles will depend on the law of force, initial positions and initial velocities.* When studying planetary motion, we will consider m_2 (or mass of sun M) as origin O and the relative vector $\vec{r}$ ($= \vec{r}_1 - \vec{r}_2$) of m_1 (or mass of planet m) with respect to m_2 (or M) will be taken as the position vector $\vec{r}$ ($= \vec{r}_1 - \vec{r}_2$) of the reduced mass μ.

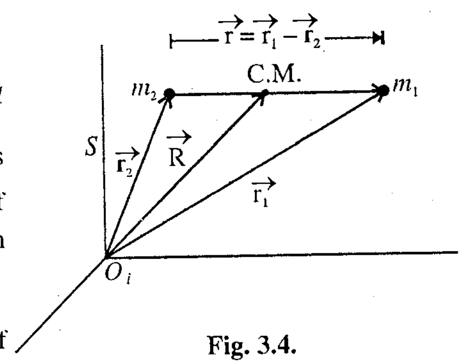

Fig. 3.4.

In **fig. 3.4, the position vector $\vec{R}$ of centre of mass of**

m_1 and m_2 masses, situated at distances $\vec{r}_1$ and $\vec{r}_2$ respectively from O_i, is given by

$$\vec{R} = \frac{m_1 \vec{R}_1 + m_2 \vec{R}_2}{m_1 + m_2} \quad ...(2)$$

Also, $$\vec{r} = \vec{r}_1 - \vec{r}_2 \quad ...(3)$$

Solving eqs. (1) and (2), we get

$$\vec{r}_1 = \vec{R} + \frac{m_2 \vec{r}}{m_1 + m_2}$$

and $$\vec{r}_2 = \vec{R} - \frac{m_1 \vec{r}}{m_1 + m_2} \quad ...(4)$$

But $$\mu = \frac{m_1 m_2}{m_1 + m_2},$$

hence $$\vec{r}_1 = \vec{R} + \frac{\mu}{m_1}\vec{r} \quad \text{and} \quad \vec{r}_2 = \vec{R} - \frac{\mu}{m_2}\vec{r} \quad ...(5)$$

In eq. (5) we have obtained the position $\vec{r}_1$ and $\vec{r}_2$ of the two particles in the inertial frame (S) in terms of the position vector of the centre of mass and relative vector $\vec{r}$.

Differentiating eq. (2) with respect to t, we get the velocity of the centre of mass (C.M.), *i.e.*,

$$\vec{V} = \frac{d\vec{R}}{dt} = \frac{m_1 d\vec{r}_1/dt + m_2 d\vec{r}_2/dt}{m_1 + m_2} \quad \text{or} \quad \vec{V} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} \quad ...(6)$$

Also, $$\vec{F}_{12} = m_1 \frac{d^2 \vec{r}_1}{dt^2} \quad \text{and} \quad \vec{F}_{21} = m_2 \frac{d^2 \vec{r}_2}{dt^2}$$

As the two masses m_1 and m_2 are interacting mutually, Newton's third law of motion is applicable. so that

$$\vec{F}_{12} + \vec{F}_{21} = 0 \quad \text{or} \quad m_1 \frac{d^2 \vec{r}_1}{dt^2} + m_2 \frac{d^2 \vec{r}_2}{dt^2} = 0 \quad \text{or} \quad \frac{d}{dt}\left(m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt}\right) = 0$$

Integrating $$m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} = \text{a constant or } m_1 \vec{v}_1 + m_2 \vec{v}_2 = \text{a constant} \quad ...(7)$$

Hence from eq. (6), the velocity of centre of mass is given by

$$\vec{V} = \frac{d\vec{R}}{dt} = \text{a constant} \quad ...(8)$$

Thus the centre of mass of two-body system moves with constant velocity. If we attach an inertial frame to the centre of mass (C.M.), then such a frame of reference is called **centre of mass-frame of frame** (or **C.M. frame of reference**). In this centre of mass-frame of reference, the velocity of centre of mass is zero (*i.e.*, $\vec{V} = 0$). (It is usually convenient to study the two-body problem in C.M. frame). The position vectors of the two particles in the C.M. frame will be obtained as [by putting $\vec{R} = 0$ in eq. (5)] :

$$\vec{r}_1 = \frac{\mu}{m_1}\vec{r} \quad \text{and} \quad \vec{r}_2 = -\frac{\mu}{m_2}\vec{r} \quad ...(9)$$

It will be proper to discuss about a specific case. If the force $F = \mu\omega^2 r$, then the equivalent particle of mass μ will move on a circle of radius r with respect to O (or m_2) [**fig. 3.5**]. However in the inertial frame, attached to the centre of mass, both the particles will move in circular orbits of radius r_1 and r_2 with the same angular velocity [**fig. 3.6**], where the distance between m_1 and m_2 is r and $r = r_1 + r_2$.

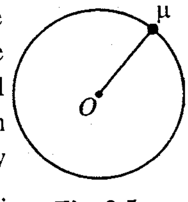

Fig. 3.5.

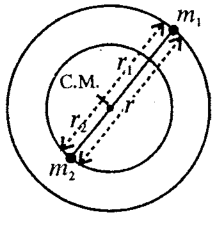

Fig. 3.6.

3·5. Motion of Two-Body System Under Gravitational Force

If two bodies of masses m_1 and m_2 are moving under mutual gravitational force, the equation of motion for equivalent one-body problem will be obtained by putting $-\frac{G m_1 m_2}{r^2}\hat{\mathbf{r}}$ in place of $\vec{F}$ in $\vec{F} = \frac{\mu d^2 \vec{r}}{dt^2}$, *i.e.*,

$$\mu \frac{d^2 \vec{r}}{dt^2} = -\frac{G m_1 m_2}{r^2}\hat{\mathbf{r}} \qquad ...(1)$$

or

$$\frac{d^2 \vec{r}}{dt^2} = -\frac{G(m_1 + m_2)}{r^2}\hat{\mathbf{r}} \qquad ...(2)$$

where $\vec{r}$ is the vector distance of m_1 from m_2 and $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is the reduced mass of the two bodies.

Thus the two-body problem under gravitational force has been reduced to one-body problem, where we have to study about the motion of a body of mass μ. In order to obtain the orbit in two-body problem, we have to solve only one-body problem.

In the circular path of a satellite, the centrifugal force on μ mass is balanced by gravitational force, *i.e.*,

$$\mu\omega^2 r = \frac{GMm}{r^2} \quad \text{or} \quad \omega^2 r = \frac{G(M+m)}{r^2} \qquad ...(3)$$

where $m_1 = m$ is the mass of the satellite and $m_2 = M$ is the mass of the earth.

Thus, $$\text{period } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{r^3}{G(M+m)}} \qquad ...(4)$$

If the mass m is negligible in comparison to M, then

$$T = 2\pi\sqrt{\frac{r^3}{GM}} \qquad ...(5)$$

Thus when a satellite is moving around the earth and its mass is not negligible in comparison to the mass of the earth, we will get correct results by replacing $(M + m)$ for M. This correction may be negligible for a satellite of small mass. However this correction may not be negligible for moon, revolving around earth, and heavy planets around sun.

Theorem 1. *Prove that the angular momentum of two-particle system under inverse square law force is conserved.*

Proof : If the angular momentum of a system is $\vec{J}$, the torque $\vec{T}$ acting on it is given by

$$\vec{T} = \frac{d\vec{J}}{dt} = \vec{r} \times \vec{F} \qquad ...(i)$$

Here, $$\vec{F} = -\frac{K}{r^2}\hat{\mathbf{r}} = -\frac{K}{r^3}\vec{r} \qquad \left(\because \hat{\mathbf{r}} = \frac{\vec{r}}{r}\right)$$

Hence $$\frac{d\vec{J}}{dt} = \vec{r} \times \left(-\frac{K}{r^3}\vec{r}\right) = -\frac{K}{r^3}\vec{r} \times \vec{r} = 0$$

On integration, $$\vec{J} = \text{a constant} \qquad ...(ii)$$

Thus the angular momentum of a system under inverse square law of force is conserved. In two-body problem, the value of angular momentum is $\mu r^2\omega$ = constant. This can be proved as follows :

Remember that $\vec{F} = \mu\frac{d^2\vec{r}}{dt^2}$, hence in equivalent one-body problem angular moment $\vec{J}$ $(= \vec{r} \times \mu\,\vec{v})$ will be conserved. Thus

$$\vec{J} = \vec{r} \times \mu\,\vec{v} = \text{constant} \qquad ...(iii)$$

But $$\vec{v} = \dot{r}\,\vec{r} + r\omega\hat{\theta} \text{ (in plane polar coordinates)}$$

where $\hat{\mathbf{r}}$ and $\hat{\theta}$ are unit vectors, mutually perpedicular and $\dot{r} = dr/dt$ and $\omega = d\theta/dt$.

In magnitude, $$v = \sqrt{\dot{r}^2 + r^2\omega^2}$$

Hence $$\vec{J} = \vec{r} \times \mu(\dot{r}\hat{\mathbf{r}} + r\omega\hat{\theta}) = \mu\dot{r}\,\vec{r} \times \hat{\mathbf{r}} + \mu r\omega\,\vec{r} \times \hat{\theta} \qquad ...(iv)$$

But $$\vec{r} \times \hat{\mathbf{r}} = \vec{r} \times \vec{r}/r = 0 \text{ and } \vec{r} \times \hat{\theta} = r\hat{\mathbf{r}} \times \hat{\theta} = r\hat{\mathbf{z}},$$

where $\hat{\mathbf{z}}$ is an unit vector, in direction perpendicular to the plane of $\hat{\mathbf{r}}$ and $\hat{\theta}$.

Hence $$\vec{J} = \mu r^2\omega\,\hat{\mathbf{z}} = \text{constant}$$

i.e., in magnitude, $$J = \mu r^2\omega = \text{constant.} \qquad ...(v)$$

Theorem 2. *Prove that the total energy of a system of two particles is conserved under inverse square force.*

Proof : The equation of motion of a system under inverse square law of force is given by

$$\mu\frac{d^2\vec{r}}{dt^2} = -\frac{K}{r^2}\hat{\mathbf{r}} \qquad ...(i)$$

This force is radial and we know that the radial acceleration in plane polar coordinates is

$$\ddot{r} - r\dot{\theta}^2 = \frac{d^2r}{dt^2} - r\omega^2.$$

Therefore, $$\mu\frac{d^2\vec{r}}{dt^2} = \mu\left(\frac{d^2r}{dt^2} - r\omega^2\right)\hat{\mathbf{r}} = -\frac{K}{r^2}\hat{\mathbf{r}}$$

Thus, $$\mu\frac{d^2r}{dt^2} - \mu r\omega^2 = -\frac{K}{r^2} \qquad ...(ii)$$

But $\omega = \dfrac{J}{\mu r^2}$ (as proved above) and $F = -\dfrac{K}{r^2} = -\dfrac{dU}{dr}$

$$\therefore \quad \mu\frac{d^2r}{dt^2} - \frac{J^2}{\mu r^3} = -\frac{dU}{dr} \quad \text{or} \quad \mu\frac{d^2r}{dt^2} = -\frac{d}{dr}\left(\frac{1}{2}\frac{J^2}{\mu r^2} + U\right)$$

Multiplying by $\dfrac{dr}{dt}$ on both sides,

$$\mu\frac{d^2r}{dt^2}\frac{dr}{dt} = -\frac{d}{dr}\left(\frac{1}{2}\frac{J^2}{\mu r^2}+U\right)\frac{dr}{dt} \text{ or } \frac{d}{dt}\left[\frac{1}{2}\mu\left(\frac{dr}{dt}\right)^2\right] = -\frac{d}{dt}\left(\frac{1}{2}\mu r^2\omega^2+U\right) \qquad (\because J=\mu r^2\omega)$$

or
$$\frac{d}{dt}\left[\frac{1}{2}\mu\left(\frac{dr}{dt}\right)^2+\frac{1}{2}\mu r^2\omega^2+U\right] = 0 \qquad \text{...(iii)}$$

On integration
$$\frac{1}{2}\mu(\dot{r}^2+r^2\omega^2)+U = \text{a constant, say } E \qquad \text{...(iv)}$$

But $\dot{r}^2+r^2\omega^2 = v^2$, hence
$$\frac{1}{2}\mu v^2 + U(r) = E, \text{ a constant} \qquad \text{...(v)}$$

where E is the total energy (kinetic energy + potential energy) of the system.

Thus *the total energy of a system under inverse square force remains constant.*

The gravitational force between celestial bodies, like sun and earth, is an inverse-square-force. We plan to study about the motion of celestial bodies under this inverse-square-force.

Ex. 1. *Calculate the reduced mass of hydrogen molecule* (H_2). *The mass of hydrogen atom is* 1.7×10^{-27} *kg.* **(Bhopal 2003; Jabalpur 2003, 1996)**

Sol. A molecule of hydrogen consists of two hydrogen atoms, of same mass m. The reduced mass of hydrogen molecule is given by

$$\mu = \frac{m\times m}{m+m} = \frac{m}{2}$$

Here, $m = 1.7 \times 10^{-27}$ kg, $\therefore$ $\mu = \frac{1}{2} \times 1.7 \times 10^{-27}$ kg $=$ **8.5×10^{-28} kg.**

Ex. 2. *The maximum and minimum distances of a comet from the sun are* 1.4×10^{12} *m and* 7×10^{10} *m respectively. If its velocity nearest to the sun is* 6×10^4 *m/sec, what is its velocity when farthest.* **(Agra 1977; Kanpur 88)**

Sol. From the law of conservation of angular momentum for the comet,

$$\mu v_1 r_1 = \mu v_2 r_2 \;\therefore\; v_1 = \frac{v_2 r_2}{r_1} = \frac{6\times10^4\times7\times10^{10}}{1.4\times10^{12}} = \mathbf{3000\ m/sec.}$$

Ex. 3. *Two masses* m_1 *and* m_2 *are initially at rest at infinite distance apart. They approach each other due to gravitational attraction. Find their speed of approach at the instant they are at d distance apart.* **(Gorakhpur 1981)**

Sol. Initially, potential energy $U_1 = 0$, kinetic energy $K_1 = 0$.

So that initial total energy = 0

$$\text{Reduced mass of the system } \mu = \frac{m_1 m_2}{m_1+m_2}$$

For two-body problem, one can use equivalent one-body problem in which one mass μ is considered to be moving. If v is the relative velocity then the total final energy at d separation is

$$= \frac{1}{2}\mu v^2 - \frac{Gm_1m_2}{d}$$

From the law of conservation of mechanical energy

$$\frac{1}{2}\mu v^2 - \frac{Gm_1m_2}{d} = 0 \quad\text{or}\quad v^2 = \frac{2Gm_1m_2}{\mu d} \quad\text{or}\quad v = \sqrt{\frac{2G(m_1+m_2)}{d}}$$

EXERCISE 3·1

1. Two masses 10 gm and 90 gm are connected by a weightless rod of length 10 cm. Calculate the reduced mass of the system.
[**Ans.** 0.09 kg.]

2. If m is the mass of hydrogen atom, find the reduced mass of H_2, HD and D_2, where D stands for deuterium with mass $2m$. **(Rewa 1998)**
[**Ans.** $m/2$, $2m/3$, m].
3. The distance between two protons of hydrogen molecule is 0.7×10^{-8} cm. Calculate the moment of inertia of the hydrogen molecule about an axis passing through the mid point on the internuclear axis and perpendicular to its length. (Mass of proton = 1.6×10^{-27} kg-m^2.) **(Gorakhpur 1980)**
[**Ans.** 3.92×10^{-48} kg-m^2.]
4. Initially a small mass m and relatively heavy mass M are at infinite separation. If mass M is considered to be fixed, what will be the velocity of m relative to M ? If both bodies are allowed to move freely, what will be the relative velocity ?

[**Ans.** $\sqrt{\frac{2GM}{d}}$; $\sqrt{\frac{2G(M+m)}{d}}$.]

[**Hint :** In the first condition, $\frac{1}{2}mv^2 = \frac{2GMm}{d}$ and in the second condition $\frac{1}{2}\mu v^2 = \frac{GMm}{d}$.]

3·6. Kepler's Laws of Planetary Motion (Motion of Celestial Bodies)

Planets move around the sun under gravitational force in different orbits. Earth is a planet among the nine planets of the sun namely, (i) Mercury, (ii) Venus, (iii) Earth, (iv) Mars, (v) Jupiter, (vi) Saturn, (vii) Uranus, (viii) Neptune and (ix) Plato. Kepler's experimental work regarding the motion of celestial bodies is of exterme importance. After taking observations for a period of more than 20 years on planets and doing a lot of calculational work, Kepler announced three laws related to planetary motion in 1619. This achievement of Kepler that the orbits of planets around the sun are elliptical is one of the greatest discoveries in the human history. Kepler's three laws are the following :

(1) The law of elliptical orbits : *Every planet moves in an elliptical orbit around the sun, the sun being at one of the foci.*

According to this law, the path of a planet around sun is in the form of an ellipse and the sun is at one of its foci. Different planets of the sun move in different orbits about it.

(2) The law of areas : *The radius vector, drawn from the sun to a planet, sweeps out equal areas in equal times, i.e., the areal velocity of the radius vector is constant.*

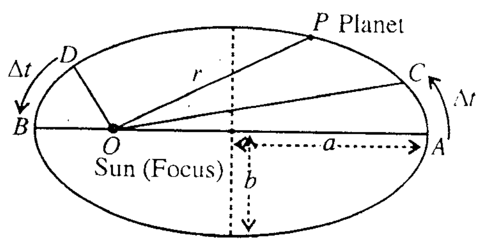

Fig. 3.7.

In (**fig. 3.7**), the line drawn from the sun O upto the planet traverses equal areas OAC and ODB in the same time Δt so that the areal speed of the radius vector remains constant. Obviously at different points of the ellipse the linear speed of the planet is different. When the planet is at more distance from the sun, it moves small distance AC in time Δt and when the planet is close to the sun, it crosses more distance DB in the same time Δt. This is why the linear speed of the planet is less for greater distance and it is more for smaller distance.

(3) The harmonic law : *The square of the period of revolution of the planet around the sun is proportional to the cube of the semi-major axis of the ellipse.*

If the period of revolution of the planet about the sun is T and the semi-major axis of the ellipse is a, then from Kepler's third law,

$$T^2 \propto a^3 \qquad \text{or} \qquad T^2 = ka^3$$

Thus according to this law, the period of revolution of the planet at a more distance relative to the closer planet will be more. If for one planet, the period of revolution is T_1 and semi-major axis is a_1 and the corresponding quantities are T_2 and a_2 for another planet, then

$$T_1^2 = ka_1^3 \qquad \text{and} \qquad T_2^2 = ka_2^3$$

So that
$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3} \qquad \text{or} \qquad \frac{T_1}{T_2} = \left(\frac{a_1}{a_2}\right)^{\frac{3}{2}}$$

Newton's law of universal gravitation : From Kepler's laws we get some knowledge regarding the planetary motion on the basis of experimental observation, but this was Newton who gave a theoretical explanation of planetary motion. According to Newton, gravitational force acts between the sun and a planet. Newton studied thoroughly the Kepler's laws of planetary motion. He announced his famous law of universal gravitation with the help of the three Kepler's laws and his laws of motion. According to Newton's law of universal gravitation, *if two masses m_1 and m_2 are placed at a distance r, then the force of gravitational attraction is given by*

$$\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{r}$$

where G is called gravitational constant.

As Newton's law of gravitation is valid for large inter-planetary distances and small distances on earth, this is why this law is said to be **universal.** This law does not hold true for intermolecular distances of the order of 10^{-7} cm and very large orbital velocities. The gravitational force, which is an inverse square law force $(-K/r^2)$, is an important example of central force.

If we consider the motion of a planet around the sun, then it will be a two-body problem and this can always be reduced to one-body problem in the form

$$\mu\frac{d^2\vec{r}}{dt^2} = -\frac{Gm_1m_2}{r^2}\hat{r}$$

where $\mu = \dfrac{m_1m_2}{m_1+m_2}$ is the reduced mass of planet-sun system.

In the next section, starting from this equation, we shall study about the the motion of celestial bodies and present theortical explanation of Kepler's three laws of planetary motion.

3·7. Mathematical Explanation of Kepler's Laws and their Deduction

If we study the motion of the system of sun and a planet, like earth, under gravitational force, it is a two-body problem which can always be reduced to one-body problem. Suppose that the masses of the sun and the planet are M and m respectively, then the equation of equivalent one-body problem will be

$$\mu\frac{d^2\vec{r}}{dt^2} = -\frac{GMm}{r^2}\hat{r} \qquad ...(1)$$

where $\vec{r}$ is the vector distance from M to m.

If we put $GMm = K$, then in general under inverse square force $(-K/r^2)$, the equation of equivalent one-body problem will be

$$\mu\frac{d^2\vec{r}}{dt^2} = -\frac{K}{r^2}\hat{r} \qquad ...(2)$$

Here we plan to discuss the solution of the problem specifically for gravitational force. The same solution will be valid for any inverse square force $(F = -K/r^2)$, where GMm is replaced by K. In case of two charges Q_1 and Q_2, $K = \dfrac{Q_1Q_2}{4\pi\varepsilon_0}$.

Conservation of angular momentum and areal velocity—Deduction of Kepler's second law : *As the gravitational force on the planet due to sun is a central force, total angular momentum of system is conserved.* Proof of this statement is as follows :

Here, $$\vec{F} = -\frac{GMm}{r^2}\hat{r}$$

Torque, $$\vec{T} = \frac{d\vec{J}}{dt} = \vec{r}\times\vec{F} = \vec{r}\times\left(-\frac{GMm}{r^2}\right)\hat{r}$$

$$= \vec{r}\times\left(-\frac{GMm}{r^3}\right)\vec{r} = 0 \qquad \left(\because \hat{r} = \frac{\vec{r}}{r} \text{ and } \vec{r}\times\vec{r} = 0\right)$$

Hence $\vec{J}$ = a constant ...(3)

Thus for sun-planet system similar to one-body problem, we have

$$\vec{J} = \vec{r} \times \mu \vec{v} = \text{a constant} \quad ...(4)$$

Hence the angular momentum of sun-planet system under gravitational force will be conserved. As $\vec{J}$ is constant, its direction will be perpendicular to the plane of $\vec{r}$ and $\vec{v}$. This means that the motion of the sun-planet system is in a plane under gravitational force.

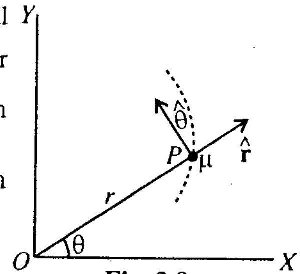

Fig. 3.8.

In plane polar coordinates, the position vector of the mass μ (or m) from origin O (or M) will be given by $\vec{r} = r\hat{\mathbf{r}}$ and its velocity will be

$$\vec{v} = \dot{r}\hat{\mathbf{r}} + r\omega\hat{\theta}$$

where $\hat{\mathbf{r}}$ and $\hat{\theta}$ are radial and transverse unit vectors respectively and $\dot{r} = \frac{dr}{dt}$ and $\omega = \dot{\theta} = \frac{d\theta}{dt}$

Hence, $\vec{J} = \vec{r} \times \mu(\dot{r}\hat{\mathbf{r}} + r\omega\hat{\theta})$ *i.e.,* $\vec{J} = \mu r\omega(\vec{r} \times \hat{\theta})$ $[\because \vec{r} \times \hat{\mathbf{r}} = 0]$

If we take a unit vector $\hat{\mathbf{z}}$ (along $\vec{J}$) perpendicular to the plane of motion, then on taking $\hat{\mathbf{r}} \times \hat{\theta} = \hat{\mathbf{z}}$ and hence $\vec{r} \times \hat{\theta} = r\hat{\mathbf{r}} \times \hat{\theta} = r\hat{\mathbf{z}}$, we have

$$\vec{J} = \mu r^2 \omega \hat{\mathbf{z}} = \text{a constant} \quad ...(5a)$$

In magnitude, $J = \mu r^2 \omega$ = a constant ...(5b)

In motion when the position vector $\vec{r}$ changes to $\vec{r} + \Delta\vec{r}$, the area $\Delta\vec{S}$ crossed by the radial vector is

$$\Delta\vec{S} = \frac{1}{2}\vec{r} \times \Delta\vec{r} \quad ...(6)$$

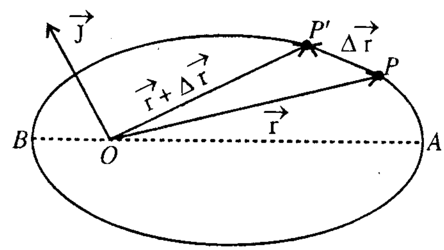

Fig. 3.9.

This area is crossed in Δt time, hence on dividing by Δt on both sides of eq. (6) and on taking the limit $\Delta t \to 0$, we obtain

$$\frac{d\vec{S}}{dt} = \frac{1}{2}\vec{r} \times \frac{d\vec{r}}{dt} = \frac{1}{2}\vec{r} \times \vec{v} \quad ...(7)$$

But from eq. (4) $\vec{r} \times \vec{v} = \vec{J}/\mu$, the areal velocity ($d\vec{S}/dt$) of the planet around the sun is given by

$$\frac{d\vec{S}}{dt} = \frac{\vec{J}}{2\mu} = \text{a constant} \quad ...(8)$$

In magnitude, $$\frac{dS}{dt} = \frac{J}{2\mu} = \frac{r^2\omega}{2} \quad ...(9)$$

Thus, *the rate of crossed area by the line, drawn from the sun upto the planet, i.e., areal velocity of the radius vector remains constant. This is the* **Kepler's second law of planetary motion.**

Differential equation of planetary orbit : Here the equation of motion is

$$\mu\frac{d^2\vec{r}}{dt^2} = -\frac{GMm}{r^2}\hat{\mathbf{r}} \quad ...(10)$$

In polar coordinates, the acceleration $\vec{a}$ is given by

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

where $(\ddot{r} - r\dot{\theta}^2)$ is the radial acceleration (along $\hat{\mathbf{r}}$) and $2\dot{r}\dot{\theta} + r\ddot{\theta}$ is the transverse acceleration (along $\hat{\theta}$).

Hence, force, $$\vec{F} = \mu\vec{a} = \mu\ (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta} \quad ...(11)$$

From eqs. (10) and (11),

$$\mu(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + \mu(r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta} = -\frac{GMm}{r^2}\hat{\mathbf{r}} \quad ...(12)$$

On equating the coefficients of $\hat{\mathbf{r}}$ and $\hat{\theta}$ of eq. (12), the equation of radial motion is obtained as

$$\mu(\ddot{r} - r\omega^2) = -\frac{GMm}{r^2} \qquad (\because \dot{\theta} = \omega.) \quad ...(13)$$

and $$r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 \quad ...(14)$$

Eq. (14) can be written as $2\dot{r}\dot{\theta} + r\ddot{\theta} = \frac{1}{r}\frac{d}{dt}(r^2\dot{\theta}) = 0$

whence $r^2\dot{\theta} = r^2\omega =$ a constant or $\frac{1}{2}r^2\omega =$ areal velocity = constant ...(15)

which is the same equation as (9).

But $\omega = \frac{J}{\mu r^2}$ [from eq. (5b)], eq. (13) assumes the form

$$\mu\ddot{r} - \frac{J^2}{\mu r^3} = -\frac{GMm}{r^2} \quad ...(16)$$

This equation is used to study the motion of the planets, satellites, double stars etc.

It will be convenient that eq. (16) is solved in the form r as a function of θ, *i.e.*, $r = r(\theta)$. Now

$$\dot{r} = \frac{dr}{dt} = \frac{dr}{d\theta}\frac{d\theta}{dt} = \frac{dr}{d\theta}\dot{\theta} = \frac{J}{\mu r^2}\frac{dr}{d\theta} \qquad \left(\because \omega = \frac{J}{\mu r^2}\right)$$

and $$\ddot{r} = \frac{d}{dt}\left(\frac{dr}{dt}\right) = \frac{d}{dt}\left(\frac{J}{\mu r^2}\frac{dr}{d\theta}\right) = \frac{d}{d\theta}\left(\frac{J}{\mu r^2}\frac{dr}{d\theta}\right)\frac{d\theta}{dt} = \frac{J}{\mu r^2}\frac{d}{d\theta}\left(\frac{J}{\mu r^2}\frac{dr}{d\theta}\right)$$

Let $u = \frac{1}{r}$, hence $$\frac{du}{d\theta} = -\frac{1}{r^2}\frac{dr}{d\theta}.$$

Therefore, $$\ddot{r} = -\frac{J^2u^2}{\mu^2}\frac{d}{d\theta}\left(\frac{du}{d\theta}\right) = -\frac{J^2u^2}{\mu^2}\frac{d^2u}{d\theta^2}$$

Hence eq. (16) is

$$-\frac{J^2u^2}{\mu}\frac{d^2u}{d\theta^2} - \frac{J^3u^3}{\mu} = -GMmu^2 \quad \text{or} \quad \frac{d^2u}{d\theta^2} + u = \frac{\mu GMm}{J^2} \quad ...(17)$$

This is the equation of orbit for planetary motion around the sun

Solution of equation of orbit and deduction of Kepler's first law : In eq. (17), right hand side has constant quantity and hence there will be a simple solution of this equation. Eq. (17) can be written as

$$\frac{d^2}{d\theta^2}\left(u-\frac{\mu GMm}{J^2}\right)+\left(u-\frac{\mu GMm}{J^2}\right)=0$$

or
$$\frac{d^2u'}{d\theta^2}+u'=0 \qquad ...(18)$$

where $u' = u - \dfrac{\mu GMm}{J^2}$.

The solution of eq. (18) is (similar to the equation of simple harmonic motion)

$$u' = A\cos(\theta+\phi) \quad \text{or} \quad u-\frac{\mu GMm}{J^2} = A\cos(\theta+\phi) \qquad ...(19)$$

where A is a constant.

Putting $u = \dfrac{1}{r}$ in eq. (19), we get

$$\frac{1}{r}-\frac{\mu GMm}{J^2} = A\cos(\theta+\phi) \quad \text{or} \quad \frac{J^2/\mu GMm}{r} = 1+\frac{J^2A}{\mu GMm}\cos(\theta+\phi)$$

or
$$\frac{l}{r} = 1 + e\cos(\theta+\phi) \qquad ...(20)$$

where $l = J^2/\mu GMm$ and $e = J^2A/\mu GMm$.

Thus, the path of the planet will be ellipse, but in the condition $e < 1$ or $J^2A/\mu GMm < 1$. This establishes **Kepler's first law**, according to *which a planet moves around the sun in an elliptical orbit and the sun is at one focus of the orbit.*

Now, we want to express e in terms of total energy E of the system which is constant for motion.

$$\text{Total energy, } E = \frac{1}{2}mv^2 - \frac{GMm}{r}.$$

At maximum and minimum distances of the planet from the origin, the velocity vector is perpendicular to $\vec{r}$. This means that at these points $v = r\,d\theta/dt = r\omega = J/\mu r$ and therefore

$$E = \frac{J^2}{2\mu r^2}-\frac{GMm}{r}.$$

If the minimum and maximum distances of the planet from the origin are r_{min} and r_{max} respectively, then

$$E = \frac{J^2}{2\mu}\left(\frac{1}{r_{min}}\right)^2-\frac{GMm}{r_{min}} = \frac{J^2}{2\mu}\left(\frac{1}{r_{max}}\right)^2-\frac{GMm}{r_{max}} \qquad ...(21)$$

Usually, we take $\phi = \pi$, so that from (20)

$$\frac{l}{r} = 1 - e\cos\theta \qquad ...(22)$$

From eq. (20),
$$\frac{l}{r_{max}} = 1 - e \quad \text{and} \quad \frac{l}{r_{min}} = 1 + e \qquad ...(23)$$

where r_{min} and r_{max} are the minimum and maximum distances of the planet from the origin, corresponding to $\theta = \pi$ and $\theta = 0$ or $\cos\theta = -1$ and $\cos\theta = 1$ respectively.

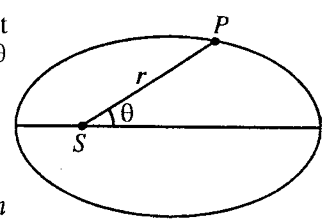

Fig. 3.10.

From eq. (23),
$$e = \frac{r_{max}-r_{min}}{r_{max}+r_{min}}. \qquad ...(24)$$

On substituting from eqs. (23) in eq. (21) and using $e = J^2A/\mu GMm$ and $l = J^2/\mu GMm$, we get

$$e = \left(1 + \frac{2EJ^2}{\mu G^2 m^2 M^2}\right)^{\frac{1}{2}}; \quad E = -\frac{\mu G^2 m^2 M^2}{2J^2}(1 - e^2) \qquad ...(25)$$

Thus, if $e < 1$, the total energy E is negative and so the planet is bound and the orbit is closed in the form of an ellipse. The orbit will be circle for $e = 0$. So that for circular orbit

$$E = -\mu G m^2 M^2 / 2J^2 \qquad ...(26)$$

In case $e = 1$ and $e > 1$, total energy E will be positive and the path will be parabola and hyperbola respectively. Remember that in solving the two-body problem of the sun and a planet (*e.g.*, earth), we have neglected the effect of other planets.

Deduction of Kepler's third law : According to the property of an ellipse,

$$\text{latus rectium, } l = \frac{b^2}{a} \qquad ...(27)$$

and $\qquad$ area of ellipse $= \pi ab \qquad ...(28)$

where a and b are the semimajor and semiminor axes respectively.
But $l = J^2/\mu Gm$, hence

$$\frac{b^2}{a} = \frac{J^2}{\mu GMm}, \text{ whence } b^2 = \frac{J^2 a}{\mu GMm} \qquad ...(29)$$

If T is the period of revolution of the planet around the sun, then

$$T = \frac{\text{Area of ellipse}}{\text{Areal velocity}} = \frac{\pi ab}{J/2\mu} = \frac{2\pi\mu ab}{J}$$

because from eq. (9), areal velocity $= J/2\mu$.

Hence
$$T^2 = \frac{4\pi^2\mu^2 a^2 b^2}{J^2} = \frac{4\pi^2\mu^2 a^2}{J^2} \cdot \frac{J^2 a}{\mu GMm} \qquad \text{[using eq. (29)]}$$

or
$$T^2 = \frac{4\pi^2 \mu a^3}{GMm} \qquad ...(30)$$

But $\mu = mM/(M + m)$, hence

$$T^2 = \frac{4\pi^2 a^3}{G(M+m)} \quad \text{or} \quad T^2 = ka^3 \qquad ...(31)$$

Since $k = \dfrac{G\pi^2}{G(M+m)}$ remains constant,

$$T^2 \propto a^3. \qquad ...(32)$$

This is **Kepler's third law**, according to which *the square of period of revolution of a planet around the sun is directly proportional to the cube of semi-major axis of the ellipse.*

In eq. (32), the constant of proportionality (k) depends on the mass of the planet, hence this constant will be different for different planets. Thus we obtain a more correct statement of the third law, *i.e., the square of the periods of various planets are proportional to the cube of their respective semi-major axes with different proportionality constants, because for each planet, proportionality constant is dfferent.* If we neglect the mass (m) of a planet compared to the mass of the sun (M), we obtain the same proportionality constant for all the planets *i.e.*,

$$T^2 = \frac{4\pi^2}{GM} a^3 \qquad ...(33)$$

This is an approximate relation and in fact, this is Kepler's third law of planetary motion. For example, $m/M = 0.1\%$ for Jupiter and $m/M = 3 \times 10^{-4}\%$ for earth. This means that proportionality constant in (31) is different for different planets. However, in case of the Bohr model of atom, all revolving electrons are identical in mass and hence μ and k are the same for all orbits in an atom. This means that proportionality constant is the same for different electrons in different orbits. Therefore Kepler's third law is very much true for the electron orbits in the Bohr atom.

Theorem 1. (a) *If the minimum and maximum distances of a planet from the sun are r_1 and r_2 respectively, then*

$$\frac{r_2}{r_1} = \frac{1+e}{1-e}.$$

(b) *If the maximum and minimum velocities of a planet around sun are v_1 and v_2 respectively, then*

$$\frac{r_2}{r_1} = \frac{v_1}{v_2} \quad \textit{and} \quad e = \frac{v_1 - v_2}{v_1 + v_2} = \frac{r_2 - r_1}{r_2 + r_1}$$

Proof : (a) Let the equation of elliptical orbit of a planet be

$$\frac{l}{r} = 1 + e\cos\theta$$

We see that at $\theta = 0$, the value of r is minimum (r_1) and at $\theta = \pi$, the value of r is maximum (r_2), *i.e.*,

$$\frac{l}{r_1} = 1 + e \quad \text{and} \quad \frac{l}{r_2} = 1 - e.$$

Dividing the two equations,

$$\frac{r_2}{r_1} = \frac{1+e}{1-e}.$$

(b) At the maximum distance r_2 (point A) and at the minimum distance r_1 (point B), the velocity vector $\vec{v}$ is perpendicular to $\vec{r}$. Therefore, in the expression of the velocity $\vec{v} = \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\theta}$, radial component will be not present, *i.e.*, $\dot{r} = 0$.

Hence at minimum and maximum distances, $\vec{v} = r\omega\,\hat{\theta}$ and in value $v = r\omega$.

If at r_1 and r_2, the velocities are v_1 and v_2 respectively then from the law of conservation of angular momentum,

$$\mu r_1^2\omega_1 = \mu r_2^2\omega_2 \quad \text{or} \quad r_1^2\omega_1 = r_2^2\omega_2$$

But $v_1 = r_1\omega_1$ and $v_2 = r_2\omega_2$, hence

$$r_1 v_1 = r_2 v_2 \quad \text{or} \quad \frac{r_2}{r_1} = \frac{v_1}{v_2} \qquad ...(i)$$

As r_2 is maximum and r_1 is minimum, hence the value of the velocity at minimum distance r_1 will be maximum and at maximum distance r_2 it will be minimum. But $\frac{r_2}{r_1} = \frac{1+e}{1-e}$, hence

$$\frac{r_2}{r_1} = \frac{1+e}{1-e} = \frac{v_1}{v_2}; \quad \therefore\ e = \frac{v_1 - v_2}{v_1 + v_2} = \frac{r_2 - r_1}{r_2 + r_1} \qquad ...(ii)$$

Theorem 2. *Prove that in an elliptical orbit, total energy depends only on the semi-major axis of the orbit.*

Proof : The apsidal (turning) distances r_1 and r_2 are known as **perihelion** and **aphelion** and are given by

$$\frac{l}{r_1} = 1 + e \quad \text{or} \quad r_1 = \frac{l}{1+e} \qquad ...(i)$$

and

$$\frac{l}{r_2} = 1 - e \quad \text{or} \quad r_2 = \frac{l}{1-e} \qquad ...(ii)$$

The semi-major axis (a) of the ellipse is one-half the sum of these two apsidal distances *i.e.*,

$$a = \frac{r_2 + r_1}{2} = \frac{1}{2}\left[\frac{l}{1+e} + \frac{l}{1-e}\right] = \frac{l}{1-e^2} \qquad \text{...(iiia)}$$

But $l = J^2/\mu K$, ($K = GMm$), then

$$a = -\frac{J^2}{\mu K} \cdot \frac{\mu K^2}{2EJ^2} = -\frac{K}{2E} \left[\text{from eq. (25), (Art 3.7), } 1 - e^2 = -\frac{2EJ^2}{\mu K^2}\right] \qquad \text{...(iiib)}$$

or

$$E = -\frac{K}{2a} \qquad \text{...(iiic)}$$

Thus *in case of an elliptical orbit, the total energy depends solely on the major axis.*

3·8. Newton's Deductions from Kepler's Laws

Newton deduced his famous law of universal gravitation from Kepler's laws with the help of his laws of motion.

Let **fig. 3.11** represent a planet P moving in an elliptical orbit around the sun, the sun being at one of the foci S of the ellipse. Let P be the position of the planet at any time t and Q at the time ($t + dt$). Then the area swept out by the radius vector SP in time dt is equal to the area ΔSPQ *i.e.*, area swept out $= \frac{1}{2} SP \times PQ$ (as PQ is infinitesimal small and is considered perpendicular to SP).

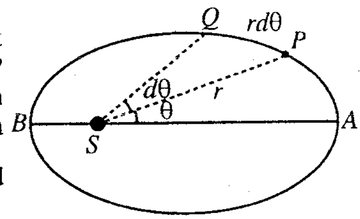

Fig. 3.11.

Here, $SP = r$ and $PQ = r\, d\theta$

Hence, area swept out $= \frac{1}{2} r^2 d\theta$.

Therefore, the radial velocity of the radius vector $= \frac{1}{2} r^2 \frac{d\theta}{dt}$.

According to Kepler's second law, this must be a constant. Let it be equal to $h/2$. Then

$$r^2 \frac{d\theta}{dt} = h. \qquad \text{...(1)}$$

The planet is moving on a curved path, hence it is obvious from Newton's first law of motion that a force is acting upon it along the radius PS. Consequently an acceleration is acting on the planet in the direction of the force. Resolving this acceleration into its two components, along and perpendicular to the radius vector, we have

(i) component a_r, along the radius vector, *i.e.*, the radial acceleration of the planet is given by

$$a_r = \frac{d^2 r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2 \qquad \text{...(2)}$$

and (ii) component a_t, perpendicular to the radius vector, *i.e.*, the transverse acceleration of the planet is given by

$$a_t = \frac{1}{r}\frac{d}{dt}\left(r^2 \frac{d\theta}{dt}\right)$$

But from eq. (1), $r^2 \frac{d\theta}{dt} = $ a constant.

Hence $\quad a_t = 0$.

Thus, the planet has no transverse acceleration and only the radial acceleration is acting on it, *i.e.*, *the force on the planet is directed towards the sun.*

From eq. (1), we have

$$\frac{d\theta}{dt} = \frac{h}{r^2} = hu^2, \qquad \text{...(3)}$$

where $u = 1/r$.

Now, $$\frac{dr}{dt} = \frac{d}{dt}\left(\frac{1}{u}\right) = -\frac{1}{u^2}\cdot\frac{du}{dt} = -\frac{1}{u^2}\frac{du}{d\theta}\cdot\frac{d\theta}{dt} = -\frac{1}{u^2}\frac{du}{d\theta}\cdot hu^2 = -h\frac{du}{d\theta}$$

$$\therefore \quad \frac{d^2r}{dt^2} = -h\frac{d^2u}{d\theta^2}\frac{d\theta}{dt} = -h^2u^2\cdot\frac{d^2u}{d\theta^2} \qquad ...(4)$$

Substituting the values of $d\theta/dt$ and d^2r/dt^2 from eq. (3) and (4) in eq. (2), we get

$$a_r = -h^2u^2\left(u + \frac{d^2u}{d\theta^2}\right) \qquad ...(5)$$

Let the equation of the elliptical orbit be

$$\frac{l}{r} = 1 + e\cos\theta \quad \text{or} \quad lu = 1 + e\cos\theta \qquad ...(6)$$

where l is the latus rectum and e the eccentricity of the ellipse.

Differentiating eq. (6) twice with respect to θ, we have

$$l\frac{d^2u}{d\theta^2} = -e\cos\theta \qquad ...(7)$$

Adding the eq. (6) and eq. (7), we get

$$l\left(u + \frac{d^2u}{d\theta^2}\right) = 1, \qquad \therefore \quad u + \frac{d^2u}{d\theta^2} = \frac{1}{l}.$$

Substituting this value of $u + \dfrac{d^2u}{d\theta^2}$ in eq. (5), we get

$$a_r = -\frac{h^2u^2}{l} = -\frac{h^2}{l}\cdot\frac{1}{r^2}$$

Let $\quad h^2/l = k$ (a constant).

Then $$a_r = -\frac{k}{r^2} \qquad ...(8)$$

or $$a_r \propto -\frac{1}{r^2} \qquad ...(9)$$

Thus, *the acceleration and hence the force acting on the planet is inversely proportional to the square of its distance from the sun*. Negative sign indicates that the force is one of attraction.

Let a and b be the semi-major and semi-minor axes of the elliptical orbit of the planet. The periodic time t of the planet, *i.e.*, the time for one revolution round the sun is given by

$$T = \frac{\text{area of the ellipse}}{\text{areal velocity}} = \frac{\pi ab}{\frac{1}{2}r^2 d\theta/dt} = \frac{2\pi ab}{h}$$

$$\therefore \quad T^2 = \frac{4\pi^2a^2b^2}{h^2} \qquad ...(10)$$

But $b^2/a = l$, the latus rectum of the ellipse.

$$\therefore \quad T^2 = \frac{4\pi^2 l}{h^2}a^3 = \frac{4\pi^2}{k}a^3 \qquad \left[\because\ k = \frac{h^2}{l}\right] \qquad ...(11)$$

Now, in accordance with the Kepler's third law $T^2 \propto a^3$ for every planet. Thus $4\pi^2/k$ is a constant and therefore k is constant for every planet. In other words, *k is independent of the nature of the planet.*

Newton compared the motion of a falling body at the earth's surface with the motion of the moon round the earth. If α be the acceleration of the moon towards the earth, then

$$\alpha = \frac{v^2}{R} = \left(\frac{2\pi R}{T}\right)^2 \times \frac{1}{R} = \frac{4\pi^2 R}{T^2}, \qquad \left[\because \; v = R\omega = \frac{2\pi R}{T}\right] \quad ...(12)$$

where R is the distance from the centre of the earth to the centre of the moon and T the time of one revolution of the moon round the earth.

But $R = 3.8 \times 10^8$ m and $T = 27.3$ days $= 2.36 \times 10^6$ sec

$$\therefore \qquad \alpha = \frac{4 \times (3.14)^2 \times 3.8 \times 10^8}{(2.36 \times 10^6)^2} = 2.7 \times 10^{-3} \text{ cm/sec}^2. \quad ...(13)$$

The ratio of the acceleration α of the moon towards the centre of the earth to the acceleration g of a falling body at the earth's surface is

$$\frac{\alpha}{g} = \frac{7.7 \times 10^{-3}}{9.81} = \frac{1}{60^2} \text{ approx.} \quad ...(14)$$

But the distance between the centre of the earth and the moon is 60 times the radius of the earth. Hence the attractive force of the earth is inversely proportional to the square of the distance of the body from the centre, *i.e.*, the law of attraction of falling bodies at the surface of the earth or of the moon towards the earth is the same as that of the attraction of the planets towards the sun.

Newton realised from this that the attraction of the earth for bodies of its surface or of the sun for planets is not a special property of them, but it is a common property for all bodies, whether large or small, *i.e.*, the **attraction is universal and mutual.**

If m and M be the masses of a planet and the sun respectively, then from eq. (8` he attractive force of the sun on the planet is given by

$$F = -\frac{km}{r^2} \quad ...(15)$$

But the attraction is mutual, the attractive force of the planet on the sun is similarly given by

$$F' = \frac{KM}{r^2} \quad ...(16)$$

where we have assumed positive direction from the sun to the planet.

According to Newton's third law of motion, action is equal and opposite to reaction, hence

$$F = -F' \qquad \text{or} \qquad \frac{km}{r^2} = \frac{KM}{r^2}$$

$$\text{or} \qquad \frac{k}{M} = \frac{K}{m} = G, \text{ (say)}, \quad \therefore \quad k = MG$$

Substituting the value of k in eq. (15), we get

$$F = -\frac{GMm}{r^2} \quad ...(17)$$

Thus, *the force of attraction between the sun and the planet is directly proportional to the product of their masses.*

From the above deductions, Newton enunciated his famous **law of universal gravitation,** which can be written for any two masses m and M in the vector form

$$\vec{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}} \quad ...(18)$$

where $\hat{\mathbf{r}} = \vec{r}/r$ is a unit vector from one particle to the other.

Ex. 1. *The eccentricity of earth's orbit is* 0.0167. *In the earth's orbit, find the ratio of maximum and minimum velocities.* **(Agra 1991)**

Sol. If the maximum and minimum velocities of a planet or earth around sun and e is the eccentricity of the orbit, then

$$\frac{v_1}{v_2} = \frac{1+e}{1-e}$$

Hence $$\frac{v_1}{v_2} = \frac{1+0.0167}{1-0.0167} = \mathbf{1.03}$$

Ex. 2. *Assume that the moon revolves round the earth in an orbit of radius* 3.84×10^5 km. *If the* mass of earth is 6×10^{24} kg and the periodic time of moon is 27.17 days, find the mass of the moon.

(Raipur 1997)

Sol. If the masses of the moon and the earth are m and M respectively, then for equivalent one-body problem

$$\mu\omega^2 r = \frac{GMm}{r^2} \qquad \text{...(i)}$$

where $\mu = \dfrac{mM}{m+M}$ and r is the distance between m and M.

If $T = 2\pi/\omega$ is the period of revolution, then $\omega = 2\pi/T$. Hence from (i)

$$\frac{mM}{m+M}\cdot\left(\frac{2\pi}{T}\right)^2 r = \frac{GMm}{r^2} \qquad \text{or} \qquad m + M = \frac{4\pi^2 r^3}{GT^2}$$

whence $$m = \frac{4\pi^2 r^3}{GT^2} - M = \frac{4\pi^2 \times (3.84 \times 10^8)^3}{6.67 \times 10^{-11} \times (27.3 \times 24 \times 60 \times 60)^2} - 6 \times 10^{24}$$

$$= 6.01775 \times 10^{24} - 6 \times 10^{24} = \mathbf{1.775 \times 10^{22}\ kg.}$$

Ex. 3. *A planet moves around the sun in an elliptical orbit. Its minimum and maximum distances from the sun are* r_1 *and* r_2 *respectively, then prove that the total energy of system is* $E = -\dfrac{GMm}{(r_1+r_2)}$ *and its angular momentum is* $J = \sqrt{\dfrac{2GMm}{(r_1+r_2)}}$, *where M is the mass of the sun.*

Sol. From theorem 2, $a = \dfrac{r_1+r_2}{2}$ and $a = -\dfrac{GMm}{2E}$

Hence $$E = -\frac{GMm}{2a} = -\frac{GMm}{r_1+r_2} \qquad \text{...(i)}$$

Total energy of the system at maximum and minimum distances remains the same and constant. Hence

$$E = \frac{1}{2}\mu v_1^2 - \frac{GMm}{r_1} = \frac{1}{2}\mu v_2^2 - \frac{GMm}{r_2}$$

But $$J = \mu r_1 v_1 = \mu r_2 v_2 \qquad \text{[See Theorem 1]}$$

$\therefore$ $$E = \frac{J^2}{2\mu r_1^2} - \frac{GMm}{r_1} = -\frac{GMm}{r_1+r_2}$$

or $$\frac{J^2}{2\mu r_1^2} = \frac{GMm}{r_1} - \frac{GMm}{r_1+r_2} = \frac{GMm\, r_2}{r_1(r_1+r_2)}$$

Hence, $$J = \sqrt{\frac{2GMr_1r_2}{r_1+r_2}} \qquad \text{...(ii)}$$

Ex. 4. *A particle of mass m moves under the action of central force whose potential is* $V(r) = Kmr^3$ $(K > 0)$, *then*

(i) *For what kinetic energy and angular momentum will the orbit be a circle of radius R about the origin ?*

(ii) *Calculate the period of circular motion.*

(Agra 1992)

Sol. $V(r) = Kmr^3$

Therefore, $F = -\dfrac{\partial V}{\partial r} = -3Kmr^2$

(i) For circular motion, $F = \dfrac{mv^2}{r} = -3Kmr^2$

Kinetic energy $\frac{1}{2}mv^2 = -\frac{3}{2}Kmr^3$

Angular momentum $= mvr = mr\,(-3Kr^3)^{1/2}$ [because $v^2 = -3Kr^3$]

(ii) Since $v = r\omega = \dfrac{2\pi r}{T}$, therefore from eq. $v^2 = -3Kr^3$, we have

$$\left[\frac{2\pi r}{T}\right]^2 = -3Kr^3 \quad \text{or} \quad \frac{4\pi^2}{T^2} = -3Kr, \text{ whence } T = \frac{2\pi}{\sqrt{-3Kr}}.$$

Ex. 5. *A particle of mass m is observed to move in a spiral orbit given by the equation $r = C\theta$, where C is a constant. Is it moving in a central force field ? If it is so, find the force law.*

Sol. The differential equation of the orbit is given by

$$\frac{d^2u}{d\theta^2} + u = -\frac{m\,f(1/u)}{J^2u^2} \quad \text{...(i)}$$

Since $u = 1/r$ and $r = C\theta$ (given), hence $u = \dfrac{1}{C\theta}$ ...(ii)

Differentiating it, we get

$$\frac{du}{d\theta} = -\frac{1}{C\theta^2} \quad \text{and} \quad \frac{d^2u}{d\theta^2} = \frac{2}{C\theta^2} = 2C^2u^3 \quad \text{...(iii)}$$

Substituting in eq. (i), we get $2C^2u^3 + u = -\dfrac{m}{J^2u^2}f\left(\dfrac{1}{u}\right)$

Therefore, $$f\left(\frac{1}{u}\right) = -\frac{J^2}{m}(u^3 + 2C^2u^5) \text{ or } f(r) = -\frac{J^2}{m}\left[\frac{1}{r^3} + \frac{2C^2}{r^5}\right] \quad \text{...(iv)}$$

Thus the particle is moving under a central force field and the force law is given by eq. (iv).

EXERCISE 3.2

1. If the mass of the sun is assumed to be (i) infinite, (ii) finite, find the equation of motion of the planet and the period of revolution in both the conditions. Suppose that the planet of mass m is moving at a distance r from the sun of mass M.

   [**Ans.** (i) $\ddot{\vec{r}} = -\dfrac{GM}{r^2}\hat{\mathbf{r}}$, $T = 2\pi\sqrt{\dfrac{r^3}{GM}}$; (ii) $\ddot{\vec{r}} = -\dfrac{G(M+m)}{r^2}\hat{\mathbf{r}}$, $T = 2\pi\sqrt{\dfrac{r^3}{G(M+m)}}$.]

2. The maximum and minimum velocities of a satellite are v_{max} and v_{min} respectively. Prove that the eccentricity of the orbit of the satellite is

   $$e = \frac{v_{max} - v_{min}}{v_{max} + v_{min}}.$$ **(Agra 1998, 95, 93)**

3. The eccentricity of the earth's orbit is 0.0125. Calculate the ratio of the maximum and minimum speeds of the earth. **(Agra 1998, 95)**

   [**Ans.** 1.025.]

4. A satellite has its largest and smallest orbital speeds v_{max} and v_{min} respectively. If the satellite has a period T, then show that it moves in an elliptical orbit having major axis with length $\dfrac{T}{2\pi}\sqrt{v_{max}v_{min}}$.

5. A planet of mass m is revolving around sun in an elliptical orbit. Its minimum and maximum distances from the sun are r_1 and r_2 respectively. Show that the period of revolution is

$$T = \pi\sqrt{\frac{(r_1 + r_2)^3}{2gR^2}}, \text{ where } g \text{ is acceleration due to gravity.}$$

6. A particle of mass m describes an elliptical orbit about a centre of attractive force at one of its focus given by $-k/r^2$, where k is a constant. Show that the speed v of the particle at any point of the orbit is given by

$$v^2 = \frac{k}{m}\left[\frac{2}{r} - \frac{1}{a}\right], \text{ where } a \text{ is the semi-major axis.}$$

[**Hint** : $E = T$ (K.E.) $+ V = \frac{1}{2}mv^2 - \frac{k}{r}$.]

7. If a planet were suddenly stopped in its orbit, supposed to be circular, prove that it will fall into the sun in a time which is $\sqrt{2}/8$ times of its period.

8. A particle of mass m moves in a central force field defined by $F = -\frac{k\vec{r}}{r^4}$. Show that if E is the total energy supplied to the particle, then its speed is given by

$$v = \sqrt{\frac{k}{mr^2} + \frac{2E}{m}}.$$

9. If the orbit described under a central force be given by $r = a(1 + \cos\theta)$ with centre at the origin, find the law of force. **(Rohilkhand 1984)**

[**Ans.** $f(r) \propto 1/r^4$.]

10. A particle describes a conic $r = \frac{p}{1 + e\cos\theta}$, where p and e involve constant quantities. Show that the force under which particle is moving is a central force. Deduce the force law. **(Agra 1992)**

[**Ans.** $F = -\frac{k}{r^2}$.]

11. A double star consists of two stars of masses m and $2m$ at a distance d apart. They rotate about their centre of mass. Find the period of revolution.

[**Ans.** $2\pi d\sqrt{\frac{d}{3Gm}}$.]

12. A double star is moving about its centre of mass under gravitational force. If the total mass of this star is M and period of revolution T, then show that the distance d between them is given by

$$d = \left[\frac{GM}{(2\pi/T)^2}\right]^{1/3}$$

QUESTIONS

Long Answer Questions

1. What is a central force ? Show that the angular momentum of a particle under central force is conserved and its direction is perpendicular to the plane of the orbit. **(Raipur 1999; Indore 97; Bhopal 93)**

2. Explain what do you understand by central forces ? Give examples. When a particle moves under the action of a central force, prove that its angular momentum is conserved and the motion takes place in a plane. **(Kanpur 1984, 80)**

3. Earth moves in an orbit around sun under its gravitational force. Show that its orbit lies in a plane and that the vector joining the earth and the sun sweeps out equal areas in equal times. **(Lucknow 1996)**
4. What do you understand by the reduced mass ? How is the two-body problem reduced to a single-body problem ? **(Jabalpur 2003, 1995; Bhopal 2000; Rewa 2001, 2000; Ujjain 1998; Bilaspur 96; Indore 99; Raipur 95; Gwalior 99)**
5. Show that the motion of two particles of masses m_1 and m_2 under a central force can be expressed in terms of motion of a single particle of reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$. Hence explain the concept of reduced mass. **(Gwalior 2001, 05; Bhopal 1995; Bilaspur 95; Indore 98; Raipur 98; Sagar 97)**
6. Two mass points P and Q attract each other by a central force. No external force acts on the system. Describe the motion of the system and its centre of mass. **(Raipur 1999; Indore 93; Jabalpur 98)**
7. A particle moves under central force. Show that its orbit lies in a plane and the radius vector from the centre of force to the particle sweeps area at constant rate. **(Agra 2001)**
8. What do you mean by an inverse square force ? Establish the equation of motion of a particle moving under an inverse square force. Show that the orbit is elliptical if the force is attractive. Find the period of revolution of the particle. **(Bhopal 1998; Rewa 95; Jabalpur 93; Indore 98)**
9. Write the Kepler's laws of planetary motion. Prove that the line joining a planet and the sun crosses equal areas in equal times. **(Jabalpur 1999)**
10. What do you mean by inverse square force ? Show that the equation of orbit of a planet around the sun under inverse square force is elliptical. **(Ujjain 2003; Indore 1998)**
11. State Kepler's laws of planetary motion and show how they can be deduced from Newton's law of gravitation. **(Avadh 1996; Delhi 92, 90; Mysore 80)**
12. Calculate the equation of the orbit of a particle moving in an inverse square law field. Find the period of revolution for a closed orbit. **(Allahabad 1980)**
13. Show that the shape of path of a body moving under an inverse square force depends on its total energy and angular momentum. **(Bhopal 2000, 1996; Sagar 94; Jabalpur 94)**
14. Write down the laws relating to the motion of celestial bodies and show that the path of a planet around the sun is elliptical. **(Jabalpur 2004; Bilaspur 99; Gwalior 2003, 1998)**
15. State Kepler's laws of planetary motion and deduce them. Use them to obtain the law of gravitation. **(Bhopal 2004, 01, 99; Sagar 2003, 1999, 98; Rewa 2003, 99; Jabalpur 99; Gwalior 99; Raipur 97; Indore 96, 99; Ujjain 99; Bilaspur 99)**
16. What are Kepler's laws of planetary motion ? Show how the universal law of gravitation of Newton has been derived from these laws ? **(Agra 1978; Allahabad 79; Delhi 94; Gorakhpur 84; Purvanchal 96, 93)**
17. Write short notes :
    (i) Reduced mass **(Indore 1999; Raipur 99)**
    (ii) Kepler's laws of planetary motion. **(Raipur 1999)**
    (iii) Inverse square force **(Raipur 1999)**

Short Answer Questions

1. What do you mean by inverse square force ? **(Bilaspur 1998)**
2. What is meant by the reduced mass of a two-particle system ? Expalin its meaning with an example. **(Sagar 2003; Bilaspur 1996; Ujjain 94; Gwalior 97)**
3. Calculate the reduced mass of HCl molecule in kg. (Mass of hydrogen atom = 1.7×10^{-27} kg.; atomic wt . of Cl = 35.5). **(Jabalpur 1995; Gwalior 93; Ujjain 92)**
   [**Ans.** 1.65×10^{-27} kg.]
4. Write down the Kepler's laws of planetary motion. Prove that the square of time period of revolutions of a planet is proportional to the cube of the semi-major axis of ellispe. **(Indore 2004, 03)**
5. What are central forces ? Give two examples of such forces. **(Kanpur 2005)**

6. Show that the angular momentum of a particle is conserved when moving under central force. **(Agra 2003; Kanpur 2005; Lucknow 2005)**
7. Show that the real velocity of a particle moving under an inverse square force always remains constant. **(Gwalior 2004; Ujjain 1998; Indore 95; Sagar 97; Bhopal 97)**
8. Show that the angular momentum and total energy of a body moving under an inverse square force remains conserved. **(Rewa 2001; Bhopal 2000, 1996; Gwalior 98, 97)**
9. State Kepler's three laws of planetary motion. **(Agra 2003; Kanpur 2005; Ujjain 1999; Bilaspur 96; Osmania 2005)**
10. State Kepler's laws of planetary motion and discuss the importance of third law. **(Kanpur 2004; Indore 1999)**
11. A particle of mass m is moving in a circle of radius r under a centripetal force $- k/r^2$ where k is a constant. Calculate the total energy of the particle. **(Ujjain 1995)**
    [**Ans.** $- k/2r$.]
12. The period of revolution of jupiter around the sun is 11.86 years. Considering the orbits of jupiter and earth to be nearly circular, compare the distances of jupiter and earth from the sun.
    [**Ans.** 5.2 : 1.]
13. The distances of two planets from the sun are in the ratio 5 : 2, then what is the ratio of their period of revolution ? **(Rewa 1998; Bilaspur 97)**
14. The radius vector drawn from the sun to a planet sweeps out equal areas in equal times. **(Lucknow 2004)**

Objective Type Questions

1. If m_H is the mass of hydrogen atom, then the reduced mass of the hydrogen molecule is :
   (a) $\frac{m_H}{2}$ (b) $2m_H$ (c) $\frac{m_H}{4}$ (d) $4m_H$ **(Agra 2004)**
2. When a particle moves under the action of a central force, the quantity that remains conserved is :
   (a) linear momentum (b) angular momentum (c) both (a) and (b) (d) none of these
3. Mass of an oxygen atom is m. The reduced mass of O_2 will be :
   (a) m (b) $2m$ (c) $m/2$ (d) $4m$
4. The mass of an electron is m_e, then what will be the reduced mass of positronium ?
   (a) $2m_e$ (b) m_e (c) $m_e/2$ (d) $3m_e$.
5. Which of the following remains conserved under the action of inverse square force ?
   (a) angular momentum (b) linear momentum (c) both (a) and (b) (d) none of these
6. The shape of the orbit of a planet depends on :
   (a) angular momentum (b) total energy (c) both (a) and (b) (d) none of these
7. If r_1 and r_2 are the minimum and maximum distances of a planet from the sun, then the eccentricity of the orbit of the planet will be :
   (a) $e = \frac{r_1}{r_2}$ (b) $e = \frac{r_1 + r_2}{r_1 - r_2}$ (c) $e = \frac{r_2 - r_1}{r_1 + r_2}$ (d) $e = \frac{r_1 - r_2}{r_1 + r_2}$
8. The vector distance $\vec{r}$ and vector velocity $\vec{v}$ of a planet relative to the sun under gravitational force :
   (a) are perpendicular to each other (b) may be at a different angle to $\pi/2$
   (c) are along the same direction (d) make an angle π with each other.

ANSWERS

1. (a), **2.** (b) **3.** (c), **4.** (c), **5.** (a), **6.** (c), **7.** (c), **8.** (b).

4

Gravitational Field and Potential

4·1. Newton's Law of Universal Gravitation

In the universe, any particle (which has mass) attracts every other particle with a force, known as **gravitational force.** According to **Newton's law of universal gravitation,** the force of attraction between any two particles is directly proportional to the product of their masses and inversely proportional to the square of distance between the two particles. If two particles of masses m_1 and m_2 are separated by a distance r, then in view of this law the gravitational force between the two particles is given by

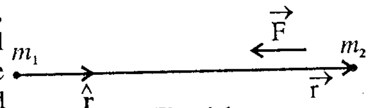

Fig. 4.1.

$$\vec{F} = -G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}} = -G\frac{m_1 m_2}{r^3}\vec{r} \qquad ...(1)$$

where $\vec{r}$ is the position vector from m_1 to m_2, $\hat{\mathbf{r}} = \vec{r}/r$ and G the constant of gravitation. The negative sign signifies the force between the two gravitating particles is that of attraction.

Magnitude of the force $F = Gm_1m_2/r^2$.

Value of $G = 6.67 \times 10^{-8}$ dyne-cm^2/gm^2 (C.G.S. units) $= 6.67 \times 10^{-11}$ newton-m^2/kg^2 (S.I. units)

The extreme importance of the gravitational force may be seen by the following examples :

(i) We are bound to live on earth due to the force of gravitation. Moreover, the gravitational forces keep the atmosphere (containing life providing agency oxygen) to retain on the earth.

(ii) The earth and other planets are bound to move around the sun (which is the most important source of light and heat for us) only due to gravitational forces. Remember that the life is possible on the earth due to the presence of the sun.

Similarly, a number of examples may be seen in everyday life showing the importance of gravitational forces.

4·2. Gravitational Field and Potential

Gravitational Field : The region around a particle or body in which any other particle experiences a gravitational force is called **gravitational field.** *The intensity of gravitational field at a point due to a mass is defined as the force experienced by unit mass, placed at that point.* It is, however, supposed that the introduction of the unit mass at the point does not change the configuration of the field. If M be the mass of a gravitating particle, the intensity of gravitational field at a distance $\vec{r}$ [**fig. 4.2**] from it is given by

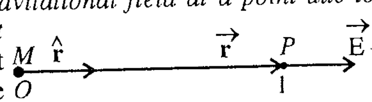

Fig. 4.2.

$$\vec{E} = -\frac{GM \times 1}{r^2}\hat{\mathbf{r}} \quad \text{or} \quad \vec{E} = -\frac{GM}{r^2}\hat{\mathbf{r}} \qquad ...(1)$$

where $\hat{\mathbf{r}} = \vec{r}/r$ is a unit vector along $\vec{r}$. Here the negative sign indicates that the direction of force is opposite to $\hat{\mathbf{r}}$ *i.e.*, towards the mass M.

If we place a mass m at a distance $\vec{r}$ from M, then the gravitational force on m is given by

$$\vec{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}} \qquad ...(2)$$

From eqs. (1) and (2), we get

$$\vec{F} = m\vec{E} \qquad ...(3)$$

or

$$\vec{E} = \frac{\vec{F}}{m} \qquad ...(4)$$

Thus, *the intensity of gravitational field is equal to the force per unit mass.* Normally the intensity of field is said to be **field** only. Sometimes intensity of gravitational field is said to be **gravitational attraction.**

Gravitational potential : The **gravitational potential** *at a point in a gravitational field of a body is the amount of work, required to be done on a unit mass in bringing it from infinity (i.e., the state of zero potential) to that point.* In other words, *the potential at a point is equal to the* **potential energy per unit mass.** Generally gravitational potential is represented by V and its value at infinity is taken to be zero. Hence the gravitational potential at a distance $\vec{r}$ from a mass M is given by

$$V = \frac{U_{(r)}}{m} = -\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r} \qquad ...(5)$$

where $U_{(r)}$ is the potential energy of a mass m placed at $\vec{r}$ and $\vec{E}$ is the intensity of the field at that distance.

Remember that the gravitational force is a conservative force and the force on unit mass is $\vec{E}$ and the definition (5) is according to (eq. 4) of Art 2.5

Gravitational potential due to a mass M : Eq. (5) is used to find the gravitational potential at a distance $\vec{r}$ due to a point mass M. The intensity of the field $\vec{E}$ at a distance $\vec{r}$ from mass M is given by

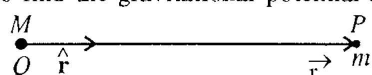

Fig. 4.3.

$$\vec{E} = -\frac{GM}{r^2}\hat{\mathbf{r}}$$

Hence

$$V = -\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r} = \int_{\infty}^{\vec{r}} \frac{GM}{r^2}\hat{\mathbf{r}} \cdot d\vec{r} = \int_{\infty}^{\vec{r}} \frac{GM}{r^2} dr$$

or

$$V = -\frac{GM}{r} \qquad ...(6)$$

If a mass m is placed at a distance $\vec{r}$ from M, then the **gravitational potential energy** of the system of M and m masses will be

$$U = -\int_{\infty}^{\vec{r}} \vec{F} \cdot d\vec{r} = -\int_{\infty}^{\vec{r}} -\frac{GMm}{r^2}\hat{\mathbf{r}} \cdot d\vec{r} = -\frac{GMm}{r} \qquad ...(7)$$

Therefore, from (6) and (7)

$$U = mV \qquad \text{or} \qquad V = U/m \qquad ...(8)$$

This is the *relation between the gravitational potential energy and gravitational potential.*

The **potential difference between two points** at distances $\vec{r}_1$ and $\vec{r}_2$ in a gravitational field is given by

$$V_2 - V_1 = -\int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r} \qquad ...(9)$$

Relation between the intensity of gravitational field ($\vec{E}$) and gravitational potential (V) : From the definition of gravitational potential,

$$V = -\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r} = \int_{\infty}^{\vec{r}} (E_x dx + E_y dy + E_z dz) \qquad ...(10)$$

where $\vec{E} = E_x\hat{\mathbf{i}} + E_y\hat{\mathbf{j}} + E_z\hat{\mathbf{k}}$ and $d\vec{r} = dx\hat{\mathbf{i}} + dy\hat{\mathbf{j}} + dz\hat{\mathbf{k}}$; E_x, E_y, E_z are the components of $\vec{E}$ along X, Y, Z axes respectively. Hence,

$$V = -\int E_x\, dx - \int E_y dy - \int E_z dz \qquad ...(11)$$

Differentiating V partially with respect to x, y and z, we get

$$\frac{\partial V}{\partial x} = -E_x, \quad \frac{\partial V}{\partial y} = -E_y, \quad \frac{\partial V}{\partial z} = -E_z,$$

i.e.,

$$E_x = -\frac{\partial V}{\partial x}, \; E_y = -\frac{\partial V}{\partial y}, \; E_z = -\frac{\partial V}{\partial z}$$

Hence

$$\vec{E} = E_x\hat{\mathbf{i}} + E_y\hat{\mathbf{j}} + E_z\hat{\mathbf{k}} = -\hat{\mathbf{i}}\frac{\partial V}{\partial x} - \hat{\mathbf{j}}\frac{\partial V}{\partial y} - \hat{\mathbf{k}}\frac{\partial V}{\partial z}$$

or

$$\vec{E} = -\vec{\nabla}V = -\text{ grad } V \qquad ...(12)$$

For one dimension,

$$E = -\frac{dV}{dx} \text{ or } E = -\frac{dV}{dr} \qquad ...(13)$$

4·3. Gravitational Potential Energy of a System of Masses

The gravitational potential energy of a system of two masses is defined as the amount of work done by an external agent to assemble the system. Thus if there are two masses m_1 and m_2 at a distance r_{12} apart, then the gravitational potential energy of the system is

$$U = -\int_{\infty}^{r_{12}} \vec{F}\cdot d\vec{r} = \int_{\infty}^{r_{12}} \frac{Gm_1m_2}{r^2}\hat{\mathbf{r}}\cdot d\vec{r} = \int_{\infty}^{r} \frac{Gm_1m_2}{r^2}dr = -\frac{Gm_1m_2}{r} \qquad ...(1)$$

The potential energy belongs to the configuration as a whole. It is not meaningful to assign a certain fraction of it to one of the charges of the system.

In case, a system consists of three particles of masses m_1, m_2 and m_3, then the gravitational potential energy of the system is given by

$$U = -\frac{Gm_1m_2}{r_{12}} - \frac{Gm_1m_3}{r_{13}} - \frac{Gm_2m_3}{r_{23}} \qquad ...(2)$$

which is the sum of potential energy of the three pairs of masses in the sum.

Now if we have a system of n masses, m_1, m_2, ..., m_n, then the gravitational potential energy of the system will be obtained by summing the potential energy for all the pairs in the system *i.e.*,

$$U = -\frac{Gm_1m_2}{r_{12}} - \frac{Gm_1m_3}{r_{13}} - \cdots\cdots\cdots - \frac{Gm_im_j}{r_{ij}} = -G\sum_{\substack{\text{all pairs}\\ i\neq j}} \frac{m_im_j}{r_{ij}} \qquad ...(3)$$

or

$$U = -G\sum_{i>j}^{i=n}\sum_{j=1}^{j=n-1}\frac{m_im_j}{r_{ij}} = -\frac{G}{2}\sum_{i=1}^{i=n}\sum_{j=1\neq i}^{j=n}\frac{m_im_j}{r_{ij}} \qquad ...(4)$$

4·4. Equipotential Surface

An equipotential surface is the surface, at all the points of which the potential is constant. The potential due to a point mass m at a distance r is

$$V = -\frac{Gm}{r}$$

and it is constant on a spherical surface of radius r. This is the equipotential surface. The intensity of the field $\vec{E}$ at r is given by

$$\vec{E} = -\vec{\nabla}V = -\frac{dV}{dr}\hat{\mathbf{r}}$$

where $\hat{\mathbf{r}} = \vec{\mathbf{r}}/r$ is along OA [**fig. 4.4**].

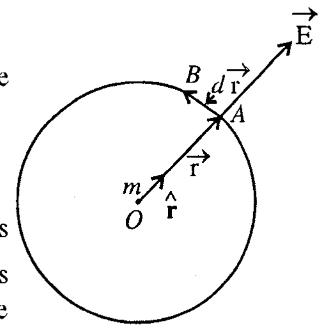

Fig. 4.4.

Now if a unit mass moves a small distance $\vec{AB} = d\vec{\mathbf{r}}$ on the surface [**fig. 4.4.**], then

$$\vec{\mathbf{E}}\cdot d\vec{\mathbf{r}} = -\frac{dV}{dr}\hat{\mathbf{r}}\cdot d\vec{\mathbf{r}} = 0$$

because vector $\hat{\mathbf{r}}$ is perpendicular to the vector $d\vec{\mathbf{r}}$ at the surface. This means that for equipotential surface, the intensity of the field $\vec{\mathbf{E}}$ is perpendicular to the surface at any point A. If a unit mass is moved from one point to the other on an equipotential surface, the amount of work done is equal to zero (because $\int_A^B \vec{\mathbf{E}}\cdot d\vec{\mathbf{r}} = V_B - V_A = 0$).

If we have a mass m' at A, then force on the mass is given by

$$\vec{\mathbf{F}} = -\vec{\nabla}U = -\vec{\nabla}(m'V) = -m'\vec{\nabla}V = m'\vec{\mathbf{E}}.$$

Thus the force is also directed perpendicular to the equipotential surface at A. Hence, if any mass is moved on an equipotential surface, no work will be done.

4·5. Gravitational Potential and Field due to a Thin Spherical Shell

The point, at which the gravitational potential is to be considered may lie (i) outside the shell, (ii) on the surface of the shell or (iii) inside the shell.

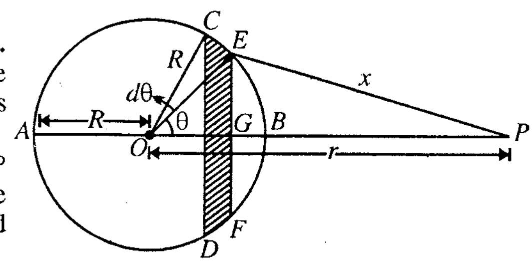

Fig. 4.5.

(i) **Point outside the shell :** Let P be a point [**fig. 4.5**] at which the potential is to be determined and O the centre of a spherical shell of radius R. The point P lies at a distance r from the centre O of the shell.

Draw two planes CD and EF, perpendicular to OP and close to each other. Thus these planes cut out a slice of the shell in the form of a ring $CDFE$ of radius EG and width CE.

If $\angle EOB = \theta$ and $\angle COE = d\theta$, then

radius of the ring $EG = OE \sin\theta = R\sin\theta$

and width of the ring $CE = OE\, d\theta = R\, d\theta$

$\therefore$ Area of the ring = circumference × its width [**fig. 4.6**]

$= 2\pi R\sin\theta \times Rd\theta$

$\therefore$ Mass of the ring $= 2\pi R\sin\theta \times Rd\theta \times \sigma = 2\pi R^2\sigma\sin\theta\, d\theta$

where σ is the mass per unit area of the surface of the shell.

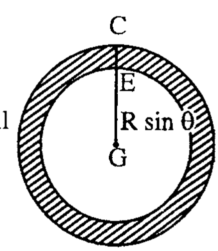

Fig. 4.6

Since every point of the ring is at the same distance x from the point P, the small potential dV at P due to the ring is given by

$$dV = -G\cdot\frac{\text{mass of the ring}}{x} = -\frac{G\cdot 2\pi R^2\sigma\sin\theta\, d\theta}{x}$$

In triangle OEP, $EP^2 = OE^2 + OP^2 - 2\,OE\cdot OP\cdot\cos\theta$

or $x^2 = R^2 + r^2 - 2Rr\cos\theta$

Differentiate x with respect to θ, $2x\,dx = 2Rr\sin\theta\, d\theta$ or $\sin\theta\, d\theta = x\,dx/Rr$

$$\therefore \quad dV = \frac{G\cdot 2\pi R^2\sigma x\,dx}{Rrx} = -\frac{G\cdot 2\pi R\sigma}{r}dx$$

Integrating this equation for the whole shell between the limits $PB = r - R$, and $PA = r + R$, we get the potential V due to the whole shell at P, *i.e.*,

$$V = \int_{r-R}^{r+R} -G\cdot\frac{2\pi R\sigma}{r}dx = -G\cdot\frac{2\pi R\sigma}{r}[x]_{r-R}^{r+R} = -G\cdot\frac{2\pi R\sigma}{r}2R = -G\cdot\frac{4\pi R^2\sigma}{r}$$

If M is the mass of the shell, then $M = 4\pi R^2\sigma$. Hence

$$V = -\frac{GM}{r} \qquad ...(1)$$

Thus, *the potential at a point outside a thin spherical shell is the same, as if its whole mass were concentrated at its centre.*

Field : The intensity of the gravitational field at the point P is given by

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left(-\frac{GM}{r}\right) = -\frac{GM}{r^2} \qquad ...(2)$$

This is the same as due to a mass M placed at O.

Force on a point mass m. If a mass m is placed at the point P, then the potential energy will be given by

$$U = mV = -\frac{GMm}{r} \qquad ...(3)$$

and the force acting on the mass due to the shell is given by

$$F = -\frac{dU}{dr} = -\frac{GMm}{r^2} \qquad ...(4)$$

Thus, we see that *the spherical shell acts at points outside as if all its mass M were concentrated at the centre of the shell.*

(ii) Point on the surface of the shell : If we consider the point P to be situated at B, *i.e.*, on the surface of the shell, then we can get the potenial there by integrating the above expression for dV, between the limits $x = 0$ and $x = 2R$. Hence, the potential at the point P lying on the surface of the shell, is given by

$$V = \int_0^{2R} -G\cdot\frac{2\pi R\sigma}{r}dx = -G\cdot\frac{2\pi R\sigma}{r}[x]_0^{2R} = -G\cdot\frac{4\pi R^2\sigma}{r}$$

Here $r = R$ and $4\pi R^2\sigma = M$,

$$\therefore \qquad V = -\frac{GM}{R} \qquad ...(5)$$

Field : The gravitational field at the point P, when it lies on the surface of the shell, is

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left(-\frac{GM}{r}\right) = -\frac{GM}{r^2}$$

Here, $r = R$, $\quad \therefore \quad E = -\dfrac{GM}{R^2}$ ...(6)

Thus, *the gravitational attraction of a shell at a point on its surface is the same as would be if its whole mass were concentrated at its centre.*

(iii) Point inside the shell : Let the point P lie inside the shell at a distance r from O [**fig. 4.7**]. Proceeding exactly in the same way as before, to find the potential at P due to the ring $CDFE$, we get

$$dV = -\frac{G\cdot 2\pi R\sigma}{r}dx$$

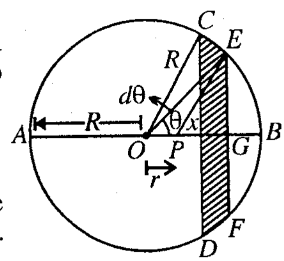

Fig. 4.7.

To obtan the potential V at P due to the whole shell, therefore, we integrate the expression for dV between the limits $BP = R - r$ and $AP = R + r$. Thus

$$V = \int_{R-r}^{R+r} -G\cdot\frac{2\pi R\sigma}{r}dx = -G\cdot\frac{2\pi R\sigma}{r}[x]_{R-r}^{R+r}$$

$$= -G\cdot\frac{2\pi R\sigma}{r}2r = -G\cdot\frac{4\pi R^2\sigma}{R}$$

or $\qquad V = -\dfrac{GM}{R} \qquad [\because M = 4\pi R^2\sigma] \quad ...(7)$

This value is the same as the potential at the surface of the shell. Thus *the potential at any point inside a spherical shell is constant and equal to the value of the potential on its surface.*

Field : The gravitational field at P is given by

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left(-\frac{GM}{R}\right) = 0 \qquad \left[\because \frac{GM}{R} \text{ is a constant}\right] \quad ...(8)$$

Thus, *the intensity of the field at any point inside the spherical shell is zero.*

Force on a point mass m : The potential energy, when a mass m is placed at any point P, is given by

$$U = mV = -\frac{GMm}{R} \text{ (for } r < R) \quad ...(9)$$

and the force acting on the mass is given by

$$F = -\frac{dU}{dR} = 0 \text{ (for } r < R) \quad ...(10)$$

Thus, *if a mass is placed at any point inside a thin spherical shell, then the force acting on it is always zero.* This is a very special property of an inverse square law force. Outside the shell, the force varies as $1/r^2$.

A graph between V and r drawn on the basis of the above results is shown in **fig. 4.8 (a)**. This shows that the potential is constant at all the points inside the shell, but decreases when $r > R$. **Fig. 4.8 (b)** represents the variation of the intensity of the field with distance r.*

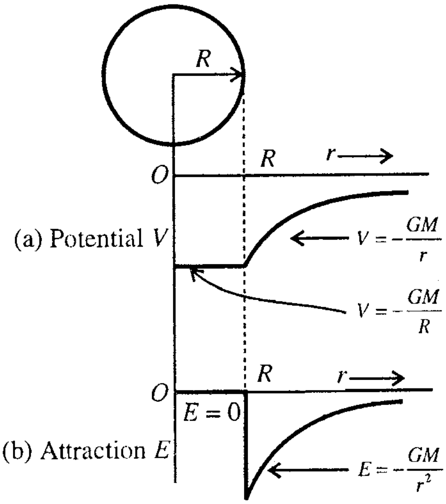

Fig. 4.8.

4·6. Gravitational Potential and Field due to a Solid Sphere

(i) Point outside the sphere : Consider a point P outside the sphere at a distance r from its centre O. We have to determine the potential at P.

Imagine the sphere to be divided into a large number of thin spherical shells concentric to the sphere, of masses m_1, m_2, ... etc. [**fig. 4.9**].

Potential at P due to each individual shell will be

$$= -\frac{G \times \text{mass of the shell}}{r}$$

$\therefore$ Potential at P due to all spherical shells of masses m_1, m_2, m_3, ... etc. is

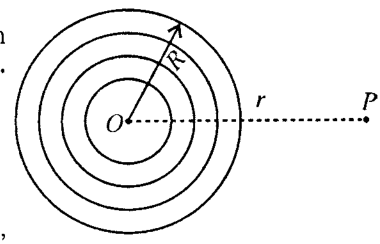

Fig. 4.9.

$$V = -\frac{Gm_1}{r} - \frac{Gm_2}{r} - \frac{Gm_3}{r} - ......$$

$$= -\frac{G}{r}(m_1 + m_2 + m_3 + ...).$$

Now, clearly $m_1 + m_2 + ... = M$, the mass of the sphere,

$$\therefore \quad V = -\frac{GM}{r} \qquad \text{(for } r > R) \qquad ...(1)$$

Thus, *in the case of a solid sphere, the potential at a point lying outside sphere is the same as if its whole mass were concentrated at the centre.*

When the point P lies on the **surface of the sphere**, obviously the potential will be

$$V = -\frac{GM}{R} \qquad [\because r = R, \text{ the radius of the sphere}]$$

* U-r graph [*i.e.*, the graph between the potential energy (U) and the distance of the mass from the centre (r)] and F-r graph (*i.e.*, the graph between the force on the mass m and its distance from the centre) are exactly of the same shape as V-r and E-r graphs respectively.

Intensity of the field : The intensity of the gravitational field at the point r outside the sphere is given by

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left(-\frac{GM}{r}\right) = -\frac{GM}{r^2} \quad (\text{for } r > R) \qquad ...(2)$$

The field at a point P, when it lies on the **surface of the sphere** is

$$E = -\frac{GM}{R^2} \qquad [\because \text{ Here } r = R]$$

Force on a mass m. If a point mass m is placed at a distance r (> R), then its potential energy due to the solid sphere is given by

$$U = mV = -\frac{GMm}{r} \qquad ...(3)$$

and the force, acting on the mass m, is

$$F = -\frac{dU}{dr} = -\frac{GMm}{r^2} \qquad ...(4)$$

(ii) Point inside the sphere : Let the point P lie inside the sphere at a distance r from its centre O **[fig. 4.10]**. The radius of the sphere is R and the density of its material ρ.

With O as centre and radius $OP = r$, we draw a sphere. Then the point P lies on the surface of the solid sphere of radius r and inside the spherical shell of internal radius r and external radius R.

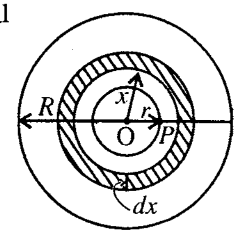

Fig. 4.10.

Volume of the inner solid sphere = $\frac{4}{3}\pi r^3$;∴ Mass of the sphere = $\frac{4}{3}\pi r^3 \rho$, where ρ is the density of the sphere.

Hence potential V_1 at P due to the inner solid sphere

$$V_1 = -\frac{G\cdot\frac{4}{3}\pi r^3\rho}{r} \quad \text{or} \quad V_1 = -\frac{4}{3}\pi r^2 \rho G \qquad ...(5)$$

To determine the potential at P due to the outer spherical shell let us imagine it to be divided into a number of thin concentric shells and consider one such shell of radius x and infinitesimally small thickness dx.

Now, the surface area of the spherical shell = $4\pi x^2$; Volume of the shell = $4\pi x^2\, dx$

∴ Mass of the shell = $4\pi x^2\, dx\, \rho$

Remembering that the potential at any point within a spherical shell is the same as on the surface, we have potential at P due to the shell = $-\frac{G 4\pi r^2 dx \rho}{x}$ = $-4\pi\rho G\cdot x dx$. The potential V_2 at the point P due to shell of internal radius r and external radius R is obtained by integrating this expression between the limits $x = r$ and $r = R$. Thus,

$$V_2 = \int_r^R -4\pi\rho G\cdot x dx = -4\pi\rho G\left[\frac{x^2}{2}\right]_r^R = -4\pi\rho G\left(\frac{R^2}{2} - \frac{r^2}{2}\right) = -2\pi G\,(R^2 - r^2)\,\rho \qquad ...(6)$$

Hence the total potential at the point P is

$$V = V_1 + V_2 = -\frac{4}{3}\pi r^2 \rho G - 2\pi\rho G\,(R^2 - r^2) = -2\pi\rho G\left(\frac{2}{3}r^2 + R^2 - r^2\right) = -\frac{2}{3}\pi\rho G\,(3R^2 - r^2)$$

$$= -\frac{4}{3}\pi R^3 \rho G \frac{(3R^2 - r^2)}{2R^3}$$

But $\frac{4}{3}\pi R^3 \rho = M$, the mass of the sphere,

$$\therefore \qquad V = -GM\left(\frac{3R^2 - r^2}{2R^3}\right) \qquad ...(7)$$

Field : The **gravitational attraction** or the intensity of the field at the point P is given by

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left[-GM\left(\frac{3R^3 - r^2}{2R^3}\right)\right] = -\frac{GM}{2R^3}\cdot 2r = -\frac{GM}{R^3}r \qquad ...(8)$$

$$\therefore \qquad E \propto r. \qquad ...(9)$$

Thus, *the intensity of the gravitational field at a point inside a solid sphere is proportional to its distance from the centre and is zero at its centre.*

Force on mass m **:** The potential energy, when a mass m is placed at any point r inside the sphere, is given by

$$U = mV = -GMm\left(\frac{3R^3 - r^2}{2R^3}\right) \quad \text{(for } r < R\text{)} \quad ...(10)$$

and the force acting on the mass is given by

$$F = -\frac{dU}{dr} = -\frac{GMm}{R^3}r \qquad \text{(for } r < R\text{)} \quad ...(11)$$

Thus, *the force on a point mass m, placed inside a sphere, is proportional to its distance from the centre and is zero at the centre.*

The graph drawn between V and r [drawn on the basis of eq. (1) and (7)] shows that in case of a solid sphere the potential continuously increases upto the centre of the sphere [**fig. 4.11(a)**]. The graph between E and r represents that the maximum value of intensity occurs at the surface of the sphere and its value decreases on proceeding towards the centre, becoming ultimately zero at the centre [**fig. 4.11(b)**].

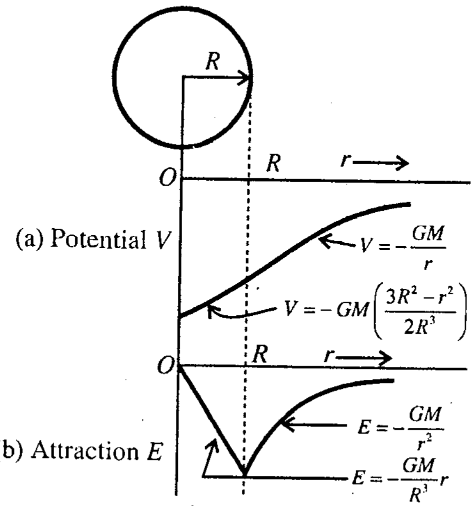

Fig. 4.11.

4·7. Gravitational Field and Potential due to a Thick Spherical Shell

(i) Point outside : Consider a thick spherical shell of inner radius R_1 and outer radius R_2 with mass M. Similar to the case of sphere, we consider the thick spherical shell, composed of a number of thin spherical shells of masses m_1, m_2, ... etc. [**fig. 4.12**]. So that the gravitational potential at a point P at distance r, outside of the spherical shell, is given by

$$V = -\frac{Gm_1}{r} - \frac{Gm_2}{r} - .... = -\frac{G}{r}(m_1 + m_2 + ...)$$

or $$V = -\frac{GM}{r}$$

where $m_1 + m_2 + ... = M$ is the mass of the thick spherical shell.

Intensity of field :
$$E = -\frac{dV}{dr} = -\frac{GM}{r^2} \qquad ...(2)$$

(ii) Point on the surface : At $r = R_2$, $V = -\frac{GM}{R_2}$ and $E = -\frac{GM}{R_2^2}$ ...(3)

(iii) Point inside the spherical shell : Let us consider the thick spherical shell to be divided into a number of concentric thin spherical shells. If we take a thin spherical shell of radius x and thickness dx, then similar to the case of solid sphere, the gravitational potential dV at a point P at r inside the shell is given by [**fig. 4.13**]

$$dV = -\frac{G4\pi x^2 dx\rho}{x} = -4\pi G\rho x dx$$

where ρ is the density of the material of the shell.

The potential V at the point P due to the thick spherical shell is obtained by integrating from R_1 to R_2, *i.e.*,

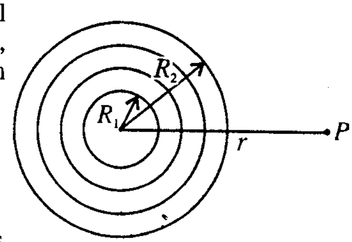

Fig. 4.12.

$$V = \int_{R_1}^{R_2} -4\pi G\rho x dx = -2\pi G (R_2^2 - R_1^2)\rho \qquad ...(4)$$

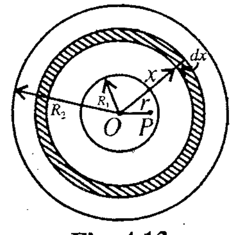

Fig. 4.13.

But $\rho = M\Big/\left[\frac{4\pi}{3}(R_2^2 - R_1^2)\right]$, $\therefore$ $V = -\frac{3}{2}GM\left(\frac{R_2^2 - R_1^2}{R_2^3 - R_1^3}\right) \qquad ...(5)$

Intensity of field $E = -\frac{dV}{dr} = 0 \qquad ...(6)$

(iv) Point between two surfaces : The point P lies at a distance r from the centre O between the two spherical surfaces of radii R_1 and R_2. Point P is on the surface of the shell of radii R_1 and r, due to which the potential is V_1 (say) and P is inside the spherical shell of radii r and R_2, due to which the potential is V_2 (say). So that

$$V_1 = -\frac{G \times \frac{4}{3}\pi(r^3 - R_1^3)\rho}{r} \text{ and } V_2 = -2\pi G(R_2^2 - r^2)\rho$$

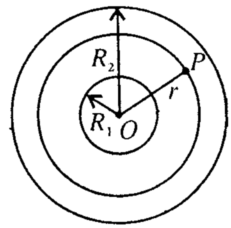

Fig. 4.14.

$$\therefore \qquad V = V_1 + V_2 = -\frac{4}{3}\pi G\rho\left[\frac{(r^3 - R_1^3)}{r} + \frac{3}{2}(R_2^2 - r^2)\right]$$

or
$$V = -\frac{2}{3}\pi G P\left(3R_2^2 - \frac{2R_1^2}{r} - r^2\right)$$

or
$$V = -\frac{GM}{2}\frac{(3R_2^2 - r^2 - 2R_1^2/r)}{(R_2^3 - R_1^3)} \qquad ...(7)$$

Hence,
$$E = -\frac{dV}{dr} = -\frac{GM}{r^2}\left(\frac{r^3 - R_1^3}{R_2^3 - R_1^3}\right) \qquad ...(8)$$

4·8. Velocity of Escape

It is a thing of common experience that when a body is projected in the upward direction, it returns back due to the gravitational pull of the earth on it. Now, if the body is projected upwards with such a velocity which will just take the body beyond the gravitational field of the earth, then it will never come back. This velocity of the body is called the **velocity of escape**.

Let us consider a body of mass m lying at a distance r (> R) from the centre of the earth (or planet or sun). The force with which the earth attracts the body is given by

$$F = -\frac{GMm}{r^2}$$

where M is the mass of the earth.

If this mass be moved through a distance dr away from the earth, then the small work done on the body is

$$dW = \frac{GMm}{r^2}dr.$$

In order that the body may not return on the earth, sufficient work must be done on it to impart the body so much kinetic energy that it moves to infinity. Therefore the work done in moving the body from the surface of the earth to infinity is

$$W = \int_R^\infty \frac{GMm}{r^2}dr = \left[-\frac{GMm}{R}\right]_R^\infty = \frac{GMm}{R} \qquad ...(1)$$

Evidently, a body of K.E. greater than this (GMm/R) would escape completely from the earth. Hence the minimum velocity v of projection of the body to escape into space is given by the relation,

$$\frac{1}{2}mv^2 = \frac{GMm}{R} \quad \text{or} \quad v = \sqrt{\frac{2GM}{R}} \qquad ...(2a)$$

But $$g = GM/R^2*, \qquad \therefore \quad v = \sqrt{2gR} \qquad \text{...(2b)}$$

Expression (2a) and (2b) are for the **escape velocity** of a body from earth.

If at the surface of the earth, we take $g = 9.8$ m/sec^2 and its radius $R = 6.4 \times 10^6$ m, then escape velocity

$$v = \sqrt{2 \times 9.8 \times 6.4 \times 10^6} = 11.12 \times 10^3 \text{ m/sec} = 11.2 \text{ km/sec.} \qquad \text{...(3)}$$

Hence, if a body is projected in the upward direction with a velocity 11.2 km/sec or more than this, it will never return to the earth. Escape velocity is the same for all bodies of different masses.

If the body is situated initially at a distance r (> R) from the centre of earth, then for escaping of the body, the kinetic energy needed, will be

$$\frac{1}{2}mv^2 = \frac{GMm}{r}$$

So that the escape velocity $$v = \sqrt{\frac{2GM}{r}} = \sqrt{\frac{2gR^2}{r}} \qquad \text{...(4)}$$

It may be noted that a particle, having velocity equal to or more than escape velocity, will escape even if it has not been projected vertically upwards.

From the expression for escape velocity $v = \sqrt{2GM/R}$, it is clear that if the particle is projected from the surface of some other planet or satellite (*i.e.*, moon), the escape velocity will depend on its **mass** and **radius**. For moon

$$v = \sqrt{\frac{2 \times 6.6 \times 10^{-11} \times 7.34 \times 10^{22}}{1.74 \times 10^6}} = 2.3722 \text{ km/sec.} \qquad \text{...(5)}$$

which is nearly $\frac{1}{5}$th the value at the earth's surface.

At ordinary temperatures, the root mean square (R.M.S.) velocity for hydrogen is nearly 2 km/sec and for O_2 and CO_2, it is nearly 0.5 km/sec. This is much below the escape velocity near the surface of the earth, hence the atmosphere is retained on earth's surface. But there may be some leakage of hydrogen (obtained from the dissociation of the vapour of water) because some of the gas molecules have much higher velocity than the mean value. The molecules of oxygen, obtained from the dissociation of water vapour, will be left in earth's atmosphere. Probably, due to this cause, *an excess of oxygen is found in the earth's atmosphere*, while the proportion of hydrogen is much higher in the rest of the universe. The escape velocity for the moon is nearly 2 km/sec, hence the molecules of hydrogen can easily escape out of the moon's atmosphere. Further, the molecules of other gases, having high velocity, will also leak from its atmosphere. *This explains the rarity of atmosphere at moon's surface.*

Thus, the gaseous atmosphere cannot be retained on smaller planets or satellites, which have escape velocity much lower.

Ex. 1. *Two bodies of masses M_1 and M_2 are placed distant d apart. Show that at the position, where the gravitational field due to them is zero, the potential is given by*

$$V = -\frac{G}{d}(M_1 + M_2 + 2\sqrt{M_1 M_2}).$$ **(Avadh 1996; Rajasthan 68)**

Sol. Let the gravitational field be zero at a point distant r from the mass M_1. Then, the distance of the same point from the mass $M_2 = d - r$. At the point, under consideration, the field due to M_1 and M_2 is zero, *i.e.*,

$$-\frac{GM_1}{r^2} = -\frac{GM_2}{(d-r)^2} \qquad \text{or} \qquad \frac{d-r}{r} = \sqrt{\frac{M_2}{M_1}}$$

Adding 1 on both sides, we have

* If a body of mass m is situated on the surface of the earth where the value of acceleration due to gravity is g, then $mg = GMm/R^2$ or $g = GM/R^2$.

$$\frac{d}{r} = \frac{\sqrt{M_2}+\sqrt{M_1}}{\sqrt{M_1}} \quad \text{or} \quad r = \frac{d\sqrt{M_1}}{\sqrt{M_1}+\sqrt{M_2}}; \therefore\ d-r = d - \frac{d\sqrt{M_1}}{\sqrt{M_1}+\sqrt{M_2}} = \frac{d\sqrt{M_2}}{\sqrt{M_1}+\sqrt{M_2}}$$

Therefore, potential at the point is

$$V = -\frac{GM_1}{r} - \frac{GM_2}{(d-r)} = -\frac{GM_1}{d\sqrt{M_1}}(\sqrt{M_1}+\sqrt{M_2}) - \frac{GM_2}{d\sqrt{M_2}}(\sqrt{M_1}+\sqrt{M_2})$$

$$= -\frac{G}{d}(M_1+\sqrt{M_1M_2}+M_2+\sqrt{M_1M_2}) = -\frac{G}{d}(M_1+M_2+2\sqrt{M_1M_2}).$$

Ex. 2. *Why do we call sometimes g as the intensity of gravity ? Show that the earth's surface potential is equal to gR, where R is the radius of the earth.* **(Allahabad 1990)**

Sol. If the earth is assumed to be a uniform sphere of radius R, then intensity of gravitational field at any point at its surface is given by

$$E = -\frac{GM}{R^2}, \text{ where } M \text{ is the mass of the earth.}$$

But the force with which the earth attracts a body of mass m lying at its surface is mg, hence

$$mg = -\frac{GMm}{R^2} \quad \text{or} \quad g = -\frac{GM}{R^2} = E.$$

Thus the gravitational intensity at any point on the earth's surface is g and consequently g is termed as intensity of gravity.

Potential at any point on earth's surface $= -\dfrac{GM}{R} = -\dfrac{GM}{R^2}R = gR.$

Ex. 3. *Find the gravitational potential at the surface of the earth from the following data : Radius of earth $R = 6.37 \times 10^6$ m; mean density of the earth $\rho = 5500$ kg/m^3 and $G = 6.66 \times 10^{-11}$ S.I. unit.* **(Raipur 1999)**

Sol.
$$V = -\frac{GM}{R} = -\frac{G}{R}\cdot\frac{4}{3}\pi R^3\rho = -\frac{4}{3}G\pi R^2\rho$$

$$= -\frac{4}{3}\times 6.66\times 10^{-11}\times 3.142\times(6.37\times 10^6)^2\times 5500 = \mathbf{-\ 6.2 \times 10^7\ J/kg.}$$

Ex. 4. *Assuming the radius of earth = 6.37×10^6 m, mean density = 5.5×10^3 kg/m^3, G = 6.66×10^{-11} nm^2/kg^2, calculate the gravitational potential at the surface of earth. Hence find the velocity to project a body of mass 5.0 kg from the earth's surface so as to reach a height equal to the radius of earth above the earth's surface.* **(Raipur 1993)**

Sol. As above, $V = -\dfrac{GM}{R} = -\ 6.2 \times 10^7$ J/kg

Suppose a mass m is projected from the surface at P of the earth so that it reaches at a height h (=R) at Q [**fig. 4.15**]. From the law of conservation of energy,

Total mechanical energy at P = Total mechanical energy at Q

$$-\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{R+R} + 0 \qquad \text{(At } Q\text{, K.E. = 0)}$$

Hence,
$$\frac{1}{2}mv^2 = \frac{GMm}{R} - \frac{GMm}{2R} = \frac{GMm}{2R} \quad \text{or} \quad v^2 = \frac{GM}{R}$$

$\therefore$
$$v = \sqrt{\frac{GM}{R}} = \sqrt{6.2\times 10^7} = \mathbf{7.9 \times 10^3\ m/s}$$

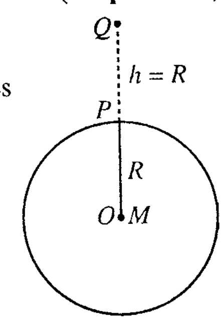

Fig. 4.15.

Ex. 5. *If a mass 50 kg is raised to a height 2R from the earth's surface, calculate the change in potential energy. ($g = 9.8$ m/sec^2; $R = 6.4 \times 10^6$ m)* **(Allahabad 1979)**

Sol. The potential energy of a mass m distance r from a solid sphere of mass M is given by

$$U_{(r)} = \text{Potential} \times m = -\frac{GMm}{r} \qquad \left[\because\ V = -\frac{GM}{r}\right]$$

(referring zero potential energy at infinite distance).

But $g = GM/R^2$, where R is the radius of earth,

$$\therefore \quad U_{(r)} = -\frac{gR^2 m}{r}$$

Difference of potential energy of mass m between the points at r_1 and r_2 is

$$U_{(r_2)} - U_{(r_1)} = -\frac{gR^2 m}{r_2} + \frac{gR^2 m}{r_1}$$

Here $r_1 = R$ and $r_2 = 2R + R = 3R$

$$\therefore \quad U_{(r_2)} - U_{(r_1)} = -\frac{gRm}{3} + gRm = \frac{2}{3} gmR$$

$$= \frac{2}{3} \times 9.8 \times 50 \times 6.4 \times 10^6 = \mathbf{2.1 \times 10^9 \text{ joules.}}$$

Ex. 6. *What is the potential energy of a mass of* 1 *kg on the surface of the earth, referred to zero potential energy at infinite distance ? Calculate also its potential energy at a distance of* 10^5 *km from the centre of the earth.*

Sol. In case of earth (or solid shere) the potential energy is given by

$$U_{(r)} = -\frac{GMm}{r}$$

(i) For the surface of the earth, $r = R$

$$\therefore \quad U_{(r)} = \frac{GMm}{R} = -\frac{(6.67 \times 10^{-11}) \times 5.98 \times 10^{24} \times 1}{6.37 \times 10^6} = \mathbf{-6.37 \times 10^7 \text{ J.}}$$

(ii) $r = 10^5$ km $= 10^8$ m,

$$U_{(r)} = -\frac{(6.67 \times 10^{-11}) \times 5.98 \times 10^{24} \times 1}{10^8} = \mathbf{-3.98 \times 10^6 \text{ J.}}$$

Ex.7. *Calculate the mass of sun from the following data—*
Distance between sun and earth $= 1.49 \times 10^{11}$ *m;* $G = 6.66 \times 10^{-11}$ *S.I. unit;* 1 *year* = 365 *days.*

(Raipur 1996; Indore 94)

Sol. If m and M are the masses of earth and sun respectively and the earth is revolving about the sun in a circular path of radius r with angular velocity ω, then

$$\frac{GMm}{r^3} = m\omega^2 r \quad \text{or} \quad M = \frac{\omega^2 r^2}{G}$$

Here, angular velocity ω of earth around the sun is given by $\omega = \dfrac{2\pi}{365 \times 24 \times 60 \times 60}$ rad/sec,
$r = 1.49 \times 10^{11}$ m and $G = 6.66 \times 10^{-11}$ S.I. unit

$$\therefore \quad M = \left(\frac{2\pi}{365 \times 24 \times 60 \times 60}\right) \times \frac{(1.49 \times 10^{11})^3}{6.66 \times 10^{-11}} = \mathbf{1.972 \times 10^{30} \text{ kg}}$$

Ex. 8. *A mass m is taken from the surface of a sphere of radius a and mass M to its centre. Calculate the necessary work done.* **(Rewa 1993)**

Sol. Work done $= U_2 - U_1$ = Potential energy at the centre – Potential energy at the surface

$$= m(V_2 - V_1) = m\left[-\frac{3}{2}\frac{GM}{a} + \frac{GM}{a}\right] = -\frac{GMm}{2a}.$$

Ex. 9. *A body from the surface of the earth is projected with velocity* v_0*. Prove that the velocity at a height h is given by*

$$v^2 = v_0^2 - \frac{2gh}{1+(h/R)}.$$

(Bundelkhand 2004; Ujjain 1995)

Sol. From the law of conservation of energy, Mechanical energy at earth's surface = Mecanical energy at height h

$$\frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{h+R}$$

or
$$v^2 = v_0^2 + \frac{2GM}{h+R} - \frac{2GM}{R} = v_0^2 - \frac{2GMh}{R(h+R)}$$

or
$$v^2 = v_0^2 - \frac{2GMh}{R^2(1+h/R)} = v_0^2 - \frac{2gh}{1+(h/R)} \qquad \left[\because g = \frac{GM}{R^2}\right]$$

Ex. 10. *If a straight tunnel is made between two places of the earth, then find the component of gravitational force along the tunnel at a distance x from the middle point of the tunnel.*

Sol. If O is the centre of the earth and C is the middle point of the tunnel, then

$$\cos OPC = \frac{CP}{OP} = \frac{x}{r}, \text{ where } r = PO.$$

The gravitational force at P along $PO = \frac{GMm}{R^3}r$

Its component along tunnel $= \frac{GMm}{R^3}r \cos OPC$

$$= \frac{GMmr}{R^3}\cdot\frac{x}{r} = \frac{mgx}{R} \qquad \left[\because g = \frac{GM}{R^2}\right]$$

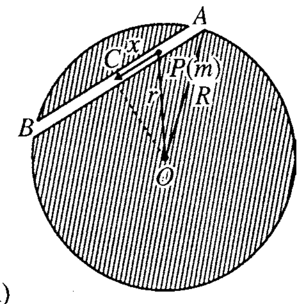

Fig. 4.16.

Ex. 11. *For the earth-sun system, calculate* (i) *the kinetic energy and* (ii) *the work which will have to be done in doubling the radius of the orbit of the earth.*

Sol. The gravitational potential at the earth due to the sun = $-GM_s/r$

$\therefore$ Potential energy of the earth $= -\frac{GM_sM_e}{r}$

K.E. of the earth $= \frac{1}{2}M_ev^2$.

In case of earth, the necessary centripetal force for its rotation is provided by gravitational attraction, *i.e.*,

$$\frac{M_ev^2}{r} = \frac{GM_sM_e}{r^2},$$

$$\therefore \quad \frac{1}{2}M_ev^2 = \frac{1}{2}\frac{GM_sM_e}{r}$$

$$\therefore \quad \text{Total energy of the earth} = -\frac{GM_sM_e}{r} + \frac{1}{2}\frac{GM_sM_e}{r} = -\frac{GM_sM_e}{2r}.$$

Evidnetly, negative sign indicates that the earth is bound with the earth with this energy, *i.e.*,

$$\text{The binding energy} = \frac{GM_sM_e}{2r} = \frac{1}{2}M_ev^2 = \frac{1}{2}M_e \times r^2\omega^2$$

$$= \frac{1}{2} \times 5.98 \times 10^{24} \times \left(\frac{1.5\times10^{11}\times2\times3.14}{365\times24\times60\times60}\right)^2 = \mathbf{2.7 \times 10^{33}\ joules.}$$

$$\left[\because r = 1.5 \times 10^{11} \text{ m and } \omega = \frac{2\pi}{365\times24\times60\times60} \text{ rad/sec.}\right]$$

When the radius of the earth's orbit is doubled, then the final binding energy will be

$$= \frac{GM_sM_e}{2(2r)} = \frac{1}{2}\frac{GM_sM_e}{2r} = \frac{2.7}{2}\times10^{33} = \mathbf{1.35 \times 10^{33}\ joules.}$$

Therefore the amount of work done in doubling the radius of the orbit

$$= 2.7 \times 10^{33} - 1.35 \times 10^{33} = \mathbf{1.35 \times 10^{33}\ joules.}$$

EXERCISE 4.1.

1. Two lead spheres of 20 cm and 2 cm in diameter are placed with their centres 100 cm apart. Calculate the force of attraction between the spheres. Given the radius of the earth as 6.37×10^6 m and its mean density 5,530 kg/m^3 (sp. gravity of lead = 11.5). **(Meerut 1968)**
[**Ans.** 16×10^{-11} newton.]

2. The force of gravitational attraction between two masses m_1 and m_2 at a distance r is given by $\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}}$. What is the potential energy at this separation. Calculate the necessary work done if the distance between them is increased from $r = r_0$ to $r = r_0 + l$.
[**Ans.** $-\frac{GMm}{r}$; $W = \frac{Gm_1m_2l}{r_0(r_0+l)}$.]
[**Hint** : $W = -\frac{Gm_1m_2}{r_0+l} + \frac{Gm_1m_2}{r_0}$.]

3. Two masses of 1 kg and 4 kg are placed at a distance of 12 cm. A third mass of 1 kg is placed on the line in between the two masses so that it experiences no resultant force. What is the position ? Calculate the binding energy of the system and energy needed in removing the third mass alone.
[**Ans.** 4 cm from 1 kg mass; 7.2×10^{-11} joules ; 5×10^{-11} joules.]

4. Three identical bodies each of mass m are located at the vertices of an equilateral triangle of side a. At what speed must they move if they all revolve under the influence of one another's gravity in a circular orbit circumscribing the triangle, while still preserving the equilateral triangle ? What is the energy required to separate them to ∞ ?
[**Ans.** $\sqrt{GM/a}$; $3Gm^2/a$.]

5. Calculate the energy required to bring a body of a mass 1 kg from the centre of earth to its surface. Given : $G = 6.67 \times 10^{-11}$ nm^2/kg^2, $R = 6.4 \times 10^6$ m, $\rho = 5.5 \times 10^3$ kg/m^{-3}. **(Ujjain 1995)**
[**Ans.** 2×10^7 J]

6. What is the potential energy of a mass of 1 kg at a distance of 10^6 km from the centre of earth ? (zero pot ential energy at ∞).
[**Ans.** -3.98×10^5 J.]

7. Calculate the amount of work done to move 1 kg mass from the surface of the earth to a point 10^5 km from the centre of the earth.
[**Ans.** 5.83×10^7 joules.]

8. If for the case of the moon $R_m = 1.7 \times 10^6$ m and $M_m = 7.3 \times 10^{22}$ kg, find (i) the gravitational acceleration at the surface of the moon, and (ii) the escape velocity from the moon.
[**Ans.** (i) 1.70 m/sec^2; (ii) 2.4×10^3 m/sec.]

9. Deduce an expression for the gravitational potential due to a sphere at an external point. Hence calculate the amount of work required to send a body of mass m from earth's surface to a height (i) $R/2$, (ii) $10R$, and (iii) $1000R$, where R is the radius of the earth. Express the result in m, R and g. **(Rajasthan 1967)**
[**Ans.** (i) $\frac{1}{3}gRm$; (ii) $(10/11)gRm$; (iii) $(1000/1001)gRm$.]

10. A mass m is raised to a height nR, where R is the radius of the earth and n is an integer. Calculate the change in its potential energy.
[**Ans.** $\frac{n}{n+1}mgR$.]

11. With what velocity should a body of mass m be projected vertically upwards so that it may be able to reach a height nR above earth's surface ?
[**Ans.** $\sqrt{\frac{2n}{n+1}gR}$.]

12. (a) A meteorite is initially at rest at a distance equal to six times the radius of the earth from the centre of the earth. Calculate the speed with which it strikes with the earth.
[**Ans.** 10.2 km/sec.]
(b) What should be the velocity given to a body to attain the height of one-half the radius of earth ?
[**Ans.** 6.47 km/sec.] **(Agra 2006)**
13. If a body is to be projected vertically upwards from earth's surface to reach a height of $10R$, how much velocity should be given ? **(Kanpur 1984)**
[**Ans.** 10.6×10^3 m/s]
14. A rocket is fixed with 90% of escape velocity. Find the maximum height reached by the rocket.
[**Ans.** $h = 4.2R$, R = radius of earth.] **(Bundelkhand 2004)**
15. Compare the gravitational potentials at the centre and at the surface of a solid sphere.
[**Ans.** 3 : 2] **(Rewa 1999; Bhopal 93)**
16. Compare the gravitational potentials on the surfaces of two planets of same density, but of diameters in the ratio 1 : 2. **(Gwalior 2001; 1995)**
[**Ans.** 1 : 4]
17. A uniform spherical shell is of mass 5 kg and radius 10 cm. Calculate the gravitational potential at the surface and intensity of gravitational field at the centre of shell. **(Gwalior 1994)**
[**Ans.** 3.33×10^{-9} J/kg; zero.]
18. The weight of a man on earth's surface is 60 kg. If the radius of earth is doubled, how much would he weigh (i) if the mass of the earth remains constant, (ii) if the average density of the earth remains constant ?
[**Ans.** (i) 15 kg; (ii) 120 kg.]
19. A man can jump vertically 1.5 metres on the earth's surface. Calculate the maximum radius of a sphere of the same mean density as the earth from whose gravitational field he could escape by jumping. (radius of earth = 6.41×10^6 m).
[**Ans.** 3.103 km.]
20. A uniform sphere has a radius of 1 cm. Calculate the percentage increase in its weight when a second uniform sphere of radius 25 cm and density 10,400 kg m^{-3} is brought immediately below and nearly touching it.
[**Ans.** 6.7×10^{-6} %.]
21. A spherical hollow sphere is made in a metallic sphere of radius r such that its surface touches the outside surface of the metallic sphere. The mass of the sphere before hollowing was M. Calculate the force of attraction on a point mass m at P due to this sphere [**fig. 4.17**].

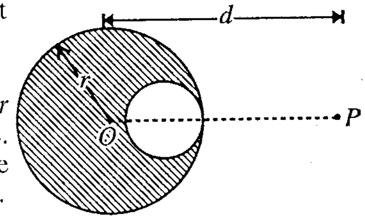

Fig. 4.17.

[**Ans.** $-GM\left(\dfrac{1}{d^2} - \dfrac{1}{2(2d-r)^2}\right)$.]
22. Prove that the least velocity with which a particle must be projected from the surface of a planet of radius R and mean density ρ, in order that it may escape completely is $v = R\sqrt{8\pi\rho G/3}$.
23. Show that two equal spheres, radius r and density equal to the mean desnsity of the earth, starting from rest at a great distance apart will collide with a common speed equal to $r\sqrt{2gR}$, where g is the intensity of gravity and R is the radius of the earth.
24. Two concentric spherical shells, of masses M_1 and M_2, having uniform density, are situated as shown in **fig. 4.18**. Calculate the force on a particle of mass m when (i) the particle is located at $r = a$; (ii) the particle is located at $r = b$ and (iii) the particle is located at $r = c$. The distance r has been measured from the centre of the shells.

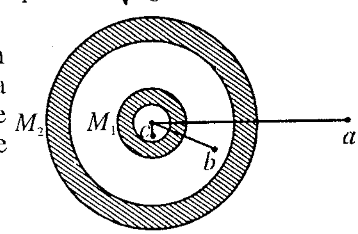

Fig. 4.18.

[**Ans.** (i) $\dfrac{G(M_1+M_2)m}{a^2}$; (ii) $\dfrac{GM_1m}{b^2}$; (iii) zero.]

25. An artificial satellite is moving in a circular orbit around the earth with a speed equal to half the magnitude of escape velocity from the earth. Determine the height of the satellite above the earth's surface. If the satellite stopped suddenly in its orbit and allowed to fall freely onto the earth, find the speed with which it hits the surface of the earth.
(Meerut 1995)

[**Ans.** $h = R_e$; $v = \sqrt{GM/R_e}$.]

26. A satellite of mass M_s is orbitting the earth in a circular orbit of radius R. It starts losing energy slowly at a constant rate C due to friction. If M_e and R_e denote the mass and radius of the earth respectively, show that the satellite falls on the earth at time

$$t = \frac{GM_s M_e}{2C}\left(\frac{1}{R_e} - \frac{1}{R}\right).$$ **(Avadh 1996)**

4·9. Gravitational Self-energy

The potential energy of a body itself is called its **self-energy**. In other words, *the* **self-energy** *of a body is defined as the amount of work done in assembling the body from infinitesimal elements, which are distributed initially at infinite separations.*

The self-energy of a system of two or more particles is the potential energy of that system. Now, let us find an expression for the gravitational **self-energy of a solid sphere.**

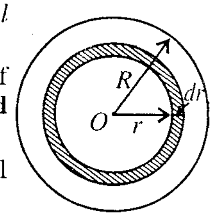

Fig. 4.19

Consider a uniform sphere of mass M and radius R. First consider a spherical core of radius r, then its mass is $\frac{4}{3}\pi r^3 \rho$, where ρ is the density of the material.

Now, if a thin layer of thickness dr is made to surround the core [**fig. 4.19**], then the mass of the layer = volume of the layer × density = surface area of the shell × thickness × $\rho = 4\pi r^2 dr\rho$.

In surrounding the core by the mass $4\pi r^2 dr\rho$, the change in potential energy of the system will be given by

$$dU = (\text{potentital})\ (\text{mass}) = -G\frac{\left(\frac{4}{3}\pi r^3 \rho\right)(4\pi r^2 dr\rho)}{r}$$

$$\left[\because \text{potential on the surface of the spherical core} = \frac{G \times \text{mass of the core}}{r}\right]$$

$$= -\frac{G}{3}(4\pi\rho)^2 r^2 dr. \qquad ...(1)$$

This is the change in self-energy when the radius of the sphere increases from r to $r + dr$. Hence the total self-energy of the sphere, when the radius is increased from 0 to R, is obtained as

$$U_s = \int_0^R -\frac{G}{3}(4\pi\rho)^2 r^4 dr = -\frac{G}{3}(4\pi\rho)^2 \frac{R^5}{5} = -\frac{3G}{5R}\left(\frac{4}{3}\pi R^3 \rho\right)^2$$

or $$U_s = -\frac{3GM^2}{5R} \qquad ...(2)$$

where $\frac{4}{3}\pi R^3 \rho = M$ is the mass of the sphere.

Ex. 1. *Calculate the gravitational self-energy of earth. How many calories of heat must have been produced during the gravitational condensation of the earth from the dust particles or gas cloud ? (Radius of the earth* = 6.37×10^6 *m, M = mass of the earth* = 5.98×10^{24} kg; $G = 6.67 \times 10^{-11}$ *S.I. units;* 4.2 *joule* = 1 *calorie)*

Sol. The gravitational self-energy of the earth is given by

$$U_s = -\frac{3}{5}\frac{GM^2}{R} = -\frac{3}{5}\cdot\frac{(6.67\times10^{-11})(5.98\times10^{24})^2}{6.37\times10^6} = -\mathbf{2.3 \times 10^{32}\ joules.}$$

If H be the amount of heat produced in calories during the gravitational condensation of the earth from the gas cloud or dust particles, we have

$$H = \frac{W}{J} = \frac{2.3 \times 10^{32}}{4.2} = \mathbf{5 \times 10^{31} \text{ calories.}}$$

Ex. 2. *If an isolated sphere of radius r and uniform density contracts at a rate dr/dt, find the energy changes involved. Calculate the rate of energy radiated by the sun in contracting at the rate of 1 km/ year in radius. (Mass of sun = 2×10^{30} kg, Radius of sun = 7×10^8 m, G = 6.67×10^{-11}* S. I. Units.)

Sol. Gravitational self-energy of a sphere of uniform density is given by

$$U_s = -\frac{3}{5}\frac{GM^2}{r}$$

$\therefore$ Rate of change of energy, $\frac{dU_s}{dt} = \frac{dU_s}{dr}\cdot\frac{dr}{dt} = \frac{3}{5}\frac{GM^2}{r^2}\cdot\frac{dr}{dt}$.

At this rate, the energy is radiated in the form the electromagnetic radiations.

Here, $$\frac{dr}{dt} = \frac{10^3}{365 \times 24 \times 60 \times 60} \qquad (\because 1 \text{ km} = 10^3 \text{m})$$

$\therefore$ Rate of energy radiated $= \frac{3}{5}\frac{GM^2}{r^2}\cdot\frac{dr}{dt} = \frac{3}{5}\frac{(6.67 \times 10^{-11})(2 \times 10^{30})^2}{(7 \times 10^8)^2} \times \frac{10^3}{365 \times 24 \times 60 \times 60}$

$$= \mathbf{1.6 \times 10^{28} \text{ J/sec.}}$$

Ex. 3. *If a body or meteor of 200 kg falls on the earth, by how much amount does the self-energy of the system of the earth and body increase or decrease ? What is the potential energy loss by the system ? If it originally started with zero velocity, what will be its velocity when it strikes the earth ? (Mass of the earth = 6×10^{24} kg and radius of the earth = 6.37×10^6m.)*

Sol. As the body is falling towards earth, the work has been done by the body *i.e.*, the self-energy of the system of the earth and the body will decrease by an amount = $-GMm/r$ where m is mass of the body, M the mass of the earth and r the distance of the body from the centre of the earth.

Decrease in the self-energy of the system (at $r = R$) $= \frac{6.67 \times 10^{-11} \times 6 \times 10^{24} \times 200}{6.37 \times 10^6} = \mathbf{12.4 \times 10^9 \text{ J.}}$

Also, the potential energy lost by the system = 12.41×10^9 joules as it is equal to the decrease in self-energy.

If v is the velocity when the body strikes the earth, then gain in its kinetic energy = $\frac{1}{2}mv^2$

But, Loss in P.E. = Gain in K.E.

$\therefore$ $$\frac{1}{2}mv^2 = 12.4 \times 10^9, \quad \therefore \quad v = \sqrt{\frac{2 \times 12.4 \times 10^9}{200}} = \mathbf{1.1 \times 10^4 \text{m/sec.}}$$

EXERCISE 4.2.

1. For the earth-sun system calculate the self-energy ($M_e = 6.0 \times 10^{24}$ kg and distance of earth from sun = 1.5×10^{11} joules.)
   [**Ans.** -5.4×10^{33} joules.]
   [**Hint** : Self-energy = Potential energy = $-GM_sM_e/r = M_ev^2$]
2. What is the gravitational self-energy of the sun ? Mass of the sun = 2×10^{30} kg, Radius of the sun = 7×10^8 m.
   [**Ans.** -2×10^{43} joules.]
3. Show that n particles of mass m separated from one another by the same effective distance r, have a gravitational self-energy equal to $-\frac{1}{2}Gn(n-1)m^2/r$. **(Kanpur 1983; Rohilkhand 79)**

4. If a body of mass 50 kg falls on the earth, by how much does the self-energy of the earth and the body increase or decrease ? What is the potential energy lost by the system ? If it initially started with zero velocity, what will be its velocity when it strikes the earth ?
[**Ans.** Decreases -3.1×10^9 joules; -3.1×10^9 joules; 1.1×10^4 m/sec.]

5. The gravitational self-energy of a uniform sphere of mass M and radius R is $-(3/5)\, GM^2/R$. Explain what is meant by this ? What happens if the sphere contracts in radius by a small amount a ?

[**Ans.** Release of energy equal to $\frac{3}{5}\frac{GM^2 a}{R^2}$ occurs.] **(Agra 1971)**

6. Calculate the decrease in the radius of the sun per year if the amount of solar energy received on the earth is 2 cal/cm^2/sec. Assume that the radiation from the sun arises only on account of its decrease in gravitational potential energy due to contraction.
($M_s = 2 \times 10^{30}$ kg, $R_s = 7 \times 10^8$ m and $R_{es} = 1.5 \times 10^{11}$ m.)
[**Ans.** 2.29 km/year.]

4·10. Gauss's Law of Gravitation

Flux of gravitational field : We consider a plane area ΔS. If this area is present in a uniform gravitational field $\vec{E}$, then the quantity

$$\Delta\phi = E\Delta S \cos\theta \qquad ...(1)$$

is called flux of gravitational field through the surface ΔS (fig. 4.20). Here θ is the. angle between the vector area $\Delta\vec{S}$ and $\vec{E}$. In vector form area $\Delta\vec{S}$ is along the normal to the surface. Hence

$$\Delta\phi = \vec{E}\cdot\Delta\vec{S} \qquad ...(2)$$

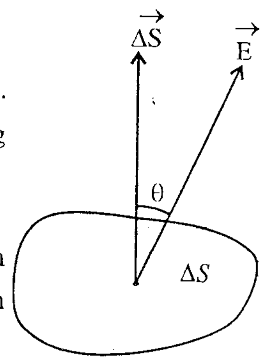

Fig. 4.20.

Now, if $\vec{E}$ is the intensity of the gravitational field at an infinitesimal area $d\vec{S}$ around point P of a closed surface due to a mass m, then the total flux through the closed surface is

$$\phi = \oint_S \vec{E}\cdot d\vec{S} \qquad ...(3)$$

Gauss's Law : *It states that the normal flux of gravitational field across a closed surface, which has a unique outward drawn normal at every point is $-4\pi G$ times the total mass enclosed by that surface, i.e.,*

$$\oint_S \vec{E}\cdot d\vec{S} = -4\pi G\Sigma m$$

where Σm is the total mass inside the closed surface.

Proof : Let a particle of mass m be situated at a point O [**fig. 4.21**]. Suppose that any closed surface S is surrounding the point O. Let BC an infinitesimal small area $d\vec{S}$ around a point P situated on the surface, where the intensity of the field is $\vec{E}$. If r is the distance of P from O, then

$$E = -Gm/r^2$$

The flux though the area $d\vec{S}$ is given by

$$d\phi = \vec{E}\cdot d\vec{S}$$

$$= EdS\cos\theta = -\frac{Gm}{r^2}dS\cos\theta$$

The infinitesimal soild angle $d\Omega$, made by area dS at O, is given by

$$d\Omega = \frac{dS\cos\theta}{r^2}$$

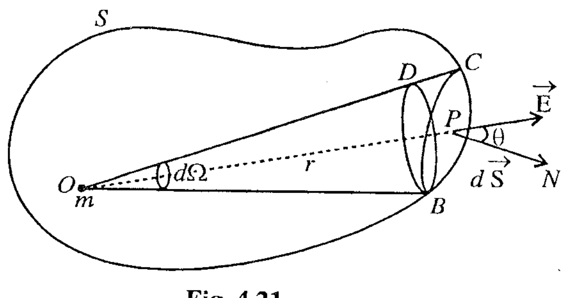

Fig. 4.21.

Hence $d\phi = -\ Gm d\Omega$ ...(4)

Hence the gravitational flux ϕ through the closed surface is

$$\phi = \oint_s \vec{E} \cdot d\vec{S} = -Gm \int d\Omega = -\ 4\pi Gm \quad ...(5)$$

If there are a number of masses m_1, m_2, ... present inside the closed surface S, then the total flux through the surface will be :

$$\phi = \oint_s \vec{E} \cdot d\vec{S} = -\ 4\pi Gm_1 - 4\pi Gm_2 - ....$$

or

$$\oint_s \vec{E} \cdot d\vec{S} = -\ 4\pi G \Sigma m \quad ...(6)$$

Hence Gauss's law is proved.

Note on : Solid angle—Consider a spherical surface of radius r. Solid angle $\Delta\Omega$ is defined as **[fig. 4.22]**

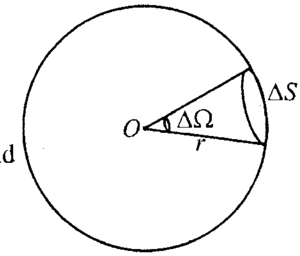

Fig. 4.22.

$$\Delta\Omega = \frac{\Delta S}{r^2} \quad ...(7)$$

where ΔS is a small surface on the spherical surface.

The unit of solid angle is sterdian. As the total area of spherical surface is $4\pi r^2$, then the solid angle subtended at the centre O is

$$\Omega = \frac{4\pi r^2}{r^2} = 4\pi \quad ...(8)$$

Now, suppose dS is an infinitesimal small area at a distance r from O **[fig. 4.23]**. If $\hat{\mathbf{n}}$ is the unit vector along the normal PN on dS (BC), then vector area will be

$$d\vec{S} = dS\,\hat{\mathbf{n}}.$$

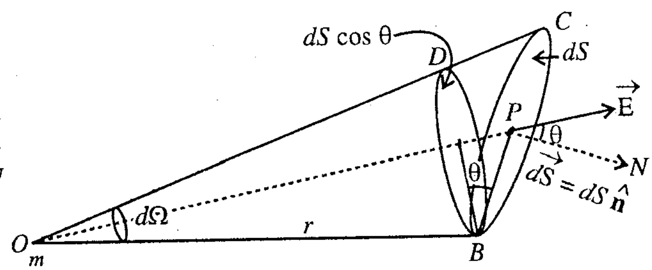

Fig. 4.23.

Solid angle subtended by this surface at O is

$$d\Omega = \frac{\text{area } BD}{r^2} = \frac{dS \cos\theta}{r^2} \quad ...(9)$$

where θ is the angle between area BD (spherical surface) and area BC or dS.

4·11. Simple Applications of Gauss's Law

(1) Intensity of the Field due to a Sphere :

(i) Point outside it : Let O be the centre of a uniform sphere of radius R and mass M. Let P be the point, at which the intensity is to be determined. Draw a sphere with centre O and radius $OP = r$. Then, by the symmetry, the intensity of the field at every point on the surface of the sphere will have the same value, say E, and will be everywhere normal to the surface.

Total normal flux of gravitational field

$$\oint_s \vec{E} \cdot d\vec{S} = \oint_s E dS = 4\pi r^2 E = -\ 4\pi GM$$

[according to Gauss's law]

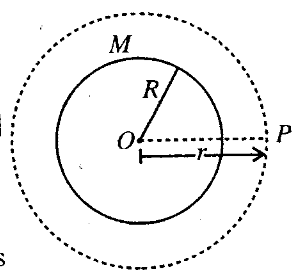

Fig. 4.24.

$$\therefore \quad E = -\frac{GM}{r^2}, \quad ...(1)$$

which has been already proved.

The negative sign indicates that the direction of the attraction is towards O.

(ii) Point inside it : Let the point P lie inside the sphere at a distance r from its centre. Here, the intensity of the field is determined by the mass enclosed within the sphere of radius r. Applying Gauss's law, we have

$$\oint_s \vec{E}\cdot d\vec{S} = -4\pi Gm \quad \text{or} \quad 4\pi r^2 E = -4\pi G\left(\tfrac{4}{3}\pi r^3 \rho\right)$$

where ρ is the density of the material in the sphere.

$$\therefore \qquad E = \tfrac{4}{3}\pi G\rho r \quad \text{or} \quad E \propto r \qquad \text{...(2)}$$

Thus, *the attraction or intensity of the field at a point inside a solid sphere varies as the distance of the point from the centre of the sphere and is zero at the centre.*

(iii) Point on its surface : If the point P lies at the surface of the sphere, then

$$E_{(r=R)} = -\tfrac{4}{3}\pi G\rho R = -\frac{GM}{R^2} \qquad \text{...(3)}$$

This is identical with the value of the field, if the point P at the surface is considered to be outside the sphere. Thus at the surface of the sphere, the intensity of the field is continuous.

(2) Intensity of the Field due to a Spherical Shell :

(i) Point outside it—Let P be a point at a distance r from the centre of a spherical shell of mass M and radius R. Draw a sphere with centre O and radius r. If F be the intensity at P, then by the symmetry of the sphere drawn,

total normal flux $= -4\pi r^2 E = -4\pi GM$ [according to Gauss's theorem]

$$\therefore \qquad E = -\frac{GM}{r^2} \qquad \text{...(4)}$$

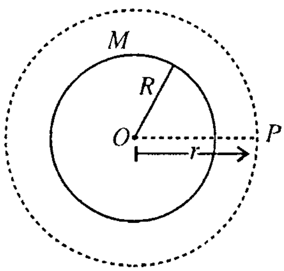

Fig. 4.25.

Thus, the field is the same as if the mass of the shell were concentrated at its centre.

(ii) Point inside it : If the point P, at which the intensity is to be determined, is taken inside the shell at a distance r from the centre, then the surface of the drawn sphere of radius r will not enclose any mass. Hence,

$$4\pi r^2 E = 0 \quad \text{or} \quad E = 0. \qquad \text{...(5)}$$

4·12. Poisson's Equation

Suppose that in a volume τ, there is a mass m and ρ is the density at a point inside the mass. If we consider a volume element $d\tau$ at that point, then obviously

$$m = \int_V \rho\, d\tau \qquad [\because \rho = dm/d\tau \text{ or } m = \int dm = \int \rho\, d\tau] \qquad \text{...(1)}$$

If surface S is enclosing volume τ, then from Gauss's theorem,

$$\int_s \vec{E}\cdot d\vec{S} = \int_V \text{div}\,\vec{E}\, d\tau \qquad \text{...(2)}$$

But from Gauss's law.

$$\int_s \vec{E}\cdot d\vec{S} = -4\pi Gm = -4\pi G\int_V \rho\, d\tau \qquad \text{...(3)}$$

From eqs. (2) and (3),

$$\int_V (\vec{\nabla}\cdot\vec{E})\, d\tau = -\int_V (4\pi G\rho)\, d\tau$$

As the volume has been taken completely arbitrary, hence

$$\vec{\nabla}\cdot\vec{E} = -4\pi G\rho$$

But $\qquad \vec{E} = -\vec{\nabla}V$, where V is the potential.

Hence $\qquad -\vec{\nabla}\cdot\vec{\nabla}V = -4\pi G\rho$

or $\qquad \nabla^2 V = 4\pi G\rho \qquad \text{...(4)}$

This is **Poisson's equation in gravitation.**

Remember that $\vec{\nabla}.\vec{\nabla} = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}\right)\cdot\left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}\right)$ or $\nabla^2 = \frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$ is called **Laplacian operator.**

If $\rho = 0$, then from eq. (4), we have

$$\nabla^2 V = 0 \quad ...(5)$$

This is **Laplace equation.**

QUESTIONS

Long Answer Questions

1. Explain the meaning of gravitational field and gravitational potential. Establish a relationship between them. What are their units ? **(Bhopal 2004, 03; Indore 98; Jabalpur 1999, 98; Ujjain 98; Bilaspur 99; Raipur 95; Rewa 98, 97; Sagar 97)**
2. Deduce expression for the gravitational potential at a point due to point mass. **(Bhopal 1993; Raipur 95)**
3. Obtain expressions for gravitational potential and gravitational field (attraction) due to a uniform spherical shell at a point : (i) inside, (ii) on the surface, and (iii) outside the shell. Draw graphs to show the variations.**(Bundelkhand 2004; Bhopal 2003; Raipur 2003; Gwalior 2004, 99; Agra 1994; Indore 99; Jabalpur 98; Ujjain 97; Sagar 99; Bilaspur 97; Rewa 97; Kanpur 95)**
4. Deduce expressions for gravitational potential due to a uniform solid sphere at a point : (i) inside, (ii) on the surface, and (iii) outside the sphere. Also find the gravitational field at these points, and hence prove that the gravitational potential at the centre of sphere is $\frac{3}{2}$ times that on its surface. **(Agra 2002; Gwalior 2004; Bhopal 1999; Sagar 97; Bilaspur 99; Ujjain 99; Jabalpur 96; Rewa 98; Indore 98, 97; Raipur 98)**
5. What do you mean by the terms (i) Gravitational Field, (ii) Gravitational Potential ? Calculate the gravitational potential due to a soild sphere at a point : (i) outside the sphere, (ii) on the surface of the sphere, (iii) inside the sphere. Calculate the gravitational field of the sphere at these points also. Show that the gravitational potential at the centre of the sphere is one and a half times more than that on its surface. **(Agra 2006, 01, 1995, 93; Gorakhpur 95; Bundelkhand 95; Kanpur 90, 83)**
6. Deduce the expression for gravitational potential for a soild sphere at a point inside it and show that the potential is maximum at the centre of the sphere. Plot gravitational potential against distance from the centre of the sphere. **(Allahabad 1990)**

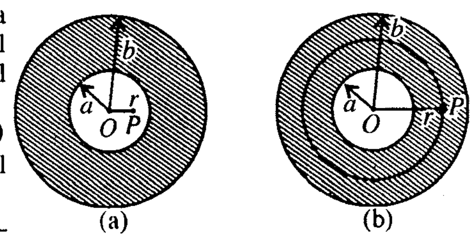

Fig. 4.26.

7. If a and b are the inner and outer radii of a thick spherical shell (hollow sphere), show that
   (i) the potenital at a point inside the shell is $-2\pi\rho G\,(b^2 - a^2)$, where ρ = density of the material of the shell. Hence show that the force on a mass, placed at an internal point is zero.
   (ii) the intensity of the field at a point, lying between the two surfaces, is given by
   $$E = -\frac{GM}{r^2}\cdot\frac{r^3-a^3}{b^3-a^3}.$$ **(Kumaun 1995)**
8. Obtain the gravitational potential and field due to a hollow spherical shell at a point : (i) inside, (ii) at the surface of it (iii) outside it. **(Agra 2005)**
9. Find the gravitational potential and attraction due to a spherical shell bounded by spheres of radii R_1 and R_2 at a point : (i) outside the shell and (iii) inside the shell. **(Kanpur 2004)**
10. State and prove Gauss's law of gravitation. **(Agra 2004)**

11. Deduce an expression for gravitational self-energy of a solid shpere. **(Agra 2004)**
12. What do you understand by the gravitational self-energy ? Find an expression for the gravitational self-energy of a soild sphere. **(Rewa 2003; Gorakhpur 1995)**

Short Answer Questions

1. Explain the terms gravitational field and gravitational potential. **(Agra 2005)**
2. Define gravitational potential and potential energy.
3. The gravitational potential energy of a body on earth's surface is – 6.4×10^6 joules, explain the meaning of this statement.
   [**Ans.** 6.4×10^6 joules of energy required to send a body out of the gravitational field of the earth from its surface.]
4. Why the gravitational potential energy has negative value ? **(Gwalior 1997)**
5. A body enters in the earth's gravitational field—the potential energy of the system (earth-body) will increase or decrease.
6. What is the ratio of gravitational potentials at the centre and the surface of a spherical shell ?
7. Draw a graph showing the variation of gravitational potential with distance for a solid sphere.
8. What is the value of gravitational force on a mass m at the centre of a solid sphere ?
   [**Ans.** Zero.]
9. If a body of mass m be projected vertically upward from earth's surface (radius of earth = R) to reach a height of $10R$, how much kinetic energy be given to the body ?
   [**Ans.** $\frac{10}{11} mgR$]
10. Show that the intensity of the gravitational field can be expressed as $\vec{E} = -\text{grad } V$, where V is the gravitational potential ?
11. Show that the gravitational field due to a uniform solid sphere inside it is directly proportional to the distance of point from the centre of sphere. **(Bilaspur 1989; Gwalior 97, 94)**
12. Show that the intensity of gravitational field is proportional to the distance from the centre of the sphere for a point inside it. **(Kanpur 1980; Kumaun 95)**
13. Prove that the gravitational potential at the centre of a soild sphere will be 3/2 times the potential at its surface. **(Bundelkhand 1995; Kanpur 80)**
14. What is the ratio of gravitational potential at the surface and centre of sphere ?
    [**Ans.** 2/3] **(Bhopal 1993)**
15. The atmosphere on the moon's surface is rarer. Why ? **(Kanpur 1995)**
16. If R is radius of the earth and g the acceleration due to gravity, what is the expression for the mass of the earth ?
    [**Ans.** $M = gR^2/G$]
17. What should be the numerical value of the angular velocity of rotation of the earth in order to make effective acceleration due to gravity equal to zero at the equator of the earth ?

Objective Types Questions

1. If g is the acceleration due to gravity on the earth's surface, the gain in the potential energy by an object of mass m raised from the surface of the earth to a height equal to the radius R_e of the earth is : **(Jabalpur 1989)**
   (a) $\frac{1}{2} mgR_e$ (b) $2mgR_e$ (c) mgR_e (d) $\frac{1}{4} mgR_e$
2. A body of mass one kg is brought to point P from infinity. Initially the body was at rest but at point P it moves with a velocity of 2 m/s. If the work done on the body is 3 joule then the potential energy at point P is :
   (a) – 3 joule/kg (b) – 2 joule/kg (c) – 5 joule/kg (d) none of these
3. A satellite is revolving round a planet (universal gravitational constant G and density ρ) in a circular orbit, then the time period of revolution of satellite near the surface of the planet is : **(Bhopal 1990)**

(a) $\sqrt{G\rho}$ (b) $\sqrt{3G\rho/\pi}$ (c) $\sqrt{G/3\pi\rho}$ (d) $\sqrt{3\pi/G\rho}$

4. If V is the gravitational potential on the surface of a thin spherical shell, then the potential at the centre of the shell is : **(Rewa 1993)**
(a) zero (b) V (c) $V/2$ (d) infinite

5. The ratio of the gravitational potential at the surface of a solid sphere to that at the centre is : **(Agra 2005)**
(a) 1 : 1 (b) 1 : 2 (c) 2 : 3 (d) 3 : 2

6. The ratio of the gravitational potentials at the centre and on the surface of a solid sphere is : **(Rewa 1999)**
(a) $\frac{1}{2}:1$ (b) 1 : 1 (c) $\frac{3}{2}:1$ (d) 2 : 1

7. The value of gravitational potentials energy is : **(Bundelkhand 2004)**
(a) always positive (b) always negative
(c) positive or negative (d) always zero

8. The nature of gravitational field is : **(Bhopal 1990)**
(a) conservative field (b) nonconservative field
(c) pseudo field (d) solenoidal field

9. The gravitational potential on the earth's surface is : **(Rewa 1997)**
(a) -6.2×10^7 joule/kg (b) 11.2×10^7 joule/kg
(c) 6.2×10^7 joule/kg (d) none of these

10. The value of gravitational field at the centre of the sphere is : **(Bundelkhand 2004)**
(a) GM/r (b) $-GM/r^2$ (c) zero (d) Mr/G

11. The escape velocity for a single stage rocket is : **(Rewa 1999, 93)**
(a) 11.2 km/sec (b) 8 km/sec (c) 9.8 km/sec (d) none of these

12. The value of the gravitational field within a soild sphere is :
(a) zero (b) constant
(c) directly proportional to the distance from its centre
(d) inversely proportional to the distance from its centre

13. The orbital velocity of a satellite near earth surface is v. If orbital velocity is increased by 41.4%, then the satellite will go in space : **(Jabalpur 1989)**
(a) no (b) yes
(c) revolve in some other orbit (d) return on the earth

14. The angular momentum of a satellite, having mass 'm', energy 'E' and path of its radiusr', is : **(Rewa 1993)**
(a) $(2mr^2E)^{1/2}$ (b) $2mr^2/E$ (c) $2mE/r^2$ (d) mr^2/E

15. If a body is projected from the earth's surface so as to reach a height nR above the surface, the increase in its potential energy will be : **(Agra 2006)**
(a) mg R (b) mg nR (c) mg $nR/(n+1)$ (d) mg$(n+1)R/n$

ANSWERS

1. (a), **2.** (c) **3.** (d), **4.** (b), **5.** (c), **6.** (c), **7.** (b), **8.** (a), **9.** (a), **10.** (c), **11.** (a), **12.** (c), **13.** (b), **14.** (b), **15.** (c).

5

Conservation of Linear and Angular Momentum

5·1. Introduction

If a body of mass m is moving with velocity $\vec{v}$, its momentum is given by $\vec{p} = m\vec{v}$ and according to Newton's second law of motion, the external force $\vec{F}$ acting on it is equal to the rate of change of momentum, *i.e.*,

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}) \qquad ...(1)$$

In Newtonion mechanics, m is independent of time or velocity, then

$$\vec{F} = m\, d\vec{v}/dt = m\, d^2\vec{r}/dt^2 \qquad ...(2)$$

Here we assume that the body is in the form of a point particle which has mass but has no size. As in translational motion, all the particles of a body move in a similar way, it makes no difference that we consider the body as a point particle or extended object. But for example, when a body rotates in the state of motion, it will not be correct to assume the body as a point particle. However, in this state of motion also, there is a point of the body whose motion in the influence of external forces can be considered as a point particle. This point is called **centre of mass** of the body. In this chapter, we plan to discuss the centre of mass of a system of particles or rigid body and its motion in addition to the principles of conservation of linear and angular momentum.

5·2. System of Particles and Concept of Rigid Body

In a system, there may be one, two or more particles. The discussion for a single particle can be generalized for a system of particles. We can assume a body to be composed of a number of small particles and thus a body can be considered as a system of particles. If on applying an external force on a body, the relative positions of its different particles does not change, then such a body is called a **rigid body**. However, if on applying the external forces its volume or shape is changed so that the relative positions of the particles of the body are changed, then that body is said to be **deformed**. In fact in nature, no body is perfectly rigid and every body can be deformed more or less by applying proper external forces. A solid body in which on applying external forces the deformation is negligible, is assumed as a rigid body. Leaving rubber etc., common solid bodies can be assumed as rigid bodies. For convenience, we will say a rigid body simply a body.

Suppose that there is a system of particles. This system may be in the form of a rigid body. There are forces of mutual interaction between every particle and other particles. These forces are called **internal forces**. In the system, the force on ith particle will be given by

$$\vec{F}_i = \vec{F}_i^{\,e} + \sum_{j=1}^{n} F_{ij} \qquad ...(1)$$

where $\vec{F}_i^{\,e}$ is the external force on the ith particle due to external source of the system; $\vec{F}_{ij}$ is the force on ith particle due to jth particle of the system and the force on the ith particle due to all the other particles of the system (j = 1 to n) is represented by the sign of summation in eq. (1) and $\vec{F}_{ii}$, the force on the ith particle due to itself is naturally zero.

According to Newton's second law of motion,

$$\vec{F}_i = \frac{d\vec{p}_i}{dt} = m_i \frac{d\vec{v}_i}{dt} = m_i \frac{d^2\vec{r}_i}{dt^2} \quad ...(2)$$

where m_i is the mass of ith particle.

Now, if we take the sum for all the particles of the system, the resultant force $\vec{F}$ with the help of eq. (1) will be obtained, *i.e.*,

$$\vec{F} = \sum_{i=1}^{n} \vec{F}_i = \sum_{i=1}^{n} \vec{F}_i^e + \sum_{i=1}^{n} \sum_{\substack{j=1 \\ i \neq j}}^{n} F_{ij} \quad ...(3)$$

But $\sum_i \vec{F}_i = \sum_i m_i \frac{d^2\vec{r}_i}{dt^2} = \frac{d^2}{dt^2} \sum_i m_i \vec{r}_i$ and $\sum_i \vec{F}_i^e = \vec{F}^e$ is the total external force. Also according to Newton's third law of motion, two particles apply equal and opposite forces on each other, *i.e.*,

$$\vec{F}_{ij} = -\vec{F}_{ji} \quad \text{or} \quad \vec{F}_{ij} + \vec{F}_{ji} = 0 \quad \text{or} \quad \sum_{i=1}^{n} \sum_{\substack{j=1 \\ i \neq j}}^{n} \vec{F}_{ij} = 0$$

Hence

$$\vec{F} = \frac{d^2}{dt}(\sum_i m_i \vec{r}_i) = \vec{F}^e \quad \text{or} \quad \vec{F}^e = \frac{d^2}{dt^2}(\sum_i m_i \vec{r}_i) \quad ...(4)$$

Thus the resultant force $\vec{F}$ on the total system is equal to the total external force acting on it and the internal forces in the system mutually cancel the effect of each other.

5·3. Centre of Mass and Its Motion

The centre of mass of a system of particles or body is that point at which the total mass of a body can be assumed to be concentrated. For the regular geometrical shape of the bodies, the centre of mass is that point about which the body is symmetrical. For the objects of irregular shape, it depends on the distribution of mass.

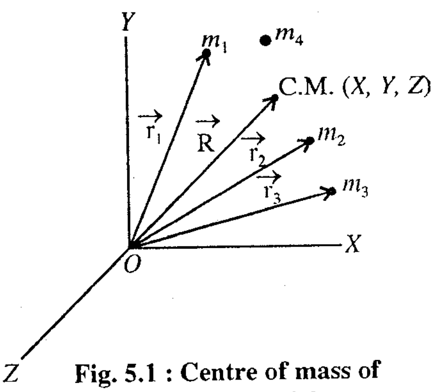

Fig. 5.1 : Centre of mass of a system of particles.

Let us consider a system of N particles, having masses m_1, m_2,..., m_N, and instantaneous position vectors $\vec{r}_1, \vec{r}_2, ... \vec{r}_N$ respectively in a frame of reference XYZ. The **centre of mass** of this system at this instant is the point whose position vector $\vec{R}$ is given by

$$M\vec{R} = m_1 \vec{r}_1 + m_2 \vec{r}_2 + ... + m_N \vec{r}_N \quad ...(1)$$

where M is the total mass of the system, *i.e.*,

$$M = \Sigma m_i = m_1 + m_2 + ... + m_N$$

Therefore,

$$\vec{R} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + ... + m_N\vec{r}_N}{m_1 + m_2 + ... + m_N} = \frac{\sum_{i=1}^{N} m_i \vec{r}_i}{\sum_{i=1}^{N} m_n} \quad ...(2)$$

The sum $\Sigma m_i \vec{r}_i$ is called the **first moment of mass for the system.**

If the coordinates of centre of mass of the system are (X, Y, Z), then

$$\vec{R} = X\hat{i} + Y\hat{j} + Z\hat{k} \quad ...(3)$$

But $\vec{r}_i = x_i\hat{i} + y_i\hat{j} + z_i\hat{k}$ and then comparing (2) and (3), we get

$$X = \frac{m_1x_1 + m_2x_2 + ... + m_Nx_N}{m_1 + m_2 + ... + m_N} = \frac{\Sigma m_i x_i}{\Sigma m_i}; \quad Y = \frac{m_1y_1 + m_2y_2 + ... + m_Ny_N}{m_1 + m_2 + ... + m_N} = \frac{\Sigma m_i y_i}{\Sigma m_i}$$

$$Z = \frac{m_1 z_1 + m_2 z_2 + \ldots + m_N z_N}{m_1 + m_2 + \ldots + m_N} = \frac{\Sigma m_i z_i}{\Sigma m_i} \qquad \ldots(4)$$

Special case : If the centre of mass is situated at the origin of the frame of reference, then $\vec{R} = 0$ and from eq. (2), we have

$$\Sigma m_i \vec{r}_i = 0 \qquad \ldots(5)$$

Hence, *the sum of moments of all the particles about the centre of mass is equal to zero.*

Motion of centre of mass : From eq. (1),

$$M\vec{R} = m_1 \vec{r}_1 + m_2 \vec{r}_2 + \ldots + m_N \vec{r}_N$$

Differentiating eq. (1) with respect to time t, we get

$$M\frac{d\vec{R}}{dt} = m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} + \ldots + m_N \frac{d\vec{r}_N}{dt}$$

Now, writing $\vec{V}$ for the velocity $d\vec{R}/dt$ of the centre of mass and $\vec{v}_1, \vec{v}_2, \ldots$ for the velocities of the particles, we have

$$M\vec{V} = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \ldots + m_N \vec{v}_N$$

or

$$M\vec{V} = \Sigma m_i \vec{v}_i = \vec{P} \qquad \ldots(6)$$

or

$$\vec{V} = \frac{\Sigma m_i \vec{v}_i}{M} = \frac{m_1 \vec{v}_1 + m_1 \vec{v}_2 + \ldots + m_N \vec{v}_N}{m_1 + m_2 + \ldots + m_N} \qquad \ldots(7)$$

This equation gives the **velocity of the centre of mass.**

Middle term of eq. (6) is the total linear momentum of the system ($\vec{P}$). Thus *the total linear momentum of a system is equal to the product of the total mass and velocity of the centre of mass.* In absence of external forces, the total lonear momentum ($\Sigma m_i \vec{v}_i$) remains constant, hence from eq. (6), we have

$$M\vec{V} = \text{a constant} \quad \text{or} \quad \vec{V} = \text{a constant} \qquad \ldots(8)$$

This is a remarkable property of the centre of mass that its **velocity remains constant in the absence of external forces.** For example, when a radioactive nucleus, moving with constant velocity, decays into a number of particles, the particles move in various directions with different velocities, but the velocity of their centre of mass remains the same. However, if initially the centre of mass is at rest, it will remain at the same position afterwards.

If the eq. (6) is again differentiated with respect to time t, we get

$$\frac{d\vec{P}}{dt} = M\frac{d\vec{V}}{dt} = \sum_N \frac{d}{dt}(m_i \vec{v}_i) = \sum_N \vec{F}_i$$

where $\vec{F}_i$ is the force acting on any particle i of the system. Hence, the total force $\sum_N \vec{F}_i$ on all the particles of the system will be equal to the external force $\vec{F}_{ext}$, because taking all the particles of the system, the forces of their mutual interaction cancel together (Newton's third law), *i.e.*,

$$\vec{F}_{ext} = \frac{d\vec{P}}{dt} = M\frac{d\vec{V}}{dt} = m\,\vec{a}_{cm} \qquad \ldots(9)$$

Thus, the acceleration of the centre of mass is due to only the external forces and is given by Newton's second law of motion. Thus, *the centre of mass of a system of particles moves as if it were a* *particle of mass equal to the total mass of the system, subjected to the external forces applied to the system.*

Centre of gravity : The gravitational force on a body is the vector sum of the gravitational forces acting on the individual elements or particles (atoms) of the body. In place of considering all the individual elements of the body, we may say : *'The gravitational force $\vec{F}_G$ acting on a body effectively acts at a single point, called the centre of gravity of the body.'* The word effectively implies that the forces on the individual elements are somehow switched off and the force $\vec{F}_G$ is turned on, the net force and net torque on the body will not change.

If $\vec{g}$ is the acceleration due to gravity and it is the same for all elements of the body, then the centre of gravity of the body coincides with the centre of mass. Normally when we consider small bodies on the surface of the earth (or above), the centre of mass and centre of gravity are taken to be the same. However, if a body is very much extended so that $\vec{g}$ is different at its different elements, then the centre of gravity and centre of mass for the body will not coincide. Further the centre of mass is defined independently of any gravitational effect and the concept of centre of mass will remain meaningful where gravitational force may not be considered.

5·4. Centre of Mass of a System of Two Particles

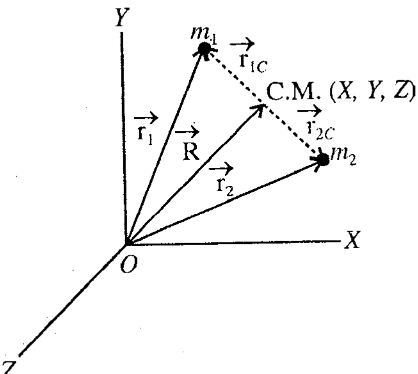

Fig. 5.2 : Centre of mass of a system of two particles.

If at an instant, the position vectors of two particles are $\vec{r}_1$ and $\vec{r}_2$ in a reference system, then the centre of mass of this two particle-system is given by

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \qquad ...(1)$$

where m_1 and m_2 are the masses of the two particles.

Hence, the coordinates of the centre of mass of two particle system are

$$X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, \quad Y = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}, \quad Z = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2} \qquad ...(2)$$

From eq. (1),

$$m_1 \vec{r}_1 + m_2 \vec{r}_2 = m_1 \vec{R} + m_2 \vec{R} \quad \text{or} \quad m_1(\vec{r}_1 - \vec{R}) + m_2(\vec{r}_2 - \vec{R}) = 0$$

i.e.,

$$m_1 \vec{r}_{1C} + m_2 \vec{r}_{2C} = 0 \quad \text{or} \quad m_1 \vec{r}_{1C} = -m_2 \vec{r}_{2C} \qquad ...(3)$$

where $\vec{r}_{1C} = \vec{r}_1 - \vec{R}$ and $\vec{r}_{2C} = \vec{r}_2 - \vec{R}$ are the position vectors of m_1 and m_2 respectively with respect to the centre of mass (C.M.) [**fig. 5.2**].

From eq (3), it is clear that the position vectors of m_1 and m_2 are in opposite directions relative to C.M. Hence, the centre of mass exists on the line joining the two particles. From eq. (3) in magnitude,

$$m_1 |\vec{r}_{1C}| = m_2 |\vec{r}_{2C}| \quad \text{or} \quad \frac{|\vec{r}_{1C}|}{|\vec{r}_{2C}|} = \frac{m_2}{m_1} \qquad ...(4)$$

Hence, **the centre of mass divides internally the line joining two particles in inverse ratio of their masses.**

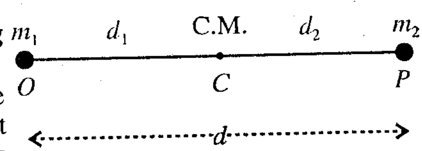

Fig. 5.3.

For simplicity, the two particles of masses m_1 and m_2 are taken at a distance d from one-another. If we take the origin O at m_1 and X-axis along the line joining m_1 to m_2, then the coordinates of m_1 and m_2 will be (0, 0, 0) and (d, 0, 0) respectively (**fig. 5.3**).

The coordinates (X, Y, Z) of the centre of mass of this two particle system will be given by

$$X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \frac{m_2 d}{m_1 + m_2},\ Y = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = 0,\ Z = 0 \qquad ...(5)$$

Thus the coordinates of the centre of mass are $\left(\frac{m_2 d}{m_1 + m_2}, 0, 0\right)$. If m_1 and m_2 are at distances d_1 and d_2 respectively from the centre of mass C, then

$$d_1 = \frac{m_2 d}{m_1 + m_2} \text{ and } d_2 = d - d_1 = d - \frac{m_2 d}{m_1 + m_2} \text{ or } d_2 = \frac{m_1 d}{m_1 + m_2} \qquad ...(6)$$

Therefore, $$\frac{d_1}{d_2} = \frac{m_2}{m_1} \text{ or } \frac{\text{The distance of } m_1 \text{ from C.M.}}{\text{The distance of } m_2 \text{ from C.M.}} = \frac{m_2}{m_1} \qquad ...(7)$$

Centre of mass of two or several groups of particles : Consider two groups of particles with n_1 and n_2 particles in a system of particles. So that the total number of particles in the system are $n_1 + n_2$. Let the centre of mass of first group of particles be C_1 and that of second be C_2. The centre of mass of both groups or system will be given by

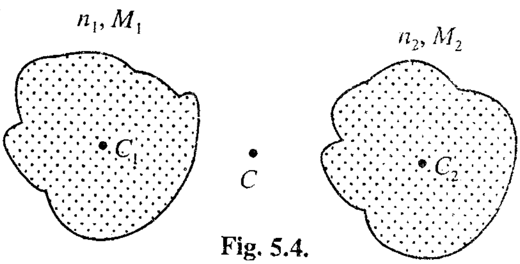

Fig. 5.4.

$$\vec{R} = \frac{\sum_{i=1}^{n_1+n_2} m_i \vec{r}_i}{\sum_{i=1}^{n_1+n_2} m_i} = \frac{\sum_{i=1}^{n_1} m_i \vec{r}_i + \sum_{i=1}^{n_2} m_i \vec{r}_i}{\sum_{i=1}^{n_1} n_1 + \sum_{i=1}^{n_2} n_2}$$

If the position vectors of C_1 and C_2 are $\vec{R}_1$ and $\vec{R}_2$ respectively, then

$$\vec{R}_1 = \frac{\sum_{i=1}^{n_1} m_i \vec{r}_i}{\sum_{i=1}^{n_1} m_i} = \frac{\sum_{i=1}^{n_1} m_i \vec{r}_i}{M_1} \text{ or } \sum_{i=1}^{n_1} m_i \vec{r}_i = M_1 \vec{R}_1 \qquad ...(8)$$

Similarly $$\sum_{i=1}^{n_2} m_i \vec{r}_i = M_2 R_2 \qquad ...(9)$$

where M_1 and M_2 are the masses of two groups of particles.

Hence $$\vec{R} = \frac{M_1 \vec{R}_1 + M_2 \vec{R}_2}{M_1 + M_2} \qquad ...(10)$$

However, this is also the centre of mass of two particles of masses M_1 and M_2 placed at the points C_1 and C_2 respectively. Thus, *if we know the centres of mass of parts of the system and their masses, we can get the combined centre of mass by treating the parts as point particles placed at their respective centres of mass.*

5·5. Centre of Mass of a Continuous Body

A rigid body may be considered as a system of closely packed particles, having continuous distribution of mass. The centre of mass of such a body will be given on replacing the sign of summation (Σ) by the sign of integration in the equation. The centre of mass of (X, Y, Z) a system of particle is given by.

$$X = \frac{\sum_i m_i x_i}{\sum_i m_i},\ Y = \frac{\sum_i m_i y_i}{\sum_i m_i},\ Z = \frac{\sum_i m_i z_i}{\sum_i m_i} \qquad ...(1)$$

If the system of particles is in the form of body of continuous distribution of mass, then we take an infinitesimal small mass dm at (x, y, z) and write the centre of mass (X, Y, Z) of the body as follows:

$$X = \frac{\int x\,dm}{\int dm} = \frac{1}{M}\int x\,dm;\; Y = \frac{\int y\,dm}{\int dm} = \frac{1}{M}\int y\,dm;\; Z = \frac{\int z\,dm}{\int dm} = \frac{1}{M}\int z\,dm. \quad ...(2)$$

where the integration is taken on proper limits.

Determination of centre of mass of a body : The position of centre of mass of a symmetrical body having uniform density, can easily be known by symmetry. Hence, the centre of mass of a thin uniform rod is at its mid-point; of a circular disc or sphere is at its centre and so on.

When a body has only one axis of symmetry it may be taken as X-axis and a suitable point on this axis is taken as origin. With the help of the Integral Calculus, the position of the centre of mass on the X-axis, the axis of symmetry, may be determined.

(1) Centre of mass of a thin uniform rod : Let the X-axis lie along the length OA of the rod and O be the origin [**fig. 5.5**]. If M and l represent its mass and length respectively, then mass per unit length $= M/l$.

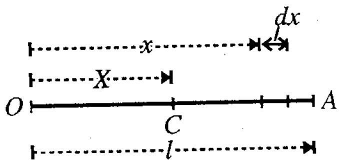

Fig. 5.5 : Centre of mass of a thin rod.

Consider a small element dx at a distance x from the origin, then its mass

$$dm = (M/l)dx$$

Hence, the distance of the centre of mass C of the rod from O is

$$X = \frac{\int x\,dm}{\int dm} = \frac{\int_0^l x(M/l)\,dx}{M} = \frac{(M/l)\left[x^2/2\right]_0^l}{M} = \frac{l}{2} \quad ...(3)$$

Hence, *the centre of mass of the rod is situated in the middle.*

(2) Centre of mass of a thin triangular plate : Let the triangular plate ABC be divided into a very large number of narrow strips, similar to KL, each parallel to BC. The centre of mass of each strip lies at its middle point hence the centre of mass of the entire plate lies on the median AD. Similary, it will also lie on the medians CF and BE. [**fig. 5.6**]. Thus, the centre of mass of the triangular plate lies at the point of intersection G of the three medians and is called the centroid of the triangle. The point G divides each median internally in the ratio 1 : 2, so that

$$\frac{DG}{AG} = \frac{1}{2} \text{ or } \frac{DG + AG}{AG} = \frac{1+2}{2} \text{ or } \frac{AD}{AG} = \frac{3}{2}$$

$$\therefore \quad AG = \frac{3}{2}AD.$$

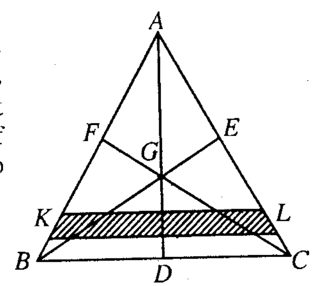

Fig. 5.6 : Centre of mass of a triangular plate.

(3) Centre of mass of a uniform semicircular wire : Let M be the mass and R the radius of a uniform semicircular wire. We take its centre as the origin O, the line joining the ends as the X-axis, and the Y-axis in the plate of the wire [**fig. 5.7**]. The centre of mass must be in the plane of the wire, *i.e.*, in the X-Y plane. Now, we take a small length dl in a direction θ with OX axis. This dl subtends an angle $d\theta$ at O. The mass dm of this small length $dl = Rd\theta$ will be given by

$$dm = (M/\pi R)Rd\theta$$

where $M/\pi R$ is the mass per unit length of the wire.

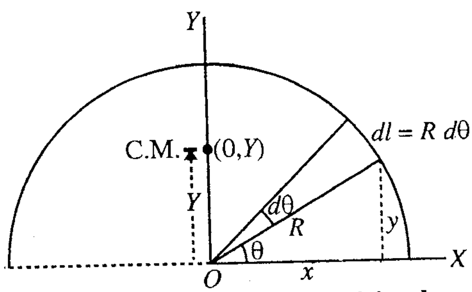

Fig. 5.7 : Centre of mass of a semicircular wire.

Hence $$X = \frac{\int x\,dm}{M} = \frac{\int_0^\pi R\cos\theta\,(M/\pi R)\,Rd\theta}{M} = 0$$

and $$Y = \frac{\int y\,dm}{M} = \frac{\int_0^\pi R\sin\theta\,(M/\pi R)\,Rd\theta}{M} = \frac{2R}{\pi} \quad ...(4)$$

Thus the centre of mass of the given semi-circular wire of radius R is $(0, 2R/\pi)$. Obviously by and looking at the symmetry of the semicircular wire, the centre of mass will be on the Y-axis at a distance $2R/\pi$ from O.

(4) Centre of mass of a uniform semicircular plate : Consider a semicircular plate of radius R and mass M. We take the origin at the centre of the semicircular plate, the X-axis along the straight edge and the Y-axis in the plane of the plate. Now we draw a simicircle of radius r on the plate with the centre at the origin. We increase r to $r + dr$ and draw another semicircle with the same centre. Consider the shaded part of the plate between the two semicircles of radii r and $r + dr$. This part my be considered as semicircular wire. **[fig. 5.8]**, whose centre of mass will be at $(0, 2r/\pi)$ and mass $dm = (2M/\pi R^2)\,\pi r dr$, where $M/(\pi R^2/2)$ is the mass per unit area of the plate and $\pi r dr$ is the area of the shaded part.

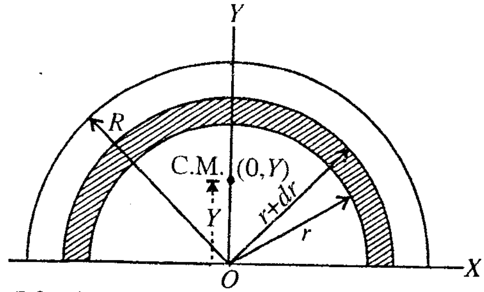

Fig. 5.8 : Centre of mass of a semicircular plate.

Due to symmetry, X-coordinate of the semicircular plate will be at $X = 0$ and Y-coordinate will be calculated as follows :

$$Y = \frac{\int y\,dm}{M} = \frac{1}{M}\int_0^R \left(\frac{2r}{\pi}\right)\left(\frac{2M}{R^2} r dr\right) = \frac{4R}{3\pi} \quad ...(5)$$

Thus the centre of mass of the given semicircular plate of radius R is at $(0, 4R/3\pi)$.

(5) Centre of mass of right circular cone : The **fig. 5.9** represents a right circular cone, having circular base AB of radius R and centre C. Let O, the apex of the cone, be the origin and OC, the X-axis. Here, OC is the axis of symmetry **[fig. 5.9]**.

Consider two parallel planes perpendicular to X-axis and at distances x and $x + dx$ from O. The portion of the cone lying between two planes is a circular disc.

Radius of the disc $r = x \tan\phi$, where ϕ is the semi-vertical angle of the cone. If h be the height of the cone, then

$$\tan\phi = R/h \;;\; \therefore\; r/x = R/h \;\text{ or }\; r = xR/h$$

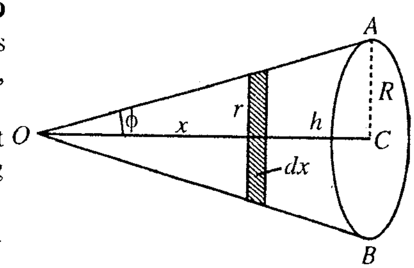

Fig. 5.9.

Area of the disc $= \pi r^2 = \dfrac{\pi x^2 R^2}{h^2}$; Volume of the disc $= \dfrac{\pi x^2 R^2}{h^2}\cdot dx$

$\therefore$ Mass of the disc $dm = \dfrac{\pi x^2 R^2 \rho}{h^2}\cdot dx,$

where ρ is the density of the material of the cone. Hence, the centre of mass of the cone will lie on OC at a distance X from O, given by

$$X = \frac{\int x\,dm}{\int dm} = \frac{\int_0^h x\cdot\frac{\pi x^2 R^2\rho}{h^2}\cdot dx}{\int_0^h \frac{\pi x^2 R^2 \rho}{h^2}\cdot dx} = \frac{\int_0^h x^3\,dx}{\int_0^h x^2\,dx} = \frac{\left[x^4/4\right]_0^h}{\left[x^3/3\right]_0^h} = \frac{h^4}{4}\times\frac{3}{h^3} = \frac{3h}{4}$$

Hence, *the centroid of a soild cone is situated at a distance* 3/4 *of its vertical height measured from its vertex.*

(6) Centre of mass of a soild hemisphere : Fig. 5.10 represents a hemisphere of radius R. Let O, the centre of the circular section, be the origin and OC be the axis of symmetry along X-axis. Consider the portion of the hemisphere, lying between two parallel planes, each perpendicular to OC, at distances x and

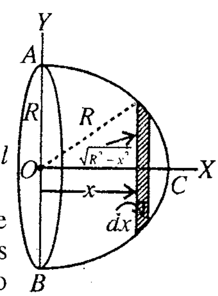

Fig. 5.10.

$x + dx$ from O. This portion will be a circular disc of radius $\sqrt{R^2 - x^2}$.

Face area of the disc $= \pi\,(R^2 - x^2)$; $\therefore$ Volume of the disc $= \pi\,(R^2 - x^2)\,dx$.

If ρ be the density, then its mass $dm = \pi\,(R^2 - x^2)\,\rho\,dx$

The centre of mass of the hemisphere will lie on OC at a distance X from O, given by

$$X = \frac{\int x\,dm}{\int dm} = \frac{\int_0^R \pi(R^2 - x^2)\,x\rho\,dx}{\int_0^R \pi(R^2 - x^2)\,\rho\,dx} = \frac{\left[R^2x^2/2 - x^4/4\right]_0^R}{\left[R^2x - x^3/3\right]_0^R} = \frac{R^4/2 - R^4/4}{R^3 - R^3/3} = \frac{R^4}{4} \times \frac{3}{2R^3} = \frac{3R}{8}.$$

5·6. Centre of Mass Frame of Reference and Laboratory Frame of Reference

No external force acts on an isolated system. Hence the centre of mass of an isolated system moves with constant velocity in an inertial system. *An inertial frame attached with the centre of mass of an isolated system of particles is called the* **centre of mass frame of reference** *or* **C-frame of reference.** The centre of mass of the system will be at rest with respect to the *C*-frame. We know that if $\vec{V}$ is the velocity of the centre of mass in an inertial frame, then the total linear momentum of a system of particles is given by

$$\vec{P} = \Sigma m_i \vec{v}_i = M\vec{V} \qquad ...(1)$$

where $\Sigma m_i = M$ is the mass of all the particles of the system. But in the *C*-frame, velocity of centre of mass is zero, *i.e.*, $\vec{V} = 0$. Hence in *the centre of mass frame of reference, the total linear momentum of the system* $(\Sigma m_i \vec{v}_{ic})$ *is always zero i.e.,*

$$\Sigma m_i \vec{v}_{ic} = \Sigma m_i(\vec{v}_i - \vec{V}) = 0 \qquad ...(2)$$

where $\vec{v}_{ic} = \vec{v}_i - \vec{V}$ = velocity of *i*th particle relative to the velocity of centre of mass.

For this reason, the *C*-frame is also called the **zero-momentum frame.** The centre of mass of a system in absence of external forces moves with constant velocity in an inertial frame, so the centre of mass frame of reference for the isolated system is an **intertial frame.** For a system of particles, which is acted by external forces, the *C*-frame becomes accelerated, *i.e.*, non-inertial and is not of much use.

Centre of mass frame of reference or *C*-frame is very much useful for the analysis of many experiments, which are difficult to be analyzed in the laboratory frame. If we attach a frame of reference with the laboratory (*i.e.*, the laboratory is at rest in this frame), then such a frame of reference is called **laboratory frame of reference** or ***L*-frame.**

Suppose in the *L*-frame, a particle of mass m_1 is moving along *X*-axis with velocity $\vec{u}$ towards another particle of mass m_2 at rest. The velocity of the centre of mass of this two particle system will be given by

$$\vec{V} = \frac{m_1\vec{u} + m_2 \times 0}{m_1 + m_2} = \frac{m_1\vec{u}}{m_1 + m_2} \qquad ...(3)$$

Hence the velocities of the two particles in the *C*-frame will be

$$\vec{u}_{1C} = \vec{u} - \vec{V} = \vec{u} - \frac{m_1\vec{u}}{m_1 + m_2} = \frac{m_2\vec{u}}{m_1 + m_2} \qquad ...(4)$$

and $$\vec{u}_{2C} = 0 - \vec{V} = -\frac{m_1\vec{u}}{m_1 + m_2} \qquad ...(5)$$

Also the momenta of the two particles in *C*-frame will be

$$\vec{p}_{1C} = m_1 \vec{u}_{1c} = \frac{m_1 m_2 \vec{u}}{m_1 + m_2} \text{ and } \vec{p}_{2C} = m_2 \vec{p}_{2C} = -\frac{m_1 m_2 \vec{u}}{m_1 + m_2}$$

Hence $\vec{p}_{1C} = -\vec{p}_{2C}$ or $|\vec{p}_{1C}| = |\vec{p}_{2C}|$ or $m_1 u_{1C} = m_2 u_{2C}$ ...(6)

Thus in *C*-frame the magnitude of the linear momenta of the two particles are equal, while in *L*-frame the linear momenta of the two particles are different, *i.e.*, $m_1 u$ and zero.

Ex. 1. *The mass and position vectors of two particles are respectively* 6 *kg,* $(6\hat{i}+7\hat{j})$ *and* 2 *kg.* $(2\hat{i}+10\hat{j}-8\hat{k})$ *m. Find the position vector of centre of mass.* **(Ujjain 1994)**

Sol. Position vector of the centre of mass of two-particle system is given by

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} = \frac{6(6\hat{i}+7\hat{j}) + 2(2\hat{i}+10\hat{j}-8\hat{k})}{6+2} = \frac{40\hat{i}+62\hat{j}-16\hat{k}}{8} = \mathbf{5\hat{i}-7{\cdot}75\hat{j}-2\hat{k}\ m.}$$

Ex. 2. *If the centre of mass of three particles of masses* 10, 20 *and* 30 *gm be at the point* (1, 1, 1) *m, then where should a fourth particle of* 40 *gm be placed so that the combined centre of mass be at* (0, 0, 0) ? **(Bilaspur 1999; Ujjain 99, 90; Indore 93, 90; Raipur 94)**

Sol. $$X = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3} \text{ or } 1 = \frac{0.01x_1 + 0.02\,x_2 + 0.03\,x_3}{0.01+0.02+0.03}$$

or $$x_1 + 2x_2 + 3x_3 = 6 \qquad ...(i)$$

Suppose 40 gm mass is placed at a point (x_4, y_4, z_4) so that the combined C.M. is at (0, 0, 0). Now,

$$0 = \frac{0.01x_1 + 0.02\,x_2 + 0.03\,x_3 + 0.04\,x_4}{0.01+0.02+0.03+0.04}$$

or $$x_1 + 2x_2 + 3x_3 + 4x_4 = 0 \qquad ...(ii)$$

Subtracting eq. (i) from eq. (ii), we have

$$4x_4 = -6 \quad \text{or} \quad x_4 = -3/2$$

In the same way, $y_4 = -3/2$ and $z_4 = -3/2$.

Therefore 40 gm mass should be placed at **(– 3/2, –3/2, –3/2) m.**

Ex. 3. *Two particles of masses* 100 *gm and* 300 *gm have at a given time position* $(2\hat{i}+5\hat{j}+13\hat{k})$ *m and* $(-6\hat{i}+4\hat{j}-2\hat{k})$ *m respectively, and velocities* $(10\hat{i}-7\hat{j}-3\hat{k})$ *and* $(7\hat{i}-9\hat{j}+6\hat{k})$ *m/sec respectively. Deduce* (i) *the instantaneous position of the centre of mass,* (ii) *the velocity of the second particle in a frame of reference of travelling with the centre of mass.* **(Kanpur 2005)**

Sol.

(i) $$\vec{R} = \frac{m_1 r_1 + m_2 r_2}{m_1 + m_2} = \frac{0.1(10\hat{i}-7\hat{j}-13\hat{k})+0.3(-6\hat{i}+4\hat{j}+2\hat{k})}{0.1+0.3} = \frac{\mathbf{-16\hat{i}+17\hat{j}+7\hat{k}}}{\mathbf{4}}\ \mathbf{m.}$$

(ii) $$\vec{V} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{0.1(10\hat{i}-7\hat{j}-3\hat{k})+0.3(7\hat{i}-9\hat{j}+6\hat{k})}{0.1+0.3} = \frac{\mathbf{31\hat{i}-34\hat{j}+15\hat{k}}}{\mathbf{4}}\ \mathbf{m.}$$

∴ Velocity of the second particle in the centre of mass frame of reference

$$= \vec{v}_2 - \vec{V} = 7\hat{i}-9\hat{j}+6\hat{k} - \frac{31\hat{i}-34\hat{j}+15\hat{k}}{4} = \frac{\mathbf{-3\hat{i}-2\hat{j}+9\hat{k}}}{\mathbf{4}}\ \mathbf{m/sec.}$$

Ex. 4. *The velocities of three particles of masses* 20 *gm,* 30 *gm and* 50 *gm are* $10\hat{i}, 10\hat{j}$ *and* $10\hat{k}$ *cm/sec respectively in an inertial frame. Find the velocity of centre of mass of the system. What are the velocities of the three particles in that frame which is moving with the centre of mass. Assume the axes of the two frames parallel mutually.*

Sol. $\vec{V} = \dfrac{m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3}{m_1+m_2+m_3} = \dfrac{20(10\hat{i})+30(10\hat{j})+50(10\hat{k})}{20+30+50} = 2\hat{i}+3\hat{j}+5\hat{k}$ cm/sec.

Hence the velocities of the particles in the centre of mass frame of reference are

$$\vec{v}_{1C} = \vec{v}_1 - \vec{V} = 10\hat{i} - (2\hat{i}+3\hat{j}+5\hat{k}) = \mathbf{8\hat{i} - 3\hat{j} - 5\hat{k}} \text{ cm/sec.}$$

$$\vec{v}_{2C} = \vec{v}_2 - \vec{V} = 10\hat{j} - (2\hat{i}+3\hat{j}+5\hat{k}) = \mathbf{-2\hat{i} + 7\hat{j} - 5\hat{k}} \text{ cm/sec.}$$

$$\vec{v}_{2C} = \vec{v}_3 - \vec{V} = 10\hat{k} - (2\hat{i}+3\hat{j}+5\hat{k}) = \mathbf{-2\hat{i} - 3\hat{j} + 5\hat{k}} \text{ cm/sec.}$$

Ex. 5. *A boat of mass M = 1400 kg and length L = 20 m is floating on a lake and is stationary. A man of mass m = 60 kg moves from one end of the boat to the other end of the boat. Assuming that the water offers no resistance to the motion of the boat, find the displacement of the boat as a result of man's trip from one end to the other end.* **(Meerut 1995, 93)**

Sol. Initially, let the man of mass m be at a distance x_1 from the shore and the boat's centre of mass at x_2. The centre of mass C of the system is at a distance X from the shore, given by

$$X = \frac{mx_1 + Mx_2}{m+M} \qquad ..(i)$$

When the man moves to the other end of the boat farther the shore, the centre of mass of the boat is displaced by d. However the centre of mass of the total system will remain at the same place in absence of external forces. Thus

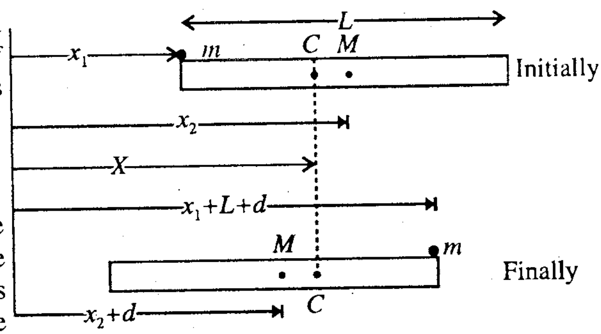

Fig. 5.11.

$$X = \frac{m(x_1 + L + d) + M(x_2 + d)}{m+M} \qquad ...(ii)$$

Subtracting eq. (i) from (ii), we get

$$0 = m(L+d) + Md \quad \text{or} \quad d = -\frac{mL}{m+M}$$

Hence, the boat will be displaced towards the shore by the distance, given by

$$= \frac{mL}{m+M} = \frac{60 \times 20}{60+1400} = \frac{1200}{1460} = \mathbf{0.82\ m}$$

EXERCISE 5·1

1. The distance between the nuclei of CO gas molecule is 1.13 Å. Determine the position of centre of mass of the molecule relative to the carbon atom. The atomic weight of carbon and oxygen atoms are 12 and 16 respectively. **(Meerut 1973)**
   [Ans. 0.6457]
2. Locate the centre of mass of a system of three particles masses 1 kg, 2 kg and 3 kg placed at the corners of an equilateral triangle of 1 metre side. **(Agra 1990, 86; Kumaun 95; Meerut 83)**
   [Ans. $(7/12, \sqrt{3}/4, 0)$m, 1 kg as origin.]
3. If the centre of mass of three particles of masses 1, 2 and 3 gm be at the point (1, 1, 2) then where should a fourth particle of mass 5 gm be placed so that the combined centre of mass be at (0, 0, 0)?
   [Ans. (–1.2, –1.2, –2.4) cm.]
4. Three particles of masses 7, 4 and 10 gm are located at (1, 5, –3), (2, 5, –7) and (3, 3, –1) cm respectively. Find the position vector of their centre of mass.
   [Ans. 15/7, 85/21, – 59/21 cm.]
5. The centre of mass of a system of three particles of masses 4.0 kg, 8.0 kg and 12.0 kg is at (1, 1, 1).

Where should another paticle of mass 3.0 kg be kept so that centre of mass of entire system may come at (0, 0, 0) ? **(Indore 1993; Ujjain 90)**
[**Ans.** (–8, –8, –8).]

6. The centre of mass of three particles of masses 10, 20 and 30 gm is at (1, –2, 3) cm. Where the particle of mass 40 gm is to be placed so that the new centre of mass is at (1, 1, 1) cm ? **(Agra 1975)**
[**Ans.** (1, 5.5, –2) cm.]

7. In an inertial frame of reference find out the velocity of the centre of mass of a system of three particles of masses 20 g, 30 g and 50 g moving with velocities $(\hat{i}+2\hat{j})$, $(\hat{j}-2\hat{k})$ and $(\hat{i}+3\hat{k})$ respectively. Also find the velocity of each particle in a frame that moves with the centre of mass and has coordinate axes parallel to the axes of the first frame. **(Agra 1997)**
[**Ans.** $\vec{V} = 0.7\hat{i}+0.7\hat{j}+0.9\hat{k}$; $0.3\hat{i}+1.3\hat{j}-0.9\hat{k}$; $-0.7\hat{i}+0.3\hat{j}-2.9\hat{k}$; $0.3\hat{i}-0.7\hat{j}+2.1\hat{k}$.]

8. Two masses constrained to move in a horizontal plane collide. If m_1 = 85 gm, m_2 = 200 gm, and initially $\vec{v}_1 = 6.4\hat{i}$ m/sec, $\vec{v}_2 = (-6\hat{i}-2\hat{j})$ m/sec, find : (i) the velocity of the centre of mass, (ii) the total linear momentum and (iii) the velocities in the reference frame in which the centre of mass is at rest.
[**Ans.** (i) $\vec{V} = (-2.8\hat{i}-1.4\hat{j})$ m/sec; (ii) $(0.796\hat{i}-0.4\hat{j})$ kg-m/sec; (iii) $(9.2\hat{i}+1.4\hat{j})$ m/sec; $(3.9\hat{i}-0.6\hat{j})$ m/sec.]

9. Three interacting particles of 20, 30 and 50 gm have each a velocity of 10 m/sec magnitude but along the positive directions of the *X*-axis, *Y*-axis and *Z*-axis respectively in a cartesian frame. What will be the velocity of the first particle, when due to forces of interaction the second particle stops moving and the velocity of the third particle has become $(6\hat{j}+8\hat{k})$ m/sec. What are their initial velocities in the centre of mass frame ? **(Agra 1991)**
[**Ans.** $(10\hat{i}+5\hat{k})$ m/sec; $(8\hat{i}-3\hat{j}-5\hat{k})$, $(-2\hat{i}+7\hat{j}-5\hat{k})$, $(-2\hat{i}-3\hat{j}+5\hat{k})$ m/sec]

10. Three particles of masses 3, 4 and 5 gm are placed at (–2, 1), (4, 2) and (2, –3) metres respectively in *X-Y* plane. The forces acting on them are $-6\hat{i}$ newtons, $8\hat{j}$ newtons and $12\hat{k}$ newtons respectively. Find the acceleration of the centre of mass of this system.
[**Ans.** 0.83 m/sec^2 at 53° 6′ with *X*-axis.]

11. Two particles *A* and *B*, initially at rest, are 1 metre apart. *A* has a mass 0.1 kg and *B* a mass of 0.3 kg. Both particles attract each other with a constant force of 10^{-2} newton. No external force acts on the system. Describe the motion of centre of mass. At what distance from the position of *A* do the particles collide ?
[**Ans.** Centre of mass remains at rest; 0.75 metre.]

12. A boy, weighing 10 kg is standing on a flat so that he is 20 m from the shore. He walks 8 m on the boat towards shore and then halts. The boat weighs 40 kg and assume that there is no friction between it and water. How far is he from the shore at the end of this time ?
[**Ans.** 13.6 m.]
[**Hint** : The centre of mass of (boat + boy) remains stationary.]

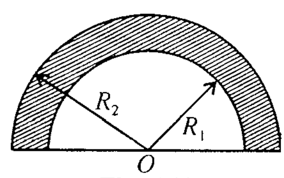

Fig. 5.12.

13. Find the centre of mass of a uniform plate having semicircular inner and outer radii R_1 and R_2 **[fig. 5.12]**.
[**Ans.** $\dfrac{4(R_1^2+R_1R_2+R_2^2)}{[3\pi(R_1+R_2)]}$ above the centre.]

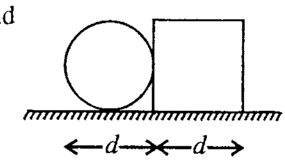

Fig. 5.13.

14. A circular plate of diameter *d* is kept in contact with a square plate of edge *d* as shown in **[fig. 5.13]**. The density of the material and the thickness are the same everywhere. What is the position of the centre of mass ?
[**Ans.** At a distance $4d/(d+\pi)$ from the centre of the disc at right hand side.]

5·7. Conservation of Linear Momentum

If a force $\vec{F}$ is acting on a particle of mass m, then according to Newton's second law of motion, we have

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}) \qquad ...(1)$$

where $\vec{p} = m\vec{v}$ is the linear momentum of the particle. Momentum is a vector quantity.

If the external force acting on the particle is zero, then

$$\vec{F} = d\vec{p}/dt = 0 \text{ or } \vec{p} = m\vec{v} = \text{a constant} \qquad ...(2)$$

Thus, in *absence of an external force, the linear momentum of a particle remains constant.* This is **principle of conservation of linear momentum** for a single particle.

Conservation of linear momentum for a system of two particles : Suppose that in an isolated system there are two particles of masses m_1 and m_2. No external force is acting on the system. However both partcles are exerting interaction forces on each other as in the collision of two particles. We assume that these forces satisfy Newton's third law of motion and we say that these forces are *Newtonian type.* This is the property of the Newtonain forces that the force $\vec{F}_{21}$ which first particle (1) exerts upon the other (2) is the negative of the force $\vec{F}_{12}$ exerted by second (2) on first (1), *i.e.*,

$$\vec{F}_{21} = -\vec{F}_{12} \qquad \text{or} \qquad \vec{F}_{12} + \vec{F}_{21} = 0 \qquad ...(3)$$

If at any instant t, the velocities of the particles are $\vec{v}_1$ and $\vec{v}_2$, then their momenta will be $\vec{p}_1 = m_1\vec{v}_1$ and $\vec{p}_2 = m_2\vec{v}_2$ respectively. Now, according to Newton's second law of motion,

$$\vec{F}_{12} = \frac{d\vec{p}_1}{dt} = \frac{d}{dt}(m_1\vec{v}_1) \quad \text{and} \quad \vec{F}_{21} = \frac{d\vec{p}_2}{dt} = \frac{d}{dt}(m_2\vec{v}_2)$$

Hence eq. (3) assumes the form

$$\frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} = 0 \qquad \text{or} \qquad \frac{d}{dt}(\vec{p}_1 + \vec{p}_2) = 0$$

i.e.,

$$\vec{p}_1 + \vec{p}_2 = \text{a constant} \qquad \text{or} \qquad m_1\vec{v}_1 + m_2\vec{v}_2 = \text{a constant} \qquad ...(4)$$

Thus *in absence of external forces on a system of two particles, their total linear momentum is conserved.*

In a collision of two particles, if the velocities of the particles before the collision are $\vec{v}_1$ and $\vec{v}_2$ and after the collision are $\vec{v}_1'$ and $\vec{v}_2'$ respectively, then according to principle of conservation of linear momentum,

$$m_1\vec{v}_1 + m_2\vec{v}_2 = m_1\vec{v}_1' + m_2\vec{v}_2' \qquad ...(5)$$

This is to be remembered that in an isolated system of two particles, the linear momentum of individual particle may change, but their total linear momentum is always conserved.

Conservation of linear momentum of a system of *n* particles : Now, let us consider a system of n particles whose masses are $m_1, m_2, ..., m_n$. The system can be a **rigid body** in which the particles are in fixed positions with respect to one another, or it can be a **collection of particles** in which there may be all kinds of internal motion.

Suppose that the particles of the system are interacting with each other and are also acted by external forces. If at an instant $\vec{p}_1 = m_1\vec{v}_1$, $\vec{p}_2 = m_2\vec{v}_2$,....., $\vec{p}_n = m_n\vec{v}_n$ are the momenta of the particles of masses $m_1, m_2, \ldots, m_n$ respectively then the total momentum $(\vec{P})$ of the system is the vector sum of the momenta of individual particles, *i.e.*,

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \ldots + \vec{p}_n \qquad \ldots(6a)$$

or
$$\vec{P} = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \ldots + m_n \vec{v}_n \qquad \ldots(6b)$$

Differentiating it with respect to time t, we have

$$\frac{d\vec{P}}{dt} = \frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} + \ldots + \frac{d\vec{p}_n}{dt}$$

or
$$\frac{d\vec{P}}{dt} = \vec{F}_1 + \vec{F}_2 + \ldots, + \vec{F}_n \qquad \ldots(7)$$

where $\vec{F}_1, \vec{F}_2, \ldots, \vec{F}_n$ represent the forces acting on the particles of masses $m_1, m_2, \ldots, m_n$ respectively.

These forces include external and internal forces both. But according to Newton's third law, the internal forces exist in pairs of equal and opposite forces, they balance each other and so do not contribute any thing to the total force. Hence the right hand side sum in eq. (5) represents the resultant force $\vec{F}_{ext}$ only due to the *external forces* acting on all the particles of the system. The internal forces cannot change the total momentum of the system, because being equal and opposite they produce equal and opposite changes in the momentum. Hence, if we want to change the total momentum of the system of the particles, it is necessary to apply external forces on that system. Then the sum of external forces is

$$\vec{F}_{ext} = \frac{d\vec{P}}{dt} = \frac{d}{dt}(\vec{p}_1 + \vec{p}_2 + \ldots + \vec{p}_n)$$

If the resultant external force is zero, then

$$\frac{d\vec{P}}{dt} = 0 \quad \text{or} \quad \vec{P} = \text{a constant.} \quad i.e., \ \vec{P} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \ldots + \vec{p}_n = \text{a constant} \qquad \ldots(8)$$

Thus *if the resultant external force acting on a system of particles is zero, the total linear momentum of the system remains constant.* This simple but quite general result is called the **law of conservation of linear momentum** for a system of particles. It is to be noted that the momenta of individual particles may change, but their sum, *i.e.*, the total linear momentum remains unaltered in the absence of external forces.

We know that if $\vec{V}$ is the velocity of centre of mass, then the total linear momentum $\vec{P}$ of the system is

$$\vec{P} = M\vec{V}$$

where $M = m_1 + m_2 + \ldots$ is the total mass of the system. Hence in absence of external forces, the conservation of linear momentum of a system implies

$$\vec{V} = \text{a constant} \qquad \ldots(9)$$

Hence, **in absence of external forces, the velocity of centre of mass remains constant.**

The law of conservation of momentum is a fundamental and exact law of nature. No violation of it has ever been found and it has been thoroughly checked by all kinds of experiments.

5·8. Examples of Conservation of Linear Momentum

According to the principle of conservation of linear momentum, in absence of external forces the total linear momentum a system is conserved. We are giving below some examples as an application of this principle.

(1) Firing of bullet from a gun : In the beginning when the bullet and the gun are in the rest position, total linear momentum of the system (gun + bullet) is zero. As the bullet is fired, the bullet after coming out of the gun moves with a velocity $\vec{v}$ (say) so that the bullet acquires a forward momentum $m\vec{v}$, where m is the mass of the bullet. In view of the principle of conservation of linear momentum, the total linear momentum of the system, gun + bullet, must remain zero even after the firing of the bullet. Hence the gun should acquire an equal backward momentum, resulting a backward push on the man. If M is the mass of the gun and $\vec{V}$ its velocity, then from the law of conservation of linear momentum

$$m\vec{v} + M\vec{V} = 0 \quad \text{or} \quad \vec{V} = -m\vec{v}/M \qquad ...(1)$$

Therefore, the gun recoils back with a velocity $\vec{V} = -m\vec{v}/M$ in a direction opposite to the velocity $\vec{v}$ of the bullet. As $m << M$, $V << v$, *i.e.*, the velocity of the gun is much smaller in comparison to the velocity of the gun.

(2) Disintegration of a nucleus or blasting of a bomb : Radioactive decay of a nucleus is another example of recoil velocity. Let a nucleus emit an α-particle. If m and M are the masses of the α-particle and the remaining nucleus and their velocities are $\vec{v}$ and $\vec{V}$ respectively after the decay, then the total linear momentum will be $m\vec{v} + M\vec{V}$. If initially the composite nucleus is at rest, then according to the law of conservation of linear momentum,

$$m\vec{v} + M\vec{V} = 0$$

Hence the recoil velocity of the nucleus $\vec{V} = -\dfrac{m\vec{v}}{M}$ ...(2)

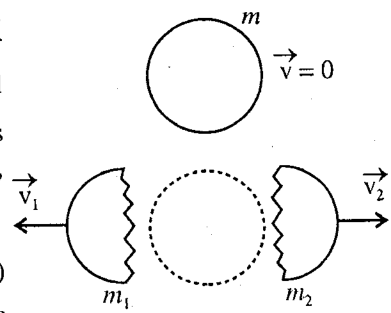

Fig. 5.14.

Similarly if a body or a bomb, which is at rest initially, explodes into two (or more) parts, then from the law of conservation of linear momentum **(fig. 5.14)**.

$$m_1\vec{v}_1 + m_2\vec{v}_2 = 0 \qquad ...(3)$$

where the body breaks into two parts after the explosion.

If initially the velocity of the body or bomb of mass m is $\vec{v}$, then from the law of conservation of momentum

$$m\vec{v} = m_1\vec{v}_1 + m_2\vec{v}_2 \qquad ...(4)$$

(3) Motion of two masses connected by a spring : Suppose that two bodies of masses m_1 and m_2 are connected by a spring and are at rest on a horizontal frictionless table. When the masses are pulled outwardly, the spring extends in length and force of tension develops in the spring. Now, if the bodies are left free, then there is no external force on the system. As the total system is at rest in the beginning (at the instant just released), its total linear momentum must remain zero afterwards also. Thus the two masses will move in the opposite directions because only then the total momentum is zero.

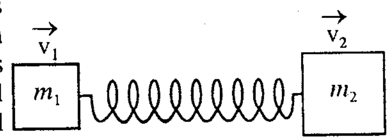

Fig. 5.15.

If m_1 and m_2 are the masses of the two bodies and their respective velocities are $\vec{v}_1$ and $\vec{v}_2$, then according to the principle of conservation of linear momentum,

$$m_1\vec{v}_1 + m_2\vec{v}_2 = 0 \quad \text{or} \quad \vec{v}_2 = -\frac{m_1\vec{v}_1}{m_2} \qquad ...(5)$$

Obviously the velocities of the two bodies are in opposite direction.

If the magnitudes of the velocities are v_1, v_2, then

$$\frac{v_2}{v_1} = \frac{m_1}{m_2} \qquad ...(6)$$

This equation will be correct at all instants after the release of the two bodies. The kinetic energy of the bodies will be $K_1 = \frac{1}{2}m_1v_1^2$ and $K_2 = \frac{1}{2}m_2v_2^2$. Hence the ratio of their kinetic energies is given by

$$\frac{K_2}{K_1} = \frac{\frac{1}{2}m_2v_2^2}{\frac{1}{2}m_1v_1^2} = \frac{m_2^2v_2^2}{m_1^2v_1^2}\cdot\frac{m_1}{m_2}$$

But $m_1v_1 = m_2v_2$ from eq. (6), hence

$$\frac{K_2}{K_1} = \frac{m_1}{m_2} \qquad ...(7)$$

From this equation, it is clear that if the mass m_1 has smaller mass, then its kinetic energy will be greater than that of greater mass m_2.

Fraction f_1 of the kinetic energy (K_1) of m_1 with respect to the total kinetic energy ($K_1 + K_2$) of the system is given by

$$f_1 = \frac{K_1}{K_1+K_2} = \frac{\frac{1}{2}m_1v_1^2}{\frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2} = \frac{\frac{1}{2}m_1v_1^2}{\frac{1}{2}m_1v_1^2+\frac{1}{2}m_1^2v_1^2/m_2} = \frac{1}{1+\frac{m_1}{m_2}} \qquad (\because m_1v_1 = m_2v_2)$$

Thus,
$$f_1 = \frac{m_2}{m_1+m_2} \qquad ...(8a)$$

Similarly fraction of kinetic energy of the second mass is

$$f_2 = \frac{m_1}{m_1+m_2} \qquad ...(8b)$$

Hence if $m_1 < m_2$, fraction of kinetic energy of m_1 will be greater when compared to that of m_2.

Clearly *the kinetic energy of the moving spring-mass system changes with time, but the division of this energy in the two bodies will not depend on time and will remain constant with lower mass, having greater energy.*

Eqs. (7) and (8) are vaild for a body falling in the gravitational field of the earth. If m_1 is the mass of the body and m_2 is the mass of the earth, then in their centre of mass frame of reference, the body acquires approximately the total energy ($f_1 = 1$), while the earth acquires extremely small energy ($f_2 \approx 0$). This is the cause the total energy of a body falling in the gravitational field of the earth is assumed to be conserved and the kinetic energy of the earth is not considered in the calculation.

(4) Collision of two particles : Let us consider two particles of masses m_1 and m_2, moving with velocities $\vec{u}_1$ and $\vec{u}_2$ respectively before the collision*. If their velocities after the collision are $\vec{v}_1$ and $\vec{v}_2$, then according to the law of conservation of momentum we have

$$m_1\vec{u}_1 + m_2\vec{u}_2 = m_1\vec{v}_1 + m_2\vec{v}_2 \qquad ...(9)$$

Here, the terms 'before' and 'after' the collision mean that the initial (and final) positions are widely separated so that the interaction forces between the particles become effectively zero. Hence the patential energy before and after the collision remains the same.

Two particle collisions are of two types :

(i) Elastic collision : This is the collision in which the kinetic energy of the system is conserved alongwith the conservation of linear momentum *i.e.*,

$$\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2 \qquad ...(10)$$

* When two particles approach each other, their mutual interaction changes the motion continuously and in consequence to it, their momenta are changed. Such a process is called **collision.** From the physics point of view a collision must not necessarily be a physical contact of the colliding particles. For example, an electron (or proton) approaching an atom is subjected to electrostatic forces and therefore collision takes place.

According to the kinetic theory of gases, such elastic collisions occur between the molecules of a gas. Such elastic collision also occurs sometimes between electrons, protons and atoms.

(ii) Inelastic collision : In an elastic collision, the linear momentum is conserved but the kinetic energies before and after the collision are not equal, *i.e.*,

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 \neq \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

In inelastic collisions some part of the initial kinetic energy is converted in the form of heat or excitation energy.

The final velocities $\vec{v}_1$ and $\vec{v}_2$ of the particles cannot be calculated uniquely only by the knowledge of their initial velocities and conservation laws of momentum and energy. The reason lies in the fact that any velocity has three components and hence for the determination of $\vec{v}_1$ and $\vec{v}_2$, six independent equations are required. But equation (9) gives three equations in component form and equation (10) gives only one equation. In case, if after the collision the direction of movement of one particle is also known, then we can calculate $\vec{v}_1$ and $\vec{v}_2$ completely.

5·9. Elastic Collision–Deflection of a Moving Particle with a Particle at Rest (*Study in C-Frame*)

Let us consider a particle of mass m_1, colliding elastically with another particle of mass m_2, which is initially at rest in the laboratory frame of reference.* Suppose in this frame, the initial velocity of m_1 is $\vec{u}_1$ and after the collision its final velocity be $\vec{v}_1$ in a direction making an angle θ_1, with its initial path. This angle θ_1, through which the particle of mass m_1 has been deflected by collision, is called **scattering angle**. Now, if we take the direction of $\vec{u}_1$ along *X*-axis and the plane containing $\vec{u}_1$ and $\vec{v}_1$ as *X-Y* plane, then the final velocity $\vec{v}_2$ of m_2 will have no *Z*-component, because $m_1\vec{u}_1$ and $m_1\vec{v}_1$ have no *Z*-components. If $\vec{v}_2$ makes an anlge θ_2 with the initial path of m_1, we have the following equations for the conservation of momentum and energy :

$$m_1u_1 = m_1v_1 \cos\theta_1 + m_2v_2 \cos\theta_2 \qquad ...(1)$$

$$m_1v_1 \sin\theta_1 = m_2v_2 \sin\theta_2 \qquad ...(2)$$

$$\frac{1}{2}m_1u_1^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad ...(3)$$

Solving these three equations simultaneously, we can know any quantity of interest. But this calculation work is a little tedious and quite lengthy. If the above problem is viewed in the centre of mass frame of reference, then its solution is simpler and more informative.

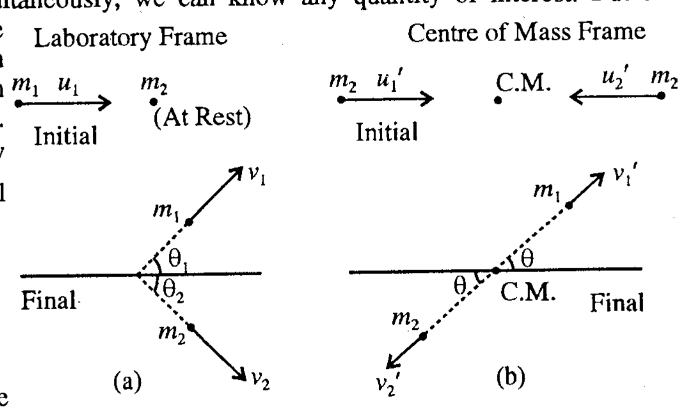

Fig. 5.16.

In the laboratory frame, the velocity $\vec{V}$ of the centre of mass of the system will be given by

$$(m_1 + m_2)\vec{V} = m_1\vec{u}_1 + m_2 \times 0$$

or
$$\vec{V} = \frac{m_1\vec{u}_1}{m_1 + m_2} \qquad ...(4)$$

The centre of mass frame of reference moves with this velocity $\vec{V}$ relative to the laboratory frame *i.e.*, centre of mass (C.M.) is at rest in the centre of mass frame of reference. In this frame, the initial and final velocities of the two particles are given by

* In case of scattering of α-particles and Compton effect, initial velocity of m_2 is zero in the laboratory frame.

$$\vec{u}_1' = \vec{u}_1 - \vec{V};\ \vec{u}_2' = -\vec{V} \text{ and } \vec{v}_1' = \vec{v}_1 - \vec{V};\ \vec{v}_2' = \vec{v}_2 - \vec{V} \quad ...(5)$$

The centre of mass is at rest in this frame, hence the particles should have equal and opposite momenta, *i.e.*,

$$m_1 \vec{u}_1' = -m_2 \vec{u}_2';\ m_1 \vec{v}_1' = -m_2 \vec{v}_2'. \quad ...(6)$$

This can be proved as follows :

$$m_1 \vec{u}_1' = m_1(\vec{u}_1 - \vec{V}) = m_1\left(\vec{u}_1 - \frac{m_1 \vec{u}_1}{m_1 + m_2}\right) = \frac{m_1 m_2 \vec{u}_1}{m_1 + m_2}$$

and

$$-m_2 \vec{u}_2' = m_2 \vec{V} = m_2 \frac{m_1 \vec{u}_1}{m_1 + m_2} = \frac{m_1 m_2 \vec{u}_1}{m_1 + m_2} \text{ i.e., } m_1 \vec{u}_1' = -m_2 \vec{u}_2'$$

Similarly, we can prove that $m_1 \vec{v}_1' = -m_2 \vec{v}_2'$.

It means that the final velocities of both particles are making the same angle (θ) with the initial direction [**fig. 5.16(b)**] and in magnitude

$$m_1 u_1' = m_2 u_2' \text{ and } m_1 v_1' = m_2 v_2' \quad ...(7)$$

In this frame, according to the law of conservation of energy, we have

$$\frac{1}{2} m_1 u_1'^2 + \frac{1}{2} m_2 u_2'^2 = \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2 v_2'^2 \quad ...(8)$$

Substituting the values u_2' and v_2' from eq. (7) in eq. (8), we have

$$\frac{1}{2} m_1 u_1'^2 + \frac{1}{2} m_2\left(\frac{m_1^2 u_1'^2}{m_2^2}\right) = \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2\left(\frac{m_1^2 v_1'^2}{m_2^2}\right)$$

or

$$\frac{1}{2} m_1 u_1'^2\left(1 + \frac{m_1}{m_2}\right) = \frac{1}{2} m_1 v_1'^2\left(1 + \frac{m_1}{m_2}\right)$$

whence

$$u_1' = v_1' \quad ...(9)$$

$\therefore$

$$u_2' = v_2' \quad \text{[from eq. (7)]} \quad ...(10)$$

Thus we see that in the centre of mass frame of reference, the magnitudes of the particle velocities are not altered by an elastic collision. Also in this frame the scattering angle θ can have any value. But in the laboratory frame, the scattering angle θ_1 cannot have all values, as shown below :

$$\tan\theta_1 = \frac{v_1 \sin\theta_1}{v_1 \cos\theta_1} = \frac{v_1' \sin\theta}{v_1' \cos\theta + V} \quad ...(11)$$

[Because from eq. (5), we have $\vec{v}_1 = \vec{v}_1' + \vec{V}$.

In component form, $\quad v_1 \sin\theta_1 = v_1' + \sin\theta$

and $\quad v_1 \cos\theta_1 = v_1' \cos\theta + V$, as $\vec{V}$ is along X-axis.]

Now, from eq. (4), we have

$$V = \frac{m_1 u_1}{m_1 + m_2} = \frac{m_1(u_1' + V)}{m_1 + m_2} \text{ or } (m_1 + m_2)V = m_1(u_1' + V);\ \therefore V = \frac{m_1 u_1'}{m_2} = \frac{m_1 v_1'}{m_2} \quad [\because u_1' = v_1']$$

Substituting this value of V in eq. (11), we get

$$\tan\theta_1 = \frac{v_1' \sin\theta}{v_1' \cos\theta + m_1 v_1' / m_2} \text{ or } \tan\theta_1 = \frac{\sin\theta}{\cos\theta + m_1/m_2} \quad ...(12)$$

In case, if $m_1 > m_2$ or $m_1/m_2 > 1$. the denominator can never be zero, *i.e.*,

$$\tan\theta_1 < \infty \text{ or } \theta_1 < \pi/2.$$

Thus, we see that *if a heavy particle is incident on a light particle initially at rest, the heavy particle will not bounce backward as a result of collision.*

If $m_1 = m_2$, then taking $\cos\theta = -1$, the denominator becomes zero. Hence θ_1 can have values upto $\pi/2$. If $m_1 < m_2$ or $m_1/m_2 < 1$, $\tan\theta_1$ can have also negative values and hence in such a case any value of θ_1 is allowed.

Ex. 1. *An electron is accelerated by a potential difference of 20 volts. What is its momentum ?*

Sol. $p = mv = \sqrt{2m(\frac{1}{2}mv^2)} = \sqrt{2m(eV)}$ [$\because \frac{1}{2}mv^2$ = change × potential difference = eV]

$$= \sqrt{2\times 0.9\times 10^{-30}\times 1.6\times 10^{-19}\times 20} = \mathbf{2.4 \times 10^{-24}\ kg\text{-}m/sec.}$$

Ex. 2. *A nucleus, initially at rest, decays radioactively by emitting an electron of momentum 1.73 MeV/c and at right angles to the direction of electron a neutrino with momentum 1 MeV/c. In what direction does the nucleus recoil ? What is its momentum in MeV/c and in S.I. units. If the mass of the residual nucleus is 3.9×10^{-25} kg, calculate its kinetic energy in electron volts.*

Sol. Initial momentum = 0

Assuming the direction of the electron as X-axis, we have

momentum of the electron = $1.73\hat{\mathbf{i}}$ MeV/c, momentum of the neutrino = $1\hat{\mathbf{j}}$ MeV/c,

momentum of the nucleus $\vec{p} = mv$ MeV/c.

From the law of conservation of momentum, we have

$$1.73\hat{\mathbf{i}} + 1\hat{\mathbf{j}} + m\vec{v} = 0 \quad \text{or} \quad m\vec{v} = -(1.73\hat{\mathbf{i}} + 1\hat{\mathbf{j}})\ \text{MeV/c.}$$

$\therefore$ Its magnitude = $mv = \sqrt{(1.73)^2 + 1^2}$ = **2 MeV/c approx.**

$$= \frac{2\times 10^6}{3\times 10^8}\times 1.6\times 10^{-19} = \mathbf{10.66 \times 10^{-22}\ kg\text{-}m/sec.}$$

Direction cosines of $\vec{p}$ are $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}, 0\right)$, *i.e.*, direction of the recoil of nucleus makes angle of 210° with X-axis.

Now, the speed of the residual nucleus = $\dfrac{10.66\times 10^{-22}}{3.9\times 10^{-25}}$ m/sec.

$\therefore$ K.E. of the residual nucleus = $\frac{1}{2}mv^2 = \frac{1}{2}(3.9\times 10^{-25})\left(\dfrac{10.66\times 10^{-22}}{3.9\times 10^{-25}}\right)^2$ J

$$= \frac{1}{2}\frac{(3.9\times 10^{-25})}{1.6\times 10^{-19}}\left(\frac{10.66\times 10^{-22}}{3.9}\right)^2 \text{eV} = \mathbf{9.1\ electron\text{-}volts.}$$

Ex. 3. *A bomb in fight explodes into two fragments when its velocity is $(10\hat{\mathbf{i}} + 2\hat{\mathbf{j}})$ m/sec. If the smaller mass M flies with velocity $(20\hat{\mathbf{i}} + 50\hat{\mathbf{j}})$ m/sec, deduce the velocity of the larger mass 3M. Deduce also the velocities of the fragments in the centre of mass reference frame, and show these in a diagram.* **(Agra 1971)**

Sol. Total mass of the bomb = $3M + M = 4M$

Initial velocity of the bomb = $10\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ m/sec.

$\therefore$ Initial momentum of the system = $4M(10\hat{\mathbf{i}} + 2\hat{\mathbf{j}})$

Let $\vec{v}$ be the velocity of the heavier fragment. So that the final momentum = $M(20\hat{\mathbf{i}} + 5\hat{\mathbf{j}}) + 3M\vec{v}$

From the law of conservation of momentum, we have

$$M(20\hat{\mathbf{i}} + 5\hat{\mathbf{j}}) + 3M\vec{v} = 4M(10\hat{\mathbf{i}} + 2\hat{\mathbf{j}})$$

whence $$\vec{v} = \tfrac{1}{3}(20\hat{i} - 42\hat{j}) \text{ m/sec.}$$

There is no external force, hence the centre of mass moves with constant velocity, *i.e.*, $\vec{V} = (10\hat{i} + 2\hat{j})$ m/sec. Further the velocities of the lighter and heavier masses in the centre of mass frame of reference are

$$20\hat{i} + 50\hat{j} - (10\hat{i} + 2\hat{j}) = (10\hat{i} + 48\hat{j}) \text{ m/sec.}$$

and $\tfrac{1}{3}(20\hat{i} - 42\hat{j}) - (10\hat{i} + 2\hat{j}) = \tfrac{1}{3}(-10\hat{i} - 48\hat{j})$ **m/sec.**

respectively.

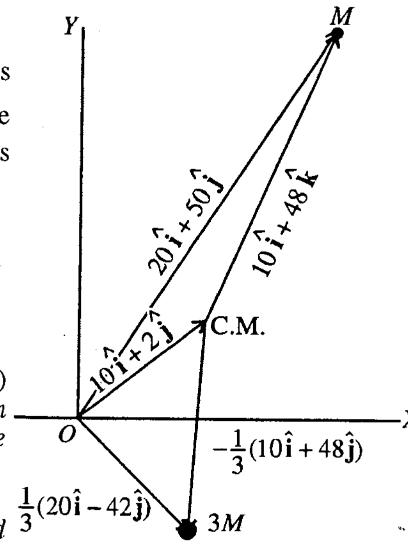

Fig. 5.17.

Ex. 4. ***Elastic head-on collision between two particles*** : (a) *Show that the relative velocity between the particles after an elastic head-on collision is equal and opposite to the relative velocity before the collision.*
(Agra 2003)

(b) *In an elastic head-on collision assuming the second particle at rest, prove that transference of energy is maximum when their mass ratio is unity.*

Sol. (a) Let us consider an elastic head-on collision of two particles of masses m_1 and m_2, which are moving originally along the same line with velocities u_1 and u_2 respectively. After the collision if their final velocities are v_1 and v_2, then from the laws of conservation of energy and momentum we have

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad \text{...(i)}$$

and $$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \qquad \text{...(ii)}$$

From eqs. (i) and (ii), we have

$$m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2) \qquad \text{...(iii)}$$

and $$m_1(u_1 - v_1) = m_2(v_2 - u_2) \qquad \text{...(iv)}$$

Dividing eq. (iii) by eq. (iv), we have

$$u_1 + v_1 = v_2 + u_2 \qquad \text{...(v)}$$

or $$u_1 - u_2 = -(v_1 - v_2) \qquad \text{...(vi)}$$

Thus, the relative velocity between the particles after an elastic collision is equal and opposite to the relative velocity before the collision.

(b) When $u_2 = 0$, then from eq. (vi) we have $v_1 = v_2 - u_1$. Substituting in eq. (i), we get

$$\frac{1}{2}m_1u_1^2 = \frac{1}{2}m_1(v_2 - u_1)^2 + \frac{1}{2}m_2v_2^2 \quad \text{or} \quad 0 = m_1v_2^2 - 2m_1v_2u_1 + m_2v_2^2$$

$$\therefore \quad v_2[v_2(m_1 + m_2) - 2m_1u_1] = 0$$

whence either $v_2 = 0$, which is impossible or $v_2 = 2m_1u_1/(m_1 + m_2)$.

Hence fraction of kinetic energy transferred to the second particle is

$$= \frac{\frac{1}{2}m_2v_2^2}{\frac{1}{2}m_1u_1^2} = \frac{\frac{1}{2}m_2(2m_1u_1/m_1 + m_2)^2}{\frac{1}{2}m_1v_1^2} = \frac{4m_1m_2}{(m_1 + m_2)^2} = \frac{4m_2/m_1}{(1 + m_2/m_1)^2} \qquad \text{...(vii)}$$

When $m_2/m_1 = 1$, the fraction of transferred kinetic energy $= \dfrac{4}{(1+1)^2} = 1.$

Thus, when the mass ratio is unity, whole of the kinetic energy is transferred to the second particle. In case $m_1 > m_2$ or $m_1 < m_2$, the value of the fraction of transferred kinetic energy will be less than 1.

Ex. 5. ***Collision of particles which stick together (Inelastic collisition)*** : *Two particles of masses m_1 and m_2 stick together on collision. If m_2 is at rest and m_1 is moving along X-axis with velocity* $\vec{u}_1$, *then calculate the following :*

(a) *What is the velocity of the combined particle ?*

(b) *What is the ratio of the kinetic energy after the collision to the initial kinetic energy ?*

(c) *Describe the motion before and after the collision in the reference frame in which the centre of mass is at rest.*

Sol. (a) Let us consider the collision of a particle of mass m_1, moving with velocity $\vec{u}_1$ with another stationary particle of mass m_2. If on collision both the particles are attached and have composite velocity $\vec{v}$, then from the law of conservation of momentum we have

$$m_1 \vec{u}_1 = (m_1 + m_2)\vec{v}$$

So that the velocity of the composite particle is

$$\vec{v} = \frac{m_1 \vec{u}_1}{m_1 + m_2} \quad \text{...(i)}$$

(b) Kinetic energy before the collision $K_i = \frac{1}{2} m_1 u_1^2$

Kinetic energy after the collision $K_f = \frac{1}{2}(m_1 + m_2)v^2$

$$\therefore \quad \frac{K_f}{K_i} = \frac{\frac{1}{2}(m_1+m_2)v^2}{\frac{1}{2}m_1u_1^2} = \frac{1}{2}(m_1+m_2)\frac{m_1^2u_1^2}{(m_1+m_2)^2} \times \frac{1}{\frac{1}{2}m_1u_1^2} = \frac{m_1}{m_1+m_2} \quad \text{...(ii)}$$

Evidently $K_f < K_i$ and so the kinetic energy is lost in the collision and this lost energy appears in the internal excitations (heat etc.) of the composite system.*

(c) Velocity of the centre of mass

$$\vec{V} = \frac{m_1 \vec{u}_1}{m_1 + m_2}$$

In the centre of mass frame of reference, the velocity of the first particle is

$$\vec{u}_1' = \vec{u}_1 - \vec{V}_1 = m_2 \vec{u}_1/(m_1 + m_2)$$

and that of second particle is

$$\vec{u}_2' = \vec{u}_2 - \vec{V} = -\vec{V} = -m_1 \vec{u}_1/(m_1 + m_2).$$

On colliding, the particles stick together, the new combined particle has mass $(m_1 + m_2)$ and must be at rest in the centre of mass frame of reference. Relative to the laboratory frame, the new particle has the velocity $\vec{V}$, which is exactly the same as the velocity of the combined particle $\vec{v}$.

Ex. 6. *A particle of mass M, moving with a velocity u, makes a head-on collision with a particle of mass m initially at rest so that their final velocities V and v are along the same line. Assuming an elastic collision prove that*

$$v = \frac{2u}{1 + m/M}.$$

If the particles coalesce on colliding, find the final common velocity and the loss in kinetic energy.

Sol. From the law of conservation of momentum, we have

$$Mu = MV + mv \quad \text{...(i)}$$

From the law of conservation of energy

* When a meteorite of mass m strikes and sticks to the earth of mass M, essentially all the kinetic energy of the meteorite appears as heat in the earth, because $m << m + M$.

$$\frac{1}{2}Mu^2 = \frac{1}{2}MV^2 + \frac{1}{2}mv^2 \text{ or } u^2 = V^2 + \frac{m}{M}v^2 \quad ...(ii)$$

Substituting the value of $V^2 \left[= (u - mv/M)^2\right]$ from eq. (i) in eq. (ii), we have

$$u^2 = u^2 + \frac{m^2v^2}{M^2} - \frac{2muv}{M} + \frac{m}{M}v^2 \text{ or } 0 = \left(\frac{m}{M}+1\right)v - 2u, \therefore v = \frac{2u}{1+m/M}.$$

In the second case, $Mu = (m + M)V$ or $V = Mu/(m + M)$

$$\text{Loss of energy} = \frac{1}{2}Mu^2 - \frac{1}{2}(m+M)V^2 = \frac{1}{2}Mu^2 - \frac{1}{2}(m+M)\frac{M^2u^2}{(m+M)^2} = \frac{1}{2}Mu^2 \cdot \frac{m}{m+M}.$$

Ex. 7. *Two particles, having the positions* $\vec{r}_1 = (3\hat{i} + 5\hat{j})\,m$ *and* $\vec{r}_2 = -(5\hat{i} + 3\hat{j})\ m$ *move with velocities* $\vec{v}_1 = (4\hat{i} + 3\hat{j})$ *m/sec and* $\vec{v}_2 = (a\hat{i} + 7\hat{j})$ *m/sec. Find :*

(i) *the value of a, if they collide,* (ii) *when and where the collision will take place ?*

Sol. The two particles will collide after a time t, if they have the same position vectors, *i.e.*,

$$\vec{r}_1 + \vec{v}_1 t = \vec{r}_2 + \vec{v}_2 t, \ \therefore\ (\vec{r}_1 - \vec{r}_2) + (\vec{v}_1 - \vec{v}_2)\,t = 0 \quad ...(i)$$

Taking cross product with $(r_1 - r_2)$, we have

$$(\vec{r}_1 - \vec{r}_2) \times (\vec{v}_1 - \vec{v}_2) = 0 \qquad [\because (\vec{r}_1 - \vec{r}_2) \times (\vec{r}_1 - \vec{r}_2) = 0] \ ...(ii)$$

$$\therefore\ (3\hat{i} + 5\hat{j} + 5\hat{i} + 3\hat{j}) \times (4\hat{i} + 3\hat{j} - a\hat{i} - 7\hat{j}) = 0 \quad \text{or} \quad 8(\hat{i} + \hat{j}) \times [(4 - a)\hat{i} - 4\hat{j}] = 0$$

or $\quad -4\hat{k} - (4 - a)\hat{k} = 0$ or $4 + 4 - a = 0$ or $a = 8$.

From eq. (i), $$t = \frac{(\vec{r}_1 - \vec{r}_2)}{(\vec{v}_1 - \vec{v}_2)} = -\frac{8(\hat{i} + \hat{j})}{-4(\hat{i} + \hat{j})} = \mathbf{2\ sec.}$$

Place of collision $= \vec{r}_1 + \vec{v}_1 t = 3\hat{i} + 5\hat{j} + (4\hat{i} + 3\hat{j})2 = (11\hat{i} + 11\hat{j})$ **m.**

Ex. 8. *Three interacting particles of masses* 100, 200 *and* 400 *gm have each a velocity of* 20 *m/sec magnitude along the positive direction of X-axis, Y-axis and Z-axis respectively. Calculate the velocity of the first particle, when due to forces of interaction the third particle stops moving and the velocity of the second particle is* $(10\hat{j} + 5\hat{k})$ *m/sec.*

Sol. Initial momentum $= m_1\vec{u}_1 + m_2\vec{u}_2 + m_3\vec{u}_3 = 2\hat{i} + 4\hat{j} + 8\hat{k}$ **m/sec.**

When the third particle stops moving due to their own interactions, then final momentum $= m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3 = 0.1\,\vec{v}_1 + 0.2(10\hat{j} + 5\hat{k})$

According to the law of conservation of momentum, we have

$$0{\cdot}1\,\vec{v}_1 + 2\hat{j} + \hat{k} = 2\hat{i} + 4\hat{j} + .8\hat{k},\ \therefore\ \vec{v}_1 = (20\hat{i} + 20\hat{j} + 70\hat{k}) \text{ m/sec.}$$

Ex. 9. *Two objects of masses* $m_1 = 200$ *gm and* $m_2 = 500$ *gm possess velocities* $\vec{v}_1 = 10\hat{i}$ *m/sec. and* $\vec{v}_2 = (3\hat{i} + 5\hat{j})$ *m/sec just prior to a collision during which they become permanently attached to each other. Calculate* (a) *the velocity of the centre of mass.* (b) *the final momentum of the combination in the laboratory frame,* (c) *the initial and final momenta in the centre of mass frame,* (d) *what fraction of the initial total kinetic energy is associated with the motion after collision ?*

Sol. (a) $$\vec{V} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} = \frac{0.2(10\hat{i}) + 0.5(3\hat{i} + 5\hat{j})}{0.2 + 0.5} = \frac{35\hat{i} + 35\hat{j}}{7} \text{ m/sec.}$$

(b) Final momentum $= (m_1 + m_2)\vec{V} = 0.7\vec{V} = 3.5\hat{i} + 2.5\hat{j}$.

(c) In the centre of mass frame of reference, the centre of mass of the system remains at rest. It means that in the centre of mass frame the initial momentum of the system is zero and hence final momentum of the system will be also zero.

(d) $$K_i = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}(0.2)(10)^2 + \frac{1}{2}(0.5)[3^2 + 5^2]$$

$$= 10 + 8.5 = 18.5 \text{ joules.}$$

Now employing the law of conservation of momentum

$$m_1\vec{u}_1 + m_2\vec{u}_1 = (m_1 + m_2)\vec{v},$$

where $\vec{v}$ is the final velocity of the combined particle in the laboratory system. Note that $\vec{V}$ is idential to $\vec{V}$.

$$\therefore \quad K_f = \frac{1}{2}(m_1 + m_2)v^2 = \frac{1}{2} \times 0.7[5^2 + (25/7)^2] = 13.2 \text{ joules}$$

$$\therefore \quad \frac{K_f}{K_i} = \frac{13.2}{18.5} = \mathbf{0.72.}$$

Ex. 10. *A sand bag of mass* 10 *kg is suspended with a* 3 *metres long weightless string. A bullet of mass* 200 *gm is fired with speed* 20 *m/sec into the bag and stays in the bag. Calculate* (i) *the speed acquired by the bag,* (ii) *the maximum displacement of the bag,* (iii) *energy converted to heat in the collision.*

Sol. Mass of the bag $M = 10$ kg; Mass of the bullet $m = 0.2$ kg;

Velocity of the bullet $u = 20$ m/sec.; Velocity of the bag $v = ?$

From the law of conservation of momentum

$$mu = (m + M)\,v \quad \therefore \quad v = \frac{mu}{m+M} = \frac{0.2 \times 20}{10.2} = \mathbf{0.392 \text{ m/sec.}} \qquad \text{...(i)}$$

After collision, the bag will oscillate like a pendulum. Let the maximum displacement of the bag be x.

Restoring force at this displacement is

$$F = -Cx = -mg'\sin\theta = -mg\cdot x/l$$

where l is the length of the string.

$$\therefore \quad C = \frac{mg}{l} = \frac{10.2 \times 9.8}{3}$$

At this maximum displacement x, the entire kinetic energy $\left[\frac{1}{2} \times 10.2 \times (0.392)^2\right]$ of the system (bag + bullet) is changed into potential form, *i.e.*, $\int_0^x Cx\,dx = \frac{1}{2}Cx^2$.

$$\therefore \quad \frac{1}{2}Cx^2 = \frac{1}{2} \times 10.2 \times (0.392)^2$$

or $$x = \sqrt{\frac{10.2 \times (0.392)^2 \times 3}{10.2 \times 9.8}} = 0.392 \times 0.54 = \mathbf{0.21 \text{ metre.}} \qquad \text{...(ii)}$$

$$\text{Energy converted into heat} = \frac{1}{2}mu^2 - \frac{1}{2}(m+M)v^2 = \frac{1}{2} \times 0.2 \times 20^2 - \frac{1}{2} \times 10.2 \times (0.392)^2$$

$$= 40 - 0.8 = \mathbf{39.2 \text{ joules.}} \qquad \text{...(iii)}$$

[**Ballistic pendulum :** This pendulum consists of a block of wood, suspended by strings from a rigid support [**fig. 5.18**] and is used to determine the velocity of a bullet, fired from a gun. By measuring the height to which the block rises, the speed of the bullet can be determined.]

Ex. 11. *A bullet of mass m and speed v is allowed to make a completely inelastic collision with a block of much greater mass M* [**fig. 5.18**]. *If the block rises to a height h, find the speed of the bullet. Find the loss in energy due to collision.*

Sol. If due to inelasitc collision the speed of the bullet + block is V, then from the law of conservation of momentum,

$$mv = (m + M)\,V \qquad \text{...(i)}$$

After the collision the kinetic energy of the bullet + block is given by

$$E = \frac{1}{2}(m+M)V^2$$

Due to this energy the block will rise up and will acquire instantaneous rest position after rising a height h. In this process, the kinetic energy will convert into potential energy, *i.e.*,

$$\frac{1}{2}(m+M)V^2 = (m+M)gh \quad \text{or} \quad V = \sqrt{2gh}$$

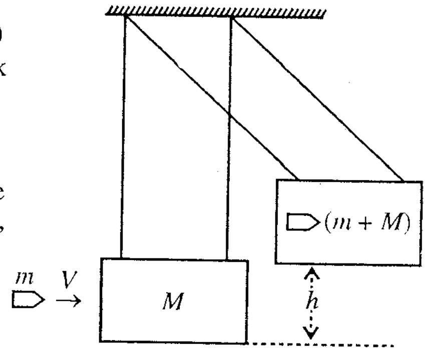

Fig. 5.18.

Substituting the value of V in (i), we get

$$mv = (m + M)\sqrt{2gh},$$

whence

$$v = \frac{m+M}{m}\sqrt{2gh} \qquad \text{...(ii)}$$

Loss of energy $= \frac{1}{2}mv^2 - \frac{1}{2}(m+M)V^2 = \frac{1}{2}m\left(\frac{m+M}{m}\right)^2 2gh - \frac{1}{2}(m+M)2gh = \boldsymbol{(m + M)\,(M/m)gh}$

EXERCISE 5·2

1. A 50 kg man standing on a surface of negligible friction kicks forward a 25 gm stone lying at his feet so that it acquires a speed of 2 metres/sec. What velocity does the man acquire as a result ?
[**Ans.** -10^{-3} m/sec.]

2. Two bodies of masses 2.0 kg and 3.0 kg are moving with velocities 6 m/s and 4 m/s respectively in mutually perpendicular directions. Find the total linear momentum. If they collide and stick together, find the velocity and direction of motion of the composite body.
[**Ans.** 16.97 kg-m/s, 3.39 m/s in direction at an angle 135° with the direction of motion of body of mass 2.0 kg.]

3. A meteor of 2 kg mass moving with a velocity of 12 km/sec explodes above the atmosphere into two pieces which move along the same path with 15 km/sec and 11 km/sec velocities. Find the mass of each piece and the energy set free by explosion.
[**Ans.** 0.5 kg and 1.5 kg; 3×10^6 joules.]

4. A bomb of mass 300 kg explodes in flight at the instant when its velocity is $\vec{v}_0 = (10\hat{i} + 5\hat{j})$ m/sec. It splits in two fragments of mass ratio 1 : 2 and the larger mass is observed to have velocity $\vec{v}_2 = (100\hat{i} + 50\hat{j} + 10\hat{k})$ m/sec just after the explosion. Deduce :

(i) velocity $\vec{v}_1$ of the smaller fragment just after the explosion.

(ii) kinetic energy produced in the explosion. **(Agra 1970 S)**

[**Ans.** (i) $(-170\hat{i} - 85\hat{j} - 20\hat{k})$ m/sec.; (ii) 0.0409 joule.]

5. An α-particle is emitted from a uranium nucleus (U^{238}) in a radioactive decay. The speed and kinetic energy of α-particle are 1.404×10^7 m/sec and 4.212 MeV respectively. Calculate the recoil speed and kinetic energy of the residual nucleus. **(Kumaun 1995)**
[**Ans.** -2.4×10^5 m/sec., 0.072 MeV.]

6. In a radioactive decay of a nucleus, an electron and a neutrino with momentum 1.28×10^{-22} and 6.4×10^{-23} kg-m/sec are emitted at right angles to each other. What is the momentum of the recoiling nucleus ? If the mass of the residual nucleus is 5.8×10^{-26} kg, calculate its recoil kinetic energy.
[**Ans.** $\approx 1.4 \times 10^{-22}$ kg-m/sec in a direction $\tan^{-1}\frac{1}{2}$ with the electron velocity, 1.1 eV.]
7. A stationary body of mass 3 kg explodes into three equal pieces. Two of the pieces fly off at right angles to each other, one with $2\hat{i}$ m/sec and the other with $3\hat{j}$ m/sec. If the explosion takes in 10^{-5} sec., find the average force acting on each piece during the explosion.
[**Ans.** $2 \times 10^5 \hat{i}$ newton; $3 \times 10^5 \hat{j}$ newton; $-(2\hat{i}+3\hat{j}) \times 10^5$ newton.]
8. A 5 kg object with a speed of 30 m/s strikes a steel plate at an angle of 45° and rebounds at the same speed and same angle. What is the change of the linear momentum in magnitude and direction ? **(Lucknow 1993)**
[**Ans.** $150\sqrt{2}$ kg-m/sec. perpendicular to steel plate.]
9. A gun weighing M fires a shell of mass m horizontally and the energy of explosion is such that it can raise the shell to a vertical height h. Show that the speed of the recoil of the gun is
$$[2m^2\, gh/M(m + M)]^{1/2}$$
10. A bullet of mass M is fired with a velocity of 50 m/sec at an angle θ with the horizontal. At the highest point of its trajectory, it collides head-on with a bob of mass $3M$ suspended by a massless string of length 1/3 m and gets embedded in the bob. After the collision, the string moves through an angle of 120°. Find (i) the angle θ, (ii) the vertical and horizontal coordinates of the initial position of the bob with respect to the point of firing of the bullet. [g = 10 m/sec^2].
[**Ans.** (i) $\theta = \cos^{-1}$ (0.8); (ii) 45 m, 120 m.]
11. A stream of liquid set at an angle θ is directed against a plane surface. The liquid after hitting the surface, spreads over it. Calculate the pressure on the surface. Assume the density of the liquid ρ and its speed v.
[**Ans.** $v^2\rho \cos^2 \theta$.]
12. Originally two particles are at the positions x_1 = 5 cm, y_1 = 0 and x_2 = 0, y_2 = 10 cm with $\vec{v}_1$ $= -4 \times 10^4 \hat{i}$ cm/sec and $\vec{v}_2$ is along the negative direction of Y-axis. (i) What must be the value of $\vec{v}_2$, if they collide ? (ii) What is the velocity of the first particle relative to the second ?
[**Ans.** (i) -8×10^2 m/sec; (ii) 4×10^2 $(2\hat{j}-\hat{i})$ m/sec.]
13. A particle of mass m_1, moving with speed v, makes an elastic head-on collision with another particle of mass m_2 $(m_2 > m_1)$. If the particle m_2 has initially equal but opposite momentum to that of m_1, show that the velocity of the first particle after the collision is $-v$ and if m_2 is at rest in the condition $m_2 >> m_1$, the loss of kinetic energy of m_1 is $4m_1/m_2$.
14. Two blocks of masses 300 gm and 200 gm are moving toward each other along a horizontal frictionless surface with velocities 50 cm/sec and 100 cm/sec respectively. (a) If the blocks collide and stick together, find their final velocity. (b) Calculate the loss of kinetic energy during the collision. (c) Find the final velocity of each block if the collision is completely elastic.
[**Ans.** (a) 10 cm/sec; (b) 0.14 joule; (c) –70 cm/sec, 80 cm/sec.]
15. A bullet of mass 2 gm, travelling at 500 m/sec., is fired into a ballistic pendulum of mass 1 kg suspended from a cord 1 m long. The bullet penetrates the pendulum and emerges with a velocity of 100 m/sec. Through what vertical height the pendulum rise ?
[**Ans.** 3.3 cm.]
16. A bullet of mass m, moving horizontally with speed v, passes through a pendulum bob of mass M and emerges with a velocity $v/2$. If the length of the pendulum is l, show that the minimum value of v, which will make the pendulum bob to swing through a complete circle is $(2M/m)\sqrt{5gl}$.
17. A rifle bullet of 0.1 kg strikes and embeds itself in a block of 0.9 kg which rests on a horizontal

frictionless table newton and is attached to a spring. The impact compresses the spring through 0.1 metre. If a force of 1 N is required to compress the spring 1/100 m, calculate the initial velocity of the bullet. What is the velocity of the block just after impact ?

[**Ans.** 3.33 m/sec; 0.333 m/sec.]

18. A steel bob of 1 kg mass is fastened to a cord 1 m and is released when the cord is horizontal [**fig. 5.19**]. At the bottom of its path, the bob strikes a 5 kg block initially at rest on a frictionless surface. If the collision is elastic, calculate the speed of the bob and speed of the block just after the collision.

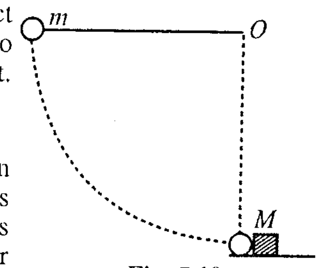

Fig. 5.19.

[**Ans.** Bob –2.953 m/sec; block 1.477 m/sec.]

19. A shell of mass 0.02 kg is fired by a gun of mass 100 kg. If the muzzle speed of the shell is 80 m/sec, calculate the recoil speed of the gun. **(Lucknow 1999)**

[**Ans.** 1.6×10^{-2} m/sec.]

5·10. Motion of a Rocket

Rocket works on the **principle of jet propulsion.** From the word 'jet' we mean a high velocity stream of fluid issuing out of a nozzle or orifice, as for example a jet of steam, a jet of gas etc. The method of driving a body or machine through the agency of jet is called **jet propulsion.** The principle of jet propulsion depends on the law of conservation of momentum, according to which the momentum of jet, emerging from a body in the backward direction is equal and opposite to the momentum of the body in the forward direction.

In a rocket, the fuel and the oxidiser are burnt in the combustion chamber. The heat of combustion makes the pressure very high inside the chamber so that a jet of hot gases emerges from the tail of the rocket with a very high exhaust velocity and in consequence the rocket moves in the forward direction [**fig. 5.20**].

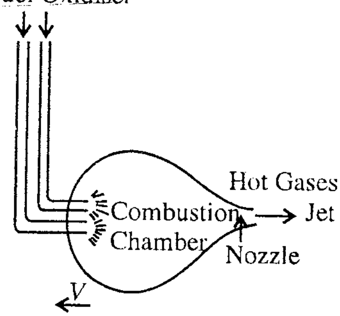

Fig. 5.20.

According to the type of fuel used, rockets are classified as (a) liquid fuel rockets, (b) solid fuel rockets. In the former type of rockets, liquid hydrogen or liquid paraffin is employed as fuel and liquid oxygen, hydrogen peroxide (H_2O_2) or nitric acid as oxidiser. In the solid fuel rockets, the fuel (*e.g.*, gunpowder) itself contains the oxidiser and therefore a separate combustion chamber is not needed.

Let the mass of the rocket at any instant t be M and its velocity be V in the laboratory frame (an inertial frame). If a mass m of the hot gases emerges out per second with exhaust velocity $-v$ relative to the rocket, then the rate of change of mass of the rocket is $dM/dt = -m$. Now, assuming v and V parallel, the velocity of the gases in the laboratory frame will be $-v + V$ and the momentum of the burnt fuel, ejected per second, will be $m(-v + V)$. Hence the force on the rocket is

$$F = -m(-v+V) = \frac{dM}{dt}(-v+V).$$

According to Newton's second law, $F = \dfrac{d}{dt}(MV)$

$$\therefore \qquad \frac{d}{dt}(MV) = \frac{dM}{dt}(-v+V) \quad \text{or} \quad M\frac{dV}{dt} + V\frac{dM}{dt} = -v\frac{dV}{dt} + V\frac{dM}{dt}$$

or

$$M\frac{dV}{dt} = -v\frac{dM}{dt} \qquad ...(1)$$

This equation gives the instantaneous force on the rocket.

From eq. (1), we get

$$dV = -\frac{dM}{M}v$$

Integrating it, we have

$$V = -v \log_e M + C, \text{ where } C \text{ is a constant of integration.}$$

If initially mass of the rocket is M_0 and velocity V_0, then

$$V_0 = -v \log_e M_0 + C \quad \text{or } C = v \log_e M_0 + V_0$$

$$\therefore \quad V = -v \log_e M + v \log_e M_0 + V_0$$

or
$$V = V_0 + v \log_e (M_0/M) \quad \text{...(2)}$$

Hence, to achieve a high final velocity of the rocket it is necessary that (i) the value of exhaust velocity v should be large and (ii) its final mass M should be much less than its initial mass M_0.

The magnitude of exhaust velocity depends on the nozzle design and the amount of heat, produced by fuel. Theoretically, the limit of exhaust velocity is equal to the root mean square velocity of the gas molecules at the temperature of the combustion chamber. Its practical limit is nearly 2 km/sec, because it becomes difficult to attain temperature more than 3000°C.

Practically the ratio of the initial and final masses M_0/M of the rocket has also an upper limit. In case of a **single stage rocket** (which has only one engine), the final mass M is mainly due to the fuel containers and it cannot be smaller than 10% of M_0. In the solid fuel rockets, final mass M may be smaller than this; but in such a case this advantage is upset by smaller exhaust velocity v which affects in consequence the final velocity V.

If $M_0/M = 10$, initial velocity $V_0 = 0$ and exhaust velocity $v = 2$ km/sec, then final velocity,

$$V = 0 + 2 \log_e (10) = 2 \times 2.3 = 4.6 \text{ km/sec.}$$

This velocity is much less than the escape velocity (11 km/sec) or the orbital velocity near the surface of the earth (8 km/sec). Hence such a large velocity cannot be obtained with the help of a one stage rocket. Employing two or three stage rockets, higher final velocities can easily be obtained.

In a **two stage rocket,** two rockets are connected in series. First stage rocket is large and heavy and second stage rocket is comparatively small and light. When the rocket is sent in space, the first stage rocket is used first and when its fuel is burnt, it gets detached and is discarded. Now, the second stage rocket starts on with the initial velocity V_0, imparted to it by the first stage. Hence according to eq. (2), the final velocity will be nearly twice the velocity, achieved by only one stage rocket. It will be slightly less than the double because for first stage the ratio M_0/M will be somewhat smaller. It is due to the fact that in M, the final mass of the first stage, the entire mass of the second stage remains included.

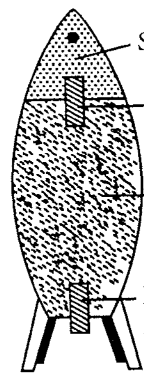

Fig. 5.21 : Two stage rocket.

In the above description, we have neglected the weight (Mg) *of the rocket. If the rocket moves vertically upwards, the force on the rocket is* $\frac{dM}{dt}(-v+V) - Mg$ *in the upward direction.* Thus

$$\frac{d}{dt}(MV) = \frac{dM}{dt}(-v+V) - Mg$$

On simplifying, we get
$$M\frac{dV}{dt} = -v\frac{dM}{dt} - Mg \quad \text{...(3)}$$

Integrating it, we get
$$V = -v \log_e M - gt + C$$

When $t = 0$, $M = M_0$ and $V = V_0$, $\therefore C = V_0 + v \log_e M_0$

whence
$$V = V_0 + v \log_e \frac{M_0}{M} - gt \quad \text{...(4)}$$

where t is the time required for burning of the fuel.

It is clear from eq. (4) that if the time t is small, then due to the attraction of the earth the reduction in the value of V is negligible. But if the fuel burns very slowly, the rocket may not be able at all to rise or it may keep hovering without attaining any velocity. Hence it is necessary that the fuel is burnt rapidly in the first part of its journey. But the fuel should not be burnt very rapidly, as it results disadvantageous in two ways : (i) if the rocket acquires a very high velocity before it has moved above the atomsphere, there will be a greater wastage of energy due to air resistance, (ii) due to its larger acceleration, the weight of the passengers may increase very large.

The body of the rocket is made cylindrical and elongated like the shape of the cigar and its nose is sharply pointed so as to reduce the air pressure on its individual parts to the very minimum. Its frame is made of a heat resisting material and its velocity during the first part of its flight through the denser layers of the air is kept sufficiently low.

Ex. 1. *A machine gun is mounted on a trolly placed on a horizontal frictionless surface. The gun fires the bullets of each of mass* 10 *gm at the rate of* 10 *per second. If the mass of the system (gun + trolly) is* 100 *kg at the instant t and the velocity of the bullets with respect to trolly is* 1000 *m/sec, find the acceleration of the trolly. What is the average thrust produced on the system ?*

Sol. Momentum of a bullet $= \frac{10}{1000} \times 100$

$= 10$ kg-m/sec,

Momentum imparted per second $= 10 \times 10$

$= 100$ kg-m/sec^2.

The thrust experienced by the system is equal in magnitude to the rate at which momentum is imparted to the bullets.

$\therefore$ Force or thrust on the system = **100 newtons** at the instant t.

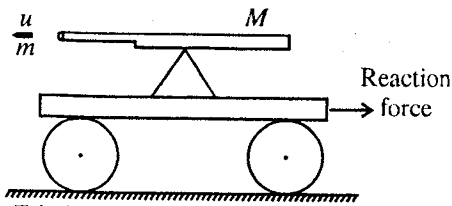

Fig. 5.22.

Hence the acceleration of the trolly $= \frac{\text{Force}}{\text{mass}} = \frac{100}{100} = $ **1 m/sec^2**.

Ex. 2. *Neglecting the attraction of the earth show that*

$$V = V_0 - v \log_e (1 - mt/M_0)$$

where V is the velocity acquired in t sec, v is the exhaust velocity and m is the constant rate of burning its fuel. **(Avadh 1996)**

Sol. $V = V_0 + v \log_e (M_0/M)$

As the rate of burning of the fuel is m, in t seconds mt mass will come out of the rocket *i.e.*, at this instant mass of the rocket $M = M_0 - mt$,

$$\therefore \quad V = V_0 + v \log_e \left(\frac{M_0}{M_0 - mt}\right) \text{ or } V = V_0 - v \log_e \left(\frac{M_0 - mt}{M_0}\right)$$

or $$V = V_0 - v \log_e (1 - mt/M_0).$$

Ex. 3. *A* 5000 *kg rocket is set for vertical firing. If the exhaust speed is* 500 *metres/sec, how much gas must be ejected per second to supply the thrust needed* (i) *to overcome the weight of the rocket,* (ii) *to give the rocket an initial upward acceleration of* 19.6 *m/sec*^{2} ?

Sol. (i) At any instant, the net upward force on the rocket is

$$F = M\frac{dV}{dt} = -v\frac{dM}{dt} - Mg$$

Now, just to overcome the weight of the rocket, the net force $F = 0$. So that

$$\frac{dM}{dt} = -\frac{Mg}{v} = -\frac{5000 \times 9.8}{500} = -98 \text{ kg/sec.}$$

Hence the gas should be ejected at the rate of **98 kg/sec.**

(ii) If a is the upward acceleration of the rocket, then

$$-v\frac{dM}{dt} - Mg = Ma \quad \text{or} \quad \frac{dM}{dt} = -\frac{M}{v}(g+a)$$

or $$\frac{dM}{dt} = -\frac{5000 \times (19.6 + 9.8)}{500} = \mathbf{-294 \text{ kg/sec.}}$$

Thus, to give the rocket an initial upward acceleration of 19.6 m/sec^2, the gas must be ejected at the rate of **294 kg/sec.**

Ex. 4. *If the two stage rocket weighs 100 kg and 10 kg separately and contains 800 kg and 90 kg of fuel separately, find the final velocity that can be achieved with an exhaust velocity of 4.5 km/sec. (Given : $\log_e 10 = 2.3$ and $\log_{10} 2 = 0.30$).* **(Avadh 1996)**

Sol. The velocity of a rocket at any time t is given by

$$V = V_0 + v \log_e (M_0/M)$$

In the first case, when the first stage rocket ends, $V_0 = 0$, $v = 4.5$ km/sec,

$M_0 = 100 + 800 + 10 + 90 = 1000$ kg and $M = 200$ kg

$$\therefore \quad V_1 = 0 + 4.5 \log_e (1000/200) = 4.5 \log_e 10 \log_{10} (10/2)$$
$$= 4.5 \times 2.3 \times (\log_{10} 10 - \log_{10} 2)$$
$$= 4.5 \times 2.3 \times (1 - 0{\cdot}3) = 4.5 \times 2.3 \times 0{\cdot}7 = 7.245 \text{ km/sec.}$$

Now, this is the initial velocity for the second stage rocket and therefore the final velocity of the rocket at the end of second stage will be

$$V = 7.245 + 4{\cdot}5 \times 2.3 \log_{10}\left(\frac{90+10}{10}\right)$$
$$= 7.245 + 4{\cdot}5 \times 2.3 = 7{\cdot}245 + 10.35 = \mathbf{17.595 \text{ km/sec.}}$$

Ex. 5. *A rocket starts vertically upward with speed v_0. Show that its speed v at a height h is given by*

$$v_0^2 - v^2 = \frac{2gh}{1+h/R},$$

where R is radius of the earth, and g is acceleration due to gravity at earth's surface. Hence deduce an expression for maximum height reached by a rocket fired with speed 90% of escape velocity.

(Agra 1984; Avadh 96; Purvanchal 93; Bundelkhand 95)

Sol. As the rocket rises upwards the value of g changes. Evidently the velocity at a height h will be given by

$$v^2 = v_0^2 - 2\int_0^h g'dh$$

At the earth's surface, $\quad g = GM/R^2$

and at height h, $\quad g' = \dfrac{GM}{(R+h)^2} = \dfrac{GM}{R^2(1+h/R)^2} = \dfrac{g}{(1+h/R)^2}.$

$$\therefore \quad v^2 = v_0^2 - 2\int_0^h \frac{g\,dh}{(1+h/R)^2} \text{ or } v^2 = v_0^2 - 2g\left[-\frac{R}{1+h/R}\right]_0^h = v_0^2 + 2g\left[\frac{R}{1+h/R} - R\right]$$

or $\quad v^2 = v_0^2 - \dfrac{2gh}{1+h/R}$ or $v_0^2 - v^2 = \dfrac{2gh}{1+h/R}.$

The escape velocity is given by $\sqrt{2gR}$. Therefore in this case $v_0 = 0.9\sqrt{2gR}$, and at the highest point reached by rocket, $v = 0$. Now substituting the values in the above relation, we get

$$0.81 \times 2gR = \frac{2gh/R}{R+h} \quad \text{or} \quad 0.81 = \frac{h}{R+h}$$

whence $h \approx 4.3\ R$. This is the expression for the maximum height attained by the rocket.

EXERCISE 5·3

1. A machine gun fires 50 gm bullets at a speed of 1000 m/sec. The gunner holding the machine gun in his hands, can exert an average force of 180 newtons against the gun. Calculate the number of bullets he can fire per minute.
   [**Ans.** 220 per minute.]
2. An empty rocket weight 5000 kg and contains 40,000 kg fuel. If exhaust of the fuel is 2.0 km/sec, calculate the maximum velocity gained by the rocket. Dedcuce the formula used. **(Agra 1970 S)**
   [**Ans.** 4.29 km/sec.]

3. What will be the ratio of the rocket velocity to the exhaust velocity at the instant that its mass is reduced to $1/20^{th}$ of its initial mass, its initial velocity beign zero. **(Agra 2006)**
[**Ans.** 2.996]
4. A rocket of mass 8000 kg is fired vertically. If exhaust speed is 800 m/s, find the rate of exhaust so that the thrust on the rocket can (i) balance the weight, and (ii) produce an acceleration of $3g$ in it.
[**Ans.** (i) 98 kg/sec; (ii) 392 kg/sec.] **(Garwal 1995)**
5. A two stage rocket consists of 6,000 kg and 500 kg stages and have fuel 40,000 kg and 3,500 kg respectively. If the exhaust velocity is 1.8 km/sec., calculate the speed obtained by last rocket.
[**Ans.** 6.62 km/sec.] **(Meerut 1989)**
6. If the two stages of a rocket weights 100 kg and 10 kg separately and contain 800 kg and 90 kg fuel separately, what is the final velocity that can be achieved with an exhaust velocity of 1.5 km/sec. (Take $\log_e 10 = 2.3$ and $\log_{10} 2 = 0.30$).
[**Ans.** 5.9 km/sec.]
7. If M is the mass of a rocket at any time, then the acceleration of a rocket moving vertically is $\left(\frac{mv}{M} - g\right)$, where m is the mass of the gas ejected per second with v velocity. What is the weight of a body inside the rocket, whose normal weight is W ?
[**Ans.** Wmv/Mg.]
8. A rocket of mass 20 kg has 180 kg of fuel. The exhaust velocity of the fuel is 1.6 km/sec. Calculate the minimum rate of consumption of fuel so that the rocket may rise from the ground. Also calculate the ultimate vertical speed gained by the rocket when the rate of consumption of the fuel is (i) 2 kg/sec. (ii) 20 kg/sec. **(Agra 1977; Bundelkhand 95; Rohilkhand 80)**
[**Ans.** 1.225 kg/sec; (i) 2.8 km/sec; (ii) 3.6 km/sec.]

[**Hint :** $\because M\dfrac{dV}{dt} = mv - Mg$

For lifting of the rocket from the ground it is necessary that at $t = 0$,

$mv - M_0 g > 0$ or $m > \dfrac{M_0 g}{v}$; $\therefore$ $m > \dfrac{200 \times 9.8}{1.6 \times 1000}$ or $m > 1.225$ kg/sec,

i.e., minimum rate of consumption of fuel so that the rocket may just rise = **1.225 kg/sec.**

Now, $$V_{max} = v \log_e \frac{M_0}{M} - gt$$

(i) $t = \dfrac{\text{mass of the fuel}}{m} = \dfrac{180}{2} = 90$ sec.; (ii) $t = 180/20 = 9$ sec. and hence get V_{max}.]

5·11. Angular Momentum and Torque—Conservation of Angular Momentum

Angular momentum : *The angular momentum of a particle is defined as the moment of its linear momentum.*

Suppose that a particle of mass m is moving with velocity $\vec{v}$ in the plane of the paper. Its linear momentum will be $m\vec{v}$. If $\vec{r}$ is the position vector of the particle P from a point O in the plane of the paper, then the moment of the linear momentum, *i.e.*, angular momentum of the particle about O in magnitude will be given by

$$J = mv(ON) = mvr_0$$

or $$J = mvr \sin\theta = pr \sin\theta \qquad ...(1)$$

where θ is the angle between the vectors $\vec{r}$ and $\vec{p}$ **[fig. 5.23]**.

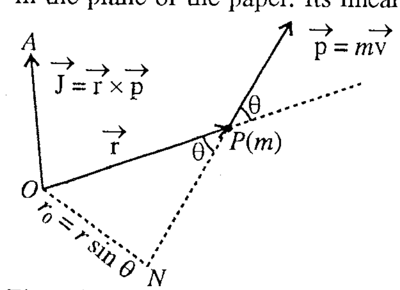

Fig. 5.23 : Angular momentum of a particle.

In vector form, the angular momentum is given by

$$\vec{J} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v} \qquad ...(2)$$

where $\vec{p} = m\vec{v}$, is the linear momentum in an inertial frame in which the reference point is stationary.

Angular momentum is a vector quantity whose magnitude ($pr \sin\theta$) is equal to the product of linear momentum and its perpendicular distance from the reference point; its direction is perpendicular to both $\vec{r}$ and $\vec{p}$ (or $\vec{v}$). The component of $\vec{J}$ along any line or axis passing through the fixed reference point is often called the angular momentum of the particle about this axis. S.I. units of the angular momentum are kg-m^2/sec or J-sec.

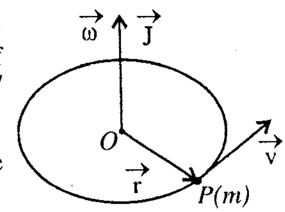

Fig. 5.24 : Circular motion.

If the particle P is in circular motion and the reference point O is the centre, then its linear velocity $\vec{v}$ will be given by $\vec{v} = \vec{\omega} \times \vec{r}$, where $\vec{\omega}$ is the angular velocity of the particle in the perpendicular direction of the plane of the circle. Hence in circular motion, the angular momentum of the particle is given by

$$\vec{J} = \vec{r} \times m\vec{v} = m[\vec{r} \times (\vec{\omega} \times \vec{r})] = m[\vec{\omega}(\vec{r}\cdot\vec{r}) - \vec{r}(\vec{r}\cdot\vec{\omega})] = mr^2\vec{\omega} \quad ...(3)$$

because $\vec{\omega}$ and $\vec{r}$ are mutually perpendicular so that $\vec{\omega}\cdot\vec{r} = 0$.

Thus, *in the circular motion of a particle, its angular momentum (about O) and angular velocity are in the same direction and the angular momentum is given by*

$$\vec{J} = mr^2\vec{\omega} \quad ...(4)$$

Remember that the moment of inertia of a particle about the axis of rotation (direction of $\vec{\omega}$) is equal to the product of the mass (m) of the particle and square of the perpendicular distance (r^2) from that axis, *i.e.*, $I = mr^2$ and then

$$\vec{J} = I\vec{\omega} \quad ...(5)$$

Torque : Consider a force $\vec{F}$, acting on the point P of a body which can rotate it about a point O. The body may be in the form of a particle at P. The torque or moment of the force (in magnitude) is equal to the product of the magnitude of the force F and its perpendicular distance $r \sin\theta$, *i.e.*, $T = Fr \sin\theta$. In vector form, the torque about the point O, is represented as

$$\vec{T} = \vec{r} \times \vec{F} \quad ...(6)$$

where $\vec{r}$ is the vector distance of the point P relative to O. The magnitude of the torque is $Fr \sin\theta$ and its direction is perpendicular to the plane containing $\vec{r}$ and $\vec{F}$.

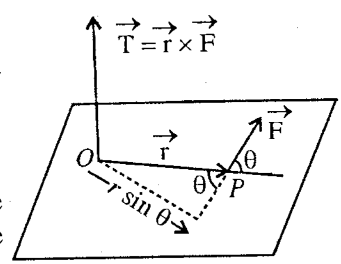

Fig. 5.25 : Torque.

Relation between Angular Momentum and Torque : Differentiating eq. (2) with respect to t, we get

$$\frac{d\vec{J}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt}$$

or

$$\frac{d\vec{J}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt} = \vec{r} \times \vec{F} \qquad \left[\because \frac{d\vec{r}}{dt} \times \vec{p} = \vec{v} \times m\vec{v} = 0\right]$$

where $\vec{F} = d\vec{p}/dt$ is the force applied on the particle.

The vector product of $\vec{r}$ and $\vec{F}$ is called **torque** or **moment of force** about the reference point and is represented by $\vec{T}$. Thus,

$$\vec{T} = \frac{d\vec{J}}{dt} = \vec{r} \times \vec{F} \quad ...(7)$$

Hence, **the torque is equal to the rate of change of angular momentum** ($\vec{J}$). Its unit is newton-metre.

Now, if the torque acting on the particle is zero, then from eq. (7)

$$\vec{T} = d\vec{J}/dt = 0 \quad \text{or} \quad \vec{J} = \text{constant} \qquad \text{...(8)}$$

Thus, *in absence of the external torque on a particle, its angular momentum remains constant.* For a rotating particle, this is the **law of conservation of angular momentum.** Remember that

$\vec{T} = \vec{r} \times \vec{F} = 0$ in three cases : (i) $\vec{F} = 0$, (ii) $\vec{r} = 0$ and (iii) $\vec{r} \parallel \vec{F}$.

5·12. Conservation of Angular Momentum for a System of Particles

If a number of particles are moving independently or attached to one another in the form of a rigid body, then the angular momentum of the system about any point is defined as the vector sum of the angular momenta of the separate particles about the same point.

Now, if $\vec{J}_1, \vec{J}_2, ...$ are the angular momenta of various particles of a system about a given point, the total angular momentum $\vec{J}$ of the system about the point is given by

$$\vec{J} = \vec{J}_1 + \vec{J}_2 + ... = (\vec{r}_1 \times m_1 \vec{v}_1) + (\vec{r}_2 \times m_2 \vec{v}_2) + ... \text{ or } \vec{J} = \sum_{n=1}^{N} (\vec{r}_n \times m\vec{v}_n) \text{ or } \sum_{n=1}^{N} (\vec{r}_n \times \vec{p}_n) \qquad \text{...(1)}$$

where $n = 1, 2, ..., N$ are the particles in the system.

$$\therefore \qquad \text{Torque, } \vec{T} = \frac{d\vec{J}}{dt} = \sum_{n=1}^{N} \vec{r}_n \times \frac{d}{dt}(m_n \vec{v}_n) = \sum_{n=1}^{N} \vec{r}_n \times \vec{F}_n \qquad \text{...(2)}$$

In this summation, the moments of the internal forces of mutual interaction balance each other, because each pair of equal and opposite collinear forces (action and reaction) will have equal and opposite moments about the reference point so that the addition of these two moments is zero. Hence, *the rate of change of angular momentum of the system about any point is the sum of the torques about that point of all the external forces acting on the system.* Thus $\vec{T}$ in eq. (2) represents the external torque on the system.

In a free system left to itself, the external torque $\vec{T} = 0$. Then

$$\frac{d\vec{J}}{dt} = 0 \quad \text{or} \quad \vec{J} = \text{constant. } i.e., \ \vec{J} = \vec{J}_1 + \vec{J}_2 + ... = \text{constant} \qquad \text{...(3)}$$

Thus, *the total angular momentum of a system of particles is constant if the resultant external torque acting on the system is zero.* This is **the principle of conservation of angular momentum.**

In the absence of an external torque, total angular momentum ($\vec{J} = \vec{J}_1 + \vec{J}_2 + ...$) of the system remains constant, although the angular momenta of the particles may mutually interchange.

Angular momentum ($\vec{J}$) and angular momentum about centre of mass (J_{cm}) : The total angular momentum $\vec{J}$ of a system of particles can be expressed in a convenient and important form by using the velocity of the centre of mass and velocities of the particles relative to the centre of mass. If $\vec{R}$ and $\vec{V}$ are the position vector and velocity of the centre of mass and $\vec{r}_{nc}$ and $\vec{v}_{nc}$ those of a particle of mass m_n relative to the centre of mass, then the position vector and velocity of the particle with respect to the reference point are given by

$$\vec{r}_n = \vec{R} + \vec{r}_{nc} \quad [\because \ \vec{r}_{nc} = \vec{r}_n - \vec{R}\,] \text{ and } \vec{v}_n = \vec{V} + \vec{v}_{nc}$$

Hence the total angular momentum for a system of N particles is

$$\vec{J} = \sum_{n=1}^{N} m_n(\vec{R}+\vec{r}_{nc})\times(\vec{V}+\vec{v}_{nc})$$

$$= \Sigma m_n (\vec{R}\times\vec{V}) + \Sigma m_n (\vec{R}\times\vec{v}_{nc}) + \Sigma m_n (\vec{r}_{nc}\times\vec{V}) + \Sigma m_n (\vec{r}_{nc}\times\vec{v}_{nc})$$

$$= \vec{R}\times\vec{V}\Sigma m_n + \vec{R}\times(\Sigma m_n \vec{v}_{nc}) + (\Sigma m_n \vec{r}_{nc})\times\vec{V} + \Sigma m_n(\vec{r}_{nc}\times\vec{v}_{nc}) \quad ...(4)$$

But $\vec{r}_{nc} = \vec{r}_n - \vec{R}$ or $m_{nc}\vec{r}_{nc} = m_n\vec{r}_n - m_n\vec{R}$

or $\Sigma m_{nc}\vec{r}_{nc} = \Sigma m_n\vec{r}_n - \Sigma m_n\vec{R} = \Sigma m_n\vec{r}_n - M\vec{R}$ ...(5)

where $M = \Sigma m_n$ = mass of all the particles of the system.

But according to the property of centre of mass, we know that

$$M\vec{R} = m_1\vec{r}_1 + m_2\vec{r}_2 + ... = \Sigma m_n\vec{r}_n$$

Hence from eq. (5), we obtain $\Sigma m_n\vec{r}_{nc} = 0$ ...(6)

Also, $\vec{v}_{nc} = \vec{v}_n - \vec{V}$ or $m_n\vec{v}_{nc} = m_n\vec{v}_n - m_n\vec{V}$

or $\Sigma m_n\vec{v}_{nc} = \Sigma m_n\vec{v}_n - \Sigma m_n\vec{V} = \Sigma m_n\vec{v}_n - M\vec{V}$ ...(7)

The velocity of centre of mass is given by

$$M\vec{V} = m_1\vec{v}_1 + m_2\vec{v}_2 + ... = \Sigma m_n\vec{v}_n$$

Thus from eq. (7), we have $\Sigma m_n\vec{v}_{nc} = 0$

Hence, from eq. (4), we have

$$\vec{J} = \vec{R}\times M\vec{V} + \Sigma(\vec{r}_{nc}\times m_n\vec{v}_{nc}) \quad ...(9)$$

In this equation $\Sigma(\vec{r}_{nc}\times m_n\vec{v}_{nc})$ represents the angular momentum of the system about the centre of mass, say $\vec{J}_{cm}$, and hence

$$\vec{J} = \vec{J}_{cm} + \vec{R}\times M\vec{V} \quad ...(10)$$

Now, $M\vec{V} = \vec{P}$ is the total linear momentum of the system. The quantity $\vec{R}\times\vec{P}$ or $\vec{R}\times M\vec{V}$ is the angular momentum of the centre of mass about the reference piont. Thus

$$\vec{J} = \vec{J}_{cm} + \vec{R}\times\vec{P} \quad ...(11)$$

Thus, *the angular momentum of a system of particles about a point or origin (situated in inertial frame) is equal to the sum of angular momentum of the system about the centre of mass and the angular momentum of the centre of mass about the reference point.*

5·13. Examples of Conservation of Angular Momentum

(1) Motion under central force – Areal velocity : A central force, acting on a particle, depends only upon the magnitude of its distance from a fixed centre. If $\vec{r}$ is the instantaneous position vector of the particle relative to the fixed centre, then the central force is represented by the relation,

$\vec{F} = f(r)\hat{\mathbf{r}}$, where $f(r)$ is a scalar function of distance r and $\hat{\mathbf{r}} = \vec{r}/r$.

The torque acting on the particle is

$$\vec{T} = d\vec{J}/dt = \vec{r}\times\vec{F} = \vec{r}\times f(r)\hat{\mathbf{r}} = f(r)\left[\vec{r}\times\vec{r}/r\right] = 0 \qquad [\because \vec{r}\times\vec{r} = 0]$$

So that $\vec{J} = \vec{r} \times m\vec{v} = \text{constant}$ ...(1)

Thus, *the angular momentum of a particle, moving under central force, is conserved.* Moreover, if $\vec{J}$ is constant, it should always be perpendicular to the plane containing $\vec{r}$ and $\vec{v}$. This means that *the path of the particle in the influence of a central force lies in a plane.*

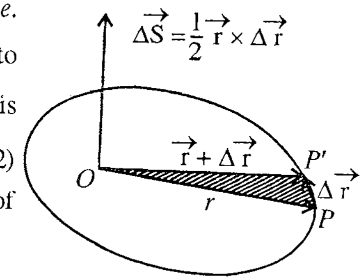

Fig. 5.26.

Now, let O be the centre of force, when the vector $\vec{r}$ changes to $\vec{r} + \Delta\vec{r}$, the vector area $\Delta\vec{S}$ swept by the radius vector in this time is

$$\Delta\vec{S} = \frac{1}{2}\vec{r} \times \Delta\vec{r} \qquad ...(2)$$

This area is swept in Δt time, therefore dividing both the sides of this equation by Δt and taking limit as $\Delta t \to 0$, then

$$d\vec{S}/dt = \frac{1}{2}\vec{r} \times \vec{v} = \vec{J}/2m \qquad ...(3)$$

This expression gives the *areal velocity* ($d\vec{S}/dt$) of the particle. But $\vec{J}$ is constant for central forces, so from relation (3) we see that *the areal velocity remains constant, when the particle moves in the influence of a central force.*

(2) Planetary or Satellite motion : The planets move in elliptical orbits about the sun at a focus.* The gravitational force on the planet is directed towards the centre of the sun. In other words, this force is a central force. Consequently, the angular momentum ($\vec{J} = \vec{r} \times m\vec{v}$) of the planet will be constant and hence the orbit of motion will be **plane.**

For planetary motion, the constancy of $\vec{J}$ means that the areal velocity $d\vec{S}/dt = (\vec{J}/2m)$ is constant. Thus, *the rate of sweeping out of the area by a radial vector from sun to the planet is a constant* This is known as **Kepler's second law of planetary motion.** Similarly, artificial satellites move in elliptical orbits around the earth and the angular momentum is conserved.**

In planetary motion, for the conservation of angular momentum ($\vec{r} \times m\vec{v}$), the planet must move faster at the point of closest approach to the sun than at the farthest point. The reason lies in the fact that at these points the angular momentum is mvr, (because at these positions $\vec{r}$ is perpendicular to $\vec{v}$) and for its constancy the shorter r is associated with larger v. If r_1, r_2 are the distances of closest and farthest points of the planet respectively and v_1, v_2 are the corresponding velocities, then

$$mv_1r_1 = mv_2r_2 \quad \text{or} \quad v_1r_1 = v_2r_2 \qquad ...(4)$$

(3) Scattering of protons or α-particles by a heavy nucleus : Suppose that a proton of charge e and mass m is moving towards a heavy nucleus. The charge on the nucleus is Ze, where Z is the atomic number. Electrostatic force of repulsion will act on the nucleus so that its path will be hyperbola [**fig. 5.27**]. The perpendicular distance p between the nucleus A and the foot B of the perpendicular drawn from A over the initial direction of the proton is called the **impact parameter**. If the velocity of the proton at infinite distance is v_0, its initial kinetic energy is $\frac{1}{2}mv_0^2$ and the angular momentum about the nucleus A is mv_0p. If the *distance of closest approach* of the proton from the nucleus is d, then at this distance kinetic energy $= \frac{1}{2}mv_c^2$, angular momentum $= mv_cd$ and potential energy $= Ze^2/d$.

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* As more correctly speaking, the focus of the ellipse should be taken at the centre of mass of the sun and the planet. As the sun is much heavier than the planet, the centre of mass is very near to the centre of the sun.

** In case if the artificial satellite comes close to the earth, its speed decreases due to the earth's atmosphere and its angular momentum decreases. But the angular momentum of the earth increases in this process and hence the total angular momentum of the system (earth and satellite) remains constant.

The electrostatic force is a central force, hence for the conservation of angular momentum and energy, we have

$$mv_0 p = mv_c d \qquad ...(5)$$

and

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_c^2 + \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{d} \qquad ...(6)$$

Substituting the value of v_c from eq. (5) in eq. (6), we have

$$\frac{1}{4\pi\varepsilon_0}\cdot\frac{Ze^2}{d} = \frac{1}{2}mv_0^2\left\{1-\left(\frac{p}{d}\right)^2\right\} \qquad ...(7)$$

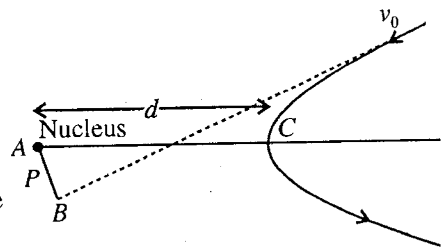

Fig. 5.27. Scattering of protons of a heavy nucleus.

from which d, the distance of closest approach of the proton, can be calculated.

(4) Angular acceleration accompanying contraction : Let a particle of mass m rotate with velocity v_0 at one end of string of length r_0. This string passes through a narrow tube and is pulled by a force. The force on the particle in the form of the tension of the string is radial so that on shortening the string by a force, the value of the torque [= force × perpendicular distance from the centre = force × 0 = 0], acting on the particle, remains zero. Therefore the angular momentum must remain constant as the string is shortened. If v be the velocity when the string is shortened to r, then for the conservation of the angular momentum, we have

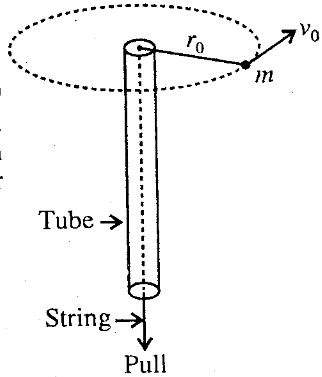

Fig. 5.28.

$$mv_0 r_0 = mvr, \quad \text{whence } v = r_0 v_0 / r \qquad ...(8)$$

Kinetic energy of the particle at $r_0 = \frac{1}{2}mv_0^2$

Kinetic energy of the particle at $r = \frac{1}{2}mv^2 = \frac{1}{2}m(r_0 v_0/r)^2$

Hence, increase in kinetic energy $= \frac{1}{2}mv_0^2\,[(r_0/r)^2 - 1] \qquad ...(9)$

In shortening the string from r_0 to r, the external force (pull) has to do work on the particle against the centrifugal force* and this work done appears in the form of the increase in the kinetic energy of the particle. This energy comes from some other external source of energy or from potential energy. The string will not be shortened, if this energy is not available.

In case, if a rotating particle at the end of a string winds up freely on a smooth fixed stick of finite diameter, then on shortening the string the particle moves nearer to the stick. But here no external energy is supplied, hence the kinetic energy of the particle should remain constant, *i.e.*, $\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2$ or $v = v_0$. Thus, in shortening r, the linear velocity of the particle is not changed, but the angular velocity ($\omega = v/r$) increases. In such a case the angular momentum about the axis of the stick decreases. The path of the particle will be in the form of spiral and the tension of the string will always be normal to the tangent to the path, hence the linear speed will remain constant.

(5) Shape of galaxy : A galaxy is a group of very large number of stars (10^{12}) with huge amounts of free gas. A galaxy is separated from other galaxies by large distances. Its shape is usually like a convex lens rather than spherical. According to present views, galaxies have been formed by the gravitational condensation of very large quantities of gases.

Now, let us consider a very large mass of gas in spherical form endowed initially with some angular

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* Centrifugal force $F = \frac{mv^2}{r} = \frac{mk^2}{r^3}$ [$\because v_0 r_0 = vr = k$, a constant; so $v = k/r$]

Hence the work done against the centrifugal force is

$$= \int_{r_0}^{r} -F\,dr = \int_{r_0}^{r} -\frac{mk^2}{r^3}\,dr = \left[\frac{mk^2}{2r^2}\right]_{r_0}^{r} = \frac{mk^2}{2}\left[\frac{1}{r^2}-\frac{1}{r_0^2}\right] = \frac{mv_0^2 r_0^2}{2}\left[\frac{1}{r^2}-\frac{1}{r_0^2}\right] = \frac{mv_0^2}{2}\left[\left(\frac{r_0}{r}\right)^2 - 1\right].$$

This is the expression for the increase in kinetic energy when the string is shortened. Hence the work done against the centrifugal force in shortening the string appears in the form of increase in kinetic energy.

momentum about any axis, say, $\vec{J}$ * [**fig. 5.29a**]. The gas (or *stars*) condense under the gravitational attraction, hence there should be only mutual interaction forces between the molecules of the gas or stars. In consequence, the total angular momentum $(\vec{J})$ of the system must be conserved.

When the gas contracts, the conservation of angular momentum requires an increase of kinetic energy.** This is supplied by the gravitational potential energy of the gas itself. Further contraction (*i.e.*, decrease in the distance r between the particle and the axis of rotation) will not take place, when the gravitational attractive force (GMm/r^2) is equal to the centrifugal force (mv^2/r).

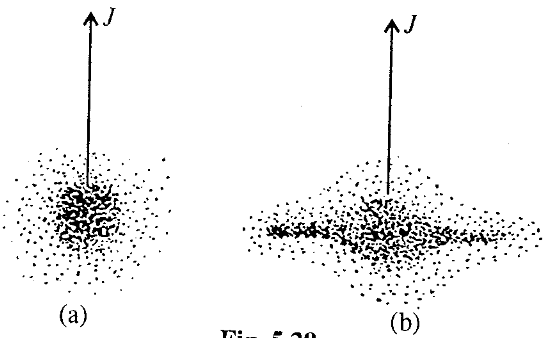

Fig. 5.29.

As the angular momentum remains constant, *i.e.*, $mvr = k$, a constant or $v = k/mr$,

$$\therefore \text{ centrifugal force} = \frac{mv^2}{r} = \frac{k^2}{mr^3} = \frac{K}{r^3} \qquad ...(10)$$

where $k^2/m = K$, another constant.

It is clear that if now the distance r between the particle and the axis of rotation decreases, the centrifugal force increases very rapidly and opposes the contraction. But the cloud of gas or of stars is able to collapse under gravitational attraction in the direction parallel to the axis of rotation, without changing the value of the angular momentum. Hence the galaxy becomes flattened and ultimately acquires a convex lens like shape [**fig. 5.29(b)**].

(6) Later on we will see that if a system is rotating with angular velocity $\vec{\omega}$, then its angular momentum about the axis of rotation is represented as

$$\vec{J} = I\vec{\omega} \qquad ...(11)$$

where I is the momentum of inertia of the system about the axis of rotation. Hence for conservation of angular momentum $I\omega$ should remain constant.

A simple but well known example for the demonstration of angular momentum is given here. A man stands on a turn table, holding his arms extended [**fig. 5.30(a)**] with a dumb-bell in each hand. The turn table is free to rotate. If someone sets the man in slow rotation and he then drops his hands to his sides, he will rotate much more rapidly [**fig. 5.30(b)**]. The total mass of the rotating system remains constant. Hence, a decrease in the distance of the dumb-bells from the axis of rotation results a corresponding decrease in the moment of inertia of the system about the axis of rotation, because the moment of inertia of the system depends upon the distribution of its mass about the axis. Since the angular momentum of the system remains constant, the angular velocity of rotating system is increased due to the decrease in the moment of inertia of the system.

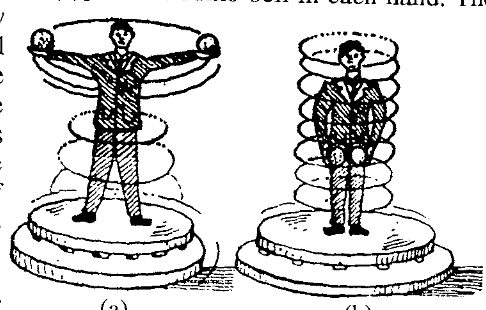

Fig. 5.30.

Similarly, a circus acorbat, a diver or a skater performing a pirouette on the toe of one skate takes advantage of this priniciple.

5·14. Spin and Orbital Angular Momentum

We know that the earth moves in its orbit around the sun and simultaneously it rotates about its own axis. In the first case the angular momentum of the earth is due to its **orbital motion** and is directed

* Till now we do not know (i) where the gas came from and (ii) why a given mass of gas should have an angular momentum. Masses without angular momentum will condense as spheres.

** As $J = mvr$ = constant hence for decrease of r, velocity, *i.e.*, kinetic energy of the particle should increase.

perpendicular to the plane of the orbit. In the second case, the angular momentum is due to the **spinning** of the earth and is along the axis of rotation of the earth. These are called **orbital angular momentum** and **spin angular momentum** respectively. The total angular momentum of the earth about the sun is the vector sum of these two quantities, whose directions are inclined at 23½° to each other. Similarly, other planets of the solar system have the orbital and spin angular momenta. The sun itself is spinning about its axis and hence has spin angular momentum. *The total angular momentum of the solar system is equal to the vector addition of all these angular momenta.* The planets, which are massive and have large radii of orbit, contribute the major portion to the total angular momentum of the system; for example contribution of the planet Jupiter is nearly 60% and that of Saturn nearly 20%. The rotation of the sun about an axis through its centre (more correctly the centre of mass of the solar system) accounts for only about 2% of the total angular momentum in the solar system.

For the earth, spin angular momentum $= I\omega = \left(\frac{2}{5}mR^2\right)\omega$ *, ...(1)

where m is the mass of earth, R its radius and $\omega = \dfrac{2\pi}{24 \times 60 \times 60}$ rad/sec is its angular velocity about the axis of rotation.

Its orbital angular momentum $= mr^2\omega'$ ...(2)

where r is the distance between the centres of the earth and the sun and $\omega' = 2\pi/(365 \times 24 \times 60 \times 60)$ radian/sec is its angular velocity about the sun.

Ex. 1. *Calculate the angular momentum of a satellite of mass m which moves in a circular orbit of radius r referred to the centre of the orbit. Express the result in terms of G, M, m and r, where M is the mass of earth.* **(Lucknow 1996)**

Sol. Angular momentum $\vec{J} = \vec{r} \times mv$ or $J = mrv \sin\theta$

Here, for circular orbit $\theta = \pi/2$ and so $\sin\theta = 1$, $\therefore J = mrv$

Its direction is perpendicular to the plane of motion.

Also, $\dfrac{mv^2}{r} = \dfrac{GMm}{r^2}$ or $(mvr)^2 = GMm^2r$, $\therefore J = (GMm^2r)^{1/2}$.

Ex. 2. *A 100 gm stone is revolved at the end of a 50 cm long string at the rate of 2 revolutions per second. Determine its angular momentum. If after 25 sec it is making only one revolution per second, find the mean torque.*

Sol. In case of circular motion,

$\therefore \quad \vec{J} = mr^2\vec{\omega} = 0.1 \times (0.5)^2 \times (2\pi \times 2) =$ **0.314 J-sec. ⊥ to the plane of motion.**

Mean torque $= \Delta J/\Delta t = mr^2\Delta\omega/\Delta t = 0.1 \times (0.5)^2 \times 2\pi/25 =$ **6.25×10^{-3} n/m along $\vec{J}$**.

Ex. 3. *Compare the value of the angular momentum of the moon in its orbit about the earth with the value of the angular momentum of rotation of the moon about its own axis.*

Sol. Angular momentum of the moon about the earth is $J_1 = mvr = mr^2\omega$

Here, mass of the moon $m = 7.35 \times 10^{22}$ kg, radius of moon's orbit $r = 3.84 \times 10^8$ m and angular velocity about the earth $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{2.36 \times 10^6}$ rad/sec, where $T = 2.36 \times 10^6$ sec is the time of one revolution of the moon round the earth.

$$\therefore \quad J_1 = 7.35 \times 10^{22} \times (3.84 \times 10^8)^2 \times \frac{2\pi}{2.36 \times 10^6} = \mathbf{2.89 \times 10^{34}\ J\text{-sec.}}$$

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* Here, $I = \frac{2}{5}mR^2$ is the moment of inertia of earth or solid sphere about the axis of rotation. Detailed description has been given in chapter 8.

Now we know that the moon always keeps the same side facing the earth. Therefore, it must make one revolution about its axis in the time it makes one revolution about the earth. Hence, the period (and thus angular frequency) for revolution about its axis is identical to its period of revolution about the earth. Thus the angular momentum J_2 of the moon about its axis is

$$J_2 = I\omega = \frac{2}{5}mR^2\omega \text{, where } R \text{ is the radius of moon}$$

$$= \frac{2}{5} \times 7.35 \times 10^{22} \times (1.74 \times 10^6)^2 \times \frac{2\pi}{2.36 \times 10^6} = 2.37 \times 10^{29} \text{ J-sec., } \therefore \frac{J_1}{J_2} = \mathbf{1.22 \times 10^5}.$$

Ex. 4. *A particle of mass 20 gm moving in a circle at 4 cm radius with constant speed of 10 cm/sec. What is its angular momentum about* (i) *the centre of the circle,* (ii) *a point on the axis of the circle and at 3 cm distance from its centre ? Which of these will be in the same direction ?*

Sol. (i) $J = mvr = 0.02 \times 0.1 \times 0.04 = \mathbf{8 \times 10^{-5}}$ **J-sec.**

Its direction is perpendicular to the plane of the circle.

(ii) $J = mvr = 0.02 \times 0.1 \times 0.05 = \mathbf{10^{-4}}$ **J-sec.**

Its direction is perpendicular to the plane of the position vector (OA) and instantaneous velocity of the particle, *i.e.*, its direction is OB which is changing with time. In case (i) the direction of J remains the same.

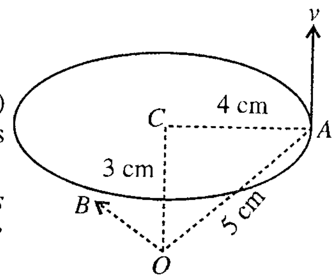


Fig. 5.31.

Ex. 5. *An electron moves about a proton in circular orbit of radius 0.5 Å. Calculate (i) the orbit angular momentum of the electron about the proton and (ii) the total energy in joules and electron-volts.*

Sol. (i) For a circular orbit of the electron, the angular momentum is given by

$$J = |\vec{J}| = |\vec{r} \times m\vec{v}| = mvr \quad [\because \vec{r} \text{ and } \vec{v} \text{ are } \perp \text{ to each other for a circular orbit.}]$$

Here the necessary centripetal force for the rotation of the electron in the circular path is provided by the coulombian force between the electron and the proton, *i.e.*,

$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} \text{ or } m^2v^2r^2 = \frac{1}{4\pi\varepsilon_0}me^2r \quad \text{...(i)}$$

$$\therefore J = mvr = \sqrt{me^2r/4\pi\varepsilon_0} = [9 \times 10^9 \times 9 \times 10^{-31} \times (1.6 \times 10^{-19})^2 \times (0.5 \times 10^{-10})]^{1/2}$$

$$= \mathbf{1.02 \times 10^{-34}} \textbf{ J-sec.}$$

(ii) Kinetic energy of the electron, $K = \frac{1}{2}mv^2 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{2r}$ $\left[\because \text{From relation (i), } mv^2 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r}\right]$

Potential energy of the electron, $U = -\frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r}$

$\therefore$ Total energy of the elctrcon $= K + U = \frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{2r} - \frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r} = -\frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{2r}$

$$= -9 \times 10^9 \frac{(1.6 \times 10^{-19})^2}{2 \times 0.5 \times 10^{-10}} = -23.04 \times 10^{-19} \text{ joule}$$

$$= -\frac{23.04 \times 10^{-19}}{1.6 \times 10^{-19}} = \mathbf{14.4 \text{ eV.}}$$

Ex. 6. *Determine the impact parameter of a 2 MeV α-particle whose distance of closest approach to a gold nucleus (Z = 79) is 2×10^{-11} cm.*

Sol. Here, charge on the α-particle = $2e$ and kinetic energy $\frac{1}{2}mv_0^2 = 2 \times 10^6 \times 1.6 \times 10^{-19}$ J

But $\quad \dfrac{1}{4\pi\varepsilon_0}\cdot\dfrac{Ze(2e)}{d} = \dfrac{1}{2}mv_0^2\left\{1-\left(\dfrac{p}{d}\right)^2\right\}$

$\therefore \quad 1-\left(\dfrac{p}{d}\right)^2 = \dfrac{1}{4\pi\varepsilon_0}\cdot\dfrac{2Ze^2}{d}\times\dfrac{2}{mv_0^2} = 9\times 10^9\,\dfrac{2\times 79\times(1.6\times 10^{-19})^2}{2\times 10^{-13}\times 3.2\times 10^{-13}} = 0.57$

whence, $p = d\sqrt{1-0.57} = 2\times 10^{-13}\sqrt{0.43} = \mathbf{1.3\times 10^{-13}}$ **m.**

Ex. 7. *A particle of mass m rotates at one end of a string of length r_1 whose other end passes through a narrow tube. If the particle makes two revolutions per second when the length of the string is 10 cm, how many revolutions will it make in a second, when the string is shortened to 5 cm ? What will be the result if the string gets shortened by winding round the tube ?*

Sol. For the conservation of angular momentum in circular motion

$$mv_1r_1 = mv_2r_2 \quad \text{or} \quad \omega_1 r_1^2 = \omega_2 r_2^2 \qquad [\because v = r\omega]$$

But $\quad \omega = 2\pi\times$ number of revolutions/sec $= 2\pi n$

$\therefore \quad n_1r_1^2 = n_2r_2^2 \quad \text{or} \quad r_2 = n_1\left(\dfrac{r_1}{r_2}\right)^2 = 2\left(\dfrac{10}{5}\right)^2 =$ **8 rev/sec.**

In this case, as $r_1 > r_2$, $\therefore$ $v_2 > v_1$' *i.e.*, final kinetic energy $\left(\frac{1}{2}mv_2^2\right)$ of the particle is greater than its initial kinetic energy.

If the string is shortened by winding round the tube, then the kinetic energy of the particle will remain constant.

Now, $\quad \frac{1}{2}mv_1^2 = \frac{1}{2}mv_2^2$ or $v_1 = v_2$ or $r_1\omega_1 = r_2\omega_2$ or $r_1n_1 = r_2n_2$

$\therefore \quad n_2 = n_1r_1/r_2 = 2\times\frac{10}{5} =$ **4 rev/sec.**

Ex. 8. *A light weight man holds two heavy dumb-bells with outstretched arms while standing on a turn table. A slow push is gven to the system so that he rotates at a rate of 1 revolution per minute. Then he pulls the dumb-bells in towards his chest. If the dumb-bells are originally 60 cm from his axis of rotation and are pulled into 10 cm from the axis of rotation, then find the frequency of revolution. Neglect the angular momentum of the man in comparison to that of the dumb-bells.*

Sol. If M is the mass of each dumb-bell, then the initial angular momentum of the system,

$$J_1 = 2M\omega_1 r_1^2 = 2M(2\pi n_1)r_1^2$$

where the angular momentum of the man has been left out.

Final angular momentum, $J_2 = 2M(2\pi n_2)r_2^2$

$\therefore$ For the conservation of angular momentum $J_1 = J_2$

i.e., $\quad 2M\,(2\pi n_1)r_1^2 = 2M\,(2\pi n_2)r_2^2$ or $n_2 = n_1r_1^2/r_2^2$

Here $\quad n_1 = 1$ rev. per minute, $r_1 = 0.6$ m and $r_2 = 0.1$ m.

$\therefore \quad n_2 = 1\times(0.6)^2/(0.1)^2 = 36$

Thus, finally the system will rotate at a rate of **36 revolutions per minute.**

EXERCISE 5·4

1. A particle of mass m moves along a circular path of radius r with velocity $\vec{v}$. What is its instantaneous angular momentum about a point on the plane of the orbit at a distance $\vec{x}$ from the centre of the axis of rotation ? **(Allahabad 1979)**

[**Ans.** $m(\vec{r}-\vec{x})^2\vec{\omega}$, $\omega = v/r$.]

2. What is the angular momentum of a particle whose linear momentum is $\vec{p}$ and position vector is $\vec{r}$ about a point whose position vector is $\vec{r}'$.
   [Ans. $(\vec{r}-\vec{r}')\times\vec{p}$.]
3. Show that the components of $\vec{J}$ along X, Y and Z axes are given by
$$J_x = yp_z - zp_y;\ J_y = zp_x - xp_z;\ J_z = xp_y - yp_x.$$
   If the particle moves only in X-Y plane, show that the resultant angular vector has z-component only.
4. Show that the angular momentum of a particle about a fixed point is equal to the product of its mass and double the area $(\frac{1}{2}r^2\omega)$ described by the rotating line in unit time and has a direction perpendicular to this area.
5. Exress total angular momentum in terms of kinetic, potential and total energy of an earth's satellite of mass m in a circular orbit of radius r.
   [Ans. $J = (2mr^2K)^{1/2} = (-mr^2U)^{1/2} = (-2mr^2E)^{1/2}$.]
6. A particle of $\frac{1}{2}$ gm mass starts from the point (1, 0, 2) m at a time $t = 0$. If the force acting on it be $2\hat{i} + (1+6t)\hat{j}$ newton, calculate the torque and angular momentum about the point (2, 2, 4) m at time $t = 1$ sec.
   [Ans. $\vec{T} = (14\hat{i} - 4\hat{j} + 5\hat{k})$ N-m; $\vec{J} = (8\hat{i} - 4\hat{j} - 2\hat{k})\,10^{-3}$ kg-m^{2}s^{-1}.]
   [Hint : $\dfrac{d^2(\vec{r}-\vec{r}_0)}{dt^2} = \dfrac{\vec{F}}{m} = 2[2\hat{i} + (1+6t)\hat{j}]$ and integrate to get $\vec{v}$ and $\vec{r}$.]
7. A particle of mass m, moving in a circular orbit of radius r, has angular momentum L about the centre. Calculate the kinetic energy of the particle in terms of L, m and r. **(Agra 1995)**
   [Ans. $L^2/2mr^2$.]
   [Hint : For circular motion, $L = mr^2\omega$, $\therefore$ K.E. $= \frac{1}{2}mv^2 = \frac{1}{2}m(r\omega)^2 = \frac{m}{2}\left(\frac{L}{mr}\right)^2 = \frac{L^2}{2mr^2}$.]
8. A 500 gm stone is revolved at the end of a 40 cm long string at the rate of 2 rev/sec. What is its angular momentum ? If after 30 sec, it makes only $\frac{1}{2}$ rev/sec; calculate the mean torque applied.
   [Ans. 1 joule-sec about the centre of circle; 0.025 N-m.]
9. A particle of mass m moves along a path given by $\vec{r} = a\hat{i}\cos\omega t + b\hat{j}\sin\omega t$. Find (a) the torque, (b) the angular momentum about the origin. **(Kanpur 1996)**
   [Ans. (a) 0; (b) $2m\,ab\,\omega\hat{k}$.]
10. Calculate the mass of our galaxy. Given the velocity of sun about the centre of the galaxy = 3×10^5 m/sec and the distance of sun from the axis of the galaxy = 3×10^{20} m ($G = 7 \times 10^{-11}$ S.I. units).
   [Ans. 4 × 10^{41} kg.]
   [Hint : $\dfrac{GMm}{r^2} = m\dfrac{v^2}{r}$ or $M = \dfrac{v^2r}{G}$, $\therefore M = \dfrac{(3\times10^5)^2\times3\times10^{20}}{7\times10^{-11}} = 4\times10^{41}$ kg.]
11. Determine the angular momentum of a 1 MeV neutron about a nucleus, if the impact parameter is 10^{-12} m.
   [Ans. 1.9 × 10^{-32} kg-m^2/sec.]
12. The distance of closest approach of a positive particle of charge e to the nucleus of atomic number 80 is 2×10^{-11} cm. Its initial kinetic energy is 2 MeV, obtain the value of the impact parameter for it. **(Kanpur 1990)**
   [Ans. 1.69 × 10^{-13} m.]
13. A neutron of energy 1 MeV passes a proton at such a distance that the angular momentum of the neutron relative to the proton approximately equals 10^{-33} J-sec. What is the distance of closest approach ? (Neglect the energy of interaction between the two particles).
   [Ans. 4 × 10^{-14} m.]

14. The maximum and minimum distances of a comet from sun are 2×10^{12} m and 8×10^{10} m respectively. If the speed of the comet at the nearest point is 60 km/sec, calculate the speed at the farthest point. **(Delhi 1994)**
[Ans. 2.4 km/sec.]
15. In the dumb-bell example of page 146, what is the initial and final kinetic energy of the dumb-bell = 20 kg.
[Ans. Initial K.E. = 284 joules, final K.E. = 10,220 joules. The man has to do work in pulling the dumb-bells towards him. The kinetic energy of the system will increase by this amount of work.]

QUESTIONS

Long Answer Questions

1. Explain the meaning of centre of mass. Find out the expression of position vector and velocity of centre of mass of a system of particles. **(Agra 1991)**
2. Explain the concept of centre of mass. How does it differ from centre of gravity ? Derive formula for its position. On what factors does the position of centre of mass of a body depend ?
(Rewa 2000; 1997; Gwalior 90; Raipur 97; Indore 92; Bhopal 97; Bilaspur 99)
3. What is the concept of centre of mass ? Determine the position vector of a system of n particles. **(Sagar 1999)**
4. Prove that the centre of mass of two point particles lies on the line joining them and the ratio of distances of centre of mass from the particles is equal to the inverse ratio of their masses.
(Bhopal 2001, 1989; Gwalior 2003, 1998; Raipur 99, 96; Indore 98)
5. (a) If the total momentum of a system is conserved, then prove that the centre of mass moves with uniform velocity or remains at rest. **(Ajmer 1989)**
(b) Show that the total linear momentum of a system of particles about the centre of mass is zero. **(Bhopal 2000; Agra 1993; Raipur 97)**
(c) Show that the acceleration of the centre of mass of a system of particles is only due to external forces. **(Rajasthan 1984)**
(d) Show that the centre of mass of two particles lies on the line joining them.
(e) An isolated system of point masses is subjected to the internal forces of interaction only. Show that the centre of mass of the system moves with a constant velocity in an inertial frame. **(Agra 1997)**
6. Write down the position vector of centre of mass of a system of n particles and hence prove that the velocity of centre of mass of the system remains constant if no external force acts on the system.
(Agra 1993, 85; Kanpur 83; Indore 98; Bilaspur 90; Bhopal 92; Ujjain 94; Sagar 99; Gwaliior 99; Allahabad 84)
7. A system of N particles is acted upon by an external force $\vec{F}$. Show that the centre of mass of the system moves like a particle of mass equal to the total mass of the system and acted upon by the total external force $\vec{F}$. **(Bhopal 1991, 90)**
8. What is linear momentum ? Prove that if the resultant of external forces acting on a n-particle system, then the total linear momentum of the system is conserved. **(Indore 1994)**
9. State and prove the law of conservation of linear momentum for a system of n particles. **(Rewa 1992; Jabalpur 99)**
10. In the laboratory frame, a particle of mass m_1 collides with velocity v_1 with another stationary particle of mass m_2. If the velocity of their centre of mass is v_c, find the velocities of these particles after the collision in the centre of mass frame. **(Indore 1994)**
11. Explain the meaning of elastic and in-elastic collisions. Describe the elastic collision of two particles in centre of mass frame. **(Raipur 1998)**

12. Prove that if two perfectly elastic particles of unequal masses but having equal and opposite momentum collide, the magnitudes of their velocities will remain unchanged by the collision though the directions may be changed.
13. Prove that in a collision of a particle of mass m_1, moving with velocity v_1 in the L-frame (laboratory frame), with a particle of mass m_2 at rest in the L-frame, the angles at which the first particle moves after the collision relative to its initial velocity are given by $\tan\theta = \sin\phi/(\cos\phi + 1/A)$, where $A = m_2/m_1$ and angles θ and ϕ refer to the L and C-frames respectively.

    Show also that the maximum value of θ for arbitrary A is given by $\tan\theta = \sqrt{1-A^2}$. Discuss the situation when A is larger than 1 and when it is smaller than 1.
14. Prove that if the internal kinetic energy of a system of two particles is K_{cm}, the magnitudes of the velocities of the particles relative to the centre of mass are
$$v_1 = [2m_2K_{cm}/m_1(m_1 + m_2)]^{1/2} \text{ and } v_2 = [2m_1K_{cm}/m_2(m_1 + m_2)]^{1/2}.$$
15. Show that in a head-on elastic collision between two particles the transference of energy is maximum when their mass ratio is unity. **(Agra 1986; Bundelkhand 95)**
16. Deduce expression for the acceleration and final velocity of a rocket. **(Garwal 1995)**
17. Describe the principle of rocket motion and establish the following expression for the final velocity acquired by the rocket $V = V_0 + v \log_e (M_0/M)$,

    where the symbols have their usual meaning. On this basis, explain how is two stage rocket superior over a single stage rocket ? **(Kanpur 2005; Meerut 2000; Sagar 1997; Jabalpur 97; Agra 95)**
18. Prove that the velocity acquired by a rocket in time t is given as
$$V = V_0 - v \log_e (1 - mt/M_0),$$
    where V_0 is the initial velocity of rocket, M_0 is the initial mass of rocket, v is the escape velocity of exhaust gases and m is the rate of consumption of fuel. What will be the value of V, if the attraction due to earth is not neglected ? **(Bhopal 1995)**
19. What is the working principle of a rocket ? Derive the differential equation representing rate of gain of speed by a rocket. How the velocity of the rocket is increased so as to escape from the earth's attracting field ? What is the advantage of two stage rocket in comparison to single stage rocket ? **(Agra 1985)**
20. Define angular momentum $\vec{J}$ and torque $\vec{T}$. Using Newton's second law in an inertial frame of reference to show that $\vec{T} = d\vec{J}/dt$. **(Kanpur 1984, 80)**
21. What is meant by angular momentum of a particle ? Show that the rate of change in angular momentum of the particle is equal to the external torque acting on it. Hence derive the law of conservation of angular momentum. **(Gwalior 2001; Bilaspur 2000; Sagar 2000; Bhopal 96; Jabalpur 96; Ujjain 97; Garwal 95; Bundelkhand 95; Delhi 89)**
22. Show that the angular momentum $\vec{J}$ of a system of particle can be expressed in the form
$$\vec{J} = \vec{J}_{cm} + \vec{R}\times\vec{P} \quad (\text{or } \vec{J} = \vec{J}_{cm} + \vec{R}\times M\vec{V})$$
    where $\vec{J}_{cm}$ = angular momentum about the centre of mass, $\vec{R}$ = position vector of centre of mass and $\vec{P} = M\vec{V}$ total linear momentum. **(Allahabad 1979; Gorakhpur 95; Kanpur 80; Jabalpur 98; Raipur 91)**
23. Define 'angular momentum' of a particle about a point. Explain about its magnitude and direction. Derive a relation between the torque applied on a rotating body and its angular momentum. Hence state the law of conservation of angular momentum. **(Agra 1997)**
24. Show that in the central force field, the angular momentum of a particle is conserved. **(Jabalpur 1994; Allahabad 84; Gorakhpur 95; Kanpur 83; Mysore 80)**

25. Show that when angular momentum of a particle is constant, (i) the motion is planer and (ii) the areal velocity is constant. **(Mysore 1980)**

26. (a) Show that conservation of angular momentum applied to planetary motion leads to the law of constant areal velocity. **(Agra 1971; Kanpur 84)**

    (b) Prove that in the planetary motion the areal velocity is constant. When the planet is closest to the sun, its velocity is maximum of its path. Why ?

27. A proton of speed u travels towards an atomic nucleus such that if it is undefected the nearest approach would be b. Show that the actual nearest approach b' is given by

$$\frac{1}{4\pi\varepsilon_0}\cdot\frac{Ze^2}{b'} = \frac{1}{2}mu^2\left[1-\left(\frac{b}{b'}\right)^2\right],$$

where Z is atomic number of the nucleus and m is the mass of proton. **(Agra 1977)**

28. What are spin and orbital angular momenta ? What do you understand by the angular momentum of the solar system ?

29. A man rotates on a turn table with an angular speed ω. He is holding equal masses at arms length. Without moving his arms, he drops the two masses. What change does occur in his angular speed? Is the angular momentum conserved ? Explain.

30. The moon revolves about the earth and rotates about its own axis in such a way that we always see the same side of the moon. (i) How are the spin and the orbital parts of the angular momentum of the moon with respect to the earth related ? (ii) By how much its spin angular momentum have to change so that we may see all the moon's surface during the course of a month ? **(Rajasthan 1984)**

    [**Hint :** (i) $\dfrac{J_{spin}}{J_{orbital}} = \dfrac{2R_m}{5r}$, where R_m is the lunar (moon's) radius and r the distance between earth and moon.

    (ii) Increase or decrease by one-half of its present value.]

31. A stone, attached to one end of a string is revolved around a stick so that the string winds upon the stick and get shortened. Explian whether any of the followig is conserved :

    (i) angular momentum, (ii) linear momentum, (iii) energy.

    [**Hint :** Value of linear momentum does not change and hence kinetic energy remains constant.]

32. Show that a gain in kinetic energy of a particle on shortening its circular orbit is indeed supplied by the work done against centrifugal force acting on the particle. **(Kanpur 1990)**

33. Write the law of conservation of angular momentum. Discuss the shape of the galaxy on the basis of this law. **(Rewa 2001; Ujjain 1995; Bilaspur 89; Agra 82; Kanpur 90)**

34. Write short notes on the following :

    (i) Centre of mass of N-particles. **(Sagar 1997)**

    (ii) Laboratory and centre of mass frame of reference. **(Agra 2003)**

    (iii) Elastic and inelastic collisions.

Short Answer Questions

1. Differentiate between a rigid body and a deformable body. **(Indore 1992)**
2. Explain the meaning of centre of mass **(Indore 2000; Agra 2006)**
3. Find the centre of mass of a two-particle system. **(Rewa 2000)**
4. What do you mean by the centre of mass frame of reference ?
5. Show that in absence of external forces, the velocity of centre of mass remains constant. **(Agra 2006, 1993; Allahabad 84)**
6. Define angular momentum. Write the unit of angular momentum.
7. Write the necessary conditions for an elastic collision. **(Raipur 1998)**
8. Distinguish between elastic and inelastic collisions. **(Lucknow 2005, 1997)**

9. What are elastic and inelastic collisions. Give examples. **(Kanpur 2002)**

10. Explain the laboratory and centre of mass frames. **(Rewa 1994; Ujjain 94; Jabalpur 96)**

11. Show that if a heavy particle is incident on a light particle, initially at rest, the heavy particle will not bounce backward as a result of collision.

12. What is a two stage rocket ? What is the advantage of a two stage rocket over a single stage rocket ? **(Agra 1970)**

13. Show that the rocket speed is equal to the exhaust speed when the ratio M_0/M is e. Show also that the rocket speed is twice the exhaust speed when $M_0/M = e^2$.

14. On what principle working process of rocket depends ? Discuss. **(Kanpur 2005)**

15. If $\vec{v}_c$ is the velocity of any particle of mass m relative to the centre of mass of a number of particles, then show that $\Sigma m v_c$ is zero whether any force acts on the particles or not.

16. If only external forces can change the state of motion of the centre of mass of a body, how does it happen that the internal forces of brakes can bring a car to rest ?

17. A light and a heavy body have equal translational kinetic energies. Which one has the greater momentum ?

18. Can a sail-boat be propelled by air blown at the sails from a fan attached to the same boat ?

19. The value of initial kinetic energy will be greater or less than final kinetic energy in inelastic collision and why ?

20. The masses of two particles are 1 kg and 4 kg. If the magnitudes of their kinetic energies are equal, then what will be the ratio of their linear momenta ?
[**Ans.** 1/2]

21. A charged particle of mass 'm' and charge 'q' is accelerated by a potential difference of V volt. What will be the value of its linear momentum ? **(Rewa 1993)**

22. When a bullet is fired by a gun, prove that the distances travelled by the bullet and the gun are inversely proportional to their masses. **(Rewa 1993)**

[**Ans.** A bullet of mass 'm' is fired by a gun of mass 'M'. If the velocity of the bullet is 'v' then according to the law of conservation of linear momentum, the gun is pushed back by a velocity $V = \frac{mv}{M}$. The ratio of that distances travelled by the bullet and gun in time 't' is $\frac{S}{s} = \frac{Vt}{vt} = \frac{(mv/M)t}{vt} = \frac{m}{M}$.]

23. A light body and a heavy body have the same kinetic energy, which one will have greater momentum ?
[**Ans.** Heavy body.]

24. A bullet is fired from a rifle. If the rifle recoils freely, determine whether the kinetic energy of the rifle is greater than, equal to or less than that of the bullet.
[**Ans.** Less.]

25. What is the law of conservation of angular momentum ? What are spin and orbital angular momenta ?

26. Show that torque $\vec{\tau} = \frac{d\vec{J}}{dt} = \vec{r} \times \vec{F}$. **(Lucknow 2005)**

27. Prove that $\text{K.E.} = \frac{J^2}{2mr^2}$, where J = Angular momentum. **(Agra 2004)**

28. Show that the time rate of change of angular momentum of a particle about a given point equals the torque acting on the particle. **(Agra 2005)**

29. What conclusion can you draw about the masses of projectile and target body in case of head-on collision (a) if projectile rebounds, (b) if projectile stops, (c) if target body flys ahead of projectile.
[**Ans.** (a) $m_p < m_t$; (b) $m_p = m_t$; (c) $m_p > m_t$.]

30. In head on collision, a particle with an initial speed v_0 strikes a stationary particle of same mass. Find the velocities v_1 and v_2 of the two particles respectively after the collision, if half of the original kinetic energy is lost. **(Agra 2006)**

[**Ans.** $v_1 = v_2 = v_0/2$]

31. If a ball of mass m moving with velocity v strikes head-on elastically with a number of balls of same masses at rest in a line, only one ball from the other side moves with the same velocity (v). Explain, why not two balls move simultaneously with velocity ($v/2$) ?

Objective Type Questions

1. The total linear momentum about the centre of mass is :
(a) never zero (b) always zero (c) sometimes zero (d) infinite **(Agra 2005)**

2. The centre of mass of a body lies :
(a) outside the material of the body (b) within material of a body
(c) on the surface of a body
(d) may lie inside, outside or on the surface of a body **(Kanpur 2004)**

3. A body of mass m strikes a wall with speed v and returns with the same speed. Its momentum will be changed by :
(a) mv (b) $2mv$ (c) $-mv$ (d) 0

4. A ball of mass 50 gm strikes a plane surface at an angle of incidence of 45°. If the speed of the ball is 2 m/s and it is reflected with the same angle and speed, its momentum will be changed by :
(a) zero (b) 0.14 kg-m/s (c) 0.1 kg-m/s (d) 0.2 kg-m/s

5. If a bomb blasts in four parts, which quantity will be conserved ?
(a) momentum (b) kinetic energy (c) potential energy (d) both (b) and (c)

6. Two bodies of masses m and $4m$ are moving with the same kinetic energy. The ratio of their linear momenta will be :
(a) 4 (b) 1 (c) 1/2 (d) 1/4

7. A body, composed of several particles, is called rigid body if the distance between any of two particles of it :
(a) remains constant always
(b) remains constant only when the body moves linearly
(c) remains constant only when the body is in rest position
(d) remains constant only when the body is in rotatory motion **(Bhopal 1989)**

8. A uranium nucleus is changed into thorium nucleus by emitting an α-particle. If the velocity of emitted α-particle is 1.5×10^7 m/s, the velocity of thorium nucleus is :
(a) 2.4×10^7 m/s (b) 24×10^5 m/s
(c) 0.24×10^5 m/s (d) 245×10^5 m/s **(Bundelkhand 2001)**

9. Which of the Newton's law is involved in Rocket propulsion ?
(a) Law of gravitation (b) First law of motion (c) Second law of motion (d) Third law of motion **(Kanpur 2001)**

10. A bullet of mass m makes a completely inelastic collision with a block of much greater mass M. If the block rises to a height h, the initial speed of the bullet is :
(a) $\sqrt{2gh}$ (b) $\frac{M+m}{m}\sqrt{2gh}$ (c) $\frac{m}{M+m}\sqrt{2gh}$ (d) $\frac{M+m}{M}\sqrt{2gh}$

11. All the particles of a body are situated at a distance d from the origin. The distance of the centre of mass of the body from the origin will be :
(a) $= d$ (b) $\leq d$ (c) $> d$ (d) $\geq d$

12. A bullet hits a block kept at rest on a smooth horizontal table and gets embedded into it. Which of the following physical quantities for the block does not change ?

(a) linear momentum (b) kinetic energy

(c) gravitational potential energy (d) temperature

13. For a particle moving under central force, the physical quantity about the centre of force is conserved :

(a) angular acceleration (b) linear acceleration

(c) linear momentum (d) none of these **(Rewa 1998, 94)**

14. The total linear momentum of a system of particles remains always constant if :

(a) a conservative force is acting on the system (b) a non-conservative force is acting on the system

(c) no resultant force is acting on the system (d) variable force is acting on the system

15. In a perfect inelastic collision, the initial and final kinetic energies are K_1 and K_2 respectively. Which of the following is true ?

(a) $K_1 < K_2$ (b) $K_1 > K_2$ (c) $K_1 = K_2$ (d) $K_2 = 0$ **(Rewa 1994)**

16. A particle of mass m, moving with velocity v along east strikes completely inelastically with a particle of same mass moving with velocity v along north. The new particle of mass $2m$ will move along north-east with the velocity :

(a) v (b) $v/2$ (c) $v/\sqrt{2}$ (d) none of these

(Rewa 1993)

17. In an elastic collision :

(a) Potential energy is conserved (b) Kinetic energy is conserved

(c) Total energy is conserved (d) Total linear momentum remains zero.

(Rewa 1992)

18. In a head on collision, a particle with initial speed v strikes a stationary particle of the same mass. Taking the collision to be elastic, their velocities after collision will be :

(a) 0, 0 (b) $v/2$, $v/2$ (c) 0, v (d) v, 0. **(Agra 2006)**

19. A roket consumes 100 kg of fuel per second exhansiting with a speed of 5 km/s. Thrust on the rocket is :

(a) -0.5×10^5N (b) -5×10^5N (c) -5×10^6N (d) -5×10^2N.

(Agra 2006)

ANSWERS

1. (b), **2.** (d) **3.** (b), **4.** (b), **5.** (a), **6.** (c), **7.** (a), **8.** (a), **9.** (d), **10.** (b), **11.** (b), **12.** (c), **13.** (d), **14.** (c), **15.** (b), **16.** (b), **17.** (a), **18.** (c), **19.** (b).

6

Dynamics of a Rigid Body

6·1. Rotation of a Rigid Body

A rigid body is a system of particles in which the relative positions of the particles remain fixed under the application of external forces. This means that a rigid body maintains its shape during motion.

Let a rigid body rotate about a fixed axis. By a fixed axis we mean an axis that is at rest in some inertial frame of reference and does not change direction relative to that frame. **Fig. 6.1** shows a rigid body which rotates about a fixed axis AB. Consider a particle P of the body. Let its position be P_0 at time $t = 0$. Draw a perpendicular P_0O to the axis of rotation. At time t, the partcle moves to the position P. If we denote $\angle P_0OP = \theta$, then it is said that the particle P has rotated through an angle θ. In fact, all other particles of the rigid body have also rotated through the same angle θ and we say that the whole body has rotated through an angle θ. This θ is said to be the **angular position** of the rigid body at time t. Hence the rotation of a rigid body is measured by angle θ of OP line from its initial position. Let θ be positive when measured anti-clockwise.

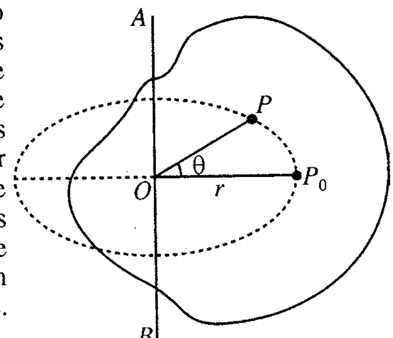


Fig. 6.1.

Angular velocity : If θ_1 is the angular position of P at time t_1 and θ_2 at time t_2, then the **average angular velocity** in the time interval $t_2 - t_1 = \Delta t$ is defined as

$$\omega_{av} = \frac{\theta_2 - \theta_1}{t_2 - t_1} = \frac{\Delta\theta}{\Delta t} \qquad ...(1)$$

where $\theta_2 - \theta_1 = \Delta\theta$.

The **instantaneous angular velocity** ω is the limit of ω_{av} as Δt approaches zero, *i.e.*, the derivative of θ with respect to t. Thus

$$\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} \qquad ...(2)$$

Normally when we say angular velocity, we mean the instantaneous angular velocity. As the body is rigid, all lines like OP rotate the same angle in the same time and hence the angular velocity ω is the characteristic of the entire body.

Angular acceleration : When there is a change in the angular velocity, we say that it has angular acceleration. If ω_1 and ω_2 are the instantaneous angular velocities at times t_1 and t_2 respectively, then the average angular acceleration is defined as

$$\alpha_{av} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t} \qquad ...(3)$$

The **instantaneous angular acceleration** is defined as the limiting value of $\Delta\omega/\Delta t$ as $\Delta t \to 0$. Thus

$$\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} \qquad ...(4)$$

Normally the term angular acceleration is used for instantaneous angular acceleration. The units of angular velocity and angular acceleration are radian/sec and radian/sec^2 respectively.

Rotation with constant angular velocity : If the body rotates through equal angles in equal time intervals, we say that it rotates with uniform angular velocity. In such a case,

$$\omega = d\theta/dt = \text{constant and } \theta = \omega t \qquad ...(5)$$

Rotation with constant angular acceleration : Let a rigid body rotate with constant angular acceleration α. From eq. (4), we have

$$\alpha = \frac{d\omega}{dt} \quad \text{or} \quad d\theta = \alpha\, dt \qquad ...(6)$$

If ω_0 and ω be the angular velocities at times $t = 0$ and $t = t$ respectively, then integrating (6) we obtain

$$\int_{\omega_0}^{\omega} d\omega = \alpha\int_0^t dt \quad \text{or } \omega - \omega_0 = \alpha t$$

Thus
$$\omega = \omega_0 + \alpha t \qquad ...(7)$$

Further from eq. (2)

$$\omega = \frac{d\theta}{dt} \quad \text{or} \quad d\theta = \omega dt = (\omega_0 + \alpha t)dt \qquad ...(8)$$

If θ_0 and θ be the angular positions at $t = 0$ and $t = t$ respectively, then integrating (8), we get

$$\int_{\theta_0}^{\theta} d\theta = \int_0^t (\omega_0 + \alpha t)\, dt \quad \text{or} \quad \theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2$$

Thus
$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2 \qquad ...(9)$$

If $\theta_0 = 0$ at $t = 0$, then

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2 \qquad ...(10)$$

Substituting the value of t [$= (\omega - \omega_0)/\alpha$] from eq. (7) in eq. (10), we get

$$\theta = \omega_0\left(\frac{\omega - \omega_0}{\alpha}\right) + \frac{1}{2}\alpha\left(\frac{\omega - \omega_0}{\alpha}\right)^2$$

or
$$2\alpha\theta = 2\omega_0(\omega - \omega_0) + (\omega - \omega_0)^2 = -\omega_0^2 + \omega^2$$

Thus
$$\omega^2 = \omega_0^2 + \alpha\theta \qquad ...(11)$$

Eqs. (7), (10) and (11) for rotational motion with constant angular acceleration are similar to equations of motion in straight line motion, *i.e.*,

$$v = v_0 + at,\; s = v_0 t + \frac{1}{2}at^2 \text{ and } v^2 = v_0^2 + 2as \qquad ...(12)$$

Relation between the linear motion of a particle of a rigid body and its rotation : Let a point P of the rigid body rotate about a fixed axis AB as in **fig. 6.1**. On the rotation of the body, this point P moves in a circle of radius r. If the body or the particle P rotates through an angle θ about the axis AB, then the linear distance moved by the particle along the circle is given by

$$s = r\theta \qquad ...(13)$$

The linear speed along the tangent is

$$v = \frac{ds}{dt} = r\frac{d\theta}{dt} = r\omega \qquad ...(14)$$

and the linear acceleration along the tangent is

$$a_t = \frac{dv}{dt} = r\frac{d\omega}{dt} = r\alpha \qquad ...(15)$$

For different particles of the rigid body, the radius r of their circles of rotation has different values, but the angular velocity ω and angular acceleration α are same for all the particles. Therefore, the linear speed and the tangential acceleration of various particles of the body are different.

The **radial (centripetal) acceleration** a_r of a particle moving with speed v in a circle of radius r is given by

$$a_r = \frac{v^2}{r} = \frac{r^2\omega^2}{r} = \omega^2 r$$

The resultant acceleration $\vec{a}$ of the particle P can be obtained by adding $\vec{a}_r = a_r\hat{r}$ and $\vec{a}_\theta = a_t\hat{\theta}$ vectorially, *i.e.*, $\vec{a} = a_r\hat{r} + a_t\hat{\theta}$ where $\hat{r}$ is unit vector along $\vec{r} = \vec{OP}$ and $\hat{\theta}$ is unit vector perpendicular to $\vec{r}$ **[fig. 6.2]**.

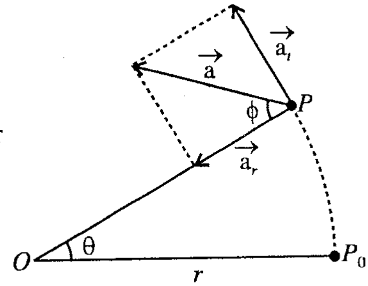


Fig. 6.2.

$$\text{Its magnitude } a = \sqrt{a_r^2 + a_t^2} = \sqrt{\left(\frac{v^2}{r}\right)^2 + \left(\frac{dv}{dt}\right)^2}$$

and
$$\tan\phi = \frac{a_t}{a_r} = \left(\frac{dv}{dt}\right)\bigg/\left(\frac{v^2}{r}\right).$$

6·2. Rotational Dynamics – Inertia Tensor

Angular momentum and moment of inertia : Let us consider a rigid body rotating with an angular velocity $\vec{\omega}$ about an axis AB, passing through a fixed point O (in an inertial frame). The direction of the angular velocity $\vec{\omega}$ will be along the axis of rotation. All the particles of the body will move in circular paths about the axis AB **[fig. 6.3]**.

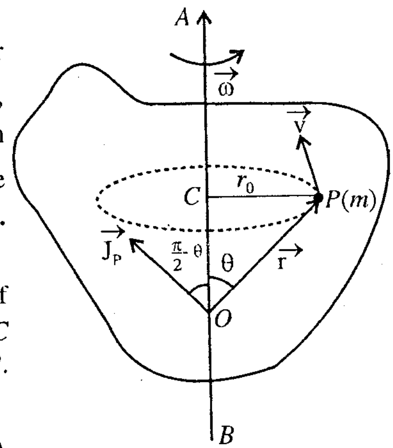


Fig. 6.3.

Let at any instant $\vec{r}$ be the position vector of a particle P of the body. If the length of the perpendicular drawn from P on AB is $PC = r_0$, then C will be the centre of the circle, described by the point P.

Speed of the particle $P = \omega r_0 = \omega r \sin PCO$

$\therefore$ Velocity of the particle P, $\vec{v} = \vec{\omega} \times \vec{r}$ ...(1)

The direction of this velocity $\vec{v}$ at any instant will be perpendicular to the position vector $\vec{r}$ and tangential to the circular path.

The angular momentum of the particle P about the point O is given by

$$\vec{J}_p = \vec{r} \times m\vec{v}$$

whose direction is perpendicular to $\vec{r}$ and $\vec{v}$. Evidently this direction will not coincide in general to the angular velocity vector $\vec{\omega}$.

Hence the angular momentum $\vec{J}$ of the entire body about the point O will be obtained by summing over $\vec{J}_p$ for all the particles of the body, *i.e.*,

$$\vec{J} = \Sigma m\,\vec{r} \times \vec{v} = \Sigma m\,\vec{r} \times (\vec{\omega} \times \vec{r})$$

whose direction, in general, will not be along $\vec{\omega}$.

Now,
$$\vec{J} = \Sigma m\{(\vec{r}\cdot\vec{r})\vec{\omega} - (\vec{r}\cdot\vec{\omega})\vec{r}\} \quad ...(2)$$

or
$$\vec{J} = \Sigma\{mr^2\vec{\omega} - (\omega r\cos\theta)\vec{r}\} \quad ...(3)$$

where $\theta = \angle POC$, is the angle between the vectors $\vec{r}$ and $\vec{\omega}$.

Hence $$\vec{J} = \Sigma m r^2[\vec{\omega} - (\omega \cos\theta / r^2)\,\vec{r}\,] \qquad ...(4)$$

From this equation, it is clear that for the entire body, in general, the angular momentum $\vec{J}$ and angular velocity $\vec{\omega}$ are are not in the same direction.*

We write $\vec{r}$, $\vec{\omega}$ and $\vec{J}$ in the cartesian components :

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}\,,\ \vec{\omega} = \omega_x\hat{i} + \omega_y\hat{j} + \omega_z\hat{k} \text{ and } \vec{J} = J_x\hat{i} + J_y\hat{j} + J_z\hat{k}.$$

From eq. (2), we have

$$\vec{J} = \Sigma m[(\omega_x\hat{i} + \omega_y\hat{j} + \omega_z\hat{k})\,r^2 - \{(x\hat{i} + y\hat{j} + z\hat{k})\cdot(\omega_x\hat{i} + \omega_y\hat{j} + \omega_z\hat{k})\}\,(x\hat{i} + y\hat{j} + z\hat{k})] \qquad ...(5)$$

where $\vec{\omega} = \omega_x\hat{i} + \omega_y\hat{j} + \omega_z\hat{k}$, $r = x\hat{i} + y\hat{j} + z\hat{k}$ and $r^2 = x^2 + y^2 + z^2$.

If J_x, J_y, J_z are the components of $\vec{J}$ along X, Y, Z axes respectively, then

$$J_x = \Sigma\, m\,[(r^2\omega_x - (x\omega_x + y\omega_y + z\omega_z)\,x] \qquad ...(6a)$$

or $$J_x = \omega_x\,\Sigma\, m\,(r^2 - x^2) - \omega_y\,\Sigma\, mxy - \omega_z\,\Sigma\, mxz$$

Similarly, $$J_y = -\,\omega_x\,\Sigma\, mxy + \omega_y\,\Sigma\, m\,(r^2 - y^2) - \omega_z\,\Sigma\, myz \qquad ...(6b)$$

and $$J_z = -\,\omega_x\,\Sigma\, mxz - \omega_y\,\Sigma\, myz + \omega_z\,\Sigma\, m\,(r^2 - z^2) \qquad ...(6c)$$

Eqs. (6) can be written as

$$J_x = I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z;\ J_y = I_{yx}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z;\ J_z = I_{zx}\omega_x + I_{zy}\omega_y + I_{zz}\omega_z \qquad ...(7)$$

where $$I_{xx} = \Sigma\, m\,(r^2 - x^2) = \Sigma\, m\,(y^2 + z^2);\ I_{yy} = \Sigma\, m\,(r^2 - y^2) = \Sigma\, m\,(x^2 + z^2);$$

$$I_{zz} = \Sigma\, m\,(r^2 - z^2) = \Sigma\, m\,(x^2 + y^2).$$

$$I_{xy} = -\,\Sigma\, mxy = I_{yx};\ I_{xz} = -\,\Sigma\, mxz = I_{zx};\ I_{yz} = -\,\Sigma\, myz = I_{zy}. \qquad ...(8)$$

The quantity I_{xx} is called **moment of inertia of the body about X-axis.** Similarly I_{yy} and I_{zz} define the moments of inertia of the body about Y- and Z- axes respectively. The quantities I_{xy}, I_{xz}, ... etc. which involve the sums of products of coordinates, are called **products of inertia.**

Any component of the angular momentum vector may be written as

$$J_\alpha = \sum_\beta I_{\alpha\beta}\omega_\beta \qquad ...(9)$$

where α, $\beta = x$, y, z.

Eq. (7) may be written in the matrix notation as

$$\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} \qquad ...(10a)$$

or $$\vec{J} = I\,\vec{\omega} \qquad ...(10b)$$

The nine elements I_{xx}, I_{xy},....., I_{zz} of the (3 × 3) matrix may be regarded as components of a single entity I. This entity I is called **inertia tensor**. Since $I_{xy} = I_{yx}$ etc., $\vec{I}$ is a **symmetric tensor.** If we denote x, y, z by x_1, x_2, x_3 respectively, then in general any element of the inertial tensor is given by

---

* For example, for a particle of mass m, the angular momentum will be given by $\vec{J}_p = mr^2[\vec{\omega} - (\omega\cos\theta/r^2)\,\vec{r}\,]$

Obiviously $\vec{J}_p$ and $\vec{\omega}$ are not in the same direction.

$$I_{\alpha\beta} = I_{\beta\alpha} = \sum_{i=1}^{N} m_i\left[\delta_{\alpha\beta}r_i^2 - x_{i\alpha}x_{i\beta}\right] \quad ...(11)$$

where α, β = 1, 2, 3.

In eqs. (7) and (8), the matrix elements are in the suitable form, if the rigid body is composed of discrete particles. In case of a continuous body, the summation sign is replaced by mass or volume integration. Thus the diagonal element I_{xx} is

$$I_{xx} = \int (r^2 - x^2)\,dm = \int \rho(\vec{r})(r^2 - x^2)\,dV = \int \rho(\vec{r})(y^2 + z^2)\,dV \quad ...(12a)$$

where dm is the mass of an infinitesimal volume element dV and $\rho(\vec{r})$ is the mass density at $\vec{r}$ within dV.

The off-diagonal element I_{xy} is

$$I_{xy} = -\int xy\,dm = -\int \rho(\vec{r})xy\,dV \quad ...(12b)$$

In general, the matrix element $I_{\alpha\beta}$ can be written as

$$I_{\alpha\beta} = \int \rho(\vec{r})(r^2\delta_{\alpha\beta} - x_\alpha x_\beta)\,dV \quad ...(13)$$

6·3. Principal Axes and Principal Moments of Inertia

From eq. (2) or eq. (10) [Art. 6.2], it is obvious that in general $\vec{J}$ and $\vec{\omega}$ are not in the same direction. However, it can be proved that whatever the shape of a body may be, there are three mutually perpendicular axes, such that the angular momentum of the body is along the angular velocity, when the body is rotated about that axis. The moments of inertia about these axes are denoted by I_1, I_2 and I_3 and are called **principal moments of inertia** with corresponding axes as **principal axes.** Thus if a body is rotating about a principal axis, then $\vec{J}$ and $\vec{\omega}$ are in the same direction *i.e.*,

$$\vec{J} = I\vec{\omega} \quad ...(1)$$

where I is scalar, being moment of inertia ($I = \Sigma m r_0^2$) about this axis.

In eq. (1) the angular momentum $\vec{J}$ and angular velocity $\vec{\omega}$ are in the same direction along the principal axis. Therefore each will have three components along X, Y, Z axes of an arbitrary coordinate system fixed in the body. Thus $\vec{J} = J_x\hat{\mathbf{i}} + J_y\hat{\mathbf{j}} + J_z\hat{\mathbf{k}} = I(\omega_x\hat{\mathbf{i}} + \omega_y\hat{\mathbf{j}} + \omega_z\hat{\mathbf{k}})$, where $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ are unit vectors along X-, Y-, Z- axes respectively. Hence

$$J_x = I\omega_x,\ J_y = I\omega_y,\ J_z = I\omega_z \quad ...(2)$$

Equating the expressions of J_x, J_y and J_z in eqs. (7) [Art. 6.2] and (2), we get

$$I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z = I\omega_x;\ I_{xy}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z = I\omega_y;\ I_{xz}\omega_x + I_{yz}\omega_y + I_{zz}\omega_z = I\omega_z$$

or

$$\begin{aligned} (I_{xx} - I)\,\omega_x + I_{xy}\,\omega_y + I_{xz}\,\omega_z &= 0 \\ I_{xy}\,\omega_x + (I_{yy} - I)\,\omega_y + I_{yz}\,\omega_z &= 0 \\ I_{xz}\,\omega_x + I_{yz}\,\omega_y + (I_{zz} - I)\,\omega_z &= 0 \end{aligned} \quad ...(3)$$

or

$$\begin{pmatrix} I_{xx}-I & I_{xy} & I_{xz} \\ I_{xy} & I_{yy}-I & I_{yz} \\ I_{xz} & I_{yz} & I_{zz}-I \end{pmatrix}\begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} = 0 \quad ...(4)$$

For these equations to have nontrivial solutions, the determinant of the coefficients must varnish

i.e.,

$$\begin{vmatrix} I_{xx}-I & I_{xy} & I_{xz} \\ I_{xy} & I_{yy}-I & I_{yz} \\ I_{xz} & I_{yz} & I_{zz}-I \end{vmatrix} = 0 \quad ...(5)$$

This is called **secular** or **characteristic equation** and its solutions are called **eigen values.** Eq. (5) is cubic in I, the three solutions $I = I_1, I_2, I_3$ are the **three principal moments of inertia.**

The direction of a principal axis is determined by substituting for I equal to one of the three roots, say $I = I_1$, in eq. (3) and determine the ratios $\omega_x : \omega_y : \omega_z$ or $\frac{\omega_y}{\omega_x} = \lambda_1, \frac{\omega_z}{\omega_x} = \lambda_2$ (say).

Thus
$$\vec{\omega}_1 = \omega_x \hat{\mathbf{i}} + \lambda_1 \omega_x \hat{\mathbf{j}} + \lambda_2 \omega_x \hat{\mathbf{k}}$$

$$\therefore \qquad \hat{\omega}_1 = \frac{\hat{\mathbf{i}} + \lambda_1 \hat{\mathbf{j}} + \lambda_2 \hat{\mathbf{k}}}{\sqrt{1+\lambda_1^2+\lambda_2^2}} \qquad ...(6)$$

This determines the direction of $\vec{\omega}_1$ or **the direction of the principal axis corresponding to I_1.** Similarly if we substitute I_2 or I_3, we may find the direction of the corresponding principal axis by using eq. (3). The magnitude of angular velocity is arbitrary and we are free to take its any value.

If is to be remembered that X, Y, Z is any general coordinate system fixed in the body and the coordinate system with principal axes as its axex X_0, Y_0, Z_0 has, in general, inclined axes with the X, Y, Z system. If $\hat{\mathbf{i}}_0, \hat{\mathbf{j}}_0, \hat{\mathbf{k}}_0$ be the unit vectors along principal axes, then the angular momentum $\vec{J}$ along an arbitrary direction can be expressed as

$$\vec{J} = J_1 \hat{\mathbf{i}}_0 + J_2 \hat{\mathbf{j}}_0 + J_3 \hat{\mathbf{k}}_0$$

$$\vec{J} = I_1 \omega_1 \hat{\mathbf{i}}_0 + I_2 \omega_2 \hat{\mathbf{j}}_0 + I_3 \omega_3 \hat{\mathbf{k}}_0 \qquad ...(7a)$$

or
$$\begin{pmatrix} J_1 \\ J_2 \\ J_3 \end{pmatrix} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} \qquad ...(7b)$$

where $\vec{\omega} = \omega_1 \hat{\mathbf{i}}_0 + \omega_2 \hat{\mathbf{j}}_0 + \omega_3 \hat{\mathbf{k}}_0$. Thus the inertia tensor I_0 is diagonal, when we consider the coordinate system fixed in the body along the principal axes, *i.e.*,

$$I_0 = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}.$$

6·4. Symmetric Bodies

In serveral cases, a body has some regular shape and the principal axes may be found by determining the axis of symmetry of the body. The axis of symmetry is one of the principal axis. For example, in case of a circular solid cylinder, its axis of cylindrical symmetry is one of the principal axes. Suppose the axis of cylindical symmetry is the Z-axis, then the contribution from (x, y, z) is cancelled by that from $(-x, -y, z)$ because $I_{xz} = -\Sigma mxz = 0$ and $I_{yz} = -\Sigma myz = 0$ and $J_z = I_{zz}\omega_z = I_3\omega_z$, Thus Z-axis (or axis of cylindrical symmetry) is the principal axis. Obviously the other two principal axes are in the X-Y plane with $I_1 = I_2$. In general, a rigid body is said to be symmetric, if two of its principal moments of inertia are equal. It may happen that all the three principal moments of inertia are equal, *i.e.*, $I_1 = I_2 = I_3$. This is the case for a cube or a sphere with origin at the centre. Such bodies may be called totally symmetric.

6·5. Moment of Inertia and Angular Momentum about an Axis

Now, we consider the magnitude (J_0) of the component of $\vec{J}$ along the axis of rotation, *i.e.*, $\vec{\omega}$. The angular momentum $\vec{J}$ [Art. 6.2, eq. (3)] of the rigid body is given by

$$\vec{J} = \Sigma[mr^2\vec{\omega} - (\omega r \cos\theta)\vec{r}] \qquad ...(1)$$

In eq. (1), we see that the first quantity in the right hand side is along $\vec{\omega}$ and in the second quantity, component of $\vec{r}$ along $\vec{\omega}$ will be $r \cos\theta$ **[fig. 6.3]**.

Hence the magnitude of the component (J_0) of the angular momentum $\vec{J}$ *along the axis of rotation* will be given by

$$J_0 = \Sigma m \{r^2\omega - (\omega r \cos\theta) r \cos\theta\} = \Sigma m r^2 \omega (1 - \cos^2\theta) = \Sigma m r^2 \omega \sin^2\theta = \omega \Sigma m r_0^2, \quad ...(2)$$

where $r_0 = r \sin\theta$ is the distance PC and ω is the same for all particles of the body.

In eq. (2) the sum $\Sigma m r_0^2$ is called the **moment of inertia of the body about the axis of rotation** and it is represented by the letter I. Thus, if m_1, m_2, etc. are the masses of various particles of the body and r_1, r_2, etc. are the corresponding perpendicular distances of these particles from the axis of rotation AB, then

$$I = \Sigma m r_0^2 = m_1 r_1^2 + m_2 r_2^2 + ... \text{ or } I = \Sigma m r^2 \quad ...(3)$$

Hence *the moment of intertia of a body about an axis is defined as the sum of the products of the masses of different particles, supposed to be constituting the body and the square of their respective distances from the axis of rotation.* In **C.G.S. system,** the unit of moment of inertia is **gm-cm²** and in **S.I. system** it is **kg-m²** and the dimensional formula is $[ML^2]$.

When the body is of continuous and, homogeneous structure, the summation is replaced by integration and then

$$I = \int r^2\, dm = \int r^2 \rho dV \quad ...(4)$$

where $dm = \rho dV$ represents the mass of an infinitesimally small element of volume dV of the body, situated at a distance r from the axis of rotation and ρ is the density.

Evidently, the moment of inertia of a body about an axis, depends upon the following :

(i) the mass of the body, and (ii) the distribution of its mass about the given axis.

It is to be noted that the mass (translational inertia) of a body remains constant, but the moment of inertia of a body can be varied because changing the distance of the mass from the axis of rotation alters the value of moment of inertia.

To have an illustration, let us compare the moment of inertia of solid cicular disc with that of a wheel, having the same mass and the same external radius. In the two cases the translational inertia (mass) is clearly the same. However, the wheel possesses considerably large moment of inertia than the disc, the axis of rotation for each has been taken to pass through their respective centres and perpendicular to their planes. The reason obviously lies in the fact that on an average, the particles, constituting the wheel, are situated at a greater distance from the axis of rotation.

It should be noted that if a part whose moment of inertia is I_1, then evidently the moment of inertia of the remaining part will be

$$I_2 = I - I_1. \quad ...(5)$$

where I, I_1 and I_2 are about the same axis.

Substituing for $\Sigma m r_0^2 = I$ in (2), the angular momentum about the axis of rotation (the component of angular momentum $\vec{J}$ (along the axis OA) is

$$\vec{J}_0 = I\vec{\omega} \text{ or } J_0 = I\omega \quad ...(6)$$

Hence **the angular momentum of a rigid body about on axis is equal to the product of moment of inertia about that axis and angular velocity.**

If we have a body with axial symmetry and the body is allowed to rotate about the axis of symmetry, then the component of $\vec{J}_p$ perpendicular to OA will be cancelled by an equal component of the angular momentum of another particle on the opposite side of the dotted circle. Therefore in such a case *the total angular momentum* $\vec{J}$ *of the body will be along the axis of rotation, i.e.,*

$$\vec{J} = I\vec{\omega} \quad \text{or} \quad J = I\omega.$$

For example, if a sphere rotates about its diameter, $\vec{J}$ and $\vec{\omega}$ are always parallel.

Torque about a point is defined as the rate of change of angular momentum about the same point, *i.e.*,

$$\vec{T} = \frac{d\vec{J}}{dt} \qquad \text{...(7)}$$

This is the **general equation** for a rotating body.

For a body rotating about its axis of symmetry

$$\vec{T} = \frac{d\vec{J}}{dt} = \frac{d}{dt}(I\vec{\omega}) = I\frac{d\vec{\omega}}{dt} \qquad \text{...(8)}$$

where $d\vec{\omega}/dt$ is the **angular acceleration.**

If the axis of rotation and axis of symmetry of the body are not one, $\vec{J}$ and $\vec{\omega}$ may not be along the same direction.

The torque about the axis of rotation is

$$T_0 = \frac{dJ_0}{dt} = \frac{d}{dt}(I\omega) \qquad \text{...(9)}$$

In case, the axis of rotation is fixed relative to the body, I will be constant. So that, the torque about the axis of rotation is

$$T_0 = I\,(d\omega/dt) \qquad \text{...(10)}$$

If $d\omega/dt = 1$, $T_0 = I$. Thus, *the moment of inertia of a body about an axis is equal to the torque, producing unit angular acceleration in it about that axis.*

In absence of external torque ($T_0 = 0$), *the angular momentum about the axis of rotation is conserved i.e.,*

$$I\omega = \text{constant} \qquad \text{...(11)}$$

Radius of gyration : Whatever may be the shape of a body, it is always possible to find a distance from the axis of rotation at which whole mass of the body could be assumed to be concentrated and even then its moment of inertia about that axis remains unaltered. For this, if the whole mass M of a body be supposed to be concentrated at a distance k from the axis of rotation, then clearly

$$I = Mk^2 = \Sigma mr^2$$

or

$$k = \sqrt{\frac{I}{M}} = \sqrt{\frac{\Sigma mr^2}{M}} \qquad \text{...(12)}$$

This quantity k is called **radius of gyration** of the body about the axis of rotation. Thus, *the radius of gyration of a body, rotating about a given axis, is equal to the distance from the axis of rotation, the square of which, when multiplied by the total mass of the body, will give the moment of inertia of the body about that axis.*

Physical significance of moment of inertia : Newton's first law of motion implies that a body maintains its state of rest or of uniform translatory motion, unless acted upon by some external force. This inertness or inability of a body to change its position of rest or of uniform motion by itself, is called **inertia.** Inertia is a fundamental property of matter and by virtue of which a body opposes an external force tending to change its state of rest or linear motion. Greater the mass of the body, greater is the force required to produce a given linear acceleration in it. Thus, the mass of a body is taken to be a measure of its translational intertia.

Similarly, the state of rotation of a rigid body about an axis remains unaltered, unless an external torque acts upon it. This means that a body, free to rotate about an axis, opposes the external torque, tending to change in its state of rest or of uniform rotation about that axis. This inertia of the body for rotational motion gives an idea about a physical quantity, called its **moment of inertia*** about the axis of rotation and is found to depend not only upon its mass but also on the actual distribution of the mass relative to the axis of rotation. Greater the moment of inertia of a body, greater is the torque required to produce a **desired angular acceleration in it**. Thus in case of rotatory motion, the quantities force and mass of translatory motion are replaced by torque and moment of inertia respectively. This is the physical significance of moment of inertia.

Based on this concept, the moment of inertia of a particle of mass m about an axis is defined as the product of its mass and square of the distance r from that axis, *i.e.*,

$$I = mr^2 \qquad ...(13)$$

If the masses of the constituting particles, of a rigid body are m_1, m_2,...... and r_1, r_2,....... are the corresponding distances from the axis of rotation, then the moment of inertia of the body about that axis is given by

$$I = \Sigma mr^2 = m_1 r_1^2 + m_2 r_2^2 + ... \qquad ...(14)$$

6·6. Rotational Kinetic Energy of a Rigid Body

Let a rigid body be rotating about an axis passing through a fixed point in the body with an angular velocity $\vec{\omega}$. The velocity $\vec{v}$ of the particle of mass m of the body is given by

$$\vec{v} = \vec{\omega} \times \vec{r}$$

The kinetic energy of this particle is given by

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\,\vec{v}\cdot\vec{v} \qquad ...(1)$$

Total kinetic energy of the entire body is given by

$$T = \Sigma\frac{1}{2}mv^2 = \Sigma\frac{1}{2}m\,\vec{v}\cdot\vec{v} = \frac{1}{2}\Sigma\,\vec{v}\cdot m\,\vec{v} = \frac{1}{2}\Sigma(\vec{\omega}\times\vec{r})\cdot m\,\vec{v}$$

$$= \frac{1}{2}\Sigma\,\vec{\omega}\cdot(\vec{r}\times m\,\vec{v}) \qquad [\because (\vec{A}\times\vec{B})\cdot\vec{C} = \vec{A}\cdot(\vec{B}\times\vec{C})]$$

$$= \frac{1}{2}\vec{\omega}\cdot\Sigma(\vec{r}\times m\,\vec{v}) \qquad \text{[because } \vec{\omega} \text{ is the same for all particles.]}$$

But $\vec{J} = \Sigma(\vec{r}\times m\,\vec{v})$ about the fixed point, hence

$$T = \frac{1}{2}\vec{\omega}\cdot\vec{J} \qquad ...(2)$$

This is the expression for *rotational kinetic energy* of a rigid body.

This is a scalar and can be written in the form

$$T = \frac{1}{2}\vec{\omega}\cdot\vec{J} = \frac{1}{2}\omega_x J_x + \frac{1}{2}\omega_y J_y + \frac{1}{2}\omega_z J_z$$

$$= \frac{1}{2}\omega_x(I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z) + \frac{1}{2}\omega_y(I_{xy}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z)$$

$$+ \frac{1}{2}\omega_z(I_{xz}\omega_x + I_{yz}\omega_y + I_{zz}\omega_z)$$

$$= \frac{1}{2}I_{xx}\omega_x^2 + \frac{1}{2}I_{yy}\omega_y^2 + \frac{1}{2}I_{zz}\omega_z^2 + I_{xy}\omega_x\omega_y + I_{xz}\omega_x\omega_z + I_{yz}\omega_y\omega_z \qquad ...(3)$$

* The inertia for rotation is given the name 'moment of inertia' on the analogy of the moment of the couple which this inertia (rotational) opposes.

The kinetic energy may be written in the compact form as

$$T = \frac{1}{2} \sum_{\alpha,\beta=1}^{3} I_{\alpha\beta}\omega_\alpha\omega_\beta \qquad ...(4)$$

In case, I_1, I_2, I_3 are the principal moments of inertia, the angular momentum $\vec{J}$ and angular velocity $\vec{\omega}$ are related by eq. (7) [Art. 6.3], *i.e.*,

$$J_1 = I_1\omega_1,\ J_2 = I_2\omega_2,\ J_3 = I_3\omega_3$$

where J_1, J_2, J_3 are the components of $\vec{J}$ along three principal axes.

Hence the kinetic energy in a coordinate system of principal axes is given by

$$T = \frac{1}{2}\vec{\omega}\cdot\vec{J} = \frac{1}{2}\omega_1 J_1 + \frac{1}{2}\omega_2 J_2 + \frac{1}{2}\omega_3 J_3 = \frac{1}{2}\sum_{\alpha=1}^{3} I_\alpha\omega_\alpha^2 \qquad ...(5)$$

We may express the kinetic energy of rotation in the usual form, used in elementary mechanics, as

$$T = \frac{1}{2} I\omega^2 \qquad ...(6)$$

where I is the moment of inertia about the axis of rotation. The proof is as follows :

We know that the component (J_0) of angular momentum $\vec{J}$ along the axis of rotation, *i.e.*, along $\vec{\omega}$ is given by

$$J_0 = I\omega$$

where $I = \Sigma m r_0^2$ is the moment of inertia about the axis of rotation.

Hence the moment of inertia about the axis of rotation is given by

$$K = \frac{1}{2}\vec{\omega}\cdot\vec{J} = \frac{1}{2}\omega J_0 \qquad (\because J\cos\theta = J_0)$$

i.e.,

$$K = \frac{1}{2} I\omega^2 \qquad ...(7)$$

6·7. Euler's Equations of Motion for a Rigid Body

If a rigid body is rotating under the action of a torque $\vec{T}$ with one point fixed, then the torque is expressed as

$$\vec{T} = \left[\frac{d\vec{J}}{dt}\right]_s \qquad ...(1)$$

where $\vec{J}$ is the angular momentum and its time derivative refers to the space coordinate system, (the axes of such a coordinate system are in fixed directions in space), represented by subscript s, because the equation holds in an inertia frame.

The body coordinate system (which is fixed in the rigid body) is rotating with an instantaneous angular velocity $\vec{\omega}$. The time derivatives of angular momentum $\vec{J}$ in the body coordinate and space coordinate systems are related as

$$\left[\frac{d\vec{J}}{dt}\right]_s = \left[\frac{d\vec{J}}{dt}\right]_b + \vec{\omega}\times\vec{J} \qquad ...(2)$$

Thus,

$$\vec{T} = \frac{d\vec{J}}{dt} + \vec{\omega}\times\vec{J} \qquad ...(3)$$

where we have dropped the body subscript because we shall represent the physical quantities of right hand side in the body coordinate system.

We choose principal axes for body set of axes. If I_1, I_2 and I_3 are the principal moments of inertia, then

$$\vec{J} = I_1\omega_1\hat{\mathbf{i}} + I_2\omega_2\hat{\mathbf{j}} + I_3\omega_3\hat{\mathbf{k}} \qquad ...(4)$$

where $\vec{\omega} = \omega_1\hat{\mathbf{i}} + \omega_2\hat{\mathbf{j}} + \omega_3\hat{\mathbf{k}}$ is the angular velocity with components ω_1, ω_2 and ω_3 along the principal axes.

As the principal moments of inertia and body base vectors $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ are constants in time with respect to the body coordinate system, we find that in the body coordinate system, using (4) the time derivative of $\vec{J}$ is

$$\frac{d\vec{J}}{dt} = I_1\dot{\omega}_1\hat{\mathbf{i}} + I_2\dot{\omega}_2\hat{\mathbf{j}} + I_3\dot{\omega}_3\hat{\mathbf{k}} \qquad ...(5)$$

Substituting in (3), we obtain

$$\vec{T} = I_1\dot{\omega}_1\hat{\mathbf{i}} + I_2\dot{\omega}_2\hat{\mathbf{j}} + I_3\dot{\omega}_3\hat{\mathbf{k}} + (\omega_1\hat{\mathbf{i}} + \omega_2\hat{\mathbf{j}} + \omega_3\hat{\mathbf{k}}) \times (I_1\omega_1\hat{\mathbf{i}} + I_2\omega_2\hat{\mathbf{j}} + I_3\omega_3\hat{\mathbf{k}}) \qquad ...(6)$$

Writing $\vec{T} = T_1\hat{\mathbf{i}} + T_2\hat{\mathbf{j}} + T_3\hat{\mathbf{k}}$, we can obtain the x, y, z components of the torque $\vec{T}$ as

$$T_1 = I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3 \qquad ...(7a)$$
$$T_2 = I_2\dot{\omega}_2 + (I_1 - I_3)\omega_3\omega_1 \qquad ...(7b)$$
$$T_3 = I_3\dot{\omega}_3 + (I_2 - I_1)\omega_1\omega_2 \qquad ...(7c)$$

Eqs. (7) are known as **Euler's equations** for the motion of a rigid body with one point fixed under the action of a torque.

In case a rigid body is rotating about a fixed axis, say principal Z-axis, then

$$\omega_1 = \omega_2 = 0 \text{ and } \omega_3 = \omega$$

Therefore, from eq. (7) we have the equations of motion as

$$T_1 = T_2 = 0$$

and

$$T_3 = I_3\dot{\omega} \quad or \quad T = I\frac{d\omega}{dt} \qquad ...(8)$$

where we have put $T_3 = T$ and $I_3 = I$ corresponding to Z-axis.

Instantaneous angular momentum about Z-axis is

$$J_3 = I_3\omega_3 \text{ or } J = I\omega \qquad ...(9)$$

and instantaneous rorational kinetic energy is

$$T = \frac{1}{2}\vec{\omega}\cdot\vec{J} = \frac{1}{2}I\omega^2 \qquad ...(10)$$

Torque-free motion of a rigid body : When a rigid body is not subjected to any net torque, the Euler's equations of motion of the body with one point fixed reduce to

$$I_1\dot{\omega}_1 = (I_2 - I_3)\,\omega_2\omega_3 \qquad ...(11a)$$

$$I_2\dot{\omega}_2 = (I_3 - I_1)\,\omega_3\omega_1 \qquad ...(11b)$$

$$I_3\dot{\omega}_3 = (I_1 - I_2)\,\omega_1\omega_2 \qquad ...(11c)$$

In case the body is not subjected to any net forces or torques, its centre of mass is either at rest or moves with uniform velocity. Obviously we may discuss the rotational motion of the rigid body in a reference system in which the centre of mass is stationary and choose the centre of mass as fixed point and origin for the principal axes in the body. In such a case, we obtain from eq. (11) two integrals of motion, describing the kinetic energy and angular momentum as constant in time.

If we multiply eqs. (11) by ω_1, ω_2, ω_3 respectively and then add, we obtain

$$I_1\omega_1\dot{\omega}_1 + I_2\omega_2\dot{\omega}_2 + I_3\omega_3\dot{\omega}_3 = (I_2 - I_3 + I_3 - I_1 + I_1 - I_2)\,\omega_1\omega_2\omega_3 = 0$$

or
$$\frac{d}{dt}\left(\frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 + \frac{1}{2}I_3\omega_3^2\right) = 0$$

or
$$\frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 + \frac{1}{2}I_3\omega_3^2 = \frac{1}{2}\vec{\omega}\cdot\vec{J} = \text{constant} \qquad ...(12)$$

which is the **principle of conservation of total rotational kinetic energy** in absence of external torque.

As
$$\vec{T} = \frac{d\vec{J}}{dt} = 0,\ \vec{J} = I_1\omega_1\hat{\mathbf{i}} + I_2\omega_2\hat{\mathbf{j}} + I_3\omega_3\hat{\mathbf{k}} = \text{constant} \qquad ...(13)$$

describes another constant of motion, representing **the principle of conservation of angular momentum.**

Ex. 1. *A grind stone starts from rest and has constant angular acceleration of* 3 *rad/sec*^{2}. *Find* (a) *the angular displacement,* (b) *the angular speed of the grindstone* 2 *sec. later.* **(Meerut 1988)**

Sol. (a) Angular displacement, $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

Here, $\omega_0 = 0$, $\alpha = 3$ rad/sec^2 and $t = 2$ sec.

$\therefore$
$$\theta = 0 + \frac{1}{2}\times 3 \times 2^2 = \mathbf{6\ radian}$$

(b)
$$\omega = \omega_0 + \alpha t$$

$\therefore$ At $t = 2$ sec., angular speed, $\omega = 0 + 3 \times 2 =$ **6 rad/sec.**

Ex. 2. *A* 12 *cm long weightless stick with masses* 5 *gm and* 10 *gm attached to its two ends is rotating about an axis passing through its centre of mass and inclined at* 30° *to its length. If it makes two revolutions per second, find* (a) *the angular velocity vector;* (b) *the linear velocity of each mass and its angular momentum vector about the centre of mass;* (c) *the total angular momentum vector of the system about the centre of mass;* (d) *the angular momentum about the axis of rotation and* (e) *the moment of inertia about this axis.*

Sol. If the distance of centre of mass from 5 gm mass is x, then

$$0.005\ x = 0.01\ (0.12 - x) \quad \text{or} \quad x = 0.08 \text{ m}$$

In **fig. 6.4** the centre of mass of the system has been represented by O. About the axis of rotation, A and B masses will describe circles of radius AC and BD respectively. Here, $AC = AO \sin 30° = 0.08 \times \frac{1}{2} = 0.04$ m and $BD = 0.04 \sin 30° = 0.02$ m.

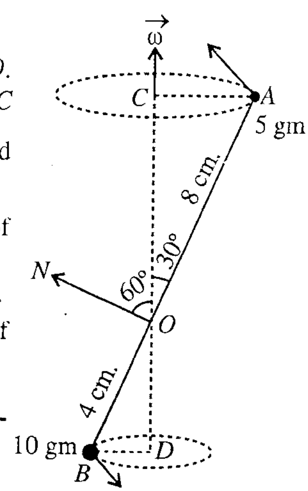


Fig. 6.4.

(a) Angular velocity $\omega = 2\pi n = 2\pi \times 2 =$ **12.5 rad/sec** along the axis of rotation.

(b) Linear velocity of 5 gm $v = \omega \times AC = 12.5 \times 0.04 =$ **0.5 m/sec.**

This velocity is perpendicular to OA, hence the angular momentum of mass A about $O = m\,\vec{r} \times \vec{v}$.

Its magnitude $= mrv = 0.05 \times OA \times 0.5 = 0.005 \times 0.08 \times 0.5 =$ $\mathbf{2 \times 10^{-4}}$ **kg-m^2/sec.**

Its direction is perpendicular to $\vec{v}$ and $\vec{OA}$, *i.e.*, it is along a line perpendicular to AB, making an angle 60° with the axis of rotation.

Similarly, linear velocity of 10 gm mass $= 12.5 \times 0.02 =$ **0.25 m/sec.**

Its angular momentum about O has the magnitude $0.01 \times 0.04 \times 0.25 = 10^{-4}$ kg-m^2/sec and direction perpendicular to AB, inclined at an angle 60° with the axis of rotation. At any instant both the angular momenta are along the same direction.

(c) Total angular momentum = $2 \times 10^{-4} + 10^{-4} =$ **3×10^{-4} kg-m²/sec along a direction perpendicular to** AB.

(d) The component of total angular momentum along the direction of axis of rotation is the angular momentum about this axis, *i.e.*,

$$= 3 \times 10^{-4} \cos 60° = 3 \times 10^{-4} \times \frac{1}{2} = \mathbf{1.5 \times 10^{-4}\ kg\text{-}m^2/sec.}$$

(e) Moment of inertia about the axis of rotation

$$= \Sigma mr^2 = 0.005 \times (0.04)^2 + 0.01 \times (0.02)^2 = \mathbf{1.2 \times 10^{-5}\ kg\text{-}m^2.}$$

The angular momentum about the axis of rotation (d) can also be expressed as $J = I\omega = 1.2 \times 10^{-5} \times 12.5 = 1.5 \times 10^{-4}$ kg-m²/sec the same result as in (d).

Ex. 2. *Two point masses, each of mass m, are connected by a massless rigid rod of length 2a, forming a dumb-bell. This dumb-bell is rotating with constant angular velocity ω about an axis which makes an angle θ with the rod. Find the magnitudes and the directions of the angular momentum and the torque, applied to the system. How are the results affected, if an identical dumb-bell is symmetrically fixed with the first dumb-bell and the system is rotated with an angular velocity ω about the same axis.*

Sol. Let the dumb-bell [**fig. 6.5**] rotate with an angular velocity $\vec{\omega}$ about the axis AOB in an inertial coordinate system. The angular momentum $\vec{J}$ of the two masses is

$$\vec{J} = \vec{J}_1 + \vec{J}_2 = m\vec{r}_1 \times (\vec{\omega} \times \vec{r}_1) + m\vec{r}_2 \times (\vec{\omega} \times \vec{r}_2)$$

As shown in **fig. 6.5**, $\vec{J}_1$ and $\vec{J}_2$ point in the same direction, but the total angular momentum $\vec{J}$ of the system is not along $\vec{\omega}$. The magnitude of $\vec{J}$ is

$$J = ma^2\omega \sin\theta + ma^2\omega \sin\theta = 2ma^2\omega \sin\theta = I\omega \sin\theta$$

where $I = 2ma^2$ is the moment of inertia of the dumb-bell about an axis perpendicular to the length of the connecting rod.

Since $\vec{J}$, which is continuously changing direction, is not constant and hence to maintain the motion, a torque $\vec{T}$ is acting on the system. The torque $\vec{T}$ is given by

$$\vec{T} = \frac{d\vec{J}}{dt} = \dot{\vec{J}}$$

where $\vec{T} = \dot{\vec{J}}$ is a vector in the direction in which the tip of the vector $\vec{J}$ is moving. Obviously

$$\dot{\vec{J}} = \vec{\omega} \times \vec{J}$$

in analogy to the relation, $\dot{\vec{r}} = \vec{\omega} \times \vec{r}$.

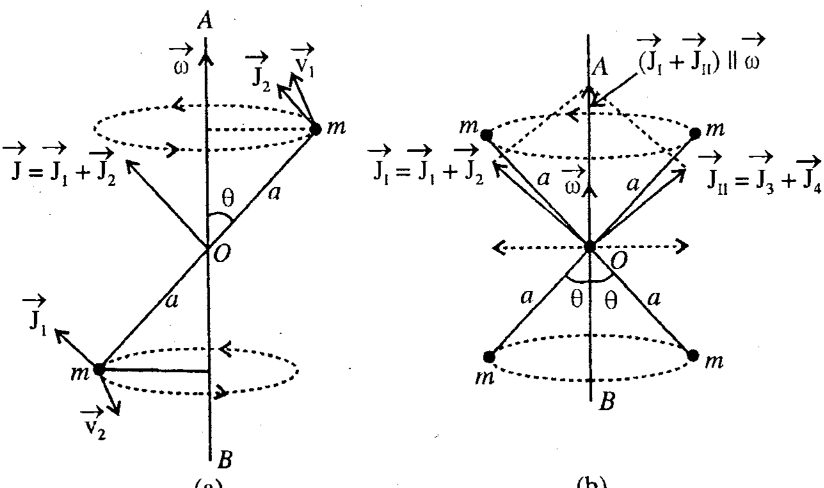


Fig. 6.5.

Hence the magnitude of the applied torque is

$$T = |\vec{J}| = \omega J \sin(90 - \theta) = \omega J \cos\theta = 2ma^2\omega^2 \sin\theta \cos\theta$$

and its instantaneous direction is perpendicular to the plane of ω and $\vec{J}$. **Fig. 6.5(b)** shows the case when the two identical dumb-bells, given in the problem, are moving symmetrically. In this case the perpendicular components of the angular momenta of the two dumb-bells will cancel and the components parallel to $\vec{\omega}$ add up so that $\vec{J}$ and $\vec{\omega}$ are in the same direction. If there are no resisting forces to the rotating system, the system, once rotated, will rotate indefinitely and in this case there is no need of the external torque.

Ex. 4. *Calculate the inertia tensor for the system of four point masses* 1 *gm,* 2 *gm,* 3 *gm and* 4 *gm, located at the points* (1, 0, 0), (1, 1, 0), (1, 1, 1) *and* (1, 1, –1) *cm.*

Sol. $$I_{xx} = \sum_{i=1}^{4} m_i(y_i^2+z_i^2) = 1\times 0 + 2\times 1 + 3\times 2 + 4\times 2 = 16 \text{ gm-cm}^2$$

Similarly, $$I_{xy} = \sum_i^4 m_i(x_i^2+z_i^2) = 17 \text{ gm-cm}^2 \text{ and } I_{zz} = \sum_i m_i(x_i^2+y_i^2) = 19 \text{ gm-cm}^2$$

Also $$I_{xy} = I_{yx} = -\sum_{i=1}^{4} m_i x_i y_i = -[0 + 2\times 1\times 1 + 3\times 1\times 1 + 4\times 1\times 1] = -9 \text{ gm-cm}^2$$

Similarly, $$I_{xz} = I_{zx} = -\sum_i m_i x_i z_i = 1 \text{ gm-cm}^2 \text{ and } I_{yz} = I_{zy} = -\sum_i m_i y_i z_i = 1 \text{ gm-cm}^2$$

Thus the inertia tensor $\boldsymbol{I}$ is $\boldsymbol{I} = \begin{pmatrix} \mathbf{16} & \mathbf{-9} & \mathbf{1} \\ \mathbf{-9} & \mathbf{17} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{19} \end{pmatrix}$.

Ex. 5. *Consider a rectangular parallelopiped of uniform density* ρ, *mass M with sides a, b and c. For origin O at one corner, find the moments and products of inertia of the parallelopiped by taking the coordinate axes along the edges. If a = b = c (case of a cube), determine the inertia tensor.*

Sol. $$I_{xx} = \int \rho(y^2+z^2)\,dV = \int_0^c\int_0^b\int_0^a \rho(y^2+z^2)\,dx\,dy\,dz$$

$$= \rho a\int_0^c\int_0^b (y^2+z^2)\,dy\,dz = \rho a\int_0^c\left[\frac{b^3}{3}+z^2 b\right]dz$$

$$= \rho a\left[\frac{b^3 c}{3}+\frac{c^3 b}{3}\right] = \frac{\rho abc}{3}(b^2+c^2) = \frac{M}{3}(b^2+c^2), \text{ where } M = \rho abc.$$

Similarly, $$I_{yy} = \frac{M}{3}(c^2+a^2),\ I_{zz} = \frac{M}{3}(a^2+c^2)$$

Also $$I_{xy} = -\int_0^c\int_0^b\int_0^a \rho\, x\, y\,dx\,dy\,dz = -\rho c\int_0^b\int_0^a xy\,dx\,dy = -\rho c\frac{a^2}{2}\frac{b^2}{2} = -\frac{M}{4}ab = I_{yx}$$

Similarly, $$I_{xz} = I_{zx} = -\tfrac{1}{4}Mac \text{ and } I_{yz} = I_{zy} = -\tfrac{1}{4}Mbc$$

For a cube $a = b = c$, then

$$I_{xx} = \tfrac{2}{3}Ma^2 = I_{yy} = I_{zz}$$

and $$I_{xy} = -\tfrac{1}{4}Ma^2 = I_{yx} = I_{xz} = I_{zx} = I_{yz} = I_{zy}$$

Therefore, $$I = \frac{Ma^2}{12}\begin{pmatrix} 8 & -3 & -3 \\ -3 & 8 & -3 \\ -3 & -3 & 8 \end{pmatrix}.$$

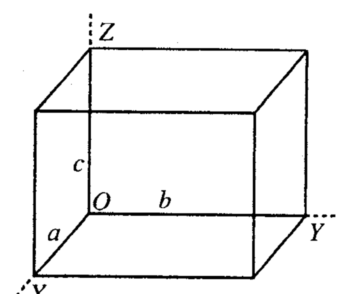


Fig. 6.6.

Ex. 6. *Consider a homogeneous cube of density ρ mass M and side a. Taking origin O at one corner and axes along the edges of the cube, determine the inertia tensor, the principal axes and their associated moments of inertia.*

Sol. : The inertia tensor I for the cube under consideration [**fig. 6.7**] can be evaluated as above, *i.e.*,

$$I = \frac{Ma^2}{12}\begin{pmatrix} 8 & -3 & -3 \\ -3 & 8 & -3 \\ -3 & -3 & 8 \end{pmatrix} = \begin{pmatrix} 8\lambda & -3\lambda & -3\lambda \\ -3\lambda & 8\lambda & -3\lambda \\ -3\lambda & -3\lambda & 8\lambda \end{pmatrix} \text{ with } \lambda = \frac{Ma^2}{12}.$$

To find the the principal moments of inertia, we solve the secular equation

$$\begin{pmatrix} 8\lambda-I & -3\lambda & -3\lambda \\ -3\lambda & 8\lambda-I & -3\lambda \\ -3\lambda & -3\lambda & 8\lambda-I \end{pmatrix} = 0$$

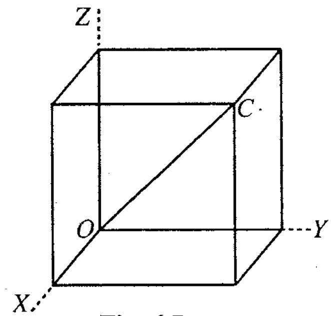


Fig. 6.7.

Subtract the second row from first, we obtain

$$\begin{pmatrix} 11\lambda-I & -(11\lambda-I) & 0 \\ -3\lambda & 8\lambda-I & -3\lambda \\ -3\lambda & -3\lambda & 8\lambda-I \end{pmatrix} = 0$$

or
$$(11\lambda-I)\begin{pmatrix} 1 & -1 & 0 \\ -3\lambda & 8\lambda-I & -3\lambda \\ -3\lambda & -3\lambda & 8\lambda-I \end{pmatrix} = 0$$

or $$(11\lambda - I)\,[(8\lambda - I)^2 - (3\lambda)^2 - 1\{(3\lambda)^2 + 3\lambda\,(8\lambda - I)\}] = 0$$

or $$(11\lambda - I)\,(22\lambda^2 - 13\lambda I + I^2) = 0$$

or $$(11\lambda - I)\,(11\lambda - 1)\,(2\lambda - I) = 0$$

Therefore, $I_1 = I_2 = 11\lambda$ and $I_3 = 2\lambda$ or $I_1 = I_2 = \frac{11}{12}Ma^2$ and $I_3 = \frac{1}{6}Ma^2$ which are the expressions for the principal moments of inertia.

Since the two roots are identical, $I_1 = I_2$, the principal axis associated with the root I_3 is the axis of symmetry. The moment of inertia tensor, when considered principal axes, is

$$I = \frac{Ma^2}{12}\begin{pmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Now we evaluate the directions of the principal axes associated with the three roots. First we evaluate the direction of the principal axis associated with the root I_3. For this, we substitute $I = I_3 = Ma^2/6 = 2\lambda$ in (4) [Art. 6.3]

$$(8\lambda - 2\lambda)\omega_x - 3\lambda\omega_y - 3\lambda\omega_z = 0$$
$$-3\lambda\omega_x + (8\lambda - 2\lambda)\omega_y - 3\lambda\omega_z = 0$$
$$-3\lambda\omega_x - 3\lambda\omega_y + (8\lambda - 2\lambda)\omega_z = 0$$

or $$2\omega_x - \omega_y - \omega_z = 0,\ -\omega_x + 2\omega_y - 2\omega_z = 0,\ -\omega_x - \omega_y + 2\omega_z = 0$$

From which, we obtain

$$\frac{\omega_x}{\omega_y} = 1,\ \frac{\omega_y}{\omega_z} = 1,\ \frac{\omega_z}{\omega_x} = 1,\ i.e.,\ \omega_x : \omega_y : \omega_z = 1 : 1 : 1$$

If we take unit vector $\hat{\omega}$ along ω corresponding to $I_3 = Ma^2/6$, then

$$\hat{\omega} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$$

which is obviously along the diagonal OC of the cube [$\vec{r} = a(\hat{i} + \hat{j} + \hat{k})$]. As $I_1 = I_2$, axes associated with I_1 and I_2 have any mutually perpendicular directions in a plane perpendicular to this diagonal.

EXERCISE 6·1

1. A heavy fly wheel rotating on its axis is slowing down because of friction in its bearings. Starting with initial angular velocity ω_0, at the end of first minute its velocity is $0.9\ \omega_0$. Assuming constant frictional forces, find its angular velocity at the end of second minute. **(Meerut 1983)**
[**Ans.** $0.8\ \omega_0$]

2. Two particles of masses 2 gm and 4 gm are attached to the two ends of a 6 cm long stick of negligible weight. Calculate the moment of inertia of the system about an axis passing through its centre of mass and perpendicular to the stick. If the stick makes two revolutions per second about this axis, find the linear velocity vector for each particle and hence find the magnitude and direction of the total angular momentum about the centre of mass. Is it in the same direction as that of the angular velocity ?
[**Ans.** 4.8×10^{-6} kg-m^2, 0.25 and 0.5 m/sec; 6×10^{-5} kg-m^2/sec in the direction of angular velocity.]

3. (a) Consider a homogeneous cube of density ρ, mass M and side a. For origin O at one corner and axes along the edges of the cube, determine the inertia tensor..

$$\left[\textbf{Ans. } I = \frac{Ma^2}{12}\begin{pmatrix} 8 & -3 & -3 \\ -3 & 8 & -1 \\ -3 & -3 & 8 \end{pmatrix}\right]$$

(b) Determine the inertia tensor, if the origin O is the centre of mass in the above problem and the axes are parallel to the edges. Could you correlate the inertia tensors in the two cases ?

$$\left[\textbf{Ans. } I = \frac{Ma^2}{6}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix};\ \text{yes.}\right]$$

[**Hint :** $I_{xx} = I^c_{xx} + M\,(Y^2 + Z^2) = \dfrac{Ma^2}{6} + M\left[\left(\dfrac{a}{2}\right)^2 + \left(\dfrac{a}{2}\right)^2\right] = \dfrac{Ma^2}{6} + \dfrac{Ma^2}{2} = \dfrac{2}{3}Ma^2$

and $I_{xy} = I^c_{xy} - MXY = \dfrac{Ma^2}{4}$.]

4. Prove that the principal moments of inertia for a system consisting of two particles of masses m_1 and m_2 connected by a massless rigid rod of length a are $I_1 = I_2 = m_1 m_2 a^2/(m_1 + m_2)$ and $I_3 = 0$.

5. A rigid body consists of three particles of masses 2, 1, 4 gm located at (1, –1, 1), (2, 0, 2), (–1, 1, 0) cm respectively. Find the angular momentum of the body if it is rotated about the origin with angular velocity $\vec{\omega} = 3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$. Find also the principal moments of inertia and directions of the principal axes of the system.

[**Ans.** $\vec{J} = -6\hat{\mathbf{j}} + 42\hat{\mathbf{k}}$; $I_1 = 18$, $I_2 = 13 - \sqrt{73}$, $I_3 = 13 + \sqrt{73}$ gm. cm^2; $\hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\frac{1}{6}(1+\sqrt{73})\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\frac{1}{6}(1-\sqrt{73})\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$.]

6. (a) Find the moments of inertia and products of inertia of a uniform square plate of side a and mass M about the X, Y, Z axes, shown as in **fig.6.8.**

(b) In the above problem, find also the principal moments of inertia and the directions of the principal axes for the plate.

[**Ans.** (a) $I_{xx} = I_{yy} = \frac{1}{3}Ma^2$, $I_{zz} = \frac{2}{3}Ma^2$, $I_{xy} = I_{yx} = -\frac{1}{4}Ma^2$,

$I_{xz} = I_{zx} = I_{yz} = I_{zy} = 0$.

(b) $I_1 = \frac{1}{12}Ma^2$, $I_2 = \frac{7}{2}Ma^2$, $I_3 = \frac{2}{3}Ma^2$; $\hat{\mathbf{i}} + \hat{\mathbf{j}}, \hat{\mathbf{i}} - \hat{\mathbf{j}}, \hat{\mathbf{k}}$.]

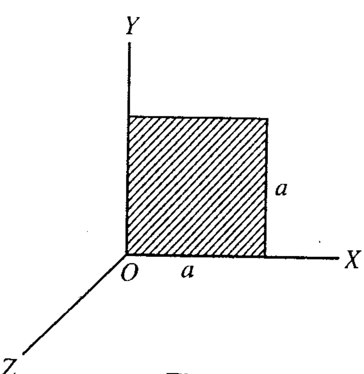


Fig. 6.8.

7. Find the principal moments of inertia at the centre of a uniform rectagular plate of sides a and b.
[**Ans.** $I_1 = \frac{1}{12}Ma^2$, $I_2 = \frac{1}{12}Mb^2$, $I_3 = \frac{1}{12}M(a^2+b^2)$.]

8. Find the moments of inertia and products of inertia of the uniform

solid sphere $x^2 + y^2 + z^2 = a^2$ in the first octant.

[**Ans.** $I_{xx} = I_{yy} = I_{zz} = \frac{2}{5}Ma^2$, $I_{xy} = I_{yz} = I_{zx} = -2Ma^2/5\pi$]

9. Determine the principal moments of inertia of a uniform cylinder of radius a and height h.

[**Ans.** $I_1 = I_2 = \frac{1}{12}M(3a^2+h^2)$, $I_3 = \frac{1}{2}Ma^2$.]

10. If T be the kinetic energy, $\vec{G}$ be the external torque about the instantaneous axis of rotation and $\vec{\omega}$ the angular velocity, then prove that $\frac{dT}{dt} = \vec{G}\cdot\vec{\omega}$.

[**Hint :** According to Euler's equations,

$$G_1 = I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3 \text{ or } I_1\dot{\omega}_1 = G_1 + (I_2 - I_3)\omega_2\omega_3 \text{ etc.}$$

$$T = \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 + \frac{1}{2}I_3\omega_3^2$$

$$\therefore \quad \frac{dT}{dt} = \omega_1(I_1\dot{\omega}_1) + \omega_2(I_2\dot{\omega}_2) + \omega_3(I_3\dot{\omega}_3)$$

$$= \omega_1[G_1 + (I_2 - I_3)\omega_2\omega_3 + \omega_2[G_2 + (I_3 - I_1)\omega_2\omega_3] + \omega_3[G_3 + (I_1 - I_2)\omega_2\omega_3]$$

$$= \omega_1 G_1 + \omega_2 G_2 + \omega_3 G_3 = \vec{G}\cdot\vec{\omega}.]$$

11. From Euler's equations of motion for a rigid body, having no external torque about a fixed point, prove that

$$T = \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 + \frac{1}{2}I_3\omega_3^2 = \text{constant and } \vec{J} = I_1\omega_1\hat{\mathbf{i}} + I_3\omega_3\hat{\mathbf{j}} + I_3\omega_3\hat{\mathbf{k}} = \text{constant}$$

where the terms have standard meaning.

6.8. Determination of Moment of Inertia about an Axis

In the case of a continuous, homogeneous body of a difinite geometrical shape, the following procedure is adopted for the calculation of mement of inertia of a body about a given axis :

(1) First, an expression is obtained for the moment of inertia of an infinitesimal mass of the body about the given axis. This expression is equal to the product of infinitesimal mass dm and square of its distance x from the axis.

(2) Then, the said expression is integrated over the limits, depending upon the shape of the body, concerned.

Thus, the moment of inertia of a body about a given axis, in general, is given by

$$I = \int x^2\, dm \qquad \text{...(1)}$$

where the limits of integration are according to the shape of the body.

(3) Whenever necessary, full use of theorems of perpendicular and parallel axes is made. These theorems are discussed below :

(i) Theorem of parallel axes. According to this theorem, *the moment of inertia of a body about any axis is equal to the sum of moment of inertia about a parallel axis through its centre of mass and the product of the mass of the body and the square of the distance between two axes, i.e.,*

$$I = I_{cm} + Ma^2 \qquad \text{...(2)}$$

where I = M.I. of the body about any axis, I_{cm} = M.I. of the body about a parallel axis passing through its centre of mass, M = mass of the body, and a = the distance between the two axes.

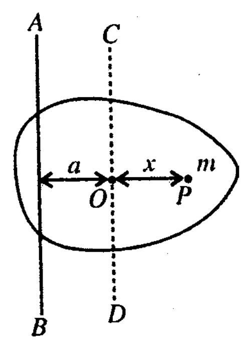


Fig. 6.9.

Proof : Let AB be the axis, in the plane of the paper, about which we have to determine the moment of inertia of the body. In **fig. 6.9**, CD is an axis parallel to AB and at a distance a from it. Axis CD passes through the centre of mass O of the body.

Let us consider a particle of the body of mass m at P, at a distance x from CD. Now,

$$\text{moment of inertia of } m \text{ about } AB = m(x+a)^2$$

$\therefore$ Moment of inertia of the whole body about AB axis, $I = \Sigma m\,(x+a)^2$

or
$$I = \Sigma mx^2 + \Sigma ma^2 + 2\Sigma mxa$$

If I_{cm} is the moment of inertia of the body about the axis CD, passing through its centre of mass, then $I_{cm} = \Sigma mx^2$. Also, $\Sigma ma^2 = a^2\Sigma m = Ma^2$, because the distance a between the two axes is constant and $\Sigma m = M$ is the mass of the body.

$\therefore$
$$I = I_{cm} + Ma^2 + 2a\Sigma mx$$

But Σmx is the sum of moments of all the particles about the axis CD, which passes through its centre of mass. So the algebraic sum of all moments about it is zero.*

Hence
$$I = I_{cm} + Ma^2.$$

(ii) Theorem of perpendicular axes. This theorem states that *the moment of inertia of a plane lamina about an axis perpendicular to its plane is equal to the sum of moments of inertia of the lamina about two axes at right angles to each other, in its own plane, and intersecting each other at the point where the perpendicular axis passes through it.*

If I_x and I_y be the moments of inertia of a plane lamina about OX and OY axes, which lie in the plane of the lamina and mutually perpendicular to each other, intersecting at the point O, then the moment of inertia I about an axis, which is passing through O and perpendicular to the plane of the lamina [**fig. 6.10**] is given by

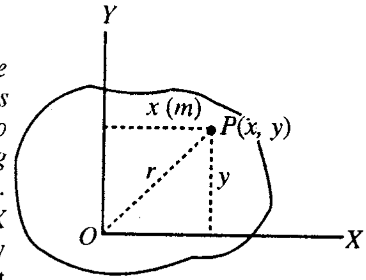


Fig. 6.10.

$$I = I_x + I_y \qquad \text{...(3)}$$

Proof : Let us consider a particle P of the lamina of mass m. The distance of the particle P from OX and OY axes are y and x respectively and the distance of P from O is r. Then, obviously,

$$I = \Sigma mr^2.$$

But $I_x = \Sigma my^2$ and $I_y = \Sigma mx^2$, $\therefore\ I_x + I_y = \Sigma my^2 + \Sigma mx^2 = \Sigma m(x^2+y^2) = \Sigma mr^2 = I.$

6·9. Calculation of Moment of Inertia of Some Rigid Bodies of Regular Shape

(1) Moment of inertia of a thin uniform rod : (i) *About an axis passing through the centre of mass and perpendicular to its length.* Let PQ [**fig. 6.11**] be a thin uniform rod of length l and mass M and AB the axis of rotation passing through the centre of mass O of the rod and perpendicular to its length.

Therefore, mass per unit length $= M/l$.

Consider a small element of thickness dx at a distance x from O. Hence the mass of the element is $(M/l)\,dx$ and its $M.I.$ about the axis through O is

$$= \frac{M}{l}dx\cdot x^2.$$

The moment of inertia of the whole srod about AB axis will be the sum of the moments of inertia of all such elements lying between $x = -l/2$ (at P) and $x = l/2$ (at Q) *i.e.*,

$$I = \int_{-l/2}^{l/2}\frac{M}{l}x^2\cdot dx = \frac{2M}{l}\int_0^{l/2}x^2\cdot dx = \frac{2M}{l}\left[\frac{x^2}{3}\right]_0^{l/2}$$
$$= \left[\frac{2M}{l}\frac{(l/2)^2}{3} - 0\right]$$

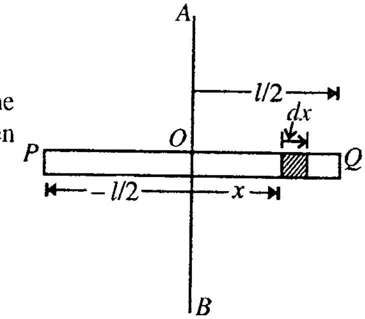


Fig. 6.11.

* For centre of mass, $X = \Sigma mx/M$. Here the distance x have been measured from the line passing through the centre of mass. So $X = 0$. This means $\Sigma mx = 0$.

or $$I = \frac{Ml^2}{12}. \quad ...(1)$$

(ii) *About an axis passing through one end of the rod and perpendicular to its length.* In this case the axis AB passes through the end P of the rod **[fig. 6.11]**.

If I be the moment of inertia of the rod about the axis AB, then according to the theorem of parallel axes

$$I = I_{cm} + Ma^2$$

Here $$I_{cm} = Ml^2/12 \text{ and } a = l/2,$$

$$\therefore \quad I = \frac{Ml^2}{12} + \frac{Ml^2}{4} \text{ or } I = \frac{Ml^2}{3} \quad ...(2)$$

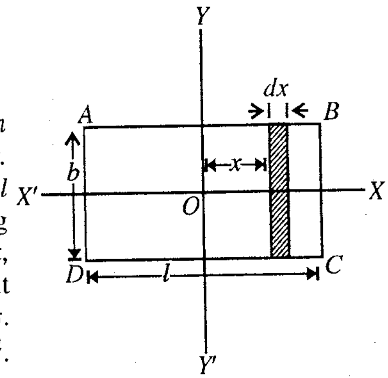


Fig. 6.12.

(2) Moment of inertia of a rectangular lamina : (i) *About an axis perpendicular to its plane and pasing through its centre of mass.* **Fig. 6.12** represents a rectangular lamina $ABCD$ of mass M, length l and breadth b. Let YY' be an axis parallel to its side AD and passing through its centre of gravity G. Consider a small strip of width dx, distant x from YY' and also parallel to it. Let σ be the mass per unit area of the lamina, then clearly the mass of the strip is equal to $b \cdot dx \cdot \sigma$.

Therefore, moment of inertia of the strip about $YY' = \sigma b \cdot dx \cdot x^2$.

In order to calculate the moment of inertia of the whole lamina about YY' the above expression is integrated between the limits $x = -l/2$ and $x = + l/2$.

Thus, $$I_y = \int_{-l/2}^{l/2} \sigma b x^2 \, dx = 2\sigma b \int_0^{l/2} x^2 \, dx = 2\sigma b \left[\frac{x^2}{3}\right]_0^{l/2} = 2\sigma b \cdot l^3/24 = Ml^2/12 \; [\because M = \sigma bl] \quad ...(3)$$

Similarly, it can be proved that the moment of inertia of the lamina about the axis XX', an axis parallel to AB or DC and passing through its centre of mass O, will be given by

$$I_x = Mb^2/12. \quad ...(4)$$

Now, according to the theorem of perpendicular axes, the moment of inertia of the rectangular lamina about an axis perpendicular to its plane and passing through its centre of mass will be expressed as

$$I = I_x + I_y = \frac{Mb^2}{12} + \frac{Ml^2}{12} = M\left(\frac{l^2 + b^2}{12}\right) \quad ...(5)$$

(3) Moment of inertia of a solid uniform bar of rectangular cross-section : (i) *About an axis perpendicular to its length and passing through its centre of mass.* Consider a solid uniform bar of length l, breadth b and thickness d, having mass M. Imagine that the bar is composed of a number of thin rectangular laminas, each of them has been placed one above the other as shown in the **fig. 6.13.** Let the mass of the each lamina be m. The length and breadth of each lamina are l and b respectively. The moment of inertia of a lamina about the axis YY' perpendicular to its plane and passing through its centre will be

$$m\frac{l^2 + b^2}{12}.$$

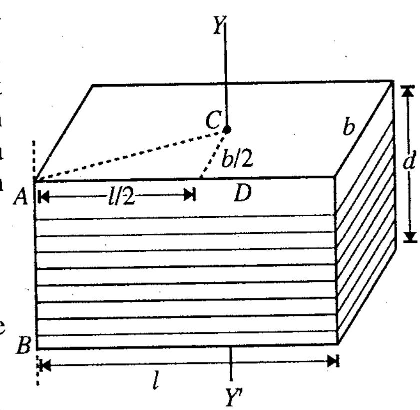


Fig. 6.13.

Therefore, the moment of inertia of all the laminas, *i.e.*, of the bar the desired axis ($YY'2$) is

$$I_y = \Sigma m . \frac{l^2 + b^2}{12} = M\frac{l^2 + b^2}{12} \quad (\because \Sigma m = M) \; ...(6)$$

(ii) *About an axis perpendicular to the length of bar and passing through one of its corners.* Applying the theorem of parallel axes, the moment of inertia of the bar about an axis AB through one of its corners and perpendicular to its length is given by

$$I = I_{cm} + Ma^2$$

Here $I_{cm} = I_y = M \cdot (l^2 + b^2)/12$ and $a^2 = AC^2 = AD^2 = (l/2)^2 + (b/l)^2 = (l^2 + b^2)/4$

$$\therefore \quad I = M\frac{l^2+b^2}{12} + M\frac{l^2+b^2}{4} \quad I = M\left[\frac{l^2+b^2+3l^2+3b^2}{12}\right] \text{ or } I = M\left[\frac{l^2+b^2}{3}\right] \quad ...(7)$$

(4) Moment of inertia of a hoop or thin circular ring : (i) *About an axis through its centre and perpendicular to its plane.* Let M be the mass and R the radius of the circular ring or hoop.

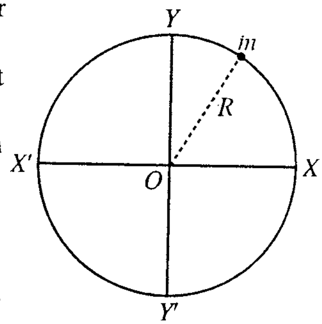


Fig. 6.14.

Consider a particle of mass m of the ring. Its moment of inertia about an axis through O and perpendicular to its plane is mR^2.

Therefore, the moment of inertia I of the entire ring about the given axis will be given by

$$I = \Sigma mR^2 = R^2\Sigma m \quad \text{or} \quad I = MR^2 \qquad [\because \Sigma m = M] \quad ...(8)$$

(ii) *About a diameter.* Obviously, the moment of inertia of a hoop is the same about any diameter. Let I be the moment of inertia about a diameter XX' then it will be the moment of inertia about a perpendicular diameter YY' (**fig. 6.14**). By the application of the theorem of perpendicular axes

$$I_z = I_x + I_y$$

Here $I_z = MR^2$ and $I_x = I_y = I \therefore MR^2 = I + I$ or $I = MR^2/2$. ...(9)

(5) Moment of inertia of a circular disc : (i) *About an axis through its centre and perpendicular to its plane.* Let M be the mass of the disc and R, its radius [**fig. 6.15**], so that its mass per unit area $= M/\pi R^2$.

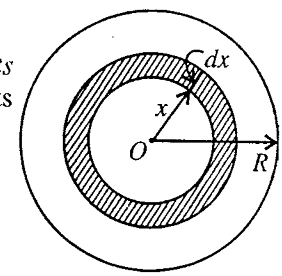


Fig. 6.15.

Consider a concentric elementary ring of radius x and thickness dx. Then,

Area of the ring = circumference × thickness = $2\pi x \cdot dx$

$$\text{Mass of the ring} = \frac{M}{\pi R^2} 2\pi x\, dx = \frac{2M \cdot x\, dx}{R^2}$$

Moment of inertia of this ring about an axis through O and perpendicular to its plane (*i.e.*, palne of the disc)

$$= \frac{2M \cdot x\, dx}{R^2} . x^2 = \frac{2Mx^3}{R^2} \cdot dx$$

Therefore, the moment of inertia of the whole disc about this axis is

$$I = \frac{2M}{R^2}\int_0^R x^3\, dx = \frac{2M}{R^2}\left[\frac{x^4}{4}\right]_0^R = \frac{2M}{R^2} \times \frac{R^4}{4}$$

$$I = \frac{MR^2}{2} \quad ...(10)$$

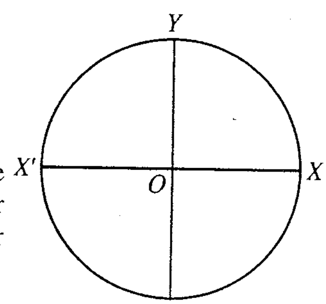


Fig. 6.16.

(ii) *About a diamter.* Clearly the moment of inertia of the disc is the same about any diameter. Let I be the moment of inertia about any diameter XX', then it will be the moment of inertia about a perpendicular diameter YY'. By the application of the theorem of perpendicular axes,

$$I_z = I_x + I_y$$

Here $I_z = MR^2/2$ and $I_x = I_y = I$, $\therefore \frac{MR^2}{2} = I + I$ or $I = \frac{MR^2}{4}$ ...(11)

Note : M.I. of an angular disc of inner and outer radii R_1 and R_2 respectively about an axis passing through its centre and perpendicular to its plane will be given by

$$= \int_{R_1}^{R_2} \frac{M}{\pi(R_2^2 - R_1^2)} 2\pi x dx \cdot x^2$$

$$= \frac{2M}{(R_2^2 - R_1^2)}\left[\frac{x^4}{4}\right]_{R_1}^{R_2} = \frac{2M}{(R_2^2 - R_1^2)} \cdot \frac{(R_2^4 - R_1^4)}{4}$$

$$= \frac{M}{2}(R_2^2 + R_1^2) \qquad ...(12)$$

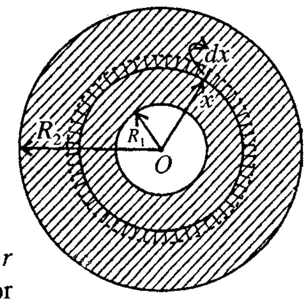


Fig. 6.17.

(6) Moment of inertia of a solid cylinder : (i) *About its own axis or axis of cylindrical symmetry.* A solid cylinder is just a thick circular disc, or it can be supposed to be composed of a large number of circular discs placed one over the other. Let the mass of each such disc be m and its radius equal to the radius R of the cylinder. Then, the moment of inertia of each disc about the axis of cylinder is $mR^2/2$. Thus the moment of inertia of the cylinder about its own axis is

$$I = \Sigma m \frac{R^2}{2} = \frac{R^2}{2} \Sigma m = \frac{MR^2}{2} \qquad ...(13)$$

[$\because$ Σm = M, the mass of the cylinder.]

(ii) *About on axis perpendicular to its geometrical axis and passing through its centre of mass.* Let M be the mass of the cylinder, R its radius and l its length, then its mass per unit length = M/l.

Let YY' be the axis of rotation through its centre and perpendicular to its own (geometrical) axis XX'. Consider a thin disc D of thickness dx at a distance x from the centre of the cylinder O [**fig. 6.18**].

Mass of the disc = $(M/l)dx$

Moment of inertia of this disc about any diameter = $(M/l)dx \times R^2/4$.

By the application of the principle of the parallel axes,

Moment of inertia of this disc about a parallel axis YY' is $\frac{M}{l} \cdot dx \cdot \frac{R^2}{4} + \frac{M}{l} \cdot dx \cdot x^2$.

The moment of inertia of the cylinder about the axis YY' is obtained by integrating the above expression between the limits $x = -l/2$ and $x = l/2$, *i.e.*,

$$I = \frac{M}{l}\int_{-l/2}^{l/2}\left[\frac{R^2}{4} + x^2\right]dx = \frac{2M}{l}\left[\frac{R^2 x}{4} + \frac{x^3}{3}\right]_0^{l/2}$$

$$= \frac{M}{l}\left[\frac{R^2}{4} \cdot \frac{l}{2} + \frac{1}{3}\left(\frac{l}{2}\right)^3\right]$$

$$= \frac{M}{l}\left[\frac{R^2 l}{4} + \frac{l^3}{12}\right] = M\left(\frac{R^2}{4} + \frac{l^2}{12}\right) \qquad ...(14)$$

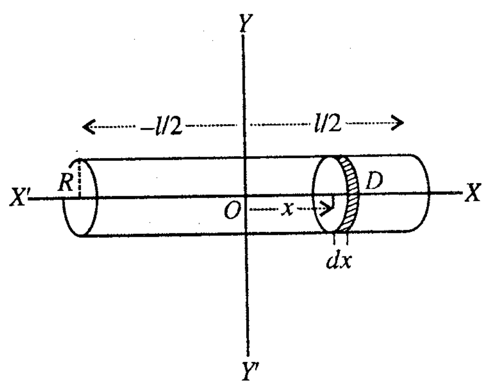


Fig. 6.18.

Note : M.I. of a hollow cylinder of inner and outer radii R_1 and R_2 respectively *about the axis of cylindrical symmerty* will be given by

$$= \frac{M}{l} dx \frac{R_1^2 + R_2^2}{4} + \frac{M}{l} dx \cdot x^2$$

$$\therefore \qquad I = \frac{M}{l}\int_{-l/2}^{l/2}\left[\frac{R_1^2 + R_2^2}{4} + x^2\right]dx$$

$$= \frac{2M}{l}\left[\frac{R_1^2+R_2^2}{4}x+\frac{x^3}{3}\right]_0^{l/2}$$

or $$I = M\left[\frac{R_1^2+R_2^2}{4}+\frac{l^2}{12}\right] \quad ...(16)$$

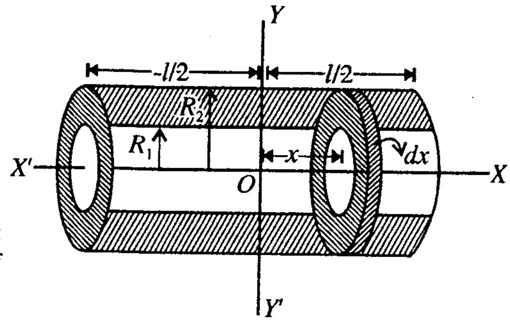


Fig. 6.19.

(7) Moment of inertia of a sphere : (i) *About a diameter.* Let **fig. 6.20** represent the section of a sphere of mass M and radius R, through its centre O. Let XX' be the diameter about which the moment of inertia is required to be found. If ρ be the density of the sphere, then its mass

$$M = \frac{4}{3}\pi R^3\rho.$$

Let us consider a very thin circular disc of thickness dx at a distance x from the centre O. Then radius of the thin disc

$$y = \sqrt{R^2-x^2}$$

Volume of this disc $= \pi(R^2 - x^2)\,dx$ and mass $= \pi(R^2 - x^2)dx\rho$

Moment of inertia of the disc about XX'

$$= \frac{\text{its mass} \times (\text{radius})^2}{2}$$

$$= \frac{\pi\rho(R^2-x^2)dx \times (R^2-x^2)}{2}$$

$$= \frac{1}{2}\pi\rho(R^4-2R^2x^2+x^4)dx.$$

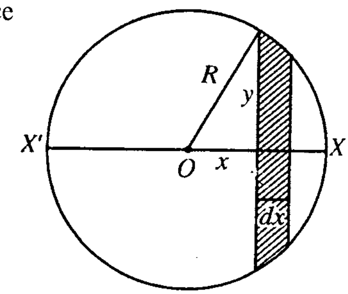


Fig. 6.20.

Hence the moment of inertia of the sphere about XX', can easily by obtained by integrating the above expression between the limits $x = -R$ to $x = R$. Then,

$$I = \int_{-R}^{R}\frac{\pi\rho}{2}(R^4-2R^2x^2+x^4)\,dx = 2\times\frac{\pi\rho}{2}\int_0^R(R^4-2R^2x^2+x^4)\,dx$$

$$= \pi\rho\left[R^4x-2R^2\cdot\frac{x^3}{3}+\frac{x^5}{5}\right]_0^R = \pi\rho\left[R^5-\frac{2R^5}{3}+\frac{R^5}{5}\right]$$

$$= \pi\rho\cdot\frac{8R^5}{15} = \frac{2}{5}\times\frac{4}{3}\pi R^3\rho\times R^2 \qquad \left[\because M = \frac{4}{3}\pi R^3\rho\right]$$

or $$I = \frac{2}{5}MR^2 \quad ...(17)$$

(ii) *About a tangent.* Any tangent to the sphere at any point is parallel to one of its diameters and is at a distance equal to R from the centre. Now, by the application of the theorem of parallel axes, the moment of inertia of the sphere about the tangent is

$$I = \frac{2}{5}MR^2+MR^2 = \frac{7}{5}MR^2 \quad ...(18)$$

(8) Moment of inertia of a spherical shell : (i) *About a diameter.* Let **fig. 6.21** represent the section of a shell of mass M and radius R through its centre O. Then,

the surface area of the shell $= 4\pi R^2$

$\therefore$ Mass per unit area $= M/4\pi R^2$.

Let us consider a thin circular ring lying between two parallel planes AB and CD perpendicular to XX'. AB and CD planes lie at distances x and $x + dx$ respectively from the centre O of the shell. The radius of the circular ring is $y = \sqrt{R^2-x^2}$ and the width AC (not dx).

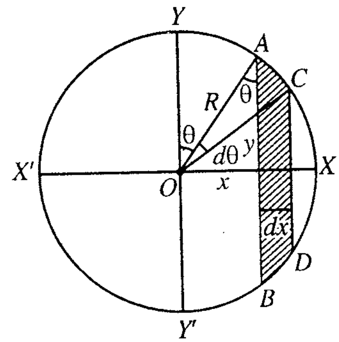


Fig. 6.21.

Join OA and OC and let $\angle YOA = \theta$ and $\angle AOC = d\theta$, then

$$x = R\sin\theta \text{ and } AC = Rd\theta$$

Differentiating $x = R\sin\theta$, we get

$$dx = R\cos\theta\, d\theta$$

Area of the ring $= 2\pi y\cdot AC = 2\pi R\cos\theta\cdot Rd\theta.$

$$\text{Its mass} = 2\pi\cdot R\cos\theta\cdot Rd\theta \times \frac{M}{4\pi R^2} = \frac{M}{2R}dx$$

$\therefore$ The moment of inertia of the thin ring about the axis XX'

$$= \text{mass} \times (\text{radius})^2 = \frac{M}{2R}dx \times y^2$$

$$= \frac{M}{2R}(R^2 - x^2)dx \qquad [\because R^2 - x^2 + y^2]$$

Hence, the moment of inertia of the whole shell about the diameter XX' is given by

$$I = \int_{-R}^{R}\frac{M}{2R}(R^2 - x^2)\,dx = \frac{M}{2R}\int_0^R (R^2 - x^2)\,dx = \frac{M}{R}\left[R^2x - \frac{x^3}{3}\right]_0^R = \frac{M}{R}\left[R^3 - \frac{R^3}{3}\right]$$

or $$I = \frac{2}{3}MR^2 \qquad ...(19)$$

(ii) *About a tangent.* Any tangent of the spherical shell is at a distance R from a parallel diameter. Since moment of inertia of the shell about any diameter is the same, hence applying the theorem of parallel axes, the moment of inertia of the shell about a tangent is given by

$$I = \frac{2}{3}MR^2 + MR^2 \quad \text{or } I = \frac{5}{3}MR^2 \qquad ...(20)$$

(9) Moment of inertia of a hollow sphere or a thick shell : (i) *About a diameter.* A hollow sphere is just a solid sphere from the inside to which a smaller concentric solid sphere has been removed. Let M be the mass of the hollow sphere and R_1 and R_2 be its inner and outer radii respectively [**fig. 6.22**].

$$\text{Volume of the hollow sphere} = \frac{4}{3}\pi(R_2^3 - R_1^3)$$

$$\therefore \text{ Mass per unit volume of the sphere} = \frac{3M}{4\pi(R_2^3 - R_1^3)}.$$

Consider the hollow sphere to be divided into thin elementary concentric shells. Consider one such shell of radius x and thickness dx, then

$$\text{Volume of the elementary shell} = 4\pi x^2\cdot dx$$

$$\text{Mass of the elementary shell} = \frac{3M}{4\pi(R_2^3 - R_1^3)} \times 4\pi x^2\cdot dx$$

$$= \frac{3M}{(R_2^3 - R_1^3)}\cdot x^2 dx$$

Moment of inertia of the elementary shell about a diameter

$$= \frac{2}{3} \times \frac{3M}{(R_2^3 - R_1^3)}\cdot x^2 dx \times x^2$$

$$= \frac{2M}{(R_2^3 - R_1^3)}\cdot x^4 dx$$

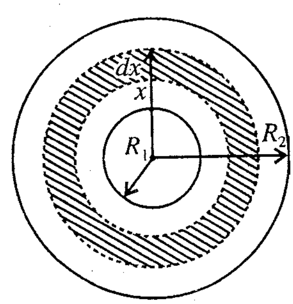


Fig. 6.22.

$\therefore$ Moment of inertia of the hollow sphere about a diameter is

$$I = \int_{R_1}^{R_2}\frac{2M}{(R_2^3 - R_1^3)}\cdot x^4 dx = \frac{2M}{(R_2^3 - R_1^3)}\int_{R_1}^{R_2} x^4 dx$$

$$= \frac{2M}{(R_2^3 - R_1^3)}\left[\frac{x^5}{5}\right]_{R_1}^{R_2} = \frac{2M}{(R_2^3 - R_1^3)} \times \frac{1}{5}(R_2^5 - R_1^5)$$

$$I = \frac{2}{5}M\left[\frac{R_2^5 - R_1^5}{R_2^3 - R_1^3}\right] \qquad ...(21)$$

It should be noted that the above method may be applied to calculate the moment of inertia of a solid sphere by putting the limits of integration, $R_1 = 0$ and $R_2 = R$. Then $I = \frac{2}{5}\frac{MR^5}{R^3} = \frac{2}{5}MR^2$.

(ii) *About a tangent.* Obviously, the tangent to the sphere, at any point, will be parallel to one of its diameters and at a distance equal to its external radius R_2 from its centre. Thus, by the theorem of parallel axes, we have

$$I = \frac{2}{5}M\left[\frac{R_2^5 - R_1^5}{R_2^3 - R_1^3}\right] + MR_2^2. \qquad ...(22)$$

Ex. 1. *A circular metal hoop of mass* 1 *kg and radius* 0.2 *metre makes* 10 *revolutions per sec about its centre. The axis of rotation being normal to the plane of the hoop.*

(a) *What is the moment of inertia about this axis ?*

(b) *What is the angular momentum about the same axis ?*

(c) *Calculate the torque, which will increase the angular momentum by* 25% *in* 10 *sec.*

(d) *If this torque is applied by a force acting tangentially on the rim of the hoop, find the value of this force.*

(Jabalpur 1992)

Sol. (a) $I = MR^2 = 1 \times (0.2)^2 =$ **0.04 kg-m²**.

(b) Here, $\omega = 2\pi n = 2 \times 3.14 \times 10 = 63$ rad/sec²

$\therefore$ $J = I\omega = 0.04 \times 63 =$ **2.5 n-m.**

(c) In 10 seconds the angular momentum increases by 25%.

$\therefore$ In 1 sec the angular momentum will increase $= \frac{2.5}{10} \times \frac{25}{100} = 0.0625$ n-m

$\therefore$ Torque = Rate of change of angular momentum = **0.0625 n-m.**

(d) If F is the force, then $F \cdot R = 0.0625 = 0.0625/0.2 =$ **0.312 newton.**

Ex. 2. *The radius of a solid sphere is* 5 *cm. Find the radius of gyration about its* (i) *diameter and* (ii) *tangent.*

(Rewa 1999; Gwalior 94)

Sol. (i) M.I. about a diameter : $I = \frac{2}{5}MR^2 = MK_1^2$

$\therefore$ $K_1 = R\sqrt{\frac{2}{5}} = 5\sqrt{\frac{2}{5}}$ cm = **3.16 cm.**

(ii) M.I. about a tangent : $I = \frac{7}{5}MR^2 = MK_2^2$

$\therefore$ $K_2 = R\sqrt{\frac{7}{5}} = 5\sqrt{\frac{7}{5}}$ = **5.92 cm.**

Ex. 3. *Four spheres each of diameter* 2a *and mass* m *are placed with their centres on the four corners of a square of side* b. *Calculate the moment of inertia of the system about any side of the square.*

(Agra 1988; Bilaspur 91; Bhopal 90; Kanpur 78)

Sol. The four spheres of same mass m and radius a are placed at the corners of the square $PQRS$ **[fig. 6.23]**.

M.I. *of* P and Q spheres about axis PQ

$$= \frac{2}{5}ma^2 + \frac{2}{5}ma^2 = \frac{4}{5}ma^2 \quad \text{...(i)}$$

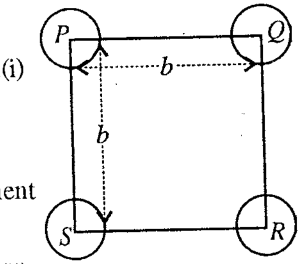


Fig. 6.23.

Similarly, M.I. of R and S spheres about axis $RS = \frac{4}{5}ma^2$.

Since the axis PQ is parallel to SR are at a distance b, therefore the moment of inertia of S and R spheres about the axis PQ

$$= \frac{4}{5}ma^2 + 2mb^2 \quad \text{...(ii)}$$

Therefore, total M.I. of the system about the axis PQ is the sum of (i) and (ii), *i.e.*,

$$= \frac{4}{5}ma^2 + \frac{4}{5}ma^2 + 2mb^2 = \frac{8}{5}ma^2 + 2mb^2.$$

Ex. 4. *A uniform thin bar of mass 6 kg and length 2.4 m is bent to make an equilateral hexagon. Calculate its moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon.* **(Kanpur 1988)**

Sol. M.I. of a regular hexagon of side a about an axis passing through the centre of mass and perpendicular to the plane of hexagon is given by

$$I = 6\left(\frac{ma^2}{12} + m \cdot OP^2\right) = 6\left(\frac{ma^2}{12} + m \cdot \frac{3a^2}{4}\right) = 5ma^2$$

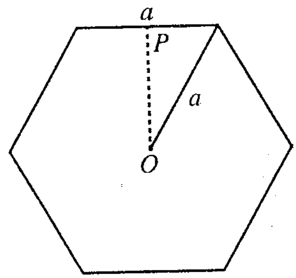


Fig. 6.24.

$[\because OP = a \cos 30° = a(\sqrt{3}/2)]$

where m is the mass of side a.

Here $\quad m = \frac{6}{6} = 1$ kg and $a = \frac{2.4}{6} = 0.4$ m.;

$\therefore \quad I = 5 \times 1 \times (0.4)^2 =$ **0.8 kg-m².**

Ex. 5. *A grindstone has a moment of inertia of 50 S.I. units. A constant couple is applied and the grindstone is found to have speed of 150 revolutions per minute, 10 sec after starting from rest. Find the couple.*

Sol. The speed of revolutions per sec $= \frac{150}{60} = \frac{5}{2}$.

$\therefore$ Angular velocity acquired $\omega = 2\pi \times$ number of revolutions per sec $= 2\pi \times \frac{5}{2} = 5\pi$ radians/sec.

This angular velocity is attained in 10 sec starting from rest, therefore the angular acceleration $d\omega/dt$ is given by

$$\frac{d\omega}{dt} = \frac{5\pi}{10} = \frac{\pi}{2} \text{ rad/sec}^2.$$

$\therefore \quad$ Couple $= I \times d\omega/dt = 50 \times \pi/2 =$ **78.5 newton-metre.**

Ex. 6. *A circular disc of radius $r = 3$ cm has been removed from a large disc of radius $R = 10$ cm. The centre of the hole, so made, is at $a = 5$ cm distance from the centre of the original disc. If the mass of the remaining disc is 182 gm, calculate its moment of inertia about an axis passing through the two centres. What is its M.I. about an axis passing through O and perpendicular to its plane.*

Sol. Area of the remaining disc $= \pi R^2 - \pi r^2 = \pi[(0.1)^2 - (0.03)^2] = 91 \times 10^{-4}\ \pi\text{m}^2$.

$\therefore$ Mass per unit area $= 0.182/(91\pi \times 10^{-4}) = 20/\pi$ kg/m²

$\therefore$ Mass of the original disc $= \frac{20}{\pi} \times$ its area $= \frac{20}{\pi} \times \pi \times (0.1)^2 = 0.2$ kg

and $\quad$ Mass of the removed disc $= \frac{20}{\pi} \times \pi \times (0.3)^2 = 0.018$ kg.

M.I. of the original disc about its diameter

$$= \frac{MR^2}{4} = 0.2 \times \frac{(0.1)^2}{4} = 5 \times 10^{-4} \text{ kg-m}^2$$

M.I. of the removed disc about its diameter

$$= 0.018 \times (0.03)^2/4 = 4.05 \times 10^{-6} \text{ kg-m}^2$$

M.I. of the remaining disc about the common diametrical line OC

$$= 5 \times 10^{-4} - 4.05 \times 10^{-6} = \mathbf{4.96 \times 10^{-4} \text{ kg-m}^2}.$$

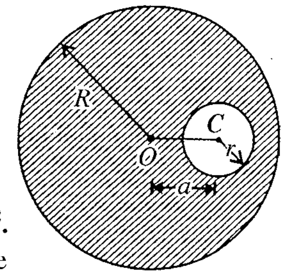


Fig. 6.25.

M.I. of the original disc about an axis through O and perpendicular to its plane

$$= MR^2/2 = 0.2 \times (0.1)^2/2 = 10^{-3} \text{ kg-m}^2.$$

M.I. of the removed material about an axis through O and perpendicular to its plane

$$= I_{cm} + ma^2 = \frac{mr^2}{2} + ma^2 = 0.018 \times \frac{(0.03)^2}{2} + 0.018 \times (0.05)^2$$

$$= 5.31 \times 10^{-5} \text{ kg-m}^2.$$

$\therefore$ M.I. of the remained disc about an axis through O and perpendicular to its plane

$$= 10^{-3} - 5.31 \times 10^{-5} = \mathbf{9.469 \times 10^{-4} \text{ kg-m}^2}.$$

Ex. 7. *Two spheres, each of* 100 *gm mass and* 5 *cm diameter, are joined by a weightless rod so that their centres are* 10 *cm apart. Calculate the moment of inertia of the system about* (i) *the line PQ joining their centres and* (ii) *a line which bisects PQ and is perpendicular to it. Locate also the position of centre of mass.*

Sol. (i) The line PQ, which joins the centres of the two spheres, is along their diameters. Hence the moment of inertia of the system about PQ is

$$I = \frac{2}{5}MR^2 + \frac{2}{5}MR^2 = \frac{4}{5}MR^2 = \frac{4}{5} \times 0.1 \times \left(\frac{0.05}{2}\right)^2 = \mathbf{5 \times 10^{-5} \text{ kg-m}^2}$$

(ii) If x be the distance of centre of mass from the centre of one (P) sphere, then

$$x = \frac{m_1x_1 + m_2x_2}{m_1 + m_2} = \frac{0 + 0.1 \times 0.1}{0.1 + 0.1} = 0.05 \text{ m}.$$

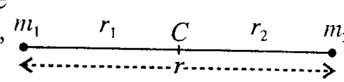
Hence, the centre of mass bisects the line PQ.

The moment of inertia of the system about an axis passing through the centre of mass and perpendicular to PQ is given by

$$I = 2[M \times (0.05)^2 + \frac{4}{5}MR^2 = 2 \times 0.1 \times (0.05)^2 + \frac{4}{5} \times 0.1 \times \left(\frac{0.05}{2}\right)^2 = \mathbf{5.5 \times 10^{-4} \text{ kg-m}^2}.$$

Ex. 8. *In a diatomic molecule, two atoms of masses m_1 and m_2 are situated at a distance r. Determine the moment of inertia of the molecule about that axis which is perpendicular to the line joining the two atoms and passes through its centre of mass.* **(Agra 2006; Indore 1996)**

Sol. Moment of inertia of a diatomic molecule : A diatomic molecule, in general, consists of two atoms, separated by some distance. Let the masses of these atoms be m_1 and m_2, separated by a distance r **[fig. 6.26]**. We know that the mass of each atom is concentrated on the nucleus, hence r must be the inter-nuclear distance. If r_1 is the distance of the centre of mass C of the molecule from m_1, then according to the property of the centre of mass, we have



$$(m_1 + m_2)r_1 = m_2r \qquad \text{(Taking } m_1 \text{ as the origin)}$$

Fig. 6.26.

or
$$r_1 = \frac{m_2r}{m_1 + m_2}, \quad \therefore \quad r_2 - r_1 = \frac{m_1r}{m_1 + m_2}$$

Hence the moment of inertia of the system about an axis passing through C and perpendicular to the line joining the masses is

$$I = m_1 r_1^2 + m_2 r_2^2 = m_1\left(\frac{m_2 r}{m_1+m_2}\right)^2 + m_2\left(\frac{m_1 r}{m_1+m_2}\right)^2 \quad \text{or} \quad I = \frac{m_1 m_2}{m_1+m_2} r^2 = \mu r^2$$

where $\mu = \dfrac{m_1 m_2}{m_1+m_2}$ is the **reduced mass** of the system.

As the axis of rotation is perpendicular to the line joining the two particles, the angular moentum will be along the direction of the rotational axis.

Ex. 9. *The internuclear distance between the two protons of hydrogen molecule is 0.74 Å. Calculate the moment of inertia of the molecule about an axis passing through its centre of mass and perpendicular to the line joining the two atoms.* ($m_p = 1.7 \times 10^{-27}$ kg.) **(Sagar 1993, 91; Agra 70)**

Sol. $I = \dfrac{m_1 m_2}{m_1+m_2} r^2 = \dfrac{1.7\times10^{-27}\times1.7\times10^{-27}}{2\times1.7\times10^{-27}}\times(0.74\times10^{-10})^2 = \mathbf{4.6 \times 10^{-48}\ kg\text{-}m^2}$.

Ex. 10. *Two spheres of the same mass are made of different metals of different densities. Which sphere will have greater moment of inertia about a diameter ?* **(Gwalior 1993)**

Sol. If r_1 and r_2 are the radii and ρ_1 and ρ_2 are the densities of the respective spheres, then in case the two spheres with same masses, we have

$$m_1 = m_2 \text{ i.e., } \frac{4}{3}\pi r_1^3 \rho_1 = \frac{4}{3}\pi r_2^3 \rho_2 \quad \text{or} \quad \frac{r_2^3}{r_1^3} = \frac{\rho_1}{\rho_2} \text{ and } I_1 = \frac{2}{5}m_1 r_1^2,\quad I_2 = \frac{2}{5}m_2 r_2^2$$

Hence
$$\frac{I_2}{I_1} = \frac{m_2 r_2^2}{m_1 r_1^2} = \frac{r_2^2}{r_1^2} = \left(\frac{r_2^3}{r_1^3}\right)^{2/3} = \left(\frac{\rho_1}{\rho_2}\right)^{2/3}$$

Thus if $\rho_1 > \rho_2$, then $I_2 > I_1$, *i.e.*, low density sphere will have more moment of inertia about a diameter.

EXERCISE 6·2

1. Find the radius of gyration of a disc of mass 100 gm and radius 5 cm about an axis passing through its centre of mass and perpedndicular to its plane.
   [**Ans.** 0.035 m.]
2. A uniform solid sphere of mass 1 kg and radius 0.2 metre makes 4 revolutions per second about a diameter. How much torque is required to increase its angular momentum about this axis by 10% in 20 seconds ?
   [**Ans.** 0.002 n-m.]
3. A uniform thin bar of mass 3 kg and length 0.9 metre is bent to make an equilateral triangle. Calculate the moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of the triangle. **(Agra 1998, 95; Kanpur 80)**
   [**Ans.** 0.045 kg-m^2.]
4. A uniform thin bar of mass 3 kg and length 2 metres is bent to make a square. Calculate the moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of the square. **(Kanpur 1979)**
   [**Ans.** 0.333 kg-m^2.]
5. Three masses, each of mass 2 kg are placed at the vertices of an equilateral triangle of side 10 cm. Calculate the moment of inertia and radius of gyration of the system about an axis perpendicular to the plane of the triangle and passing through (a) a vertex, (b) mid-point of one side, and (c) the centre of mass.
   [**Ans.** (a) 0.04 kg-m^2, 0.081 m; (b) 0.025 kg-m^2, 0.064 m; (c) 0.02 kg-m^2, 0.057 m.]

6. Two spheres each of 200 gm mass and 5 cm radius are joined by a rod of negligible weight so that their centres are separated by 8 cm distance. Find the moment of inertia of the system about (i) the line joining their centres and (ii) a line which bisects the line joining their centres and is perpendicular to it.

[**Ans.** (i) 4×10^{-4} kg-m^2; (ii) 10.4×10^{-14} kg-m^2.]

7. A flywheel of mass 500 kg and diameter 1 metre revolves about its axis. Its frequency of revolution is increased by 18 in 5 seconds. Calculate the torque applied. (Assume flywheel in the form of disc.)
[**Ans.** 1.414×10^3 S.I. units.]

8. A disc of mass 100 gm and radius 10 cm has a projection on its circumference. The mass of the projection is negligible. A 20 gm bit of putty with a velocity of 5 cm/sec strikes the projection and sticks fast. Find the angular velocity given to the disc.
[**Ans.** 0.143 rad/sec.]

9. The ratio of the radii of a solid cylinder and a solid sphere is 1 : 2. If their moments of inertia about their own axes are equal, find the ratio of their masses. **(Gwalior 2001)**
[**Ans.** 16 : 5.]

10. (a) What must be the relation between length e and radius r of a cylinder, if the moment of inertia of the cylinder about its axis is to be the same as its moment of inertia about the equatorial axis. **(Agra 1994)**

(b) If the moment of inertia of a cylinder about an equatorial axis is to be a minimum, determine the ratio l/R, when the mass of the cylinder is kept constant. **(Agra 1991; Gorakhpur 61; Kanpur 79; Meerut 95)**

[**Ans.** (a) $R = l/\sqrt{3}$; (b) $l/R = \sqrt{3/2}$.]

[**Hint :** $$I = M\left(\frac{R^2}{4} + \frac{l^2}{12}\right) \text{ and } M = \pi R^2 l\rho$$

Now, $$\frac{dM}{dR} = \pi\rho\left(2Rl + R^2\frac{dl}{dR}\right) = 0 \quad [\because M \text{ is constant}] \text{ or } \frac{dl}{dR} = -\frac{2l}{R}$$

But $$\frac{dI}{dR} = 0,\ i.e.,\ M\left(\frac{2R}{4} + \frac{2l}{12}\cdot\frac{dl}{dR}\right) = 0 \text{ or } \frac{R}{2} + \frac{l}{6}\times\left(-\frac{2l}{R}\right) = 0, \text{ where } \frac{l}{R} = \sqrt{\frac{3}{2}}.]$$

11. Show that the radius of gyration of a circular plate of radius a about an axis perpendicular to its plane through the circumference is given by $k = a\sqrt{3/2}$.

12. Obtain expression for rational kinetic energy of a rigid body in terms of (i) angular momentum and angular velocity, (ii) angular velocity and moment of inertia. **(Gorakhpur 1995)**

[**Ans.** K.E. $= \frac{1}{2}J\omega = \frac{1}{2}I\omega^2$].

13. If the diameter of earth suddenly contracts to its half, how is the length of a day affected ?
[**Ans.** 6 hrs] **(Kanpur 1998; Gwalior 93)**

[**Hint :** From the law of conservation of angular momentum, $I_1\omega_1 = I_2\omega_2$, or $I_1(2\pi/T_1) = I_2(2\pi/T_2)$.

Hence $$T_2 = T_1\left(\frac{I_2}{I_1}\right), \text{ where } I_1 = \frac{2}{5}MR_1^2 \text{ and } I_2 = \frac{2}{5}MR_2^2 \qquad (M = \text{mass of earth})$$

As $$R_2 = \frac{1}{2}R_1,\ \therefore\ \frac{I_2}{I_1} = \left(\frac{R_2}{R_1}\right)^2 = \frac{1}{4} \text{ and } T_2 = T_1 \times \frac{1}{4} = \frac{24}{4} = 6 \text{ hrs. } (\because T_1 = 24 \text{ hrs.})]$$

14. The moment of inertia of a reel of thread about its axis is Mk^2. If the loose end of the thread is held in hand and the reel is allowed to unroll itself, while falling down under the action of gravity, show that it falls down with an acceleration $ga^2/(a^2 + k^2)$, where a is the radius of the reel.

(Nagpur 1956)

15. A thin uniform disc of radius 25 cm and mass 1 kg has a hole of radius 5 cm in it. If the centre of the hole be at a distance of 10 cm from the centre of the disc, calculate the moment of inertia about an axis perpendicular to its plane and passing through the centre of the hole. **(Gorakhpur 1965)**
[**Ans.** 0.0412 kg-m^2.]

16. A uniform flat disc of mass M and radius R rotates about a horizontal axis through its centre with an angular speed ω. A piece of mass m breaks of the edge of the disc at an instant such that it moves, vertically above the point at which it broke off. How high above the point does it rise ? Calculate the final angular momentum and energy of the broken disc [**fig. 6.27**]. **(Ajmer 1998)**
[**Ans.** $h = \sqrt{R^2\omega^2/2g}$; $J = (M - 2m)R^2\omega/2$; $K = (M - 2m)R^2\omega^2/4$.]

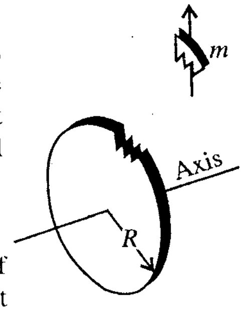


Fig. 6.27.

17. The flat surface of a hemisphere of radius r is cemented to one flat surface of a cylinder of radius r and length L. If the total mass is M, calculate the moment of inertia of the combination about the axis of the cylinder. Assume the material of the hemisphere and the cylinder to be the same. **(Kanpur 1983, 80)**

$$\left[\textbf{Ans.}\ \left\{Mr^2\left(\frac{L}{2} + \frac{2}{15}r\right)\right\} \Big/ \left(L + \frac{2}{3}r\right).\right]$$

18. A thin rod AB of mass m and length l is lying on a horizontal frictionless table. A particle of mass m moves with speed v perpendicular to the rod and collide non-elastically with the end B of the rod [**fig. 6.28**]. (a) Calculate the velocity of the centre of mass before and after the collision. (b) Calculate the angular momentum of the system just before the collision and angular velocity about the centre of mass just after the collision.
[**Ans.** (a) $v/2$ in each case; (b) $mvl/4$ and $6v/5l$.]

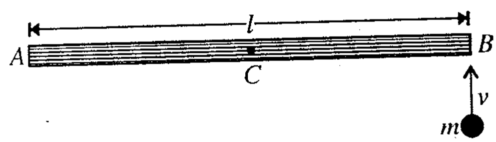


Fig. 6.28

6·10. Kinetic Energy of a Rotating Body about an Axis

The kinetic energy of a rotating body about an axis depends not only upon its mass and angular velocity but also depends upon the position of the axis and the distribution of mass about that axis.

Suppose a body of mass M is rotating about an axis AB. Assume that the body is composed of a very large number of small particles [**fig. 6.29**]. Consider a particle P of elementary mass m at a distance r from the axis. If ω is the angular velocity, the linear velocity of the particle P will be $r\omega$.

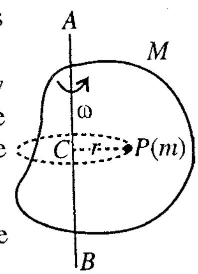


Fig. 6.29.

Therefore, the kinetic energy of ratation of the particle P is $\frac{1}{2}mr^2\omega^2$. Hence the kinetic energy (K.E.) of rotation of the whole body is given by

$$E = \Sigma\frac{1}{2}mr^2\omega^2 = \frac{1}{2}\omega^2\Sigma mr^2$$

since the angular velocity ω is the same for all the constituent particles of the body.

Now, $\Sigma mr^2 = I$, the moment of inertia about the axis AB,

$$\therefore \quad \textit{K.E. of rotation } E = \frac{1}{2}I\omega^2 \qquad ...(1)$$

If $\omega = 1$, *obviously* $\qquad I = 2 \times E$.

Hence, *the moment of inertia of a body may also be defined as twice its kinetic energy of rotation, the angular velocity of the body being unity.* ...(2)

But $\quad$ angular momentum $J = I\omega$

where J is along the axis of rotation

$$\therefore \quad J = I\sqrt{2E/I} = \sqrt{2EI} \text{ or } E = J^2/2I \qquad ...(3)$$

This is the relation between the rotational kinetic energy and angular momentum about the same axis.

Rolling bodies : Let us calculate the kinetic energy of a rolling body. When a body with circular symmetry (*e.g.*, cylinder, wheel, disc, sphere etc.) rolls on a plane surface, its motion is a combination of translation and rotation. At any instant the axis normal to the plane of the diagram through the point of contact P is the axis of rotation. If the speed of the centre of mass is V (relative to an observer fixed on the surface), the instantaneous angular speed about an axis through P is $\omega = V/R$, where R is the radius of the body. Therefore, at that instant all the particles of the rigid body are moving with the same angular speed* ω about the axis through P and the motion of the body is equivalent to a **pure rotation**. Thus, the total kinetic energy is

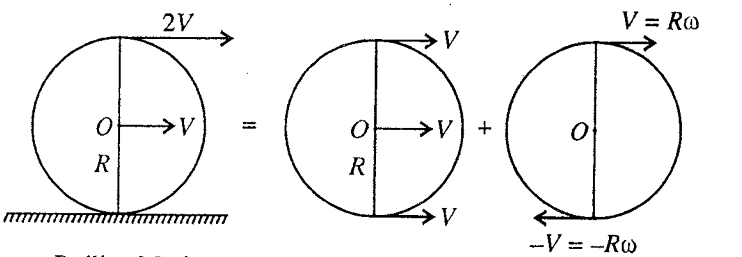


Fig. 6.30.

$$E = \frac{1}{2} I_p \omega^2,$$

where I_p is the moment of inertia about the axis through P.

From the theorem of parallel axes,

$$I_p = I_{cm} + MR^2 \qquad ...(4)$$

where I_{cm} is the moment of inertia of the body of mass M about a parallel axis through O.

$$\therefore \quad E = \frac{1}{2} I_{cm}\omega^2 + \frac{1}{2} MR^2\omega^2 = \frac{1}{2} I_{cm}\omega^2 + \frac{1}{2} MV^2 \qquad ...(5)$$

If k is the radius of gyration of the body about a parallel axis through O, then $I = Mk^2$ and therefore total kinetic energy

$$E = \frac{1}{2} Mk^2 \frac{V^2}{R^2} + \frac{1}{2} MV^2 = \frac{1}{2} MV^2 \left(\frac{k^2}{R^2} + 1\right) \qquad ...(6)$$

Alternatively, the motion of the cylinder can be considered as the sum of the translational motion with speed V and pure rotational motion about the centre C with angular speed $\omega = V/R$ **[fig. 6.30]**. So that, the total kinetic energy is

$$E = \frac{1}{2} MV^2 + \frac{1}{2} I_{cm}\omega^2 = \frac{1}{2} MV^2 \left(1 + \frac{k^2}{R^2}\right) \qquad ...(7)$$

Ex. 1. *What is the angular momentum of a particle whose rotational kinetic energy is* 18 *joules, if the angular momentum vector coincides with the axis of rotation and its moment of inertia about the axis is* 0.01 *kg-m*^{2}. **(Agra 1985)**

Sol. If the angular momentum vector coincides with the axis of rotation, then the angular momentum is given by

$$J = I\omega$$

K.E. of rotation, $E = \frac{1}{2} I\omega^2 = J^2/2I$

$$\therefore \quad J = \sqrt{2EI} = \sqrt{2 \times 18 \times 0.01} = \mathbf{0.6 \ kg\text{-}m^2/sec.}$$

* It is to be noted that at any instant different particles of the body have different linear speeds. The point P is at rest ($v = 0$) instantaneously, the centre of mass has speed $V = R\omega$ and the highest point of circumference has speed $2R\omega$ relative to P.

Ex. 2. *A circular disc of radius 0.1 m and mass 1 kg is rotating at the rate of 10 revolutions a second about an axis at right angles to its plane and passing through its centre. Find the work that must be done to increase the rate of revolutions to 20 per second.*

Sol. M.I. of the disc about the given axis is

$$I = \frac{MR^2}{2} = \frac{1 \times (0.1)^2}{2} = 5 \times 10^{-3} \text{ kg-m}^2$$

Initial angular velocity $\omega_1 = 2\pi \times$ number of revolutions per second $= 2\pi \times 10$ rad/sec.

Final angular velocity $\omega_2 = 2\pi \times 20$ rad/sec.

The work done in increasing the rate of revolutions = Final K.E. – Initial K.E.

$$= \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2 = \frac{1}{2} \times 5 \times 10^{-3}[(40\pi)^2 - (20\pi)^2]$$

$$= \frac{1}{2} \times 5 \times 10^{-3} \times 1200 \times \pi^2 = \mathbf{29.5 \text{ joules.}}$$

Ex. 3. *A solid sphere of mass 100 gm and radius 2.5 cm rolls without sliding with a uniform velocity of 10 cm per second along a straight line on a smooth horizontal table. Calculate its total energy.* **(Agra 1984)**

Sol. Total K.E. $= \frac{1}{2}Mv^2(1 + k^2/R^2)$.

For a sphere, $I = Mk^2 = \frac{2}{5}MR^2$, $\therefore$ $k^2 = \frac{2}{5}R^2$ or $k^2/R^2 = \frac{2}{5}$

$\therefore$ Total K.E. $= \frac{1}{2}Mv^2\left(1 + \frac{2}{5}\right) = \frac{7}{10}Mv^2 = \frac{7}{10} \times 0.1 \times (0.1)^2 = \mathbf{7 \times 10^{-4} \text{ joules.}}$

Ex. 4. *A flywheel in the form of a solid circular disc of mass 5000 kg and radius 1 metre is rotating, making 120 revolutions per minute. Compute the kinetic energy and the angular impulse, if the flywheel is brought to rest in 2 seconds; friction is to be neglected.*

Sol. M.I. of the flywheel about its axis

$$= \frac{MR^2}{2} = \frac{5000 \times 1^2}{2} = 2500 \text{ kg-m}^2$$

Angular speed, $\omega = 2\pi n$, where n = No. of revolutions/sec.

$= 2\pi \times 120/60 = 4\pi$ rad/sec.

$\therefore$ K.E. $= \frac{1}{2}I\omega^2 = \frac{1}{2} \times 2500 \times (4\pi)^2 = \mathbf{1.97 \times 10^5 \text{ joules.}}$

Angular impulse = Change in angular momentum $= I\omega$

$= 2500 \times 4\pi = \mathbf{3.14 \times 10^4}$ **S.I. units.**

6·11. Motion of a Body Rolling Down an Inclined Plane

If a body rolls down an inclined plane without slipping, it has both translatory and rotatory motions. The body rolls down, hence it suffers vertical descent causing to its loss of potential energy. At the same time the body acquires linear and angular velocities and, therefore, it gains kinetic energy of translation and rotation. If there would have been no loss of energy through friction etc., the loss of potential energy must be equal to the gain in kinetic energy.

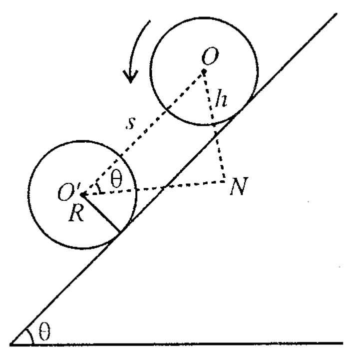


Fig. 6.31.

Let a body of mass M and radius R be rolling down a plane inclined at an angle θ to the horizontal. If the body acquires an angular speed ω and its centre of mass acquires a linear speed V, after covering a distance s along the inclined plane or a vertical height h $(= s \sin \theta)$, then

loss of potential energy in travelling a vertical height $h = Mgh$.

$$\text{gain in K.E.} = \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}MV^2 = \frac{1}{2}MV^2\left(\frac{k^2}{R^2}+1\right),$$

where k is the radius of gyration about an axis through the centre of mass and parallel to the plane.

According to the law of conservation of energy,

loss in P.E. of the body = gain in K.E. by the body *i.e.*,

$$Mgh = \frac{1}{2}MV^2\left(\frac{k^2}{R^2}+1\right)$$

$$\therefore \quad V^2 = \frac{2gh}{1+k^2/R^2} = \frac{2gs\sin\theta}{1+k^2/R^2} \qquad [\because h = s\sin\theta] \quad ...(1)$$

If the centre of mass of the body acquires an acceleration a on covering the distance s along the plane after starting from rest, then $V^2 = 2as$. Therefore,

$$2as = \frac{2gs\sin\theta}{1+k^2/R^2}. \text{ So that, } a = \frac{g\sin\theta}{1+k^2/R^2} \quad ...(2)$$

Thus, it is obvious from the above equation that for a given inclination θ, acceleration of the body is inversely proportional to $(1 + k^2/R^2)$. Hence, it follows that lesser the value of k as compared to R the greater is the acceleration of the body rolling down the plane and lesser the time of descent. These effects will be reversed when k is larger. Again, as the expression (2) for acceleration does not contain the mass of the body, the acceleration will be independent of the mass. The magnitude of acceleration and so the time of descent for two bodies of the same mass, size and appearance but of different radius of gyration k (depending upon their internal structure) are different. This difference in times of descent provides an excellent means to distinguish between two spheres of the same mass, size and appearance when one of them is hollow and the other solid.

For identification of the two spheres, one hollow and the other solid but of the same masses (M) and same external radii (R), we allow them to roll down an inclined plane from the same height simultaneously. The moment of inertia of a solid sphere of mass M and radius R about the axis of rotation (diameter) is

$$I = \frac{2}{5}MR^2 = Mk_1^2, \quad \therefore \quad k_1^2 = \frac{2}{5}R^2 \quad ...(3)$$

where k_1 is the radius of gyration about the axis of rotation.

If the external and internal radii of the hollow sphere are R and R_1 respectively, then its moment of inertia about the axis of rotation is

$$I = \frac{2}{5}M\cdot\frac{R^5-R_1^5}{R^3-R_1^3} = Mk_2^2$$

where k_2 is the radius of gyration of the hollow sphere about the axis of rotation.

$$\therefore \quad k_2^2 = \frac{2}{5}\cdot\frac{R^5-R_1^5}{R^3-R_1^3} = \frac{2}{5}\cdot\frac{R^5(1-R_1^5/R^5)}{R^3(1-R_1^3/R^3)}$$

or

$$k_2^2 = \frac{2}{5}R^2\cdot\frac{1-(R_1^5/R^5)}{1-(R_1^3/R^3)} \quad ...(4)$$

From eq. (4) and eq. (3), we have

$$\frac{k_2^2}{k_1^2} = \frac{1-(R_1^5/R^5)}{1-(R_1^3/R^3)} \quad ...(5)$$

But $R > R_1$, hence $\quad \dfrac{R_1^5}{R^5} < \dfrac{R_1^3}{R^3}$ or $1 - \dfrac{R_1^5}{R^5} > 1 - \dfrac{R_1^3}{R^3}$

or $$\frac{1-R_1^5/R^5}{1-R_1^3/R^3} > 1, \therefore \qquad k_2^2 > k_1^2. \qquad ...(6)$$

Thus the value of k^2 for a hollow sphere is greater than that for a solid sphere, although the masses and external radii of the two spheres are the same. Hence it is obvious from the relation (2) that the acceleration of the solid sphere must be greater than that of the hollow one. Therefore, the solid sphere will cover a larger distance on the inclined plane than the hollow shere in the same interval of time, *i.e.*, if the two spheres are allowed to roll down an inclined plane simultaneously, the solid sphere will reach down the plane first.

Ex. 1. *A sphere, a disc, a spherical shell, a ring and a cylinder are allowed to roll down simultaneously on an inclined plane from the same height without slipping. Prove that the sphere reaches down first, then the disc and the cylinder, next the shell and in the last the ring.*

Sol. Let M and R be the mass and radius of each body, rolling down an inclined plane at an angle θ with the horizontal. Since the different bodies have different radius of gyration (k), they will be accelerated by different amount and hence will require different time intervals in travelling a certain distance on an inclined plane.

(i) Solid sphere : The moment of inertia of a solid sphere about the axis of rotation (or diameter) is given by

$$I = \frac{2}{5}MR^2 = Mk^2, \text{ where } k \text{ is the radius of gyration.}$$

$$\therefore \qquad k^2/R^2 = 2/5$$

$$\therefore \qquad \text{Acceleration } a_1 = \frac{g\sin\theta}{1+(k^2/R^2)} = \frac{g\sin\theta}{1+2/5} = \frac{5}{7}g\sin\theta.$$

(ii) Disc : $I = MR^2/2 = Mk^2$ or $\dfrac{k^2}{R^2} = \dfrac{1}{2}$, $\therefore\ a_2 = \dfrac{g\sin\theta}{1+\frac{1}{2}} = \dfrac{2}{3}g\sin\theta$

(iii) Spherical shell : $I = \dfrac{1}{2}MR^2 = Mk^2$ or $\dfrac{k^2}{R^2} = \dfrac{2}{3}$, $\therefore\ a_3 = \dfrac{g\sin\theta}{1+\frac{2}{3}} = \dfrac{3}{5}g\sin\theta$

(iv) Ring : $I = M\dfrac{R^2}{2} = Mk^2$ or $\dfrac{k^2}{R^2} = \dfrac{1}{2}$, $\therefore\ a_4 = \dfrac{g\sin\theta}{1+1} = \dfrac{1}{2}\sin\theta$

(v) Cylinder : $I = MR^2/2 = Mk^2$ or $\dfrac{k^2}{R^2} = \dfrac{1}{2}$ $\therefore\ a_5 = \dfrac{g\sin\theta}{1+\frac{1}{2}} = \dfrac{2}{3}g\sin\theta = a_2$

Obviously, $a_1 > a_2\ (= a_5) > a_3 > a_4$, hence, if a sphere, a disc, a cylinder, a spherical shell and a ring are allowed to roll down an inclined plane simultaneously from the same height without slipping, the sphere will reach down first, then the disc and cylinder, then the shell and in the last ring.

Ex. 2. *A solid disc* (a) *rolls* (b) *slides from rest down an inclined plane. Neglect friction and compare the velocities in both cases when the disc reaches the bottom of incline.*

Sol. Here $\qquad I = MR^2/2 = Mk^2, \therefore\ k^2 = R^2/2$

The acceleration of the disc, when it rolls down over a plane inclined to an angle θ to the horizontal, is given by

$$a = \frac{g\sin\theta}{1+k^2/R^2} = \frac{g\sin\theta}{1+R^2/2R^2} = \frac{2g\sin\theta}{3}.$$

If s be the length of the inclined plane, then the velocity V of the disc at the bottom is given by

$$V^2 = 2\times\frac{2g\sin\theta}{3}\times s \qquad ...(i)$$

since the disc starts from rest.

In case, the disc slides through a distance s, the acceleration a' will be given by

$$a' = g \sin\theta$$

Hence in second case the velocity V' is given by

$$V'^2 = 2g \sin\theta \times s \quad \text{...(ii)}$$

Dividing (ii) by (i) and taking square root, we get

$$\frac{V}{V'} = \sqrt{\frac{2}{3}}$$

This is the required comparison of velocities in two cases.

Ex. 3. *A uniform sphere of mass* 2 *kg and radius* 10 *cm is released from rest on an inclined plane which makes* 30° *angle with the horizontal. Deduce its*

(i) *angular acceleration,*

(ii) *linear acceleration along the plane,*

(iii) *kinetic energy as it travels* 2 *metres along the plane.* **(Agra 1970 S)**

Sol. Linear acceleration along the plane

$$a = \frac{g \sin\theta}{1+k^2/R^2} = \frac{9.8 \times 1/2}{1+2/5} = \mathbf{3.5\ m/sec^2} \quad [\because \text{ for sphere } k^2 = \tfrac{2}{5}R^2]$$

Linear velocity $V = R\omega$, so $\dfrac{dV}{dt} = R\dfrac{d\omega}{dt} = a$, linear acceleration.

$\therefore$ Angular acceleration $= a/R = 3.5/0.1 =$ **35 rad/sec²**.

Gain in K.E. = Loss in P.E. in vertical descent h $(= s \sin\theta)$

$$= Mgs \sin\theta = 2 \times 9.8 \times 2 \times \tfrac{1}{2} = \mathbf{19.6\ joules.}$$

EXERCISE 6.3

1. A sphere of mass 500 gm, diameter 4 cm rolls without slipping with a velocity of 10 cm/sec. Calculate its total energy. **(Kanpur 1978)**
   [**Ans.** 3.5×10^{-3} joules.]
2. Calculate the angular mometnum of a body whose rotational kinetic energy is 10 joules, if the angular momentum vector coincides with the axis of rotation and its moment of inertia about this axis is 8 gm × cm².
   [**Ans.** 4×10^{-3} kg-m²/sec.]
3. A metre stick is held vertically with one end on the ground and is then allowed to fall. Calculate the speed of the other end when it touches the ground, assuming that the end on the ground does not slip.
   [**Ans.** 5.42 m/sec.]
   [**Hint :** $\frac{1}{2}I\omega^2 = Mg \times 0.5$.]
4. Two spheres each of mass 1 kg and radius 7 cm are connected to the axle of the governor of an engine. They are revolving with a speed of 200 revolutions per second and their centres remain at a distance of 14 cm from the axis of rotation. Calculate their energy.
   [**Ans.** 3.406×10^4 joules.]
5. A wheel of radius 6 cm is mounted so as to rotate about a horizontal axis through its centre. A string of negligible mass, wrapped round its circumference, carries a mass of 200 gm, attached to its free end. When let fall, the mass descends through 100 cm in the first 5 sec. Calculate the angular acceleration of the wheel and its moment of inertia.
   [**Ans.** (4/3) rad/sec²; 0.0875 kg-m².]
6. A flywheel has a mass 15 kg and radius of gyration 0.15 m and is rotating 1800 rev/min, what torque and power are necessary so that the said angular speed is attained in 5 sec.
   [**Ans.** 12.7 n-m; 1197 watts.]

7. A thin plane disc of radius r and mass m, initially at rest, is set into rotation with angular velocity ω. Calculate the work done in each case if the axis of rotation is
(i) through its centre and perpendicular to its plane,
(ii) through a point on its edge and perpendicular to its plane,
(iii) a diameter **(Lucknow 1980)**

[**Ans.** (i) $\frac{1}{4}MR^2\omega^2$; (ii) $\frac{3}{4}MR^2\omega^2$; (iii) $\frac{1}{8}MR^2\omega^2$.]

8. A body of radius R and mass M is rolling horizontally without slipping with speed v. It then rolls up a hill to a maximum height h. If $h = 3v^2/4g$, (a) calculate the moment of inertia of the body, (b) what might the body be ? **(Agra 1987)**

[**Ans.** (a) $MR^2/2$; (b) cylinder or disc.]

9. A disc of mass M and radius R is pivoted about a horizontal axis through its centre and a small body of the same mass M is attached to the rim of the disc. If the disc is released from rest with the small body at the end of a horizontal radius, calculate the angular speed when the small body is at the bottom.

[**Ans.** $\sqrt{4g/3R}$ rad/sec.]

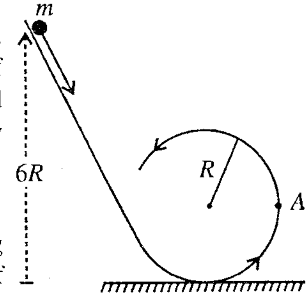


Fig. 6.32.

10. A small solid marble of mass m and radius r rolls without slipping along the loop track as shown in **fig. 6.32**. If it starts from rest at a height of $6R$ above the bottom, what is the force (horizontal and vertical components) acting on it at the point A ?

[**Ans.** (50/7) mg and mg.]

11. A diatomic molecule of similar atoms has a mass 5.3×10^{-26} kg and moment of inertia 1.94×10^{-46} kg-m^2 about an axis through the centre of mass and perpendicular to the line joining the atoms. If such a molecule has a mean speed of 500 m/sec and its rotational kinetic energy is 2/3 that of translational, find its average angular speed.

[**Ans.** 6.7×10^{12} rad/sec.]

12. A stick of mass M and length l lies on a frictionless horizontal table. A small ball of mass m, moving with speed v as shown in **fig. 6.33** collides elastically with the stick : (a) What quantities are conserved ? (b) What must be the mass m of the ball so that it remains at rest just after the collision ?

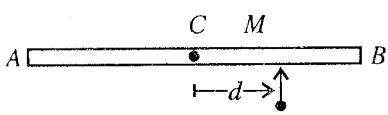


Fig. 6.33.

[**Ans.** (a) Linear momentum, angular momentum and mechanical energy, (b) $Ml^2/(12d^2 + l^2)$.]

13. A solid sphere rolls, without sliding from rest, down an inclined plane. Find the velocity after it has suffered a vertical drop of 2 metres. (g = 10 m/sec^2.)

[**Ans.** 5.34 m/sec^2.]

14. The mass of a ring is 9 kg. If it (a) slides without rolling with a velocity of 10 m/sec, (b) rolls without sliding with this velocity, compare the kinetic energies in two cases.

[**Ans.** 2.]

15. A solid spherical ball rolls down a plane inclined at 30° to the horizontal. Calculate the distance travelled in 4 seconds after starting from rest. (g = 9.8 m/sec^2.)

[**Ans.** 28 metres.]

16. Calculate the ratio of rotational to translational kinetic energy for a solid cylinder, rolling down a plane without slipping. **(Kanpur 1990)**

[**Ans.** 1 : 2.]

17. A cylinder of mass 100 gm, and diameter 25 cm, rolls from rest without slipping down a smooth

inclined plane of gradient 1 in 10 and 10 metres long. Calculate the total kinetic energy and its energy due to rotation when it reaches the bottom.
[**Ans.** 0.98 joule; 0.327 joule.]

18. A sphere, a uniform spherical shell and a thin ring all of the same radius, are released from rest at the top of an inclined plane. If the sphere reaches the bottom with speed 0.2 m/sec, calculate the speed with which the other two reach the bottom.
[**Ans.** 0.183 m/sec^2, 0.167 m/sec.]

19. A disc starting from rest rolls down an inclied plane of 1 in 100 in 5.33 sec. If the distance moved is 100 cm, find the acceleration due to gravity.
[**Ans.** 9.81 m/sec^2.]

20. A small sphere of mass m and radius r rolls without slipping on the inside of a large hemisphere of radius R. The axis of symmetry of the hemisphere is vertical. It starts at the top from rest. (a) What is its kinetic energy at the bottom ? What fraction is rotational and what translational ? (b) What normal force does the small sphere exert on the hemisphere at the bottom ?
[**Ans.** (a) $K = mg\ (R - r)$; 2/7 rotational and 5/7 translational; (b) 17/7 mg.]

6·12. Flywheel and its Moment of Inertia

A flywheel is a large heavy wheel with a long cylindrical axle passing through its centre. Its shape is such that its most of the mass is concentrated at the rim. Its centre of gravity lies on its axis of rotation so that it can come to rest in any desired position.

A flywheel, whose moment of inertia is to be determined, is mounted on two ball bearings to reduce friction. Its axle is supported in a horizontal position, at a convenient height from the floor. A small loop is made at one end of a fine cord and it is slipped on to a tiny peg P on the axle. Almost the whole length of the cord is wound round the axle and to its other free end a known mass m is attached **[fig. 6.34]**.

Theory and method : When the mass is allowed to fall under gravity, the cord is unwound and the flywheel rotates about its own axis. When the mass covers a distance h equal to the length of the cord wound round the axle, the cord is detached from the peg.

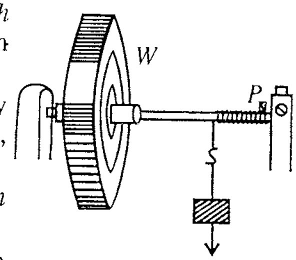


Fig. 6.34.

Let v be the velocity acquired by the mass m, ω the angular velocity acquired by the wheel and n_1 the rotations performed by it upto the instant, when the cord just leaves the axle.

The loss of potential energy in dropping mass m through a distance h $= mgh$.

This potential energy is used up in giving the kinetic energy to the system and against functional work done, *i.e.*,

(1) Gain of kinetic energy by the dropping mass $= \frac{1}{2}mv^2$.

(2) Gain of rotational kinetic energy by the flywheel $= \frac{1}{2}I\omega^2$,
where I is the moment of inertia of the flywheel about the axle.

(3) Work done against force of friction $= n_1F$,
where F is the amount of work done per revolution against the force of friction.

Thus, from the law of conservation of energy

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + n_1F. \qquad \text{...(1)}$$

When the cord is detached from the axle, the flywheel rotates with the maximum angular velocity ω but due to frictional force acting on its bearings the wheel comes to rest after performing, say, n_2 rotations. The whole rotational kinetic energy of the wheel is used up in doing work against friction, hence

$$n_2F = \frac{1}{2}I\omega^2. \qquad \text{...(2)}$$

Substituting the value of F obtained from eq. (2) in eq. (1), we get

$$mgh = \frac{1}{2}mv^2 + I\omega^2 + \frac{1}{2}n_1 \cdot I\omega^2/n_2$$

$$= \frac{1}{2}mr^2\omega^2 + \frac{1}{2}I\omega^2(1 + n_1/n_2), \qquad [\because v = r\omega]$$

where r is the radius of the axle.

Now the energy equation can be written as

$$2mgh = mr^2\omega^2 + I\omega^2(1 + n_1/n_2); \quad \therefore \quad I = \frac{2mgh - mr^2\omega^2}{\omega^2(1 + n_1/n_2)}. \qquad ...(3)$$

At the time, when the cord leaves off the axle, the wheel possesses the maximum agular velocity ω and after making n_2 rotations in a time t seconds, its velocity is reduced to zero. Assuming that the friction is steady during this time, the average angular velocity during these t seconds is $(\omega + 0)/2 = \omega/2$. Thus,

$$\frac{\omega}{2} = \frac{2\pi n_2}{t} \quad \text{or} \quad \omega = \frac{4\pi n_2}{t}. \qquad ...(4)$$

Substituting the value of ω from eq. (4) in eq. (3), we get

$$I = \frac{2mgh - mr^2 . \dfrac{(16\pi^2 n_2{}^2)}{t^2}}{\dfrac{16\pi^2 n_2{}^2}{t^2} \times \dfrac{n_2 + n_1}{n_2}} \quad \text{or} \quad I = \frac{m(ght^2 - 8\pi^2 n_2{}^2 r^2)}{8\pi^2 n_2 (n_1 + n_2)} \qquad ...(5)$$

In experiment, a cord is wound round the axle and a known mass m is attached to it. The length of the thread h wound round the axle is already measured by a metre scale and its length is so adjusted that when the mass is about to reach the floor, the other end of the cord leaves off the axle.

For measuring the number of rotations, there is the arrangement of automatic counter. If it is not available, then a chalk mark is made on the circumference of the wheel to count the number of rotations. The mass m is allowed to fall under gravity and the number of rotations n_1 made by the flywheel during the descent of the mass are counted. These rotations n_1 are obviously equal to the number of turns of the cord on the axle at the very start. When the thread leaves off the axle, a stop-clock is started. The time t and in this time interval the number of rotations n_2 of the wheel before coming to rest are noted.

Hence by substituting the values of various quantities in the right hand side of the equation (5), the moment of inertia I of the flywheel about the axle as axis can be calculated.

Use of flyweel in stationary engine : A flywheel is essentially a heavy wheel with a long axle. Stationary engines are provided with flywheel of large moment of inertia and use is made of their kinetic energy of rotation to obtain uniform motion with a non-uniform drive.

In the case of a stationary engine the force acting on the piston is not steady, but becomes zero twice in a cycle as the piston occupies two dead positions per cycle. Thus the engine exerts the driving force on the machine, coupled to it, in jerks.

A flywheel with a suitable large moment of inertia helps the engine to transmit its driving power to the machine steadily and smoothly. The flywheel stores sufficient energy when the piston is acted upon by maximum force and later on this stored energy is delivered to the machine when the force acting on the piston decreases to a minimum and thus the motion of the machine is rendered smooth and steady.

The flywheel helps in another way. If the machine is suddenly thrown into gear, the flywheel prevents its sudden stoppage on account of the power stored in it. A part of the kinetic energy of the flywheel is used to keep the machine in motion.

The larger the moment of inertia of the flywheel, the greater is its kinetic energy of rotation. This is achieved by concentrating the mass of the flywheel on the rim and making its spokes strong enough to withstand the large centrifugal force.

Mass of a flywheel is usually concentrated at the rim : If the moment of inertia of a flywheel is large, its kinetic energy of rotation will also be large. The larger kinetic energy of rotation of a flywheel is required in stationary engines. The moment of inertia of a mass varies as the square of its distance from the axis of rotation, hence the mass must be situated at a greater distance, possible from the axis to have its moment of inertia more. Thus to get the large moment of inertia of the flywheel, its mass is usually made to concentrate at its rim.

Ex. 1. *A cord is wound round the horizontal axle (whose radius is* 1.2 *cm) of a flywheel and a mass of* 1 *kg is attached to the free end of the cord. Starting from rest, the mass is released from the axle after falling through* 100 *cm. After the mass is released, the flywheel is observed to make* 14 *turns in* 5.8 *sec before coming to rest. Calculate* (a) *the kinetic energy of the mass at the moment of release;* (b) *the moment of inertia of the flywheel.*

Sol. Let I be the moment of inertia of the flywheel, v, the velocity acquired by the mass m descending through a distance h. Then, in accordance with the law of conservation of energy,

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + Nw,$$

or
$$2mgh = mv^2 + I\omega^2 + 2Nw \qquad ...(i)$$

where ω is the angular velocity of the flywheel, N the number of revolutions made by it before the mass gets detached from the axle, and w, the work done per revolution, against friction.

If the wheel makes n rotations before coming to rest *after the mass has been detached from the axle,* clearly, work done against friction during these n rotations, *i.e.*, nw, must be equal to the *K.E. of rotation of the wheel* which is all used up in doing this work. So that,

$$nw = \frac{1}{2}I\omega^2, \text{ whence } w = \frac{1}{2}\frac{I\omega^2}{n}$$

Substituting this value of w in relation (i) above, we have

$$2mgh = mv^2 + I\omega^2 + 2N\frac{1}{2}\frac{I\omega^2}{n} \quad \text{or} \quad 2mgh - mv^2 = I\omega^2(1 + N/n),$$

whence,
$$I = \frac{2mgh - mv^2}{\omega^2(1+N/n)} = \frac{2mgh - mr^2\omega^2}{\omega^2(1+N/n)} = \frac{(2mgh/\omega^2) - mr^2}{(1+N/n)} \qquad ...(ii)$$

because $v = r\omega$, where r is the *radius of the axle.*

Now, the average angular velocity of the wheel = $(\omega + 0)/2 = \omega/2$.

But since the wheel makes n rotaions or describes an angle $2\pi n$ in time t before it comes to rest, its *average angular velocity* is clearly also equal to $2\pi n/t$. We, therefore, have

$\omega/2 = 2\pi n/t$, whence $\omega = 4\pi n/t$.

Substituting this value of ω in relation (ii) for I above, we have

$$I = \frac{m\{(2ght^2/16\pi^2n^2) - r^2\}}{(1+N/n)} = m\left(\frac{n}{n+N}\right)\left(\frac{ght^2}{8\pi^2n^2} - r^2\right)$$

$$= \frac{m}{n+N}\left(\frac{ght^2}{8\pi^2 n} - nr^2\right) \qquad ...(iii)$$

Here, $m = 2$ kg, $h = 1$ m, $r = 1.2 \times 10^{-2}$ m.

$$N = \frac{\text{length of the cord}}{\text{circumference of the axle}} = \frac{1^*}{2 \times \pi \times 1.2 \times 10^{-2}} = 13.25$$

$n = 14$ and $t = 5.8$ sec.

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* Because the length of the thread or the cord just unwound as the mass reaches the floor, we take h to be equal to the length of the thread or the cord.

Putting these values in expression (iii) above, we have

$$I = \frac{2.0}{14+13.25}\left[\frac{980\times 1\times(5.8)^2}{8\pi\times 14} - 14(1.2\times 10^{-2})^2\right] = \mathbf{0.0275\ kg\text{-}m^2},$$

and K.E. of the mass at the time of release from the axle

$$= \frac{1}{2}mv^2 = \frac{1}{2}mr^2\omega^2 = \frac{1}{2}\times 2\times(1.1\times 10^{-2})^2 \times (4\pi n/t^2)$$

$$= \frac{1}{2}\times 2\times(1.2\times 10^{-2})^2\times(30.34)^2 \qquad \left[\because \frac{4\pi n}{t} = \frac{4\pi\times 14}{5.8} = 33.34\right]$$

$$= \mathbf{0.1325\ joules.}$$

Ex. 2. *A wheel of radius 6 cm, is mounted so as to rotate about a horizontal axis through its centre. A string of negligible mass, wrapped round its circumference carries a mass of 200 gm., attached to its free end. When let fall, the mass descends through 100 cm., in the first 5 sec. Calculate the angular acceleration of the wheel and its moment of inertia.* **(Bombay; Gorakhpur 1971)**

Sol. Let the linear velocity and linear acceleration acquired by the flywheel in 5 sec. be v and a respectively, in travelling through a distance of 100 cm. Obviously, this velicity v and acceleration f will be the same as those of the descending mass after 5 sec.

The distance s travelled by a particle in t sec with initial velocity zero is given by

$$s = \frac{1}{2}at^2$$

Here, s = 100 cm,; f = ?; t= 5 sec.

Therefore, $$100 = \frac{1}{2}a\times 5^2 \text{ or } a = 8 \text{ cm/sec}^2.$$

$$\text{Angular acceleration} = \frac{\text{linear acceleration}}{\text{radius of the wheel}} \qquad \left[\because \frac{dv}{dt} = \frac{d(r\omega)}{dt} = r\frac{d\omega}{dt}\right]$$

But, radius of wheel $r = 6$ cm.

$$\therefore \quad \text{Angular acceleration} = \frac{8}{6} = \frac{4}{3} \text{ rad/sec}^2.$$

Also, $$v = u + at, \quad \therefore \quad v = 0 + 8\times 5 = 40 \text{ cm/sec.}$$

Let I be the moment of inertia of the flywheel. And if a particle of mass m descends through a distance h, then

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \quad \text{or} \quad mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\times\frac{v^2}{r^2}$$

Here, m = 200 gm, v = 40 cm/sec., h = 100 cm and r = 6 cm.

$$\therefore \quad 200\times 980\times 100 = \frac{1}{2}\times 200\times 40\times 40 + \frac{1}{2}I\times 40\times 40/36$$

or $$I/9 = 980\times 100 - 800 = 100\times 972$$

$$\therefore \quad I = 100\times 972\times 9 = \mathbf{8.748\times 10^5\ gm\times cm^2}.$$

EXERCISE 6·4

1. A rope is wrapped around a flywheel 1 metre in radius and a body of mass 5 kg hangs from its one end. If the wheel is free to rotate through a horizontal axis perpendicular to its length, calculate the angular acceleration of the wheel assuming its mass to be 10 kg. **(Utkal 1966)**
[**Ans.** 9.8/sec^2.]
2. A flywheel of mass 500 kg and diameter 1 metre revolves about an axle. Its frequency of revolution is increased by 18 in 5 seconds. Calculate the torque applied. **(Banaras)**
[**Ans.** 1.422×10^4 kg. wt-cm.]

QUESTIONS

Long Answer Questions

1. A rigid body is rotating about an axis through the origin. Deduce relations connecting the components of total angular momentum with the components of the angular velocity.
2. What is inertia tensor ? What are moments of inertia and products of inertia ? Prove that the inertia tensor is symmetric.
3. What do you mean by principal axes and principal moments of inertia coefficients ? **(Kanpur 2004)**
4. (a) Obtain Euler's equations of motion for a rotating rigid body. **(Agra 2005, 04; Osmania 2005)**
   (b) Using these equations, explain conservation of energy and angular momentum. **(Osmania 2005)**
5. Show that the angular momentum $\vec{J}$ of a rotating rigid body is given by $\vec{J} = I\omega$, where ω is the angular velocity. Show that I is the tensor of second rank. **(Meerut 1982)**
6. Show that the kinetic energy of a rotating rigid body can be expressed as $T = \frac{1}{2}\vec{J}\cdot\vec{\omega}$
   Hence show that if I is the moment of inertia about the axis of rotation, prove that the kinetic energy can be expressed as $T = \frac{1}{2}I\omega^2$.
7. Show that the kinetic energy of a rotating rigid body in a coordinate system of principal axes is given by $T = \frac{1}{2}(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2)$
8. Define moment of inertia and explain its physical significance. **(Kanpur 2005; Sagar 01; Bhopal 1999; Ujjain 96)**
9. What is moment of inertia of a body ? State its unit and dimensions. State the factors on which moment of inertia of a body depends. Explain its physical significance. **(Sagar 2001; Bilaspur 1995; Ujjain 99; Indore 95; Jabalpur 93; Rewa 94)**
10. Obtain the angular momentum of a rotating rigid body about a fixed axis and hence define moment of inertia about the axis of rotation. **(Allahabad 1979)**
11. Deduce the fundamental equation of motion of a rigid body about fixed axis and show that it is rotational analogue to $\vec{F} = d\vec{P}/dt$ in the case of translation motion. **(Kanpur 1983)**
12. Establish a relation between the torque applied and the angular acceleration produced in a body about a given axis. What is the effect of the torque on the rotation of the body ? **(Kanpur 1995)**
13. Show that the moment of inertia of a body about an axis is equal to that torque which produces a unit angular acceleration in that body. **(Indore 1994)**
14. What is moment of inertia and radius of gyration ? Explain its physical significance. **(Bhopal 1999; Ujjain 96)**
15. Obtain an expression for the rotational kinetic energy E of a body rotating uniformly about an axis and hence establish its relationship with angular momentum l, given as $E = l^2/2I$. **(Rewa 1995)**
16. Establish an expression for the kinetic energy of a rotating body and give the relation between rotational kinetic energy and angular momentum about the same axis. **(Agra 1983)**
17. (a) Show that for a rigid body the angular momentum about the axis of rotation is equal to the product of moment of inertia about that axis and angular speed.
    (b) Prove that the torque acting on a rigid body about a fixed axis is equal to the product of angular acceleration and moment of inertia about the same axis.
18. Show that the moment of inertia of a system of particles about an axis parallel to the axis passing through the centre of the system is given by
$$I = \Sigma m_i r_i^2 + R^2 \Sigma m_i$$
    where r_i is the distance of the ith particle of mass m_i from the axis through the centre of mass and R is the distance between the two axes. **(Gwalior 1991; Kanpur 78)**

19. Determine the moment of inertia of a uniform thin circular ring of mass m, radius R about an axis (i) passing through its centre of gravity and perpendicular to its plane, (ii) its diameter, (iii) its tangent. **(Rewa 2001; Ujjain 1999)**
20. Find the moment of inertia of a uniform, thin circular disc of mas m and radius R about its diameter and about an axis perpendicular to its plane but passing through its centre. **(Rewa 2001; Ujjain 1999; Gwalior 96)**
21. Derive an expression for the moment of inertia of a uniform rectangular lamina about an axis passing through its centre of gravity and perpendicular to its plane. **(Ujjain 1997; Sagar 91; Bilaspur 90; Raipur 93; Jabalpur 99; Indore 96)**
22. State and prove the theorem of parallel axes. Hence derive the moment of inertia of a rectangular bar about an axis passing through one of its ends and through its centre. **(Agra 1978)**
23. Derive an expression for the moment of inertia of a rectangular bar about an axis passing through its centre of gravity and perpendicular to its length. **(Delhi 1992; Kanpur 90)**
24. Find the moment of inertia of (i) a sphere, and (ii) a spherical shell about a diameter and a tangent. What will be their moment of inertia about an axis at a distance x from the diameter ? **(Gwalior 2000; Raipur 1999; Agra 95, 93; Alld. 84; Delhi 92; Garwal 95; Gorakhpur 95; Kanpur 84; Raj. 82)**
25. (a) Derive an expression for the moment of inertia of a solid cylinder about an axis through its centre of mass and perpendicular to its own axis. **(Ujjain 2000; Bilaspur 2000; Agra 1994, 91, 87; Garwal 95; Purvanchal 95)**
    (b) Find the moment of inertia of a cylinder about a line perallel to its axis and touching its surface. **(Kanpur 1983)**
26. Define moment of inertia. Obtain an expression for the moment of inertia of an angular disc with internal and external radii R_1 and R_2 respectively about (i) its own axis and (ii) a tangential axis in the plane of the disc.
    Give the significance of radius of gyration and obtain its value in the above two cases. **(Rajasthan 1994)**
27. Deduce an expression for the moment of inertia of a thick spherical shell about its diameter. **(Gwalior 1997; Sagar 97)**
28. The flat surface of a hemisphere of radius r is cemented to one flat surface of a cylinder of radius r and length L. If the total mass is M, show that M.I. of the combination about the axis of the cylinder will be
$$Mr^2\left(\frac{L}{2}+\frac{4r}{15}\right)\Big/\left(L+\frac{2r}{3}\right).$$
(Agra 1995; Bundelkhand 95)
29. What is radius of gyration ? Obtain a formula for the acceleration of a body rolling down an inclined plane. **(Garwal 1995)**
30. Derive an expression for the acceleration of a body rolling down an inclined plane, without slipping and hence show that the acceleration of a solid sphere is $\frac{5}{7} g \sin \theta$, where θ is the angle which the plane makes with the horizonal. **(Sagar 1989, 92; Bhopal 96)**
31. Determine the kinetic energy of a solid sphere of mass M rotating and moving with speed v. **(Gwalior 1990)**
32. Write short notes on the following :
    (i) Radius of gyration, (ii) Inertia tensor, (iii) Theorem of parallel axes **(Gwalior 2000)**
    (iv) Theorem of perpendicular axes, (v) Flywheel **(Agra 2003, 1995)**

Short Answer Questions

1. What are the factors on which the moment of inertia of a body rotating about an axis depends.

2. What is radius of gyration ? Write its unit.

3. State and prove the theorem of parallel axes related to the moment of inertia.
(Agra 2006, 04; Bhopal 1999, 98; Ujjain 98)

4. State and prove the theorem of perpendicular axes. **(Kanpur 2005; Agra 04; Bhopal 01; Jabalpur 1997)**

5. A uniform thin rod of length $2a$ and mass m is bent at its middle point so that the bent part makes an angle θ with each other. Show that the moment of inertia of the system about an axis passing through O and perpendicular to the plane AOB [**fig. 6.35**] is independent of the angle θ and has the value $ma^2/3$.

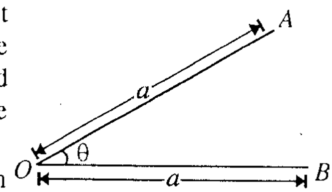
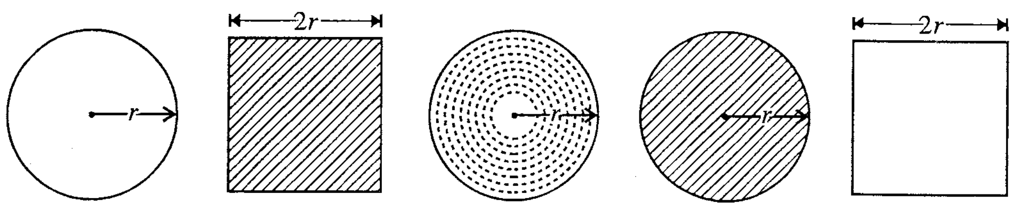


Fig. 6.35.

6. If two circular discs of the same mass and thickness are made from metals having different densities, which disc, either, will have the larger moment of inertia about its central axis ?

7. Five bodies are shown in **fig. 6.37**. Each body has the mass M, and their dimensions are shown in **fig. 6.37**. Which one has the largest. M. I. about a perpendicular axis through the centre of mass ? Which is the smallest ? Write also the moment of inertia in different cases.



(a) Ring (b) Cube (c) Cylinder (d) Sphere (e) Square frame

Fig. 6.36.

[**Ans.** Square frame (largest); Sphere (smallest); (a) Mr^2, (b) $\frac{2}{3}Mr^2$, (c) $Mr^2/2$, (d) $\frac{2}{5}Mr^2$, (e) $\frac{4}{3}Mr^2$.]

8. A body of mass m slides down on a smooth inclined plane, having angle θ with the horizontal, and reaches the bottom with a speed v. If the same mass were in the form of a ring which rolls down on an inclined plane (having the same angle θ with the horizontal), what will be the speed of the ring at the bottom ?

[**Ans.** $v/\sqrt{2}$.]

9. Two solid balls, one of iron and other of marble, are allowed to roll down on an inclined plane from rest and same height, which ball reach first ?

[**Ans.** Simultaneously.]

10. If two circular discs of the same mass and thickness are made from metals having different densities, which disc will have smaller moment of inertia about its central axis ?

[**Ans.** The disc of more density]

11. A ballet-dancer stretches her hands out for slowing down. On what principle does it depend ?

12. A body is rotating about an axis. Is it necessary that an external torque is acting ?

[**Ans.** No.]

13. Why does a wheel have more mass towards rim ?

14. A disc of metal is melted and recast in the form of a solid sphere. What will happen to the moment of inertia about a vertical axis passing through the centre. (Take the disc in horizontal plane)

15. A solid ball and a cylinder roll down an inclined plane without slipping, which reaches the bottom first ? Discuss.

16. What is the moment of inertia of a uniform semicircular wire of mass m and radius r about a line perpendicular to the plane of the wire though the centre ?
[Ans. mr^2.]

17. The centre of a wheel is rolling on a plane surface moves with a speed v. What is the speed of a particle on the rim of the wheel at the same level as the centre ?
[Ans. $\sqrt{2}v$.]

18. Two metal spheres of different densities and equal masses are made. Which sphere, either will have more moment of inertia about the diameter ? **(Ujjain 2000)**

19. What must be the relation between length l and radius R of a cylinder, if the moment of inertia of the cylinder about its axis is to be the same as its moment of inertia about the equatorial axis ?

[**Ans.** $l = \sqrt{3}R$; In the given condition $\frac{1}{2}mR^2 = m\left[\frac{l^2}{12}+\frac{R^2}{4}\right]$ or $\frac{R^2}{4} = \frac{l^2}{12}$ or $l = \sqrt{3}R$.]

20. The moment of inertia of a thin circular ring about an axis through its centre and perpendicular to its plane is 200 gm × cm. Calculate its moment of inertia about its any diameter. **(Sagar 1996)**
[**Ans.** 100 gm × cm^2]

21. A disc of mass M and radius R is cut about diameter AB and a semi-circular disc of double thickness is formed by putting one part over the other. Calculate the moment of inertia about AB axis.
[**Ans.** $MR^2/4$] **(Gwalior 1989)**

22. If a solid sphere of moment of inertia I about its diameter is converted into a disc of moment of inertia I about its axis, find the ratio of their radii. **(Gwalior 1989)**

[**Ans.** $\sqrt{5}/2$, $I = \frac{2}{5}mR_1{}^2 = \frac{1}{2}mR_2{}^2$.]

23. Two particles of mass 1 kg each are revolving about an axis at perpendicular distances of 1 m and 2 m; calculate the radius of gyration.
[**Ans.** 1.58 m.]

24. Two spheres of same mass and same external radius are exactly similar in appearance. One of them is hollow while the other is solid. Explain clearly how can you identify them ?

25. A solid sphere rolls down two different inclined planes of the same height but different inclines. Will it reach the bottom with the same speed in each case ? Will it take longer to roll down one incline than the other ? If so, which one and why ?

26. Show that the ratio of rotational to translational kinetic energy for a solid cyliner rolling down a plane without slipping is 1 : 2.

27. A solid cylinder of mass M rolls without slipping down a plane of length l inclined at an angle θ with the horizontal. Show that the velocity of the centre of mass of the cylinder at the bottom of the plane is $(2/\sqrt{3})(gl \sin\theta)^{1/2}$. If, however, the cylinder slides down the plane (without rolling) what is the final velocity ?

28. A circular hoop of radius R starts rolling down a smooth inclined plane without slipping. Show that its acceleration down the plane is $\frac{1}{2}g \sin\theta$. **(Rohilkhand 1980)**

29. A spherical shell and a solid sphere roll down an inclined plane. Show that their acceleration will be in the ratio of 21 : 25. **(Rajasthan 1983)**

30. A body of mass m and radius r rolling on a horizontal surface with velocity v (without slipping), rises to a maximum height h on a hill. If $h = 3v^2/4g$, determine the moment of inertia of the body.

[**Ans.** $\frac{1}{2}mr^2$.]

31. A solid body of radius r and mass m rolls on a plane surface with angular velocity ω. Find its K.E. **(Lucknow 2004)**

32. Find an expression for the acceleration of a body rolling down an indined plane without slipping. Hence show that it is independent of mass of body. **(Kanpur 2004)**

Objective Questions

1. Dimensional formula of moment of inertia is :

(a) $[MLT^2]$ (b) $[ML^2T]$ (c) $[ML^2T^0]$ (d) $[M^0L^2T]$

(Rewa 1997)

2. A point particle of mass 1.0 gm is revolving on a circle of diameter 8 cm. Its moment of inertia is :

(a) $2 \text{ gm} \times \text{cm}^2$ (b) $4 \text{ gm} \times \text{cm}^2$ (c) $8 \text{ gm} \times \text{cm}^2$ (d) $16 \text{ gm} \times \text{cm}^2$

(Bhopal 1991)

3. A ring, a disc, a solid sphere and a spherical shell has same mass and radius which one has least moment of inertia about its geometrical axis ?

(a) ring (b) solid sphere (c) disc (d) spherical shell

(Kanpur 2004)

4. A cylinder of mass 80 kg and radius 2 m is rolling on a table without sliding. If its velocity if 5 m/s then its total K.E. is :

(a) 1500 joule (b) 700 joule (c) 1000 joule (d) zero

(Rewa 2000)

5. The moment of inertia of a disc (mass M, radius R) about an axis passing through its centre and perpendicular to its plane is :

(a) $\frac{2}{5}MR^2$ (b) $\frac{1}{2}MR^2$ (c) $\frac{1}{5}MR^2$ (d) MR^2

(Rewa 1999)

6. Moment of Inertia × Angular velocity is equal to :

(a) couple of force (b) force (c) angular momentum (d) work done

(Bhopal 1991)

7. The K.E. of a body rotating about its axis with an angular velocity of 1 radian/sec is E. The moment of inertia of the body about the same axis is :

(a) E (b) $2E$ (c) $E/2$ (d) $\sqrt{E}$ **(Gwalior 1990)**

8. Moment of inertia of a solid sphere and spherical shell of equal masses about their diameters will be :

(a) more for solid sphere (b) less for solid sphere

(c) the same (d) more for any sphere **(Gwalior 1990)**

9. The boy is standing on a rotating table with dumb-bells in his hands. He suddenly withdraws his hands to his chest. The angular velocity of the table will :

(a) decrease (b) increase (c) remain same (d) become zero

(Agra 2003)

10. The moment of inertia of any disc about the axis, parallel to its diameter and passing through its circumference is :

(a) $\frac{1}{4}mR^2$ (b) $\frac{5}{4}mR^2$ (c) $\frac{1}{2}MR^2$ (d) $\frac{3}{2}MR^2$

(Agra 2005)

11. Which one of the following bodies of equal mass, has maximum moment of inertia about an axis passes through the centre of gravity and perpendicular to its plane ?

(a) disc of radius a (b) ring of radius a

(c) square lamina of arm $2a$ (d) four rods of length $2a$ forming a square

(Jabalpur 1990)

12. A solid sphere, a disc, a spherical shell and a ring of equal mass and diameter are released from the same height at the same time on an inclined plane. All roll down without slipping. Which one will reach the bottom first ?

(a) disc (b) sphere (c) ring (d) spherical shell

(Bhopal 1990)

13. The moment of inertia of a body rotating with unit angular velocity is euqal to :

(a) twice its kinetic energy of rotation (b) half of its kinetic energy of rotation

(c) its kinetic energy of rotation (d) four times its kinetic energy of rotation

(Bundelkhand 2001)

14. We have two spheres of same mass one of which is spherical shell and another a solid. They have same moment of inertia about their respective diameters. The ratio of their radii is given by :

(a) 5 : 7 (b) 3 : 5 (c) $\sqrt{3}:\sqrt{5}$ (d) $\sqrt{3}:\sqrt{7}$

(Kanpur 2001)

15. Four solid spheres each of diameter $2a$ and mass m are placed with their centres on the four corners of a square of side b. The moment of inertia about one side of square is :

(a) $\frac{8}{5}ma^2 + 2mb^2$ (b) $\frac{8}{5}ma^2 + mb^2$ (c) $\frac{8}{5}ma^2 + 4mb^2$ (d) $\frac{8}{5}ma^2 + \frac{1}{2}mb^2$

(Agra 2005)

16. If a rigid body is rotating with an angular velocity $\vec{\omega}$ about an instantaneous axis through a fixed point in the body, the anguar momentum vector $\vec{J}$ about the same point :

(a) will be always in the direction of $\vec{\omega}$. (b) may be in the direction of $\vec{\omega}$.

(c) may have different direction to that of $\vec{\omega}$. (d) will be always perpendicular to $\vec{\omega}$.

17. A sphere of mass M and radius r slips on a rough horizontal surface. At some instant, it has horizontal velocity v and rotational velocity $v/2r$. The translational velocity after the sphere starts pure rolling is :

(a) v (b) $6v/7$ (c) zero (d) $v/2$

18. If I_1, I_2 and I_3 represent the principal moments of inertia of a rigid body and $\vec{\omega} = (\omega_1, \omega_2, \omega_3)$ is the angular velocity with components along the three principal axes :

(a) the z-component of the torque acting on the body in general is $T_3 = I_3\dot{\omega}_3$

(b) the z-component of the torque acting on the body in general is $T_3 = I_3\dot{\omega}_3 + (I_2 - I_1)\omega_1\omega_2$

(c) for torque free motion of the rigid body, in general we have $I_3\omega_3 =$ constant

(d) for torque free motion of the rigid body, in general we have $I_3\dot{\omega}_3 = (I_1 - I_2)\omega_1\omega_2$.

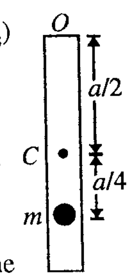


Fig. 6.37.

19. A particle of mass m is attached to a thin rod of length l and mass $4m$. The distance of the particle from the centre of mass of the rod is $a/4$. The moment of inertia of the combination about an axis passing through O normal to the rod **[fig. 6.37]** is :

(a) $\frac{64}{48}ma^2$ (b) $\frac{91}{48}ma^2$ (c) $\frac{27}{48}ma^2$ (d) $\frac{51}{48}ma^2$

ANSWERS

1. (c),	**2.** (d),	**3.** (b),	**4.** (a),	**5.** (b),	**6.** (c),	**7.** (b),	**8.** (b),
9. (b),	**10.** (b),	**11.** (d),	**12.** (b),	**13.** (b),	**14.** (c),	**15.** (a),	**16.** (b), (c)
17. (b),	**18.** (b),(d)	**19.**(b).					

7

Simple Harmonic Motion

7·1. Oscillatory Motion

Oscillatory or vibratory phenomenon is very common in nature. In physics, familiar examples of this phenomenon are : (i) oscillations of a simple pendulum, (ii) oscillations of spring-mass system, (iii) vibrations of a tuning fork, (iv) vibrations of a guitar string etc. The terms oscillation and vibration have practically the same meaning and both are used for the same purpose. In general usage, an oscillation is possibly slower, but we shall not consider any such distinction. The vibratory behaviour is responsible for all kinds of wave phenomena. In sound waves, particles of the medium and in electromagnetic waves, electric and magnetic field vectors perform oscillatory motions. In fact, the study of oscillatory phenomena is of extreme importance in physics.

The feature that the oscillatory phenomena have in common is periodicity. If a particle moves such that it retraces its path regularly after equal intervals of time, its motion is said to be **periodic**. This inerval of time, required to complete one round trip (cycle) of motion is called **period**. Now, if a particle in periodic motion moves back and forth over the same path about an equilibrium position, we call the motion **vibratory** or **oscillatory**.

The time taken for the system to undergo one complete oscillation is the period T of the system. Since the system executes $1/T$ vibrations per unit time, this quantity is called the frequency of vibration (ν), given by

$$\nu = \frac{1}{T}$$

The dimensions of frequency are $(\text{time})^{-1}$. We sometimes express frequency in cycles (or vibrations) per second. One cycle per second (cps) is known as one hertz (Hz). The **hertz** is the SI unit of frequency.

7·2. Potential Energy Curve and Equilibrium

In general, if the potential energy (U) is a continuous function of position, then a graph may be plotted to show the variation of potential energy with the position of the particle. Such a graph is known as **potential energy curve. Fig. 7.1** shows a possible potential energy curve $U - x$ for a particle constrained to move along X-axis.

In fact, the nature of the potential energy curve is very useful in understanding the motion of a particle, even without solving its equation of motion. We can discuss a complex force more conveniently in terms of potential energy curve, because potential energy is a scalar quantity and therefore easy to handle than the force itself which is a vector. The force on the particle can be found by using the relation $F = - dU/dx$.

In **fig. 7.1**, the slope (dU/dx) of the potential energy curve at any point gives the force ($F = - dU/dx$), acting on the particle placed at that point. The slope (dU/dx) is positive, whenever U increases with the increase in x (*e.g.*, for the part AB and CD of the curve) and negative whenever U decreases with the increase in x (*e.g.*, for the part GA and BC). Therefore, the force $F(= - dU/dx)$ is negative (*i.e.*, towards negative X-axis) or directed to the left whenever the potential energy is increasing with x, and positive or directed to the right whenever the potential energy is decreasing with x [**fig. 7.1**]. This means that *the force, acting on the particle at any point, tries to bring the particle to the region of lower potential energy.*

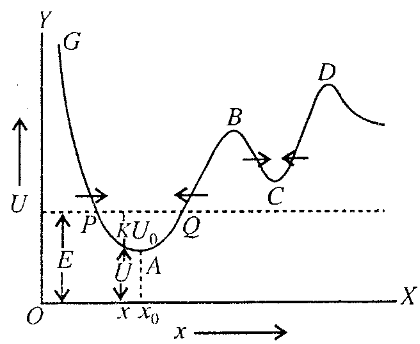


Fig. 7.1 : Potential energy curve.

Stable, unstable and neutral equilibrium : A system is said to be in **stable equillibrium**, if a small displacement of the system from the rest position (by giving a little energy to it) results in a small bounded motion about the equilibrium position. In case, small displacement of the system from the equilibrium position results in an unbounded motion, it is in an **unstable equilibrium**. Further, if the system on displacement has no tendency to move about or away the equilibrium position, it is said to be in **neutral equilibrium**. An example of stable equilibrium is a pendulum in the rest position and that of an unstable equilibrium is an egg standing on one end. A coin or a book placed flat anywhere on a table is in neutral equilibrium.

In the **fig. 7.1**, at the points A, B, C and D, the potential energy U is minimum or maximum, the value of dU/dx is zero. Therefore, if a particle is placed at such a point with **zero** velocity, it will experience no force ($F = -\,dU/dx = 0$) and *so it will remain at rest.* These points of the curve are known as the **positions of equilibrium.**

In the figure, B and D are the points corresponding to maximum potential energy. A particle at rest at such a point will remain at rest. However, if the particle is displaced even the slightest distance from this point the force $F = -\,dU/dx$ will tend to push the particle farther away from the equilibrium position. Hence the points B and D are the **positions of unstable equilibrium.**

In the figure, A and C are the points corresponding to minimum potential energy. A particle at rest at such a point will remain at rest. If the particle is displaced slightly in either direction by giving a little energy, the force $F = -\,dU/dx$ will tend to take it back towards the equilibrium position. Hence the points A and C are the **positions of stable equilibrium.**

In case, if U is constant in a region, then the slope dU/dx and hence the force ($F = -\,dU/dx$), acting on a particle placed at any point of that region, is zero. In such a region, if a particle is displaced from one point to the other, it will remain there without experiencing any restoring force. So we call the region of constant potential energy as the region of *neutral equilibrium.*

7·3. Potential Well and Oscillatory Motion

In **fig. 7.2(a)** the potential energy has a minimum at the point O ($x = 0$) and therefore the point O is the position of stable equilibrium, Suppose the particle is displaced from the point O (where the potential energy is U_0) by giving some energy so that its total energy E, represented by the horizontal line PQ, is greater than U_0. The particle will oscillate under the action of restoring force ($F = -\,dU/dx$) between the turning points P and Q (or A and B) about the point O. Note that, in general, the force versus displacement

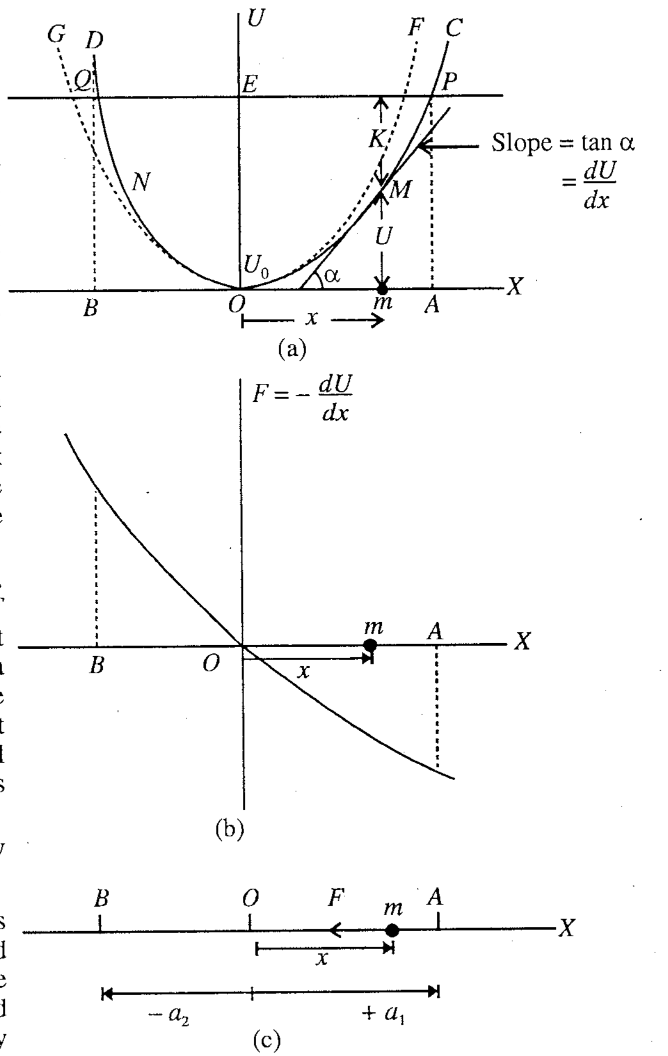


Fig. 7.2 : Small amplitude and anharmonic oscillations : (a) Potential energy curve [$U - x$] with a minimum at O, (b) Force versus displacement curve [$F - x$], (c) Particle displacement between turning points $x = +\,a_1$ and $x = -\,a_2$; Observe that in (a) for small x, parabola (GOF) coincides with the potential energy curve (DOC).]

curve [**fig. 7.2(b)**] is not a straight line and the turning points are not situated symmetrically about O [**fig. 7.2(c)**]. During an oscillation, there is mutual transference of energy between kinetic and potential forms with the condition that the total energy E (= $K + U$) remains constant, presuming that no frictional] forces are acting on the system. The oscillating particle is confined in the region between the points P and Q. Such a region is called **bounded region** or **potential well** and if the particle is limited to this region, it is said to be in the **bounded state.** Hence the bounded region is present near a point of stable equilibrium. The difference of potential energy corresponding to the top and the bottom of the potential well is the **binding energy** for that potential well. If a particle is in a potential well, it will remain there in the bounded state in the condition that its total energy E or $U + K$ is less than the potential energy corresponding to the top point (C or D).

Harmonic and anharmonic oscillations : The potential function U can be expanded in a Taylor's series about the point of stable equilibrium 0 *i.e.,*

$$U = U_0 + \left(\frac{dU}{dx}\right)_0 x + \frac{1}{2!}\left(\frac{d^2U}{dx^2}\right)_0 x^2 + \frac{1}{3!}\left(\frac{d^3U}{dx^3}\right)_0 x^3 + ... \qquad ...(1)$$

where U_0, $\left(\frac{dU}{dx}\right)_0$, etc. are the values at $x = 0$. As the point O is the point of minimum potential energy, the value of $\left(\frac{dU}{dx}\right)_0$ should be zero and $\left(\frac{d^2U}{dx^2}\right)_0$ should be positive quantity, say C. Putting C_1, C_2 etc., in place of $\left(\frac{d^3U}{dx^3}\right)_0$, $\left(\frac{d^4U}{dx^4}\right)_0$ ...., we get

$$U = U_0 + \frac{C}{2!}x^2 + \frac{C}{3!}x^3 + ... \qquad ...(2)$$

If we assume the minimum energy $U_0 = 0$, then we have

$$U = \frac{1}{2}Cx^2 + \frac{1}{6}C_1x^3 + \frac{1}{24}C_2x^4 + ... \qquad ...(3)$$

The force acting on the particle is given by

$$F = -\frac{dU}{dx} = -Cx - \frac{C_1x^2}{2} - \frac{C_2x^2}{6} - .... \qquad ...(4)$$

If C_1, C_2 .... etc, are zero, then

$$F = -Cx \text{ and } U = \frac{1}{2}Cx^2 \qquad ...(5)$$

In such a case, if a graph is plotted between U and x, it will represent a parabola *i.e., the potential well is* **porabolic** *in shape* [GOF in **fig.7.2(a)**]. Here, the force acting on the particle is proportional to the displacement in a direction opposite to it. The constant $C = \left(\frac{d^2U}{dx^2}\right)_0$ is a positive quantity because $x = 0$ is the point of minimum potential energy. This positive constant C is called the **force constant.** Actually, this is the case of a harmonic oscillator, for which essentially a parabolic potential function should exist.

In case of oscillations of small amplitude, the displacement x is small so that terms containing higher powers of x are negligible. Thus from eqs. (3) and (4), we have $U = \frac{1}{2}Cx^2$ and $F = -Cx$. *Hence for small amplitude oscillations, the potential well may be treated as parabolic and the oscillations are simple hormonic type.* **Fig. 7.2** shows that for small x, parabola coincides to the general potential well and near $x = 0$, $F - x$ curve is close to a straight line.

If C_1, C_2, ... are not zero and the amplitude is not small, the potential energy well is not parabolic and in general the oscillations of the system will be **anharmonic**. Such an oscillating system is called **anharmonic oscillator**. Eqs. (3) and (4) represent expressions for potential energy and force for a particle, executing anharmonic oscillations. For example, if the amplitude is not small, the oscillations of a mass attached to a spring or simple pendulum are anharmonic.

7·4. Nature of Simple Harmonic Motion—Oscillations of a Model Spring-Mass System

Any system has the capability of vibrations. These vibrations may be simple or complex. The oscillations of a system corresponding to a displacement versus time graph to be sinusoidal are called simple harmonic oscillations. In order to understand the basic features of oscillatory phenomenon, we consider a spring-mass arrangement as a model system **[fig. 7.3]**. The arrangement consists of a mass m attached to one end of a massless spring whose other end is fixed to a rigid wall. The system lies on a frictionless horizontal table and O is the equilibrium position of the mass m. Suppose O be the origin at $x = 0$ and the line along the length of the spring be X-axis.

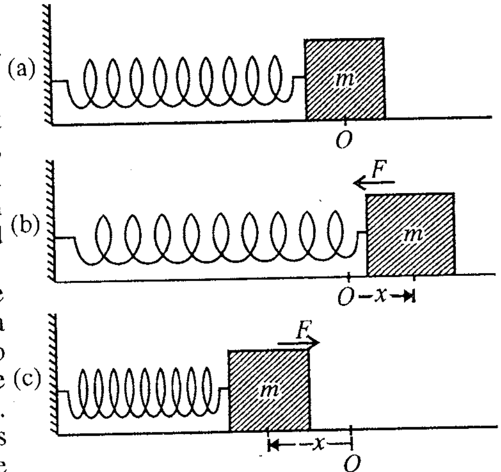


Fig. 7.3 : Spring-mass system : (a) system in equilibrium, (b) restoring force during extension, (c) restoring force during compression.

Restoring force in the spring-mass system : Let the spring be extended by pulling the mass m. Consequently a restoring force is developed in the spring which tries to bring back the mass towards the equilibrium position. We observe that more the extension, more is the restoring force. In case if the spring be compressed, the spring force acts again towards the equilibrium position. In both cases, the direction of the restoring force is opposite to the displacement. According to Hooke's law for small displacement the restoring force F is given by

$$F = -Cx \quad \text{or} \quad \vec{F} = -C\vec{x} \qquad ...(1)$$

where C is some positive constant, known as **force constant** or **spring factor.** Thus *force constant is the force exerted by the spring per unit extension* and its S.I. unit is newton/metre (N/m).

Potential energy of the spring-mass system : Let the spring be extended (or compressed) slowly by the application of an external force F_e *e.g.*, by hand such that it counterbalances the restoring force *i.e.*, $F_e = -F = Cx$. Work will be done on the system by us and this work will become its potential energy. If the potential energy of the system in the undisplaced position is assumed to be zero, then the potential energy at the displacement x is obtained to be

$$U = \int_0^x F_e dx = -\int_0^x F\,dx = \int_0^x Cx dx = \frac{1}{2}Cx^2 \qquad ...(2)$$

If a graph is plotted between U and x, it is a *parabola*. Thus the potential energy of the spring-mass system is the minimum (zero) in the equilibrium position and it increases symmetrically proportional to the square of the displacement on either side of the origin O **[fig. 7.4]**.

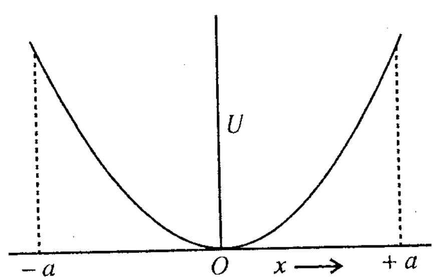


Fig. 7.4.

The point O, where the potential energy is minimum, is the **position of stable equilibrium,** because at $x = 0$, $F = -dU/dx = -Cx$ *i.e.*, force $F = 0$ and if the particle is displaced towards either side of O, the restoring force tries to bring back the system in the equilibrium position. Obviously the particle is capable to have oscillatory motion about the position of stable equilibrium. In fact, this is the necessary condition for oscillatory motion about a point that the potential energy of the particle is minimum at that point.

Oscillations of the model spring-mass system : If the mass, after displacing to a maximum (say $x_{max} = a$) is released, it moves towards the equilibrium position under the action of the restoring force. Kinetic energy and consequently momentum is developed in the system which is maximum at the equilibrium position. The mass overshoots the origin and starts to displace the other side. Now the mass is acted on by a restoring force in the opposite direction; its velocity decreases and kinetic energy changes to potential form. At the maximum displacement in the opposite side ($x_{max} = -a$), the entire energy is in the potential form and the mass is instantaneously at rest. The restoring force in this position accelerates the mass back towards the equilibrium position where it overshoots again. As the time elapses, the direction of the displacement alternates continually and the system executes vibrations. In absence of external influence, oscillations of the system are called **free** or **natural** oscillations. One may note that when the model spring-mass system is executing oscillations, the restoring force acting on the mass is proportional to its displacement from the equilibrium position in a direction opposite to it. In fact, we define below such an oscillating system as a harmonic oscillator and its oscillations to be simiple harmonic. We shall establish the differential equation of such an oscillating system and find that its solution gives the displacement versus time graph to be simusoidal in form which is the characteristic of the simple harmonic motion.

It is to be noted that oscillatory phenomenon results from the two intrinsic properties of the system : (i) elasticity of the spring or spring factor, and (ii) inertial property of the mass or mass factor. The mass tries to come back to its equilibrium position due to spring factor, while the inertial property of the mass makes it to overshoot the equilibrium position. When we discuss various vibrating systems as examples of harmonic oscillator, we will observe that a pair of quantities analogous to spring and mass factors exists in each case.

7·5. Simple Harmonic Motion and Harmonic Oscillator

Simple harmonic motion is a particular type of periodic motion and is very common in nature. *A system is said to execute* **simple harmonic motion** *when it vibrates periodically in such a way that at any instant the restoring force acting on it is proportional to its displacement from its position of stable equilibrium in its path and is always directed towards that position.*

A system executing simple harmonic motion is called **harmonic oscillator.**

Let us consider a particle of mass m, executing simple harmonic motion about the equilibrium position O with maximum displacement a on either side [**fig. 7.5(a)**]. At any instant t, if the displacement of the particle is x, then its acceleration will be d^2x/dt^2. Now, according to the definition of simple harmonic motion,

Restoring force (F) $\propto$ – displacement (x)

or $$m\frac{d^2x}{dt^2} \propto -x$$

or $$m\frac{d^2x}{dt^2} = -Cx \quad ...(1)$$

where C is the force constant. Here, the negative sign indicates that the force on the particle is directed opposite to x increasing.

The potential energy U of the particle at the displacement x is given by

$$U = -\int_0^x F\,dx$$

$$= \int_0^x Cx\,dx = \frac{1}{2}Cx^2 \quad ...(2)$$

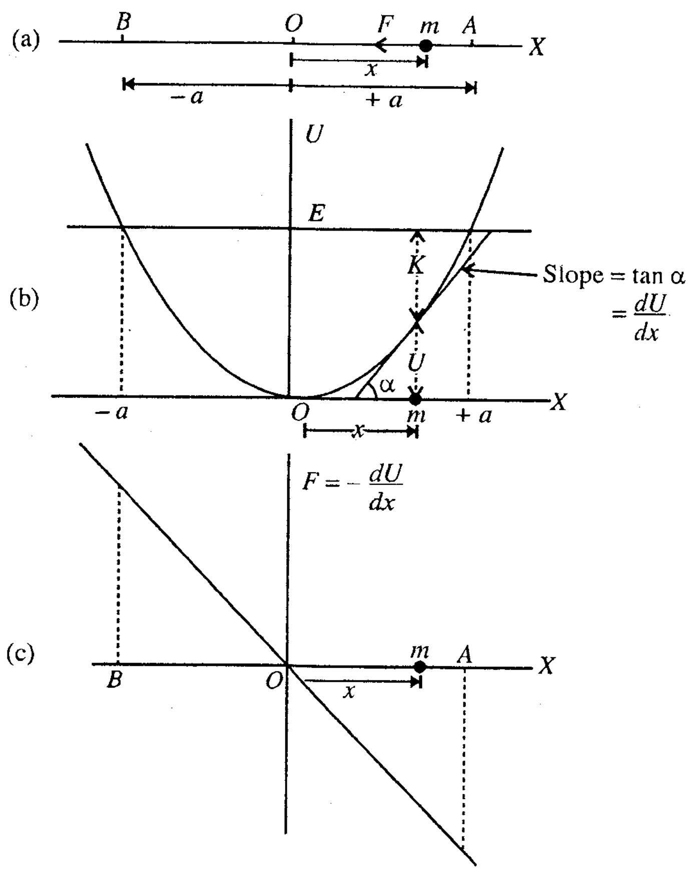


Fig. 7.5 : Simple harmonic motion : (a) Particle displacement between turning points $x = +a$ and $x = -a$, (b) Parabolic potential energy curve, (c) Force versus displacement curve (straight line).

The variation of force (F) and potential energy (U) of the particle, executing simple harmonic motion, with displacement (x) are shown in **fig. 7.5(c)** and **fig. 7.5(b)** respectively. Since U varies as x^2, the potential energy will be the same at the maximum displacement $x_{max} = +a$ or $-a$. This maximum displacement a is the **amplitude** of motion.

Thus the simple harmonic motion is characterized with the following properties :

(1) It is a periodic motion with a certain periodic time T.

(2) The restoring force on the particle is proportional to the displacement from the equilibrium position in a direction opposite to it [**fig. 7.5(c)**].

(3) The potential energy of the oscillating particle varies as the square of the displacement [**fig. 7.5(b)**].

(4) The truning points are situated at equal distances from the equilibrium position.

7·6. Differential Equation of Simple Harmonic Motion and its Solution

For simple harmonic oscillations, we have

$$\frac{d^2x}{dt^2} + \frac{C}{m}x = 0 \qquad ...(1)$$

Let us put $\omega^2 = C/m$. Then

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \qquad ...(2)$$

Eq. (2) is called the **differential equation of simple harmonic motion.**

In order to find the displacement of the particle in S.H.M., we solve eq. (2). Multiplying eq. (2) by $2dx/dt$, we obtain

$$\frac{2dx}{dt}.\frac{d^2x}{dt} + \omega^2 \cdot 2x\frac{dx}{dt} = 0$$

Integrating it, we obtain

$$\left(\frac{dx}{dt}\right)^2 + \omega^2 x^2 = K, \text{ where } K \text{ is a constant of integration.} \qquad ...(3)$$

The velocity of the oscillating particle is zero at the maximum value of the displacement *i.e.*, at $x = x_{max} = a$ (say), $dx/dt = 0$ *i.e.*, $0 + \omega^2 a^2 = K$ or, $K = \omega^2 a^2$.

Substituting the value of K in eq. (3), we get

$$\left(\frac{dx}{dt}\right)^2 + \omega^2 x^2 = \omega^2 a^2 \quad \text{or} \quad \left(\frac{dx}{dt}\right)^2 = \omega^2(a^2 - x^2)$$

or
$$\frac{dx}{dt} = \pm\omega\sqrt{a^2 - x^2} \qquad ...(4)$$

If we take positive root, then eq. (4) can be written as

$$\frac{dx}{\sqrt{a^2 - x^2}} = \omega dt \quad \text{or} \quad \sin^{-1}\left(\frac{x}{a}\right) = \omega t + \phi$$

or
$$x = a \sin(\omega t + \phi) \qquad ...(5)$$

where ϕ is a constant of integration. If we take negative root, then

$$-\frac{dx}{\sqrt{a^2 - x^2}} = \omega dt \quad \text{or} \quad \cos^{-1}\left(\frac{x}{a}\right) = \omega t + \varphi$$

or
$$x = a \cos(\omega t + \varphi) \text{ where } \varphi \text{ is a constant of integration.} \qquad ...(6)$$

Eq. (5) and eq. (6), represent sine and cosine forms of the solutions and both are valid solutions of the differential equation (2) of the harmonic oscillator. The two solutions become identical, if the two constants of integration ϕ and φ are related as $\varphi = \phi - \pi/2$. It is to be seen from eqs. (5) and (6) that in simple harmonic motion, the displacement is represented as a sine or cosine function of time and if a graph is plotted between displacement (x) and time (t), it is called a **sinusoidal graph [fig. 7.6]**. In other words, *the motion represented by sinusoidal graph, is known as simple harmonic motion.* In our later analysis, we shall use the sine form of the solution [eq. (5)].

Amplitude and phase. When $\sin(\omega t + \phi) = 1$ or -1, then displacement has the maximum value $a\ (= x_{max})$ on either side of the equilibrium position $x = 0$. Hence a is the **amplitude** of vibration.

The term $(\omega t + \phi)$, occurring in the equation (5) for displacement x, is called the **phase** of vibration. The phase tells us the position of the oscillating particle at any time t. At $t = 0$, the value of the phase is ϕ and therefore ϕ is called **initial phase** or **phase constant.** *This gives the information regarding the initial position from where we started to measure the displacement.* Suppose at $t = 0$, the particle is at $x = x_0$. So that from eq. (5), we have

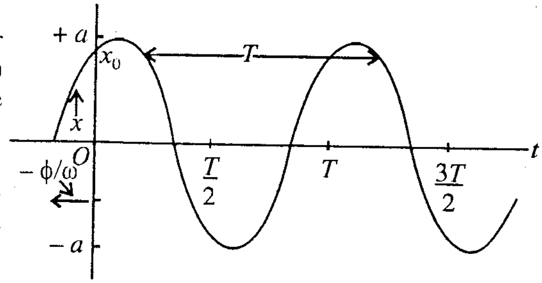


Fig. 7.6 : Simple harmonic motion, $x = a \sin(\omega t + \phi)$, of period T, amplitude a and phase constant ϕ.

$$x_0 = a \sin \phi \qquad ...(7)$$

One may note that eq. (2) will be satisfied by eq. (5) for any finite value of a and ϕ. These constants a and ϕ in fact are found from the boundary conditions for a particular simple harmonic motion.

If we measure the time, when the particle is at the equilibrium position *i.e.*, at $t = 0$, $x = 0$, then from eq. (5), we obtain

$$\sin \phi = 0 \quad \text{or} \quad \phi = 0$$

In this situation, eq. (5) is obtained to be

$$x = a \sin \omega t \qquad ...(8)$$

For $\phi = 0$, we have drawn the displacement-time graph in **fig. 7.7(a).**

For $\phi = \frac{\pi}{4}, \frac{\pi}{2}$ and π the graphs are shown in **fig. 7.7(b), (c)** and **(d)** respectively. In all cases, the graphs, representing SHM's, have exactly the same shape, but there is the shifting of the origin along the time axis. If **fig. 7.7(b)** and **fig. 7.7(c)** represent two separate simple harmonic vibrations, then $\pi/4$ is the phase difference between them and the later is leading in phase. The phase difference for the **fig. 7.7(a)** and **fig. 7.7(d)** is π and the two vibrations are said to be in opposite phase.

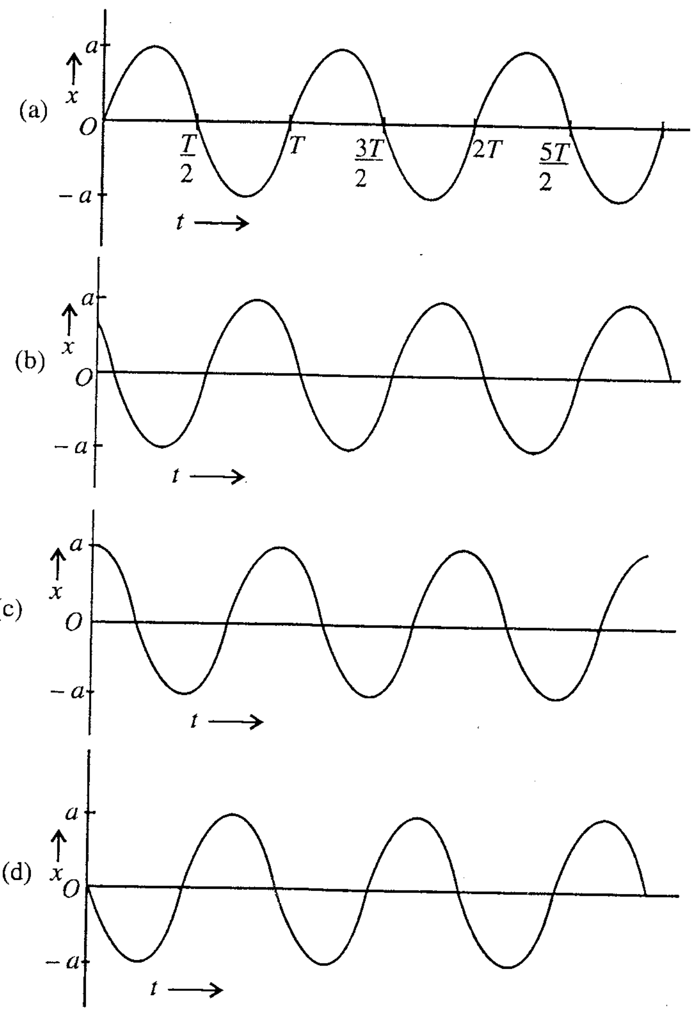


Fig. 7.7 : Displacement-time graphs for S.H.M. represented by $x = a \sin(\omega t + \phi)$: (a) $\phi = 0, x_0 = 0$, (b) $\phi = \pi/4, x_0 = a/\sqrt{2}$, (c) $\phi = \pi/2, x_0 = a$ and (d) $\phi = +\pi, x_0 = 0$.

Period and frequency : Now, let us find the **physcial significance** of the constant ω. If the time t in eq. (5) is increased by $2\pi/\omega$, the function becomes

$$x = a \sin[\omega(t + 2\pi/\omega) + \phi] \quad \text{or}$$

$x = a \sin [\omega(t + 2\pi + \phi)$ or $x = a \sin (\omega t + \phi)$.

That is, the displacement of the particle is the same after a time $2\pi/\omega$. Therefore, $2\pi/\omega$ is the **period** (T) of the motion, *i.e.*,

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{C}} \qquad \left(\because \omega^2 = \frac{C}{m}\right) \qquad ...(9)$$

The number of vibrations per second is called the frequency (υ) of the oscillator and is given by

$$\upsilon = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{m}{C}} \qquad ...(10a)$$

$$\therefore \quad \omega = 2\pi\upsilon = 2\pi/T \qquad ...(10b)$$

The quantity ω is called the **angular frequency.**

Velocity and acceleration : The displacement of the mass executing S.H.M. is given by

$$x = a \sin (\omega t + \phi)$$

Its velocity (v) is given by

$$v = \frac{dx}{dt} = \omega a \cos (\omega t + \phi) \qquad ...(11a)$$

or $$v = \omega a\sqrt{1 - \sin^2(\omega t + \phi)} \qquad ...(11b)$$

$$= \omega a\sqrt{1 - x^2/a^2} = \omega\sqrt{a^2 - x^2}$$

Its acceleration is

$$\frac{dv}{dt} = \frac{d^2x}{dt^2} = -\,\omega^2 \sin (\omega t + \phi) \qquad ...(12a)$$

or $$\frac{dv}{dt} = \frac{d^2x}{dt^2} = -\,\omega^2 x \qquad ...(12b)$$

The variations of displacement (x), velocity (dx/dt) and acceleration (d^2x/dt^2) for $\phi = 0$ with time t are given in **fig. 7.8**. Eqs. (11b) and (12b) give the dependence of velocity and acceleration on the displacement (x). One may observe that the velocity and acceleration lead the displacement in phase by $\pi/2$ and π respectively.

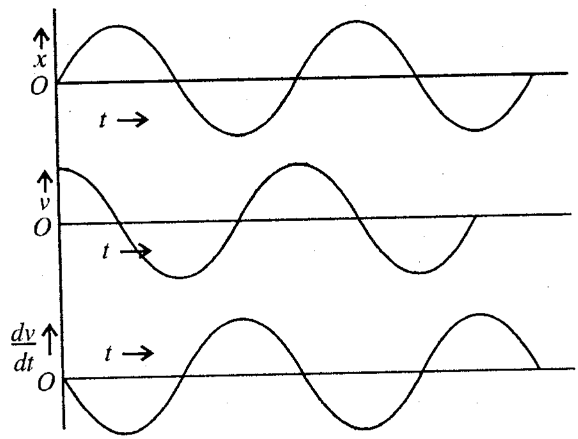


Fig. 7.8 : Variation of displacement (x), velocity (v) and acceleration, with time t.

Alternative Method (Use of Complex numbers) : With the help of the complex numbers we can often solve easily the complicated oscillatory problems. Here, we will make use of complex numbers in solving the differential equation of harmonic oscillator *i.e.*,

$$\frac{d^2x}{dt^2} + \omega^2 x = 0, \text{ where } \omega = \sqrt{\frac{C}{m}}.$$

Let the solution of this equation be $x = Ae^{\alpha t}$

Now, substituting the value of $\frac{d^2x}{dt^2}$ and x in the above equation, we have

$$A\alpha^2 e^{\alpha t} + \omega^2 A e^{\alpha t} = 0, \text{ or } \alpha^2 + \omega^2 = 0, \text{ as in general } Ae^{\alpha t} \neq 0.$$

$$\therefore \quad \alpha = \pm\sqrt{-\omega^2} = \pm i\omega, \text{ where } i = \sqrt{-1}$$

$$\therefore \quad x = A_1 e^{i\omega t} \text{ or } x = A_2 e^{-i\omega t}$$

Hence the general solution of the equation will be given by

$$x = A_1 e^{i\omega t} + A_2 e^{-i\omega t}$$

where A_1 and A_2 are constants, which can be determined from the initial conditions.

Now, $\quad x = A_1 (\cos \omega t + i \sin \omega t) + A_2 (\cos \omega t - i \sin \omega t)$

$$= (A_1 + A_2) \cos \omega t + (A_1 i + A_2 i) \sin \omega t.$$

Let $\quad (A_1 + A_2) = a \sin \phi$ and $(A_1 i + A_2 i)\ a \cos \phi$. Then,

$$x = a \sin \phi \cos \omega t + a \cos \phi \sin \omega t$$

or
$$x = a \sin (\omega t + \phi)$$

and
$$v = dx/dt = a\omega \cos (\omega t + \phi) = \omega\sqrt{a^2 - x^2}\ .$$

7·7. Potential Energy and Kinetic Energy in Simple Harmonic Motion

In simple harmonic motion, the oscillating system possesses two types of energy :

(i) **Potential energy,** which is due to its displacement from the mean position.

(ii) **Kinetic energy,** which is due to its velocity.

Hence, at any instant the total energy of the oscillator will be the sum of these two energies. As the system is conservative (*i.e.*, no dissipative or frictional forces are acting), the total mechanical energy E $(= K + U)$ must be conserved.

At the displacement x, the expressions for the potential energy (U), kinetic energy (K) and total energy (E) are given by

$$U = \frac{1}{2}Cx^2 = \frac{1}{2}m\omega^2x^2 \qquad [\because \omega^2 = C/m] \qquad \text{...(1a)}$$

or
$$U = \frac{1}{2}m\omega^2a^2 \sin^2 (\omega t + \phi) \qquad [\because x = a \sin (\omega t + \phi)] \qquad \text{...(1b)}$$

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(a^2 - x^2) = \frac{1}{2}C(a^2 - x^2) \qquad \text{...(2a)}$$

or
$$K = \frac{1}{2}m\omega^2[a^2 - a^2 \sin^2(\omega t + \phi)] = \frac{1}{2}m\omega^2a^2 \cos^2(\omega t + \phi) \qquad \text{...(2b)}$$

and
$$E = U + K = \frac{1}{2}Cx^2 + \frac{1}{2}C(a^2 - x^2) = \frac{1}{2}Ca^2$$

$$= \frac{1}{2}m\omega^2a^2 = 2m\pi\upsilon^2a^2 \qquad \text{...(3)}$$

Thus, we see that the total mechanical energy is constant, as we expect and has the value $\frac{1}{2}Ca^2$, *i.e.*, the total energy is proportional to the square of the **amplitude**. Hence, if the system is once oscillated the motion will continue for indefinite period without any decrease in amplitude, provided no damping (frictional) forces are acting on the system.

It is clear from the expression $U = \frac{1}{2}Cx^2$ that if we plot a graph between the potential energy U and displacemetn x, we get a parabola having vertex at $x = 0$ [**fig. 7.9**]. The curve for the total energy E (constant) is the horizontal line. The particle cannot go beyond the points where this line intersects the potential energy curve because U can never be larger than E. These points are **turning points** of the motion and correspond to the maximum displacement. At these positions (*i.e.*, at $x = \pm a$) the total energy of the oscillator is wholly potential (*i.e.*, $E = U = \frac{1}{2}Ca^2$) but the kinetic

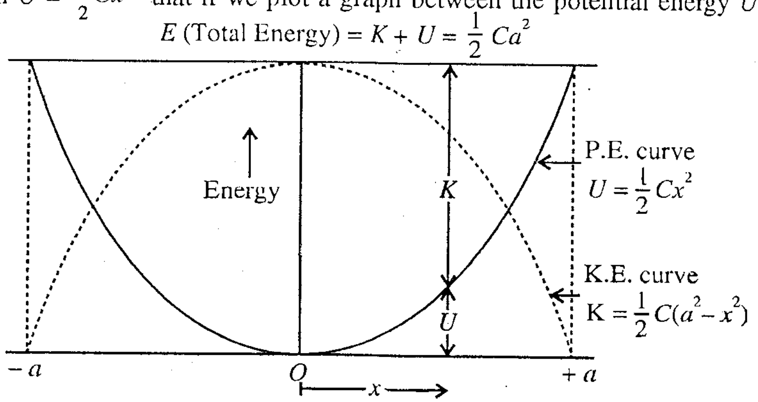


Fig. 7.9 : Variation of potential energy (U), kinetic energy (K) and total energy (E) with displacement (x) in simple harmonic motion.

energy is zero and so the amplitude of motion is $a = \pm\sqrt{2E/C}$. At the equilibrium position, the potential energy is zero, but the kinetic energy has the mximum value $E = \frac{1}{2}mv_{\max} = \frac{1}{2}Ca^2 = E$ with maximum velocity $v_{\max} = \sqrt{2E/m}$. At other intermediate points the energy is partly kinetic and partly potential but their sum (total energy) is always $\frac{1}{2}Ca^2$.

Average values of kinetic and potential energies : The time average of a quantity $Q(t)$ over a time interval T is given by

$$Q_{av} = \frac{1}{T}\int_0^T Q(t)\,dt \qquad \text{...(4)}$$

Hence the average kinetic energy for a period T is given by

$$K_{av} = \frac{1}{T}\int_0^T K dt = \frac{1}{T}\int_0^T \frac{1}{2}mv^2\,dt = \frac{1}{T}\int_0^T \frac{1}{2}ma^2\omega^2\cos^2(\omega t+\phi)\,dt \qquad \text{[from eq. (2)]}$$

$$= ma^2\frac{\omega^2}{2T}\int_0^{T=2\pi/\omega}\frac{1}{2}[1+\cos(2\omega t+2\phi)]\,dx = \frac{1}{4}\frac{ma^2\omega^2}{2\pi/\omega}.\frac{2\pi}{\omega} = \frac{1}{4}ma^2\omega^2 = \frac{1}{4}Ca^2 \qquad \text{...(5)}$$

Similarly, $$U_{av} = \frac{1}{T}\int_0^T U\,dt = \frac{1}{T}\int_0^T \frac{1}{2}Cx^2\,dt = \frac{1}{T}\int_0^T \frac{1}{2}Ca^2\sin^2(\omega t+\phi)\,dt = \frac{1}{4}Ca^2 \qquad \text{...(6)}$$

From eqs. (5) and (6), we get

$$K_{av} = U_{av} \qquad \text{...(7)}$$

Also, $$U_{av} + K_{av} = \frac{1}{2}Ca^2 = E = E_{av} \qquad \text{...(8)}$$

because the total energy is constant over the period.

This is an important characteristic of harmonic oscillator that *its average kinetic energy is equal to the average potential energy.*

Ex. 1. Fig. 7.10 *shows the potential energy diagram of a system like a diatomic moelcule, potential energy U being plotted against internuclear distance r. Deduce values of the following :*

(i) *Equilibrium internuclear distance* r_0.

(ii) *Binding energy* E_b.

(iii) *Force constant near the bottom of potential well.* **(Agra 1970)**

Sol. (i) r_0 **= 2 Å**

(ii) E_b = **– 4 electron volts.**

(iii) Near the bottom of the potential well, it is parabolic and hence the potential energy is given by

$$U = \frac{1}{2}Cx^2, \text{ whence } C = 2U/x^2.$$

Taking U between – 4 eV and – 2 eV *i.e.*, U = 2 eV and $x = \frac{1}{2}$ Å, then

$$C = \frac{2\times 2}{\left(\frac{1}{2}\right)^2} = 16 \text{ eV/Å}^2$$

$$= \frac{16\times 1.6\times 10^{-19}}{(10^{-10})^2} = \mathbf{256\ J/m^2}.$$

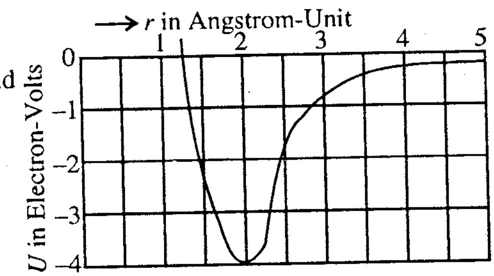


Fig. 7.10.

Ex. 2. *A particle of mass m in the potential energy field* $U = U_0 - Px + Qx^2$. *Find the expression for the force. Calculate force constant. At what point does the force vanishes ? Is this a point of stable equilibrium ? If yes, find out the period.*

Sol. $$F = -\,dU/dx = +\,P - 2Qx. \qquad \text{...(i)}$$

Evidently, this is a linear restoring force and the force constant $C = 2Q$. ...(ii)

The force vanishes, where $dU/dx = 0$ *i.e.*, $P - 2Qx = 0$ or $x = \dfrac{P}{2Q}$. ...(iii)

Now, $d^2U/dx^2 = +2Q$ *i.e.*, if Q is positive, at $x = P/2Q$, there is the minima of potential energy curve and hence this point is in stable equilibrium.

Here $$F = P - 2Qx \quad \text{or} \quad m\frac{d^2x}{dt^2} + 2Qx = P$$

or $$\frac{d^2x}{dt^2} + \frac{2Q}{m}\left(x - \frac{P}{2Q}\right) = 0 \quad \text{or} \quad \frac{d^2x'}{dt^2} + \frac{2Q}{m}x' = 0, \text{ where } x' = x - \frac{P}{2Q}. \qquad \text{...(iv)}$$

Eq. (iv) represents a simple harmonic motion whose period is given by

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{2Q}} \qquad \text{...(v)}$$

Ex. 3. *A particle of mass* 0.1 *kg lies in a potential field* $U = 5x^2 + 10$ *joule/kg. Write the differential equation of motion and deduce the frequency of oscillation.*

Sol. Potential energy of 0.1 kg, $U = 0.1\,(5x^2 + 10)$ joule.

$$\therefore \text{ Force} = -\frac{dU}{dx} = -x, \quad \therefore \quad m\frac{d^2x}{dt^2} = -x \quad \text{or} \quad 0.1\frac{d^2x}{dt^2} = -x \text{ i.e., } \frac{d^2x}{dt^2} + 10x = 0$$

This is the differential equation of S.H.M. whose frequency is $\upsilon = \sqrt{10}/2\pi$ = **0.5 Hz.**

Ex. 4. *A particle is moving with simple harmonic motion in a straight line, when the distance of the particle from the equilibrium position has the value* x_1 *and* x_2 *the corresponding values of velocity are* u_1 *and* u_2. *Show that the period is* $T = 2\pi\left[\dfrac{x_2^2 - x_1^2}{u_1^2 - u_2^2}\right]^{1/2}$ **(Agra 1991; Kanpur 83)**

Sol. Let the equilibrium position of the particle be O. If P, Q are the two positions of the particle at distances x_1 and x_2, from O respectively [**fig. 7.10**], then velocity u_1 of the particle at the position P is given by

$$u_1 = \omega\sqrt{a^2 - x_1^2} \qquad \text{...(i)}$$

The velocity u_2 at position Q is

$$u_2 = \omega\sqrt{a^2 - x_2^2} \qquad \text{...(ii)}$$

Squaring (i) and (ii) and subtracting, we get

$$u_1^2 - u_2^2 = \omega^2(x_2^2 - x_1^2) \quad \text{or} \quad \omega = \sqrt{\frac{u_1^2 - u_2^2}{x_2^2 - x_1^2}}$$

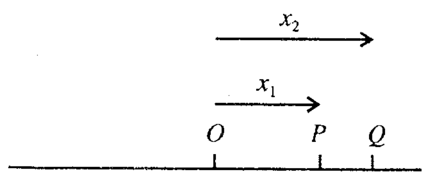


Fig. 7.11.

$$\therefore \qquad \text{time period } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{u_1^2 - u_2^2}}.$$

Ex. 5. *A particle of mass* 2 *gm moves along the X-axis and is attracted towards origin by a force* 8×10^{-3} *x newton. If is is initially at rest at* $x = 10$ *cm, find* (i) *the differential equation of motion,* (ii) *the position of the particle at any time,* (iii) *the velocity of the particle at any time,* (iv) *amplitude and frequency of vibration.*

Sol. (i) $$\text{Force } m\frac{d^2x}{dt^2} = -8 \times 10^{-3}x \quad \text{or} \quad 2 \times 10^{-3}\frac{d^2x}{dt^2} = -\,8 \times 10^{-3}x \quad \text{or} \quad \frac{d^2x}{dt^2} + 4x = 0$$

This is the differential equation of motion.

(ii) Let the solution of the equation be $x = a \sin (2t + \phi)$ and hence $\dfrac{dx}{dt} = 2a \cos (2t + \phi)$

When $t = 0$, $x = 0.1\text{m}$, $\therefore 0.1 = a \sin \phi$

When $t = 0$, $dx/dt = 0$, $\therefore$ $0 = 2a \cos \phi$, so that $a = 0$ or $\phi = \pi/2$; but $a \neq 0$, $\therefore \phi = \pi/2$

Therefore $0.1 = a \sin \pi/2$ or $a = 0.1$ m, $\therefore x = 0.1 \sin (2t + \pi/2) = 0.1 \cos 2t$

This equation gives the position of the particle at any time t.

(iii) Now, $v = dx/dt = -0.2 \sin 2t$.

(iv) Amplitude = **0.1 m**; Frequency, $\nu = \omega/2\pi = 2/2\pi = 1/\pi =$ **0.3 Hz.**

Ex. 6. *A body having a mass of 4 gm executes S.H.M. The force acting on the body, when displacement is 8 gm, is 24 gm wt. Find the period. If the maximum velocity is 500 cm/sec, find the amplitude and maximum acceleration.* ($g = 9.81$ *m/sec*^{2}).

Sol. Force acting on the body at 8 cm

$$\text{displacement} = m\omega^2 x = 4 \times 10^{-3} \times \omega^2 \times 8 \times 10^{-2} = 24 \times 10^{-3} \times 9.81$$

whence $$\omega = \sqrt{6 \times 981/8} = 27.12 \text{ rad/sec.}$$

$$\therefore \quad \text{Period } T = \frac{2\pi}{\omega} = \frac{2 \times 3.14}{27.12} = \mathbf{0.2315 \text{ sec.}}$$

The maximum velocity of the body $v = \omega a = 0.5$ m/sec.

$$\therefore \quad a = \frac{v}{\omega} = \frac{0.5}{27.12} = \mathbf{0.1844 \text{ m.}}$$

The maximum acceleration $= \omega^2 a = 735.75 \times 0.1844 =$ **135.67 m/sec²**.

Ex. 7. *A vertical U-tube of uniform cross-section contains water upto height 30 cm. Show that if the water on one side is depressed and then released, its motion up and down the two sides of the tube is simple harmonic and calculate its time period.*

Sol. Let PQ be the initial level of water in the U-tube upto a height $h = 0.3$ m.

When the level of water in tube A is depressed through a distanc y, it rises at the same time by the same amount in the tube B. Now the difference between the two levels P' and $Q' = 2y$ **[fig. 7.12]**.

The weight of this column of length $2y$ will now act on the mass of the water in the tube, as a result it will oscillate up and down.

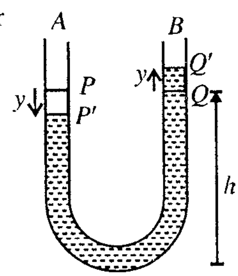


Fig. 7.12.

The weight of the column of length $2y = 2y \times A \times \rho g$ newton where A is cross-section of the tube and ρ the density of the liquid.

$\therefore$ Force acting on the mass of water,

$$F = -2yA\rho g \text{ newton} \qquad \text{...(i)}$$

But $$\text{force} = m(d^2y/dt^2) = 2hA\rho(d^2y/dt^2) \qquad \text{...(ii)}$$

Hence $$2hA\rho\left(\frac{d^2y}{dt^2}\right) = -2yA\rho g \quad \text{or} \quad \frac{d^2y}{dt^2} + \frac{g}{h} \times y = 0.$$

Thus, the motion of water column is simple harmonic, whose time period is given by

$$T = 2\pi\sqrt{\frac{h}{g}} = 2\pi\sqrt{\frac{0.30}{9.81}} = \mathbf{1.098 \text{ sec.}}$$

Ex. 8. *If the earth were a homogeneous sphere of radius R, and a straight hole were bored in it through its centre. Show that a particle dropped into the hole will execute a S.H.M. and find out its time period.* **(Agra 2006; Gwalior 2004; Jabalpur 03; Bundelkhand 04; Jodhpur 1967)**

Sol. If the mass of the earth is M and R its radius, then the acceleration due to gravity at the surface of the earth given by

$$g = GM/R^2, \text{ where } G \text{ is gravitational constant.}$$

If the density of the earth is ρ, then $M = \frac{4}{3}\pi R^3 \cdot \rho$.

$$\therefore \qquad g = \frac{4}{3}\pi R\rho \cdot G. \qquad ...(i)$$

If g' be the value at depth h below the surface of the earth, then

$$g' = \frac{4}{3}\pi(R-h)\rho \cdot G. \qquad ...(ii)$$

Dividing eq. (ii) by eq. (i), we have $\frac{g'}{g} = \frac{R-h}{R}$ or $g' = \frac{g}{R}(R-h)$.

If we measure the distance from the centre of the earth, putting $R - h = x$, acceleration of the particle, $\frac{d^2x}{dt^2} = g' = -\frac{g}{R}x$

Negative sign shows that the acceleration acts towards the centre of the earth, *i.e.*, opposite to x increasing.

$$\therefore \qquad \frac{d^2x}{dt^2} + \frac{g}{R}x = 0.$$

Thus, the particle, dropped into the hole, will execute S.H.M. Its periodic time is given by

$$T = 2\pi\sqrt{\frac{R}{g}}.$$

Ex. 9. *Two cities on the surface of the earth are joined by a straight smooth underground tunnel of length 640 km. A body is released into the tunnel from one city. How much time will it take to reach the other city ? Derive the formula used.* $G = 6.67 \times 10^{-11}$ *S.I. units and* $\rho = 5{,}520\ kg/m^3$. *Calculate the velocity when the body would be nearest to the centre of the earth in its journey.* **(Bombay 1967)**

Sol. Suppose that A and B are the two cities, which are connected by a straight tunnel of length $AB = 640$ km.

Let O be the centre of the earth and R its radius. OC is the perpendicular, drawn from the centre O on AB, where C is the mid-point of the tunnel AB. Let a body of mass m be released into the tunnel from its end A and let it be at P, distant x from C at any instant t **[fig. 7.13]**.

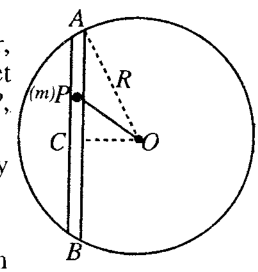


Fig. 7.13.

If M is the mass of the earth, then on its surface the weight (mg) of the body is given by

$$mg = GMm/R^2.$$

The weight of the body at P, distant OP (= r) from the centre of the earth is

$$mg' = \frac{GM'm}{r^2} = \frac{G\frac{4}{3}\pi r^3\rho m}{r^2} = \frac{G\cdot\frac{4}{3}\pi R^3\rho m r}{R^3} = \frac{GMm}{R^2}\cdot\frac{r}{R} \qquad [\because M' = \tfrac{4}{3}\pi r^3\rho \text{ and } M = \tfrac{4}{3}\pi R^3\rho]$$

The direction of this force is PO, whose component along PC

$$= \frac{GMm}{R^2}\cdot\frac{r}{R}\cos OPC = \frac{GMm}{R^2}\cdot\frac{r}{R}\cdot\frac{x}{r} = \frac{GMm}{R^3}x$$

Hence the equation of motion of mass m is

$$m\frac{d^2x}{dt^2} + \frac{GMm}{R^3}x = 0 \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{GM}{R^3}x = 0.$$

The equation represents a S.H.M. of period $T = 2\pi\sqrt{\frac{R^3}{GM}}$ *i.e.*, this is the time in which the body will reach from A to B and again from B to A. Hence, time taken by the body to pass through the tunnel is

$$\frac{T}{2} = \pi\sqrt{\frac{R^3}{GM}} = \pi\sqrt{\frac{R^3}{G\cdot\frac{4}{3}\pi R^3\rho}} = \sqrt{\frac{3\pi}{4\rho\cdot G}} = \sqrt{\frac{3\times 3.14}{4\times 5520\times 6.62\times 10^{-11}}} = \mathbf{2528\ sec.}$$

$$\text{Velocity at } C, (v) = \omega a = \sqrt{\frac{GM}{R^3}}\cdot\frac{AB}{2} = \sqrt{G\cdot\tfrac{4}{3}\pi\rho}\cdot\frac{AB}{2}$$

$$= \sqrt{6.67\times 10^{-11}\times\tfrac{4}{3}\times 3.14\times 5520}\cdot\frac{640}{2}\times 10^3 = \mathbf{3.972\times 10^2\ m/sec.}$$

Ex. 10. *The total energy of a particle executing a S.H.M. of period 2π second is 10.24×10^{-4} joule. The displacement of the particle at $\pi/4$ sec. is $8\sqrt{2}$ cm. Calculate the amplitude of the motion and the mass of the particle.*

Sol. The displacement of a particle, executing S.H.M. is given by

$$x = a\sin\pi t = a\sin 2\pi t/T \qquad [\because \omega = 2\pi/T]$$

$$\therefore \quad 8\sqrt{2}\times 10^{-2} = a\sin\frac{2\pi}{2\pi}\times\frac{\pi}{4} = a\sin\frac{\pi}{4} = \frac{a}{\sqrt{2}}$$

$\therefore$ Amplitude of vibration $a = 0.16$ m.

If m be the mass of the particle, then its total energy is given by

$$E = \frac{2m\pi^2a^2}{T^2} = 10.24\times 10^{-4} \quad\text{or}\quad m = \frac{10.24\times 10^{-4}\times T^2}{2\pi^2\times a^2}$$

Here $\qquad T = 2\pi$ sec and $a = 0.16$ m, $\quad\therefore\quad m = \dfrac{10.24\times 10^{-4}\times 2\pi\times 2\pi}{2\times\pi^2\times 0.16\times 0.16} = \mathbf{0.8\ kg.}$

Ex. 11. *If the potential energy of a harmonic oscillator in its resting position is 5 joules and the total energy is 9 joules, when the amplitude is 1 metre, what is the force constant ? If its mass is 2 kg, what is the period ?*

Sol. Potential energy in rest position = 5 joules.

At the maximum displacement (equal to amplitude), the potential energy = 9 joules.

$\therefore$ Gain in potential energy = 9 − 5 = 4 joules.

The gain in energy at the maximum displacement should be equal to $\frac{1}{2}Cx_{max}^2 = \frac{1}{2}C\times(1)^2 = C/2$ joules. Thus,

$$C/2 = 4 \quad\text{or}\quad C = \mathbf{8\ joules/m.}$$

$$\therefore \qquad \text{Period } T = 2\pi\sqrt{m/C} = 2\pi\sqrt{2/8} = \mathbf{3.14\ sec.}$$

Ex. 12. *If $U = U_0 \cos cx$, then draw potential-energy curve. Show that the positions of stable equilibrium can occur at $x = (2n + 1)$, where n is an integer. If the total energy of the particle is E, show that its maximum kinetic energy at the centre of a potential well is $E + U_0$. If the total energy is $-0.5\ U_0$, find the amplitude of oscillations and period.*

Sol. The potential energy curve with a constant c multiplied to x as abscissa is plotted in **fig. 7.14.**

Now, $\quad U = U_0\cos cx, \quad \dfrac{dU}{dx} = -U_0c\sin cx$

and $\quad \dfrac{d^2U}{dx^2} = -U_0c^2\cos cx.$

For maxima or minima, $\dfrac{dU}{dx} = -U_0c\sin cx = 0$

or $cx = r\pi$ ($\because U_0c \neq 0$), where $r = 0, 1, 2, 3, \ldots\ldots$.

The quantity d^2U/dx^2 is negative for even integers and positive for odd integers. Hence for the minima in the U-x curve, r should be odd integer *i.e.*, $r = (2n + 1)$. Therefore

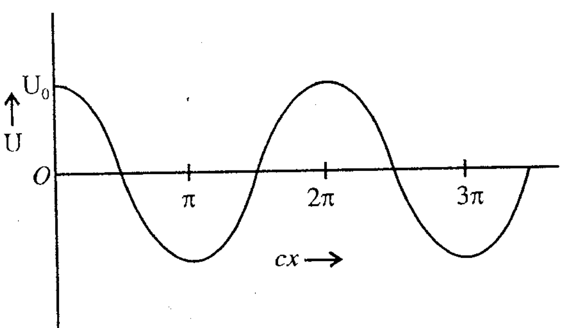


Fig. 7.14.

$$cx_0 = (2n + 1)\pi \quad \text{or} \quad x_0 = (2n + 1)\frac{\pi}{c}$$

At the centre of a potential well $U = -U_0$. Then

$$U + K = E \quad \text{or} \quad -U_0 + K = E,$$

$$\therefore \qquad K = E + U_0$$

Proved.

Maximum potential energy when oscillating, $U_{max} = E = -0.5\, U_0 = -U_0/2$. This will occur at x', given by

$-\frac{U_0}{2} = U_0 \cos cx'$ or $cx' = \frac{2\pi}{3}$ or $x' = \frac{2\pi}{3c}$ (taking the value near to the 1st minimum as in **fig. 7.14**)

Hence the amplitude is given by $a = |x' - x_0| = \left|\frac{2\pi}{3c} - \frac{\pi}{c}\right| = \frac{\pi}{3c}$, where $x_0 = \frac{\pi}{c}$ for the first minimum.

Force constant for small amplitude oscillations, $k = \left.\frac{d^2U}{dx^2}\right|_{x=x_0} = U_0 c^2$

$$\text{Period } T = 2\pi\sqrt{\frac{m}{U_0 c^2}}.$$

Ex. 13. *Draw graphs for the variation of potential energy, kinetic energy and total energy with time for simple harmonic motion. Show that the period for potential or kinetic energy is one-half of the period of vibration.*

Sol. Potential energy, $U = \frac{1}{2}Ca^2 \sin^2(\omega t + \phi)$

Kinetic energy, $K = \frac{1}{2}Ca^2 \cos^2(\omega t + \phi)$

Total energy, $E = U + K = \frac{1}{2}Ca^2$

The graphs for the variation of U, K and E with time with $\phi = 0$ are shown in **fig. 7.15.**

If $U(t)$ is equal to $U(t + \frac{\pi}{\omega})$, it means that the period of potential energy is $\frac{\pi}{\omega}$ *i.e.*, one-half that of vibration.

$$\begin{aligned} U(t + \tfrac{\pi}{\omega}) &= \tfrac{1}{2}Ca^2 \sin^2[\omega(t + \tfrac{\pi}{\omega}) + \phi] \\ &= \tfrac{1}{2}Ca^2 \sin^2(\omega t + \phi + \pi) \\ &= \tfrac{1}{2}Ca^2 \sin^2(\omega t + \phi) \\ &= U(t) \end{aligned}$$

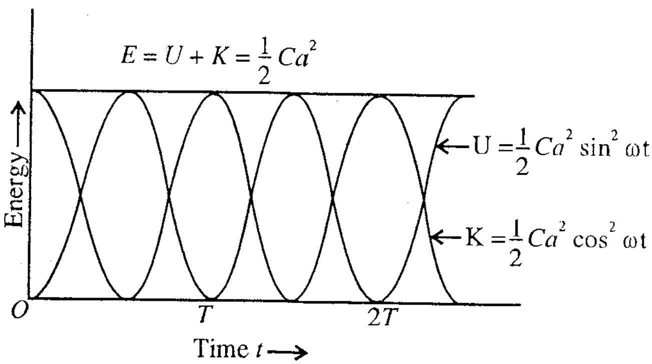


Fig. 7.15 : Variation of U, K and E with time.

Similar is the case for the period of kinetic energy.

Ex. 14. *Prove that in simple harmonic motion when the average is taken with respect to the position over one cycle, the average potential energy equals* $\frac{1}{6}Ca^2$ *and the average kinetic energy equals* $\frac{1}{3}Ca^2$ *Explain physically why this result is different to that when the average is taken with respect to time.*

Sol.

$$U = \frac{1}{2}Cx^2 \quad \text{and} \quad K = \frac{1}{2}C(a^2 - x^2)$$

$$U_{av} = \frac{1}{a}\int_0^a \frac{1}{2}Cx^2 dx = \frac{C}{2a}\left(\frac{x^3}{3}\right)_0^a = \frac{1}{6}Ca^2$$

$$K_{av} = \frac{1}{a}\int_0^a \frac{1}{2}C(a^2 - x^2)\,dx = \frac{C}{2a}\left[a^2x - \frac{x^3}{3}\right]_0^a = \frac{1}{3}Ca^2$$

When the average is taken with respect to the position, one can see from **fig. 7.9** that the area between U-x curve and X-axis is one half to that between K-x curve and X-axis. However when the average is taken with respect to the time, area between U-t curve [**fig. 7.15**] and t-axis is exactly equal to that between K-t curve and t-axis. Hence the results in two cases are different.

EXERCISE 7·1

1. **Fig. 7.16** shows the potential energy curve for a mass 0.1 kg, having abscissa in metres and ordinate in joules. (a) State the possible positions around which the particle can oscillate, (b) What is the maximum energy of oscillation in each case ? (c) State the interval of total energy for which the particle can oscillate about : (i) only the smaller r-value and (ii) both the r-values. (d) Find the minimum value of total energy so that the motion of the particle will be unbounded. When $r \to \infty$, calculate the speed of the particle.
[**Ans.** (a) 1.5 m and 4.5 m; (b) 9.5 joules and 4 joules; (c) (i) -7.5 joules $< E < +2$ joules, (ii) -3 joules $< E < +1$ joule; (d) 2 joules, $= 6.3$ m/sec.]

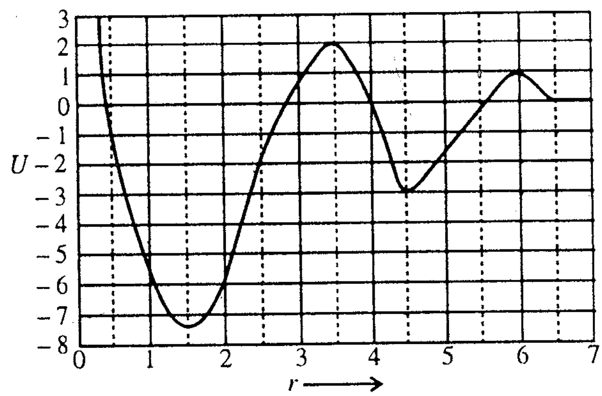


Fig. 7.16.

2. **Fig. 7.17** shows the potential energy curve of a system in arbitrary units for U and r. State the possible r-values about which the system can oscillate; also the maximum energy of oscillations in each case. If the system just escapes from the potential well at smaller r, with what kinetic energy does it go to $r \to \infty$? **(Agra 1971)**
[**Ans.** $r_1 = 2$ units and $r_2 = 4.5$ units; $E_1 = 3.5$ units and $E_2 = 1.5$ units; $K = 1$ unit.]

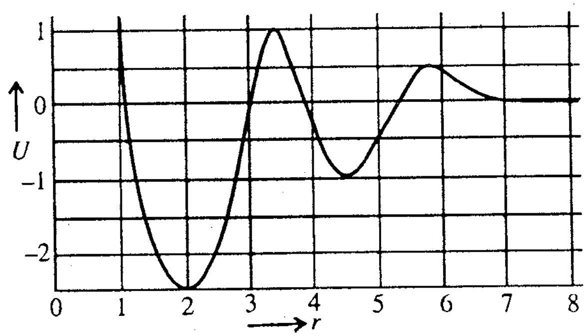


Fig. 7.17.

3. The potential energy function for the force between two atoms in a diatomic molecule can approximately be expressed as [**fig. 7.18**]
$U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where a and b are positive constants, and x is the distance between the atoms.
(i) For what values of x, $U(x)$ is equal to zero ?
(ii) For what values of $x, U(x)$ is minimum ?
(iii) Calculate the force between the two atoms and plot $F(x)$. Show that the two atoms repel each other for x less than x_0 and attract each other for x greater than x_0. What is the value of x_0 ?
(iv) Assuming that one of the atoms remains stationary and the other moves along X-axis, describe the possible motions.
(v) Calculate the dissociation energy D (the energy required to break the molecule into atoms) of the molecule. **(Agra 1972)**

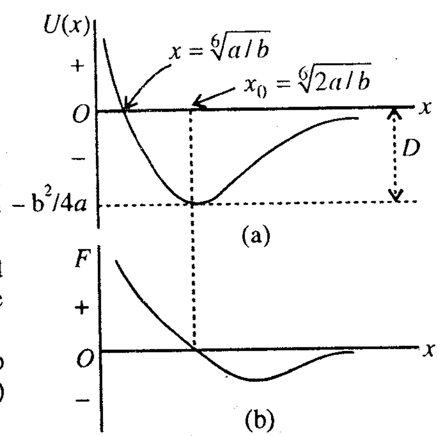


Fig. 7.18.

[**Ans.** (i) $x = \left(\frac{a}{b}\right)^{1/6}$ and $x = \infty$; (ii) $\left(\frac{2a}{b}\right)^{1/6}$; (iii) $F_{(x)} = \frac{12a}{x^{13}} - \frac{6b}{x^7}$, $x_0 = \left(\frac{2a}{b}\right)^{1/6}$; (iv) Oscillatory motion about the point $x = (2a/b)^{1/6}$; (v) $b^2/4a$ or more.]

4. A particle experiences a force associated with the potential energy $U = 3x^2 - x^3$. Find the positions of stable and unstable equilibrium.

   [**Ans.** $x = 0$, stable equilibrium; $x = 2$, unstable equilibrium.]

5. Let the potential energy of a particle be expressed as

$$U = A - \frac{B}{x} + \frac{C}{x^2},$$

   where A, B and C are positive constants. Find the force constant for small ocillations. For what values of the total energy will the motion be bounded ? Find the period of small oscillations.

   [**Ans.** Force constant $= \frac{B^4}{8C^3}$; Total energy must be less than A; $T = 2\pi\sqrt{\frac{8mC^3}{B^4}}$.]

6. The Yukawa potential is given by $U_{(r)} = -\frac{r_0}{r}U_0 e^{-r/r_0}$

   This gives a fairly accurate description of the interaction between the nucleons (*i.e.*, neutrons, protons). The constant $r_0 = 1.5 \times 10^{-15}$ m (nearly) and $U_0 = 50$ MeV (nearly). Now, (a) calculate the corresponding expression for the force of attraction; (b) for showing the short range nature of this force, find the ratio at $r = 4r_0$ to the force at $r = r_0$. **(Kumaun 1995)**

   [**Ans.** (a) $F = -\frac{r_0 U_0}{r} e^{-r/r_0}\left[\frac{1}{r_0} + \frac{1}{r}\right]$; (b) $\frac{F_{(4r_0)}}{F_{(r_0)}} = \frac{1}{126}$ nearly.]

7. A particle of mass 10 gm is present in a potential field $V = 50x^2 + 100$ erg/gm. Find the vibration frequency. **(Agra 1991)**

   [**Ans.** 1.6 vibrations/sec.]

8. A particle of mass m kg lies in a potential field given by $V = 200x^2 + 300$ J/kg. Find the frequency and time period.

   [**Ans.** 3.2 Hz; 0.31 sec.]

9. The displacement equation for a particle in simple harmonic motion at an instant t is,

$$x = 0.01 \sin 100\,\pi(t + 0.005) \text{ m}$$

   Calculate (i) amplitude; (ii) time period; (iii) maximum velocity and (iv) displacement at start. **(Ujjain 2004; Bhopal 03)**

   [**Ans.** (i) 0.01 m; (ii) 0.02 sec.; (iii) 3.14 m/s; (iv) 0.01 m.]

10. A particle of mass 10 gm moves under a potential $V(x)$ given by $V(x) = 8 \times 10^5 x^2$ J/kg, where x is in metres. Deduce the time-displacement relation when the total energy is 800 joules.

    $\left[\textbf{Ans. } \frac{1}{\sqrt{10}} \sin(400\sqrt{10}\, t + \phi)\right]$ **(Agra 1979 S)**

11. If the potential energy curve $U = -\frac{A}{x} + \frac{B}{x^2}$, where A and B are positive constants, find the position of stable equilibrium and the force constant for small oscillations. Find the period of small oscillations.

    [**Ans.** $2B/A$; $A^4/8B^3$; $T = 4\pi B\sqrt{2mB/A^2}$]. **(Kumaun 1995)**

12. A particle makes S.H.M. along a straight line and its velocity when passing through points 3 cm and 4 cm from the centre of its path is 16 cm/sec and 12 cm/sec respectively. Find (a) the amplitude; (b) the time period of motion. **(Rohilkhand 1979)**

    [**Ans.** 0.05 m; 1.571 sec.]

13. Prove that the velocity of a particle in S.H.M. at a distance $\sqrt{3}/2$ of its amplitude from the centre is half of its velocity at the central position.

14. A test tube weighs 6 gm and has an external diameter of 23 cm is floated in water by placing 10 gm mercury at the bottom of the tube. The tube is depressed by a small amount and then released. Find its time period.

    [**Ans.** 0.453 sec.]

15. A block (cuboid) of density ρ and dimensions a, b, c is floating in water with side a vertical. If it is pushed down and then released, show that resulting motion is simple harmonic, having period $2\pi\sqrt{a\rho/1000g}$.

16. A light elastic string is suspended vertically from a point and carries mass at its free end which stretched it through l cm. Show that the vertical oscillations of the system are simple harmonic type and the time period of oscillation is equal to that of simple pendulum of length l cm.

17. A smooth straight tunnel is bored through the moon along any line taken at random. Show that if a smooth ball is dropped inside the tunnel, it executes simple harmonic motion. In how much time, will it reach one end of the tunnel from the other end ? Given mass of the moon = 7.35×10^{22} kg, radius of the moon = 1.75×10^{4} m and $G = 6.67 \times 10^{-11}$ S.I. units.

[**Ans.** 3.284 sec.]

[**Hint :** $T = 2\pi\sqrt{R/g} = 2i\sqrt{R^3/GM}$;Time from one end to the other end = $T/2$.]

18. A particle is dropped down in a deep hole which extends to the centre of the earth. Calculate its velocity at a depth of one kilometer from the surface of the earth. Assume that g = 10 m/sec and radius of the earth = 6400 kilometers. **(Gorakhpur 1965)**

[**Ans.** 141.4 m/sec.]

19. If a smooth tunnel is bored through the earth along one of its diameters, then calculate the time of one oscillation of the ball. ($R = 6.4 \times 10^3$ km and g = 9.8 metres/sec^2). **(Agra 2006)**

[**Ans.** 84 minutes.]

20. A particle of mass m can move in a horizontal frictionless pipe (situated on the earth's surface) under the influence of earth's gravitational force [**fig. 7.19**]. Assuming that x is very small compared with R, prove that the particle executes S.H.M. of period $T = 2\pi\sqrt{R/g}$.

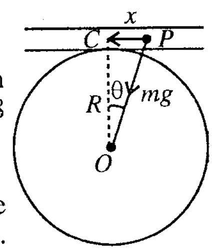


Fig. 7.19.

21. A particle of mass m is dropped from a height h ($h << R$) above a hole, drilled along a diameter in the earth as shown in **fig. 7.20**. (a) Find the speed of the particle, when it is passing through the centre of the earth. (b) Would the motion be simple harmonic ? (c) Would the motion be periodic ? Explain.

[**Ans.** (a) $\sqrt{\frac{GM}{R} + 2gh}$; (b) No; (c) Yes.]

22. A particle in S.H.M. has velocities u_1 and u_2 when displacements from the mean positions are x_1 and x_2 respectively. Calculate period, amplitude and maximum speed of the particle. **(Kanpur 2005)**

[**Ans.** $T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{u_1^2 - u_2^2}}$, $a = \sqrt{\frac{u_1^2 x_2^2 - u_2^2 x_1^2}{u_1^2 - u_2^2}}$, $v = \sqrt{\frac{u_1^2 x_2^2 - u_2^2 x_1^2}{x_2^2 - x_1^2}}$.]

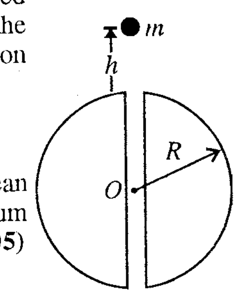


Fig. 7.20.

23. If the potential energy of a hormonic oscillator in its resting position is 10 joules and the average kinetic energy is 5 joules, what is its total energy at any instant ?

[**Ans.** 20 joules.]

7·8. Examples of Simple Harmonic Oscillations

Eearlier we have discussed the properties of harmonic oscillator by considering the oscillations of model spring-mass system. The oscillatory phenomenon results due to (i) the spring factor and (ii) inertial property of the mass. The system must also possess a position of stable equilibrium and when disturbed, the system executes oscillations about this position under the action of restoring force and inertia. Now, we plan to discuss a number of physical systems in which the mathematics of harmonic oscillator, developed earlier, will be applied. We shall see that these physical systems appear to have little in common, but all of them possess a pair of quantities analogous to spring and mass factors of which the former (spring) stores potential energy and the later (mass) is capable of possessing kinetic energy. Further in each case, a differential equation for the oscillating system similar to the equation of harmonic oscillator is obtained *i.e.*,

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \qquad ...(1)$$

From such an equation, one can draw the information regarding the period, frequency etc. of the system.

7·9·1. Spring and Mass System

The system consists of a massless spring, one of its ends is connected to a mass m and the other end is fixed to a rigid support. Here, the mass and spring are on a smooth horizontal surface [**fig. 7.21**]. If a force is applied on the mass to stretch or compress the spring and then released, the motion of the mass is simple harmonic. If x is the displacement of mass m at any time, then the restoring force exerted by the spring is

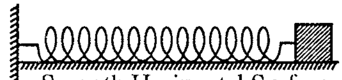


Fig. 7.21.

$$F_x = -Cx, \text{ where } C \text{ is force constant.}$$

Also, $\qquad$ force $= m(d^2x/dt^2)$

Therefore, equation of motion is

$$m\frac{d^2x}{dt^2} = -Cx \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{C}{m}x = 0 \qquad ...(1)$$

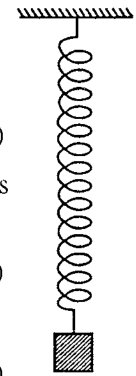


Fig. 7.22.

This equation represents that the motion of spring-mass system is simple harmonic. Its periodic time and frequency is given by

$$T = 2\pi\sqrt{\frac{m}{C}}\ ;\ n = \frac{1}{2\pi}\sqrt{\frac{C}{m}} \qquad ...(2)$$

The solution of the equation of motion is

$$x = a \sin(\omega t + \phi), \text{ where } \omega = \sqrt{\frac{C}{m}} \qquad ...(3)$$

Now, suppose the spring-mass system is hanging vertically from a rigid support and it is oscillating in earth's gravitational field.

The question arises : Is there any effect of gravity on the oscillations of the hanging spring-mass arrangement ? For this purpose, first we consider a spring of negligible mass, suspended vertically from a rigid support S [**fig. 7.23(a)**]. To the free end, now we attach an object of mass m. Due to the weight mg of the object, the spring extends and suppose that the equilibrium position is attained at an extension x_0 of the spring [**fig. 7.23 (b)**]. At equilibrium, the spring force $(-Cx_0)$ balances the weight (mg) of the object and the net force is zero *i.e.,*

$$mg - Cx_0 = 0 \qquad ...(4)$$

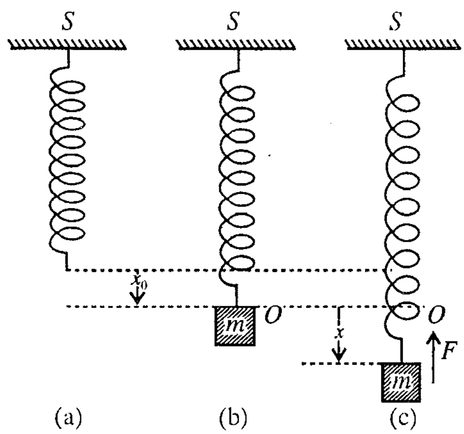


Fig. 7.23 : (a) Spring hanging from a rigid support S; (b) Equilibrium position of the mass m, attached to the free end of the spring : $mg - Cx_0 = 0$; (c) Oscillating mass under the restoring force $F = -Cx$.

Now if mass is displaced downwards through distance x from this equilibrium position O, the total spring force will be $-C(x_0 + x)$. Hence the net force is $mg - C(x_0 + x) = -Cx$, which tends to restore the object to the equilibrium position O [**fig. 7.23(c)**]. Thus the restoring force is $F = -Cx$ and the equation of motion is

$$m\frac{d^2x}{dt^2} = -Cx \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{C}{m}x = 0 \qquad ...(5)$$

This is the standard differential equation of simple harmonic motion whose frequency of vibration is given by eq. (2). Thus the gravity has no effect on the force constant or frequency of oscillation. It is to be noted that the oscillations are vertical due to the vertical tension in the spring, caused by the weight of the object.

Mass attached between two springs : Let us consider two identical springs of relaxed length l_0, attached to rigid supports S_1 and S_2. A mass m, which can slide over a frictionless surface [**fig. 7.24(a)**], is connected to rigid supports through the two springs [**fig. 7.24(b)**]. The system is in equilibrium when the mass is at the position O and the length of each spring is l. Therefore each spring has a force of tension $C(l - l_0)$ at equilibrium. Now if the mass m is displaced and left free, it will oscillate. Suppose that at any time t, the displacement of mass m is x [**fig. 7.24(c)**].

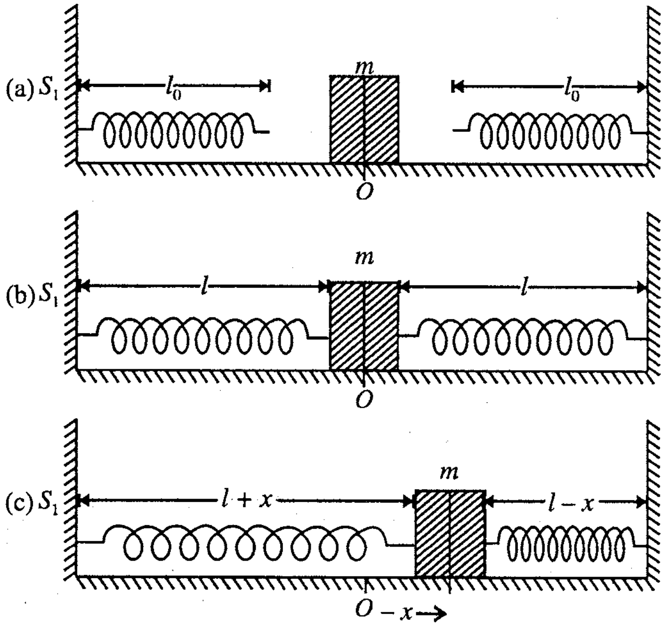


Fig. 7.24 : (a) Spring in the relaxed position; (b) Springs attached to mass *m* so that the equilibrium position of the mass is *O*; (c) Oscillating mass *m*.

Force exerted by left hand side spring = $- C(l + x - l_0)$ towards O

Force exerted by right hand side spring = $- Cx$ towards O

Total restoring force $F = - C(l - l_0 + 2x)$

Hence the equation of motion of the mass m is

$$m\frac{d^2x}{dt^2} = - C(l - l_0 + 2x) \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{2C}{m}\left(x + \frac{l - l_0}{2}\right) = 0 \qquad ...(6)$$

Let $$x + \frac{(l - l_0)}{2} = x' \quad \text{and} \quad \omega^2 = \frac{2C}{m}.$$

Then $$\frac{d^2x'}{dt^2} + \omega^2 x' = 0 \qquad ...(7)$$

This is identical to the differential equation (1) of the harmonic oscillator. The frequency of oscillation is given by

$$n = \frac{1}{2\pi}\sqrt{\frac{2C}{m}} \qquad ...(8)$$

Ex. 1. *A* 1.6 *kg-wt. extends a spring* 8 *cm from its unstretched position. The mass is replaced by a body of* 50 *gm. The mass is pulled and then released, find the period of oscillation.* **(Mysore 1980)**

Sol. $$F = - Cx, \quad \text{or} \quad C = - F/x$$

$$\therefore \quad C = \frac{1.6g}{0.08} = \frac{1.6 \times 9.8}{0.08} = \mathbf{196\ N/m}$$

$$\therefore \quad \text{Period, } T = 2\pi\sqrt{\frac{m}{C}} = 2\pi\sqrt{\frac{50 \times 10^{-3}}{196}} = \mathbf{0.11\ sec.}$$

Ex. 2. (a) *Figure below shows a mass M resting on a smooth table between two firm supports A, B and controlled by two massless springs. If the mass M is* 30 *gm, and the force constants of the two springs are* 2 *and* 1 *N/m, deduce* (i) *the frequency of small oscillations of M,* (ii) *the enrgy of oscillations*

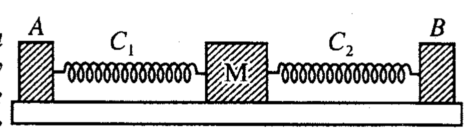


Fig. 7.25.

for amplitude 0.5 *cm.*(**Agra 1969**)

(b) *If the two springs are connected as shown in* **fig 7.26**, *find the period of oscillation.*

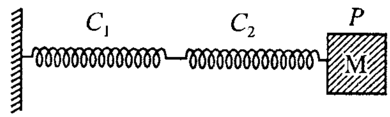


Fig. 7.26.

Sol. (a) If the mass M is displaced through a distance x towards left or right, the restoring forces in the two springs are

$$F_1 = -C_1x \text{ and } F_2 = -C_2x$$

acting on the mass along the same direction. So that, the total restoring force on the mass M is $F = -(C_1 + C_2)x$. Thus, the force constant of the system is $C = C_1 + C_2 = 2 + 1 = 3$ N/m.

(i) Mass $M = 30$ gm $= 0.03$ kg

$$\therefore \quad \text{Frequency, } \upsilon = \frac{1}{2\pi}\sqrt{\frac{C}{M}} = \frac{1}{2\pi}\sqrt{\frac{3}{0.03}} = \frac{5}{\pi} = \frac{5}{3.14} = \mathbf{1.6\ (nearly).}$$

(ii) $$\text{Energy } E = \text{Max. P.E.} = \frac{1}{2}Cx_{max}^2 = \frac{1}{2}\times 3\times(0.5\times 10^{-2})^2 = \mathbf{3.75 \times 10^{-5}\ J.}$$

(b) When the springs of force constant C_1 and C_2 are connected as shown in **fig. 9.26**, the displacement x of mass M produces the same restoring force F in the two springs. If the two springs are extended through x_1 and x_2, then

$$F = -Cx_1 = -Cx_2$$

and therefore $$x = x_1 + x_2 = -\frac{F}{C_1} - \frac{F}{C_2} = -F\left(\frac{C_1+C_2}{C_1C_2}\right) \quad \therefore \quad F = -\frac{C_1C_2}{C_1+C_2}x$$

Hence force constant, $$C = \frac{C_1C_2}{C_1+C_2} = \frac{2\times 1}{2+1} = \frac{2}{3}$$

$$\therefore \quad \text{Period } T = 2\pi\sqrt{\frac{M}{C}} = 2\pi\sqrt{\frac{0.03\times 3}{2}} = \mathbf{1.33\ sec.}$$

Ex. 3. (i) *In* **fig. 7.27(a), 7.27(b)** *and* **7.27 (c)**, *three combinations of two springs of force constants k_1 and k_2 are shown. Show that the period of oscillation in different cases is given by* (a) $2\pi\sqrt{\frac{m}{k_1+k_2}}$, (b) $2\pi\sqrt{\frac{m}{k_1+k_2}}$;

(c) $2\pi\sqrt{m\left(\frac{1}{k_1}+\frac{1}{k_2}\right)}$.

(ii) *If $k_1 = k_2$, show that the values of ω_0^2 in case of* **fig. 7.27(a)** *and* **fig. 7.27(c)** *are in the ratio* 1 : 4.

(iii) *If $m = 30$ gm, $k_1 = 20\ Nm^{-1}$ and $k_2 = 10\ Nm^{-1}$, deduce the frequency of small oscillations and the energy of oscillation for amplitude* 1.0 *cm for* **fig. 7.27(b)**.

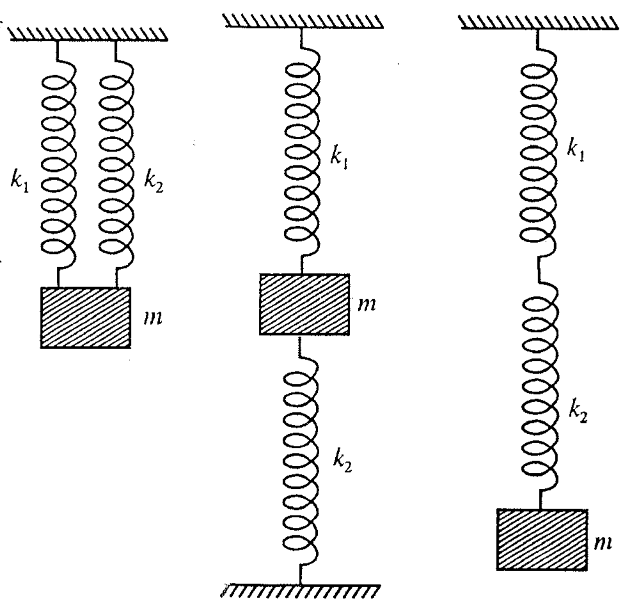


Fig. 7.27 : Oscillations of a mass attached to two springs.

Sol. (i) (a) In this arrangement, both springs will be extended by the same length x and will exert force on m, given by

$$F = -k_1x - k_2x.$$

$$\therefore \quad m\frac{d^2x}{dt^2} + (k_1+k_2)x = 0$$

Period, $T = 2\pi\sqrt{\frac{m}{k_1+k_2}}$.

(b) In this case, if the mass is displaced to left or right by distance x, the restoring forces $F_1 = -k_1x$ and $F_2 = -k_2x$ will act on m along the same direction. Hence the total restoring force on the mass m is

$$F = -k_1x - k_2x = -(k_1 + k_2)x$$

$\therefore$ Period, $T = 2\pi\sqrt{\dfrac{m}{k_1 + k_2}}$.

(c) In this case, the two springs of force constants k_1 and k_2 are connected in series. When the mass m is displaced by the distance x, then the same restoring force will act in the springs, extending them by x_1 and x_2 due to different force constants *i.e.*,

$$F = -k_1x_1 = -k_2x_2 \text{ and } x = x_1 + x_2 = -\frac{F}{k_1} - \frac{F}{k_2} \text{ or } x = -\left(\frac{1}{k_1} + \frac{1}{k_2}\right)F \text{ or } F = -\frac{1}{\dfrac{1}{k_1} + \dfrac{1}{k_2}}x$$

Hence $T = 2\pi\sqrt{m\left(\dfrac{1}{k_1} + \dfrac{1}{k_2}\right)}$.

(ii) $k_1 = k_2 = k$ (say)

Case (a) $\omega^2 = \dfrac{2k}{m}$; **Case (b)** $\omega^2 = \dfrac{k}{2m}$; $\therefore$ $\dfrac{\omega^2}{\omega'^2} = \mathbf{4}$

(iii) $\upsilon = \dfrac{1}{2\pi}\sqrt{\dfrac{k_1 + k_2}{m}} = \dfrac{1}{2\pi}\sqrt{\dfrac{20 + 10}{30 \times 10^{-3}}} = \mathbf{5.03\ Hz.}$

$$E = \frac{1}{2}ka^2 = \frac{1}{2} \times 30 \times (10^{-2})^2 = \mathbf{1.5 \times 10^{-3}\ J.}$$

Ex. 4. Spring having mass : *Show that the frequency of oscillation of a body of mass M suspended from a uniform spring of force constant C and mass m is given by* $n = \dfrac{1}{2\pi}\sqrt{\dfrac{C}{M + m/3}}$.

Sol. If l is the length of the spring, then mass per unit length = m/l.

Now, let us consider a length ds of the spring at a distance s (measured along the length) below the fixed upper end. The mass of this small element is $(m/l)\ ds$.

If the velocity of the lowest point of the spring is v, then evidently the velocity of the length ds at a distance s from rigid support will be $(s/l)v$.

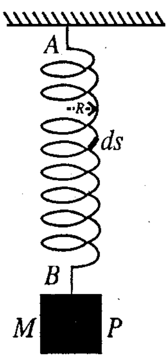


Fig. 7.28.

$\therefore$ Kinetic energy of the element $= \dfrac{1}{2}\left(\dfrac{m}{l}ds\right)\left(\dfrac{vs}{l}\right)^2$

$\therefore$ K.E. of the whole spring $= \displaystyle\int_0^l \frac{1}{2}\frac{mv^2}{l^3}s^2 ds = \frac{1}{2}\frac{mv^2}{l^3}\cdot\frac{l^3}{3} = \frac{mv^2}{6}$

Kinetic energy of the mass M at the lower end of the spring $= \dfrac{1}{2}Mv^2$.

Total kinetic energy of the system $= \dfrac{1}{2}Mv^2 + \dfrac{1}{6}mv^2 = \dfrac{v^2}{2}\left(M + \dfrac{m}{3}\right)$

If x is the displacement of the mass from its mean position, then $v = \dfrac{dx}{dt}$

and total K.E. $= \left(M + \dfrac{m}{3}\right)\left(\dfrac{dx}{dt}\right)^2$.

Potential energy of the spring $= \displaystyle\int_0^x Cx\,dx = \frac{1}{2}Cx^2$.

From the law of conservation of energy

$$\frac{1}{2}\left(M + \frac{m}{3}\right)\left(\frac{dx}{dt}\right)^2 + \frac{1}{2}Cx^2 = \text{a constant.}$$

Differentiating it with respect to time and dividing by dx/dt, we have

$$\left(M + \frac{m}{3}\right)\frac{d^2x}{dt^2} + Cx = 0 \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{2}{(M + m/3)}x = 0$$

This is the equation of simple harmonic motion, whose frequency is

$$\upsilon = \frac{1}{2\pi}\sqrt{\frac{C}{(M + m/3)}}.$$

EXERCISE 7·2

1. An object of mass 2.5 kg is hanging from a massless spring. Another object of mass 400 gm, hung below the first object, stretches the spring 2 cm farther. If the object of 400 gm is removed and the first object is set into vertical oscillations, determine the period of oscillation.
[**Ans.** 0.71 sec.]

2. A body of mass 4.9 kg when suspended from a spring, oscillates with a period 0.5 sec. By what amount will the spring be contracted on the removal of the body ?
[**Ans.** 0.062 m.]

3. One end of a vertical spring is attached to a fixed support and to the lower end a mass m is suspended. If by suspending the mass, the lower end of the spring is stretched through x_0 distance, then find the spring constant and period of oscillation.
[**Ans.** mg/x_0; $2\pi\sqrt{x_0/g}$.]
[**Hint :** $mg = Cx_0$.]

4. (a) Two springs are joined and connected to mass m as shown in **fig. 7.26**. If the springs separately have force constants C_1 and C_2, show that the frequency of oscillation is $n = \frac{1}{2\pi}\sqrt{\frac{C_1C_2}{(C_1+C_2)m}}$.
(b) If the two massless springs are attached to a mass m and to fixed supports as shown in **fig. 7.25**, then show that the frequency of oscillation is given by $n = \frac{1}{2\pi}\sqrt{\frac{C_1+C_2}{m}}$.

5. In the in **fig. 7.25** M is a mass which is placed between two rigid supports A and B, and is controlled by two massless springs. If the mass of M is 50 gms and the force constants of springs are 3000 and 2000 dynes/cm, deduce :
(i) the 'frequency' of small oscillations of the mass M.
(ii) the energy of vibration of amplitude 0.4 cm.
(iii) the velocity of mass M when it passes through its mean position. **(Agra 1978)**
[**Ans.** (i) 1.59 Hz; (ii) 4×10^{-5} joules; (iii) 0.04 m/sec.]

6. A mass, suspended from a spring, vibrates in S.H.M. Show that the ratio of kinetic to potential energy is 3 : 1, when the displacement is one-half the amplitude.

7. When a mass 40 kg is placed on a platform P [**fig. 7.29**] its centre of gravity lowers 0.2 cm. Calculate the force constant of the springs connected to P. If the mass of the platform is 500 kg, find the period of oscillation : (a) when the mass has not been placed and (b) when the mass has been placed on the platform.
[**Ans.** 1.96×10^5 N/m; (a) 0.317 sec; (b) 0.336 sec.]

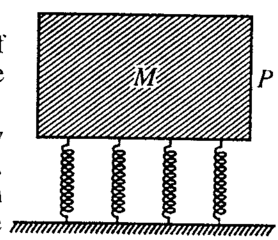


Fig. 7.29.

8. The spring factors for two springs are $C_1 = 1$ N/m and $C_2 = 3$ N/m and each of them is 1 meter long in its unstretched position. Both of the springs are connected to a mass P of mass 1 kg and their other ends are free [**fig. 7.30**]. If the distances of each free end from the fixed supports is 0.5 m, calculate the length of each spring in equilibrium state, when the free ends are attached to rigid supports. What is the period of oscillation ?
[**Ans.** $l_1 = 1.75$ m, $l_2 = 1.25$ m; 3.14 sec.]

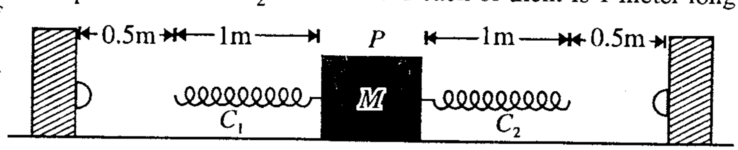


Fig. 7.30.

7·9·2. Angular Oscillations — Simple and Compound Pendulums

In case of spring-mass system, the mass oscillates about an equilibrium position along a straight line. When we observe the oscillations of a pendulum or balance wheel in a watch, the system rotates back and forth about an equilibrium position under the action of restoring torque. In these rotational systems, the restoring torque and moment of inertia replace restoring force and mass of spring-mass system respectively. If the restoring torque in the system is proportional to the angular displacement and directed opposite to it, the system executes **angular simple harmonic vibrations** and a differential equation analogous to the equation of harmonic oscillator can be established. We consider here the restoring torques due to gravity (*e.g.*, in simple and compound pendulums) and due to elasticity (*e.g.*, in torsional pendulum).

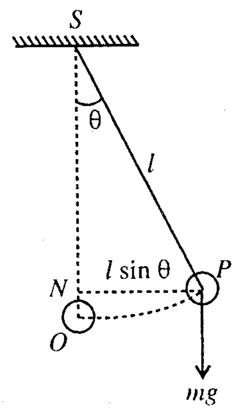


Fig. 7.31 : Simple pendulum.

(1) Simple Pendulum

A simple pendulum consists of a small heavy (ideally point) bob of mass m, suspended by means of an inextensible massless string of length l to a rigid support S [**fig. 7.31**]. If the bob is drawn to one side from its equilibrium position O and then left free, it starts to oscillate in a vertical plane. Let θ be the angular displacement at any time t. Due to the weight mg of the bob, a restoring torque (*i.e.*, moment of force) τ is acting on the system and is given by $\tau = -mgl \sin\theta$.

This is the trorque about the point S and bears negative sign because its direction is opposite to θ increasing. If I be the moment of inertia of the bob about the point S and $d^2\theta/dt^2$ be the angular acceleration at the angular displacement θ, then torque can be expressed as $\tau = I\dfrac{d^2\theta}{dt^2}$

Therefore, $$I\frac{d^2\theta}{dt^2} = -mgl\sin\theta \qquad ...(1)$$

If the bob is considered a point mass m, then its moment of inertia about S is given by

$$I = ml^2 \qquad ...(2)$$

Hence $$ml^2\frac{d^2\theta}{dt^2} = -mgl\sin\theta \quad \text{or} \quad \frac{d^2\theta}{dt^2} + \frac{g}{l}\sin\theta = 0$$

or $$\frac{d^2\theta}{dt^2} + \omega^2\sin\theta = 0, \quad \text{where } \omega = \sqrt{\frac{g}{l}}, \qquad ...(3)$$

Taylor's expansion of $\sin\theta$ is

$$\sin\theta = \theta - \frac{\theta^3}{\lfloor 3} + \frac{\theta^5}{\lfloor 5} - ..... \qquad ...(4)$$

For sufficiently small θ (in radians), we can neglect all terms in the expansion except the first term. In such a case,

$$\sin\theta \approx \theta \qquad ...(5)$$

Thus for small angular amplitude, eq. (3) is obtained to be

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \qquad ...(6)$$

Eq. (6) is completely analogous to the differential equation of the harmonic oscillator. The solution of the equation is given by

$$\theta = \theta_m \sin(\omega t + \theta) \qquad ...(7)$$

where θ_m is the maximum angular displacement and ϕ is the phase constant.

The period of oscillation of the simple pendulum is given by

$$T = \frac{2\pi}{\omega} \quad \text{or} \quad T = 2\pi\sqrt{\frac{l}{g}} \qquad ...(8)$$

(2) Compound Pendulum

A compound pendulum is a rigid body capable of oscillating freely in a vertical plane about a horizontal axis under the action of gravity. Let **fig. 7.32** represent the vertical section of a rigid body through the centre of gravity G. It is capable of oscillating about a horizontal axis through S, the centre of suspension, where the vertical section through G meets the horizontal axis. In equilibrium, the centre of gravity G of the body lies vertically below S. Let the distance $SG = l$. Now if the body is displaced to one side and left free, it oscillates about the equilibrium position. If θ be the angular displacement at any instant t, then the restoring torque about S is given by

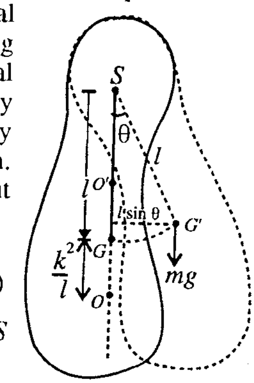


Fig. 7.32 : Compound pendulum.

$$I\frac{d^2\theta}{dt^2} = -\, mgl \sin\theta \qquad \text{...(9)}$$

where I is the moment of inertia of the body about the horizontal axis through S and mg is the weight of the body.

For sufficiently small angular displacement θ, eq. (9) assumes the form

$$\frac{d^2\theta}{dt^2} + \frac{mgl}{I}\theta = 0 \qquad \text{...(10)}$$

This equation represents that the compound pendulum is executing a simple harmonic motion whose period of oscillation is given by

$$T = 2\pi\sqrt{\frac{I}{mgl}} \qquad \text{...(11)}$$

If I_g be the moment of inertia of the rigid body about a parallel axis through its centre of gravity G, then by the application of the theorem of parallel axis, we have

$$I = I_g + ml^2 = mK^2 + ml^2 = m(K^2 + l^2)$$

where K is the radius of gyration of the body about a parallel axis through G.

Hence the expression of periodic time of the pendulum becomes

$$T = 2\pi\sqrt{\frac{m(K^2+l^2)}{mgl}} \quad \text{or} \quad T = 2\pi\sqrt{\frac{\frac{K^2}{l}+l}{g}} \qquad \text{...(12)}$$

Here, the periodic time is the same as that of a simple pendulum of length $\left(\frac{K^2}{l}+l\right)$, say L. This length L is called the **length of an equivalent simple pendulum.** If we take a point O on the line SG produced such that $SO = \frac{K^2}{l}+l$, the length of an equivalent simple pendulum, then the point O is called **centre of oscillation.**

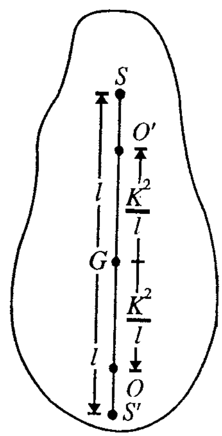


Fig. 7.33.

Interchangeability of centres of suspension and oscillation : The distance of the centre of oscillation from the centre of gravity $= K^2/l = l'$ (say).

Thus, the expression for the time period of the compound pendulum can be written as

$$T = 2\pi\sqrt{\frac{l'+l}{g}} \qquad \text{...(13)}$$

If the pendulum is inverted and made to oscillate about the centre of oscillation, then its time period T' will be given by

$$T' = 2\pi\sqrt{\frac{(K^2/l')+l'}{g}}$$

But $\quad K^2/l = l' \quad \text{or} \quad K^2 = ll'$

$\therefore$

$$T' = 2\pi\sqrt{\frac{(ll/l')+l'}{g}} \quad \text{or} \quad T = 2\pi\sqrt{\frac{l+l'}{g}} \qquad \text{...(14)}$$

From eqs. (13) and (14), we get

$$T = T' \qquad ...(15)$$

Hence, *the time periods of oscillation about the centre of suspension and the centre of oscillation are equal, i.e., they are interchangeable.*

Condition for maximum and minimum time periods ***(Variation of T with l)*** **:** The period of oscillation of a compound pendulum, is given by

$$T = 2\pi\sqrt{\frac{(K^2/l)+l}{g}} \qquad ...(16)$$

Obviously, as l is varied, time period T also alters, if other factors remain constant. Relation (16) shows **the variation of time period with length.**

Eq. (16) can be written as

$$T^2 = \frac{4\pi^2}{g}\left(\frac{K^2}{l}+l\right).$$

Differentiating it with respect to l, we get

$$2T\frac{dT}{dl} = \frac{4\pi^2}{g}\left(-\frac{K^2}{l^2}+1\right). \qquad ...(17)$$

For minimum and maximum periods $dT/dl = 0$, *i.e.*, $-K^2/l^2 + 1 = 0$ or $l = \pm K$

If we calculate the value of d^2T^2/dl^2 and put $l = K$, the quantity d^2T^2/dl^2 comes out to be positive. Hence for this value of l, time period of compound pendulum has minimum value. Condition for the period to be **minimum** is

$$l = K \qquad ...(18)$$

Thus, *the period of oscillation of the pendulum is minimum, when the distance between the centre of suspension and the centre of gravity is equal to the radius of gyration of the pendulum about a parallel axis through the centre of gravity.*

Now, if we put $l = 0$ in eq. (16), the value of T becomes infinite, *i.e.*,

$$T = \text{a maximum} \qquad ...(19)$$

In other words, *the period of oscillation of the compound pendulum is maximum, when the axis of suspension passes through centre of gravity of the pendulum.*

Four points, collinear with the C.G., about which the time period is the same : The time period of a compound pendulum, is given by

$$T = 2\pi\sqrt{\frac{K^2+l^2}{lg}} \qquad ...(20)$$

where K is the radius of gyration of the compound pendulum about an axis passing through its centre of gravity and perpendicular to the plane of the oscillation and l the distance between the centre of suspension and its centre of gravity.

Squaring eq. (20), we get

$$T^2 = 4\pi^2\left(\frac{K^2+l^2}{lg}\right) \quad \text{or} \quad l^2 + K^2 = \frac{lgT^2}{4\pi^2} \quad \text{or} \quad l^2 - \frac{gT^2}{4\pi^2}\cdot l + K^2 = 0 \qquad ...(21)$$

This is a quadratic equation in l and gives two values of l. If l_1 and l_2 be two values, then

$$l_1 + l_2 = \frac{gT^2}{4\pi^2} \qquad ...(22)$$

and

$$l_1 l_2 = K^2. \qquad ...(23)$$

Both l_1 and l_2 are positive, because their sum and also product both are positive. From this we conclude that for a particular value of T there are two points distant l_1 and l_2 from the centre of gravity of the pendulum, about which the pendulum may be suspended and yet we have the same time period.

If $l_1 = l$, then $\qquad l_2 = K^2/l$ $\qquad$ [from eq. (23)]

Now, if two circles are drawn with centre G and radii l and K^2/l [**fig. 7.34**], the period of oscillation about any point of suspension on either circle is the same. Hence there are infinite number of points in a compound pendulum for which the time period has the same value. If a straight line through G is drawn, it will cut these circle in four points S, O', O, S'. Hence *in all there are four points collinear with the C. G. of the pendulum, about which the time periods are equal.*

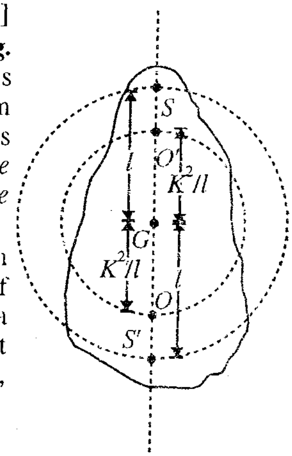


Fig. 7.34 : Compound Pendulum.

Obviously, if S is the centre of suspension, O is the centre of oscllation or vice-versa; similarly if S' is the centre of suspension, $O'$is the centre of oscillation or vice-versa. Consider a pair of such points which (a) lie on a straight line passing through G, (b) lie on opposite side of G, and (c) lie at unequal distances from G. In **fig. 7.34**, S and O or S' and O' are such points, then

$$SO = SG + GO = l + \frac{K^2}{l} = \frac{l^2+K^2}{l}.$$

Similarly, $\qquad S'O' = \dfrac{l^2+K^2}{l}.$

Substituting in eq. (20), we get

$$T = 2\pi\sqrt{\frac{SO}{g}} = 2\pi\sqrt{\frac{S'O'}{g}} = 2\pi\sqrt{\frac{L}{g}}. \qquad ...(24)$$

where $SO = S'O' = L$ is the *length of an equivalent simple pendulum.*

Bar Pendulum—Determination of the Value of g

One of the convenient and simple form of a compound pendulum, used to determine the value of g, is the bar pendulum. It is just a uniform metal bar PQ about one metre long, having holes drilled at equal distances along its length symmetrically about its centre of gravity [**fig. 7.35**]. A knife edge can be inserted into any desired hole and is kept on a horizontal support. Now the bar can be made to oscillate in a vertical plane. In an actual experiment, the bar is allowed to oscillate about the horizontal knife edge which is inserted in each hole in turn, starting from P to G. The period of oscillation corresponding to each hole is obtained by noting the time for twenty five oscillations. The bar is then inverted and similarly the periods are obtained about different holes from G to Q.

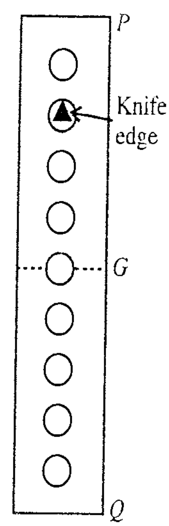


Fig. 7.35 : Bar pendulum.

Now if a graph is plotted between the distance of the centre of suspension from one end of the bar and periodic time, it consists of two curves, symmetrical about a line which cuts X-axis at G' [**fig. 7.36**]. This point corresponds to the centre of gravity G of the bar about which the period is infinite. For a particular value of period T, if we draw a straight line parallel to X-axis, it cuts the two curves at the points A, B, C and D. Obviously if A is the centre of suspension, then C is the corresponding point of oscillation and AC is the length ($L = l + \frac{K^2}{l} = AG'' + G''C$) of equivalent simple pendulum. Similar is the case for the points B and D and length BD. The mean of the distances AC and BD gives the length L of equivalent simple pendulum. Now one can use the relation $T = 2\pi\sqrt{\frac{L}{g}}$ to determine the value of g.

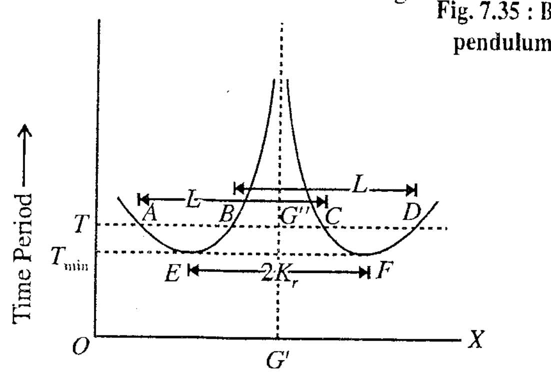


Fig. 7.36 : Graph showing the variation of time period with distance for a bar pendulum.

Points of superiority of a compound pendulum over a simple pendulum : (1) A simple pendulum is an ideal concept only and cannot be realized in actual practice. In the case of a compound pendulum the length of an equivalent simple pendulum can be found easily and accurately. Thus the value of g can be determined accurately with compound pendulum.

(2) The distance between the two positions of knife edges can be measured accurately. In the case of a simple pendulum both the point of suspension and point of oscillation are indefinite and hence the true length of the pendulum can hardly be measured correctly.

(3) The compound pendulum vibrates as a whole and thus there is no possibility of a lag as it may present between the bob and the thread in the case of a simple pendulum.

(4) Due to its large mass and moment of inertia, the compound pendulum keeps on vibrating for a fairly long time and thus the time period can be measured accurately by finding the time for a large number of vibrations. In a simple pendulum the vibrations die out quickly as the mass of the bob is comparatively very small.

(3) Torsional Pendulum

If one end of a fairly thin and long wire is clamped to a rigid support and the other end is attached to the centre of a heavy body (e.g., disc or sphere), then this arrangement is called the **torsional pendulum [fig. 7.37]**.

If the disc or sphere be turned in the horizontal plane to twist the wire and then released, it executes torsional vibrations of a definite period about the wire as axis.

Now, if the disc is rotated through an angle θ, then the restoring torque, set up in the wire, will be $-C\theta$, where $C\ (= \eta\pi r^4/2l)$ is the torsional rigidity or couple per unit twist of the wire. Here η is the modulus of rigidity of the material of the wire and the wire has length l with radius r.

This torque produces an angular acceleration $d^2\theta/dt^2$ in the disc or sphere. If I be the moment of inertia of the disc $(I = \frac{1}{2}MR^2)$ or sphere $(I = \frac{2}{5}MR^2)$ about the wire as axis, then the torque is $I\dfrac{d^2\theta}{dt^2}$. Hence the equation of motion is

$$I\frac{d^2\theta}{dt^2} + C\theta = 0 \quad \text{or} \quad \frac{d^2\theta}{dt^2} + \frac{C}{I}\theta = 0 \qquad ...(25)$$

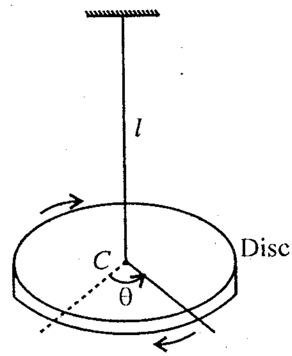


Fig. 7.37 : Torsional pendulum : Restoring torque is opposite to θ increasing.

This equation represents a simple harmonic motion of period

$$T = 2\pi\sqrt{\frac{I}{C}}. \qquad ...(26)$$

It may be noted that in the derivation of this formula no approximation is required. Hence the period of torsional oscillations remains the same for large amplitude oscillations, provided that the elastic limit of the suspension wire has not been exceeded.

The solution of eq. (25) is of the form

$$\theta = \theta_0 \sin(\omega t + \phi) \qquad ...(27)$$

where θ_0 = angular amplitude, $\omega = \sqrt{C/I}$ and ϕ = initial phase.

Inertia table is a modification of torsional pendulum and is employed to determine the moment of inertia of the bodies.

Eq. (26) may be used to determine the moment of inertia $I\ (= CT^2/4\pi^2)$ of the suspended symmetrical body if the torsional rigidity $C\ (= \eta\pi r^4/2l)$ is known.

Ex. 1. *How is the period of a simple pendulum affected, when the support is* (i) *moved horizontally with an acceleration f,* (ii) *moved vertically upward with an acceleration f,* (iii) *moved downward with an* *acceleration f,* (iv) *allowed to fall freely ?*

Sol. Period of simple pendulum $T = 2\pi\sqrt{\frac{l}{g}}$.

(i) When the support is moved horizontally with an acceleration f, the effective force on the bob is $mg' = m\sqrt{g^2+f^2}$.

Period $$T_1 = 2\pi\sqrt{\frac{l}{(g^2+f^2)^{1/2}}}$$ *i.e.*, the period will decrease.

(ii) When the support is moved vertically upwards with an acceleration f, the effective acceleration due to gravity will be $g + f$. Then

$$T_2 = 2\pi\sqrt{\frac{l}{g+f}}$$ *i.e.*, the period will decrease.

(iii) When the support moves vertically downwards with an acceleration f,

$$T_3 = 2\pi\sqrt{\frac{l}{g-f}}$$ *i.e.*, the period will increase.

(iv) When the support falls freely, *i.e.*, $f = g$, the effective force on the bob is $mg' = m(g - f) = 0$. In this case, the pendulum will not vibrate.

Ex. 2. *Using conservation of energy, show that the angular speed dθ/dt of a simple pendulum is given by*

$$\frac{d\theta}{dt} = \left[\frac{2}{ml^2}\{E - mgl(1-\cos\theta)\}\right]^{1/2}$$

where E is total energy of osicllation, L and m are length and mass of the pendulum, and θ is angular displacement from the vertical. **(Agra 1993, 84, 70)**

Sol. The potential energy of the simple pendulum at any position P relative to the lowest position O of the bob **[fig. 7.31]** is obviously given by

$$U = mg(l - l\cos\theta),$$

because the bob is displaced through a vertical height $(l - l\cos\theta)$.

The kinetic energy of the pendulum at the position P is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m(l\,d\theta/dt)^2 \qquad [\because v\ (= r\omega) = l\,d\theta/dt]$$

$\therefore$ Total energy, $E = K + U = \frac{1}{2}ml^2(d\theta/dt)^2 + mgl(1-\cos\theta)$

which should be a constant, according to the law of conservation of energy.

$$\therefore \qquad \frac{d\theta}{dt} = \text{m}\left[\frac{2}{ml^2}\{E - mgl(1-\cos\theta)\}\right]^{1/2}.$$

Ex. 3. *A simple pendulum of length l and mass m is oscillating with a maximum angular displacement θ_0 radians. Show that the tension in the string at the angular displacement θ is mg (3 cos θ – 2 cos θ_0).*

Sol. Let the bob P of mass m be oscillating about the point S. At the angular displacement θ of the pendulum, the weight mg can be resolved into two components $mg\cos\theta$ and $mg\sin\theta$ **[fig. 7.38]**. The latter component provides the restoring force for simple harmonic motion and the former component reduces the tension T of the string. So that the necessary centripetal force for the rotation of the pendulum in a circular path of radius l at the angular displacement θ is

$$T - mg\cos\theta = mv^2/l \qquad \text{...(i)}$$

where v is the velocity of the bob at P.

Total energy of the particle = $mg\ ON$ (at P_0 relative to O) = $mgl(1 - \cos\theta_0)$

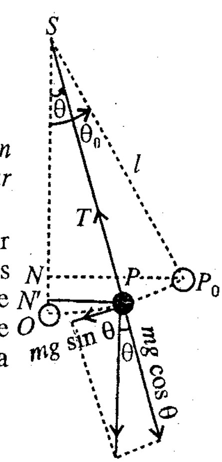


Fig. 7.38.

$$= mgl(1 - \cos\theta) + \frac{1}{2}mv^2 \text{ (at } P).$$

$$\therefore \qquad mgl(1-\cos\theta_0) = mgl(1-\cos\theta)+\frac{1}{2}mv^2$$

or $\qquad 2mg\cos\theta - 2mg\cos\theta_0 = mv^2/l \qquad$ ...(ii)

$\therefore$ From eq. (i) and (ii), we get

$$T - mg\cos\theta = 2mg\cos\theta - 2mg\cos\theta_0$$

whence $\qquad T = 3mg\cos\theta - 2mg\cos\theta_0 = mg\,(3\cos\theta - 2\cos\theta_0).$

Ex. 4. *Derive an expression for the small angle oscillations of a simple pendulum of length L comparable with the radius of earth R. What is the period of a pendulum of an infinite length on the surface of the earth ?*

Sol. As shown in **fig. 7.39,** the restoring force on the bob is

$$m\frac{d^2x}{dt^2} = -\,mg\sin(\theta + \alpha)$$

where θ and α both are small. So that $\sin(\theta + \alpha) = \theta + \alpha$ and $\theta = x/L$ and $\alpha = x/R$ and therefore,

$$\frac{d^2x}{dt^2} + g\left(\frac{1}{L} + \frac{1}{R}\right)x = 0.$$

Thus, the motion of the pendulum is simple harmonic of period

$$T = \frac{2\pi}{\sqrt{g\left(\frac{1}{L}+\frac{1}{R}\right)}} = 2\pi\sqrt{\frac{lR}{g(L+R)}}$$

When $\frac{1}{L} << \frac{1}{R}$ or practically L is tending to infinite length,

$$T = \frac{2\pi}{\sqrt{\left(\frac{1}{\infty}+\frac{1}{R}\right)}} = 2\pi\sqrt{\frac{R}{g}}.$$

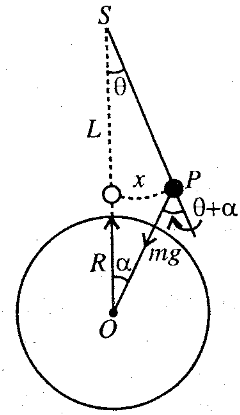


Fig. 7.39.

Ex. 5. *A simple pendulum possesses* 0.05 *joule energy when its length is* 2 *m and the amplitude of motion of the bob is* 4 *cm. Calculate its energy (1) when the length remains the same but the amplitude is doubled, (2) the amplitude remains the same but the length is made one-half.*

Sol. $E = \frac{1}{2}m\omega^2a^2$; Here, $E = 0.05$ J, $\omega^2 = g/l = g/2$ and $a = 4 \times 10^{-2}$ m

$$\therefore \qquad 0.5 = \frac{1}{2}m(g/2)(4\times10^{-2})^2$$

(1) When $a' = 2 \times 4 \times 10^{-2}$ m, $E' = \frac{1}{2}m(g/2)(2\times4\times10^{-2})^2$

$$\therefore \qquad \frac{E'}{0.05} = 4 \quad \text{or} \quad E' = \mathbf{0.2\ J}$$

(2) Here $l' = 2/2 = 1$ m,

$$\therefore \qquad E'' = \frac{1}{2}m(g/2)(4\times10^{-2})^2$$

$$\therefore \qquad \frac{E''}{0.05} = 2 \quad \text{or} \quad E'' = \mathbf{0.1\ J}$$

Ex. 6. *If a mass m is suspended from the end of a string of length l and a conical pendulum is formed, show that the angular velocity ω of the mass is related to the breaking strength T of the string by the relation,* $\omega = \sqrt{T/ml}$. **(Punjab)**

Sol. In the figure, the conical pendulum has been shown. The mass m rotates in a circle of radius r with an angular velocity ω. The necessary centripetal force ($m\omega^2r$) required to move this mass in a circle with the given velocity is provided by the horizontal component of tension or breaking strength T of the string, *i.e.*,

$$T \sin\theta = m\omega^2 r, \quad \therefore \quad \omega = \sqrt{\frac{T\sin\theta}{mr}} = \sqrt{\frac{T\sin\theta}{ml\sin\theta}} \quad [\because\ r = l\sin\theta]$$

or $$\omega = \sqrt{\frac{T}{ml}}.$$

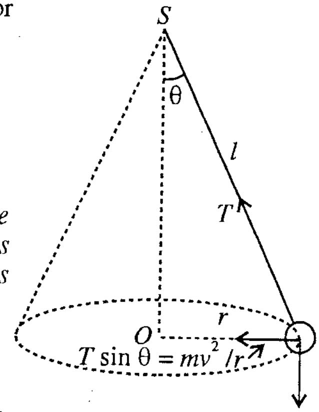


Fig. 7.40.

Ex. 7. *A disc of metal of radius R with its plane vertical can be made to swing about a horizontal axis passing through any one of a series of holes bored along a diameter. Show that the minimum period of oscillation, is given by*

$$T = 2\pi\sqrt{\frac{1.414R}{g}}.$$

Sol. The vibrating disc will act as compound pendulum, whose time period is given by

$$T = 2\pi\sqrt{\frac{(K^2/l) + l}{g}}$$

where K is the radius of gyration of the disc and l the distance between the centre of gravity and the centre of suspension.

The condition for the time period to be mininum is $l = K$.

$$\therefore \qquad T = 2\pi\sqrt{\frac{2K}{g}}$$

For disc, $\frac{1}{2}MR^2 = MK^2 \quad \therefore \quad K = \frac{R}{\sqrt{2}}$, where R is the radius of the disc.

$$\therefore \qquad T = 2\pi\sqrt{\frac{2\times R}{\sqrt{2}g}} = 2\pi\sqrt{\frac{1.414R}{g}}.$$

This is the required expression.

Ex. 8. *A uniform thin rod of length* 120 *cm and width* 6 *cm is swinging in a vertical plane as pendulum about a point A at some distance from one end. If the time of swing is minimum, find the distance of A from the end of the rod.* **(Punjab)**

Sol. Moment of inertia of a uniform thin rod of length l and width b about the axis passing through C.G. and perpendicular to the plane of vibration is given by

$$I = M\left(\frac{l^2+b^2}{12}\right) = MK^2, \text{ where } K \text{ is the radius of gyration.}$$

Here, $l = 120$ cm $= 1.2$ m; $b = 6$ cm $= 0.06$ m.

$$\therefore \qquad K^2 = \frac{l^2+b^2}{12} = 0.1203 \text{ m}^2.$$

If the time of swing is minimum, then distance between point of suspension and C.G. $= K = \sqrt{0.1203}\text{ m} = 0.347$ m.

Therefore, distance from one end of the rod = 0.60 – 0.347 = **0.253 m.**

Ex. 9. *A body of mass* 200 *gm oscillates about a horizontal axis at a distance of* 20 *cm from its centre of gravity. If the length of the equivalent simple pendulum be* 35 *cm find its moment of inertia about the axis of suspension.* **(Patna)**

Sol. The oscillating body acts as compound pendulum, whose time period is given by

$$T = 2\pi\sqrt{\frac{(K^2/l)+l}{g}}.$$

The length $\frac{K^2}{l} + l = L$ is the length of the equivalent simple pendulum.

Here, $l = 0.2$ m, $L = 0.35$ m, $\therefore\ 0.35 = \dfrac{K^2}{0.20} + 0.2$, *i.e.*, $K^2 = 0.03$.

The moment of inertia I of the body about the axis of suspension is

$$I = mK^2 + ml^2 = m(K^2 + l^2) = 0.2\ (0.03 + 0.2^2) = \mathbf{0.014\ kg\text{-}m^2}.$$

Ex. 10. *A circular disc of radius 20 cm oscillates as a pendulum about a point on its circumference. Calculate the period of oscillation.* **(Bihar 1964)**

Sol. Let the disc of radius 20 cm be suspended from the point P of its circumference.

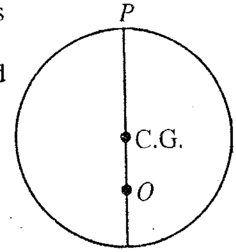


Fig. 7.41.

The moment of inertia of the disc, about an axis passing through its C.G. and perpendicular to its plane will be given by

$$I = MK^2 = \frac{1}{2}M \times (0.2)^2 \text{ or } K^2 = 0.02$$

The distance of centre of oscillation from the point of suspension is

$$L = PO = (K^2/l) + l = 0.3 \text{ m}.$$

$$\therefore \text{Period of oscillation, } T = 2\pi\sqrt{\frac{L}{g}} = 2 \times 3.14 \times \sqrt{\frac{0.3}{9.8}} = \mathbf{1.098\ sec.}$$

Ex. 11. *A compound pendulum is formed by suspending a heavy ring of radius 490.5 cm from a string. What is the length of the string when the period is minimum ? Find this period.* **(Rajasthan)**

Sol. The time period of oscillation of a compound pendulum, is given by

$$T = 2\pi\sqrt{\frac{(K^2/l) + l}{g}} \qquad \text{...(i)}$$

Here, a heavy ring is suspended with the help of a string. If the weight of the string is negligible with respect to the heavy ring, C.G. of the system will lie at the centre of the ring.

From eq. (i) the time period will be minimum, when $K = l$

But for the ring, $\quad I = MR^2 = MK^2$ or $R = K$.

where R is the radius of the ring (consider all the mass is concentrated at the rim).

$$\therefore \qquad R = l$$

i.e., the distance between centre of suspension and centre of ring is R. In other words, for minimum time period, the length of the string must be zero and ring must be suspended to any point of the circumference. Putting in (i) $K = l = R$, we get

$$T_{\min} = \sqrt{\frac{(l^2/l) + l}{g}} = 2\pi\sqrt{\frac{2l}{g}} = 2\pi\sqrt{\frac{2R}{g}}.$$

Here, $R = 490.5$ cm and $g = 981$ cm/sec^2.

$$\therefore \qquad T_{\min} = 2 \times 3.14 \times \sqrt{\frac{2 \times 490.5}{981}} = \mathbf{6.28\ sec.}$$

Ex. 12. *A thin uniform bar of length 120 cm is made to oscillate about an axis through its end. Find the period of oscillation and other points about which it can oscillate with the same period.*

(Poona 1963; Punjab 53; Raipur 65)

Sol. The moment of inertia of a thin bar of length a and mass M, is given by

$$I = \frac{Ma^2}{12} = MK^2, \text{ where } K \text{ is the radius of gyration.}$$

$$\therefore \qquad K^2 = \frac{a^2}{12} = \frac{120 \times 120}{12} = 1200.$$

The time period of compound pendulum, is given by

$$T = 2\pi\sqrt{\frac{(K^2/l) + l}{g}} = 2\pi\sqrt{\frac{(1200/60) + 60}{980}} \qquad [\because\ l = SG = 60 \text{ cm}]$$

$$= 2 \times 3.14 \times \frac{2}{7} = \mathbf{1.796\ sec.}$$

Let L be the length of equivalent simple pendulum, then

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$\therefore \qquad L = \frac{gT^2}{4\pi^2} = \frac{980}{(3.14)^2} \times \frac{(2\times 3.14\times 2)^2}{49} = 80 \text{ cm} = \mathbf{0.8\ m.}$$

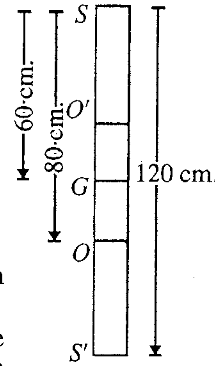


Fig. 7.42.

Therefore, there is another point O at a distance of 80 cm, from S or 20 cm from C.G. about which time period will be the same.

If the pendulum is inverted and suspended from S', time period will be the same due to symmetry. Corresponding to S', there will be a point O at a distance of 20 cm from C.G.

Hence the bar will have same time periods about four points S, S', O, O' whose distance from S are 0, 120, 80 and 40 cm respectively.

Ex. 13. *A soild sphere of mass* 2 *kg and radius* 0.05 *metre is suspended from a wire. Find the period of oscillation, if the torque required to twist the wire is* 4×10^{-3} *newton-m/radian.*

(Bilaspur 2003; Sagar 03)

Sol. Moment of inertia of the sphere about any diameter is

$$I = \frac{2}{3}MR^2 = \frac{2}{3}\times 2\times 0.05 \times 0.05 = 0.002 \text{ kg} \times \text{m}^2.$$

$$\therefore \text{Period of oscillation, } T = 2\pi\sqrt{\frac{I}{C}} = 2\pi\times\sqrt{\frac{0.002}{4\times 10^{-3}}} = \mathbf{4{\cdot}44\ sec.}$$

Ex. 14. *A body is executing torsional oscillations about a wire of torsional rigidity C. If the total energy of oscillation is E and the moment of inertia of the body about the axis of rotation is I, show that the angular velocity* ($d\theta/dt$) *at the displacement* θ *is given by*

$$\frac{d\theta}{dt} = 2\pi\sqrt{\frac{C}{L}}\sqrt{\frac{2E}{C}-\theta}.$$

(Bundelkhand 2004)

Sol. Let C be the couple per unit twist. If the wire is twisted through an angle θ, then the restoring couple is $-C\theta$. Now if the wire is twisted to angle $\theta + d\theta$, then the amount of work done is

$$dW = C\theta d\theta$$

Hence the total amount of work done in twisting the wire through an angle θ is

$$W = \int_0^\theta C\theta\, d\theta = \frac{1}{2}C\theta^2$$

At the angular displacement θ, this work done becomes the potential energy of the system *i.e.,*

$$U = \frac{1}{2}C\theta^2.$$

The instantaneous angular displacement θ of a body, executing torsional oscillation is given by

$$\theta = \theta_0 \sin(\omega t + \phi), \text{ where } \omega = \sqrt{\frac{C}{I}}$$

So that $$\frac{d\theta}{dt} = \theta_0\omega\cos(\omega t + \phi)^*$$

K.E. at the angular displacement θ is

$$K = \frac{1}{2}I\left(\frac{d\theta}{dt}\right)^2 = \frac{1}{2}I\theta_0^2\omega^2\cos^2(\omega t+\phi)$$

$$= \frac{1}{2}I\theta_0^2\omega^2\left(1-\frac{\theta^2}{\theta_0^2}\right) = \frac{1}{2}I\omega^2(\theta_0^2-\theta^2) = \frac{1}{2}C(\theta_0^2-\theta^2)$$

* Here, ω is not the actual angular velocity but governing the phase of motion.

$\therefore$ Total energy, $E = K + U = \frac{1}{2}C(\theta_0^2 - \theta^2) + \frac{1}{2}C\theta^2 = \frac{1}{2}C\theta_0^2$

$\therefore$ Angular amplitude $\theta_0 = \sqrt{\frac{2E}{C}}$

Now, $\frac{d\theta}{dt} = \theta_0\sqrt{1 - \frac{\theta^2}{\theta_0^2}} = \sqrt{\frac{C}{I}}\sqrt{\theta_0^2 - \theta^2} = \sqrt{\frac{C}{I}}\sqrt{\frac{2E}{C} - \theta^2}$.

Ex. 15. *A sphere of moment of inertia 3355 gm-cm² when suspended by a thin wire executes torsional oscillations of period 4.3 seconds. Calculate the maximum angular velocity of the sphere and its average potential energy when the amplitude is π/2 radians.*

Sol. Let the equation for the angular displacement of the sphere be

$$\theta = \theta_0 \sin(\omega t + \phi)$$

$\therefore$ Angular velocity $d\theta/dt = \theta_0\omega \cos(\omega t + \phi)$

Its maximum value $= \theta_0\omega = 2\pi n\theta_0$.

Here $\theta_0 = \pi/2$ radians and frequency $n = 1/4.3$ per sec.

$\therefore$ Max. angular velocity $= 2 \times 3.14 \times \frac{1}{4.3} \times \frac{3.14}{2}$ = **2.3 rad/sec.**

Average potential energy $= \frac{1}{4}I\omega^2\theta_0^2$ [In place of $\frac{1}{2}m\omega^2a^2$]

$$= \frac{1}{4} \times 3355 \times 10^{-7} \times \left(\frac{2\pi}{4.3}\right)^2 \times \left(\frac{\pi}{2}\right)^2 = \mathbf{4.45 \times 10^{-4}\ J.}$$

EXERCISE 7·3

1. A simple pendulum has 0.1 joule energy when its length is 2 metres and the amplitude of motion of bob is 3 cm. What will be its energy when length is not changed but amplitude is 6 cm, (ii) amplitude is 3 cm but length is changed to 1 metre ?
   [**Ans.** 0.4 joule, 0.2 joule.]
2. Show that the maximum tension in the string of a simple pendulum, when the angular amplitude θ_m is small, is $mg(1 + \theta_m^2)$. At what position of the pendulum is the tension a maximum ?
   [**Ans.** At the equilibrium position ($\theta = 0$).] **(Kanpur 1984)**
3. A simple pendulum of length l and mass m is suspended in a car that is travelling with a constant speed v around a circle of radius R. If the pendulum undergoes small oscillations about its equilibrium position, what will be its frequency of oscillation ?
   $\left[\textbf{Ans. } \frac{1}{2\pi}\sqrt{\frac{(g^2 + v^2/R^2)^{1/2}}{l}}.\right]$
4. A sphere of radius R is suspended by a string from a fixed point, so that the distance from the centre of sphere to the point of suspension is ($l >> R$). Show that the period of the pendulum is
   $$T = 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{R^2}{5l^2}\right).$$
   If $R = 0.05$ m and $l = 1$ m, what is the length of the equivalent simple pendulum ?
   [**Ans.** 1.001 m.]
5. A body of mass 200 gm oscillates about a horizontal axis at a distance of 20 cm from its centre of gravity. If the length of the equivalent simple pendulum be 35 cm, find its moment of inertia about the axis of suspension.
   [**Ans.** 1.4×10^{-4} kg × m².]
6. A thin uniform bar of length 1.2 m is made to oscillate about an axis through its end. Find the period of oscillation and other points about which it can oscillate with the same period. **(Poona 1969; Punjab)**
   [**Ans.** 1.796 sec; Points are situated at distance 0.4, 0.8, 1.2 m from one end.]

7. (i) Assuming small amplitude oscillations, what would be the frequency of a simple pendulum of length 2 metres, in an elevator accelerating upward at a rate of 2.0 metres/sec^2 ?

(ii) What would be its frequency in free fall ?

[**Ans.** (i) 0.38 Hz; (ii) Zero.]

8. Prove that a uniform rod of length L hanging as a compound pendulum from a pivot at one end has the same frequency as a simple pendulum of length $2L/3$. for oscillations of small amplitude. **(Delhi 1989)**

9. A thin rod of length 60 cm is used as a compound pendulum. Find the position of a point such that the period about it is a minimum.

[**Ans.** 17.3 cm from C.G.]

10. A disc of radius r oscillates as a pendulum about a horizontal axis perpendicular to the plane of the disc through a point $r/2$ from its centre. Calculate the period of oscillation. **(Bombay 1970, 64)**

[**Ans.** $T = 2\pi\sqrt{3r/2g}$.]

11. A metal disc of radius R is made to oscillate about an axis passing through a point on its edge. Find (a) the equivalent length of simple pendulum, and (b) the minimum time period of oscillation.

[**Ans.** $3R/2$; $T = 2\pi\sqrt{3R/2g}$.]

12. Calculate the time of oscillation of a uniform square lamina of side 30 cm about an axis perpendicular to its plane and passing through one of its corners. (g = 9.8 m/sec^2).

[**Ans.** $T = \pi\sqrt{13/98}$.]

13. A uniform circular rod with a radius of 2 cm oscillates when suspended from a point on its axis at a distance of 4 cm from one end. If the length of the rod is 1 metre, find the point or points from which, it suspended, the periodic time would remain unaltered. **(Bombay)**

[**Ans.** At 31.87 cm, 68.13 cm and 96 cm from the same end.]

14. A uniform rod AB of mass 100 gm and length 120 cm can swing in a vertical plane about A as pendulum. A particle of mass 200 gm is attached to the rod at a distance x from A. Find x such that the period of vibration is a minimum. **(Gorakhpur 1964; Madras 51)**

[**Ans.** x = 2.748 cm.]

15. A rod of length 1 metre and mass 0.5 kg is fixed at one end and is initially hanging vertical. The other end is now raised until it makes an angle of 60° with respect to the vertical. How much work is required ? **(Lucknow 1978)**

[**Ans.** 1.225 joule.]

16. A solid cylinder of 2 cm radius weighing 200 gm is rigidly connected, with its axis vertical, to the lower end of a fine wire. The period of oscillation of the cylinder under the influence of the torsion of wire, is 2 seconds. Calculate the couple necessary to twist it through four complete turns.

[**Ans.** 9.9×10^{-3} N-m.] **(Allahabad 1959)**

17. A torsion pendulum consists of a rectangular wood block 8 cm × 12 cm × 3 cm and with a mass 0.3 kg suspended by means of a wire passing through its centre in such a way that the shortest side is vertical. If the period of torsional oscillations is 2.4 sec, find the torsion constant of the wire.

[**Ans.** 3.565×10^{-3} N-m/rad.]

18. A solid spherical bob of radius 0.05 m and mass 3.5 kg is suspended from a wire of torsional rigidity 6×10^{-3} N-m/rad to form a torsion pendulum. Find the period. Find also the period, in case if a hollow sphere of the same mass and same material but with external radius 6 cm were used.

[**Ans.** 4.8 sec; 6.62 sec.]

19. The balance wheel of a watch vibrates with an angular amplitude of π radians and a period of 0.5 second. Determine (i) the maximum angular speed of the wheel, (ii) the angular speed of the wheel at the displacement $\pi/2$ radians, and (iii) the angular acceleration of the wheel at the displacement $\pi/4$ radians.

[**Ans.** (i) 39 rad/sec., (ii) 34 rad/sec., (iii) 120 rad/sec^2.]

7·9·3. Bifilar Suspension

If a heavy and uniform bar or rod be suspended horizontally by means of two equal, vertical, flexible and inelastic threads, equidistant from the centre of gravity of the bar, this arrangement constitutes a bifilar suspension. On displacing the bar in its own plane horizontally and then releasing, it executes a S.H.M. under gravity about the vertical axis. We shall consider a Bifilar suspension with parallel threads.

Let a uniform rod AB of weight mg be suspended in a horizontal position by two equal, vertical and parallel threads of CA and DB, each of length l and $2a$ distance apart.

Let AB be the equilibrium position of the rod. Now, if the rod is displaced about the vertical axis through O through an angle θ to the position $A'B'$, the suspension threads take the positions CA' and DB'. The suspension threads CA' and DB' make an angle ϕ with their original positions CA and DB respectively **[fig. 7.43]**.

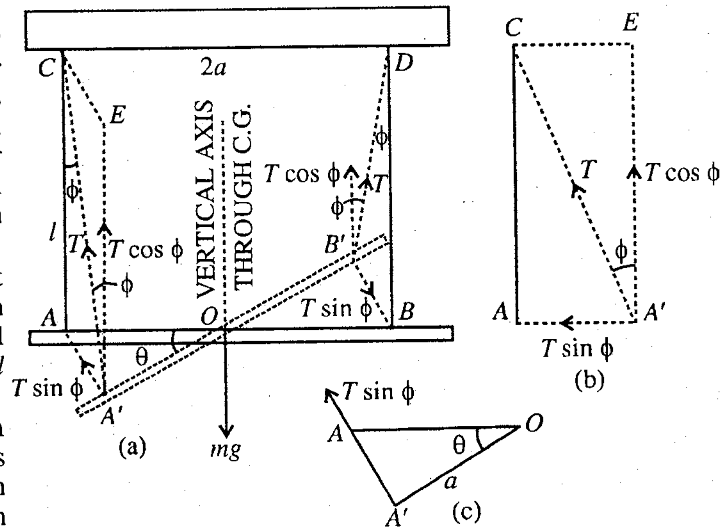


Fig. 7.43 : Bifilar suspension.

Let T be the tension in each thread. It can be resolved into two components :

(i) $T \cos \phi$ along vertical direction, and

(ii) $T \sin \phi$ acting horizontally along $A'A$ or $B'B$.

Evidently, the vertical components support the weight, *i.e.*, $2T \cos \phi = mg$ or $T \cos \phi = mg/2$

As ϕ is small, $\cos \phi = 1$ and $\sin \phi = \phi$, $\therefore \quad T = mg/2$

Also, $\quad AA' = AC \cdot \phi = OA \cdot \theta$ or, $l \cdot \phi = a \cdot \theta$ or $\phi = a \cdot \theta / l$.

The horizontal component of the tension T, acting along $A'A$, is

$$T \sin \phi = T\phi = \frac{mg}{2} \cdot \phi = \frac{mga}{2l} \cdot \theta$$

The two horizontal components (each $T \sin \phi$) along $A'A$ and $B'B$ are equal, opposite and parallel, hence constitute a couple, whose moment is given by

$$T \sin \phi \cdot 2a = \frac{mga}{2l} \cdot \theta \cdot 2a = \frac{mga^2\theta}{2l}.$$

If I be the moment of inertia of the rod about the vertical axis through O and $\frac{d^2\theta}{dt^2}$ its angular acceleration, then equation of motion is

$$\frac{Id^2\theta}{dt^2} + \frac{mga^2}{l} \cdot \theta = 0 \quad \text{or} \quad \frac{d^2\theta}{dt^2} + \frac{mga^2}{l \cdot I} \cdot \theta = 0$$

This is the equation of S.H.M. whose **periodic time,** is given by

$$T = 2\pi \sqrt{\frac{l \cdot I}{mga^2}} = \frac{2\pi}{a} \sqrt{\frac{l \cdot I}{mg}}.$$

But $I = mK^2$, where K is the radius of gyration of the rod about the vertical axis through O. Therefore,

$$T = \frac{2\pi}{a} \sqrt{\frac{mK^2 \cdot l}{mg}} = \frac{2\pi K}{a} \sqrt{\frac{l}{g}}, \quad \text{whence,} \quad g = \frac{4\pi^2 K^2 l}{a^2 T^2}.$$

Thus, the value of g can be determined easily.

7·9·4. Helmholtz Resonator

Let us consider a flask of volume V with a narrow neck of length l and area of cross-section A such that the volume V is much large when compared to the volume lA of the neck [**fig. 7.44**]. Such a system is sometimes called Helmholtz resonator because the air in it can oscillate at a natural frequency of its own and can show resonance with the frequency of incident sound wave which is equal to its natural frequency.

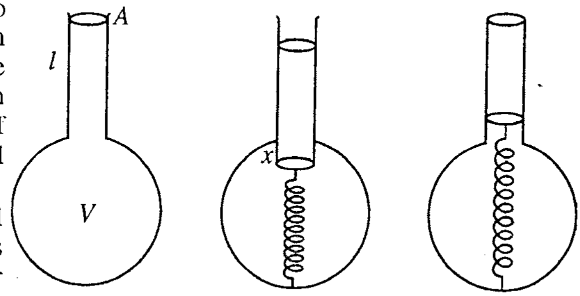


Fig. 7.44 : Oscillations in Helmholtz resonator : The air in the neck behaves like the mass and the air in the bulb like the spring.

Let us calculate an expression for the natural frequency of the resonator. The resonator receives the exciting waves through the neck. We may consider the free vibrations in which the air in the neck of the flask moves in and out like a solid piston which alternately compresses and rarefies the air inside the flask. The air in the neck behaves like the mass and the air in the bulb like the spring of the mechanical oscillator.

Suppose the air within the neck moves through a small distance x inwardly from the equilibrium position. The change in the volume of the air in the bulb is $\Delta V = xA$ and the resulting increase in pressure p is given by

$$p = -E_\gamma \frac{\Delta V}{V} = -E_\gamma \frac{Ax}{V} \qquad ...(1)$$

because the coefficient of elasticity $E_\gamma = \dfrac{\text{Stress}}{\text{Strain}} = \dfrac{p}{-\Delta V/V}$ and E_γ is the adiabatic elasticity of the gas (as the changes during compressions and rarefactions being adiabatic). This excess of pressure p inside the air in the bulb over the atmospheric air provides restoring force F opposite to the movement of the air piston :

$$F = pA = -\frac{E_\gamma A^2}{V} x. \qquad ...(2)$$

The mass of the air piston is $lA\rho$, ρ being the density of air. Using Newton's second law of motion, we get

$$lA\rho \frac{d^2x}{dt^2} = -\frac{E_\gamma A^2}{V} x \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{E_\gamma A}{Vl\rho} x = 0 \qquad ...(3)$$

This equation is identical to the differential equation of the harmonic oscillator. Hence the motion of the air of the neck executes simple harmonic motion and the natural frequency of the oscillator is given by

$$\upsilon = \frac{1}{2\pi}\sqrt{\frac{E_\gamma A}{Vl\rho}} = \frac{v}{2\pi}\sqrt{\frac{A}{Vl}} \qquad ...(4)$$

where $v = \sqrt{E_\gamma / \rho}$ is the speed of sound.

Now if a sound wave is allowed to fall on the open end of the flask, then it will resonate only if its natural frequency is equal to the frequency of the incident sound wave. If a compound note is sounded near a resonator and if it contains the component to which the resonator can resonate, it will accept and reinforce the component. If we have a large number of resonators having different known natural frequencies, then we can find the frequencies of different components of the note by just noting the resonating resonators. Helmholtz devised special type of such resonators to analyse complex sounds and hence they are known after his name as Helmholtz resonators. An audible note is heard when a cork is rapidly withdrawn from a tightly stoppered bottle. This is due to the adiabatic oscillations of the air in the neck.

7·9·5. Charge Oscillations in *L-C* Circuit

Let us consider a coil of self-inductance L and negligible resistance connected in series with a capacitor of capacitance C [**fig. 7·45**]. Initially the condenser is charged by a battery to a charge $+ Q_0$ on the right plate. Next the connection of the battery is cut off with the help of a key and the condenser is discharged through the inductance L.

Let Q be the charge at any time t on the capacitor C in the circuit. The potential difference across the capacitor C is $V = Q/C$. We take V to be positive when the right plate of the condenser is positively charged and represent the current direction by the arrow in **fig. 7.45**. Obviously the voltage across the capacitor is the same as the induced voltage $L\,di/dt$ across the inductance L, where di/dt is the rate of change of the current in the circuit. Thus

$$V = \frac{Q}{C} = L\frac{di}{dt} \qquad ...(1)$$

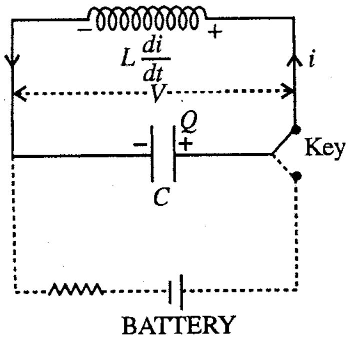


Fig. 7.45 : L-C. circuit.

But $i = -dQ/dt$ (the charge is decreasing on the right plate and the current is flowing as shown),

$$L\frac{d^2Q}{dt^2} + \frac{Q}{C} = 0 \qquad ...(2)$$

Eq. (2) is analogous to the differential equation of simple harmonic motion of spring-mass system. We can think Q as the charge displacement, L as the charge inertia and $1/C$ as the force constant. Here the electromotive force (Q/C) across the capacitance plays the role of restoring force in the mechanical oscillator.

Eq. (2) can be written as

$$\frac{d^2Q}{dt^2} + \frac{1}{LC}Q = 0 \text{ or } \frac{d^2Q}{dt^2} + \omega^2 Q = 0 \qquad ...(3)$$

This is identical to equation of the harmonic oscillator. Thus the charge on the condenser is oscillatory and the frequency of oscillation is given by

$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \qquad ...(4)$$

The solution of eq. (3) is expressed as

$$Q = Q_0 \sin(\omega t + \phi) \qquad ...(5)$$

Thus we find that the charge on the condenser oscillates between the values $+Q_0$ and $-Q_0$ and its frequency depends on L and C. Differentiating eq. (5) with respect to t, we get the instantaneous current i, *i.e.*,

$$i = -dQ/dt = -Q_0\omega \cos(\omega t + \phi)$$

or $$i = i_0 \sin\left(\omega t + \phi - \frac{\pi}{2}\right) \qquad ...(6)$$

where $$Q_0\omega = i_0.$$

Thus the current lags in phase by angle $\pi/2$ to the charge.

Practically there is always some resistance in L-C circuit and its effect will be considered later. Now let us calculate the energy stored in the inductance L and capacitance C at any instant t. The current is build up from 0 to i in time t and the energy E_L stored in the inductance L is obtained by integrating the instantaneous power with respect to time *i.e.*,

$$E_L = \int_0^t Vi\,dt = \int_0^t \frac{di}{dt} i\,dt = \int_0^I Li\,di = \frac{1}{2}Li^2$$

The potential energy stored at the instant t in the capacitor is

$$E_C = \frac{1}{2}\frac{Q^2}{C}$$

Therefore the total energy in the circuit at any time t is

$$E = E_L + E_C = \frac{1}{2}Li^2 + \frac{1}{2}\frac{Q^2}{C} \qquad ...(7)$$

This equation for total energy is similar to the equation for mechanical oscialltor ($E = \frac{1}{2}mv^2 + \frac{1}{2}Kx^2$). As Q and i vary with time, the inductance and capacitor exchange energy periodically similar to the energy exchanges in the spring and mass system.

7·9·6. Oscillations of a Magnet

Let a bar magnet be suspended freely in the magnetic field of the earth by a thread from a rigid support. In the equilibrium position, it will stay in the north-south direction (magnetic meridian). If the magnet is deflected from the equilibrium position and then left free, it will execute angular oscillations.

Let us consider a bar magnet of magnetic moment M, suspended freely in the earth's magnetic field B. Suppose the bar magnet is displaced through an angle and then released, it will execute angular oscillations. If, at any instant t, the angular displacement is θ from the direction of the magnetic field **[fig. 7.46]**, then the restoring torque acting on it is given by

$$\tau = -MB \sin\theta \qquad ...(1)$$

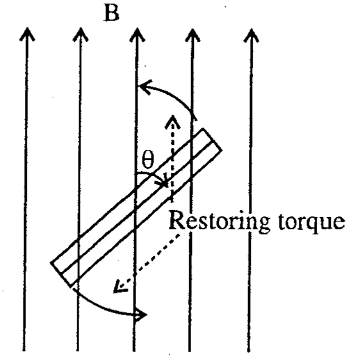


Fig. 7.46 : Oscillations of a freely suspended bar magnet of magnetic moment M in the earth's magnetic field B.

If I be the moment of inertia of the bar magnet about the suspension thread and $d^2\theta/dt^2$ the instantaneous acceleration, the equation of motion is

$$I\frac{d^2\theta}{dt^2} = -MB\sin\theta \qquad ...(2)$$

For sufficiently small θ, $\sin\theta = \theta$. Then from (2)

$$\frac{d^2\theta}{dt^2} + \frac{MB}{I}\theta = 0 \qquad ...(3)$$

This equation represents a simple harmonic motion whose period of vibration is given by

$$T = \sqrt{\frac{I}{MB}} \qquad ...(4)$$

If the length and width of the magnet are l and b respectively and its mass is m, then its moment of inertia about the axis of suspension is $m\left(\frac{l^2+b^2}{12}\right)$.

7·9·7. Oscillation of Two Particles Connected by a Spring

Let us consider two masses m_1 and m_2 connected by a weightless spring. The masses are free to vibrate only along the axis of the spring. Such an oscillating system is called **two body harmonic oscillator.** When due to some reason one of the masses is slightly displaced from its equilibrium position, this produces an extension or compression in the spring. Due to spring action, it exerts a linear restoring force on the masses so that both of the masses vibrate about their mean equilibrium positions and the centre of mass of the system remains stationary.

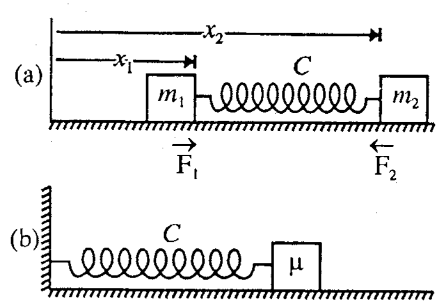


Fig. 7.47 : (a) Two-body oscillator : $F_2 = -Cx = -F_1$, $x = (x_2 - x_1) - r_0$, r_0 = unstretched length of the spring of force constant C; (b) Equivalent single-body oscillator of mass $\mu\left(=\frac{m_1m_2}{m_1+m_2}\right)$ attached to a rigid wall by a spring of the same force constant C.

Suppose that two masses m_1 and m_2 are connected by a massless spring of force constant C. The masses are constrained to move along the axis of the spring **[fig. 7.47]**. Let r_0 be the normal length of the spring. Let x_1 and x_2 be the coordinates of the two ends of the spring at any instant t. Therefore the instantaneous length of the spring is $r = x_2 - x_1$. The change in length is thus obtained to be

$$x = r - r_0 = (x_2 - x_1) - r_0 \quad ...(1)$$

For $x > 0$, $x = 0$ and $x < 0$, the spring is extended, normal and compressed respectively. Suppose that at any instant, the spring is increased in length *i.e.*, $x > 0$. The spring is exerting the same force (Cx) in magnitude on the two masses but the force F_1 $(= Cx)$, acting on m_1, is opposite to the force F_2 $(= - Cx)$ on m_2 *i.e.*,

$$F_1 = Cx \text{ and } F_2 = - Cx \quad ...(2)$$

Applying Newton's second law, we obtain the equations of motion of the two masses as follows :

$$m_1 \frac{d^2x_1}{dt^2} = Cx \quad \text{and} \quad m_2 \frac{d^2x_2}{dt^2} = - Cx$$

$$\therefore \quad \frac{d^2x_1}{dt^2} = \frac{C}{m_1}x \quad \text{and} \quad \frac{d^2x_2}{dt^2} = -\frac{C}{m_2}x$$

On subtraction

$$\frac{d^2x_2}{dt^2} - \frac{d^2x_1}{dt^2} = -\frac{C}{m_2}x - \frac{C}{m_1}x \quad \text{or} \quad \frac{d^2(x_2 - x_1)}{dt^2} = -\left(\frac{1}{m_1} + \frac{1}{m_2}\right)Cx$$

or

$$\frac{d^2x}{dt^2} = -\frac{C}{\mu}x \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{C}{\mu}x = 0 \quad ...(3)$$

where $\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$ or $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is the **reduced mass** of the system. We have used $\frac{d^2(x_2 - x_1)}{dt^2}$ $= \frac{d^2x}{dt^2}$ because r_0 is constant.

Eq. (3) is identical in form to the equation for simple harmonic oscillator, developed for one-body oscillations. The frequency of oscillation of the system is given by

$$\nu = \frac{1}{2\pi}\sqrt{\frac{C}{\mu}} \quad ...(4)$$

Thus the system has the same frequency of oscillation as a single object of mass μ, connected by a similar spring of force constant C to a rigid wall. Hence the two-body oscillator **[fig. 7.47(a)]** is equivalent to one-body oscillator **[fig. 7.47(b)]**. However in **[fig. 7.47(a)]** x is the relative displacement of the two masses from their equilibrium positions rather than the displacement of a single mass from its equilibrium position and μ is the reduced mass tather than the mass of single object. The potential energy of the two-body harmonic oscillator, when the length of the spring changes by x, is given by

$$U = \int_0^x Cx\,dx = \tfrac{1}{2}Cx^2 = \tfrac{1}{2}C(r - r_0)^2 \quad ...(5)$$

Vibrations of a diatomic molecule : A diatomic molecule, *e.g.*, HCl molecule, is an example of two-body system which can oscillate along the axis of symmetry. The atoms in a diatomic molecule are coupled through electromagnetic forces and the bonding between them may be imagined like a spring. Thus we may consider a diatomic molecule as a system of two masses connected by a spring.

In a diatomic molecule, the binding forces are electromagnetic in origin which are composed of attractive and repulsive parts. The equilibrium separation r_0 between the two atoms of the molecule is that at which the attractive and repulsive forces exactly balance each other. If the separation is slightly increased, the two atoms try to come to the equilibrium distance under resultant attractive force (restoring force) and if the separation is slightly decreased, they repel each other to have equilibrium configuration. Thus the vibrations of the molecule may take place, if the system is disturbed by giving some energy.

Fig. 7.48(a) represents a typical potential energy (U–r) curve for a diatomic molecule. The potential energy U is assumed to be zero at infinite separation of atoms and it is negative for moderate distances, measured at atomic scale. At a given distance, this U represents the amount of energy required to separate the atoms completely (with no kinetic energy). For separations smaller than r_0, U increases and becomes large positive as the atoms push each other strongly at very small separations. The potential energy is minimum, say $-U_0$, at the equilibrium separation r_0. The region of the bottom of the potential energy curve is bounded and for small $|r - r_0|$, a parabolic potential well may be fitted to the bottom of U-r curve. This means that if the system is disturbed by giving some energy, then for small $|r - r_0|$ values, the system will execute simple harmonic vibrations. For slight changes in equilibrium separation, one can write the equation for potential energy curve as follows :

$$U = -U_0 + \frac{1}{2}C(r - r_0)^2 \qquad ...(6)$$

The force acting at the separation r is given by

$$F = -\frac{dU}{dr} = -C(r - r_0) \qquad ...(7)$$

A graph between the force F and the seperation r, derived from the U - r curve of **fig. 7.47(a)**, is shown in **[fig. 7.47(b)]**. It is linear for small separations $|r - r_0|$. The frequency of small amplitude oscillations is given by

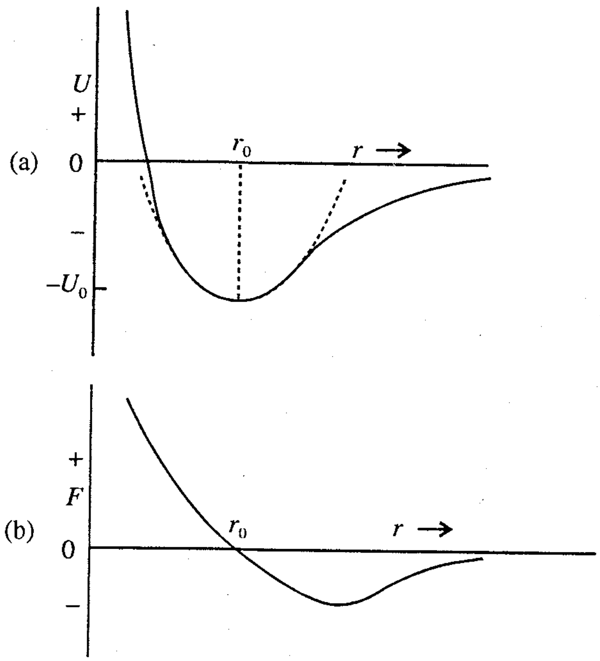


Fig. 7.48 : Potential energy ($U - r$) curve for a diatomic molecule : (a) Minimum potential energy $-U_0$ occurs at equilibrium internuclear separation r_0; (b) Variation of force F ($= -dU/dr$) with internuclear separation r.

$$\upsilon = \frac{1}{2\pi}\sqrt{\frac{C}{\mu}} \qquad ...(8)$$

where μ is the reduced mass of the diatomic molecule.

Ex. 1. *A vessel has neck of* 2 *cm diameter and* 2.5 *cm length. If the capacity of the vessel is* 2 *litres, calculate the wavelength to which this system will resonate.*

Sol. Frequency, $\upsilon = \frac{v}{2\pi}\sqrt{\frac{A}{Vl}}$

$\therefore$ Wavelength, $\lambda = \frac{v}{\upsilon} = 2\pi\sqrt{\frac{Vl}{A}} = 2\pi\sqrt{\frac{2.5\times10^{-2}\times2\times10^{-3}}{\pi\times(0.01)^2}}$ = **2.5 m.**

Ex. 2. *What is the frequency of electrical oscillations in an inductance of* 10 *millihenry in series with a* 2 μf *capacitance. If the maximum potential difference across the condenser is* 5 *volts, find the energy of the oscillating system.* **(Rewa 2004; Bhopal 04)**

Sol. $\upsilon = \frac{1}{2\pi\sqrt{LC}}$

Here $L = 10 \times 10^{-3}$ henry and $C = 2 \times 10^{-6}$ farad.

$\therefore$ $\upsilon = \frac{1}{\sqrt{10\times10^{-3}\times2\times10^{-6}}}$ = **1.1 × 10³ Hz.**

Energy, $E = \frac{1}{2}CV^2 = \frac{1}{2}\times 2\times10^{-6}\times 5^2$ = **25 × 10⁻⁶ joules.**

Ex. 3. *Show that the total energy in the L–C circuit, remains constant with time. What is the expression obtained for the total energy ?*

Sol.
$$E = \frac{1}{2}Li^2 + \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}LQ_0{}^2\omega^2\cos^2(\omega t + \phi) + \frac{1}{2}\cdot\frac{Q_0{}^2}{C}\sin^2(\omega t + \phi)$$

But
$$\omega^2 = \frac{1}{LC}, \quad \therefore \; E = \frac{1}{2}\frac{Q_0{}^2}{C}.$$

Hence the total energy remains constant with time. The expression for total energy is given above.

Ex. 4. *Two bar magnets of same dimensions and same material are oscillated freely one by one in the earth's magnetic field. Compare the magnetic moments of the two megnets.*

Sol. Here,
$$T_1 = 2\pi\sqrt{\frac{I}{M_1 B}} \quad \text{and} \quad T_2 = 2\pi\sqrt{\frac{I}{M_2 B}}$$

$$\therefore \quad \frac{T_2}{T_1} = \sqrt{\frac{M_1}{M_2}} \quad \text{or} \quad \frac{M_1}{M_2} = \frac{T_2^2}{T_1^2}.$$

Ex. 5. *Two masses of* 10 *gm and* 90 *gm are connected by a spring of length* 10 *cm and force constant* 10^3 *N/m*^{2}*, calculate the frequency of oscillation.*

Sol.
$$\nu = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{C}{\mu}}$$

Here $C = 10^3$ N/m and
$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{0.01 \times 0.09}{0.01 + 0.09} = 0.009 \text{ kg}.$$

$$\therefore \quad \nu = \frac{1}{2\pi}\sqrt{\frac{10^3}{0.009}} = \mathbf{53\ Hz.}$$

Ex. 6. *Consider two discs of masses 25 gm and 50 gm (as shown in figure) with massless spring of force constant* 10^3 *N/m. Calculate the frequency of oscillation of the spring* (i) *when the system is resting on a table, and* (ii) *when the table is removed and the system is falling freely.*

Sol. In the first case, one disc is resting on the table and therefore the other disc of mass 25 gm will vibrate.

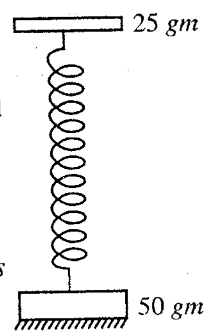


Fig. 7.49.

$$\therefore \quad \text{Frequency, } \nu = \frac{1}{2\pi}\sqrt{\frac{C}{m}} = \frac{1}{2\pi}\sqrt{\frac{10^3}{0.025}} = \frac{10^2}{\pi} = \mathbf{31.8\ Hz.}$$

In the second case, when the system is falling freely, both discs will bibrate and hence

$$\nu = \frac{1}{2\pi}\sqrt{\frac{C}{\mu}} = \frac{1}{2\pi}\sqrt{\frac{10^3 \times 3}{0.05}} = \mathbf{39\ Hz.} \left(\because \mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{0.025 \times 0.05}{0.025 + 0.05} = \frac{0.05}{3}\text{kg}\right)$$

Ex. 7. *The potential energy of a diatomic molecule at a separation r of its atoms is represented as*

$$U(r) = -\frac{e^2}{4\pi\varepsilon_0 r} + \frac{C}{r^9}$$

where the first and second terms are the contributions due to attractive and repulsive parts respectively; (e = electronic charge, ε_0 *= permittivity and C = constant). Find an expression for the force F. Show that the value of constant C is* $e^2 r_0{}^8/36\pi\varepsilon_0$ *and the force constant C is* $2e^2/\pi\varepsilon_0 r_0{}^3$. *where* r_0 *is equilibrium separation. What is the frequency of small amplitude oscillations ?*

Sol.
$$U(r) = -\frac{e^2}{4\pi\varepsilon_0 r} + \frac{C}{r^9}$$

$$\therefore \quad F = -\frac{dU}{dr} = -\frac{e^2}{4\pi\varepsilon_0}\cdot\frac{1}{r^2} + \frac{9C}{r^{10}}$$

At the equilibrium separation $r = r_0$, force vanishes *i.e.*,

$$\frac{e^2}{4\pi\varepsilon_0}\cdot\frac{1}{r_0^{\,2}} - \frac{9C}{r_0^{\,10}} = 0 \quad \text{or} \quad C = \frac{e^2 r_0^{\,8}}{36\pi\varepsilon_0}$$

Now, $$\left|\frac{d^2U}{dr^2}\right|_{r=r_0} = \left|-\frac{e^2}{4\pi\varepsilon_0}\cdot\frac{2}{r^3} + \frac{90C}{r_0^{\,11}}\right|_{r=r_0} = \left|-\frac{e^2}{4\pi\varepsilon_0}\cdot\frac{2}{r_0^{\,3}} + \frac{90}{r_0^{\,3}}\cdot\frac{e^2}{36\pi\varepsilon_0}\right| = \frac{2e^2}{\pi\varepsilon_0 r_0^{\,3}}$$

which is positive and hence $r = r_0$ is the separation at stable equilibrium.

$$\text{Force constant, } C = \frac{2e^2}{\pi\varepsilon_0 r_0^{\,3}}$$

$$\text{Frequency, } \nu = \frac{1}{2\pi}\sqrt{\frac{C}{\mu}} = \frac{1}{2\pi}\sqrt{\frac{2e^2}{\pi\varepsilon_0 r_0^{\,3}}\cdot\frac{m_1+m_2}{m_1 m_2}}$$

where $\mu = \dfrac{m_1 m_2}{m_1 + m_2}$ is the reduced mass of the diatomic molecule.

Ex. 8. *Suppose that for* HCl *molecule, if r_0 is equal to* 1.3 Å, *find the value of the force constant and frequency of small amplitude oscillations.* (m_H = 1.67 × 10^{-27} kg and m_{Cl} = 35 m_H)

Sol. $$C = \frac{2e^2}{\pi\varepsilon_0 r_0^{\,3}} = \frac{8e^2}{4\pi\varepsilon_0 r_0^{\,3}} = \frac{8\times(1.6\times10^{-19})^2\times9\times10^9}{(1.3\times10^{-10})^3} = \mathbf{838.96\ N\text{-}m}$$

$$\nu_0 = \frac{1}{2\pi}\sqrt{\frac{C}{\mu}} = \frac{1}{2\pi}\sqrt{\frac{838.96\times36}{35\times1.67\times10^{-27}}} = \mathbf{11.4\times10^{13}\ Hz.}$$

where $$\mu = \frac{m_H m_{Cl}}{m_H + m_{Cl}} = \frac{1.67\times10^{-27}\times35\times1.67\times10^{-27}}{1.67\times10^{-27}+35\times1.67\times10^{-27}} = \frac{35\times1.67\times10^{-2}}{36}.$$

Ex. 9. *The following diagram shows a typical potential energy curve for a system like a diatomic molecule. Two cases a and b are shown. Now,*

(i) *Read off the equilibrium internuclear distance and binding energy in the two cases.*

(ii) *What is the significance of the fact that the well for case* (b) *is much narrower than the for case* (a) ?

Sol. (i) **Fig. 7.50** represents the variation of the total potential energy of a diatomic molecule with the inter-nuclear distance *r*. As in the equilibrium position, the potential energy of the system is minimum, hence for the curve, (a) the equilibrium inter-nuclear distance is nearly 3 Å and for curve (b) it is nearly 4 Å.

If we neglect the zero point energy, the binding energy is numerically equal to the minimum potential energy. Hence for the curve (a) the binding energy is nearly **7 eV** and for the curve (b) it is **8 eV**.

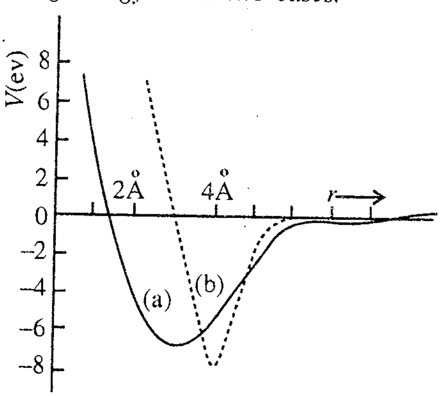


Fig. 7.50.

(ii) The potential well for case (b) is much narrower than for case (a). It signifies that in the former case d^2U/dr^2 has numerically larger value and hence in case (b) force constant C has the greater value. For the same reduced mass, frequency $\left(\frac{\nu}{8} = \frac{1}{2\pi}\sqrt{\frac{C}{\mu}}\right)$ will be higher in the case (b).

EXERCISE 7·4

1. A flask has a neck of radius 1 cm and length 2.5 cm. If the capacity of the flask is 2 litres, determine the frequency to which the system will resonate. (speed of sound in air = 350 m $\sec^{-1}$).
[**Ans.** 34.9 Hz.]

2. A simple Helmholtz resonator is made by blowing air into the mouth of bottle. The volume of the bottle is 2000 cc and the neck has a cross-section area 10 sq. cm and a length of 5 cm. Calculate the frequency of the sound which is generated. The velocity of sound in air is 344 m/sec.
[**Ans.** 173.3 Hz.]
3. A coil of inductance 16 henry carries a steady current of 10 amp. If the circuit is broken by a switch across which a condenser of capacity 4 μF is placed, calculate the frequency of the oscillating circuit.
[**Ans.** 19.9 Hz.]
4. Calculate the frequency of electrical oscillations in an inductance of 20 millihenry in series with a capacitor of 1 μF capacitance. If the maximum potential difference across the condenser is 10 volts, calculate the energy of oscillation.
[**Ans.** 1125 Hz; 5×10^{-5} joule.]
5. A condenser of capacity 20 μF and an inductance of 10 mH are joined in series. Calculate the frequency of electrical oscillations. If maximum potential difference across the condenser is 10 volts, calculate the energy in the oscillating system. **(Rohilkhand 1984)**
[**Ans.** 3.5×10^2 Hz; 10^{-3} joules.]

[Two Body Harmonic Oscillator]

6. A disc of mass 60 gm, attached to the lower end of a spring of force constant 100 gm wt/cm, is resting on a table. A 30 gm disc is attached to the upper end of the spring. Find the compression of the spring due to the weight of the upper mass. If the table is suddenly removed, calculate the amplitude and frequency of oscillations of the heavier mass as the system falls freely.
[**Ans.** 3 mm., 1 mm., 11 Hz.]
7. (a) If two discs of masses 100 gm and 200 gm are connected by a massless spring of force constant 100 N-m^{-1}, find the frequency of the spring when the heavier disc is resting on the table and the system is vertical.
(b) If the table is removed and the system falls freely then what is the frequency of vibration ?
[**Ans.** (a) 5 Hz; (b) 6.1 Hz.]
8. Two masses of 2 kg and 4 kg are joined by a spring of force constant 133 N/m and are placed on a frictionless horizontal table. Find the frequency of two-body oscillations produced when the masses are slightly drawn apart and released. What is the ratio of their kinetic energies ?
[**Ans.** 50/π Hz; 2 : 1.]
[**Hint :** The force acting on the two masses are always equal and opposite, hence their instantaneous momenta are also equal and opposite *i.e.*, $m_1v_1 = m_2v_2$ and therefore

$$\frac{K_1}{K_2} = \frac{\frac{1}{2}m_1v_1^2}{\frac{1}{2}m_2v_2^2} = \frac{(m_1^2v_1^2)/m_1}{(m_2^2v_2^2)/m_2} = \frac{m_2}{m_1} = \frac{4}{2} = 2.]$$

9. Show that the kinetic energy of two-body oscillator is given by $K = \frac{1}{2}\mu v^2$, where μ is the reduced mass and v (= $v_1 - v_2$) is the relative velocity.
10. Two bodies of masses $3m$ and $4m$ are joined by a massless spring of length l and force constant C and the system lies on a frictionless table. A particle of mass m, moving with speed v along the length of the spring, collides and strikes to the body of mass $3m$. What is the amplitude and period of vibration of the spring ?

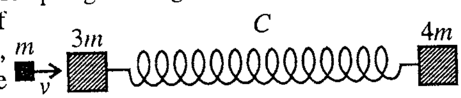


Fig. 7.51.

$\left[\textbf{Ans. } \frac{v}{8}\sqrt{\frac{m}{2C}}; 2\pi\sqrt{\frac{2m}{C}}.\right]$
11. The vibration frequency of H^1Cl^{35} molecule is 8990×10^{10} Hz. Deduce the vibration frequency of H^2Cl^{35} molecule, assuming force constant to be the same in two cases.
[**Ans.** 6446×10^{10} Hz.]
12. An HCl molecule is known to vibrate at a fundamental frequency 8.7×10^{13} Hz. What is the effective force constant C for the coupling forces between the atoms ?
[**Ans.** 485 N-m.]

13. From spectroscopic measurements, the fundamental vibration angular frequency molecule is 7.55×10^{14} rad./sec. Deduce (i) force constant for HF molecule, and (ii) work done in stretching the atoms by 0.5 Å apart. [$m_H = 1.66 \times 10^{-27}$ kg, $m_F = 19 \times 1.66 \times 10^{-27}$ kg. Internuclear distance $r_0 = 1$ Å.]

[**Ans.** (i) 898.9 N-m^{-1}, (ii) 1.12×10^{-18} J.] **(Agra 1978)**

14. In the electrical circuit shown in **fig. 7.52,** the key is closed for a long time. If the switch is opened at $t = 0$, determine the current in the *L-C* circuit and the charge on the capacitor at any time.

[**Ans.** $i = 5$ amp $\cos \omega t$ and $Q = (5 \times 10^{-4}$ Coul.) $\sin \omega t$.]

[**Hint :** Take $i = i_0 \cos \omega t$, then $Q = \int_0^t i\,dt = \frac{i_0}{\omega} \sin \omega t$.]

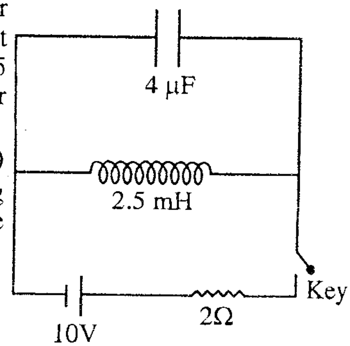


Fig. 7.52.

QUESTIONS

Long Answer Questions

1. What do you understand by the potential energy curve ? What are the positions of stable equilibrium and why ? What is neutral equilibrium ?
2. What is a potential well ? Show that for small swing oscillations, the potential well may be treated as parabolic ? What is the significance of a parabolic potential well ? Analyse mathematically the oscillations in one dimensional potential well. **(Garwal 1995)**
3. Explain potential well and potential energy curve. Show that small oscillations of any system about an equilibrium position will always be simple harmonic. Discuss the factors which govern the frequency of these harmonic oscillations. **(Agra 1970; Kanpur 95; Rajasthan 83)**
4. Show that the potential well for the oscillations of low amplitude can be considered to be parabolic and the oscillations near the position of stable equilibrium of parabolic potential well are simple harmonic. **(Ujjain 2003)**
5. What is a harmonic oscillator ? Establish its differential equation and find the expressions for its velocity, displacement and period. **(Agra 1990, 86, 85; Gwalior 2003; Bhopal 03; Indore 03)**
6. Show that a harmonic oscillator moves according to the law $x = A \sin(\omega t + \phi)$, where A and ϕ are constants. **(Agra 2005)**
7. Show that the solution of the equation $\frac{d^2x}{dx^2} + \omega^2 x = 0$ is of the form $x = a \sin(\omega t + \phi)$.

   Hence prove that for initial displacement x_0 and initial velocity v_0, the amplitude $a = \sqrt{x_0^2 + v_0^2 / \omega^2}$ and initial phase $\phi = \tan^{-1}(\omega x_0 / v_0)$. **(Kanpur 1995)**
8. Show that for a harmonic oscillator, mechanical energy remains constant and it is proportional to the square of the amplitude. **(Agra 1998; Rohilkhad 79; Raipur 2003; Gwalior 2004)**
9. Show that for a harmonic oscillator, the average potential energy is equal to the average kinetic energy. **(Agra 1998)**
10. Prove that in simple harmonic motion when the average is taken with respect to the position over one cycle, the average potential energy equals $Ca^2/6$ and the average kinetic energy equals $Ca^2/3$. Explain physically why this result is different to that when the average is taken with respect to time ?
11. Prove that
    (a) For a harmonic oscillator, when the displacement is half the amplitude, potential energy is 1/4 of the total energy.
    (b) At $a/\sqrt{2}$ displacement the potential energy is equal to the kinetic energy.
    (c) Free harmonic oscillations continue for infinite time. **(Kanpur 1998)**
12. (a) A body of mass M connected with a massless horizontal spring of force constant C is set into simple harmonic motion. Derive an expression for its time period.
    (b) If the spring has a mass m, then deduce the new expression for time period.
    (Kanpur 1984; Agra 88)

13. Show that the motion of a mass attached at the free end of a massless spring suspended by a rigid support is simple harmonic. Establish the expression for the time period of oscillations. **(Sagar 2003; Gwalior 03)**
14. Write down the differential equations for the following harmonic oscillators :
    (i) Mass on a spring, (ii) Torsion pendulum, (iii) *L-C* circuit.
    Write down the formulae for the period and frequency of the above harmonic oscillators. **(Agra 1991)**
15. Show that oscillations of the following systems can be represented by a single equation and hence discuss their equivalence :
    (i) Mass on a spring, (ii) Torsion pendulum, (iii) *L-C* circuit, (iv) Simple pendulum. **(Agra 1992)**
16. What is a simple pendulum ? Show that the oscillations of a simple pendulum for small amplitude are simple harmonic. Deduce the expression for the time period of its oscillations. **(Jabalpur 2004)**
17. Discuss the working of a compound pendulum and prove that the centre of suspension and oscillation are mutually interchangeable. Calculate the periodic time and show that there are four points in it about which the periods are the same. Find the condition for maximum and minimum time period of compound pendulum. **(Kanpur 2005; Purvanchal 1995)**
18. What is a compound pendulum ? Write the differential equation of its motion and deduce an expression for its time period. Also show that the time period of pendulum with respect to its four points remains the same. **(Gwalior 2004; Jabalpur 04, 03; Rewa 03; Indore 03, Raipur 03; Bilaspur 03; Bhopal 04; Ujjain 03)**
19. Describe with complete theory the bar pendulum method to determine g in the laboratory. **(Purvanchal 1993)**
20. Write down the differential equation for the oscillations of torsional pendulum and deduce the expressions for its time period. **(Jabalpur 2004; Ujjain 04)**
21. What is torsional pendulum ? How will you determine the moment of inertia of a body by a torsional pendulum ? **(Agra 1998)**
22. Write down the differential equations for the oscillations of *L-C* circuit and deduce the expression for its time period. **(Agra 2004; Gwalior 03; Bhopal 03; Bilaspur 03)**
23. Show that in a circuit containing an inductance and a capacitance, the discharge is oscillatory and undamped. How does the energy change its form in this system ? **(Kanpur 1980; Rohilkhand 84)**
24. What is Helmholtz resonator ? Write down the differential equation for its oscillations and deduce expressions for its time period. **(Jabalpur 2004; Bilaspur 03; Raipur 03)**
25. What is a Helmholtz resonator ? Why is it called resonator ? Deduce an expression for the frequency of the Helmholtz resonator. What are the uses of this resonator ?
26. Obtain the differential equations for oscillations of bifilar system and hence find their time period. **(Indore 2004; Bilaspur 03)**
27. What is bifilar suspension ? Show that it executes simple harmonic oscillations. How will you determine the value of g with its help ?
28. What is a two body harmonic oscillator ? Prove that the oscillations of the two particles connected by a spring are simple harmonic.
29. Show that the motion of two particles attached at the end of a massless spring is simple harmonic. Obtain the expression for the frequency of oscillations. **(Indore 2004)**
30. The force constant for a spring is C and a mass m is suspended to it. The spring is cut in half and the same mass m is suspended from one of the halves. Does the frequency of vibration remain the same before and after the spring cut ? How are the frequencies related ?
31. If the mass of the spring is taken into account, explain how this will change the expression for the period of oscillation of a (massless) spring-mass system ?
32. Suppose we are given with a block of unknown mass and a spring of unknown force constant. Can you predict the period of this block spring system simply by measuring the elongation of the spring produced by attaching the block to it ? If x_0 is the extension, then find the period.
    [**Ans.** $T = 2\pi\sqrt{x_0/g}$.]

33. (a) Show that the period of oscillation of a ring of radius R and mass M about an axis through its rim is $T = 2\pi\sqrt{2R/g}$.

(b) Show that a particle dropped in a tunnel across the earth would perform simple harmonic motion. **(P.C.S., U.P. 1979)**

34. An angular ring, of internal and external radii a and b, is allowed to oscillate in its own vertical palne about an axis perpendicular to the plane of the ring and passing through a point on its outer edge. Show that the period of oscillation is $2\pi\sqrt{(a^2+2b^2)/bg}$.

Short Answer Questions

1. What is a potential well ? **(Kanpur 2005)**

2. What is a potential well ? What will be the motion of a particle in a potential well ? **(Ujjain 2003)**

[**Hint :** The motion of a particle in a potential well in general will be anharmonic, but the small amplitude oscillations will be simple harmonic.]

3. Define stable, unstable and neutral equilibrium in terms of potential energy. Explain with one example for each. **(Delhi 1989)**

4. Differentiate between the periodic motion and simple harmonic motion, giving two examples of each. **(Sagar 2003)**

[**Ans.** If a moving particle retraces its path regularly after an interval of time, its motion is said to be periodic. Examples : (i) uniform circular motion of a particle, (ii) motion of a planet on elliptical path, (iii) motion of a simple pendulum etc. Simple harmonic motion is a particular type of periodic motion, in which at any instant the restoring force is proportional to displacement and directed opposite to it. Examples : motion of simple pendulum, spring-mass system, bar pendulum etc.]

5. A particle of mass 0.1 kg lies in a potential field $V = 50x^2 + 100$ J/kg. What is the force on the particle ? Write the differential equation of motion.

[**Ans.** Force $F = -10x$ newton; $\frac{d^2x}{dt^2} + 100x = 0$.]

6. Find the frequency of oscillation in Q. 5.

[**Ans.** $5/\pi$.]

7. The equation of a particle executing simple harmonic motion is

$$x = (1\text{ m})\sin\left[(\pi s^{-1})t + \frac{\pi}{3}\right].$$

Write down the amplitude, time period and maximum speed.

[**Ans.** $a = 1$ m, $T = 2s$, $v_m = 5\pi$ m/s.]

8. A particle executing S.H.M. comes to rest at the extreme positions. Is the resultant force on the particle zero at these postions ? Explain.

[**Ans.** No.]

9. Two particles execute simple harmonic motions of the same amplitude and frequency on parallel lines side by side. They are cross one another when moving in opposite directions each time their displacment is half their amplitude. What is the phase difference between them ?

[**Ans.** Phase difference = $2\pi/3$.]

10. In S.H.M. when the displacement is one-half the amplitude, what fractions of the total energy are kinetic and potential ? **(Rewa 2004)**

[**Ans.** $U = E/4$; $K = 3E/4$.]

11. At what displacement the energy half kinetic and half potential ? **(Lucknow 1980)**

[**Ans.** $\frac{a}{\sqrt{2}}$.]

12. Show that in a compound pendulum, the centre of suspension and centre of oscillation are interchangeable. **(Delhi 1998)**

13. Show that in simple harmonic motion the displacement of the particle is the same after a time $2\pi/\omega$. **(Kanpur 2005)**

14. A small insect moves with constant speed in a vertical circle, when sun is just up. Does its shadow formed by the sun on a horizontal plane move in a simple harmonic motion. Explain.
[**Ans.** Yes.]
15. A particle moves on X-axis according to the equation $x = A + B \sin \omega t$. Is it S.H.M. ? If yes, what is the amplitude ?
[**Ans.** B.]
16. The motion of a particle is given by $x = A \sin \omega t + B \cos \omega t$. Is it S.H.M. ? If yes, what is the amplitude ?
[**Ans.** Yes, $a = \sqrt{A^2+B^2}$.]
17. A spring-mass system oscillates in a car. If the car accelerates on a horizontal road, does its frequency increase, decrease or remain the same ? Explain.
[**Ans.** Remain the same.]
18. Express the kinetic and potential energies as a function of total energy of harmonic oscillator when its displacement half the amplitude. **(Agra 2005)**

[**Ans.** $K = \frac{3}{4}E, U=\frac{1}{4}E$.]

19. Given $x = 2.5 \sin \left(\frac{2\pi t}{3}+\frac{\pi}{2}\right)$, then find the values of angular velocity, maximum linear velocity, initial phase and time period. **(Agra 2004)**
[**Ans.** 2.09 radian/sec, 5.22 units, $\pi/2$, 3 sec.]
20. Obtain an equation for simple harmonic oscillations of charge in a L-C circuit.
[**Ans.** $\frac{d^2Q}{dt^2}+\frac{1}{LC}Q = 0$ with $T = 2\pi\sqrt{LC}$.] **(Agra 2004)**
21. Find average kinetic energy of a simple harmonic oscillator. **(Kanpur 2005)**
22. Find an expression for the frequency of a diatomic molecule whose nuclei have masses m_1 and m_2 and force constant K. **(Kanpur 2005)**

Objective Type Questions

1. An example of stable equilibrium is :
(a) a hanging spring-mass system in the stationary position (b) a pendulum in the rest position,
(c) an egg standing on one end (d) a book placed flat anywhere on a table.
2. In simple harmonic motion, the displacement of a particle in one time period is :
(a) a (b) $2a$ (c) $4a$ (d) 0
3. A particle executes simple harmonic motion. The amplitude of its oscillation is 'a'. The P.E. of the particle is maximum when displacement is :
(a) $\pm a$ (b) 0 (c) $\pm a/2$ (d) $\pm a/\sqrt{2}$
4. The time period of a LC circuit is : **(Agra 2005)**
(a) $\frac{2\pi}{\sqrt{LC}}$ (b) $2\pi\sqrt{LC}$ (c) $2\pi\sqrt{\frac{L}{C}}$ (d) $2\pi\sqrt{\frac{C}{L}}$
5. A pendulum is suspended from the roof of a car. Time period of the pendulum is 'T' when the car is at rest. If the car accelerates with an acceleration a, then the time period of the pendulum 'T' will be :
(a) more than T (b) equal to T (c) less than T (d) can not say
6. The amplitude of a particle executing simple harmonic motion is 4 cm. At what point of displacement from equilibrium its energy will be half K.E. and half P.E. ?
(a) 1 cm (b) $\sqrt{2}$ cm (c) 3 cm (d) $2\sqrt{2}$ cm
7. A particle moves such that its acceleration is given by $a = -\mu^2 x$. The period of oscillation is : **(Kanpur 2004)**
(a) $2\pi\mu$ (b) $2\pi/\mu$ (c) $2\pi/(\mu)^{1/2}$ (d) $2(\pi/\mu)^{1/2}$

8. Two bodies A and B of equal mass are suspended by massless springs of spring constants k_1 and k_2 respectively. The maximum velocities of the bodies are the same when they are oscillating vertically. The ratio of amplitudes of A and B is :

(a) $\frac{k_1}{k_2}$ (b) $\sqrt{\frac{k_1}{k_2}}$ (c) $\frac{k_2}{k_1}$ (d) $\sqrt{\frac{k_2}{k_1}}$

9. The length of a simple pendulum is L and the mass of the bob is m. It is oscillating about a verticle line in a plane within the angular limits $-\phi$ and $+\phi$. For angular displacement $|\theta| < \phi$, the tension in the string is T and the velocity of the bob is v. In this condition :

(a) $T \cos \theta = mg$ (b) $T - mg \cos \theta = mv^2/L$

(c) the tangential acceleration of the bob $= g \sin \theta$ (d) $T = mg \cos \theta$

10. The total energy of a simple pendulum is E. When the displacement is half of the amplitude, its kinetic energy will be :

(a) E (b) $3E/4$ (c) $E/2$ (d) $E/4$ **(Kanpur 2001)**

11. A simple harmonic motion is represented by the equation : $y = 10 \sin \left(10t + \frac{\pi}{6}\right)$. Its frequency is :

(Purvanchal 2001)

(a) $5/\pi$ (b) $16/\pi$ (c) $\pi/5$ (d) $\pi/10$

12. The total energy in the simple harmonic motion is :

(a) ka^2 (b) $\frac{1}{2}ka^2$ (c) $1/ka$ (d) None of these

(Purvanchal 2001)

13. Equation of motion of a particle is $x = a \sin \omega t + b \cos \omega t$. The motion of the particle is :

(a) anharmonic motion (b) S.H.M. of amplitude $(a + b)$

(c) S.H.M. of amplitude $(a + b)/2$ (d) S.H.M. of amplitude $\sqrt{a^2 + b^2}$

14. A particle is moving along X-axis according to the equation $x = x_0 \sin^2 \omega t$. Its motion is :

(a) anharmonic (b) simple harmonic motion of amplitude x_0

(c) simple harmonic motion of time period π/ω (d) simple harmonic motion of amplitude $2x_0$

15. When a body is hanged on a spring, then the length of the spring is increased by 20 cm. If the oscillations are obtained by stretching the spring, then its time period will be :

(a) $2\pi/980$ (b) $2\pi/\sqrt{980}$ (c) $2\pi/\sqrt{7}$ (d) $2\pi/7$

16. If the effective length of a pendulum is infinite. Its time period will be :

(a) 1 hr (b) 0 (c) ∞ (d) 84.6 min.

(Bundelkhand 2004)

ANSWERS

1. (a)(b), **2.** (d) **3.** (a), **4.** (b), **5.** (c), **6.** (d), **7.** (b), **8.** (d), **9.** (b), **10.** (b), **11.** (a), **12.** (b), **13.** (d), **14.** (c), **15.** (d), **16.** (d).

8

Superposition of Simple Harmonic Motions

8·1. Introduction

In various fields of physics, there are several important physical phenomena *e.g.*, interference, diffraction, polarisation, stationary waves, modulation, beats etc. where the displacement of a point may be visualized as a superposition of two or more linear simple harmonic motions. There may be cases where a point may be vibrating under the influence of two or more simple harmonic vibrations and the resultant motion may be much complicated. Our eardrum vibrates, in general, under a complicated combination of harmonic vibrations. In the present unit, first we discuss the principle of superposition and then we apply this principle to various situations where two (or more) simple harmonic motions are acting on a particle simultaneously along the same line or perpendicularly.

8·2. Principle of Superposition

Let two (or more) simple harmonic motions be acting simultaneously on a particle (or point). At any instant, if x_1 be the displacement of the particle due to one simple harmonic motion and x_2 due to the other, then according to the principle of superposition, the resultant displacement of the particle is the vector sum of the two displacements, *i.e.*,

$$x = x_1 + x_2 \qquad ...(1)$$

We know that the differential equation for a simple harmonic motion is

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \qquad ...(2)$$

This is a **linear homogeneous equation** of second order. Such equation has the characteristic property that the *sum of any two solutions of it is itself a solution, i.e.,* if x_1 and x_2 are the solutions of eq. (2), then the sum $x = x_1 + x_2$ of these solutions also satisfies eq. (2). Thus, if x_1 and x_2 are the solutions of eq. (2), then

$$\frac{d^2x_1}{dt^2} = \omega^2 x_1 \qquad \text{and} \qquad \frac{d^2x_2}{dt^2} = \omega^2 x_2 \qquad ...(3)$$

The addition of the two equations gives

$$\frac{d^2(x_1+x_2)}{dt^2} = \omega^2(x_1 + x_2) \quad \text{or} \quad \frac{d^2x}{dt^2} = \omega^2 x \qquad ...(4)$$

This means that $x = x_1 + x_2$ satisfies eq. (2) and hence *the differential equation of simple harmonic motion is a linear homogeneous equation and follows the principle of superposition.* In the present work, we shall study only those oscillations which satisfy linear equations and are in accordance with the principle of superposition. It is to be mentioned that the differential equations of damped and driven harmonic oscillators are linear differential equations.

In case of nonlinear equation

$$\frac{d^2x}{dt^2} = -k_1x - k_2x^2 - k_3x^3 - .... \qquad ...(5)$$

(e.g., for anharmonic oscillator), the superposition of two solutions does not satisfy the equation itself. In case of simple pendulum, the differential equation has the form

$$\frac{d^2\theta}{dt^2} + \omega^2 \sin\theta = 0 \qquad [\because \omega = \sqrt{g/l}\,] \qquad ...(6)$$

But $\sin \theta = \theta - \frac{\theta^3}{\lfloor 3} - \frac{\theta^5}{\lfloor 5} - .....$

Hence $$\frac{d^2\theta}{dt^2} = -\omega^2\theta + \frac{\omega^2}{6}\theta^3 - \frac{\omega^2}{120}\theta^5 - .... \qquad ...(7)$$

Thus the differential equation for the motion of simple pendulum is nonlinear and in such a case, the principle of superposition will be not applicable. However for small θ, θ^3, θ^5 etc. terms can be neglected and then the equation.

$$\frac{d^2\theta}{dt^2} = -\omega^2\theta \quad \text{or} \quad \frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \qquad ...(8)$$

is the linear equation, respresenting simple harmonic motion, and for such small amplitude oscillations of simple pendulum, the principle of superposition is applicable. Large amplitude oscillations of simple pendulum are anharmonic and represented by nonlinear equation (7). In the present chapter, we shall apply the principle of superposition for compounding two simple harmonic motions which are in a straight line or mutually perpendicular. However this principle is applicable for the superposition of many linear simple harmonic motions.

8·3. Superposition of Two Simple Harmonic Motions of Same Frequency (or Equal Periods) along a Line

Let two simple harmonic motions of same frequency ($\upsilon = \omega/2\pi$) [equal periods (= $2\pi/\omega$] but of different amplitudes and phases be simultaneously acted on a particle in the same straight line. Let the two component vibrations be represented by

$$x_1 = a_1 \sin(\omega t + \phi_1) \qquad ...(1)$$

and $$x_2 = a_2 \sin(\omega t + \phi_2) \qquad ...(2)$$

where a_1 and a_2 are the amplitudes and ϕ_1 and ϕ_2 the epochs or initial phases of the component vibrations. The phase difference between them is $(\phi_1 - \phi_2)$.

According to the principle of superposition, the resultant displacement of the particle is given by

$$x = x_1 + x_2 \qquad ...(3)$$

Thus, $$x = a_1 \sin(\omega t + \phi_1) + a_2 \sin(\omega t + \phi_2)$$
$$= a_1[\sin \omega t \cos \phi_1 + \cos \omega t \sin \phi_1] + a_2[\sin \omega t \cos \phi_2 + \cos \omega t \sin \phi_2]$$
$$= \sin \omega t[a_1 \cos \phi_1 + a_2 \cos \phi_2] + \cos \omega t[a_1 \sin \phi_1 + a_2 \sin \phi_2] \qquad ...(4)$$

Now, suppose

$$a \cos \theta = a_1 \cos \phi_1 + a_2 \cos \phi_2 \qquad ...(5)$$

and $$a \sin \theta = a_1 \sin \phi_1 + a_2 \sin \phi_2 \qquad ...(6)$$

Then $$x = a \sin \omega t \cos \theta + a \cos \omega t \sin \theta \quad \text{or} \quad x = a \sin(\omega t + \theta) \qquad ...(7)$$

where a is the resultant amplitude of oscillation, $(\omega t + \theta)$ is its phase.

Eq. (7) represents that **the resultant motion is simple harmonic** whose frequency (= $\omega/2\pi$) is the same as that of any component vibration, but the amplitude a and phase θ are different. Squaring and adding eqns. (5) and (6) we obtain

$$a^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos(\phi_1 - \phi_2)$$

or $$a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos(\phi_1 - \phi_2)} \qquad ...(8)$$

which tells us the **amplitude of resultant simple harmonic motion.**

Dividing eq. (6) by eq. (5),

$$\tan \theta = \frac{a_1 \sin \phi_1 + a_2 \sin \phi_2}{a_1 \cos \phi_1 + a_2 \cos \phi_2} \quad \text{or} \quad \theta = \tan^{-1} \frac{a_1 \sin \phi_1 + a_2 \sin \phi_2}{a_1 \cos \phi_1 + a_2 \cos \phi_2} \qquad ...(9)$$

This angle θ gives the **phase of the resultant motion.**

Let us consider some special cases :

(a) If $\phi_1 - \phi_2 = 0$ or $2n\pi$, where n is an integer, the two vibrations are in the **same phase** and the resultant amplitude has the maximum value, *i.e.*,

$$a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2} = a_1 + a_2 \quad ...(10)$$

(b) If $\phi_1 - \phi_2 = \pi$ or $(2n + 1)\pi$, where n is any interger, the two vibrations are in the **opposite phase** and the resultant amplitude has the minimum value *i.e.*,

$$a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2} = a_1 - a_2 \quad ...(11)$$

If in addition $a_1 = a_2$, the resultant amplitude $a = 0$, *i.e.*, the particle remains in rest position.

8·4. Interference

If two waves of the same frequency move in a medium along the same direction, then due to superposition the resultant displacement or intersity is maximum at some points and minimum at some other points. This phenomenon is called **interference.** The interference is said to be **constructive** at the points of maximum intensity and **destructive** at the points of minimum intensity.

Let two simple harmonic progressive waves of the same frequency ($= \omega/2\pi$) move in a medium along positive direction of X-axis. For these waves, the wave velocity and wave number ($k = 2\pi/\lambda$) is the same. If due to these waves the displacements at a point are y_1 and y_2 respectively then the equations of these harmonic waves can be represented as

$$y_1 = a_1 \sin (kx - \omega t) \quad ...(1)$$

and

$$y_2 = a_2 \sin (kx - \omega t + \phi) \quad ...(2)$$

where ϕ is the phase difference between the two waves.

According to the principle of superposition, the resultant displacement at that point is given by

$$\begin{aligned} y = y_1 + y_2 &= a_1 \sin (kx - \omega t) + a_2 \sin (kx - \omega t + \phi) \\ &= a_1 \sin (kx - \omega t) + a_2 \sin (kx - \omega t) \cos \phi + a_2 \cos (kx - \omega t) \sin \phi \\ &= \sin (kx - \omega t) (a_1 + a_2 \cos \phi) + \cos (kx - \omega t) (a_2 \sin \phi) \quad ...(3) \end{aligned}$$

Let

$$a_1 + a_2 \cos \phi = a \cos \theta \quad ...(4)$$

$$a_2 \sin \phi = a \sin \theta \quad ...(5)$$

Then

$$y = a [\sin (kx - \omega t) \cos \theta + \cos (kx - \omega t) \sin \theta]$$

or

$$y = a \sin (kx - \omega t + \theta) \quad ...(6)$$

Thus the resultant wave is also simple harmonic whose amplitude is a and phase difference θ relative to first wave. Squaring and adding eqs. (4) and (5), we get

$$a^2 \cos^2 \theta + a^2 \sin^2 \theta = (a_1 + a_2 \cos \phi)^2 + a_2^2 \sin^2 \phi \quad \text{or} \quad a^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos \phi$$

$$\therefore \quad a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos\phi} \quad ...(7)$$

Also,

$$\tan \theta = \frac{a_2 \sin\phi}{a_1 + a_2 \cos\phi} \quad ...(8)$$

As the intensity is directly proportional to the square of amplitude, the resultant intensity is given by

$$I \propto a^2 \quad \text{or} \quad I = ka^2 \quad ...(9)$$

Hence

$$I = k(a_1^2 + a_2^2 + 2a_1a_2 \cos \phi) \quad \text{[from eq. (7)]} \quad ...(10)$$

or

$$I = ka_1^2 + ka_2^2 + 2ka_1a_2 \cos \phi$$

or

$$I = I_1 + I_2 + 2\sqrt{I_1I_2} \cos \phi \quad ...(11)$$

where $I_1 = ka_1^2$ and $I_2 = ka_2^2$ are the two different intensities of the interfering waves. We see from eqs. (7) and (10) that the resultant amplitude and resultant intensity at a point depends on the phase difference ϕ.

Case I : For **constructive interference,** the resultant amplitude or the intensity at a point is maximum for which

$$\cos \phi = +1 \text{ or } \phi = 0 \text{ or } 2n\pi, \text{ where } n \text{ is an integer.} \quad ...(12)$$

Hence for constructive interference

$$a^2 = a_1^2 + a_2^2 + 2a_1a_2 \quad \text{or} \quad a^2 = (a_1 + a_2)^2 \quad \text{or} \quad a = a_1 + a_2 \quad ...(13)$$

When

$$a_1 = a_2 = A, \text{ then } a = 2A \quad ...(14)$$

Resultant intensity $I_{max} = I_1 + I_2 + 2\sqrt{I_1I_2} = (\sqrt{I_1} + \sqrt{I_2})^2 = k(a_1 + a_2)^2 \quad ...(15)$

When $a_1 = a_2 = A$ and $I_0 = kA^2$ (for each wave), then $I_{max} = 4kA^2 = 4I_0 \quad ...(16)$

Thus at the points where the two interfering waves meet in the same phase *i.e.*, the phase difference between them is $\phi = 2n\pi$, $n = 0, 1, 2, ...$, the resultant amplitude or intensity is maximum. This interference is called **constructive.**

Case II : For **destructive interference,** the resultant amplitude or intensity is minimum. For minimum amplitude or intensity,

$$\cos\phi = -1 \text{ or } \phi = \pi \text{ or } \phi = (2n+1)\pi, \text{ where } n \text{ is an integer.} \quad ...(17)$$

Hence for destructive interference

$$a^2 = a_1^2 + a_2^2 - 2a_1a_2 \text{ or } a^2 = (a_1 - a_2)^2 \text{ or } a = |a_1 - a_2| \quad ...(18)$$

If $a_1 = a_2$, then $a = 0$ ...(19)

Resultant intensity $I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2} = (\sqrt{I_1} - \sqrt{I_2})^2$ or $I_{min} = k(a_1 - a_2)^2$ ...(20)

If $a_1 = a_2$, then $I_{min} = 0$ ...(21)

Thus at the points where the two interfering waves meet in opposite phase *i.e.*, phase difference $\phi = (2n+1)\pi$, $n = 0, 1, 2, ...$, the resultant amplitude or intensity is minimum. Such interference is called **destructive.**

Conditions for Constructive and Destructive Interference

We know that the path difference Δ between the two interfering waves is related to the phase difference ϕ as

$$\phi = \frac{2\pi}{\lambda}\Delta \text{ or } \Delta = \frac{\lambda}{2\pi}\phi \quad ...(22)$$

Hence for constructive interference,

$$\phi = 2n\pi \text{ or } \Delta = n\lambda, \text{ where } n = 0, 1, 2, ..., \quad ...(23)$$

For destructive interference,

$$\phi = (2n+1)\pi \text{ or } \Delta = (2n+1)\lambda/2, \text{ where } n = 0, 1, 2, ... \quad ...(24)$$

The phenomenon of interference takes place both for transverse and longitudinal waves. This means that this phenomenon occurs both for light and sound waves.

Conditions for Distinct and Sustained Interference

(1) Two sources of waves must be coherent *i.e.*, the sources should emit the waves of same frequency and the phase difference between them should remain constant.

(2) The amplitudes of the two interfering waves should be equal or nearly equal ($a_1 \approx a_2$).

(3) Two interfering waves must propagate in the same direction.

(4) The displacements due to two waves must be along a straight line.

Average internsity and conservation of energy : The resultant intensity at a point due to two interfering waves is given by

$$I = k(a_1^2 + a_2^2 + 2a_1a_2\cos\phi) \quad ...(25)$$

Average intensity for the phase difference $\phi = 0$ to $\phi = 2\pi$ is given by

$$I_{av} = \frac{\int_0^{2\pi} I\,d\phi}{\int_0^{2\pi} d\phi} = \frac{\int_0^{2\pi} k(a_1^2 + a_2^2 + 2a_1a_2\cos\phi)\,d\phi}{2\pi}$$

$$= \frac{k}{2\pi}[a_1^2\phi + a_2^2\phi + 2a_1a_2\sin\phi]_0^{2\pi}$$

$$= \frac{k}{2\pi}(a_1^2 + a_2^2)2\pi = k(a_1^2 + a_2^2) \quad ...(26)$$

But $ka^2 = I_1$ and $ka_2^2 = I_2$, hence $I_{av} = I_1 + I_2$ ...(27)

Hence for interferences the average intensity is equal to the sum of intensities due to two waves. The value of maximum intensity $I_{max} = k(a_1^2 + a_2^2 + 2a_1a_2)$ is more by $2ka_1a_2$ and minimum intensity $I_{min} = k(a_1^2 + a_2^2 - 2a_1a_2)$ is less by $-2ka_1a_2$ from average intensity. As the intensity is equal to the energy falling on unit area, hence $2ka_1a_2$ energy is transferred at the points of maximum from the points of minimum intensity. This implies that in interference, the distribution of the total energy of the two waves occurs at different points, but the average energy is equal to the sum of the energy of the two waves. Hence **there is the conservation of energy in the phenomenon of interference.**

8·5. Composition of Two Rectangular Simple Harmonic Motions of the Same Frequencies–Lissajous Figures

When a particle is vibrating simultaneously under two simple harmonic motions at right angles to each other, the resultant motion of the particle is called Lissajous Figures. The nature of the motion, or the curve traced out, depends upon the amplitudes, frequencies, and the phase difference of the two component simple harmonic motions. We shall consider below some simple cases of the composition of two rectangular vibrations.

Let the two simple harmonic motions of the same frequencies be acting along the axis of X and axis of Y respectively. If their displacements at any time t be x and y, then they can be represented by

$$x = a \sin(\omega t + \phi) \quad \text{...(1)}$$

$$y = b \sin \omega t \quad \text{...(2)}$$

where a and b are the amplitudes of the component vibrations having same frequency $\omega/2\pi$ (or period $2\pi/\omega$) and the first vibration is ahead of the second by a phase angle ϕ, *i.e.*, the phase difference between the two vibrations is ϕ.

From eq. (1), we have $\quad x = a[\sin \omega t \cos \phi + \cos \omega t \sin \phi] \quad$ ...(3)

But from eq. (2) $\quad \sin \omega t = \dfrac{y}{b}$ and $\cos \omega t = \sqrt{1-\dfrac{y^2}{b^2}}$

Substituting the values of $\sin \omega t$ and $\cos \omega t$ in (3), we get

$$\frac{x}{a} = \frac{y}{b}\cos \phi + \sin \phi \sqrt{1-\frac{y^2}{b^2}} \quad \text{or} \quad \left(\frac{x}{a}-\frac{y}{b}\cos \phi\right)^2 = \sin^2 \phi \left(1-\frac{y^2}{b^2}\right)$$

Thus $$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{2xy}{ab}\cos \phi = \sin^2 \phi \quad \text{...(4)}$$

This expression represents the resultant motion, which is in general an ellipse inclined to the axes of co-ordinates.

Special Cases :

(a) *When* $\phi = 0$, the relation (4) becomes

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{2xy}{ab} = 0 \quad \text{or} \quad \left(\frac{x}{a}-\frac{y}{b}\right)^2 = 0 \text{ or } \pm\left(\frac{x}{a}-\frac{y}{b}\right) = 0. \quad \text{...(5)}$$

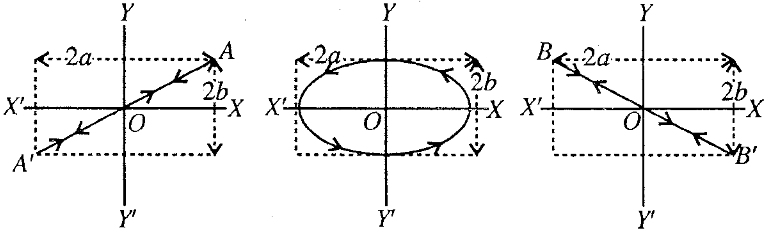
This equation represents a pair of coincident straight line $y = bx/a$, passing through the origin and lying in the first and third quadrants [**fig. 8.1(a)**].

This motion along a straight line represents the case of *linearly polarized light* in optics.

(b) *When* $\phi = \pi/2$, the expression (4) is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1 \quad \text{...(6)}$$

This equation represents an ellipse, whose axes are coincident with the X-and Y-axes [**fig. 8.1(b)**]. The semi-major and semi-minor axes of the ellipse are a and b. Further as the time t starts to increase from zero, x begins to decrease from its maximum positive value and y simultaneously begins to go positive from $y = 0$ position. Thus



(a) $\phi = 0$ (b) $\phi = \pi/2$ (c) $\phi = \pi$

Fig. 8.1.

the ellipse is described in the anticlockwise direction [**fig. 8.1(b)**]. The same ellipse is obtained for $\phi = 3\pi/2$ or $-\pi/2$, but now the motion is in clockwise direction. Thus when the phase difference ϕ between the two rectangular simple harmonic motions is $+\pi/2$, the resultant motion is *elliptically polarized.*

Now, if $a = b$, the ellipse degenerates in the circle

$$x^2 + y^2 = a^2 \qquad \text{...(7)}$$

Hence, *two simple harmonic motions at right angles to each other of equal amplitudes but with phases differing by $\pi/2$, are equivalent to a uniform circular motion, the radius of the circle being equal to the amplitude of either simple harmonic motion.* This case corresponds to *circularly polarized light* in optics.

Conversely, *a uniform circular motion can be resolved into two simple harmonic motions at right angles to each other, their amplitudes being equal, but their phases differing $\pi/2$.*

(c) *When* $\phi = \pi$, eq. (4) is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{2xy}{ab} = 0 \text{ or } \left(\frac{x}{a} + \frac{y}{b}\right)^2 = 0 \qquad \text{...(8)}$$

which represents a pair of coincident straight lines $y = -bx/a$, passing through the origin and lying in the second and fourth quadrants as BOB' [**fig. 8.1(c)**].

For any other value of ϕ, except discussed above, the path will be an ellipse, inclined to the co-ordinate axes.

8·6. Composition of Two Rectangular S.H.Ms. of Nearly Equal Frequencies

If the two component vibrations are exactly of equal frequencies (or periods), the elliptical paths, referred to above, remain perfectly steady. But, when the two frequencies differ slightly from each other, there comes about a gradual but progressive change in the relative phase (ϕ) of the two vibrations and consequently the shape of the ellipse slowly undergoes a change. All these elliptic path lie within a rectangle of sides $2a$ and $2b$.

Now, starting with the phase in agreement, *i.e.,* $\phi = 0$, the ellipse coincides with the diagonal $\frac{x}{a} - \frac{y}{b} = 0$ of the rectangle.

As ϕ increases from 0 to $\pi/2$, the ellipse opens out to the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, passing through intermediate oblique positions.

When ϕ increases from $\pi/2$ to π, the ellipse closes up again and finally coincides with the other diagonal $\frac{x}{a} + \frac{y}{b} = 0$ of the rectangle [**fig. 8.2**].

As the phase difference changes from π to 2π, the reverse process takes place, until the ellipse again coincides with the first diagonal. All these changes have been represented in **fig. 8.2**.

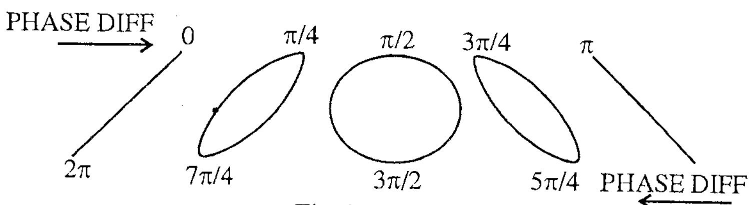


Fig. 8.2.

It should be noted that *the frequency of the complete cycle is equal to the difference of the frequencies of the two component simple harmonic vibrations.*

8·7. Composition of Two Rectangular S.H.M.'s of Frequencies in the Ratio 2 : 1

Let the two component rectangular simple harmonic vibrations be represented along X-axis and Y-axis as

$$x = a \sin (2\omega t + \phi) \qquad \text{...(1)}$$

and $\quad y = b \sin \omega t \qquad \text{...(2)}$

where ϕ is the phase angle by which the first motion is ahead of the second. Clearly the frequency of (1) vibrations is double to that of (2).

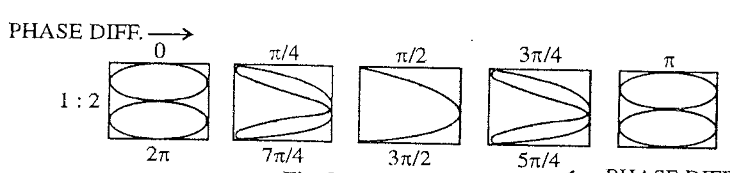


Fig. 8.3.

The resultant motion, which can be obtained by eleminating ωt from the equations (1) and (2), in general will be a curve, having two loops for any value of phase difference and amplitudes.

Here, we will consider only two special cases, given below :

(a) *When* $\phi = 0$, we get

$$x = a \sin 2\omega t = 2a \sin \omega t \cos \omega t$$

or $$\frac{x}{a} = 2 \sin \omega t \cos \omega t \quad \text{or} \quad \frac{x}{a} = \frac{2y}{b}\sqrt{1-\frac{y^2}{b^2}} \quad \text{or} \quad \frac{x^2}{a^2} = \frac{4y^2}{b^2}\left(1-\frac{y^2}{b^2}\right) = -\frac{4y^2}{b^2}\left(\frac{y^2}{b^2}-1\right)$$

Thus, $$\frac{x^2}{a^2}+\frac{4y^2}{b^2}\left(\frac{y^2}{b^2}-1\right) = 0 \qquad ...(3)$$

This equation represents the figure of '8' **[fig. 8.3]**.

(b) *When* $\phi = \pi/2$, we have

$$x = a \sin (2\omega t + \tfrac{\pi}{2}) = a \cos 2\omega t = a(1-2 \sin^2 \omega t)$$

$$\therefore \quad \frac{x}{a} = 1 - \frac{2y^2}{b^2} \qquad \left[\because \sin^2 \omega t = \frac{y^2}{b^2}\right]$$

or $$\frac{2y^2}{b^2} = 1 - \frac{x}{a} = -\frac{x-a}{a}, \quad \therefore \quad y^2 = -\frac{b^2}{2a}(x-a) \qquad ...(4)$$

This is the equation of a parabola with vertex $(a, 0)$.

If the frequencies of the two rectangular vibrations depart slightly from the ratio 2 : 1, the form of the curve slowly varies as ϕ changes **[fig. 8.3]**.

Lissajous Figures

When a particle oscillates simultaneously under the action of two simple harmonic motions at right angles to each other, the curve traced by the resultant motion of the particle is called **Lissajous figure** after the name of J.A. Lissajous who made an extensive study of such motions. The shape of the curve depends upon the amplitudes, frequencies and phase difference of the component vibrations. We have seen that if the frequencies of the two S.H.Ms. at right angles are equal, the Lissajous figure, in general, is an ellipse and has a definite shape for a particular phase difference **[fig. 8.1]**. For $\phi = 0$ or π, it is a straight line and for $\phi = \frac{\pi}{2}$ and $a_1 = a_2$, it is a circle.

When the two frequencies of component perpendicular vibrations are in the ratio 2 : 1, the Lissajous figures are relatively complex. It has the shape of figure '8' for $\phi = 0$ or π and that of parabola for $\phi = \frac{\pi}{2}$. If the component frequencies are in the complicated ratio, the Lissajous figures are odd and complicated.

8·8. Experimental Demonstration of Lissajous Figures

(1) Optical method : For this purpose, two electrically maintained tuning forks A and B are set up so that the vibrations of one take place in a vertical plane and those of the other in a horizontal plane. Each fork has a small mirror attached to the side of a prong. Light from a small powerful source, rendered convergent with the help of a convex lens, is allowed to fall on the mirror M of the vertical fork A. The light, reflected from M_1, strikes the mirror M_2 of the horizontal fork B, from where it is directed towards the screen S. The position of the lens is so adjusted that it produces a sharp image on the screens **[fig. 8.4]**.

If now only the vertical fork is vibrated, the spot of light on the screen executes a vertical S.H.M. so rapidly, that only a bright vertical line is observed on the screen. If the horizontal fork alone vibrates, a bright horizontal line is seen. Thus these motions represent two simple harmonic vibrations at right angles to each other. Now, if both the forks vibrate simultaneously, the two rectangular vibrations of the spot of light are compounded and the spot describes the path of the resultant motion on the screen. If the frequencies of the forks bear some simple ratio to each other, a Lissajous figure will be seen on the screen. For example, if the frequencies are in the ratio 2 : 1, a figure of '8' may be obtained. In case if the ratio between the frequencies is approximately two, it will gradually pass through the cyclic of changes of figures as shown in **fig. 8.3**. This method is usually employed for the comparison of the frequencies of two forks.

(2) By cathode ray oscillograph : Lissajous figures can be produced easily by means of a cathode ray oscilloscope (CRO). Different alternating sinusoidal voltages are applied at X and Y deflection plates of the CRO and the electron beam traces the resultant effect on the flourescent screen. When the applied voltages have the same frequency, we can obtain the various curves of **fig. 8.2**, by adjusting the phases and amplitudes. By using the CRO, we can also obtain easily the Lissejous figures for the component frequencies for the ratio 1 : 2 or any other ratio.

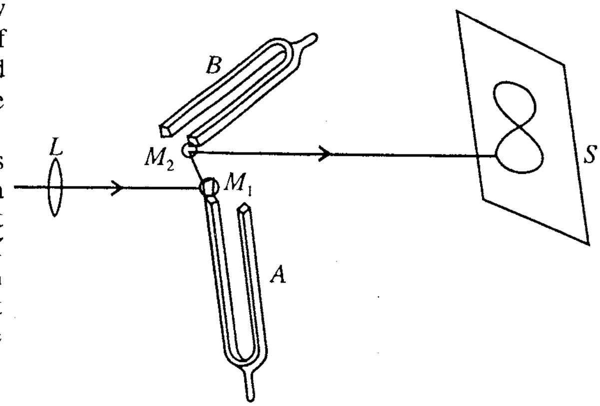


Fig. 8.4.

8·9. Uses of Lissajous Figures

(1) Lissajous figures are used in obtaining beautiful designs for printing in cloth industry.

(2) Lissajous figures find useful applications in the acoustical measurements, *e.g.*, determintaion of unknown frequency. If the frequencies of vibration of the two rectangular vibrations are not exactly equal or in the ratio 2 : 1, the form of the Lissajous figure undergoes a gradual progressive change. The time, taken by the curve to undergo a complete cycle of changes, enables us to find the frequency of one vibration, if the frequency of the other is known. Let the frequency of one vibration be n and that of the other be slightly different n', the frequencies being nearly in the ratio 1 : 1. If t be the time in which the cycle undergoes a complete change, then the frequency of the other will be $n' = n \pm 1/t$.

If the frequencies are nearly in the ratio 2 : 1, then $n' = 2n \pm 1/t$.

To find whether n' is greater of less than n or $2n$ (as the case may be) a small piece of wax is attached to the unknown (*e.g.*, the prong of the fork) and again the time of one cycle of changes is determined. If this time increases, the unknown frequency is higher and if it decreases, the unknown frequency is lower than n or $2n$.

(3) One practical application of Lissajous figures is that they enable us to know by inspection the period ratio of its constituent vibrations. It is evident that if the horizontal and vertical lines are drawn on a Lissajous figure and if they cut it m and n times respectively, the required period ratio of the two vibrations is $m : n$. *i.e.*,

$$\frac{\text{Frequency along } Y\text{-direction } (\nu_2)}{\text{Frequency along } X\text{-direction } (\nu_1)} = \frac{m}{n}$$

$$= \frac{\text{Number of intersection points of the Lissajous figure by a parallel line to } X\text{-axis}}{\text{Number of intesection points of the figure by a parallel line to } Y\text{-axis}}$$

Ex. 1. *A simple pendulum is made to oscillate under two simple harmonic motions, given by* $x = 5 \cos \omega t$ *and* $y = 6 \cos (\omega t + \phi)$.

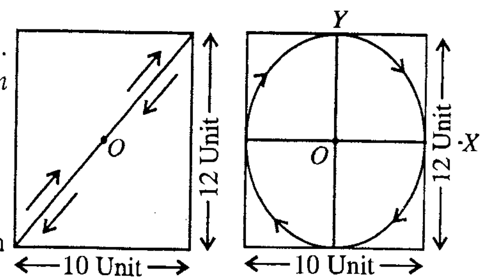
Find the resultant path of the bob, when $\phi = 0$ *and* $\frac{\pi}{2}$. *Draw a plot of the path in each case, indicating the direction of movement of the bob.*

Sol. For $\phi = 0$, $x = 5 \cos \omega t$ and $y = 6 \cos \omega t$

$\therefore \qquad - = - \quad$ or $y = 1.2x$

In this case the path of the bob is straight line as shown in **fig. 8.5(a)**.

For $\qquad \phi = -, \; x = 6 \cos \omega t$



(a) (b)

Fig. 8.5.

and $$y = -6 \sin \omega t$$

$\therefore$ $$\sin^2 \omega t + \cos^2 \omega t = \frac{y^2}{6^2} + \frac{x^2}{5^2} \text{ or } \frac{x^2}{25} + \frac{y^2}{36} = 1$$

Now the path of the bob is along the ellipse in clockwise direction as shown in **fig. 8.5(b)**.

Ex. 2. *The ratio of intensities of two waves is 1 : 9. If these waves interfere, compare the maximum and minimum intensities.* **(Rewa 2003; Bhopal 03; Gwalior 1997)**

Sol. $I_1 = ka_1^2$ and $I_2 = ka_2^2$; $\quad \therefore \frac{I_2}{I_1} = \frac{a_2^2}{a_1^2}$

Here $\frac{I_2}{I_1} = \frac{9}{1}$, $\quad \therefore \frac{a_2^2}{a_1^2} = \frac{9}{1}$ or $\frac{a_2}{a_1} = \frac{3}{1}$

Now $$\frac{I_{max}}{I_{min}} = \frac{(a_2 + a_1)^2}{(a_2 - a_1)^2} = \frac{\left(\frac{a_2}{a_1} + 1\right)^2}{\left(\frac{a_2}{a_1} - 1\right)^2} = \frac{\left(\frac{3}{1} + 1\right)^2}{\left(\frac{3}{1} - 1\right)^2} = \frac{4^2}{2^2} = \mathbf{\frac{4}{1}}.$$

Ex. 3. *The ratio of intensities of the two waves is β. Show that in their interference pattern,*

$$\frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{2\sqrt{\beta}}{\beta+1}.$$ **(Bundelkhand 2004; Agra 1989; Gorakhpur 86)**

Sol. $I_1 = k_1 a_1^2$ and $I_2 = k_2 a_2^2$; $\therefore \frac{I_2}{I_1} = \frac{a_2^2}{a_1^2}$

Also $I_{max} = k(a_2 + a_1)^2$ and $I_{min} = k(a_2 - a_1)^2$

Here $$\frac{I_2}{I_1} = \beta \text{ i.e., } \frac{a_2^2}{a_1^2} = \beta \text{ or } \frac{a_2}{a_1} = \sqrt{\beta}$$

$\therefore$ $$\frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{(a_2 + a_1)^2 - (a_2 - a_1)^2}{(a_2^2 + a_1^2) + (a_2^2 - a_1^2)} = \frac{4a_1 a_2}{2(a_1^2 + a_2^2)}$$

$$= \frac{2(a_1 a_2 / a_1^2)}{(a_1^2 + a_2^2)/a_1^2} = \frac{2(a_2/a_1)}{1 + (a_2/a_1)^2} = \frac{2\sqrt{\beta}}{1+\beta}$$ **Proved.**

Ex. 4. *Two vibrations of right angles to one another are described by the equations*

$$x = 10 \cos (5\pi t) \text{ and } y = 10 \cos \left(5\pi t + \frac{\pi}{3}\right)$$

Construct the Lissajous figures of the resulting motion. **(Meerut 1995)**

Sol. $$x = 10 \cos (5\pi t) \text{ and } y = 10 \cos \left(5\pi t + \frac{\pi}{3}\right)$$

Now, $$y = 10 \cos 5\pi t \times \cos\frac{\pi}{3} - 10 \sin (5\pi t) \times \sin\frac{\pi}{3}$$

or $$y = \frac{x}{2} - \frac{\sqrt{3}}{2}\sqrt{10^2 - x^2} \qquad \left[\because \sin 5\pi t = \sqrt{1 - (\cos 5\pi t)^2}\right]$$

or $$(2y - x)^2 = 3(100 - x^2) \text{ or } x^2 + 4y^2 - 4xy = -3x^2 + 300$$

or $$x^2 + y^2 - xy = 75$$

whence, $$\frac{x^2}{(5\sqrt{3})^2} + \frac{y^2}{(5\sqrt{3})^2} - \frac{xy}{(5\sqrt{3})(5\sqrt{3})} = \mathbf{1}.$$

Thus the resulting motion is an oblique ellipse.

Ex. 5. *Two tuning forks produce a Lissajou's figure. The figure changes in form from a parabola to a figure of eight and again to a parabola, the whole cycle occupying 6 seconds. If the frequency of one fork is 100, find possible frequencies of the other.* **(Punjab 1964; Rajasthan 55; Sagar 64)**

Sol. The figure changes from a parabola to a figure of '8' and then again to a parabola. Evidently then the component frequencies are nearly in the ratio 1 : 2 since the frequency of one tuning fork is 100, that of the other may be nearly 200 or 50.

The same figure being repeated in 6 sec., suggests that in 6 seconds the other fork completes one vibration more or less than double or half the vibrations performed by the first fork. Therefore, in one second the other fork will complete $\frac{1}{6}$ vibration more or less than double or half the vibrations made by the first fork. Obviously, therefore, the other fork will in one second, complete $200 \pm \frac{1}{6}$ or $50 \pm \frac{1}{6}$ vibrations.

Hence the possible frequencies of the other fork are $\mathbf{200 \pm \frac{1}{6}}$ and $\mathbf{50 \pm \frac{1}{6}}$.

Ex. 6. *Two tuning forks whose frequencies are approximately in the ratio 2 : 1 are employed to produce Lissajous figures, and it is observed that the figure goes through a cycle of change in 15 sec. On loading slightly the fork of the upper pitch, the figure goes through a cycle of changes in 10 sec. If the fork of lower pitch has a frequency of 300 Hz. calculate the frequency of the other fork before and after loading.*

Sol. Since the frequency of the fork of lower pitch is 200 Hz, that of second may approximately be 600 Hz. Before loading the second fork, the figure goes through a cycle of changes in 15 sec., *i.e.*, the frequency of the second fork may be equal to $600 \pm \frac{1}{15}$.

On loading the second fork, the time of the cycle of changes is reduced to 10 sec. Hence the frequency of the second fork after loading may be $600 \pm \frac{1}{10}$.

But on loading the frequency of the fork must decrease. Therefore, the frequency of the second fork before loading $= 600 - \frac{1}{10} = \mathbf{599\frac{14}{15}}$ **Hz.** and after loading $= 600 - \frac{1}{10} = \mathbf{599\frac{9}{10}}$ **Hz.**

Ex. 7. *Show that a linear simple harmonic motion can be obtained by the superposition of two equal and opposite circular motions of the same angular frequency.*

Sol. Let us consider two equal and opposite circular motions of the same angular frequency ω [**fig. 8.6**]. At any time t, the position vectors of the rotating point in the two cases are given by

$$\vec{OP_1} = a \cos \omega t\, \hat{\mathbf{i}} + a \sin \omega t\, \hat{\mathbf{j}}$$

$$\vec{OP_2} = a \cos \omega t\, \hat{\mathbf{i}} - a \sin \omega t\, \hat{\mathbf{j}}$$

The superposition of two vectors gives

$$\vec{OP_1} + \vec{OP_2} = 2a \cos \omega t\, \hat{\mathbf{i}}$$

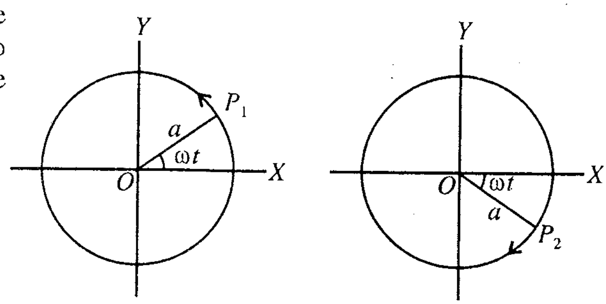


Fig. 8.6.

Thus the resultant displacement due to the two equal and opposite circular motions is along X-axis *i.e.*, straight line and describes a S.H.M. of angular frequency ω.

Ex. 8. *In a cathode ray oscilloscope, the electrons are deflected by two mutually perpendicular electric fields according to the displacement-time relations, given by*

$$x = 4 \sin (2\pi\upsilon t + \pi/6) = 4 \sin (2\pi\upsilon t)$$

Determine the resultant path followed by the electrons.

Sol. $$\frac{x^2}{4^2} + \frac{y^2}{4^2} - \frac{2xy}{4\times4}\cos\frac{\pi}{6} = \sin^2\frac{\pi}{6}$$ [See eq. (4), Art. 8.5] or $$\frac{x^2}{16}+\frac{y^2}{16}-\frac{2xy}{16}\cdot\frac{\sqrt{3}}{2} = \frac{1}{4} \quad \text{or} \quad x^2 + y^2 - \sqrt{3}xy - 4 = 0$$

Thus the resultant path is an ellipse.

Ex. 9. *Lissajous figure in the case of two vibrating forks is a parabola prove that the frequencies are in the ratio 1 : 2.*

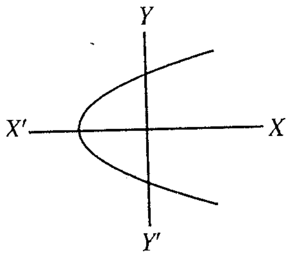


Fig. 8.7.

Sol. Suppose that one fork of frequency ν_1, is vibrating along X-axis and other of frequency υ_2 along Y-axis. Then if the parabola cuts the X-axis m times and Y-axis n times, then

$$\frac{\upsilon_2}{\upsilon_1} = \frac{m}{n} = \frac{1}{2} = 1 : 2.$$

EXERCISE 8·1

1. Two simple harmonic motions are represented by the equations
$$y_1 = 10 \sin (3\pi t + \pi/4) \text{ and } y_2 = 5[\sin (3\pi t) + \sqrt{3} \cos (3\pi t)].$$
Compare their amplitudes of oscillations **(Meerut 1994)**
[Ans. 1 : 1.]
2. Two simple harmonic vibrations along a line are represented by
$$x_1 = \sin (20\pi t + \pi/6) \text{ cm} \quad \text{and} \quad x_2 = 4 \sin (20\pi t + \pi/3) \text{ cm}.$$
Determine the amplitude and frequency of resultant vibration.
[Ans. 6.7 cm; 10 Hz.]
3. The ratio of intensities of two coherent sources is 81 : 1. Find the ratio of maximum and minimum intensities. **(Agra 1987; Garwal 80)**
[Ans. 25 : 16.]
4. The ratio of the intensities of two sources is 9. Find the ratio $\frac{I_{max} - I_{min}}{I_{max} + I_{min}}$. **(Agra 2003)**
[Ans. 3/5.]
5. The ratio of maximum and minimum intensities due to superposition of two waves is 4 : 1. Compare the amplitudes of the two waves.
[Ans. 3 : 1.]
6. Two forks are used to produce Lissajous figures. It is found that the figure completes its cycle in 10 sec. If the frequency of A (known to be slightly longer than that of B) is 200. Calculate the frequency of B.
[Ans. 199.9 Hz.]
7. Two tuning forks are vibrated perpendicularly and a beam of light is reflected from the mirrors, attached to the prongs of the forks. The elliptical figures are produced on the screen and they change in shape as time progresses. The figures complete a cycle of changes in 10 seconds. If the frequency of one fork is 250 (slightly less than that of the other), determine the frequency of the other.
[Ans. 250.1 Hz.]
8. Two tuning forks P and Q of frequencies close to each other are used to obtain Lissajous figures and it is observed that the figure goes through a cycle of changes in 20 seconds. Now if P is loaded slightly with wax, the figure goes through a cycle of changes in 10 seconds. If the frequency of Q is 300 Hz, what is the frequency of P before and after loading.
[Ans. 299.95 Hz; 299.9 Hz.]
9. Two rectangular simple harmonic vibrations of frequency ratio 2 : 1 are represented as follows :
$$x = a_1 \cos (2\omega t + \phi) \text{ and } y = a_2 \cos \omega t$$
Find the resultant motion, when (i) $\phi = 0$; (ii) $\pi/2$ and (iii) π.
[Ans. (i) Parabola; (ii) Figure of '8'; (iii) Parabola oppositely directed to that of (i).]

8·10. Superposition of Two Simple Harmonic Motions of Different Frequencies along the Same Line—Modulation and Beats

In physics, there are a number of phenomena in which the motion is a superposition of two simple harmonic motions, having different angular frequencies ω_1 and ω_2. For example, when two tuning forks of different frequencies vibrate simultaneously, they emit sinusoidal waves. The oscillatory motion of the eardrum is, in fact, a superposition of two simple harmonic motions due to the two waves. To simplify mathematically, it is assumed that the two harmonic motions have the same amplitude a so that the phase difference $[(\omega_2 t + \phi_2) - (\omega_1 t + \phi_1) = (\omega_2 - \omega_1)t + \phi_2 - \phi_1]$ between the two vibrations is changing continuously with time. Since the initial phase difference $(\phi_2 - \phi_1)$ does not play any significant role in this case, we assume that the two oscillations have zero initial phase and are represented as

$$x_1 = a \sin \omega_1 t \text{ and } x_2 = a \sin \omega_2 t \quad \text{...(1)}$$

The resultant motion is obtained as a superposition of the two motions *i.e.*,

$$x = x_1 + x_2 = a \sin \omega_1 t + a \sin \omega_2 t \quad \text{...(2)}$$

Modulation

Eq. (2) can be written as

$$x = 2a \cos\left(\frac{\omega_1 - \omega_2}{2}\right)t \sin\left(\frac{\omega_1 + \omega_2}{2}\right)t \quad \text{...(3)}$$

which represents an oscillatory motion with angular frequency $\frac{\omega_1 + \omega_2}{2}$ and amplitude $2a \cos\left(\frac{\omega_1 - \omega_2}{2}\right)t$.

We define an **average angular frequency** ω_{av} and a **modulated angular frequency** ω_{mod} by

$$\omega_{av} = \frac{\omega_1 + \omega_2}{2} \text{ and } \omega_{\text{mod}} = \frac{\omega_1 - \omega_2}{2} \quad \text{...(4)}$$

and a **modulation amplitude** $a_{\text{mod}}(t)$ by

$$a_{\text{mod}}(t) = 2a \cos\left(\frac{\omega_1 - \omega_2}{2}\right)t = 2a \cos \omega_{\text{mod}} t \quad \text{...(5)}$$

Therefore the superposed oscillation is given by

$$x = a_{\text{mod}}(t) \sin \omega_{av}(t) \quad \text{...(6)}$$

The superposition of two simple harmonic motions of different frequencies is shown in **fig. 8.8**. The figure also shows the variation of the modulation amplitude with time.

We see from eq. (5) that the modulation amplitude varies with time with a frequency, given by

$$\upsilon_{\text{mod}} = \frac{\omega_{\text{mod}}}{2\pi} = \frac{(\omega_1 - \omega_2)/2}{2\pi} = \frac{\upsilon_1 - \upsilon_2}{2} \quad \text{...(7)}$$

In a cycle, the modulation amplitude in eq. (6) is $2a$, 0, $-2a$ and 0, as the phase angle $(\omega_{\text{mod}} t)$ becomes 0, $\pi/2$, π and $3\pi/2$ respectively. However the intensity (I) is proportional to the square of the amplitude, it will be maximum two times and also minimum two times in a cycle of the modulation amplitude. For example, in case of two sound waves (of equal amplitudes), the ear will hear the maximum or minimum sound at double the frequency of the modulated amplitude. *This fluctuation in the intensity of sound is called* **beats**. The frequency of the beats is given by

$$\upsilon_{\text{beat}} = 2\upsilon_{\text{mod}} = \upsilon_1 - \upsilon_2 \quad \text{...(8)}$$

Mathematically, we obtain this result as follows :

$$I \propto a^2_{\text{mod}}(t) = 4a^2 \cos^2 \omega_{\text{mod}}(t) = 2a^2 (1 + \cos 2\omega_{\text{mod}} t) \quad \text{...(9)}$$

Therefore $a^2_{\text{mod}}(t)$ or I oscillates at angular frequency $2\omega_{\text{mod}}$ *i.e.*, the frequency of the beats is twice the modulation frequency. However one can hear the beats when $|\upsilon_1 - \upsilon_2|$ is less than about 10 Hz.

Ex. 1. *Two vibrations along the same line are represented by the equations*

$$x_1 = a \sin 20\pi t \textit{ and } x_2 = a \sin 22\pi t.$$

Deteremine the beat frequency and draw a graph of the resultant motion over one beat period.

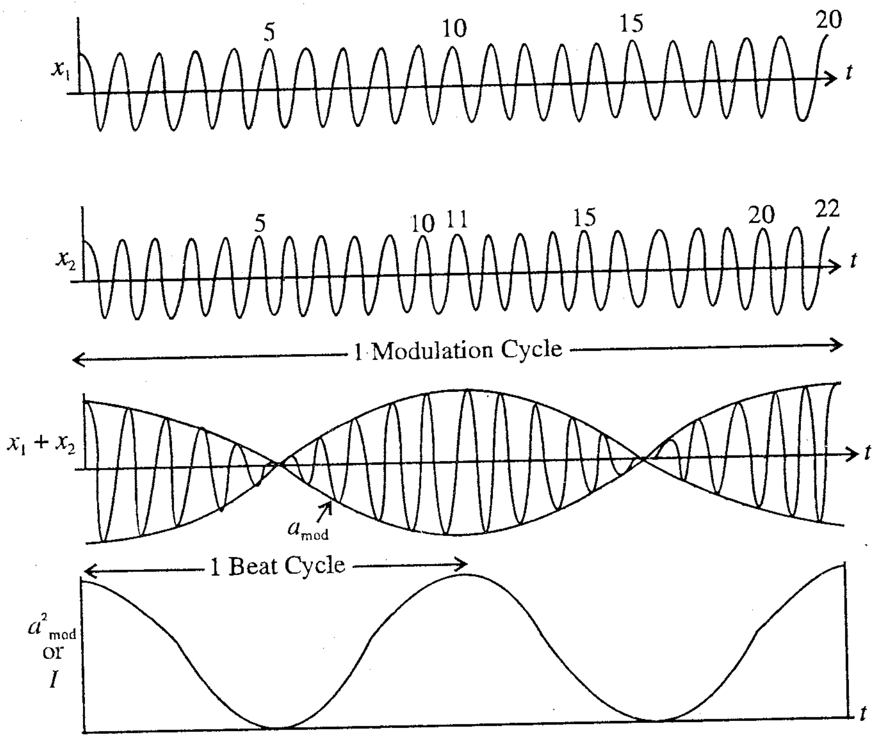


Fig. 8.8 : Modulation and formation of beats.

Sol. Here, $\nu_1 = 10$ Hz and $\nu_2 = 11$ Hz. Beat frequency = $|\nu_1 - \nu_2|$ = **1 Hz.**

$\therefore$ For graph see **fig. 8.8.**

QUESTIONS

Long Answer Questions

1. Two simple harmonic motions of the same period are acting on a particle along one line. Discuss the resultant motion. **(Ujjain 1997; Lucknow 82)**
2. What is the principle of superposition ? Deduce an expression for the phase and amplitude of the resultant motion obtained due to the superposition of two linear simple harmonic motions of same frequency ($= \omega/2\pi$), amplitudes a_1 and a_2 and with ϕ_1 and ϕ_2 initial phases. **(Indore 2004; Bhopal 04; Ujjain 03)**
3. What do you understand by interference ? Write down the necessary conditions for the distinct and sustained interference and show that the law of conservation of energy is obeyed in the phenomenon of interference. **(Jabalpur 2003; Rewa 04)**
4. The displacements of two mutually perpendicular simple harmonic oscillations of same frequency are $x = a \sin(\omega t + \phi)$ and $y = b \sin \omega t$
   Obtain the expression for the resultant oscillation obtained due to their superposition. What will be the resultant path if $\phi = 0°$, $\pi/2$ and π ? **(Ujjain 2004; Jabalpur 04)**
5. Discuss the superposition of two mutually perpendicular simple harmonic vibrations of the same frequency. What happens in the case of different frequencies ? **(Agra 2005)**
6. What are Lassajous figures ? Two simple harmonic motions on a particle are acting simultaneously. $x = a_1 \sin \omega t$ and $y = a_2 \cos(\omega t + \phi)$. Show that in general the particle will move on an elliptical path. When will the path be circular ? **(Gorakhpur 1983)**

7. Explain Lissajous figures when two simple harmonic oscillations of the frequency ratio 1 : 1 and phase difference varying between 0 and π, superpose. **(Rewa 2003, 04; Indore 03)**
8. When two mutually perpendicular simple harmonic motions of amplitudes and time periods in the ratio 1 : 1, but phase difference (i) 90°, (ii) 45° superpose, deduce the expression for the figures obtained. **(Gwalior 2004)**
9. If two mutually perpendicular simple harmonic oscillations represented by equations $x = a \sin (2\omega t + \phi)$ and $y = b \sin \omega t$., superpose simultaneously on a particle, deduce the equation for the resultant path of the particle. What will be the shape of the path if $\phi = 0°$, $\pi/2$ and π ? **(Indore 2004; Ujjain 03)**
10. Two perpendicular simple harmonic motions, whose amplitudes are different but the frequencies are in the ratio 2 : 1, are superposed together. Discuss those conditions in which the phase difference is 0, $\pi/2$ and π. **(Ujjain 1979, 74; Kanpur 73)**
11. What are Lissajous figures ? Establish the expression for the resultant motion obtained due to superposition of two mutually perpendicular simple harmonic motions of time period in ratio 1 : 2. **(Bundelkhand 2004; Gwalior 03; Bhopal 03; Sagar 03; Bilaspur 03)**
12. What are Lissajous figures ? Obtain an expression for resultant figures. If the ratio of frequencies of two SHMs is 2 : 1 and are superimposed normally, then describe the resultant Lissajous figures. **(Agra 2004)**
13. What are Lissajous figures ? How will you demonstrate them ? Discuss their uses.
14. Two simple harmonic motions, given by $x_1 = a_1 \cos \omega_1 t$ and $x_2 = a_2 \cos \omega_2 t$, are acting on a particle. Show that the amplitude of resultant displacement is $a = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos(\omega_1 - \omega_2)}$.

Short Answer Questions

1. What is principle of superposition ?
2. Show that in case of the differential equation of simple harmonic motion, the sum of any two solutions of it is itself a solution.
3. What are linear and nonlinear equations ?
4. What is interference ? What are the conditions for constructive and destructive interference ? **(Bhopal 2004; Ujjain 04)**
5. State the necessary conditions for the interference of waves ?
6. Discuss the energy conserved in the phenomenon of interference ?
7. What are Lissajous figures and what are their uses ? **(Kanpur 2005; Ujjain 03)**
8. If the Lissajous figure has the shape of letter '8', what is the ratio of component frequencies ? [**Ans.** 2 : 1.]
9. What is the shape of Lissajous figure, when two mutually perpendicular simple harmonic motions of same amplitude and frequency are superposing on each other with a phase difference of $\pi/2$. [**Ans.** Circle.]

Objective Type Questions

1. If two simple harmonic motions of the same frequency, having amplitudes 3 cm and 4 cm, are acting on a particle along a line and they have phase difference ϕ, the resultant motion due to superposition will have amplitude :
   (a) 3 cm (b) 4 cm (c) 5 cm (d) more than 5 cm
2. Two simple harmonic motions, having equal amplitudes and frequencies, are acting on a particle along the same direction. If the resultant amplitude is equal to the amplitude of one simple harmonic motion, the phase difference between them is :
   (a) 0 (b) $\pi/2$ (c) $2\pi/3$ (d) π
3. When two simple harmonic motions of equal amplitudes and equal frequencies are acting perpendicularly on a particle, the resulting motion is along a circular path. The phase difference between the two vibrations is :
   (a) 0 (b) $\pi/4$ (c) $\pi/2$ (d) π

4. When two simple harmonic motions of nearly equal frequencies and amplitudes are acting on a particle simultaneously along a line, the displacement-time curve :

(a) is a sine curve (b) is a saw tooth curve

(c) possesses constant amplitude (d) possesses variable amplitude

5. Interference is possible :

(a) only by longitudinal wave (b) only by transverse wave

(c) by longitudinal as well as transverse waves (d) none of these **(Ujjain 1991)**

6. Due to the superposition of two perpendicular simple harmonic motions of the same frequency, the Lissajous figure for $\pi/2$ phase difference will be obtained as :

(a) straight line (b) figure of '8' (c) ellipse (d) parabola

ANSWERS

1. (c), **2.** (c) **3.** (a), **4.** (d), **5.** (c), **6.** (c).

9

Coupled Oscillators and Normal Modes

9·1. Introduction

So far we have discussed about an isolated, single oscillating system such as simple pendulum, spring-mass system etc. The characteristic feature of such an oscillating system is a single frequency of vibration. On the other hand the wave motion such as along a string or on the surface of a pond is regarded as oscillations of a large number of particles, each connected or coupled to its neighbours by some sort of elastic medium. If one end of a string is disturbed, this causes to oscillate the neighbouring particles through the coupling medium and in turn these influence further particles and so on. In this way, a wave moves in a medium, but the question arises how the coupling affects the motion of an oscillator due to the other. In order to simplify and understand the problem, in this chapter first we discuss the motion of two coupled oscillators in some detail by taking specifically the example of a system of two pendulums connected by a spring [**fig. 9.1**]. Later on we also discuss some other examples of two coupled oscillators. Next the analysis is extended to discuss the motion of N-coupled oscillators by taking the example of N beads attached equidistantly on a string under tension. This analysis leads us finally into a discussion of wave motion in a continuous medium.

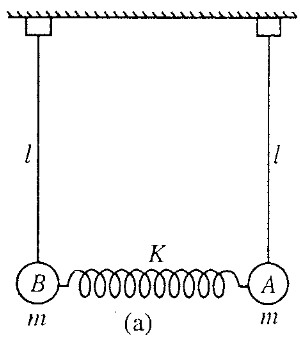


Fig. 9.1 : Two coupled pendulums in equilibrium.

We shall see that when we go from a single oscillator to deal the problem of two coupled oscillators, the analysis results in some interesting and surprising new features. It is found that the motion of two coupled oscillators, in general, is much compolicated and none of the oscillators executes simple harmonic motion. However, for small amplitude oscillations, we are able to express the general motion as a **superposition** of two independent simple harmonic motions, both going on simultaneously. These two simple harmonic motions are called as **normal modes** or **simply modes**. The system may be made to oscillate only in one mode or the other by having suitable initial conditions. In a particular mode of vibration, each of the oscillators performs simple harmonic motion of the same frequency and the two oscillators pass their equilibrium positions simultaneously. Further we see that a system of N-coupled oscillators has exactly N independent modes of vibration and the general motion of the system can be expressed as the superposition of N modes. Each mode has its own frequency and wavelength. Finally considering exceedingly large number of particles and allowing the interparticle distance to approach zero, we find the system as continuous and its motion is dealt as **waves**.

9·2. Oscillations of Two Coupled Pendulums

Let us consider two indentical pendulums A and B whose bobs are connected by a light spring of force constant K. The relaxed length of the spring is equal to the distance between the two bobs in the equilibrium position [**fig. 9.1**]. In the present analysis, we consider small amplitude oscillations, restricted to the plane in equilibrium configuration. Thus the system of two coupled pendulums, under consideration, has two degrees of freedom*. Now let us observe what happenss when we displace the bob A to one side with the bob B holding fixed and then release them simultaneously [**fig. 9.2**]. It is observed that the

* The **degree of freedom** of a system of particles is determined by the number of coordinates defining their positions at a time. An oscillating pendulum in a vertical plane has one degree of freedom as its position can be described by one coordinate only. Two coordinates are required to determine the position of two oscillating coupled pendulums in a plane, and hence it has two degrees of freedom. N particles if constrained to move along one direction (or one dimension), have N degrees of freedom and in general for three independent directions (or three dimensions), a system of N particles possesses $3N$ degrees of freedom.

pendulum A oscillates with a continuous decrease in amplitude and pendulum B starts to oscillate with

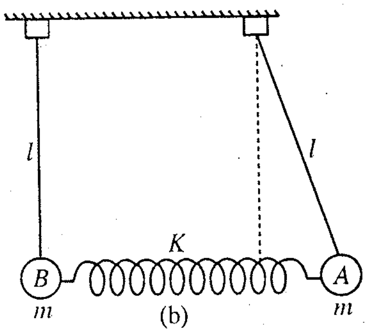


Fig. 9.2 : Holding the bob B, bob A is displaced and then the system is left free to oscillate.

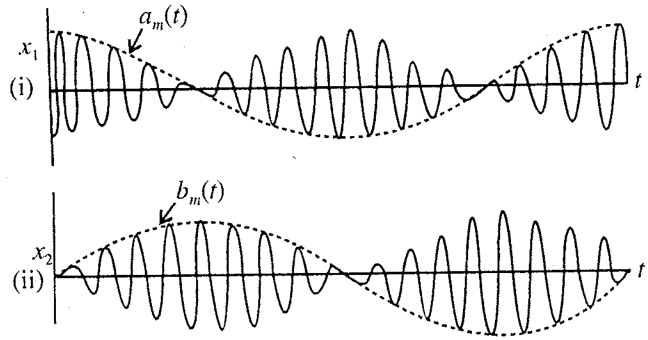


Fig. 9.3 : Displacement time curves for two coupled pendulums : upper figure (i) for A and lower figure (ii) for B.

a gradual increase in the amplitude. Next we see that the two pendulums acquire equal amplitudes. The process of decrease of amplitudes of A and increase of amplitude of B continues until the amplitude of A becomes almost zero and that of B becomes the same as that was initially given to A (for free oscillations).* The process is now reversed and the motion of B is gradually transferred back to A. We observe that the mutual transference of motion between A and B goes on continuously and periodically. In **fig 9.3**, we show the dsplacement curves for the two pendulums.

9·2·1. Theory for Two Coupled Pendulums

Let x_1 and x_2 be the displacements of the bobs A and B respectively at any time t so that the change in the length of the spring is $x_1 - x_2$. Consequently the spring exerts a force of magitude $K(x_1 - x_2)$ on A or B [**fig. 9.4**]. The bobs A and B are also acted by forces $-mgx_1/l$ and $-mgx_2/l$ due to gravity (for small amplitude oscillations)**. Hence the forces acting on the bobs A and B are given by

$$F_1 = -mg\frac{x_1}{l} - K(x_1 - x_2)$$

and

$$F_2 = -mg\frac{x_2}{l} - K(x_1 - x_2)$$

Now, writing $F_1 = d^2x_1/dt^2$ and $F_2 = d^2x_2/dt^2$ the equations of motion of the bobs A and B are obtained as

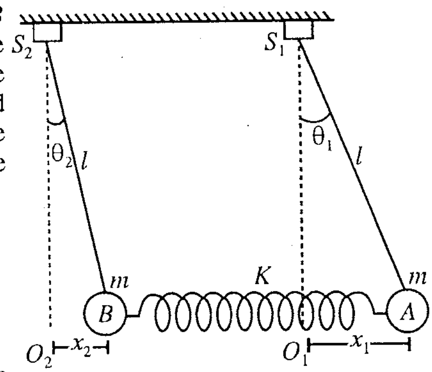


Fig. 9.4 : Two coupled pendulums in oscillation.

$$m\frac{d^2x_1}{dt^2} = -mg\frac{x_1}{l} - K(x_1 - x_2) \qquad \text{...(1a)}$$

and
$$m\frac{d^2x_2}{dt^2} = -mg\frac{x_2}{l} + K(x_1 - x_2) \qquad \text{...(1b)}$$

where obviously for $x_1 - x_2 > 0$, the spring is elongated and the spring force acts against the displacement of A but favours the displacement of B.

Writing $\omega_0^2 = g/l$ and $\omega_s^2 = K/m$, we get

$$\frac{d^2x_1}{dt^2} + \omega_0^2 x_1 + \omega_s^2(x_1 - x_2) = 0 \qquad \text{...(2a)}$$

* Experimentally, the maximum amplitude of A or B will decrease with time as damping forces are acting on the system.

** Force due to gravity on bob $A = -mg \sin\theta_1 = -mgx_1/l$;
Force due to gravity on bob $B = -mg \sin\theta_2 = -mgx_2/l$.

and $$\frac{d^2x_1}{dt^2}+\omega_0{}^2x_2-\omega_s{}^2(x_1-x_2) = 0 \qquad \text{...(2b)}$$

Eq. (2a) can be solved for x_1, if we know x_2 and conversely for the second equation. In fact these are the **coupled differential equations** which are to be solved simultaneously.

Adding eqs. (2a) and (2b), we get

$$\frac{d^2}{dt^2}(x_1+x_2)+\omega_0{}^2(x_1+x_2) = 0 \qquad \text{...(3)}$$

Subtracting eq. (2b) from eq. (2a), we obtain

$$\frac{d^2}{dt^2}(x_1-x_2)+(\omega_0{}^2+2\omega_s{}^2)(x_1-x_2) = 0 \qquad \text{...(4)}$$

We define two new coordinates

$$q_1 = \frac{x_1+x_2}{2} \text{ and } q_2 = \frac{x_1-x_2}{2} \qquad \text{...(5)}$$

Then eqs. (3) and (4) assume the form

$$\frac{d^2q_1}{dt^2}+\omega_1{}^2q_1 = 0 \qquad \text{...(6)}$$

and $$\frac{d^2q_2}{dt^2}+\omega_2{}^2q_2 = 0 \qquad \text{...(7)}$$

where $\omega_0 = \omega_1$ and $(\omega_0{}^2 + 2\omega_s{}^2)^{1/2} = \omega_2$.

We see that the motion of the coupled system is now described by two uncoupled differential equations (6) and (7); each of the equation describes a simple harmonic motion of single frequency (ω_1 or ω_2) in terms of single coordinate (q_1 or q_2).

Now if $q_2 = 0$, we obtain from (5), $x_1 = x_2$ and $q_1 = x_1 = x_2$. In such a case, eq. (6) takes the form

$$\frac{d^2x_1}{dt^2}+\omega_1{}^2x_1 = 0 \text{ or } \frac{d^2x_2}{dt^2}+\omega_1{}^2x_2 = 0 \qquad \text{...(8)}$$

i.e., each pendulum vibrates with an angular frequency $\omega_1 = \sqrt{g/l}$ and the two pendulums vibrate in the same phase with equal displacements.

If $q_1 = 0$, we obtain from (5), $x_1 = -x_2$ and $q_2 = x_1 = -x_2$ and eq. (7) assumes the form

$$\frac{d^2x_1}{dt^2}+\omega_2{}^2x_1 = 0 \text{ or } \frac{d^2x_2}{dt^2}+\omega_2{}^2x_2 = 0 \qquad \text{...(9)}$$

In this case, (i) the instantaneous displacement of one pendulum is opposite and equal to that of the other and the two pendulums are oscillating in the opposite phase; (ii) each of the pendulum vibrates with an angular frequency

$$\omega_2 = \left(\omega_0^2 + 2\omega_s^2\right)^{1/2} = \sqrt{\frac{g}{l}+\frac{2K}{m}} \qquad \text{...(10)}$$

We note that when $q_2 = 0$, q_1 coordinate describes the motion and when $q_1 = 0$, q_2 coordinate describes the motion. In fact, *these are the two specific independent ways of oscillations of the system in which each of the pendulum vibrates in simple harmonic motion with the same frequency. These two independent ways of oscillations of the two coupled pendulums are called* **modes.**

9·2·2. Normal Modes and Normal Coordinates

Solutions of eqs. (6) and (7) can be written as

$$q_1 = a_1 \cos(\omega_1 t + \phi_1) \qquad \text{...(11)}$$

$$q_2 = a_2 \cos(\omega_2 t + \phi_2) \qquad \text{...(12)}$$

These two simple harmonic motions, obtained after decoupling the coupled equations, are called

normal modes or simply **modes.** Each **mode** of vibration has its **normal frequency** (or **mode frequency**), ω_1 or ω_2, and **normal coordinate** (or **mode coordinate**), q_1 or q_2. With suitable initial conditions, we can make the system to oscillate in one mode or the other. The amplitude and phase constant for Mode 1 are a_1 and ϕ_1, and for Mode 2, a_2 and ϕ_2 respectively.

We see that the eqs. (6) and (7) have exactly the same form as the differential equation for simple harmonic motion and hence the methods, developed for simple harmonic motion are applicable here. It is to be kept in mind that the normal coordinates q_1 and q_2 do not measure the displacements similar to ordinary coordinates x_1 and x_2 but even then they specify completely the system at any instant. One can use eqs. (5) to obtain x_1 and x_2 from q_1 and q_2 or vice-versa.

9·2·3. Superposition of Modes

The general motion of the two pendulums is expressed by the coordinates x_1 and x_2, which are given by making use of the eqs. (5) as follows :

$$x_1 = q_1 + q_2 \text{ and } x_2 = q_1 - q_2 \qquad ...(13)$$

With the help of eqs. (11) and (12), we get

$$x_1 = a_1 \cos(\omega_1 t + \phi_1) + a_2 \cos(\omega_2 t + \phi_2) \qquad ...(14)$$

and

$$x_2 = a_1 \cos(\omega_1 t + \phi_1) - a_2 \cos(\omega_2 t + \phi_2) \qquad ...(15)$$

We find that the displacement of any pendulum is a linear combination or **superposition** of two modes q_1 and q_2, oscillating simultaneously. Eqs. (14) and (15) describe the general motion of the system and the constants of motion are obtained by the initial conditions. When the system is oscillating in Mode 1, $q_2 = (x_1 - x_2)/2 = 0$ or $x_1 = x_2$ and $q_1 = (x_1 + x_2)/2 = x_1 = x_2$. Therefore, in this case $x_1 = a_1 \cos(\omega_1 t + \phi_1)$ and $x_2 = a_1 \cos(\omega_1 t + \phi_1)$, *i.e.*, both pendulums are oscillating in the same phase [**fig. 9.5(a)**]. If the system is oscillating in Mode 2, $q_1 = (x_1 + x_2)/2 = 0$ or $x_1 = -x_2$ & $q_2 = (x_1 - x_2)/2 = x_1 = -x_2$. Therefore, now $x_1 = a_2 \cos(\omega_2 t + \phi_2)$ and $x_2 = -a_2 \cos(\omega_2 t + \phi_2)$, *i.e.*, the two pendulums are oscillating in the opposite phase or at a phase difference of π, [**fig. 9.5(b)**]. Thus, *in Mode* 1, *the two coordinates oscillate with the same frequency and phase constant, having the amplitude ratio* 1. *In Mode* 2, x_1 *and* x_2 *have the same frequency and phase constant, but the amplitude ratio is* –1.

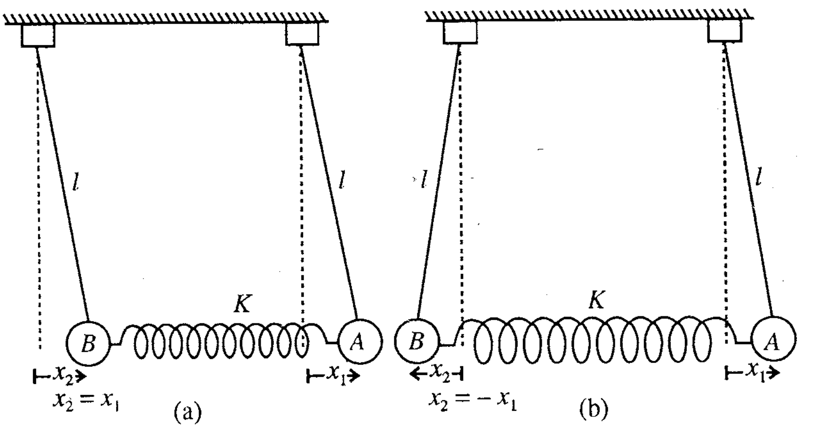


Fig. 9.5 : Oscillations of two coupled identical pendulums : (a) Mode 1 with lower frequency (ω_1) and $x_1 = x_2$; (b) Mode 2 with higher frequency (ω_2) and $x_1 = -x_2$. In Mode 1, two pendulums are in same phase and in Mode 2, they are in opposite phase.

We note that when the system oscillates in Mode 1 or Mode 2, both of the pendulums (or x_1 and x_2 coordinates) are executing simple harmonic motion. In all other states of vibration, the displacement of each pendulum (x_1 or x_2), given by eqs. (14) and (15), depends on both mode frequencies (ω_1 and ω_2) and hence the motion is no longer simple harmonic.

9·2·4. Oscillations of Two Coupled Pendulums for Given Initial Conditions

Now we apply the analysis to the motion of two coupled pendulums, shown in [**fig. 9.2**] in the beginning. The initial conditions for the system at $t = 0$ are as follows :

$$x_1 = 2a,\ dx_1/dt = 0,\ x_2 = 0 \text{ and } dx_2/dt = 0 \qquad ...(16)$$

From eqs. (14) and (15), we get

$$2a = a_1 \cos\phi_1 + a_2 \cos\phi_2 \text{ and } 0 = a_1 \cos\phi_1 - a_2 \cos\phi_2$$

whence $a_1 \cos\phi_1 = a$ and $a_2 \cos\phi_2 = a$ ...(17)

Differentiating eqs. (14) and (15) with respect to t, we obtain

$$\frac{dx_1}{dt} = -a_1\omega_1 \sin(\omega_1 t + \phi_1) - a_2\omega_2 \sin(\omega_2 t + \phi_2)$$

and

$$\frac{dx_2}{dt} = -a_1\omega_1 \sin(\omega_1 t + \phi_1) + a_2\omega_2 \sin(\omega_2 t + \phi_2)$$

Applying initial conditions (16), we obtain

$$a_1\omega_1 \sin\phi_1 + a_2\omega_2 \sin\phi_2 = 0 \text{ and } a_1\omega_1 \sin\phi_1 - a_2\omega_2 \sin\phi_2 = 0$$

or

$$a_1\omega_1 \sin\phi_1 = 0 \text{ and } a_2\omega_2 \sin\phi_2 = 0.$$

As a_1, a_2, ω_1 and ω_2 are not equal to zero,

$$\phi_1 = \phi_2 = 0.$$

Hence from eqs. (17), we have

$$a_1 = a \text{ and } a_2 = a.$$

Thus, for the given boundary conditions, the normal mode solutions (11) and (12) assume the form

$$q_1 = a\cos\omega_1 t \text{ and } q_2 = a\cos\omega_2 t \qquad ...(18)$$

and the instantaneous displacements of two pendulums [eqs. (14) and (15)] are obtained to be

$$x_1 = a\cos\omega_1 t + a\cos\omega_2 t \qquad ...(19a)$$

$$x_2 = a\cos\omega_1 t - a\cos\omega_2 t \qquad ...(19b)$$

9·2·5. Modulation and Phenomenon of Beats

The displacements x_1 and x_2 of the pendulums A and B, by using eqs. (19a) and (19b) can be written as

$$x_1 = 2a\cos[(\omega_2 - \omega_1)t/2]\cos[(\omega_2 + \omega_1)t/2] \qquad ...(20a)$$

and

$$x_2 = 2a\sin[(\omega_2 - \omega_1)t/2]\sin[(\omega_2 + \omega_1)t/2] \qquad ...(20b)$$

These equations are similar to equations for **modulated oscillations** or **beats.** Writing $(\omega_2 + \omega_1)/2 = \omega_a$, the average frequency of two modes, and $(\omega_2 - \omega_1)/2 = \omega_m$, the modulation frequency, we obtain

$$x_1 = (2a\cos\omega_m t)\cos\omega_a t = a_m(t)\cos\omega_a t \qquad ...(21)$$

and

$$x_2 = (2a\sin\omega_m t)\sin\omega_a t = b_m(t)\sin\omega_a t \qquad ...(22)$$

where $a_m(t) = 2a\cos\omega_m t$ and $b_m(t) = 2a\sin\omega_m t$, are the **modulation amplitudes**. From eqs.(21) and (22), we find that each of the pendulum executes sinusoidal oscillations of angular frequency ω_a, modulated in amplitude. If we plot the displacement-time graphs for the two pendulums they are obtained as shown in **fig. 9.3**. At $t = 0$, the amplitude of the pendulum A is maximum while that of B is zero. As the time increases, the amplitude of A decreases from the maximum value ($2a$) and the amplitude of B increases to a maximum ($2a$). The trends are then reversed and the amplitude of A builds up to a maximum value again while that of B reduces to zero. For free vibrations, the process repeats indefinitely. This explains the observed motion of the two coupled pendulums.

9·2·6. Energy of Two Coupled Pendulums

We assume the system to execute fast oscillations so that the modulation amplitude [$a_m(t)$ of pendulum A or $b_m(t)$ of pendulum B] remains constant over an oscillation of a pendulum. Thus during one cycle, we can think a pendulum as a harmonic oscillator of average angular frequency ω_a with constant amplitude $a_m(t)$ or $b_m(t)$. Then similar to the simple harmonic motion, the energies of the A and B pendulums are obtained as

$$E_1 = \frac{1}{2}m\omega_a^2 a_m^2(t) = 2ma^2\omega_a^2\cos^2\omega_m(t) \qquad ...(23)$$

and

$$E_2 = \frac{1}{2}m\omega_a^2 b_m^2(t) = 2ma^2\omega_a^2\sin^2\omega_m(t) \qquad ...(24)$$

Total energy of both pendulums

$$E = E_1 + E_2 = 2ma^2\omega_a^2 \qquad ...(25)$$

Thus, **the total energy E of both pendulums remains constant.** The energy difference of the pendulums is

$$E_1 - E_2 = 2ma^2\omega_a^2 (\cos^2 \omega_m t - \sin^2 \omega_m t) = E \cos 2\omega_m t = E \cos (\omega_2 - \omega_1)t \qquad ...(26)$$

Using eqs. (25) and (26), we get

$$E_1 = \frac{E}{2}[1 + \cos(\omega_2 - \omega_1)t] \text{ and } E_2 = \frac{E}{2}[1 - \cos(\omega_2 - \omega_1)t] \qquad ...(27)$$

At $t = 0$ *i.e.*, in the beginning, $E_1 = E$ and $E_2 = 0$, so that pendulum A possesses the total vibration energy. As the time t increases from zero, the energy of A pendulum decreases and that of B pendulum increases, keeping the total energy E of the system constant. When phase angle $(\omega_2 - \omega_1)\,t$ becomes $\pi/2$, the energy of two pendulums becomes equal (*i.e.*, $E/2$).

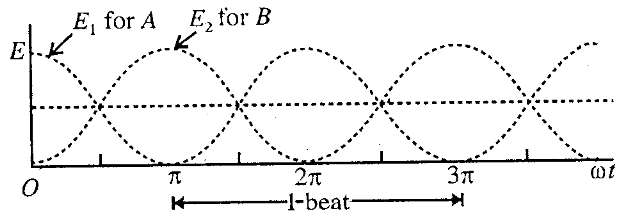


Fig. 9.6 : Energy of two coupled pendulums.

Further when $(\omega_2 - \omega_1)\,t = \pi$, $E_1 = 0$ and $E_2 = E$. At this instant, the pendulum B possesses the entire vibration energy. As the phase angle further increases from π to 2π, pendulum A acquires the total energy with no energy with B. The process goes on. Thus the total vibration energy flows back and forth between the two pendulums in a time interval T **[fig. 9.6]**, given by

$$(\omega_2 - \omega_1)\,T = 2\pi \text{ or } T = 2\pi/(\omega_2 - \omega_1) \qquad ...(28)$$

Thus, if the energy (or amplitude) of a pendulum is maximum at a certain instant, it will be maximum again after a time interval T, given by eq. (28). This time interval T is the **beat period.**

Theorem : (a) *Show that the expressions for the kinetic and potential energy of two coupled pendulums in ordinary coordinates are :*

$$K.E. = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 \text{ and } P.E. = \frac{1}{2}\frac{mg}{l}x_1^2 + \frac{1}{2}\frac{mg}{l}x_2^2 + \frac{1}{2}K(x_2 - x_1)^2$$

(b) *Using $q_1 = (x_1 + x_2)/2$ and $q_2 = (q_1 - x_2)/2$ for the definition of normal coordinates q_1 and q_2, obtain these expressions in the following form :*

$$K.E. = (\dot{q}_1^2 - \dot{q}_2^2) \text{ and } P.E. = m\omega_1^2 q_1^2 + m\omega_2^2 q_2^2$$

If the normal coordinates are defined as

$$q_1 = \sqrt{\frac{m}{2}}(x_1 + x_2) \text{ and } q_2 = \sqrt{\frac{m}{2}}(x_1 - x_2),$$

show that the expressions for K.E. and P.E. are obtained in the form :

$$K.E. = \frac{1}{2}\dot{q}_1^2 + \frac{1}{2}\dot{q}_2^2 = \text{ and } P.E. = \frac{1}{2}\omega_1^2 q_1^2 + \frac{1}{2}\omega_2^2 q_2^2.$$

Proof : (a) Let the system of two coupled pendulums be allowed to oscillate so that at any instant t, x_1 and x_2 are the displacements from the equilibrium positions O_1 and O_2 respectively **[fig. 9.7]**. If θ_1 and θ_2 be the angular displacements at this instant, then the potential energy of the system is given by

$$P.E. = mgl(1 - \cos\theta_1) + mgl(1 - \cos\theta_2) + \frac{1}{2}K(x_1 - x_2)^2$$

where the potential energy in the equilibrium configuration is assumed to be zero. For small amplitude oscillations,

$$1 - \cos\theta_1 = 1 - (1 - \theta_1^2/2) = \theta_1^2/2 = x_1^2/2l^2$$

and similarly $1 - \cos\theta_2 = x_2^2/2l^2$,

where $\qquad \theta_1 = x_1/l$ and $\theta_2 = x_2/l$.

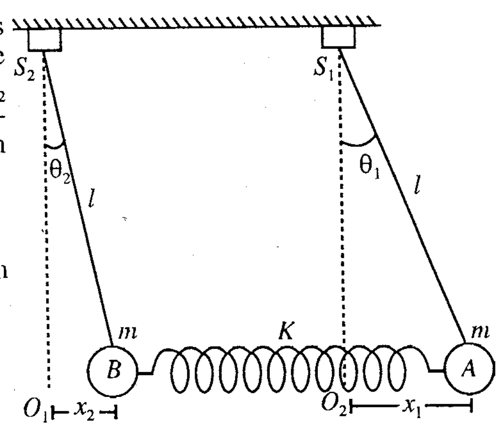


Fig. 9.7.

Thus, $$\text{P.E.} = \frac{1}{2}\frac{mg}{l}x_1^2 + \frac{1}{2}\frac{mg}{l}x_2^2 + \frac{1}{2}K(x_1 - x_2)^2 \quad \text{...(i)}$$

At the displacements x_1 and x_2, the speeds of A and B pendulums will be $dx_1/dt = x_1$ and $dx_2/dt = x_2$ respectively. Therefore, kinetic energy

$$\text{K.E.} = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 \quad \text{...(ii)}$$

(b) Now, $q_1 = (x_1 + x_2)/2$ and $q_2 = (x_1 - x_2)/2$

$\therefore$ $x_1 = q_1 + q_2$ and $x_2 = q_1 - q_2$;

so that $\dot{x}_1 = \dot{q}_1 + \dot{q}_2$ and $\dot{x}_2 = \dot{q}_1 - \dot{q}_2$

Now, $$\text{K.E.} = \frac{1}{2}m(\dot{q}_1 + \dot{q}_2)^2 + \frac{1}{2}m(\dot{q}_1 - \dot{q}_2)^2 = m(\dot{q}_1^2 + \dot{q}_2^2) \quad \text{...(iii)}$$

$$\text{P.E.} = \frac{1}{2}m\omega_0^2(\dot{x}_1^2 + x_2^2) + \frac{1}{2}m\omega_s^2(x_1 - x_2)^2$$
$$= m\omega_0^2(q_1^2 + q_2^2) + 2m\omega_s^2 q_2^2 = m\omega_1^2 q_1^2 + m\omega_2^2 q_2^2 \quad \text{...(iv)}$$

If the normal coordinates are defined as

$$q_1 = \sqrt{\frac{m}{2}}(x_1 + x_2) \quad \text{and } q_2 = \sqrt{\frac{m}{2}}(x_1 - x_2)$$

$$\therefore \quad x_1 = \frac{1}{\sqrt{2m}}(q_1 + q_2) \text{ and } x_2 = \frac{1}{\sqrt{2m}}(q_1 - q_2) \quad \text{...(v)}$$

Substituting (v) in (i) and (ii), we get

$$\text{K.E.} = \frac{1}{2}\dot{q}_1^2 + \frac{1}{2}\dot{q}_2^2 \quad \text{...(vi)}$$

$$\text{P.E.} = \frac{1}{2}\omega_1^2 q_1^2 + \frac{1}{2}\omega_2^2 q_2^2 \quad \text{...(vii)}$$

The advantage of defining the normal coordinates in the form (v) is that the P.E. does not contain any cross term in q_1 and q_2. Also K.E. and P.E. expressions do not contain explicitly any masses and stiffnesses, which are the properties of the individual parts of the system but not of the modes. Further the expressions for K.E. and P.E. look like the ordinary expressions of K.E. and P.E. of one coordinate oscillator on the removal of mass factor.

Ex. 1. *Two identical simple pendulums, each of length* 0.5 *m, are connected by a light spring between the two bobs. The mass of each bob is* 0.1 *kg and the force constant of the spring is* 2 *N-m. When one pendulum is clamped, find the period of other pendulum. When the clamp is removed, find the periods of two normal modes of the system.* ($g = 9.8\ ms^{-2}$).

Sol. Equation of motion of pendulum A (when B is clamped) [**fig. 9.1**] :

$$m\frac{d^2x}{dt^2} = -mg\frac{x}{l} - Kx \quad \text{or} \quad \frac{d^2x}{dt^2} + \left(\frac{g}{l} + \frac{K}{m}\right)x = 0$$

Therefore, period of A, $$T = \frac{2\pi}{\sqrt{\frac{g}{l} + \frac{K}{m}}} = \frac{2\pi}{\sqrt{\frac{9.8}{0.5} + \frac{2}{0.1}}} = \mathbf{0.99\ sec.}$$

For Mode 1, $$T_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{\sqrt{g/l}} = \frac{2\pi}{\sqrt{9.8/0.5}} = \mathbf{1.42\ sec.}$$

For Mode 2, $$T_2 = \frac{2\pi}{\omega_2} = \frac{2\pi}{\sqrt{\frac{g}{l} + \frac{2K}{m}}} = \frac{2\pi}{\sqrt{\frac{9.8}{0.5} + \frac{2\times 2}{0.1}}} = \mathbf{0.75\ sec.}$$

Ex. 2. *Obtain normal coordinate and ordinary coordinate solutions for a system of two coupled pendulums for the following initial conditions :*

(1) At $t = 0 : x_1 = a, \dfrac{dx_1}{dt} = 0, x_2 = a$ and $\dfrac{dx_2}{dt} = 0,$

(2) At $t = 0 : x_1 = a, \dfrac{dx_1}{dt} = 0, x_2 = -a$ and $\dfrac{dx_2}{dt} = 0,$

(3) At $t = 0 : x_1 = 0, \dfrac{dx_1}{dt} = 0, x_2 = 2a$ and $\dfrac{dx_2}{dt} = 0.$

Sol. $x_1 = q_1 + q_2 = a_1 \cos(\omega_1 t + \phi_1) + a_2 \cos(\omega_2 t + \phi_2)$ ...(i)

$x_2 = q_1 - q_2 = a_1 \cos(\omega_1 t + \phi_1) - a_2 \cos(\omega_2 t + \phi_2)$ ...(ii)

$$\frac{dx_1}{dt} = -a_1\omega_1 \sin(\omega_1 t + \phi_1) - a_2\omega_2 \sin(\omega_2 t + \phi_2) \quad ...(iii)$$

$$\frac{dx_2}{dt} = -a_1\omega_1 \sin(\omega_1 t + \phi_1) + a_2\omega_2 \sin(\omega_2 t + \phi_2) \quad ...(iv)$$

(1) Substituting for initial conditions, we get

$a = a_1 \cos\phi_1 + a_2 \cos\phi_2, \quad a = a_1 \cos\phi_1 - a_2 \cos\phi_2$

and $0 = a_1\omega_1 \sin\phi_1 + a_2\omega_2 \sin\phi_2, \; 0 = a_1\omega_1 \sin\phi_1 - a_2\omega_2 \sin\phi_2$

Therefore, $a_1 \cos\phi_1 = a, a_2 \cos\phi_2 = 0, a_1\omega_1 \sin\phi_1 = 0, a_2\omega_2 \sin\phi_2 = 0$

Since $a_1, a_2, \omega_1, \omega_2$ are not equal to zero; $\phi_1 = \phi_2 = 0, a_1 = a, a_2 = 0$

$\therefore \quad x_1 = a\cos\omega_1 t, x_2 = a\cos\omega_1 t$ and $q_1 = a\cos\omega_1 t, q_2 = 0$, where $\omega_1 = \sqrt{g/l}$.

In the given condition, the system is oscillating only in one mode q_1 and the pendulums are oscillating in the same phase.

(2) Substituting the initial conditions in eqs. (i) to (iv), we get $\phi_1 = 0, \phi_2 = 0, a_1 = 0, a_2 = -a$

$\therefore \quad x_1 = -a\cos\omega_2 t, x_2 = a\cos\omega_2 t$ and $q_1 = 0, q_2 = -a\cos\omega_2 t$, where $\omega_2 = \sqrt{\dfrac{g}{l} + \dfrac{2K}{m}}$.

The system is now oscillating only in mode q_2 and the pendulums are oscillating in the opposite phase.

(3) Here $\phi_1 = \phi_2 = 0, a_1 = a, a_2 = -a$

$\therefore \quad x_1 = a\cos\omega_1 t - a\cos\omega_2 t, x_2 = a\cos\omega_1 t + a\cos\omega_2 t$

and $q_1 = a\cos\omega_1 t, q_2 = -a\cos\omega_2 t$

In this case, the system is oscillating in both modes q_1 and q_2 simultaneously.

EXERCISE 9·1

1. Two identical simple pendulums are connected by a massless spring attached to the two bobs. Each bob has mass of 1 kg, the force constant of the spring being 0.8 N/m. When one pendulum is clamped, the period of other is found to be 1.25 s. Determine the periods of the two modes of the system with the clamp removed.
[**Ans.** 1.23 sec; 1.28 sec.]
2. Determine the normal mode frequencies of a pair of coupled pendulums, if the two pendulums are of different masses M and m with the same length l. Given that the pendulum of mass M is started oscillating with amplitude a, find the maximum amplitude of the other pendulum in the subsequent oscillations.
[**Ans.** $\omega_1^2 = \dfrac{g}{l}, \omega_2^2 = \dfrac{g}{l} + \dfrac{k}{M}; \dfrac{2Ma}{M+m}$.]
3. Two identical simple pendulums are coupled to each other by a spring. When both pendulums are in-phase and out-of-phase, the periods are found T_1 and T_2 respectively. In case, one pendulum is kept at rest initially and the other is allowed to oscillate, T_3 is found to be the time interval between two rest positions of either pendulum. Show that
$$\omega_3 = \omega_2 - \omega_1, \omega_n = 2\pi/T_n \, (n = 1, 2, 3).$$

9·3. General Analytical Approach for Mode Solutions of Two Coupled Oscillators

Consider two coupled equations in coordinates ψ_1 and ψ_2 in the form :

$$\frac{d^2\psi_1}{dt^2} + C_{11}\psi_1 + C_{12}\psi_2 = 0 \quad ...(1)$$

and

$$\frac{d^2\psi_2}{dt^2} + C_{21}\psi_1 + C_{22}\psi_2 = 0 \quad ...(2)$$

where C_{ij} (i = 1, 2 and j = 1, 2) are constant coefficients. Our interest is to know the solutions for normal modes, when C_{ij} are general constants, not necessary as in equations [eqs. (2a) and (2b), Art 9.2] of two coupled pendulums. Consider the system be oscillating in only one normal mode. It means that both the coordinates x_1 and x_2 execute simple harmonic motions with the same normal frequency and phase constant with different amplitudes, in general. Therefore, we take the solutions of the form :

$$\psi_1 = a \cos(\omega t + \phi) \text{ and } \psi_2 = b \cos(\omega t + \phi) \quad ...(3)$$

Differentiating ψ_1 and ψ_2 twice with respect to t, we obtain

$$\frac{d^2\psi_1}{dt^2} = -\omega^2 a \cos(\omega t + \phi) = -\omega^2\psi_1 \quad ...(4)$$

and

$$\frac{d^2\psi_2}{dt^2} = -\omega^2 b \cos(\omega t + \phi) = -\omega^2\psi_2 \quad ...(5)$$

Substituting these values in eqs. (1) and (2), we get

$$-\omega^2\psi_1 + C_{11}\psi_1 + C_{12}\psi_2 = 0$$

and

$$-\omega^2\psi_2 + C_{21}\psi_1 + C_{22}\psi_2 = 0$$

or

$$(C_{11} - \omega^2)\psi_1 + C_{12}\psi_2 = 0 \quad ...(6)$$

and

$$C_{21}\psi_1 + (C_{22} - \omega^2)\psi_2 = 0 \quad ...(7)$$

From eqs. (6) and (7), we have

$$\frac{\psi_2}{\psi_1} = -\frac{C_{11} - \omega^2}{C_{12}} \text{ and } \frac{\psi_2}{\psi_1} = -\frac{C_{21}}{C_{22} - \omega^2} \quad ...(8)$$

So that $(C_{11} - \omega^2)(C_{22} - \omega^2) - C_{12}C_{21} = 0$ or $\begin{vmatrix} C_{11} - \omega^2 & C_{12} \\ C_{21} & C_{22} - \omega^2 \end{vmatrix} = 0$ ...(9)

Eq. (9) can also be obtained directly from eqs. (6) and (7) because the determinant of coefficients of the linear homogeneous equations vanishes.

Eq. (9) is a quadratic equation in ω^2 and has two solutions, say $\omega^2 = \omega_1^2$ and $\omega^2 = \omega_2^2$. We consider only positive square root values, *i.e.*, ω_1 and ω_2, because negative square root values have no physical meaning. Thus, when the system is oscillating in a single mode, it can oscillate in two ways : (1) with frequency ω_1 in Mode 1, and (2) with frequency ω_2 in Mode 2. When we put $\omega^2 = \omega_1^2$ in eq. (8), we obtain the ratio of ψ_1 and ψ_2 coordinates for Mode 1, *i.e.*,

$$\left(\frac{\psi_2}{\psi_1}\right)_{\text{Mode 1}} = \left(\frac{b}{a}\right)_{\text{Mode 1}} = \frac{b_1}{a_1} = \frac{\omega_1^2 - C_{11}}{C_{12}} \quad ...(10)$$

Similarly for Mode 2, we have

$$\left(\frac{\psi_2}{\psi_1}\right)_{\text{Mode 2}} = \left(\frac{b}{a}\right)_{\text{Mode 2}} = \frac{b_2}{a_2} = \frac{\omega_2^2 - C_{11}}{C_{12}} \quad ...(11)$$

where a_1, b_1 are the amplitudes of two oscillators in Mode 1 and a_2, b_2 in Mode 2. Thus in view of eqs. (3), we can write the two coordinates ψ_1 and ψ_2 as follows :

In Mode 1 : $\psi_1 = a_1 \cos(\omega_1 t + \phi_1)$ and $\psi_2 = b_1 \cos(\omega_1 t + \phi_1)$ ...(12)

In Mode 2 : $\psi_1 = a_2 \cos(\omega_2 t + \phi_2)$ and $\psi_2 = b_2 \cos(\omega_2 t + \phi_2)$ ...(13)

The differential equations are linear and hence the sum (superposition) of two solutions is also a solution, *i.e.*,

$$\psi_1 = a_1 \cos(\omega_1 t + \phi_1) + a_2 \cos(\omega_2 t + \phi_2) \quad \text{...(14)}$$

$$\psi_2 = b_1 \cos(\omega_1 t + \phi_1) + b_2 \cos(\omega_2 t + \phi_2) \quad \text{...(15)}$$

The coordinates ψ_1 and ψ_2 in eqs. (14) and (15) express the general motion of the system and are obtained as superposition of the two normal modes. The constant a_1, a_2, ϕ_1 and ϕ_2 in eq. (14) can be obtained by applying initial conditions. The constants in eq. (15) are automatically fixed because ψ_1, ψ_2 have been already determined and b_1, b_2 are known in view of eqs. (10) and (11).

9·4. Normal Mode Frequencies and Solutions for the System of Two Coupled Pendulums : Using General Approach

First establish the differential equations for a system of two coupled pendulums [eqs. (2) of Art. 9.2], given by

$$\frac{d^2x_1}{dt^2} + (\omega_0^2 + \omega_s^2)x_1 - \omega_s^2 x_2 = 0 \quad \text{...(1)}$$

and
$$\frac{d^2x_2}{dt^2} - \omega_s^2 x_1 + (\omega_0^2 + \omega_s^2)x_2 = 0 \quad \text{...(2)}$$

where $\omega_0^2 = g/l$ and $\omega_s^2 = K/m$.

Here $C_{11} = \omega_0^2 + \omega_s^2$, $C_{12} = C_{21} = -\omega_s^2$ and $C_{22} = \omega_0^2 + \omega_s^2$.

Let the system be vibrating in one mode with solutions of the form :

$$x_1 = a \cos(\omega t + \phi) \text{ and } x_2 = b \cos(\omega t + \phi) \quad \text{...(3)}$$

The mode frequencies are obtained from the equation

$$\begin{vmatrix} C_{11}-\omega^2 & C_{12} \\ C_{21} & C_{22}-\omega^2 \end{vmatrix} = 0 \quad \text{...(4)}$$

Substituting for C_{11}, C_{12}, C_{21} and C_{22} in eq. (4), we get

$$\begin{vmatrix} \omega_0^2 + \omega_s^2 - \omega^2 & -\omega_s^2 \\ -\omega_s^2 & \omega_0^2 + \omega_s^2 - \omega^2 \end{vmatrix} = 0 \text{ or } (\omega_0^2 + \omega_s^2 - \omega^2)^2 - \omega_s^4 = 0$$

So that
$$\omega^2 = \omega_0^2 = \frac{g}{l} = \omega_1^2 \text{ and } \omega^2 = \omega_0^2 + 2\omega_s^2 = \frac{g}{l} + \frac{2K}{m} = \omega_2^2$$

i.e.,
$$\omega_1 = \sqrt{\frac{g}{l}} \text{ and } \omega_2 = \sqrt{\frac{g}{l} + \frac{2K}{m}} \quad \text{...(5)}$$

These are the **frequencies of the two normal modes** of vibration of the two coupled pendulums.

For Mode 1, $\omega_1^2 = \omega_0^2$ and hence

$$\frac{b_1}{a_1} = \frac{\omega_1^2 - C_{11}}{C_{12}} = \frac{\omega_0^2 - \omega_0^2 - \omega_s^2}{-\omega_s^2} = 1 \text{ i.e., } b_1 = a_1 \quad \text{...(6)}$$

For Mode 2,
$$\omega_2^2 = \omega_0^2 + 2\omega_s^2$$

and hence
$$\frac{b_2}{a_2} = \frac{\omega_2^2 - C_{11}}{C_{12}} = \frac{\omega_0^2 + 2\omega_s^2 - \omega_0^2 - \omega_s^2}{-\omega_s^2} = -1 \text{ i.e., } b_2 = -a_2 \quad \text{...(7)}$$

Thus, is Mode 1, $x_1 = a_1 \cos(\omega_1 t + \phi_1)$ and $x_2 = a_1 \cos(\omega_1 t + \phi_1)$ or $x_1 = x_2$ ...(8)

and in Mode 2, $x_1 = a_2 \cos(\omega_2 t + \phi_2)$ and $x_2 = -a_2 \cos(\omega_2 t + \phi_2)$ or $x_1 = -x_2$ ...(9)

Oscillations of two coupled pendulums in two modes are shown in **fig. 9.5**.

Now, the general motion can be expressed as

$$x_1 = a_1 \cos(\omega_1 t + \phi_1) + a_2 \cos(\omega_2 t + \phi_2) \quad \text{...(10)}$$

and
$$x_2 = a_1 \cos(\omega_1 t + \phi_1) - a_2 \cos(\omega_2 t + \phi_2) \quad \text{...(11)}$$

which are the equations (14) and (15) for a system of coupled pendulums.

9·5. Longitudinal Oscillations of Two Coupled Masses

We consider two identical masses m, connected to each other by a massless spring of force constant K and to two fixed walls by two massless springs of force constant K' [**fig. 9.8**]. In **fig 9.8(a)**, the system is in equilibrium and we assume that in this position, the springs do not exert forces. Let the system be displaced from equilibrium position and allowed to execute longitudinal oscilla-tions [**fig. 9.8(b)**]. If x_1 and x_2 be the displacements of 1st and 2nd masses, then the equations of motion of the two masses are :

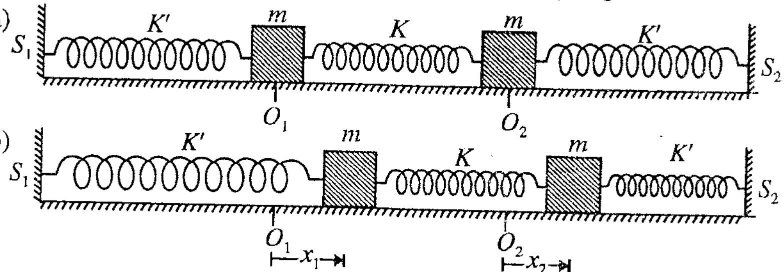


Fig. 9.8 : Longitudinal oscillations of two coupled masses : (a) system in equilibrium, (b) system in oscillation.

$$m\frac{d^2x_1}{dt^2} = -K'x_1 - K(x_1 - x_2) \quad ...(1)$$

$$m\frac{d^2x_2}{dt^2} = -K'x_2 + K(x_1 - x_2) \quad ...(2)$$

These equations can be written as

$$\frac{d^2x_1}{dt^2} + (\omega_0^2 + \omega_s^2)x_1 - \omega_s^2 x_2 = 0 \quad ...(3)$$

$$\frac{d^2x_2}{dt^2} + \omega_s^2 x_1 + (\omega_0^2 + \omega_s^2)x_2 = 0 \quad ...(4)$$

where $\omega_0^2 = K'/m$ and $\omega_s^2 = K/m$.

Eqs. (3) and (4) are identical to eqs. (1) and (2) of general approach *i.e.*,

$$\frac{d^2\psi_1}{dt^2} + C_{11}\psi_1 + C_{12}\psi_2 = 0 \text{ and } \frac{d^2\psi_2}{dt^2} + C_{21}\psi_1 + C_{22}\psi_2 = 0$$

So that here, $C_{11} = \omega_0^2 + \omega_s^2$, $C_{12} = C_{21} = -\omega_s^2$ and $C_{22} = (\omega_0^2 + \omega_s^2)$

If the system is vibrating in one mode with solutions of the form

$$x_1 = a\cos(\omega t + \phi) \text{ and } x_2 = b\cos(\omega_2 t + \phi) \quad ...(5)$$

Then the normal mode frequencies are obtained from the equation

$$\begin{vmatrix} C_{11} - \omega^2 & C_{12} \\ C_{21} & C_{22} - \omega^2 \end{vmatrix} = 0 \text{ or } \begin{vmatrix} \omega_0^2 + \omega_s^2 - \omega^2 & -\omega_s^2 \\ -\omega_s^2 & \omega_0^2 + \omega_s^2 - \omega^2 \end{vmatrix} = 0$$

or $$(\omega_0^2 + \omega_s^2 - \omega^2)^2 - \omega_s^4 = 0$$

whence $\omega^2 = \omega_0^2 = \omega_1^2$ (say) and $\omega^2 = \omega_0^2 + 2\omega_s^2 = \omega_2^2$ (say)

Thus $$\omega_1 = \omega_0 = \sqrt{\frac{K'}{m}} \text{ and } \omega_2 = \sqrt{\omega_0^2 + 2\omega_s^2} = \sqrt{\frac{K'}{m} + \frac{2K}{m}} \quad ...(6)$$

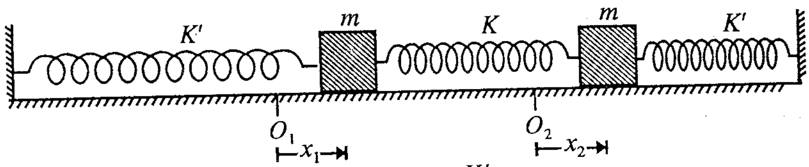
These are the two normal mode frequencies of the two coupled masses.

In Mode 1, $$\frac{b_1}{a_1} = \frac{\omega_1^2 - C_{11}}{C_{12}} = \frac{\omega_0^2 - \omega_0^2 - \omega_s^2}{-\omega_s^2} = 1 \; i.e.,\; b_1 = a_1 \quad ...(7)$$

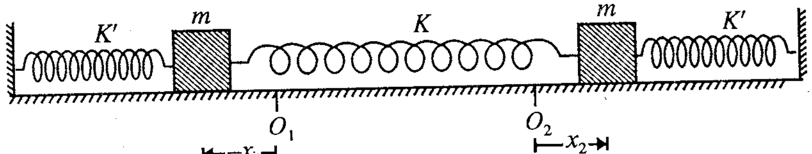
and Mode 2, $$\frac{b_2}{a_2} = \frac{\omega_2^2 - C_{11}}{C_{12}} = \frac{\omega_0^2 + 2\omega_s^2 - \omega_0^2 - \omega_s^2}{-\omega_s^2} = -1 \; i.e.,\; b_2 = -a_2 \quad ...(8)$$

Thus in Mode 1, $x_1 = a_1 \cos(\omega_1 t + \phi_1)$ and $x_2 = a_1 \cos(\omega_1 t + \phi_1)$ or $x_1 = x_2$ ...(9)

and in Mode 2, $x_1 = a_2 \cos(\omega_2 t + \phi_2)$ and $x_2 = -a_2 \cos(\omega_2 t + \phi_2)$ or $x_1 = -x_2$ ...(10)



(a) Mode 1 : $x_2 = x_1$, $\omega_1^2 = \dfrac{K'}{m}$ (Lower Frequency)



(b) Mode 2 : $x_2 = -x_1$, $\omega_2^2 = \dfrac{K'}{m} + \dfrac{2K}{m}$ (Higher Frequency)

Fig. 9.9 : Normal modes of two coupled masses.

When the system oscillates only in Mode 1, $x_1 = x_2$ and when only in Mode 2, $x_1 = -x_2$. Mode 1 with lower frequency and Mode 2 with higher frequency are shown in **fig. 9.9(a)** and **fig. 9.9(b)** respectively. The general motion of the system of two coupled masses is obtained by the superposition of the two modes *i.e.*,

$$x_1 = a_1 \cos(\omega_1 t + \phi_1) + a_2 \cos(\omega_2 t + \phi_2) \quad \text{...(11)}$$

and

$$x_2 = a_1 \cos(\omega_1 t + \phi_1) - a_2 \cos(\omega_2 t + \phi_2) \quad \text{...(12)}$$

9·6. Transverse Oscillations of a String with Two Beads

We condider two beads of equal masses (m) are attached to a flexible light elastic string, fixed at its two ends. The fixed ends are at $x = 0$ and $x = 3d$ and the two beads lie at $x = d$ and $x = 2d$ [**fig. 9·10**]. The tension in the string is T_0 and remains the same at all the times. We consider transervse oscillations of the system in the plane of the paper. If at any time t, the displacements of two masses are y_1 and y_2 from their equilibrium positions respectively, the restoring forces acting on the two masses are given by

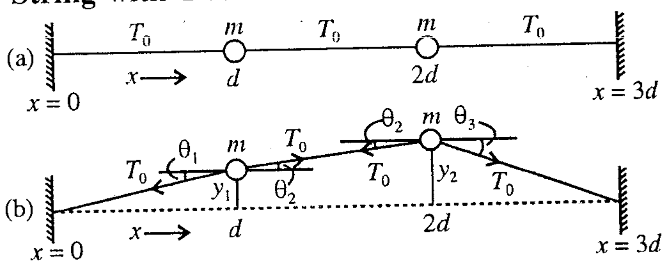


Fig. 9.10 : Transverse oscillations of two beads on a light string : (a) system in equilibrium, (b) system in oscillation with instantaneous displacements y_1 and y_2.

$$F_1 = -T_0 \sin\theta_1 + T_0 \sin\theta_2 \text{ and } F_2 = -T_0 \sin\theta_2 - T_0 \sin\theta_3$$

For small displacements y_1 and y_2,

$$\sin\theta_1 = \frac{y_1}{d}, \qquad \sin\theta_2 = \frac{y_2 - y_1}{d} \text{ and } \sin\theta_3 = \frac{y_2}{d}.$$

because for small θ, $\theta = \sin\theta = \tan\theta$.

Writing $F_1 = md^2y_1/dt^2$ and $F_2 = md^2y_2/dt^2$, the equation of motion of two masses are :

$$m\frac{d^2y_1}{dt^2} + \frac{T_0}{d}y_1 - \frac{T_0}{d}(y_2 - y_1) = 0 \quad \text{...(1)}$$

and

$$m\frac{d^2y_2}{dt^2} + \frac{T_0}{d}y_2 + \frac{T_0}{d}(y_2 - y_1) = 0 \quad \text{...(2)}$$

or

$$\frac{d^2y_1}{dt^2} + 2\omega_0^2 y_1 - \omega_0^2 y_2 = 0 \quad \text{...(3)}$$

$$\frac{d^2 y_2}{dt^2} - \omega_0^2 y_1 + 2\omega_0^2 y_2 = 0 \qquad ...(4)$$

where $\omega_0^2 = T_0/md$.

Eqs. (3) and (4) are identical to eqs. (1) and (2) of general approach *i.e.*,

$$\frac{d^2 y_1}{dt^2} + C_{11}\psi_1 + C_{12}\psi_2 = 0 \text{ and } \frac{d^2 y_2}{dt^2} + C_{21}\psi_1 + C_{22}\psi_2 = 0$$

So that here $C_{11} = C_{22} = 2\omega_0^2$ and $C_{12} = C_{21} = -\omega_0^2$.

If the system is vibrating in one mode with solutions of the form :

$$x_1 = a\cos(\omega_1 t + \phi) \text{ and } x_2 = b\cos(\omega_2 t + \phi) \qquad ...(5)$$

then the normal mode frequencies are obtained from the equation

$$\begin{vmatrix} C_{11} - \omega^2 & C_{12} \\ C_{21} & C_{22} - \omega^2 \end{vmatrix} = 0 \text{ or } \begin{vmatrix} 2\omega_0^2 - \omega^2 & -\omega_0^2 \\ -\omega_0^2 & 2\omega_0^2 - \omega^2 \end{vmatrix} = 0$$

or $\quad (2\omega_0^2 - \omega^2)^2 - \omega_0^4 = 0 \quad \text{or} \quad 2\omega_0^2 - \omega^2 = \pm\,\omega_0^2$

Thus $\quad \omega^2 = \omega_0^2 = \omega_1^2 \text{ (say) and } \quad \omega^2 = 3\omega_0^2 = \omega_2^2 \text{ (say)}$

i.e., $\quad \omega_1 = \omega_0 = \sqrt{T_0/md} \text{ and } \omega_2 = \omega_0\sqrt{3} = \sqrt{3T_0/md} \qquad ...(6)$

These are the **two normal mode frequencies** of the coupled system.

Thus $\omega_2/\omega_1 = \sqrt{3}$ *i.e., the* **normal modes of vibration have the frequency ratio** $\sqrt{3}:1$.

Also in Mode 1, $\dfrac{b_1}{a_1} = \dfrac{\omega_1^2 - C_{11}}{C_{12}} = \dfrac{\omega_0^2 - 2\omega_0^2}{-\omega_0^2} = 1$ *i.e.*, $b_1 = a_1$ $\qquad$ ...(7)

and in Mode 2, $\dfrac{b_2}{a_2} = \dfrac{\omega_2^2 - C_{11}}{C_{12}} = \dfrac{3\omega_0^2 - 2\omega_0^2}{-\omega_0^2} = -1$ *i.e.*, $b_2 = -a_2$ $\qquad$ ...(8)

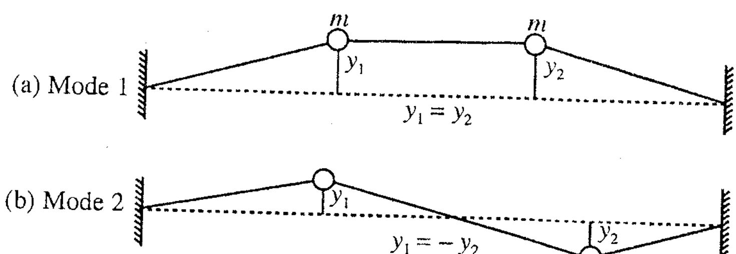


Fig. 9.11 : Transverse oscillations of two beads on string : (a) Mode 1; (b) Mode 2.

Thus in Mode 1, $x_1 = a_1\cos(\omega_1 t + \phi_1)$ and $x_2 = a_1\cos(\omega_1 t + \phi_1)$ *i.e.*, $x_1 = x_2$ $\qquad$ ...(9)

and in Mode 2, $x_1 = a_2\cos(\omega_2 t + \phi_2)$ and $x_2 = -a_2\cos(\omega_2 t + \phi_2)$ *i.e.*, $x_1 = -x_2$ $\qquad$ ...(10)

The general motion is obtained by the superposition of the two modes *i.e.*,

$$x_1 = a_1\cos(\omega_1 t + \phi_1) + a_2\cos(\omega_2 t + \phi_2)$$

and

$$x_2 = a_1\cos(\omega_2 t + \phi_1) - a_2\cos(\omega_2 t + \phi_2) \qquad ...(11)$$

9·7. Two Inductively Coupled *LC* Circuits

We consider the inductance and capacitance of one circuit to be L_1 and C_1 and those of another circuit L_2 and C_2 respectively. If M be the mutual inductance, then the potential difference across C_1 and C_2 are given by

$$\frac{Q_1}{C_1} = -L_1\frac{dI_1}{dt} + M\frac{dI_2}{dt} \text{ and } \frac{Q_2}{C_2} = -L_2\frac{dI_2}{dt} + M\frac{dI_1}{dt} \qquad ...(1)$$

where dI_1/dt and dI_2/dt are the rate of change of currents in two circuits, Q_1 and Q_2 are instantaneous charges on the capacitors, so that $I_1 = dQ_1/dt$ and $I_2 = dQ_2/dt$. Therefore the equations can be written as

$$L_1 \frac{d^2Q_1}{dt^2} + \frac{Q_1}{C_1} = M\frac{d^2Q_2}{dt^2}$$

and
$$L_2 \frac{d^2Q_2}{dt^2} + \frac{Q_2}{C_2} = M\frac{d^2Q_1}{dt^2} \qquad ...(2)$$

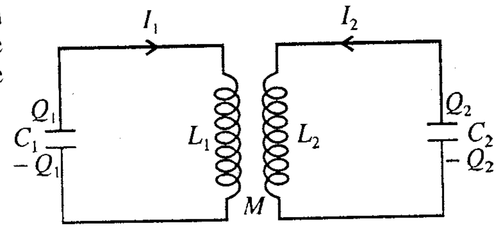


Fig. 9.12 : Coupled L-C Circuit.

In order to obtain the normal modes of charge oscillations, we assume that there exists a normal mode of angular frequency ω and phase constant ϕ. Therefore,

$$Q_1 = a\cos(\omega t + \phi) \text{ and } Q_2 = b\cos(\omega t + \phi) \qquad ...(3)$$

Hence,
$$\frac{d^2Q_1}{dt^2} = -\omega^2 a_1 \cos(\omega + \phi) = -\omega^2 Q_1 \text{ and } \frac{d^2Q_2}{dt^2} = -\omega^2 Q_2 \qquad ...(4)$$

Substituting in the differential equations (2), we obtain

$$-\omega^2 Q_1 + \frac{Q_1}{L_1C_1} = -\frac{\omega^2 M}{L_1}Q_2 \text{ and } -\omega^2 Q_2 + \frac{Q_2}{L_2C_2} = -\frac{\omega^2 M}{L_2}Q_1 \qquad ...(5)$$

Writing
$$\omega_p^2 = \frac{1}{L_1C_1} \text{ and } \omega_s^2 = \frac{1}{L_2C_2}, \text{ we get}$$

$$(\omega_p^2 - \omega^2)Q_1 = -\frac{\omega^2 M}{L_1}Q_2 \text{ or } \frac{Q_1}{Q_2} = -\frac{\omega^2}{\omega_p^2 - \omega^2}\frac{M}{L_1} \qquad ...(6)$$

and
$$(\omega_s^2 - \omega^2)Q_2 = -\frac{\omega^2 M}{L_2}Q_1 \text{ or } \frac{Q_1}{Q_2} = -\frac{\omega_s^2 - \omega^2}{\omega^2}\frac{L_2}{M} \qquad ...(7)$$

Equating (6) and (7), we obtain

$$\frac{\omega^2}{\omega_p^2 - \omega^2}\frac{M}{L_1} = \frac{\omega_s^2 - \omega^2}{\omega^2}\frac{L_2}{M} \text{ or } (\omega_p^2 - \omega^2)(\omega_s^2 - \omega^2) = \frac{M^2\omega^4}{L_1L_2} \qquad ...(8)$$

Coupling coefficient for the two circuits is defined as $\mu = \dfrac{M}{\sqrt{L_1L_2}}$, *i.e.*, $\mu^2 = \dfrac{M^2}{L_1L_2}$ ...(9)

We assume the two circuits to be identical. Therefore,

$$\omega_p^2 = \omega_s^2 = \frac{1}{LC} = \omega_0^2 \text{(say)} \; (L_1 = L_2,\, C_1 = C_2) \qquad ...(10)$$

Hence, eq. (8) assumes the form

$$(\omega_0^2 - \omega^2)^2 = \mu^2\omega^4 \text{ or } \omega_0^2 - \omega^2 = \pm\mu\omega^2 \text{ or } \omega^2(1 \pm \mu) = \omega_0^2$$

Thus,
$$\omega = \pm\frac{\omega_0}{\sqrt{1 \pm \mu}} \qquad ...(11)$$

The acceptable normal mode frequencies are positive, hence

$$\omega_1 = \frac{\omega_0}{\sqrt{1+\mu}} \text{ and } \omega_2 = \frac{\omega_0}{\sqrt{1-\mu}} \qquad ...(12)$$

These are the angular frequencies of the normal modes of two identical LC circuits coupled inductively. For weak coupling ($\mu << 1$), $\omega_1 = \omega_2 = \omega_0 = 1/\sqrt{LC}$ and the two circuits behave independently. In case of strong coupling, the two normal mode frequencies will be different.

Ex. 1 *Two equal masses (m) are connected to each other with the help of a spring of force constant K and then the upper mass is connected to a rigid support by an identical spring as shown in* **fig. 9.13** *The system is is allowed to oscillate in the vertical direction. Show that the frequencies of two normal* *modes are* $(3\pm\sqrt{5})K/2m$ *and the ratios of the amplitudes of two masses in the two modes are* $(\frac{1}{2}\sqrt{5}\pm1)$.

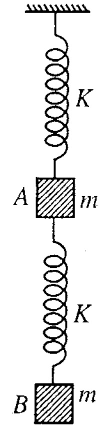


Fig. 9.13.

Sol. Equations of motion of masses A and B are :

$$m\frac{d^2x_1}{dt^2} = -K(x_1-x_2) \text{ and } m\frac{d^2x_2}{dt^2} = -Kx_2-K(x_2-x_1)$$

or $$\frac{d^2x_1}{dt^2}+\frac{K}{m}x_1-\frac{K}{m}x_2 = 0 \text{ and } \frac{d^2x_2}{dt^2}-\frac{K}{m}x_1+\frac{2K}{m}x_2 = 0$$

Using eq. (9) of Art 9.6 the normal mode frequencies are given by

$$\begin{vmatrix} \frac{K}{m}-\omega^2 & -\frac{K}{m} \\ -\frac{K}{m} & \frac{2K}{m}-\omega^2 \end{vmatrix} = 0 \text{ or } \omega^4-\frac{3K}{m}\omega^2+\frac{K^2}{m^2} = 0$$

Therefore, $$\omega^2 = \frac{3K}{2m}\pm\frac{K\sqrt{5}}{2m}.$$

Thus, $\omega_1^2 = (3-\sqrt{5})\frac{K}{2m}$ for slower mode and $\omega_2^2 = (3+\sqrt{5})\frac{K}{2m}$ for faster mode.

Amplitude ratio for the slower frequency mode,

$$\frac{b_1}{a_1} = \frac{\omega_1^2-\frac{K}{m}}{-K/m} = \frac{(3-\sqrt{5})\frac{K}{2m}-\frac{K}{m}}{-K/m} = \frac{1}{2}(\sqrt{5}-1)$$

Amplitude ratio for the faster frequency mode,

$$\frac{b_2}{a_2} = \frac{\omega_2^2-\frac{K}{m}}{-K/m} = \frac{(3+\sqrt{5})\frac{K}{2m}}{-K/m} = -\frac{1}{2}(\sqrt{5}+1).$$

Thus, in the faster mode the two masses are vibrating oppositely with the amplitude ratio $\frac{1}{2}(\sqrt{5}+1)$.

Ex. 2 *A system of two coupled oscillators is shown if* **fig. 9.14.** *Find expressions for normal mode frequencies. If the force constant of the middle spring is the geometric mean of the side springs, what are the modified expressions for the normal mode frequencies ?*

Sol. Equations of motion of masess are :

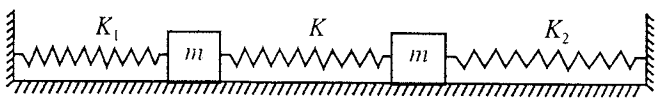


Fig. 9.14.

$$m\frac{d^2x_1}{dt^2} = -K_1x_1-K(x_1-x_2)$$

and $$m\frac{d^2x_2}{dt^2} = -K_2x_2+K(x_1-x_2)$$

or $$\frac{d^2x_1}{dt^2}+\left(\frac{K_1+K}{m}\right)x_1-\frac{K}{m}x_2 = 0 \text{ and } \frac{d^2x_2}{dt^2}-\frac{K}{m}x_1+\left(\frac{K_2+K}{m}\right)x_2 = 0$$

Normal mode frequencies (ω) are given by

$$\begin{vmatrix} \frac{K_1+K}{m}-\omega^2 & -\frac{K}{m} \\ -\frac{K}{m} & \frac{K_2+K}{m}-\omega^2 \end{vmatrix} = 0 \text{ or } (K_1+K-m\omega^2)(K_2+K-m\omega^2)-K^2 = 0$$

$$m^2\omega^4-m\omega^2(2K+K_1+K_2)+(K_1+K)(K_2+K)-K^2 = 0$$

$$\therefore \quad \omega^2 = \frac{m(2K+K_1+K_2)\pm m\sqrt{(2K+K_1+K_2)^2-4(K_1K_2+KK_1+KK_2)}}{2m^2}$$

or
$$\omega^2 = \frac{2K + K_1 + K_2 \pm \sqrt{4K^2 + (K_1 - K_2)^2}}{2m}$$

Thus,
$$\omega^2 = \frac{1}{m}\left(K + \frac{K_1 + K_2}{2}\right) \pm \frac{1}{m}\sqrt{K^2 + \left(\frac{K_1 - K_2}{2}\right)^2}$$

Hence,
$$\omega_1^2 = \frac{1}{m}\left(K + \frac{K_1 + K_2}{2}\right) + \frac{1}{m}\sqrt{K^2 + \left(\frac{K_1 - K_2}{2}\right)^2}$$

and
$$\omega_2^2 = \frac{1}{m}\left(K + \frac{K_1 + K_2}{2}\right) - \frac{1}{m}\sqrt{K^2 + \left(\frac{K_1 - K_2}{2}\right)^2}$$

where ω_1 and ω_2 are the frequencies of normal modes.

When $K^2 = K_1K_2$, the modified expressions are :

$$\omega_1^2 = \frac{1}{m}(K + K_1 + K_2),\ \omega_2^2 = \frac{K}{m}.$$

Ex. 3 *Two inductively coupled identical circuits, each having a natural frequency* 600 *Hz, have coupling coefficient* 0.44. *Determine the two normal mode frequencies.*

(Rewa 2004; Gwalior 03; Sagar 03)

Sol.
$$\omega_1 = 2\pi\nu_1 = \frac{\omega_0}{\sqrt{1+\mu}} = \frac{2\pi\nu_0}{\sqrt{1+\mu}}$$

or
$$\nu_1 = \frac{\nu_0}{\sqrt{1+\mu}} = \frac{600}{\sqrt{1+0.44}} = \textbf{500 Hz.}$$

Also
$$\nu_2 = \frac{\nu_0}{\sqrt{1-\mu}} = \frac{600}{\sqrt{1-0.44}} = \textbf{802 Hz.}$$

EXERCISE 9·2

1. If an object of mass m is suspended to a rigid support with the help of a spring of force constant K, it vibrates with a frequency 2 Hz. Now two indentical objects A and B, each of mass m, are joined together by a spring of force constant K' and then they are connected to rigid supports S_1 and S_2 by two indentical springs, each of force constant K [fig 9.15]. Next, if A is clamped, B vibrates with a frequency 2.5 Hz. Calculate the frequencies of the two modes of vibration. Find also the ratio K'/K of the two force constants.

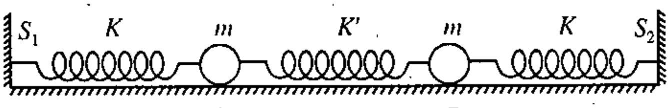


Fig. 9.15.

[**Ans.** 2 Hz, 2.92 Hz; 9/16.]

2. Two indentical masses are connected with two indentical massless springs and placed on a horizontal frictionless surface as shown in **fig. 9.16.** Considering only motion along the line joining their centres, find the ratio of the normal mode frequencies of the two masses.

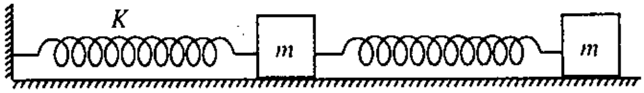


Fig. 9.16.

[**Ans.** $\left(\frac{3+\sqrt{5}}{3-\sqrt{5}}\right)^{1/2}$.]

3. Two identical LC circuits each having a natural frequency of 500 Hz are coupled inductively in such

a way that their coupling coefficient is 0.5. Find the frequencies of the two normal modes of the coupled circuits.
[**Ans.** 408 Hz, 707 Hz.]

9·8. *N*-Coupied Oscillators (Transverse Oscillations)

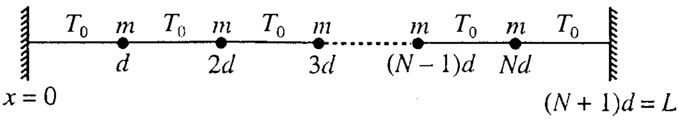


Fig. 9.17(a) : System of *N* beads on a light string in equilibrium.

We have so far discussed the coupled oscillations of two-body systems with two degrees of freedom. Such a system can oscillate in two normal modes. Now the analysis of two coupled oscillators is extended to investigate the motion of an array of *N*-copuled oscillators along one dimension. It will be seen that such a system with *N* degrees of freedom can vibrate in *N* independent normal modes.

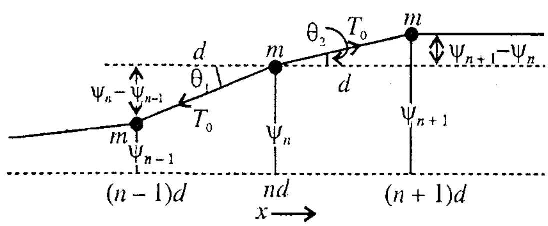


Fig. 9.17(b) : Transrevse oscillations of a beaded string.

Consider a flexible light elastic string to which *N* identical beads, each of mass *m*, are attached at equal distance. The string is fixed at $x = 0$ and $x = L$ and the beads are situated at $x = d$, $2d,\ldots,Nd$ so that $L = (N + 1)d$ [**fig 9.17(a)**]. The tension in the string is T_0, present at all points of the string at all times. Let us consider only transverse oscillations of the system, confined to the plane of the paper. We denote the displacement of the *n*th mass from its equilibrium position at any time *t* by $\psi_n(t)$. In any general oscillation of the system [**fig. 9.17(b)**], suppose the displacements of the nearest neighbouring beads, situated at $x = (n-1)d$ and $x = (n+1)d$, at the instant *t* are $\psi_{n-1}(t)$ and $\psi_{n+1}(t)$ respectively. The force F_n on the *n*th mass towards the equilibrium position is given by

$$F_n = -T_0 \sin\theta_1 + T_0 \sin\theta_2$$

If the displacements ψ_n are small in comparison to the interparticle distance *d*, then

$$\sin\theta_1 = \frac{\psi_n - \psi_{n-1}}{d} \text{ and } \sin\theta_2 = \frac{\psi_{n+1} - \psi_n}{d}$$

because for small θ, θ = tan θ = sin θ.

Hence, $$F_n = -T_0\left(\frac{\psi_n - \psi_{n-1}}{d}\right) + T_0\left(\frac{\psi_{n+1} - \psi_n}{d}\right) \quad \ldots(1)$$

If $d^2\psi_n/dt^2$ is the acceleration of the *n*th mass, then

$$F_n = m\frac{d^2\psi_n}{dt^2}$$

and hence the equation of motion of the *n*th mass is

$$m\frac{d^2\psi_n}{dt^2} = -\frac{T_0}{d}(2\psi_n - \psi_{n-1} - \psi_{n+1}) \quad \ldots(2)$$

$$\frac{d^2\psi_n}{dt^2} + 2\omega_0^2\psi_n - \omega_0^2(\psi_{n-1} + \psi_{n+1}) = 0 \quad \ldots(3)$$

where we have put $\omega_0^2 = T_0/md$. For $n = 1, 2, 3, \ldots, N$, there are *N* differential equations of second order.

Normal mode of *N*-beaded string : We consider that the system is oscillating in a single mode so that each bead is executing a simple harmonic motion of the same frequency ω and phase constant ϕ. Thus the displacements of various masses in this mode of vibration can be represented as

$$\psi_1 = a_1 \cos(\omega t+\phi);\psi_2 = a_2 \cos(\omega t+\phi);\ \psi_3 = a_3\cos(\omega t+\phi);\ \ldots\ldots;\psi_{n-1} = a_{n-1}\cos(\omega t+\phi);$$

$$\psi_n = a_n \cos(\omega t+\phi);\ \psi_{n+1} = a_{n+1}\cos(\omega t+\phi);\ldots;\ \psi_N = a_N\cos(\omega t+\phi). \qquad ...(4)$$

Substituting for ψ_{n-1}, ψ_n and ψ_{n+1} in eq. (3), we obtain

$$-a_n\omega^2\cos(\omega t+\phi)+2\omega_0^2 a_n\cos(\omega t+\phi)-\omega_0^2[a_{n-1}\cos(\omega t+\phi)+a_{n+1}\cos(\omega t+\phi)] = 0 \qquad ...(5)$$

In general $\cos(\omega t+\phi)$ is not zero. Dividing eq. (5) by $\cos(\omega t+\phi)$, we get.

$$(-\omega^2+2\omega_0^2)a_n - \omega_0^2(a_{n-1}+a_{n+1}) = 0 \qquad ...(6)$$

or

$$\frac{a_{n-1}+a_{n+1}}{a_n} = \frac{-\omega^2+2\omega_0^2}{\omega_0^2} \qquad ...(7)$$

For a particluar value of ω, the right hand side of eq. (7) is constant and consequently the ratio of left hand side of eq. (7) must be constant and independent of n. At the ends of the string $a_0 = a_{N+1} = 0$. Therefore, we want to find an expression for a_n so that eq. (7) is satisfied with the boundary conditions at the ends of the string. We assume that the amplitude of the nth mass, situated at a distance nd from the fixed end, is represented as

$$a_0 = a \sin knd \qquad ...(8)$$

where a and k are constants for the mode under consideration. Thus the value of left hand side of eq. (7) in view of eq. (8) is

$$\frac{a_{n-1}+a_{n+1}}{a_n} = \frac{\sin k(n-1)d+\sin k(n+1)d}{\sin knd} = \frac{2\sin knd\cos kd}{\sin knd} = 2\cos kd. \qquad ...(9)$$

which is constant and independent of n. Thus the assumption (8) gives the desired constancy of the right hand side of eq. (7). To find the value of k we use from the boundary conditions, *i.e.*, at $x = 0$ and $x = (N+1)d$, $a_0 = a_{N+1} = 0$. Substituting these boundary conditions in eq. (8), we get

$$a_0 = a \sin 0 = 0$$

and

$$a_{N+1} = a\sin k(N+1)d = 0 \text{ or } \sin k(N+1)d = 0$$

i.e.,

$$k(N+1)d = r\pi \text{ or } k = \frac{r\pi}{(N+1)d} \qquad ...(10)$$

where $r = 1, 2, 3, \ldots, N$. We exclude $r = 0$ value because for $r = 0$, $k = 0$, *i.e.*, the amplitude of any mass equal to zero at all times which means, there is no possible motion. The reason for ignoring, the values of r greater than N will be clear later on.

Substituting the value of k from eq. (10) in (8), we get

$$a_n = a\sin\frac{nr\pi}{N+1} \qquad ...(11)$$

Therefore, left hand side of eq. (9) is obtained to be

$$\frac{a_{n-1}+a_{n+1}}{a_n} = 2\cos\frac{r\pi}{N+1} \qquad ...(12)$$

Using eq. (12) , eq. (7) assumes the form

$$\frac{-\omega^2+2\omega_0^2}{\omega_0^2} = 2\cos\frac{r\pi}{N+1}$$

whence,

$$\omega^2 = 2\omega_0^2\left[1-\cos\frac{r\pi}{N+1}\right] \text{ or } \omega^2 = 4\omega_0^2\sin^2\frac{r\pi}{2(N+1)} \qquad ...(13)$$

i.e.,

$$\omega = 2\omega_0\sin\frac{r\pi}{2(N+1)} \quad \text{or } \omega = 2\omega_0\sin\frac{kd}{2} \qquad ...(14)$$

We have taken here positive square root only because negative frequencies have no physical meaning.

The relation (14) gives normal mode frequency for a particular value of r. We call this frequency ω as ω_r, the frequencey of rth mode of vibration, *i.e.*,

$$\omega_r = 2\omega_0 \sin\frac{r\pi}{2(N+1)} = 2\sqrt{\frac{T_0}{md}}\sin\frac{r\pi}{2(N+1)} \qquad ...(15)$$

Different values of r give different mode frequencies (ω_r) of the system.

When all the beads are oscillating in the rth mode, the actual displacement of the nth bead can be written as

$$\psi_n = a\sin\frac{nr\pi}{N+1}\cos(\omega_r t + \phi) \qquad ...(16)$$

Now the general motion of the system can be obtained as the susperposition of various modes for different values of r. At $t = 0$, if we take the particle displacement to be maximum. then we have $\phi = 0$. With this initial condition, eq. (16) is read as

$$\psi_n = a\sin\frac{nr\pi}{N+1}\cos\omega_r t \qquad ...(17)$$

Number of normal modes : *It can be shown that the string with N beads can vibrate transversely in N normal modes with frequencies* ω_r, *where* $r = 1, 2, ...,$ N. In order to prove this statement, we draw a graph between ω_r as ordinate and $\frac{r\pi}{2(N+1)}$ as abscissa **[fig 9.18]**. As we increase the value of r from 1 to N, we get N frequencies on a sine curve and when $r = N + 1$, abscissa is $\pi/2$ and $\omega_r = 2.\omega_0 = \omega_m$ a maximum value, but there is no possible motion corresponding to this value because $a_n = a\sin\frac{nr\pi}{N+1} = 0$ for $r = N + 1$.

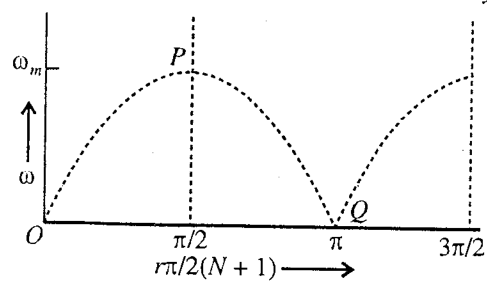


Fig. 9.18 : Graph between mode number [multiplied by $p/2(N+1)$] and mode frequency.

When $r > N + 1$, say $r = N + p$, then we shall get only the duplication of frequencies and physically we shall not get any new motion. suppose $N + 1 \geq p > 1$, then

$$\omega_{N+p} = \omega_m \sin\frac{(N+p)\pi}{2(N+1)} = \omega_m \sin\left[\pi - \frac{(N-p+2)\pi}{2(N+1)}\right] = \omega_m \sin\frac{[N-(p-2)]\pi}{2(N+1)} = \omega_{N-(p-2)} \qquad ...(18)$$

Thus, the frequency of $(N + p)$th mode is equal to the frequency of $N - (p - 2)$th mode. Obviously $N - (p - 2)$th mode has its frequency in the range from ω_1 to ω_N. For example, when $p = 2$ or $r = N+2$, we have

$$\omega_{N+2} = \omega_m \sin\frac{N\pi}{2(N+1)} = \omega_N \qquad ...(19)$$

Similarly we can show that $\omega_{N+3} = \omega_{N-1}$, $\omega_{N+4} = \omega_{N-2}$ and so on.

When $p = N + 2$ or $r = 2N + 2$, $\omega_{2N+2} = 0$ and there is no possible motion again. This range in the **fig. 9.18** corresponds from $\pi/2$ (for $r = N + 1$) to π (for $r = 2N + 2$) with the repetition of frequencies of OP curve on PQ. For next range of $N + 1$ values of r, the frequencies will again be duplicated. Therefore, it is sufficient to consider the values of r from $r = 1$ to $r = N$ and ignore $r > N + 1$. Thus, there are only N normal modes of vibration of a N-beaded string.

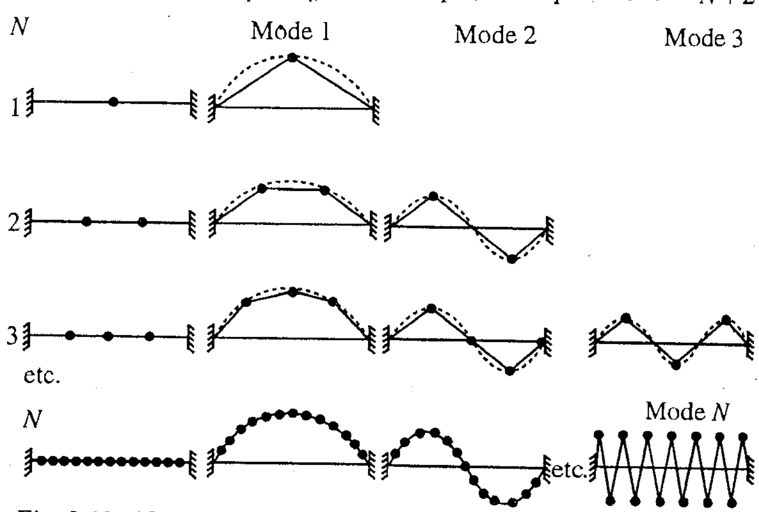


Fig. 9.19 : Normal mode of a string with N particles : String with 1, 2, 3, 4 ......, particles have 1, 2, 3, 4, ..., N modes of variation respectively.

We show in **fig. 9.19** the normal or independent modes of vibration for a string, equipped with N = 1, 2, 3, 4, ... beads. For N = 1, there is one normal mode of vibration. For n = 2, 3, ..., these are two three, ... modes of vibration respectively. For N beads, there are N independent modes of vibration. In any case, the segments of the string between any two consecutive beads remain straight due to tension and at an instant, all the beads lie on a sine curve.

Representation of various modes : Now we discuss various modes for sufficient number of particles. The particle displacements in the rth mode are given by eq. (17)

$$\psi_n = \left(a \sin \frac{nr\pi}{N+1}\right) \cos \omega_r t \qquad ...(20)$$

where $n = 1, 2, 3, ..., N$ represents various beads on the sting. For $r = 1$, all the beads are vibrating with the same frequency ω_1 in Mode 1 and the amplitude $\left(a \sin \frac{n r \pi}{N+1}\right)$ is varying with particle number n. The value of the amplitude increases as n changes from $n = 1$ to onwards and becomes nearly maximum when n = N/2. Further it decrases and becomes zero at $n = N + 1$ or the fixed bead. **[fig.9.20(a)]**. Thus in Mode 1, all the masses execute simple harmonic motions in phase with frequency ω_1 with different amplitudes in a loop.

For $r = 2$, the vibrational frequency of all the beads is ω_2. The amplitude $\left(a \sin \frac{2n\pi}{N+1}\right)$ first increases and then decreases with particle number n. It is zero in the middle (when N is odd, there is a bead at the centre at rest) and again a similar variation in amplitude occurs. Now the beaded string vibrates in two loops as shown in **[fig. 9.20(b)]**. The frequency of vibration ω_2 in Mode 2 is approximately twice that of ω_1 in Mode 1 $\left[\omega_1 = 2\omega_0 \sin \frac{\pi}{2(N+1)} \text{ and } \omega_2 = 2\omega_0 \sin \frac{2\pi}{2(N+1)}\right]$.

In Mode 3 (for $r = 3$), the frequency of vibration is approximately three times to that in Mode 1 and the beaded string vibrates in 3 loops **[fig. 9.20(c)]** and so on. In this mode of vibration, the beads of first loop will be out of phase to those in the 2nd looop and in phase with those in 3rd loop. The distance over which one complete cycle occurs defines wavelength λ. Hence for Mode 1, $\lambda_1/2 = L$ or $\lambda_1 = 2L$, for Mode 2, $\lambda_2 = L$ or $\lambda_2 = 2L/2$, for Mode 3, $\lambda_3 = 2L/3$ and for Mode r, $\lambda_r = 2L/r$.

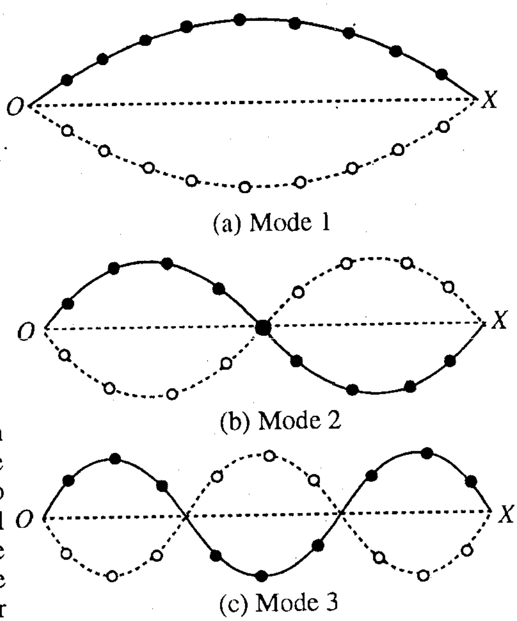


Fig. 9.20 : First three modes of a beaded string.

Now let us relate the constant k for Mode r to the wavelength. From eq. (10),

$$k = \frac{r\pi}{(N+1)d} = \frac{r\pi}{L} = \frac{2\pi}{2L/r} = \frac{2\pi}{\lambda_r} \qquad [\because (N+1)d = L] \quad ...(21)$$

The wave number ($\bar{\nu}$) is defined as the inverse of the wavelength (λ), *i.e.*, $\bar{\nu} = 1/\lambda$. Therefore,

$$k = \frac{2\pi}{\lambda} = 2\pi\bar{\nu} \qquad ...(22)$$

This k is called a **angular wave number.**

We can write eq. (14) for the frequency in terms of wave number (k) or wavelength (λ) as follows :

$$\omega = 2\omega_0 \sin kd/2 = \omega_m \sin kd/2 \text{ or } \omega = \omega_0 \sin \pi d/\lambda \qquad ...(23)$$

where $\omega_m = 2\omega_0 = 2\sqrt{T/md}$. This relation between ω and k is called **dispersion relation.**

9·9. Longitudinal Oscillations of *N*-Coupled Masses

We consider here the longitudianl oscillations of a system of N-coupled masses. In **fig. 9.21**, we have shown a system of N identical masses (m) and $N + 1$ identical springs. each of force constant K. The free ends of the system are rigidly fixed at S_1 and S_2.

The spring-mass system is allowed to oscillate longitudinally. If ψ_{n-1}, ψ_n and ψ_{n+1}, are the displace -ments of $(n-1)$th, nth and $(n+1)$th masses respectively **[fig. 9.22]**, the equation of motion of nth mass is given by

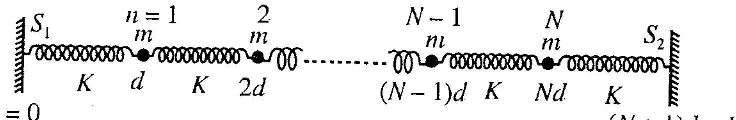


Fig. 9.21 : System of N masses and $N+1$ springs.

$$m\frac{d^2\psi_n}{dt^2} = -K(\psi_n - \psi_{n+1}) - K(\psi_n - \psi_{n-1})$$

or
$$\frac{d^2\psi_n}{dt^2} + 2\omega_0^2\psi_n - \omega_0^2(\psi_{n-1} + \psi_{n+1}) = 0 \quad ...(1)$$

where $\omega_0^2 = K/m$.

Eq. (1) is identical to eq. (3) of N-coupled transverse oscillators and hence the displacement ψ_n of the nth mass in the rth (or kth) mode of vibration is given by [using eq. (17), of last article] :

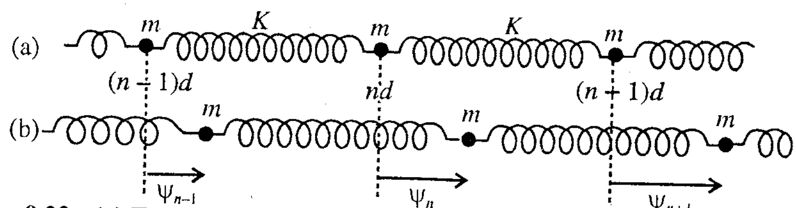


Fig. 9.22 : (a) Equilibrium position, (b) Displaced instantaneous position.

$$\psi_n = a \sin nkd \cos \omega t \quad ...(2)$$

where $k = \dfrac{r\pi}{(N+1)d} = \dfrac{r\pi}{L}$ with $r = 1, 2, 3, ..., N$ and $(N+1)d = L$.

The dispersion relation for the system is obtained to be

$$\omega = \omega_m \sin\frac{kd}{2} = \omega_m \sin\frac{\pi d}{\lambda} \quad ...(3)$$

where $\omega_m = 2\omega = 2\sqrt{K/m}$. In the limit $\lambda >> d$,

$$\omega = \sqrt{\frac{Kd^2}{m}} \cdot k \quad ...(4)$$

which is the (non-dispersive) dispersion relation in the continuous approximation.

9·10. The Wave Equation

When the number of beads attached on the string become very large and the distance between two consecutive beads (*i.e.*, d) tends to zero, the system becomes a massive continuous string. It is shown below that the coupled oscillations of the beaded string become **waves** in the continuous limit.

The equation of motion of the nth bead on the string is given by [eq. (2) Art. 9.8],

$$m\frac{d^2\psi_m}{dt^2} = -\frac{T_0}{d}(2\psi_n - \psi_{n-1} - \psi_{n+1}) \quad ...(1)$$

This can be written in the following form :

$$\frac{d^2\psi_n}{dt^2} = \frac{T_0}{m}\left[\frac{\psi_{n+1} - \psi_n}{d} - \frac{\psi_n - \psi_{n-1}}{d}\right] \quad ...(2)^*$$

* In case of longitudinal oscillations of N-coupled masses [from eq. (1) of Art. 9·9]

$$\frac{d^2\psi_n}{dt^2} = \frac{K}{m}\left[\frac{\psi_{n+1} - \psi_n}{d} - \frac{\psi_n - \psi_{n-1}}{d}\right]$$

and proceeding as for the case of the string, one can obtain the wave equation (6) : $\dfrac{d^2\psi}{dt^2} = v^2\dfrac{d^2\psi}{dx^2}$

where $v = \sqrt{K/\rho}$ is the wave velocity.

Now let the distance between two consecutive beads be made small. We denote d by δx and consider the limit $\delta x \to 0$ in the condition that the masses make a continuous string. In such a case,

$$\frac{d^2\psi_n}{dt^2} = \frac{T_0}{m}\left[\frac{\psi_{n+1}-\psi_n}{\delta x} - \frac{\psi_n - \psi_{n-1}}{\delta x}\right]$$

or

$$\frac{d^2\psi_n}{dt^2} = \frac{T_0}{m}\left[\left(\frac{\delta\psi}{\delta x}\right)_{n+1} - \left(\frac{\delta\psi}{\delta x}\right)_n\right] \quad ...(3)$$

If the position of the nth bead be represented by x, then in the limit $\delta x \to 0$, we have

$$\frac{d^2\psi_n}{dt^2} = \frac{T_0}{m}\left[\left(\frac{\delta\psi}{\delta x}\right)_{x+dx} - \left(\frac{\delta\psi}{\delta x}\right)_x\right] \quad ...(4)$$

Since ψ is also changing with x, the value of $d\psi/dx$ at $x + dx$ will be given by

$$\left(\frac{d\psi}{dx}\right)_{x+dx} = \frac{d}{dx}(\psi + d\psi) = \frac{d\psi}{dx} + \frac{d^2\psi}{dx^2}\cdot dx$$

$$\therefore \quad \frac{d^2\psi}{dt^2} = \frac{T_0}{m}\frac{d^2\psi}{dx^2}\cdot dx \text{ or } \frac{d^2\psi}{dt^2} = \frac{T_0}{\rho}\frac{d^2\psi}{dx^2} \quad ...(5)$$

where $m/dx = \rho$, the mass per unit length of the string. Eq. (5) represents the **wave equation**. The quantity T_0/ρ has the dimension of the square of velocity and hence writing $v^2 = T_0/\rho$, we get the wave equation as

$$\frac{d^2\psi}{dt^2} = v^2\frac{d^2\psi}{dx^2} \quad ...(6)$$

The particle displacement in the rth mode is given by [eq. (20) or (17) of Art. 9·8]

$$\psi_n = a \sin\frac{nr\pi}{N+1}\cos\omega_r t = a\sin\frac{nr\pi d}{L}\cos\omega_r t \qquad [(N+1)d = L] \quad ...(7)$$

If we drop the suffix n to represent the diplacement ψ of a point mass of the continuous string at a distance $x = nd$ from the fixed end $x = 0$, then

$$\psi = a\sin\frac{r\pi x}{L}\cos\omega_r t \quad ...(8a)$$

or

$$\psi = a\sin kx\cos\omega_r t \quad ...(8b)$$

where $k = r\pi/L$.

If fact, this is the equation of standing wave in a string, fixed at its both ends.

Let us investigate the form of the dispersion relation when N becomes very large. If we consider only first few modes, *i.e.*, r is small ($r << N$), then

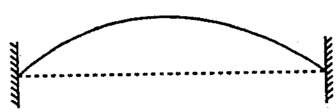
$$\omega = \omega_m \sin\frac{kd}{2} = 2\sqrt{\frac{T_0}{md}}\cdot\frac{kd}{2}$$

$$= \sqrt{\frac{T_0}{m/d}}\,k = \sqrt{\frac{T_0}{\rho}}\,k \quad \text{or} \quad \omega = vk \quad ...(9)$$

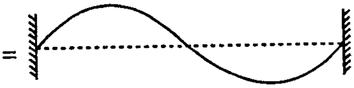
where we have put $\sin\theta = \theta$ for small θ, as $\theta = \frac{kd}{2} = \frac{r\pi}{2(N+1)}$ is small for $r << N$.

Such a relation is found between ω and k for waves, moving in a continuous medium and correspond to the familiar relation, given by

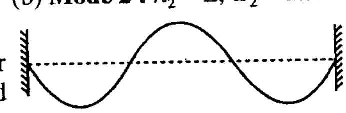
$$v = \nu\lambda \quad ...(10)$$



(a) **Mode 1 :** $\lambda_1 = 2L$, $\omega_1 = 2\pi v/\lambda_1 = \pi v/L$



(b) **Mode 2 :** $\lambda_2 = L$, $\omega_2 = 2\pi v/\lambda_2 = 2\pi v/L = 2\omega_1$



(c) **Mode 3 :** $\lambda_3 = 2L/3$, $\omega_3 = 2\pi v/\lambda_3 = 3\pi v/L = 3\omega_1$

Fig. 9.23.

because $\omega = vk$ or $2\pi\nu = v.\ 2\pi/\lambda$ or $\nu = v/\lambda$

If N is very large, various modes are represented by eq. (8) and at any instant, first few modes (for $r = 1, 2, 3$) are shown in **fig. 9.23**. In fact, these are the various modes of vibration of a stretched string, fixed at its ends (with $\lambda_r = 2L/r$). The frequency in the Mode 1 (for $r = 1$) is given by

$$\omega_1 = \pi v/L \qquad \text{...(11)}$$

This is called **fundamental mode of vibration** of the string.

For Mode 2, $$\omega_2 = 2\pi v/L = 2\omega_1 \qquad \text{...(12)}$$

For Mode 3, $$\omega_3 = 3\pi v/L = 3\omega_1 \qquad \text{...(13)}$$

and so on. Thus if we consider first few modes, then in general, we have

$$\omega_r = r\omega_1 \qquad \text{...(14)}$$

In fact, we know that such a relation holds for a stretched string.

Ex. 1. (a) *If ω_1, ω_2 and ω_3 are the frequencies of first three modes of vibration of a N-beaded string, calculate the ration ω_2/ω_1 and ω_3/ω_1 for $N = 3$, 10 and 100 beads. Does the ratio depend on the number of beads ? What do you conclude for a string with large number of beads ?*

(b) *If $N = 100$, calculate ω_{99}/ω_1. What do you conclude from the value of this ratio ?*

Sol. : (a) $$\omega_r = 2\omega_0 \sin\frac{r\pi}{2(N+1)}$$

$$\therefore \qquad \frac{\omega_2}{\omega_1} = \frac{\sin\dfrac{2\pi}{2(N+1)}}{\sin\dfrac{\pi}{2(N+1)}} \quad \text{and} \quad \frac{\omega_3}{\omega_1} = \frac{\sin\dfrac{3\pi}{2(N+1)}}{\sin\dfrac{\pi}{2(N+1)}}$$

(1) For $N = 3$, $$\frac{\omega_2}{\omega_1} = \frac{\sin 2\pi/8}{\sin\pi/8} = 1.85 \text{ and } \frac{\omega_3}{\omega_1} = \frac{\sin 3\pi/8}{\sin\pi/8} = 2.42$$

(2) For $N = 10$, $$\frac{\omega_2}{\omega_1} = \frac{\sin 2\pi/22}{\sin\pi/22} = 1.975 \text{ and } \frac{\omega_3}{\omega_1} = \frac{\sin 3\pi/22}{\sin\pi/22} = 2.915$$

(3) For $N = 100$, $$\frac{\omega_2}{\omega_1} = \frac{\sin 2\pi/202}{\sin\pi/202} = 2 \text{ and } \frac{\omega_3}{\omega_1} = \frac{\sin 3\pi/202}{\sin\pi/202} = 3$$

We find that the ratios ω_2/ω_1 and ω_3/ω_1 depend on the number of beads and approach to 2 and 3 respectively, when bead number becomes large. In such a case $\omega_r/\omega_1 = r$ or $\omega_r = r\omega_1$. In fact, these are the ratios for a continuous massive string. We conclude that a large number beaded light string behaves like a continuous massive string for first few modes.

(b) $$\frac{\omega_{99}}{\omega_1} = \frac{\sin 99\pi/200}{\sin\pi/200} \simeq 64$$

This as such less than 99. Hence, if the mode number is large, the relation $\omega_r = r\omega_1$ does not hold correct.

Ex. 2. *Set up differential equations for each mass of a system of 3 identical masses, attached to a light string with tension T_0.*

Find out the ratio of ampliudes a_1, a_2 and a_3 for the three modes of vibration of **fig.9.24**. *Determine the normal mode frequencies. Use eq.* (17) [Art. 9.8] *to draw the sine curves for the instantaneous displacements in the different modes of vibration.*

Sol. Using eq. (3) [Art. 9.8], we get $(\psi_0 = \psi_4 = 0)$

$$\frac{d^2\psi_1}{dt^2} + 2\omega_0^2\psi_1 - \omega_0^2\psi_2 = 0$$

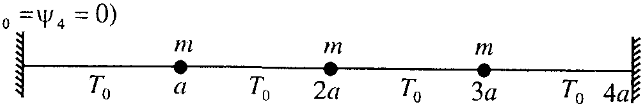


Fig. 9.24.

$$\frac{d^2\psi_2}{dt^2}+2\omega_0^2\psi_2-\omega_0^2(\psi_1+\psi_3) = 0$$

$$\frac{d^2\psi_3}{dt^2}+2\omega_0^2\psi_3-\omega_0^2\psi_2 = 0$$

where $\omega_0^2 = T_0/md$.

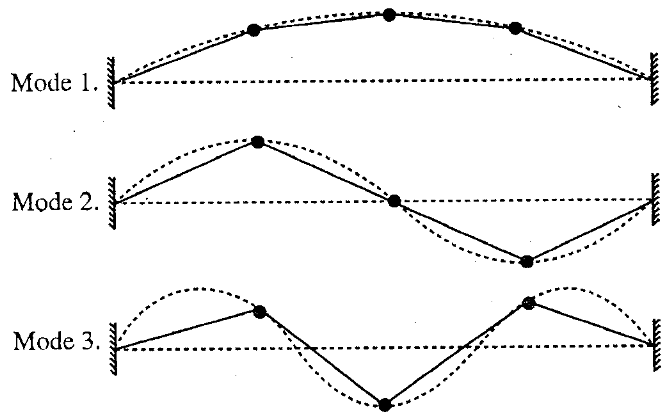


Fig. 9.25.

In Mode r, $$\frac{a_n}{a_{n-1}} = \frac{\sin\dfrac{nr\pi}{N+1}}{\sin\dfrac{(n-1)r\pi}{N+1}} = \frac{\sin nr\pi/4}{\sin(n-1)r\pi/4}$$

In Mode 1, $$\frac{a_2}{a_1} = \frac{\sin\pi/2}{\sin\pi/4} = \sqrt{2}$$

and $$\frac{a_3}{a_1} = \frac{\sin 3\pi/4}{\sin\pi/4} = 1$$

In Mode 2, $$\frac{a_2}{a_1} = \frac{\sin\pi}{\sin\pi/2} = 0 \quad \text{and} \quad \frac{a_3}{a_1} = \frac{\sin 3\pi/2}{\sin\pi/2} = -1$$

In Mode 3, $$\frac{a_2}{a_1} = \frac{\sin 3\pi/2}{\sin 3\pi/4} = -\sqrt{2} \quad \text{and} \quad \frac{a_3}{a_1} = \frac{\sin 9\pi/4}{\sin 3\pi/4} = 1$$ [see figure 9.25]

Normal mode frequencies are given by eq. (13) [Art. 9·8], *i.e.*,

$$\omega_r^2 = 4\omega_0^2\sin^2\frac{r\pi}{2(N+1)} = 4\omega_0^2\sin^2\frac{r\pi}{8}$$

$$\therefore \quad \omega_1^2 = 4\omega_0^2\sin^2\frac{\pi}{8} = (2-\sqrt{2})\,\omega_0^2 \qquad \left[\because \sin^2\frac{\pi}{8} = \frac{\sqrt{2}-1}{2\sqrt{2}}\right]$$

$$\omega_2^2 = 4\omega_0^2\sin^2\frac{\pi}{4} = 2\omega_0^2 \text{ and } \omega_3^2 = 4\omega_0^2\sin^2\frac{3\pi}{8} = (2+\sqrt{2})\,\omega_0^2 .$$

Ex. 3 *Consider a system of 4 beads of same mass, attached to a light string. Show that the mode frequency for $r = 9$ is equal to the mode frequency for $r = 1$. Represent graphically the instantaneous displacements in Mode 1 and in Mode 9. What do you conclude from this ?*

Sol. We know that, $\omega_r = \omega_m \sin\dfrac{r\pi}{10}$ $\left[\because \omega_r = \omega_m \sin\dfrac{r\pi}{2(N+1)}\right]$

For $r = 1$, $\omega_1 = \omega_m \sin\pi/10$ and for $r = 9$, $\omega_9 = \omega_m \sin 9\pi/10$.

But, $\omega_9 = \omega_m \sin(\pi - \pi/10) = \omega_m \sin\pi/10$

Thus, $\omega_9 = \omega_1$

Particle displacements in Mode 1 are shown by solid curve and those in Mode 9 by dotted curve [**fig. 9.26**]. We conclude that it is sufficient to consider Mode 1 and to discard Mode 9 for a 4-beaded string. In fact, this system can oscillate in 4 independent modes of vibration for $r = 1$, 2, 3 and 4, ignore $r > 5$ modes.

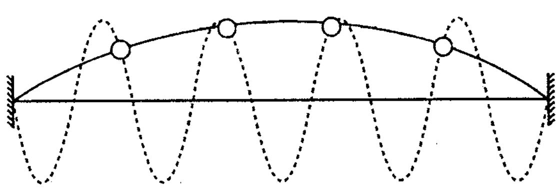


Fig. 9.26.

EXERCISE 9·3

1. Three equal masses m are equally spaced along a string of length $4a$. The tension in the string is T. Show that the three normal mode frequencies and the corresponding normal mode displacements are :

$$\omega_1 = \omega_0\sqrt{1-1/\sqrt{2}},\, x_1 = x_3 = x_2/\sqrt{2};\ \omega_2 = \omega_0,\ x_1 = -x_3 \text{ and } x_2 = 0;$$

$$\omega_3 = \omega_0\sqrt{1+1/\sqrt{2}},\, x_1 = x_3 = -x_2/\sqrt{2}, \text{ where } \omega_0 = \sqrt{2T/ma}$$

2. A system of N identical masses is coupled by $N + 1$ springs each of force constant K **[fig. 9.27]**. The free ends of the system are rigidly fixed at S_1 and S_2. The spring-mass system is allowed to oscillate longitudinally. If x_{n-1}, x_n and x_{n+1} are the displacements of $(n-1)$th, nth and $(n+1)$th masses respectively **[fig. 9.28]**, show that the equation of motion of nth mass is

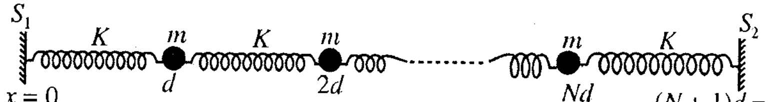


Fig. 9.27.

$$\ddot{x} + 2\omega_0^2 x_n - \omega_0^2(x_{n-1} + x_{n+1}) = 0, \text{ where } \omega_0^2 = K/m.$$

Further show that the displacement in the rth or kth mode of vibration can be given by

$x_n = a \sin kd \cos \omega t$,

where $k = r\pi/l$.

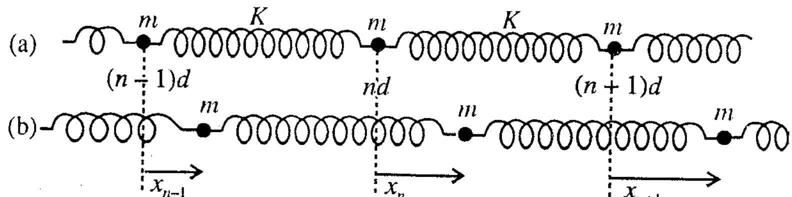


Fig. 9.28.

Find the dispersion relation for the system.

[**Ans.** $\omega = 2\omega_0 \sin kd/2$.]

QUESTIONS

Long Answer Questions

1. What are normal modes and normal coordinates of a coupled system ? Illustrate with suitable example. **(Gorakhpur 1988)**
2. Discuss the motion of two pendulums of equal string lengths and masses with their bobs connected by a spring. Set up the differential equations and solve them to obtain the normal modes and frequencies.
3. Show that in two coupled pendulums the energy bounces back and forth from one pendulum to the other with a frequency equal to the difference between the two mode frequencies.
4. Show that the total energy of the two coupled pendulums is equal to the sum of energies associated with the two modes.
5. Two identical pendulums are connected together by a massless spring of force constant K. Find its angular frequency when they oscillate (i) in same phase, (ii) in opposite phase. **(Indore 2003)**
6. The differential equations for a two coupled oscillators are :

$$\frac{d^2\psi_1}{dt^2} + a_1\psi_1 + b_1\psi_2 = 0 \text{ and } \frac{d^2\psi_2}{dt^2} + a_2\psi_1 + b_2\psi_2 = 0.$$

Determine the normal mode solutions and frequencies.

7. A uniform massless string fixed at both ends is beaded with N identical particles of mass m with a constant spacing a between two consecutive particles. Show that the frequencies of the normal modes

of transverse oscillations are obtained to be

$$\omega_n = 2\sqrt{\frac{T}{ma}}\sin\left[\frac{n\pi}{2(N+1)}\right],$$

where T is the tension and $n = 1, 2, ..., N$.

Draw the normal mode configurations for $N = 3, 4$, particles.

8. Two identical masses m are connected by three similar springs of force constant K. The free ends of the springs are rigidly fixed. Determine the normal mode solutions and frequencies along the line joining the centres of the masses. Show that the two normal modes have frequency ratios $\sqrt{3}:1$.
9. Taking the case of two coupled spring and mass systems, discuss the normal mode analysis. **(Agra 2006)**
10. What is meant by n coupled oscillator ? Show that a wave propagates in it as a result of longitudinal motion. **(Bhopal 2004)**
11. Obtain wave equation considering the case of N-coupled oscillators. What is its physical significance ? **(Osmania 2005)**
12. Show that the transverse coupled oscillations of a beaded string under tension becomes waves in the continuous limit.
13. Write short notes on the following :
(i) Two coupled oscillators; (ii) Normal modes; (iii) Normal frequencies.

Short Answer Questions

1. What do you understand by normal modes of a coupled system ? Explain. **(Agra 2004)**
2. Establish the equation of motion for a system of two simple pendulums coupled together. **(Kanpur 2005)**
3. What are coupled oscillators ? Give two examples. **(Osmania 2005)**
4. Could a system of two coupled pendulums oscillate in a single mode ? Could a pendulum of the system oscillate under the combined effect of both of the modes ?
5. In case of two coupled pendulums when only one mode is present, show that (i) each pendulum executes simple harmonic motion, (ii) both pendulums oscillate with the same frequency, and (iii) both pendulums pass through their equilibrium positions simultaneously.
6. Any of the two coupled pendulums oscillate always in simple harmonic motion. Whether is it true or false ? Explain.
7. In case of normal mode of two coupled identical sprning-mass systems, both masses vibrate with the frequency of single spring mass system. Explain. **(Agra 2005)**
8. What do you understand by modulation and phenomenon of beats in case of a two-body coupled oscillator ?
9. What is the advantage of discussing the motion of coupled oscillators in the normal modes ?
10. N-identical beads are attached at equal distances on a string under tension. Draw diagrams of various modes of transverse vibrations of the system.
11. If in **fig. 9.8**, the two masses are weakly coupled ($C << C'$), show that the ratio of the difference of mode frequencies to the lower mode frequency is nearly C/C'.

Objective Type Questions

1. In case of two coupled identical pendulums, oscillating in a plane :
(a) each pendulum always execute simple harmonic motion
(b) two pendulums may execute simple harmonic motions
(c) the general motion can be expressed as a superposition of two simple harmonic motions of the same frequency

(d) the general motion can be expressed as a superposition of two simple harmonic motions of different frequencies

2. Two coupled pendulums are oscillating only in one mode. Which statement is valid ?
   (a) $x_1 = a_1 \cos \omega_1 t$ and $x_2 = a_2 \sin \omega_1 t$
   (b) $x_1 = a_1 \cos \omega_1 t$ and $x_2 = a_2 \cos \omega_2 t$
   (c) $x_1 = a_1 \cos \omega_1 t$ and $x_2 = a_2 \cos \omega_1 t$
   (d) $x_1 = a_1 \cos (\omega_1 t + \pi/2)$ and $x_2 = a_2 \cos (\omega_1 t + \pi)$
3. In case of two coupled identical pendulums, in general :
   (a) the potential energy is a homogeneous quadratic function, when expressed in terms of actual displacements
   (b) the potential energy is not a homogeneous quadratic function, when expressed in terms of actual displacements
   (c) the potential energy is a homogeneous quadratic function, when expressed in terms of normal coordinates
   (d) the potential energy is not a homogeneous quadratic function, when expressed in terms of normal coordinates
4. For a system of 5-beaded string, the ratio of the angular frequencies ω_3/ω_2 will be equal to :
   (a) $3:2$ (b) $2:\sqrt{3}$ (c) $2:3$ (d) $2:5\sqrt{3}$
5. For a system of 5-beaded string, the ratio of amplitudes of 2nd and 4th particles in the second mode of vibrations is given by :
   (a) $1:2$ (b) $-1:1$ (c) $\sqrt{3}:2$ (d) $2:1$

ANSWERS

1. (b), (d) **2.** (c) **3.** (b), (c) **4.** (b) **5.** (c)

10

Damped Harmonic Oscillator

10·1. Introduction

A particle will perform simple harmonic motion with a costant amplitude (or energy) of vibration, provided the restoring force on it is proportional to the displacement ($F = - Cx$). In actual practice, we observe that if a physical system, once oscillated, its oscillations die out with the passage of time. For example, if we oscillate a pendulum, the amplitude of vibration does not remain constant but decreases gradually with time and ultimately the pendulum comes to rest. In fact, when a body moves in a medium, frictional foces act on the body in a direction opposite to its movement and the work done against the frictional forces results in the decrease of its mechanical energy. This is why, an oscillating pendulum loses energy and consequently its amplitude of oscillation becomes smaller and smaller as the time elapses.

In the presence of frictional forces, the motion of the oscillator is said to be **damped** by friction and the vibrating system is called **damped harmonic oscillator**. The frictional or damping force on the oscillator arises from the retarding action of the medium. The magnitude of this force is some function of the velocity of the moving body. At ordinary speeds, the damping force (F_d) is found proportional to the velocity (v) in a direction opposite to it *i.e.*, $F_d = - \gamma v$, where γ is a constant called the damping coefficient. It depends on the size of the body.

10·2. Theory of Damped Harmonic Oscillator

In the presence of frictional forces, arising due to the viscosity of the medium, the oscillatory system is called **damped harmonic oscillator**. The free oscillations of a physical system in the presence of frictional forces are callled **damped harmonic motions**. When damping is taken into account, then a harmonic oscillator is acted by the following forces :

(1) A restoring force $- Cx$ at the displacement x due to the spring action, and

(2) A damping force $- \gamma v$, where $v = dx/dt$ is the velocity of the oscillator at the displacement x and γ is the damping coefficient. Thus the resultant force on the oscillating body is given by

$$F = - Cx - \gamma \frac{dx}{dt}$$

Using Newton's second law, the equation of motion of the damped harmonic oscillator is given by

$$m\frac{d^2x}{dt^2} = - Cx - \gamma . \frac{dx}{dt} \qquad ...(1)$$

where m is the mass of the oscillating body.

Dividing by m, eq. (1) can be written as

$$\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + \omega_0^2 x = 0 \qquad ...(2)$$

where $2b = \gamma/m$ and $\omega_0^2 = C/m$.. Here $\omega_0 = \sqrt{C/m}$ is the natural angular frequency of the oscillator in absence of the damping force. The constant $b = \gamma/2m$ is called **damping constant**. Also if we put $m/\gamma = \tau = 1/2b$, then τ is called **relaxation time** or **time constant**.

Equation (2) is called the **differential equation of damped harmonic oscillator**.

Solution of the differential equation : Let the solution of eq. (2) be

$$x = Ae^{\alpha t} \qquad ...(3)$$

Substituting the values of x, dx/dt and d^2x/dt^2 in eq. (2), we have

$$(\alpha^2 + 2b\alpha + \omega_0^2)e^{\alpha t} = 0 \text{ or } \alpha^2 + 2b\alpha + \omega_0^2 = 0$$

$$\therefore \qquad \alpha = -b \pm \sqrt{b^2 - \omega_0^2}$$

Hence the solutions of the eq. (2) are

$$x = A_1 e^{(-b+\sqrt{b^2-\omega_0^2})t} \text{ and } x = A_2 e^{(-b-\sqrt{b^2-\omega_0^2})t}$$

Thus the most general solution of eq. (2) is

$$x = A_1 e^{(-b+\sqrt{b^2-\omega_0^2})t} + A_2 e^{(-b-\sqrt{b^2-\omega_0^2})t}$$

i.e.,
$$x = e^{-bt}(A_1 e^{(\sqrt{b^2-\omega_0^2})t} + (A_2 e^{(-\sqrt{b^2-\omega_0^2})t}) \qquad ...(4)$$

where A_1 and A_2 are constants, depending upon the initial conditions of motion.

The quantity $\sqrt{b^2 - \omega_0^2}$ is imaginary, real or zero, it depends on the relative values of b and ω_0. If $b < \omega_0$, then $\sqrt{b^2 - \omega_0^2}$ is imaginary and it is called **underdamped case.** If $b > \omega_0$ or $b = \omega_0$, the situations are known as **overdamped** and **critically damped** respectively.

(i) Underdamped case/($b < \omega_0$) *i.e.*, case of damped harmonic oscillations : If the damping is so low that $b < \omega_0$, then $\sqrt{b^2 - \omega_0^2} = \sqrt{-1\,(\omega_0^2 - b^2)} = i\sqrt{\omega_0^2 - b^2} = i\,\omega$, where $i = \sqrt{-1}$ and $\omega = \sqrt{\omega_0^2 - b^2}$, which is a real quantity. Now, from eq. (4), we have

$$x = e^{-kt}[A_1 e^{i\omega t} + A_2 e^{-i\omega t}] \qquad = e^{-kt}[A_1(\cos\omega t + i\sin\omega t) + A_2(\cos\omega t - i\sin\omega t)]$$

or
$$x = e^{-kt}[(A_1 + A_2)\cos\omega t + i(A_1 - A_2)\sin\omega t]$$

As x is real quantity, $(A_1 + A_2)$ and $i\,(A_1 - A_2)$ must be real quantities. Clearly, A_1 and A_2 are complex quantities. If $A_1 + A_2 = a_0 \sin\phi$ and $i\,(A_1 - A_2) = a_0\cos\phi$, then

$$x = a_0 e^{-bt}\sin(\omega t + \phi) \qquad ...(5)$$

This equation represents a **damped harmonic motion.** This motion is oscillatory or ballistic whose **periodic time** is given by

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}} \qquad ...(6)$$

Thus, the effect of damping is to increase the periodic time. If $b = 0$, *i.e.*, no damping, then $T = 2\pi/\omega_0$. But in actual cases, the effect of damping on periodic time is usually negligible except few extreme cases.

The **amplitude** of the damped harmonic oscillations [eq. (5)] is given by

$$a = a_0 e^{-bt} = a_0 e^{-t/2\tau} \qquad ...(7)$$

where a_0 is the amplitude in the absence of damping. In the presence of damping, the amplitude decreases exponentially with time. In **fig 10.1**, time displacement curve is shown for damped harmonic motion. As the maximum value of $\sin(\omega t + \phi)$ is alternately +1 and –1, obviously the time displacement curve of the vibrating body lies entirely between the curves $a = a_0 e^{-bt}$ and $a = -a_0 e^{-bt}$, shown by the dotted lines.

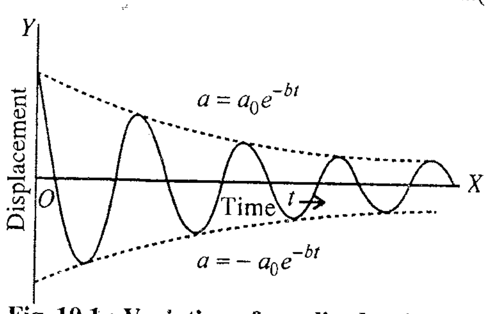


Fig. 10.1 : Variation of amplitude with time.

Logarithmic decrement : The time interval between two successive maximum displacements (*i.e.*, amplitudes) of left hand and right hand sides is $T/2$. If a_n and a_{n+1} are the two successive amplitudes, then

$$a_n = a_0\, e^{-bt} \text{ and } a_{n+1} = a_0\, e^{-b(t+T/2)}$$

Therefore,
$$\frac{a_n}{a_{n+1}} = \frac{e^{-bt}}{e^{-b(t+T/2)}} = e^{bT/2} \qquad ...(8)$$

The quantity $e^{bT/2}$, say d, is a constant of motion, representing reduction in the amplitude and hence we call it **decrement.**

Now, $$d = e^{bT/2} = \frac{a_n}{a_{n+1}} ; \therefore \log_e d = \log_e \frac{a_n}{a_{n+1}} = \frac{bt}{2} = \lambda \text{ (say)} \quad ...(9)$$

We call λ as the **logarithmic decrement** which is equal to the natural logarithmic of the ratio of two successive amplitudes of vibration.

If a_1 is the first maximum displacement, occurring after $T/4$ time, then

$$a_1 = a_0 e^{-bT/4} \text{ or } a_0 = a_1 e^{bT/4} \quad ...(10)$$

From which one can determine the amplitude a_0 in absence of damping.

(ii) Overdamped case : If the damping is so high $b > \omega_0$, then $\sqrt{b^2 - \omega_0^2}$, say β, is a real quantity and then from eq. (4), we have

$$x = (A_1^{-(b-\beta)t} + A_2^{-(b+\beta)t}) \quad ...(11)$$

As $b > \beta$, both quantities of right hand side decrease exponentially with time and the motion is **non-oscillatory**. Such a motion is called **dead beat** or **aperiodic** and its main application is in dead beat galvanometers.

(iii) Critically damped case : If b = ω_0, then from eq. (4), we have

$$x = (A_1 + A_2)e^{-bt} = Ce^{-bt}, \text{ where } C = A_1 + A_2$$

In this equation, there is only one constant, hence it does not provide us the solution of differential equation (2) of second order.

Now, suppose $\sqrt{b^2 - \omega_0^2} = h$, which is a small quantity. Hence from eq. (11), we have

$$x = e^{-bt}[A_1e^{ht} + A_2e^{-ht}] = e^{-bt}[A_1(1 + ht + ...) + B(1 - ht + ...)]$$

Neglecting the small terms, containing h^2 and higher powers of h, we get

$$x = e^{-bt}[(A_1 + A_2) + h(A_1 - A_2)t] \text{ or } x = e^{-bt}(P + Qt) \quad ...(12)$$

where $P = A_1 + A_2$ and $Q = h(A_1 - A_2)$

Also $$\frac{dx}{dt} = e^{-bt}Q + e^{-bt}(-b)(P + Qt)$$

If initially at $t = 0$, the displacement of the particle is $x = x_0$ and there the velocity is v_0, then

$$x_0 = P \text{ and } v_0 = Q - kP = Q - kx_0 \text{ or } Q = v_0 + kx_0$$

$$\therefore \quad x = [x_0 + (v_0 + bx_0)t]e^{-bt} \quad ...(13)$$

This equation represents that initially the displacement increases due to the factor $[x_0 + (v_0 + kx_0)t]$. But as time elapses, the exponential term becomes relatively more important and the displacement returns continuously from the maximum value to zero and the oscillatory motion does not just occur. Such a motion is called **critically damped** or **just aperiodic.**

10·3. Power Dissipation of Damped Harmonic Oscillator

When a body oscillates in a medium (*e.g.*, oscillations of simple pendulum in air), the frictional or damping forces act on the body in a direction opposite to its movement. In this process work is done by the body in overcoming the frictional forces. This results in a continuous decrease in the mechanical energy of the oscillating body and thereby in dimnishing the amplitude of oscillation with time. Finally the vibrating body comes to rest, if left to itself for a sufficiently long time.

The total energy of a damped harmonic oscillator decreases with time. We are interested to find the rate at which the energy is dissipated *i.e.*, the power dissipated, in the weak damping limit ($b << \omega_0$).

At any instant t, the displacement of a damped harmonic oscillator is given by

$$x = a_0e^{-bt}\sin(\omega t + \phi) \quad ...(1)$$

$\therefore$ Velocity of the particle $dx/dt = a_0 e^{-bt}[-b\sin(\omega t + \phi) + \omega\cos(\omega t + \phi)]$ ...(2)

$\therefore$ Kinetic energy of vibration,

$$K = \frac{1}{2}ma_0^2e^{-2bt}\{b^2\sin^2(\omega t + \phi) + \omega^2\cos^2(\omega t + \phi) - 2b\omega\sin(\omega t + \phi)\cos(\omega t + \phi)\} \quad ...(3)$$

Potential energy, $$U = \frac{1}{2}Cx^2 = \frac{1}{2}m\omega_0^2x^2 = \frac{1}{2}ma_0^2 e^{-2bt}\{\omega_0^2\sin^2(\omega t + \phi)\} \quad ...(4)$$

The average total energy for a period will be the sum of time average of kinetic and potential energy. If the amplitude of oscillation does not change much in one cycle of motion, then the factor e^{-2bt} may be taken as constant. Now, we are left with the averages of $\sin^2(\omega t + \phi)$, $\cos^2(\omega t + \phi)$ and $2\sin(\omega t + \phi)\cos(\omega t + \phi)$, whose average values for a period are $\frac{1}{2}, \frac{1}{2}$ and 0 respectively.

Hence the average kinetic energy, $K_{av} = \frac{1}{2} m a_0^2\, e^{-2bt}\left\{b^2 \cdot \frac{1}{2} + \omega^2 \cdot \frac{1}{2}\right\}$

or $$K_{av} = \frac{1}{4} m a_0^2\, \omega^2\, e^{-2bt} \qquad \text{[For low damping, } b^2 << \omega_0^2] \quad ...(5)$$

Average potential energy, $U_{av} = \frac{1}{4} m a_0^2\, e^{-2bt}\, \omega^2$ [For low damping $\omega \approx \omega_0$] ...(6)

$\therefore$ Average total energy, $E = K_{av} + U_{av} = \frac{1}{2} m a_0^2\, \omega^2\, e^{-2bt} = E_0\, e^{-2bt}$ or $E_0\, e^{-t/\tau}$* ...(7)

$\therefore$ Average power dissipation, $P = -dE/dt = m a_0^2\, \omega^2\, b e^{-2bt} = 2bE$ or E/τ ...(8)

This dissipation of energy is caused by the damping force $\gamma\, dx/dt$. Hence *if we calculate the negative value of the average rate of doing work by this force, i.e., $<\gamma\,(dx/dt)^2>$, then we will get exactly the same expression for power dissipation, as obtained above.* The proof is as follows :

Average rate of doing work by the damping force $= <F\frac{dx}{dt}> = -\gamma <\left(\frac{dx}{dt}\right)^2>$

$$= -2bm\left(\frac{1}{2} a_0^2 \omega^2 e^{-2bt}\right) = -2bE = -E/\tau \quad ...(9)$$

Hence, **negative of the average rate of doing work =** $\frac{E}{\tau}$ **= power dissipation.**

This loss of energy in damped harmonic oscillations appears in the form of heat in the oscillating system and dissipated in the medium.

Alternative treatment : In case of harmonic oscillator, we know that the total energy E of the oscillator is proportional to the square of the amplitude. In such a condition, we may consider that around time t, the oscillations are well described over some cycles by a simple harmonic motion, in which the total energy $E(t)$ is given by

$$E = \frac{1}{2} C a^2 = \frac{1}{2} C a_0^2\, e^{-2bt} \quad ...(10)$$

Writing $E_0 = \frac{1}{2} C a_0^2$, the initial value of the energy, we have

$$E = E_0\, e^{-2bt} \text{ or } E = E_0\, e^{-t/\tau} \quad ...(11)$$

The energy of the oscillator after one period (T) is

$$E_T = E_0\, e^{-2b(t+T)} = E\, e^{-2bT} \quad ...(12)$$

Hence the average power dissipation is given by

$$P_{av} = \frac{\text{Energy loss in one period}}{\text{Periodic time}} = \frac{E - E_T}{T} = \frac{E - Ee^{-2bT}}{T} = \frac{E - E(1-2bt)}{T} \text{ (as damping is small)}$$

$$= 2bE = E/\tau \quad ...(13)$$

10·4. Quality Factor Q

When damping is small, the term Q, known as **quality factor**, is very widely used for various oscillating systems. The quality factor Q of an oscillating system is defined as the product of natural angular frequency (ω_0) (in absence of damping) and time constant (τ) of the oscillator *i.e.*,

$$Q = \omega_0 \tau \quad ...(1)$$

Evidently Q is a *dimensionless quantity.*

For a lightly damped harmonic oscillator, $\omega \approx \omega_0$. Therefore

$$Q = \omega \tau \quad ...(2)$$

* The average energy contains time t means that this is the value of the average energy over one cycle at about the time t.

But, $$Q = \omega\tau = \frac{2\pi}{T}\cdot\frac{E}{P_{av}} = \frac{2\pi E}{P_{av}T} \qquad \left[\because \omega = \frac{2\pi}{T} \text{ and } P_{av} = \frac{E}{\tau}\right]$$

or $$Q = 2\pi \frac{\textbf{average energy stored}}{\textbf{average energy loss per period}} \qquad ...(3)$$

Thus Q-factor of an oscillating system is equal to 2π times the ratio of the average energy stored to the average energy loss per period.

Now, for a mechanical oscillator, $\omega_0 = \sqrt{C/m}$ and relaxation time, $\tau = m/\gamma$, then from (1)

$$Q = \sqrt{\frac{C}{m}}\cdot\frac{m}{\gamma} = \frac{\sqrt{mC}}{\gamma} \qquad ...(4)$$

where γ is damping constant.

Thus high value of Q implies low value of the damping for the oscillating system. This Q is a measure of the extent to which an oscillator is free from damping. In case of an **undamped oscillator,** $\gamma = 0$, and *hence Q is* **infinite.**

The energy of the oscillation ($E = E_0 e^{-t/\tau}$) decreases to e^{-1} of its initial value in the time τ and in this time the oscillator performs $\omega_0\tau/2\pi = Q/2\pi$ oscillations.

10·5. Examples of Damped Harmonic Oscillator

All actual harmonic oscillators are subject to damping forces due to friction *e.g.*, an oscillating pendulum, spring-mass system experiences air resistance. In any case of interest, one can establish a differential equation, similar to the equation for damped harmonic oscillator, and discuss the solution. Here we shall discuss specifically two examples of practical interest : (1) oscillations of the coil of a suspension type galvanometer and (2) charge oscillations in an *LCR* circuit.

(1) **Suspension type galvanometer :** In a suspension type galvanometer, a coil is suspended in a magnetic field, provided by a horse-shoe magnet. When a charge or current is passed through the galvanometer for a short time and left free, the coil performs mechanical angular oscillations.

If θ is the angular displacement of the system at any instant t, the coil experiences the following couples :

(1) A restoring couple $-C\theta$, where C is the restoring couple per unit angle of twist, and

(2) A damping couple $-\gamma\, d\theta/dt$ due to air friction and electromagnetic damping.*

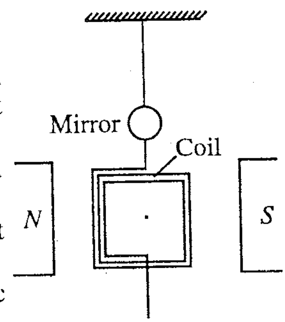


Fig. 10.2.

Hence the differential equation of motion of the coil is

$$I\frac{d^2\theta}{dt^2} = -\gamma\frac{d\theta}{dt} - C\theta \qquad ...(1)$$

where I is the moment of inertia of galvanometer coil about the suspension fibre.

We can write eq. (1) in the standard form of the differential equation for damped harmonic oscillator :

$$\frac{d^2\theta}{dt^2} + 2b\frac{d\theta}{dt} + \omega_0^2\theta = 0 \qquad ...(2)$$

where $2b = \gamma/I$ and $\omega_0^2 = C/I$.

The motion of the coil, represented by eq. (2) is oscillatory for low damping *i.e.*, for $b < \omega_0$. *A low damped galvanometer is called* **ballistic galvanometer**. Hence for a ballistic galvanometer, the value of the moment of inertia I should be large and the damping should be small. The electromagnetic damping is reduced by winding the coil over a nonconducting frame of bamboo, ivory etc. so that the induced currents, opposing the motion, are minimised.

* When the coil of the galvanometer rotates in the magnetic field, an induced current is set up in the coil, which opposes the motion of the coil. This electromagnetic damping is found proportional to $d\theta/dt$.

The solution of eq. (2) is of the form, *i.e.*,

$$\theta = \theta_0 \, e^{-bt} \sin(\omega t + \phi) \qquad \text{...(3)}$$

where $\theta_0 \, e^{-bt}$ is the amplitude of angular oscillations and θ_0 is the angular amplitude in the absence of damping.

The period of oscillations is given by

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}} = \frac{2\pi}{\sqrt{\frac{C}{I} - \frac{\gamma^2}{4I^2}}} \qquad \text{...(4)}$$

Determination of logarithmic decrement and correction for damping : In a ballistic galvanometer, when the charge is passed, its coil is deflected to a certain angle and then starts to oscillate. The succesive maximum angular displacements (or amplitudes) to the left and right hand sides, which are called **throws**, become less and less at the passage of the time. If $\theta_1, \theta_2, \theta_3, \ldots, \theta_n, \theta_{n+1}, \ldots$ are the successive throws of the galvanometer coil, then the **logarithmic decrement** is given by.

$$\lambda = \log_e \frac{\theta_n}{\theta_{n+1}} = \frac{bT}{2} \qquad \text{...(5)}$$

The proof is as follows :

Suppose θ_n and θ_{n+1} are the successive throws of the galvanometer coil, then

$$\theta_n = \theta_0 \, e^{-bt} \text{ and } \theta_{n+1} = \theta_0 \, e^{-b(t + T/2)}$$

where θ_0 is the throw in absence of damping and T is the time of oscillation.

Therefore, $$\frac{\theta_n}{\theta_{n+1}} = \frac{e^{-bt}}{e^{-b(t-T/2)}} = e^{bT/2}; \; \therefore \; \log_e \frac{\theta_n}{\theta_{n+1}} = \frac{bT}{2} = \lambda \qquad \text{...(6)}$$

From which damping constant $b = \gamma/2I = 2\lambda/T$ can be calculated.

However, in order to avoid error in observing two successive throws θ_n and θ_{n+1}, we observe θ_n, nth throw, and θ_{n+m}, $(n + m)$th throw, of the galvanometer coil. Now, the logarithmic decrement λ can be determined from the following relation :

$$\lambda = \frac{1}{m} \log_e \frac{\theta_n}{\theta_{n+m}} \qquad \text{...(7)*}$$

Also, if θ_1 is the first throw, occurring after time $T/4$, then

$$\theta_1 = \theta_0 \, e^{-bT/4} \text{ or } \theta_0 = \theta_1 \, e^{bT/4} = \theta_1 \, e^{\lambda/2}$$

For small λ, $$e^{\lambda/2} = 1 + \lambda/2; \; \therefore \; \theta_0 = \theta_1(1 + \lambda/2) \qquad \text{...(8)}$$

Thus by observing the first throw of the galvanometer coil one can make the correction for the damping and dtermine the throw (θ_0) in absence of the damping.

(2) *LCR* circuit : Consider a coil of self inductance L and resistance R, connected to a capacitor C [**fig. 10·3**]. In the beginning, the condenser is charged to a charge q_0 on the right plate by a battery

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* The decrement d for two successive throws is given by

$$d = \frac{\theta_n}{\theta_{n+m}} = e^{\lambda}$$

This quantity is the same for any two successive throws.

Therefore, $$\frac{\theta_n}{\theta_{n+1}} = \frac{\theta_{n+1}}{\theta_{n+2}} = \frac{\theta_{n+2}}{\theta_{n+3}} = \ldots\ldots = \frac{\theta_{n+m-2}}{\theta_{n+m-1}} = \frac{\theta_{n+m-1}}{\theta_{n+m}} = e^{\lambda}$$

Multiplying all the m quantities, we get

$$\frac{\theta_n}{\theta_{n+1}} \times \frac{\theta_{n+1}}{\theta_{n+2}} \times \ldots\ldots \times \frac{\theta_{n+m-2}}{\theta_{n+m-1}} \times \frac{\theta_{n+m-1}}{\theta_{n+m}} = e^{m\lambda}$$

or $$\frac{\theta_n}{\theta_{n+m}} = e^{m\lambda} \text{ or } \log_e \frac{\theta_n}{\theta_{n+m}} = m\lambda, \; \therefore \; \lambda = \frac{1}{m} \log_e \frac{\theta_n}{\theta_{n+m}}$$

and then it is discharged through the inductance L and capacitance C by switching off the battery circuit and connecting the LCR circuit by a key. At any instant, if q be the charge on the capacitor C, then the potential difference across the capacitor C is

$$V = \frac{q}{C} = L\frac{di}{dt} + Ri \text{ or } L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0 \qquad \left(\because i = -\frac{dq}{dt}\right) \qquad ...(9)$$

Here L, R, $1/C$ play the role of mass (m), damping (γ) and force constant (K) respectively.

Eq. (9) can be written as

$$\frac{d^2q}{dt^2} + 2b\frac{dq}{dt} + \omega_0^2 q = 0 \qquad ...(10)$$

where $b = R/2L$ and $\omega_0^2 = 1/LC$.

Eq. (10) is exactly like the differential equation for the damped harmonic oscillator. The discharge will be oscillatory, when

$$\omega_0^2 > b^2 \quad \text{or} \quad \frac{1}{LC} > \frac{R^2}{4L^2}.$$

The solution of eq. (10) is of the form

$$q = q_0\, e^{-(R/2L)t} \sin(\omega t + \phi) \qquad ...(11)$$

We note that the charge amplitude, $q_0 e^{-(R/2L)t}$, decays exponentially with time. The decay in charge amplitude is caused by the resistance R, because in absence of R (*i.e.*, $R = 0$), the charge amplitude will remain constant (q_0). Hence in a LCR circuit, resistance is only the dissipative element.

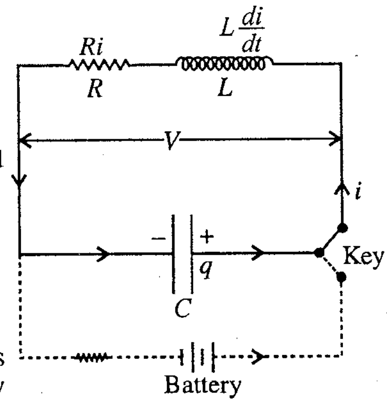


Fig. 10.3 : *LCR* circuit.

The frequency of the oscillatory discharge is given by

$$\nu = \frac{\omega}{2\pi} = \frac{\sqrt{\omega_0^2 - b^2}}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \qquad ...(12)$$

When $R = 0$, we obtain the case of harmonic charge oscillations and the frequency of oscillations, $\nu_0 = 1/2\pi\sqrt{LC}$ for LC circuit. Thus the presence of resistance in the circuit decreases the frequency.

***Q*-Factor in *LCR* circuit :** For low resistance circuit,

$$Q = \omega_0\tau = \frac{\omega_0}{2b} = \frac{L\omega_0}{R} \qquad ...(13)$$

For pure LC circuit $R = 0$, hence $Q = \infty$.

Now, if we differentiate eq. (11) with respect to t and place $-dq/dt = i$, then on arranging the differentiated terms, we get

$$i = i_0 e^{-bt} \sin(\omega t + \theta) \qquad ...(14)$$

where I_0 is the maximum value of the circuit.

Cases when $\omega_0^2 \le b^2$ or $\frac{1}{LC} \le \frac{R^2}{4L^2}$

In case $\omega_0^2 < b^2$ or $\frac{1}{LC} < \frac{R^2}{4L^2}$, the circuit is overdamped and the discharge will be **non-oscillatory.**

Further, the case, when $\omega_0^2 = b^2$ or $\frac{1}{LC} = \frac{R^2}{4L^2}$ represents **critical damping** or **just aperiodic.**

Ex. 1 *The amplitude of vibration of a spring-mass system decreases from* 10 *cm, to* 2.5 *cm in* 200 *seconds. Find the damping constant and relaxation time. In how much time does the amplitude falls to* 1/*e times the initial value ?*

If the ocillator performs 100 *oscillations in the given time, compare the periods with and without damping. What do you conclude ?*

Sol. The amplitude at an instant t of a damped oscillator is given by

$$a = a_0 e^{-bt} \text{ or } e^{bt} = \frac{a_0}{a} \text{ or } b = \frac{1}{t}\log_e \frac{a_0}{a}$$

Here, t = 200 sec., a_0 = 10 cm and a = 2.5 cm,

$$\therefore \qquad b = \frac{1}{200}\log_e \frac{10}{2.5} = \mathbf{6.93 \times 10^{-3}\ sec^{-1}}; \ \tau = \frac{1}{2b} = \frac{1}{2 \times 6.93 \times 10^{-3}} = \mathbf{72\ sec.}$$

Now, for $\frac{a}{a_0} = \frac{1}{e}$, $\qquad t = \frac{1}{b}\log_e e = \frac{1}{b} = \frac{1}{6.93 \times 10^{-3}} = \mathbf{144\ sec.}$

$$\text{Period, } T = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}} = \frac{200}{100} = 2 \text{ sec.}$$

Now, $$2 = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}}, \ \therefore \omega_0^2 = \pi^2 + b^2 = \pi^2 + (6.93 \times 10^{-3})^2 \text{ or } \omega_0 = 3.1416$$

$$\therefore \qquad T_0 = 1.9999 \text{ sec} \simeq \mathbf{2\ sec.}$$

Note that the change in the time period due to damping is negligible.

Ex. 2. *A damped vibrating system starting from rest reaches a first amplitude of* 500 *mm, which reduces to* 50 *mm after* 100 *oscillations, each of period* 2.3 *seconds. Find the damping constant, relaxation time and correction for the first displacement for damping.*

(Gwalior 2004; Indore 03; Rewa 04; Sagar 1999; Meerut 94)

Sol. The amplitude of damped vibrations is given by

$$a = a_0 e^{-bt}$$

Time period, T = 2.3 sec.

First amplitude of the system will be at $t = T/4$ and the amplitude after 100 oscillations *i.e*, 201th amplitude will occur at $(100T + T/4)$ time, hence

$$a_1 = a_0 e^{-bT/4} \text{ and } a_{201} = a_0 e^{-b(100T + T/4)};$$

$$\therefore \qquad \frac{a_{201}}{a_1} = e^{-100bT} \text{ or } \frac{50}{500} = e^{-100 \times 2.3b} \text{ or } 10^{-1} = e^{-230b} \quad \text{or} \quad \log_e 10 = 230b,$$

$\therefore$ Damping constant, $\quad b = 2.3/230 = \mathbf{0.01\ sec^{-1}}$

$$\therefore \qquad \text{Relaxation time, } \tau = \frac{1}{2b} = \frac{1}{2 \times 0.01} = \mathbf{50\ sec.}$$

$$\because \qquad a_1 = a_0 e^{-bT/4}, \quad \therefore \quad 500 = a_0 e^{-0.01(2.3/4)}$$

or $$a_0 = 500\, e^{0.023/4} \text{ or } \log_{10} a_0 = \log_{10} 500 + \frac{0.023}{4}\frac{1}{2.3}$$

or $$\log_{10} a_0 = 2.699 + 0.0025 = 2.7015, \text{ whence } a_0 = \mathbf{502.9\ mm.}$$

Thus in absence of damping, the amplitude of motion would have been **502.9 mm.**

Ex. 3. *The differential equation for a certain system is*

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + \omega_0^2 x = 0$$

If $\omega_0 >> k$, *find the time in which*

(i) *amplitude falls to* 1/*e times the initial value.*

(ii) *energy of the system falls to* 1/*e times the initial value,*

(iii) *energy falls to* $1/e^4$ *times the initial value.*

Sol. (i) The given differential equation represents the equation of motion of damped harmonic oscillator, whose amplitude of vibration is given by

$$a = a_0 e^{-bt}$$

When amplitude falls to $1/e$ times the initial value a,

$$a_0 e^{-bt} = a_0/e \text{ or } e^{bt} = e^1 \quad \text{or} \quad bt = 1, \ \therefore \quad t = \mathbf{1/b\ sec.}$$

(ii) Energy, $E_t = E_0\, e^{-2bt} = E_0/e \quad \therefore \quad e^{2bt} = e$ or $2bt = 1$ or $t =$ **1/2b sec.**

(iii) $E_t = E_0\, e^{-2bt} = E_0/e^4$ or $2bt = 4$ or $t =$ **2/b sec.**

Ex. 4. *A damped harmonic oscillator has its first amplitude of 20 cm. It reduces to 2 cm after 100 oscillations, each of period 4.6 sec. Calculate the logarithmic decrement and damping constant. Find the number of oscillations in which the amplitude drops to 50%.*

Sol. $$\lambda = \frac{1}{m}\log_e \frac{a_n}{a_{n+m}} = \frac{1}{100}\log_e \frac{20}{2} = \frac{1}{100}\log_e 10 = \frac{2.3}{100}\log_{10} 10 = \mathbf{2.3 \times 10^{-2}}.$$

$$\lambda = \frac{bT}{2}, \quad \therefore \quad b = \frac{2\lambda}{T} = \frac{2\times 2.3\times 10^{-2}}{4.6} = \mathbf{10^{-2}\ sec^{-1}}.$$

$$m = \frac{1}{\lambda}\log_e \frac{a_n}{a_{n+m}} = \frac{1}{2.3\times 10^{-2}} \times 2.3 \times \log_{10} 2 = 10^2 \times 0.3 = \mathbf{30}.$$

Ex. 5. *The frequency of a tuning fork is 300 Hz. If its quality factor Q is 5×10^4, find the time after which its energy becomes 1/10 of its initial value. What is the value of relaxation time for the oscillator ?* **(Bhopal 1997; Bilaspur 97; Rewa 95; Sagar 90)**

Sol. $E = E_0\, e^{-2bt} = E_0 e^{-t/\tau}$. When energy becomes 1/10 of its initial value,

$$E_0/10 = E_0 e^{-t/\tau} \text{ or } 10 = e^{t/\tau} \text{ or } t/\tau = \log_e 10 = 2.3, \ \therefore\ t = 2.3\tau$$

But $$\tau = \frac{Q}{\omega} = \frac{Q}{2\pi n} = \frac{5\times 10^4}{2\times 3.14\times 300}, \ \therefore t = \frac{2.3\times 5\times 10^4}{2\times 3.14\times 300} = \mathbf{61.1\ sec.}$$

Ex. 6. *A box of 100 gm is attached to one end of a spring whose other end is fixed to a rigid support. When a mass of 900 gm is placed inside the box the system performs 4 vibrations per second and the amplitude falls from 2 cm to 1 cm in 15 seconds. Calculate* (i) *the force constant,* (ii) *the relaxation time,* (iii) *Q of the system and* (iv) *Q when only 100 gm (empty box) is with the spring.* ($\log_e 10 = 2.3$)

Sol. (i) Frequency of vibration of the system, $\nu = 4\ \text{sec}^{-1}$,

$\therefore$ $$\omega_0 = 2\pi\nu = 2 \times 3.14 \times 4 = 25 \text{ rad./sec.} \qquad \text{(if the damping is small)}$$

But, $$\omega_0 = \sqrt{C/m} \text{ or } C = m\,\omega_0^2 = 1 \times 25^2 = \mathbf{625\ N\text{-}m.}$$

(ii) At any instant t, the amplitude of the system is given by

$$a = a_0\, e^{-bt} \text{ or } 0.01 = 0.02\, e^{-15b} \text{ or } \frac{1}{2} = e^{-15b} \text{ or } 15b = \log_e 2$$

$\therefore$ $$b = \frac{\log_e 2}{15} = \frac{2.3\times 0.3}{15} = \mathbf{0.046\ sec^{-1}}$$

$\therefore$ Relaxation time, $$\tau = \frac{1}{2b} = \frac{1}{2\times 0.046} = \mathbf{11\ sec.}$$

(iii) $$Q = \omega_0\tau = 25 \times 11 = \mathbf{275}.$$

Damping force $= \gamma\, dx/dt$, where $\gamma = 2\,bm = 2 \times 0.046 \times 1$.

(iv) When the box is empty, *i.e.*, the mass is 100 gm, the damping force $\gamma\, dx/dt$ and restoring force Cx remain unchanged. Now

$$\omega_0 = \sqrt{\frac{C}{m'}} = \left(\frac{625}{0.1}\right)^{\frac{1}{2}} = 79 \text{ rad./sec. and } \tau = \frac{1}{2b} = \frac{m'}{\gamma} = \frac{0.1}{2\times 0.046\times 1} = 1.1 \text{ sec.}$$

$\therefore$ $$Q = \omega_0\tau = 79 \times 1.1 = \mathbf{87}.$$

Ex. 7. *If the suspension of a galvanometer coil exerts a restoring couple 5×10^{-5} N-m/rad, the period is 6.28 sec and the amplitude decreases to 1/10 of its original value in 92 seconds, find the value of I, γ and Q factor.*

Sol. Period, $$T = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}} = 6.28 \text{ or } \omega_0^2 - b^2 = 1 \qquad \text{...(i)}$$

Now, $\dfrac{a}{a_0} = e^{-bt}$, $\therefore$ here $\dfrac{1}{10} = e^{-92b}$ or $\log_e 10 = 92b$; $\therefore b = 2.3/92 = 1/40$.

Substituting this value of b in relation (i), we have $\omega_0^2 - (1/40)^2 = 1$ or $\omega_0 = 1$ nearly.

Now, $\omega_0^2 = C/I$, $\therefore 1 = 5 \times 10^{-5}/I$, $\therefore I =$ $\mathbf{5 \times 10^{-15}\ kg \times m^2}$;

and $$\frac{\gamma}{I} = 2b \text{ or } \frac{\gamma}{5\times 10^{-5}} = 2 \times \frac{1}{40}, \; \therefore \gamma = \mathbf{2.5 \times 10^{-6}} \textbf{ N-m sec.}$$

$$\therefore \quad Q = \frac{\omega_0}{2b} = \frac{1}{2\times 1/40} = \mathbf{20}.$$

Ex. 8. *A condenser of 2 microfarad capacity is discharged through a 1 ohm resistance and 2 henry inductance. Show that the system will act as damped harmonic oscillator. Calculate the frequency and quality factor of LC circuit and find the time in which the amplitude of oscillations is reduced to 1%.* **(Indore 1998)**

Sol. $$\nu = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

Here, $C = 2 \times 10^{-6}$ farad, $L = 2$ henry and $R = 1$ ohm.

$$\therefore \quad \nu = \frac{1}{2\pi}\sqrt{\frac{1}{2\times 2\times 10^{-6}} - \frac{1}{16}} = \frac{10^3}{2\pi \times 2} = \mathbf{79.5\ Hz}.$$

As resistance is low, so

$$Q = \frac{L\omega}{R} = \frac{2\times 2\pi}{1} \times \frac{10^3}{2\pi \times 2} = \mathbf{10^3}$$

The amplitude, $q = q_0 e^{-R/2L}$

As amplitude decreases to 1% of its initial value q_0, we have

$$\frac{q_0}{100} = q_0 e^{-t/4} \text{ or } e^{t/4} = 100 \text{ or } \frac{t}{4} = \log_e 100 = 2.3 \times 2 \text{ or } t = \mathbf{18.4\ sec.}$$

Ex. 9. *In a circuit, $L = 2$ mh and $C = 5\ \mu F$. If $R = 1\ \Omega$, $40\ \Omega$, and $100\ \Omega$, discuss the behaviour of the circuit in each case. Find the frequency of oscillation and Q-factor in the relevant cases.*

Sol. Here, $L = 2 \times 10^{-3}$ h and $C = 5 \times 10^{-6}$ F

$$\therefore \quad \frac{1}{LC} = \frac{1}{2\times 10^{-3}\times 5\times 10^{-6}} = 10^8$$

Case I : $R = 1\ \Omega$, $\therefore$ $$\frac{R^2}{4L^2} = \frac{1}{4\times(2\times 10^{-3})^2} = 6.25 \times 10^4$$

$\therefore \frac{1}{LC} > \frac{R^2}{4L^2}$ and hence the **circuit is oscillatory.**

$$\nu = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \mathbf{1.59\ KHz}.$$

and $$Q = \frac{\omega_0 L}{R} = \frac{2\pi \times 1.59\times 10^3 \times 2\times 10^{-3}}{1} = \mathbf{19.98}$$

Case II : $R = 40\ \Omega$, $\therefore$ $$\frac{R^2}{4L^2} = \frac{40\times 40}{4\times(2\times 10^{-3})^2} = 10^8$$

Hence $$\frac{1}{LC} = \frac{R^2}{4L^2} \text{ (or } \omega_0^2 = b^2)$$

This is the case of **cirtical damping.**

Case III : $R = 100\ \Omega$, $\therefore$ $$\frac{R^2}{4L^2} = \frac{100^2}{4\times(2\times 10^{-3})^2} = 6.25 \times 10^8$$

Therefore $$\frac{R^2}{4L^2} > \frac{1}{LC} \text{ (or } b^2 > \omega_0^2)$$

This is the case of **overdamping.**

EXERCISE 10·1

1. The time period of pendulum is 2 sec. If the angular retardation due to air is 0.04 time angular velocity of pendulum and original amplitude is 1°, calculate the amplitude after 10 oscillations.
[**Ans.** $40'$ approx.]
2. A simple pendulum has a period of 2 sec. and amplitude of 2°. After 10 complete oscillations its amplitude is reduced to 1.5°. Find the relaxation time and damping constant. **(Agra 1989)**
[**Ans.** 34.76 sec; 0.01438 sec^{-1}.]
3. The quality factor of a tuning fork is 6×10^4 and its frequency is 512 Hz. In what time its energy will become 1/10 of its initial value ?
[**Ans.** 42.94 sec.]
4. The differential equation of an oscillating system is expressed as : $\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + n^2x = 0$.
If $b << n$, find the time in which : (i) the amplitude of oscillation falls to $1/e$ of its initial value, (ii) the energy falls to $1/e$ of its initial value. **(Bilaspur 1996; Sagar 96)**
[**Ans.** (i) $(1/b)$ sec. (ii) $(1/2b)$ sec.]
5. The amplitude of an oscillator of frequency 200 hertz reduces to 1/10th of its initial amplitude after 2000 oscillations. Calculate the quality factor, relaxation time and damping constant.
(Bilaspur 2003; Raipur 03;Indore 1997; Rewa 94)
[**Ans.** 2.73×10^3; 2.17 sec.]
6. The quality factor of a tuning fork of frequency 384 Hz is 10^4. How much time will be taken in reducing the energy $1/e$ of its initial value ? How many oscillations take place in this duration ? **(Sagar 1998)**
[**Ans.** 4.1 sec.; 1574.47.]
7. The period of vibration of a galvanometer coil about the suspension fibre is 4 seconds. The amplitude of vibration decreases to 1/10 of its original value in 46 seconds, find the value of I, γ and Q factor. Take the restoring couple per unit twist $C = 10^{-5}$ N-m/rad.
[**Ans.** $I = 2.45 \times 10^{-5}$ kg-m^2, $\gamma = 2.45 \times 10^{-6}$ N-m-sec, $Q = 6.4$.]
8. Q of a oscillatory circuit is 100 and the frequency is 200 Hz. Calculate the change in frequency when the circuit resistance is reduced to zero.
[**Ans.** 2.5 Hz (increase).]
9. Show that the damping reduces the frequency of an oscillator by $12.5/Q^2$ per cent, where Q is the quality factor.
[**Hint :** Angular frequency of undamped oscillator $= \omega_0$
Angular frequency of damped oscillator $= \omega = \sqrt{\omega_0^2 - b^2}$

$$\therefore \frac{\omega}{\omega_0} = \frac{\sqrt{\omega_0^2 - b^2}}{\omega_0} = \sqrt{\frac{\omega_0^2 - b^2}{\omega_0^2}} = \left(1 - \frac{b^2}{\omega_0^2}\right)^{\frac{1}{2}} = \left(1 - \frac{1}{4Q^2}\right)^{\frac{1}{2}} \qquad \left[\because Q = \omega_0 \tau = \frac{\omega_0}{2b} \text{ or } \frac{b}{\omega_0} = \frac{1}{2Q}.\right]$$

As b is small in comparison of ω_0, hence using binomial expansion

$$\frac{\omega}{\omega_0} = 1 - \frac{1}{8Q^2} \text{ or } 1 - \frac{\omega}{\omega_0} = \frac{1}{8Q^2} \text{ or } \frac{\omega_0 - \omega}{\omega_0} = \frac{1}{8Q^2}$$

$\therefore$ Percentage decrease in frequency due to damping $= \frac{\omega_0 - \omega}{\omega_0} \times 100\% = \frac{100}{8Q^2}\% = 12.5\%$.]
10. The quality factor of a sonometer wire is 2×10^3. On plucking it makes 240 vibrations per second. Calculate the time in which the amplitude decreases to half the initial value.
[**Ans.** 1.8 sec.]

11. A spherical ball of radius 0.5 cm and mass 0.4 gm oscillates in water under the action of a spring of force constant $K = 5 \times 10^{-2}$ N/m. The coefficient of viscosity for water is 0.1 S.I. units. Find the number of oscillations that will occur in the time for the amplitude to drop to 1/2 of initial value. What is Q-factor of the oscillator ?
[**Ans.** 15.8; 66.95.]
12. A box of mass 0.2 kg is attached to one end of a spring whose other end is fixed to a rigid support. When a mass of 0.8 kg is placed inside the box, the system performs 4 oscillations per second and the amplitude falls from 2 cm to 1 cm in 30 seconds. Calculate (i) the force constant; (ii) the time constant; (iii) Q-factor; and (iv) Q-factor, when only 0.2 kg (empty box) is with the spring.
[**Ans.** (i) 625 N/m; (ii) 22 sec.; (iii) 550; (iv) 243.]
13. The viscous force on a sphere of radius r moving with velocity v in a medium of coefficient of viscosity η is $-6\pi\eta rv$. Pendulum consists of aluminium sphere of radius 0.005 m suspended from a string of length 1 meter. Determine how the air viscosity affects the amplitude and period. Density of Al = 2,650 kg/m^3, coefficieint of viscosity $\eta = 1.78 \times 10^{-5}$ S.I. units.

[**Ans.** $a = a_0 e^{-0.000605\,t}$ and $T = \dfrac{2\pi}{\sqrt{g/l - k^2}} = \dfrac{2\pi}{\sqrt{9.8 - 3.65 \times 10^{-7}}} = \dfrac{2\pi}{\sqrt{9.8}}$.

Thus, the viscosity of the air practically does not affect the period, although it affects amplitude.]
14. Give the theory of oscillations in series LCR circuit with small damping. Deduce the frequency and quality factor for a circuit with $L = 2$ mh, $C = 5$ μF and $R = 0.2$ ohm. **(Indore 1998; Agra 71)**
[**Ans.** 1.59×10^3 Hz; 100.]
15. In a LCR circuit $L = 10$ mh, $C = 1$ μF and $R = 0.1$ ohm. How long a time does the charge oscillation take to decay half amplitude ? What is the quality factor for the circuit ?
[**Ans.** 0.14 sec; 1000.]
16. In a LCR circuit, $L = 0.3$ henry and $C = 30$ μF. Find the maximum value of the resistance which will make the series circuit to be oscillatory. **(Jabalpur 1997)**
[**Ans.** 200 Ω]
17. A condenser of 3 microfarad capacity is discharged through a 2 ohm resistance and 4 henry inductance. Show that the discharge is oscillatory. Calculate the frequency and Q-factor of the circuit. Find the time in which the charge amplitude becomes 1% of the initial value.
[**Ans.** The discharge is oscillatory; 18.4 sec.]

[**Hint.** : If $\dfrac{1}{LC} > \dfrac{R^2}{4L^2}$, the discharge is oscillatory.]

QUESTIONS

Long Answer Questions

1. What is meant by a damped harmonic oscillator ? Write the differential equation for it and find its solution. **(Gwalior 2003; Indore 03, Bhopal 04; Jabalpur 04, 1998; Ujjain 04, 99; Osmania 2005)**
2. Solve the differential equation of damped harmonic oscillator with all cases and discuss in detail the underdamped case. **(Agra 1994, 86, 85)**
3. What do you understand by the relaxation time, damping constant and quality factor of a damped harmonic oscillator ? Obtain their expressions. **(Jabalpur 2004)**
4. Obtain the expressions of average total energy and average power loss for a damped harmonic oscillator. **(Raipur 2003; Bhopal 04)**
5. If the damping constant $b << \omega_0$, prove that

(a) the average total energy $E = \frac{1}{2} m\omega^2 a_0^2 e^{-2bt}$.

(b) average power dissipation $P = E/\tau$.

(c) average rate of doing work by the damping force $-\gamma\, dx/dt$ is equal to the average power dissipation (*i.e.* to E/τ). **(Bundelkhand 2004)**

6. A particle is oscillating under a damping force. Show that the average power loss (P) is given by $P = E/\tau$, where E is the average energy and τ is the relaxation time. What happens to the dissipated energy ? **(Ajmer 1980)**
7. A pendulum consists of bob of radius r suspended from a string of length l. Find how the air viscosity affects the amplitude and its period ?

   [**Hint :** We know that damping force $= -6\pi\eta rv = -6\pi\eta rl\,(d\theta/dt)$

   $\therefore$ Eq. of motion $ml\ddot{\theta} + 6\pi\eta rl\dot{\theta} + mg\theta = 0$ or $\ddot{\theta} + (6\pi\eta rl/m)\dot{\theta} + (g/l)\theta = 0$

   $$\therefore \quad b = \frac{3\pi\eta r}{m} = \frac{3\pi\eta r}{(4/3)\pi r^3\rho} = \frac{9\eta}{4r^2\rho} \text{ and } \omega_0 = \sqrt{\frac{g}{l}}$$

   Hence amplitude, $a = a_0 e^{-bt}$ and $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{\sqrt{\omega_0^2 - b^2}}$.]
8. Show that when damping is low, a galvanometer coil executes oscillatory motion. What is a ballistic galvanometer ? If the first throw of the galvanometer is known, then how will you make correction for the effect of damping ?
9. Show that the discharge of a condenser through an inductance and resistance (not high) is oscillatory. Find an expression for its frequency.
10. Describe all cases of LCR circuit for damped vibration and calculate the frequency of the circuit. **(Kanpur 2005)**

Short Answer Questions

1. Write the differential equations for the following cases of damped harmonic oscillator : (i) mechanical, (ii) electrical. **(Rajasthan 1984)**

   [**Hint :** (i) $m\dfrac{d^2x}{dt^2} + \dfrac{\gamma\,dx}{dt} + Cx = 0$. (ii) $\dfrac{d^2q}{dt^2} + R\dfrac{dq}{dt} + \dfrac{q}{C} = 0$.]
2. Define damping constant and relaxation time.
3. How does the displacement change with time in case of a low damped harmonic oscillator ? Draw displacement versus time graph in such a case.
4. What is logarithmic decrement ? **(Osmania 2005)**
5. Discuss the overdamped case of an oscillator. Where is this case realized in practice ?
6. What do you expect regarding the period of damped harmonic oscillator to increase or decrease relative to the period of the oscillator in absence of damping ?

   [**Ans.** The damping forces resist the motion of the oscillator. Hence the period is expected to increase.]
7. The differential equation for damped harmonic oscillator is $\dfrac{d^2x}{dt^2} + 2b\dfrac{dx}{dt} + \omega_0^2 x = 0$

   If $\omega_0 > b$ find the time in which amplitude falls to $1/e$ times the initial value. **(Kanpur 2000)**

   [**Ans.** $1/b$ sec.]

   [**Hint :** $a = a_0 e^{-bt}$ or $\dfrac{a_0}{a} = \dfrac{1}{e} = e^{-bt}$ or $e^{-1} = e^{-bt}$ or $bt = 1$ or $t = 1/b$ sec.]
8. Does the damping increase or decrease the period of harmonic oscillations ? Explain.
9. What is meant by 'Q' of an oscillator ? How is it affected by damping ? **(Osmania 2005; Raj. 1984)**
10. The value of quality factor is high for small damping. Explain this statement.
11. Show that the energy of oscillation in a damped harmonic oscillator decreases to e^{-1} of its initial value in the time τ and in this time, the oscillator performs $Q/2\pi$ oscillations.

    [**Note :** This is why τ is sometimes called decay time.]

    [**Hint :** $E = E_0\, e^{-t/\tau}$. In time $t = \tau$, $\dfrac{E}{E_0} = e^{-1}$ or $\dfrac{E}{E_0} = \dfrac{1}{e}$

    $$Q = \omega_0\tau \text{ or } \tau = \frac{Q}{\omega_0},\ T = \frac{2\pi}{\omega} = \frac{2\pi}{\omega_0} \text{ (for low damping)}$$

    $$\therefore \quad \text{Number of oscillations in } \tau \text{ time} = \frac{\tau}{T} = \frac{Q}{\omega_0}\frac{\omega_0}{2\pi} = \frac{Q}{2\pi}.]$$

12. Show that for low damping a galvanometer coil executes S.H.M. **(Kanpur 2005)**

13. Write down the expression for quality factor in case of an oscillating LCR circuit. How does the inductance influence the quality factor ?

Objective Type Questions

1. At small velocities the damping force is :
(a) directly proportional to the square of velocity.
(b) directly proportional to square root of velocity.
(c) directly proportional to velocity.
(d) inversely proportional to velocity.

2. Consider the damped harmonic oscillator governed by the law $\frac{d^2x}{dt^2}+2k\frac{dx}{dt}+\omega_0^2 x = 0$
The condition for overdamping is :
(a) $k < \omega_0$ (b) $k = \omega_0$ (c) $k > \omega_0$ (d) $k = \omega_0^2$ **(Agra 2005, 04)**

3. If the damping of oscillatory motion is zero, then its relaxation time will be :
(a) zero (b) infinite (c) m/k (d) ω **(Agra 2004)**

4. In case of underdamped oscillations, the amplitude of oscillation of an object with time :
(a) decreases (b) increases
(c) remains constant (d) may decrease or increase

5. The relation of the energy of damped harmonic oscillator is :
(a) $E = E_0 e^{-t/\tau}$ (b) $E = E_0 e^{-2t/\tau}$ (c) $E = E_0 e^{-t/2\tau}$ (d) $E = E_0 e^{t/\tau}$

6. $\frac{d^2y}{dt^2}+2b\frac{dy}{dt}+\omega_0^2 y = 0$ is the equation of a damped harmonic oscillator. The oscillation, will be underdamped, if :
(a) $b > \omega_0$ (b) $b = \omega_0$ (c) $b < \omega_0$ (d) $b >> \omega_0$ **(Rewa 1994)**

7. The relation between quality factor Q and relaxation time τ of an oscillator is :
(a) $Q = \omega/\tau$ (b) $Q = \tau/\omega$ (c) $Q = \omega\tau$ (d) $Q = 1/\omega\tau$ **(Kanpur 2004)**

8. The meaning of high quality factor of an oscillator is :
(a) damping is more (b) damping is infinite
(c) damping is zero (d) damping is small

9. The quality factor of L-C-R damped oscillator of zero resistance is given by :
(a) $R/L\omega$ (b) ω/RL (c) $L\omega/R$ (d) $L\omega R$ **(Rewa 1997**

10. If the quality factor of the oscillator is large, it implies that :
(a) damping is high (b) damping is low **(Bundlkhan 2004)**
(c) damping is infinite (d) damping is zero.

11. The amplitude of damped simple harmonic oscillator :
(a) remains constant
(b) decreases exponentially
(c) suddenly becomes zero without any oscillation
(d) increases exponentionally **(Gwalior 1991)**

Answers

1. (c)	**2.** (c)	**3.** (b)	**4.** (a)	**5.** (a)	**6.** (c)	**7.** (c)	**8.** (d)	**9.** (c)	**10.** (b)	**11.** (b)

11

Driven Harmonic Oscillator and Resonance

11·1. Introduction

In the last chapter, we have seen that the free oscillations of a physical system die out with time due to the damping forces. The oscillations of the system can be maintained, if an external periodic force is applied to it. In this chapter, we plan to investigate the behaviour of an oscillating system under the application of a harmonic periodic force and study the response* of the system as a function of frequency of the force. In fact this problem is of great practical importance in various branches of physics. If we apply a harmonically oscillating force to a damped harmonic oscillator at a frequency (p), which is, in general, different to the natural frequency of the oscillator, in the beginning we obtain the transient response *i.e.*, in the initial stages, the oscillator tends to vibrate with its natural frequency (ω), while the impressed periodic force tries to impose its own frequency (say p) upon it and hence a sort of tussle takes place between the two. When the transient effects have died away, the oscillator ultimately acquires the steady state and vibrates at the frequency (ω) of the applied harmonic force and not at its natural frequency (ω). Such oscillations of the system at the frequency of the applied periodic force are called **forced** or **driven oscillations.** The external applied harmonic force is called the **driver** and the oscillating system itself is called the **driven** or **forced oscillator**. The driving harmonic force delivers periodic impulses to the oscillator so that the loss of energy in doing work against the damping forces is recovered. Consequently in the steady state, the oscillator vibrates with constant amplitude. If we vary slowly the driving frequency (p), we obtain the **resonant response** at a frequency (p_r) which is equal or close to the natural frequency (ω) of the oscillator. In this chapter, first we shall establish the differential equation of driven harmonic oscillator and obtain the steady state solution. We shall then study the response of the oscillator at various frequencies of the driving force and discuss in detail the resonant response. Next we shall deal with the driven LCR circuit as an important example of forced oscillator. Finally, we shall discuss the transient state and the interesting phenomenon of transient beats between the driving force and the free oscillations.

11·2. Theory of Driven (Forced) Harmonic Oscillator

If an external periodic force is applied on a damped harmonic oscillator, then the oscillating system is called **driven** *or* **forced oscillator** *and its oscillations are called* **forced** *(or* **driven***)* **vibrations.** *The applied periodic force is called* **driver.** *Let the periodic force represented by* $F = F_0 \sin pt$*, where* F_0 *is the maximum value of the force of frequency* $p/2\pi$. Hence the differential equation of motion is

$$m\frac{d^2x}{dt^2} = -\gamma\frac{dx}{dt} - Cx + F$$

where m is the mass of the oscillating particle, $-\gamma\, dx/dt$ is the damping force and $-Cx$ is the restoring force. Now, the above equation can be written as

$$\frac{d^2x}{dt^2} + \frac{\gamma}{m}\frac{dx}{dt} + \frac{C}{m}x = \frac{F_0}{m}\sin pt$$

or

$$\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + \omega_0^2 x = f_0 \sin pt \qquad ...(1)$$

where $2b = \gamma/m = 1/\tau$, $\omega_0 = \sqrt{C/m} = 2\pi\nu_0$ = the natural angular frequency of the system in the absence of damping and driven force, and $f_0 = F_0/m$.

* We can mean by response either the displacement (x) or velocity (dx/dt).

Eq. (1) represents the *differential equation of motion for the driven harmonic oscillator.*

Solution of the Differential Equation of the Driven Harmonic Oscillator : Steady State Solution

Let the seteady state solution of the driven harmonic oscillator be

$$x = a \sin (pt - \theta) \qquad ...(2)$$

because the amplitude (a) of the forced oscillations remains constant and the frequency of vibrations is equal to the ($p/2\pi$) of the force. Here θ represents the phase difference between the force and the resultant displacement of the system. Now, from eq. (3), we have

$$\frac{dx}{dt} = pa \cos (pt - \theta) \text{ and } \frac{d^2x}{dt^2} = -p^2 a \sin (pt - \theta)$$

Substituting these values in eq. (1), we get

$$-p^2 a \sin (pt - \theta) + 2bpa \cos (pt - \theta) + \omega_0^2 a \sin (pt - \theta) = f_0 \sin (pt - \theta + \theta)$$

or $\quad a(\omega_0^2 - p^2) \sin (pt - \theta) + 2bpa \cos (pt - \theta) = f_0 \cos \theta \sin (pt - \theta) + f_0 \sin \theta \cos (pt - \theta) \qquad ...(3)$

This relation holds for all the values of t. Hence when $(pt - \theta) = 0$ or $2m\pi$, then from eq. (3), we have

$$2bpa = f_0 \sin \theta \qquad ...(4)$$

and when $(pt - \theta) = \pi/2$ or $(2m + 1)\pi/2$, then

$$(\omega_0^2 - p^2)a = f_0 \cos \theta \qquad ...(5)$$

Squaring eqs. (4) and (5) and on adding, we get

$$4b^2p^2a^2 + (\omega_0^2 - p^2)^2a^2 = f_0^2 (\sin^2\theta + \cos^2\theta) \text{ or } [4b^2p^2 + (\omega_0^2 - p^2)]a^2 = f_0^2$$

$$\therefore \quad a = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2p^2}} = \frac{F_0/m}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2p^2}} \qquad ...(6)$$

Dividing eq. (4) by eq. (5), we get

$$\tan \theta = \frac{2bp}{\omega_0^2 - p^2} \text{ or } \theta = \tan^{-1}\left(\frac{2bp}{\omega_0^2 - p^2}\right) \qquad ...(7)$$

Substituting these values in eq. (2), we get

$$x = \frac{F_0/m}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2p^2}} \sin (pt - \theta) \qquad ...(8)$$

where θ is given by eq. (7)

Thus the amplitude of forced oscillations not only depends on force amplitude (F_0) and mass of the oscillating body (m) but also on the driving frequency (p), natural frequency (ω_0) of the oscillator (in absence of damping) and the damping constant (b). The phase of the displacement lags that of the harmonic force by angle θ which is also depending on p, ω_0 and b. The reader may note that we may sometime use word frequency for angular frequency. Further note that the steady state solution has the frequency of the driving force and its amplitude is constant.

In case of undamped system, $b = 0$ so that from eq. (7), $\theta = 0$ and therefore from eq. (8)

$$x = \frac{F_0/m}{\omega_0^2 - p^2} \sin pt \qquad ...(9)$$

Thus *for a simple harmonic oscillator,* when a periodic force $F = F_0 \sin pt$ is applied, *the driven force and displacemet* (x) *are in the same phase* (as $\theta = 0$). Hence we conclude that **phase lag is essentially a consequence of damping.**

Variation of Amplitude (a) and Phase (θ) with the Driving Frequency (p)

We consider below the amplitude (a) and phase lag (θ) of the oscillator at different driving frequencies in the case of low damping.

Case I : Low driving frequency ($p << \omega_0$) : When the driving frequency (p) is much less than the natural frequency (ω_0) of the oscillator (in absence of damping), we obtain from eq. (6)

$$a \to \frac{f_0}{\omega_0^2} = \frac{F_0/m}{C/m} = \frac{F_0}{C} \text{ and } \tan\theta \to 0 \text{ or } \theta \to 0 \qquad ...(10)$$

Thus at the low driving frequency, the amplitude does not depend on the mass (m) of the oscillating system or damping but it is controlled by the spring factor (C). In this limit, the periodic force displaces slowly the system against the restoring force of the spring and the displacement is said to be in phase with the driving force.

Case II : Resonant response : Now, *if the driving frequency is increased,* the value of the amplitude a increases. To find the condition for maximum amplitude (response), we differentiate eq. (6) with respect to p and put $da/dp = 0$ *i.e.,*

$$\frac{da}{dp} = \left(\frac{F_0}{m}\right)\left(-\frac{1}{2}\right)\frac{2(\omega_0^2 - p^2)(-2p) + 8b^2 p}{[(\omega_0^2 - p^2)^2 + 4b^2p^2]^{\frac{3}{2}}} = 0 \text{ or } 2b^2 - \omega_0^2 + p^2 = 0$$

Hence $$p = \sqrt{\omega_0^2 - 2b^2} = p_r \text{ (say)} \qquad ...(11)$$

For this value of p, d^2a/dp^2 is found to be negative. Hence the amplitude a is maximum at $p = p_r$, given by eq. (11).

At $p_r = \sqrt{\omega_0^2 - 2b^2}$, a is maximum, say a_{max} amplitude, and from eq. (6),

$$a_{max} = \frac{f_0}{2b\sqrt{\omega_0^2 - b^2}} \qquad ...(12)$$

Thus the amplitude of the oscillating system is maximum at a particular value of the driving frequency (p_r).

This phenomenon in which the amplitude of the driven oscillator is maximum at a certain driving frequency (p_r) is called **amplitude resonance** *and this frequency is called as* **amplitude.**

Resonant frequency : The value of this angular frequency ($p_r = \sqrt{\omega_0^2 - 2b^2}$) is less than the natural frequency of undamped or damped vibrations (ω_0 or $\omega = \sqrt{\omega_0^2 - b^2}$) of the oscillator. However, we are considering low damping so that $p_r \approx \omega_0$ and then

$$a_{max} = \frac{f_0}{2b\omega_0} = \frac{f_0\tau}{\omega_0} \qquad ...(13)$$

Thus in case of zero damping ($b = 0$), the amplitude will be infinity. But this is not possible as the frictional forces acting on the oscillating system, are never zero.

For low damping, the ratio of the response (amplitude) at resonance to the response at low or zero driving frequency is given by

$$\frac{\text{Response (amplitude) at resonance (for low damping)}}{\text{Response (amplitude) at zero driving frequency}} = \frac{f_0\tau/\omega_0}{f_0/\omega_0^2} = \omega_0\tau = Q \qquad ...(14)$$

Thus *the response at resonance is high for a system with high Q-factor.*

At $p = \omega_0$, we obtain from eq. (7)

$$\tan\theta = \infty \text{ or } \theta = \frac{\pi}{2} \text{ and } a = \frac{f_0}{2b\omega_0} \qquad ...(15)$$

Thus at $p = \omega_0$, the amplitude is less than a_{max} ($= f_0/2b\sqrt{\omega_0^2 - b^2}$).

However for low damping, $p_r \approx \omega_0$ and then amplitude is maximum, given by eq. (15).

Case III : High driving frequency ($p >> \omega_0$) : When the driving frequency is much high,

$$\tan\theta \to -\frac{2b}{p} \to -0 \text{ or } \theta \to \pi \text{ and } a = \frac{f_0}{p^2} \text{ nearly or } a = \frac{F_0}{mp^2} \qquad ...(16)$$

Thus in the high frequency limit, the response decreases rapidly as $1/p^2$ and tends to zero. In this limit, the oscillating mass responds like a free object, shaking rapidly in opposite phase ($\theta = \pi$) to the impressed force of high frequency.

The variation of amplitude a with the frequency p of the driving force is shown in **fig. 11.1(a)**. As the angular frequency is increased from the low or zero frequency limit, the amplitude a increases continuously and becomes maximum at a frequency p_r, which is just before and close to ω_0. This is **amplitude resonance.** Now, if the driving frequency is further increased, the amplitude (a) decreases regularly and at very high frequencies approaches to very small value. When the damping is greater, the resonant frequency (p_r) is less than ω_0 but in case of low damping the resonant frequency of the force is nearly equal to the natural frequency (ω_0) of the oscillator. It is also clear from these curves that for low damping the height of the peak becomes greater and when the damping is zero (in the ideal case) the height of the peak (*i.e.*, the maximum amplitude) rises to infinity.

Regarding the phase, we note from our analysis that the displacement x always lags in phase θ relative to the driving force; the value of θ is zero at low frequencies, increases with frequency p and passes $\pi/2$ at $p = \omega_0$ and finally becomes π at high frequencies [**fig. 11.1(b)**].

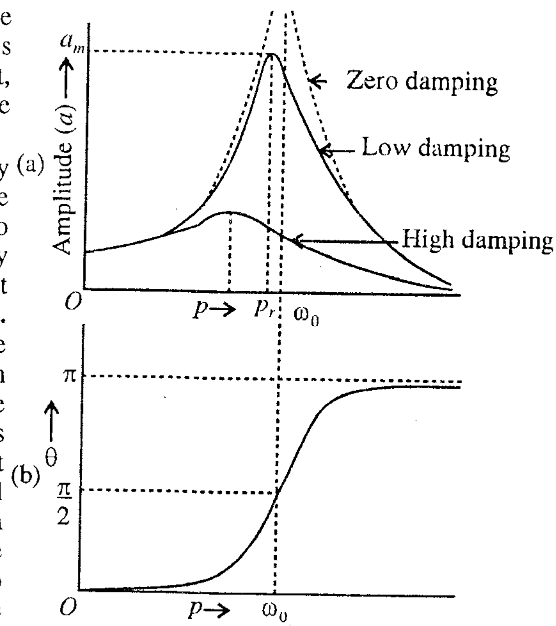


Fig. 11.1 : Forced oscillations : (a) Variation of amplitude (a) of the oscillating system with the driving frequency; (b) Phase lag (θ) of displacement relation to the driving force as a function of driving frequency (p).

11·3. Resonance and Sharpness of Resonance—Half Width of Resonance Curve

In case of driven harmonic oscillator, if we study the variation of amplitude of the oscillating system with the driving frequency, we find that at a particular frequency, called the resonance frequency (p_r), the amplitude is maximum. This phemomenon is called **amplitude rsonance.** The amplitude resonance occurs at the frequency, given by

$$p_r = \sqrt{\omega_0^2 - 2b^2} \qquad \text{...(1)}$$

where ω_0 is frequency of undamped oscillator and b is damping constant.

At this frequency, the amplitude of the oscillating system is given by

$$a_{\max} = \frac{f_0}{2b\sqrt{\omega_0^2 - b^2}} = \frac{F_0 / m}{2b\sqrt{\omega_0^2 - b^2}} \qquad \text{...(2)}$$

where F_0 is the amplitude of the periodic force $F = F_0 \sin pt$ and m is the mass of the oscillator.

If the damping is small $b << \omega_0$, then the resonant frequency is equal to the frequency of the oscillator *i.e.*,

$$p_r = \omega_0 \text{ and } a_{\max} = \frac{F_0}{2bm\omega_0} = \frac{F_0\tau}{m\omega_0} \qquad \text{...(3)}$$

However at zero or low driving frequency, the amplitude is given by

$$a_0 = \frac{f}{\omega_0^2} = \frac{F_0}{m\omega_0^2} \qquad \text{...(4)}$$

Thus,
$$\frac{a_{\max}}{a_0} = \frac{F_0\tau / m\omega_0^2}{F_0 / m\omega_0^2} = \omega_0\tau = Q$$

i.e., Quality factor, $Q = \dfrac{a_{max}}{a_0} = \dfrac{\text{Amplitude at resonance}}{\text{Amplitude at zero (or low) driven frequency}}$ ...(5)

Obviously the amplitude (response) will be more for a high Q-oscillating system.

The amplitude of the oscillating system decreases from the maximum value A_{max} with the change (decrease or increase) of the frequency of the impressed force. In **fig.11.1**, for different values of damping, curves have been drawn between the driven frequency and the amplitude.

The term **sharpness of resonance** *refers to the rate of fall in amplitude with the change of force frequency on each side of the resonant frequency.* When the damping is low, the response (amplitude) falls of very rapidly on either side of the resonant frequency and we say that the resonance is **sharp**. On the other hand, for high damping the response falls off very slowly on either side of the resonant frequency and the resonance is said to be **flat**. Thus, the resonance is flat or sharp according to the damping for the oscillating system is large or small. Familiar examples of a flat or sharp resonance are the resonance of an air column and a sonometer wire with a tuning fork respectively. Due to its large damping the air column responds with tuning fork over a fairly wide range in the neighbourhood of resonance. Thus in this case it is actually difficult to get an exact point of resonance and resonance is said to be **flat.** But in case of sonometer wire, the damping is small so that the resonance is **sharp** and hence to have the resonant point, the length of wire is adjusted very precisely.

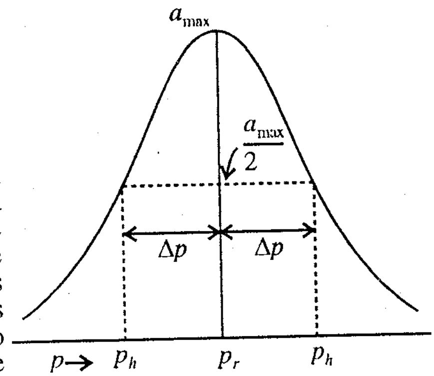


Fig. 11.2 : Amplitude resonance curve and half width (Δp)

Half width of amplitude resonance curve : If p_h is the value of angular frequency when the amplitude falls to half the value at resonance, then the change in p is called **half width** of the resonance curve, *i.e.*,

$$\text{Half width, } \Delta p = |p_h - p_r| \quad \text{...(6)}$$

At resonance,

$$a_{max} = \frac{f_0}{2b\sqrt{\omega_0^2 - b^2}} = \frac{f_0}{\sqrt{4b^2\omega_0^2 - 4b^4}} \quad \text{...(7)}$$

At the frequency p_h, the amplitude is $a_{max}/2$.

$$\therefore \quad \frac{a_{max}}{2} = \frac{f_0}{\sqrt{(\omega_0^2 - 2b^2 - p_h^2)^2 + 4b^2\omega_0^2 - 4b^4}} \quad \text{[using eq. (6), Art. 11.2] ...(8)}$$

From (7) and (8), we have

$$(\omega_0^2 - 2b^2 - p_h^2)^2 + 4b^2\omega_0^2 - 4b^2 = 4(4b^2\omega_0^2 - 4b^4)$$

But,

$$p_r^2 = \omega_0^2 - 2b^2$$

$$\therefore \quad (p_r^2 - p_h^2)^2 = 3(4b^2\omega_0^2 - 4b^4) = 3(4b^2p_r^2 + 4b^4) \quad \text{...(9)}$$

If damping is small, $4b^4$ is negligible and then we have

$$p_r^2 - p_h^2 = \pm\sqrt{3}\,(2bp_r) \text{ or } p_h^2 = p_r^2 \mp \sqrt{3}\,(2bp_r)$$

or

$$p_h = p_r\left(1 \mp \sqrt{3}\,\frac{2b}{p_r}\right)^{1/2} = p_r\left(1 \mp \frac{\sqrt{3}b}{p_r}\right) = p_r \mp \sqrt{3}b$$

$$\therefore \quad \textbf{Half width, } \Delta p = |p_h - p_r| = \sqrt{3}b \quad \text{...(10)}$$

This is the expression for half width of the amplitude resonance curve.

$$\textbf{Full width} = 2\Delta p = 2\sqrt{3}b \quad \text{...(11)}$$

But,

$$Q = \omega_0 p = \frac{\omega_0}{2b} \text{ or } b = \frac{\omega_0}{2Q},$$

hence

$$\text{Half width, D}p = \sqrt{3}\,\frac{\omega_0}{2Q} \quad \text{...(12)}$$

Thus for high *Q-oscillating system the half width of the amplitude resonance curve is small*

11·4. Quality Factor and Effect of Damping

The quality factor of an oscillating system with damping is defined as

$$Q = \omega_0 \tau = \frac{\omega_0}{2b} \qquad \text{...(1)}$$

where τ is relaxation time and b is damping constant.

The quality factor can be interpreted in terms of amplitude of the oscillating system because [eq. (14), Art. 11·2].

$$\frac{\text{The amplitude at resonance (for low damping)}}{\text{The at amplitude at low frequency of the driving force } (p \to 0)} = \frac{f_0 \tau / \omega_0}{f_0 / \omega_0^2} = \omega_0 \tau = Q \qquad \text{...(2)}$$

Thus the Q-factor can be defined as the ratio of amplitude at resonance to the amplitude at low frequencies.

In case of driven oscillator,

$$a = \frac{F_0 / m}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} \text{ and } \theta = \tan^{-1}\left(\frac{2bp}{\omega_0^2 - p^2}\right) \qquad \text{...(3)}$$

Let, $a_0 = \dfrac{f_0}{\omega_0^2} = \dfrac{F_0}{m\omega_0^2}$ at low frequencies so that a and θ can be expressed

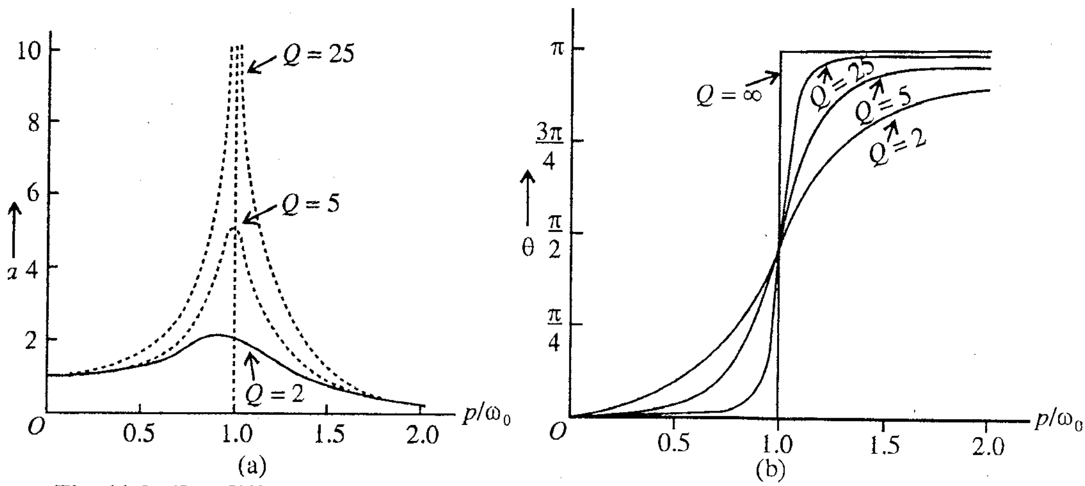


Fig. 11.3 : For different Q (a) Variation of amplitude (a) with driving frequency(p); (b) Variation of phase lag (θ) with driving frequency (p).

$$a = a_0 \frac{\omega_0/p}{[\{(\omega_0/p) - (p/\omega_0)\}^2 + 1/Q^2]^{\frac{1}{2}}} \text{ and } \tan\theta = \frac{1}{(\omega_0/p) - (p/\omega_0)} \qquad \text{...(4)}$$

Using eq. (4), amplitude (a) and phase lag (θ) versus driving frequency (p) curves are shown in **fig. 11.3** for different values of Q. The amplitude acquires maximum value (a_m) at the amplitude resonant frequency (p_r). Using eq. (11) and (12) of Art. 11.2, we can express a_m and p_r in terms of Q as follows (when damping is considered) :

$$\frac{a_{max}}{a_0} = \frac{Q}{\sqrt{1 - (1/4Q^2)}} \text{ and } \frac{p_r}{\omega_0} = \sqrt{1 - \frac{1}{2Q^2}} \qquad \text{...(5)}$$

For different values of Q, each curve has a maximum value a_m, provided $Q \leq 1/\sqrt{2}$ because clearly from eq. (5), then there are no oscillations. We find that when Q is high (or damping is small), the value of a_{max} is high and for small Q (or high damping), it is small. The amplitude resonant frequency p_r is less than ω_0, differing from it more for smaller Q. However we consider, generally, lightly damped or high Q systems, so that $a_{max}/a_0 = Q$ and $p_r/\omega_0 = 1$ nearly.

11·5. Velocity Resonance

The velocity of the driven oscillator is

$$v = \frac{dx}{dt} = \frac{f_0 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2p^2}} \cos(pt - \theta) \qquad \text{[from eq. (8), Art. 11.2]}$$

or
$$v = v_0 \sin(pt - \theta + \pi/2) \qquad \text{...(1)}$$

where
$$v_0 = \frac{f_0 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2p^2}} = \frac{f_0}{\sqrt{(p - \omega_0^2/p)^2 + 4b^2}}$$ is the *velocity amplitude*

and
$$\theta = \tan^{-1}\left(\frac{2bp}{\omega_0^2 - p^2}\right).$$

Thus the velocity amplitude v_0 varies with the driven frequency p [**fig.11.4**]. When $p = 0$, $v_0 = 0$ and when $p = \omega_0$, v_0 attains the maximum value. Hence at the frequency $p = \omega_0 = \sqrt{C/m}$ of the impressed force, the velocity has the maximum value and we call it **velocity resonance.** At velocity resonance,

$$\theta = \tan^{-1}\left(\frac{2bp}{0}\right) = \frac{\pi}{2},$$

$\therefore$ Velocity phase constant $= -\theta + \pi/2 = -\pi/2 + \pi/2 = 0$.

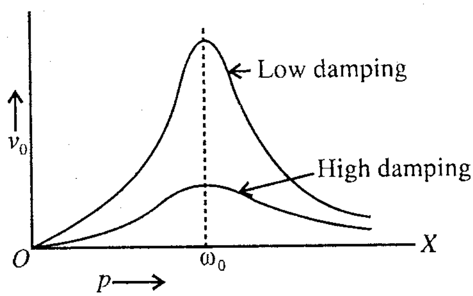


Fig. 11.4 : Variation of velocity amplitude with driven frequency (p).

This means that at velocity resonance, the velocity of the oscillator is in phase with the applied force. This is the most favourable situation for transfer of energy from the applied force to the oscillator, because the rate of work done on the oscillator by the impressed force is Fv, which is always positive for F and v in phase.

When $p \gtrless \omega_0$, the velocity amplitude is smaller than at $p = \omega_0$.

11·6. Power Absorption and Resonance Width

An oscillating physical system loses energy in doing work against damping forces. In a driven harmonic oscillator, this loss of energy is supplied by the applied harmonic force and hence the oscillations of the system are maintained at constant amplitude. One will be interested in knowing the rate at which the energy must be supplied to maintain the vibrations. The rate of supplying the energy or power absorption by the oscillating system at any instant t is given by

$$P = \frac{dW}{dt} = F\frac{dx}{dt} = Fv \qquad \text{...(1)}$$

Now,
$$F = mf_0 \sin pt \text{ and } v = \frac{dx}{dt} = \frac{pf_0 \cos(pt - \theta)}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2p^2}}$$

Hence at any time the energy absorbed by the oscillating system is

$$P = F\frac{dx}{dt} = mf_0 \sin pt \cdot \frac{pf_0 \cos(pt - \theta)}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2p^2}} \qquad \text{...(2)}$$

Hence the average power absorbed is

$$P_{av} = \left(F\frac{dx}{dt}\right)_{av} = \frac{mf_0^2 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2p^2}} [\sin pt \cos(pt - \theta)]_{av} \qquad \text{...(3)}$$

Now the average of sin pt cos $(pt - \theta)$ for one period T

$$= \frac{1}{T}\int_0^T \sin pt \cos (pt - \theta)\, dt = \frac{1}{T}\int_0^T \frac{1}{2}\{\sin(2pt-\theta)+\sin\theta\}dt = \frac{\sin\theta}{2}$$

$$= \frac{1}{2}\frac{2bp}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2p^2}} \qquad \text{[from eq. (7); Art 11.2]}$$

$$\therefore \qquad P_{av} = \frac{1}{2}mf_0^2 \frac{2bp^2}{(\omega_0^2 - p^2)^2 + 4b^2p^2} \qquad ...(4)$$

Replacing $\frac{f_0^2 p^2}{(\omega_0^2 - p^2)^2 + 4b^2p^2}$ by v_0^2, we have

$$P_{av} = mbv_0^2 = \frac{mv_0^2}{2\tau} \text{ or } \frac{\gamma v_0^2}{2} \qquad ...(5)$$

We see from eq. (4) that when $p = \omega_0$, the absorbed average power is maximum, say P_m, given by

$$P_m = \frac{1}{4}\frac{mf_0^2}{b} = \frac{1}{2}mf_0^2\tau. \qquad ...(6)$$

Thus *the power absorbed is maximum at the frequency of velocity resonance and not at the frequency of amplitude resonance*. A graph between P_{av} versus p for *a* particular value of damping constant b (or $Q = \omega_0\tau = \omega_0/2b$) is shown in **fig.11.5**.

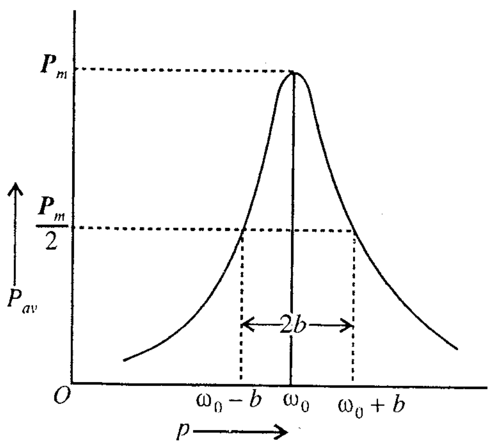


Fig. 11.5 : Average power absorbed (P_{av}) as a function of driving frequency : Full width of the resonance curve $2\Delta p = 2b$ or $1/\tau$; $Q = \omega_0\tau$.

When steady state is attained, the average power dissipated* should be the same as the average power supplied by the driving force.

Resonance width : If p_h is the value of the driving frequency when the average power falls to half the value at resonance, the *half-width of the power resonance* curve is given by

$$\Delta p = |p_h - \omega_0| \qquad ...(7)$$

The power drops to its half the maximum value, when

$$\frac{1}{2}P_m = (P_{av})_{p_h}$$

$$\frac{1}{2}\frac{mf_0^2}{4b} = \frac{1}{2}mf_0^2 \frac{2bp_h^2}{(\omega_0^2 - p_h^2)^2 + 4b^2p_h^2}$$

or $$8b^2p_h^2 = (\omega_0^2 - p_h^2)^2 + 4b^2p_h^2$$

or $$\omega_0^2 - p_h^2 = \pm 2bp_h$$

or $$\omega_0 - p_h = \pm\frac{2bp_h}{\omega_0 + p_h} = \pm\frac{2b}{1 + \omega_0/p_h} \approx \pm b \text{ or } \frac{1}{2\tau} \qquad \text{[taking } \omega_0/p_h = 1 \text{ approx.]}$$

$\therefore$ **Half width** of the power versus frequency curve

$$\Delta p = |\omega_0 - p_h| = \frac{1}{2\tau} = \frac{\omega_0}{2Q} \qquad [\because Q = \omega_0\tau] \qquad ...(8)$$

The quantity $2\Delta p$ is called the **band-width** or **full-width** at the half maximum power and is given by

---

* Instantaneous power dissipated = (force) $dx/dt = 2bm(dx/dt)^2 = 2bm \cdot \frac{f_0^2 p^2}{(\omega_0^2 - p_0^2)^2 + 4b^2p^2}\cos^2(pt - \theta)$

Average force dissipated $= \frac{1}{2}mf_0^2 \frac{2bp^2}{(\omega_0^2 - p^2)^2 + 4b^2p^2}$. $\qquad [\because \{\cos^2(pt - \theta)_{av}\} = \frac{1}{2}]$

$$2\Delta p = \frac{\omega_0}{Q} = \frac{1}{\tau} \text{ or } 2b \qquad ...(9)$$

Thus the resonance full-width for forced oscillations is equal to the inverse of decay time (τ) for free oscillations.

Also, $$Q = \omega_0\tau = \frac{\omega_0}{2\Delta p} = \frac{\text{Frequency at resonance}}{\text{Full width at half maximum power}} \qquad ...(10)$$

Alternatively, $$Q = 2\pi \frac{\text{Average energy stored}}{\text{Average energy dissipated per cycle}} \qquad ...(11)$$

At any instant, stored energy of oscillator = K.E + P.E.

Time averaged stored energy, $E = \langle\text{K.E.}\rangle + \langle\text{P.E.}\rangle = \frac{1}{2} \langle m\left(\frac{dx}{dt}\right)^2\rangle + \langle\frac{1}{2}Cx^2\rangle$

$$= \frac{1}{2}mp^2a^2 \langle\cos^2(pt-\theta)\rangle + \frac{1}{2}m\omega_0^2a^2 \langle\sin^2(pt-\theta)\rangle$$

$$= \frac{1}{4}mp^2a^2 + \frac{1}{4}m\omega_0^2a^2 = \frac{1}{4}m(p^2+\omega_0^2)a^2 \qquad ...(12)$$

At $p \approx \omega_0$, $$E = \frac{1}{2}m\omega_0^2a^2$$

Average energy dissipated per sec. $= \langle\gamma \frac{dx}{dt}\rangle = \gamma\, p^2a^2 \langle\cos^2(pt-\theta)\rangle = \frac{m}{2\tau}p^2a^2$ ...(13)

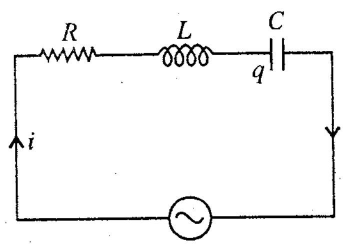
Average energy dissipated per cycle (at $p \approx \omega_0$) $= \frac{m\omega_0^2a^2}{2\tau} \cdot \frac{2\pi}{\omega_0}$ ...(14)

$$\therefore \quad Q = 2\pi \frac{\frac{1}{2}m\omega_0^2a^2}{\frac{m\omega_0^2a^2}{2\tau} \cdot \frac{2\pi}{\omega_0}} = \omega_0\tau = \frac{\omega_0}{2\Delta p} \qquad ...(15)$$

Sharpness of resonance and quality factor : We may use the term, sharpness of resonance, to refer the rapid rate of fall in power with the frequency on each side of the resonant frequency (ω_0). Thus for small band-width ($2\Delta p$) or high Q system, the resonance, is said to be **sharp**. For low Q system, the band-width will be more and the power falls off slowly. The resonance in such a case is said to be **flat**. Thus Q is a measure of the sharpness of the resonance.

11·7. Driven LCR Circuit

Driven LCR circuit is an example of the driven harmonic oscillator in which the oscillations are sustained by an alternating electromotive force. Consider a circuit containing the inductance L, resistance R and capacity C in series. An alternating E.M.F. $E = E_0 \sin pt$ is applied to this circuit, where E_0 is the peak value **[fig.11.6]**. If at any instant t, q is the charge on the condenser C, then the potential difference across its plates is q/C. If the current in the circuit at this instant is i, then the self-induced E.M.F. in the inductance is $-L\, di/dt$. According to Ohm's law, the potential difference between the ends of the resistance is RI. Hence



$E = E_0 \sin pt$

Fig. 11.6.

Total potential difference $= Ri + \frac{q}{C} =$ total E.M.F. $= E - L\,di/dt$,

i.e. $$L\frac{di}{dt} + Ri + \frac{q}{C} = E \qquad ...(1)$$

or $$\frac{d^2q}{dt^2}+\frac{R}{L}\frac{dq}{dt}+\frac{q}{LC}=\frac{E_0}{L}\sin pt. \qquad \left[\because i=\frac{dq}{dt}\right] \qquad ...(2)$$

This equation is similar to the equation of driven harmonic oscillator having R/L in place of $2b$, $1/LC$ for $\omega_0^{\ 2}$ and E_0/L for f_0. Hence the steady state solution of the above equation is

$$q=\frac{E_0/L}{\sqrt{\left(\frac{1}{LC}-p^2\right)^2+\left(\frac{pR}{L}\right)^2}}\sin(pt-\theta) \text{ with } \theta=\tan^{-1}\frac{pR/L}{\frac{1}{LC}-p^2} \qquad ...(3)$$

Differentiating q in eq. (3) with respect to t, we can find the current $i = dq/dt$ in the circuit. Practically, we require the knowledge of the current in the circuit*.

$$i=\frac{E_0p/L}{\sqrt{\left(\frac{1}{LC}-pL\right)^2+R^2}}\cos(pt-\theta) \text{ or } i=\frac{E_0}{\sqrt{R^2+\left(pL-\frac{1}{pC}\right)^2}}\sin(pt-\phi) \qquad ...(4a)$$

where $$\phi=\theta-\frac{\pi}{2}=\tan^{-1}\frac{pL-1/pC}{R} \quad \left[\because \tan\phi=-\cot\theta=\frac{p^2-1/LC}{pR/L}=\frac{pL-1/pC}{R}\right] \qquad ...(4b)$$

which represents that the current i lags is phase ϕ from the E.M.F. E.

The quantity $\sqrt{R^2+(pL-1/pC)^2}$ is called the **impedance** of the circuit and is represented by Z. Its unit is Ohm. Hence

$$i=\frac{E_0}{Z}\sin(pt-\phi) \text{ or } i=i_0\sin(pt-\phi) \qquad ...(5)$$

where $i_0 = E_0/Z$ is the peak value of the current.

The quantity $(pL - 1/pC)$ is called the **reactance** of the circuit and is denoted by X. Therefore

$$\text{Impedance, } Z=\sqrt{R^2+X^2}=\sqrt{(\text{Resistance})^2+(\text{Reactance})^2} \qquad ...(6)$$

Now, the current amplitude $i_0=\frac{E_0}{Z}=\frac{E_0}{\sqrt{R^2+X^2}}$ is smaller for the larger value of Z. The expression

for $$Z=\sqrt{R^2+(pL-1/pC)^2} \qquad ...(7)$$

depends upon the frequency (p) of the applied alternating E.M.F. We will consider below the important cases, when the driven frequency is increased. We denote $\omega_0 = 1/\sqrt{LC}$ as the undamped frequency of the circuit.

Case I : $p<<\omega_0$, the term $-1/pC$ in impedance is **dominant** and is very large and $Z=\sqrt{R^2+(pL-1/pC)^2}\approx 1/pC$.

Therefore, the current amplitude is (when $p\to 0$)

$$i_0=\frac{E_0}{1/pC} \text{ and } \phi=-\frac{\pi}{2} \qquad ...(8)$$

Thus, for low values of p, the current amplitude has a small value. On increasing the driven frequency (p) gradually, the magnitude of reactance $(pL - 1/pC)$ and hence the impedance Z decreases resulting in the increase of the current amplitude.

---

* **A shorter procedure for solution of eq.** (1) : Let the solution be $i = i_0 \sin(pt - \phi)$. Hence $q=\int i\,dt=\frac{i_0}{p}\cos(pt-\phi)$

Substituting the values of di/dt, i and q in eq. (1) and proceeding as before, the values of i and ϕ can be calculated.

Case II : When $p = \omega_0 = 1/\sqrt{LC}$, the reactance

$$X = pL - \frac{1}{pC} = \frac{L}{\sqrt{LC}} - \frac{\sqrt{LC}}{C} = \sqrt{\frac{L}{C}} - \sqrt{\frac{L}{C}} = 0 \quad ...(9)$$

and so the impedance Z is minimum and is equal to R. In consequence the current amplitude $i_0 = E_0/Z = E_0/R$ becomes maximum. This is the case of **electrical resonance.** If $R = 0$, then the current amplitude at resonance becomes infinite. The **resonant frequency** of the circuit is given by

$$\nu_r = \frac{p}{2\pi} = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \quad ...(10)$$

In such a case, $\phi = 0$ and $i = \frac{E_0}{R} \sin pt$ ...(11)

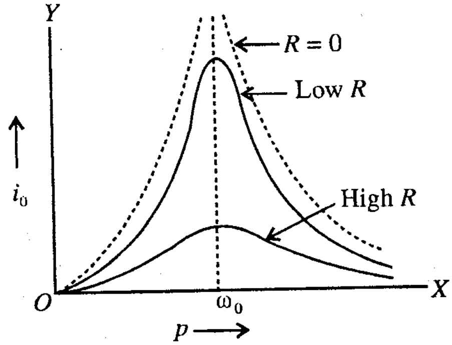


Fig. 11.7.

Thus at *resonance the current and the applied E.M.F. are in the same phase.*

Case III : When $p >> \omega_0$, the term $L\omega$ in the expression for Z becomes dominant and very large and then.

$$i_0 = E_0/pL \text{ and } \phi = \pi/2 \quad ...(12)$$

The variation of I_0 with p is shown in **fig. 11.7** for different values of R. The smaller value or R, the sharper is the resonance.The variation of phase difference ϕ between the current and the E.M.F. is shown in **fig. 11.8.** The current lags in phase for $p < \omega_0$ and exceeds in phase for $p > \omega_0$.

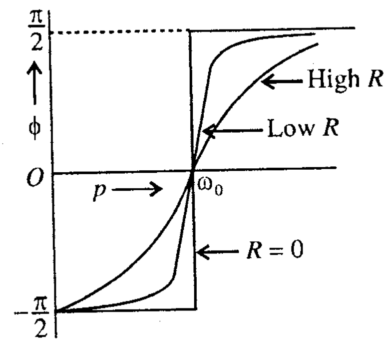


Fig. 11.8.

Power in A.C. circuits : The power of an electrical circuit is defined as the rate of its doing work and at any time it is measured by the product of the current and the potential difference. If at any instant i is the current and E, the E.M.F., then the instantaneous power delivered by the source to the circuit is

$$P = Ei. \quad ...(13)$$

Now, let us consider the power relations by taking LCR circuit. The instantaneous power delivered by the source to the circuit

$$P = Ei = E_0 \sin pt \; i_0 \sin (pt - \phi) = \frac{E_0 i_0}{2} [\cos \phi - \cos (2pt - \phi)] \quad ...(14)$$

where, the average value of $\cos (2pt - \phi)$ for one cycle is zero, hence the average power of an A.C. circuit is given by

$$P_{av} = \frac{E_0 i_0}{2} \cos \phi = \frac{E_0}{\sqrt{2}} \frac{i_0}{\sqrt{2}} \cos \phi = E_{rms} \, i_{rms} \cos \phi \quad ...(15)$$

where, $E_{rms} = \frac{E_0}{\sqrt{2}}$ and $i_{rms} = \frac{i_0}{\sqrt{2}}$.

This $\cos \phi$ in eq. (15) is called the **power factor** of the circuit and it will be given by

$$\cos \phi = \frac{R}{\sqrt{R + (pL - 1/pC)^2}} = \frac{R}{Z} \quad \text{[From eq. (4b)]}$$

At resonance, $\phi = 0$ so the average power delivered to the circuit is maximum and then

$$(P_{av})_{max} = (P_{av})_{res} = \frac{1}{2}E_0 i_0 = \frac{1}{2}\frac{E_0^2}{R} = \frac{1}{2}i_0^2 R \quad ...(16)$$

But,
$$i_{rms} = \frac{i_0}{\sqrt{2}} = \frac{1}{\sqrt{2}} \cdot \frac{E_0}{Z} = \frac{E_{rms}}{Z}$$

$\therefore$ Power in A.C. circuit $= (i_{rms} Z).\ i_{rms} R/Z = i_{rms}^{\ 2} R$ ...(17)

Thus in an A.C. circuit, the power appears only in the form of heat.

Analogous to mechanical oscillator, we can show easily that

$$Q = \omega_0 \tau = \frac{L\omega_0}{R} = \frac{\omega_0}{2\Delta p} = \frac{\text{Frequency at resonance}}{\text{Full width at half maximum power}}. \quad ...(18)$$

***Q*-Factor :** Similar to mechanical oscillator, the quality factor is given by

$$Q = \omega_0 \tau = \frac{\omega_0 L}{R}, \text{ where } \omega_0 = \frac{1}{\sqrt{LC}} \quad ...(19)$$

Full width or the band-width of the power resonance curve is given by

$$2\Delta p = \frac{1}{\tau} = \frac{R}{L} = \frac{\omega_0}{Q} \quad ...(20)$$

Thus
$$Q = \frac{\omega_0}{2\Delta\omega} = \frac{\text{Frequency at resonance}}{\text{Full width at half maximum power}} \quad ...(21)$$

The full width $2\Delta\omega$ is small for high Q. Circuits with Q of the order of 10^2 to 10^4 are quite common. The resonant circuit in a radio receiver with Q of the order of 10^2 to 10^3 may select a particular station and discard the others.

11·8. Transient State and Steady State

In this chapter we have so far studied the steady state behaviour of an oscillating system, acted by a harmonic driving force. In fact, in a practical situation, the driving force is first switched on at some instant on a damped harmonic oscillator. The frequency (p) of driving force in general different to the natural frequency (ω) of the damped harmonic oscillator. In the beginning, the oscillator tries to vibrate with its natural frequency (ω) and the applied (driven) periodic force tends to impose its own frequency (p) upon it. Hence for some time, the oscillating system is in the **transient state** and the free oscillations of frequency ω die out and ultimately the system starts oscillating with a constant amplitude with the frequency (p) of the driven harmonic force. This is called the **steady state.** In the transient state, the oscillations of the system take place under the superposition of two oscillations of frequenclies $\omega = \sqrt{\omega_0^2 - b^2}$ and p.

We consider the system to be at rest with no initial displacement at $t = 0$, the time when the driving force is switched on. We are interested in knowing the nature of the transient motion just after the instant $t = 0$.

Mathematically, the problem is to find the general solution of the differential equation of the driven harmonic oscillator :

$$\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + \omega_0^2 x = f_0 \sin pt \qquad [f_0 = F_0/m] \quad ...(1)$$

The steady state solution is not the general solution. According to the therory of differential equations, the general solution $x(t)$ of eq. (1) is the superposition of two solutions $x_1(t)$ and $x_2(t)$, given by

(1) $x_1(t) = a \sin(\omega t - \theta)$, which is the steady state solution (particular solution) of eq. (1).

(2) $x_2(t) = a_0\, e^{-bt} \sin(\omega t + \phi)$, which is the general solution of the differential equation,

$$\frac{d^2x}{dt^2}+2b\frac{dx}{dt}+\omega_0^2 x=0.$$

Thus the general solution of eq. (1) is

$$x = x_1 + x_2 \quad \text{or} \quad x = a \sin(pt-\theta) + a_0 e^{-bt}\sin(\omega t+\phi) \qquad ...(2)$$

We impose the initial conditions that at $t = 0$, $x = 0$ and $dx/dt = 0$ on the general solution x, given by eq. (2) and its time derivative dx/dt. Now,

$$\frac{dx}{dt} = pa\cos(pt-\theta) + a_0 e^{-bt}[-b\sin(\omega t+\phi)+\omega\cos(\omega t+\phi)] \qquad ...(3)$$

Imposing the, initial conditions on eqs. (2) and (3), we obtain

$$0 = -a\sin\theta + a_0 \sin\phi \text{ and } 0 = pa\cos\theta - a_0 b\sin\phi + a_0\omega\cos\phi \qquad ...(4)$$

We are interested here in the case in which the *damping is weak and the driving frequency is close to the resonant frequency i.e.*, $b << \omega$ and $p \approx \omega$. In such a case, the conditions (4) become

$$a_0 \sin\phi = a\sin\theta \text{ and } a_0\cos\phi = -a\cos\theta \qquad ...(5)$$

Dividing $\quad \tan\phi = -\tan\theta$ or $\phi = \pi - \theta$. So that $a_0 = a$ $\qquad$ ...(6)

Hence the general solution is

$$x = a\sin(pt-\theta) - a_0 e^{-bt}\sin(\omega t-\theta) \qquad ...(7)$$

Now, first let us consider the case of zero damping ($b = 0$) or ideal oscillator, then from eq. (9) [Art. 11.2],

$$a = \frac{F_0}{m(\omega_0^2-p^2)},\ \theta = 0 \text{ and } \omega = \omega_0 \qquad ...(8)$$

Hence eq. (7) takes the form

$$x = \frac{F_0}{m(\omega_0^2-p^2)}(\sin pt - \sin\omega_0 t) \text{ or } x = \frac{2F_0}{m(\omega_0^2-p^2)}\cos\frac{p-\omega_0}{2}\sin\frac{p+\omega_0}{2}t \qquad ...(9)$$

or

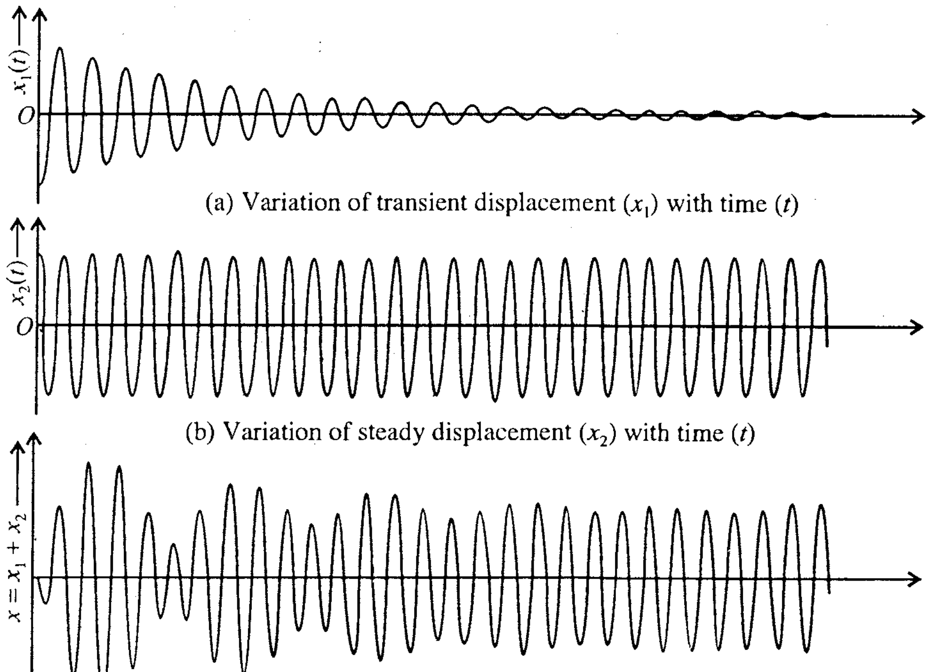
$$x = a_m(t)\sin\omega_a t \qquad ...(10)$$

where $\quad a_m(t) = \dfrac{2F_0}{m(\omega_0^2-p^2)}\cos\omega_m t$ with $\omega_m = \dfrac{\omega_0-\omega}{2}$ and $\omega_a = \dfrac{p+\omega_0}{2}$

This is the case when we were studying the superposition of two simple harmonic motions of two slightly different frequencies and phenomenon of modulation and beats.

Thus the beats occur between the natural and driven oscillations at a frequency $\omega_0 - p$ and in case of zero damping, phenomenon of beat goes on continuously.

Next, we consider the case, when the damping is present *i.e*, the case of damped harmonic oscillator alongwith the applied harmonic force we see from eq. (7) that at $t = 0$, the displacement is a suerposition of two harmonic terms with the driving frequency (p) close to the natural frequency (ω). Thus the initial motion consists of beats between the free and driven oscillations. If there were no damping, these beats would continue forever, as we have seen above. However due to damping the second term of eq. (7) dimnishes with time, resulting in weak beats. These are called transient beats and the oscillator is said to be in the **transient state**. After a sufficiently long time compared to the decay time τ, the **steady state solution** becomes dominant [*i.e.* first term of eq. (7)] and the oscillator executes vibrations of constant amplitude (with no beats) with the frequency of driven harmonic force **[fig. 11.9.]**. In the beginning, due to damping, the oscillator adjusts gradually its phase relative to the driving harmonic force and the relative phase is stabilized in the steady state.



(a) Variation of transient displacement (x_1) with time (t)

(b) Variation of steady displacement (x_2) with time (t)

(c) Variation of resultant displacement (x) with time (t)

Fig. 11.9 : Variation of displacement of a driven oscillator (for low damping) with time from starting at $t = 0$.

Ex. 1. *The amplitude of forced vibrations is given by*

$$a = \frac{f}{\sqrt{(\omega_0^2 - p^2)^2 + p^2/\tau^2}}.$$

If $Q = 40$, calculate the value of A/A_{max} for $p/\omega_0 = 0.99$. **(Kanpur 2005)**

Sol. $Q = \omega_0\tau = 50$ or $\tau = 50/\omega_0$

and

$$a = \frac{f}{\sqrt{(\omega_0^2 - p^2)^2 + p^2/\tau^2}} = \frac{f}{\omega_0^2\sqrt{\left(1-\dfrac{p^2}{\omega_0^2}\right)+\dfrac{1}{2500}\dfrac{p^2}{\omega_0^2}}}$$

For small damping, at resonance $a_{max} = \dfrac{f}{\omega_0^2 \times 1/50} = \dfrac{50f}{\omega_0^2}$

When $p/\omega_0 = 0.99$, $\dfrac{a}{a_{max}} = \dfrac{1}{50}\dfrac{1}{\sqrt{(1-0.99)^2+\dfrac{1}{2500}(0.99)^2}} = \mathbf{0.71}.$

Ex. 2. *In case of a forced harmonic oscillator, the amplitude of vibrations increases from* 0.02 *mm at very low frequencies to a value* 5 *mm at the frequency* 100 *Hz. Find* (i) *Q-factor of the system,* (ii) *damping constant b and relaxation time and* (iii) *half width of amplitude resonance curve.*

(Jabalpur 1997; Sagar 95)

Sol. (i) $$Q = \omega_0\tau = \frac{\omega_0}{2b} = \frac{a_{max}}{a \text{ at zero or very low frequency}} = \frac{5}{0.02} = \mathbf{250.}$$

(ii) Resonant frequency = 100 Hz, $\therefore\ \omega_0 = 2\pi \times 100$

Hence $$\tau = \frac{Q}{\omega_0} = \frac{250}{2\pi \times 100} = \mathbf{0.4\ sec.} \text{ and } b = \frac{1}{2\tau} = \frac{1}{2\times 0.4} = \mathbf{1.25\ per\ sec.}$$

(iii) Half width $\Delta p = \sqrt{3}\times 1.25 = \mathbf{2.2\ rad/sec.}$

Ex. 3. *If a constant torque of* 8×10^{-5} *N-m produces an angular displacement of* $2°$ *in a torsion pendulum whose moment of inertia is* 10^{-3} *kg* $\times$ *m*^{2}*, calculate the frequency of the periodic external torque which will produce resonance.*

Sol. Angular displacement = $2° = 2 \times \pi/180$ radian; applied couple = 8×10^{-5} N-m

Restoring couple for unit displacement $= \dfrac{8\times10^{-5}}{2\pi} \times 180$ N-m/rad.

At resonance, frequency of the torque = approx. natural frequency

$$= \frac{1}{2\pi}\sqrt{\frac{C}{I}} = \frac{1}{2\pi}\sqrt{\frac{8\times10^{-5}\times180}{2\pi\times10^{-3}}} = \mathbf{0.24\ Hz.}$$

Ex. 4. *In case of a driven harmonic oscillator, the amplitude of vibrations increases from* 0.1 *mm at very low frequencies to a maximum value of* 5 *mm at the frequency* 200 *Hz. Find Q-factor, damping constant b and relaxation time.*

Sol. $$Q = \frac{\text{Amplitude at resonance}}{\text{Amplitude at zero or very low frequency}} = \frac{5}{0.1} = \mathbf{50.}$$

Here, resonant frequency $\nu_0 = 200$ Hz, $\therefore\ \omega_0 = 2\pi \times 200$

Therefore, $$\tau = \frac{Q}{\omega_0} = \frac{50}{2\pi\times200} = \mathbf{0.04\ sec.} \text{ and } b = \frac{1}{2\tau} = \mathbf{12.6\ per\ sec.}$$

Ex. 5. *Find the average energy stored in a* 0.1 *kg mass, attached to a spring. The mass is oscillating with an amplitude of* 5 *cm in resonance with a harmonic force of frequency* 30 *Hz. If the quality factor of the oscillator is* 100, *calculate the power losses.*

Sol. At resonance, $p = \omega_0$, the average energy stored

$$= \frac{1}{2} m\,\omega_0^2 a^2 = \frac{1}{2} \times 0.1\,(2\pi \times 30)^2 \times (5 \times 10^{-2})^2 = \mathbf{4.44\ J.}$$

Now, $$Q = \frac{\text{Average energy stored}}{\text{Average energy dissipated per period}}$$

Period, $T = 1/30$ sec; $\therefore$ Average energy dissipated in 1/30 sec. $= \dfrac{2\pi\times4.44}{100} = \mathbf{0.279\ J.}$

$\therefore$ Average energy dissipated per sec. or power losses $= 30 \times 0.279 = \mathbf{8.37\ J.}$

Ex. 6. *An A. C. supply of frequency* 10000 *and* 110 *volts is connected across a circuit containing a resistance of* 10 *ohms, an inductance of* 10^{-3} *henry and a capacity of* 1 μ*F. Find the value of the current. What must be the value of the capacity in order that the current may be maximum.* **(Agra 1998)**

Sol. Impedance, $$Z = \sqrt{R^2 + (pL - 1/pC)^2} = \sqrt{R^2 + (2\pi nL - 1/2\pi nC)^2}$$

$$= \sqrt{10^2 + \left(2\pi\times10000\times10^{-3} - \frac{1}{2\pi\times10000\times10^{-6}}\right)} = 48 \text{ ohms approx.}$$

$\therefore$ Current = e.m.f./Z = 110/48 = **2.293 amp.**

Maximum current will flow when the total resistance is zero and resonance is produced. Hence the maximum current will be when $pL - 1/pC = 0$ *i.e.*, when

$$C = \frac{1}{Lp^2} = \frac{1}{L \times 4\pi^2 n^2} = \frac{1}{10^{-3} \times 4 \times \pi^2 \times (10^4)^2} = \mathbf{0.253 \times 10^{-6}\ farad.}$$

Ex. 7. *An A.C. potential of* 1×10^5 sec^{-1} *frequency and* 1.0 *volt amplitude is applied to a series LCR circuit. If* R = 2.0 *ohm,* L = 0.50 *milihenry, calculate the value of C to secure resonance. Also calculate the rms value of current and peak potenial difference across the condenser.* **(Agra 1970)**

Sol. Capacity, required to produce resonance, is given by

$$C = \frac{1}{4\pi^2 n^2 L} = \frac{1}{4 \times (3.14)^2 \times (10^5)^2 \times 0.5 \times 10^{-3}} = \mathbf{5 \times 10^{-9}\ farad.}$$

$$I_{rms} = \frac{E_{rms}}{R} = \frac{E_0}{R\sqrt{2}} = \frac{\mathbf{1}}{\mathbf{2\sqrt{2}}}\ \textbf{ampere} \qquad \text{[at resonance } Z = R]$$

Peak potential differential across the condenser = peak current × reactance across condenser

$$= \frac{E_0}{R} \times \frac{1}{pC} = \frac{1}{2} \times \frac{1}{10^5 \times 5 \times 10^{-9}} = \mathbf{10^3\ volts.}$$

EXERCISE 11·1

1. Write the equation of motion of a simple harmonic undamped oscillator to which a force $F = F_0 \cos pt$ is applied. Verify that its solution is

$$x = \frac{F_0}{m(\omega_0^2 - p^2)} \cos pt.$$

Discuss the resonance in this case.

2. In a forced harmonic oscillator, the amplitude of vibration increases from 0.01 mm at very low frequencies to a maximum value 5 mm at the frequency 100 c.p.s. Calculate the quality factor, damping constant and half width of the resonance curve.
[**Ans.** 500; 1.25/sec.; 2.2 rad./sec.]

3. If Q = 50, find the values of A/A_{max}, when p/ω_0 = 0.98 and 0.97.
[**Ans.** 0.45, 0.32.]

4. A mass of 1 kg is suspended from a linear spring with a force constant $C = 10^3$ N/m, and damping coefficient $\gamma = 50 \times 10^{-3}$ N-sec/m. The spring is driven by an external force $F = F_0 \sin pt$, where F_0 = 2.5 newton and p is twice the natural frequency of the system. (i) Calculate the amplitude of the resulting motion. (ii) How much is the displacement shifted in phase from the driving force ?
[**Ans.** (i) 8.3×10^{-4} m; (ii) –179.9°.]

5. For a high Q-system, show that the width of the amplitude resonance curve is nearly $\sqrt{3}b$, where the full-width is measured between those frequencies where $a = a_m/2$.

6. Calculate the average energy stored in a 20 gm mass attached to a spring and vibrating with an amplitude 1 cm in resonance with a periodic force whose frequency is 20 Hz. If the quality factor of the oscillator be 160, how much energy being dissipated per second ?
[**Ans.** 0.016 J; 0.0125 J.]

7. A condenser of 0.5 μF capacity is discharged through a 2 ohm resistance and a 2 henry inductance. Find the freqency of electrical oscillations and the time in which the amplitude of oscillations is reduced to 0.1%.
[**Ans.** 160 Hz; 14 sec.]
[**Hint.** : $Q_0/1000 = Q_0 e^{-Rt/2L} = Q_0 e^{-t/2}$ or $e^{t/2}$ = 1000.]

8. A torsional pendulum with moment of inertia 900 gm-cm^2 and torsion constant 4×10^{-3} N-m/rad executes resonant vibrations under the influence of an external torque. Calculate the frequency of the applied torque. What should be the Q of the oscillator so that the amplitude of oscillation may reduce to 20%, when the frequency of the applied torque increased by 0.2% ?
[**Ans.** 1.06 Hz; 187.]
9. A circuit contains a resistance of 2 ohm, an inductance of 0.005 henry and a capacity of 600 μF in series with a potential difference of 100 volts at driven frequency 50 Hz. Find the current, the power factor and the power.
[**Ans.** 26.5 amp (r.m.s value) ahead of the driven voltage by $61°54'$; $\cos \phi = 0.471$; $p = 1114$ watts.]
10. In an experiment on forced oscillations the frequency of a sinusoidal driving force is changed while its amplitude is kept constant. It is found that the amplitude of vibration is 0.10 mm at very low frequency of the driver and goes upto a maximum of 0.5 mm at driving frequency 200 sec^{-1}. Calculate (i) Q of the system, (ii) relaxation time τ, and (iii) half width of resonance curve.
(Agra 1970 S)
[**Ans.** (i) 500; (ii) 0.4 sec; (iii) 2.16 rad/sec.]
11. Calculating the resonating frequency and Q of the circuit containing 0.02 μF capacity, 80 mh inductance and 0.25 ohm resistance.
[**Ans.** 4×10^3 Hz; 8×10^3.]
12. Find the resonating frequency and Q of a circuit containing 0.04 μF capacity, 40 mH inductance and 0.2 ohm resistance. What is the band-width of the power resonance curve ?
[**Ans.** $v_0 = 3.98 \times 10$ Hz; $Q = 5000$; $\Delta p = 5$ rad./sec.]
13. A condenser of 0.5 μF capacity is discharged through a 2 ohm resistance and a 2 henry inductance. Find the frequency of electrical oscillations and the time in which the amplitude of oscillations is reduced to 0.1%.
[**Ans.** 160 Hz; 14 sec..]
14. A resistance of 10 ohms is joined in series with an inductance of 0.5 henry. What capacitance should be put in series with the combination to obtain maximum current ? What will be the potential difference across the resistance, inductance and capacity ? The current is being supplied by a 200 volts and 50 cycles A.C. mains.
[**Ans.** 20.27×10^{-6} farad; 20 amp; 200 volts; 3141.6 volts, 3141.6 volts.]
15. A coil of 3 mH is given. Calculate the capacity which will make the resonance frequency 1000 kilo Hz.
[**Ans.** 8.85 peko-farad.]
16. LCR circuit has $R = 10$ ohms, $L = 200$ micro-henry and $C = 500$ peko-farad. Determine the angular frequency separation between the half power points and quality factor for the circuit.
[**Ans.** 5×10^4 rad./sec.; 63.3.]

QUESTIONS

Long Answer Questions

1. What are free, damped and forced oscillators ? Obtain the differential equations for the forced oscillator and solve it. **(Bhopal 2003)**
2. Obtain the differential equations for a forced oscillator and solve it for the steady state.
3. If the frequency of the impressed force on an oscillator is changed gradually, then what changes will take place in the response (amplitude) ? Draw a graph, representing this variation. Prove that the displacement always lags in phase from the driven force. Show by means of a graph the relation between the frequency of the force and this phase difference.

4. The amplitude A of forced vibrations in a system is given by

$$A = \frac{F/m}{[(\omega_0^2 - \omega^2)^2 + (\omega/\tau)^2]^{1/2}} \cdot \text{(Proof not required).}$$

Symbols have their usual meanings. Show that in the limit $\omega << \omega_0$ the response is independent of the mass, while in the limit $\omega >> \omega_0$ the response is independent of the spring constant of the system.

5. Give the theory of dirven harmonic oscillator. Show that at resonance the displacement always lags behind the driving force in phase by $\pi/2$, while the velocity at resonance is always in phase with the driving force. **(Kanpur 1995; Agra 93)**

6. Describe the transient and steady states of a forced oscillaror. **(Gwalior 2003)**

7. Write down the differential equation for a forced harmonic oscillator and explain the significance of each term in it. Obtain solution of this equation and discuss it. Write the condition of resonance and explain the sharpness of resonance. **(Indore 2004; Jabalapur 03)**

8. Deduce an expression for the velocity amplitude of a driven harmonic oscillator. Show that for very high, equal and very low driving frequencies in comparison to the natural frequency, this amplitude is governed by the inertia factor, the resistance factor and the spring factor respectively. **(Agra 1972)**

9. For a driven harmonic oscillator of natural frequency $\omega/2\pi$ described by

$$m\ddot{x} + r\dot{x} + kx = F \sin pt,$$

deduce an expression for velocity amplitude x_0. Show that for the case $p >> \omega$, $p = \omega$ and $p << \omega$, this amplitude is governed by inertia factor, the resistance factor and the spring factor respectively.

10. Explain the power absorption, quality factor and resonance of a forced oscillator. **(Gwalior 2004; Bhopal 03)**

11. Write down the expression for the amplitude of a forced oscillator. Explain the resonance half power points on the amplitude-frequency curve. Show that the quality factor $Q = \frac{\omega_r}{2\Delta\omega}$, where the symbols have their usual meanings. **(Sagar 2003)**

12. Prove that for the forced oscillations of a damped oscillator the average power of the applied force is equal to the average power dissipated by the damping force.

13. What do you understand by resonance ? Explain the sharpness of resonance with the help of theoretical results of forced oscillations and deduce the expression for its quality factor. **(Bundelkhand 2004)**

14. Show that in a driven oscillator, the maximum power is absorbed at the frequency of velocity resonance and not at the frequency of amplitude resonance. **(Agra 2006)**

15. Show that if a condenser is discharged through an inductance and resistance, it behaves like a damped oscillator. When does the above circuit behave like a forced oscillator ? Explain with diagram. **(Sagar 2003)**

16. Explain the details of the following terms in case of driven harmonic oscillator :
(i) Quality factor, (ii) Sharpness of resonance, and (iii) Velocity resonance. **(Agra 1991)**

17. Write short notes on the following :
(i) Sharpness of resonance curve, **(Kanpur 2005; Indore 03; Sagar 03)**
(ii) Quality factor, **(Indore 2003)**
(iii) Forced harmonic oscillator. **(Indore 2003)**

Short Answer Questions

1. Establish the differential equations of driven harmonic oscillator.

2. What are transient and steady states ? Explain.
3. What is the phase difference between the force and the resulting displacement, when the driving frequency is very low ? Explain your answer.
4. What is amplitude resonance ? Is it true or false : At amplitude resonance, the driven frequency is equal to the natural frequency of the oscillator.
   [**Ans.** False.]
5. What is the sharpness of resonance ? Explain on what factors does it depend ?
6. What do you mean by velocity resonance ?
7. What is half-width of resonance curve ? For small damping how does it depend on damping constant ?
8. What is electrical resonance ?
9. Show that for small damping, the ratio of the amplitude at resonance to the amplitude at zero driven frequency is equal to quality factor.
10. Show that at resonance for low damping the phase difference between the force and resultant displacement of the oscillator is $\pi/2$.

Objective Type Questions

1. Steady state, a forced oscillator vibrates at the :
   (a) driving frequency.
   (b) natural frequency of the oscillator.
   (c) average frequency of the driver and free oscillator.
   (d) frequency slightly less than the natural frequency of the oscillator.
2. Vibrations of the diaphragm of a microphone are :
   (a) free oscillations (b) damped oscillations
   (c) resonant oscillations (d) forced oscillations
   (Bundelkhand 2004; Jabalpur 1999)
3. Sharper the resonance, the band-width is :
   (a) larger (b) smaller (c) moderate (d) none of these
   (Kanpur 2004)
4. Power absorbed in a driven oscillator is maximum at the :
   (a) amplitude resonance.
   (b) velocity resonance.
   (c) highest possible driving frequency.
   (d) frequency where the amplitude drops to $1/e$ of its maximum value.
5. For electrical LCR circuit, the quality factor is :
   (a) $\omega L/R$ (b) $R/\omega L$ (c) $\omega R/L$ (d) ωLR
   where ω is the frequency at resonance.
6. In the condition of resonance, the phase difference between the steady displacement and the driving force is :
   (a) zero (b) $-\pi/2$ (c) π (d) $+\pi/2$

ANSWERS

1. (a), **2.** (d) **3.** (b), **4.** (b), **5.** (a), **6.** (b)

12

Motion of Charged Particles in Electric and Magnetic Fields

12·1. Charged Particle in an Electric Field $\vec{E}$ — $\vec{E}$ as an Accelertaing Field

Electric force on a charge : According to Coulomb's law, a point charge q_2, situated at a vector distance $\vec{r}$ from another point charge q_1, experiences the electric force $\vec{F}$, given by

$$\vec{F} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2} \hat{\mathbf{r}} \qquad ...(1)$$

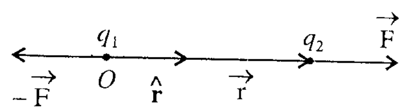


Fig. 12.1 : Electric force on a charge.

In magnitude,

$$\vec{F} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2}$$

where $\hat{\mathbf{r}} = \vec{r}/r$ is a unit vector along $\vec{r}$, ε_0 is the permittivity of free space with value $\varepsilon_0 = 8.854 \times 10^{-12}$ Coul^2/N-m^2 and $1/4\pi\varepsilon_0 = 9 \times 10^9$ N-m^2/Coul^2. **Fig. 12.1** shows that if force $\vec{F}$ acts on q_2, then $-\vec{F}$ will act on q_1.

Electric field : When a charged particle is so situated that an electric force acts upon it, we say that it is present in an **electric field.** The intensity of the electric field $\vec{E}$ is defined by the relation

$$\vec{F} = q\vec{E} \qquad ...(2)$$

where q is the charge on which we observe the force $\vec{F}$.

Thus *the intensity of electric field $\vec{E}$ is defined as the force per unit positive charge at the position of the charged particle.*

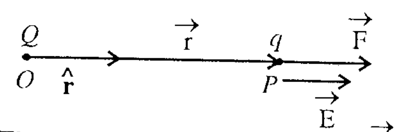


Fig. 12.2 : Electric field intensity $\vec{E}$ at a point P.

In **fig. 12.2,** if Q is the charge at the orgin O, then force on the charge q is

$$\vec{F} = \frac{1}{4\pi\varepsilon_0} \frac{Qq}{r^2} \hat{\mathbf{r}} \qquad ...(3)$$

Hence from (2) and (3), we have

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}} \qquad ...(4)$$

S.I. unit of the electric field $\vec{E}$ is *N/coulomb* or *volt/m.*

$\vec{E}$ as an accelerating field : If a particle, having charge q, is in an uniform electric field $\vec{E}$, then it will experience a force $\vec{F} = q\vec{E}$. According to Newton's second law of motion, the acceleration of the charge is given by

$$\vec{a} = \frac{\vec{F}}{m} = \frac{q\vec{E}}{m} \qquad ...(5)$$

where m is the mass of the charged particle.

Gain in kinetic energy by a charged particle in an electric field : If a positive charge q is accelerated from rest by the electric field $\vec{E}$, then the work done on the charged particle moving from one point to another, appears in the form of change in its kinetic energy (*work energy theorem*) *i.e.*,

$$\int_1^2 \vec{F} \cdot d\vec{r} = \frac{1}{2}mv^2 \qquad ...(6)$$

where v is the speed acquired by the particle in the electric field.

But, $$\int_1^2 \vec{F} \cdot d\vec{r} = \int_1^2 q\vec{E} \cdot d\vec{r} = q\int_1^2 \vec{E} \cdot d\vec{r}$$

By definition of potential difference between the two points

$$V_1 - V_2 = \int_1^2 \vec{E} \cdot d\vec{r} = V \text{ (say)}; \therefore \int_1^2 \vec{F} \cdot d\vec{r} = qV \qquad ...(7)$$

Thus from (6) and (7), we have

$$\frac{1}{2}mv^2 = qV \qquad ...(8)$$

Gain in kinetic energy = Charge × Potenital difference

Thus *a positively charged particle gains kinetic energy in moving from a state of higher potential to a state of lower potential and loses kinetic energy in moving from a state of lower potential to that of higher potential.* The reverse case occurs for a negatively charged particle.

The energy of a charged particle is usually measured in electron volts. *When an electron moves through a potential difference of* 1 *volt, then the energy gained by it is* **one electron volt** *i.e.*,

$$1 \text{ eV} = \text{elctronic charge} \times 1 \text{ volt potential difference}$$

or $$\mathbf{1 \ eV = 1.6 \times 10^{-19} \times 1 = 1.6 \times 10^{-19} \ joule} \qquad ...(9)$$

The other multiples of electron volt, namely million electron volt (MeV) and billion electron volt (BeV), are also used.

$$\mathbf{1 \ MeV = 10^6 \ eV \ ; \ 1 \ BeV = 10^9 \ eV.}$$

For constant $\vec{E}$, $\int_1^2 \vec{F} \cdot d\vec{r} = q\vec{E} \cdot \int_1^2 d\vec{r} = q\vec{E} \cdot (\vec{r}_2 - \vec{r}_1) = q\vec{E} \cdot \vec{d}$

where $\vec{r}_2 - \vec{r}_1 = \vec{d}$ is the displacement of the particle. If $\vec{E}$ and $\vec{d}$ are along the same direction, then $q\vec{E} \cdot \vec{d} = qEd$ and hence

$$\int_1^2 \vec{F} \cdot d\vec{r} = qEd \qquad ...(10)$$

Using (7), we have $qEd = qV$ or $Ed = V$ or $V = \frac{E}{d}$ ...(11)

Useful data : Charge on electron or proton = 1.6×10^{-19} coulomb; Mass of electron = 0.91×10^{-30} kg; Mass of proton = 1.67×10^{24} kg.

Motion of a charged particle in uniform accelerating electric field $\vec{E}$: If a particle of mass m and charge q is accelerated in an uniform electric field $\vec{E}$, then its acceleration $\vec{a} = d\vec{v}/dt$ is given by

$$\vec{a} = \frac{\vec{F}}{m} \text{ or } \frac{d\vec{v}}{dt} = \frac{q\vec{E}}{m} \qquad ...(12)$$

By integrating eq. (12), we obtain

$$\vec{v} = \frac{q\vec{E}}{m}t + \vec{C} \qquad ...(13)$$

where $\vec{C}$ is the constant of integration.

When $t = 0$, $\vec{v} = \vec{u}$ (initial velocity of the particle), $\therefore \vec{u} = \vec{C}$

$\therefore$ Equation (13) is $\vec{v} = \frac{q\vec{E}}{m}t + \vec{u}$ or $\frac{d\vec{r}}{dt} = \frac{q\vec{E}}{m}t + \vec{u}$ ...(14)

Again integrating eq. (14), we get

$$\vec{r} = \frac{q\vec{E}}{2m}t^2 + \vec{u}t + \vec{C_1} \quad ...(15)$$

When $t = 0$, suppose the position vector of the particle $\vec{r}$ is $\vec{r_0}$; then $\vec{C_1} = \vec{r_0}$

Therefore $$\vec{r} = \frac{q\vec{E}}{2m}t^2 + \vec{u}t + \vec{r_0} \quad ...(16)$$

If $\vec{u}$ and $\vec{r_0}$ are along the direction of the field $\vec{E}$, then the position vector $\vec{r}$ will have the directions of $\vec{E}$. It is to be noted that in general, eq. (16) is applicable in its vector form whatever may be the directions of $\vec{r_0}$, $\vec{u}$ and $\vec{E}$. If the initial position vector of the particle is $\vec{r_0}$ $(\vec{OP})$ and at $t = 0$ the velocity is $\vec{u}$ (along PQ), then at any instant t, its displacement $\vec{r}$ will be equal to the vector sum of $\vec{r_0}$, $\vec{u}t$ and $q\vec{E}t^2/2m$, where the field is constant in magnitude and its direction is along $\vec{QR}$ **[fig. 12.3]**.

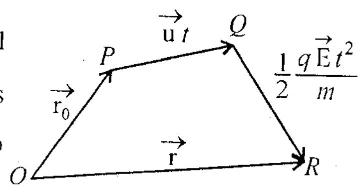


Fig. 12.3.

If the particle is at rest $(\vec{u} = 0)$ and initially at the origin *i.e.*, $\vec{r_0} = 0$, then

$$\vec{r} = \frac{1}{2}\frac{q\vec{E}}{m}t^2 \quad ...(17)$$

where obviously the displacement is along $\vec{E}$.

12·2. Electron Gun

An electron gun is an arrangement of electrodes, which produces a narrow and intense beam of electrons. It consists of (i) an *indirectly heated cathode, C,* (ii) a *control grid G,* (iii) *a focussing anode A_1*, and (iv) *an accelerating anode A_2* **[fig.12.4]**. The control grid G is held at negative potential with respect to cathode C, heated by filament F, so that the electrons emitted from the cathode experience a force of repulsion from the grid and are obtained in the form of a narrow beam. This thin beam of electrons is accelerated by two cylindrical anodes A_1 and A_2. Focussing anode focuses the electron beam by controlling the positive potential in it. The potential of accelerating anode A_2 (about 10,000 volts) is much higher than that of focussing anode. Hence this anode A_2 accelerates the narrow beam to very high velocity. Thus the electron gun assembly produces a narrow and fast beam of electrons. If no deflecting plates are present, the electrons move in straight line after emerging from the accelerating anode to the centre of the screen where they produce a bright spot.

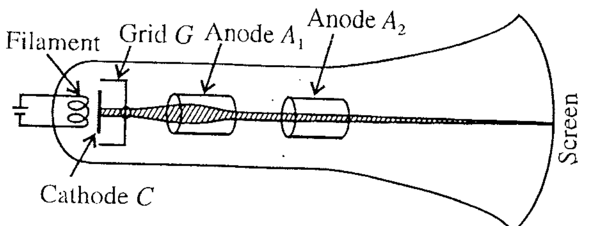


Fig. 12.4 : Electron Gun.

If the electrons, are accelerated through a potential difference V, then the kinetic energy of the electrons is given by

$\frac{1}{2}mv^2 = eV$; so that the velocity of an electron, $v = \sqrt{2eV/m}$.

12·3. Discharge Tube

If in a glass tube containing air we establish a high potential difference (≈ 10 kilo volt) across the positive and negative electrodes and reduce the pressure (1 mm or less), then the electric conduction starts. The electric flow in the tube at low pressures is called **electric discharge** and the glass tube with the said arrangement is called **discharge tube [fig. 12.5]**. At ordinary pressure and low voltage, no discharge takes place in a gas between the two eelctrodes.

When the gas pressure in a discharge tube is kept about 10^{-3} mm of mercury and a potential difference of 10 kilo-volt is applied between its electrodes by an induction coil, some invisible rays move towards anode perpendicular to the cathode and the walls of the glass tube show a bright fluorescence. The colour of the fluorescence depends on the composition of the glass. These invisible rays, emitted by the cathode and producing fluorescence on the glass walls, are called **cathode rays.** In the glass tube any gas is taken and the electrodes of any metal are used, such rays are always emitted from the cathode. From experiments, it is known that these rays consist of negatively charged particles, each of mass 9.1×10^{-31} kg and charge 1.6×10^{-19} coulomb. These particles are called **electrons.** Thus the *cathode rays are streams of fast moving electrons.*

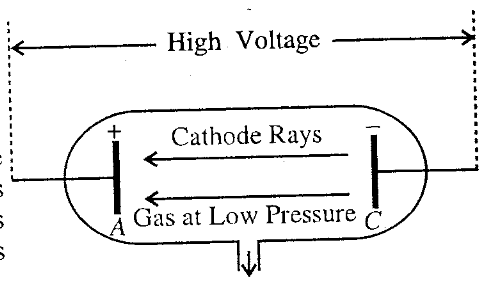


Fig. 12.5 : Discharge Tube.

Electric discharge tubes are used in making fluorescent tubes, sodium lamp, neon and helium tubes.

12·4. Linear Accelerator

In several experiments, high energy α-particles, protons etc. are needed. The energy of these charged particles is increased by accelerating them. If we allow a charged particle in a vacuum tube to pass through accelerating potential differences successively, its energy can be increased in sufficiently high amount. The arrangement used for this purpose is called linear accelerator. In an accelerator, positive particles, such as α-particles, protons etc., are passed through an alternating electric field of radio frequency so that the energy of a particle increases successively.

Construction : In a linear accelerator, the arrangement consists of several coaxial cylindrical electrodes 1, 2, 3, 4, ... in series along the axis of a vacuum tube. These cylindrical electrodes are known as drift tubes. The length of one after another tube increases successively. Alternate cylindrical tubes 1, 3, 5, ... are connected to one end of a radio frequency oscillator and rest of the alternate tubes 2, 4, 6,.... are connected to the other end. This is why two nearby tubes have potentials of opposite sign. A positively charged particle, whose energy is to be increased, moves along the axis of the tubes through the aperture *A*. The lengths of the cylindrical tubes are arranged in such a way so that the distance travelled by the particle in one half cycle of the electric field is equal to the distance between two successive tubes and for the rest of the half cycle the distance moved by the particle is equal to the length of the next hollow cylinder. Thus in the space in between any two nearby tubes and through each tube, the particle moves for the same time and the speed of the charged particle increases regularly.

Principle of working : Suppose a positive ion (say α-particle) leaves the aperture *A* and is accelerated during the half cycle when tube 1 is negative with respect to *A*. In the cylindrical tube, the velocity of the particle does not increase. The frequency of the alternating electric field is so arranged that as the particle comes out of the tube 1, tube 1 becomes positive and tube 2 negative. Hence the positively charged particle in the space in between two electrodes is accelerated

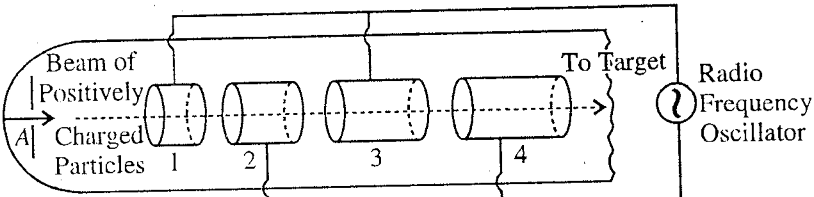


Fig. 12.6 : Linear accelerator.

towards the tube 2 and the energy of the particle increases. The length of the tube 2 is such that at the instant when the positively charged particle comes out of it, tube 2 becomes positive and tube 3 negative. This results again in the acceleration and thereby in increasing the energy of the particle. This process continues and finally we get the particles of sufficiently high energy.

Remember that as the velocity of the charged particle increases, the lengths of the successive tubes and the distance between two electrodes increases so as to cross each of them in equal time.

Let the maximum value of the alternating voltage between any two nearby electrodes be V volt. If there are n electrodes, there will be n spaces between n electodes, where we have included one space in between aperture A and electrode 1. When the particles having charge q, passes through the potential difference V between the two electrodes, the kinetic energy of the charged particle increases by qV. Thus, when the charged paticle crosses through n spaces between the electrodes, its kinetic energy will increase by nqV and hence the final kinetic energy of the paricle will be given by

$$\frac{1}{2}m_n v_n^2 = nqV + K \qquad \text{...(1)}$$

where m is the mass of the particle and K is its kinetic energy when it leaves the aperture A.

Hence the final velocity of the particle will be

$$v_n = \sqrt{\frac{2(nqV+K)}{m}} \qquad \text{...(2)}$$

If we want to determine the length L_p of the pth electrode, then the velocity corresponding to pth electrode will be

$$v_p = \sqrt{\frac{2(pqV+K)}{m}}$$

Hence,

$$L_p = v_p \frac{T}{2} = \frac{v_p}{2f} = \frac{1}{2f}\sqrt{\frac{2(pqV+K)}{m}} \qquad \text{...(3)}$$

from which the length of the pth cylindrical electrode can be obtained.

Thus, if we want to have a chraged particle of higher energy from a linear accelerator, then according to eq. (1),

(1) the number of cylindrical electrodes (n) should be large.

(2) the maximum value of the alternating potential difference (V) should be high,

and (3) the initial kinetic energy of the particle (K) should be high.

We see that in a linear accelerator, the lengths of the cylindrical tubes increases and hence their number cannot be increased more than a limit. If we want to decrease the lengths of the tubes, then according to eq. (3) the value of radio frequency should be largest possible.

First of all in 1931, by using a linear accelerator Sloan and Lawfence obtained positive ions of 1.26 MeV energy. In 1947, Alverz et al. obtained a beam of protons of 32 MeV energy by such an accelerator. The length of the accelerator in the experiment was 13 metre. In Stanford University, an electron beam of 350 MeV energy was obtained by using an accelerator of length 65 metre.

12·5. Electric Field as Deflecting Field (Motion of a Charged Particle in a Uniform Transverse Electic Field)

When a charged particle enters in a uniform electric field in a direction perpendicular to the field, it gets deflected perpendicularly to its initial velocity.

Suppose that a beam of charged particles (*e.g.*, electrons) is moving along X-axis through a pair of parallel plates [**fig. 12.7**]. Let the potential difference between the plates be V. Hence, if d is the distance between the plates then the intensity of the field along Y-axis is $E_y = V/d$. As the force qE_y on the charged particle remains perpendicular to its initial velocity v_x, so v_x remains constant. If the electron travels a distance x along X-axis in the field, covering this distance, it requires a time $t = x/v_x$. In this time, the transverse velocity, gained by particle, is

$$\frac{dy}{dt} = v_y = 0 + at = \frac{qE_y}{m}\cdot\frac{x}{v_x} \quad [\because \text{ initially } u_y = 0] \qquad \text{...(1)}$$

where $a = qE_y/m$ is the acceleration of the paricle.

Hence the displacement along Y-axis is

$$y = \frac{1}{2}\frac{qE_y}{m}t^2 = \frac{qE_y}{2m}\left(\frac{x}{v_x}\right)^2 \text{ or } y = \frac{qE_y}{2mv_x^2}\cdot x^2 \quad ...(2)$$

Here, the quantity $\frac{qE_y}{2mv_x^2}$ is constant, say A, then

$$y = Ax^2 \quad ...(3)$$

which represents a parabola. Thus, *the path of a charged particle in a transverse uniform electric field is parabola.*

The directrion of motion of the particle is changed due to the field. At any instant t if the inclination of the direction of motion to X-axis be ϕ, then

$$\tan\phi = \frac{dy}{dx} = \frac{qE_y}{mv_x^2}x \quad ...(4)$$

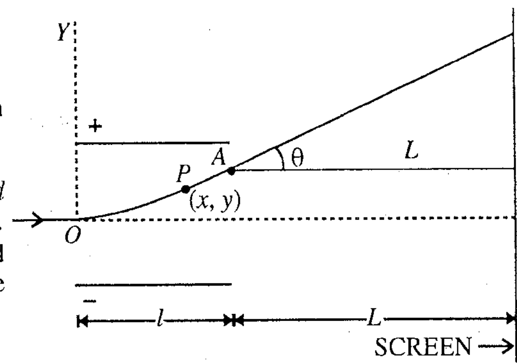


Fig. 12.7 : Motion of a charged particle in a transverse uniform electric field.

But the particle remains in the field for a distance l, the length of the plates, hence due to the entire field the displacement XB (say y_1) along Y-axis will be obtained by substitution $x = l$ in eq. (2) *i.e.*,

$$y_1 = \frac{qE_y l^2}{2mv_x^2} \quad ...(5)$$

After traversing the field, the direction of the particle is changed through the angle θ *i.e.*,

$$\tan\theta = \frac{qE_y l}{mv_x^2} \quad \text{[substituting in eq. (4) } \phi = \theta \text{ and } x = l] \quad ...(6)$$

Now, if we want to know the displacement of the particle on the screen, placed at a distance L from the end A of the plate, then it is to be noted that as soon as the particle emerges from the field, no force acts on it. Hence, the particle will travel towards the screen along the tangent drawn at the point, where the electron just comes out of the field. Now the total displacement on the screen is

Y = displacement between the plates (XB) + displacement outside the plates (BC)

$$= \frac{qE_y l^2}{2mv_x^2} + L\tan\theta = \frac{qE_y l^2}{2mv_x^2} + L\cdot\frac{qE_y l}{mv_x^2} = \frac{qE_y}{mv_x^2}\left(\frac{l^2}{2} + Ll\right) \quad ...(7)$$

This principle has its applications in cathode ray oscillograph, television picture tubes etc.

12·6. Cathode Ray Oscilloscope (CRO)

Cathode ray oscilloscope is an instrument in which the variations of potential difference are seen on a screen with the help of a beam of electrons, producing a bright spot on the screen. Cathode ray tube is the principal part of the oscilloscope.

Cathode ray tube : It consists of a glass tube of conical shape evacuated to a very low pressure of the order of 10^{-5} mm of mercury. The tube is equipped with three parts : electron gun, deflection plates assembly and fluorescent screen.

(1) Electron gun : It is a device to produce a narrow and accelerated beam of electrons. The electron gun consists of four parts : (i) cathode, (ii) control grid, (iii) focussing anode and (iv) accelerating anode. The control grid is held at a negative potential with respect to cathode, heated by a filament F. The electrons emitted from cathode are repelled by the grid and are obtained in the form of a narrow beam. This beam of electorns is accelerated by two cylindrical anodes which have increasing potentials with respect to the cathode. First anode focusses the beam and the second anode accelerates the beam to a high velocity. Thus, we get a narrow beam of fast moving electrons from the electron gun and it produces a bright spot of light on the screen.

(2) Deflection plates assembly : Beyond the electron gun, the deflection of the beam may be done by two sets of deflecting plates : One set is the vertical deflection (Y_1Y_2) plates and the other set is the horizontal deflection (X_1X_2) plates. As shown in **fig. 12.8**, the vertical deflecting plates (Y_1Y_2) are fixed horizontally in the tube. The electron beam can be made to move up-down vertically on the screen by applying proper potential to these plates. The horizontal deflecting plates (X_1X_2) are mounted in a vertical plane so that an application of potential can make the electron beam right and left horizontally on the screen.

(3) **Fluorescent screen :** The screen is the inside face of the tube and it is coated with a fluorescent material. When the electron beam strikes the screen, a bright spot of light is produced at the place of collision.

Working : The alternating potential whose waveform is to be seen on the screen, is applied across the Y-plates. This electric field makes the bright spot to move up and down in the vertical plane. Because of persistent of vision, a bright vertical line is seen on the screen. Now a special varying potential is applied across the X-plates. This potential increases linearly with time to a maximum and then suddenly

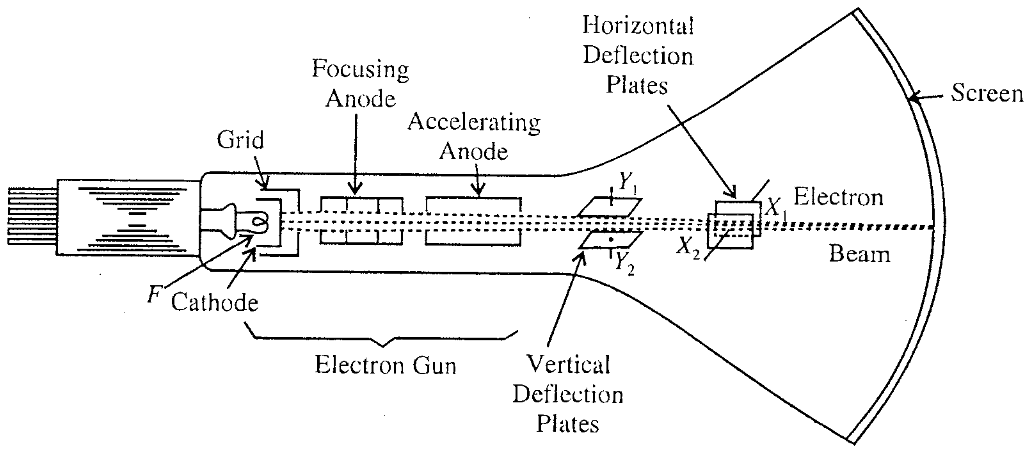


Fig. 12.8 : Cathode ray Oscilloscope.

falls to zero, then again increases similarly, again falls to zero suddenly and so on. This potential is called **'time-base potential'** and is produced by a a special electrical circuit. This time-base potential causes the spot of light to sweep linearly across the screen in the horizontal direction and then flies back to the initial position for the next sweep. This results in a horizontal line to appear on the screen which is called the **'time-base line'**. If the unknown varying potential has been applied to the Y-plates, the spot of light moves up and down due to the unknown potential and at the sane time the spot moves horizontally. Therefore, the waveform of the unknown potential is seen on the screen.

If we want to compare the intensities of two electric fields, one field is applied to X-plates and the other to Y-plates. Now the coordinates of the bright spot on the screen are proportional respectively to the horizontal and vertical deflecting fields and hence one can compare the intensities of two electric fields.

Sensitivity : The sensitivity of a cathode ray oscilloscope (CRO) is defined by the deflection of the light spot on the screen due to an application of 1 volt potential difference between the deflecting plates. For example, if a potential difference of 100 volts. is applied on the vertical deflecting plates and this results in a vertical shift of 5 mm of the light spot, then the sensitivity S of the CRO is given by

$$S = \frac{5 \text{ mm}}{100 \text{ volts}} = 0.05 \text{ mm/volt.}$$

In general, Sensitivity, $S = \dfrac{\text{Displacement of the spot on the screen}}{\text{Applied potential difference between } X_1 \text{ and } X_2 (\text{or } Y_1 \text{ and } Y_2) \text{ plates}}$ ...(1)

If V is the potential difference between the vertically deflecting plates and y is the displacement on the screen, then sensitivity,

$$S = \frac{y}{V} \qquad ...(2)$$

An expression for vertical deflection y of electrons due to a potential difference V has been deduced in Art. 12.5 *i.e.*,

$$y = \frac{eE_y}{mv_x^2}\left(\frac{l^2}{2} + Ll\right) \qquad ...(3)$$

where $E_y = V/d$, d = distance between the plates, l = length of the plates, L = the distance from the end of the plates to the screen and v_x = velocity of the electrons, coming out of the electron gun. e and m are the charge and mass of an electron respectively.

If the electrons have been accelerated to v_x velocity by a potential difference of V_a volt in the electron gun, then

$$\frac{1}{2}mv_x^2 = eV_a \text{ or } v_x^2 = eV_a/m \qquad ...(4)$$

Hence
$$y = \frac{Vl}{2V_a d}\left(\frac{l}{2} + L\right) \qquad ...(5)$$

Hence Sensitivity,
$$S = \frac{y}{V} = \frac{l}{2V_a d}\left(L + \frac{l}{2}\right) \qquad ...(6)$$

Hence the **sensitivity** of a CRO is (i) *directly proportional to the length l of the deflecting plates,* (ii) *inversely proportional to the separation d between the plates,* (iii) *inversely proportional to the accelerating voltage* V_a *and* (iv) *directly proportional to the distance* $(L + l/2)$.

Uses of CRO : Cathode ray oscilloscope is a very useful equipment in the study of electronics. It is used in different rays in the laboratories, medical science and industries. Some important uses are the following :

(1) To study the waveform of an unknown alternating potential.

(2) To determine the amplitude of an alternating voltage or current.

(3) To determine the frequency of an alternating potential. For this purpose, an alternating potential of known frequency is applied on horizontal plates and the alternating potential is applied to vertical plates. By changing the known frequency, Lissajous figure is obtained on the screen. If a horizontal line cuts the Lissajous figure (*e.g.*, of shape 8) m times and vertical line n times then,

$$\frac{\nu_v}{\nu_h} = \frac{m}{n}$$

where ν_v is unknown frequency and ν_h is the known frequency. If both the frequencies are unknown then the two frequencies can be compared by above procedure.

(4) To measure inductance, frequency, phase angle, power factor, hysteresis loss etc. in the laboratory.

(5) It is extremely useful in radar and television.

(6) In medical science, it is used to study the heart-beats and heart diseases.

Ex. 1. *Charge 4e is placed at the point A (1, 2, 3) cm. Find the electric field intensity at the point B (4, 5, 10) cm.*

Sol.
$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} = \frac{1}{4\pi\varepsilon_0}\frac{4e}{r^2}\hat{r}$$

Here, $\vec{r} = \vec{r}_B - \vec{r}_A = (4\hat{i} + 5\hat{j} + 10\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$ cm $= 3\hat{i} + 3\hat{j} + 7\hat{k}$ cm

$= 10^{-2}(3\hat{i} + 3\hat{j} + 7\hat{k})$ m

$\therefore \quad r = 10^{-2}\sqrt{9+9+49} = 10^{-2}\sqrt{67}$ m; $\therefore \hat{r} = \frac{\vec{r}}{r} = \frac{3}{\sqrt{67}}\hat{i} + \frac{3}{\sqrt{67}}\hat{j} + \frac{7}{\sqrt{67}}\hat{k}$

Hence $\vec{E} = 9 \times 10^9 \frac{4e}{(67)^{3/2} \times 10^{-4}} (3\hat{i} + 3\hat{j} + 7\hat{k}) = 1.05 \times 10^{-7}(3\hat{i} + 3\hat{j} + 7\hat{k})$ **volt/m.**

Ex. 2. *Calculate the value of the electric field intensity which will give a proton an acceleration equal to the acceleration due to gravity.* **(Kanpur 1990)**

Sol. Force $= eE = mg. \quad \therefore E = \frac{mg}{e} = \frac{1.67 \times 10^{-30} \times 9.8}{1.6 \times 10^{-19}} =$ **1.02 × 10^{-9} volt/m.**

Ex. 3. *In a cathode-ray tube, the distance between cathode and anode is 50 cm. and the potential difference is 50,000 V. If an electron leaves the cathode from rest, and there is no obstruction in its path, then with what velocity will it strike the anode ? Mass of electron is 9.0 × 10^{-31} kg and its charge is 1.6 × 10^{-19} C.*

Sol. If an electron of mass m and charge e passes through a potential difference V, then its kinetic energy will be given by

$$\frac{1}{2}mv^2 = eV \text{ or } v = \sqrt{2eV/m}$$

Here, $e = 1.6 \times 10^{-19}$ Coul., $V = 50{,}000$ V and $m = 9.0 \times 10^{-31}$ kg.

Hence, $v = \sqrt{\frac{2 \times (1.6 \times 10^{-19}) \times 50{,}000}{9.0 \times 10^{-31}}} =$ **1.33 × 10^8 ms^{-1}.**

Ex. 4. *A positive ion of charge q and mass m, accelerated by a potential difference V, passes with a constant velocity through the space between two grids d distance apart. Show that transit time between the grids is* $d/(2qV/m)^{1/2}$.

Sol. Work done by the electric field when the ion moves through a potential difference = Charge × Potential differenc = qV.

This becomes the kinetice energy of the particle *i.e.*,

$$\frac{1}{2}mv^2 = qV, \therefore v = \sqrt{2qV/m.}$$

If t be the transit time between the grids d distance apart, then

$$t = \frac{d}{v} = \frac{d}{(2qV/m)^{1/2}}.$$

Ex. 5. *What must be the initial velocity of electrons which can just strike a conductor at* – 1200 *volts potential.*

Sol. Potential of the conductor = – 1200 volts.

The electrons, having negative charge, are repelled by a conductor at negative potential. In order that an electron just reaches the conductor, it should have the kinetic energy given by

$$\frac{1}{2}mv^2 = eV, \therefore v = \sqrt{\frac{2eV}{m}} = \sqrt{\frac{2 \times (1.6 \times 10^{-19}) \times 1200}{9.0 \times 10^{-31}}} = \mathbf{2.05 \times 10^7 \ m/sec.}$$

Ex. 6. *If an alpha particle is accelerated by a potential difference of* 1000 *volts, find its kinetic energy in electron volts and velocity in m/sec. [Charge an a particle = 2e and its mass = 4 × 1.6 × 10^{-27} kg.]*

Sol. Kinetic energy, $\frac{1}{2}mv^2 = qV = 2eV = 2 \times 1.6 \times 10^{-19} \times 1000$ joule

$$= \frac{2 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-19}} \times 1000 \text{ eV} = \mathbf{2000 \ eV.}$$

Velocity, $v = \sqrt{\frac{4eV}{m}} = \sqrt{\frac{4 \times 1.6 \times 10^{-19} \times 1000}{4 \times 1.6 \times 10^{-27}}} = \sqrt{10^{11}} =$ **3.1 × 10^5 m/sec.**

Ex. 7. *A beam of electrons, in which the velocity of each electron is 3.0 × 10^7 m/sec is deflected by 1.8 mm in travelling a distance 10 cm in an electric field of intensity 1.8 × 10^3 volt/metre, in the normal direction. Calculate the specific charge e/m of electron.* **(Bilaspur 2003)**

Sol. Deflection, $y = \frac{1}{2}at^2 = \frac{1}{2}\frac{qE}{m}\left(\frac{l}{v}\right)^2$; $\therefore \frac{e}{m} = \frac{2yv^2}{El^2}$

where $a = \frac{qE}{m} = \frac{em}{m}$ and time of travel $t = \frac{\text{length of the plates}}{\text{velocity}} = \frac{l}{v}$.

Here, $y = 1.8$ mm $= 1.8 \times 10^{-3}$ m, $E = 1.8 \times 10^3$ volt/m; $l = 10$ cm $= 0.1$ m, $v = 3.0 \times 10^7$ m/s

Hence $$\frac{e}{m} = \frac{2yv^2}{El^2} = \frac{2\times(1.8\times10^{-3})\times(3.0\times10^7)^2}{(1.8\times10^3)\times(0.1)^2} = \mathbf{1.8 \times 10^{11}\ Coul/kg.}$$

Ex. 8. *Initially the velocity of an electron is* $\vec{u} = 10^4\hat{i}$ *m/sec and the position vector is* $\vec{r}_0 = 1\hat{j}$ *m. Find the position and velocity vectors after* 10^{-8} *sec. in an electric field of* $300\,\hat{i}$ *volt/m intensity.*

Sol. $$\vec{v} = \vec{u} + \vec{a}t = \vec{v} + \frac{\vec{F}}{m}t = \vec{u} - \frac{eE}{m}t = 10^4\,\hat{i} - \frac{(1.6\times10^{-19})(300\hat{i})}{0.91\times10^{-30}}\times10^{-8} = \mathbf{-5.17\times10^5\,\hat{i}\ m/sec.}$$

and $$\vec{r} = \vec{r}_0 + \vec{u}t - \frac{1}{2}\frac{eE}{m}t^2 = 1\hat{j} + 10^4\hat{i}\times10^{-8} - \frac{1}{2}\cdot\frac{1.6\times10^{-19}\times300\hat{i}(10^{-8})^2}{0.91\times10^{-30}} = \mathbf{\hat{j} - 0.25\times10^{-2}\,\hat{i}\ m.}$$

Ex. 9. *An electron beam travels through a potential difference of* 1000 *volts. Find its velocity. If this electron beam is passed through the deflecting plates of* 2 *cm length lying* 0.5 *cm apart, find the potential difference between them so as to deflect the beam through* 5°. *Find also the transit time of an electron through the deflecting plates.*

Sol. If an electron, passing through a potential difference V, acquires a velocity v, then

$$\frac{1}{2}mv^2 = qV \therefore v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2\times1.6\times10^{-19}\times1000}{9.1\times10^{-31}}} = \mathbf{1.9\times10^7\ m/sec.}$$

$$\tan\theta = \frac{v_y}{v_x} = \frac{qE_y}{m}\frac{l}{v_x^2}.\ \text{If } \theta \text{ is small, } \tan\theta = \theta,\ \therefore E_y = \frac{\theta m v_x^2}{lq}.$$

Here, $$E_y = \frac{V}{d} = \frac{V}{0.5\times10^{-2}} = 200\,V.$$

$$\therefore \quad V = \frac{E_y}{200} = \frac{\theta m v_x^2}{200lq}\ \text{volts} = 5\times\frac{\pi}{180}\times\frac{9.1\times10^{-31}\times(1.9\times10^7)^2}{200\times2\times10^{-2}\times1.6\times10^{-19}} = \mathbf{44.8\ volts.}$$

Transit time, $$t = \frac{l}{v_x} = \frac{2\times10^{-2}}{1.9\times10^7} = \mathbf{1.05\times10^{-9}\ sec.}$$

Ex. 10. *The sensitivity of a cathode ray oscilloscope is* 0.03 *mm/volt. On apply an unknown potential difference on its horizontal plates, the spot on the screen displaced* 6 *mm horizontally. Calculate the unknown potenital difference.*

Sol. Sensitivity $= \dfrac{\text{Deflection}}{\text{Potential difference between plates}}$

Here, sensitivity $= 0.03$ mm/volt, deflection $= 3$ mm

$\therefore$ Potential $= \dfrac{\text{Deflection}}{\text{Sensitivity}} = \dfrac{6\text{ mm}}{0.03\text{ mm/volt}} = \mathbf{200\ volt.}$

EXERCISE 12·1

1. An electron is moving with a velocity of 4×10^6 ms^{-1}. Calculate its kinetic energy in eV. **(Ujjain 2003)**
[**Ans.** 45.5 eV.]
2. Calculate the kinetic energy in joule and velocity in m/sec. of an electron of 100 electron volt energy.
[**Ans.** 1.6×10^{-17} J; 6×10^6 m/sec.]
3. A force of $(5\,\hat{i} - 2\,\hat{j})\times10^{-5}$ newton is acting on a particle of mass 5 gm. If the initial velocity of

the particle is $(4\,\hat{i} + 10\,\hat{k})$ cm/sec. and initial position is (0, 3, 0) cm, find the displacement and the position of the particle after 15 seconds.

[**Ans.** $(1.72\,\hat{i} - 0.42\,\hat{j} + 1.5\,\hat{k})$ m; $(1.72\,\hat{i} - 0.45\,\hat{j} + 1.5\,\hat{k})$ m.]

4. What is the order of magnitude of the ratio of electrostatic and the gravitational forces between two electrons ? ($G = 6.67 \times 10^{-11}$ S.I. units). **(Agra 1990, 84)**
[**Ans.** $\approx 10^{42}$.]

$$\left[\textbf{Hint : } F_1 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}, F_2 = \frac{Gm^2}{r^2}.\right]$$

5. What is the force on a proton of charge 1.6×10^{-19} coulomb in an electric field of 3×10^4 volts/cm ?
[**Ans.** 4.8×10^{-13} newton.]

6. Calculate the intensity of electric field which will give a proton an acceleration equal to $2g$. **(Agra 1990)**
[**Ans.** 2.05×10^{-7} volt/m.]

7. Determine in volt per metre the intensity of electric field which can just balance the weight of (i) an electron, (ii) an oil drop of mass 0.001 milligram containing one extra electron.
[**Ans.** (i) 5.51×10^{-11} volt/m; (ii) 6×10^{10} volt/m.]
[**Hint :** $eE = mg$.]

8. In a cathode-ray rube, an electron starting from rest from cathode reaches the anode with a velocity of 10^7 ms^{-1}. What is the potential difference between the electrodes ?
[**Ans.** 280 volts.]

9. The distance between the plates of a cathode-ray oscilloscope is 6.25 mm and a potential difference 3.125 volts is set up between them. What will be the acceleration of an electron due to the electric field produced between the plates ?
[**Ans.** 8.91×10^{16} m/sec.]

10. Determine the electric field in newton/Coul. and volt/cm necessary just to balance (i) a proton, (ii) an α-particle. Take the mass of α-particle 4 times that of proton and its charge two times that of proton.
[**Ans.** (i) 1.02×10^{-7} newton/Coul; 1.02×10^{-9} volt/cm.
(ii) 2.05×10^{-7} newton/Coul; 2.05×10^{-9} volt/cm.]

11. Calculate the gain in kinetic energy by an (i) electron, (ii) α-particle in moving from the plate, having potential 0 volt, to the plate having potential 100 volts.
[**Ans.** (i) 100 eV; (ii) – 200 eV or loss in kinetic energy = 200 eV.]

12. What must be the initial velocity of an electron which can just strike a conductor at – 300 volts potential.
[**Ans.** 1.025×10^7 m/s]

13. A particle charged with two electrons, is accelerated by a potential difference of 1000 volts. After crossing the field, it travels 0.025 cm in 10^{-8} sec. in field free space, find its mass.
[**Ans.** 10.2×10^{-23} kg.]

14. Adjoining figure shows a particular charge distribution $\frac{1}{2}q, -q, \frac{1}{2}q$. Show that for $r >> a$ the electric field at P is given by $\dfrac{3Q}{4\pi\varepsilon_0 r^4}$, where $Q = qa^2$.

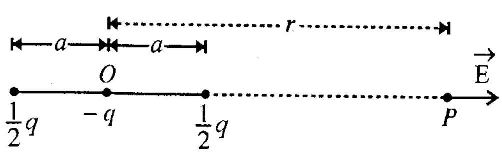


Fig. 12.9.

15. If (i) an electron, (ii) proton travels in an electric field of intensity 1 volt/cm, find the velocity acquired and distance travelled in 1 peco second (1 peco second = 10^{-12} second).
[**Ans.** (i) 18 m/sec, 0.9×10^{-11} m; (ii) 0.01 m/sec, 5×10^{-15} m.]

16. A proton of mass 1.6×10^{-27} kg is travelling with a velocity $5 \times 10^6\,\hat{i}$ m/s. Calculate the transverse

deflection in travelling a length $0.1\,\hat{i}$ m when a uniform electric field $20000\,\hat{j}$ volt/m is switched on in the space where the proton is travelling. Proton charge is 1.6×10^{-19} coulomb. **(Agra 1997)**
[**Ans.** 0.4 mm.]

17. If an electron of energy 10^{-17} joules moves through the deflecting plates of length 2 cm and transverse electric field of strength 0.01 stat volts/cm, find vector velocity and the angle $(\vec{v}, x)$ as it leaves the region between the plates. (1 stat volt = 300 volts.)
[**Ans.** $4.5 \times 10^6\,\hat{i} + 2.3 \times 10^5\,\hat{j}$ m/sec; 2.7°.]

18. In a cathode ray oscilloscope, the deflecting plates are of length 2 cm and are separated from each other by 0.5 mm. An electron accelerated through a potential difference of 1000 volts, enters in between the deflecting plates. Calculate the potential difference between the plates so that the electron is deflected by 0.25 mm from its path.
[**Ans.** 1.25 volts.]

19. A cathode ray oscillograph has deflecting plates of length 2 cm and separation 0.5 cm. Calculate the potential difference (in volts) between the plates which will cause angular deviating of 0.04 radian in an electron beam of speed 8.0×10^8 cm/sec. ($e/m = 1.67 \times 10^{11}$ Coul./kg.) **(Agra 1971)**
[**Ans.** 3.84 volts.]

20. An electron enters the region between the plates of a parallel plate capacitor at a point equidistant from either plate. The capacitor plates are 2×10^{-2} m apart and 10^{-1} m long. A potential difference of 300 V is kept across the plates. Assuming that the initial velocity of the electron is parallel to the capacitor plates, calculate the largest value of the velocity of the electron so that it does not fly out of the capacitor at the other end. (Given : $e = 1.6 \times 10^{-19}$ C and $m_e = 9.1 \times 10^{-31}$ kg).
(Roorkee 1997)

[**Ans.** 3.63×10^7 m/sec.]

12·7. Charged Particle in a Constant Uniform Magnetic Field

When a charged particle moves through a magnetic field, a force acts on the particle. This force is called **Lorentz force.**

Suppose that a charge q is moving in a magnetic field $\vec{B}$ with velocity $\vec{v}$. Then the force acting on the particle is given by

$$\vec{F} = q(\vec{v} \times \vec{B}) \qquad ...(1)$$

where the force is in newtons, q in coulombs, $\vec{v}$ in m/sec and $\vec{B}$ in tesla. If q is measured in e.s.u., $\vec{v}$ in cm/sec and $\vec{B}$ in gauss, q will be replaced by q/c and the force will be in dynes *i.e.*,

$$\vec{F} = \frac{q}{c}(\vec{v} \times \vec{B}) \qquad ...(2)$$

In view of eq. (1) following cases may arise :

(1) If the charged particle is at rest *i.e.*, $\vec{v} = 0$, then

$$\vec{F} = 0 \qquad ...(3)$$

Thus *if a charged particle is not moving in a magnetic field, no force will act on it and the particle will remain at rest.*

(2) If the charged particle is moving along or parallel to the magnetic field, then $\vec{v}$ and $\vec{B}$ are in the same direction. Hence

$$\vec{v} \times \vec{B} = 0;\ \text{So that from (1)}\ \vec{F} = 0 \qquad ...(4)$$

Thus *if a charged particle is moving along the direction of the magnetic field, no foce will act on the particle and it will move with constant velocity along a straight line.*

(3) **Transverse magnetic field** *i.e.*, $\vec{v} \perp \vec{B}$: If the charged particle is moving with velocity $\vec{v}$ in a direction perpendicular to the magnetic field $\vec{B}$, then there is angle $\theta = \pi/2$ between $\vec{v}$ and $\vec{B}$. Hence the magnetic force on the particle is $F = q|\vec{v} \times \vec{B}| = qvB \sin\theta = qvB \sin \pi/2 = qvB$. The direction of this force is perpendicular to both $\vec{v}$ and $\vec{B}$, given by right hand screw rule.

Now, suppose the particle is moving with velocity v in a direction perpendicular to X-axis (or Y-axis) and enters the magnetic field $\vec{B}$ at apoint P_0. The direction of $\vec{B}$ is perpendicular to the plane (X–Y plane) of the paper in the downward direction, shown by (×) in **fig 12.10**. As soon as the charged particle will enter the magnetic field, a force qvB will start to act perpendicular to velocity towards O. As the force is acting perpendicular to the velocity, its speed will remain constant and the path of the particle will be circular. The necessary centripetal force (mv^2/ρ), to rotate the particle on a circular path, is given by the magnetic force (qvB) *i.e*,

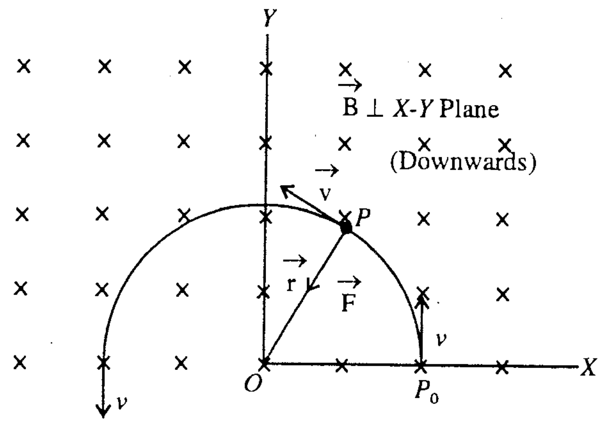


Fig. 12.10 : Motion of charge in transverse magnetic field.

$$\frac{mv^2}{\rho} = qvB \qquad \text{...(5)}$$

where m is the mass of the particle and ρ is the radius of circular path. The direction of the force is towards O, the centre of the circular path.

Here the force $\vec{F}$ is perpendicular to $\vec{v}$, hence *the work done on the charged particle by the force is zero i.e., its kinetic energy will remain constant.*

This is because $\vec{F} \cdot \vec{v} = 0$ and hence

$$W = \int \vec{F} \cdot d\vec{r} = \int \vec{F} \cdot \frac{d\vec{r}}{dt} dt = \int (\vec{F} \cdot \vec{v}) dt = 0 \qquad \text{...(6)}$$

From eq. (5)

$$\rho = mv/qB \qquad \text{...(7)}$$

The radius of curvature of the particle moving in a transverse magnetic field is called **gyro-radius** or **cyclotron radius.**

Here, the **angular velocity** of the particle in circular motion is given by

$$\omega = v/\rho = qB/m \qquad \text{...(8)}$$

Hence **period** of circular motion, $T = 2\pi/\omega = 2\pi m/qB$. ...(9)

and **frequency** of circular motion, $\nu = 1/T = \omega/2\pi = qB/2\pi m$ ...(10)

This is called **gyro-frequency** or **cyclotron frequency.**

Clearly the cyclotron frequency does not depend on the velocity or energy of the particle. However at **relativistic speeds**, mass of the particle ($m = m_0/\sqrt{1 - v^2/c^2}$) varies with velocity and hence at these speeds, the cyclotron frequency depends on the velocity or kinetic energy of the particle.

(4) If the moving particle makes an angle θ with the direction of the magnetic field *i.e.*, the angle between $\vec{v}$ and $\vec{B}$ is θ, then the velocity $\vec{v}$ can be resolved into two components **[fig. 12.11]**.

(i) $v_1 = v \cos\theta$, which is along the direction of $\vec{B}$; due to this magnetic force $[q(\vec{v}_1 \times \vec{B}) = 0]$ is zero and the particle will move along $\vec{B}$ with this speed v_1.

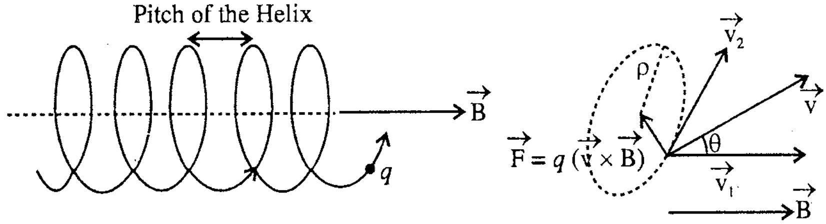


Fig. 12.11 : Motion of a charge in a magnetic field.

(ii) $v_2 = v \sin\theta$, which is perpendicular to the magnetic field $\vec{B}$; due to this, a force qv_2B $[\because q(\vec{v} \times \vec{B}) = qv_2B]$ will act on the particle in a direction perpendicular to $\vec{v}_2$ and hence the path of the particle will be circular.

Thus the charged particle due to $\vec{v}_1$ velocity component will move along the direction of the magnetic field $\vec{B}$ and due to **velocity component** $\vec{v}_2$, it will move on a circular path. Hence the path of the particle is **helix** (or spring like) with its axis along $\vec{B}$ **[fig. 12.11]**.

The necessary centripetal force for rotation on a circular path will be given by the magnetic force qv_2B *i.e.*,

$$mv_2^2/\rho = qv_2B; \therefore \rho = mv_2/qB \qquad ...(11)$$

This is the expression for **gyro-radius** or **cyclotron radius.**

The angular velocity of circular motion (in circular or helical path) is given by the following :

$$\omega = v_2/r = qB/m \qquad ...(12)$$

The period in which the charged particle completes one circle is given by

$$T = 2\pi/\omega = 2\pi m/qB \qquad ...(13)$$

The number of revolutions per second performed by a charged particle in a magnetic field is called **gyro-frequency** or **cyclotron frequency**. Hence the cyclotron frequency is

$$n = 1/T = 2\pi/\omega = qB/2\pi m \qquad ...(14)$$

Pitch of the helix : This is the distance travelled by the charged particle along $\vec{B}$ in the time in which it completes one circle. Thus,

$$\text{Pitch of the helix} = v_1T = \frac{2\pi m}{qB} v_1 \qquad ...(15)$$

[In Gaussian CGS system of units $\rho = \dfrac{mv_2c}{qB}$, $\omega = \dfrac{qB}{mc}$, $T = \dfrac{2\pi mc}{qB}$, $n = \dfrac{qB}{2\pi mc}$ and pitch $= \dfrac{2\pi mc}{qB} v_1$]

The circular or helical paths of the charged particles become visible within a cloud chamber or bubble chamber, where a magnetic field is applied.

Alternative method – Solution of the equation of motion of charged particle in a magnetic field : If a charge q is moving in a magnetic field $\vec{B}$ with velocity $\vec{v}$, then the force acting on the particle is given by

$$\vec{F} = q(\vec{v} \times \vec{B})$$

If m is the mass of the charged particle, the equation of motion is

$$m\frac{d\vec{v}}{dt} = q(\vec{v} \times \vec{B}) \qquad ...(1)$$

If we assume X-axis along the direction of the magnetic field, then $\vec{B} = B\hat{i}$. Writing $\vec{v}$ in its three mutually perpendicular components, eq. (1) will have the form

$$m\frac{d}{dt}(v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} + v_z\hat{\mathbf{k}}) = q\,(v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} + v_z\hat{\mathbf{k}}) \times (B\,\hat{\mathbf{i}})$$

or
$$\hat{\mathbf{i}}\,\frac{dv_x}{dt} + \hat{\mathbf{j}}\,\frac{dv_y}{dt} + \hat{\mathbf{k}}\,\frac{dv_z}{dt} = \frac{qB}{m}(v_z\hat{\mathbf{j}} - v_y\hat{\mathbf{k}}) \quad ...(2)$$

Separating the components, we have

$$\frac{dv_x}{dt} = 0 \quad ...(3a), \qquad \frac{dv_y}{dt} = \frac{qB}{m}v_z \quad ...(3b), \qquad \text{and } \frac{dv_z}{dt} = -\frac{qB}{m}v_y \quad ...(3c)$$

It is clear from the eq. (3a) that there is no acceleration in the direction of magnetic field. Integrating eq. (3a), we get

$$v_x = \text{a constant.} \quad ...(4)$$

Hence *in the direction of the magnetic field the velocity of the particle remains constant.* Let it be v_1.

Substituting the value of v_z from eq. (3b) in eq. (3c), we have

$$\frac{d^2v_y}{dt^2} = -\left(\frac{qB}{m}\right)^2 v_y \text{ or } \frac{d^2v_y}{dt^2} = -\,\omega^2 v_y \quad ...(5)$$

where $\omega = qB/m$.

Multiplying both sides of eq. (5) by $2dv_y/dt$, we have

$$2\frac{dv_y}{dt}\frac{d^2v_y}{dt^2} = -\,\omega^2\, 2\frac{dv_y}{dt}v_y$$

On integration, $\left(\dfrac{dv_y}{dt}\right)^2 = -\,(\omega v_y)^2 + C$, where C is the constant of integration.

This equation can be written in the form $\dfrac{dv_y}{\sqrt{C-(\omega v_y)^2}} = dt$

Intergrating it, we get

$$\frac{1}{\omega}\sin^{-1}\left(\frac{\omega v_y}{\sqrt{C}}\right) = t + C_1, \text{ where } C_1 \text{ is another constant of intergration}$$

or
$$v_y = \frac{\sqrt{C}}{\omega}\sin(\omega t + \omega C_1)$$

Writing $\sqrt{C}/\omega = A$ and $\omega C_1 = \phi$, we obtain

$$v_y = A\sin(\omega t + \phi) \quad ...(6)$$

where the constants A and ϕ can be determined directly from the initial boundary conditions.

Substituting the value of v_y in eq. (3b), we get

$$\frac{d}{dt}[A\sin(\omega t + \phi)] = \omega v_z \quad \text{or} \quad v_z = A\cos(\omega t + \phi) \quad ...(7)$$

Squaring eqs. (6) and (7) and then on adding, we get.

$$A^2 = v_y^2 + v_z^2$$

If v_2 is the velocity of the particle in the Y-Z plane, then

$$v_y^2 + v_z^2 = v_2^2, \ \therefore\ v_2 = A$$

Now, $v_y = dy/dt = v_2\sin(\omega t + \phi)$ and $v_z = dz/dt = v_2\cos(\omega t + \phi)$.

Integrating both equations, we get.

$$y = -\frac{v_2}{\omega}\cos(\omega t + \phi) \quad ...(8a), \qquad \text{and} \qquad z = \frac{v_2}{\omega}\sin(\omega t + \phi) \quad ...(8b)$$

Squaring eqs. (8a) and (8b) and then on adding, we have

$$y^2 + z^2 = v_2^2/\omega^2 \qquad ...(9)$$

This equation represents a circular motion in Y-Z plane with angular (frequency) velocity $\omega = qB/m$. *The radius of this circle is*

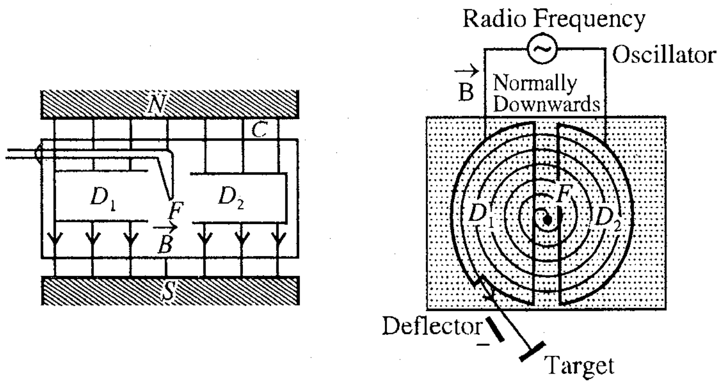
$$\rho = \frac{v_2}{\omega} = \frac{mv_2}{qB} \qquad ...(10)$$

This is the expression for the **cyclotron** or **gyro-radius**. Thus the motion of the charged particle will consist of a circular motion in Y-Z plane, added over a uniform linear motion along X-axis *i.e.*, the path of the particle will be helix.

12·8. Cyclotron

In order to produce high energy charged particles, in 1932 E.O. Lawrence invented a device, known as **cyclotron.**

Construction : The vertical and horizontal sections of the cyclotron are shown in **fig.12.12**. The apparatus consists of a flat cylindrical vacuum chamber C in which there are two semicylindrical hollow electrodes D_1 and D_2. Due to their shape of D they are called **dees**. These dees have their diametrical edges parallel and slightly separated from one another. The electrodes (dees) are connected to the terminals of a radio frequency oscillator so that a high frequency (10 to 15 MHz) alternating potential difference is applied between the dees. The chamber C is placed horizontally in between N and S poles of a powerful electromagnet. This electromagnet produces a field of the order of 15000 oersted. In between the dees, a filament F is placed which can be heated electrically. When electric current flows through this filament, the electrons are emitted. These electrons ionise the gas, which is filled inside the chamber at low pressures and so the positive ions are produced. For example, protons are produced from hydrogen gas, deutrons from deuterium gas and α-patricles from helium gas. When the ions acquire high energy within dees, they come out of the chamber through a thin aluminium window. The positive ions are deflected from their paths by a plate, called the deflector, having constant negative potential relative to D_1 and then they are allowed to fall on the target.



(a) Vertical section (b) Horizontal section

Fig. 12.12 : Cyclotron.

Working principle and theory : If a positive ion is produced near the filament F, at that instant it is attracted towards the negative D (say D_1). Electric field is ineffective, when the charged particle is inside the dee. Hence the ion starts to move in the horizontal plane. Since the magnetic field is acting vertically, the ion, before coming to the gap between the dees, moves on a circular path. Then the ion comes back in the gap. If the frequency of the applied alternating potential difference and the intensity of the magnetic field are so adjusted that the ion completes a semicircle in the half period of the alternating potential difference, then it will reach in the gap at the instant when second dee (D_2) is negative. Thus the particle will be again accelerated and the same process will be repeated after each half cycle. Evidently, the potential difference is in the condition of **resonance** with the circular motion of the ions. Hence the requirement for the periodic acceleration of the particle is

Frequency of the electric field = Cyclotron frequency

i.e., $$n = qB/2\pi m \qquad ...(1)$$

When the particle crosses the gap, it is accelerated and so its velocity is increased. Consequently, the radius of its path ($\rho = mv/qB = v/\omega$) increases at the same angular velocity. This process continues, until the radius of the path acquires a maximum value R, which is equal to the radius of the dees, when it comes out the cyclotron. The particle emerges out with **maximum kinetic energy** or **speed**, given by the relation :

$$R = \frac{mv_{max}}{qB} \qquad ...(2)$$

i.e.,

$$v_{max} = \frac{qBR}{m} \text{ and K.E.} = \frac{1}{2}mv_{max}^2 = \frac{1}{2}\frac{q^2B^2R^2}{m} \qquad ...(3)$$

Its **momentum**, $p_{max} = mv_{max} = qBR$ ...(4)

Thus, to get the particles of high energy it is essential that the radius of the final circular path *i.e.*, cyclotron radius (R) and the value of the magnetic field (B) should be large. It is to be noted that in the cyclotron, the energy of the particle is independent of the potential difference applied to the dees. When the potential difference is small, the particle will acquire the same energy through a large number of revolutions. But when it is large, the particle will acquire this energy in few turns.

However to get the particles of very high energies, the velocity becomes comparable to the velocity of light and so the mass of the particle starts to change according to the relation $m = m_0/\sqrt{1 - v^2/c^2}$. In consequence, the particle will fall out of the step with radio frequency and will not cross the gap when alternating potential difference is maximum. This sets a limit to the energy, which a particle can have inside the cyclotron. The electron cannot be accelerated to high energies by a cyclotron because due to its small mass it will have a large velocity for a comparatively small kinetic energy and then its mass will increase.

12·9. Curvature of Tracks for Energy Determination of Nuclear Particles

In a cloud chamber or bubble chamber, different kinds of moving charged particles produce different types of tracks. If we want to determine the momentum and energy of a charged particle, then the cloud chamber is kept in a strong magnetic field. Suppose that in a beam of charged particles, the charge of each particle is q and initially each particle is moving with a velocity v along a straight line. Now if a magnetic field is switched on perpendicular to the motion of the particles, then a charged particle will move on a path of r radius and produce a curved track in the cloud chamber. The necessary centripetal force for rotation is provided by the magnetic force $q\,|\vec{v} \times \vec{B}| = qvB$ *i.e.*,

$$qvB = \frac{mv^2}{r} \quad \text{or} \quad mv = qBr \qquad ...(1)$$

Thus momentum of particle, $p = mv = qBr$ ...(2)

Kinetic energy of particle, $E = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{q^2B^2r^2}{2m}$ ...(3)

Therefore, if the radius r of the circular path of a charge particle is more, its kinetic energy will be more. If we know, the charge q, mass m, magnetic field B and radius r of the curved track, we can find the kinetic energy of the charged particle by using eq. (3).

Remember that the above equation is correct at non-relativistic speeds. At relativistic speeds the kinetic energy is given by

$$E = \sqrt{p^2c^2 + m_0^2c^4} - m_0c^2 \qquad ...(4)$$

where $p = qBr$, m_0 = rest mass of the particle and c = speed of light.

12·10. Deflection of a Charged Particle in a Transverse Magnetic Field and Fast CRO

Consider a charged particle of charge q and mass m. Suppose it is moving initially with velocity v along X-asis. When it enters in a transverse magnetic field $\vec{B}$ (perpendicular to X-Y plane or plane of the paper downward), the charged particle will be acted by a magnetic force $F = qvB$ perpendicular to its velocity along OY. The acceleration of the particle along Y-direction is given by

$$a = \frac{\text{Force}}{\text{Mass}} = \frac{qvB}{m} \qquad ...(1)$$

Due to this force, the particle will start to move on a circular path of radius $\rho = mv/qB$ and will be deflected upwards. If the deflection of the charge paricle is small during its motion in the magnetic field in comparison to ρ, the raduis of circular path, then we may assume the acceleration a to be constant along Y-direction. If the particle moves a length of the magnetic field region, then the time of travel in the magnetic field t, is given by

$$t = l / v \qquad ...(2)$$

In this duration, the particle will be deflected along Y-direction, given by

$$y_1 = \frac{1}{2}at^2 \text{ or } y_1 = \frac{1}{2}\frac{qvB}{m}\frac{l^2}{v^2} = \frac{qBl^2}{2mv} \qquad ...(3)$$

When the charged particle emerges out of the magnetic field, no magnetic force will act on it and it moves along a straight line QT. If v_y is the velocity developed in the Y-direction in time t, then

$$v_y = at = \frac{al}{v} = \frac{qBl}{m} \text{ and } \tan\theta = \frac{v_y}{v_x} = \frac{qBl}{mv} \qquad ...(4)$$

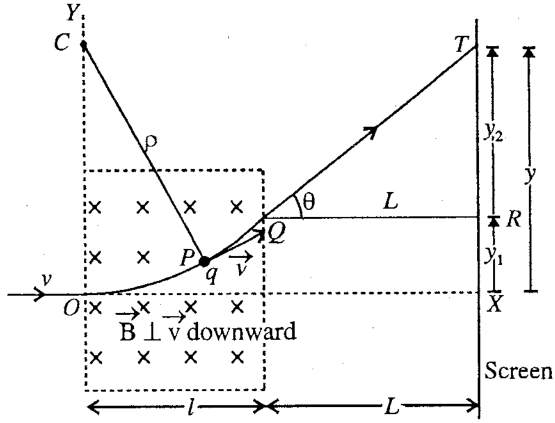


Fig. 12.13 : Deflection of a charged particle in transverse magnetic field.

If the distance of the screen is L from Q, the point of emergence [**fig. 12.13**], then the additional displacement along Y-axis :

$$y_2 = L\tan\theta = \frac{LqBl}{mv} \qquad ...(5)$$

Hence the total displacemet $XT = Y$ on the screen is given by

$$y = y_1 + y_2 \text{ or } y = \frac{qBl^2}{2mv} + \frac{qBlL}{mv} = \frac{qBl}{mv}\left(\frac{l}{2} + L\right) \qquad ...(6)$$

Hence the deflection is directly proportional to the magnetic field B. This principle is used in fast CRO and in Thomson method of finding e/m for positive ions.

Fast CRO : In case of a cathode ray oscilloscope, if the electron beam, obtained from the electron gun, is deflected by magnetic field instead by electric field, we may obtain much larger deflection and hence much larger picture on the screen even by employing a short cathode ray tube. Such an arrangement, using magnetic field for deflection of electron beam, is called **fast CRO**. As shown above, the deflection produced by a magnetic field B on the screen is given by $y = \frac{qBl}{mv}\left(\frac{l}{2} + L\right)$ instead $y = \frac{qBl}{mv^2}\left(\frac{l}{2} + L\right)$ for the case of electric field. Hence the deflection, produced by magnetic field is large because the presence of v in the denominator of the expression of y instead of v^2 in that of electric field.

12·11. 180° Magnetic Focusing

For the separation of different types of ions, 180° magnetic focusing is also a method used in mass spectrographs.

Let a beam of charged particles enter a region through a slit in which a uniform magnetic field of intensity $\vec{B}$ is present. The field $\vec{B}$ is perpendicular to the beam or the plane of the paper [**fig. 12.14**]. In this field, any charged particle of the beam will move along a circular path of radius $\rho = mv/qB$, where v is the component of velocity in the plane of the paper *i.e.*, normal to $\vec{B}$. Now, if we examine the beam after 180° of motion, we find that it is spread out in the plane of the motion, because the various particles, having different masses and velocities describe the circular paths of different radii of curvature. Thus *we can obtain a beam of particles having closely* **equal momenta** *after* 180° *deflection,* provided that the particles have the same charge. Therefore, such

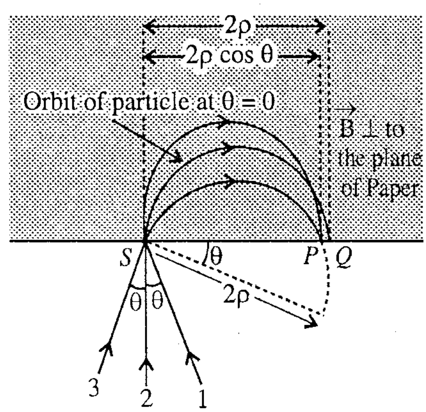


Fig. 12.14 : 180° magnetic focusing.

an arrangement can be used as **momentum selector**. This is also one advantage of this method that particles of equal momenta, but moving through an entrance slit at slightly different angles, are brought to an approximately common focus on the screen after a deflection of 180°.

If the initial path of the particle (2) is normal to the screen, the particle after describing circular path will reach at Q and SQ will be equal to 2ρ, the diameter of the circle. If the initial path is (1) or (3), making an angle θ to (2), then it will reach at P after describing an arc of circle, whose diameter is inclined to SQ at an angle θ. The width of the focused image is

$$PQ = SQ - SP = 2\rho - 2\rho \cos\theta = 2\rho\,(1 - \cos\theta) = \rho\theta^2 \quad ...(1)$$

$$\left[\because \cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} ..., \text{ neglecting } \theta^4 \text{ and higher terms}\right]$$

For small value of θ, θ^2 will be very small, hence the width PQ is small. The image of S at Q will be a line, because the velocity along the magnetic field, remains constant. If the slit S is parallel to the magnetic field, then the image lines of different points of the slit coincide and a narrow line image is obtained on the screen.

To determine the relative abundances of the isotopes of an element, a second slit is placed near the point Q in a mass spectrometer. A mixture of such ions, accelerated by an adjustable potential difference, is passed through the receiving slit S. Only those ions will pass through the second slit at Q, for which

$$\frac{mv}{q} = B\rho \qquad \left(\because \frac{mv^2}{\rho} = qBv\right) \quad ...(2)$$

These ions then send a current through a sensitive galvanometer. If the accelerating potential is gradually changed evidently the ions of different m/q will pass through the slit at Q in succession and can be detected by the deflection of the galvanometer. This deflection is directly proportional to the relative number of various types of ions. Thus, with the help of this method, we can determine the relative abundances of the isotopes of an element.

12·12. Magnetic Focusing–Magnetic Lens

If an electron beam is focussed with the help of a magnetic field, it is called **magnetic focusing.** Suppose the electrons from an electron gun are accelerated through a potential differenc V along the X-axis. The velocity of these electrons is given by

$$\frac{1}{2}mv^2 = eV \text{ or } v = \sqrt{2eV/m} \quad ...(1)$$

These electrons emerge from a narrow hole at O into a uniform magnetic field $\vec{B}$, acting along the X-axis. However, some of the electron emerging from the hole make slightly divergent angles *i.e.*, electron beam coming out of the hole of electron gun is little divergent, as shown in **fig. 12.15**. This can be shown that this electron beam will be focussed at a point P on the X-axis because of the magnetic field and hence it is called **magnetic focusing.**

Suppose that the velocity $\vec{v}$ of an electron at O, makes a small angle θ with the direction of the magnetic field $\vec{B}$. Those electrons which are moving along X-axis, there will be no effect of magnetic field as $\vec{v}$ is parallel to $\vec{B}$ *i.e.*, $\vec{v} \times \vec{B} = 0$ and hence $\vec{F} = q\,(\vec{v} \times \vec{B}) = 0$.

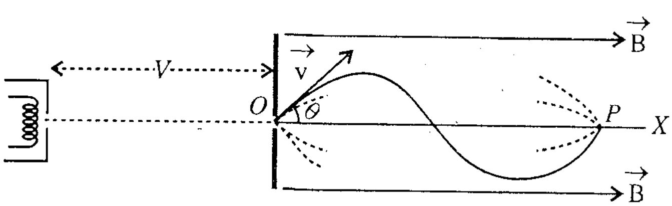


Fig. 12.15 : Magnetic focusing (magnetic lens).

The component of this velocity $\vec{v}$ along X-axis is $v_1 = v\cos\theta \simeq v$ and the perpendicular component is $v_2 = v\sin\theta$. Since this v_2 is perpendicular to the magnetic field, a magnetic force ev_2B will act on the electron and will provide the necessary centripetal force to move it on a circular path of radius r. If the radius of the circular path is r, then

$$\frac{mv_2^2}{r} = ev_2B \text{ or } v_2 = \frac{eBr}{m} \quad ...(2)$$

The time required to move once on the circular path,

$$T = \frac{2\pi r}{v_2} = \frac{2\pi rm}{eBr} = \frac{2\pi m}{eB} \qquad ...(3)$$

As the velocity of the particle along X-axis is $v_1 \simeq v$ approximately, hence the electron will move along X-axis with a circular motion. Thus the path of the electron will be **helix.** The pitch of this helix, say l, will be the distance moved by the electron in time T *i.e.*,

$$l = v_1 T = \frac{v 2\pi m}{eB} = \frac{2\pi m}{eB}\sqrt{\frac{2eV}{m}} = \sqrt{\frac{8\pi^2 mV}{eB^2}} \qquad \left[\because v = \sqrt{\frac{2eV}{m}}\right] \qquad ...(4)$$

We see from (3) that the time period is independent of the angle θ, hence all the electrons, emerging from the hole O (or electron gun) at small angel θ will come together or focus at a point P after moving a distance l. The magnetic field will act on the divergent electron beam similar to convergent lens focusses a divergent light beam. This device, used to focus a divergent beam of electron, is called **magnetic lens.** By changing B, the focal length of the magnetic lens can be changed.

In the electron gun, the focusing of electron beam is done with the help of electrostatic potential. However the electron beam coming out of the electron gun has a tendency to diverge due to mutual repulsive forces between the electrons. Hence magnetic focusing has the advantage over the electrostatic focusing that it focusses all the electrons to a single point and makes the beam intense. Magnetic focusing is used in television where we need sharp focusing on the screen.

Ex. 1. *An electron of energy* 10 *eV is revolving in a circular path in a magnetic field* 1×10^{-4} *weber/m². Calculate* : (i) *speed of electron,* (ii) *radius of circular path and* (iii) *frequency of the electron* ($m_e = 9.0 \times 10^{-31}$ *kg*) **(Rewa 2004; Indore 2003; Sagar 2003; Raipur 2003)**

Sol. (i) Here, Kinetic energy of electron, $\frac{1}{2}mv^2 = 10\ eV = 10 \times 1.6 \times 10^{-19}$ joule

Hence, speed of electron, $v = \sqrt{\dfrac{2\times10\times1.6\times10^{-19}}{9.0\times10^{-31}}}$ = **1.9 × 10⁶ m/s**

(ii) Radius of circular path, $\rho = \dfrac{mv}{qB} = \dfrac{(9.0\times10^{-31})\times(1.9\times10^{6})}{(1.6\times10^{-19})\times(1\times10^{-4})}$ = **0.11 m.**

(iii) Frequency of electron, $n = \dfrac{v}{2\pi r} = \dfrac{1.9\times10^{6}}{2\times3.14\times0.11}$ = **2.75 × 10⁶ per sec.**

Ex. 2. *An electron of energy* 10 *eV is moving in a circular path of radius* 0.106 *m in a plane normal to the uniform magnetic field. Calculate the intensity of magnetic field.* **(Ujjain 2003)**

Sol. $\frac{1}{2}mv^2 = 10\ eV,\ \therefore\ v = \sqrt{\dfrac{20\times1.6\times10^{-19}}{9.1\times10^{-31}}}$ = **1.9 × 10⁶ m/sec.**

Now, $\rho = \dfrac{mv}{eB},\ \therefore\ B = \dfrac{mv}{e\rho} = \dfrac{9.1\times10^{-31}\times1.9\times10^{6}}{1.6\times10^{-19}\times0.106}$ = **10⁻⁴ tesla.**

Ex. 3. *An electron of velocity* $\vec{v} = (2\hat{i} + 3\hat{j})\,10^6$ *m/sec enters a region of uniform magnetic field* $\vec{B} = 500\,\hat{i}$ *gauss, so that its path becomes helical. Now,*

(i) *In what direction does the axis of the helix lie ?*

(ii) *Calculate the radius of the helix.*

(iii) *Calculate the number of rotations as the electron advance* 10 *cm along the axis of the helix.* **(Agra 1970 S)**

Sol. Velocity of the electron $\vec{v} = (2\hat{i} + 3\hat{j})\,10^6$ m/sec.

Magnetic field $\vec{B} = 500\,\hat{\mathbf{i}}$ gauss $= 5 \times 10^{-2}\,\hat{\mathbf{i}}$ tesla

$\therefore \quad v_1 = 2 \times 10^6\,\hat{\mathbf{i}}$ m/sec. and $v_2 = 3 \times 10^6\,\hat{\mathbf{j}}$ m/sec.

(i) The axis of the helix line along $\vec{B}$ *i.e.,* along X-axis.

(ii) Radius of the helix, $\rho = \dfrac{mv_2}{qB} = \dfrac{9.1 \times 10^{-31} \times 3 \times 10^6}{1.6 \times 10^{-19} \times 5 \times 10^{-2}} = \mathbf{3.4 \times 10^{-4}}$ **m.**

(iii) Distance travelled in 1 rotation = pitch of the helix.

$$= \frac{2\pi m}{qB} v_1 = \frac{2\pi \times 9.1 \times 10^{-31} \times 2 \times 10^6}{1.6 \times 10^{-19} \times 5 \times 10^{-2}}$$

$\therefore$ No. of rotations in 0.1 m distance $= \dfrac{0.1 \text{ m}}{\text{pitch of the helix}} = \dfrac{0.1 \times 1.6 \times 10^{-19} \times 5 \times 10^{-2}}{2\pi \times 9.1 \times 10^{-31} \times 2 \times 10^6} \approx \mathbf{72}.$

Ex. 4. *If x is the distance between the point where an electron enters and leaves transverse magnetic field, show that the deflection θ is equal to xqB/mv radians and the displacement y of the beam just as it leaves the field is given by $(r - y)^2 + x^2 = r^2$, where r is the radius of curvature of the path.*

(Kanpur 1981)

Sol. At the point Q, where the electron leaves the magnetic field, the electron will go along the tangent at that point [**fig. 12.16**]. This tangent makes an angle θ with the initial direction of the electron *i.e.,* θ is the angle of deflection and evidently the same angle θ will be subtended by the arc of the circle, described in the magnetic field, at the centre of curvature. Now the distance moved along X-axis is

$$x = r \sin \theta \qquad \ldots\text{(i)}$$

Distance moved along Y-axis is

$$y = r - r \cos \theta$$

or

$$r - y = r \cos \theta \qquad \ldots\text{(ii)}$$

Squaring and adding (i) and (ii), we get

$$(r - y)^2 + x^2 = r^2 (\cos^2 \theta + \sin^2 \theta) = r^2 \qquad \ldots\text{(iii)}$$

From relation (i), if θ is small, $\sin \theta = \theta = x/r$

But $\quad r = \dfrac{mv}{qB}, \quad \therefore \theta = \dfrac{xqB}{mv}. \qquad \ldots\text{(iv)}$

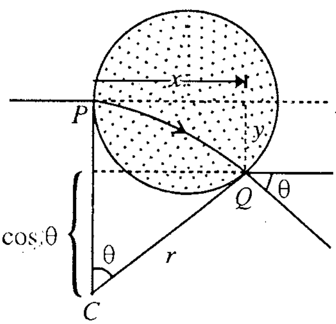


Fig. 12.16.

Eqs. (iii) and (iv) give us the desired relations.

Ex. 5. *For a proton initially travelling with velocity $v = 5 \times 10^8$ cm/sec., calculate the following:*

(i) *Force experienced in a magnetic field $\vec{B} = 2000\,\hat{\mathbf{j}} + 4000\,\hat{\mathbf{k}}$ gauss.*

(ii) *Acceleration experienced in an electric field $\vec{E} = 200\,\hat{\mathbf{i}} + 100\,\hat{\mathbf{j}}$ volts/cm.*

(iii) *Transverse deflection in travelling a length $L_x = 10$ cm in an electric $E_y = 200$ volts/cm.*

(Agra 1969)

Sol. (i) Force in magnetic field

$$\vec{F} = q(\vec{v} \times \vec{B}) = q(5 \times 10^6)\,\hat{\mathbf{i}} \times (0.2\,\hat{\mathbf{j}} + 0.4\,\hat{\mathbf{k}})$$

$$= 1.6 \times 10^{-19} \times 5 \times 10^6 \times 0.2\,(\hat{\mathbf{k}} - 2\,\hat{\mathbf{j}}) = 1.6 \times 10^{-13}\,(\hat{\mathbf{k}} - 2\,\hat{\mathbf{j}})\text{ N.}$$

Therefore the magnitude of the force, $F = 1.6 \times 10^{-13}\sqrt{5} = \mathbf{3.57 \times 10^{-13}}$ **newton.**

Its direction cosines are $\left(0, -\dfrac{2}{\sqrt{5}}, \dfrac{1}{\sqrt{5}}\right)$.

(ii) In electric field, $m\vec{a} = q\vec{E}$

$$\therefore \quad \vec{a} = \frac{q\vec{E}}{m} = \frac{1.6\times10^{-19}}{1.67\times10^{-27}} \times (200\hat{i} + 100\hat{j}) \times 10^2 = \frac{1.6}{1.67} \times 10^{12} \times \sqrt{5}\left(\frac{2\hat{i}}{\sqrt{5}} + \frac{\hat{j}}{\sqrt{2}}\right)$$

Magnitude of, $\vec{a}$ = **2.14×10^{12} m/sec^2**; Its direction cosines are $\left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0\right)$.

(iii) Transverse deflection, $y = \frac{1}{2}at^2 = \frac{1}{2}\left(\frac{qE_y}{m}\right)\left(\frac{l}{v_x}\right)^2$

$$= \frac{1}{2} \times \frac{1.6\times10^{-19}}{1.67\times10^{-27}} \times 200 \times 10^2 \times \left(\frac{0.1}{5\times10^6}\right)^2 = \mathbf{3.83 \times 10^{-4}\ m.}$$

Ex. 6. *If the maximum value of the radio frequency voltage be* 10 *kilo volts, find the number of revolutions a proton has to make within the cyclotron before achieving one-fifth velocity of light.*

(Agra 2004; Jabalpur 04; Kanpur 1978)

Sol. Each time, when the proton crosses that gap between the dees, it gains qV energy. If the proton performs n revolutions within the cyclotron to achieve $c/5$ velocity, then the energy gained by the proton is

$$2nqV = \frac{1}{2}mv^2 \text{ or } n = \frac{mv^2}{4qV} \text{ where } v = \frac{c}{5} = \frac{3\times10^8}{5} = 6 \times 10^7 \text{ m/sec.}$$

Here, $m = 1.67 \times 10^{-27}$ kg; $v = 6 \times 10^7$ m/sec; $q = 1.6 \times 10^{-19}$ coulomb; $V = 10000$ volts.

$$\therefore \quad n = \frac{1.67\times10^{-27}\times(6\times10^7)^2}{4\times1.6\times10^{-19}\times10000} = \mathbf{939}.$$

Ex. 7. *Calulate the frequencies of the radio frequency oscillator in a cyclotron of* 1.5 *tesla field when accelerating protons. If the radius of the dees is* 50 *cm, find the energy of emergent protons.*

(Gwalior 2004; Bilaspur 03)

Sol. $$n = \frac{qB}{2\pi m}$$

For proton, $$n_1 = \frac{1.6\times10^{-19}\times1.5}{2\times3.1\times12.67\times10^{-27}} = \mathbf{24\ MHz}$$

Energy of a proton $= \frac{1}{2}mv^2 = \frac{1}{2}mR^2\omega^2$ $\quad [\because v = R\omega]$

$$= \frac{1}{2}mR^2(2\pi n)^2 = 2\pi^2 mR^2 n_1^2$$

(where R = maximum radius of the path of proton = radius of dees.)

$= 2 \times (3.14)^2 \times 1.67 \times 10^{-27} \times (0.5)^2 \times (24 \times 10^6)^2 = 45 \times 10^{-13}$ joules

$= 45 \times 10^{-13}/1.6 \times 10^{-19}$ eV = **2.8×10^7 eV.**

Ex. 8. *A cyclotron accelerating protons has dees of* 40 *cm radius and a radio frequency supply of* 12 *megahertz at* 10000 *volts maximum value. Calculate* (i) *the strength of the magnetic field,* (ii) *the speed and kinetic energy acquired by the proton and* (iii) *maximum number of revolutions and time spent by a proton within the dees.*

Sol. (i) $B = \frac{m}{q}\cdot\omega = \frac{m}{q}2\pi n = \frac{1.67\times10^{-27}\times3\times10^8\times2\times3.14\times12\times10^6}{1.6\times10^{-19}} = \mathbf{0.79\ tesla.}$

(ii) Velocity of the proton, acquired within the dees, is

$$v = \frac{qBR}{m} = \frac{1.6\times10^{-19}\times0.79\times0.4}{1.6\times10^{-27}} = \mathbf{3 \times 10^7\ m/sec.}$$

Since $v = c/10$, the relativistic correction (1/2%) is negligible.

$\therefore$ Energy of the proton $\frac{1}{2}mv^2 = \frac{1}{2}\times 1.67\times 10^{-27}\times(3\times 10^7)^2 = 7.5\times 10^{-13}\ J = \mathbf{4.7\times 10^6\ eV}$.

(iii) When a proton crosses the gap between the dees, it gains energy qV joules or V electron volts. It crosses the gap twice in each revolution and hence gain in energy $2V$ eV. If the proton makes N revolutions inside the dees, then

$$2NV\text{ eV} = 4.7\times 10^6\text{ eV},\ \therefore N = \frac{4.7\times 10^6}{2V} = \frac{4.7\times 10^6}{2\times 10000} = \mathbf{235}.$$

$$\therefore \quad \text{Time spent within dees} = \frac{235}{n} = \frac{235}{12\times 10^6} = \mathbf{1.95\times 10^{-5}\ sec.}$$

Ex. 9. *An electron is accelerated by a potential difference of* 10^6 *volts and then it enters a transverse magnetic field of strength* 1000 *gausses. Find the radius of the circular path of the electron by using classical mechanics and relativistic mechanics. Calculate also cyclotron frequency in each case.*

Sol. Kinetic energy $= \frac{1}{2}mv^2 = 10^6$ eV. $= 10^6\times 1.6\times 10^{-19}$ J

$$\therefore \quad \text{Classically, speed of the electron, } v = \sqrt{\frac{2\times 10^6\times 1.6\times 10^{-19}}{9.1\times 10^{-13}}} = \mathbf{5.93\times 10^8\ m/sec.}$$

[Note that this speed is greater than the speed of light $c = 3\times 10^8$ m/sec. So it is impossible in view of theory of special relativity.]

$$\therefore \quad \text{Radius of electron path classically, } \rho = \frac{mv}{qB} = \frac{9.1\times 10^{-31}\times 5.9\times 10^8}{1.6\times 10^{-19}\times 0.1} = \mathbf{0.0337\ m.}$$

$$\text{Cyclotron frequency classically, } n = \frac{qB}{2\pi m} = \frac{1.6\times 10^{-19}\times 0.1}{2\times 3.14\times 9.1\times 10^{-31}} = \mathbf{2.8\times 10^9\ Hz.}$$

Relativistic Kinetic Energy $= (m - m_0)\,c^2 = 10^6$ eV

$$\therefore \quad m = m_0 + \frac{10^6\text{eV}}{c^2} = 9.1\times 10^{-31} + \frac{10^6\times 1.6\times 10^{-19}}{(3\times 10^8)^2}$$
$$= 9.1\times 10^{-31} + 17.7\times 10^{-31} = 26.8\times 10^{-31}\text{ kg.}$$

$$\text{But,}\quad m = \frac{m_0}{\sqrt{1-v^2/c^2}} \text{ or } \frac{v^2}{c^2} = 1 - \left(\frac{m_0}{m}\right)^2;\ \therefore \frac{v}{c} = \sqrt{1-\left(\frac{9.1\times 10^{-31}}{26.8\times 10^{-31}}\right)} = 0.94 \text{ or } v = 0.94c$$

$$\therefore \text{ Radius of path relativistically, } \rho = \frac{mv}{qB} = \frac{26.8\times 10^{-31}\times 0.94\times 3\times 10^8}{1.6\times 10^{-19}\times 0.1} = \mathbf{0.0473\ m.}$$

$$\text{Relativistic gyro frequency, } n = \frac{qB}{2\pi m} = \frac{1.6\times 10^{-19}\times 0.1}{2\times 3.14\times 26.8\times 10^{-31}} = \mathbf{9.5\times 10^8\ Hz.}$$

Ex. 10. *An ion beam enters a magnetic field and suffers* 180° *deflection in a circular path, with* SP = 60 *cm* **[fig. 12.14]**. *If the incident beam has a cone of semi-angle* 2°, *deduce the width of the focal line around P.*

Sol. Here, $\theta = 2° = 2\pi/180$ radian

$$\therefore \quad \text{Width of the focal line} = \rho\theta^2 = \frac{0.6}{2}\left(\frac{2\pi}{180}\right)^2 = \mathbf{3.656\times 10^{-4}\ m.}$$

Ex. 11. *A beam of consisting is* U^{235} *and* U^{238} *positive ions suffers* 180° *magnetic focusing by a uniform magnetic field of* 1 *tesla. If the radius of curvature of* U^{238} *ions is* 1 *metre, calculate the distance on the screen between the two ions when* (a) *their energies are same,* (b) *their velocities are the same.*

(Indore 2004)

Sol. If a charge q moves on a circular path, then

$$qvB = \frac{mv^2}{\rho}.$$

$\therefore$ For U^{235} ions, $qv_1B = \dfrac{m_1v_1^2}{\rho_1}$ or $\dfrac{m_1v_1}{\rho_1} = qB$ ...(i)

and for U^{238} ions, $qv_2B = \dfrac{m_2v_2^2}{\rho_2}$ or $\dfrac{m_2v_2}{\rho_2} = qB$ ...(ii)

From eqs. (i) and (ii), we have

$$\frac{m_1v_1}{\rho_1} = \frac{m_2v_2}{\rho_2} \quad \text{or} \quad \rho_1 = \rho_2 \times \frac{m_1v_1}{m_2v_2} \qquad \text{...(iii)}$$

(a) When their energies are same : $\frac{1}{2}m_1v_1^2 = \frac{1}{2}m_2v_2^2$; $\therefore \dfrac{v_1}{v_2} = \sqrt{\dfrac{m_2}{m_1}}$

Therefore, using eq. (iii), we have

$$\rho_1 = \rho_2 \times \frac{m_1}{m_2} \times \sqrt{\frac{m_2}{m_1}} = \rho_2 \times \sqrt{\frac{m_1}{m_2}} = 1 \times \sqrt{\frac{235}{238}} = \mathbf{0.9937\ m.}$$

Hence the distance on the screen between the two ions

$$= 2(\rho_2 - \rho_1) = 2 \times (1 - 0.9937) = \mathbf{0.0126\ m.}$$

(b) When their velocities are the same *i.e.*, $v_1 = v_2$, then from eq. (iii) we get $v_1 = v_2$, then

$$\rho_1 = \frac{m_1}{m_2}\rho_2 = \frac{235}{238} \times 1 = \mathbf{0.9874\ m.}$$

Hence the distance between the two ions = $2(\rho_2 - \rho_1) = 2 \times (1 - 0.9874) = \mathbf{0.0252\ m.}$

EXERCISE 12·2

1. An electron is accelerated through a potential difference of 12000 volts. On entering in a uniform magnetic field of intensity 10^{-3} tesla, find the radius of path of electron. **(Bhopal 2004; Ujjain 2004)**

   [**Ans.** 36.7 cm.]

2. An electron is moving with a velocity of $(9\hat{\mathbf{i}} + 7\hat{\mathbf{j}} - \hat{\mathbf{k}})$ cm/sec through a magnetic field of intensity 1000 $(\hat{\mathbf{i}} + \hat{\mathbf{j}})$ gauss. How much is the force acting on it ? **(Kanpur 1984)**

   [**Ans.** 3.92×10^{-22} newton; direction cosines $\left(-\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)$.]

3. Find the gyro-radius and cyclotron frequency of (i) a proton, (ii) an electron travelling with a velocity of 10 cm/sec in a field of 10000 gauss. **(Agra 1980)**

   [**Ans.** (i) 0.01m, 1.52×10^7 Hz; (ii) 5×10^{-6} m, 2.8×10^{10} Hz.]

4. An alpha particle, having the mass of 4 protons and charge of two protons, describes a circular path of 10 cm radius in a field of 2000 gauss. Calculate its velocity.

   [**Ans.** 9.6×10^5 m/sec.]

5. What is the velocity and kinetic energy of a proton, which undergoes in circular path of radius 100 cm under a magnetic field of 100 gauss ? ($m_p = 1.6 \times 10^{-27}$ kg.)

   [**Ans.** 10^6 m/sec, 8×10^{-16} joules.]

6. A charged particle is moving in magnetic field of intensity 10^{-2} Wb m^{-2} with a velocity of 10^7 ms^{-1} in a circular path of radius 0.6 cm. Find out the specific charge (e/m) of the particle.

   [**Ans.** 1.66×10^{11} Coul/kg.]

7. A proton moves with a velocity 10^7 m/sec, inclined at 30° angle to the direction of the magnetic field of intensity 10^4 gauss. Calculate the following :

   (i) Lorentz force, (ii) Radius of helix,

(iii) Pitch of the helical path, (iv) Frequency of revolutions.

[**Ans.** (i) 8×10^{-13} newton; (ii) 0.05 m; (iii) Pitch $\left(= \frac{2\pi m}{qB} v \cos 30°\right) = 0.53$ m.,
(iv) Frequency $= 1.6 \times 10^7$ Hz..]

8. A particle having charge q coulomb, mass m kg moves with velocity $(v_1 \hat{\mathbf{i}} + v_2 \hat{\mathbf{j}})$ m/sec in a magnetic field $10^4 B \hat{\mathbf{i}}$ gauss. Find the expression for these terms : (i) Lorentz force, (ii) Radius of helix, (iii) Pitch of the helical path, and (iv) Frequency of solutions. **(Agra 1991)**
[**Ans.** (i) $F = qv_2 B \hat{\mathbf{k}}$; (ii) $\rho = mv_2/qB$; (iii) pitch $= 2\pi m v_1/qB$; (iv) $n = qB/2\pi m$.]

9. An electron and a proton are accelerated by an electric field of 1 statvolt/cm acting for 10 cm; they then enter a uniform magnetic field of 1 weber/m^2 in a plane perpendicular to the field.
(i) What is the cyclotron frequency of each particle ?
(ii) What is the radius of circular orbit of each particle ? **(Kanpur 1980)**
[**Ans.** (i) 2.9×10^{10} Hz, 1.5×10^7 Hz; (ii) $r_e = 1.8 \times 10^{-4}$ m, $r_p = 7.7 \times 10^{-3}$ m.]

[**Hint :** $r_e = \frac{v_e}{\omega_e}$ and $v_e = \sqrt{2as} = \sqrt{\frac{2eEs}{m}}$ and $\omega_e = \frac{eB}{m}$.]

10. What is the energy of a proton when it emerges out from the cyclotron after making 50 revolutions in it, if the voltage across the dees is 20 KV. **(Agra 1982; Kanpur 80)**
[**Ans.** 2×10^6 eV.]

11. A proton in a cyclotron gains energy 10^3 eV while crossing the gap within the dees. What will be the energy of the proton in joules when it comes out after making 150 revolutions within the dees? **(Agra 1991)**
[**Ans.** 4.8×10^{-4} joules.]

12. A proton starting from rest is accelerated in a cyclotron to a final speed 3×10^7 m/sec. How much work is done on the proton by electric froce of the cyclotrón which accelerates it. The relativistic effect may be neglected.
[**Ans.** 4.5×10^6 eV.]

13. Calculate the radius of a cyclotron orbit of a proton travelling with $c/6$ velocity in a magnetic field of 10^4 gauss, neglecting the relativistic correction. **(Rewa 2004; Rohilkhand 1979)**
[**Ans.** 0.52 m.]

14. A proton is travelling with a velocity 2.0×10^7 m/s along east in the horizontal plane and enters in a magnetic field of intensity 0.34 weber/m^2 directed along north in the horizontal plane. Calculate: (i) the magnitude and direction of force on proton, (ii) the radius of path of proton and (iii) the lateral displacement of proton in the time in which it travels a distance 0.02 m along east.
[**Ans.** (i) 1.09×10^{-12} N upwards in the vertical plane; (ii) 0.625 m; (iii) 3.2 cm.]

15. The radius of dee of a cyclotron is 0.9 m and it works in a uniform magnetic field of intensity 0.8 tesla. Calculate the cyclotron frequency and the maximum energy acquired for (i) proton; (ii) deutron, $(m_d = 2m_p)$ **(Indore 2004)**
[**Ans.** (i) 12.2 MHz, 24.8 MeV; (ii) 6.1 MHz, 1.4 MeV.]

16. A stream of protons and deutrons in a vacuum chamber enters a uniform magnetic field. Both protons and deutrons have been subjected to the same accelerating potential; hence the kinetic energies of the particles are the same. If the ion stream is perpendicular to the magnetic field and protons move in a circular path of radius 15 cm.; find the radius of the path traversed by the deutrons. Given that mass of a deutron is twice that of proton. **(Agra 1971)**
[**Ans.** 0.212 m.]

17. A cyclotron is used to accelerate protons with a radio frequecny 10 MHz with a voltage amplitude 1.5×10^4 volts. If the radius of dees is 30 cm, calculate (i) the final velocity acquired, (ii) the energy of emerging protons, (iii) the minimum number of revolutions that a proton makes before coming out of the cyclotron. Find also the time interval for which the proton remains inside dees. (Neglect relativistic effects).

[**Ans.** (i) 7.48×10^7 m/sec; (ii) 29.2 MeV; (iii) 973 revolutions; Time = 9.73×10^{-5} sec.]

18. A proton, with kinetic energy 10 eV, moves on a circular path in a uniform magnetic field. What will be the kinetic energies of (i) an α-particle and (ii) a deutron, moving on the same circular path in the same field.

[**Ans.** (i) 10 eV; (ii) 5 eV.]

19. A 50 kilovolt alternating potential difference is applied to the dees, whose radius is 38.1 cm. of a cyclotron by an oscillator. The deutron whose mass is 2 amu acquires energy of 4 MeV. Calculate following :

(i) Magnetic field strength; (ii) Frequency of oscillator;

(iii) Number of revolutions which the deutron has to make the cyclotron to gain energy.

(1 amu = 1.6×10^{-24} gm.) **(Kanpur 1979)**

[**Ans.** (i) 10.5 tesla; (ii) 8.3 MHz, (iii) 40.]

20. If an electron is moving with 2.7×10^8 m/sec. velocity in a direction perpendicular to a 200 gauss magnetic field, find the radius of curvature of the path (i) classically and (ii) on the basis of relativistic mass. ($m = 0.91 \times 10^{-31}$ kg; $e = 1.6 \times 10^{-19}$ coulomb)

[**Ans.** (i) 0.077 m; (ii) 0.177 m.]

21. A particle carries a charge of 4×10^{-9} coulomb. When it moves with a velocity v_1 of 3×10^4 m/sec at 45° above X-axis in the X-Y plane, a uniform magnetic field exerts a force F_1 along Z-axis. When the particle moves with a velocity v_2 of 2×10^4 m/sec. along Z-axis there is a force F_2 of 4×10^{-5} newton exerted on it along X-axis. What is the magnitude and direction of other magnetic field ?

[**Ans.** 0.5 tesla downward parallel to Y-axis.]

22. If a beam of protons has initial divergence of 5° from the mean path, calculate the width of the focal line obtained by 180° magnetic focusing with 50 cm radius of curvature of the path. If the speed of the protons is 10^9 cm/sec, calculate the strength of the magnetic field.

[**Ans.** 3.8×10^{-3} m; 0.2088 tesla.]

23. A charge q is accelerated by a potential difference V volt and allowed to enter a magnetic field $\vec{B}$. In the field it moves in a semicircle, striking a photographic plate a distance x from the entry slit. Show that the mass m is given by $m = \frac{B^2 q}{V} x^2$.

12·13. Charged Particle in Parallel Magnetic and Electric Transverse Fields

Suppose the magnetic field $\vec{B}$ and electric field $\vec{E}$ are along Y-axis and a particle of charge $+q$ is moving along negative Z-axis with a velocity $\vec{v}$ [**fig. 12.17**].

Therefore, $\vec{B} = B\hat{j}$, $\vec{E} = E\hat{j}$ and $\vec{v} = -v\hat{k}$.

Force produced by magnetic field = qBv along X-axis

$$\therefore \quad \text{Acceleration of the particle, } a = \frac{d^2x}{dt^2} = \frac{qBv}{m} \qquad ...(1)$$

We may assume the acceleration to be constant, if the change in magnitude and direction of the velocity due to the two fields is small.

Now, $v_x = dx/dt = at$ [$\because$ initially $dx/dt = 0$]

and $$x = \frac{1}{2}at^2 \qquad ...(2)$$

If the length of the magnetic field region be l, then the time of travel in the magnetic field is $t = l/v$.

Hence the velocity of the particle after emergence from magnetic field $v_x = a \cdot l/v$ and its displacement

$$x_1 = \frac{1}{2}\left(\frac{l}{v}\right)^2 = \frac{1}{2}\frac{qBl^2}{mv} \qquad ...(3)$$

If the screen is placed at a distance d from the point where the magnetic field ends, then it will reach the screen in d/v time.

So it is further displaced through

$$x_2 = v_x\left(\frac{d}{v}\right) = a\left(\frac{l}{v}\right)\left(\frac{d}{v}\right) = \frac{qBld}{mv} \qquad ...(4)$$

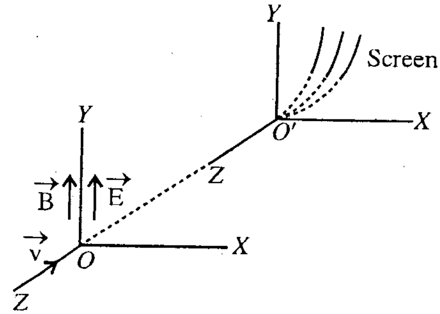


Fig. 12.17 : Thomson's parabolas.

Therefore total displacement on the srceen

$$x = x_1 + x_2 = \frac{qB}{mv}\left(\frac{l^2}{2} + ld\right) \qquad ...(5)$$

The electric field is along Y-axis, hence the acceleration is

$$a' = \frac{d^2y}{dt^2} = \frac{qE}{m}$$

Solving as above, here the total displacement is

$$y = y_1 + y_2 = \frac{qB}{mv^2}\left(\frac{l'^2}{2} + l'd'\right) \qquad ...(6)$$

where l' is the length of the electric field region and d' the distance of the screen from the end of electric field.

Eliminating v from (5) and (6), we have

$$\frac{x^2}{y} = \frac{q}{m}\frac{B^2}{E}\frac{(\frac{1}{2}l^2 + ld)^2}{(\frac{1}{2}l'^2 + l'd')} = A \text{ (say)} \qquad ...(7)$$

Evidently this represents a parabola $x^2 = Ay$. The *particles of same q/m or e/m but of different velocities will lie on different points of a parabloa.* Thomson used such parabolas to determine e/m for positive ions. As he obtained experimentally these parabolas first of all, they are called Thomson's parabolas.

If q/m is eliminated from (5) and (6), then

$$\frac{y}{x} = \frac{E}{vB}\left[\frac{\frac{1}{2}l'^2 + ld'}{\frac{1}{2}l^2 + ld}\right] \text{ or } y = mx \qquad ...(8)$$

This equation represents a straight line. Hence *all the particles with same velocity fall on a straight line passing through the centre of the screen, whatever for them q and m may be.*

Determination of the ratio of masses of atoms : For this purpose, we draw a line $pqnrs$ parallel to X-axis through the parabolas [**fig. 12.18**] for a constant value of y. If for m and m' ionic (atomic) masses, the intersection points provide pn and qn as x-coordinates. Hence from eq. (7)

$$\frac{(pn)^2}{(qn)^2} = \frac{m'}{m} \qquad ...(9)$$

while in each case the ionic charge (q) remains the same. Hence by measuring the lengths pn and qn, we can obtain the ratio of

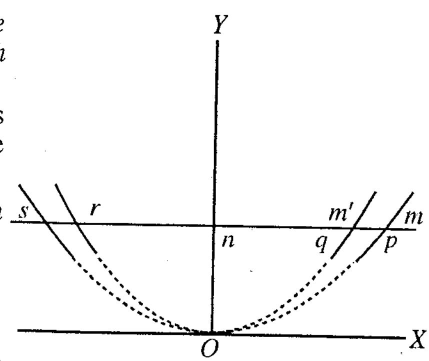


Fig. 12.18 : Thomson's parabolas for different positive ions.

atomic masses. Remember that the atoms have been used in obtaining the Thomson's parabolas in the form of ions and the ionic and atomic masses for a particular element are nearly the same.

12·14. Positive Rays

If in a cathode ray tube we have a perforated cathode facing the anode, some rays pass through the cathode to the opposite side. These rays are called *positive rays* because they travel from anode towards cathode *i.e.*, opposite to cathode rays. In fact, these rays are positive ions in motion, produced by the collision of cathode rays with gas atoms between the anode and cathode. The positive ions are attracted by the cathode and pass through it with velocity, forming the positive rays.

The positive rays are deflected by electric and magnetic fields, the deflection being in opposite direction to that of cathode rays indicating thereby their positive nature. These rays depend upon the nature of the gas filled inside the discharge tube.

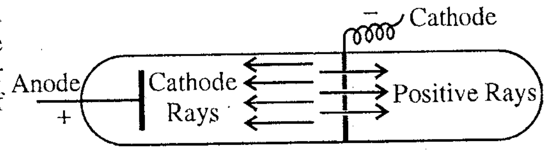


Fig. 12.19 : Positive rays.

Thomson in his experiment allowed the positive rays to pass between the poles of an electro-magnet and then to fall on a fluorescent screen. A strong electric field was also applied simultaneously parallel to the magnetic field. He could get simultaneously several parabolas on the screen. Each parabola was obtained for particular type of ions. The positive ions, produced inside the discharge tube, acquire any velocity from zero to a certain maximum value depending on the potential difference inside the tube because they are formed from cathode to anode. Thus the portions of the parabolas corresponding to very high velocities are not obtained on the screen. These parts of the parabolas are shown by dotted lines in **fig. 12.18**.

12·15. Discovery of Isotopes

If two or more atoms of an element have the same atomic number but different atomic masses, they are said to be isotopes of that element. These isotopes of an element have the same chemical properties.

In his experiment, when Thomson passed the positive ions of neon gas through the parallel electric and magnetic fields, he obtained two parabolas on the screen, displaced from each other by a small distance [**fig. 12.18**]. For a possible explanation, Thomson proposed that the neon gas was containing a mixture of two types of atoms of atomic masses approximately 20 and 22 in the form of two isotopes (using atomic weight of hydrogen atom as unity). By density measurement, the atomic weight of neon was found to be 20.2 and the cause of this result was mentioned that the neon gas is the result of mixing of these isotopes in different amounts. Finally this result was found to be true by the experiment of Aston. For the first time by the experiments of Thomson it was shown that there are **two permanent isotopes of neon.** In 1910, Soddy used first of all the word 'isotope' for the temporary isotopes, found in natural radioactivity. As the time passed, isotopes of various elements were discovered. For example, oxygen has three isotopes of atomic masses 16, 17 and 18 and chlorine two isotopes of atomic masses 35 and 37.

12·16. Elements of Mass Spectrography

The equipment which is used to find the atomic masses of the elements is called **mass spectrography.** The spectrograph focusses different lines of mass spectrum. In 1912, Thomson made the simplest mass spectrograph in which positive ions are focussed on the screen after passing through parallel electric and magnetic fields. A parabola is formed on the screen by the positive ions of same q/m. As discussed above, uning the positions of these parabolas the masses of different isotopes can be compared and if we know the atomic mass of one isotope, the mass of other isotope can be calculated. In fact in 1919, Aston prepared such a mass spectrograph in which by adjusting the electric and magnetic fields, all the positive ions of the same q/m were focussed at the same point. Modern equipments depend on mass doublet technique, in which the difference of masses of two ions having the same atomic number, can be determined with accuracy.

12·17. Charged Particle in Crossed Electric and Magnetic Fields

If a charged particle moves through an electric field $\vec{E}$ and magnetic field $\vec{B}$, then the force acting on it is given by

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B}) = q(\vec{E} + \vec{v} \times \vec{B}) \quad ...(1)$$

[In Gaussian system, $\vec{F} = q[\vec{E} + \frac{1}{c}(\vec{v} \times \vec{B})]$

We want to discuss the motion of a charged particle in crossed (*i.e.*, mutually perpendicular) electric and magnetic fields. For this, let us consider a uniform electric field $\vec{E}(= E\hat{j})$ along Y-axis and a uniform magnetic field $\vec{B}(= B\hat{k})$ along Z-axis*. Hence

$$\vec{F} = q[E\hat{j} + (v_x\hat{i} + v_y\hat{j} + v_z\hat{k}) \times B\hat{k}]$$

or
$$m\frac{d\vec{v}}{dt} = qE\hat{j} + qB(v_y\hat{i} - v_x\hat{j}) \qquad ...(2)$$

or
$$\hat{i}\frac{dv_x}{dt} + \hat{j}\frac{dv_y}{dt} + \hat{k}\frac{dv_z}{dt} = \left(\frac{qE}{m} - \frac{qBv_x}{m}\right)\hat{j} + \frac{qBv_y}{m}\hat{i}$$

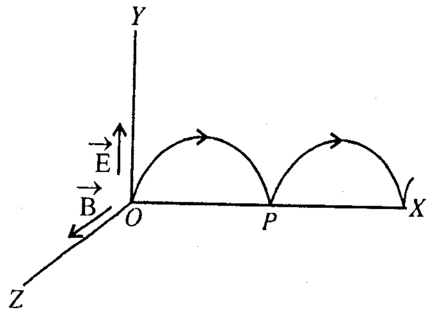


Fig. 12.20 : Motion of a charged particle in crossed electric and magnetic fields.

Hence the cartesian components of acceleration are given by

$$\frac{dv_x}{dt} = \frac{qBv_y}{m} \quad ...(3a), \qquad \frac{dv_y}{dt} = \frac{qE}{m} - \frac{qBv_x}{m} \quad ...(3b), \quad \text{and} \quad \frac{dv_z}{dt} = 0 \quad ...(3c)$$

From eq. (3c), we have $\qquad v_z = \text{a constant} \qquad ...(4a)$

This equation represents that the components of velocity along the direction of the magnetic field remains constant in the combined effect of electric and magnetic fields. If this component of velocity is zero, motion of the particle will be entirely in the X-Y plane.

Substituting the value of v_x from eq. (3b) in eq. (3a), we get $-\frac{m}{qB}\cdot\frac{d^2v_y}{dt^2} = \frac{qBv_y}{m}$

Now, writing $\omega = qB/m$, we have $\qquad \frac{d^2v_y}{dt^2} + \omega^2 v_y = 0$

This equation can easily be solved similarly as we solved the equation in Article 12.7. Thus,

$$v_y = A\sin(\omega t + \phi) \qquad ...(4b)$$

where $\omega = qB/m$ is the angular frequency of the particle and it is 2π times the cyclotron frequency.

Substituting this value of v_y in eq. (3b), we get

$$v_x = \frac{E}{B} - A\cos(\omega t + \phi) \qquad ...(4c)$$

The value of the constant A and ϕ can be determined by the initial values of v_x and v_y. Thus, for simplicity if we assume that initially at $t = 0$, the particle is in the state of rest, *i.e.*, $v_x = v_y = v_z = 0$, from eq., (4b), $\phi = 0$, and from eq. (4c), $A = E/B$.

Therefore, $v_y = \frac{dy}{dt} = \frac{E}{B}\sin\omega t$; $v_x = \frac{dx}{dt} = \frac{E}{B}[1 - \cos\omega t]$, and $v_z = \frac{dz}{dt} = 0 \qquad ...(5)$

Now, integrating these equations, we have

$$x = \frac{E}{B}\left[t - \frac{1}{\omega}\sin\omega t\right] + C_1,\ y = -\frac{E}{B\omega}\cos\omega t + C_2 \text{ and } z = C_3$$

If at $t = 0$, $x = y = z = 0$, then $C_1 = C_3 = 0$ and $C_2 = E/B\omega$ and hence

$$x = \frac{E}{B\omega}(\omega t - \sin\omega t),\ y = \frac{E}{B\omega}(1 - \cos\omega t), \text{ and } z = 0 \qquad ...(6)$$

Eqs. (5) and (6) represent that the motion of the particle will be cycloidal and head of the cycloid

* If initially the charged particle is in rest position, the particle will start the motion upwardly with an acceleration $q\vec{E}/m$. As soon as the particle will acquire velocity, it will experience a force $q(\vec{v} \times \vec{B})$ due to the magnetic field so that the particle after moving on a curved path, will return in the rest position at the point P of X-axis. Again the new cycle of its motion starts and the path of the particle will be **cycloid.**

will move with constant velocity along the X-axis. It is clear from eqns. (5) and (6) that the motion of the particle consists of (a) *a constant velocity E/B along X-axis* and (b) *a clockwise circular motion of angular velocity ω in X-Y plane.*

If an observer moves along X-axis with a velocity E/B, then from eq. (4) the components of the velocity relative to the frame of this observer are given by

$$v_x = -A\cos(\omega t + \phi),\ v_y = A\sin(\omega t + \phi), \text{ and } v_z = \text{a constant} \qquad ...(7)$$

This shows that in this moving frame the particle moves along a circular or helical path accordingly as $v_z = 0$ or finite

Thus, in such a frame the particle does not experience any force due to the electric field and it appears that only the magnetic field is acting on the particle.

12·18. Velocity Selector

Let us consider that a charged particle is moving in crossed electric and magnetic fields of intensities $\vec{E}$ and $\vec{B}$ respectively.

If E and B are acting along Y- and Z- axes respectively, the force qE due to the electric filed is in the upward direction along Y-axis. Now, if $v_y = 0$ (*i.e.*, no magnetic force is acting on the particle perpendicular to Y-axis) and the magnetic field is so adjusted that the force qBv_x in the downward direction due to it is balanced by the force in the upward direction (qE), then

$$qE = qBv_x \qquad ...(1)$$

Substituting $v_y = 0$ and $\vec{F} = 0$ in eq. (2) [Art. 12.17], $v_x = \dfrac{E}{B}$

[In Gaussian CGS system $qE = \dfrac{qBv_x}{c}$ or $v_x = \dfrac{cE}{B}$.]

This means that *only those particles, whose velocity along Y-axis is zero, and is equal to $v_x = E/B$, will move undeflected along X-axis with constant velocity v_x*. Hence with the help of this system, we can have particles of desired velocity by adjusting E and B. Such a system is called **velocity selector.**

Now, if we place a slit on X-axis parallel to Z-axis and allow to start charged particles of different velocities, the crossed electric and magnetic fields will act as velocity selector and only the particle of velocity $v_x = E/B$ will pass through the slit. It will be better if we place two or more parallel slits on X-axis, because some accelerated particles will also get through a single slit, in case the point P of **fig. 12.20** coincides with it.

The arrangement, used in a velocity selector, has been shown in **fig. 12.21**. A narrow beam of charged particles, defined by slit S_1, passes through crossed electric and magnetic fields. The direction of magnetic field is perpendicular to the plane of the paper. This field is generally applied with the help of a suitable electromagnet. Placing two plates parallel to X-axis, a high potential difference is applied between them. The plates are so oriented that the direction of the electric field is parallel to the plane of the paper (*i.e.*, normal to the magnetic field) and perpendicular to the direction of the beam. The two fields are adjusted in such a way that the deflection, produced by one is balanced by the other. Only those particles, whose velocities are such that they do not experience any force normal to their direction of travel, will pass through the slits S_2 and S_3 and other particles will be deflected to the sides.

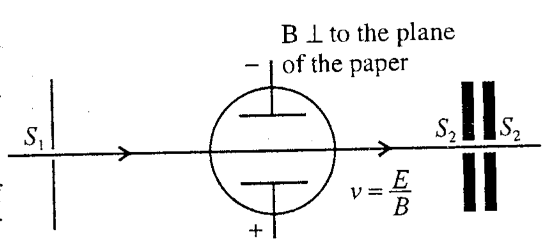


Fig. 12.21 : Velocity selector

Note that this process of velocity selection is independent of the mass and charge of the particles. Therefore, through a given combination of crossed electric and magnetic fields, all particles of speed $v = E/B$, irrespective of their charges and masses, pass undeflected.

Ex. 1. *Ions travelling horizontally in vacuum pass through a region containing vertical electric and magnetic fields and fall on a screen. If some particltes having e/m 9.67×10^7 Coul./kg are displaced horizontally 2 m and vertically 1 cm, what must be the e/m of the particles which are displaced* *horizonally by* 0.5 *cm and vertically* 2 *cm* ?

Sol. If x and y are the horizontal and vertical displacements on the screen, when a beam passes through parallel electric and magnetic fields,

$$\frac{x^2}{y} = \frac{e}{m}\cdot\frac{B^2}{E}\cdot A$$

where A is constant, depending on the geomery of the apparatus.

In the 1st case, $$\frac{(2\times10^{-2})^2}{0.01} = 9.67\times10^7\times\frac{B^2A}{E} \quad \text{...(i)}$$

In the 2nd case, $$\frac{(0.5\times10^{-2})^2}{0.02} = \frac{e}{m}\times\frac{B^2}{E}A \quad \text{...(ii)}$$

Dividing eq. (ii) by eq. (i), we get

$$\frac{0.25}{4}\times\frac{1}{2} = \frac{e}{m\times9.67\times10^7};\ \therefore\ \frac{e}{m} = \frac{9.67\times10^7}{32} = \mathbf{3.02\times10^6\ Coul./kg.}$$

Ex. 2. *A beam of electrons passes through a region of electric field* E_y = 200 *volts/cm and an overlapping transverse magnetic field* B_z = 1000 *gauss. Find the velocity of those electrons which pass undeflected. Will α-particles of the same velocity be deflected ?*

Sol. In crossed electric and magnetic fields, the velocity of the electrons, which pass undeflected, is given by

$$v_x = E/B$$

Here, $E_y = 200\times100 = 2\times10^4$ volts/m and $B_z = 0.1$ tesla.

$$\therefore\qquad v_x = \frac{2\times10^4}{0.1} = \mathbf{2\times10^5\ m/sec.}$$

From the expression of the velocity it is clear theat the charge and mass of the particle do not appear in the equation, hence α-particles of the same velocity (v_x) will **not be deflected.**

Ex. 3. *A proton is moving with a velocity of* $(3\hat{i}+\hat{j})\ 10^5$ *m/sec in an electric field of intensity* $(\hat{i}+3\hat{j}+3\hat{k})$ *e.s.u and a magnetic field of* $(2000\hat{j}+3000\hat{k})$ *gauss. Find the direction and magnitude of the total force acting on it.* (Unit of intensity 1 e.s.u. = 3×10^4 volt/m)

Sol. $$\vec{F} = q\vec{E} + q\vec{v}\times\vec{B} = 1.6\times10^{-19}(\hat{i}+3\hat{j}+3\hat{k})\times3\times10^4$$
$$+\ 1.6\times10^{-19}(3\hat{i}+\hat{j})\ 10^5\times(2000\hat{j}+3000\hat{k})\times10^{-4}$$
$$= 4.8\times10^{-15}\left[(\hat{i}+3\hat{j}+3\hat{k})+\frac{6\hat{k}-9\hat{j}+3\hat{i}}{3}\right] = 4.8\times10^{-15}(\hat{i}+3\hat{j}+3\hat{k}+\hat{i}-3\hat{j}+2\hat{k})$$
$$= 4.8\times10^{-15}(2\hat{i}+5\hat{k})$$

$$\therefore\qquad F = \sqrt{2^2+5^2}\times4.8\times10^{-15} = \mathbf{25.48\times10^{-15}\ N};\ \text{Its direction cosines}\ \left(2/\sqrt{29},\ 0,\ 5/\sqrt{29}\right).$$

EXERCISE 12·3

1. In Thomson's spectrograph, with only the electric field of intensity 10^4 volt/m, the deflection obtained on screen is 1 cm. The length of electric field is 5 cm and the screen is at distance 15 cm from the end of electric field. If the velocity of ion while entering the electric field is 10^6, calculate its specific charge (q/m).

   [**Ans.** 1.14×10^4 coul./kg.]

2. A positively charged ion beam, containing singly ionized helium atoms (He^+) and α-particle of different speeds, is moving along negative Z-axis. When the beam passes through small regions of parallel electric and magnetic field along Y-axis, two parabolas are traced on the photopraphic plate. If the smaller parabola has 2 cm x-displacement for a certain y-displacement, calculate the x-displacement on the bigger parabola for the same y-displacement. To what type of particles, the smaller parabola belong.

   [**Ans.** 0.0282 m; singly ionized helium.]

3. In a positive ray apparatus singly ionized and doubly ionized particles form identical parabolas when the magnetic fields are respectively 5000 and 7500 gauss. Compare the masses of two particles, assuming the electric field to be the same. **(Kanpur 1995)**
[**Ans.** 1 : 2.25.]
4. A positive ion beam moving in the X-direction enters a region in which there is an electric field $E_y = 3000$ volts/cm and magnetic field $B_z = 300$ gauss. Deduce the speed of those ions which may pass undeflected through the region. What will happen to ions which are (i) faster, (ii) slower than these ?
[**Ans.** 10^7 m/sec. (i) For the ions, having more speed than 10^7 m/sec the magnetic force along negative Y-axis is greater than the electric force towards positive Y-axis and hence the particles will be deflected towards negative Y-axis, (ii) For the. ions, having smaller speeds, the ions will be deflected along positive Y-axis.]
5. A charged particle moves along X-direction in which there is an electric field of 3000 volts/cm intensity along Y-axis and a perpendicular magnetic field 300 gauss along Z-direction. What is velocity when net force on the electron is zero ?
[**Ans.** $v_x = 10^7$ m/sec.]
6. A magnetic field of intensity 10000 gauss is crossed with the electric field between two parallel plates 2 cm apart and having a potential difference of 600 volts. Find the velocity of charged particles, which can travel undeflected in direction perpendicular to both.
[**Ans.** 3×10^4 m/sec..]
7. A particle having mass m and charge e is released from the origin in which electric and magnetic fields are given by $\vec{B} = -B\hat{j}$ and $\vec{E} = E\hat{k}$. Find the speed of the particle as a function of z-coordinate.
[**Ans.** $\sqrt{2qEz/m}$.] ·
8. Calculate the following for a proton, which is moving with a velocity $v_x = 4 \times 10^6$ m/sec. :
(a) A force experienced by it in magnetic field of strength $\vec{B} = 2500\,\hat{j} - 5000\,\hat{i}$ gauss.
(b) An accelertation experienced by it in a unifrom electric field of strength $\vec{E} = 500\,\hat{i} - 300\,\hat{j}$ volts/cm
(c) The transverse displacement to move a distance $l_x = 8$ cm in an electric field $E_y = 200$ volts/cm. What will be direction cosines in each case (a) and (b) ? **(Kanpur 1996)**
[**Ans.** (a) Force $= 3.58 \times 10^{-13}\left(2\,\hat{j}/\sqrt{5} + \hat{k}/\sqrt{5}\right)$N, with magnitude 3.6×10^{-13} N and direction cosines $\left(0,\ 2/\sqrt{5},\ 1/\sqrt{5}\right)$. (b) Acceleration $= 5.59 \times 10^{12}\left(5\,\hat{i}/\sqrt{34} - 3\,\hat{j}/\sqrt{34}\right)$ m/s^2 with magnitude. $5{\cdot}59 \times 10^{12}$ m/s^2 and direction cosines $\left(5/\sqrt{34},\ -3/\sqrt{34},\ 0\right)$; (c) Transverce displacement = 0.58 mm.]

QUESTIONS

1. A positively charged particle (q_0) of mass m is placed stationary at a point P whose position vector is $\vec{r}_0$. At time $t = 0$ a uniform electric field $\vec{E}$ is switched on in the space containing the point P. Derive a relation for the position vector $\vec{r}$ of the particle at a time t. **(Agra 1997)**
[**Ans.** $\vec{r} = \vec{r}_0 + \frac{1}{2}\frac{q}{m}\vec{E}t^2$.]
2. Show the an electric field behaves like an accelerating field and the energy acquired by charged particle in an electric field is equal to the product of charge and potential difference. Hence define electron-volt. **(Bhopal 2004; Bilaspur 03)**
3. What is an electron gun ? Expalin its working. **(Jabalpur 2004; Bhopal 04)**
4. Explain the construction and principle of linear accelerator. Deduce expression for the energy acquired by the charged particle in it in terms of the apparatus constants. **(Bhopal 2004; Sagar 04)**

5. (a) If a charged particle is moving through a transverse uniform electric field, show that the path of the particle will be parabola inside this field.
   Find out an expression for the direction of emergence of the particle from the field with the initial direction of motion. **(Agra 1990, 81; Rajasthan 83)**
   (b) Calculate the displacement on the screen of an electron beam by a transverse electric field acting for a length l. Where is this principle employed ? **(Kanpur 1978; Rohilkhand 79)**
6. Show that when a charged particle enters an electric field in direction normal to the field, it gets deflected and the deflection produced is directly proportional to the intensity of field. In which device is this principle used ? **(Bhopal 2003)**
7. A charged particle enters a transverse uniform electric field. Show that the path of particle the field is parabolic. Calculate the displacement of the particle obtained on screen in the electric field of length l if the screen is at a distance L from the terminus of electric field. **(Gwalior 2004; Indore 03; Ujjain 03)**
8. Discuss the theory and working of the cathode ray oscillograph and show how will you compare the intensities of two electric fields. **(Kanpur 1995)**
9. Describe the construction and working of a cathode ray oscilloscope with a proper diagram. State few uses of it. **(Gwalior 2003; Ujjain 04)**
10. What is meant by the sensitivity of a cathode ray oscilloscope ? Deduce its expression and state the factors on which it depends ? **(Indore 2003)**
11. Discuss the motion of a charged particle in a uniform and constant magnetic field. **(Kanpur 1995)**
12. If a charged particle enters in a transverse uniform magnetic field, then prove that the path of the particle in this field is circular. Find its time period and the displacement on the screen of the particle in the magnetic field of length l if the distance of the screen from the end of the magnetic field is L. **(Ujjain 2003; Sagar 03)**
13. Prove that when an ion with charge q and moving with a velocity v enters a magnetic field of strength B at an angle θ from B, it executes a helical path. Show that the pitch of the helix is given by $2\pi mv \cos\theta/Bq$. **(Kanpur 1995; Rohikhand 1980; Jabalpur 2004)**
14. A charged particle of mass M and charge q is moving in a constant magnetic field B. Show that
   (a) the component of velocity of the particle along the axis of the magnetic field is constant.
   (b) for a uniform circular motion in a plane perpendicular to the axis of the magnetic field, the angular velocity of the particle ω is given by $\omega = qB/M$. **(Rewa 2004; Agra 1972; Kanpur 79)**
15. Write down the equation of motion of a charged particle in a magnetic field and solve it to show that :
   (i) the path of the particle in the plane perpendicualr to the applied magnetic field is a circle.
   (ii) the gyrofrequency is independent of the velocity or energy of the particle. **(Rajasthan 1984)**
16. Describe the principle of cyclotron. Explain its construction and working and hence deduce an expression for the maximum kinetic energy acquired by the particle. **(Gwalior 2003; Rewa 03, 04; Indore 04)**
17. Explain the construction, theory and working of cyclotron. How is this device used for producing high energy charged particles ? Establish the frequency of cyclotron in resonance condition. **(Agra 1995, 86)**
18. Explain the construction and working of cyclotron with the help of a labelled diagram. Obtain expression for the maximum kinetic energy acquired by the particle. Discuss its limitations. Does the energy of paricle depend on the voltage applied ? Explain. **(Sagar 2003; Jabalpur 03; Indore 03; Bhopal 03; Gwalior 04)**
19. Describe the Thomson's parabolic method of determination of specific charge q/M of positive ions, with the proper diagram. Deduce the formula used. **(Ujjain 2004)**
20. A beam containing ions of various charges q, masses m and speed v (along X-axis) enters a region of uniform electric field E and magnetic field B (both along Y-axis). The ions are received on a screen after travelling a distance L. Show that :
   (i) all ions of a given q/m value fall along a parabola irrespective of their speeds.

(ii) all ions of a given speed fall along a straight line irrespective of their q/m value.
(Raipur 2003; Agra 1970 S; Kanpur 83)

21. What are isotopes ? Discuss the discovery of isotopes from the analysis of positive ions in Thomson's parabolic method ?
22. Discuss the motion of a charged particle in mutually perpendicular electric and magnetic fields. Hence explain the principle of velocity selector. **(Rajasthan 1982)**
23. Prove that the motion of a charged particle in crossed electric and magnetic fields is cycloidal. How will the motion appear in a frame moving with velocity E/B along X-axis.
(Raipur 2003; Rewa 03; Agra 04; Kanpur 1984, 80)
24. Explain the working of a velocity selector. Discuss the motion of a charged particle in mutually perpendicular magnetic field $\vec{B}$ and electric field $\vec{E}$. Also state the paths of particle if its initial velocity is E/B. **(Bilaspur 2004; Agra 04)**
25. Explain the principle of a velocity selector for charged particles, using crossed electric and magnetic fields. **(Agra 2004, 1984; Kanpur 95, 78)**
26. Write short nores on the following :
    (i) Cathode ray oscilloscope **(Indore 2004, 03; Raipur 03)**
    (ii) Electron gun **(Indore 2003)**
    (iii) Magnetic lens
    (iv) Discovery of isotopes **(Indore 2004)**
    (v) Principle of mass spectograph **(Gwalior 2003)**

Short Answer Questions

1. Prove the equation $v^2 - u^2 = \frac{2qE}{m} \cdot (\vec{r} - \vec{r}_0)$, where the terms have usual meanings.
2. Deduce expression for acceleration of a charged particle in an uniform electric field. **(Bhopal 2004)**

   [**Ans.** $\vec{a} = q\vec{E}/m$.]
3. What is eV ? It is the unit of which physical quantity ?
4. Prove that the path of charged particle in a transverse uniform electric field is parabolic.
5. Show that if a charge is accelerated by a potential difference, then the change in kinetic energy of the particle is equal to the product of charge and potential difference.
6. Discuss the principle of electron gun.
7. What is Lorentz force ? Is it conservative ?
8. Of the three vectors in the equation $\vec{F} = q(\vec{v} \times \vec{B})$, which pairs are always at right angles to each other ? Which may have any angle between them ?
9. A positively charged particle projected towards east is deflected towards north by a magnetic field. What may be the direction of the field ?
   [**Ans.** Downward.]
10. Prove that when an electric charge moves in a magnetic field, no work is done. **(Agra 1985)**
11. When an electron and a proton are projected with the same speed perpendicular to a magnetic field which one will have smaller frequency ?
    [**Ans.** Proton.]
12. Show that the kinetic energy of a charged particle moving in a constant uniform magnetic field remains unchanged. **(Rewa 2003; Agra 1980; Kanpur 95)**
13. Discuss the principle of cyclotron. **(Agra 2005, 1986)**
14. Explain 180° magnetic deflection related to dynamics of charged particles. **(Agra 2006, 05)**
15. What do you understand by the phenomenon of 180° magnetic deflection, related to the dynamics of a charged particle ? **(Agra 2004)**

16. Show that the cyclotron frequency of a given kind of ions is independent of the energy. State the limiting condition, if any. **(Agra 1970)**

17. Why can cyclotron not be used to accelerate neutrons ? **(Agra 1995, 82)**

18. Electric discharge does not take place at 'very' low pressure in gases; Why ?

[**Ans.** At low pressure of the order of 10^{-4} mm or below, then the number of atoms in the gas is so much reduced that sufficient positive ions are not available. Hence the discharge current does not flow.]

19. A beam of protons is deflected sideways. Could this deflection be caused (i) by an electric field ? (ii) by a magnetic field ? If either could be possible, how would you be able to say which was present ?

20. Let the room, in which you are seated, be filled with a uniform magnetic field with $\vec{B}$ pointing upwards. If at the centre of the room two electrons are suddenly projected horizontally with the same speed but in opposite directions, discuss their motions. If one particle is an electron and one a positron, then also discuss their motions.

21. Time-base voltage is applied between X-deflecting plates of a cathode-ray oscillograph. In what direction is the deflection produced in the electron-beam emitted from the electron-gun of the oscillograph ? How does this deflection change with time ? How does the scintillation spot move on the screen of oscillograph ?

[**Ans.** The electron-beam emitted from the electron-gun is deflected in the horizontal direction due to the time-base voltage. This deflection increases linearly with time, and then suddenly falls to zero. This results in the scintillation spot on the screen to move horizontally from one side to the other, and to return very rapidly in its original position. This process repeats again and again.]

22. Show that the radius of curvature of a charged paritcle moving at right angles to a magnetic field is proportional to its momentum.

23. A proton and an α-particle enter in a uniform magnetic field, in direction normal to the field. Compare the period of revolution for them. **(Gwalior 2003)**

[**Ans.** $T = \dfrac{2\pi m}{qB}$, $\therefore$ $\dfrac{T_1}{T_2} = \left(\dfrac{m_1/q_1}{m_2/q_2}\right) = \dfrac{1}{4} \times \dfrac{2}{1} = \dfrac{1}{2}$ [$\because$ m_2 = mass of α-particle = 4 × mass of proton = $4m_1$ and $q_2 = 2q_1$].]

Objective Type Questions

1. If a charged particle is moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ and an electric field $\vec{E}$, then the force acting on it will be :

(a) $q\vec{E}$ (b) $q(\vec{v} \times \vec{B})$ (c) mv^2/ρ (d) $q[\vec{E} + (\vec{v} \times \vec{B})]$.

2. A strong argument of the partricle-nature of cathode rays is their ability to :

(a) produce fluorescence (b) travel through vacuum

(c) get deflected by electric and magnetic fields (d) cast shadow.

3. The energy which an electron acquires when accelerated through a potential difference of 1 volt is called :

(a) 1 joule (b) 1 electron-volt (c) 1 erg (d) 1 watt.

4. The path of an electron in a uniform travsverse electric field is :

(a) straight line (b) circle (c) parabola (d) hyperbola.

5. The unit of electric field intensity is :

(a) newton/coulomb (b) volt/metre (c) newton-coulomb (d) volt-meter

6. A charged particle of charge 0.5 coulomb is accelerated by a potential difference of 2000 volt. The kinetic energy of the particle will be :

(a) 1000 joule (b) 1000 kilo-watt-hour (c) 100 eV (d) zero.

7. If a charged particle, which is at rest, experiences electromagnetic force then :

(a) electric field should not be zero (b) magnetic field should not be zero

(c) electric field may be zero or not (d) magnetic field may be zero or not.

8. When a positively charged particle is projected towards east, it is deflected towards north by the magnetic field. The direction of the magnetic field is :
(a) towards west (b) towards south (c) upwards (d) downwards

9. An electron of energy 10 eV moves on a circular path in a magnetic field of strength 1×10^{-4} wb/m^2, then the radius of its path will be :
(a) 1.1 m (b) 1.0 m (c) 11m (d) 0.11m

10. 2 joule work has to be done in taking a charge of 200 coulomb from one point to other. If the distance between the points is 0.2 m, then the potential difference between these points is :
(a) 0.4 volt (b) 0.2 volt (c) 8 volt (d) 0.1 volt.

11. If the magnetic field in a cyclotron is made half and the radius of the dee is doubled, then maximum K.E. becomes :
(a) eight times (b) four times (c) double (d) no change.

12. Kinetic energy of a charged particle accelerated by a cyclotron does not depend on :
(a) magnetic field (b) mass of the particle
(c) potential difference between dees (d) radius of dee.

13. An electron, starting from rest, accelerated by 200 volt. acquires a velocity 8.4×10^6 m/sec. e/m of electron is :
(a) 1.76×10^9 Coul/kg (b) 1.76×10^{11} Coul/kg (c) 1.76×10^{13} Coul/kg (d) 1.76×10^{18} Coul/kg

14. K.E. of a charged particle in a magnetic field :
(a) remains constant (b) goes on increasing due to force
(c) may remain constant or change (d) goes on decreasing due to force.

15. A charged particle is travelling in a uniform magnetic field and a parallel uniform electric field. At some instant, the velocity of the particle is normal to both fields. The path of the particle will be :
(a) straight line (b) circle
(c) helix of uniform pitch (d) helix of varying pitch.

16. The velocity of the undeflected ions coming out of the crossed electric and magnetic fields E and B is :
(a) E/B (b) B/E (c) EB (d) $E-B$. **(Agra 2005)**

17. If a charged particle passes through a region, where electric and magnetic fields $\vec{E}$ and $\vec{B}$ are present, without acceleration :
(a) $\vec{E}$ and $\vec{B}$ are perpendicular to each other (b) $\vec{v}$ is perpendicular to $\vec{E}$
(c) $\vec{v}$ is perpendicular $\vec{B}$ (d) $|\vec{E}| = v|\vec{B}|$.

18. A charged particle moves in the X-direction through a region in which there is an electric field $E_y = 3000$ V/cm and a perpendicular magnetic field $B_z = 300$ gauss. The speed of the particle V_0 for no deffiction is :
(a) 10^5 m/s (b) 10^6 m/s (c) 10^7 m/s (d) 10^8 m/s **(Agra 2006)**

19. In Q. 18, what will happen to particles which are faster than the no diffection speed V_0 ?
(a) defection in the $-Y$ direction (b) defiction in the $-Z$ direction
(c) deflection in the $+$Y direction (d) deflection in the $+$Z direction **(Agra 2006)**

20. In Q. 18, what will happen to particles which are slower than the no diffection speed V_0 ?
(a) defection in the $-Y$ direction (b) defiction in the $-Z$ direction
(c) deflection in the $+$Y direction (d) deflection in the $+$Z direction **(Agra 2006)**

21. An electrion emerges from a cyclotron after making 100 revolution inside it. If the potential difference across the dees be 1KV, its energy will be : **(Agra 2006)**
(a) 0.5×10^5 eV (b) 1×10^5 eV (c) 2×10^5 eV (d) 4×10^5 eV

ANSWERS

1. (d)	**2.** (c)	**3.** (b)	**4.** (c)	**5.** (a), (b)	**6.** (a)	**7.** (a)	**8.** (d)
9. (d)	**10.** (d)	**11.** (d)	**12.** (c)	**13.** (b)	**14.** (a)	**15.** (d)	**16.** (a)
17. (a), (b)	**18.** (c)	**19.** (a)	**20.** (c)	**21.** (c)			

13

Elasticity

13·1. Deformations of a Body and Elasticity

A body is said to be **rigid**, if the relative positions of its constituent particles remains unchanged, when external forces act upon it. Actually, no body is perfectly rigid and every body can be deformed more or less, by the suitable application of the forces. These applied forces are called **deforming forces.** When there are **small deformations** in a body due to the applied forces, it regains its original shape or size after the removal of the forces. This property of a material body is called **elasticity** *i.e.*, **Elasticity** *is that property of matter by virtue of which a body regains the original shape or size on the removal of the deforming forces.*

Elastic limit : *If the deforming forces are applied on a body upto a certian limit, and the body regains its original shape or size after the removal of the forces, then such a limit of eleasticity is called* **elastic limit.**

For example, if an iron wire is clamped at one end in the roof and by attaching a hanger on the other end, the wire is loaded, it will extend in length due to the applied weight (force). If the applied weight is not much, the wire will acquire its original length on the removal of the weight. However if the wire is loaded so much that the elastic limit is crossed, then wire will not regain its original length and in fact, it will be permanently extended in length or it will break. Different materials have different limits of eleasticity.

It must be noted here that the deformation in the shape or size of a body is the result of equal and opposite forces or couples in equilibrium.

Perfectly elastic and plastic bodies : *If the body recovers completely its original condition as soon as the deforming forces are removed,. it is said to be* **perfectly elastic,** *while if it completely retains its modified form, it is said to be* **perfectly plastic**. There are, however, no perfectly elastic or plastic bodies and actual bodies lie between the two extremes. The nearest approach to a perfectly elastic body is a quartz fiber and to a perfectly plastic body is putty.

Homogeneous and isotropic bodies : In this chapter, we plan to study the elastic property of the **homogeneous** and **isotropic** materials. *A material is said to be homogeneous and isotropic, if its composition is uniform and its properties are the same at all points and in all directions inside it.* In these materials, not only elastic property is the same in all directions, but other properties like thermal expansion, heat conduction, refraction etc. are also identical. Nearly all the fluids *i.e.*, liquids and gases are the examples of homogeneous and isotropic materials. Solid metals, like copper, iron , gold etc. in the form of wire or body are considered to be isotropic in view of elasticity.

Solids, for example wood and crystals, are not isotropic. Such materials with different properties in different directions are called **non-isotropic** or **anisotropic.** For example in cubic cystals, the elastic property is anisotropic.

13·2. Stress and Strain

Stress : When a system of external forces act on a body, relative displacements of its various particles take place and consequently the body is deformed. For such deformation, the body offers resistance and internal forces of reaction are developed inside the body. These internal forces, so developed, are equal in magnitude and opposite to the deforming forces and equilibrium is established. The forces of internal reaction tend to restore the body to its original form. On account of the restoring forces, the body recovers its shape or size, when the deforming forces are removed.

The restoring force per unit area, set up inside the body, is called **stress** *and it is measured by the magnitude of the deforming force acting on unit area of the body within the elastic limit.* Thus, if a force F is acting on a area, A, then

$$\text{stress} = \frac{F}{A}$$

Since stress is the force per unit area, its dimensional formula is $ML^{-1}T^{-2}$ and S.I. unit is N/m^2. There are two types of stress :

(1) Normal stress : If the stress is normal to the surface, it is called the **normal stress.** The stress is always normal in the case of a change in length of the wire or in the case of the change in the volume of a body. When there is the change in length, the stress is said to be **tensile.** The normal stress is of two types: compressive and expansive (*i.e.*, tensile) accordingly as a decrease or increase in volume takes place.

(2) Tangential or shearing stress : When the stress is tangential to the surface, it is called a **shearing** or *tangential stress.* Such a stress is produced, when one face of a body is kept fixed and a tangential deforming force is applied on the other face. If the force producing the stress is inclined at an angle normal to the surface, the stress will have the **normal** and **tangential** components.

Strain : When a system of forces or couples acting upon a body causes relative displacements of its various parts, a change in length, volume or shape is produced. The body is then said to be **strained.**

The change produced in the dimension of a body by a system of forces or couples, in equilibrium, is called **strain** *and it is measured by the ratio of change in length to the original length* **(longitudinal strain)** *or change in volume to the original volume* **(volume strain).** *If there is a change in shape, the strain, called as* **shearing strain**, *is measured by the angle through which a line originally perpendicular to the fixed face is turned.*

There are three types of strain :

(1) Longitudinal strain : *If the length of a wire is changed due to deforming forces, then the change per unit length is called* **longitudinal strain.** Let L be the original length of a wire and l be the change in its length due to applied forces. Then,

$$\text{longitudinal strain, } \alpha = \frac{l}{L}$$

(2) Volume strain : *If the volume of a material is changed by the deforming forces, then the change in per unit volume is called* **volume strain.** Let V be original volume of a material and v be the change in volume by the application of deforming forces, then

$$\text{volume strain, } \gamma = \frac{v}{V}$$

(3) Shearing strain : *If one face of a body is fixed and the deforming force on another face is so applied that its shape changes, then the strain is called* **shearing strain** *and it is measured by the angle* (θ) *in between a line perpendicular to the fixed face in the initial position and final position of the line on the application of deforming force.*

Thus, the strain is just a ratio or an angle, it is a dimensionless quantity and has no units.

13·3. Hooke's Law

This is the basic law of elasticity and was established by Robert Hooke in 1678. This law states that within elastic limit, the stress is proportional to the corresponding strain, *i.e.*,

$$\text{Stress} \propto \text{Strain or Stress} = E \times \text{Strain } i.e., E = \frac{\text{Stress}}{\text{Strain}}$$

The constant of proportionality E is called **modulus of elasticity**. Its value is independent of the magnitude of the stress or the strain and depends only on the nature of the material and its other conditions. Modulus of elasticity has the same dimensional formula as that of stress, *i.e.*, $ML^{-1}T^{-2}$, since strain is a dimensionless quantity.

Corresponding to three types of strain, we have three modulii of elasticity, namely, (i) **Young's modulus**, corresponding to longitudinal strain, (ii) **Bulk modulus**, corresponding to volume strain, and (iii) **Modulus of rigidity**, corresponding to shearing strain.

13·4. Elastic Behaviour of a Wire

(1) **Behaviour of a wire under increasing load :** If a wire is clamped at one end and loaded at the other gradually, the stress increases continually until the wire breaks down. Now, if we plot a graph between the stress on the Y-axis and strain on the X-axis, in general, a curve of the form, shown in **fig. 13.1** is obtained for ductile metals. In the figure the part OP of the curve is a straight line which shows that upto the point P, stress is proportional to strain, *i.e.*, Hooke's law is obeyed upto P. The point P is called the **limit of proportionality** (or proportional limit).

If the stress is further increased, a point P' known as the **elastic limit** of the material is reached. This point lies near the point P and upto this point, the wire takes back its original length, when the load is removed. It should be noted that, in general, for the part PP' of the curve, stress must not necessarily be proportional to strain. In case, if a given material obeys Hooke's law in the part PP' of the curve, the points P and P' corresponding to proportional limit and elastic limit coincide with each other.

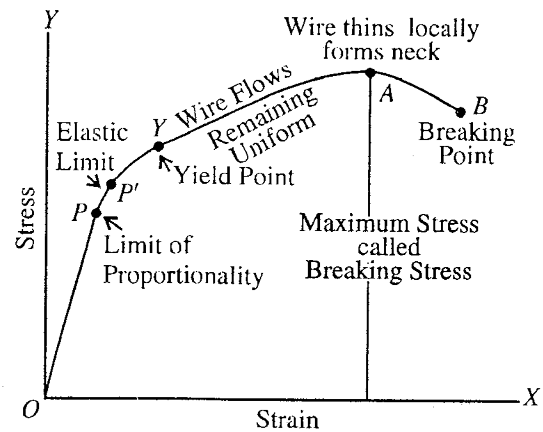


Fig. 13.1 : Behaviour of a wire under increasing load.

On increasing the load beyond elastic limit, the stress-strain curve takes a bend. Now, at any point, if the wire is unloaded, it does not regain its original length and gets a permanent stretch, which we call **permanent set**. If the wire is now loaded, an entirely new stress-strain curve will represent its behaviour.

If the load is still further increased, a point Y is reached, where the strain is much greater for a small increase in the load. This point Y is called yield-point and the corrresponding stress being the **yield-stress**. Beyond the point Y, the extension increases rapidly without an increase in the load, *i.e.*, the material of the wire flows beyond Y. This is known as **plastic-flow.** As the wire becomes thin, the stress (load per unit area) becomes considerably greater and the wire cannot support the same as before and the wire is to be prevented from being broken, the load must be diminished.

After crossing the yield-point, first the wire becomes thin. This thinning of the wire no longer remains uniform and the diameter of a section decreases considerably, *i.e.*, the specimen wire exhibits a local constriction. Now, the wire shows a phenomenon, known as 'necking'. Immediately, as this occurs, the stress decreases automatically and the portion AB of the curve is obtained; ultimately a point B, known as **breaking-point**, is reached, at which the wire breaks. The stress corresponding to this point B is called **breaking stress**. Here, the stress corresponding to A is called the **ultimate strength** or **tensile strength** of the given material. The tensile strength of a material is defined as the ratio of maximum load to which the specimen wire may be subjected by slowly increasing load to the original cross-sectional area of the wire. The ultimate strength of a material is, however, measured by the stress causing the test specimen to break.

(2) **Elastic after effect :** When a fine fibre of quartz is twisted within elastic limit and then released, it comes back to its original form as soon as the twisting force is removed. Almost all substances take some time to recover their original condition when deforming forces are removed. This delay in recovering back the original condition by the test specimen, after the removal of the deforming force, is called **elastic after effect** for that material.

Elastic after effect is different for different materials. Glass shows this effect to a marked degree, but quartz, phosphor bronze, silver and gold are few exceptions to this effect, which regain their original condition immediately after the removal of deforming forces. On accound of much less magnitude of elastic after effect, quartz, phosphor bronze and silver are employed as the suspensions in quadrant electrometer, moving coil galvanometer and in Boy's experiment for the determination of G, etc.

(3) **Elastic fatigue :** Lord Kelvin found that the rate of decay of torsional oscillations of a wire, being allowed to vibrate continuously for some days, is much greater than when the wire is fresh one. He said that this happens due to **elastic fatigue**. The wire gets tired and continue vibrating with difficulty. If it is allowed to rest for some time, its initial condition is recovered.

If a wire is loaded repeatedly or in other words subjected to a large number of cycles of stresses, it gets tired or ruptured due to gradual fracture of the material and hence loses its strength for some time. Now, if the wire is allowed to rest for some time, the fracture inside the material is removed and the material regains its strength. Thus, *the elastic fatigue may be defined as the loss of the strength of the material for some time, caused by the repeated stresses in it.*

13·5. Elastic Modulii of Isotropic Materials

(1) **Young's modulus :** *Within the elastic limit, the ratio of longitudinal stress to the corresponding strain, produced, is called* **Young's modulus** *for the material of a body and is denoted by the letter 'Y'.*

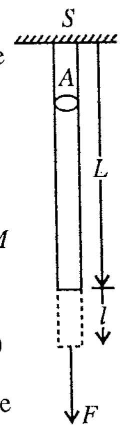


Fig. 13.2.

Let L be the original length of a wire of cross-section A and it elongates l by the application of a force F. Then in equilibrium,

$$\text{stress} = F/A \text{ and strain} = l/L$$

$$\therefore \quad \text{Young's modulus, } Y = \frac{F/A}{l/L} = \frac{F \cdot L}{A \cdot l} \qquad ...(1)$$

If the radius of a wire is r, then area of cross section of the wire is $A = \pi r^2$. Now, if M kg mass is suspended at the lower end of the wire, then $F = Mg$ and

$$Y = \frac{MgL}{\pi r^2 l} \qquad ...(2)$$

We see that by the application of more force (Mg), if the change in length (l) of the wire is less, the value of Young's modulus (Y) of the material of the wire will be more.

Steel (or iron) is more elastic than rubber : When same force is applied on a steel wire and a rubber piece, rubber is extended much more than steel and ordinarily it is said that rubber is more elastic than steel. However it is not true in the language of physics. If we consider two wires of steel and rubber of the same length (L) and same radius (r) and load each of them by same weight (Mg), then the change (l_r) in length of the rubber wire will be more than that (l_s) for steel wire.

$$\text{Hence} \qquad Y_r = \frac{MgL}{\pi r^2 l_r} \text{ and } Y_s = \frac{MgL}{\pi r^2 l_s}$$

$$\therefore \qquad \frac{Y_s}{Y_r} = \frac{l_r}{l_s} \text{ i.e., } Y_s > Y_r.$$

Hence the value of the Young's modulus or coefficient of elasticity of steel (or iron) is more than that of rubber *i.e.*, **steel** (or **iron**) **is more elastic than rubber**.

Force exerted on the two clamped ends of a rod on cooling or heating : If a rod of length L and area of cross-section A is clamped at both the ends and is cooled, then its length should decrease. However the ends are fixed and hence the length even on cooling remains the same. Clearly the clamped ends exert force of tension on the rod *i.e.*, if the rod would have not been clamped, the length L of the rod might have decreased to $L - l$. If the rod of length L is cooled by $\Delta\theta$°C, then from the definition of coeffficient of linear expansion, we have

$$\alpha = \frac{l}{L\Delta\theta} \qquad [\because L = L_0(1+\alpha\theta)] \text{ or } \Delta L = L_0\,\alpha\Delta\theta \text{ or } l = L\alpha\Delta\theta\ ] \quad ...(1)$$

Hence $$\text{strain} = \frac{l}{L} = \alpha\Delta\theta$$

$$\therefore \text{ Young's modulus of rod, } Y = \frac{\text{Stress}}{\text{Strain}} = \frac{\text{Force}}{\text{Area} \times \alpha\Delta\theta} = \frac{\text{Force of tension}}{A\ \alpha\Delta\theta}$$

Hence force of tension at the clamped ends of the rod = $YA\alpha\Delta\theta$ ...(2)

Similarly, when a rod clamped at both ends, is heated, corresponding to the elongation in its length there is extra force exerted by the clamped (fixed) ends on the rod and the length of the rod remains the same. If the rod is heated by $\Delta\theta$°C, then the force exerted by the clamped ends on the rod is

$$F = YA\alpha\Delta\theta \quad ...(3)$$

(2) **Bulk modulus :** When a solid or a fluid (liquid or gas) is subjected to change in pressure, its volume changes, but the shape remains unchanged. The force per unit area, applied normally and uniformly to the surface of the body (or pressure) gives the **stress** and the change per unit volume, the **strain**, Now, *within the limit of elasticity the ratio of uniform and normal stress on the surface of a body to the volumetric strain is called* **bulk modulus** *of the material of the body and it is denoted by the letter 'K'.*

Let V be the original volume of a body and by the application of a pressure p, let the volume be changed to $(V - v)$. Then

$$\text{stress} = p \text{ and volume strain} = v/V$$

$$\therefore \quad \text{bulk modulus, } K = \frac{p}{-v/V} = -\frac{pV}{v} \quad ...(4)$$

The negative sign indicates here, that by the increase of pressure a decrease in volume occurs.

The reciprocal of bulk modulus of a substance is called its **compressibility**, *i.e.*,

$$\text{compressibility} = \frac{1}{K} = -\frac{v/V}{P} \quad ...(5)$$

Isothermal and adiabatic elasticities (Bulk moduli) of a gas : Depending upon the type of change, either isothermal or adiabatic, there are two types of elasticities, possessed by a gas, *viz.*, (i) isothermal elasticity, and (ii) adiabatic elasticity.

(i) **Isothermal elasticity :** Isothermal elasticity of a gas is its elasticity, when a change in the volume of the gas takes place with pressure, the temperature remaining constant. It is denoted by E_i.

Let P and V be the initial pressure and volume of a given amount of a perfect gas. If the temperature remains constant, then from Boyle's law

$$PV = \text{constant}$$

Defferentiating it, we get

$$V \cdot dP + P \cdot dV = 0; \quad \text{whence } P = \frac{dP}{-dV/V} = E_i, \text{ isothermal elasticity.} \quad ...(6)$$

Hence, *the isothermal elasticity of a perfect gas is equal to its original pressure.*

(ii) Adiabatic elasticity : Adiabatic elasticity of a gas is the elasticity when the change in volume of a gas taken place with pressure, but no heat is allowed to enter or leave the gas. It is denoted by E_a.

Let P and V be the pressure and the volume of a given amount of a perfect gas. If the gas is subjected to an adiabatic change, then

$$PV^{\gamma} = \text{constant}$$

where γ is the ratio of two specific heats of the gas.

Differentiating this equation, we get

$$P\gamma V^{\gamma-1}\, dV + V^{\gamma} dP = 0 \text{ or } \gamma P dV + V dP = 0$$

$$\therefore \qquad \gamma P = \frac{dP}{-dV/V} = E_a, \text{ adiabatic elasticity.} \qquad ...(7)$$

$$\text{But } P = E_i, \therefore \qquad \frac{E_a}{E_i} = \gamma. \qquad ...(8)$$

Thus, *the ratio of two elasticities of a perfect gas is equal to the ratio of its two specific heats.*

(3) Modulus of rigidity or Shear modulus : *Within the elastic limit, the ratio of tangential stress to the shearing strain is called modulus of rigidity of the material of a body and is denoted by the letter* 'η'.

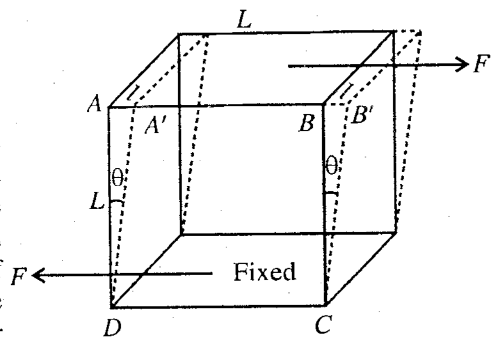


Fig. 13.3.

In this case, the shape of a body changes, but its volume remains unchanged. Consider a cube of side L, whose lower edge CD is fixed. A tangential force F is applied in the directrion shown over the upper face of area A $(= L^2)$. A force of reaction of the same magnitude F acts on the lower fixed face in the opposite direction, as shown in **fig. 13.3**. These two equal and opposite forces, with the lines of action different, constitute a couple. Due to this couple, face AD of the cube is displaced to the position $A'D$ and face BC to the position $B'C$ through an angle θ, known as the angle of shear or shearing strain. The rotation of the cube by the above mentioned couple is prevented by an opposite couple of equal moment, provided by the constraints.

Thus, $\qquad$ shearing strain $= \theta = \tan\theta = l/L$, if θ is small

where $AA' = BB' = l$ and $AD = BC = L$.

$$\text{Shearing stress} = \frac{\text{Tangential force}}{\text{Area of the face}} = \frac{F}{A}$$

$$\text{Modulus of rigidity, } \eta = \frac{F/A}{\theta} = \frac{F}{A\cdot\theta} = \frac{F\cdot L}{Al} \qquad ...(9)$$

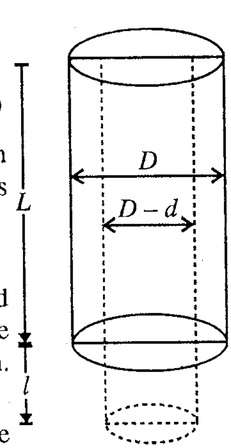


Fig. 13.4.

This is a relation exactly similar to the one for Young's modulus, but here in place of linear stress, there is a tangential stress and l, a displacement at right angles L occurs instead along it.

In S.I. system, the units for Y, K and η are newton/m^2.

(4) Poisson's ratio : When a wire is stretched along its length, it is elongated along with a contraction in its diameter. Thus, the length of the wire increases in the direction of the forces, whereas a contraction occurs in the perpendicular direction. This fact is not only true for wire, but for all other bodies under strain.

The ratio of change in dimension to the initial dimension, perpendicular to the direction of the forces, is called **lateral strain**. *Within elastic limit, the lateral strain,* *is proportional to the longitudinal strain, i.e, the ratio of lateral strain and longitudinal strain is a constant for the material of a body. This constant is known as* **Posson's ratio.**

If β and α are the lateral and longitudinal strains respectively, then

$$\text{Poisson's ratio, } \sigma = \frac{\beta}{\alpha}$$

Consider a wire of length L and diameter D. Let the length of the wire be increased by l under the action of two equal and opposite forces and let a corresponding decrease in diameter be d. **[fig.13.4]**. Then,

Longitudinal strain, $\alpha = l/L$ and Lateral strian, $\beta = d/D$.

$$\therefore \quad \text{Poisson's ratio, } \sigma = \frac{\beta}{\alpha} = \frac{d/D}{l/L} = \frac{L}{l} \cdot \frac{d}{D} \qquad ...(10)$$

Expressing in differential form,

$$\sigma = \frac{L}{D}\frac{dD}{dL} \qquad ...(11)$$

Poisson's ratio is a pure number and has no units and dimensions since it is the ratio of two strains.

13·6. Shearing Stress Equivalent to an Equal Linear Tensile Stress and an Equal Compressive Stress at Right Angles to Each Other

Consider the front view of a cube $ABCD$ of side L **[fig. 13.5.]**. The base DC of the cube is fixed and a tangential force F is applied on the upper face AB.

Now, shearing stress $= F/L^2$.

If there is only the force F acting on the upper face AB, then the cube should move as a whole in the direction of the force. But the cube is fixed at the lower face DC, hence an equal and opposite force acts on the face DC. These two forces, therefore, constitute a couple which tends to rotate the cube in the clockwise direction.

The cube, however, does not rotate, hence the face DC which is fixed exerts an equal and opposite couple by exerting forces (say F') on the faces AD and CB shown in the figure.

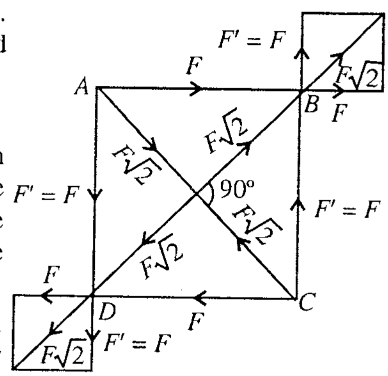


Fig. 13.5.

Now, moment of clockwise couple $= F \times AD$.

and the moment of the anti-clockwise couple $= F' \times AB$.

But as the cube is in equilibrium,

$$\therefore \qquad F \times AD = F' \times AB \text{ or } F = F' \qquad [\because AB = AD = L]$$

Thus, a tangential force F applied to the face AB gives rise to equal tagential forces acting along all the faces in suitable directions, as shown in fig.13.5.

The forces acting along the faces AB and CB give rise to a resultant force $F\sqrt{2}$ along DB and forces acting along the face AD and CD also give rise to a resultant force $F\sqrt{2}$ along BD. Hence, an outward pull is exerted along the diagonal BD at the corners B and D, resulting in its extension.

Similarly, at each of the corners A and C a resultant force $F\sqrt{2}$ acts inwards along the diagonal AC, resulting in its contraction.

Thus, a tangential force F acting on one of the face of the cube is equivalent to a force of extension $F\sqrt{2}$ acting outwards along the diagonal BD and a force of compression $F\sqrt{2}$ acting inwards along the diagonal AC.

If a plane is drawn perpendicular to the plane of the paper through the diagonals AC or BD in the cube, its area will be

$$= L \times L\sqrt{2} = L^2\sqrt{2}.$$

Hence, linear tensile stress along $BD = \frac{F\sqrt{2}}{L^2\sqrt{2}} = \frac{F}{L^2}$

Similarly, linear compressive stress along $AC = \frac{F\sqrt{2}}{L^2\sqrt{2}} = \frac{F}{L^2}$

Evidently, F/L^2 is the shearing stress over the face AB of the cube. Thus, *a* **shearing stress** *is equivalent to an equal linear tensile stress and an equal compressive stress at right angles to each other.*

13·7. Relations Connecting Various Elastic Constants

The elastic constants are dependent to each other, since any change in the size and shape of a body may be obtained by first changing the size of the body only and then by changing the shape only. Thus, expressions can be derived, showing inter-relations connecting them.

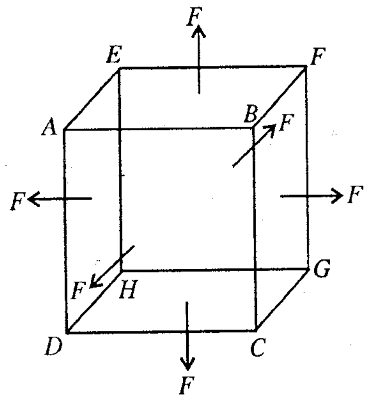


Fig. 13.6.

(i) **Relations connecting** Y, K **and** σ $[Y = 3K(1 - 2\sigma)]$: Let $ABCDEFGH$ represent a cube of unit side. Let us consider a force F, which acts normally and uniformly on each of its six faces in the outward direction. Since each force F is acting on unit area, these forces are representing stresses.

If α is the increase per unit length per unit tension along the direction of the force then the elongation produced in each of the edges, namely AB, BF and BC, will be $F\alpha$. If β is the contraction produced per unit length per unit stress perpendicular to the edges, then contraction produced perpendicular to each of the edge, namely AB, BF and BC, will be $F\beta$. Thus the side AB (similarly BF and BC) will increase by $F\alpha$ but simultaneously contract by $F\beta + F\beta$ due to the two other pairs of tensile stresses.

Hence, the sides of the cube become

$$AB = 1 + F\alpha - F\beta - F\beta = 1 + F(\alpha - 2\beta)$$
$$BF = 1 + F\alpha - F\beta - F\beta = 1 + F(\alpha - 2\beta)$$
$$BC = 1 + F\alpha - F\beta - F\beta = 1 + F(\alpha - 2\beta).$$

Hence, the final volume of the cube is

$$AB \times BF \times BC = [1 + F(\alpha - 2\beta)]^3 = 1 + 3F(\alpha - 2\beta) \quad \text{[according to Binomial expansion]}$$

The terms containing higher powers of α and β have been neglected since α and β are very small quantities.

As the initial length of a side of the cube is 1, its initial volume = 1

$\therefore$ Change in volume = Final volume − Initial volume

$$= 1 + 3F(\alpha - 2\beta) - 1 = 3F(\alpha - 2\beta).$$

$$\text{Volume strain} = \frac{3F(\alpha - 2\beta)}{1} = 3F(\alpha - 2\beta)$$

$$\therefore \text{ Bulk modulus, } K = \frac{\text{Normal stress}}{\text{Volumetric strain}} = \frac{F}{3F(\alpha - 2\beta)}$$

or
$$K = \frac{1}{3(\alpha - 2\beta)} \text{ or } K = \frac{1/\alpha}{3(1 - 2\beta/\alpha)}.$$

But α is the increase per unit length per unit stress, hence clearly

Stress = 1 (as the area of any edge of the cube is unity)

and Linear strain $= \alpha/1 = \alpha$, $\quad \therefore Y = 1/\alpha$

Also $\beta/\alpha = \sigma$ = Poisson's ratio.

Therefore,
$$K = \frac{Y}{3(1 - 2\sigma)} \text{ or } Y = 3K(1 - 2\sigma) \qquad ...(1)$$

(ii) Relation connecting Y, η and σ [$\eta = Y/[2(1+\sigma)]$] : Let $ABCD$ represent the front face of a cube of side L. A tangential force F is applied on its upper face AB and the bottom face DC is fixed. As a result of this force the cube is sheared to $A'B'CD$ through an angle θ [**fig. 13.7**], Then, shearing strain is

$$\theta = \frac{AA'}{AD} = \frac{BB'}{BC} = \frac{l}{L},$$

where the displacement $AA' = BB' = l$.

Shearing stress,

$$T = \frac{F}{\text{Area of the upper face of the cube}} = \frac{F}{L^2}$$

$\therefore$ Coefficient of rigidity, $\eta = \dfrac{T}{\theta}$.

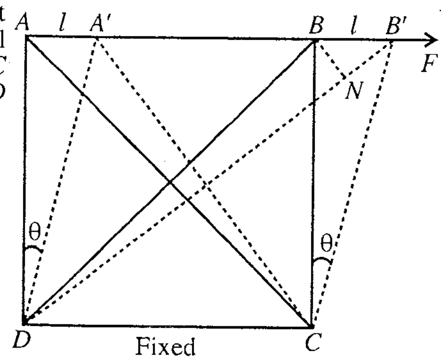


Fig. 13.7.

But a shearing stress along AB is equivalent to a tensile stress along DB and an equal compressive stress along AC at right angles to each other.

Let α and β be the longitudinal and lateral strains per unit stress respectively. Then extension along diagonal DB due to tensile stress $= DB \cdot T \cdot \alpha$

and extension along diagonal DB due to compression stress along $AC = DB \cdot T \cdot \beta$

Total extension along $DB \; = DB \cdot T \cdot (\alpha + \beta) = L\sqrt{2}\,T(\alpha + \beta)$.

Let us draw a perpendicular BN on DB'. Then increase in the length of diagonal DB is nearly equal to NB'. As θ is very small, therefore $\angle AB'C$ is nearly $90°$ and $\angle BB'N = 45°$.

Thus, $\quad NB' = BB' \cos 45° = \dfrac{BB'}{\sqrt{2}} = \dfrac{l}{\sqrt{2}}$; $\;\therefore\; L\sqrt{2}\,T(\alpha + \beta) = \dfrac{l}{\sqrt{2}}$ or $T\dfrac{L}{l} = \dfrac{1}{2(\alpha+\beta)}$.

But $\quad T \cdot \dfrac{L}{l} = \dfrac{T}{l/L} = \dfrac{T}{\theta} = \eta$; $\therefore \eta = \dfrac{1}{2(\alpha+\beta)} = \dfrac{1}{2\alpha(1+\beta/\alpha)}$

But $\quad \beta/\alpha = \sigma$ and $Y = \dfrac{\text{Stress}}{\text{Longitudinal strain}} = \dfrac{1}{\alpha}$ [$\because$ α is the longitudinal strain per unit stress]

Therefore, $$\eta = \frac{Y}{2(1+\sigma)} \qquad ...(2)$$

(iii) Relation connecting Y, K and η [$Y = 9K\eta/(\eta + 3K)$] : From relation (1),

$$1 - 2\sigma = Y/3K$$

From relation (2), $\quad 2 + 2\sigma = Y/\eta$

Adding the above two equations, we get

$$3 = \frac{Y}{\eta} + \frac{Y}{3K} = Y\left(\frac{1}{\eta} + \frac{1}{3K}\right) = Y\left(\frac{3K+\eta}{3\eta K}\right)$$

Thus $$Y = \frac{9\eta K}{3K + n} \qquad ...(3a)$$

This may be written as

$$\frac{9}{Y} = \frac{3K+\eta}{\eta K} \text{ or } \frac{9}{Y} = \frac{3}{\eta} + \frac{1}{K} \qquad ...(3b)$$

(iv) Relation connecting K, η and σ [$\sigma = (3K - 2\eta)/(2\eta + 6K)$] : From relations (1) and (2), we have

$$Y = 3K(1 - 2\sigma) \text{ and } Y = 2\eta(1 + \sigma) \qquad ...(4)$$

Equating the two values of Y, we have

$$3K(1 - 2\sigma) = 2\eta(1 + \sigma) \text{ or } 3K - 6K\sigma = 2\eta + 2\eta\sigma$$

or $$\sigma(2\eta + 6K) = 3K - 2\eta \; i.e., \; \sigma = \frac{3K - 2\eta}{2\eta + 6K} \qquad ...(5)$$

Limiting values of σ : Equating the values of Y in (4), we have

$$3K(1 - 2\sigma) = 2\eta(1 + \sigma) \quad ...(6)$$

where K and η are essentially positive quantities. Therefore,

(i) *if Poisson's ratio is a positive quantity,* then both right hand side and left and side of above equation must be positive. This is possible when

$$1 - 2\sigma > 0 \text{ or } \sigma < \frac{1}{2} \text{ or } 0.5. \quad ...(7)$$

(ii) *if σ is a negative quantity,* left hand side expression of above equation will be positive. Then, the right hand side expression must also be positive. Hence

$$1 + \sigma > 0 \text{ or } \sigma > -1 \quad ...(8)$$

Thus, the theoretical limiting values of σ are -1 and 0.5, *i.e.,*

$$-1 < \sigma < 0.5 \quad ...(9)$$

13·8. Work Done Per Unit Volume in Deforming a Body – Internal Energy due to Strain.

In order to deform a body, work has to be done. This energy, so spent is stored up in the body, and is called **elastic potential energy** or **internal energy** of the strained body or simply **strain energy.**

We will calculate below, the amount of work done, in three principal cases of strain.

(i) **Longitudinal strain :** When a wire of length L and cross-section A is stretched by a force F, acting along its length, then in equilibrium

$$Y = \frac{F}{A}\frac{L}{l} \text{ (where } l \text{ is the increase in length.); } \therefore F = \frac{YAl}{L}$$

For a further small increase in length dl, the small work done, is given by

$$dW = Fdl = \frac{YAl}{L}dl$$

Hence, the total work done in increasing the length of the wire by an amount l is

$$W = \int_0^l \frac{YA}{L} l\, dl = \frac{YAl^2}{2L} = \frac{Fl}{2} \qquad \left[\because \frac{YAl}{L} = F\right] \quad ...(1)$$

This amount of work done is stored in the elongated wire as elastic potential energy or internal potential energy.

The volume of the wire = AL.

$$\therefore \quad \text{Work done per unit volume} = \frac{YAl^2}{2L \times AL} = \frac{1}{2}\frac{Yl^2}{L^2} = \frac{1}{2}\cdot\frac{Yl}{L}\cdot\frac{l}{L} \quad ...(2)$$

But $$\frac{Yl}{L} = \frac{F}{A} = \text{Stress and } \frac{l}{L} = \text{Strain.}$$

$$\therefore \text{ Strain energy per unit volume} = \frac{1}{2} \times \text{Stress} \times \text{Strain.} \quad ...(3)$$

(ii) **Volume strain :** Consider a body of volume V compressed by an amount v by the application of a uniform pressure p acting normal to its whole surface. Then by the definition of bulk modulus, we have

$$K = \frac{p}{v/V} = \frac{pV}{v} \text{ or } p = K\frac{v}{V}$$

If the surface area of the body be A, then the compressive force acting on the body is

$$F = pA = KAv/V$$

The small work done for a movement dx in the direction of F, that is normal to the surface, is given by

$$dW = Fdx = KAvdx/V.$$

But $\quad Adx = dv$ is the infinitesimal change in volume.

Hence $\quad dW = \frac{K}{V} v\, dv$

$\therefore$ Total work done for a change of volume v is

$$W = \int_0^v \frac{K}{V} v dv = \frac{K}{V}\frac{v^2}{2} \qquad ...(4)$$

Hence, the work done or strain energy per unit volume

$$= \frac{W}{V} = \frac{1}{2}\frac{K}{V^2}v^2 = \frac{1}{2}\cdot\frac{Kv}{V}\cdot\frac{v}{V} \qquad ...(5)$$

$$= \frac{1}{2} \times \text{Stress} \times \text{Strain}. \qquad ...(6)$$

(iii) Shearing strain : Consider the front view of a cube $ABCD$ of side L. A tangential force F is applied on the upper face. The lower edge of the cube is fixed. As a result of the force F, the upper plane of the cube is displaced through $AA' = BB' = l$. Now, for a further dislacement dl of the plane, the amount of work done, is given by

$$dW = F \cdot dl$$

Thus, the total amount of work done for the displacement l is

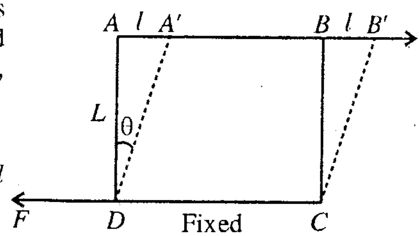


Fig. 13.8.

$$W = \int_0^l F \cdot dl$$

But $\quad \eta = \frac{F}{A\theta} = \frac{F}{L^2} \times \frac{L}{l}$ or $F = \eta \cdot L \cdot l \qquad [\because A = L \times L \text{ and } \theta = l/L]$

$$\therefore \quad W = \int_0^l \eta \cdot L \cdot l \cdot dl = \frac{1}{2}\eta \cdot L \cdot l^2 \qquad ...(7)$$

Volume of the cube $= L^3$

$$\therefore \quad \text{Work done per unit volume} = \frac{1}{2}\cdot\frac{\eta \cdot L \cdot l^2}{L^3} = \frac{1}{2}\cdot\frac{\eta l}{L}\cdot\frac{l}{L} \qquad ...(8)$$

Here $\eta l/L$ = Stress and $l/L = \theta$ = Strain.

$$\therefore \quad \text{Work done or Strain energy per unit volume} = \frac{1}{2} \times \text{Stress} \times \text{Strain}. \qquad ...(9)$$

Ex. 1. *An iron wire of radius 1 mm is fixed at its two ends in its normal state at 30°C temperature. When the temperature becomes 20°C, determine the tension in the rod ($Y = 2.2 \times 10^{11}$ N/m² and $\alpha = 11 \times 10^{-6}$/°C for iron).*

Sol. Let the tension developed in the wire be T, when it is cooled from 30°C to 20°C. Hence

$$\text{Stress} = T/\pi\,(10^{-3})^2$$

Now, $\alpha = \frac{L_2 - L_1}{L_1(\theta_2 - \theta_1)}$ or Strain $= \frac{L_2 - L_1}{L_1} = \alpha\,(\theta_2 - \theta_1) = 11 \times 10^{-6} \times (30 - 20) = 11 \times 10^{-5}$.

$\therefore \quad Y = \frac{\text{Stress}}{\text{Strain}} = \frac{T}{\pi \times (10^{-3})^2} \times \frac{1}{11 \times 10^{-5}} = 2.2 \times 10^{11}$; $\therefore T = \pi \times 2.2 \times 11 =$ **72.5 newton.**

Ex. 2. *Determine the Poisson's ratio of rubber if due to pulling a rubber thread, its change in volume is negligible in comparison to its change in length.*

Sol. If the rubber thread has the length l and radius r, then volume of the thread is

$$V = \pi r^2 l = \text{constant (given)}$$

Now,
$$\delta V = \pi\,(2r\,\delta r l + r^2\,\delta l) = 0$$

or
$$2l\,\delta r + r\delta l = 0 \text{ or } -\frac{\delta r}{r} = \frac{\delta l}{2l}$$

Hence,
$$\text{Poisson's ratio, } \sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{-\delta r/r}{\delta l/l} = \frac{1}{2} = \mathbf{0.5.}$$

Ex. 3. *Young's modulus of a metal is 2×10^{11} N/m^2 and Poisson's ratio is 0.25. Calculate the bulk modulus.* **(Rewa 2003)**

Sol.
$$K = \frac{Y}{3(1-2\delta)} = \frac{2\times10^{11}}{3(1-2\times0.25)} = \frac{2\times10^{11}}{3\times0.5} = \mathbf{1.33 \times 10^{11}}\ N/m^2$$

Ex. 4. *Find the Poisson's ratio for steel, if Young's modulus is 2×10^{11} N/m^2 and modulus of rigidity is 8×10^{10} N/m^2.* **(Sagar 1995; Jabalpur 92)**

Sol. $\eta = \dfrac{Y}{2(1+\sigma)}$, whence $\sigma = \dfrac{Y}{2\eta} - 1 = \dfrac{2\times10^{11}}{2\times8\times10^{10}} - 1 = 1.25 - 1 = \mathbf{0.25.}$

Ex. 5. *A wire of length 5 m and diameter 1 mm is stretched by 0.3 mm by applying a load of 10.0 kgf. Calculate the internal energy of the strained wire. Take $g = 9.8$ m/s^2.* **(Jabalpur 2003)**

Sol. Strain energy or internal energy of the strained wire = $\frac{1}{2}Fl$.

Here $F = 10$ kgf $= 10 \times 9.8$ N, $l = 0.3$ mm $= 0.3 \times 10^{-3}$ m.

$\therefore$ Internal energy of the strained wire = $\frac{1}{2} \times (10 \times 9.8) \times (0.3 \times 10^{-3}) = \mathbf{1.47 \times 10^{-2}\ J.}$

EXERCISE 13·1

1. A steel wire of length 2.0 m and cross-section 1×10^{-6} m^2 is held between rigid supports with a tension 200 newton. If the middle point of the wire is pulled 5 mm sideways, calculate the change in tension. Also calculate the change in tension if temperature changes by 5°C. [For steel $Y = 2.2 \times 10^{11}$ N-m^2 and $\alpha = 8.8 \times 10^{-6}$/°C] **(Agra 1970)**

   [**Ans.** 2.75 newton; 8.8 newton.]

2. Two wires A and B are made of same material. The length of the wire A is half the length of wire B and diameter of wire A is the double the diameter of wire B. If same weight is suspended from each of them, compare the increase in their lengths.

   [**Ans.** $l_a/l_b = 1/8$.]

3. The breaking stress of a material is 10^9 N/m^2 and its density is 3×10^3 kg/m^3. Calculate the length of its wire which breaks due to its own weight when suspended.

   [**Ans.** 3.4×10^4 m.]

4. The Young's modulus and bulk modulus for streel are respectively 2×10^{11} N/m^2 and 11×10^{10} N/m^2. Calculate the Poisson's ratio. **(Jabalpur 1994)**

   [**Ans.** $\sigma = 0.197$.]

5. The elastic limit of an elevator metal cable is 2.8×10^8 N-m^2. What should be the maximum upward acceleration of the elevator of mass 2×10^6 gm supported by a cable of cross-sectional area 3 cm^2 so that the stress may not exceed one-fourth of the elastic limit.

   [**Ans.** 0.7 ms^{-2}.]

6. A sheet of India rubber of 10 cm square and 2 cm thick has one face fastened to a vertical wall and to the other face a piece of wood is cemented. When a load of 30 kg is hung from the wood, the wood is found to be finaily lowered by 0.03 cm. Find the coefficient of rigidity of the rubber and the energy stored per cubic meter of the rubber.

[**Ans.** 1.96×10^6 N-m^2; 221 Jm^{-3}.]

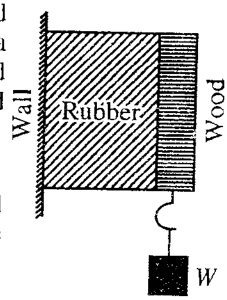


Fig. 13.9.

7. A vertical rubber cord, fixed at its upper end, is elongated 10 cm, when a load is applied gradually at its lower end. Calculate the maximum extension of the cord when the load is applied suddenly, explaining the method of calculation. Determine also the period of subsequent oscillations.

[**Ans.** 0.2 m; 0.63 sec.]

[**Hint :** In equilibrium for a load Mg, $Mg - Cx_0 = 0$ or $x_0 = Mg/C$, where C = force constant. However if a load Mg is applied suddenly, the loss of gravitational potential energy of the mass upto a maximum extension, say x, will be in the form of instantaneous elastic potential energy $\frac{1}{2}Cx^2$ *i.e.*,

$$Mgx = \frac{1}{2}Cx^2 \quad \text{or} \quad x = 2Mg/C = 2x_0$$

i.e, maximum extension is double the extension (x_0) in equilibrium. In fact, when the load Mg is suddenly applied, elastic potential energy in the cord and kinetic energy in the mass, suspended, *are developed due to the loss of gravitational* P.E. At x_0, the kinetic energy is

$$Mgx_0 - \frac{1}{2}Cx_0^2 = Cx_0^2 - \frac{1}{2}Cx_0^2 = \frac{1}{2}Cx_0^2. \qquad [\because Mg = Cx_0]$$

In fact, this is the oscillation energy due to which the mass will oscillate with a period $T = 2\pi\sqrt{M/C} = 2\pi\sqrt{x_0/g}$.]

8. Find the formula for the work done in stretching a wire and apply it to find the elastic energy stored up in wire, originally 5 metres long and 1 mm in diameter, which has been stretched by 3/10 mm due to a load of 10 kg. [Take $g = 9.8$ ms^{-2}.]

[**Ans.** 0.0147 joules.]

9. A steel wire of length 2 m is stretched through 2 mm. The cross-sectional area of the wire is 4 mm. Calculate the elastic potential energy stored in the streteched wire. Y for steel = 2×10^{11} N/m^2. **(Meerut 1995)**

[**Ans.** 0.8 joule.]

10. A wire 4 metres long, 0.3 mm in diameter is stretched by a force of 8000 gm wt. If the extension in the length amounts to 1.5 mm, calculate the energy strored in the wire. **(Punjab 1962)**

[**Ans.** 5.88×10^{-3} J.]

11. A wire of length 2 m and area of cross-section 10^{-6} m^2 is stretched to elongate by 10^{-4} m. Calculate the internal energy of the wire. ($Y = 2 \times 10^{11}$ N/m^2) **(Gwalior 2003; Indore 1994)**

[**Ans.** 5×10^{-4} J.]

13·9. Angle of Twist and Angle of Shear

Let us consider a cylindrical rod or wire of length l and radius r clamped at the upper end and twisted by a couple at the lower free end [**fig. 13.10**]. By doing so, each circular cross-section of the rod rotates through an angle. This angle is called the **angle of twist** and is proportional to the distance of cross-section from the clamped end. If the angle of twist at the free end is θ, then clearly.

$$r\theta = l\phi = BB', \text{ where } \phi \text{ is the angle of shear.} \qquad ...(1)$$

From **fig. 13.10 (a)**, it is clear that ϕ is constant for the cylindrical surface of radius r,

In any section normal to the axis at a distance, say l', from the fixed end, let the angle of twist be θ', then

$$r\theta' = l'\phi \qquad ...(2)$$

Dividing eq. (2) by eq. (1), we get $\frac{\theta'}{\theta} = \frac{l'}{l}$. Hence for $l' < l$, $\theta' < \theta$.

Thus, we find that the angle of twist, which is a constant in any section normal to the axis at a given distance from the fixed end, goes on decreasing as the distance from the fixed end is decreased. At the fixed end, $l = 0$, hence $\theta = 0$.

Fig. 13.10(b) shows the shearing of any two coaxial cylindrical layers of radius r and r'. The angle of twist θ at the lower free end is a constant. For the outer cylindrical surface.

$$l\phi = r\theta \qquad ...(3)$$

and for the inner cylindrical layer

$$l\phi' = r'\theta \qquad ...(4)$$

As $r' < r$, $\quad \therefore \quad \phi' < \phi$.

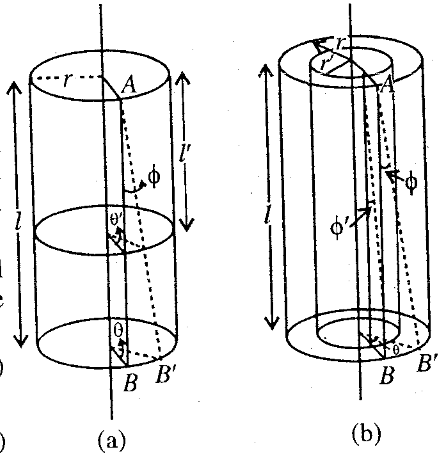


Fig. 13.10 : Angle of twist and angle of shear.

Hence we see that the angle of shear, a constant for any cylindrical surface goes on decreasing as the radius of the surface decreases and that for $r = 0$, $\phi = 0$. It is a maximum on the surface of the cylinder.

13·10. Twisting Couple on a Cylindrical Rod or Wire

Let us consider a cylindrical rod of length l and radius r. The upper end of the rod is kept fixed and a twisting couple is applied at its lower end in a plane perpendicular to its length. Due to elasticity of the material for the rod, a restoring couple is set up in it to resist or oppose the twisting couple. In the position of equilibrium, the two couples balance each other. Let us calculate the value of this couple.

Imagine the cylinder to consist a large number of thin coaxial cylindrical shells and consider one such shell of radius x, thickness dx and length l. As the rod is twisted, a line AB (parallel to its axis) on the surface of the cylindrical shell takes up the position AB' through an angle $BAB' = \theta$. The angle ϕ is the angle of shear. To understand this, let this cylindrical shell were cut along AB before twisting and flattened out. it will form a rectangular plate $ABCD$, but after twisting it acquires the shape of a parallelogram $AB'C'D$ **[fig. 13.12]**. Clearly this is a case of shear and angle ϕ is the angle of shear. If θ is the angle of twist at the lower end of the rod, then

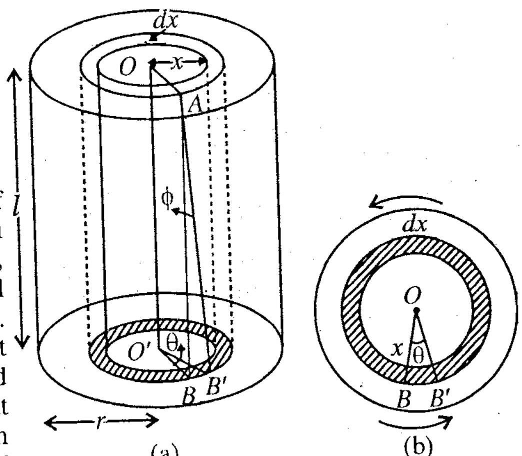


Fig. 13.11 : Twisting of a cylinder.

$$BB' = AB\cdot\phi = OB\cdot\theta \text{ or } l\phi = x\cdot\theta$$

$$\therefore \qquad \phi = x\theta/l \qquad ...(1)$$

Let F be the tangential force acting on the base of this thin cylindrical shell producing a shear. Then

$$\text{Tangential stress} = \frac{F}{\text{Area of the base of the shell}}$$

But, area of the base of the shell

$$= \text{Circumference} \times \text{thickness} = 2\pi x \cdot dx$$

$$\therefore \qquad \text{Tangential stress} = \frac{F}{2\pi x . dx}.$$

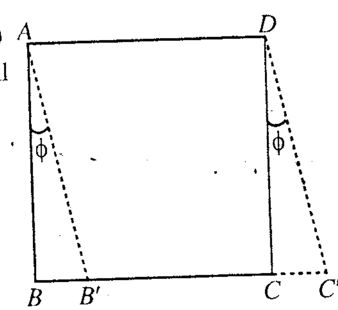


Fig. 13.12.

If η is the modulus of rigidity of the material of the rod, then

$$\eta = \frac{\text{Tangenital stress}}{\text{Shearing strain}}$$

$$\therefore \qquad \eta = \frac{F}{2\pi\, x{\cdot}dx\phi} = \frac{F}{2\pi\, x{\cdot}dx} \cdot \frac{l}{x\theta} \quad \text{or} \quad F = \frac{2\pi\eta\theta}{l} \cdot x^2 dx$$

The moment of this force about the axis of the rod $= \dfrac{2\pi\eta\theta \cdot x^2 dx \cdot x}{l} = \dfrac{2\pi\eta\theta}{l} x^3 dx$ ...(2)

This is equal to the couple required to twist the shell of radius x and thickness dx through an angle θ. Integrating the above expression between the limits $x = 0$ and $x = r$, we get the total twisting couple, *i.e.*,

$$\text{Twisting couple on the rod} = \int_0^r \frac{2\pi\eta\theta}{l} \cdot x^3 dx = \frac{2\pi\eta\theta}{l} \frac{r^4}{4} = \frac{\eta\pi r^4}{2l} \qquad ...(3)$$

If $\theta = 1$, then twisting couple per unit twist, say C, is

$$C = \frac{\eta\pi r^2}{2l} \qquad ...(4)$$

This C is called the **torsional rigidity** of the material of the wire. Since twisting couple is numerically equal to the restoring couple, C is called the **restoring couple per unit twist.**

Twisting couple for a hollow cylindrical rod : If the cylindrical rod is hollow, having internal and external radii r_1 and r_2 respectively, then the couple required to twist the rod through an angle θ will be obtained by intergrating the expression (2) between the limits $x = r_1$ to $x = r_2$, *i.e.*,

$$\text{Twisting couple} = \int_{r_1}^{r_2} \frac{2\pi\eta\theta}{l} \cdot x^3 dx = \frac{\pi\eta}{2l}(r_2^4 - r_1^4)\theta. \qquad ...(5)$$

Hence the couple required to twist the hollow rod through 1 radian, *i.e.*, torsional rigidity is given by

$$C' = \frac{\pi\eta}{2l}(r_2^4 - r_1^4)$$

Comparison of torsional rigidities for hollow and solid cylindrical rods : Now, if we have two cylinders, one hollow and other solid, of the same material, length and mass, the couples required to twist them through 1 radian are C' and C respectively.

$$\frac{C'}{C} = \frac{r_2^4 - r_1^4}{r^4} = \frac{(r_2^2 + r_1^2)(r_2^2 - r_1^2)}{r^4} \qquad ...(6)$$

But the two cylinders have the same mass and are made of the same material.

$$\therefore \qquad \pi(r_2^2 - r_1^2)l\rho = \pi r^2 l\rho \quad \text{or } r_2^2 - r_1^2 = r^2 \qquad ...(7)$$

$$\text{Hence} \qquad \frac{C'}{C} = \frac{r_2^2 - r_1^2}{r^2} \qquad ...(8)$$

From eq. (7) $\qquad r_2^2 = r^2 + r_1^2$

Adding r_1^2 on both sides, we have

$$r_2^2 + r_1^2 = r^2 + 2r_1^2,\ i.e.,\ r_2^2 + r_1^2 > r^2 \quad \text{or} \quad \frac{r_2^2 - r_1^2}{r^2} > 1$$

$$\therefore \qquad \frac{C'}{C} > 1 \quad \text{or} \quad C' > C \qquad ...(9)$$

This means that *if two cylinders (one hollow and other solid) of same length, mass and material are given, then to twist the cylinders through the same angle, more torque will be applied for the hollow cylinder than the soild one, i.e,* a greater couple is required to twist the hollow cylinder. Thus, **a hollow shaft can transmit larger torques for the same angle of twist.**

A shaft is a cylindrical rod, employed for transmitting rotational motion to other machines. To one end of the shaft an external torque is applied so that the shaft itself gets twisted till the restoring couple is balanced by the external couple.

13·11. Work Done in Twisting a Wire or Rod

Let a wire or rod of length l and radius r be fixed at the upper end. At the lower end of the wire, a couple is applied so that a twist of an angle θ is produced at this end.

If C be the couple per unit twist of the wire then the couple, required to twist the wrie through an angle θ, will be $C\theta$. Now, the work done in twisting the wire further through a small angle $d\theta$ is given by

$$dW = C\theta \cdot d\theta$$

Hence, the total work done in twisting the wire through the angle θ is

$$W = \int_0^{\theta} C\theta \cdot d\theta = \frac{1}{2}C\theta^2 = \frac{\eta\pi r^4\theta^2}{4l} \qquad \left[\because C = \frac{\eta\pi r^4}{2l}\right] \quad ...(10)$$

The energy, spent in doing this work, is stored up in the wire and is known as **strain energy.**

13·12. Determination of Modulus of Rigidity By Torsional Pendulum

A torsional pendulum consists of a long thin specimen wire, whose one end is attached to a rigid support, to the other free end of the wire, a heavy disc is fixed in the centre, The disc is turned in the horizontal plane and then released, it executes torsional vibrations, whose time period, is given by.

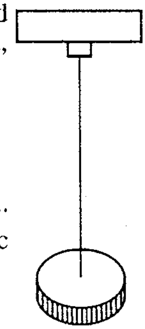


Fig. 13.13.

$$T = 2\pi\sqrt{\frac{I}{C}} = 2\pi\sqrt{\frac{I \cdot 2l}{\eta\pi r^4}} \qquad \left[\because C = \frac{\eta\pi r^4}{2l}\right]$$

where η is the modulus of rigidity of the material of the wire, r its radius and l its length.

If R is the radius of the disc and M its mass, then the moment of inertia I of the disc about the wire as axis is $I = MR^2/2$.

Thus, $$T = 2\pi\sqrt{\frac{MR^2}{2}\cdot\frac{2l}{\eta\pi r^4}}\,; \qquad \therefore \eta = \frac{4\pi l MR^2}{T^2 r^4}$$

The time period T is measured with the help of a stop-watch. The radius r of the wire occurs in fourth power in the formula, hence it is measured accurately by a screw gauge at several places of the wire in two perpendicular directions and mean value of r is calculated. The length of the wire is measured by a metre scale. Thus, by substituting the value of various quantities in the above formula, η for the material of the wire can be calculated.

Ex. 1. *A gold wire, 0.32 mm in diameter, elongates by 1 mm when stretched by a force of 330 gm-wt and twists through 1 radian when equal and opposite torques of 145×10^{-7} N-m are applied at its ends. Find the value of Poisson's ratio for gold.* **(Bombay 1968; Mysore 84)**

Sol. When a wire of radius r and length L is elongated by l with a force F, the Young's modulus of its material is given by

$$Y = \frac{F}{\pi r^2}\cdot\frac{L}{l}$$

Here, $F = Mg = 0.33 \times 9.81$ newton, $r = 1.6 \times 10^{-4}$ m, $l = 10^{-3}$m

$$\therefore \qquad Y = \frac{0.33\times9.81\times L}{\pi\times(1.6\times10^{-4})^2\times10^{-3}} \text{ N/m}^2 \qquad ...(i)$$

The torque, applied to twist the wire by 1 radian, is given by

$$C = \frac{\eta\pi r^4}{2L} = 145 \times 10^{-5} \text{ N-m}, \therefore \eta = \frac{145\times10^{-7}\times2L}{\pi\times(1.6\times10^{-4})^4} \qquad ...(ii)$$

$$\therefore \qquad \text{Poisson's ratio, } \sigma = \frac{Y}{2\eta} - 1 = \frac{0.330\times9.81\times L\times\pi\times(1.6\times10^{-4})^4}{\pi\times(1.6\times10^{-4})^2\times10^{-3}\times2\times145\times2L} - 1 = 1.429 - 1 = \mathbf{0.429}.$$

Ex. 2. *A circular rod of Young's modulus* 2.04×10^6 *kg-wt per sq. cm, Poisson's ratio* 0.4, *length* 1 *metre and cross-section* 0.95 *sq. mm is stretched by a weight of* 10 *kg. Find the extension and diminution in cross-section.*

Sol. When a wire of length L and cross-section A is elongated through l by a force F, the Young's modulus of its material is given by.

$$Y = \frac{F \times L}{A \times l} \qquad \therefore\ l = \frac{F \times L}{Y \times A} \qquad \text{...(i)}$$

Here, $F = mg = 10 \times 9.81$ m., $L = 1$ m, $A = 0.95 \times 10^{-6}$ m^2, $Y = 2.04 \times 10^6$ kg-wt per sq. cm $= 2.04 \times 10^6 \times 9.81 \times 10^4$ N/m^2.

Hence the value of extension l is given by

$$l = \frac{10 \times 9.81 \times 1}{2.04 \times 10^6 \times 9.81 \times 10^4 \times 0.95 \times 10^{-6}} = \mathbf{5.16 \times 10^{-4}\ m}$$

Poisson's ratio σ is given by

$$\sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{\delta r / r}{l / L}$$

where r is the radius of the wire and δr the diminution produced in it.

$$\therefore \qquad \delta r = \sigma \times \frac{l}{L} \times r \qquad \text{...(ii)}$$

Cross-section area of the wire $A = \pi r^2 = 0.95 \times 10^{-6}$ sq. m.

$$\therefore \qquad r = \sqrt{0.95 \times 10^{-6} / \pi} = \mathbf{5.499 \times 10^{-4}\ m.}$$

Poisson's ratio for the material $\sigma = 0.4$.

Substituting the value of various quantities in (ii) , we get

$$\delta r = 0.4 \times \frac{5.16 \times 10^{-4}}{1} \times 5.499 \times 10^{-4} = 11.35 \times 10^{-8}\ \text{m.}$$

$$\because \qquad A = \pi r^2,\ \therefore\ \delta A = 2\pi r \cdot \delta r$$

where δA is the diminution in cross-section area A.

Therefore $\qquad \delta A = 2 \times \pi \times 5.499 \times 10^{-4} \times 11.35 \times 10^{-8} = \mathbf{3.92 \times 10^{-10}\ sq.m.}$

Ex. 3. *A circular bar one metre long and of* 8 *mm diameter is rigidly clamped at one end in a vertical position. A couple of magnitude* 2.5 *N-m, is applied at the other end. As a result a mirror fixed at this end deflects a spot of light by* 15 *cm on a scale one metre away. Calculate the modulus of rigidity of the bar.*

Sol. When the mirror turns through an angle θ, the reflected ray turns through an angle 2θ. As the spot of light is reflected through 15 cm on a scale 100 cm away, then

$$\tan 2\theta = 0.15/1 \quad \text{or} \quad 2\theta = 0.15\ (\text{if } \theta \text{ is small}),\ \therefore\ \theta = 0.075\ \text{radian.}$$

The mirror is fixed at the lower end of the bar, therefore its lower end is deflected by 0.075 radian. Now, to twist the bar by 0.075 radian, the couple required is

$$C\theta = \frac{\eta \pi r^4}{2l} \times 0.075\ \text{N-m.}$$

Here, $l = 1$ m., $r = 4 \times 10^{-3}$ m and $C\theta = 2.5$ N-m.

$$\therefore\ \frac{\eta\pi \times (4 \times 10^{-3})^4 \times 0.075}{2 \times 1} = 2.5 \quad \text{or } \eta = \frac{2.5 \times 2 \times 1}{3.14 \times (4 \times 10^{-3})^4 \times 0.075} = \mathbf{8.29 \times 10^{10}\ N/m^2.}$$

Ex. 4. *An iron wire of* 1 *mm radius,* 100 *cm length is twisted through* 18°. *Where is the shearing stress is maximum ? Find the maximum angle of shear. If modulus of rigidity of iron is* 8×10^{10} *N/m*^{2}, *what is the torsional couple ?*

Sol. $\eta = \dfrac{\text{Shearing stress}}{\text{Shearing strain }(\phi)}$, $\quad\therefore$ Shearing stress $= \eta\phi$

Hence the shearing stress will have maximum value when ϕ is maximum. But ϕ is maximum at the surface of the wire, the reason is given below :

The strain at distance x (radially) will be $\phi = x\theta/l$. It has the maximum value, when $x = r$, the radius of the wire, *i.e.*,

$$\phi_{max} = r\theta/l, \text{ at the surface of the wire.}$$

Here, $r = 10^{-3}$ m, $\theta = 18°$ and $l = 1$ m.

$$\therefore \quad \phi_{max} = 10^{-3} \times 18°/1 = 0.018°$$

$$\therefore \quad \text{Torsional couple, } C\theta = \frac{\eta\pi r^4}{2l}\theta = \frac{8\times10^{10}\times3.14\times(10^{-3})^4}{2\times1}\times 18\times\frac{\pi}{180} = \mathbf{3.8 \times 10^{-2}\ N\text{-}m.}$$

Ex. 5. *Two cylinders A and B of radii r and 2r are soldered coaxially. The free end of A is clamped and the free end of B is twisted by an angle θ. Calculate twist at the junction, taking the material of the cylinders to be the same and lengths equal.*

Sol. Let T be the couple applied at the free end of B and θ' the twist at the junction. The couple T produces a relative twist θ' between the ends of the cylinder A and relative twist $(\theta - \theta')$ between the ends of the cylinder B. Thus, if η be the modulus of rigidity for the material of the cylinders and l the length of each cylinder, we have

$$T = \frac{\eta\pi r^4\theta'}{2l} = \frac{\pi\eta(2r)^4(\theta-\theta')}{2l}; \;\therefore\quad \theta' = 16\,(\theta - \theta') \text{ or } \theta' = \mathbf{16\theta/17.}$$

Ex. 6. *Two solid cylinders of the same material having lengths l, 2l and radii r, 2r respectively are joined coaxially. Under a couple applied between the free ends, the shorter cylinder shows a twist 30°. Calculate the twist of the longer cylinder.*

Sol. $$\text{Torque, } T = \frac{\pi\eta r^4\theta_1}{2l} = \frac{\pi\eta(2r)^4\theta}{2(2l)} \text{ or } \theta_1 = 8\theta \;\therefore\; \theta = \theta_1/8 = 30/8 = \mathbf{3.75°}.$$

Ex. 7. *A thin walled circular tube of mean radius 10 cm and thickness 0.05 cm is melted up and recast into a solid rod of the same length. Compare the torsional rigidities in two cases.*

Sol. Mean radius of the hollow tube $= 10 \times 10^{-2}$ m. Its thickness $= 0.05 \times 10^{-2}$ m.

Extermal radius, $r_1 = 10 + 0.05/2$ cm $= 10.025 \times 10^{-2}$ m.

Internal radius, $r_2 = 10 - 0.05/2$ cm $= 9.975 \times 10^{-2}$ m.

Now, if r be the radius of recast solid rod having the same length and same mass as the hollow one, then

$$\pi\,(r_1^2 - r_2^2)l\rho = \pi r^2 l\rho$$

where l is the length of the rod and ρ the density of the material.

$$\therefore \quad r_1^2 - r_2^2 = r^2$$

or $\quad r^2 = (10.025 \times 10^{-2})^2 - (9.975 \times 10^{-2})^2 = 0.05 \times 20 \times 10^{-4} = 10^{-4}$ or $r = 10^{-2}$ m.

Torsional rigidity of the hollow rod, $C_1 = \eta\pi\,(r_1^4 - r_2^4)/2l$

Torsional rigidity of solid rod, $C_2 = \eta\pi r^4/2l$

$$\therefore \quad \frac{C_1}{C_2} = \frac{r_1^4 - r_2^4}{r^4} = \frac{(10.025)^2 + (9.975)^2}{(1)^2} = \frac{\mathbf{200}}{\mathbf{1}}.$$

Ex. 8. *A body suspended symmetrically from the lower end of a wire, 100 cm long and 1.22 mm in diameter, oscillates about the wire as axis with a period of 1.25 sec. If the modulus of rigidity of the material of the wire 8.0×10^{10} N/m², calculate the M.I. of the body about the axis of rotation.*

(I.A.S.)

Sol. $T = 2\pi\sqrt{I/C}$ $\therefore$ $I = CT^2/4\pi^2$ ....(i)

where I is M.I. of the body about the axis of rotation and $C = \eta\pi r^4/2l$

Here, $\eta = 8 \times 10^{10}$ N/m^2, $l = 1$ m, $r = 0.61 \times 10^{-2}$ m, $T = 1.25$ sec

$$\therefore \quad C = \frac{8\times10^{10}\times3.14\times(0.61\times10^{-2})^4}{2\times1} \text{ N-m}$$

Substituting the values of C and T in (i), we get

$$I = \frac{8\times10^{10}\times3.14\times(0.61\times10^{-2})^4}{2\times1}\times\frac{(1.25)^2}{4\times(3.14)^2} = \mathbf{8.889\times10^{-4}\ kg\text{-}m^2.}$$

Ex. 9. *Find the amount of work done in twisting a steel wire of radius 1 mm and length 25 cm through an angle of 45°, the modulus of rigidity of steel being 8×10^{11} dyne/cm^2.* **(Allahabad 1990)**

Sol. The work done in twisting a wire from zero to θ angle, is given by

$$W = \int_0^\theta C\theta\, d\theta = \frac{1}{2}C\theta^2 \qquad \text{...(i)}$$

Here, $$C = \frac{\eta\pi r^4}{2l} = \frac{8\times10^{10}\times3.14\times(10^{-4})^4}{2\times0.25} \text{ and } \theta = 45\times\frac{\pi}{180} = \frac{\pi}{4} \text{ radian.}$$

$$\therefore \quad \text{Work done, } W = \frac{1}{2}C\theta^2 = \frac{1}{2}\times\frac{8\times10^{10}\times3.14\times(10^{-3})^4}{2\times0.25}\times\left(\frac{\pi}{4}\right)^2 = \mathbf{0.1547\ joules.}$$

EXERCISE 13·2

1. One end of a wire of 1 mm radius and 50 cm length is twisted through an angle of 45°. Calculate the angle of shear on its surface. **(Agra 1980)**
[**Ans.** 0.09°]

2. A solid rod of circular section of radius 3 cm is recast into a hollow rod of outer radius 5 cm., length and mass remaining unchanged. By what factor does the couple per unit twist change by doing so ? **(Kanpur 1990)**
[**Ans.** 17/8.]
[**Hint :** $\frac{C'}{C} = \frac{r_2^2 + r_1^2}{r^2} = \frac{2r_2^2 - r^2}{r^2}$ [$\because r_2^2 = r^2 + r_1^2$] $= \frac{2\times5^2-4^2}{4^2} = \frac{17}{8}$.]

3. Two cylindrical rods of length l and $2l$ and radii $2r$ and r respectively are welded coaxially. One end being fixed, the other is twisted by 110°. Calculate the twist of the thicker rod. **(Rohilkhand 1979)**
[**Ans.** 3.33°.]

4. The couple per unit twist for a certain solid cylinder of radius r is 100 N-m. Calculate the contribution to this couple due to the central part upto radius $r/4$, and due to the outermost part between radii $3r/4$ and r. **(Agra 1970)**
[**Ans.** 0.39 N-m; 68.4 N-m.]

5. One end of a wire 2 mm in diameter and 50 cm in length is twisted through 0.8 raidan. Calculate the shearing strain at the surface of the wire. **(Agra 1971)**
[**Ans.** 1.6×10^{-3} rad.]

6. One end of a rod of 4 cm radius and 60 cm length is twisted through an angle of 30°, calculate the angle of shear at its surface. Also calculate the angle of shear at the surface of a thin coaxial cylindrical layer which divides the mass of the rod into two equal parts.
[**Ans.** 2°, 1.413°.]

7. A wire of 1 mm radius and 2 m length is twisted through 90°. Calculate the angle of shear at the surface, at the axis of the wire and at a point midway between the axis and the surface. If the modulus of rigidity of the material 5×10^{10} N/m^2, what is the torsional couple ? **(Agra 1977)**
[**Ans.** 0.045°, 0°, 0.0225°; 0.0616 N-m.]

8. The restoring couple per unit twist in a solid cylinder of 3 cm radius is 9×10^8 dyne-cm. What will be the restoring couple per unit twist in a hollow cylinder of the same mass and length but external radius 5 cm ?

   [**Ans.** 410 N-m.]

9. A metal bar soldered to the lower end of a wire in the middle, vibrates torsionally. How will the time of oscillation be changed by halving (a) the linear dimensions of the wire, (b) the linear dimensions of the bar, (c) the linear dimensions of both bar and wire.

   [**Ans.** (a) $1/\sqrt{2}$ times; (b) 1/2 times; (c) $1/2\sqrt{2}$ times.]

10. When a certain mass is attached to the end of vertical wire 0.6 mm in diameter, the extension produced is 1.12 mm and period of torsion oscillation is 15 sec. If the radius of gyration of the mass about the axis of the wire is 3 cm, calculate the ratio of Young's modulus to the rigidity modulus.

    [**Ans.** 2.49.]

11. A metal disc of 10 cm radius and mass 1 kg is suspended in a horizontal plane by a vertical wire attached to its centre. If the diameter of the wire is 1 mm, length 2 metre and the period of torsional vibrations 4 seconds, find the rigidity of the wire.

    [**Ans.** 12.57×10^{10} N-m^{-2}]

12. A uniform circular disc is suspended by a steel wire and the system is allowed to vibrate torsionally. The periodic time is 4 seconds. Find the period, if (i) the length of the wire is reduced to one half, and (ii) two particles each having a mass equal to 1/4 times the mass of the disc are placed on diametricly opposite points on the circumference of the disc. **(Bombay 1967)**

    [**Ans.** (i) 2.8 sec; (ii) 5.6 sec.]

13. A cylindrical metal bar having length 24 cm and diameter 4 cm is suspended by a wire 50 cm long such that the axis of the bar is horizontal. It is observed that the arrangement makes 100 torsional oscillations in 235.9 sec. Determine the coefficient of rigidity of the material of the wire. Density of the material of the bar = 9,000 kg-m^{-3}, and radius of wire = 0.1 cm. **(Punjab 1968)**

    [**Ans.** 3×10^{10} N-m^{-2}.]

14. The couple per unit twist for a uniform cylinder of length L, radius r, rigidity η is given by $\pi\eta r^4/2l$. Calculate the couple per unit twist for a hollow cylinder of same length and same amount of material and external radii $2r$. **(Rajasthan 1967)**

    [**Ans.** $\frac{7}{2}\cdot\frac{\pi\eta r^4}{l}$.]

13·13. Bending of a Beam

Beam : A rod (or bar) of uniform rectangular or circular cross-section, whose length is very large in comparison to its thickness or radius is called a beam.

A beam may be considered as made up of a number of thin plane (or cylindrical) layers, collected parallel (or coaxial) to each other. Also, each layer is regarded as to consist of a number of parallel longitudinal fibres, which are called the **longitudianl filaments.**

Neutral surface : When a beam is clamped horizontally at one end and loaded at the other, it undergoes bending. The filaments (or fibres) of outward side are lengthened and subjected to tension, while those of the inner side are shortened and compressed. In between these two portions, there is a layer or surface in which the filaments are neither elongated nor shortened. Such a surface is called neutral surface. In **fig 13.14**, $EFGH$ is the neutral surface.

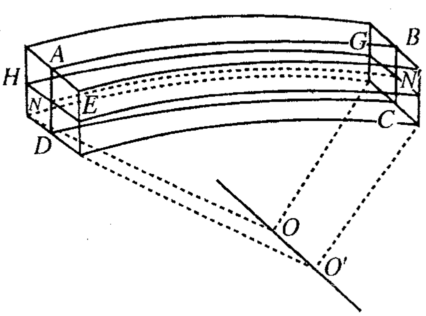


Fig. 13.14.

Plane of bending : As shown in **fig. 13.14**, the beam has a plane of symmetry $ABCD$ paralled to its length. This plane intersects every transverse section in a straight line, which is an axis of symmetry for that section.

If the bending is uniform, all longitudinal filaments are bent into circular arcs in planes parallel to the plane of symmetry $ABCD$. Then, plane $ABCD$ may be called the **plane of bending**. The centres of curvature of the longitudinal filaments lie on a straight line perpendicular to the plane of bending. This straight line may be called as the axis of bending. In the **fig. 13.14**, line OO' is the axis of bending.

Neutral axis : The line of intersection of the plane of bending with the neutral surface (both are perpendicular to each other) is called the neutral axis. In **fig. 13.14**, line NN' is the neutral axis.

Bending moment : When a horizontal beam is fixed at one end and loaded at the other, a bending is produced due to the moment of the load. In this position the beam remains in equilibrium, provided the limit of elasticity has not been exceeded.

Let $ABCD$ represent a section of the beam fixed rigidly in the wall at AD with the other end BC, loaded with a weight W, as shown in **fig. 13.15.** Such a beam is called **cantilever.**

Let us consider the equilibrium of a portion $BCFE$ of the beam, cut by a transverse plane EF across it. A force W* acts downwards at the end BC, hence an equal and opposite reaction force equal to it is acting vertically along FE. These two equal and opposite forces constitute a couple. This clockwise couple which bends the beam, is called bending couple and the moment of this couple is called the **bending moment.**

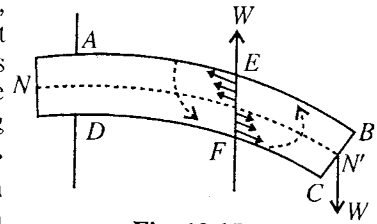


Fig. 13.15.

But the beam is in equilibrium, therefore, there must also be an equal and opposite couple to balance it. The only other forces which act on $EBCF$ are obviously the forces due to stretching and compression of the filaments of the beam passing across EF. The filaments in the upper portion of the beam are elongated and thus, they must be in tension, the portion of the filaments to the left of EF exerts a pulling force on the portion to the right of EF; similarly a portion of the filaments in the lower half of the beam, to the left of EF, exerts a pushing force on the portion to the right of EF, because the filaments below the neutral surface are shortened. The distribution of these forces at the section EF has been shown in **fig. 13.15** and their resultant is a couple which balances the bending couple.

The moment of this balancing couple (due to tensile and compressive stresses) is called the **moment of the resistance to bending**. Since in the position of equilibrium it is equal and opposite to the bending couple, it is referred as **bending moment of the beam.**

Expression for bending moment : Let us consider a small portion of the beam bounded by two transverse sections AB and CD close to each other. After bending as in **fig. 13.16 (b)**, AC is elongated to $A'C'$ and BD is shortened to $B'D'$. Line EF represents the neutral surface, which is neither stretched nor shortened.

Let the small portion, under consideration, be bent in the form of a circular arc, subtending an angle θ at the centre of curvature O. Let R be the radius of curvature of the neutral surface $E'F'$.

Consider a filament GH distant z from EF in the unbent position of the beam. After bending its position is shown by $G'H'$ [**fig. 13.16(b)**].

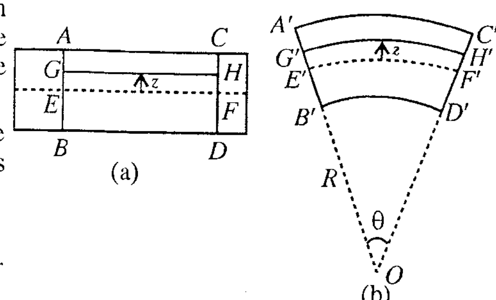


Fig. 13.16.

Now, $\quad G'H' = (R + z)\theta$

Before bending $\quad GH = EF$ and $E'F' = EF$ (for neutral surface)

$\therefore \quad GH = E'F'$

But $E'F' = R\theta$, $\therefore \quad GH = R\theta$

Change in the length of the filament $= G'H' - GH = (R + z)\theta - R\theta = z\theta$.

$$\text{Strain} = \frac{\text{Change in length}}{\text{Initial length}} = \frac{G'H' - GH}{GH} = \frac{z\theta}{R\theta} = \frac{z}{R} \qquad ...(1)$$

* The weight of the beam has been neglected in comparison to the load W.

If $PQRS$ [**fig. 13.17**] represents a section of the beam at right angles to its length and the plane of bending, then, clearly, the forces acting on the filaments are perpendicular to this section. The line MN lies on the neutral surface.

Let the breadth of the section be $PQ = b$, and its depth, $QR = d$.

The forces, producing elongations and contractions in filaments, act perpendicular to the upper and lower halves, $PQMN$ and $MNRS$ respectively, of the section $PQRS$ and their directions being opposite to each other.

Now, consider a small area δa of the section $PQRS$ about a point A, distant z from the neutral surface. The strain produced in a filament passing through this area will be z/R, as shown above.

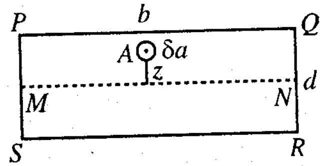


Fig. 13.17.

Now, $\quad Y = \text{stress/strain}, \quad \therefore \ \text{stress} = Y \times \text{strain}.$

Hence, $\quad$ Stress about the point $A = Y \times z/R$,

where Y is Young's modulus for the material of the beam.

$$\text{Force on the area } \delta a = Y \cdot \frac{z}{R} \cdot \delta a$$

$$\text{Moment of this force about the line } MN = Y \cdot \frac{z}{R} \cdot \delta a . z = \frac{Y \delta a z^2}{R}$$

But the moments acting on both the upper and the lower halves of the section are in the same direction, therefore, the total amount of the moments acting on the filaments in the section $PQRS$ is given by

$$\Sigma \frac{Y \delta a z^2}{R} = \frac{Y}{R} \Sigma \delta a z^2 = \frac{YI}{R} \qquad ...(2)$$

where $I = \Sigma \delta a \, z^2$ and is called the *second moment (or geometrical moment of inertia) of the sectional area* about MN and therefore, equal to aK^2, where a is the whole area of surface $PQRS$ and K, its radius of gyration about MN.

This quantity YI/R is the restoring couple or the **bending moment** of the beam, being equal and opposite to the moment of bending couple due to the load in the position of equilibrium. Thus,

$$\textbf{Bending moment} = \frac{YI}{R} \qquad ...(3)$$

The quantity YI is known as **flexural rigidity**.

In case of a beam of rectangular cross-section of breadth b and depth d, the area of cross-section is $b \cdot d$ and (radius of gyration)^2 is equal to $d^2/12$. Hence,

$$I = \text{area of cross-section} \times d^2/12 = b \cdot d \times d^2/12 = bd^3/12^* \qquad ...(4)$$

Similarly, in the case of cylindrical rod of radius r the area of cross-section is πr^2 and the square of radius of gyration of the cross-section about an axis through the centre of the section and perpendicular to the plane of bending is $r^2/4$. Thus

$$I = \pi r^2 \times r^2/4 = \pi r^4/4. \qquad ...(5)$$

In case, the rod is hollow, having r_1 and r_2 as the internal and external radii, then

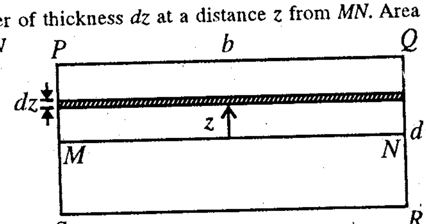
$$I = \frac{\pi}{4}(r_2^4 - r_1^4). \qquad ...(6)$$

---

* We want to know geometrical moment of inertia about the axis MN. Take a layer of thickness dz at a distance z from MN. Area of the section of this layer $= b \cdot dz$. Its geometrical moment of inertia about MN $= bdz \cdot z^2$.

$\therefore$ Geometrical M.I. of the total sectional area of the beam about MN is

$$I = \int_{-d/2}^{d/2} bdz . z^2 = 2\int_0^{d/2} bdz \cdot z^2 = 2b\left[\frac{z^3}{3}\right]^{d/2} = \frac{2b}{3}\left(\frac{d}{2}\right)^3 = \frac{bd^3}{12}.$$



Similarly, for a circular section, $\quad I = \int_0^r \frac{2\pi x \, dx \, x^2}{2} = \frac{\pi r^4}{4}.$

Fig. 13.18.

13·14. Cantilever

A horizontal beam, fixed at one end and loaded at the free end, is called a **cantilever.**

Let a beam of length l be fixed at its one end A and loaded with a load W at B. Let the end B be deflected to the position B' under the action of the load W*. Suppose the axis of X lies horizontally in the direction of the unbent beam and the axis of Y, vertically downwards. Consider a section of the beam, as at P, at a distance x from the end A. Then the moment of the bending couple due to the load W

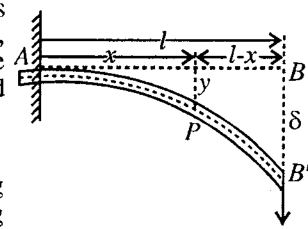


Fig. 13.19.

$$= W(l - x).$$

The cantilever is in equilibrium in the bent position, hence the bending couple must be balanced at the section under consideration by the restoring couple or bending moment.

The bending moment $= YI/R$,

where Y is the Young's modulus of the material of the beam, I geometrical moment of inertia of the section and R the radius of curvature of the neutral axis at the section.

Thus, $$\frac{YI}{R} = W(l - x) \qquad ...(1)$$

The radius of curvature R is given by the equation

$$R = \frac{\{1+(dy/dx)^2\}^{3/2}}{d^2y/dx^2} \qquad ...(2)$$

where y is the depression of the beam at a distance x from the fixed end.

Since y is small, $(dy/dx)^2$ can be neglected in comparison to 1. Hence, we get

$$R = \frac{1}{d^2y/dx^2} \qquad ...(3)$$

Substituting for R in equation (1) above, we have

$$YI.\frac{d^2y}{dx^2} = W(l - x) \qquad ...(4)$$

or $$\frac{d^2y}{dx^2} = \frac{W}{YI}(l - x) \qquad ...(5)$$

This equation may now be solved to get the value of the depression of the free end *i.e.*, at $x = l$.

Integrating equation (5), we get

$$\frac{dy}{dx} = \frac{W}{YI}\left(lx - \frac{x^2}{2}\right) + C_1 \qquad ...(6)$$

where C_1 is constant of integration, which can be known from the conditions of the problem. One such condition is that at $x = 0$, $dy/dx = 0$.

Applying this conditions in equation (6), we have $C_1 = 0$

and thus $$\frac{dy}{dx} = \frac{W}{YI}\left(lx - \frac{x^2}{2}\right) \qquad ...(7a)$$

It is to be pointed out that at P, the slope of the curve $\tan\theta = dy/dx$, θ being the angle between X-axis and tangent at P of the bent portion APB'.

Thus $$\frac{dy}{dx} = \tan\theta = \frac{W}{YI}\left(lx - \frac{x^2}{2}\right) \qquad ...(7b)$$

* We assume that the weight of the beam is negligible in comparison to the load W. Ofcourse, the depression $\delta = BB'$ is much smaller than the length of the beam, but it is shown relatively much more for clarity.

Integrating once again,

$$y = \frac{W}{YI}\left(\frac{lx^2}{2} - \frac{x^3}{6}\right) + C_2 \qquad ...(8)$$

where C_2 is another constant of integration.

But at $x = 0$, $y = 0$, hence substituting these values in eq. (8), we get $C_2 = 0$

Thus, **the expression for the depression** y **at a distance** x **from the fixed end is**

$$y = \frac{W}{YI}\left(\frac{lx^2}{2} - \frac{x^3}{6}\right) \qquad ...(9)$$

At **the free end**, $x = l$ and let $y = \delta$, then the expression for depression is

$$\delta = \frac{W}{YI}\left(\frac{l^2}{2} - \frac{l^2}{6}\right) \text{ or } \delta = \frac{Wl^3}{3YI} \qquad ...(10)$$

For a beam of rectangular cross-section of breadth b and depth d, $I = bd^3/12$ and hence

$$\delta = \frac{4Wl^3}{Ybd^3} \qquad ...(11)$$

For a beam of circular cross-section of radius r, $I = \pi r^4/4$

$$\therefore \qquad \delta = \frac{4Wl^2}{3\pi r^4 Y} \qquad ...(12)$$

13·15. Transverse Vibrations of a Loaded Cantilever

Consider a rod of length l, clamped at one end and loaded at its free end with mass M. In the equilibrium position, the deflection of the free end will be given by [eq. (10) above],

$$W = Mg = \frac{3YI}{l^3}\delta \qquad ...(1)$$

Let the mass M be slightly depressed and then released. Now, the cantilever will execute tansverse vibrations. If y be the displacement of the free end of the cantilever at any time from the equilibrium position. Then a restoring force F will act opposite to W.

Therefore, $$W - F = \frac{3YI}{l^3}(\delta + y) \qquad ...(2)$$

Subtracting eq. (13) from eq. (14), we get

$$F = -\frac{3YI}{l^3}y \qquad ...(3)$$

If d^2y/dt^2 be the acceleration produced in the mass M at that instant at the free end, then the force F is Md^2y/dt^2.

Therefore, the equation of motion is

$$M\frac{d^2y}{dt^2} + \frac{3YI}{l^3}y = 0 \quad \text{or} \quad \frac{d^2y}{dt^2} + \frac{3YI}{Ml^3}\cdot y = 0 \qquad ...(4)$$

This equation repressents a S.H.M., whose periodic time is given by

$$T = 2\pi\sqrt{\frac{Ml^3}{3YI}} \qquad ...(5)$$

13·16. Beam Supported at its Ends and Loaded in the Middle

Consider a beam supported on two knife edges K_1 and K_2 at a distance l apart in a horizontal plane. Let it be loaded at its middle point O. The reaction of each knife edge is obviously equal to $W/2$ acting upwards. The beam bends as shown in the figure and the depression is maximum in the middle.

The middle part of the beam is almost horizontal. Hence the beam may be considered as equivalent to two inverted cantilevers OA and OB. The length of each cantilever will be $l/2$, clamped at the end O and carrying an upward load $W/2$ at the other. Therefore, the depression of O below the knife edges is the elevation of the loaded end of such a cantilever from the lowest position O.

Let us consider a vertical section at P, distant x from O and consider the part PB. The moment of the deflecting couple is

$$= \frac{W}{2} \cdot PB = \frac{W}{2} \cdot \left(\frac{l}{2} - x\right)$$

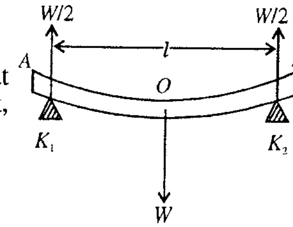


Fig. 13.20.

In the equilibrium position this deflection couple must be balanced at the section under consideration by the restoring couple or bending moment, established inside the beam, *i.e.*,

$$\frac{W}{2}\left(\frac{l}{2} - x\right) = \frac{YI}{R} \qquad \text{...(1)}$$

If y is the elevation of the section P above X-axis, then radius of curvature of the neutral axis at this section is given by

$$R = \frac{1}{d^2y/dx^2} \qquad \text{...(2)}$$

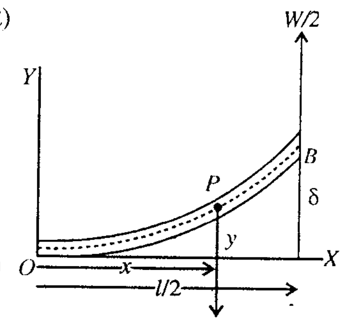


Fig. 13.21.

Hence eq. (1) is

$$\frac{W}{2}\left(\frac{l}{2} - x\right) = \frac{YId^2y}{dx^2} \quad \text{or} \quad \frac{d^2y}{dt^2} = \frac{W}{2YI}\left(\frac{l}{2} - x\right)$$

Intergrating this expression, we have

$$\frac{dy}{dx} = \frac{W}{2YI}\left(\frac{l}{2}x - \frac{x^2}{2}\right) + C_1 \qquad \text{...(3)}$$

where C_1 is the constant of integration.

At O, $x = 0$ and $dy/dx = 0$; hence $0 = 0 + C_1$ or $C_1 = 0$,

$$\therefore \qquad \frac{dy}{dx} = \frac{W}{2YI}\left(\frac{l}{2}x - \frac{x^2}{2}\right) \qquad \text{...(4)}$$

Integrating further, we have

$$y = \frac{W}{2YI}\left(\frac{l}{2}\frac{x^2}{2} - \frac{x^3}{6}\right) + C_2$$

where C_2 is another constant of integration.

Again at O, $x = 0$ and $y = 0$, $\therefore C_2 = 0$

$$\therefore \qquad y = \frac{W}{2YI}\left(\frac{l}{2}\frac{x^2}{2} - \frac{x^3}{6}\right) \qquad \text{...(5)}$$

Now, at the free end, $x = l/2$ and $y = \delta$ (say). Therefore

$$\delta = \frac{W}{2YI}\left(\frac{l}{2}\frac{l^2}{8} - \frac{l^3}{48}\right) \quad \text{or} \quad \delta = \frac{Wl^3}{48YI} \qquad \text{...(6)}$$

If b and d be the breadth and thickness of the beam respectively, then $I = bd^3/12$. Then

$$\delta = \frac{Wl^2}{4Ybd^3} \qquad \text{...(7)}$$

If the cross-section of the beam be a circle of radius r, then

$$\delta = \frac{Wl^3}{12Y\pi r^4} \qquad \text{...(8)}$$

This is the same as the depression at the middle point O. If M be the mass with which the beam has been loaded in the middle, then eq. (7) is

$$\delta = \frac{Mgl^3}{4Ybd^3} \quad \text{...(9)}$$

The use of this equation is made in the construction of **girders**. The depression of a beam at the middle is proportional directly as the cube of its length and inversely as the width and cube of the depth or thickness. Thus, in order that the girder may not bend appreciably, its span, *i.e.*, length is made small and its breadth and depth large. At the same time Young's modulus for the material of the beam must be large.

Shape of girders 'I' : When a girder is supported at its ends, its filaments above the neutral surface (or in the upper half), are compressed and those below the neutral surface (or in the lower half) are elongated. The compression or extension of the filaments is maximum at the upper or lower faces of the girder respectively, hence the stresses are maximum at these faces and they decrease, as we move towards the neutral surface from either side. Consequently, the upper and lower faces of the beam should be much stronger than its middle portions. In other words, the middle portion may be made of much smaller breadth than the upper and the lower faces. This is done by manufacturing the girders with their section in the form of **I** so that breadth at the upper and lower faces may be sufficiently larger than that of the inner parts. This saves quite a good amount of the material without sacrificing the strength of the girder.

Stiffness of a beam : *The stiffness of a beam is defined as the ratio of maximum depression of the beam to its span.* It is usually represented by the symbol $1/n$. For steel girders of large span, n must lie between 1000 and 2000 and for shorter span, between 500 and 700. For timber beams, n should be not less than 360.

13·17. Determination of Young's Modulus by Bending of a Beam

As shown above the depression of a beam of rectangular cross-section, supported at the ends and loaded in the middle, is given by

$$\delta = \frac{Mgl^3}{4Ybd^3} \text{ or } Y = \frac{Mgl^3}{4bd^3\delta}$$

Thus, to determine the Young's modulus of the material of a beam, the beam is supported horizontally and symmetrically on two knife edges, K_1 and K_2, fixed at a distance l apart, as shown in the **fig.13.22**. A hanger, supported by a knife edge is then placed on the middle of the beam and the load is applied by placing the weight on it. The depression δ of the beam, so produced, is determined by a micrometer screw in equilibrium. At the moment, when micrometer screw just touches the beam, a current starts flowing in the galvanometer G, from the battery, causing to deflect the needle of the former and the reading is noted. A series of loads are applied and observations are taken for load increasing and decreasing. A graph is then plotted between the load of mass M as abscissa [**fig. 13.23**] and the corresponding mean depression as the ordinate. The slope of the straight line, so obtained, given the values δ/M.

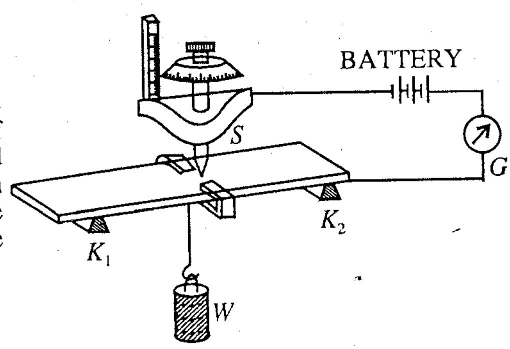


Fig. 13.22.

Now,
$$Y = \frac{Mgl^3}{4bd^3\delta} = \frac{gl^3}{4bd^3} \cdot \frac{M}{\delta}$$

Thus, knowing the values of l, b and d, Young's modulus of the material of the beam may be determined.

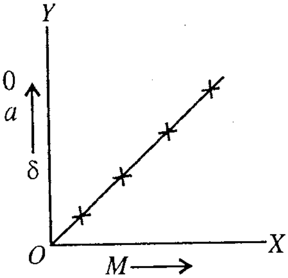


Fig. 13.23.

Ex. 1. *The depression at the free end of a cantilever of length* 1.5 *m is* 10 *cm when a certain load is suspended at this end. What is the depression at a distance* 0.5 *m from the fixed end ?*

(Bhopal 2003)

Sol. Here; l = 1.5 m, δ = 10 cm = 0.1 m, x = 0.5 m, y = ?

Depression at the free end $\delta = \frac{Wl^3}{3YI}$, $\therefore \frac{W}{YI} = \frac{3\delta}{l^3}$

Depression at a distance, x from the fixed end $y = \frac{Wx^2}{2YI}\left(l - \frac{x}{3}\right) = \frac{W}{YI}\,\frac{x^2(3l-x)}{2\times 3} = \left(\frac{3l-x}{2l^3}\right)x^2\delta$

$$\therefore \quad y = \frac{(3\times 1.5)-0.5}{2\times(1.5)^3}\times(0.5)^2\times 0.1 = \mathbf{0.0148\ m.}$$

Ex. 2. *A steel wire of* 1 *mm radius is bent to form a circle of* 10 *cm radius. What is the bending moment and the maximum stress, if* $Y = 2\times 10^{11}$ *N-m*^{2}.

Sol. Bending momet $= YI/R$

Here, $I = \pi r^4/4 = 3.14\times(10^{-3})^4/4$; $\therefore$ Bending moment $= \frac{2\times 10^{11}\times 3.14\times(10^{-3})^4}{4\times 0.1} = \mathbf{1.57\ N\text{-}m.}$

Stress $= Yz/R$

For maximum stress the value of z should be maximum, *i.e.*, $z = r$.

$$\therefore \quad \text{Maximum stress} = \frac{Yr}{R} = \frac{2\times 10^{11}\times(10^{-3})}{0.1} = \mathbf{2\times 10^9\ N\text{-}m^{-2}}.$$

Ex. 3. *A cantilever of length l and uniform cross-section shows a depression of* 2 *cm at the loaded end. What will be the depression at distance l/4, l/2 and 3l/4 from the fixed end ?*

Sol. $$y = \frac{Mgx^2}{2YI}\left(l - \frac{x}{3}\right)$$

When $x = l$, $y = 0.02$ cm,

$$\therefore \quad 0.02 = \frac{Mgl^2}{2YI}\left(l - \frac{l}{3}\right) = \frac{Mgl^3}{3YI}$$

When $x = l/4$,

$$y_1 = \frac{Mg}{2YI}\left(\frac{l}{4}\right)^2\left(l - \frac{l}{12}\right) = \frac{Mgl^2}{32YI}\cdot\frac{11}{12} = \frac{11}{128}\cdot\frac{Mgl^3}{3YI}$$

$$= \frac{11}{128}\times 0.02 = \mathbf{1.72\times 10^{-3}\ m.}$$

When $x = l/2$,

$$y_2 = \frac{Mg}{2YI}\left(\frac{l}{4}\right)^2\left(l - \frac{l}{6}\right) = \frac{Mgl^3}{8YI}\cdot\frac{5}{6} = \frac{5}{16}\cdot\frac{Mgl^3}{3YI}$$

$$= \frac{5}{16}\times 0.02 = \mathbf{6.2\times 10^{-3}\ m.}$$

When $x = \frac{3l}{4}$

$$y_3 = \frac{Mg}{2YI}\left(\frac{3l}{4}\right)^2\left(l - \frac{1}{4}\right) = \frac{Mgl^3}{2YI}\cdot\frac{27}{64} = \frac{Mgl^3}{3YI}\cdot\frac{81}{128}$$

$$= \frac{81}{128}\times 0.02 = \mathbf{1.27\times 10^{-2}\ m.}$$

Ex. 4. *A* 10 *cm wide and* 0.2 *mm thick metal sheet is bent to form a cylinder of* 10 *cm length and* 50 *cm radius. If the Young's modulus of the metal is* 1.5×10^{11} *N-m*$^{-2}$ *calculate* (i) *the stress and strain on the convert surface and* (ii) *the bending moment.*

Sol. (i) $$\text{Strain} = \frac{z}{R} = \frac{2\times 10^{-4}}{2}\times\frac{1}{0.5} = \mathbf{2\times 10^{-4}}$$ **(Agra 1991)**

[$\because$ for convex surface z = half thickness of the sheet]

Stress = $Y \times$ strain $= 1.5 \times 10^{11} \times 2 \times 10^{-4} =$ **3.0×10^{7} N-m^{-2}**

(ii) Bending moment $= \dfrac{YI}{R} = \dfrac{Y}{R} \cdot \dfrac{bd^3}{12} = \dfrac{1.5\times10^{11}}{0.5} \times \dfrac{0.1\times(2\times10^{-4})^3}{12} =$ **2×10^{-2} N-m.**

Ex. 5. *A light metal rod of length 60 cm and of radius 1 cm is clamped at one end loaded at the free end, with 5.5 kg. Calculate the depression of the free end, assuming $Y = 9 \times 10^{11}$ dynes/sq. cm. and $g = 980$ cm/sec^2.* **(I.A.S.)**

Sol. At the free end, $\delta = \dfrac{Wl^3}{3YI}$

For a circular cross-section of radius r, $I = \pi r^4/4$

$$\therefore \quad \delta = \frac{4Wl^3}{3Y\pi r^4}$$

Here, $W = 5.5 \times 9.8$ N, $l = 0.6$ m., $Y = 9 \times 10^{10}$ N/m^2. and $r = 0.01$ m.

Therefore, $$\delta = \frac{4\times5.5\times9.8\times(0.6)^3}{3\times9\times10^{10}\times3.14\times(0.01)^4} = \mathbf{5.495 \times 10^{-3}\ m.}$$

Ex. 6. *A vertical steel rod of circular cross-section of radius 1 cm is rigidly fixed in the earth. Its upper end is 3 metres from the earth. A thick string which can stand a maximum tension of 2 kg-wt. is tied at the upper end of the rod and pulled horizontally. Find how much the top will be deflected before the string snaps. (Y for steel $= 2 \times 10^{11}$ S.I. units; $g = 10$ m/sec^2).*

Sol. The lower end of a vertical rod is fixed and the upper end is pulled by a horizontal force, hence this is the case of a cantilever.

The deflection of the upper end, is given by

$$\delta = \frac{Wl^3}{3YI}$$

Here, the horizontal force $W = 2 \times 10$ N/m^2; length of the rod $l = 3$ m; value of $Y = 2 \times 10^{11}$ N/m^2.

For circular cross-section $I = \dfrac{\pi r^4}{4} = \dfrac{\pi\times(0.01)^4}{4} = \dfrac{\pi\times10^{-8}}{4}$

$$\therefore \quad \delta = \frac{2\times10\times(3)^3\times4}{3\times2\times10^{11}\times3.14\times10^{-8}} = \mathbf{0.1146\ m.}$$

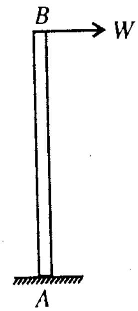


Fig. 13.24.

Ex. 7. *Compare loads required to produce equal depression for two beams, made of the same material and having the same length and weight with the only difference that while one has circular cross-section, the cross-section of the other is square.* **(Nagpur 1960)**

Sol. The depression δ at the middle point of a beam of length l of circular cross-section of radius r, is given by

$$\delta = \frac{W_1 l^3}{12Y\pi r^4}, \text{ where } W_1 \text{ is the weight loaded.} \quad ...(i)$$

If in case of rectangular cross-section beam the weight loaded be W_2, then the depression δ', is given by

$$\delta' = \frac{W_2 l^3}{4bd^3 Y}$$

since the two beams have the same length and are made of same material.

But for square cross-section $b = d$

$$\therefore \quad \delta' = \frac{W_2 l^3}{4b^4 Y} \quad ...(ii)$$

The two beams have equal weights and made of the same material, therefore their masses and hence volume must be equal, *i.e.*, $\pi r^2 l = b^2 l$ or $b^2 = \pi r^2$

Substituting the value of b^2 in (ii), we get

$$\delta' = \frac{W_2 l^3}{4\pi^2 r^4 Y} \qquad \text{...(iii)}$$

But depression is the same in both cases, *i.e.*, $\delta = \delta'$. Therefore equating (i) and (iii), we get

$$\frac{W_1 l^3}{12Y\pi r^4} = \frac{W_2 l^3}{4\pi^2 r^4 Y} \text{ or } \frac{W_1}{W_2} = \frac{3}{\pi}, \text{ which is the required ratio.}$$

Ex. 8. *A cantilever of length l is loaded at a point x_1, calculate the depression of the tip of the cantilever. If the same load is shifted to the tip, what is the depression of the point x_1 ?*

(Bombay 1970 S)

Sol. The expression for the depression y at a distance x from the fixed end, is given by

$$y = \frac{W}{YI}\left(\frac{lx^2}{2} - \frac{x^3}{6}\right) \qquad \text{...(i)}$$

Alos at x, $$\frac{dy}{dx} = \tan\theta = \frac{W}{YI}\left(lx - \frac{x^2}{2}\right) \qquad \text{...(ii)}$$

In the first case, the cantilever is loaded at $OP = x_1$, Hence the beam will not bend beyond x_1 by the load W, but the portion $P'Y$ will be tangent at P' and the total depress $\delta = XY$, will be given by

$$\delta = XX' + X'Y = PP' + P'X' \tan\theta$$

Hence $$PP' = \frac{Wx_1^3}{3YI},\ P'X' = l - x_1 \text{ and } \tan\theta = \frac{W}{YI}\left(x_1^2 - \frac{x_1^2}{2}\right) = \frac{Wx_1^2}{2YI}$$

$$\therefore \quad \delta = \frac{Wx_1^2}{3YI} + (l - x_1)\frac{Wx_1^2}{3YI} = \frac{W}{YI}\left(\frac{lx_1^2}{2} - \frac{x_1^3}{6}\right) = \frac{Wx_1^2}{2YI}\left(l - \frac{x_1}{3}\right) \qquad \text{...(iii)}$$

This is the depression of the tip of the cantilever.

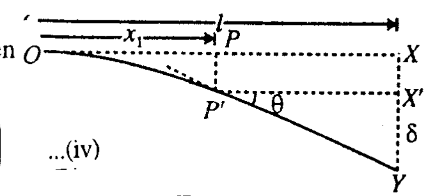


Fig. 13.25.

In the second case, the same load is shifted to the tip, then the depression at any point x_1, is given by

$$\delta' = \frac{W}{YI}\left(\frac{lx_1^2}{YI} - \frac{x_1^3}{6}\right) = \frac{Wx_1^2}{2YI}\left(l - \frac{x_1}{3}\right) \qquad \text{...(iv)}$$

We see that in two cases the depression is the same.

Ex. 9. *A steel strip is clamped horizontally from one end. On applying a 500 gm load at the free end the bending in equilibrium state is 5 cm. Calculate,* (i) *the potential energy in the strip,* (ii) *the frequency of vibration if the load is disturbed from equilibrium (neglect mass of the strip itself).*

Sol. (i) The bending is 5 cm by the load 500 gm load = 0.05 × 9.81 newton, hence force per unit bending or force constant is

$$C = \frac{0.5 \times 9.81}{0.05} = \mathbf{98.1\ N\text{-}m^{-1}}$$

$$\therefore \quad \text{Potential energy} = \int_0^x Cx\,dx = \frac{1}{2}Cx^2 = \frac{1}{2} \times 9.81 \times (0.05)^2 = \mathbf{0.12\ joules.}$$

(ii) $$\text{Frequency, } n = \frac{1}{2\pi}\sqrt{\frac{C}{M}} = \frac{1}{2\pi}\sqrt{\frac{9.81}{0.5}} = \mathbf{2.23\ sec^{-1}\ (nearly).}$$

Ex. 10. *A solid cylindrical rod of radius 6 mm bends by 8 mm under a certain load. Deduce the bending when all other things remaining the same the beam is replaced by a hollow cylindrical one with external radius 10 mm, internal 8 mm.*

Sol. If δ and δ' are the depressions of the two beams, then

$$\delta = \frac{Mgl^3}{3YI} \text{ and } \delta' = \frac{Mgl^3}{3YI'}; \ \delta' = \frac{I}{I'}\delta$$

For solid beam, $I = \pi r^4/4$ and for hollow beam, $I' = \dfrac{\pi(r_2^4 - r_1^4)}{4}$,

$$\therefore \qquad \delta = \frac{6^4}{10^4 - 8^4} \times 8 \times 10^{-3} = \mathbf{1.76 \times 10^{-3}\ m.}$$

EXERCISE 13·3

1. A cantilever of length 50 cm is depressed by 15 mm at the loaded end. Calculate the depression at distance 30 cm from the fixed end. **(Gorakhpur 1994)**

   [**Ans.** 6.48 mm.]

2. A cylindrical rod of length 1 m and radius 1 cm is uniformly bent into a circular arc of radius 10 m. Calculate the bending moment. (Young's modulus for the material of the rod = 9.1×10^{10} N-m^{-2}) **(Mysore 1980)**

   [**Ans.** 7.14 N-m.]

3. A steel rod of length 50 cm, width 2 cm and thickness 1 cm is bent in the form of an arc of radius 200 cm. Calculate (i) the stress and strain on the convex surface of the rod, and (ii) the bending moment. ($Y = 2 \times 10^{11}$ N-m^2.) **(Agra 1980)**

   [**Ans.** (i) 2×10^9 N-m^{-2}, 0.01; (ii) 167 N-m.]

4. A 10 cm wide and 0.2 mm thick metal sheet is bent to form a cylinder of 80 cm length and 50 cm radius. If the Young's modulus of the metal is 1.5×10^{11} N-m^2, calculate (i) the stress and strain on convex sufrace; (ii) bending moment. **(Kanpur 1980)**

   [**Ans.** (i) 3×10^7 N-m^{-2}, 2×10^{-4}; (ii) 0.02 N-m.]

5. A rod of rectangular section having thickness and breadth each of 0.5 cm is bent in the form of an arc having radius of curvature 1000 cm. If the Young's modulus of the material of the rod is 10^{11} N-m^{-2}, calculate (i) the stress and strain on convex sufrace. (ii) bending moment.

   [**Ans.** (i) $2{\cdot}5 \times 10^7$ N-m^{-2}, 2.5×10^{-4}; (ii) 0.52 N-m.]

6. A steel wire of 1.00 mm radius is bent in the form of a circular arc of radius 50 cm Calculate : (i) the bending moment, (ii) the maximum stress. ($Y = 2 \times 10^{11}$ N-m^{-2}) **(Agra 1976, 70 S; Delhi 92)**

   [**Ans.** (i) 0.314 N-m; (ii) 4×10^8 N-m^{-2}.]

7. The end of a given strip cantilever depresses 10 mm under a certain load. Calculate the depression under same load for another cantilever of same material 2 times in length, 2 times in width and 3 times in thickness (vertical). **(Agra 1970)**

   [**Ans.** 1.48 mm.]

8. Show that the longitudinal stress at a distance z from the neutral surface of a beam bent to a radius of curvature R is Yz/R. Where is the tensile stress maximum in a beam, supported at two ends and loaded at the middle ?

   [**Ans.** The lower surface in the middle region.]

9. One end of a 100 cm long rod is fixed horizontally. When a load of 500 gm is suspended at the other end, it is depressed through 5 mm, find the depression at 40 cm distance from the fixed end

when a 5 kg load is suspended at (i) the free end, (ii) 40 cm from the fixed end. **(Kanpur 1978)**
[**Ans.** (i) 1.04 cm; (ii) 0.3 cm.]

10. For the same mass per unit length, show that a beam of square section is stiffer than one of circular section, the deflections being in the ratio $3 : \pi$.

11. A beam one metre long, 4 cm wide, 5 mm thick is supported on two knife-edges 80 cm apart. The depression produced by a 2 kg load hung from the centre of the knife edges is 5 mm. Find Young's modulus of the material of the beam.

[**Ans.** 9.6×10^{10} N-m^{-2}.]

12. A cylindrical rod of iron AB, values 0.5 cm and length 25 cm is clamped at A so that its axis is horizontal. Find the deflections of the free end of the rod when a force of 3 kg-wt is applied : (i) at B, (ii) at the point midway between A and B. ($Y = 21 \times 10^{11}$ dyne/cm^2).

[**Ans.** (i) 0.467 cm; (ii) 0.145 cm.]

13. A cantilever of uniform cross-section and length l shows a depression δ at the loaded end. What will be the depression at a distance $l/2$ from the fixed end ? **(Agra 1987)**

[**Ans.** $\frac{5}{16}\delta$.]

14. A uniform light rod 120 cm long is clamped horizontally at one end. A weight of 100 gm is attached to the free end. Calculate the depression of a free point 90 cm distance from the clamped end. The diameter of the rod is 2 cm. ($Y = 1.013 \times 10^{11}$ dyne/cm^2, $g = 980$ cm/sec^2.) **(Delhi 1990)**

[**Ans.** 0.4492 cm.]

15. A cantilever of length 1.0 m depresses by 5 mm when a load of 0.5 kgf is suspended at its free end. Now a load of 5 kgf is suspended (i) at the free end; (ii) at distance 40 cm from the fixed end. When will be the depression at distance 40 cm from the fixed end ? **(Bundelkhand 2004)**

[**Ans.** (i) 1.04 cm; (ii) 0.32 cm.]

QUESTIONS

Long Answer Questions

1. Explain the meaning of stress and strain. State Hooke's law and hence define Young's modulus, bulk modulus, modulus of rigidity and Poisson's ratio of an isotropic and homogeneous substance. **(Bhopal 1999)**

2. Define Young's modulus, bulk modulus, modulus of rigidity and Poisson's ratio. Obtain the relations connecting these quantities. **(Agra 2006; Jablpur 2004, 03; Bundelkhand 95;Delhi 92, 90; Kanpur 90; Ujjian 93; Jablpur 2004, 03)**

3. Show that the shearing stress is equivalent to an equal linear tensile stress and an equal compressive stress at right angles to each other. Prove that $Y = 3K(1 - 2\sigma)$. **(Agra 2005; Purvanchal 1993)**

4. Show that the Young's modulus Y, modulus of rigidity η and Poissron's ratio are related by the equation $Y = 2\eta(1 + \sigma)$ or $\sigma = \frac{Y}{2\eta} - 1$ **(Agra 2004; Ujjain 04; Bundelkhand 04; Allahabad 90; Rohiljhand 84)**

5. Establish the relations :

(i) $Y = 3K(1 - 2\sigma)$ or $K = \frac{Y}{3(1-2\sigma)}$ **(Jabalpur 2003; Indore 04; Bundelkhand 04; Ujjain 03; Allahabad 1990)**

(ii) $Y = \frac{9\eta K}{\eta + 3K}$ or $\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{K}$, where the symbols have their usual meanings.

(Kanpur 2005; Gwalior 03; Jabalpur 04; Purvanchal 1994; Ajmer 88)

(iii) $\sigma = \dfrac{3K - 2\eta}{2\eta - 6K}$ **(Gwalior 2003)**

6. (a) Show that the potential energy per unit valume of a strained wire is

$$U = \frac{1}{2} \times \text{Stress} \times \text{Strain}$$ **(Gulbarga 1995; Rajasthan 83)**

(b) Find the work done in stretching a wire. **(Agra 1991)**

(c) Show that the theoretical limiting values of Poisson's ratio are –1 and 0.5.
(Jabalpur 2004; Ujjain 03; Agra 1985; Bundelkhand 95; Kanpur 84; Rohilkhand 84)

7. Deduce an expression for the couple per unit twist of a uniform solid cylinder.
(Agra 1995, 93, 91, 87; Gorakhpur 95; Kanpur 95)

8. Define 'Torsion Rigidity' of a wire. How is it related to 'Modulus of Rigidity' ? Show that a hollow rod is a better shaft than a solid one of the same material, mass and length.
(Agra 1977; Knapur 95)

9. (a) From first principles show that the couple per unit twist for a hollow cylindrical rod is $\pi\eta\ (a_2^4 - a_1^4)/2l$, with usual notations. **(Agra 1971)**

(b) Obtain an expression for couple required to twist a cylinder by one turn (*i.e.*, 360°).
(Agra 1992)

[**Ans.** couple = $\eta\pi^2 r^4/l$.]

10. Explain the meaning of angle of twist and angle of shear. Show that the angle of shear is maximum at the outer surface of a cylindrical rod. **(Indore 2003)**

11. What do you mean by the torsion in a cylinder ? Obtain an expression for the torque required to twist a uniform solid cylinder. **(Ujjain 2004; Indore 03)**

12. Drive an expression for the torsional rigidity of a uniform cylindrical rod. How much work is needed to produce θ radian twist in the cylinder ? **(Raipur 2003; Bundelkhand 04)**

13. Deduce an expression to calculate the required work done in twisting a cylinder by an angle θ. Prove that a hollow cylinder is stronger than a solid cylinder of same length and same mass.
(Agra 2004)

14. Two cylindrical steel shafts have the same mass per unit length. One is solid, while the other which is hollow has an external radius twice the internal radius. Show that the torsional rigidities are in the ratio 5 : 3 and the maximum strains produced by equal twisting couples bear the same ratio.
(Kanpur 1980)

15. What is a beam ? Explain the meaning of the following terms :
(i) Natural layer, (ii) Plane of bending. (iii) Bending moment. **(Gwalior 2004)**

16. What do you understand by terms neutral surface, plane of bending, neutral axis and bending moment ? Derive an expression for the moment of couple required to bend uniform metallic bar into an arc of a circle of small curvature. **(Kanpur 1979)**

17. What is bending moment ? Find an expression for the depression at a point distant x from the fixed end of a cantilever. **(Jabalpur 2004; Rewa 03; Indore 03; Agra 1994; Allahabad 80; Ajmer 89)**

18. What is cantilever ? Deduce an expression for the depression when the cantilever is loaded at one end, if the weight of the cantilever is negligible. **(Osmania 2005)**

19. What is bending moment ? Deduce an expression for it if the beam is of : (i) rectangular, (ii) circular, cross-section. **(Indore 2003; Bhopal 03; Bilaspur 03; Gwalior 04)**

20. What do you mean by bending moment ? Derive an expression for the depression of a uniform beam supported at its ends and loaded in the middle. **(Agra 1990l; Allahabad 84)**

21. Show that the depression at the free end of a light cantilever, when the end is loaded is proportional to the cube of its length. **(Gorakhpur 1979)**

22. Show that loads, required to produce equal depressions for two beams made of the same material and having the same length and weight with the only difference that they are with circular and rectangular cross-sections respectively, are in the rario 3 : π. **(Agra 1995; Allah. 80; Kanpur 84)**

Short Answer Questions

1. Explain steel is more elastic than rubber. **(Bundelkhand 2004)**
2. What are angle of twist and angle of shear ? How do you relate them for twisting of a cylinder ?
3. Differentiate between angle of twist and angle of shear. **(Agra 1991, 87)**
4. Why a hollow shaft is stronger than a solid one of the same mass ?
(Kanpur 2005; Sagar 03; Rajasthan 1983; Gorakhpur 84; Purvanchal 93)
5. Why girders for supporting roofs are formed in the shpe of 'I.
(Bhundelkhand 2004; Agra 1993; Lucknow 88; Gorakhpur 84; Purvanchal 93)
6. A cantilever of uniform section is more likely to break near its fixed end.
7. A beam of square cross-section is stiffer than one of circular cross-section of the same material, mass and length. **(Gorakhpur 1984)**
8. What is cantilever ? **(Sagar 2003)**
9. Show that torsional rigidity is greater for hollow cylinder than for a solid one of the same material, mass and length. **(Agra 2006)**
10. What are the theoretical limits of Poisson's ratio for an isotropic and homogeneous body ?
[**Ans.** $-1 < \sigma < 0.5$.] **(Agra 2005)**
11. Obtain a relation among the elastic coefficients Y, η and σ. **(Agra 2004)**
[**Ans.** $Y = 2\eta\,(1 + \sigma)$.]
12. Explain the term bending moment, flexural rigidity and torsional rigidity. **(Kanpur 2005)**
13. What is Poisson's ratio ? Give its limiting values. **(Kanpur 2004)**
14. Obtain in terms of bending moment M, an expression for longitudinal stress in a beam at a distance z from the neutral axis. **(Agra 2006)**
15. Two wires A and B of same material and same length are suspended vertically. The diameter of wire A is double that of wire B. (i) Compare the increase in lengths for a given load. (ii) Compare the breaking load.
[**Ans.** (i) 1 : 4; (ii) 4 : 1.]
16. Calculate the Poisson's ratio for silver. Given Young's modulus = 7.25×10^{10} N/m^2, Bulk modulus = 11×10^{10} N/m^2.
[**Ans.** 0.39.]
17. Compare loads required to produce equal depressions for two beams of same material, length and weight when one has a circular cross-section and other has square cross-section. **(Kanpur 2004)**

Objective Type Questions

1. The steel wire of length l is elongated by δl under its own weight. The fractional change in its volume is : **(Agra 2005)**

(a) $(1 + \sigma)\dfrac{\delta l}{l}$ (b) $(1 - \sigma)\dfrac{\delta l}{l}$ (c) $(1 + 2\sigma)\dfrac{\delta l}{l}$ (d) $(1 - 2\sigma)\dfrac{\delta l}{l}$

2. U is the potential energy of a wire stretched by a force F. If same wire is stretched by a force double in magnitude, then elastic potential energy in the wire is : **(Gwalior 1990)**
(a) $U/2$ (b) $2U$ (c) $4U$ (d) U^2
3. Strain energy per unit volume of a stretched wire is : **(Bhopal 1989)**

(a) $\frac{1}{2}$ Stress × Strain (b) $\frac{1}{2}$ Force × Strain (c) Stress × Strain (d) Force × Strain

(Bhopal 1989)

4. If a force F is applied normally on each face of a unit cube, the change in its length is :
(a) $F(1 + 2\sigma)/Y$ (b) $3F(1 - 2\sigma)/Y$ (c) $F(1 - \sigma)/Y$ (d) $F(1 + \sigma)/Y$
5. In the formula $Y = \dfrac{9\eta K}{AK + \eta}$, the value of A is : **(Bundelkhan 2004)**
(a) 3 (b) 2 (c) 6 (d) 1

6. The relation among Young's modulus 'Y', bulk modulus 'K' and the modulus of rigidity 'η' is: **(Rewa 1997)**

(a) $Y = \left(\frac{1}{3\eta} + \frac{1}{K}\right)$ (b) $Y\left(\frac{1}{\eta} + \frac{1}{3K}\right) = 3$ (c) $Y\left(\frac{3}{\eta} + \frac{1}{K}\right) = 3$ (d) $Y\left(\frac{1}{\eta} + \frac{3}{K}\right) = 3$

7. The correct relation is :

(a) $Y > \eta$ (b) $\sigma > -1$ (c) $\sigma = (Y/2\eta) - 1$ (d) $\sigma = Y/3K$

8. The correct relation is : **(Gwalior 1991)**

(a) $Y > \eta$ (b) $\sigma < -1$ (c) $\sigma = (Y/2\eta) - 1$ (d) $\sigma = 3K/Y$

9. The twisting couple per unit twist is **(Rewa 1991)**

(a) $\frac{\pi\eta r^4}{4l}$ (b) $\frac{\pi^2\eta r^4}{4l}$ (c) $\frac{\pi^2\eta r^4}{2l}$ (d) $\frac{\pi\eta r^4}{2l}$

10. For given load and same cross-section, the ratio of depressions for a rod of square and circular cross-sections is :

(a) $9 : \pi$ (b) $4 : \pi$ (c) $3 : \pi$ (d) $16 : \pi$

(Bhopal 1990)

11. A beam of length l, width b and thickness t is supported at its ends and loaded in the middle, then the depression at the middle point is :

(a) $\frac{Wl^3}{4Ybt^3}$ (b) $\frac{Wb^3}{4Ybt^3}$ (c) $\frac{Wl^3}{4Yb^2t^3}$ (d) $\frac{Wl^2}{4Ybt^3}$

12. The depression produced at the free end of a loaded cantilever of length l, beadth b and depth d is : **(Agra 2005)**

(a) $\frac{Wl^3}{Ybd^3}$ (b) $\frac{2Wl^3}{Ybd^3}$ (c) $\frac{Wl^3}{3Ybd^3}$ (d) $\frac{4Wl^3}{Ybd^3}$

13. Cross-sectional area of a wire is 0.5 cm^2. Young's modulus is 2×10^{11} N/m^2. What will be the magnitude of force rquired to double the length of the wire ?

(a) 10^7 N (b) 10^8 N (c) 10^{10} N (d) zero

14. The depression at the free end of a cantilever of uniform cross-sectional area and length l is δ. The depression at a distance of $l/4$ from the fixed end of the cantilever will be :

(a) 0.086 δ (b) δ (c) 0.88 δ (d) 1.5 δ

15. If the length of a wire is doubled by applying a stress of magnitude 10×10^8 N/m^2, then the Young's modulus of the material of wire is : **(Bundelkhand 2001)**

(a) 20×10^8 N/m^2 (b) 10×10^8 N/m^2 (c) 5×10^8 N/m^2 (d) 15×10^8 N/m^2

16. A steel wire of 1 mm radius is bent in the form of a circular arc of radius 50 cm. Given Young's molulus for steel is 2×10^{12} dynes/cm^2, (i) the bending moment will be **(Agra 2006)**

(a) $2\pi \times 10^6$ dyne-cm (b) $\pi \times 10^6$ dyne-cm (c) $\frac{\pi}{2} \times 10^6$ dyne-cm (d) $4\pi \times 10^6$ dyne-cm

(ii) the maximum stress will be **(Agra 2006)**

(a) 2×10^9 dyne/cm^2 (b) 4×10^9 dyne/cm^2 (c) 8×10^9 dyne/cm^2 (d) 1×10^9 dyne/cm^2

ANSWERS

1. (c) **2.** (c) **3.** (a) **4.** (b) **5.** (b) **6.** (b) **7.** (b),(c) **8.** (c) **9.** (c) **10.** (c) **11.** (a) **12.** (d)
13. (a) **14.** (a) **15.** (b). **16.** (i) (b), (ii) (b)

14

Kinematics of Moving Fluids –Viscosity

14·1. Ideal and Viscous Fluids

Liquids and gases both are called fluids. In fact, fluid is that state of matter which can flow from one place to another. *An ideal fluid is incompressible and non-viscous.* An **incompressible fluid** means that the density of a fluid remains constant with the change of pressure. If the tangential force between the two layers (which are in contact with each other) of a flowing liquid is zero *i.e.*, there is no frictional force between them, then such a liquid is called **non-viscous.** In fact, there is no ideal fluid. Mostly liquids can be considered incompressible, because on increasing the pressure, the change in their volume is extremely small. For example, in case of water when a pressure of 1 atmosphere is increased, the change in volume in comparison to its original volume is only 4.8×10^{-5} times. Similarly, there is always more or less viscous (or frictional) force between the layers of a flowing fluid. Viscous force is less in flowing gases and it is more in case of moving liquids. Here, we shall study more about liquids. Hence, **viscous fluids**, are those fluids in the layers of which a force of friction acts when flowing. The force of friction between the layers of a moving fluid is called **viscous force.**

14·2. Streamline and Turbulent Flows – Critical Velocity

When a liquid flows such that each particle of the liquid passing a point, moves along the same path and has the same velocity as its preceding particles, the flow of the liquid is called **orderly** *or* **streamline flow**. In this type of flow, the velocity at every point in the liquid remains constant in magnitude as well as in direction. In such a case, a line, along which a particle of the liquid moves and the direction of which at any point gives the direction of flow of the liquid at that point, is called a **streamline**. More correctly, *a* **streamline** *is a curve such that, at any instant, the tangent to it at a point gives the direction of the flow of the liquid at that point.* A streamline may be straight or curved, accordingly as the lateral pressure on it is the same or different. In **fig.14.1**, a streamline AB has been shown in a flowing liquid. The velocity of the liquid at P, Q and R point is v_1, v_2 and v_3 respectively. The particles which pass through the point P also pass through the points Q and R. Although in stramline flow of a liquid, the velocities v_1, v_2 and v_3 at the respective points P, Q and R may be different, yet at any point, say P, the velocity of a particle of the liquid will remain the same.

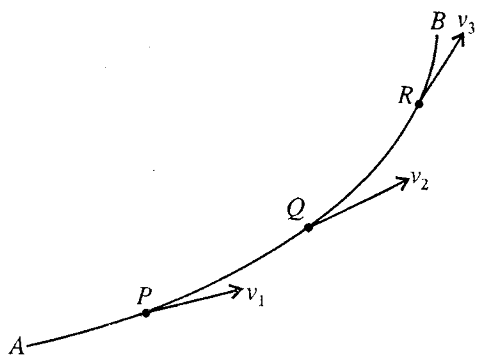
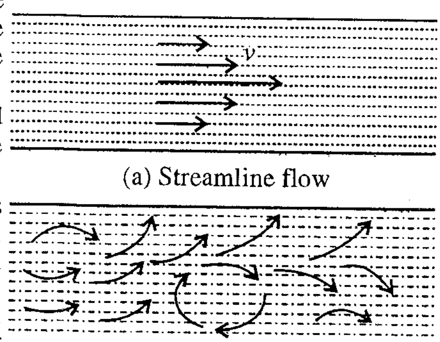


Fig. 14.1 : A Streamline in a Flowing Liquid

In a streamline flow, the energy needed to drive the liquid is used up only in overcoming the viscous drag between the layers.

The flow of the liquid remains streamline only as long as the velocity does not exceed a certain value, called its **critical velocity**. Beyond its critical velocity, the flow of the liquid ceases to be streamline or orderly, but becomes zigzag or sinuous in character and the paths and velocities of the liquid particles change continuously and haphazardly. This unsteady motion of the liquid is called **turbulent flow.** Here, most of the energy needed to drive the liquid is dissipated in the formation of eddies and whirlpools in the liquid.



(a) Streamline flow

(b) Turbulent flow

Fig. 14.2 : (a) Streamline flow and (b) Turbulent flow.

The difference between streamline and turbulent flows can be demonstrated by introducing a small jet of colouring matter at the centre into a tube through which a liquid may be allowed to flow with a gradually increasing velocity. As long as the velocity remains below its critical value, the colouring matter is confined to a thin streak along the axis of the tube. As the velocity of the flow is increased to its critical value, the colouring matter takes a zigzag path. As soon as,. this velocity exceed the critical value, the colouring matter spreads out in all directions and fills the whole tube. Now, the motion of the liquid is no longer orderly or streamlined, but it has become **turbulent.**

Critical Velocity : *The maximum velocity of a liquid upto which its flow is streamline is called the* **critical velocity.** Above this velocity, the flow ceases to be streamline and becomes turbulent.

The expression for the critical velocity of a liquid can be derived by the method of dimensions. Critical velocity, v_c, may possibly depend upon (a) η, the coefficient of viscosity, (b) ρ, the density of the liquid, and (c) r, the radius of the tube.

Therefore, $$v_c = K\eta^a\rho^b r^c, \text{ where } K \text{ is a constant.} \qquad ...(1)$$

Substituting the dimensions of various physical quantities, we get

$$[LT^{-1}] = [ML^{-1}\, T^{-1}]^a\, [ML^{-3}]^b\, [L]^c \quad \text{or} \quad [LT^{-1}] = [M^{a+b}\, L^{-a-3b+c}\, T^{-a}]$$

According to the principle of homogeneity of dimensions, we have

$$a + b = 0;\ -a - 3b + c = 1 \text{ and } -a = -1;\ \therefore\ a = 1,\ b = -1 \text{ and } c = -1$$

Hence $$v_c = \frac{K\eta}{\rho r} \qquad ...(2)$$

This constant K is called **Reynold's number**. Its values for narrow tubes is 1000 nearly, while for tubes of wide bores its value may be very much different. Thus, it is a pure numerical value and is independent of the system of units used. The flow will be steady and steamline until this number is not exceeded. After exceeding this number the flow becomes turbulent.

Thus, the **critical velocity of a liquid** is

(i) *directly proportional to the coefficient of viscosity of the liquid,*

(ii) *inversely proportional to its density, and*

(iii) *inversely proportional to the radius of the tube.*

Therefore, narrow tubes, low density and high viscosity of a liquid tend to make the liquid flow streamline whereas wider tubes, high density and low viscosity of the liquid tend to promote turbulence.

14·3. Equation of Continuity

Consider a small area A_1 at P perpendicular to the direction of steady flow of a liquid and draw the sreamlines through the boundary of A_1 so that a tube PQ is obtained. This is known as a tube of flow. Since the sides of such a tube are everywhere in the direction of flow of the liquid, no liquid crosses the sides. Actually, the liquid moves inside a tube of flow just as if the tube has solid walls. We assume, that the liquid, under consideration, is perfectly mobile and incompressible. Thus, the same amount of liquid flows across every section of the tube in a given time, *i.e.*, the volume of the liquid passing through area A_1 is equal to that passing through the area A_2. Let the velocity of flow at P be v_1 and at Q be v_2 [**fig.14.3**]. Hence,

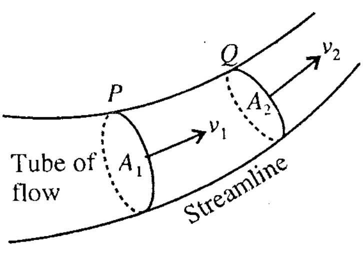


Fig. 14.3 : Tube of flow.

volume of the liquid entering per sec or rate of flow at P $= A_1v_1$*

and, volume of the liquid leaving per sec at Q $= A_2v_2$.

If ρ_1 and ρ_2 are the densities of the liquid at two sections, then

mass of the liquid entering per sec at P $= A_1v_1\rho_1$

and, mass of the liquid leaving per sec at Q $= A_2v_2\rho_2$.

If there are no sources or sinks within the volume of the liquid between the sections P and Q, then

* If the velocity of the liquid across the area A be v, then, obviously, the liquid crossing A will move forward a distance v inside the tube in 1 second. Thus, the volume of the liquid crossing the area A in one second will be equal to the volume of the cylinder of cross-sectional area A and length v, *i.e.*, the rate of flow of the liquid through A $= Av$.

the masses of the liquid crossing the areas A_1 and A_2 per second must be the sane, *i.e.*,

$$A_1 v_1 \rho_1 = A_2 v_2 \rho_2.$$

The liquid is incompressible, therefore $\rho_1 = \rho_2$. Thus, we have

$$A_1 v_1 = A_2 v_2 \qquad ...(1)$$

Thus, the volume of the liquid entering the section P is equal to that leaving the section Q. The above equation is known as **equation of continuity** for the steady flow of a perfectly mobile and incompressible liquid. This means that *where the cross-section of the tube is small, the velocity is greater and vice-versa. This explains why water at a place, where it is deep, appears to be still.*

14·4. Energy Possessed by a Liquid

A liquid, in motion, possesses three forms of energy, *viz.* (1) pressure energy, (2) kinetic energy, and (3) potential energy.

(1) **Pressure energy :** Let an incompressible liquid of density ρ be contained in a tank, provided with side tube T at a depth h below the free surface AB of the liquid in the tank [**fig. 14.4**]. At the level CD (the axis of the tube T), there is a certain pressure $p = \rho gh$. Obviously, if more liquid is to be introduced into the tank at this pressure, it must be forced in by the piston, fitted in the tube T. If α be the cross-sectional area of the piston, then the force on the pistion is $p \cdot \alpha$. Now, if the piston is pushed inward through a small distance x, a volume $\alpha \cdot x$ or a mass $\alpha x \rho$ of the liquid is forced into the tank and the amount of work done in doing so, is

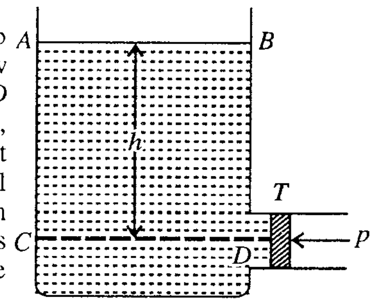


Fig. 14.4.

$$W = p\alpha x$$

This work becomes the pressure energy of the mass $\alpha x \rho$ of the liquid in the tank, beacuse it is capable of performing the same amount of work done in pushing the piston back, when escaping from the tank.

Thus, the pressure energy per unit mass of the liquid $= \dfrac{p\alpha x}{\alpha x \rho} = \dfrac{p}{\rho}$ ...(1)

Since $p\alpha x$ is the pressure energy *of* $\alpha \cdot x$ volume,

the pressure energy per unit volume of the liquid $= \dfrac{p\alpha x}{\alpha x} = p.$ ...(2)

(2) **Kinetic energy :** The liquid has inertia and, therefore, it possesses kinetic energy, when in motion. Thus, if a mass m of the liquid is flowing with a velocity v, its kinetic energy will be $\frac{1}{2} mv^2$. Hence,

Kinetic energy per unit mass $= \frac{1}{2} v^2$ ...(3)

and kinetic energy per unit volume $= \frac{1}{2} \rho v^2$ ...(4)

(3) **Potential enegy :** The liquid possesses the potential energy by virtue of its position. Thus, if a mass m be situated at a mean height h above some reference horizontal level, its potential energy will be mgh. Hence,

Potential energy per unit mass $= gh,$ ...(5)

and potential energy per unit volume $= \rho gh.$ ...(6)

14·5. Euler's Equation of Motion for Flow of Non-Viscous Liquid

Consider an infinitesimally small length PQ of a tube of flow of length dl in ideal liquid in stream-line flow [**fig. 14.5**]. Because of the infinitesimally small length, the area of cross-section dA of this portion of the tube of flow may be assumed to be uniform. If ρ be the density of the fluid, the mass of the liquid dm in portion PQ is given by

$$dm = dA\, dl\, \rho$$

and its weight $= dmg = dA\, dl\, \rho g$

where g is the acceleration due to gravity. This weight (dmg) of the liquid in PQ acts vertically downward at its centre of gravity G and makes an angle θ with the direction of the flow [**fig. 14.5**]. It can be resolved into two components :

(i) $dmg \cos\theta$ along the direction of flow and (ii) $dmg \sin\theta$ perpendicular to the direction of flow, its effect is balanced by the lateral thrust due to the adjoining tube of flow.

If p be the pressure at the face P of the small tube of flow, then the pressure on its other face will be :

$$p + dp = p + \frac{\partial p}{\partial l} dl$$

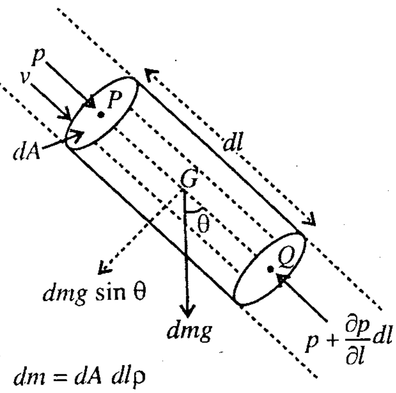


Fig. 14.5.

Hence, the force F_1 acting on face P is

$F_1 = pdA$ in the forward direction,

and the force F_2 on face Q is

$$F_2 = \left(p + \frac{\partial p}{\partial l} dl\right) dA \text{ in the backward direction.}$$

Thus the net force F acting on mass dm of the liquid in PQ is obtained to be

$$F = F_1 - F_2 = pdA - \left(p + \frac{\partial p}{\partial l} dl\right) dA + dA\, dl\, \rho\, g \cos\theta$$

or
$$F = -\frac{\partial p}{\partial l} dl\, dA + dA\, dl\, \rho\, g \cos\theta \qquad ...(1)$$

If v be the velocity of the liquid at the face P of the tube, then its acceleration will be dv/dt. According to Newton's second law of motion, the force acting on the tube PQ is

$$F = \text{mass} \times \text{acceleration} = dm\frac{dv}{dt} \text{ or } F = (dAdl\,\rho)\frac{dv}{dt} \qquad ...(2)$$

This velocity v in general is a function of distance l and time t *i.e.*,

$$v = v(l, t).$$

Hence,
$$dv = \frac{\partial v}{\partial l} dl + \frac{\partial v}{\partial t} dt$$

So that
$$\frac{dv}{dt} = \frac{\partial v}{\partial l}\frac{dl}{dt} + \frac{\partial v}{\partial t} = \frac{\partial v}{\partial l} v + \frac{\partial v}{\partial t}, \text{ where } \frac{dl}{dt} = v.$$

Substituting this in eq. (2), we obtain

$$F = dAdl\,\rho\left(v\frac{\partial v}{\partial l} + \frac{\partial v}{\partial t}\right). \qquad ...(3)$$

If h be the vertical height of the tube PQ from a chosen plane, then clearly [**fig. 14.6**].

$$\cos\theta = -\frac{\partial h}{\partial l} \qquad ...(4)$$

Hence from eqs. (1), (3) and (4), we have

$$dAdl\rho\left(v\frac{\partial v}{\partial l} + \frac{\partial v}{\partial t}\right) = -\frac{\partial p}{\partial l} dldA - dAdl\rho\, g\frac{\partial h}{\partial l}$$

or
$$\rho\left(v\frac{\partial v}{\partial l} + \frac{\partial v}{\partial t}\right) = -\frac{\partial p}{\partial l} - \rho\, g\frac{\partial h}{\partial l}$$

or
$$\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial l} + \frac{1}{\rho}\frac{\partial p}{\partial l} + g\frac{\partial h}{\partial l} = 0. \qquad ...(5)$$

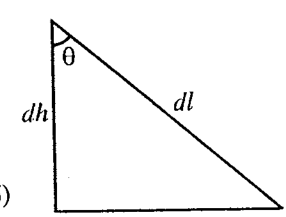


Fig. 14.6.

This is the **Euler's equation of motion** for flow of non-viscous liquid. It is an important equation, applicable to both steady and unsteady flow.

Note : In the vector form Euler's equation will have the following form :

$$\frac{\partial \vec{v}}{\partial l} + \vec{v} \cdot \vec{\nabla} v + \frac{\vec{\nabla} p}{\rho} - \frac{\vec{f}}{\rho} = 0, \text{ where } \vec{f} \text{ is the force per unit volume.} \qquad ...(6)$$

In the case of gravitational force, $\vec{f} = -\rho g \vec{\nabla} h$.

Integration of Euler's Equation and Deduction of Bernoulli's Equation—In streamline flow, the value of the velocity v at a point remains constant in time *i.e.*, $\frac{dv}{dt} = 0$. Hence from eq. (5), we obtain

$$v\frac{\partial v}{\partial l} + \frac{1}{\rho}\frac{\partial p}{\partial l} + g\frac{\partial h}{\partial l} = 0 \text{ or } dp + \rho v dv + \rho g dh = 0 \qquad ...(7)$$

Integrating it along the sreamline, we get

$$\int dp + \rho \int v dv + \rho g \int dh = 0 \text{ or } p + \frac{1}{2}\rho v^2 + \rho g h = \text{a constant} \qquad ...(8)$$

This is knwon as **Bernoulli's equation.** The sum of the terms on the left hand side of eq. (8) tells us the total energy per unit volume of the flowing liquid (preseure energy + kinetic energy + gravitational energy), which remains constant during flow of the liquid. This is the **Beruoulli's theorem.**

If we divide eq. (8) by ρ, then

$$\frac{p}{\rho} + \frac{v^2}{2} + gh = \text{a constant} \qquad ...(9)$$

which is **another form** of **Bernoulli's equation.**

On dividing (9) by g, we get

$$\frac{p}{\rho g} + \frac{v^2}{2g} + \text{h} = \text{a constant} \qquad ...(10)$$

which is yet **another form** of **Bernoulli's equation.** This means that the sum of pressure head, velocity head and gravitational head remains constant in streamline flow of an ideal liquid.

If the liquid is flowing horizontally, h is the same and hence

$$\frac{p}{\rho} + \frac{v^2}{2} = \text{a constant} \qquad ...(11)$$

If at two points in a tube of flow p_1, p_2 are the pressures and v_1, v_2 are the respective velocities, then

$$\frac{p_1}{\rho} + \frac{v_1^2}{2} = \frac{p_2}{\rho} + \frac{v_2^2}{2} \text{ or } \frac{p_1}{\rho} - \frac{p_2}{\rho} = \frac{v_2^2}{2} - \frac{v_1^2}{2} \qquad ...(12)$$

If $p_1 > p_2$, the left hand side of this equation tells us the decrease in pressure energy per unit volume during the displacement of the fluid from one point to another, while the right hand side gives us the increase in the kinetic energy per unit volume.

14·6. Bernoulli's Theorem

This theorem states that the total energy of a small amount of an incompressible liquid, flowing from one point to another without friction, remains constant throughout the displacement i.e.,

Pressure enegry + Kinetic energy + Potential energy = a constant.

When a unit mass flows through a tube **[see Art. 14·4]** :

$$\frac{p}{\rho} + \frac{1}{2}v^2 + hg = C, \text{ a constant} \qquad ...(1)$$

This relation is called **Bernoulli's equation.**

Proof of the therorem : **Fig. 14.7** represents a tube of flow between two transverse sections P and Q. Let the pressure, velocity and cross-section area at P be p_1, v_1 and A_1 and the corresponding quantities at Q be p_2, v_2 and A_2 respectively.

We may speak the steady flow as the entrance of a small amount of mass m of the liquid through the section P and the simultaneous exit of the same amount of mass m of the liquid through the section Q.

The mass m of the liquid, entering the tube through the area A_1, moves forward in a small time δt, a distance $v_1\ \delta t$ parallel to the axis of the tube. But this flow of the liquid is along the direction of the force $A_1 p_1$, acting over the area A_1. Thus, the work done on the system, is given by

$$W_1 = A_1 p_1 \cdot v_1 \delta t$$

At the section Q, the mass m of the liquid leaves the tube in the time interval δt but the flow is in a direction opposite to that of the force $A_2 p_2$, acting over the area A_2. Thus, in this case, the work done by the system, is given by

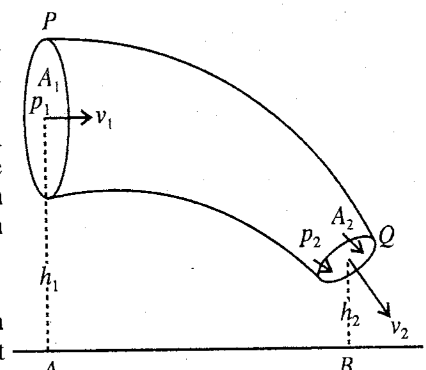


Fig. 14.7.

$$W_2 = A_2 p_2 \cdot v_2 \delta t$$

Hence, the net amount of work done by the pressure forces during the time δt on the system, is

$$W = W_1 - W_2 = (A_1 p_1 v_1 - A_2 p_2 v_2) \cdot \delta t. \qquad ...(2)$$

Let the mean heights of the sections P and Q be h_1 and h_2 respectively above a horizontal line AB. Evidently, when the mass m flows from P to Q, its gravitational potential energy changes by

$$U = mg\ (h_2 - h_1).$$

If $A_1 > A_2$, then from the equation of continuity v_2 must be greater than v_1. Thus, the gain in kinetic energy by the mass m during the flow is $K = \frac{1}{2} m\ (v_2^2 - v_1^2)$.

Hence, the total gain in the energy of the system in time δt is

$$U + K = mg\ (h_2 - h_1) + \frac{1}{2} m\ (v_2^2 - v_1^2) \qquad ...(3)$$

According to the principle of conservation of energy, the work done W on the system during the time δt must be equal to the gain in its total energy, *i.e.*, from eqs. (2) and (3), we have

$$(A_1 p_1 v_1 - A_2 p_2 v_2)\ \delta t = mg\ (h_2 - h_1) + \frac{1}{2} m\ (v_2^2 - v_1^2) \qquad ...(4)$$

But $A_1 v_1 \delta t$ represents the volume of the liquid entering the area A_1 in time δt, hence

$$A_1 v_1\ \delta t = m/\rho.$$

The same volume leaves area A_2, hence

$$A_2 v_2\ \delta t = m/\rho.$$

Thus eq. (4) can be written as

$$\frac{m}{\rho}(p_1 - p_2) = mg\ (h_2 - h_1) + \frac{1}{2} m\ (v_2^2 - v_1^2)$$

or
$$\frac{p_1}{\rho} + \frac{1}{2} v_1^2 + gh_1 = \frac{p_2}{\rho} + \frac{1}{2} v_2^2 + gh_2$$

or
$$\frac{p}{\rho} + \frac{1}{2} v^2 + gh = C, \text{ a constant.} \qquad ...(5)$$

Hence, the **Bernoulli's equation** is established.

From equation (5), the sum of the three types of energy of a perfectly mobile and incompressible liquid in motion is constant and thus, when one type of energy decreases, the other types of energy will increase and vice-versa. Hence, *the three forms of energy of a liquid are mutually convertible into each other.* Dividing eq. (5) by g, we get

$$\frac{p}{\rho g} + \frac{1}{2} \cdot \frac{v^2}{g} + h = C', \text{ another constant} \qquad ...(6)$$

The term $p/\rho g$ is called the pressure head, $\frac{1}{2}v^2/g$ the velocity head and h, the gravitational head, *i.e.*, this is the height through which the liquid must fall freely to attain the velocity v. Thus,

pressure head + velocity head + gravitational head = a constant.

Hence, in other words, **Bernoulli's theorem** states that *at all points, in the streamline flow of an incompressible liquid, the sum of the pressure head, velocity head and gravitational head is a constant.*

If the liquid is flowing horizontally, gravitational head h is a constant, so that from eq. (6), we have

$$\frac{p}{\rho g}+\frac{1}{2}\cdot\frac{v^2}{g} = \text{a constant} \quad \text{or} \quad \frac{p}{\rho}+\frac{1}{2}v^2 = \text{a constant.} \qquad ...(7)$$

Now, if in a liquid flowing horizontally, the pressure and velocity be p_1 and v_1 at a point and at another, p_2 and v_2 respectively, we have

$$\frac{p_1}{\rho}+\frac{1}{2}v_1^2 = \frac{p_2}{\rho}+\frac{1}{2}v_2^2 \qquad ...(8)$$

This relation shows that *greater velocity corresponds to diminished pressue p and vice-versa.* For example, when a liquid flows through a horizontal tube having a **constriction,** the velocity of the liquid in the constricted part is increased and, therefore, the pressure is decreased there. This principle has been applied in the design of flowing instruments, *e.g.*, venturimeter, pitot tube and in the construction of various exhaust jet pumps.

14·7. Applications of Bernoulli's Theorem

(1) Velocity of Effiux of a Liquid Torricelli's Theorem : Fig. 14·8 represents a tank, containing a liquid of density ρ. The free surface of the liquid is at a height h above the level of the orifice O in the tank. The liquid is allowed to escape through O. Both at the free surface A of the liquid and at the orifice O the pressure is atmospheric, say P. If the tank be sufficiently wide, the velocity of the liquid surface may be taken zero. If v be the **velocity of effiux** at the orifice O, then according to Bernoulli's theorem

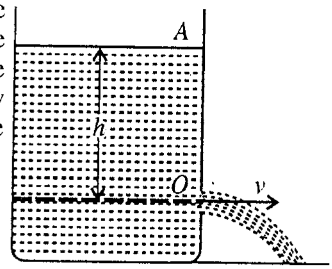


Fig. 14.8.

Total energy at A = Total energy at O.

$$\therefore \quad \frac{P}{\rho}+0+hg = \frac{P}{\rho}+\frac{1}{2}v^2+0$$

or $$v^2 = 2gh, \quad \text{whence} \quad v=\sqrt{2gh} \qquad ...(1)$$

This result was first deduced by Torricelli and hence is known as **Torriceli's theorem** or **Law of efflux** and may be stated as follows :

The velocity of efflux of a liquid through an orifice is equal to that which a body acquires in falling freely from the surface of the liquid to the orfice, since a body falling freely through a height h under the action of gravity will have the velocity v according to the relation.

$$v^2 = 2gh \quad \text{or} \quad v = \sqrt{2gh}. \qquad ...(2)$$

This is the same relation as obtained above. No liquid is perfectly free from internal friction or viscosity, hence the ideal velocity is never reached.

Maximum Horizontal Distance of Fall of Liquid

The path of liquid particles on emerging out of the orifice is **parabolic.** In a projectile motion, the horizontal and vertical motions are independent of each other. Hence, in the vertical direction, initial velocity of a liquid particle is zero and it falls with acceleration g due to gravity. If the liquid particle falls by the height $H-h$ in time t **[fig. 14.9]**, then from second equation of motion,

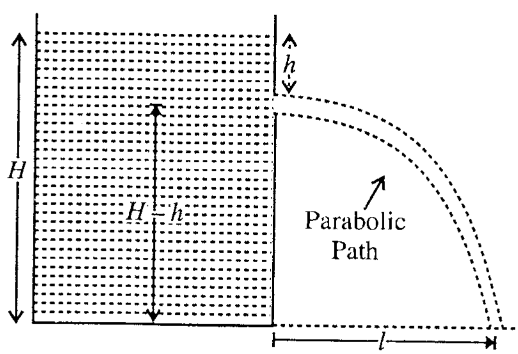


Fig. 14.9.

$$H-h=\frac{1}{2}gt^2, \text{ whence } t=\sqrt{2(H-h)/g} \qquad ...(3)$$

There is no acceleration on the liquid moving in the horizontal direction and its horizontal velocity remains constant. If in this time t, the distance travelled in the horizontal direction is l, then

$$l = vt = \sqrt{2gh}\,\sqrt{2(H-h)/g} = 2\sqrt{h/(H-h)} \qquad ...(4)$$

For the maximum value of the horizontal distance l of the falling liquid on the floor,

$$\frac{dl}{dh} = 0 \text{ or } \frac{d}{dh}\left(\sqrt{h(H-h)}\right) = \frac{H-2h}{2\sqrt{hH-h^2}} = 0$$

or

$$H - 2h = 0 \text{ or } h = \frac{H}{2} \qquad ...(5)$$

In this condition,

$$l_{\max} = 2\sqrt{\frac{H}{2}\left(H-\frac{H}{2}\right)} = H \qquad ...(6)$$

Thus, *if there is an orifice just in the middle of the wall of the container, the stream of the liquid falls at a maximum horizontal distance from the bottom of the container, being equal to the height of the liquid in the container.*

(2) **Vena Contracta :** When a liquid contained in a vessel, is allowed to flow through an orifice in the side wall, the volume of the liquid flowing per second through the orifice should be $\alpha\, v$, where α is the cross-sectional area of the orifice and v the velocity of efflux, given by the Torricelli's theorem. In fact the volume of the liquid, flowing through the orifice actually observed is always found to the smaller than, and about 0.6 of the calculated value. This deviation can not be explained by the loss of energy due to the friction at the orifice alone. Even in the case of an ideal (*i.e*, perfectly mobile and incompressible liquid), all molecules of the liquid, crossing the orifice do not move perpendicularly towards it, but they come from all directions so that the streamlines are curved near the edges of the orifice, as shown in **fig.14.10**. When the liquid, coming from the sides of the vessel, enters the orifice, it still has some lateral velocity due to inertia, and even after leaving the orifice it continues to move towards the centre of the section of the jet. This goes on until the outward pressure from the interior of the jet balances this inward tendency. Thus, *the liquid jet, contracts to a neck,* called **vena contracta,** just outside the orifice at C. At the vena contracta, the jet becomes uniform and the velocity becomes the same throughout. It is the velocity, which is given by Torricelli's theorem.

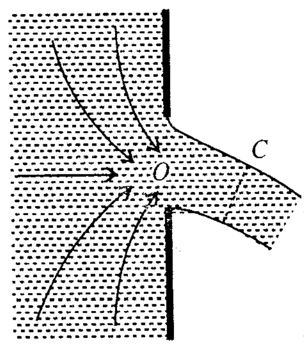


Fig. 14.10 : Vena contracta.

It is evident that the area of the jet at the vena contracta is smaller than the area of the orifice and is found to be about 0.62 times the later. Therefore, the volume of the liquid passing out orifice in unit time is equal to 0.62 $\alpha\, v$. The ratio (c) between the area of vena contracta and the orifice is called the **coefficient of contraction**. For a circular orifice the value of c is 0.62 approximately.

(3) **Venturimeter :** A venturimeter is an instrument used for measuring the rate of flow of liquids in pipes and it is a practical application of Bernoulli's theorem.

In its simplest form, a venturimeter consists two wide tubes A and C, connected by a narrow coaxial constricted tube B, called the throat [**fig. 14.11**]. To know the pressure difference between the tubes A and B, two tubes D and E are fixed in them respectively. To measure the rate of flow of liquid, the venturimeter is inserted horizontally in the pipe line and has properly designed tapers to the inlet and outlet ends to avoid turbulence and thus streamline flow is maintained.

The principle of this instrument is that when a liquid flows through a tube of varying cross-section, the velocity and pressure vary along the tube, the velocity being the greatest where the pressure is the least and vice-versa.

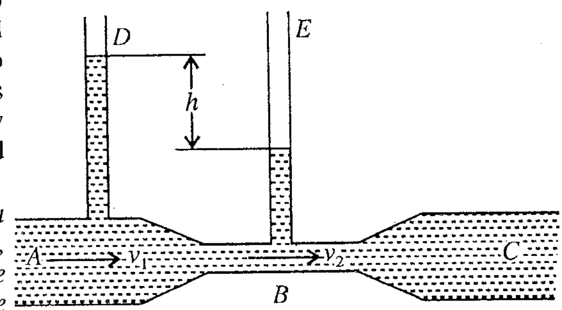


Fig. 14.11 : Venturimeter.

When the liquid flows from A to B, owing to the decrease in cross-section area, its velocity becomes greater. If v_1 be its velocity in tube A of cross-sectional area A_1 and v_2 in the tube B of crosss-sectional area A_2 and V be the volume of the liquid that passes in unit time, then according to the principle of continuity, we have

$$V = A_1 v_1 = A_2 v_2 \qquad ...(7)$$

As the mean height of the liquid in A and B is the same and its flow is horizontal, there is no change of gravitational potential energy. If p_1 be the pressure in A and p_2 in B and ρ be the density of the liquid, then according to Bernoulli's theorem

$$\frac{p_1}{\rho} + \frac{1}{2}v_1^2 = \frac{p_2}{\rho} + \frac{1}{2}v_2^2 \qquad \text{[Art. 14.6, eq. (8)]}$$

or

$$p_1 - p_2 = \frac{\rho}{2}(v_2^2 - v_1^2) \qquad ...(8)$$

It is evident from eq. (7) that as $A_1 > A_2$, $v_2 > v_1$ and therefore according to eq. (8) p_1 is greater than p_2. Consequently, the level of the liquid in D is higher than in E.

If h is the difference of levels of the liquid in the two tubes D and E, then

$$p_1 - p_2 = h\rho g$$

Hence, from eqs. (7) and (8), we have

$$h\rho g = \frac{\rho}{2}\left(\frac{V^2}{A_2^2} - \frac{V^2}{A_1^2}\right) \quad \text{or} \quad hg = \frac{V^2}{2}\cdot\frac{A_1^2 - A_2^2}{A_1^2 A_2^2}; \therefore \text{ Rate of flow } V = A_1 A_2\sqrt{\frac{2hg}{A_1^2 - A_2^2}}. \qquad ...(9)$$

Thus, the rate of flow V of the liquid through the pipe can be measured.

(4) Pitot Tube : This is another device, that is used for the measurement of the velocity or rate of flow of a liquid, flowing in pipes.

The arrangement, consists of two vertical tubes AB and CD with small apertures, situated in flowing water [**fig. 14.12**]. The plane of the aperture B is parallel to the direction of flow, so that the level of the water at A measures the pressure at B. The aperture D is facing the flow. Since the water is stopped in the plane of the aperture D, its velocity there becomes zero. Hence, the kinetic energy per unit volume is reduced from $\frac{1}{2}v^2$ to zero, where v is the velocity of the flow of water. Consequently, the pressure of water increases by an amount $\frac{1}{2}v^2$ and, therefore, the warer rises to a higher level in the tube CD than in AB. If h be the difference of levels in the two tubes, we have

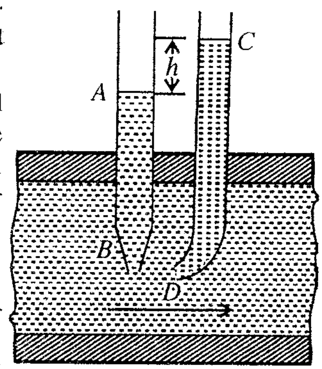


Fig. 14.12 : Pitot tube.

$$\frac{1}{2}v^2 = hg \quad \text{or} \quad v = \sqrt{2hg} \qquad ...(10)$$

Multiplying by the area of cross-section α of the pipe, we get the rate of flow V of the liquid through the pipe. Thus

$$V = \alpha v = \alpha\sqrt{2gh} \qquad ...(11)$$

(5) Filter Pump : This is a simple device, used for reducing the pressure in a vessel. Water from a tape issues out in the form of a jet from a small orifice A [**fig. 14.13**]. Due to the high velocity of the jet the pressure is lowered below that of the atmospheric and consequently the air comes from the vessel through the side tube B, if the pressure in the vessel is originally atmospheric. Thus, the withdrawal at of air through B soon lowers the presure inside the vessel.

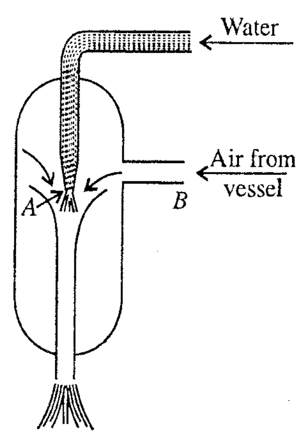


Fig. 14.13 : Filter pump.

(6) Magnus Effect : When the tennis or cricket players throw a spinning ball, it moves along a curved path in air. We call this as the **swing** of the ball.

In throwing, when the ball spins, the air around it also rotates with it [**fig. 14·14**]. If the spinning ball moves forward with a velocity v, the air ahead the ball rushes with velocity v', say, to fill up the space the evacuated space behind the ball. Hence, the resultant velocity of the air above the ball becomes $v' - v$ and that below the ball becomes $v' + v$. Therefore, the velocity of air above the ball is decreased and below to it is increased. However, according to **Bernoulli's theorem**, the pressure is more at the place where the velocity is less or vice-versa. This means that the pressure of the air above the ball is more than that below the ball. Because of this difference of pressure, a force acts on the ball and the ball traces a curved path. We call this motion of the spinning ball along a curved path as the **Magnus effect**.

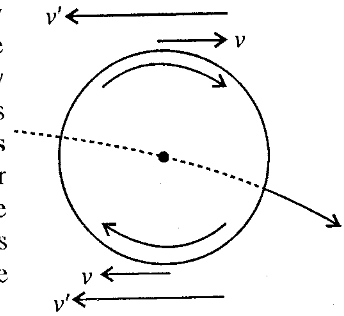


Fig. 14.14. Magnus effect

(7) **Shape of the Wings of an Aeroplane :** The shape of the wings of an aeroplane is made so that the curvature of its upper surface is greater than its lower surface. Further the shape of its front end is made round and the back end is flattened [**fig. 14.15**]. When the aeroplane is flying, we consider the flow of air above and below the wing to be streamlined. Due to the shape of wing, the air moves more rapidly over its upper surface than the lower surface. In view of Bernoullis' equation, for a flowing fluid where the velocity is decreased, the pressure is increased or vice-versa. Hence, the pressure of the air p_1, at the lower surface of the wing is more than the pressure p_2 at the upper surface of the wing. Due to this pressure difference $(p_1 - p_2)$, the wing of the aeroplane gets a resultant upward force called **lift force**. This force causes the aeroplane to rise.

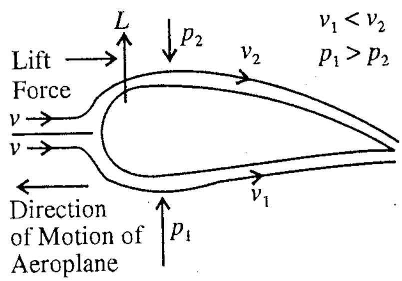


Fig. 14.15 : Shape of the wing of an aeroplane.

Ex. 1. *What is the velocity of efflux of liquid at the bottom of a vessel if the total pressure at the bottom is 2 atmosphere and density of liquid is 800 kg/m³ ?* **(Rewa 1992)**

Sol. Pressure due to liquid $= \rho g h = 2$ atmosphere – 1 atmosphere

or $\rho g h = 2\rho' g \times 0.76 - \rho' g \times 0.76 = \rho' g \times 0.76$

where ρ' is the density of mercury.

Here, $\rho = 800$ kg/m³ and $\rho' = 13{,}600$ kg/m³.

$$\therefore \quad h = \frac{13{,}600}{800} \times 0.76 = 12.92 \text{ m}.$$

Using Bernoulli's equation, (Torricelli's theorem), the velocity of efflux is

$$v = \sqrt{2hg} = \sqrt{2 \times 9.8 \times 12.92} = \mathbf{15.9 \text{ m/s}}.$$

Ex. 2. *Find the velocity of efflux of a liquid of density 2,500 kg/m³ from a tank to which the pressure of the liquid at the orifice is 9.8 × 10³ n/m² above the pressure of the atmosphere.*

Sol. If the height of liquid column above the orifice be h, then pressure at the orifice

$$= h \times 2500 \times 9.8 \text{ N/m}^2.$$

According to the given problem, pressure at the orifice is 9.8×10^3 N-m². Hence,

$$h \times 2500 \times 9.8 = 9.8 \times 10^3, \quad \therefore h = 0.4 \text{ m}$$

$$\therefore \text{ Velocity of efflux,} \quad v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 0.4} = \mathbf{2.8 \text{ m/sec}}.$$

Ex. 3. *Water flows along a horizontal pipe of which the cross-section is not constant. The pressure is 1 cm of mercury, where the velocity is 35 cm/sec. Find the pressure at a point, where the velocity is 65 cm/sec.* **(Allahabad 1982; Delhi 79; Poona 63)**

Sol. According to Bernoulli's theorem, in case liquid is flowing horizontally

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

where p_1, p_2 are the pressures at two points of the pipe and v_1, v_2 corresponding velocities.

Here, $\rho = 1000$ kg-m^{-3}; $p_1 = 0.01 \times 13600 \times 9.81$ N/m^2; $v_1 = 0.35$ m/sec; $p_2 = ?$; $v_2 = 0.65$ m/sec

$$\therefore\ 0.01 \times 13600 \times 9.81 + \frac{1}{2} \times 100 \times 0.35^2 = p_2 + \frac{1}{2} \times 1000 \times 0.65^2$$

or
$$p_2 = 136 \times 9.81 + \frac{1000}{2}(0.35^2 - 0.65^2) = 134.16 - 150 = 1184.16 \text{ N/m}^2 = \frac{1184.16}{9.81 \times 13.6}$$

$$= 8.87 \times 10^{-3} \text{ m of Hg} = \mathbf{8.87\ mm\ of\ Hg.}$$

Ex. 4. *A venturimeter is connected at the end of a pipe of water. The pressure reading of the venturimeter is 3.5×10^5 N/m^2 when the pipe is closed and 3.0×10^5 N/m^2 when the pipe is open. Calculate the velocity of flow of water in the pipe.* **(Raipur 1997; Indore 97)**

Sol. Here, $v_1 = 0$ (when the pipe is closed), $\rho = 1 \times 10^3$ kg/m^3, $p_1 = 3.5 \times 10^5$ N/m^2, $p_2 = 3.0 \times 10^5$ N/m^2

According to Bernouilli's theorem, $p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$ or $v_2^2 = \frac{2}{\rho}(p_1 - p_2) + v_1^2$

$$\therefore\quad v_2^2 = \frac{2}{1 \times 10^3} \times (3.5 \times 10^5 - 3.0 \times 10^5) = 100 \text{ or } v_2 = \mathbf{10\ m/s.}$$

Ex. 5. *A vertical tube of 8 mm diameter at the bottom has water passing through it. If the pressure be atmospheric at the bottom, where the rate of flow is 50 cc per sec, what is the pressure at a point in the tube 50 cm above the bottom, where the diameter is 4 mm ?*

Sol. Area of cross-section of tube at the bottom $= 3.14 \times (4 \times 10^{-3})^2 = 5.024 \times 10^{-5}$ sq.m

$\therefore$ Rate of flow = cross-sectional area × velocity

Hence, velocity v_1 of the water at the bottom is

$$v_1 = \frac{50 \times 10^{-6}}{5.024 \times 10^{-5}} = 0.9952 \text{ m/sec.}$$

$$\therefore\quad \text{K.E. of unit mass} = \frac{1}{2}v_1^2 = \frac{1}{2} \times 0.9952^2 = 0.4952 \text{ joules.}$$

Pressure energy per unit mass of water

$$\frac{p_1}{\rho} = 0.76 \times 9.8 \times \frac{13600}{1000} = 101.2928 \text{ joules.}$$

$$\therefore\quad \text{Total energy} = 0.4952 + 101.2928 = 101.788 \text{ joules.}$$

Let the velocity be v_2 at a height of 50 cm or 0.5 m from the bottom. Then,

$v_2 \times$ area of cross-section $= 50 \times 10^{-6}$ m^3.

$$\therefore\quad v_2 = \frac{50 \times 10^{-6}}{3.14 \times (2 \times 10^{-3})^2} = 3.9808 \text{ m/sec}$$

$\therefore$ K.E. of unit mass above 0.5 m from the bottom $= \frac{1}{2} \times (3.9808)^2 = 7.9234$ joules

P.E. of unit mass above 0.5 m from the bottom $= 0.5 \times 9.8 = 4.9$ joules

Pressure energy of unit mass above 0.5 m from the bottom $= \dfrac{x \times 13600 \times 9.8}{1000} = 133.28x$ joules

where x m is the pressure at a height of 0.5 m.

$$\therefore\quad \text{Total energy} = 12.8234 + 133.28\,x \text{ joules.}$$

Therefore, according to Bernoulli's theorem

$$12.8234 + 133.28\,x = 101.788$$

whence
$$x = 0.6675 \text{ m of Hg} = \mathbf{66.75\ cm\ of\ mercury.}$$

Ex. 6. *A pipe is running full of water. At a certain point A, it tapers from 60 cm diameter to 20 cm at B. The pressure difference between A and B is 1 m is water column. Find the rate of flow of water through the pipe.* **(Delhi 1971)**

Sol. If p_1 and p_2 be the pressures at A and B respectively and v_1, v_2 the corresponding velocities, then

$$\text{Rate of flow} = \text{velocity} \times \text{area of cross-section.}$$

The radii at A and B are 0.3 m and 0.1 cm,

$$\therefore \quad \text{Rate of flow} = v_1 \times \pi \times (0.3)^2 = v_2 \times \pi \times (0.1)^2 \text{ or } v_2 = 9v_1.$$

The average height of water remains unchanged on tapering, hence the gravitational energy remains unchanged there.

According to Bernoulli's theorem,

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2, \qquad \therefore p_1 - p_2 = \frac{1000}{2}(v_2^2 - v_1^2)$$

But $\quad p_1 - p_2 = 1 \times 1000 \times 9.81 \text{ N/m}^2;\ v_2 = 9v_1$

$$\therefore \quad 1 \times 1000 \times 9.81 = \frac{1000}{2}(81v_1^2 - v_2^2) \quad \text{or } 40v_1^2 = 9.81, \quad \therefore \ v_1 = 0.4952 \text{ m/sec.}$$

Hence, $\quad$ rate of flow $= 0.4952 \times 3.142 \times (0.3)^2 = \mathbf{0.74\ m^3/sec.}$

Ex. 7. *A fountain spreads water in air upto a height of 50 m. Calculate the maximum pressure at the base of the fountain.* **(Sagar 1999)**

Sol. Pressure will be maximum at the base of the fountain, when the water has no kinetic energy. Hence according to Bernoulli's theorem,

$$p_1 + \rho g h_1 = p_2 + \rho g h_2$$

Hence, pressure at the base, $\quad p_1 = p_2 + \rho g\,(h_2 - h_1)$

$$= (0.76 \times 13.6 \times 10^3 \times 9.8) + (1 \times 10^3) \times 9.8 \times 50$$

$$= 1.013 \times 10^5 + 4.9 \times 10^5 = \mathbf{5.913 \times 10^5\ N/m^3}.$$

Ex. 8. *A tank contains water to a height H. Calculate the range of water from an orifice at depth H/2 form the free surface of water.* **(Ujjain 1999)**

Sol. According to Bernoulli's equation (Torricelli's theorem), velocity of efflux is

$$v = \sqrt{2gH/2} = \sqrt{gH}$$

Let t be the time required by the liquid in falling $H/2$ height. Hence

$$\frac{H}{2} = \frac{1}{2}gt^2 \quad \text{or} \quad t = \sqrt{H/g}$$

$\therefore$ Range (horizontal) of the falling water from the orifice,

$$l = vt = \sqrt{gH} \times \sqrt{H/g} = H.$$

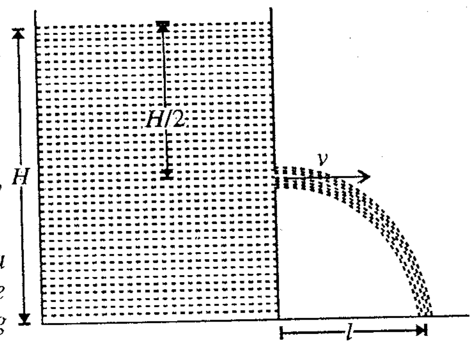


Fig. 14.16.

Ex. 9. *The diameter of a pipe at two points, where a venturimeter is connected, is 5 cm and 8 cm and the difference of levels in it is 4 cm. Calculate the volume of water flowing through the pipe per second.* **(Poona 1962)**

Sol. As, $\quad V = A_1 A_2 \sqrt{\dfrac{2hg}{A_1^2 - A_2^2}}$

Here $\quad A_1 = \pi(0.08/2)^2 = 16\pi \times 10^{-4}$ sq.m; $A_2 = \pi(0.05/2)^2 = 6.25\pi \times 10^{-4}$ sq.m and h = 0.04 m

$$\therefore \quad V = 16\pi \times 10^{-4} \times 6.25\pi \times 10^{-4} \sqrt{\frac{2 \times 9.8 \times 0.04}{[(16\pi)^2 - (6.25\pi)^2]10^{-8}}}$$

$$= 16 \times 6.25 \times \pi \times 10^{-4} \sqrt{\frac{2 \times 9.8 \times 0.04}{216.9375}} = \mathbf{18.89 \times 10^{-4}\ m^3/sec.}$$

Ex. 10. *A cylindrical vessel of cross-section* 30 *sq. cm contains water to a depth of* 15 *cm. There is a hole in the bottom of radius* 1 *mm. Find the time in which it will be emptied.*

Sol. The velocity of efflux is given by

$$v = \sqrt{2gh}$$

where h is the height of the water in the cylindrical vessel at any instant t.

Let dh be the decrease of level in time dt. Then the vloume flowing in dt second will be Adh, where A is the cross-section of the vessel. Hence, the rate of flow of water is given by

$$= -A\frac{dh}{dt} \quad \text{...(i)}$$

Negative sign indicates that as time increases, height decreases.

Also, rate of flow = velocity of efflux × area of cross-section of orifice

$$= \sqrt{2gh} \times \pi \times (10^{-3})^2 \quad \text{...(ii)}$$

Equating (i) and (ii), we have

$$-A\frac{dh}{dt} = \sqrt{2gh} \times \pi \times (10^{-3})^2 \quad \text{or } dt = -\frac{A}{\sqrt{2g} \times \pi \times 10^{-6}} \times \frac{dh}{\sqrt{h}} \quad \text{...(iii)}$$

Here, $A = 30 \times 10^{-4}\text{m}^2$; $g = 9.81\text{m/sec}^2$; initial height of water = 0.15 m; final height of water = 0

Substituting these values in (iii) and integrating between limits h = 0.15 m to h = 0, we get the time in which the vessel will be emptied *i.e.*,

$$t = -\frac{30 \times 19^{-4}}{\sqrt{2 \times 9.81 \times 3.14 \times 10^{-6}}} \times \left[2\sqrt{h}\right]_{0.15}^{0} = 167 \text{ sec.} = \textbf{2 min 47 sec.}$$

Ex. 11. *A pitot tube is fixed in a main of diameter* 15 *cm and the difference of pressure indicated by the gauge is* 4 *cm of water column; calculate the volume of water passing through the main in a minute.*

Sol. Area of cross-section $\alpha = \pi \times (0.075)^2$

Volume of liquid flowing per sec $= \alpha v = a.\ \sqrt{2gh} = \pi \times (0.075)^2 \times \sqrt{2 \times 9.8 \times 0.04}$

$\therefore$ Volume flowing per minute $= \pi \times (0.075)^2 \times 60 \times \sqrt{2 \times 9.8 \times 0.04} = \mathbf{0.9395\ m^3}$.

EXERCISE 14·1

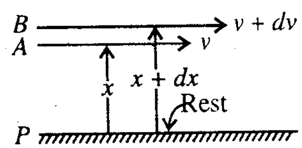
1. A tank containing water has an orifice in one vertical side. If the centre of the orifice is 10 meters below the surface level in the tank, find the velocity of efflux, assuming that there is no wastage of energy. (g = 9.8 metre/sec^2)
   [**Ans.** 14 m/sec.]
2. Calculate the velocity of efflux of water from a tank in which the pressure is 980 N/m^2 above the atmospheric pressure.
   [**Ans.** 1.4 m/sec.]
3. In a horizontal pipe water enters at one end with a velocity 0.5 m/s and leaves the other end with a velocity 0.6 m/s. Calculate the pressure difference at the ends. **(Gwalior 1993)**
   [**Ans.** 55 N/m^2.]
4. Water flows in an arrangement of two horizontal pipes of different diameters 3 cm and 6 cm respectively, connected in series. The velocity of water is 4 m/s and pressure is 2.0×10^4 N/m^2 in the first pipe. Calculate the velocity and pressure of water in the second pipe. **(Jabalpur 1995)**
   [**Ans.** 1 m/s, 2.75×10^4 N/m^2.]
5. A venturimeter whose throat is 6 cm in diameter is inserted in a water main of 10 cm diameter. If the instrument indicates a pressure difference equal to 8 cm of water, find the rate of flow of water through the main.
   [**Ans.** 3.795×10^{-3} m^3/sec.]

6. If a venturimeter is connected at two points where the diameters of the pipe are 8 cm and 5 cm and the pressure difference between two points is shown to be 6 cm of water column. Determine the rate at which the water is flowing through the pipe.
[**Ans.** 2.310×10^{-3} m/sec.]
7. A pitot tube is fixed in a main of diameter 20 cm and the difference of pressure indicated by the gauge is 5 cm of water column. Calculate the volume of water passing through the main in a minute.
[**Ans.** 1.864 m^3/min.]
8. Water enters in a horizontal pipe of non-uniform area of cross-section at one end with a velocity 40 cm/sec and leaves the other end with a velocity 60 cm/sec. If pressure of water at the entering end is 1500 N/m^2, find the pressure of water at the other end. (Density of water = 1.0×10^3 kg/m^3)
[**Ans.** 1400 N/m^2.]
9. Find the time in which a water tank of a uniform section of 100 sq. cm and fitted with water upto a height of 16 cm will be emptied through an orifice of area 0.1 sq. cm near the bottom. Take coefficient of contraction equal to 0.62
[**Ans.** 4 min 51.4 sec.]

14·8. Viscosity

When a liquid moves slowly and steadily over a fixed horizontal surface, *i.e.*, when its flow is streamline, every layer of the liquid moves parallel to the fixed surface. Its layer in contact with the fixed surface is at rest and the velocity of every other layer increases uniformly and continuously as we move upwards, *i.e.*, the greater the distance of a layer from the fixed surface, the greater its velocity. It is to be remembered that in a streamline flow, the velocity of the liquid is constant at any point.

Now consider the motion of two adjacent layers, say A and B of the liquid at distances x and $x + dx$ above the fixed horizontal surface, which are moving with velocities v and $v + dv$ respectively [**fig. 14.17**]. As the upper layer B is moving faster than the lower layer, the upper layer tends to increase the velocity of the lower layer, while the lower layer tends to decrease the velocity of the upper layer. The two layers together tend to destroy their relative motion as if there is some backward dragging force acting tangentially on the layers. Thus, to maintain the relative motion an external force must be applied to overcome this backward drag. This external force is provided by the pressure difference. In absence of any such outside force, the relative motion between the layers is destroyed and flow of the liquid ceases.



Horizontal fixed surface

Fig. 14.17.

This property by virtue of which a liquid opposes the relative motion between its different layers is known as **viscosity** *and the tangential force that tends to destroy the relative motion is called* **viscous force.** Viscosity is also called the internal friction of the liquid.

If the two layers of liquid are separated by a distance dx and are moving with a relative velocity dv, the quantity dv/dx is called velocity gradient. According to Newton, the tangential viscous force F, acting on a layer is proportional to its area A and the velocity gradient dv/dx, *i.e.*,

$$F \propto -A\frac{dv}{dx} \text{ or } F = -\eta A\frac{dv}{dx} \qquad ...(1)$$

where constant η is called the **coefficient of viscosity** of the liquid. Negative sign simply indicates that viscous force acts opposite to the direction of flow. It is obvious that the external force required to maintain the relative velocity dv between the two layers will also be equal to F, but opposite in sign*, *i.e*, $F = \eta A dv/dx$. If $A = 1$ and $dv/dx = 1$, then

$$F = \eta. \qquad ...(2)$$

Hence, *the* ***coefficient of viscosity*** *of a liquid may be defined as the tangential force per unit area of a layer, required to maintain unit velocity gradient normal to the direction of flow.*

The dimensional formula for the coefficient of viscosity (η) is $[ML^{-1}T^{-1}]$.

S.I. unit of coefficient of viscosity is **newton-sec/m²**.

* In fact, at any point of the liquid, backward dragging force is balanced by the forward external force (due to the pressure difference) so that the velocity of the liquid at that point is constant.

The C.G.S. unit of viscosity is **dyne-sec/m²** which is often called **poise**, in honour of Poiseuille, whose work on viscosity is important. The 100th part of 1 poise is called **centipoise** and 1 poise is equal to 0.1 S.I. units.

Viscosity in liquids corresponds to solid friction in the respect that it opposses the relative motion between two layers. The viscous force between two layers of a liquid depends upon (i) the area of the layers, (ii) the relative velocity between them and, (iii) the distance apart between them, while solid friction is independent of the area of contact of the solid surface and the relative velocities between them.

14·9. Flow of a Liquid through a Capillary Tube – Poiseuille's Formula

Poiseuille's Equation : Suppose a liquid be flowing through a capillary tube AB of radius r and length l, by applying a constant pressure differenc p, between the ends of the tube [**fig. 14.18**]. Poiseuille derived a formula for the volume of a liquid, flowing per second through the tube on the following **assumptions** :

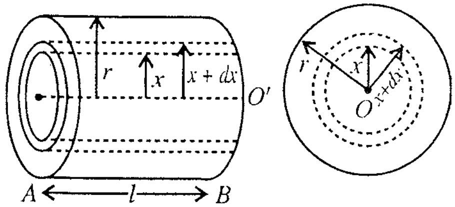


Fig. 14.18.

(1) The flow of the liquid in the tube is steady, streamline and parallel to the axis of the tube.

(2) The pressure over any cross-section at right angles to the axis of the tube is constant, *i.e.*, there is no radial flow.

(3) The velocity of the liquid layer in contact with the side of the tube is zero and increases in a regular manner as the axis of the tube is approached.

Consider a cylindrical layer of the liquid, coaxial with the tube of radius x. Let the velocity of the liquid at a distance x from the axis of the tube be v so that dv/dx is the velocity gradient.

Let us now consider the cylinder of radius x. The liquid outside this cylinder is moving less rapidly than inside it and thus exerts on the surface of the cylinder a retarding viscous force F. In accordance with the Newton's Law of viscous flow

$$F = -\eta\ A\ dv/dx \qquad \text{...(1)}$$

where, η is the coefficient of viscosity, and A the surface area of the cylinder, *i.e.*, $A = 2\pi xl$.

So that

$$F = -\ \eta \cdot 2\pi xl \cdot dv/dx \qquad \text{...(2)}$$

Since p is the difference of pressure between the two ends of the capillary tube, the force driving the liquid forward

$$= p \times \pi x^2 \qquad \text{...(3)}$$

For steady flow, there must be no acceleration of the liquid, *i.e.*, the forward force must be equal to the retarding viscous force. Hence,

$$p \times \pi x^2 = -\ \eta \cdot 2\pi\ xl \cdot \frac{dv}{dx} \text{ or } dv = -\ \frac{p}{2\eta l} \cdot x\ dx \qquad \text{...(4)}$$

Integrating, we get

$$v = -\frac{p}{2\eta l} \cdot \frac{x^2}{2} + C \qquad \text{...(5)}$$

where C is a constant of integration.

When $x = r$, $v = 0$ (since velocity of the layer in contact with fixed surface is zero),

$$\therefore \qquad 0 = -\frac{p}{2\eta l} \cdot \frac{r^2}{2} + C \quad \text{or} \quad C = \frac{p}{2\eta l} \cdot \frac{r^2}{2}$$

Hence, $v = -\dfrac{p}{2\eta l} \cdot \dfrac{x^2}{2} + \dfrac{p}{2\eta l} \cdot \dfrac{r^2}{2}$ or $v = \dfrac{p}{4\eta l}(r^2 - x^2)$ ...(6)

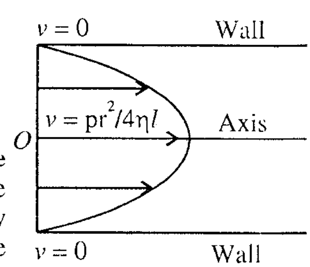


Fig. 14.19.

This expression gives us the **velocity distribution** or **velocity of the liquid at a distance x from axis of the tube**. The velocity distribution curve of the advancing liquid in the tube is **parabola [fig. 14.19]**; the velocity increases from 0 at the walls ($x = r$) of the tube to a mximum at its centre ($x = 0$).

Let us now consider that the layer, having inner radius x, has the thickness dx so that the volume dV of the liquid flowing per second through it is given by

$$dV = \text{velocity} \times \text{cross-sectional area of the cylindrical layer of thickness } dx.$$

$$= v \times 2\pi x\, dx = \frac{p}{4\eta l}(r^2 - x^2)\, 2\pi x\, dx$$

Therefore, the total volume flowing per second through the tube is

$$V = \int_0^r \frac{p}{4\eta l}(r^2 - x^2)\cdot 2\pi x\, dx = \frac{\pi p}{2\eta l}\int_0^r (xr^2 - x^3)\, dx$$

$$= \frac{\pi p}{2\eta l}\left[\frac{r^2x^2}{2} - \frac{x^4}{4}\right]_0^r = \frac{\pi p}{2\eta l}\left[\frac{r^4}{2} - \frac{r^4}{4}\right]$$

i.e.,
$$V = \frac{\pi p r^4}{8\eta l}. \qquad ...(7)$$

This, is known as **Poiseuille's formula.**

In an actual experiment, the pressure differenc p is measured by the difference of liquid levels in two liquid columns at the two ends of the tube of length l. If ρ is the density of the liquid, flowing through the tube, then. $p = h\rho g$.

Limitations of the Poiseuille's formula : (1) This formula holds only for steamlined flow of the liquid.

(2) It is not true if the layers of the liquid in contact of the walls of the tube are not at rest and thus have a relative slipping.

(3) Poiseuille's formula has been derived on the basis that the pressure difference across the tube is used to overcome the viscous forces and no part of it is spent in giving the kinetic energy to the liquid.

14·10. Determination of Coefficient of Viscosity by Using Poiseuille's Formula

Apparatus : The apparatus for the determination of coefficient of viscosity by Poiseuille's method is shown in **fig. 14.20**. A capillary tube T of uniform bore, whose radius r and length l are known, is fitted horizontally with a rubber bung at each end. B and C are two wide brass tubes. The capillary T is connected to a out-flow tube D through C and to a water vessel A through B and a rubber tube. The pressure difference between the end of the capillary tube T is recorded by the difference in the height of the liquid in the two limbs (E and F) of the manometer. The vessel A, in which the tap water is coming, is fitted with an overflow tube O, so that a constant head of water may be maintained in the system.

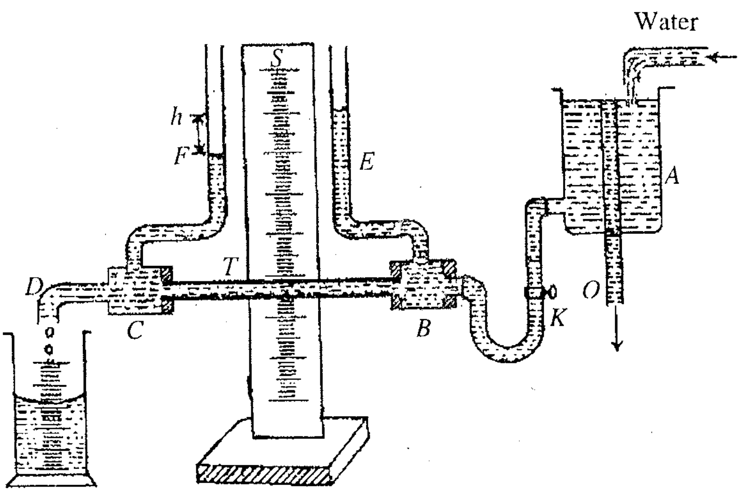


Fig. 14.20.

Method : With the help of a pinch cock K, a steady flow of water is maintained through the capillary tube such that water leaves the tube D in a slow trickle. Water is collected in a measuring flask for a known time and hence water flowing through the tube per second (V) is determined. The pressure difference P is directly read by the manometer and will be equal to $h\rho g$. Now the coefficient of viscosity is determined by the formula.

$$\eta = \frac{\pi p r^4}{8Vl}.$$

Since r is involved with 4th power in the expression, extra care is taken to measure the diameter of the capillary tube with the help of a high power microscope.

Precautions : (1) The pressure difference should be kept small to obtain steamline flow.

(2) The bore of the tube must be narrow, as the formula is not applicable for wide bore tubes. The reason lies in the fact that the wider tubes tend to promote turbulence, because the critical velocity is inversely proportional to the radius of the tube.

(3) The tube should be long enough so that any non-uniformity of flow at the inlet be negligible.

(4) The tube should be placed horizontally to avoid any effect of gravity.

14·11. Capillaries in Series and Parallel

If two capillaries of length l_1, l_2 and radii r_1, r_2 respectively are connected in series [**fig. 14.21**], then obviously the volume flowing per second through them will be the same, *i.e.*,

$$V = \frac{\pi p_1 r_1^4}{8\eta l_1} = \frac{\pi p_2 r_2^4}{8\eta l_2} \qquad ...(1)$$

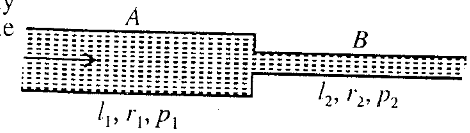


Fig. 14.21 : Capillaries in series.

where p_1 and p_2 are pressure differences across the two capillaries. Now, if p is the pressure difference across the composite tube, then

$$p = p_1 + p_2 \quad \text{or} \quad p_2 = p - p_1$$

$$\therefore \quad V = \frac{\pi p_1 r_1^4}{8\eta l_1} = \frac{\pi (p - p_1) r_2^4}{8\eta l_2} \quad \text{or} \quad \frac{p_1 r_1^4}{l_1} = \frac{p r_2^4}{l_2} - \frac{p_1 r_2^4}{l_2} \quad \text{or} \quad p_1\left(\frac{r_1^4}{l_1} + \frac{r_2^4}{l_2}\right) = \frac{p r_2^4}{l_2}$$

or
$$p_1 = \frac{p r_2^4 / l_2}{\left(\frac{r_1^4}{l_1} + \frac{r_2^4}{l_2}\right)} \qquad ...(2)$$

$$\therefore \quad V = \frac{\frac{\pi r_1^4}{8\eta l_1} \cdot \frac{p r_2^4}{l_2}}{\left(\frac{r_1^4}{l_1} + \frac{r_2^4}{l_2}\right)} = \frac{\pi p}{8\eta} \frac{r_1^4 r_2^4}{l_1 l_2 \cdot \frac{r_1^4 r_2^4}{l_1 l_2}\left(\frac{l_1}{r_1^4} + \frac{l_2}{r_2^4}\right)} = \frac{\pi p}{8\eta}\left(\frac{l_1}{r_1^4} + \frac{l_2}{r_2^4}\right)^{-1} \qquad ...(3)$$

This expression represents the volume flowing per second through two capillary tubes, connected in series.

If the two capillaries are joined in parallel, *i.e.*, the pressure difference across them is the same, then the volume of the liquid flowing per second through them [**fig. 14.22**], is given by

$$V = V_1 + V_2 = \frac{\pi p r_1^4}{8\eta l_1} + \frac{\pi p r_2^4}{8\eta l_2} = \frac{\pi p}{8\eta}\left(\frac{r_1^4}{l_1} + \frac{r_2^4}{l_2}\right) \qquad ...(4)$$

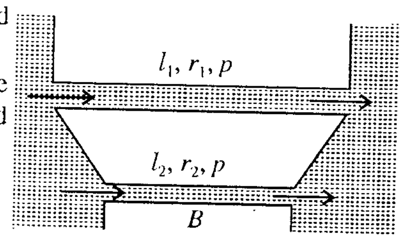


Fig. 14.22 : Capillaries in parallel.

If we put $R = 8\eta l / \pi r^4$, then $\quad V = p/R$

[From Poiseulle's equation]

For **series arrangement,** $\quad V = p/(R_1 + R_2) \quad$ [From eq. (3)] ...(5)

For **parallel arrangement,** $\quad V = p\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \quad$ [From eq. (4)] ...(6)

Hence, it is clear that here V, p, R are equivalent to current, potential difference and resistance of an electrical circuit.

In case a number of capillaries are connected in series or parallel, then

$$V = p/(R_1 + R_2 + R_3 + ...) \text{ in series arrangement,} \qquad ...(7)$$

and
$$V = p\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + ...\right) \text{ in parallel arrangement.} \qquad ...(8)$$

Ex. 1. *A flat of plate area 10 cm^2 is kept on another plate with a layer of glycerine of width 1 mm in between the plates. If the coefficient of viscosity of glycerine is 10 poise, find the force required to move the upper plate with a velocity 7 cm/sec.* **(Indore 1995)**

Sol. Here, η = 10 poise = 1 kg m^{-1} s^{-1}, $A = 10\ cm^2 = 10^{-3} m^2$,

$$\therefore \quad \frac{dv}{dx} = \frac{7\ \text{cm/sec}}{1\ \text{mm}} = \frac{7\ \text{cm/sec}}{0.1\ \text{cm}} = 70\ \text{sec}^{-1}$$

We know that the force required to move the plate will be equal and opposite to the viscous force.

Hence, $$F = \eta A \frac{dv}{dx} = 1 \times 10^{-3} \times 70 = \mathbf{0.07\ newton.}$$

Ex. 2. *A liquid is flowing through a 25 cm long tube of 1 mm internal diameter due to a pressure of 10 cm of mercury. Calculate* (i) *the volume of the liquid flowing out in one minute,* (ii) *the velocity of the liquid on the axis of the capillary and at a distance 0.1 mm, 0.2 mm and 0.5 mm from the axis of the tube.* (η = 0.05 *S.I. units.*) **(Indore 1998; Raipur 98)**

Sol. (i) We know that $V = \dfrac{\pi P r^4}{8\eta l}$

Here, $P = h\rho g = 0.1 \times 13{,}600 \times 9.8$ N/m^2, $r = 5 \times 10^{-4}$ m, η = 0.05 S.I. units, and l = 0.25 m.

$$\therefore \quad V = \frac{3.14 \times 9.1 \times 13600 \times 9.8 \times (5 \times 10^{-4})^4}{8 \times 0.05 \times 0.25} = 0.0267 \times 10^{-6}\ \text{m}^3/\text{sec}.$$

$\therefore$ Volume flowing per minute = $0.0267 \times 10^{-6} \times 60$ = $\mathbf{1.6 \times 10^{-6}}$ **m^3/min.**

(ii) Now, $$v = \frac{P}{4\eta l}(r^2 - x^2).$$

But at the axis of the tube, $x = 0$

$$\therefore \quad v = \frac{Pr^2}{4\eta l} = \frac{0.1 \times 13600 \times 9.8 \times (5 \times 10^{-4})^2}{4 \times 0.05 \times 0.25} = \mathbf{0.067\ m/sec.}$$

At x = 0.1 mm = 0.1×10^{-3} m,

$$v = \frac{h\rho g(r^2 - x^2)}{4\eta l} = \frac{0.1 \times 13600 \times 9.8 \times [(5 \times 10^{-4})^2 - (10^{-4})^2]}{4 \times 0.05 \times 0.25} = \mathbf{0.064\ m/sec.}$$

At x = 0.2 mm = 0.2×10^{-3} m, $v = \dfrac{0.1 \times 13600 \times 9.8 \times [(5 \times 10^{-4})^2 - (2 \times 10^{-4})^2]}{4 \times 0.05 \times 0.25} = \mathbf{0.056\ m/sec.}$

At $x = 0.5 \times 10^{-3}$ m (*i.e.*, at the wall of capillary), v = **0**.

Ex. 3. *Calculate the mass of water flowing in 10 minutes through a tube 0.1 cm in diameter, 40 cm long if there is a constant pressure head of 20 cm of water. The coefficient of viscosity of water is 0.00089 S.I. units.* **(Agra 1995)**

Sol. $$V = \frac{\pi P r^4}{8\eta l} = \frac{3.14 \times 0.2 \times 1000 \times 9.8 \times (5 \times 10^{-4})^4}{8 \times 0.00089 \times 0.4} = \mathbf{13.51 \times 10^{-8}\ m^3}.$$

The volume of water flowing in 10 minutes = $13.51 \times 10^{-8} \times 600 = 81.06 \times 10^{-6}$ m^3.

$\therefore$ Mass of the water flowing in the given time = $81.06 \times 10^{-6} \times 1000$ = $\mathbf{81.06 \times 10^{-3}}$ **kg.**

Ex. 4. Two tubes A and B of lengths 1 *m and* 0.5 *m have radii* 0.1 *mm and* 0.2 *mm respectively. If a liquid is passing through the two tubes, entering A at a pressure of* 80 *cm of mercury and leaving B at a pressure* 76 *cm, find the pressure at the junction of A and B.* **(Gwalior 1994; Bihar 64)**

Sol. According to Poiseuille's equation the volume of a liquid flowing per second through a tube is given by $$V = \frac{\pi P r^4}{8\eta l}$$

Here, the tubes A and B of lengths 1 m and 0.5 m have been joined at the junction O. The liquid is entering the tube A at C and leaving the tube B at D **[fig. 14.23]**. Let P' be the pressure at the junction O, then pressure difference between C and O = 0.8 – P'.

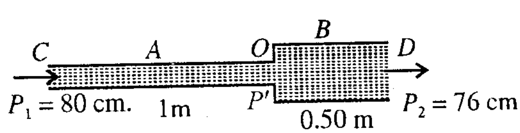


Fig. 14.23.

According to the question, the volume of liquid flowing per second through the tube A,

$$= \frac{\pi(0.8 - P') \times \rho g \times (0.1 \times 10^{-3})^4}{8\eta \times 1}$$

The volume of the liquid flowing per sec through the tube B,

$$= \frac{\pi(P' - 0.76) \times \rho g \times (0.2 \times 10^{-3})^4}{8\eta \times 0.5}$$

The two tubes A and B are joined end to end, hence the volume flowing per sec through them must be the same *i.e.*,

$$\frac{\pi(0.8 - P') \times \rho g \times (0.1 \times 10^{-3})^4}{8\eta \times 1} = \frac{\pi(P' - 0.76) \times \rho g \times (0.2 \times 10^{-3})^4}{8\eta \times 0.5}$$

or $$\frac{0.8 - P'}{2} = (P' - 0.76)\ 16, \therefore\ P' = \frac{0.76 \times 32 + 0.8}{33} = \mathbf{0.761\ m\ of\ Hg}.$$

Ex. 5. *Fig.* 14.24 *below shows two wide tubes P, Q connected by three capillaries A, B, C whose relative lengths and radii are indicated. If a pressure P is maintained across A, , deduce*

(i) *the ratio of liquid flowing through A and B.*

(ii) *the pressure across B, and across C.* **(Agra 1969 S; B'khand 93)**

Sol. (i) Volume flowing through A is

$$V_1 = \frac{\pi p r^4}{8\eta l}.$$

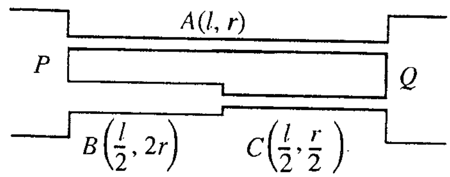


Fig. 14.24.

Volume flowing through B and C is the same, *i.e.*,

$$V_2 = \frac{\pi p_1 (2r)^4}{8\eta(l/2)} = \frac{\pi p_2 (r/2)^4}{8\eta(l/2)} \quad \text{or } 256 p_1 = p_2$$

Also, $p = p_1 + p_2$, $\therefore\ p = p_1 + 256 p_1$ or $p_1 = p/257$

$$\therefore \quad \frac{V_1}{V_2} = \frac{p}{32 p_1} = \frac{p}{32 \times p/257} = \frac{257}{32}$$

(ii) Pressure across B is $p_1 = \mathbf{p/257}$.

Pressure across C is $p_2 = p - p_1 = \mathbf{256\ p/257}$.

Ex. 6. *Three capillaries of lengths* 8L, 0.2L, *and* 2L, *with radii r,* 0.2r *and* 0.5r, *respectively are connected in series. If the total pressure acorss the system in an experiment is p, deduce the pressure across the shortest capillary.* **(Agra 1971; Allahabad 83)**

Sol. The volume of the liquid flowing per second through each capillary is the same *i.e.*,

$$V = \frac{\pi p_1 r^4}{8\eta \times 8L} = \frac{\pi p_2 (0.2r)^4}{8\eta \times 0.2L} = \frac{\pi p_3 (0.5r)^4}{8\eta \times 2L}$$

or $$\frac{p_1}{64} = \frac{p_2}{1000} = \frac{p_3}{256} \quad \text{or } p_1 = \frac{8}{125} p_2 \text{ and } p_3 = \frac{32}{125} p_2$$

But $$p_1 + p_2 + p_3 = p, \quad \therefore \quad (8/125)p_2 + p_2 + (32/125)p_2 = p$$

Therefore, $$p_2 = \frac{p}{\frac{8}{125} + 1 + \frac{32}{125}} = \frac{p}{\frac{165}{125}} = \frac{125}{165} p = \mathbf{0.758p}.$$

Ex. 7. *A vessel of cross-section* 20 *sq. cm has at the bottom a horizontal capillary tube of length* 10 *cm and internal radius* 0.5 *mm. It is initially filled with water at a height of* 20 *cm above the capillary tube. Find the time taken by the vessel to empty one-half of its contents, given that viscosity of water is* 0.01 *poise.* **(Lucknow 1986; Meerut 95; Raipur 94)**

Sol. According to Poiseuille's equation,

$$V = \frac{\pi P r^4}{8\eta l} \quad ...(i)$$

Let A be the area of cross-section of the vessel and h the height or water above the capillary tube.

$\therefore$ Pressure $P = h\rho g$.

Consider that in small time dt, the level in vessel falls by the amount dh, then the volume of water flowing out in dt second is $A \times dh$ *i.e.*, rate of flow of water is given by

$$V = -A\,(dh/dt)$$

where negative sign indicates that as time increases, height of water decreases.

Substituting for V and P in (i), we have

$$-A\frac{dh}{dt} = \frac{\pi h\rho g r^4}{8\eta l} \quad \text{or} \quad dt = -\frac{8A\eta l}{\pi\rho g r^4}\times\frac{dh}{h} \quad ...(ii)$$

Here, $A = 20 \times 10^{-4}$ sq. m, $l = 0.1$ m, $\eta = 0.001$ S.I. units, $r = 5 \times 10^{-4}$ m.

Initial height of water = 0.2m, Final height of water = 0.1 m.

Hence, substituting the values of various factors in eq. (ii) and integrating between the limits $h = 0.2$ and $h = 0.1$, we get the time taken to empty one-half of the contents which is given by

$$t = -\frac{8\times 20\times 10^{-4}\times 0.001\times 0.1}{3.14\times 9.8\times 1000\times(5\times 10^{-4})^4}\,[\log_e h]_{0.2}^{0.1}$$

$$= \frac{8\times 20\times 10^{-4}\times 10^{-4}}{3.14\times 9.8\times 1000\times(5\times 10^{-4})^4}\times \log_{10} 2 \times 2.30 \qquad [\log_{10} 2 = 0.3010]$$

$$= 576.9 \text{ sec} = \mathbf{9\ min.\ 36.9\ sec.}$$

EXERCISE 14·2

1. In a horizontal tube of length 40 cm and internal radius 1 mm, water flows from a constant level at height 15 cm. Calculate the maximum velocity of flow of water. Coefficient of viscosity of water = 0.0098 poise, $g = 980$ cm/s^2. **(Ujjain 1999)**
   [**Ans.** 0.937 m/s.]
2. Water is conveyed through a tube 8 cm in diameter and 4 km in length at the rate of 120 litres per minute. Calculate the pressure required to maintain the flow.
   Coefficient of viscosity of water = 10^{-3} S.I. units, 1 atmospheric pressure = 1.03×10^5 N/m^{-2}. **(I.A.S. 1963)**
   [**Ans.** 0.0795 atmospheric pressure.]
3. 0.0914 cc of a liquid is flowing out per second through a capillary tube of 1 mm radius. Calculate the velocity of the liquid at a point (i) on the axis of the capillary, (ii) adjacent to the wall of the capillary and (iii) at 0.5 mm distance from the axis. **(Rohilkhand 1984)**
   [**Ans.** (i) 0.02 m^{-1}; (ii) zero; (iii) 0.015 ms^{-1}.]
4. Compare the rate of flow of a liquid through $a/2$ section of the capillary with that through the entire tube. Here a is the radius of capillary. **(Bilaspur 1989)**
   [**Ans.** 7 :16.]
5. 24 gm of water flows out of capillary tube every minute, when a constant pressure gradient of 10 C.G.S. units is maintained between its ends. If the radius of the capillary bore is 0.1 cm, calculate the coefficient of viscosity of water. **(Mysore 1980)**
   [**Ans.** 7.85×10^{-4} S.I. units.]
6. A capillary tube 1 mm in diameter and 20 cm in length is fitted horizontally to a vessel kept full of alcohol of density 800 kg/m^3. The depth of the centre of the capillary tube below the surface of the alcohol is 30 cm. If the viscosity of alcohol is 0.012 C.G.S. units. Find the amount that will flow in 5 minutes.
   [**Ans.** 57.74×10^{-3} kg.]

7. A wide tube which stands vertically has been sealed into its lower end a short length of capillary tube which is horizontal. It is filled with glycerine of density 1260 kg-m^{-3} and viscosity 0.85 S.I. units to a certain mark and it takes 45 sec for the glycerine to fall a second mark by escaping through the capillary tube. With castor oil of density 870 kg m^{-3}, the time takes to fall from one mark to the other is 69 sec. Determine the viscosity of castor oil.
[**Ans.** 0.899 S.I. units.]
8. A cylindrical vessel of radius 5 cm has at the bottom a horizontal capillary tube of length 20 cm and an external radius 0.4 mm. If the vessel is filled with water, find the time taken by water to flow out so as to reduce the volume of the water in the vessel to half of its original value. (Coefficient of viscosity of water = 0.01 poise, g = 980 cm/sec^2) **(Utkal 1967)**
[**Ans.** 3 hours 6 min.]
9. A capillary of length 10 cm and diameter 2 mm is joined horizontally at the bottom of a vessel of area of cross-section 30 cm^2. If the vessel is filled with water upto 20 cm from its bottom, find the time for half of it to be empty. Coefficient of viscosity of water is 0.01 poise. Deduce the formula used. **(Raipur 1994)**
[**Ans.** 3.378 second.]
10. A horizontal capillary of 1 mm diameter and 10 cm length is connected to a tank of water and 10 cc of water flows out per minute. How much water will flow out per minute if another capillary of 0.5 mm diameter and 1 cm length is connected in series with the first capillary ?
[**Ans.** 3.8×10^{-6} m^3 min^{-1}.]
11. A liquid is flowing through two capillaries of lengths 16 cm and 54 cm and diameters 2 mm and 3 mm respectively, joined in series. If the total pressure difference across the two tubes be 8 cm of mercury, what is the pressure difference between the ends of the second tube alone ?
[**Ans.** 3.2 cm.]
12. Two capillary tubes of bore diameters 0.05 cm and 0.01 cm and of length 20 cm and 5 cm respectively are connected in series. If the pressure difference at the two extreme ends is maintained at 15 cm of water, calculate the pressure difference across the first tube.
[**Ans.** 0.095 cm of water.]
13. A horizontal capillary of diameter 2 mm and length 20 cm is connected to a tank of water and 0.2 cc of water flows our per second. Calculate the rate at which water will flow out if another capillary of length 10 cm and diameter 1 mm is connected in series with the first capillary. **(Agra 1980)**
[**Ans.** 2.2×10^{-8} m^3 s^{-1}.]
14. A horizontal capillary of 1 mm radius and 100 cm length is connected to a tank and 5 ml of water flows out per minute. What will be the rate of flow of water if another capillary of 0.5 mm radius and 10 cm length is connected in series with first capillary ? **(Agra 1991)**
[**Ans.** 1.92 ml/min.]
15. Two horizontal capillaries A and B are connected in series so that a steady stream of liquid flows through them. A is 1 mm in internal diameter and 1 metre long; B is 0.6 mm in internal diameter and 60 cm long. The liquid pressure at the entrance A is 20 cm of water above atmospheric. At the exit end, the pressure is atmospheric. What is the pressure at the junction of the tubes ?
[**Ans.** 16.4 cm of water above atmospheric.]
16. A capillary of length L and radius R is connected in series with another capillary of length L/4 and radius R/2. If the pressure difference across the combined arrangement is P, find the pressure difference across each capillary. **(Ujjain 1997)**
[**Ans.** 0.2P and 0.8P.]
17. Three capillaries of same length, but internal radii r, $2r$ and $3r$ are connected in series and liquid flows through them under streamline conditions. If the pressure across the whole system is 77 cm of water, deduce the pressure across the first capillary. **(Gorakhpur 1995)**
[**Ans.** 71.64 cm of water.]

[**Hint :** $$V = \frac{\pi P_1 (r)^4}{8\eta l} = \frac{\pi P_2 (2r)^4}{8\eta l} = \frac{\pi P_3 (3r)^4}{8\eta l}$$

But $$P_1 + P_2 + P_3 = 77 \text{ cm} = \frac{8\eta V l}{\pi}\left(\frac{1}{r^4} + \frac{1}{16r^4} + \frac{1}{81r^4}\right) \text{ and } P_1 = \frac{8\eta V l}{\pi}\cdot\frac{1}{r^4}$$

$$\therefore \quad \frac{P_1}{77} = \frac{1}{\left(1 + \frac{1}{16} + \frac{1}{81}\right)} \text{ or } P_1 = 71.64 \text{ cm of water.]}$$

18. Three capillary tubes of the same radius r but of lengths l_1, l_2 and l_3 are fitted horizontally to the bottom of a long cylinder which contains liquid at a constant head, and which can flow through these tubes. Show that the length L of a single outflow tube of the same radius r, which can replace the three capillaries, is given by

$$\frac{1}{L} = \frac{1}{l_1} + \frac{1}{l_2} + \frac{1}{l_3}.$$

19. Three capillaries of same length but internal radii $3r$, $4r$ and $5r$ are connected in series and a liquid flows through them in streamline conditions. If the pressure across the third capillary is 8.1 cm, deduce the pressure across the first capillary. **(Agra 1970 S; Bundelkhand 94)**
[**Ans.** 6.25 cm.]

14·12. Stoke's Formula

When a body falls through a fluid (liquid or gas), the layer of the fluid in contact of the body moves with its velocity. Thus, a relative motion is set up between the various layers of the fluid. The relative motion is opposed by the force of viscosity. This opposing force increases with the velocity of the body; until the viscous force acting upwards plus the upthrust of the fluid on the body are together equal to the weight of the body acting downwards, so that the resultant force on the body becomes zero. The body then attains a constant velocity called the **terminal velocity**.

Stoke has shown that if a small sphere of radius a be moving slowly through a homogeneous viscous fluid of infinite extension with a terminal velocity v, the viscous force exerted upon the sphere is given by

$$F = 6\pi\eta a v \quad \text{...(1)}$$

where η is the coefficient of viscosity of the fluid

The above expression may be derived by the method of dimensions.

Let us write

$$F = k\eta^a a^b v^c \quad \text{...(2)}$$

where k is a dimensionless constant.

Substituting the dimensions of the various quantities on both sides of the equation, we get

$$[MLT^{-2}] = [ML^{-1}T^{-1}]^a\,[L]^b\,[LT^{-1}]^c = [M^a\,L^{b+c-a}\,T^{-a-c}]$$

Comparing indices of M, L and T, we get

$$a = 1;\ b + c - a = 1 \text{ and } -a - c = -2;\ \therefore a = 1,\ b = 1 \text{ and } c = 1.$$

Stoke showed that if the extension of the fluid is infinite, then $k = 6\pi$.

Therefore, by putting the volues of a, b, c and k in (2), we have

$$F = 6\pi\eta a v \quad \text{...(3)}$$

Let the density of the spherical body be ρ. Then, its weight will be $\frac{4}{3}\pi a^3 \times \rho \times g$.

The upthrust on it due to displacement of the fluid = $\frac{4}{3}\pi a^3 \sigma g$., where σ is the density of the fluid.

When the body is moving with a constant velocity the resultant downward force will be equal to viscous force. Therefore,

$$6\pi\eta av = \frac{4}{3}\pi a^3 g(\rho - \sigma) \text{ or } \eta = \frac{2}{9}\frac{a^2 g(\rho - \sigma)}{v} \qquad ...(4)$$

In experiment, by determining the terminal velocity of the fluid, one can find the coefficient of viscosity by using the formula (4)

Determination of terminal velocity :

From eq. (4), the expression for terminal velocity is given by

$$v = \frac{2}{9}\frac{a^2 g(\rho - \sigma)}{\eta} \qquad ...(5)$$

According to this equation, the terminal velocity of a spherical body in a fluid is

(i) directly proportional to the square of radius (a^2),

(ii) directly proportional to the difference of densities ($\rho - \sigma$) of the body and fluid, and

(iii) inversely proportional to the coefficient of viscosity of the fluid.

When $\rho < \sigma$ *i.e.*, the density of spherical body (ρ) is less than the density (σ) of the medium, then v will be negative and hence the body will move up instead moving down. For example, a spherical bubble in water moves up.

Falling of the rain drops : We know that rain drops are formed by the condensation of water on the dust particles in the air. If a drop falls under gravity, its velocity increases first and hence viscous force, acting opposite to the velocity, also increases. Now, a stage is reached, when the viscous force becomes equal to the force of gravity. Therefore, the drop acquires a constant or terminal velocity which is directly proportinal to the square of the radius of the drop ($v \propto a^2$). In consequence, smaller drops fall with smaller terminal velocity and bigger ones with larger terminal velocity. Initially when the rain drops are formed, their size is very small and hence their terminal velocity is so small that a large number of them appear floating in the sky in the form of **cloud.**

Falling of a soldier with parachute : When a soldier jumps from an aeroplane with parachute, he falls, in the beginning with the acceleration due to gravity, but due to the viscosity of the air, viscous force acts in the upward direction. Now, his velocity increases, viscous force becomes more and more and ultimately the force due to gravity is balanced by the viscous force so that the soldier falls with a constant or terminal velocity.

Determination of viscosity of a very viscous liquid (castor oil) : The liquid whose viscosity is to be determined is taken in a large glass jar. A small steel sphere is dropped centrally. Now the time intervals of its passing through two successive equal distances PQ and QR are observed **[fig. 14.25]**. If they are equal, it means that the shpere has attained a constant velocity. The time t taken to travel the distant PQ or QR is noted. Now, terminal velocity $v = PQ/t$. The radius a of the sphere is measured by a travelling microscope. Now, if we know the densities ρ and σ of the steel and liquid, we can calculate η at the temperature of the liquid from the following realtion :

$$\eta = \frac{2}{9}\frac{a^2 g(\rho - \sigma)}{v}$$

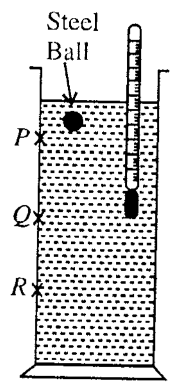


Fig. 14.25.

Ex. 1. *A steel ball of radius* 2 *mm falls in glycerine. Calculate its terminal speed. Density of steel* = 8×10^3 *kg/m*^{3}, density of glycerine = 1.2×10^3 *kg/m*^{3}, *coefficient of viscosity of glycerine* = 0.85 *kg m*$^{-1}$ *s*$^{-1}$.

(Rewa 2004)

Sol. Here, $r = 2$ mm $= 2 \times 10^{-3}$m, $\rho = 8 \times 10^3$ kg/m^3, $\sigma = 1.2 \times 10^3$ kg/m^3, $g = 9.8$ m/s^2 and $\eta = 0.85$ kg m^{-1} s^{-1}

Terminal speed,
$$v = \frac{2}{9}\frac{r^2(\rho - \sigma)g}{\eta} = \frac{2}{9} \times \frac{(2\times10^{-3})^2 \times (8\times10^3 - 1.2\times10^3)\times 9.8}{0.85}$$

$$= \mathbf{6.97 \times 10^{-2}\ m/sec.}$$

Ex. 2. *An air bubble of radius 1 cm is allowed to rise through a long cylinderical column of a viscous liquid and travels at a steady rate of 0.21 cm s^{-1}. If the density of the liquid is 1470 kg m^{-3}, find its viscosity. Assume g = 9.8 ms^{-2}, neglect the density of air.* **(Ranchi 1966; Patna 2004)**

Sol. The coefficient of viscosity is given by the relation

$$\eta = \frac{2}{9}\frac{a^2 g(\rho - \sigma)}{v}$$

Given, $a = 0.01$ m, ; $g = 9.8$ ms^{-2}, density of air bubble $\sigma = 0$,
density of medium $\rho = 1470$ kg m^{-3} and $v = 21 \times 10^{-3}$ ms^{-1}.

$$\therefore \qquad \eta = \frac{2}{9}\cdot\frac{1470 \times (0.01)^2 \times 9.8}{2.1 \times 10^{-3}} = \mathbf{152.4\ N\text{-}s/m^2}.$$

Ex. 3. *A spherical ball of radius 1×10^{-3} m and density 10^4 kg/m^3 falls freely under gravity through a distance h before entering a tank of water. If after entering the water, the velocity of the ball does not change, find h. The coefficient of viscosity of water is 9.8×10^{-4} N-s/m^2.* **(Purvalchal 1995)**

Sol. After entering the water, the velocity of the ball does not change. This means that the ball has acquired a velocity v in falling through distance h equal to terminal velocity v, given by

$$v = \sqrt{2gh}$$

As the ball is moving with constant velocity, the resultant downward force is equal to the upward viscous force, *i.e.*,

$$6\pi\eta a v = \frac{4}{3}\pi a^3 g(\rho - \sigma) \quad \text{or} \quad v = \frac{2}{9}\frac{a^2 g(\rho - \sigma)}{\eta} \quad \text{or} \quad \sqrt{2gh} = \frac{2}{9}\frac{(10^{-3})^2 \times 9.8 \times (10^4 - 10^3)}{9.8 \times 10^{-4}} = 20$$

$$\therefore \qquad \eta = \frac{400}{2g} = \frac{400}{2 \times 9.8} = \mathbf{20.4\ m}.$$

Ex. 4. *Two identical water drops fall in air with the terminal velocity of 5 ms^{-1}. If these drops combine to form a single drop, what be the new terminal velocity ?* **(Jabalpur 2003; Raipur 03)**

Sol. If the radius of the big drop is R and that of the small drop be r, then according to the problem

Volume of one big drop = Volume of two small drops or $\frac{4}{3}\pi R^3 = 2 \times \frac{4}{3}\pi r^3$, $\quad \therefore R = (2)^{1/3} r$

As the terminal velocity of drop, $v \propto$ (radius)^2

$$\therefore \frac{\text{Terminal velocity of big drop } v_2}{\text{Terminal velocity of small drop } v_1} = \frac{R^2}{r^2},\ v_2 = \left(\frac{R}{r}\right)^2 v_1 = (2)^{2/3} \times 5 = \mathbf{7.94\ m/sec.}$$

EXERCISE 14·3

1. Water drop of radius 0.001 cm is falling through air. Find its terminal velocity. Neglect the density of air. (η of air = 1.8×10^{-5} S.I. units) **(Allahabad 1990)**
[**Ans.** 12×10^{-2} ms^{-1}.]
2. A water drop of radius 1.0×10^{-5} m is falling through air. Calculate the terminal speed of drop. (For air, $\eta = 1.89 \times 10^{-5}$ kg m^{-1} s^{-1}) **(Ujjain 1994)**
[**Ans.** 1.15 cm/sec.]
3. Eight drops of water of same radius fall in air with terminal speed 5 cm/sec. If they combine to form a big drop, what will be the terminal speed of this drop ?
(Indore 1999; Ujjain 98; Bhopal 90; Gwalior 90)
[**Ans.** 0.2 m/sec.]
4. A small liquid drop falling freely in air reaches a terminal velocity v. Given that viscous drag on a spherical drop is $6\pi\eta a v$, where a is the radius of the drop, calculate the terminal velocity of a water droplet of radius 1.5×10^{-1} cm. Given : $\eta = 1.8 \times 10^{-4}$ C.G.S. units (for air) and $g = 9.81$ m/s^2. **(Rajasthan 1964)**
[**Ans.** 272 ms^{-1}.]

14·13. Variation of Viscosity with Temperature

The viscosity of a liquid decreases with rise in temperature. For example, the viscosity of water at 80°C is found to be only one-third of its value at 10°C. Several workers have tried to get a relationship between the temperature and the viscosity of liquids. No satisfactory relation has been found to exist between viscosity and temperature, but various empirical relations have been suggested from time to time. Slotte, for example, gave the formula

$$\eta_1 = \frac{\eta_0}{1+\alpha t+\beta t^2} \qquad ...(1)$$

where α and β are constants and η_0 the viscosity at 0°C.

In 1934, Andrade gave a satisfactory theory for liquids. He suggested that a temperature relation of the form $\eta = Ae^{c/T}$ applies as a first approximation, A and c being constants. This formula gives very fair agreement with experimental results. A more detailed application of Andrade's theory gives the more complex viscosity temperature relation.

$$\eta v^{1/3} = Ae^{c/vt} \qquad ...(2)$$

where v is the specific volume.

This formula agrees closely with the results for many liquids.

In the case of gases, the viscosity increases with temperature. According to kinetic theory of gases, the variation of viscosity with temperature is given by

$$\eta = a\eta_0 T^{1/2} \qquad ...(3)$$

where a is a constant, η_0 the viscosity at 0°C and T the absolute temperature at which the viscosity is η.

14·4. Viscosity of Gases

Gases possess viscosity and the definition of viscosity of [Art. **14.8**] applies to gases as well as to liquids. The most remarkable property of the viscosity of the gases is that within wide limits of pressure, the viscosity is independent of the pressure, being under ordinary circumstances the same at a pressure of a few millimeters of mercury as at atmospheric pressure.

In following respects, the viscosity of a gas differs from that of a liquid :

(1) Viscosity of a gas increases with temperature, while that of a liquid decreases.

(2) In the case of a liquid, the layer of liquid in contact of the walls of the tube remains stationary, whereas in the case of gas, slipping occurs and the gas behaves as though the radius of tube were increasesd by λ, the mean free path of its molecules.

(3) Although, at moderate pressures, the coefficient of viscosity, like that of a liquid, is independent of pressure, but it is not true at very high and very low pressures. At very high pressures, the coefficient of viscosity of a gas increases with rise of pressure and at very low pressures, as λ exceeds the linear dimensions of the vessel, the coefficient of viscosity is proportional to the pressure.

(4) In the case of a liquid flow through a capillary tube, the volume of the liquid traversing any cross-section of the tube in a given time is constant. In the case of gases, since the density varies with pressure, it is only the mass, and not the volume passing any cross-section of the tube per second, which remains constant. Hence the Poiseuille's formula, deduced for an in compressible liquid, is not valid for compressible gases.

QUESTIONS

Long Answer Questions

1. What do you mean by an ideal fluid ? Prove that in streamline flow of an ideal fluid in tube, the product of area of cross-section of tube and velocity of fluid remains constant.
(Gwalior 1998; Rewa 98; Bhopal 95)
2. Write the equation of continuity of flow of liquid and prove it. **(Gwalior 2003)**
3. What is a tube of flow ? Obtain a relation between the area of cross-section and the velocity of the liquid at any point in a tube of flow. Explain the principle and working of a venturimeter, deriving the relation required.
(Poona 1962)
4. What are the various forms of energy possessed by a liquid ? Show that the various types of energy possessed by a liquid are mutually convertible, one into another.

5. What are the different forms of energy in a flowing liquid. Show that in stream line and frictionless flow of a liquid, the total mechanical energy of the liquid remains constant at each point. **(Indore 2004; Rewa 1998)**
6. Deduce Euler's equation for the flow of non-viscous liquid. **(Gwalior 1997; Raipur 1999)**
7. Write down the Euler's equation of motion of a fluid. By integrating it show that the pressure energy and kinetic energy of the fluid are interchangeable. **(Bilaspur 2003; Gwalior 1990; Bhopal 92; Indore 98; Vikram 94; Jabalpur 99; Sagar 98; Rewa 99)**
8. State and explain the principle of continuity in the flow of an ideal liquid. Derive Euler's equation of motion for the flow and hence obtain Bernoulli's theorem from it. Explain two applications of Bernoulli's theorem. **(Gwalior 1999; Bilaspur 98; Ujjain 91; Bhopal 91; Rewa 97; Raipur 95; Indore 92, 95; Sagar 97)**
9. State and prove Bernoulli's theorem and mention its applications. **(Rewa 2004; Bhopal 03; Jabalpur 03, 1999; Indore 99; Gorakhpur 95; Allahabad 83)**
10. Give Bernoulli's theorem and describe how the passage of water through a constriction in the tube gives rise to reduction in pressure.
11. State Bernoulli's theorem and apply it to obtain an expression for the reduction of pressure when water flows through a constriction in a pipe. Describe a device, depending upon this theorem, for the measurement of liquid flow. **(Punjab 1962)**
12. State and explain Bernoulli's theorem. Show how this is applied to measure the rate of discharge through city water mains. **(Punjab 1959)**
13. State and prove the Torricelli's theorem. **(Rewa 1993; Gwalior 98; Jabalpur 98; Indore 99; Raipur 99)**
14. State and derive Bernoulli's theorem for the flow of liquid. Describe its application to the working of a venturimeter. **(Delhi 1967)**
15. State Bernoulli's theorem for ideal liquids and with its help show that the velocity of efflux of a liquid from an orifice in the wall of a vessel containing liquid upto a height h above the orifice is $\sqrt{2gh}$. **(Ujjain 2004; Bhopal 1999)**
16. The area of cross-section at two points of a venturimeter are a_1 and a_2 where the difference in pressure is equal to the height h of column of liquid. Obtain an expression for the volume of liquid flowing out per sec from the tube. **(Ujjain 1995; Sagar 95; Rewa 98; Raipur 99; Bilaspur 96; Jabalpur 97)**
17. What is viscosity ? Define coefficient of viscosity. Establish Poiseuille's equation. State the assumptions involved. **(Rewa 2004; Bhopal 04; Indore 03; Sagar 03; Agra 1993; Purvanchal 94; Kanpur 84)**
18. Define coefficient of viscosity. Describe how the coefficient of viscosity is determined for water ? **(Agra 1992)**
19. Show that the velocity porfile of viscous fluid flowing through a capillary tube of uniform circular cross-section is parabolic. **(Gorakhpur 1995; 80; Kanpur 80; Rohilkhand 84)**
20. Establish Poiseuille's formula for liquid flowing through a capillary tube. What are the limiatation of the formula ? What corrections are possible to it ? **(Kanpur 2005)**
21. Deduce an expression for the distribution of velocity of a liquid flowing through a uniform capillary of circular cross-section. What is the nature of velocity profile ? **(Agra 1995; 84)**
22. What is Newton's law of viscous flow ? Define coefficient of viscosity and critical velocity. Derive an expression for the rate of flow of liquid through capillaries in parallel. **(Kanpur 1990)**
23. Obtain an equation for the volume of viscous fluid flowing per second through a narrow tube of circular cross-section. On the basis of this, find expressions for flow through two capillaries of different radii and lengths (i) in series, (ii) in parallel. **(Kanpur 1979)**
24. Two capillaries of lengths l_1, l_2 and radii r_1, r_2 are connected in : (i) series, (ii) parallel. Find the rate of flow of liquid in each case. **(Bundelkhand 2004; Sagar 91, 95; Jabalpur 89; Gwalior 93; Raipur 96)**
25. Write down Poiseuille's formula for the rate of flow of a liquid through a capillary. From this show that if two capillaries of radii r_1 and r_2 having length l_1 and l_2 respectively are set in series, the rate of flow V is given by,

$$V = \frac{\pi P}{8\eta}\left(\frac{l_1}{r_1^4}+\frac{l_2}{r_2^4}\right)^{-1},$$

where P is pressure difference across the arrangement and η is the coefficient of viscosity.

(Agra 1994, 87; Kanpur 83; Rohilkhand 81)

26. A cylindrical vessel of constant cross-sectional area A is filled with a liquid to a depth h above an orifice in its side. A horizontal capillary tube of length l and mean internal radius a is attached to the orifice. Show that the volume of liquid in the vessel and above the side orifice will be halved in time $\frac{8\eta lA}{\pi a^4 \rho g}$ log 2, where η is the coefficient of viscosity of the liquid, g the acceleration due to gravity and ρ the density of the liquid.

27. (a) Distinguish between streamline and turbulent flows of a liquid and explain the significance of the Reynold's number. **(Agra 2006)**

(b) Derive Poiseuille's formula for the rate of steady flows of a liquid through a capillary tube of circular cross-section. Why does it fail in the case of a gas ? **(Agra 2006)**

Short Answer Questions

1. State the characteristics of an ideal liquid. **(Raipur 1999; Bhopal 93)**

[**Ans.** Ideal liquid is non-viscous and incompressible.]

2. What is the total energy of fluid at a point ? **(Rewa 1998; Bhopal 93)**

3. Distinguish between streamline and turbulent motion of a liquid. **(Ujjain 1999; Sagar 99; Bilaspur 98; Purvanchal 95; Ranchi 66)**

4. Can two streamlines cross in steady flow of a liquid ?

[**Ans.** No.]

5. Explain the equation of continuity. **(Rewa 1998)**

6. What do you mean by Reynold number ? **(Agra 2004)**

7. Write down the Euler's equation for the motion of non-viscous liquid. **(Gwalior 2003)**

8. Derive Euler's equation of fluid mechanics. **(Kanpur 2004)**

9. State and prove Bernoulli's theorem. **(Kanpur 2005)**

10. Water is flowing in a horizontal tube of non-uniform cross-section in steady state. At a point A the velocity of flow of water is five times the velocity of flow at another point B. How many times will be the diameter of the tube at A relative to that at B ?

11. There is an orifice in the wall of a container filled with water and the water is emerging out. If the diameter of the orifice is increased a little, what will be the difference in (a) velocity of efflux of water and (b) volume of the water coming out per second ?

[**Ans.** (a) No difference; (b) More volume.]

12. In order to supply the water at a distant plant, a gardner partially closes the exit hole of the pipe by putting his finger on it. Explain why this results in the water stream going to a larger distance.

13. Water is coming out slowly from a vertical pipe. We observe that as the water descends after coming out, its area of cross-section reduces. Explain this on the basis of equation of continuity.

14. Why in a river deeper water appears to be still ?

15. Explain the following :

(i) The front end of the wing of an aeroplane is round shaded, while the rear end is flattened.

(ii) A ping-pong ball placed on the stream of a fountain dances.* **(Ujjain 1994)**

(iii) As the parachute of a soldier falling from an aeroplane opens, his acceleration decreases and ultimately becomes zero. **(Sagar 2003)**

(iv) The pressure of water is reduced when it passes from the wider tube to a narrower tube.

16. Define coefficient of viscosity. State its dimensional formula and S.I. unit. How are the S.I. unit and C.G.S. unit of coefficient of viscosity related ? **(Indore 2003; Bhopal 04)**

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* As the stream of water from the fountaion rises up with a very large velocity, then according to Bernoulli's theorem, the air pressure in it decreases. When a ball placed on the stream and goes out of the stream, the outer air pushes back into the stream, where there is low pressure region. Thus the ball dances on the stream of a fountain.

17. What are the conditions to apply Poiseuille's formula ? **(Sagar 1989)**
[**Ans.** (i) Flow of liquid should be streamline., (ii) the capillary should be horizontal, and (iii) the pressure across the end of the tube must remain constant.]

18. Prove that velocity profile of a viscous liquid flowing through a capillary tube of uniform cross-section is parabolic.

19. If the flow of a liquid through a capillary tube is streamline flow, then at which place point of the tube, the velocity is : (i) maximum; (ii) minimum.

20. Two capillary tubes whose lengths and radius are l_1, r_1 and l_2, r_2 respectively have pressure difference P across their ends. If 'η' is the viscosity coefficient of the liquid, then what will be the volume of the liquid flowing per second, if they are connected in (i) series; (ii) parallel ?

21. A horizontal capillary is connected at the bottom of a vessel. The volume of liquid flowing out per sec through the capillary is Q. What will be the rate of flow of liquid, if : (i) the radius of capillary is doubled, (ii) another capillary of same length and same radius is connected in series with the first capillary ? **(Gwalior 1995)**
[**Ans.** (i) 16 Q; (ii) $Q/2$.]

22. A capillary is connected to a tank of water and the volume of water flowing through it is v m/sec. If the radius of capillary is doubled, what will be the flow rate ?
[**Ans.** $16v$.]

23. A horizontal capillary is conected to a water tank and V is the volume of water flowing per second through it. If the capillary is broken in the middle then what will be the flow rate ?
[**Ans.** $2V$.]

24. The diamater of a metallic bullet is double the diameter of other one. What will be the ratio of their terminal velocities if they are dropped in water ?

25. What is the effect of temperature on the viscosity of liquids ?

26. A small ball is released in a jar containing glycerine. After some time, the ball attains a constant speed. Explain this. **(Sagar 1997)**

27. Explain how we can determine the coefficient of viscosity of a fluid by measuring the terminal velocity of a tiny sphere falling freely through it. **(Agra 2005)**

28. Derive Stoke's formula for the velocity of a small sphere falling through a viscous liquid. **(Kanpur 2004)**

29. Does the velocity of falling rain drop continuously increase ? Explain.

Objective Type Questions

1. For an ideal fluid, the value of coefficient of viscosity is :
(a) zero (b) infinity (c) negative (d) positive
(Gwalior 1990)

2. A cylindrical container has an orifice near the bottom. The velocity of the flowing liquid out of the orifice does not depend on :
(a) area of orifice
(b) height of the liquid surface over the orifice
(c) density of the liquid
(d) acceleration due to gravity

3. Venturimeter works on :
(a) the principle of Archimedes
(b) Stoke's law
(c) Bernoulli's theorem
(d) Equation of continuity

4. Bernoulli's theorem is based on the principle of conservation of :
(a) momentum (b) mass (c) energy (d) angular momentum

5. A cylinder is filled with non-viscous liquid of density ρ to height h_0 and a hole is made at a height h_1 from the bottom of the cylinder. The velocity of the liquid coming out of the hole is :
(a) $\sqrt{2h_0 g}$ (b) $\sqrt{2g(h_0 - h_1)}$ (c) $\sqrt{g\rho h_1}$ (d) $\sqrt{g\rho h_0}$

6. A water tank on the top of a tall building feeds water in the taps on different floors. The water will rush out at highest speed from a tap on floor
(a) nearest to the tank
(b) farthest from the tank
(c) near middle of the building
(d) speed will be the same on all the floors.

7. Magnus effect is very near to the :
(a) magnetic field
(b) electric field
(c) Bernoulli's theorem
(d) magnetic effect of current

8. Flight of an aeroplane works on :
(a) Archimedes principle (b) Pascal's law
(c) Bernoulli's theorem (d) Stoke's law

9. Water is flowing in a capillary tube, If the velocity of flow of water is 6 m/s, the velocity head of water will be :
(a) 6 m (b) 60 m (c) 18 m (d) 1.8 m **(Rewa 2000)**

10. The unit of velocity head is :
(a) metre (b) kg/s (c) m/s^2 (d) m/s **(Rewa 1995, 92)**

11. The property of viscosity is found :
(a) only in liquids (b) only in solids
(c) only in solids and liquids (d) only in liquids and gases

12. The dimension of velocity gradient is :
(a) $[LT^{-1}]$ (b) $[T^{-1}]$ (c) $[L^2T^{-1}]$ (d) $[LT^{-2}]$ **(Rewa 1994)**

13. The coefficient of viscosity of a liquid is equal to the external force that acts between two successive layers of unit cross-sectional area of liquid to :
(a) balance internal frictional force between layers
(b) annul viscous force between layers
(c) maintain unit velocity gradient between layers
(d) maintain the motion of liquid between the layers **(Gwalior 1991)**

14. The property of a liquid by virtue to which it opposes the relative motion between its different layers is :
(a) elasticity (b) surface tension (c) viscosity (d) none of these

15. An object entering the earth's atmosphere at a high velocity catches fire due to :
(a) viscosity of air (b) the heat content of the atmosphere
(c) the pressure of the certain larger gases (d) the higher force of gravity **(Kanpur 2001)**

16. Rate of flow of a liquid in a capillary is :
(a) directly proportional to the pressure difference across its ends
(b) directly proportional to the cross-sectional area of the capillary
(c) directly proportional to the length of the capillary
(d) directly proportional to the coefficient of viscosity of flowing liquid **(Gwalior 1989)**

17. A siphon of glass capillary is connected to a bottle. If the viscosity coefficients of water and petrol are 0.01 and 0.02 poise, then the ratio of time taken to evacuate the bottle [if it is filled by water and petrol (relative density 0.8)] is :
(a) 5 : 2 (b) 1 : 2 (c) 8 : 2 (d) 2 : 51 **(Rewa 1992)**

18. Raindrops fall near the surface of the earth with :
(a) terminal acceleration (b) terminal velocity
(c) terminal retardation (d) varying velocity

19. The diameter of a spherical body A is half the diameter of another spherical body B. The ratio of their terminal velocities in the water is :
(a) 1 : 2 (b) 1 : 1 (c) 1 : 4 (d) 2 : 1 **(Rewa 1999)**

20. According to Stoke's law, the viscous force is :
(a) $6\pi\eta a/v$ (b) $6\pi av/\eta$ (c) $6\pi\eta v/a$ (d) $6\pi\eta av$ **(Rewa 1997)**

21. The small liquid drops falling freely in air reaches a terminal velocity v because of :
(a) weight of the drop (b) viscous force of air
(c) surface tension (d) none of these

22. The viscous force on a spherical body of radius r falling in a medium of viscosity η with speed is given by :
(a) $\pi vr\eta$ (b) $6\pi v/r\eta$ (c) $6\pi vr\eta$ (d) $\pi vr\eta/6$
(Bundelkhand 2001)

23. If the diameter of any sphere A is half of the diameter of other sphere B, then the ratio of their terminal velocities will be :
(a) 1 : 2 (b) 2 : 1 (c) 1 : 4 (d) 8 : 1
(Agra 2004; Kanpur 04)

ANSWERS

1. (a), **2.** (c) **3.** (c), **4.** (c), **5.** (b), **6.** (b), **7.** (c), **8.** (c), **9.** (d), **10.** (a), **11.** (d), **12.** (b), **13.** (c), **14.** (c), **15.** (a), **16.** (a), **17.** (d), **18.** (b), **19.** (c), **20.** (d), **21.** (b), **22.** (c), **23.** (c).

15

Surface Tension

15·1. Free Surface of a Liquid is in the State of Tension

The free surface of a liquid always behaves like a stretched elastic membrane. This means that in the free surface of a liquid there are acting forces of tension due to which the free surface has a natural tendency to acquire smallest possible area. This special property of a liquid can be illustrated by the following experiments :

(1) If a circular wire ring be dipped into a soap solution and carefully taken out, a thin soap film is formed across it. On placing a wet loop of cotton thread lightly on the film, we find that the loop lies as it is and it does not show any tendency to change its shape **[fig. 15.1 (a)]**. Now, if the film, inside the loop, is pricked with a hot sharp needle, the inside film disappears and the loop of thread at once takes up the circular form **[fig. 15.1 (b)]**, due to the outward pull (arising due to the film lying outside the loop).

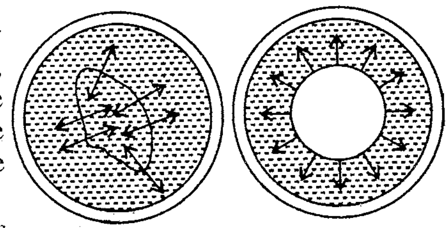


Fig. 15.1.

We know that for a given perimeter, a circle encloses maximum area. Obviously, when the thread is pulled into the circular form, the area of this film between loop and the ring is minimum possible.

(2) If we hold an ordinary camel-hair paint brush under the surface of water, its hairs will be seen separately spread out. As we take it out, the hairs come closer together in a compact mass as though they were taken closer by the contraction of some invisible memebrane.

(3) A freely suspended drop of water, formed at the end of a tap, assumes spherical shape. We know that for a given volume sphere has the least surface area. Therefore, to acquire minimum possible surface area, the drop takes the spherical shape **[fig. 15.2]**.

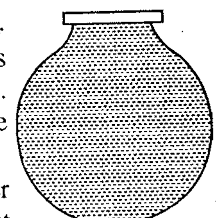


Fig. 15.2.

(4) If we place a greased needle on a piece of blotting paper and put the paper lightly on the surface of water, the blotting paper will soon sink to the bottom, but the needle is found to float on the surface although it is much heavier than water **[fig. 15.3]**. On examining the surface of water below the needle, it is seen to be slightly depressed as if the liquid surface were acting as a stretched membrane, so that the weight of the needle is balanced by the inclined upward tension forces in the membrane. If one end of the needle is pressed to pierce a hole in the surface of water, it goes slantingly downward.

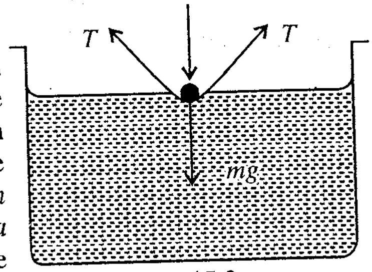


Fig. 15.3.

The above experiments clearly show that the surface of a liquid behaves as though it were covered with stretched elastic membrane, having a natural tendency to contract. Thus, there are forces of tension in the free surface of a liquid. However there is one important difference between the elastic memebrane and the surface of a liquid that in an elastic memebrane the tension increases with stretching or its surface area is increased in accordance with the Hooke's law, whereas the *tension in the liquid surface is quite independent of stretching or of the area of its surface,* unless the stretching force reduces the thickness of the film below 5×10^{-6} mm after which the tension diminishes rapidly.

15·2. Surface Tension

It is clear from above that the free surface of a liquid is in a state of tension. This pull or tension in the liquid surface is known as **surface tension.**

Let us imagine a straight line AB of length l, drawn in the free surface of a liquid, then on account of tension of the liquid surface, the molecules on the two sides of the line will have a tendency to pull away from each other [**fig. 15.4**]. The direction of the pulling force is at right angles to AB and tangential to the liquid surface. The magnitude of this force is found to be proportional to the length of the line. Thus, if F be the force acting at right angles on either side of the line AB, then

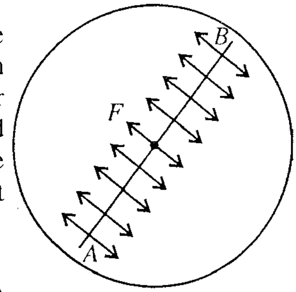


Fig. 15.4.

$$F \propto l$$

or
$$F = T \cdot l \qquad ...(1)$$

where T is a constant for a liquid at a particular temperature. This constant is called the **surface tension** of the liquid.

If $l = 1$ unit, $$F = T \qquad ...(2)$$

Thus, *surface tension of a liquid is defined as the tangential force per unit length, acting at right angles on either side of a line imagined to be drawn in the free liquid surface in equilibrium.*

At the boundary of a liquid surface, the force of surface tension acts inwards only, whereas at the other places in the surface it acts on either side of the imaginary line.

The magnitude of surface tension depends upon the nature of the liquid and its surrounding medium. If the surrounding medium, is not stated, it is supposed to be air.

S.I. unit of surface tension is **newton per metre** and its dimensional formula can be derived as follows :

$$[\text{Surface tension}] = \frac{[\text{Force}]}{[\text{Length}]} = \frac{[MLT^{-2}]}{[L]} = [MT^{-2}] \qquad ...(3)$$

C.G.S. unit of surface tension is **dyne/cm** and S.I. unit is **N/m** (1 dyne/cm = 10^{-3} N/m)

15·3. Molecular Forces

Surface tension is essentially a molecular phenomenon, hence we must be first clear about the forces that act between the molecules.

There are two kinds of molecular forces :

(i) force of adhesion or adhesive force, and (ii) force of cohesion or cohesive force.

The force of attraction between the molecules of different susbstances is called **force of adhesion,** while that between molecules of the same substance is called **force of cohesion.**

The force of cohesion is entirely different from ordinary gravitational force and does not obey the ordinary inverse square law. Probably, this force varies inversely as the eighth power of the distance between two molecules ($F \propto 1/r^8$) and therefore falls of rapidly with distance and becomes almost negligible at a distance of 10^{-9} m.

Molecular range and sphere of influence : *The maximum distance upto which the force of cohesion between two molecules can act is called the* **molecular range.** *It is of the order of* 10^{-9} *m being different for different substances. The sphere, drawn with a molecule at centre, having radius equal to molecular range, is called* **sphere of influence.** A molecule attracts and in turn is attracted by only those molecules which are enclosed within its sphere of influence.

15·4. Explanation of Surface Tension in Terms of Molecular Forces

The tendency of a liquid surface to contract to a minimum possible area suggests that it is in a state of tension, which we call as surface tension. This property of liquid sufrace has been explained on the basis of molecular theory by Laplace and is known as the **Laplace theory of molecular interpretation of surface tension.**

Now referring to **fig 15.5**, consider three molecules A, B and C of a liquid, with their spheres of influence drawn around them. Since the sphere of influence of A lies completely inside the liquid, it is equally attracted in all directions by molecules within its sphere of influence. Henece, *there is no resultant cohesive force acting on it.*

In the case of molecule B, the sphere of influence lies partly outside the liquid. The number of molecules in the upper half, attracting the molecule B in the upward direction is less than the number of molecules in the lower half attracting it in the downward direction. Hence, *a resultant downward force is acting on it.*

The sphere of influence of molecule C is exactly half outside and half inside the liquid. Hence, *this molecule is attracted in the downward direction with maximum force, perpendicular to the surface.*

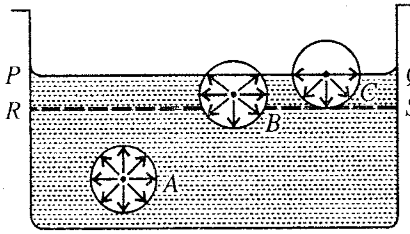


Fig. 15.5. Explanation of surface tension by molecular theory.

Now, if we draw a plane RS parallel to free surface PQ of the liquid at a distance equal to the molecular range, then the layer of the liquid between the planes PQ and RS is termed as the **surface film.** Clearly, what is true for molecules B and C is true for all other molecules in the surface film. Hence, all the molecules in the surface film are pulled downwards due to the resultant cohesive force, the magnitude of which increases from surface RS to free surface PQ.

Now, if a molecule from the interior of the liquid is brought up to the surface film, work has to be done against the downward cohesive force on it and its potential energy increases. Therefore, ***the potential energy of the molecules in the surface film is greater than the potential energy of the molecules lying below it.*** *A system in equilibrium has a tendency to acquire lowest possible potential energy. Thus, to decrease the potential energy of the molecules in the surface film, the film tries to have minimum number of molecules in it. This can be done only by decreasing the surface area of the film,* because its thickness is already fixed (being equal to molecular range). *This is the reason, why the free surface of a liquid always tends to have minimum possible area and consequently it is in the state of tension.*

15·5. Surface Energy

We have seen above that the potential energy of the molecules in the surface film is greater than that of those inside the liquid. *The excess of potential energy per unit area of the film is called its* **surface energy.**

We know that the surface of a liquid is in a state of tension and hence it has a tendency to decrease its surface area. Therefore, if the area of a liquid surface is increased, work has to be done against surface tension. This work is stored up in the surface film as part of **surface energy** of the new surface area created.

Let us consider a rectangular wire frame $PQRS$ with two parallel sides PS and QR at a distance l m apart. The wire PQ is free to move up or down. Let a soap film be formed across $PQRS$ by dipping the frame into a soap solution for a short while. Naturally there will act a force on the movable wire PQ in the upward direction due to surface tension. This force will act normally to the wire and in the plane of the film. If T be the surface tension of the liquid soap solution, then upward force on the wire PQ is given by

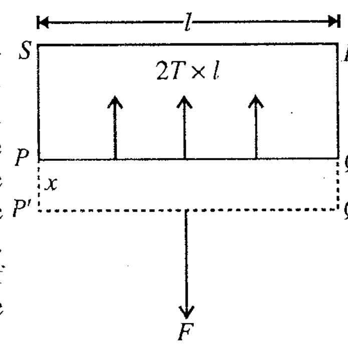


Fig. 15.6.

$$2T \times l = 2Tl \qquad ...(1)$$

where factor 2 arises, because there are two surfaces (the front and the back) of the soap film and each has a surface tension T.

Hence, to keep the wire in equilibrium, an external force $F = 2Tl$ newton must be applied (the weight of the wire PQ itself is also included in F). Now, if the movable wire PQ be pulled downward through a small distance x into the position $P'Q'$, the work done by the external force

$$W = \text{Force} \times \text{Distance} = F \times x = 2Tlx \quad ...(2)$$

This work done forms the surface energy of the increased surface area. Increase in the surface area of the film is $A = 2lx$. Hnece,

$$\text{Work done per unit area} = \frac{W}{A} = \frac{2Tlx}{2lx} = T \quad ...(3)$$

Thus, *the surface tension is defined as the amout of work done in increasing the surface area of liquid film by unity under isothermal conditions. The surface tension can thus be expressed in joules per square metre. Hence the unit of surface tension is* **N/m** *or* **J/m²**.

It has been assumed above that the temperature of the film remains constant. But when the film is stretched, molecules come to form new surface from the interior of the liquid. These molecules have greater potential energy than those inside the liquid. This excess of potential energy is supplied by the kinetic energy of the molecules. Consequently, the film gets cooled and to keep its temperature constant, heat is supplied from the surrounding medium. Thus, the surface energy of the new area, formed, is the sum of mechanical work done and heat absorbed from the atmosphere.

If E be the surface energy of the film per unit area and H the amount of heat supplied per unit change of area to keep the temperature constant, then

$$E \cdot 2lx = 2lx \cdot T + H \cdot 2lx \quad \text{or} \quad E = T + H$$

$$\therefore \quad T = E - H \quad ...(4)$$

Hence, *surface tension is equal to the mechanical part of the surface energy, which may be called as free surface energy.*

15·6. Pressure Difference across a Curved Surface

When the free surface of a liquid is plane, as in **fig. 15.7** (a), a molecule in the surface is attracted by other liquid molecules equally in all directions. Consequently, the resultant force due to surface tension is zero. But if the liquid surface is curved, there is a resultant force of surface tension which acts normally to the surface. For a convex liquid surface [**fig. 15.7(b)**], this resultant force is directed inwards towards the interior of the liquid, while for a concave liquid surface [**fig. 15.7(c)**], it acts outwards away from the surface.

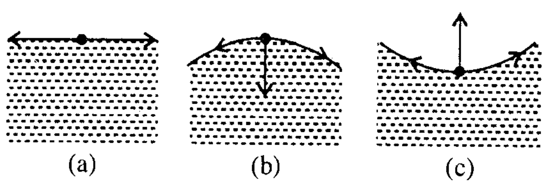


Fig. 15.7.

Hence, for the equilibrium of a curved surface, there must be a pressure difference across it and then the resultant due to surface tension will be balanced by the excess of pressure force, acting on the *concave side.*

Expression for Excess Pressure on a Curved Surface : If we have a curved liquid surface at rest, then the inward pressure on it due to surface tension must be balanced by an equal excess of pressure outward, acting on the concave side.

Let $ABCD$ be a small curvilinear rectangle of the curved liquid surface. Its side AB is of length x and radius of curvature r_1 with centre at O_1 and the side BC is of length y and radius of curvature r_2 with centre at O_2 [**fig. 15.8**]. Clearly, $AO_1 = BO_1 = r_1$ and $BO_2 = CO_2 = r_2$. Geometrically radii of curvature, such as AO_1 and BO_2, are called principal radii of curvature. Area of the rectangle $ABCD = xy$.

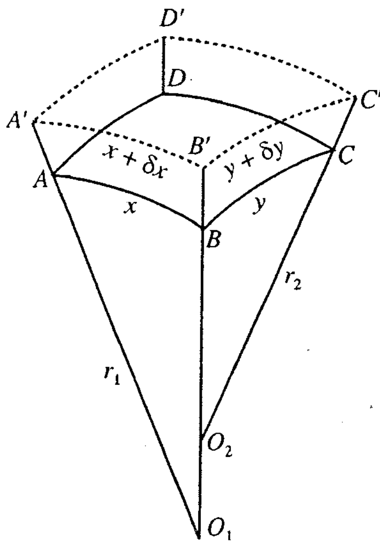


Fig. 15.8.

Let the excess of pressure on the concave side be p. When the liquid surface is at rest, the outward force on the element $ABCD$ is $= pxy$.

Suppose the surface is expanded by giving an infinitesimally small normal displacement δz (= AA') so that the element occupies a new position $A'B'C'D'$. Hence, the work done by the excess pressure p is

$$W = \text{Force} \times \text{Displacement} = pxy \times \delta z \quad \text{...(1)}$$

If $x + dz$ and $y + dy$ be the lengths of $A'B'$ and $B'C'$ respectively, then increase in the area of the liquid surface under consideration is

$$= (x + \delta x)(y + \delta x) - xy = x\delta y + y\delta x$$

The product $\delta x \delta y$ of very small quantities has been nelgected.

$\therefore$ Increase in surface energy = Surface tension × Increase in area = $T(x\delta y + y\delta x)$ ...(2)

This increase in the surface energy is equal to the work done in expanding the surface by excess of pressure, hence equating (1) and (2), we get

$$pxy\delta z = T(x\delta y + y\delta x)$$

or

$$p = T\left(\frac{1}{y}\cdot\frac{\delta y}{\delta z} + \frac{1}{x}\cdot\frac{\delta x}{\delta z}\right) \quad \text{...(3)}$$

Now from similar triangles ABO_1 and $A'B'O_1$, we have

$$\frac{A'B'}{A'O_1} = \frac{AB}{AO_1} \quad \text{or} \quad \frac{x+\delta x}{r_1+\delta z} = \frac{x}{r_1} \qquad [\because A'O_1 + AA' = r_1 + \delta z]$$

or

$$\frac{x+\delta x}{x} = \frac{r_1+\delta z}{r_1} \text{ or } 1 + \frac{\delta x}{x} = 1 + \frac{\delta z}{r_1} \text{ or } \frac{\delta x}{x} = \frac{\delta z}{r_1}, \quad \therefore \frac{1}{x}\cdot\frac{\delta x}{\delta z} = \frac{1}{r_1}. \quad \text{...(4)}$$

Similarly, from similar triangles BCO_2 and $B'C'O_2$, we have

$$\frac{1}{y}\cdot\frac{\delta y}{\delta z} = \frac{1}{r_2}. \quad \text{...(5)}$$

Substituting these values of $\frac{1}{x}\cdot\frac{\delta x}{\delta z}$ and $\frac{1}{y}\cdot\frac{\delta y}{\delta z}$ in eq. (3), we get

$$p = T\left(\frac{1}{r_1} + \frac{1}{r_2}\right) \quad \text{...(6)}$$

If one of the curvatures is **convex** and the other is **concave**, the radii of curvature r_1 and r_2 are of **opposite signs**. Thus, in such a case, we have

$$p = T\left(\frac{1}{r_1} - \frac{1}{r_2}\right) \quad \text{...(7)}$$

Combining the equations (6) and (7), we may write the general relation as

$$p = T\left(\frac{1}{r_1} \pm \frac{1}{r_2}\right) \quad \text{...(8)}$$

If instead of a liquid surface, there are two surfaces of a film or membrane, then the excess of pressure is given by

$$p = 2T\left(\frac{1}{r_1} \pm \frac{1}{r_2}\right) \quad \text{...(9)}$$

The relations (8) and (9) hold for all surfaces. A few important particular cases are given below.

(i) Case of a spherical surface : (a) In case of a single spherical surface such as that of a liquid drop or an air bubble inside the liquid, we have $r_1 = r_2 = r$ (say). Hence,

$$p = T\left(\frac{1}{r}+\frac{1}{r}\right) = \frac{2T}{r} \qquad ...(10)$$

(b) In case of a spherical film, which has two surfaces such as a soap bubble, the expression for excess pressure is

$$p = 2 \times \frac{2T}{r} = \frac{4T}{r} \qquad ...(11)$$

(ii) Case of cylindrical surface : Here $r_1 = r$ (say) and $r_2 = \infty$, hence for one surface,

$$p = T\left(\frac{1}{r}+\frac{1}{\infty}\right) = \frac{T}{r} \qquad ...(12)$$

For two surfaces, $$p = \frac{2T}{r}.$$

(iii) Case of a catenoid : If the surface is one of revolution with no difference of pressure, the surface is a catenoid. Here $p = 0$, hence, we have

$$\frac{1}{r_1} \pm \frac{1}{r_2} = 0 \qquad ...(13)$$

An example of such a surface is that of a a soap film, supported in between two parallel rings and the ends of the film are burst so that

$$p = 0. \qquad ...(14)$$

15·7. Excess Pressure inside a Spherical Drop or Bubble—Alternative Procedure

(1) Excess pressure inside a spherical drop : The surface of a spherical drop is convex **[fig. 15.9]** and therefore, the molecules on the surface experience a resultant force, acting inwardly. For the equilibrium of the spherical surface, the resultant inward force must be balanced by the excess of pressure force, acting on the concave side.

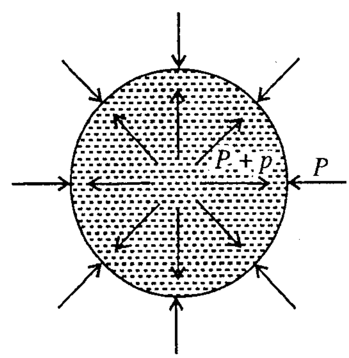


Fig. 15.9 : Spherical liquid drop.

Let r be the radius of the drop and let p be the pressure on the convex side and $P + p$ that on its concave side **[fig. 15.9]**. Therefore, excess of pressure inside the drop is p.

Now let us consider the equilibrium of one half of the drop, as shown in the **fig. 15.10**. Neglecting the weight of the drop, there are two forces acting on it :

(i) The outward force on the plane face AB, due to the excess pressure p,

$= $ Excess of pressure $\times$ Area of plane face $AB = p \times \pi r^2$

(ii) The inward force, due to surface tension, acting on the edge of the circle AB,

$= $ Surface tension of the liquid $\times$ Circumference of the circle AB

$= T \times 2\pi r$

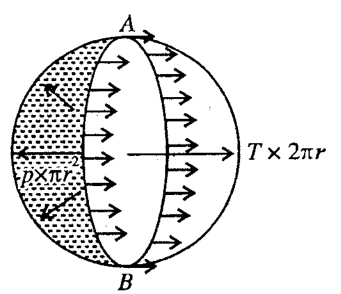


Fig. 15.10 : Equilibrium of one-half drop.

Since half of the drop, under consideration, is in equilibrium hence the outward force must be equal to inward force *i.e.*,

$$p \times \pi r^2 = T \times 2\pi r, \quad \therefore\ p = \frac{2T}{r} \qquad ...(1)$$

(2) Excess pressure inside an air bubble in a liquid : Inside a liquid, an air bubble will have only one spherical surface. There will be the excess of pressure p on the concave side [**fig. 15.11**], *i.e.*, inside the bubble and hence this case is dealt exactly in the same manner as described above. The expression for the excess of pressure, as derived above, is

$$p = \frac{2T}{r} \qquad ...(2)$$

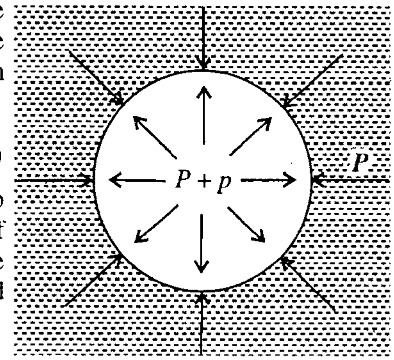


Fig. 15.11 : Air bubble in a liquid.

(3) **Excess pressure inside a soap bubble :** A spherical soap bubble has two free surfaces in contact of air. There is the excess of pressure, acting on the concave side. If p be the excess of pressure then by considering the equilibrium of half spherical bubble, outward force on the plane face AB [**fig. 15.12**] is

$$= p \times \pi r^2.$$

The inward force of surface tension on the circumference of the circle AB due to a single surface is $T \times 2\pi r$, hence the total force of surface tension due to both the surfaces is

$$= 2 \times T \times 2\pi r$$

For the equilibrium of the bubble, the outward force due to excess of pressure must be equal to the inward force due to surface tension.

Hence, $$p \times \pi r^2 = 2 \times T \times 2\pi r,$$

$$\therefore \qquad p = \frac{4T}{r} \qquad ....(3)$$

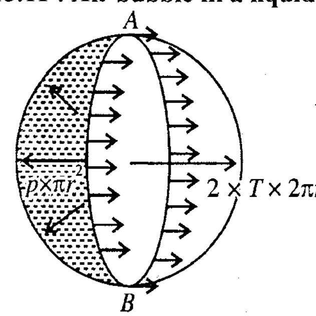


Fig. 15.12 : Soap bubble.

Thus we see that the excess pressure inside a drop or bubble is inversely proportional to its radius, so that the smaller the bubble or drop, the greater excess pressure exists inside it.

Demonstration of excess of pressure in a bubble : The fact that smaller the radius of the bubble, the greater is the excess of pressure inside required to maintain it in equilibrium, can be demonstrated by the following experiment.

Two tubes with equal circular apertures are connected to taps T_1, T_2 and T_3 as shown in **fig. 15.13**. Their lower ends are dipped in soap solution and withdrawn. With taps T_1 and T_2 open and T_3 closed, a small bubble A is formed by blowing air gently into the apparatus. Next with T_1 and T_3 open and T_2 closed, a larger bubble B is formed.

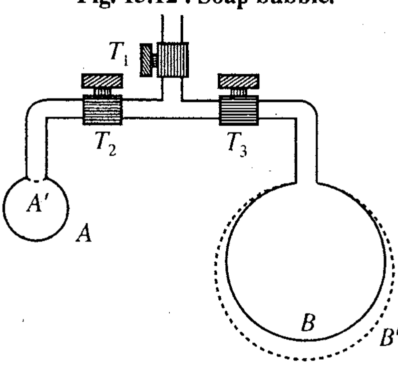


Fig. 15.13.

Closing the tap T_1, the two bubbles are put in communication with each other by opening the taps T_2 and T_3. We see that the smaller bubble A collapses gradually and the larger bubble B grows further; as shown by dotted lines, until their radii of curvature are equal*. This shows that the air has flown from the smaller to the larger bubble, and hence the smaller bubble must have had a greater pressure of air inside it than the larger one.

15·8. Measurement of Surface Tension of a Liquid by Forming a Bubble

The expression for the excess pressure inside a spherical soap bubble is given by

$$p = 4T/r. \qquad ...(1)$$

The use of this equation is made in measuring the surface tension of a liquid by forming a bubble.

* As the smaller bubble collapses and larger bubble is expanded, a certain stage is reached that the bubble A becomes hemisphere and on any further shrinkage, its radius of curvature increases. Stable equilibrium is reached when the curvature at A' is the same as the curvature of the bubble B'.

The apparatus, used for this purpose, is shown in **fig.15.14**. The end of the vertical glass tube is dipped in the liquid so that on taking out the tube a liquid film is formed at A. Next, opening the stop-cock B, a small bubble is blown on A. When a bubble of suitable size is obtained, the stop-cock is closed. The diameter of the bubble as well as the difference in level, say h, between two limbs of the manometer are measured by a travelling microscope. If r is the radius of the bubble and ρ the density of the manometer liquid, then

$$h\rho g = 4T/r. \quad \text{or} \quad T = h\rho g r/4 \qquad ...(2)$$

Therefore, by putting the measured values of different physical quantities h, ρ, g and r, one can determine the value of the surface tension T of the liquid used to form the bubble.

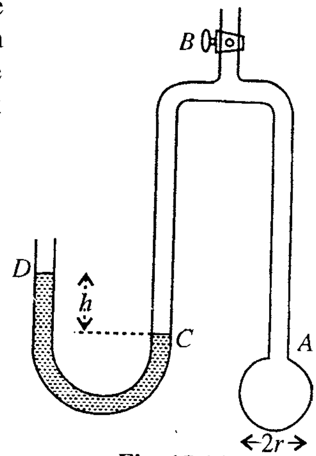


Fig. 15.14.

15·9. Film of Water between two Glass Plates

If a small drop of water is placed between two clean glass plates AB and CD and squeezed, a thin layer or film of water wets the plates over an almost circular area, say of radius R. The film, thus formed, is concave along its boundary. If d be the thickness of the film, then the radius of curvature of the concave meniscus at A or B is nearly $d/2$ [**fig. 15.15**].

Since R and $d/2$ are the radii of curvature of the free surface of water in two perpendicular directions, then the pressure inside the film is less than the atmospheric by an amount

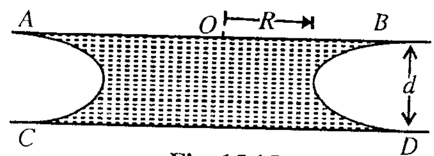


Fig. 15.15.

$$p = T\left(\frac{2}{d} - \frac{1}{R}\right) \qquad ...(1)$$

Negative sign of R indicates that the two centres of curvature of free surface lie in opposite directions. But R is very large when compared to d so that $1/R$ can be neglected in comparison to $2/d$. Hence, we have

$$p = 2T/d \qquad ...(2)$$

The excess pressure of the atmosphere on the two plates pushes them closer together. If A be the area of the film, then the force, pushing the plates together, is given by

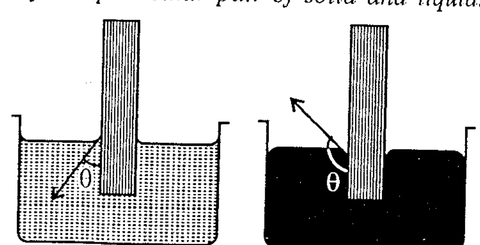
$$p = (2T/d)A \qquad ...(3)$$

Evidently, smaller the distance d between the plates, the greater is the force F pressing them together. Hence, when the thickness of the film is very small, a large force is required to draw apart the two plates normally.

15·10. Angle of Contact - Wetting

In general, when the free surface of a liquid comes in contact of a solid, it becomes curved near the place of contact. *The angle between the tangent to the liquid surface at the point of contact and the solid surface, inside the liquid, is called the angle of contact for a particular pair of solid and liquid.*

For example, if a plate of glass is dipped in water with its sides vertical, it will be noticed that water is drawn up along the plate and acquires a curved shape [**fig. 15.16 (a)**]. But if the glass is dipped in mercury, the surface becomes curved and depressed below [**fig. 15.16 (b)**]. *For pure water and very clean glass, the angle of contact is zero, but for ordinary water and glass it is about* 8°. In case of mercury and glass, its value is about 138°. In general, when the liquid has concave meniscus, the angle of contact is acute, while it is obtuse for the liquid having convex meniscus. *If the angle of contact between a liquid*



(a) Glass plate in water

(b) Glass plate in mercury

Fig. 15.16 : Angle of contact.

and solid is zero, the liquid is said to **wet the solid** *and this phenomenon is called* **wetting.** In fact the adhesive forces between the water and glass molecules are more than those cohesive forces between the molecules of water and the water wets the surface of glass.

The angle of contact depends upon the following :

(i) The nature of the liquid and solid.

(ii) The medium which exists above the free surface of the liquid.

When air is above the mercury, the angle of contact between mercury and glass is different to the angle of contact when there is a layer of water, exists above mercury.

(iii) The cleanliness and freshness of the given two surfaces in contact.

The angle of contact does not depend upon the angle of inclination of the solid to liquid surface.

15·11. Equilibrium of a Liquid Drop

Consider a liquid drop in equilibrium on the horizontal surface of a solid. Both solid and liquid are in contact of air. The common line of their contact is through P, perpendicular to the plane of the paper. Surface tension always occurs, wherever a surface boundary is present. Let T_1, T_2 and T_3 be the surface tensions for liquid air, solid air and liquid-solid respectively. Then, these quantities represent the forces per unit length of the line of contact in the direction shown in **fig. 15.17**.

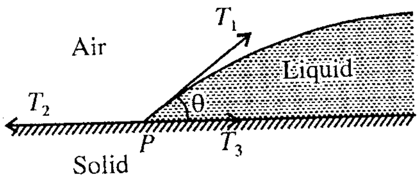


Fig. 15.17.

Let θ be the angle of contact of the liquid with the solid. Since the system is in equilibrium,

$$T_1 \cos\theta + T_3 = T_2, \qquad \therefore \qquad \cos\theta = \frac{T_2 - T_3}{T_1}$$

Thus, if T_2 is greater than T_3, $\cos\theta$ is positive and hence θ is less than 90°. If $T_2 < T_3$, $\cos\theta$ is negative and lies between 90° and 180°. This is the case with mercury on glass, where T_3 is large and θ is about 140°. Therefore, mercury collects itself into drops, when placed on a clean glass plate.

If in case, $T_2 > T_1 + T_3$, there will be no equilibrium and the liquid will spread over the solid surface. This is the case of water, when placed over a perfectly clean glass plate and the **water is said to wet the glass plate.**

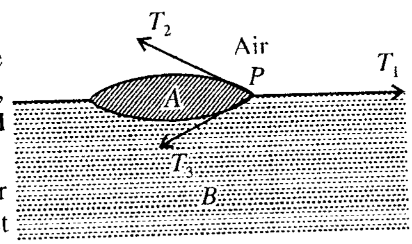


Fig. 15.18.

In case, if a drop of liquid A is placed on the surface of another immiscible and heavier liquid B [**fig. 15.18**], both being in contact of air, there are three surface tensions to consider, namely, (i) T_1, that of the surface between air and liquid B, (ii) T_2, that between air and liquid A and (iii) T_3, that between the liquids A and B.

For equilibrium, the three forces T_1, T_2 and T_3, must be such that the sum of any two must be greater than the third. Thus, a triangle can be constructed, whose sides are proportional to T_1, T_2 and T_3 and taken in order. This triangle of forces is known as **Neumann's triangle**. If this triangle is not formed, the equilibrium of the drop is not possible.

If the figure is a section at right angles to the line of contact through P, then the line of contact would be acted upon by forces T_1, T_2 and T_3 N-m^{-1}. If the drop is in equilibrium, Neumann's triangle can be constructed. In fact, we are not known with any two pure liquids for which it is possible to construct the triangle of forces. One of the three surface tensions is always greater than the sum of other two so that there is no equilibrium like that shown in the figure. Thus the lighter liquid spreads over the surface of the liquid on which it rests. For example, when a drop of pure water is placed on the surface of mercury, it spreads over it and forms a uniform thin layer, since the surface tension of mercury is about 0.55 N-m^{-1} and of water 0.075 N-m^{-1}.

If, however, mercury surface is contaminated with grease, its surface tension decreases, and then it is possible for drops of water to stand upon it. Thus, now the Neumann's triangle can be constructed.

15·12. Some Examples of Surface Tension

(1) Shape of drops : The shape, assumed by a liquid drop, is governed by the action of two forces: (i) Force of gravity, and (ii) Force of surface tension.

The condition for the shape of the drop, in equilibrium, is that its total potential energy (*i.e.*, gravitation P. E. + P. E. due to surface tension) must be minimum.

In case, if forces due to surface tension alone are acting on the drop, its potential energy will be minimum, when its surface area is the least possible. Since for a given volume, sphere has the least surface area, the liquid drop will acquire a spherical shape.

Fig. 15.19.

On the other hand, if forces of gravity alone were acting on the drop, its gravitational potential energy would be minimum, when the centre of gravity of drop is at the lowest position. Hence, the drop must spread out or flattened in order to bring the centre of gravity of the drop at the lowest position.

In the present case, both types of forces are acting on the drop simultaneously. Hence, the resultant shape of the drop is a compromise of both in such a way that the potential energy due to surface tension and gravitation is a minimum.

The gravitational potential energy (mgh) depends upon the mass of the drop and hence upon its volume and is *thus proportional to the cube of its linear dimensions,* while the potential energy due to surface tension depends upon the surface area of the drop and is thus *proportional to the square of the linear dimensions.* Consequently, in case of a *large drop,* the potential energy due to surface tension is negligible in comparison to gravitational potental energy and thus the drop becomes *flat* thereby lowering the centre of gravity and reducing its potential energy to the lowest possible. For example, a large drop of mercury is flat.

In case of a *small drop,* the potential energy due to surface tension is predominant and hence it assumes spherical shape. This explains why small drops of rain dew, oil, mercury etc. are spherical. The fact that rain drops are perfectly spherical has been proved by the shape and colouring of the rainbows and halos, since any slight deviation from this shape of rain drops would entirely change the character of the phenomena.

(2) Shape of a oil drop in a mixture of water and alcohol of same density : If drop of oil is placed in a mixture of alcohol and water of the same density as that of the oil, the weight of the drop is exactly balanced by the upthrust of the mixture and thus the effect of gravity on the drop is completely cancelled. Hence, the only force governing the shape of the drop is that due to surface tension. The potential energy of the drop due to surface tension is minimum, when its surface area is the least possible. Since a sphere has a minimum possible area for a given volume, the drop acquires the shape of a *perfect sphere.*

(3) Movement of camphor particles on water : When small particles of camphor are placed on a clean surface of water, they begin to move rapidly hither and thither in different directions. Particles of camphor are gradually dissolved in water and lower its surface tension. This causes the particles of camphor to be pulled towards the surrounding water of higher surface tension. Since, the particles dissolve more quickly at some points than at others, the force due to surface tension at all the points is not uniform. Consequently, they are drawn in various directions and thus exhibit a vigorous haphazard movement.

(4) Energy used in spraying—Coldness due to spraying : When a liquid is sprayed, then small dropllets are formed and new surface area or surface film is formed. The energy used in creating the new surface film is converted in the form of **surface energy** of the droplets. There are two sources of this energy : (i) Work done by external source and (ii) thermal energy inside the liquid.

Hence, when a liquid is sprayed, the heat energy of the water droplets is decreased and thus **coldness results due to spraying**.

15·13. Formation of Droplets from a Relatively Big Drop

If we spray a big drop of radius R into n droplets, each of radius r, then the total surface area of the droplets will be greater than that of the big drop by ΔS, where

$$\Delta S = \frac{4}{3} n\pi r^2 - \frac{4}{3} \pi R^2 \qquad ...(1)$$

Hence, the work done in spraying small droplets from a big drop is given by

$$W = \text{Surface tension} \times \text{New surface area created}$$

or
$$W = T\,(4n\pi r^2 - 4\pi R^2) \qquad ...(2)$$

Since the big drop is sprayed in the form of droplets and hence the volume of big drop is equal to the volume of sprayed droplets *i.e.*,

$$\frac{4}{3}\pi R^3 = \frac{4}{3} n\pi r^3 \text{ or } R^3 = nr^3 \text{ or } r = R/n^{1/3} \qquad ...(3)$$

This expression relates the radii of a droplet and big drop.

Now, if we coalesce n small droplets to a big drop, then the surface area of the liquid will decrease and a net amount of energy E will be obtained :

$$E = T\,(4n\pi r^2 - 4\pi R^2) \qquad ...(4)$$

15·14. Effect of Impurities on Surface Tension

Surface tension of a liquid is very sensitive to the presence of impurities on its surface, being termed as contamination. Surface tension varies widely with the degree of contamination of the liquid surface.

Surface tension of solutions is always less than that of pure solvent or liquid, but we do not know any simple law, which governs the variation of surface tension with concentration.

15·15. Variation of Surface Tension with Temperature

The surface tension of all liquids decreases linearly with rise of temperature over small temperature ranges, so that the surface tension T of a liquid at $t°$ is given by

$$T = T_0\,(1 - \alpha t)$$

where T_0 is the value of surface tension at 0°C and α the temperature coefficient for it.

Ex. 1. *Calculate :* (i) *the amount of work done and* (ii) *excess pressure inside it, in blowing a soap bubble of diameter* 2.0 *cm. Take surface tension of soap solution* = 2.9×10^{-2} *N/m.*

(Bilaspur 2003, 1994; Bhopal 03; Gwalior 95)

Sol. Here, radius of bubble $r = 2.0/2$ cm $= 10^{-2}$ m, $T = 2.9 \times 10^{-2}$ N/m.

Due to two surfaces in a soap bubble total surface area of the bubble will be given by

$$S = 2 \times 4\pi r^2 = 8\pi r^2 = 8 \times 3.14 \times (10^{-2})^2 = 2.512 \times 10^{-3} \text{ m}^2.$$

(i) Amount of work done, $W = T \times S = (2.9 \times 10^{-2}) \times (2.512 \times 10^{-3}) =$ **7.28×10^{-5} joule**

(ii) Excess pressure inside the bubble $p = \dfrac{4T}{r} = \dfrac{4 \times 2.9 \times 10^{-2}}{10^{-2}} =$ **11.6 N/m^2.**

Ex. 2. (i) *Calculate the work done against surface tension force in blowing a soap bubble of* 5 *cm radius if the surface tension of soap solution is* 0.025 *N/m.* (ii) *A soap bubble of surface tension* 0.025 *N/m has a radius of* 5 *cm. If the bubble is blown to a radius of* 10 *cm, what is the amount of work done ?*

(Gorakhpur 1960; Purvanchal 93)

Sol. (i) In blowing a soap bubble of radius 5 cm, new surface area of $2 \times 4\pi \times (0.05)^2$ (two surfaces of bubble) is formed.

$\therefore$ Work done = surface tension × surface area created $= 0.025 \times 2 \times 4\pi \times (0.05)^2 =$ **1.5714×10^{-3} J.**

(ii) Initial total surface area of the bubble $= 2 \times 4\pi \times (0.05)^2 = 0.02\pi$ sq.m.

Final total surface area of the bubble $= 2 \times 4\pi \times (0.1)^2 = 0.08\pi$ sq.m.

Increase in the surface area $= 0.08\pi - 0.02\pi = 0.06\pi \text{ m}^2$

$\therefore$ Amount of work done = surface tension × increase in surface area

$= 0.005 \times 0.06\pi = \mathbf{4.71 \times 10^{-3}}$ **joule.**

Ex. 3. *What would be the pressure inside a small air bubble of* 0.1 *mm radius, situated just below the surface of water ? Surface tension of water* = 0.072 *N/m and atmospheric prssure* = 1.013×10^5 N/m^2.

(Magadh 1963)

Sol. The bubble is situated below the surface of water, therefore it has only one surface. Hence, the excess of pressure inside the air bubble over atmospheric pressure is given by

$$p = 2T/r$$

Here, $T = 0.072$ N/m and $r = 10^{-4}$ m; $\therefore\ p = \dfrac{2 \times 0.072}{10^{-4}} = 0.0144 \times 10^5 \text{ N/m}^2$

Therefore, total pressure inside the air bubble = p + atmospheric pressure

$= 0.0144 \times 10^5 + 1.013 \times 10^5 = \mathbf{1.0274 \times 10^5\ N/m^2}$.

Ex. 4. *If the excess pressure inside a spherical bubble is balanced by that due to a column of oil (sp. gr.* 0.8*)* 2 *mm high when* r = 1.0 *cm, find the surface tension of the soap bubble.*

Sol. The excess of pressure inside a soap bubble above atmospheric pressure is given by

$$p = 4T/r$$

where T is the surface tension for soap solution and r the radius of the bubble.

Here, $r = 0.01$ m, $p = h\rho g = 2 \times 10^{-3} \times 0.8 \times 1000 \times 9.8 \text{ N/m}^2$.

$$\therefore \quad 2 \times 0.8 \times 9.8 = \frac{4T}{0.01}, \quad \text{whence} \quad T = \frac{2 \times 0.8 \times 9.8 \times 0.01}{4} = \mathbf{0.0392\ n/m.}$$

Ex. 5. *A bubble of radius* 1 *cm is blown in an atmosphere whose pressure is* 10^5 *N/m*^{2}*. If the surface tension of the liquid comprising the film in order that the radius of the bubble may be doubled. Assume isothermal conditions.*

(Lucknow 1963)

Sol. Let P N/m^2 be the pressure inside the bubble when its radius is 0.01 m. The excess pressure inside is then $(P - 10^5)$ N/m^3, which must be equal to $4T/r$ *i.e.*,

$$P - 10^5 = \frac{4 \times 0.05}{0.01} = 20 \text{ N/m}^2, \qquad \therefore\ P = (10^5 + 20) \text{ N/m}^2.$$

Consider that the pressure inside the bubble becomes P' when it expands to a radius of 2 cm under isothermal condition. From Boyle's law, we have

$$P \times \tfrac{4}{3}\pi \times (0.01)^3 = P' \times \tfrac{4}{3}\pi \times (0.02)^3$$

$$\therefore \quad P' = \frac{P}{2^3} = \frac{10^5 + 20}{8} = 12502.5 \text{ N/m}^2.$$

The surrounding atmosphere should be at a pressure P' such that the e inside the bubble is again equal to $4T/r$, where r is the new radius of the bubble, *i.e.*,

$$P' - P'' = \frac{4 \times 0.05}{0.02} = 10 \text{ N/m}^2$$

$\therefore \quad P'' = P' - 10 = 12502.5 - 10 = 12492.5 = \mathbf{1.25 \times 10^4}$ **N**

19.

Ex. 6. *A spherical air bubble is formed in water at a depth of* 1 *m from* ... *diameter of the air bubble is* 0.5 *mm and the surface tension of water is* 0.075 *N/m. Ca* ... *essure inside the bubble, if the pressure of the air over the surface of water is* 76 *cm of* ...

Sol. The pressure inside a bubble at a depth 1 m in water will be the su... three pressures:

(1) Atmospheric pressure on the surface of water $= 0.76 \times 13600 \times 9.8$ ... m^2

(2) Pressure of the water column over the bubble $= 1 \times 1000 \times 9.81 \text{ N/m}^2$

(3) Excess of pressure inside the bubble in water $= \frac{2T}{r} = \frac{2 \times 0.075}{0.25 \times 10^{-3}} = 600 \text{ N/m}^2$

$\therefore$ Total pressure inside the bubble = 101400 + 9810 + 600 = 111810 = **1.1181 × 10^5 n/m^2.**

Ex. 7. *Two sherical soap bubbles coalesce, if V is the consequent change in volume of the contained air and S the change in the total surface area, show that*

$$3PV + 4ST = 0.$$

where T is the surface tension of the soap solution and P the atmospheric pressure.

(Gorakhpur 1964; Purvancal 95)

Sol. Let r_1, r_2 be the radii of two spherical bubbles and P_1, P_2 the corresponsding pressures inside the bubble. Then

$$P_1 = P + 4T/r_1 \quad ...(i)$$

and

$$P_2 = P + 4T/r_2 \quad ...(ii)$$

where P is the atmospheric pressure and T the surface tension for soap solution.

If R be the radius of the combined bubble and p the pressure inside it, then

$$p = P + 4T/R \quad ...(iii)$$

If v_1, v_2 be the volumes of the bubbles before they coalesce and v the volume of the combined bubble, then

$$v_1 = \tfrac{4}{3}\pi r_1^3,\ v_2 = \tfrac{4}{3}\pi r_2^3 \text{ and } v = \tfrac{4}{3}\pi R^3$$

Now as the temperature remains constant during the change, we have from Boyle's law

$$P_1 v_1 = P_2 v_2 = pv.$$

Substituting the values of various quantities, we have

$$(P + 4T/r_1)\tfrac{4}{3}\pi r_1^3 + (P + 4T/r_2)\tfrac{4}{3}\pi r_2^3 = (P + 4T/R)\tfrac{4}{3}\pi R^3$$

or

$$P\left(\tfrac{4}{3}\pi r_1^3 + \tfrac{4}{3}\pi r_2^3 - \tfrac{4}{3}\pi R^3\right) + (4T/3)(4\pi r_1^2 + 4\pi r_2^2 - 4\pi R^2) = 0$$

But change in volume $= V = \frac{4}{3}\pi r_1^3 + \frac{4}{3}\pi r_2^3 - \frac{4}{3}\pi R^3$

and change in surface area $= S = 4\pi r_1^2 + 4\pi r_2^2 - 4\pi R^2$

$$\therefore \quad PV + \frac{4T}{3} \times S = 0 \text{ or } \mathbf{3PV + 4ST = 0.}$$

Ex. 8. *Two plane pieces of glass have a water drop between them which is circular and of diameter 10.0 cm. If the glass plates are 0.50 mm apart, what force perpendicular to the plates will be required to separate them ? (Surface tension of water = 0.072 N/m, angle of contact = 0°)* **(Lucknow 1961)**

Sol. The resultant force pressing the upper plates downward is given by the relation

$$F = \frac{2T}{d} \times A$$

where T is the surface tension, A the area of the plate wetted by the liquid film and d the distance between the two plates. The angle of contact is already assumed to be zero.

Here $T = 0.072$ N/m, $A = \pi \times (0.05)^2$ sq.m and $d = 0.5 \times 10^{-3}$ m.

$$\therefore \quad F = \frac{2 \times 0.072 \times 3.14 \times (0.05)^2}{0.5 \times 10^{-3}} = \mathbf{2.2608 \text{ newton.}}$$

This is the amount of force, required to separate the plates.

Ex. 9. *Calculate the work expanded in spraying a drop of water of one millimeter radius into 10^6 droplets all of the same size, the surface tension of water being 7.2×10^{-2} N/m.*

(Bhopal 2004; Rewa 04; Ujjain 03; Gwalior 01; Allahhabad 1965; I.A.S. 73; Gorakhpur 94)

Sol. When a drop of water is sprayed into a number of droplets, surface area increases and therefore, work is expanded in doing so. This work done forms a part of the surface energy of the new surface area so created.

Radius of the drop = 10^{-3} m; Surface area of the drop = $4\pi r^2 = 4\pi \times (10^{-3})^2$ sq. m

and Volume of 1 the drop = $\frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times (10^{-3})^3$ m^3

Number of droplets formed = 10^6

Volume of 10^6 droplet = $\frac{4}{3}\pi \times (10^{-3})^3$ m^3

$\therefore$ Volume of droplets = $\frac{4}{3} \times (10^{-3})^3/10^6$ m^3

Let the radius of each small droplet be r_1. Then

$$\frac{4}{3}\pi r_1^3 = \frac{4}{3}\pi\ (10^{-3}/10^2)^3 \quad \therefore r_1 = 10^{-5} \text{ m}$$

Surface area of 1 droplet = $4\pi r_1^2 = 4\pi \times (10^{-5})^2$ sq. m.

Surface area of 10^6 droplets = $10^6 \times 4\pi \times (10^{-5})^2$ sq. m. = $4\pi \times 10^{-4}$ sq.m.

Increase in surface area = $4\pi \times 10^{-4} - 4\pi \times (10^{-3})^2 = 4\pi \times 10^{-4}\ (1 - 0.01)$

$= 4 \times \frac{22}{7} \times 0.99 \times 10^{-4}$ sq. m.

Energy of area expanded = surface tension $\times$ increase in area

$= 7.2 \times 10^{-2} \times 4 \times \frac{22}{7} \times 0.99 \times 10^{-4}$ = **8.96×10^{-5} joules.**

Ex. 10 *What amount of work is needed against surface tension, when a water drop of diameter* 0.2 *cm splits into* 27000 *drops of equal volume ? Surface tension of water* = 7×10^{-2} *N/m.*

(Bundelkhand 2004; Rewa 2003; Raipur 2003;)

Sol. Here, radius of big water drop R = 0.1 cm = 1×10^{-3} m

If the radius of each small drop be r, then

Volume of 27000 small water drops = Volume of one big drop

or $27000 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$ or $r = \frac{R}{30} = \frac{1}{3} \times 10^{-4}$ m

$\therefore$ Increase in area $= 27000 \times 4\pi r^2 - 4\pi R^2 = 4\pi\ [27000 \times \left(\frac{1}{3} \times 10^{-4}\right)^2 - (10^{-3})^2]$

$= 4\pi \times 29 \times 10^{-6} = 3.64 \times 10^{-4}$ m^2

$\therefore$ Work done = surface tension $\times$ increase in area = $(7 \times 10^{-2}) \times 3.64 \times 10^{-4}$

= **2.55×10^{-5} joules.**

Ex. 11. *What amount of energy will be liberated, if* 1000 *droplets of water each* 10^{-6} *cm in diameter coalesce to form one large spherical drop ? Assume the surface tension of water to be* 0.075 *N/m.*

(Punjab 1961)

Sol. Here in the formation of large drop by coalescing a number of droplets, the surface area will decrease, hence a net amount of energy will be liberated.

Radius of each droplet = $10^{-8}/2$ m

Number of droplets = 1000

Surface area of 1000 droplets = $1000 \times 4\pi \times (10^{-8}/2)^2 = \pi \times 10^{-13}$ sq. m.

Volume of 1000 droplets = $1000 \times \frac{4}{3}\pi \times (10^{-8}/2)^3$

Let the radius of large drop be r. Then

$$\frac{4}{3}\pi r^3 = 1000 \times \frac{4}{3}\pi \times (10^{-8}/2)^2, \quad \therefore \quad r = 5 \times 10^{-8} \text{ m}$$

Surface area of large drop = $4\pi \times (5 \times 10^{-8})^2 = \pi \times 10^{-14}$ sq. m.

Hence, decrease in surface area $= \pi \times 10^{-13} - \pi \times 10^{-14}$

$= \pi \times 10^{-13} (1 - 0.1) = 9\pi \times 10^{-14}$ sq. m.

Hence, the amount of energy liberated $=$ S. T. $\times$ decrease in surface area

$= 0.075 \times 9 \times 3.142 \times 10^{-14} = \mathbf{2.12 \times 10^{-14}}$ **joules.**

EXERCISE 15·1

1. Surface tension of soap solution is 10^{-2} n/m. What will be the excess of pressure inside a soap bubble of 2 cm radius ? **(Bilaspur 1999)**
[**Ans.** 4 N/m^2.]
2. Calculate the excess of pressure in a soap bubble of 2 mm radius, if value of T is 0.03 N/m. **(Kanpur 1971)**
[**Ans.** 60 N/m^2.]
3. Calculate the excess pressure inside a soap bubble of radius 3 cm. The surface tension of the soap solution is 0.02 n/m. Find also the surface energy of soap bubble. **(I.A.S. 1977)**
[**Ans.** 2.667 N/m^2; 4.52×10^{-4} joules.]
4. The excess of pressure inside a soap bubble is equal to the 20 cm height of an oil column of density 0.8 gm/cc. If the surface tension of soap solution is 75×10^{-3} N/m, find the radius of the bubble. **(Garwal 1994)**
[**Ans.** 0.19 mm.]
5. A spherical soap bubble of radius 1.0 cm is formed inside another of radius 2.0 cm. Find the radius of a single soap bubble whose internal pressure is the same as that of the smaller bubble. Prove the formula used in the calculation. **(Delhi 1965)**
[**Ans.** 0.666 cm.]
6. Calculate the work done in blowing a soap bubble of radius 10 cm ($T = 0.08$ N/m). What additional work will be performed in further blowing it so that its radius is doubled ? **(Bombay 1962)**
[**Ans.** 7.54×10^{-3} J; 0.0226 J.]
7. What is the work done in blowing a soap bubble of 10 cm ($T = 0.03$ N-m^{-1}) ? What additional work is done in further blowing it so that its radius becomes 15 cm ? **(Allahabad 1980; Lucknow 84; Mysore 80)**
[**Ans.** 7.536×10^{-3} J; 9.42×10^{-3} J.]
8. The pressure of air inside a soap bubble of diameter 0.7 cm is 8 mm of water above the atmospheric prssure. Determine the surface tension of the soap solution. **(Gorakhpur 1962)**
[**Ans.** 0.06867 N/m.]
9. A soap bubble is spherical in shape and has a diameter of 10 cm. If the surface tension of the surface separating soap solution and air is 0.04 S.I. units; what is the excess pressure of the air in the bubble over the atmospheric pressure ? **(Poona 1968)**
[**Ans.** 3.2 N/m^2.]
10. A sphere of 7 cm diameter weights 3.5 kg in air. If it is held with its lower half immersed in water, what will be its apparent weight ? (Surface tension = 0.0725 N/m.)
[**Ans.** 3.367 kg-wt.]
11. Two soap bubbles of radii a and b coalesce to form a single bubble of radius c. If the external pressure is P, show that the surface tension is given by **(Gorakhpur 1984)**

$$(P/4)\,(c^3 - a^3 - b^3) \div (a^2 + b^2 - c^2).$$

[**Hint.** $\left(P + \frac{4T}{a}\right) \cdot \frac{4}{3}\pi a^3 + \left(P + \frac{4T}{b}\right) \cdot \frac{4}{3}\pi b^3 = \left(P + \frac{4T}{c}\right) \cdot \frac{4}{3}\pi c^3$

$$\text{or } 4T\cdot\frac{4}{3}\pi\,(a^2 + b^2 - c^2) = P\cdot\frac{4}{3}\pi\,(c^3 - a^3 - b^3),\ \therefore\ T = \frac{P}{4}\cdot\frac{(c^3 - a^3 - b^3)}{(a^2 + b^2 - c^2)}\,]$$

12. A drop of mercury of radius 1 cm is sprayed into 10^6 droplets of equal size. Assuming the temperature to remain constant, calculate the work done. (surface tension of mercury = 0.45 N/m). **(Indore 1999; Gwalior 99; Ujjain 95)**
[**Ans.** 5.65×10^{-2} J.]

13. Calculate the amount of energy needed to break a drop of water 2 mm in radius into 10^6 dropletes of equal size. (Surface tension of water = 0.073 N/m.)
[**Ans.** 3.629×10^{-4} joules.]

14. Calculate the work done in spraying a spherical drop of mercury of one millimeter radius into a million droplets of equal size, the surface tension of mercury being 0.55 N/m. **(Purvanchal 1994; Bombay 62; Delhi 71)**
[**Ans.** 6.839×10^{-4} joules.]

15. Calculate the loss of energy if 1000 drops of water of diameter 2 mm coalesce to form one large drop. Surface tension of water = 0.07 N/m. **(Allahabad 1964)**
[**Ans.** 7.913×10^{-4} joules.]

16. How much change in energy ocuurs, when a drop of water 4 mm in diameter is formed by combining one million droplets of equal size assuming the surface tension of water equal to 0.07 N/m.
[**Ans.** 3.481×10^{-4} joules.]

17. Calculate the loss of energy of 1000 drops of water each of diameter 2 mm, coalesce to form one large drop. The surface tension of water is equal to 0.072 N/m.
[**Ans.** 8.143×10^{-4} joules.]

18. Calculate the vertical force necessary to detach a horizontal plate of radius 4 cm from the surface of a liquid of surface tension 0.03 N/m, density 1000 kg/m^3, and angle of contact zero. **(Bombay 1968)**
[**Ans.** 0.07536 newton.]

QUESTIONS

Long Answer Questions

1. Define surface tension and surface energy. Show that surface tension is numerically equal to surface energy. **(Garhwal 1993)**
2. Define surface tension and surface energy. Find a relation between surface tension and the amount of work done in enlarging the surface of a liquid.
3. Explain the cause of surface tension on the basis of cohesive forces. **(Bhopal 2004; Sagar 03)**
4. Deduce expression for the pressure difference across a spherical surface due to surface tension. **(Gwalior 2004; Indore 04)**
5. Deduce an expression for the excess pressure across a curved surface of liquid. **(Ujjain 2004; Raipur 03)**
6. Show that the excess pressure acting on the curved surface of a curved membrane is given by

$$P = 2T\left(\frac{1}{r_1} + \frac{1}{r_2}\right),$$

where r_1 and r_2 are the radii of curvature and T the surface tension of the membrane. **(Bombay 1962; Punjab 68)**

7. Assuming the excess of pressure on the curved surface of a drop is to be given by

$$P = T\left(\frac{1}{r_1}+\frac{1}{r_2}\right).$$

Show that the excess of pressure inside spherical drop is twice that in cylindrical drop of the same radius. **(Mysore 1980)**

8. Deduce an expression for the excess pressure inside (i) a liquid drop, (ii) a soap bubble, (iii) an air bubble. **(Gwalior 2004)**
9. Show that the excess of pressure inside a soap bubble of radius R over the atmosphere outside it is equal to $4T/R$, where T is the surface tension of the soap solution. **(Allahabad 1980; Garwal 95; Gorakhpur 80)**

Short Answer Questions

1. Define surface tension and state its unit and dimensions. **(Bhopal 2004; Vikarm 03)**
2. Explain the meaning of surface tension and surface energy and write their units. **(Indore 2003)**
3. What do you understand by surface tension, surface energy and angle of contact ? **(Kanpur 2005)**
4. Explain surface energy. Show that surface tension is numerically equal to the surface energy. **(Kanpur 2004)**
5. Explain the phenomenon of surface tension.
6. What do you understand by adhesive and cohesive forces ?
7. What is angle of contact ? **(Ujjain 2004)**
8. What is the effect of temperature on the surface tension of liquid ?
9. Small drops of liquid are spherical in shape. Why ?
10. Explain why small drops of mercury are spherical and large drops become flat ?
11. Explain why water on a clean glass surface tends to spread out while mercury on the same surface tends to form drops ?
12. Small drops of liquid are brought into contact to form a big drop, then the surface energy of big drop will increase or decrease ? Temperature of the big drop will be greater or smaller than the temperature of small drops.
13. Find an expression for the excess pressure inside a liquid spherical drop. **(Kanpur 2004)**
14. Find an expression for the excess pressure inside a soap bubble. **(Kanpur 2005)**
15. State the effect of : (i) temperature, (ii) presence of impurities, on surface tension. **(Bhopal 2004)**
16. Explain why :
    (i) Small drops of liquid are spherical in shape ? **(Punjab 1961)**
    (ii) Small drops of mercury are spherical, while large drops of mercury are flat ?
    (iii) Particles of camphor exhibit a vigorous movement on the surface of water ? **(Punjab 1963 S)**
    (iv) A drop of oil in a mixture of alcohol and water of the same density is a perfect sphere ?
    (v) Mercury does not spread on a sheet of glass while water does. **(Gorakhpur 1969)**
    (vi) A drop of oil placed over a clean surface of water spreads out into a thin film, but floats as a globule upon a greasy surface ? **(Gorakhpur 1969)**
    (vii) Pressure difference exists between the two sides of a curved surface ?
    (viii) A great force is required to draw apart normally two glass plates enclosing a thin water film?

    or

    The small space between two parallel glass plates is filled with water film. Explain why it is much easier to separate the plates by sliding one over the other than by a direct pull ? **(Agra 1967)**

Objective Type Questions

1. Cohesive forces are directly proportional to :
   (a) r^{-15} (b) r^{-12} (c) r^{-8} (d) r^{-2}
2. The maximum distance upto which cohesive forces can act :
   (a) 1 cm (b) 10^{-2} cm (c) 10^{-7} cm (d) 10^{-4} cm
3. The unit of surface tension is :
   (a) joule/meter^2 (b) newton/meter (c) newton-meter (d) newton/meter^2
4. The excess pressure inside a spherical liquid drop of radius R and surface tension T is :
   (a) T/R (b) $2T/R$ (c) $4T/R$ (d) $2\pi T/R$
   (Agra 2005, 04)
5. The excess pressure inside a soap bubble (surface tension T and radius r) is :
   (a) T/r (b) $2T/r$ (c) $4T/r$ (d) $-T/r$. **(Rewa 1999, 93)**
6. When 10^3 small droplets combine to form one big drop, the surface energy :
   (a) decreases (b) increases (c) remains unchanged (d) will remain zero
   (Bundelkhand 2004)
7. Water wets surface of glass because the angle of contact for water-galss is :
   (a) almost zero (b) 45° (c) 90° (d) 135°
8. If more air is blown in a soap bubble, then pressure inside it will :
   (a) increase (b) decrease (c) not be present (d) become zero
9. The angle of contact of mercury-glass surface is :
   (a) 0° (b) 8° (c) 90° (d) 135° **(Kanpur 2004)**
10. Two soap bubbles have radii in the ratio 2 : 1. What is the ratio of excess pressure inside them ?
    (a) 1 : 2 (b) 2 : 1 (c) 1 : 4 (d) 4 : 1 **(Kanpur 2004)**

ANSWERS

1. (c), **2.** (c) **3.** (a), (b) **4.** (b), **5.** (c), **6.** (a), **7.** (a), **8.** (b), **9.** (d), **10.** (a).

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PART-II

Electricity, Magnetism and Electromagnetic Theory

1

Vector Algebra

1·1. Scalars and Vectors

Measurable physical quantities can be divided mainly into two classes :

(1) Scalars : *Those physical quantities, which have magnitude only but no particular direction in space, are called scalars.* Examples of scalar quantities are mass, volume, density, temperature, electric potential, charge etc.

The complete representation of a scalar quantity requires (i) a *unit of the same nature,* (ii) *a numerical value,* stating how many times the unit is contained in the quantity. For example, the mass of a body is 20 gm, we mean that the unit of measurement is a gramme and 20 such units when brought together, will comprise the mass of the body.

(2) Vectors : *Those physical quantities, which have directions as well as magnitudes, are called vectors. Moreover, a vector quantity must satisfy the parallelogram law of addition of vectors* (Art. 1.4). Familiar examples of vectors are displacement, velocity, acceleration, force, momentum, magnetic and electric field intensities etc.

The complete specification of a vector quantity requires (i) a *unit of the same kind,* disregarding direction, (ii) *a numerical value,* and (iii) *a statement of direction.* For example, the velocity of a body is 20 km/hour due east.

1·2. Representation of a Vector

A scalar quantity is represented in light face italics or by ordinary letters such as *A*. *For the representation of a vector quantity we write it by a letter bearing an arrow over its top (e.g.,* $\vec{A}$*), while in printing it is also represented in bold face type letters (e.g.,* **A***). The magnitude of a vector is represented in italics.* For example, *A* is the magnitude of a vector $\vec{A}$. The magnitude of a vector is also called mod $\vec{A}$ or $|\vec{A}|$.

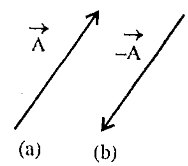


Fig. 1.1 : Representation of a vector.

Graphically a vector is represented by an arrow whose length is equal to the magnitude of the vector [**fig. 1.1 (a)**]. For this purpose, a convenient scale is chosen and the line, drawn for the magnituide of the vector, bears the arrow-head in the direction of the vector in space. **Fig. 1.1(b)** shows a vector $-\vec{A}$ which is antiparallel to $\vec{A}$ and having the same magnitude ($|\vec{A}|$).

1·3. Different Kinds of Vectors

(1) Unit vector : *A vector, whose magnitude is unity, is called a unit vector.* A unit vector in the direction of $\vec{A}$ is written as $\hat{\mathbf{A}}$ and is given by [**fig. 1.2 (a)**] :

$$\hat{\mathbf{A}} = \frac{\vec{A}}{A} \; i.e., \; \vec{A} = A\hat{\mathbf{A}} \qquad ...(1)$$

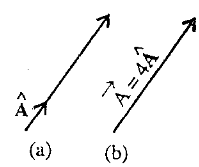


Fig. 1.2 : Unit vector.

This unit vector $\hat{\mathbf{A}}$ is read as 'A hat' or 'A caret', *e.g.,* if $|\vec{A}| = 4$, then $\vec{A} = 4\hat{\mathbf{A}}$ [**fig. 1.2 (b)**].

(2) Equality of vectors : *Two vectors* $\vec{A}$ *and* $\vec{B}$ *are said to be equal, if they have equal magnitudes and are parallel, pointing in the same direction* [**fig. 1.3**]. *Thus, if* $\vec{A}$ *and* $\vec{B}$ *are equal vectors, then*

$$\vec{A} = \vec{B} \qquad ...(2)$$

It is not necessary that the two vectors should have the same origin. Thus, if PQ and RS are two parallel lines in space and if $|\vec{A}| = |\vec{B}| = |\vec{C}|$, then

$$\vec{A} = \vec{B} = \vec{C} \qquad ...(3)$$

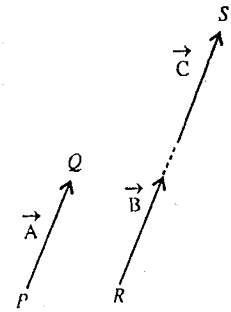


Fig. 1.3 : Equal Vectors.

Thus, all vectors, which have the same direction and the same magnitude, whatever their situation in space may be, are *equal.*

(3) **Parallel or collinear vectors :** *Those vectors, which have the same line of action or lines of action parallel to the same line are called collinear or parallel vectors* ; they need not have the same direction.

For example, $\vec{A}$ and $\lambda\vec{A}$ are collinear vectors, where λ is a scalar number. Evidently if two vectors $\vec{A}$ and $\vec{B}$ are collinear, then there exists a scalar λ such that

$$\vec{B} = \lambda\vec{A} \qquad ...(4)$$

If $\vec{A}$ and $\vec{B}$ have the same direction, then λ is a positive number and if both are oppositely directed, then λ is a negative number.

(4) **Vector obtained by the multiplication of a vector by a scalar :** By the multiplication of a vector $\vec{a}$ by a scalar n we understand a vector $\vec{A}$ whose magnitude is equal to the magnitude of the product of the magnitude of $\vec{a}$ and scalar n. The vector, so obtained, has the direction of $\vec{a}$ or the opposite direction, depending on whether the scalar n is positive or negative. Thus,

$$\vec{A} = n\,\vec{a} \qquad ...(5)$$

(5) **Polar vectors :** *Those vectors which are related to the linear or directional effect are called polar vectors.* Examples of these vectors are velocity, acceleration, momentum, force etc.

(6) **Axial vectors :** *Those vectors which are related to the rotation about an axis are called axial vectors.* Examples of such vectors are angular velocity, angular momentum, torque etc. An axial vector can be represented by an arrow similar to a polar vector whose length is a measure of its value and when looking towards the direction of the arrow, the rotation appears clockwise.

1·4. Addition and Subtraction of Vectors

Let $\vec{A}$ and $\vec{B}$ be two vectors. For obtaining their sum or resultant, $\vec{B}$ is carried parallel to itself until the beginning of $\vec{B}$ coincides with the head of $\vec{A}$. Now, the vector $\vec{V}$, drawn between the tail of $\vec{A}$ and the head of $\vec{B}$, is the sum $\vec{A}+\vec{B}$, *i.e.*,

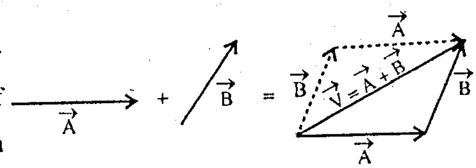


Fig. 1.4 : Addition of two vectors.

$$\vec{V} = \vec{A}+\vec{B} \qquad ...(1)$$

The same result is obtained, if the head of $\vec{B}$ is attached with the tail of $\vec{A}$, as shown by the dotted lines [**fig. 1.4**]. Then

$$\vec{A}+\vec{B} = \vec{B}+\vec{A}. \qquad ...(2)$$

Thus the sum of two vectors is the diagonal of a parallelogram of which the vectors are the two sides. Such addition of two vectors is called the **parallelogram law of vectors** and is *commutative, i.e.*, independent of which vector is taken first.

To add several vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$, first the sum of $\vec{A}$ and $\vec{B}$ is found and then the sum of $\vec{A}+\vec{B}$ and $\vec{C}$. The result will be the same if we find the sum of $\vec{A}$ and $\vec{B}+\vec{C}$ [**fig. 1.5**]. Thus

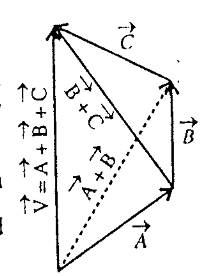


Fig. 1.5 : Addition of several vectors.

$$\vec{V} = (\vec{A}+\vec{B})+\vec{C} = \vec{A}+(\vec{B}+\vec{C}). \qquad ...(3)$$

Therefore, the sum of any number of vectors is *associative, i.e.*, the vectors may be added in any desired manner. It is not necessary that the all vecotrs should lie in the same plane.

To subtract the vector $\vec{B}$ from $\vec{A}$, we reverse the vector $\vec{B}$ and the rule of addition is applied [**fig. 1.6**]. Thus, the resultant

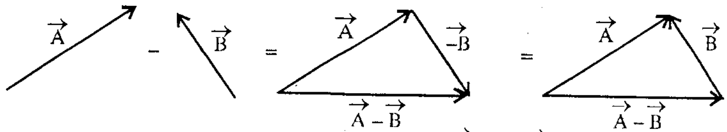


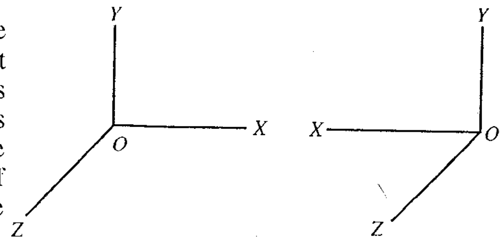
Fig. 1.6 : Subtraction of vector $\vec{B}$ from $\vec{A}$.

$$\vec{V} = \vec{A} + (-\vec{B}) = \vec{A} - \vec{B}. \qquad ...(4)$$

1·5. Right Handed and Left Handed Coordinate Systems

The coordinate systems which are generally used are rectangular or orthogonal. They may be of two types : (i) *right handed* [**fig. 1.7 (a)**], (ii) *left handed* [**fig. 1.7 (b)**]. Evidently both coordinate systems are the mirror image of each other.

In case of the right handed coordinate system, the thumb, the fore-finger and the middle finger of the right hand give the relative directions of X, Y and Z axes respectively. In such a case, a rotation from X to Y axis will appear clockwise if seen from the origin towards the positive direction of Z-axis. Fundamentally, any one of the two coordinate systems may be employed but the right handed system is commonly used.



(a) Right Handed Coordinate System

(b) Left Handed Coordinate System

Fig. 1.7.

1·6. Components of a Vector

The components of a vector $\vec{A}$ are any number of vectors whose sum is $\vec{A}$. The components, most frequently used, are those parallel to X, Y and Z axes of the right handed cartesian coordinate system and they are called *rectangular components* of the vector.

Let the origin O be one extremity of a vector $\vec{A}$. Now draw a rectangular parallelopiped with its three edges lying along the three axes such that $\vec{A}$ is the diagonal through O. If $\vec{A}_x$, $\vec{A}_y$ and $\vec{A}_z$ are the vector intercepts along X, Y and Z axes respectively [**fig. 1.8**], then

$$\vec{A} = \vec{A}_x + \vec{A}_y + \vec{A}_z. \qquad ...(1)$$

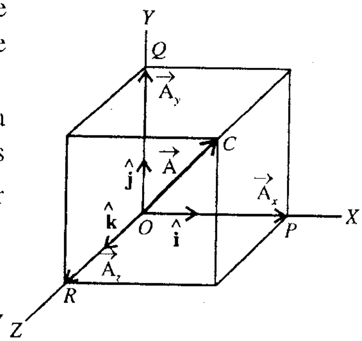


Fig. 1.8 : Components of a vector.

Let $\hat{i}$, $\hat{j}$, $\hat{k}$ represent vectors of unit magnitude along the X, Y and Z axes respectively; these unit vectors along the axes will be frequently used in the later work. Now, if A_x, A_y and A_z are the magnitudes of the vectors $\vec{A}_x$, $\vec{A}_y$ and $\vec{A}_z$ respectively, then

$$\vec{A}_x = A_x\hat{i}, \ \vec{A}_y = A_y\hat{j}, \ \vec{A}_z = A_z\hat{k},$$

Therefore, equation (1) can be written as

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k} \qquad ...(2)$$

By geometry, the magnitude of the vector $\vec{A}$ is

$$OC^2 = OP^2 + OQ^2 + OR^2$$

or $$A^2 = A_x^2 + A_y^2 + A_z^2 \text{ or } A = \sqrt{A_x^2 + A_y^2 + A_z^2} \qquad ...(3)$$

The magnitude of the component vectors are given by

$$A_x = A\cos(\vec{A}, x), \qquad A_y = A\cos(\vec{A}, y), \qquad A_z = A\cos(\vec{A}, z), \qquad ...(4)$$

Thus, $\cos(\vec{A}, x) = A_x/A$, $\cos(\vec{A}, y) = A_y/A$ and $\cos(\vec{A}, z) = A_z/A$ ...(5)

With the help of the cosines given in (5) the direction of the vector $\vec{A}$ can be found. Therefore, these are called **direction cosines.**

If we put the values of A_x, A_y, A_z from eq. (4) in eq. (3), we have

$$\cos^2(\vec{A}, x) + \cos^2(\vec{A}, y) + \cos^2(\vec{A}, z) = 1 \qquad ...(6)$$

Thus, **the sum of the square of three direction cosines is equal to unity.**

The unit vector $\hat{A}$ along $\vec{A}$ can be written as

$$\hat{A} = \frac{\vec{A}}{A} = \frac{A_x}{A}\hat{i} + \frac{A_y}{A}\hat{j} + \frac{A_z}{A}\hat{k} \qquad \text{[from eq. (2)]}$$

where A_x/A, A_y/A, and A_z/A are the direction cosines of $\vec{A}$.

$$\therefore \quad \hat{A} = \cos(\vec{A}, x)\,\hat{i} + \cos(\vec{A}, y)\,\hat{j} + \cos(\vec{A}, z)\,\hat{k} \qquad ...(7)$$

The addition of the two vectors can be written in terms of components as

$$\vec{V} = \vec{A} + \vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) + (B_x\hat{i} + B_y\hat{j} + B_z\hat{k})$$

or

$$\vec{V} = (A_x + B_x)\,\hat{i} + (A_y + B_y)\,\hat{j} + (A_z + B_z)\,\hat{k} \qquad ...(8)$$

$$\text{Its magnitude} = |\vec{V}| = \sqrt{(A_x + B_x)^2 + (A_y + B_y)^2 + (A_z + B_z)^2} \qquad ...(9)$$

Its direction cosines are $(A_x + B_x)/V$, $(A_y + B_y)/V$, and $(A_z + B_z)/V$.

1·7. Position Vector

If the coordinates of a point P be (x, y, z) relative to the origin of a cartesian coordinate system, then its distance from the origin can be expressed in the vector form as

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \qquad ...(1)$$

We call it ($\vec{r}$) the position vector of the point P [fig. 1.9 (a)]. The direction cosines of $\vec{r}$ are obviously (x/r), (y/r), (z/r).

If there are two points P and Q, having coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2), then from the origin O the position vectors $\vec{OP} = \vec{r_1}$ and $\vec{OP} = \vec{r_2}$ will be represented as

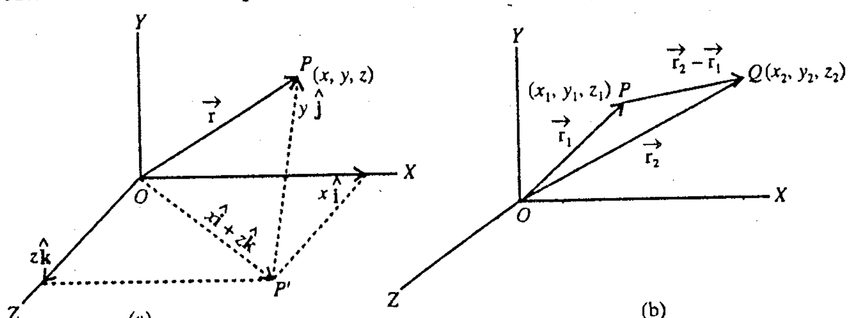


Fig. 1.9 : Position vector.

$$\vec{r_1} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k} \text{ and } \vec{r_2} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$$

$$\therefore \quad \vec{PQ} = (\vec{r_2} - \vec{r_1}) = (x_2 - x_1)\,\hat{i} + (y_2 - y_1)\,\hat{j} + (z_2 - z_1)\,\hat{k} \qquad ...(2)$$

The relative position vector $\vec{PQ} = \vec{r_2} - \vec{r_1}$ is shown in fig. 1.9(b).

1·8. Invariance of the Magnitude of a Vector under Rotation of the Coordinate system

Let us consider a vector

$$A = A_x\,\hat{i} + A_y\,\hat{j} + A_z\,\hat{k} \qquad ...(1)$$

in the coordinate system $S(x, y, z)$. Leaving the direction of this vector $\vec{A}$ unchanged, let the coordinate system be rotated about the origin O to the position $S'(x', y', z')$. There should be no difficulty in thinking of leaving the direction of $\vec{A}$ unchanged, as $\vec{A}$ may be thought to be fixed with respect to the fixed stars **[fig. 1.10]**.

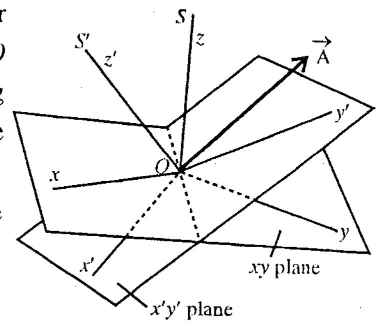


Fig. 1.10.

Now, if $\hat{i}'$, $\hat{j}'$, $\hat{k}'$ are the unit vectors along the new coordinate axes, then the vector $\vec{A}$ may be expressed as

$$\vec{A} = A_{x'}\,\hat{i}' + A_{y'}\,\hat{j}' + A_{z'}\,\hat{k}' \qquad ...(2)$$

where $A_{x'}$, $A_{y'}$, and $A_{z'}$ are the components of the vector $\vec{A}$ in the new coordinate system.

Evidently, the length of the vector $\vec{A}$ must be independent of the orientation of the coordinate system, *i.e.*, the magnitude of the vector $\vec{A}$ in the frames S and S' should be invariant. Therefore,

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{A_{x'}^2 + A_{y'}^2 + A_{z'}^2} \qquad ...(3)$$

Thus, the form of the magnitude of a vector is the same in all cartesian coordinate systems which differ by a rigid rotation of the coordinate axes.

1·9. Product of Two Vectors

In vector algebra, the product of two vectors is defined in two different ways :

(1) Scalar product : If the multiplication of two vectors gives a scalar, the product is called **scalar product.**

(2) Vector product : If the two vectors are so multiplied that the result of multiplication is a vector, the product is called **vector product.**

Both types of product are very much useful in Physics. We will discuss in the following each one of them.

1·9·1. Scalar or Dot Product of Two Vectors

The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as a scalar quantity $AB\cos\theta$, where A and B are the magnitudes of the vectors and θ is the angle between their directions.

We write the scalar product of two vectors as $\vec{A}\cdot\vec{B}$ which is read $\vec{A}$ dot $\vec{B}$ and therefore it is also called **dot product.** Thus

$$\vec{A}\cdot\vec{B} = AB\cos\theta \qquad ...(1)$$

The scalar product $AB\cos\theta$ of the vectors $\vec{A}$ and $\vec{B}$ may be considered as either the multiplication of the magnitude of $\vec{A}$ and the projection of $\vec{B}$ along $\vec{A}$ [$AB\cos\theta = A\,(B\cos\theta)$], **[fig. 1.11 (a)]** or the multiplication of the magnitude of $\vec{B}$ and the projection of $\vec{A}$ along $\vec{B}$ [$AB\cos\theta = B\,(A\cos\theta)$], **[fig. 1.11 (b)].**

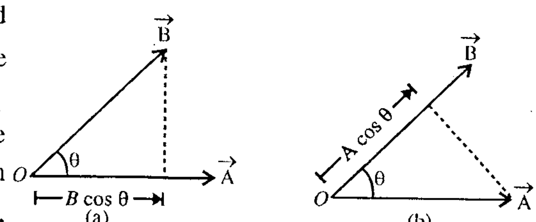


Fig. 1.11 : Scalar product.

It is clear from the definition of the scalar product that the *commutative law of multiplication* holds, *i.e.*,

$$\vec{A}\cdot\vec{B} = \vec{B}\cdot\vec{A} = AB\cos\theta \qquad ...(2)$$

If the vectors $\vec{A}$ and $\vec{B}$ are **parallel** (pointing in the same direction) to each other, then ($\theta = 0$)

$$\vec{A}\cdot\vec{B} = AB\cos\theta = AB$$

and if the vectors are similar, then $\vec{A}\cdot\vec{A} = A^2$ ...(3)

If this property is applied to the unit vectors $\hat{i}$, $\hat{j}$, $\hat{k}$, which are mutually perpendicular to each other, then

$$\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1 \qquad ...(4)$$

In case, if the two vectors are **perpendicular** to each other, *i.e.*,

$$\cos\theta = \cos\frac{\pi}{2} = 0 \text{ and then } \vec{A}\cdot\vec{B} = 0 \qquad ...(5)$$

This is the **condition for two orthogonal vectors.** If this property is applied to unit vectors $\hat{i}$, $\hat{j}$, $\hat{k}$, then

$$\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0 \qquad ...(6)$$

As a consequence of these results, we have

$$\vec{A}\cdot\vec{B} = (A_x\,\hat{i} + A_y\,\hat{j} + A_z\,\hat{k})\cdot(B_x\,\hat{i} + B_y\,\hat{j} + B_z\,\hat{k}) \text{ or } \vec{A}\cdot\vec{B} = A_x B_x + A_y B_y + A_z B_z \qquad ...(7)$$

We can also write

$$\vec{A}\cdot\vec{A} = A_x^2 + A_y^2 + A_z^2 = A^2. \qquad ...(8)$$

1·9·2. Applications of Scalar Product

(1) Work and power : If a constant force $\vec{F}$ acts on a body and produces a displacement $\vec{s}$, **[fig. 1.12]**, then the work W done by the force on the body is defined as W = (Component of force along the displacement) × (distance)

$$= (F\cos\theta)\,s = \vec{F}\cdot\vec{s} \qquad ...(9)$$

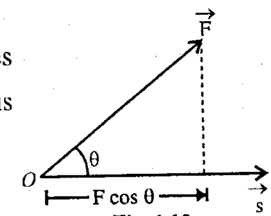


Fig. 1.12.

The rate of doing work is called *power.* Thus, power

$$P = \frac{dW}{dt} = \frac{d}{dt}[\vec{F}\cdot\vec{s}] = \vec{F}\cdot\frac{d}{dt}(\vec{s}) = \vec{F}\cdot\vec{v} \qquad ...(10)$$

where $\vec{v} = d\vec{s}/dt$ is the velocity of the body.

(2) Law of cosine : Let the two sides of a triangle be represented by the vectors $\vec{A}$ and $\vec{B}$. Then the third side may be taken as $\vec{C} = \vec{A} - \vec{B}$ **[fig. 1.13]**. Now, on taking the scalar product of each side of this expression with itself, we get

$$(\vec{A}-\vec{B})\cdot(\vec{A}-\vec{B}) = (\vec{C})\cdot(\vec{C}) \quad \text{or} \quad A^2 + \vec{A}\cdot\vec{B} - \vec{B}\cdot\vec{A} + B^2 = C^2$$

or $A^2 + B^2 - 2\vec{A}\cdot\vec{B} = C^2$ or $C^2 = A^2 + B^2 - 2AB\cos(\vec{A},\vec{B})$ ...(11)

which is the famous cosine law for triangle.

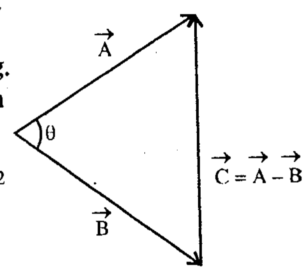


Fig. 1.13.

(3) Equation of a plane : Suppose $ABCD$ be a plane under consideration. Let a normal $\vec{p}$ be drawn from an origin O to this plane; the point O is not lying in the plane. If $\vec{r}$ be any arbitrary vector from the origin O to any point P in the plane, then the projection of $\vec{r}$ on $\vec{p}$ must be equal in magnitude to p. That is,

$$ON = OP\cos\theta \quad \text{or} \quad p = r\cos\theta \quad \text{or} \quad p^2 = p\,r\cos\theta \quad \text{or} \quad \vec{r}\cdot\vec{p} = p^2 \qquad ...(12)$$

which is the equation of the plane

Writing $\qquad \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

and $\qquad \vec{p} = p_x\,\hat{i} + p_y\,\hat{j} + p_z\,\hat{k}$,

We will have the equation of the plane as

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (p_x\,\hat{i} + p_y\,\hat{j} + p_z\,\hat{k}) = p^2$$

or
$$xp_x + yp_y + zp_z = p^2 \qquad \text{...(13)}$$

This is the usual expression in analytical gemoetry for the equation of a plane.

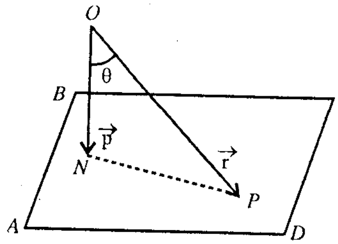


Fig. 1.14.

1·9·3. Vector or Cross Product

The vector (or cross) product of two vectors $\vec{A}$ *and* $\vec{B}$ *inclined at an angle* θ *to each other is defined as a vector whose magnitude is AB* sin θ *and direction is that of a unit vector* $\hat{\mathbf{n}}$ *perpendicular to the plane containing* $\vec{A}$ *and* $\vec{B}$. **[fig. 1.15]**.

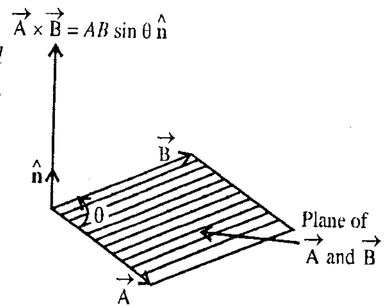


Fig. 1.15 : Vector product.

Symbolically, the vector product of $\vec{A}$ and $\vec{B}$ is denoted by $\vec{A} \times \vec{B}$ and is read as $\vec{A}$ cross $\vec{B}$ Thus, we have

$$\vec{A} \times \vec{B} = AB \sin\theta\ \hat{\mathbf{n}} \qquad \text{...(14)}$$

The direction of the product $\vec{A} \times \vec{B}$ *i.e.*, that of the unit vector $\hat{\mathbf{n}}$ is determined by the right hand screw rule. According to this rule, if a right-hand-screw* is turned from the direction of $\vec{A}$ towards that of $\vec{B}$ through the smaller angle, the direction of its advance gives the direction of $\vec{A} \times \vec{B}$ (or that of $\hat{\mathbf{n}}$). **[fig. 1.16]**.

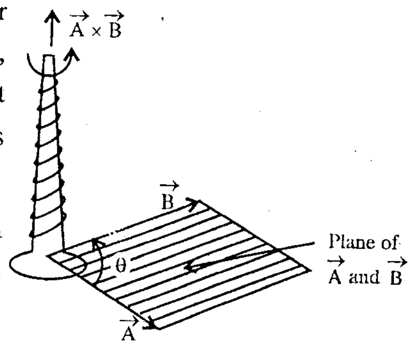


Fig. 1.16 : Motion of right hand screw.

From the definition of vector product it is evident that a change in the order of the factors in a cross product reverses the sign of the product, because the sense of rotation of the screw is reversed,

i.e.,
$$\vec{B} \times \vec{A} = -AB \sin\theta\ \hat{\mathbf{n}} = -\ \vec{A} \times \vec{B} \qquad \text{...(15)}$$

Hence, the vector product of two vectors is *not commutative* and the order of the terms should be strictly maintained. But the distributive law still holds, *i.e.*,

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C} \qquad \text{...(16)}$$

If two vectors $\vec{A}$ and $\vec{B}$ are **parallel** or **collinear**, *i.e.*,

sin θ = sin 0 = 0 and then from eq. (14), we have

$$\vec{A} \times \vec{B} = 0 \qquad \text{...(17)}$$

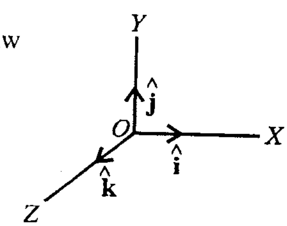


Fig. 1.17.

If we apply this property to unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$ **[fig. 1.17]**, then

$$\hat{\mathbf{i}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}} \times \hat{\mathbf{k}} = 0 \qquad \text{...(18)}$$

In case, if $\vec{A}$ and $\vec{B}$ are **perpendicular** to each other, then

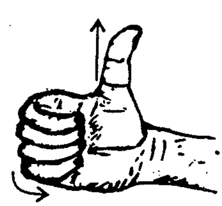


Fig. 1.18 : Right hand screw rule.

* A right hand screw is one that if one's right hand is placed **[fig. 1.18]** with the fingers pointing in the direction of rotation, the screw advances in the direction of the thumb.

$$\vec{A} \times \vec{B} = AB\,\hat{\mathbf{n}} \text{ or } \frac{\vec{A}}{A} \times \frac{\vec{B}}{B} = \hat{\mathbf{n}} \text{ or } \hat{\mathbf{A}} \times \hat{\mathbf{B}} = \hat{\mathbf{n}}$$

Applying this property to unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$, whice are mutually orthogonal, we have

$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}} = -\hat{\mathbf{j}} \times \hat{\mathbf{i}};\ \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}} = -\hat{\mathbf{k}} \times \hat{\mathbf{j}};\ \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} = -\hat{\mathbf{i}} \times \hat{\mathbf{k}} \quad ...(19)$$

If we write the vectors in terms of their rectangular components, then

$$\vec{A} = A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}} \text{ and } \vec{B} = B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}}$$

Now,

$$\vec{A} \times \vec{B} = (A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}) \times (B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}})$$

$$= (A_y B_z - A_z B_y)\hat{\mathbf{i}} + (A_z B_x - B_z A_x)\hat{\mathbf{j}} + (A_x B_y - A_y B_x)\hat{\mathbf{k}} \quad ...(20)$$

This result can be remembered easily, when written in the determinant form, *i.e.*,

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \quad ...(21)$$

1·9·4. Applications of Cross Product

(1) **Area of a parallelogram :** If two vectors $\vec{A}$ and $\vec{B}$ are respectively the sides of a parallelogram *OPRQ*, then area of the parallelogram

$$= OP\,(QN) = A\,(B \sin\theta) = AB \sin\theta = |\vec{A} \times \vec{B}| \quad ...(22)$$

Thus, *the area of the parallelogram with sides* $\vec{A}$ and $\vec{B}$ is $|\vec{A} \times \vec{B}|$. This is twice the area of the triangle *OPR* with sides $\vec{A}$ and $\vec{B}$, *i.e.*,

$$\text{Area of triangle } (OPR) = \frac{1}{2}|\vec{A} \times \vec{B}|. \quad ...(23)$$

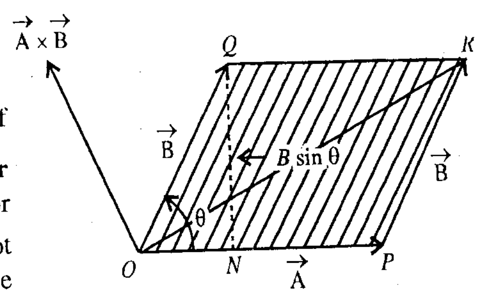


Fig. 1.19 : Area of parallelogram.

The direction of $\vec{A} \times \vec{B}$ is perpendicular to the plane of the parallelogram ; thus we may think of $\vec{A} \times \vec{B}$ as the **vector area of the parallelogram.** However, the direction of the vector area depends how we sweep the area, *i.e.*, $\vec{A} \times \vec{B}$ or $\vec{B} \times \vec{A}$. Not only the area of a parallelogram, but any plane area may be considered as a vector whose magnitude is equal to the area and direction normal to it.

(2) **Torque and angular momentum :** Consider a force $\vec{F}$, acting on the point *P* of a body which can rotate it about a point *O*. The torque or moment of the force (in magnitude) is equal to the product of the force *F* and its perpendicular distance *r* sin θ *i.e.*, $T = F\,r \sin\theta$. In vector from, the torque is represented as

$$\vec{T} = \vec{r} \times \vec{F} \quad ...(24)$$

where $\vec{r}$ is the vector distance of the point *P* relative to *O* [**fig. 1.20**]. The magnitude of the torque is $Fr \sin\theta$ and its direction is perpendicular to the plane containing $\vec{r}$ and $\vec{F}$.

If $\vec{p} = m\vec{v}$ is the linear momentum of a particle, then its moment of momentum *i.e.*, angular momentum $\vec{J}$ about a point *O* is defined as

$$\vec{J} = \vec{r} \times \vec{p} \quad ...(25)$$

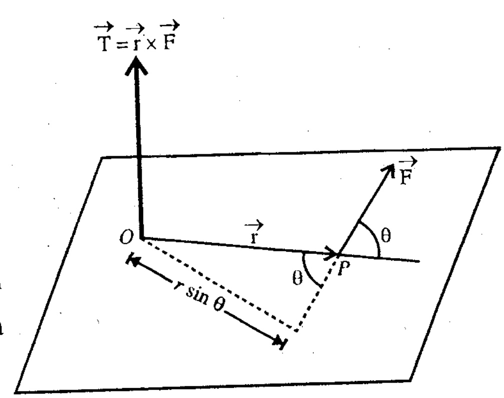


Fig. 1.20 : Torque.

where $\vec{r}$ is the distance of the particle from O. The direction of angular momentum is perpendicular to the plane of $\vec{r}$ and $\vec{p}$.

(3) Force on a moving charge in a magnetic field : If a charge q is moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$, then for the force $\vec{F}$, acting on the charged particle, experimentally following relation holds good

$$F = qvB \sin \theta$$

where θ is the angle between the direction of the field $\vec{B}$ and the velocity $\vec{v}$ of the particle and F, v, B are the magnitudes of the corresponding physical quantities. The direction of the force is perpendicular to both $\vec{v}$ and $\vec{B}$.

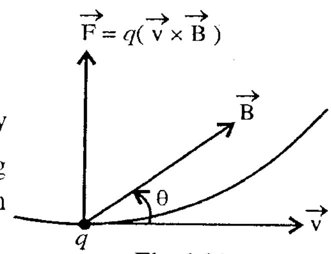


Fig. 1.21.

In vector from, the force

$$\vec{F} = q\,(\vec{v} \times \vec{B}) \qquad \text{...(26)}$$

From this relation it is evident that the force $\vec{F}$ is perpendicular to both $\vec{v}$ and $\vec{B}$. This force is called **Lorentz force.**

The above relation will hold when either S.I. units or electrostatic units (e.s.u) or electromagnetic units (e.m.u) are employed for measuring all the quantities.

S.I. unit for the measurement of the magnetic field is weber/metre^2 or tesla. An earlier unit for $\vec{B}$, still common in use, is the gauss (or oersted) :

$$1 \text{ gauss} = 10^{-4} \text{ tesla.}$$

S.I. unit of charge is coulomb. Measurement of charge is also common in e.s.u. or stat coulomb.

$$1 \text{ coulomb} = 3 \times 10^9 \text{ e.s.u.}$$

If $\vec{B}$ is given in gauss, *i.e.*, in e.m.u and q in e.s.u., then the force in dynes is given by

$$\vec{F} = \frac{q}{c}\,(\vec{v} \times \vec{B}) \qquad \text{...(27)}$$

because 1 e.s.u. charge is equal to 1/c e.m.u.

Simultaneously if the charged particle moves through an electric field $\vec{E}$, the additional force $q\vec{E}$ will act on the particle. In such a case, the Lorentz force is

$$\vec{F} = q\vec{E} + q\,(\vec{v} \times \vec{B}) \qquad \text{...(28)}$$

(4) Condition for two vectors to be parallel : As the cross product of two parallel vectors is zero, hence

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = 0 \qquad \text{...(29)}$$

$$\therefore = (A_y B_z - A_z B_y)\hat{i} - (A_x B_z - B_x A_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k} = 0$$

This is possible only when the coefficients of $\hat{i}$, $\hat{j}$ and $\hat{k}$ are separately zero *i.e.*,

$$A_y B_z - A_z B_y = 0, \; A_x B_z - A_z B_x = 0, \; A_x B_y - A_y B_x = 0$$

or $\quad A_y B_z = A_z B_y, \; A_x B_z = A_z B_x, \; A_x B_y = A_y B_x$

i.e., $\quad \dfrac{A_y}{B_y} = \dfrac{A_z}{B_z}, \; \dfrac{A_x}{B_x} = \dfrac{A_z}{B_z}, \; \dfrac{A_x}{B_x} = \dfrac{A_y}{B_y}$ *i.e.*, $\dfrac{A_x}{B_x} = \dfrac{A_y}{B_y} = \dfrac{A_z}{B_z}$ ...(30)

This, is the required condition for two vectors to be parallel.

1·9·5. Null or Zero Vector

A vector whose magnitude is zero and direction is indeterminate, is called **null** or **zero vector**. It is represented by $\vec{0}$. For example, if $\vec{A}$ and $\vec{B}$ vectors have equal magnitude but are in opposite directions (*i.e.*, $\vec{B} = -\vec{A}$), then

$$\vec{A} + \vec{B} = \vec{0} \qquad ...(31)$$

For mathematical consistency, it is convenient to have a vector of zero magnitude. Its direction has no meaning because the origin and terminus of a null vector are coincident.

When we determine the vector product of two parallel vectors, the concept of null vector is also helpfull *i.e.*, for parallel vectors $\vec{A}$ and $\vec{B}$,

$$\vec{A} \times \vec{B} = \vec{0} \qquad ...(32)$$

For any vector $\vec{A}$,

$$\vec{A} + \vec{0} = \vec{A}, \quad \text{and} \quad \vec{A} \times \vec{0} = \vec{0}, \qquad ...(33)$$

and for any number α, $\alpha\vec{0} = \vec{0}$. ...(34)

Ex. 1. *If* $\vec{r} = 3\hat{i} + \hat{j} + 2\hat{k}$, *find a unit vector along* $\vec{r}$. *Calculate also the direction cosines of the vector* $\vec{r}$. *What is its projection vector in the X-Y plane ? Calculate the magnitude of its component vectors in XZ and YZ planes.*

Sol. Magnitude of $\vec{r} = \sqrt{3^2 + 1^2 + 2^2} = \sqrt{14}$. If $\hat{r}$ is the unit vector along $\vec{r}$, then

$$\hat{r} = \frac{\vec{r}}{r} = \frac{3\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{14}}$$

Now, $\cos(\vec{r}, x) = \dfrac{3}{\sqrt{14}}$, $\cos(\vec{r}, y) = \dfrac{1}{\sqrt{14}}$ and $\cos(\vec{r}, z) = \dfrac{2}{\sqrt{14}}$

$\therefore$ Direction cosines are $\left(\dfrac{3}{\sqrt{14}}, \dfrac{1}{\sqrt{14}}, \dfrac{2}{\sqrt{14}}\right)$.

Projection vector of $\vec{r}$ in X-Y plane $= 3\hat{i} + \hat{j}$.

The component vectors of $\vec{r} = 3\hat{i} + \hat{j} + 2\hat{k}$ in the XZ and YZ planes are respectively $\vec{r}_{xz} = 3\hat{i} + 2\hat{k}$ and $\vec{r}_{yz} = \hat{j} + 2\hat{k}$.

The magnitudes of the component vectors are

$$\left|\vec{r}_{xz}\right| = \sqrt{3^2 + 2^2} = \sqrt{13} \text{ and } \left|\vec{r}_{yz}\right| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

Ex. 2. *If* $\vec{A} = \hat{i} + 2\hat{j} + \hat{k}$ and $\vec{B} = 2\hat{i} - \hat{j} + \hat{k}$, *calculate the magnitudes and direction cosines of* $(\vec{A} + \vec{B})$ and $(\vec{A} - \vec{B})$.

Sol. Here, $\vec{A} + \vec{B} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{A} - \vec{B} = -\hat{i} + 3\hat{j}$

$\therefore$ $\left|\vec{A} + \vec{B}\right| = \sqrt{3^2 + 1^2 + 2^2} = \sqrt{14}$ and $\left|\vec{A} - \vec{B}\right| = \sqrt{1^2 + 3^2} = \sqrt{10}$

$\therefore$ Direction cosines of $(\vec{A} + \vec{B})$ are $\left(\dfrac{3}{\sqrt{14}}, \dfrac{1}{\sqrt{14}}, \dfrac{2}{\sqrt{14}}\right)$ and that of $(\vec{A} - \vec{B})$ are $\left(-\dfrac{1}{\sqrt{10}}, \dfrac{3}{\sqrt{10}}, 0\right)$.

Ex. 3. *If* $\vec{A} = \hat{i} + 2\hat{j} + 4\hat{k}$ and $\vec{B} = 2\hat{i} - \hat{j} + 3\hat{k}$, *calculate their scalar product. What is the component of* $\vec{B}$ *along* $\vec{A}$ *and that of* $\vec{A}$ *along* $\vec{B}$?

Sol. Scalar product $\vec{A}\cdot\vec{B} = (\hat{i} + 2\hat{j} + 4\hat{k})\cdot(2\hat{i} - \hat{j} + 3\hat{k}) = 2 - 2 + 12 = 12.$

Now, $\vec{A} = \hat{i} + 2\hat{j} + 4\hat{k} \quad \therefore \quad A = \sqrt{1^2 + 2^2 + 4^2} = \sqrt{21}$

$\vec{B} = 2\hat{i} - \hat{j} + 3\hat{k} \quad \therefore \quad B = \sqrt{2^2 + 1^2 + 3^2} = \sqrt{14}$

$\therefore$ Unit vector along of $\vec{A}$ along $\vec{B}$ are

$$\hat{\mathbf{A}} = \frac{1}{\sqrt{21}}(\hat{i} + 2\hat{j} + 4\hat{k}) \text{ and } \hat{\mathbf{B}} = \frac{1}{\sqrt{14}}(2\hat{i} - \hat{j} + 3\hat{k})$$

The component of $\vec{A}$ along $\vec{B}$ [fig. **1.22**]

$$= (OP)\,\hat{\mathbf{B}} = A\cos\theta\,\hat{\mathbf{B}}$$

$$= \frac{AB\cos\theta}{B}\,\hat{\mathbf{B}} = \frac{\vec{A}\cdot\vec{B}}{B}\hat{\mathbf{B}}$$

$$= \mathbf{12/\sqrt{14}\ \text{along}}\ \vec{B}$$

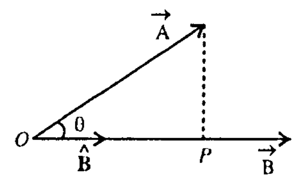


Fig. 1.22.

Similarly, the component of $\vec{B}$ along $\vec{A}$ is

$$= \frac{1}{A}(\vec{A}\cdot\vec{B})\hat{\mathbf{A}} = \frac{\mathbf{12}}{\sqrt{\mathbf{21}}}\ \textbf{along}\ \vec{A}.$$

Ex. 4. *If the non-parallel vectors* $\vec{A}$ and $\vec{B}$ *are equal in magnitude, show that vector* $(\vec{A}+\vec{B})$ *is perpendicular to the vector* $(\vec{A}-\vec{B})$. **(Agra 1994)**

Sol. If the vectors $(\vec{A}+\vec{B})$ and $(\vec{A}-\vec{B})$ are perpendicular to each other, then

$$(\vec{A}+\vec{B})\cdot(\vec{A}-\vec{B}) = 0$$

Here $|\vec{A}| = |\vec{B}|$ or $A^2 = B^2$

Now, $(\vec{A}+\vec{B})\cdot(\vec{A}-\vec{B}) = \vec{A}\cdot\vec{A} - \vec{A}\cdot\vec{B} + \vec{B}\cdot\vec{A} - \vec{B}\cdot\vec{B} = A^2 - B^2 = 0$

Hence, $(\vec{A}+\vec{B})$ **is perpendicular to** $(\vec{A}-\vec{B})$.

Ex. 5. *If* $\vec{A} = 4\hat{i} + 6\hat{j} - 3\hat{k}$ *and* $\vec{B} = 2\hat{i} + 5\hat{j} + 7\hat{k}$, *find out the angle between* $\vec{A}$ *and* $\vec{B}$.

Sol. $\vec{A} = 4\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{B} = 2\hat{i} + 5\hat{j} + 7\hat{k}$

$\therefore \quad A = \sqrt{4^2 + 6^2 + 3^2} = \sqrt{61}$ and $B = \sqrt{2^2 + 5^2 + 7^2} = \sqrt{78}$

Now, $\vec{A}\cdot\vec{B} = AB\cos\theta$

Here, $\vec{A}\cdot\vec{B} = (4\hat{i} + 6\hat{j} - 3\hat{k})\cdot(2\hat{i} + 5\hat{j} + 7\hat{k}) = 8 + 30 - 21 = 17$

$$\therefore \quad \cos\theta = \frac{\vec{A}\cdot\vec{B}}{AB} = \frac{17}{\sqrt{61}\times\sqrt{78}} \text{ or } \theta = \cos^{-1}\left(\frac{17}{\sqrt{4758}}\right) = \mathbf{75^\circ\ 43'56''}.$$

This θ is the angle between the vectors $\vec{A}$ and $\vec{B}$.

Ex. 6. *If* (l_1, m_1, n_1) *and* (l_2, m_2, n_2) *are the direction cosines of two vectors, show that the angle* θ *between them is given by* $\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$. **(Agra 1993, 87)**

Sol. Let $\vec{A}$ and $\vec{B}$ be the two vectors. Then from $\vec{A}\cdot\vec{B} = AB\cos\theta$, we have

$$\therefore \cos\theta = \frac{\vec{A}\cdot\vec{B}}{AB} = \frac{A_xB_x + A_yB_y + A_zB_z}{AB} = \frac{A_x}{A}\frac{B_x}{B} + \frac{A_y}{A}\frac{B_y}{B} + \frac{A_z}{A}\frac{B_z}{B} = l_1l_2 + m_1m_2 + n_1n_2.$$

because direction cosines are $\cos(\vec{A}, X) = A_x/A = l_1$, $\cos(\vec{B}, X) = B_x/B = l_2$ etc,

Ex. 7. *The projection velocity of a rocket is expressed as* $\vec{V} = 5\hat{v}_x + 7\hat{v}_y + 9\hat{v}_z$, where $\hat{v}_x$, $\hat{v}_y$, $\hat{v}_z$ *are unit velocity vectors along east, north and vertical directions respectively. Calculate the magnitudes of horizontal and vertical components of the velocity. Also deduce the change in angle of projection if the vertical component is doubled.* **(Agra 1969)**

Sol. $\vec{V} = 5\hat{v}_x + 7\hat{v}_y + 9\hat{v}_z$

Horizontal component = $5\hat{v}_x + 7\hat{v}_y$; its magnitude = $\sqrt{5^2+7^2} = \sqrt{\mathbf{74}}$

Vertical component = $9\hat{v}_z$, its magnitude = **9.**

In second case, vertical component is doubled, *i.e.*, $2(9\hat{v}_z) = 18\hat{v}_z$.

Then, velocity $\vec{V}' = 5\hat{v}_x + 7\hat{v}_y + 18\hat{v}_z$.

Obviously, the change in direction of the velocity will be equal to the angle between $\vec{V}$ and $\vec{V}'$. Then from $\vec{V}\cdot\vec{V}' = VV'\cos(\vec{V},\vec{V}')$, we have

$$\cos(\vec{V},\vec{V}') = \frac{\vec{V}\cdot\vec{V}'}{VV'} = \frac{25+49+162}{\sqrt{25+49+81}\times\sqrt{25+49+324}} = \frac{236}{\sqrt{155}\times\sqrt{398}} = 0.95$$

Therefore, angle $(\vec{V},\vec{V}') = \cos^{-1}(0.95) = \mathbf{18^\circ\ 12'}$.

Ex. 8. *A particle moves from position* $3\hat{i} + 2\hat{j} - 6\hat{k}$ *to* $14\hat{i} + 13\hat{j} + 9\hat{k}$ metre while a uniform force $4\hat{i} + 1\hat{j} + 3\hat{k}$ *newton acts on it. Calculate the work done by the force.* **(Agra 1969 S)**

Sol. Force $\vec{F} = 4\hat{i} + 1\hat{j} + 3\hat{k}$ newton.

Displacement $\vec{s} = (14\hat{i} + 13\hat{j} + 9\hat{k}) - (3\hat{i} + 2\hat{j} - 6\hat{k}) = 11\hat{i} + 11\hat{j} + 15\hat{k}$ metre

$\therefore$ Work done = $\vec{F}\cdot\vec{s} = (4\hat{i} + 1\hat{j} + 3\hat{k})\cdot(11\hat{i} + 11\hat{j} + 15\hat{k}) = 44 + 11 + 45 =$ **100 joules.**

Ex. 9. *Find a vector having module 6 and perpendicular to the vectors* $\vec{A} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 3\hat{k}$. *Find the sine of the angle between* $\vec{A}$ and $\vec{B}$.

Sol.
$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 2 & -2 & 3 \end{vmatrix} = 8\hat{i} - 4\hat{j} - 8\hat{k}$$

Vector $\vec{A}\times\vec{B}$ is perpendicular to $\vec{A}$ and $\vec{B}$, but its magnitude is $\left|(\vec{A}\times\vec{B})\right| = \sqrt{8^2+4^2+8^2} = 12$.

$\therefore$ A unit vector along $\vec{A}\times\vec{B}$ is $\dfrac{\vec{A}\times\vec{B}}{\left|(\vec{A}\times\vec{B})\right|} = \dfrac{8\hat{i}-4\hat{j}-8\hat{k}}{12}$

Therefore, a vector, whose magnitude is 6 and is perpendicular to $\vec{A}$ and $\vec{B}$, is

$$\vec{V} = 6\frac{(8\hat{i}-4\hat{j}-8\hat{k})}{12} = 4\hat{i} - 2\hat{j} - 4\hat{k}$$

But $\left|\vec{A}\times\vec{B}\right| = AB\sin\left|\vec{A},\vec{B}\right|$; $\therefore$ $\sin(\vec{A},\vec{B}) = \dfrac{\left|\vec{A}\times\vec{B}\right|}{AB} = \dfrac{12}{(\sqrt{9}\times\sqrt{17})} = \dfrac{\mathbf{4}}{\sqrt{\mathbf{17}}}$.

Ex. 10. *Find the area of the parallelogram determined by the vectors* $(\hat{i} + 2\hat{j} + 3\hat{k})$ *and* $(3\hat{i} - 2\hat{j} + \hat{k})$. **(Purvanchal 1995)**

Sol. Area of the parallelogram (magnitude) = $\left|\vec{A}\times\vec{B}\right| = (\hat{i} + 2\hat{j} + 3\hat{k})\times(3\hat{i} - 2\hat{j} + \hat{k})$.

$$= |(8\hat{i}+8\hat{j}-8\hat{k})| = 8|(\hat{i}+\hat{j}-\hat{k})| = 8\sqrt{1+1+1} = \mathbf{8\sqrt{3}\ \text{sq. units.}}$$

$\therefore$ Direction of area $\hat{\mathbf{n}} = \dfrac{8\hat{i}+8\hat{j}-8\hat{k}}{8\sqrt{3}} = \dfrac{\mathbf{1}}{\sqrt{\mathbf{3}}}(\hat{i} + \hat{j} - \hat{k})$.

Ex. 11. *If the vectors* $(\hat{i} + 2\hat{j} + 4\hat{k})$ *and* $(5\hat{i})$ *represent the two sides of a triangle, find the following* : (i) *area of the triangle,* (ii) *third side of the triangle, and* (iii) *a vector perpendicular to the plane of the traingle.*

Sol. (i) Area of the triangle $= \frac{1}{2}\vec{A}\times\vec{B} = \frac{1}{2}(\hat{i} + 2\hat{j} + 4\hat{k}) \times (5\hat{i}) = 10\hat{j} - 5\hat{k}$

Its magnitude $= \sqrt{10^2+5^2} = \mathbf{5\sqrt{5}\ \text{units.}}$

(ii) Third side of the triangle $\vec{A}-\vec{B} = (\hat{i} + 2\hat{j} + 4\hat{k}) - (5\hat{i}) = \mathbf{-4\hat{i} + 2\hat{j} + 4\hat{k}}$.

(iii) A vector perpendicular to the plane of the triangles is $\vec{A}\times\vec{B}$, *i.e.*, $\mathbf{20\hat{j} - 10\hat{k}}$.

Ex. 12. *A horizontal force of 5 newton is acting towards the north-east direction at a point situated 20 cm to the east of a pivot and 40 cm higher than the pivot. Calculate the magnitude of the moment of force and the direction of axis of rotaion.*

Sol. If $\hat{i}$, $\hat{j}$, $\hat{k}$ represent unit vectors along east, north and vertical directions,

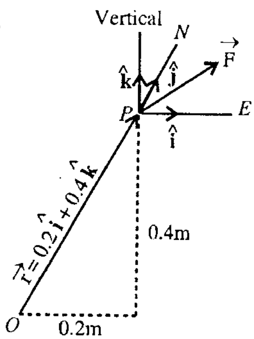


Fig. 1.23.

then the force [**fig. 1.23**], $\vec{F} = \dfrac{5}{\sqrt{2}}\hat{i} + \dfrac{5}{\sqrt{2}}\hat{j}$ newton

Distance $\vec{r} = 0{\cdot}2\hat{i} + 0{\cdot}4\hat{k}$

$\therefore$ Moment of force $= \vec{r} \times \vec{F} = \dfrac{1}{\sqrt{2}}(-2\hat{i} + 2\hat{j} + \hat{k})$

$\therefore$ Magnitude of the moment of the force $= \dfrac{1}{\sqrt{2}}\times\sqrt{9} = \mathbf{2{\cdot}116\ N\text{-}m.}$

Evidently, direction of the axis of rotation is parallel to $\vec{r} \times \vec{F}$, whose direction cosines are $(-2/3, 2/3, 1/3)$.

Ex. 13. *Find vectors of magnitude 8 which are perpendicular to the vector* $\vec{V} = \hat{i} - \hat{j} + \hat{k}$ *and parallel to the plane* $x + 2y - 3z = 15$.

Sol. Let the required vectors be represented by $\vec{A}$, having magnitude $A = 8$.

Evidently, the plane $x + 2y - 3z = 15$ is perpendicular to the vector $\vec{V}' = \hat{i} + 2\hat{j} - 3\hat{k}$

Therefore, $\vec{A}$ must be perpendicular to both $\vec{V}$ and $\vec{V}'$. Unit vectors, which are perpendiculat to $\vec{V}$ and $\vec{V}'$, will be given by

$$\hat{\mathbf{A}} = \pm\frac{\vec{V}\times\vec{V}'}{|\vec{V}\times\vec{V}'|}$$

Here, $\vec{V}\times\vec{V}' = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 1 & 2 & -3 \end{vmatrix} = \hat{i} + 4\hat{j} + 3\hat{k}$ and $|\vec{V}\times\vec{V}'| = \sqrt{1^2+4^2+3^2} = \sqrt{26}$

$$\therefore \quad \hat{\mathbf{A}} = \pm\frac{1}{\sqrt{26}}(\hat{i} + 4\hat{j} + 3\hat{k})$$

Therefore, $\vec{A} = A\ \hat{A} = \pm\dfrac{\mathbf{8}}{\sqrt{\mathbf{26}}}(\hat{i} + 4\hat{j} + 3\hat{k})$.

Ex. 14. *Find the vectors whose magnitude is* $\sqrt{12}$ *and which is perpendiculat to* $\hat{j} + \hat{k}$ *vector and lies in the plane* $2x + 3y + z = 5$.

Sol. Let the required vector be represented as

$$\vec{V} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}\,;\; V_x^2 + V_y^2 + V_z^2 = 12. \quad \text{...(i)}$$

The vector $\vec{V}$ is perpendicular to the vector $\hat{j} + \hat{k}$, hence

$$(V_x \hat{i} + V_y \hat{j} + V_z \hat{k}) \cdot (\hat{j} + \hat{k}) = 0 \quad \text{or} \quad V_y + V_z = 0 \quad \text{or} \quad V_z = -V_y \quad \text{...(ii)}$$

The vector $\vec{V}$ lies in the plane $2x + 3y + z = 5$. This means that $\vec{V}$ is perpendicular to $2\hat{i} + 3\hat{j} + \hat{k}$ *i.e.*,

$$(V_x \hat{i} + V_y \hat{j} + V_z \hat{k}) \cdot (2\hat{i} + 3\hat{j} + \hat{k}) = 0 \quad \text{or} \quad 2V_x + 3V_y + V_z = 0 \quad \text{...(iii)}$$

Subtracting eq. (ii) from (iii), we have

$$2V_x + 2V_y = 0 \quad \text{or} \quad V_x = -V_y \quad \text{...(iv)}$$

Substituting the values of V_x and V_z from eq. (ii) and (iv) in eq. (i), we get

$$3V_y^2 = 12 \quad \text{or} \quad V_y = \pm 2., \quad \therefore \quad V_x = \mp 2 \text{ and } V_z = \mp 2$$

Therefore, $\vec{V} = \pm 2(-\hat{i} + \hat{j} - \hat{k})$

Alternatively, $$\hat{n} = \pm\frac{(\hat{j}+\hat{k})\times(2\hat{i}+3\hat{j}+\hat{k})}{(\hat{j}+\hat{k})\times(2\hat{i}+3\hat{j}+\hat{k})} = \pm\frac{-2\hat{i}+2\hat{j}-2\hat{k}}{\sqrt{12}}$$

$$\therefore \quad \vec{V} = \sqrt{12}\,\hat{n} = \pm 2(-\hat{i} + \hat{j} - \hat{k}).$$

Ex. 15. *A proton of velocity* $(3\hat{i} + 2\hat{j})\,10^6$ *m/sec enters a magnetic field* $2000\hat{j} + 3000\hat{k}$ *gauss. Calculate the acceleration. Also deduce the simultaneous electric field which will keep the particle undeflected. (Mass of proton* $= 1.7 \times 10^{-27}$ *kg)* **(Agra 1970)**

Sol. $\because$ Acceleration $= \dfrac{\vec{F}}{m_p} = \dfrac{q}{m_p}(\vec{v} \times \vec{B})$ $\quad [\because F = q(\vec{v}\times\vec{B})]$

$$= \frac{1.6\times10^{-19}}{1.7\times10^{-27}}[(3\hat{i} + 2\hat{j})10^6 \times (2000\hat{j} + 3000\hat{k})\,10^{-4}]$$

$$= \mathbf{2.8 \times 10^{13}\,(2\hat{i} - 3\hat{j} + 2\hat{k})\ m/sec^2.}$$

Let $\vec{E}$ be the electric field, required to balance the force due to magnetic field. So that

$$\vec{F} = q(\vec{v}\times\vec{B}) + q\vec{E} = 0 \quad \text{or} \quad \vec{E} = -(\vec{v}\times\vec{B}) = \mathbf{-3 \times 10^5\,(2\hat{i} - 3\hat{j} + 2\hat{k})\ N/coul.}$$

EXERCISE 1.1

1. Find the vector, which must be added to the sum of two vectors $\hat{i} + 2\hat{j} + \hat{k}$ and $\hat{i} - 2\hat{j} + 2\hat{k}$ to get a resultant unit vector along Z-axis.
   [**Ans.** $-2\hat{i}$.]
2. If $\vec{A} = 2\hat{i} - 4\hat{j} - 3\hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} + \hat{k}$, find $|\vec{A}+\vec{B}|$, $|\vec{A}-\vec{B}|$ and a unit vector parallel to $(\vec{A}+\vec{B})$.
   [**Ans.** $\sqrt{17}$; $\sqrt{53}$; $\frac{1}{\sqrt{17}}(3\hat{i} - 2\hat{j} - 2\hat{k})$]
3. If $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{B} = 2\hat{i} - 3\hat{j} + \hat{k}$, calculate the magnitudes and direction cosines of $(\vec{A}+\vec{B})$ and $(\vec{A}-\vec{B})$.
   [**Ans.** $\sqrt{20}, 6$; $\left(\frac{2}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}}\right)$, (0, 1, 0)]

4. If vectors $\vec{A} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{B} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{C} = \hat{i} - 2\hat{j} + 2\hat{k}$, find the magnitude and direction cosines of vectors $(\vec{A} + \vec{B} + \vec{C})$ and $(\vec{A} - \vec{B} + \vec{C})$. **(Agra 1982)**

[**Ans.** $\sqrt{17}, \sqrt{5}$; $\left(\frac{4}{\sqrt{17}}, 0, \frac{1}{\sqrt{17}}\right), \left(0, -\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right)$]

5. If $\vec{A} = \hat{i} + \hat{j} + \sqrt{2}\,\hat{k}$, find (i) the angle between this vector and Z-axis. (ii) the magnitude of the component of this vector in the *Y-Z* plane.

[**Ans.** (i) 45°; (ii) $\sqrt{3}$ unit.]

6. If α, β be the direction cosines of a vector with respect to *X* and *Y*-axes and if *C* is its *Z*-component, find the magnitude of the vector.

[**Ans.** $C/\sqrt{1 - \alpha^2 - \beta^2}$.]

7. Show that $\vec{A} = 2\hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{B} = 2\hat{i} + 4\hat{j} + 8\hat{k}$ are mutually perpendicular. **(Rewa 2003)**

8. If $\left|\vec{A} + \vec{B}\right| = \left|\vec{A} - \vec{B}\right|$, show that $\vec{A}$ and $\vec{B}$ are perpendicular to each other. **(Agra 1990)**

9. If the vectors $(3\hat{i} - 2\hat{j} + \hat{k})$ and $(2\hat{i} + 6\hat{j} + m\hat{k})$ are perpendicular to each other, find the value of *m*.

[**Ans.** $m = 6$.]

10. If $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{B} = 3\hat{i} + 6\hat{j} + 2\hat{k}$, find the projection of $\vec{A}$ along $\vec{B}$ and the angle between the vectors. **(Garhwal 1995)**

[**Ans.** 4/7, $\theta = \cos^{-1}$ (16/21).]

11. Find the value of *n* such that vector $4\hat{i} + 3\hat{j} - 7\hat{k}$ and $5\hat{i} - 2\hat{j} - n\hat{k}$ may be orthogonal. **(Rohilkhand 1980)**

[**Ans.** $n = -2$.]

12. Show that $(\vec{A} + \vec{B})^2 - (\vec{A} - \vec{B})^2 = 4(\vec{A} \cdot \vec{B})$. **(Agra 1980)**

13. If the angle between unit vectors $\hat{a}$ and $\hat{b}$ is θ, then prove that

$$\sin\frac{\theta}{2} = \frac{1}{2}\,|\hat{a} - \hat{b}|.$$ **(Bundelhand 1995, 94)**

14. A particle on which a force $\vec{F} = -3\hat{i} - \hat{j} + 2\hat{k}$ newton acts is displaced from point (4, –3, –5) *m* to point (–1, 4, 3) *m*. Calculate the work done by the force. **(Agra 1984)**

[**Ans.** 24 joule.]

15. Two constant forces $2\hat{i} + 2\hat{j} + 5\hat{k}$ and $3\hat{i} - 5\hat{j} - 4\hat{k}$ act together on a particle which is displaced from the point $2\hat{i} + 4\hat{k}$ to the point $20\hat{i} + 17\hat{j} - 3\hat{k}$. Calculate the work done by the forces. **(Rohilkhand 1980)**

[**Ans.** 142 units.]

16. Find the sides and angles of the triangle formed by the points whose position vectors are $\hat{i} - 2\hat{j} + 3\hat{k}$, $2\hat{i} + \hat{j} - \hat{k}$ and $3\hat{i} - \hat{j} + 2\hat{k}$. **(Agra 1992; Purvanchal 95)**

[**Ans.** $\sqrt{26}, \sqrt{14}, \sqrt{6}$; 43.9°, 27.0°, 109.1°.]

17. Given two vectors such that $\vec{A} + \vec{B} = 11\hat{i} - \hat{j} + 5\hat{k}$ and $\vec{A} - \vec{B} = -5\hat{i} + 11\hat{j} + 9\hat{k}$. Find the angle included between $\vec{A}$ and $(\vec{A} + \vec{B})$, using vector method. **(Kumaun 1995)**

$$\left[\textbf{Ans.}\cos^{-1}\left(\frac{63}{\sqrt{(83)(147)}}\right) = 55.25^\circ.\right]$$

18. If $\vec{A} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{B} = 3\hat{i} + 2\hat{j} - \hat{k}$, then find : (i) $\vec{A} \cdot \vec{B}$ (ii) angle between $\vec{A}$ and $\vec{B}$, (iii) $\vec{A} \times \vec{B}$. **(Rewa 2004)**

[**Ans.** (i) -1, (ii) $\cos^{-1}(-1/84)$, (iii) $-3\hat{i} + 7\hat{j} + 5\hat{k}$.]

19. Find the unit vector which is horizontal and perpendicular to the vector $4\hat{i} + 3\hat{j} - 7\hat{k}$, where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along east, north and vertically upward directions respectively. **(Agra 1995)**

[**Ans.** $(3\hat{i} - 4\hat{j})/5$.]

20. Find unit vector which is normal to the vector $\vec{V} = x\hat{i} + y\hat{j} + z\hat{k}$ and lies in the plane $x = C$.

$$\left[\textbf{Ans.} \frac{1}{\sqrt{y^2 + z^2}}(z\,\hat{j} - y\,\hat{k}) or \frac{1}{\sqrt{y^2 + z^2}}(-z\,\hat{i} + y\,\hat{k}).\right]$$

21. Write down unit vectors which are perpendicular to the vector $l\hat{i} + m\hat{j} + n\hat{k}$ and lie (i) in the X-Y plane, (ii) in the Y-Z plane and (iii) in X-Z plane.

$$\left[\textbf{Ans.}\ \text{(i)} \frac{m\,\hat{i} - l\,\hat{j}}{\sqrt{m^2 + l^2}} \text{ and } \frac{-m\,\hat{i} + l\,\hat{j}}{\sqrt{m^2 + l^2}} \text{ and similary for (ii) and (iii).}\right]$$

$$\left[\textbf{Hint : (i) } \hat{V} = \pm \frac{\hat{k} \times (l\,\hat{i} + m\,\hat{j} + n\,\hat{k})}{|\hat{k} \times (l\,\hat{i} + m\,\hat{j} + n\,\hat{k})|} = \pm \frac{m\,\hat{i} - l\,\hat{j}}{\sqrt{m^2 + l^2}}.\right]$$

22. Determine unit vectors which are normal to X-axis and lie in the plane $3x - 2y + 3z = 15$.

[**Ans.** $\pm(3\hat{j} + 2\hat{k})/\sqrt{13}.$]

$$\left[\textbf{Hint : } \hat{V} = \pm \frac{\hat{i} \times (3\,\hat{i} - 2\,\hat{j} + 3\,\hat{k})}{|\hat{i} \times (3\,\hat{i} - 2\,\hat{j} + 3\,\hat{k})|} = \pm(3\,\hat{j} + 2\,\hat{k})/\sqrt{13}.\right]$$

23. Two particles are emitted from a common source and at a particular time have displacements

$$\vec{r}_1 = 4\,\hat{i} + 3\,\hat{j} + 8\,\hat{k}\,;\ \vec{r}_2 = 2\,\hat{i} + 10\,\hat{j} + 5\,\hat{k}.$$

(a) Sketch the position of the particles and write expression for the displacement $\vec{r}$ of particle 2 relative to particle 1.

[**Ans.** $\vec{r} = -2\hat{i} + 7\hat{j} - 3\hat{k}$.]

(b) Use the scalar product to find the magnitude of each vector.

[**Ans.** $r_1 = 9.4$; $r_2 = 11.3$; $r = 7.9$.]

(c) Calculate the projection of $\vec{r}$ on $\vec{r}_1$.

[**Ans.** -1.2.]

24. Show that the acute angle between two diagonals of a cube has cosine 1/3. What is the angle between the diagonals of the cube and one of its edge ?

[**Ans.** $\cos^{-1}(1/\sqrt{3})$.]

25. Find a vector perpendicular to the two vectors $(\hat{i} + \hat{j})$ and $(\hat{j} + \hat{k})$. **(Ujjain 1981)**

[**Ans.** $\hat{i} - \hat{j} + \hat{k}$]

26. Find the area of parallelogram whose diagonals are $(3\hat{i} + \hat{j} - 2\hat{k})$ and $(\hat{i} - 3\hat{j} + 4\hat{k})$. **(Bundelkhand 1995)**

[**Ans.** $\sqrt{74}$ unit.]

[**Hint.** $\vec{A} + \vec{B} = (3\hat{i} + \hat{j} - 2\hat{k})$ and $\vec{A} - \vec{B} = (\hat{i} - 3\hat{j} + 4\hat{k})$; find $|\vec{A} \times \vec{B}|$.]

27. Write down the condition that (i) $\vec{A}$ is perpendicular to $\vec{B}$,(ii) $\vec{A}$ is parallel to $\vec{B}$ and (iii) $\vec{A}$ is perpendicular to the plane of $\vec{B}$ and $\vec{C}$.

[**Ans.** $\vec{A}\cdot\vec{B}=0$; (ii) $\vec{A}\times\vec{B}=0$; (ii) $\vec{A}\times(\vec{B}\times\vec{C})=0$]

28. Find the area of the parallelogram determined by the vector $\hat{i}+2\hat{j}+3\hat{k}$ and $3\hat{i}-2\hat{j}+\hat{k}$. Find also the sine of the angle between them. **(Gorakhpur 1978)**

[**Ans.** $8\sqrt{3}$ sq. units; $4\sqrt{3}/7$.]

29. Find a unit vector perpendiculat to $\hat{i}+2\hat{k}$ and $\hat{i}+\hat{j}-\hat{k}$. Find also the area of the triangle with these two vectors as adjacent sides. **(Gorakhpur 1980)**

[**Ans.** $\frac{1}{\sqrt{14}}(-2\hat{i}+3\hat{j}+\hat{k})$; $\frac{1}{2}\sqrt{14}$.]

30. If the position vectors of the points *A, B, C* are $\vec{r}_1=7\hat{i}-5\hat{j}+3\hat{k}$, $\vec{r}_2=2\hat{i}+4\hat{j}-\hat{k}$ and $\vec{r}_3=-8\hat{i}+22\hat{j}-9\hat{k}$, show that $\vec{AB}$ and $\vec{BC}$ are collinear vectors.

31. Deduce the conditions that vectors $5\hat{i}+7\hat{j}-3\hat{k}$ and $2\hat{i}-b\hat{j}+c\hat{k}$ may be parallel.
(Bilaspur 2003; Agra 1978; Rohilkhand 79)

[**Ans.** $b=-14/5$; $c=-6/5$.]

32. Show that the cross product of each two of the following vectors is parallel to the third :

$$(\hat{i}+\hat{j}+\hat{k}),\ (\hat{i}-2\hat{j}+\hat{k}),\ (\hat{i}-\hat{k})$$ **(Gorakhpur 1980; Kumaun 95)**

33. Prove that (a) $(\vec{A}\times\vec{B})^2+(\vec{A}\cdot\vec{B})^2=A^2B^2$. **(Agra 1990, 86; Purvanchal 92)**

(b) $(\vec{A}+\vec{B})\times(\vec{A}-\vec{B})=2(\vec{B}\times\vec{A})$ **(Purvanchal 1996)**

34. If $\vec{A}\times\vec{B}=\vec{C}$ and $\vec{B}\times\vec{C}=\vec{A}$, show (i) that $\vec{A}$ is perpendicular to $\vec{B}$ and $\vec{B}$ is perpendicular to $\vec{C}$ and (ii) that $B=1$ and $A=C$.

35. Show that the vector area of the triangle, whose vertices are the points $\vec{A}, \vec{B}, \vec{C}$ is $\frac{1}{2}(\vec{B}\times\vec{C}+\vec{C}\times\vec{A}+\vec{A}\times\vec{B})$.

36. Show that the condition, that the three points determined by the vectors $\vec{A}, \vec{B}$ and $\vec{C}$ are collinear, is

$$\vec{A}\times\vec{B}+\vec{B}\times\vec{C}+\vec{C}\times\vec{A}=0$$ **(Rohilkhand 1980)**

37. Find $\vec{A}\times\vec{B}$, if $\vec{A}=\hat{i}+\hat{j}$ and $\vec{B}=\hat{j}+\hat{k}$. Now, calculate the scalar products of $\vec{A}\times\vec{B}$ with $\vec{A}$ and $\vec{B}$ separately and prove that it is perpendicular to both.

[**Ans.** $(\hat{i}-\hat{j}+\hat{k})$.]

38. Find the unit vector perpendicular to each of vectors $3\hat{i}-\hat{j}+\hat{k}$ and $3\hat{i}-4\hat{j}+2\hat{k}$.
(Agra 1986)

[**Ans.** $\pm\frac{2}{\sqrt{94}}(2\hat{i}-3\hat{j}-9\hat{k})$.]

39. Determine unit vectors which are perpendicular to the vector $\hat{i}-\hat{j}+\hat{k}$ and parallel to the plane $2x+3y-5z=5$.

[**Ans.** $\pm\frac{1}{\sqrt{78}}(2\hat{i}+2\hat{j}+5\hat{k})$.]

40. If $\vec{A} = \hat{i} + \hat{j} + \hat{k}$, $\vec{B} = -\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{C} = 2\hat{i} - 2\hat{j} + 3\hat{k}$, find unit vectors which are perpendicular to $\vec{A}$ and lie in the plane of $\vec{B}$ and $\vec{C}$.

[**Ans.** $\pm\frac{1}{\sqrt{222}}(-10\hat{i} + 11\hat{j} - \hat{k})$.]

41. A force given by $\vec{F} = 3\hat{i} + 2\hat{j} - 4\hat{k}$ is applied at the point (1, –1, 2). Find the moment of the force about the point (1, –2, 3). **(Gahrwal 1995; Purv. 93)**

[**Ans.** $-(2\hat{i} + 3\hat{j} + 3\hat{k})$ unit.]

42. A horizontal force of 5 newton is acting towards the north-east direction at a point situated 2 metre to the east of a point and 4 metre higher than the pivot. Calculate the magnitude of the moment of force and direction of axis of rotation.

$\left[\textbf{Ans. } 15\sqrt{2}\ \text{n-m}, \left(-\frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right).\right]$

43. Find the torque about the point $(\hat{i} + 2\hat{j} - \hat{k})\ m$ of a force $(3\hat{i} + \hat{k})$ N through the point $(2\hat{i} - \hat{j} + 3\hat{k})\ m$. **(Agra 2005)**

[**Ans.** $\vec{\tau} = -3\hat{i} + 11\hat{j} + 9\hat{k}$ N-m.]

44. Find the force on an electron, when it moves with a velocity 10^7m/sec making an angle 45° with a magnetic field of strength 10 gauss.

[**Ans.** 1.13×10^{-15} N.]

45. Calculate the magnitude and direction of the force acting on a proton moving with a velocity $(2\hat{i} + \hat{j})\ 10^6$ m/sec in a magetic field $2000\ (\hat{i} + 2\hat{j} - \hat{k})$ gauss.

$\left[\textbf{Ans. } 3.2\times 10^{-14}\sqrt{14}\ \text{N; Direction cosines}\left(-\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right).\right]$

46. (a) Given $\vec{A} = 5\hat{i} - 7\hat{j} + 3\hat{k}$, $\vec{B} = -4\hat{i} + 7\hat{j} + 8\hat{k}$, deduce the following :

(i) $(\vec{A}\times\vec{B})$ and $(\vec{B}\times\vec{A})$. (ii) Magnitude of component of $\vec{A}$ along $\vec{B}$.

(b) Deduce the unit vector which is perpendicular to both the vectors

$\vec{P} = 3\hat{i} - 7\hat{j} + \hat{k}$, and $\vec{Q} = 3\hat{i} - 4\hat{j} + 2\hat{k}$. **(Agra 1979)**

[**Ans.** (a) (i) $7(5\hat{i} + 4\hat{j} + \hat{k})$, $-7\ (5\hat{i} + 4\hat{j} + \hat{k})$; (ii) $-93\sqrt{129}$; (b) $\frac{1}{\sqrt{94}}(2\hat{i} - 3\hat{j} - 9\hat{k})$.]

1·10. Triple Products

If $\vec{A}$, $\vec{B}$ and $\vec{C}$ are three vectors then the vector product of $\vec{A}$ and $\vec{B}$ is a vector. This product of $\vec{A}$ and $\vec{B}$ can have a scalar as well as a vector product with the vector $\vec{C}$. Thus, there are two types of triple products, namely $(\vec{A}\times\vec{B})\cdot\vec{C}$ and $(\vec{A}\times\vec{B})\times\vec{C}$ which have their frequent use in physical processes.

1·10·1. Scalar Triple Product $(\vec{A}\times\vec{B})\cdot\vec{C}$

The product $(\vec{A}\times\vec{B})$ is a vector, say $\vec{V}$. Finally, the dot product $\vec{V}\cdot\vec{C}$ is a scalar, and hence such a triple product is called scalar one. The vector $(\vec{A}\times\vec{B})$ is a vector normal to the plane of $\vec{A}$ and $\vec{B}$ with a magnitude V equal to the area of the shaded parallelogram **[fig. 1.24]**. Now,

$$(\vec{A}\times\vec{B})\cdot\vec{C} = \vec{V}\cdot\vec{C} = V\,C\cos\theta = (\text{magnitude of } \vec{V}) \times (\text{component of } \vec{C} \text{ along } \vec{V})$$

= (area of the base of the parallelopiped formed by $\vec{A}, \vec{B}, \vec{C}$)
× (its perpendicular height)
= volume of the parallelopiped of edges $\vec{A}, \vec{B}, \vec{C}$. ...(1)

Thus, the scalar triple product $(\vec{A} \times \vec{B}) \cdot \vec{C}$ represents the volume of the parallelopiped whose sides are vectors $\vec{A}, \vec{B},$ and $\vec{C}$.

Any face of this parallelopiped can be taken as the base and hence its volume can be represented by the following three expressions :

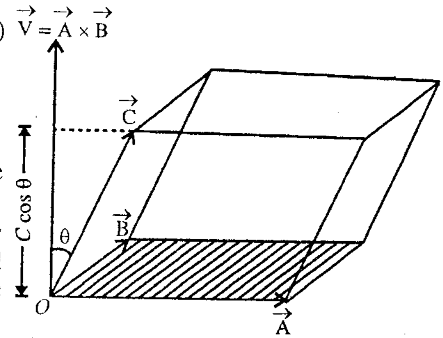


Fig. 1.24 : Scalar triple product.

$$(\vec{A} \times \vec{B}) \cdot \vec{C} = (\vec{B} \times \vec{C}) \cdot \vec{A} = (\vec{C} \times \vec{A}) \cdot \vec{B} \quad ...(2)$$

To retain the volume with positive sign, the cyclic order of $\vec{A}, \vec{B},$ and $\vec{C}$ should be maintained. As the order of terms in scalar product is meaningless,

$$\therefore \qquad \vec{C} \cdot (\vec{A} \times \vec{B}) = \vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) \quad ...(3)$$

Thus the dot and the cross in the scalar triple product may be interchanged without altering the value of the product. So, a scalar triple product is often written as $[\vec{A}\vec{B}\vec{C}]$ without showing the dot or cross.

From relation (2) and (3) we see that three vectors have six identical scalar triple products, which may be written concisely as $[\vec{A}\vec{B}\vec{C}]$ *i.e.*,

$$[\vec{A}\vec{B}\vec{C}] = \vec{A} \cdot (\vec{B} \times \vec{C}) = (A_x\,\hat{i} + A_y\,\hat{j} + A_z\,\hat{k}) \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

where $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$, $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$ and $\vec{C} = C_x\hat{i} + C_y\hat{j} + C_z\hat{k}$.

or
$$[\vec{A}\vec{B}\vec{C}] = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} \quad ...(4)$$

Coplanar Vectors : If three vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are in the same plane, then the volume of the parallelopiped, having sides $\vec{A}$, $\vec{B}$ and $\vec{C}$, must be zero *i.e.*,

$$\text{Volume } [\vec{A}\vec{B}\vec{C}] = 0 \quad \text{or} \quad \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = 0 \quad ...(5)$$

This is the **condition of coplanarity of three vectors.**

1·10·2. Vector Triple Product $\vec{A} \times (\vec{B} \times \vec{C})$

The product $(\vec{B} \times \vec{C})$ is a vector normal to the plane containing $\vec{B}$ and $\vec{C}$. Therefore, $\vec{A} \times (\vec{B} \times \vec{C})$ is a vector, say $\vec{V}$, normal to the plane containing $\vec{A}$ and $(\vec{B} \times \vec{C})$, *i.e.*, vector $\vec{V}$ lies in the plane of $\vec{B}$ and $\vec{C}$ [**fig. 1.25**].

The vector triple product may be expressed as the sum of two terms : $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}\,(\vec{A} \cdot \vec{C}) - \vec{C}\,(\vec{A} \cdot \vec{B}) \quad ...(6)$

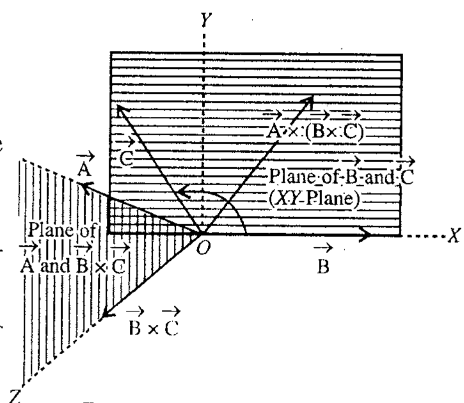


Fig. 1.25 : Vector triple product.

Proof of this relation is given below :

For our convenience, we choose X-axis along $\vec{B}$, Y-axis perpendicular to it in the plane of $\vec{B}$ and $\vec{C}$, then Z-axis will be along $\vec{B}\times\vec{C}$. Now, the three vectors in terms of their axial components are :

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k},\ \vec{B} = B_x\hat{i} \text{ and } \vec{C} = C_x\hat{i} + C_y\hat{j}$$

because the choice of the axes makes $B_y = B_z = C_z = 0$.

Now, $\quad \vec{B}\times\vec{C} = B_x C_y\,\hat{k}$ and hence $\vec{A}\times(\vec{B}\times\vec{C}) = A_y B_x C_y\,\hat{i} - A_x B_x C_y\,\hat{j}$

But, $\quad \vec{A}\cdot\vec{C} = A_x C_x + A_y C_y$ and $\vec{A}\cdot\vec{B} = A_x B_x$,

$$\therefore \quad \vec{A}\times(\vec{B}\times\vec{C}) = A_y C_y\,\vec{B} - A_x B_x C_y\,\hat{j}.$$

Now, adding and subtracting $A_x C_x B_x\,\hat{i}$ on the right hand side of the above equation, we have

$$\vec{A}\times(\vec{B}\times\vec{C}) = (A_y C_y + A_x C_x)\vec{B} - A_x C_x B_x\,\hat{i} - A_x B_x C_y\,\hat{j}$$

$$= (A_x C_x + A_y C_y)\vec{B} - A_x B_x (C_x\,\hat{i} + C_y\,\hat{j})$$

or
$$\vec{A}\times(\vec{B}\times\vec{C}) = (\vec{A}\cdot\vec{C})\vec{B} - (\vec{A}\cdot\vec{B})\vec{C} \qquad ...(7)$$

Changing the order of multiplication in a vector triple product changes its value as shown below

$$\vec{A}\times(\vec{B}\times\vec{C}) = -\vec{C}\times(\vec{A}\times\vec{B}) = (\vec{A}\cdot\vec{C})\vec{B} - (\vec{B}\cdot\vec{C})\vec{A} \qquad ...(8)$$

Another type of triple product may be considered of the form $(\vec{A}\cdot\vec{B})\vec{C}$. As the scalar product of $\vec{A}$ and $\vec{B}$ is a scalar, the final multiplication with $\vec{C}$ is a vector in the direction of $\vec{C}$.

Ex. 1. If $\vec{A} = 4\hat{i} - 5\hat{j} + 3\hat{k}$, $\vec{B} = 2\hat{i} - 10\hat{j} - 7\hat{k}$ and $\vec{C} = 5\hat{i} + 7\hat{j} - 4\hat{k}$, *calculate the following*: (i) $\vec{A}\times\vec{B}\cdot\vec{C}$ (ii) $\vec{A}\times(\vec{B}\times\vec{C})$. **(Agra 1984)**

Sol. (i)
$$\vec{A}\times\vec{B}\cdot\vec{C} = \begin{vmatrix} 4 & -5 & 3 \\ 2 & -10 & -7 \\ 5 & 7 & -4 \end{vmatrix} = 4(40 + 49) - 5(-35 + 8) + 3(14 + 50)$$

$$= 356 + 135 + 192 = \mathbf{683}.$$

(ii)
$$\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}\cdot\vec{C}) - \vec{C}(\vec{A}\cdot\vec{B})$$

$$= (2\hat{i} - 10\hat{j} - 7\hat{k})(20 - 35 - 12) - (5\hat{i} + 7\hat{j} - 4\hat{k})(8 + 50 - 21)$$

$$= -54\hat{i} + 270\hat{j} + 189\hat{k} - 185\hat{i} - 259\hat{j} + 148\hat{k}$$

$$= \mathbf{-239\hat{i} + 11\hat{j} + 337\hat{k}}.$$

Ex. 2. *A parallelopiped with one vertex at the origin has three adjacent vertices at the points* (10, –5, 3), (3, –4, 7) *and* (–5, –6, 3) *in rectangular coordinates* (x, y, z). *Calculate its volume.*

(Agra 1980)

Sol. The position vectors of the three given vertices may be expressed as $\vec{A} = 10\hat{i} - 5\hat{j} + 3\hat{k}$, $\vec{B} = 3\hat{i} - 4\hat{j} + 7\hat{k}$ and $\vec{C} = -5\hat{i} - 6\hat{j} + 3\hat{k}$. Hence the volume of the parallelopiped, whose edges are three vectors, is

$$\vec{A}\cdot(\vec{B}\times\vec{C}) = \begin{vmatrix} 10 & -5 & 3 \\ 3 & -4 & 7 \\ -5 & -6 & 7 \end{vmatrix} = \mathbf{306}.$$

Ex. 3. *Show that the law of distribution holds in case of cross product.*

Sol. According to the law of distribution in case of cross product.

$$\vec{A}\times(\vec{B}+\vec{C}) = \vec{A}\times\vec{B}+\vec{A}\times\vec{C} \quad \text{or} \quad \vec{A}\times(\vec{B}+\vec{C}) - \vec{A}\times\vec{B}-\vec{A}\times\vec{C} = 0 \qquad \text{...(i)}$$

If the law of distribution were not correct for the case of cross product, the value of the left hand side expression of eq. (i) would not have been zero. Let it be equal $\vec{X}$. Thus

$$\vec{X} = \vec{A}\times(\vec{B}+\vec{C}) - \vec{A}\times\vec{B} - \vec{A}\times\vec{C}\cdot \qquad \text{...(ii)}$$

Taking dot product with an arbitrary vector $\vec{Y}$ of both sides of eq. (ii), we have

$$\vec{X}\cdot\vec{Y} = \vec{Y}\cdot[\vec{A}\times(\vec{B}+\vec{C})] - \vec{Y}\cdot(\vec{A}\times\vec{B}) - \vec{Y}\cdot(\vec{A}\times\vec{C}).$$

But in a scalar triple product dot and cross can be interchanged without affecting the result. Therefore,

$$\vec{X}\cdot\vec{Y} = (\vec{B}+\vec{C})\cdot(\vec{Y}\times\vec{A}) - \vec{B}\cdot(\vec{Y}\times\vec{A}) - \vec{C}\cdot(\vec{Y}\times\vec{A})$$

As the distributive law holds for scalar product,

$$\therefore \qquad \vec{X}\cdot\vec{Y} = \vec{B}\cdot(\vec{Y}\times\vec{A}) + \vec{C}\cdot(\vec{Y}\times\vec{A}) - \vec{B}\cdot(\vec{Y}\times\vec{A}) - \vec{C}\cdot(\vec{Y}\times\vec{A}) = 0$$

This means that either $\vec{X} = 0$ or $\vec{Y}$ is normal to $\vec{X}$. But the vector $\vec{Y}$ is any arbitrary vector so that it may have any direction or magnitude. Therefore, the quantity $\vec{X} = 0$. Hence

$$\vec{A}\times(\vec{B}+\vec{C}) - \vec{A}\times\vec{B}-\vec{A}\times\vec{C} = 0 \quad \text{or} \quad \vec{\mathbf{A}}\times(\vec{\mathbf{B}}+\vec{\mathbf{C}}) = \vec{\mathbf{A}}\times\vec{\mathbf{B}}+\vec{\mathbf{A}}\times\vec{\mathbf{C}}.$$

Ex. 4. *If* $\vec{A}$ *and* $\vec{B}$ *are two vectors, show that the component of* $\vec{A}$ *perpendicular to* $\vec{B}$ *is given by* $\vec{B}\times(\vec{A}\times\vec{B})/\vec{B}^2$. *What is the component of* $\vec{A}$ and $\vec{B}$? **(Agra 1995, 93, 78; Bundelkhand 94)**

Sol. Component $(\vec{A}_1)$ of $\vec{A}$ along $\vec{B} = A\cos\theta\,\hat{\mathbf{B}} = (\vec{A}\cdot\hat{\mathbf{B}})\,\hat{\mathbf{B}}$

$$= (\vec{A}\cdot\vec{B})\vec{B}/B^2 = (\vec{A}\cdot\vec{B})\vec{B}/\vec{B}^2 \qquad [\because B^2 = (\vec{B}\cdot\vec{B}) = \vec{B}^2\,]$$

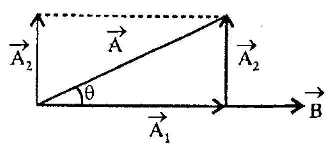


Fig. 1.26.

Component $(\vec{A}_2)$ of $\vec{A}$ perpendicular to $\vec{B}$

$$= \vec{A}-\vec{A}_1 = \vec{A}-(\vec{A}\cdot\vec{B})\vec{B}/B^2$$

$$= \frac{1}{B^2}[\vec{A}(\vec{B}\cdot\vec{B})-(\vec{A}\cdot\vec{B})\vec{B}] = \vec{\mathbf{B}}\times(\vec{\mathbf{A}}\times\vec{\mathbf{B}})/\vec{\mathbf{B}}^2.$$

Ex. 5. *Solve the following product :* $(\vec{A}\times\vec{B})\cdot(\vec{C}\times\vec{D})$ **(Agra 19982)**

Sol. $(\vec{A}\times\vec{B})\cdot(\vec{C}\times\vec{D})$ may be regarded as a scalar triple product of three vectors $\vec{A}$, $\vec{B}$ and $(\vec{C}\times\vec{D})$. Since the position of dot and cross may be interchanged in a scalar triple product without altering its value, then

$$(\vec{A}\times\vec{B})\cdot(\vec{C}\times\vec{D}) = \vec{A}\cdot[\vec{B}\times(\vec{C}\times\vec{D})]$$

$$= \vec{A}\cdot[(\vec{B}\cdot\vec{D})\vec{C}-(\vec{B}\cdot\vec{C})\vec{D}] = (\vec{\mathbf{A}}\cdot\vec{\mathbf{C}})(\vec{\mathbf{B}}\cdot\vec{\mathbf{D}})-(\vec{\mathbf{A}}\cdot\vec{\mathbf{D}})(\vec{\mathbf{B}}\cdot\vec{\mathbf{C}}).$$

Ex. 6. *Show that*

(i) $\quad (\vec{A}\times\vec{B})\times(\vec{C}\times\vec{D}) = [\vec{A}\cdot(\vec{B}\times\vec{D})]\vec{C}-[\vec{A}\cdot(\vec{B}\times\vec{C})]\vec{D}.$ **(Agra 1975; Rohilkhand 79)**

(ii) $\quad \vec{A}\times[\vec{B}\times(\vec{D}\times\vec{C})] = (\vec{A}\times\vec{C})(\vec{B}\cdot\vec{D})-(\vec{A}\times\vec{D})(\vec{B}\cdot\vec{C}).$

Sol. (i) If we put $\vec{A} \times \vec{B} = \vec{V}$, then

$$\vec{V} \times (\vec{C} \times \vec{D}) = (\vec{V} \cdot \vec{D})\vec{C} - (\vec{V} \cdot \vec{C})\vec{D}$$

$$= [(\vec{A} \times \vec{B}) \cdot \vec{D}]\vec{C} - [(\vec{A} \times \vec{B}) \cdot \vec{C}]\vec{D}.$$

As in a scalar triple product like $[(\vec{A} \times \vec{B}) \cdot \vec{D}]$, the position of dot and cross may be interchanged.

$$\therefore \quad (\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = [\vec{A} \cdot (\vec{B} \times \vec{D})]\vec{C} - [\vec{A} \cdot (\vec{B} \times \vec{C})]\vec{D}.$$

(ii) $$[\vec{A} \times [\vec{B} \times (\vec{C} \times \vec{D})] = \vec{A} \times [(\vec{B} \cdot \vec{D})\vec{C} - (\vec{B} \cdot \vec{C})\vec{D}]$$

$$= (\vec{A} \times \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \times \vec{D})(\vec{B} \cdot \vec{C}).$$

EXERCISE 1.2

1. The edges of a parallelopiped are given by the vectors $\hat{i} + 2\hat{j}$, $4\hat{j}$ and $\hat{j} + 3\hat{k}$ from the origin. Find its volume.
   [**Ans.** 12 unit.]
2. If $\vec{A} = \hat{i} - \hat{j} - 2\hat{k}$, $\vec{B} = 3\hat{i} + 5\hat{j} + 6\hat{k}$ and $\vec{C} = -\hat{i} + 4\hat{j} + m\hat{k}$, determine m such that
   (a) all the three vectors are coplanar.
   (b) the volume of the parallelopiped made by them be 12 unit.
   [**Ans.** (a) $m = 6.5$; (b) $m = 8$.]
3. Write down the expansion for $(\vec{A} \times \vec{B}) \times \vec{C}$.
   [**Ans.** $(\vec{A} \cdot \vec{C})\vec{B} - (\vec{B} \cdot \vec{C})\vec{A}$.]
4. Prove the following :
   (a) (i) $\vec{A} \times (\vec{B} \times \vec{C}) + \vec{B} \times (\vec{C} \times \vec{A}) + \vec{C} \times (\vec{A} \times \vec{B}) = 0$ **(Agra 1977; Kumaun 96; Purvanchal 95,)**
   (ii) $(\vec{A} - \vec{B}) \times (\vec{A} + \vec{B}) = 2(\vec{A} \times \vec{B})$. **(Agra 1978)**
   (iii) $\hat{i} \times (\vec{v} \times \hat{i}) + \hat{j} \times (\vec{v} \times \hat{j}) + \hat{k} \times (\vec{v} \times \hat{k}) = 2\vec{v}$
   (Agra 2005, 1994; Avadh 96; Bundelkhand 95; Kumaun 95)
   (b) Show that if $\vec{A}, \vec{B}, \vec{C}$ and $\vec{D}$ are coplanar vectors, then $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = 0$
   (c) Prove that four points $(4\hat{i} + 5\hat{j} + \hat{k})$, $-(\hat{j} + \hat{k})$, $(3\hat{i} + 9\hat{j} + 4\hat{k})$ and $4(-\hat{i} + \hat{j} + \hat{k})$ are coplanar. **(Agra 1986.)**
5. Given the three vectors $\vec{A} = 5\hat{i} - 7\hat{j} + 3\hat{k}$, $\vec{B} = -4\hat{i} + 7\hat{j} - 8\hat{k}$, $\vec{C} = 2\hat{i} - 3\hat{j}$, deduce the following : (i) $\vec{A} \cdot \vec{B} \times \vec{C}$ and $\vec{A} \times \vec{B} \cdot \vec{C}$; (ii) $\vec{A}^2 \vec{C}$ **(Agra 1970 S)**
   [**Ans.** – 14 for each, (ii) $83(2\hat{i} - 3\hat{j})$.]
6. If $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{1}{2}\vec{b}$, find the angles which the vector $\vec{a}$ makes with $\vec{b}$ and $\vec{c}$. **(Agra 2004)**
   [**Ans.** Angle between $\vec{a}$ and $\vec{b}$ = 90°, angles between $\vec{a}$ and $\vec{c}$ = 60°.]

1·11. Scalar and Vector Fields

If a physical quantity is expressed as a continuous function of the position of a point in a region of space, then such a function is called the *function of position* or *point function* and the region in which it specifies the physical quantity is called the *field*. According to the nature of the physical quantity, there are two main kinds of the field :

(i) Scalar field and (ii) Vector field.

(1) Scalar field : If a scalar quantity is given as a continuous function $\Phi(x, y, z)$ of position in a region, then we speak of a **scalar field.** The examples of scalar field are (i) distribution of temperature in a medium, (ii) distribution of electrostatic or gravitational potential etc. The field can be mapped graphically by a series of surfaces - such as isothermal or equipotential surfaces - upon each of which the function Φ (x, y, z) has a definite constant value. Such surfaces are called **level surfaces.** Thus the surface Φ (x, y, z) = constant is a level surface.

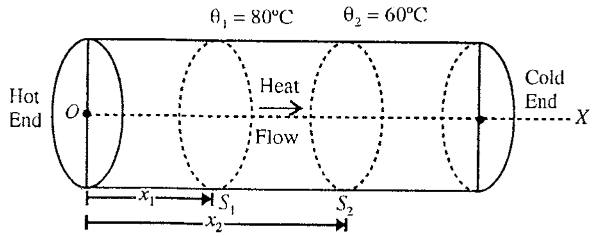


Fig. 1.27.

For example, if one end of a metallic rod is heated and the other end is kept cold, then the temperature is different at different points of the rod and hence the distribution of temperature (θ) in the material of the rod is an example of scalar field. In the steady condition, each transverse section of the rod, the value of the temperature (θ) is the same. In **fig. 1.27** at S_1 section the temperature is θ_1 and at S_2 section the temperature is θ_2. Such surfaces are called **isothermal surfaces**. In **fig. 1.27.** isothermal surfaces S_1 and S_2 are **plane surfaces.** These isothermal surfaces may be of any shape instead plane surfaces, but the condition is that at every point of a surface the temperature is the same. In **fig. 1.28** the contours of θ_1, θ_2 and θ_3 temperatures are isothermal surfaces. Similarly in **fig. 1.29**, due to a point charge Q, the surfaces of equal potential, called as **equipotential surfaces**, are S_1 and S_2. These equipotential surfaces are spherical and the value of potential at these surface is V_1 and V_2 respectively. Thus, a scalar field can be mapped by equipotential surfaces.

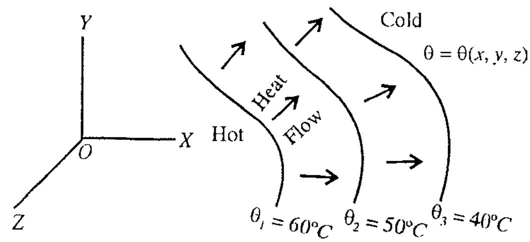


Fig. 1.28.

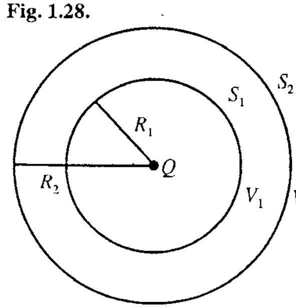


Fig. 1.29.

(2) Vector field : If a vector quantity is given as a continuous vector function $\vec{F}$ (x, y, z) of position, then we speak of a **vector field.** Examples of a vector field are (i) the distribution of velocity in a fluid, (ii) the distribution of electric or magnetic field intensity etc. *At any given point of the field, the function* $\vec{F}$ (x, y, z) *is specified by a vector of definite magnitude and direction, both of which generally change continuously from point to point throughout the field region.*

For example, the velocity $\vec{v}$ of a particle of rotating rigid body will be a vector quantity whose magnitude and direction change from point to point. Hence, the velocity $\vec{v}$ (x, y, z) of the particles of the rotating rigid body represent a vector field [**fig. 1.30**].

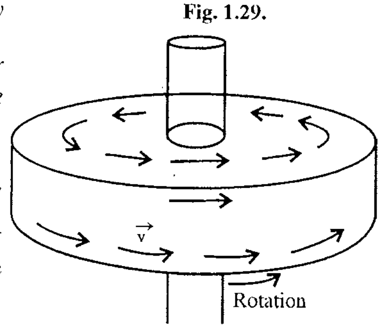


Fig. 1.30.

Some other example of vector field.

(i) Stream line motion of a liquid : A simple example of a vector field is the stream line flow of a liquid in a narrow tube. At different points of the tube, the velocity $\vec{v}$ of the particles of the liquid is different, the velocity will be less where the tube is wide and greater where the tube is narrow. [**fig. 1.31**]. This velocity of the flowing liquid in the tube can be represented by vectors at different points. Here the velocity $\vec{v}(x, y, z)$ at different points inside the tube represent a vector field.

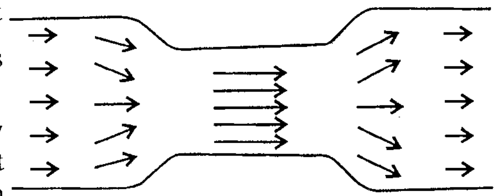


Fig. 1.31.

In stream line motion of a viscous liquid in a capillary tube, the velocity of different layers of the liquid is different and hence the velocity distribution of a flowing liquid on a transverse section of the capillary tube is an example of vector field.

(ii) Due to a point charge Q, the electrostatic field $\vec{E}$ is an example of vector field. Clearly, the direction of the electric field $\vec{E}$ at any point P is perpendicular to the equipotential surface S [**fig. 1.32**]. In between the two plates of a parallel plate condenser, the electric field $\vec{E}$, having direction from positive plate to negative plate will be uniform. Similary, the distribution of gravitational field and of magnetic field are examples of vector fields.

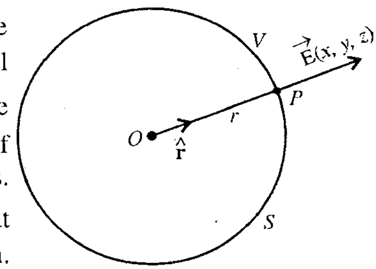


Fig. 1.32.

Flux lines : In a vector field, the vector quantity $\vec{F}(x, y, z)$ at a point (x, y, z) is represented by definite magnitude and direction. Taking any arbitrary point, proceed an infinitesimal distance in the direction of the vector at that point so as to arrive at a closely-neighbourhood point. Now, if we proceed from there in a similar way we trace out a curved line. The tangent at any point of this curved line gives the direction of the vector at that point and we call it a **vector line, line of flow** or **flux line.** For the representation of the magnitude of the vector at any point on a flux line, we draw a very small surface pependicular thereto and select a number of points per unit area upon this surface, which are numerically proportional to the magnitude of the vector. Now, flux line may be drawn through each of these points. Thus the vector field has been mapped out by flux lines. The direction of the lines is that of the vector function and their density, *i.e., the number of lines crossing per unit area perpendicular to their direction, is a measure of the magnitude of the vector.* Thus, if we consider a small area δS perpendicular to vector function $\vec{F}$, then the number of flux lines passing through that area will be $\delta\phi = F\,\delta S$. If the vector area is $\delta\vec{S}$ $(= \delta S\ \hat{n})$ and θ is the angle between $\vec{F}$ and $\delta\vec{S}$, then the number of flux lines will be $\vec{F}\cdot\delta\vec{S} = F\delta S\cos\theta$ [**fig. 1.33**]. Now if we want to know the flux lines ϕ through any surface, then

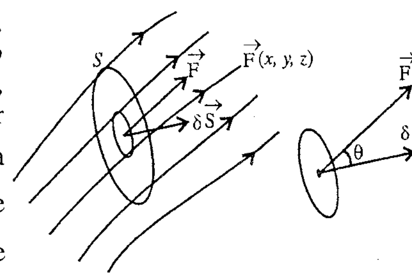


Fig. 1.33.

$$\phi = \int_S \vec{F}\cdot d\vec{S}$$

In the above discussion, the scalar or vector point functions are single valued at every point of the field.

1.12. Gradient of Scalar Field

If $\Phi(x, y, z)$ be a scalar function of position in space (*i.e.*, of the coordinate of x, y, z), then its partial derivatives along the three orthogonal axes are $\partial \Phi/\partial x$, $\partial \Phi/\partial y$ and $\partial \Phi/\partial z$. The **gradient of the scalar function** Φ is defined by

$$\text{grad } \Phi = \hat{\mathbf{i}}\frac{\partial \Phi}{\partial x} + \hat{\mathbf{j}}\frac{\partial \Phi}{\partial y} + \hat{\mathbf{k}}\frac{\partial \Phi}{\partial z}$$

In Vector Algebra, vector differential operator $\vec{\nabla}$ (read as 'del' or 'nebla') plays an important role and is defined by

$$\vec{\nabla} = \hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}$$

Operating with $\vec{\nabla}$ on the scalar function Φ, we have

$$\vec{\nabla}\Phi = \hat{\mathbf{i}}\frac{\partial \Phi}{\partial x} + \hat{\mathbf{j}}\frac{\partial \Phi}{\partial y} + \hat{\mathbf{k}}\frac{\partial \Phi}{\partial z} \quad ...(1)$$

This is the same expression as defined for the gradient of the scalar sunction Φ or grad Φ. The del operator is a vector operator and when operated on a scalar, it converts the scalar into a vector. This vector ($\vec{\nabla}\Phi$) is called the *gradient of the scalar.*

For example, the scalar quantities, such as temperature and electric potential, can be represented from point to point in space by a scalar function $\Phi(x, y, z)$ and we say that a scalar field is present in the space. The scalar field can be mapped out by level surfaces, upon each of which $\Phi(x, y, z)$ has a constant value.

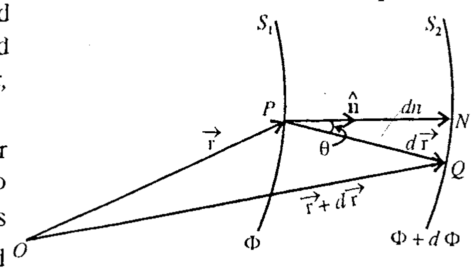


Fig. 1.34.

Consider two surfaces S_1 and S_2 very close to each other on which the values of scalar function are Φ and $\Phi + d\Phi$ respectively. Suppose P and Q are points on the two surfaces [**fig. 1.34**], whose distances from the origin O are $\vec{r}$ and $\vec{r} + d\vec{r}$ respectively so that vector distance $\vec{PQ} = d\vec{r}$. If dn represents the distance (PN) along the normal from the point P (on surface S_1) to the surface S_2, we may write.

$$dn = dr\cos\theta = \hat{\mathbf{n}}\cdot d\vec{r} \quad ...(2)$$

where $\hat{\mathbf{n}}$ is the unit normal to the surface S_1 at P.

The rate of increase of Φ at P in the direction of $\hat{\mathbf{n}}$ is greatest and is equal to $\partial \Phi/\partial n$ [**fig. 1.35**].

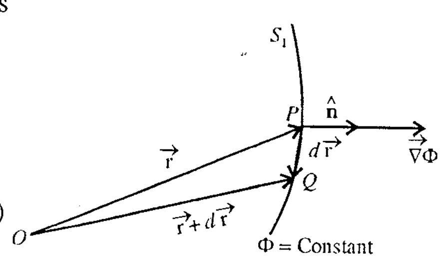


Fig. 1.35.

Therefore,

$$d\Phi = \frac{\partial \Phi}{\partial n}dn = \frac{\partial \Phi}{\partial n}\hat{\mathbf{n}}\cdot d\vec{r}$$

$$\text{But } \vec{\nabla}\Phi\cdot d\vec{r} = \left(\hat{\mathbf{i}}\frac{\partial \Phi}{\partial x} + \hat{\mathbf{j}}\frac{\partial \Phi}{\partial y} + \hat{\mathbf{k}}\frac{\partial \Phi}{\partial z}\right)\cdot(\hat{\mathbf{i}}\,dx + \hat{\mathbf{j}}\,dy + \hat{\mathbf{k}}\,dz)$$

$$= \frac{\partial \Phi}{\partial x}dx + \frac{\partial \Phi}{\partial y}dy + \frac{\partial \Phi}{\partial z}dz$$

$$\therefore \quad d\Phi = \frac{\partial \Phi}{\partial n}\hat{\mathbf{n}}\cdot d\vec{r} = \vec{\nabla}\Phi\cdot d\vec{r} \quad ...(3)$$

Therefore,

$$\vec{\nabla}\Phi = \left(\frac{\partial \Phi}{\partial n}\right)\hat{\mathbf{n}}. \quad ...(4)$$

Hence, *the magnitude of the vector $\vec{\nabla}\Phi$ is equal to the maximum rate of change Φ with respect to the space variables and has the direction of that change.*

In particular, if $d\vec{r}$ lies on the surface S_1, we have from (3)

$$d\Phi = \vec{\nabla}\Phi \cdot d\vec{r} = 0 \qquad ...(5)$$

which means that the vector $\vec{\nabla}\Phi$ is normal to the surface Φ = constant.

If S_1 and S_2 surfaces are plane, then the gradient of scalar field Φ [**fig. 1.36**] is

$$\vec{\nabla}\Phi = \frac{\partial\Phi}{\partial x}\hat{i} = \frac{d\Phi}{dx}\hat{i} \text{ or } |\vec{\nabla}\Phi| = \frac{d\Phi}{dx} \qquad ...(6)$$

where the vector $\vec{\nabla}\Phi$ is perpendicular to the surface Φ = constant *i.e.*, the vector $\vec{\nabla}\Phi$ is along X-axis and $d\Phi/dx$ is equal to the maximum rate of its change in that direction.

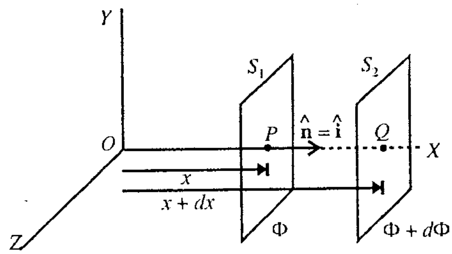


Fig. 1.36.

Physical Significance of Gradient

The gradient of scalar field is of great importance in different fields of physics. For example, if V is the electrostatic potential at a distance r from a point charge Q, then equipotential surface will be spherical [**fig. 1.37**]. The electric field $\vec{E}$ at the point P distant r from Q is related to the electrostatic potential V as

$$\vec{E} = -\vec{\nabla}V = -\text{ grad } V \qquad ...(7)$$

where the potential $V(x, y, z)$ and field $\vec{E}(x, y, z)$ represent the scalar and vector fields respectively at the point P.

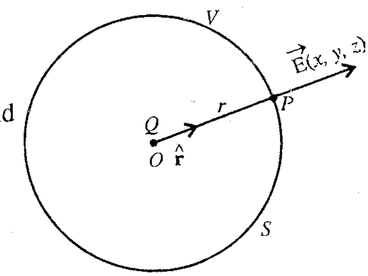


Fig. 1.37.

If $\hat{r}$ is unit vector along $\vec{OP}$, then

$$\vec{E} = -\vec{\nabla}V = -\frac{\partial V}{\partial r}\hat{r} = -\frac{dV}{dr}\hat{r} \qquad ...(8)$$

where $\vec{E}$ is along the unit vector $\hat{r}$ and obviously $\hat{r}$ is perpendicular to the equipotential surface. Here, the negative gradient of the electric potential (scalar field) is the electric field intensity $\vec{E}$ (vector field); the field intensity $\vec{E}$ at the point P is a vector which is along the direction of greatest rate of decrease of potential (i.e., along $\hat{r}$) and has the value equal to the maximum rate of change of potential (dV/dr).

Similarly, the gravitational field intensity $\vec{E}$ is related with gravitational potential V as $\vec{E} = -\vec{\nabla}V$.

Now, if q charge is present at the point P, then the potential energy will be $U(= qV)$ and the force on that charge q will be

$$\vec{F} = -\vec{\nabla}U = -\text{ grad U} = -\frac{\partial U}{\partial r}\hat{r} = -\frac{dU}{dr}\hat{r} \qquad ...(9)$$

In a parallel plate condenser its plates are at V_1 and V_2 potential $(V_1 > V_2)$, then the electric field intensity $\vec{E}$ will be uniform and in the direction perpendicular to the plates [**fig. 1.38**].

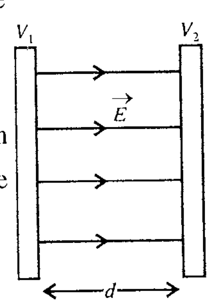


Fig. 1.38.

$$\vec{E} = -\frac{\partial V}{\partial x}\hat{i} = -\frac{V_2 - V_1}{d}\hat{i} \text{ or } E = \frac{V_1 - V_2}{d} \qquad ...(10)$$

Here, the uniform electric field intensity $\vec{E}$ is obtained by the gradient of the potential along X-axis and its value is obtained by eq. (10).

In a **potentiometer**, the potential decreases on the wire from point A to point B and the value of potential gradient is V/l, where V is the potential difference between the ends of potentiometer wire and l is the length of the wire.

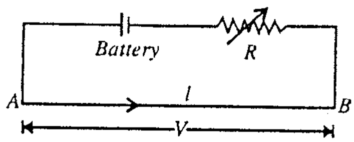


Fig. 1.39.

In **heat conductivity experiment** by Searle's apparatus, if θ_1 and θ_2 are the temperatures of S_1 and S_2 isothermal surfaces respectively, situated correspondingly at x_1 and x_2, then the thermal gradient is given by

$$\frac{d\theta}{dx} = \frac{\theta_2-\theta_1}{x_2-x_1} = -\frac{\theta_1-\theta_2}{x_2-x_1}. \quad ...(11)$$

which is negative and heat flow is in the direction of decreas of temperature.

Ex. 1. *In conduction of heat in a metallic rod under steady state, an isothermal surface is defined as $\theta = 4x - 5y + 3z$. Find the temperature field and show that the temperature field is uniform.* **(Gwalior 2004)**

Sol. Let the temperature field be $\vec{E}$. Then $\vec{E} = -\text{ grad }\theta = -\left(\hat{i}\frac{\partial\theta}{\partial x} + \hat{j}\frac{\partial\theta}{\partial x} + \hat{k}\frac{\partial\theta}{\partial x}\right)$

or
$$\vec{E} = -\left[\hat{i}\frac{\partial}{\partial x}(4x-5y+3z)+\hat{j}\frac{\partial}{\partial y}(4x-5y+3z)+\hat{k}\frac{\partial}{\partial z}(4x-5y+3z)\right]$$
$$= -\left[\hat{i}(4)+\hat{j}(-5)+\hat{k}(3)\right] = \mathbf{-4\hat{i} + 5\hat{j} - 3\hat{k}}$$

As this field is independent of x, y, z, hence it is a uniform field.

Ex. 2. *If $V = 4x^2 + 3y^2 - 9z^2$ represents the electrostatic potential at a point, find the electric field intensity at a point $(-1, 2, 3)$.* **(Jabalpur 1995; Bilaspur 89)**

Sol. Electric field intensity $\vec{E}$ is given by

$$\vec{E} = -\text{ grad } V = -\left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial x}+\hat{k}\frac{\partial}{\partial x}\right)(4x^2 + 3y^2 - 9z^2)$$
$$= -[8x\hat{i} + 6y\hat{j} - 18z\hat{k}] = -8x\hat{i} - 6y\hat{j} + 18z\hat{k}$$

At point $(-1, 2, 3)$, $\vec{E} = -8 \times -1\hat{i} - 6 \times 2\hat{j} + 18 \times 3\hat{k} = \mathbf{8\hat{i} - 12\hat{j} + 54\hat{k}}$

Ex. 3. *The potential function in an elelctric field is represented by $V = C(x^2 + y^2 + z^2)$, where C is a constant. Show that the electric field is radial.*
(Ujjain 1990)

Sol. Here, $V = C(x^2 + y^2 + z^2)$

$\therefore$ Electric field $\vec{E} = -\vec{\nabla} V = -\left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial x}+\hat{k}\frac{\partial}{\partial x}\right) C(x^2 + y^2 + z^2) = -2C(x\hat{i} + y\hat{j} + z\hat{k})$

But, $x\hat{i} + y\hat{j} + z\hat{k} = \vec{r}$, $\quad\therefore\quad \vec{E} = -2C\vec{r}$

As a field along or opposite to $\vec{r}$ is radial, henc **$\vec{E}$ is radial field.**

Ex. 4. *If $\vec{r}$ is the position vector of a point deduce the value of grad $(1/r)$.* **(Gwalior 1995)**

Sol. Position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$; its magnitude $r = (x^2 + y^2 + z^2)^{1/2}$

Hence,
$$\text{grad}\left(\frac{1}{r}\right) = \text{grad }(x^2 + y^2 + z^2)^{-1/2} = \left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial z}\right)(x^2 + y^2 + z^2)^{-1/2}$$
$$= \left[\hat{i}\frac{\partial}{\partial x}(x^2+y^2+z^2)^{-1/2}+\hat{j}\frac{\partial}{\partial y}(x^2+y^2+z^2)^{-1/2}+\hat{k}\frac{\partial}{\partial z}(x^2+y^2+z^2)^{-1/2}\right]$$

Now, $\frac{\partial}{\partial x}(x^2+y^2+z^2)^{-1/2} = \left(-\frac{1}{2}\right)(x^2+y^2+z^2)^{-3/2}\,2x = \frac{-x}{(x^2+y^2+z^2)^{3/2}}$

Similarly, $\frac{\partial}{\partial y}(x^2+y^2+z^2)^{-1/2} = \frac{-y}{(x^2+y^2+z^2)^{3/2}}$ and $\frac{\partial}{\partial z}(x^2+y^2+z^2)^{-1/2} = \frac{-z}{(x^2+y^2+z^2)^{3/2}}$

$$\therefore \quad \text{grad}\left(\frac{1}{r}\right) = \left[\hat{\mathbf{i}}\left\{\frac{-x}{(x^2+y^2+z^2)^{3/2}}\right\}+\hat{\mathbf{j}}\left\{\frac{-y}{(x^2+y^2+z^2)^{3/2}}\right\}+\hat{\mathbf{k}}\left\{\frac{-z}{(x^2+y^2+z^2)^{3/2}}\right\}\right]$$

$$= \frac{-(x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}})}{(x^2+y^2+z^2)^{3/2}} = \frac{-\vec{\mathbf{r}}}{r^3}.$$

Ex. 5. *If* $\vec{\mathbf{r}} = \hat{\mathbf{i}}x + \hat{\mathbf{j}}y + \hat{\mathbf{k}}z$, *prove that grad* $r^n = nr^{n-2}\vec{\mathbf{r}}$.

Sol. $$\text{grad } r^n = \vec{\nabla} r^n = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}\right)(x^2+y^2+z^2)^{n/2} \qquad [\because r = (x^2+y^2+z^2)^{1/2}]$$

$$= \hat{\mathbf{i}}\left(\frac{n}{2}\right)(x^2+y^2+z^2)^{(n/2)-1}2x+\hat{\mathbf{j}}\left(\frac{n}{2}\right)(x^2+y^2+z^2)^{(n/2)-1}2y$$

$$+\hat{\mathbf{k}}\left(\frac{n}{2}\right)(x^2+y^2+z^2)^{(n/2)-1}2z$$

$$= \text{n}(\hat{\mathbf{i}}x+\hat{\mathbf{j}}y+\hat{\mathbf{k}}z)(x^2+y^2+z^2)^{(n-2)/2} = n\vec{\mathbf{r}}r^{n-2}.$$

Ex. 6. *If* $S = S(x, y, z) = x^2 - x^2y + xy^2z^2$, *find grad S at the point* (2, –1, –3).

Sol. $$\text{grad } S = \vec{\nabla} S, \text{ where } \vec{\nabla} = \hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}$$

$$\therefore \quad \text{grad } S = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}\right)(x^2 - x^2y + xy^2z^2)$$

$$= \hat{\mathbf{i}}\frac{\partial}{\partial x}(x^2-x^2y+xy^2z^2)+\hat{\mathbf{j}}\frac{\partial}{\partial y}(x^2-x^2y+xy^2z^2)+\hat{\mathbf{k}}\frac{\partial}{\partial z}(x^2-x^2y+xy^2z^2)$$

$$= \hat{\mathbf{i}}(2x - 2xy + y^2z^2) + \hat{\mathbf{j}}(-x^2 + 2xyz^2) + \hat{\mathbf{k}}(2zxy^2)$$

Here, $x = 2$, $y = -1$ and $z = -3$, hence

$$\text{grad } S = \hat{\mathbf{i}}[2\times 2 - 2\times 2\times(-1) + (-1)^2\times(-3)^2] + \hat{\mathbf{j}}[-(2)^2 - 2\times 2\times(-1)\times(-3)^2]$$
$$+\hat{\mathbf{k}}[2\times(-3)\times 2\times(-1)^2]$$

$$= \mathbf{17\hat{i} - 40\hat{j} - 12\hat{k}}.$$

Ex. 7. *If* ϕ *and* ψ *are differential functions of x, y, z, prove that* grad $(\phi\psi) = \phi\vec{\nabla}\psi + \psi\vec{\nabla}\phi$.

(Gwalior 2001; Bilaspur 1998)

Sol. $$\text{grad }(\phi\psi) = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}\right)(\phi\psi)$$

$$= \hat{\mathbf{i}}\left(\phi\frac{\partial\psi}{\partial x}+\psi\frac{\partial\phi}{\partial x}\right)+\hat{\mathbf{j}}\left(\phi\frac{\partial\psi}{\partial y}+\psi\frac{\partial\phi}{\partial y}\right)+\hat{\mathbf{k}}\left(\phi\frac{\partial\psi}{\partial z}+\psi\frac{\partial\phi}{\partial z}\right)$$

$$= \phi\left(\hat{\mathbf{i}}\frac{\partial\psi}{\partial x}+\hat{\mathbf{j}}\frac{\partial\psi}{\partial y}+\hat{\mathbf{k}}\frac{\partial\psi}{\partial z}\right)+\psi\left(\hat{\mathbf{i}}\frac{\partial\phi}{\partial x}+\hat{\mathbf{j}}\frac{\partial\phi}{\partial y}+\hat{\mathbf{k}}\frac{\partial\phi}{\partial z}\right) = \phi\vec{\nabla}\psi+\psi\vec{\nabla}\phi.$$

Ex. 8. *If* $\vec{\mathbf{r}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$, *prove that* grad $f(r) = \hat{\mathbf{r}}\frac{\partial f(r)}{\partial r}$ and grad $f(r)\times\vec{\mathbf{r}} = 0$.

Sol. $$\text{grad } f(r) = \hat{\mathbf{i}}\frac{\partial f(r)}{\partial x}+\hat{\mathbf{j}}\frac{\partial f(r)}{\partial y}+\hat{\mathbf{k}}\frac{\partial f(r)}{\partial z}$$

But, $\quad \dfrac{\partial f(r)}{\partial x} = \dfrac{\partial f(r)}{\partial r}\dfrac{\partial r}{\partial x}$ and $r = \sqrt{x^2+y^2+z^2}$; so that $\dfrac{\partial r}{\partial x} = \dfrac{1}{2}\dfrac{2x}{\sqrt{x^2+y^2+z^2}} = \dfrac{x}{r}$;

$\therefore \quad \dfrac{\partial f(r)}{\partial x} = \dfrac{x}{r}\dfrac{\partial f(r)}{\partial r}$; similarly $\dfrac{\partial f(r)}{\partial y} = \dfrac{y}{r}\dfrac{\partial f(r)}{\partial r}$ and $\dfrac{\partial f(r)}{\partial z} = \dfrac{z}{r}\dfrac{\partial f(r)}{\partial r}$

$$\therefore \quad \text{grad}\, f(r) = \frac{x\hat{i}}{r}\frac{\partial f(r)}{\partial r} + \frac{y\hat{j}}{r}\frac{\partial f(r)}{\partial r} + \frac{z\hat{k}}{r}\frac{\partial f(r)}{\partial r} = (x\hat{i}+y\hat{j}+z\hat{k})\frac{1}{r}\frac{\partial f(r)}{\partial r} = \frac{\vec{r}}{r}\frac{\partial f(r)}{\partial r} = \hat{r}\frac{\partial f(r)}{\partial r}.$$

Thus, $\quad \text{grad}\, f(r) \times \hat{r} = \dfrac{\partial f(r)}{\partial r}\hat{r}\times\hat{r} = 0,$ $\quad$ [$\because$ For any vector $\vec{A}\times\vec{A} = 0$.]

Ex. 9. *Obtain an unit vector normal to the surface* $xyz = K$ (constant) *at the point* $(1, -1, -2)$.

Sol. The direction of grad ϕ is normal to the surface ϕ = constant. Hence for surface $\phi = x\,y\,z$ = constant, we have

$$\vec{\nabla}\phi = \left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial z}\right)(xyz) = yz\,\hat{i} + zx\,\hat{j} + xy\,\hat{k}$$

Its value at $(1, -1, -2)$, $\vec{\nabla}\phi = (-1)(-2)\,\hat{i} + (-2)(1)\,\hat{j} + 1(-1)\,\hat{k} = 2\hat{i} - 2\hat{j} - \hat{k}$

$$\therefore \text{ Unit normal vector } \hat{n} = \frac{\vec{\nabla}\phi}{|\vec{\nabla}\phi|} = \frac{2\hat{i}-2\hat{j}-\hat{k}}{\sqrt{2^2+(-2)^2+(-1)^2}} = \frac{2\hat{i}-2\hat{j}-\hat{k}}{3} = \frac{2}{3}\hat{i}-\frac{2}{3}\hat{j}-\frac{1}{3}\hat{k}.$$

Ex. 10. *The temperature at a point* (x, y, z) *is given by* $T = x^2 - y^2 + xyz + 270$, find in what direction and with what rate, the temperature is increasing maximum at the point $(-1, 2, 3)$.

Sol. Here, $\quad \vec{\nabla}T = (2x + yz)\,\hat{i} + (-2y + xz)\,\hat{j} + xy\,\hat{k}$

At the point $(-1, 2, 3)$, $\vec{\nabla}T = 4\hat{i} - 7\hat{j} - 2\hat{k}$

Hence, the temperature increase at the maximum rate along the vector $(4\hat{i} - 7\hat{j} - 2\hat{k})$ and the maximum rate is

$$\frac{\partial T}{\partial r} = |\vec{\nabla}T| = \sqrt{4^2+7^2+2^2} = \sqrt{69}.$$

Ex. 11. *If* $\vec{a}$ *is a constant quantity, then show that* grad $(\vec{a}\cdot\vec{r}) = \vec{a}$. **(Gwalior 2004)**

Sol. Suppose $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore \quad \vec{a}\cdot\vec{r} = a_1x + a_2y + a_3z$; where a_1, a_2 a_3 are constants.

$$\text{Hence, grad}\,(\vec{a}\cdot\vec{r}) = \hat{i}\frac{\partial}{\partial x}(a_1x+a_2y+a_3z)+\hat{j}\frac{\partial}{\partial y}(a_1x+a_2y+a_3z)+\hat{k}\frac{\partial}{\partial z}(a_1x+a_2y+a_3z)$$

$$= \hat{i}\,(a_1 + 0 + 0) + \hat{j}\,(0 + a_2 + 0) + \hat{k}\,(0 + 0 + a_3) = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} = \vec{a}.$$

EXERCISE 1.3

1. If the electrostatic potential at a point is $V = 2x^2 - 3y^2 - z^2$, find the electric field intensity at the point $(1, 2, -1)$. **(Jabalpur 1992)**
   [Ans. $\vec{E} = -4\hat{i} + 12\hat{j} + 2\hat{k}$]
2. A region is specified by the potential function $V = 2x + 3y + z$. Find the expression for potential gradient and intensity.
   [**Ans.** $2\hat{i} + 3\hat{j} + \hat{k}$; $-(2\hat{i} + 3\hat{j} + \hat{k})$]

3. The electric field intensity at a point of a closed surface is expressed as $\vec{E} = 2\hat{i} + 3\hat{j} - 2\hat{k}$ volt/metre. Calcluate the electric flux through a small element of area $(\hat{i} + \hat{j} + \hat{k})$ m^2.
[**Ans.** 3 volt-m]
[**Hint.** Electric flux $\phi = \vec{E} \cdot \Delta\vec{S} = (2\hat{i} + 3\hat{j} - 2\hat{k}) \cdot (\hat{i} + \hat{j} + \hat{k})$
$= 2 + 3 + (-2)(1) = 2 + 3 - 2 = 3$ volt-m.]

4. In conduction of heat in a metallic rod under steady state, an isothermal surface is defined as $\theta = -3x + 2y - 5z$. Find the temperature field and show that the temperature field is uniform.
[**Ans.** $3\hat{i} - 2\hat{j} + 5\hat{k}$]

5. Obtain the unit vector normal to the surface xyz = constant at a point (1, 2, –1).
[**Ans.** $\frac{1}{3}(-2\hat{i} - \hat{j} + 2\hat{k})$]

6. Obtain the maximum rate of variation of scalar function $\phi = x^2yz^2$ with the distance. What will be the unit vector at the point (–1, 1, 2) in this direction ?
[**Ans.** $\frac{1}{\sqrt{96}}(-8\hat{i} + 4\hat{j} + 4\hat{k})$]
[**Hint.** $\hat{n} = \frac{\vec{\nabla}\phi}{|\vec{\nabla}\phi|}$; Get $\vec{\nabla}\phi$ at (–1, 1, 2) first and then obtain the unit vector $\hat{n}$.]

7. If $\phi = \frac{1}{r}$, prove that $\vec{\nabla}\phi = -\frac{1}{r^2}\vec{\nabla}r$.

8. Find the value of grad $(1/r)$ and grad $(\log r)$? **(Kumaun 1996; Bundelkhand 94)**
[**Ans.** $-\vec{r}/r^3$, $\vec{r}/r^2$]

9. Prove that $\nabla^2(1/r) = 0$. **(Agra 1976)**

10. If $\phi = 3x^2y - y^3z^2$, find grad ϕ at the point (1, 1, 1).

11. If u and v are functions of (x, y, z), show that

$$\vec{\nabla}(uv) = v\vec{\nabla}u + u\vec{\nabla}v$$

1·13. Vector Intergration

The integrals, which we generally come across in Vector Algebra are the **line integral**, the **surface integral** and the **volume integral**.

(1) Line integral : Let PQ be any curve, drawn in a vector field and $d\vec{l}$ an element of length along this curve at any point B. Let $\vec{A}$ represent the vector at B in a direction making an angle θ with $d\vec{l}$ [**fig. 1.40**]. In general if $\vec{A}$ varies in magnitude and direction from point to point along the curve, the integral

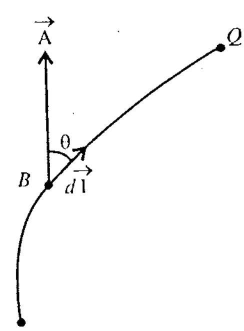


Fig. 1.40 : Line integral.

$$\int_P^Q \vec{A} \cdot d\vec{l} = \int_P^Q A\cos\theta\, dl \qquad ...(1)$$

is defined as the **line integral** of $\vec{A}$ along the curve PQ.

In terms of cartesian components,

$$\int_P^Q \vec{A} \cdot d\vec{l} = \int_P^Q (A_x dx + A_y dy + A_z dz) \qquad ...(2)$$

For example, if a force $\vec{F}$ is acting on a moving particle then the line integral $\int \vec{F} \cdot d\vec{l}$ over the path described by the particle is the work done by the force *i.e.*,

$$W = \int_P^Q \vec{F} \cdot d\vec{l} \qquad ...(3)$$

Similarly, if a unit positive charge is moved in an **electric field** $\vec{E}$ from P to Q, then the **line integral** $\int \vec{E} \cdot d\vec{l}$ represents, the **potential difference** (ΔV) between these points *i.e.*,

$$\Delta V = -\int_P^Q \vec{E} \cdot d\vec{l} \qquad ...(4)$$

We shall study later that if the vector field is conservative, then the value of the line integral along a closed path is zero *i.e.*,

$$\oint \vec{A} \cdot d\vec{l} = 0 \qquad ...(5)$$

(2) Surface Integral or Flux : Consider any element of infinitesimal area $d\vec{S}$ upon a surface S in a vector field. If $\vec{A}$ be the value of the vector at the middle point of the element, then the integral over the surface **[fig. 1.41]**

$$\iint_S \vec{A} \cdot d\vec{S} = \iint_S A \cos\theta \; dS \qquad ...(6)$$

is defined as the **surface integral** or **total flux** of vector $\vec{A}$ through the whole surface S.

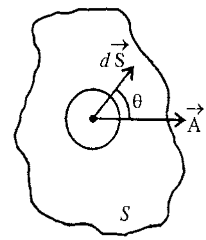


Fig. 1.41 : Surface integral.

In cartesian components,

$$\iint_S \vec{A} \cdot d\vec{S} = \iint_S (A_x dS_x + A_y dS_y + A_z dS_z) \qquad ...(7)$$

For example, suppose that $\vec{A} = \vec{v}$ represents the velocity of a moving fluid. If we draw a fixed surface S in the moving liquid, then the surface integral $\iint_S \vec{v} \cdot ds$ tells us the **rate of flow of liquid** or the **volume of the liquid flowing** per second through the surface S.

In an electric field $\vec{E}$, the **surface integral** $\iint_S \vec{E} \cdot d\vec{S}$ represents the **flux** through the surface S.

(3) Volume integral : If we consider a closed surface in space in a vector field $\vec{A}$ enclosing a volume V then

$$\iiint_V \vec{A} \, dV = \int_x \int_y \int_z (A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}) dx \; dy \; dz \qquad ...(8)$$

is defined as the **volume integral [fig. 1.43]**.

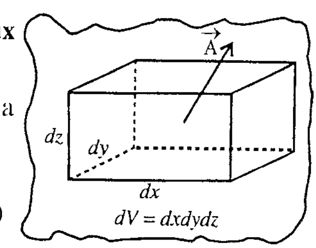


Fig. 1.42 : Volume integral.

1·14. Divergence

If $\vec{A}$ (x, y, z) is a vector field, the scalar product of the vector operator $\vec{\nabla}$ and $\vec{A}$ is a scalar and is called the **divergence of** $\vec{A}$, *i.e.*,

$$\text{div}\,\vec{A} = \vec{\nabla} \cdot \vec{A} = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial x} + \hat{\mathbf{k}}\frac{\partial}{\partial x}\right) \cdot (\hat{\mathbf{i}} A_x + \hat{\mathbf{j}} A_x + \hat{\mathbf{k}} A_x)$$

or

$$\text{div}\,\vec{A} = \vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \qquad ...(1)$$

On the basis of the explanation, given below, we can define div $\vec{A}$ as follows : **The divergence of a vector field at a point is a scalar quantity which represents the flux of that vector field diverging per unit volume from that point.**

Further, we see in the following discussion that the quantity $\vec{\nabla} \cdot \vec{A}$ at a point is the amount of flux per unit volume diverging from that point. Hence, the name divergence originated from the interpretation of $\vec{\nabla} \cdot \vec{A}$.

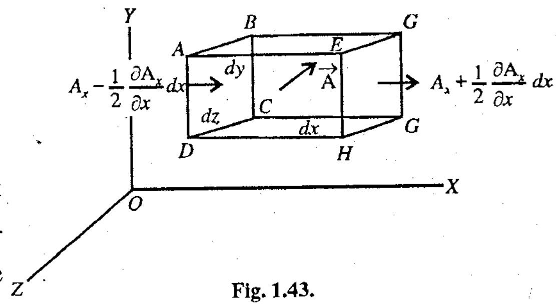


Fig. 1.43.

Physical Interpretation and Significance of div $\vec{A}$

In order to understand the meaning and physical signficance of div $\vec{A}$, we consider an example of the flow of a liquid. Here, the velocity $\vec{v}$ of the liquid represents the vector field $\vec{A}(x, y, z)$. Consider the flow of the liquid in a space. In that space, let $\vec{A}$ be the value of a vector field at the mid point of a small rectangular parallelopiped with sides dx, dy and dz. The axial component of the vector $\vec{A}$ are A_x, A_y, A_z, and in general, the value of the vector $\vec{A}$ is different at different points of the field.

On the face $ABCD$ the value of the X-component of the vector $\vec{A}$ is $A_x - \frac{1}{2}\frac{\partial A_x}{\partial x}dx$.

As the face is infinitesimally small, this value can be taken as uniform over the face ABCD. Similarly, for the face $EFGH$, the X-component is $A_x + \frac{1}{2}\frac{\partial A_x}{\partial x}dx$.

The **flux** through a face is defined as the product of the area of the face and the normal component of the vector upon it.

$$\therefore \quad \text{Flux entering the face } ABCD = \left(A_x - \frac{1}{2}\frac{\partial A_x}{\partial x}dx\right)dy\,dz$$

and

$$\text{flux leaving the face } EFGH = \left(A_x + \frac{1}{2}\frac{\partial A_x}{\partial x}dx\right)dy\,dz$$

Hence, the excess of flux leaving the parallelopiped over that entering it in the X-direction is

$$\left(A_x + \frac{1}{2}\frac{\partial A_x}{\partial x}dx\right)dy\,dz - \left(A_x - \frac{1}{2}\frac{\partial A_x}{\partial x}dx\right)dy\,dz = \frac{\partial A_x}{\partial x}dx\,dy\,dz$$

In the hydromechanical case, this represents the **net volume** of the fluid passing per second in the X-direction. Similarly, contributions parallel to Y and Z directions $\frac{\partial A_y}{\partial x}dx\,dy\,dz$ and $\frac{\partial A_z}{\partial x}dx\,dy\,dz$.

Hence the **net flux diverging from the element** is

$$d\phi = \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right)dx\,dy\,dz = (\vec{\nabla} \cdot \vec{A})\,dx\,dy\,dz = (\text{div}\,\vec{A})\,dV \qquad ...(2)$$

This quantity $d\phi$ tells us the net volume of the liquid diverging from $dV = dx\,dy\,dz$.

Hence,
$$\frac{d\phi}{dV} = \text{div}\,\vec{A} = \vec{\nabla} \cdot \vec{A} \qquad ...(3)$$

Thus $\vec{\nabla} \cdot \vec{A}$ (or div $\vec{A}$) **at a point gives the amount of flux diverging per unit volume from that point. If $\vec{A}$ is denoting the velocity of the moving fluid, then $\vec{\nabla} \cdot \vec{A}$ tells the rate at which the fluid**

is diverging from the point per unit volume. Vector funtion $\vec{A}$ can be the volume of a flowing gas and hence we can use the word fluid instead liquid.

Equation of continuity : If in a fluid, the divergence of the field (velocity) at a point is positive, then either the fluid is *expanding* and the *density at that point is decreasing* or that point is the **source of fluid**. If the divergence at a point is negative, then either the liquid is *contracting* and its *density* is *increasing at that point* or that point is **negative source** or **sink**.

If at the point (x, y, z), the velocity of the fluid is $\vec{v}$ and density is ρ, then the vector $\rho\vec{v} = \vec{A}$ represents the mass of the fluid passing through unit area per second and the net mass going out of the volume dV per second will be div $(\rho\vec{v})dV$. Due to this, the density of the fluid in that volume dV will be less and if $d\rho/dt$ is the rate of change of density with time, then $(\partial\rho/\partial t)\,dV$ will be the decrease in the mass of the fluid per second in the volume dV. Hence,

$$\text{div}\,(\rho\vec{v})dV = -\frac{\partial\rho}{\partial t}dV \quad i.e., \quad \frac{\partial\rho}{\partial t} + \text{div}\,(\rho\vec{v}) = 0 \qquad \text{...(4)}$$

This equation (4) is called the **equation of continuity.** Here, in general ρ is a function of x, y, z and time t *i.e.*, $\rho = \rho\,(x, y, z, t)$ and the fluid in general is compressible.

Thus, div $\vec{A}$ or $\vec{\nabla}\cdot\vec{A}$ at a point is the amount of flux diverging per unit volume from that point and hence the divergence of a vector field is an important concept which has physical importance in the different fields of physics. We have seen its importance in **hydrodynamics.** If $\vec{J}$ represents the current density, at a point in the charge flow, then div $\vec{J}$ gives the rate of flow of charge per unit volume at that point so that the charge density is decreasing there (for positive div $\vec{J}$) *i.e.*,

$$\text{div}\,\vec{J} = -\frac{\partial\rho}{\partial t} \quad \text{or} \quad \frac{\partial\rho}{\partial t} + \text{div}\,\vec{J} = 0 \qquad \text{...(5)}$$

This is known as *equation of continuity in charge flow.*

In Quantum Mechanics, the *divergence of the probability current density* and the *time derivative of the position probability density* is represented by an equation of continuity similar to eq. (5).

Solenoidal vector : If the flux, entering any element of field space is exactly balanced by that leaving it, the quantity div $\vec{A} = 0$. Then, there is no source or sink, nor its density is changing, *i.e.*, the fluid is incompressible. If the fluxes entering and leaving an element are equal, the lines of flow of the vector $\vec{A}$ should form either closed curves (e.g., in case of magnetic field of a current) or extend to infinity. *A vector which satisfies this condition is called* **solenoidal.**

For example, one of the Maxwell's electromagnetic field equation is

$$\text{div}\,\vec{B} = 0 \quad \text{or} \quad \vec{\nabla}\cdot\vec{B} = 0 \qquad \text{...(6)}$$

where magnetic induction $\vec{B}$ is a **solenoidal vector.**

1·15. Gauss's Theorem

According to this theorem, the volume integral of the divergence of a vector field $\vec{A}$ taken over any volume V is equal to the surface integral of $\vec{A}$ taken over the closed surface surrounding the volume. That is

$$\iiint_V (\vec{\nabla}\cdot\vec{A})\,dV = \iint_S \vec{A}\cdot d\vec{S} \qquad \text{...(1)}$$

Proof. In cartesian coordinates,

$$\vec{\nabla}\cdot\vec{A} = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial x}+\hat{\mathbf{k}}\frac{\partial}{\partial x}\right)\cdot(A_x\,\hat{\mathbf{i}} + A_y\,\hat{\mathbf{j}} + A_z\,\hat{\mathbf{k}}) = \frac{\partial A_x}{\partial x}+\frac{\partial A_y}{\partial x}+\frac{\partial A_z}{\partial x}, \text{ and } dV = dx\,dy\,dz$$

and $\vec{A} \cdot d\vec{S} = (A_x\,\hat{i} + A_y\,\hat{j} + A_z\,\hat{k}) \cdot (dS_x\,\hat{i} + dS_y\,\hat{j} + dS_z\,\hat{k}) = A_x\,dS_x + A_y\,dS_y + A_z\,dS_z$

$= A_x\,dy\,dz + A_y\,dx\,dz + A_z\,dx\,dy.$

Hence, Gauss's theorem in cartesian coordinates is

$$\iiint_V \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial x} + \frac{\partial A_z}{\partial x}\right) dx\,dy\,dz = \iint_S (A_x dS_x + A_y dS_y + A_z dS_z) \qquad ...(2)$$

Let us consider the first part of the left-hand side integral. As shown along a strip of cross-section $dy\,dz$ extending from $P_1(x_1, y, z)$ to $P_2(x_2, y, z)$, *i.e.*,

$$\iiint_V \frac{\partial A_x}{\partial x}\,dx\,dy\,dz = \iint_S [A_x]_{P_1}^{P_2}\,dy\,dz$$

$$= \iint_S [A_x(P_2) - A_x(P_1)]\,dy\,dz$$

$$= \iint_S [A_x(x_2, y, z) - A_x(x_1, y, z)]\,dy\,dz$$

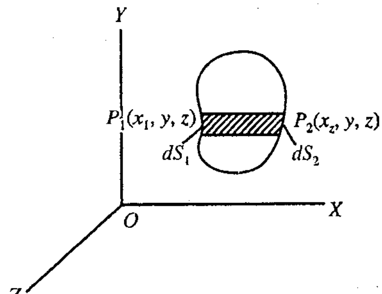


Fig. 1.44.

Evidently, at P_1, $A_x\,dy\,dz = -A_x\,dS_x$ and at P_2, $A_x\,dy\,dz = A_x\,dS_x$ (since at P_1 and P_2, the X-components of area have opposite directions).

$$\therefore \quad \iiint_V \frac{\partial A_x}{\partial x}\,dx\,dy\,dz = \iint_S A_x\,dS_x\,{}^{*} \qquad ...(3)$$

Similarly, $$\iiint_V \frac{\partial A_y}{\partial y}\,dx\,dy\,dz = \iint_S A_y\,dS_y \qquad ...(4)$$

and $$\iiint_V \frac{\partial A_z}{\partial z}\,dx\,dy\,dz = \iint_S A_z\,dS_z \qquad ...(5)$$

Hence, adding (3), (4) and (5), we have

$$\iiint_V \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right) dx\,dy\,dz = \iint_S (A_x dS_x + A_y dS_y + A_z dS_z)$$

or $$\iiint_V (\vec{\nabla} \cdot \vec{A})\,dV = \iint_S \vec{A} \cdot d\vec{S} \quad \text{or} \quad \iiint_V (\text{div}\,\vec{A})\,dV = \iint_S \vec{A} \cdot d\vec{S} \qquad ...(6)$$

This proves the Gauss's theorem of divergence.

1.16. Green's Theorem

We can make some important transformations by the use of Gauss's theorem.

According to Gauss's theorem,

or $$\iiint_V (\vec{\nabla} \cdot \vec{A})\,dV = \iint_S \vec{A} \cdot d\vec{S} \qquad ...(1)$$

Let us consider that the vector field $\vec{A}$ be the product of scalar function ϕ_1 and the gradient of another scalar function ϕ_2, *i.e.*,

$$\vec{A} = \phi_1 \nabla \phi_2 = \hat{i}\,\phi_1 \frac{\partial \phi_2}{\partial x} + \hat{j}\,\phi_1 \frac{\partial \phi_2}{\partial y} + \hat{k}\,\phi_1 \frac{\partial \phi_2}{\partial z}$$

Now, $$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \frac{\partial}{\partial x}\left(\phi_1 \frac{\partial \phi_2}{\partial x}\right) + \frac{\partial}{\partial y}\left(\phi_1 \frac{\partial \phi_2}{\partial y}\right) + \frac{\partial}{\partial z}\left(\phi_1 \frac{\partial \phi_2}{\partial z}\right)$$

---

* The reader should not confuse that we have written for

$\iiint [A_x(P_2) - A_x(P_1)]\,dy\,dz = \iint (A_x dS_x + A_x dS_x)$ only $\iint A_x dS_x$,

because if we take all the stripes then each intergral $\iint A_x ds_x$ will account for the half surface integral. Hence $\iint A_x dS_x$ account for the surface integral for X-component.

$$= \phi_1\left(\frac{\partial^2\phi_2}{\partial x^2}+\frac{\partial^2\phi_2}{\partial y^2}+\frac{\partial^2\phi_2}{\partial z^2}\right)+\frac{\partial\phi_1}{\partial x}\cdot\frac{\partial\phi_2}{\partial x}+\frac{\partial\phi_1}{\partial y}\cdot\frac{\partial\phi_2}{\partial y}+\frac{\partial\phi_1}{\partial z}\cdot\frac{\partial\phi_2}{\partial z} = \phi_1\nabla^2\phi_2+\vec{\nabla}\phi_1\cdot\vec{\nabla}\phi_2$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$ is called **Laplacian operator**

Substituting for $\vec{\nabla}\cdot\vec{A}$ in eq. (1), we get

$$\iiint_V\left(\phi_1\nabla^2\phi_2+\vec{\nabla}\phi_2\cdot\vec{\nabla}\phi_1\right)dV = \iint_S(\phi_1\,\vec{\nabla}\phi_2)\cdot d\vec{S} \qquad ...(2)$$

We call the transformation represented by eq. (2), as the **first form of Green's theorem.**

If we mutually interchange the scalar functions ϕ_1 and ϕ_2 in eq. (2), we obtain

$$\iiint_V\left(\phi_2\nabla^2\phi_1+\vec{\nabla}\phi_1\cdot\vec{\nabla}\phi_2\right)dV = \iint_S(\phi_2\,\vec{\nabla}\phi_1)\cdot d\vec{S} \qquad ...(3)$$

Subtracting eq. (3) from eq. (2), we get

$$\iiint_V(\phi_1\nabla^2\phi_2-\phi_2\nabla^2\phi_1)\,dV = \iint_S(\phi_1\nabla\phi_2-\phi_2\nabla\phi_1)\cdot d\vec{S} \qquad ...(4)$$

Eq. (4) represents the **second form of Green's theorem.** These transformations (2) and (4), obtained by the use of Gauss's theorem, find important applications in the field of electro-dynamics and hydro-dynamics.

Ex. 1. *If* $\vec{r}=x\,\hat{i}+y\,\hat{j}+z\,\hat{k}$, *find the value of* div $\vec{r}$. **(Indore 1990)**

Sol.

$$\text{div}\,\vec{r} = \vec{\nabla}\cdot\vec{r} = \left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial z}\right)\cdot(x\hat{i}+y\hat{j}+z\hat{k})$$

$$= \frac{\partial x}{\partial x}+\frac{\partial y}{\partial y}+\frac{\partial z}{\partial z} = 1+1+1 = \mathbf{3}.$$

Ex. 2. *If* $\vec{A}=x^2y\,\hat{i}-2y^3z^2\,\hat{j}+xy^2z\,\hat{k}$, *find the divergence of this vector at the point* (1, 1, 1).

Sol.

$$\text{div}\,\vec{A} = \vec{\nabla}\cdot\vec{A} = \left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial z}\right)\cdot(x^2y\hat{i}-2y^3z^2\hat{j}+xy^2z\hat{k})$$

$$= \frac{\partial}{\partial x}(x^2y)+\frac{\partial}{\partial y}(-2y^3z^2)+\frac{\partial}{\partial z}(xy^2z) = 2xy-6y^2z^2+xy^2$$

At the point (1, 1, 1), $\vec{\nabla}\cdot\vec{A} = 2(1)(1)-6(1)^2(1)^2+(1)(1)^2 = \mathbf{-3}.$

Ex. 3. (a) *If* ϕ *is a scalar function, find the value of* div (grad ϕ). **(Rewa 1998)**

(b) Prove that div grad (l/r) = 0. **(Gwalior 1996)**

Sol. (a)

$$\text{div (grad }\phi) = \vec{\nabla}\cdot(\vec{\nabla}\phi) = \left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial z}\right)\cdot\left(\hat{i}\frac{\partial\phi}{\partial x}+\hat{j}\frac{\partial\phi}{\partial y}+\hat{k}\frac{\partial\phi}{\partial z}\right)$$

$$= \frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}=\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\phi = \nabla^2\phi.$$

(b)

$$\text{div grad}\left(\frac{1}{r}\right) = \nabla^2\left(\frac{1}{r}\right)=\frac{\partial^2}{\partial x^2}\left(\frac{1}{r}\right)+\frac{\partial^2}{\partial y^2}\left(\frac{1}{r}\right)+\frac{\partial^2}{\partial z^2}\left(\frac{1}{r}\right)$$

But

$$\frac{\partial^2}{\partial x^2}\left(\frac{1}{r}\right) = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(\frac{1}{r}\right)\right]= \frac{\partial}{\partial x}\left(-\frac{1}{r^2}\frac{\partial r}{\partial x}\right)$$

$\because\ r^2 = x^2 + y^2 + z^2,\ \therefore\ 2r\dfrac{\partial r}{\partial x} = 2\dfrac{x}{r}$ or $\dfrac{\partial r}{\partial x} = \dfrac{x}{r}$

Hence, $$\frac{\partial^2}{\partial x^2}\left(\frac{1}{r}\right) = \frac{-\partial}{\partial x}\left(\frac{x}{r^3}\right) = -\frac{1}{r^3} - \frac{\partial}{\partial x}\left(\frac{1}{r^3}\right) = -\frac{1}{r^3} + \frac{3x}{r^4}\frac{\partial r}{\partial x}$$

$$= -\frac{1}{r^3} + \frac{3x}{r^4}\frac{x}{r} = -\frac{1}{r^3} + \frac{3x^2}{r^5}$$

Similarly, $$\frac{\partial^2}{\partial y^2}\left(\frac{1}{r}\right) = -\frac{1}{r^3} + \frac{3y^2}{r^5} \text{ and } \frac{\partial^2}{\partial z^2}\left(\frac{1}{r}\right) = -\frac{1}{r^3} + \frac{3z^2}{r^5}$$

$\therefore$ $$\text{div grad}\left(\frac{1}{r}\right) = -\frac{3}{r^3} + \frac{3}{r^5}(x^2 + y^2 + z^2) = -\frac{3}{r^3} + \frac{3}{r^3} = 0.$$

Ex. 4. *If* $\vec{A}$ *and* $\vec{B}$ *are vector fields, then show that*

$$\text{div}(\vec{A} + \vec{B}) = \text{div}\,\vec{A} + \text{div}\,\vec{B}.$$

(Bhopal 1996; Gwalior 94; Ujjain 89; Sagar 90)

Sol. $$\text{div}(\vec{A} + \vec{B}) = \vec{\nabla}\cdot(\vec{A} + \vec{B}) = \left[\hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}\right]\cdot[(A_x + B_x)\,\hat{\mathbf{i}} + (A_y + B_y)\,\hat{\mathbf{j}} + (A_z + B_z)\,\hat{\mathbf{k}}]$$

$$= \frac{\partial}{\partial x}(A_x + B_x) + \frac{\partial}{\partial y}(A_y + B_y) + \frac{\partial}{\partial z}(A_z + B_z)$$

$$= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} + \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}, \quad = \vec{\nabla}\cdot\vec{A} + \vec{\nabla}\cdot\vec{B}$$

$$= \text{div}\,\vec{A} + \text{div}\,\vec{B}$$

Ex. 5. *If* $\vec{r} = \hat{\mathbf{i}}\,x + \hat{\mathbf{j}}\,y + \hat{\mathbf{k}}\,z$, *find,*

(i) div $(r^3\,\vec{r})$ (ii) div $(r^n\,\vec{r})$ (iii) div $(r^{-3}\,\vec{r})$

Sol. (i) div $(r^3\,\vec{r})$

$$= \vec{\nabla}\cdot(r^3\,\vec{r}) = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}\right)\cdot\left\{r^3(\hat{\mathbf{i}}x + \hat{\mathbf{j}}y + \hat{\mathbf{k}}z)\right\} = \frac{\partial}{\partial x}(xr^3) + \frac{\partial}{\partial y}(yr^3) + \frac{\partial}{\partial z}(zr^3)$$

$$= \left\{r^3 + x\frac{\partial}{\partial x}(x^2 + y^2 + z^2)^{3/2}\right\} + \left\{r^3 + y\frac{\partial}{\partial y}(x^2 + y^2 + z^2)^{3/2}\right\} + \left\{r^3 + z\frac{\partial}{\partial z}(x^2 + y^2 + z^2)^{3/2}\right\}$$

$$= \left\{r^3 + x\cdot\frac{3}{2}(x^2 + y^2 + z^2)^{1/2}\cdot 2x\right\} + \ldots\ldots\ldots + \ldots\ldots\ldots$$

$$= 3r^3 + 3(x^2 + y^2 + z^2)^{1/2}(x^2 + y^2 + z^2) = 3r^3 + 3r^3 = \mathbf{6r^3}.$$

(ii) $$\text{div}(r^n\,\vec{r}) = \vec{\nabla}\cdot(r^n\,\vec{r}) = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}\right)\cdot\left\{r^n(\hat{\mathbf{i}}x + \hat{\mathbf{j}}y + \hat{\mathbf{k}}z)\right\} = \frac{\partial}{\partial x}(xr^n) + \frac{\partial}{\partial y}(yr^n) + \frac{\partial}{\partial z}(zr^n)$$

$\because$ $r = (x^2 + y^2 + z^2)^{1/2}$, $\therefore$ $r^n = (x^2 + y^2 + z^2)^{n/2}$

$$\therefore \text{div}(r^n\,\vec{r}) = \left(r^n + x\frac{\partial}{\partial x}r^n\right) + \left(r^n + y\frac{\partial}{\partial y}r^n\right) + \left(r^n + z\frac{\partial}{\partial z}r^n\right)$$

$$= 3r^n + x\frac{\partial}{\partial x}(x^2 + y^2 + z^2)^{n/2} + y\frac{\partial}{\partial y}(x^2 + y^2 + z^2)^{n/2} + z\frac{\partial}{\partial z}(x^2 + y^2 + z^2)^{n/2}$$

$$= 3r^n + x\frac{n}{2}(x^2 + y^2 + z^2)^{n/2-1}\,2x + y\frac{n}{2}(x^2 + y^2 + z^2)^{n/2-1}\,2y + z\frac{n}{2}(x^2 + y^2 + z^2)^{n/2-1}\,2z$$

$$= 3r^n + n(x^2 + y^2 + z^2)\,(x^2 + y^2 + z^2)^{n/2-1} = 3r^n + n(x^2 + y^2 + z^2)^{n/2}$$

$$= 3r^n + nr^n = \mathbf{(3 + n)r^n}.$$

(iii) $\operatorname{div}(r^{-3}\vec{r}) = \vec{\nabla}\cdot\frac{\vec{r}}{r^3} = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}\right)\cdot\left(\frac{\hat{\mathbf{i}}x+\hat{\mathbf{j}}y+\hat{\mathbf{k}}z}{r^3}\right) = \frac{\partial}{\partial x}\left(\frac{x}{r^3}\right)+\frac{\partial}{\partial y}\left(\frac{y}{r^3}\right)+\frac{\partial}{\partial z}\left(\frac{z}{r^3}\right)$

$= \frac{1}{r^3}-x\frac{3}{r^4}\frac{\partial r}{\partial x}+\frac{1}{r^3}-y\frac{3}{r^4}\frac{\partial r}{\partial y}+\frac{1}{r^3}-z\frac{3}{r^4}\frac{\partial r}{\partial z}$

But, $\quad r^2 = x^2+y^2+z^2, \therefore \frac{\partial r}{\partial x} = \frac{x}{r}, \frac{\partial r}{\partial y} = \frac{y}{r}, \frac{\partial r}{\partial z} = \frac{z}{r}$

$\therefore \operatorname{div}(\vec{r}/r^3) = \frac{1}{r^3}-\frac{3x^2}{r^5}+\frac{1}{r^3}-\frac{3y^2}{r^5}+\frac{1}{r^3}-\frac{3z}{r^5} = \frac{3}{r^3}-\frac{3}{r^5}(x^2+y^2+z^2) = \frac{3}{r^3}-\frac{3}{r^5}r^2 = \frac{3}{r^3}-\frac{3}{r^3} = 0.$

Ex. 6. *Show that divergence of a vector field obeying inverse square law is zero i.e., such a field is solenoidal.* **(Indore 2003; Raipur1997; Ujjain 95)**

Sol. Vector field obeying inverse square law is represented as

$$\vec{F} \propto \frac{1}{r^2}\hat{\mathbf{r}} \quad \text{or} \quad \vec{F} = \frac{K}{r^2}\hat{\mathbf{r}} = \frac{K}{r^3}\vec{r} = \frac{K}{r^3}(\hat{\mathbf{i}}\,x+\hat{\mathbf{j}}\,y+\hat{\mathbf{k}}\,z)$$

where $\vec{r} = \hat{\mathbf{i}}x+\hat{\mathbf{j}}y+\hat{\mathbf{k}}z$ and K is a constant.

$$\therefore \quad \operatorname{div}\vec{F} = \frac{\partial F_x}{\partial x}+\frac{\partial F_y}{\partial y}+\frac{\partial F_z}{\partial z} = \frac{\partial}{\partial x}\left(\frac{Kx}{r^3}\right)+\frac{\partial}{\partial y}\left(\frac{Ky}{r^3}\right)+\frac{\partial}{\partial z}\left(\frac{Kz}{r^3}\right)$$

$$= \left\{\frac{K}{r^3}+Kx\left(\frac{-3}{r^4}\right)\frac{\partial r}{\partial x}\right\}+\left\{\frac{K}{r^3}+Ky\left(\frac{-3}{r^4}\right)\frac{\partial r}{\partial y}\right\}+\left\{\frac{K}{r^3}+Kz\left(\frac{-3}{r^4}\right)\frac{\partial r}{\partial z}\right\}$$

$$= \frac{3K}{r^3}-\frac{3K}{r^4}\left\{x\frac{\partial r}{\partial x}+y\frac{\partial r}{\partial y}+z\frac{\partial r}{\partial z}\right\}$$

But $\quad r^2 = x^2+y^2+z^2$

$\therefore \quad 2r\frac{\partial r}{\partial x} = 2x$ or $\frac{\partial r}{\partial x} = \frac{x}{r}$, similarly $\frac{\partial r}{\partial y} = \frac{y}{r}$ and $\frac{\partial r}{\partial z} = \frac{z}{r}$

Thus, $\quad \operatorname{div}\vec{F} = \frac{3K}{r^3}-\frac{3K}{r^4}\left[x\frac{x}{r}+y\frac{y}{r}+z\frac{z}{r}\right] = \frac{3K}{r^3}-\frac{3K}{r^5}(x^2+y^2+z^2) = \frac{3K}{r^3}-\frac{3K}{r^5}(r^2) = 0.$

Ex. 7. *Prove that* : div grad $r^n = n(n+1)r^{n-2}$

Sol. div grad $r^n = \vec{\nabla}\cdot\vec{\nabla}r^n$

Here, $\quad \vec{\nabla}r^n = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}\right)r^n = nr^{n-1}\left(\frac{\partial r}{\partial x}\hat{\mathbf{i}}+\frac{\partial r}{\partial y}\hat{\mathbf{j}}+\frac{\partial r}{\partial z}\hat{\mathbf{k}}\right);$

But, $\quad r^2 = x^2+y^2+z^2, \therefore 2r\frac{\partial r}{\partial x} = 2x$ or $\frac{\partial r}{\partial x} = \frac{x}{r}$ similarly, $\frac{\partial r}{\partial y} = \frac{y}{r}, \frac{\partial r}{\partial z} = \frac{z}{r}$

Hence, $\quad \vec{\nabla}r^n = nr^{n-1}\left(\frac{x}{r}\hat{\mathbf{i}}+\frac{y}{r}\hat{\mathbf{j}}+\frac{z}{r}\hat{\mathbf{k}}\right) = nr^{n-2}(x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}})$

$\therefore$ div grad $r^n = \vec{\nabla}\cdot\vec{\nabla}r^n = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}\right)\cdot nr^{n-2}(x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}})$

$$= \frac{\partial}{\partial x}(nr^{n-2}x)+\frac{\partial}{\partial y}(nr^{n-2}y)+\frac{\partial}{\partial z}(nr^{n-2}z) = n(n-2)r^{n-3}\frac{\partial r}{\partial x}x+nr^{n-2}\frac{\partial x}{\partial x}+\ldots..$$

$$= nr^{n-2}+n(n-2)r^{n-3}\frac{x^2}{r}+nr^{n-2}+n(n-2)r^{n-3}\frac{y^2}{r}+nr^{n-2}+n(n-2)r^{n-3}\frac{z^2}{r}$$

$$= 3nr^{n-2}+n(n-2)r^{n-3}\left(\frac{x^2+y^2+z^2}{r}\right) = 3nr^{n-2}+n(n-2)r^{n-2}$$

$[\because r^2 = x^2+y^2+z^2]$

$$= 3nr^{n-2}+n^2r^{n-2}-2nr^{n-2} = nr^{n-2}+n^2r^{n-2} = \boldsymbol{n(n+1)r^{n-2}}.$$

Ex. 8. *If* $r = (x^2 + y^2 + z^2)^{1/2}$, *prove that* div grad $f(r) = \vec{\nabla}\cdot\vec{\nabla} f(r) = \dfrac{\partial^2 f(r)}{\partial r^2} + \dfrac{2}{r}\dfrac{\partial f(r)}{dr}$. *Hence prove that* div grad $r^n = n(n+1)r^{n-2}$.

Sol. We have proved that grad $f(r) = \dfrac{\vec{r}}{r}\dfrac{\partial f(r)}{\partial r}$ Hence,

$$\vec{\nabla}\cdot\vec{\nabla} f(r) = \vec{\nabla}\cdot\frac{\vec{r}}{r}\frac{\partial f(r)}{\partial r} = \frac{\partial}{\partial x}\left(\frac{x}{r}\frac{\partial f(r)}{\partial r}\right) + \frac{\partial}{\partial y}\left(\frac{y}{r}\frac{\partial f(r)}{\partial r}\right) + \frac{\partial}{\partial z}\left(\frac{z}{r}\frac{\partial f(r)}{\partial r}\right)$$

Its first term will be

$$\frac{\partial}{\partial x}\left(\frac{x}{r}\frac{\partial f(r)}{\partial r}\right) = \frac{1}{r}\frac{\partial f(r)}{\partial r} + x\frac{\partial}{\partial x}\left(\frac{1}{r}\frac{\partial f(r)}{\partial r}\right) = \frac{1}{r}\frac{\partial f(r)}{\partial r} + x\left(-\frac{1}{r^2}\right)\frac{\partial r}{\partial x}\frac{\partial f(r)}{\partial r} + \frac{x}{r}\frac{\partial}{\partial x}\left(\frac{\partial f(r)}{\partial r}\right)$$

$$= \frac{1}{r}\frac{\partial f(r)}{\partial r} - \frac{x}{r^2}\frac{x}{r}\frac{\partial f(r)}{\partial r} + \frac{\partial}{\partial r}\left(\frac{\partial f(r)}{\partial r}\right)\frac{\partial r}{\partial x}\frac{x}{r} = \frac{1}{r}\frac{\partial f(r)}{\partial r} - \frac{x^2}{r^3}\frac{\partial f(r)}{\partial r} + \frac{\partial^2 f(r)}{\partial r^2}\frac{x}{r}\frac{x}{r}$$

$$= \frac{1}{r}\frac{\partial f(r)}{\partial r} - \frac{x^2}{r^3}\frac{\partial f(r)}{\partial r} + \frac{x^2}{r^2}\frac{\partial^2 f(r)}{\partial r^2}$$

Similarly finding other terms and adding them, we get

$$\vec{\nabla}\cdot\vec{\nabla} f(r) = \left(\frac{1}{r}\frac{\partial f(r)}{\partial r} - \frac{x^2}{r^3}\frac{\partial f(r)}{\partial r} + \frac{x^2}{r^2}\frac{\partial^2 f(r)}{\partial r^2}\right) + \left(\frac{1}{r}\frac{\partial f(r)}{\partial r} - \frac{y^2}{r^3}\frac{\partial f(r)}{\partial r} + \frac{y^2}{r^2}\frac{\partial^2 f(r)}{\partial r^2}\right)$$
$$+ \left(\frac{1}{r}\frac{\partial f(r)}{\partial r} - \frac{z^2}{r^3}\frac{\partial f(r)}{\partial r} + \frac{z^2}{r^2}\frac{\partial^2 f(r)}{\partial r^2}\right)$$
$$= \frac{3}{r}\frac{\partial f(r)}{\partial r} - \left(\frac{x^2+y^2+z^2}{r^3}\right)\frac{\partial f(r)}{\partial r} + \left(\frac{x^2+y^2+z^2}{r^2}\right)\frac{\partial^2 f(r)}{\partial r^2} = \frac{\partial^2 f(r)}{\partial r^2} + \frac{2}{r}\frac{\partial f(r)}{\partial r}$$
$$[\because r^2 = x^2 + y^2 + z^2]$$

If $\quad f(r) = r^n$, then $\vec{\nabla}\cdot\vec{\nabla} f(r) = \nabla^2 r^n = n(n-1)r^{n-2} + \dfrac{2}{r}nr^{n-1}$

or $\quad$ div grad $r^n = \boldsymbol{n(n+1)r^{n-2}}$.

Ex. 9. If $\vec{A} = (x+3y)\,\hat{\mathbf{i}} + (y-z)\,\hat{\mathbf{j}} + (x+cz)\,\hat{\mathbf{k}}$ *is a solenoidal vector find the value of c.*

Sol. Here $\quad A_x = x + 3y,\ A_y = y - z,\ A_z = x + cz;\ \therefore\ \dfrac{\partial A_x}{\partial x} = 1,\ \dfrac{\partial A_y}{\partial y} = 1,\ \dfrac{\partial A_z}{\partial z} = c.$

Hence $\quad \vec{\nabla}\cdot\vec{A} = \dfrac{\partial A_x}{\partial x} + \dfrac{\partial A_y}{\partial y} + \dfrac{\partial A_z}{\partial z} = 1 + 1 + c = 2 + c.$

But $\vec{A}$ is a solenoidal vector, hence $\vec{\nabla}\cdot\vec{A} = 0$ *i.e.*, $2 + c = 0$ or $c = \mathbf{-2}$.

Ex. 10. *Prove that* : div $(\psi\vec{A})$ = (grad ψ) $\cdot\vec{A} + \psi$ div $\vec{A}$. **(Bilaspur 1998; Jabalpur 2003, 93)**

Sol. $\quad$ Let $\vec{A} = A_x\,\hat{\mathbf{i}} + A_y\,\hat{\mathbf{j}} + A_z\,\hat{\mathbf{k}}$, $\therefore\ \psi\vec{A} = \psi A_x\,\hat{\mathbf{i}} + \psi A_y\,\hat{\mathbf{j}} + \psi A_z\,\hat{\mathbf{k}}$

$$\text{div}\,(\psi\vec{A}) = \vec{\nabla}\cdot(\psi\vec{A}) = \frac{\partial}{\partial x}(\psi A_x) + \frac{\partial}{\partial y}(\psi A_y) + \frac{\partial}{\partial z}(\psi A_z)$$
$$= \psi\frac{\partial A_x}{\partial x} + \frac{\partial\psi}{\partial x}A_x + \psi\frac{\partial A_y}{\partial y} + \frac{\partial\psi}{\partial y}A_y + \psi\frac{\partial A_z}{\partial z} + \frac{\partial\psi}{\partial z}A_z$$
$$= \psi\left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right) + \left(\frac{\partial\psi}{\partial x}A_x + \frac{\partial\psi}{\partial y}A_y + \frac{\partial\psi}{\partial z}A_z\right)$$
$$= \psi(\vec{\nabla}\cdot\vec{A}) + \left\{\hat{\mathbf{i}}\frac{\partial\psi}{\partial x} + \hat{\mathbf{j}}\frac{\partial\psi}{\partial y} + \hat{\mathbf{k}}\frac{\partial\psi}{\partial z}\right\}\cdot(A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}) = \psi\ \text{div}\ \vec{\mathbf{A}} + (\text{grad}\ \psi)\cdot\vec{\mathbf{A}}.$$

Ex. 11. *Prove that :* $\vec{A}(\vec{\nabla}\cdot\vec{r})-(\vec{A}\cdot\vec{\nabla})\vec{r} = 2\vec{A}$. **(Rewa 1999)**

Sol. $\vec{A}(\vec{\nabla}\cdot\vec{r}) = \vec{A}\left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial z}\right)\cdot(x\hat{i}+y\hat{j}+z\hat{k}) = \vec{A}\left(\frac{\partial x}{\partial x}+\frac{\partial y}{\partial y}+\frac{\partial z}{\partial z}\right) = \vec{A}(1+1+1) = 3\vec{A}$.

and $(\vec{A}\cdot\vec{\nabla})\vec{r} = \left(A_x\frac{\partial}{\partial x}+A_y\frac{\partial}{\partial y}+A_z\frac{\partial}{\partial z}\right)(x\hat{i}+y\hat{j}+z\hat{k}) = (A_x\hat{i}+A_y\hat{j}+A_z\hat{k}) = \vec{A}$.

Hence, $\vec{A}(\vec{\nabla}\cdot\vec{r})-(\vec{A}\cdot\vec{\nabla})\vec{r} = 3\vec{A}-\vec{A} = \mathbf{2\vec{A}}$.

Ex. 12. *A closed surface S encloses a volume V. Find the value of* div $\vec{F}$ *and* $\iint_S \vec{F}.d\vec{S}$ *for a vector field* $\vec{F} = 3x\,\hat{i}+4y\,\hat{j}+2z\,\hat{k}$. **(Indore 1993)**

Sol. According to Gauss's divergence theorem, $\iint_S \vec{F}.d\vec{S} = \iiint_V (\text{div}\,\vec{F})dV$

$$\therefore \quad \text{div}\,\vec{F} = \frac{\partial F_x}{\partial x}+\frac{\partial F_y}{\partial y}+\frac{\partial F_z}{\partial z} = \frac{\partial}{\partial x}(3x)+\frac{\partial}{\partial y}(4y)+\frac{\partial}{\partial z}(2z) = 3+4+2 = 9$$

Hence, $\iint_S \vec{F}.d\vec{S} = \iiint_V 9dV = \mathbf{9V}$.

Ex. 13. *Prove that* $\iint_S \vec{r}.d\vec{S} = 3V$, *where* $\vec{r}$ *is the position vector at any point on the surface enclosing volume V.* **(Bhopal 1998, 90)**

Sol. According to Gauss's theorem, $\iint_S \vec{A}.d\vec{S} = \iiint_V (\text{div}\,\vec{A})dV$

Here, $\vec{A} = \vec{r}$ and $\text{div}\,\vec{A} = \text{div}\,\vec{r} = 3$

$$\therefore \quad \iint_S \vec{r}.d\vec{S} = \iiint_V (\text{div}\,\vec{r})dV = \iiint_V 3dV = \mathbf{3V}.$$

Ex. 14. *If* $\vec{a}$ *is a constant vector and* $\vec{r}$ *is the position vector, prove that* $\text{div}\,(\vec{a}\times\vec{r}) = 0$. **(Gwalior 1999; Raipur 92)**

Sol. Here, $\vec{a} = \hat{i}a_x+\hat{j}a_y+\hat{k}a_z$ = constant and $\vec{r} = \hat{i}x+\hat{j}y+\hat{k}z$

$$\therefore \quad \text{div}\,(\vec{a}\times\vec{r}) = \vec{\nabla}\cdot(\vec{a}\times\vec{r}) = \begin{vmatrix} \partial/\partial x & \partial/\partial y & \partial/\partial z \\ a_x & a_y & a_z \\ x & y & z \end{vmatrix} = \frac{\partial}{\partial x}(a_y z - a_z y)+\frac{\partial}{\partial y}(a_z x - a_x z)+\frac{\partial}{\partial z}(a_x y - a_y x)$$

$$= \left[0.z+a_y\frac{\partial z}{\partial x}-0.y-a_z\frac{\partial y}{\partial x}\right]+\left[0.x+a_z\frac{\partial x}{\partial y}-0.z-a_x\frac{\partial z}{\partial y}\right]+\left[0.y+a_x\frac{\partial y}{\partial z}-0.x-a_y\frac{\partial x}{\partial z}\right] = 0.$$

EXERCISE 1.4

1. If vector field $\vec{F} = 3x\hat{i}+4y\hat{j}+2z\hat{k}$, find the value of div $\vec{F}$. **(Gwalior 1998)**
   [**Ans.** 9.]
2. If $\vec{J} = (ax+3y+4z)\,\hat{i}+(x-2y+3z)\,\hat{j}+(3x+2y-z)\,\hat{k}$ is solenoidal, then find the value of a.
   [**Ans.** $a = 3$]

3. Find the divergence of vector $\vec{A} = x^2z\,\hat{i} - 2y^2z^2\,\hat{j} + xy^2z\,\hat{k}$.
[**Ans.** $2xz - 4yz^2 + xy^2$]
4. Prove that the vector field $(x + 3y)\,\hat{i} + (y - 3z)\,\hat{j} + (x - 2z)\,\hat{k}$ is solenoidal.
5. If $\vec{r} = x^2z\,\hat{i} - 2y^3z^2\,\hat{j} + xy^2z\,\hat{k}$, find div $\vec{r}$ at the point (1, 1, 1).
[**Ans.** –3]
6. If $\vec{F} = y\,\hat{i} + (x^2 + y^2)\,\hat{j} + (yz + zx)\,\hat{k}$, find div $\vec{F}$ at the point (1, 1, 1).
7. Prove that
(i) div $\vec{r} = 3$, (ii) div $\hat{r} = 2/r$, (iii) div $(r^n\,\vec{r}) = (3 + n)\,r^n$,
(iv) div $(\vec{r}/r^3) = 0$, **(Purvanchal 1995, Agra 96)**
(v) $\vec{\nabla}\cdot(\vec{a}\times\vec{r}) = 0$, **(Agra 1972)**
8. If $\vec{A} = f(r)\,\hat{r}$, find div $\vec{A}$. If $f(r) = kr^{n-1}$, find div $\vec{A}$.
[**Ans.** $3f(r) + r\,\partial f/\partial r$; $k(n + 2)r^{n-1}$]
9. If S is any closed surface enclosing a volume V, and $\vec{F} = x\,\hat{i} + 2y\,\hat{j} + 3z\,\hat{k}$, show that $\iint_S \vec{F}.d\vec{S} = 6V$. **(Kumaun 1995)**

[**Hint.** $\iint_S \vec{F}.d\vec{S} = \iiint_V (\text{div}\,\vec{F})dV$.

Calculate div $\vec{F} = \vec{\nabla}\cdot\vec{F} = 6$. Hence R.H.S. is $\int_V 6\,dV = 6V$.]

10. If $\iint_S \vec{E}.d\vec{S} = \frac{1}{\epsilon_0}\iiint_V \rho\,dV$ and $\vec{E} = -\text{grad}\,\phi$, where ρ is the charge density, $\vec{E}$ is the electric field intensity and ϕ is scalar potential, prove that by using Gauss's theorem
$$\text{div}\,\vec{E} = \rho/\epsilon_0 \quad \text{and} \quad \nabla^2\phi = -\rho/\epsilon_0$$
(Rewa 1992)
11. Prove that $\iint_S (ax\hat{i} + by\hat{j} + cx\hat{k}).\hat{n}dS = \frac{4}{3}(a + b + c)$ where S is the surface of the sphere, whose centre is the origin and radius is unity.
12. Find $\iint_S \vec{r}\cdot d\vec{S}$, where S is the surface of a cube, which is enclosed by $x = 0$, $x = a$, $y = 0$, $y = a$, $z = 0$, $z = a$.
[**Ans.** $3a^3$.]

1·17. Curl

If $\vec{A}(x, y, z)$ is a vector field, the cross product of the operator $\vec{\nabla}$ and $\vec{A}$ is a vector and is called curl or rotation of $\vec{A}$. Thus,

$$\text{curl}\,\vec{A} = \vec{\nabla}\times\vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{j}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{k}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) \qquad ...(1)$$

The curl of a vector field $\vec{A}$ at a point is related to the line integreal $\left(\oint \vec{A} \cdot d\vec{l}\right)$, taken around the boundary of an infinitesimal small plane area situated around that point and curl $\vec{A}$ is defined as **"Curl of a vector field ($\vec{A}$) at a point is a vector (curl $\vec{A}$) whose magnitude is equal to the maximum value of the line integral of that vector $\left(\oint \vec{A} \cdot d\vec{l}\right.$ per unit area along the boundary of an infinitesimal small area around that point and the direction is along that infinitesimal small area".** This definition will emerge from the following discussion.

The concept of curl is very useful in Vector Calculus and has significant and wide applications in hydrodynamics and electro-magnetism.

Physical Interpretation and Significance of curl $\vec{A}$

In order to understand the physical significance of curl of a vector field $\vec{A}$, we consider the **circulation of a fluid** around an infinitesimal small area in XY-plane. In hydrodynamics, the line integral of a vector field $\vec{A}$ along a closed curve (as the velocity $\vec{v}$ of a liquid in whirlpools and eddies) is called the **circulation** of $\vec{A}$ along that path. For convenience, we take this path as infinitesimal rectangle in XY-plane and calculate **the line integral of this vector field $\vec{A}$ around the infinitesimal rectangle.**

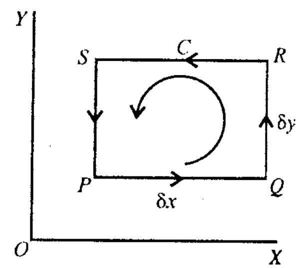


Fig. 1.45.

In **fig 1.45**, we consider an infinitesimal rectangle $PQRS$ with sides δx and δy in XY-plane. Let the X- and Y- components of $\vec{A}$ at the point P be A_x and A_y respectively. The average value of the X-component of $\vec{A}$ over the path PQ can be taken $A_x + \frac{\partial A_x}{\partial x} \cdot \frac{\delta x}{2}$ and over the path RS as $A_x + \frac{\partial A_x}{\partial x} \cdot \frac{\delta x}{2} + \frac{\partial A_x}{\delta y} \cdot \delta y$

$$\text{Therefore,} \quad \int_P^Q \vec{A} \cdot d\vec{l} + \int_R^S \vec{A} \cdot d\vec{l} = \left(A_x + \frac{\partial A_x}{\partial x} \cdot \frac{\delta x}{2}\right)\delta x - \left(A_x + \frac{\partial A_x}{\partial x} \cdot \frac{\delta x}{2} + \frac{\partial A_x}{\partial y} \cdot \delta y\right)\delta x$$

$$= -\frac{\partial A_x}{\partial y} \cdot \delta x \, \delta y \qquad \text{...(2)}$$

because the direction of the path RS is towards the negative direction of X.

$$\text{Similarly,} \quad \int_Q^R \vec{A} \cdot d\vec{l} + \int_S^P \vec{A} \cdot d\vec{l} = \frac{\partial A_y}{\partial x} \delta x \, \delta y \qquad \text{..(3)}$$

By adding (2) and (3), we get the line integral

$$\oint_{PQRS} \vec{A} \cdot d\vec{l} = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) \delta x \delta y \qquad \text{...(4)}$$

But $\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$ is the Z– component of curl $\vec{A}$ or $\vec{\nabla} \times \vec{A}$ and the infinitesimal area $\delta x \, \delta y$ may be regarded as the magnitude to the vector area $\delta S_z \hat{\mathbf{k}}$. Thus,

$$\oint_{PQRS} \vec{A} \cdot d\vec{l} = (\text{curl } \vec{A})_z \, dS_z = (\text{curl } \vec{A}) \cdot \delta S_z \hat{\mathbf{k}} \qquad \text{...(5)}$$

This equation is correct for any infinitesimal area of magnitude δS_z at P, having boundary C, *i.e.*,

$$\oint_C \vec{A} \cdot d\vec{l} = (\text{curl } \vec{A}) \cdot \delta S_z \hat{\mathbf{k}} \qquad \text{...(6)}$$

Now, of we consider any infinitesimal plane area $\delta \vec{S} = \delta S_x \hat{\mathbf{i}} + \delta S_y \hat{\mathbf{j}} + \delta S_x \hat{\mathbf{k}}$ at P oriented in any direction with a boundary C (fig. 1.46), then evidently.

$$\oint_C \vec{A}\cdot d\vec{l} = (\text{curl}\,\vec{A})\cdot \delta\vec{S} \qquad ...(7)$$

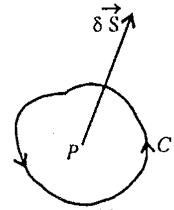


Fig. 1.46.

The maximum value of $\oint_C \vec{A}\cdot d\vec{l} = |\,\text{curl}\,\vec{A}\,|\,\delta S$.

Thus, the magnitude of curl $\vec{A}$ at a point is the maximum value of the line intergral of $\vec{A}$ along a boundary $C(\oint \vec{A}\cdot d\vec{l})$ per unit area; this boundary C is that of an infinitesimal plane area (δS) at the point under consideration.

From (7)

$$\oint_C \vec{A}\cdot d\vec{l} = (\text{curl}\,\vec{A})_x\ \delta S_x\ \hat{i} + (\text{curl}\,\vec{A})_y\ \delta S_y\ \hat{j} + (\text{curl}\,\vec{A})_z\ \delta S_z\ \hat{k} \qquad ...(8)$$

where $(\text{curl}\,\vec{A})_x = \left(\dfrac{\partial A_z}{\partial y} - \dfrac{\partial A_y}{\partial z}\right)$, $(\text{curl}\,\vec{A})_y = \left(\dfrac{\partial A_x}{\partial z} - \dfrac{\partial A_z}{\partial x}\right)$ and $(\text{curl}\,\vec{A})_z = \left(\dfrac{\partial A_y}{\partial x} - \dfrac{\partial A_x}{\partial y}\right)$. **These are the cartesian components of curl $\vec{A}$ and**

$$\text{curl}\,\vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} = \vec{\nabla}\times\vec{A} \qquad ...(9)$$

In order to understant the **physical picture of circulation of moiving fluid** (e.g. water) **at a point,** we put *a* paddle-wheel at that point [**fig.1.47**]. If the velocity of the fluid over one part of the circumference of the wheel is more than that at the other, the wheel will rotate and if we divide the circulation $\oint \vec{v}\cdot d\vec{l}$ by the area δs of the wheel, then we will obtain the value of the component of curl $\vec{v}$ along the axis of the wheel.

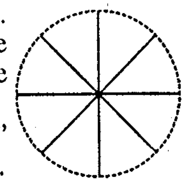


Fig. 1.47.

In a circulating fluid, the value of curl $\vec{v}$ at a point is a measure of angular velocity of the fluid close to that point.

Rotation of a rigid body : A simple example of curl of a vector is a rotating rigid body about an axis. Let a rigid body be rotating about an axis AB with an angular velocity $\vec{\omega}$. Any particle P of the body will move on a circle of radius $r\sin\theta$; the position vector of P is $\vec{r}$ with respect to the origin O on the axis AB [**fig. 1.48**].

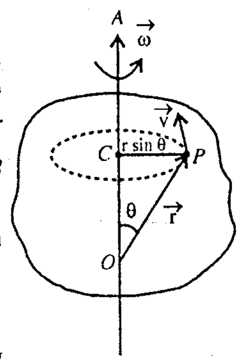


Fig. 1.48.

Therefore, the value of the linear velocity of the body will be $\omega r\sin\theta$ which can be represented in vector form as

$$\vec{v} = \vec{\omega}\times\vec{r} \qquad ...(10)$$

In general, this equation represents velocity of any particle of the rotating rigid body.

Now, we calculate $\vec{\nabla}\times\vec{v} = \vec{\nabla}\times(\vec{\omega}\times\vec{r}) \qquad ...(11)$

For this, we use formula for vector triple product

$$\vec{A}\times(\vec{B}\times\vec{C}) = (\vec{A}\cdot\vec{C})\vec{B} - (\vec{A}\cdot\vec{B})\vec{C} \qquad ...(12)$$

Remember that $\vec{\nabla}$ is not ordinary vector, but it has the properties of both vector and differential operator. Hence, it is always written before those variable quantities which can be differentiated. Therefore, using (12), we have

$$\vec{\nabla}\times(\vec{\omega}\times\vec{r}) = (\vec{\nabla}\cdot\vec{r})\vec{\omega} - (\vec{\omega}\cdot\vec{\nabla})\vec{r} \qquad ...(13)$$

But, $\quad \vec{\omega}\,(\vec{\nabla}\cdot\vec{r}) = \vec{\omega}\left(\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}\right)\cdot(x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}}) = \vec{\omega}\left(\frac{\partial x}{\partial x}+\frac{\partial y}{\partial y}+\frac{\partial z}{\partial z}\right) = 3\vec{\omega}$

and $\quad (\vec{\omega}\cdot\vec{\nabla})\,\vec{r} = \left(\omega_x\frac{\partial}{\partial x}+\omega_y\frac{\partial}{\partial y}+\omega_z\frac{\partial}{\partial z}\right)(\hat{\mathbf{i}}x+\hat{\mathbf{j}}y+\hat{\mathbf{k}}z) = \hat{\mathbf{i}}\omega_x+\hat{\mathbf{j}}\omega_y+\hat{\mathbf{k}}\omega_z = \vec{\omega}$

Substituting these values in (13) and using (11), we have

$$\vec{\nabla}\times\vec{v} = \vec{\nabla}\times(\vec{\omega}\times\vec{r}) = 3\vec{\omega}-\vec{\omega} = 2\vec{\omega} \quad \textit{i.e.}, \ \text{curl}\,\vec{v} = 2\vec{\omega} \ \text{ or } \ \vec{\omega} = \frac{1}{2}\,\text{curl}\,\vec{v} \qquad ...(14)$$

This result gives us again a clue as the name curl $\vec{v}$ (or rotation $\vec{v}$ or rot $\vec{v}$ as it is sometimes called). Hence *if a rigid body is rotating about an axis, then the curl of the linear velocity at a point is equal to two times its angular velocity.* In the case of circular motion of a fluid, an infinitesimal volume around a point at an instant can be understood as a rigid body so that the curl of the velocity of the fluid at that point is equal to two times its angular velocity, while the instantaneous axis of rotation is along curl $\vec{v}$.

If in the motion of a body we have the curl of the body, then particle of the body are moving with an instantaneous angular velocity. Such kind of motion is called **rotational** or **vortical.**

Irrotational vector - Consertative vector field : If the curl of a vector field $\vec{A}$ is zero *i.e.*,

$$\text{curl}\,\vec{A} = \vec{\nabla}\times\vec{A} = 0 \qquad ...(15)$$

Then $\vec{A}$ is called **irrotational vector** or **lamellar vector.** We give two important examples of this vector—(1) **gravitational force** and (2) **electrostatic force.** Both kinds of forces can be represented by the following expression

$$\vec{F} = \frac{k\hat{\mathbf{r}}}{r^2} = \frac{k\vec{r}}{r^3} \qquad ...(16)$$

where $\vec{r} = r\hat{\mathbf{r}}$ is the vector distance from the centre of force; for gravitational force, $k = -Gm_1m_2$ and for electrostatic forces, $k = q_1q_2/4\pi\epsilon_0$.

Here, $$\vec{\nabla}\times\vec{F} = \vec{\nabla}\times\frac{k\vec{r}}{r^3} = k\begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ x/r^3 & y/r^3 & z/r^3\end{vmatrix} = 0 \qquad ...(17)$$

where $\vec{r} = x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}}$, and $|\vec{r}| = r = (x^2+y^2+z^2)^{1/2}$ and $\frac{\partial}{\partial y}\left(\frac{z}{r^3}\right)-\frac{\partial}{\partial z}\left(\frac{y}{r^3}\right) = -\frac{3zy}{r^5}+\frac{3yz}{r^5} = 0$ etc.

If $\vec{F} = f(r)\hat{\mathbf{r}}$, which represents a central foce with $f(r)$ as a scalar function of r, then it can be proved that

$$\vec{\nabla}\times\vec{F} = \vec{\nabla}\times f(r)\hat{\mathbf{r}} = 0 \qquad ...(18)$$

Hence, *a central force is an example of irrotational vector.*

Now, if a vector field can be represented as the gradient of a scalar function ϕ *i.e.*, $\vec{A}$ = grad ϕ, then that vector field $\vec{A}$ will be irrotational, because

$$\text{curl}\,\vec{A} = \text{curl grad}\,\phi = \vec{\nabla}\times\vec{\nabla}\phi = \begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z\end{vmatrix}$$

$$= \hat{\mathbf{i}}\left[\frac{\partial^2\phi}{\partial y\partial z}-\frac{\partial^2\phi}{\partial z\partial y}\right]+\hat{\mathbf{j}}\left[\frac{\partial^2\phi}{\partial z\partial x}-\frac{\partial^2\phi}{\partial x\partial z}\right]+\hat{\mathbf{k}}\left[\frac{\partial^2\phi}{\partial x\partial y}-\frac{\partial^2\phi}{\partial y\partial x}\right] = 0 \qquad ...(19)$$

Such an irrotational vector field, which can be represented by the gradient of a scalar function is called conservative force field and the curl of conservative force field is always zero.

Examples of conservative force field are (1) gravitational field and (2) electrostatic field, which can be represented as the negative gradient of the potential V *i.e.*, for a conservative field $\vec{E}$,

$$\vec{E} = -\vec{\nabla}V \text{ and curl } \vec{E} = -\vec{\nabla}\times\vec{\nabla}V = 0 \qquad ...(20)$$

as proved above.

1·18. Stoke's Theorem

The theorem states that *the line integral of a vector $\vec{A}$ around any closed curve C is equal to the surface integral of curl $\vec{A}$ taken over any surface S of which the curve is a boundary edge, i.e.,*

$$\oint_C \vec{A}\cdot d\vec{l} = \iint_S \text{curl}\,\vec{A}\cdot d\vec{S} = \iint_S (\vec{\nabla}\times\vec{A})\cdot d\vec{S} \qquad ...(1)$$

Proof. If a vector $\vec{A}$ be a function of position, then its line integral along a closed curve C is given by

$$L = \oint_C \vec{A}\cdot d\vec{l}$$

where $d\vec{l}$ is the small element of the path. Now divide the enclosed area into two parts by a line pq so as to form two closed curved C_1 and C_2 **[fig. 1.49]**. If L_1 and L_2 are the line integrals for the two paths C_1 and C_2, then their sum $(L_1 + L_2)$ will be equal to L for the curve C. The reason lies in the fact that the part pq is traversed from p to q for the curve C_1 and from q to p for the path C_2 so that its contribution is cancelled. Thus, if we divide the area enclosed by the curve C into a number of smaller areas δS_1, δS_2 ..... bounded by curves C_1, C_2, C_3, ..., C_n, ..., then

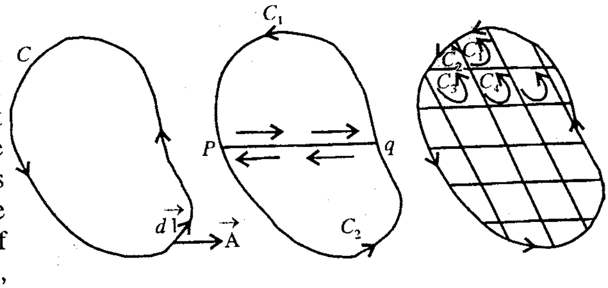


Fig. 1.49.

$$\oint_C \vec{A}\cdot d\vec{l} = \Sigma\oint_{Cn} \vec{A}\cdot d\vec{l} \qquad ...(2)$$

The curve C may be divided in such a way that small areas δS_1, δS_2 may lie in different planes. But from the definition of curl $\vec{A}$ at a point inside δS_n, we have

$$\text{curl}\,\vec{A}\cdot\delta\vec{S}_n = \oint_{Cn}\vec{A}\cdot d\vec{l}$$

Thus, $$\oint_C \vec{A}\cdot d\vec{l} = \Sigma\oint_{Cn}\vec{A}\cdot d\vec{l} = \Sigma\,\text{curl}\,\vec{A}\cdot d\vec{S}_n \;\; i.e., \oint_C \vec{A}\cdot d\vec{l} = \iint_S \text{curl}\,\vec{A}\cdot d\vec{S} \qquad ...(3)$$

This is *Stoke's theorem*. It should be noted that the boundary curve or the related surface need not to be plane one.

If the surface to which Stoke's theorem is applied is a closed surface, the length of the boundary edge is zero and then $\iint_S \text{curl}\,\vec{A}\cdot d\vec{S} = 0$ ...(4)

In case, curl $\vec{A}$ or $\vec{\nabla}\times\vec{A} = 0$ everywhere, then according to Stoke's theorem the line integral of $\vec{A}$ around any closed curve vanishes.

Ex. 1. *If* $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$, *then prove that* (i) curl $\vec{r} = 0$, (ii) curl $(\vec{r}\, r^n) = 0$.

Sol. (i) $$\text{curl}\,\vec{r} = \vec{\nabla}\times(\hat{i}x+\hat{j}y+\hat{k}z) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial z}{\partial y}-\frac{\partial y}{\partial z}\right)+\hat{j}\left(\frac{\partial x}{\partial z}-\frac{\partial z}{\partial x}\right)+\hat{k}\left(\frac{\partial y}{\partial x}-\frac{\partial x}{\partial y}\right) = \mathbf{0}$$

[$\because$ x, y, z are independent variable]

(ii) $$\text{curl}\,(\vec{r}\, r^n) = \vec{\nabla}\times(\hat{i}xr^n+\hat{j}yr^n+\hat{k}zr^n) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xr^n & yr^n & zr^n \end{vmatrix}$$

$$= \hat{i}\left[\frac{\partial(zr^n)}{\partial y}-\frac{\partial(yr^n)}{\partial z}\right]+\hat{j}\left[\frac{\partial(xr^n)}{\partial z}-\frac{\partial(zr^n)}{\partial x}\right]+\hat{k}\left[\frac{\partial(yr^n)}{\partial x}-\frac{\partial(xr^n)}{\partial y}\right]$$

$$= \hat{i}\left[znr^{n-1}\frac{\partial r}{\partial y}-ynr^{n-1}\frac{\partial r}{\partial z}\right]+\ldots..$$

$$= \hat{i}\,nr^{n-1}\left[\frac{zy}{r}-\frac{yz}{r}\right]+\ldots. \qquad \left[\because \frac{\partial r}{\partial y}=\frac{\partial\sqrt{x^2+y^2+z^2}}{\partial y}=\frac{y}{r}\right]$$

$$= nr^{n-2}[\hat{i}\,(zy-yz)+\hat{j}\,(xz-zx)+\hat{k}\,(xy-yx)] = \mathbf{0}.$$

Ex. 2. *Prove that* curl $(\vec{A}+\vec{B}) = \text{curl}\,\vec{A} + \text{curl}\,\vec{B}$. **(Sagar 1995; Ujjain 90; Gwalior 91)**

Sol. $$\text{curl}\,(\vec{A}+\vec{B}) = \vec{\nabla}\times(\vec{A}+\vec{B}) = \left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial y}\right)\times[(A_x+B_x)\hat{i}+(A_y+B_y)\hat{j}+(A_z+B_z)\hat{k}]$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (A_x+B_x) & (A_y+B_y) & (A_z+B_z) \end{vmatrix}$$

$$= \hat{i}\left[\frac{\partial}{\partial y}(A_z+B_z)-\frac{\partial}{\partial z}(A_y+B_y)\right]+\hat{j}\left[\frac{\partial}{\partial z}(A_x+B_x)-\frac{\partial}{\partial x}(A_z+B_z)\right]$$
$$+\hat{k}\left[\frac{\partial}{\partial x}(A_y+B_y)-\frac{\partial}{\partial y}(A_x+B_x)\right]$$

$$= \hat{i}\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)+\hat{j}\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)+\hat{k}\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)$$
$$+\hat{i}\left(\frac{\partial B_z}{\partial y}-\frac{\partial B_y}{\partial z}\right)+\hat{j}\left(\frac{\partial B_x}{\partial z}-\frac{\partial B_z}{\partial x}\right)+\hat{k}\left(\frac{\partial B_y}{\partial x}-\frac{\partial B_x}{\partial y}\right)$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ B_x & B_y & B_z \end{vmatrix} = \textbf{curl}\ \vec{\mathbf{A}} + \textbf{curl}\ \vec{\mathbf{B}}.$$

Ex. 3. *Prove that*

(i) curl grad $\phi = 0$. **(Agra 2006; Rewa 1998; Sagar 90; Ujjain 89; Bhopal 95; Bilaspur 95; Raipur 95)**

(ii) div curl $\vec{A} = 0$ **(Agra 2006; Bhopal 2000, 98; Indore 99, 97; Sagar 96; Ujjain 89)**

(iii) curl $(\phi\vec{A}) = \phi$ curl $\vec{A}$ + gard $\phi\times\vec{A}$ **(Indore 1997; Rewa 99, 97; Raipur 92; Jabalpur 90)**

Sol. (i) curl grad ϕ $= \vec{\nabla} \times \vec{\nabla}\phi = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial y}\right) \times \left(\hat{\mathbf{i}}\frac{\partial \phi}{\partial x} + \hat{\mathbf{j}}\frac{\partial \phi}{\partial y} + \hat{\mathbf{k}}\frac{\partial \phi}{\partial z}\right)$

$$= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \end{vmatrix} = \hat{\mathbf{i}}\left(\frac{\partial^2\phi}{\partial y \partial z} - \frac{\partial^2\phi}{\partial z \partial y}\right) + \hat{\mathbf{j}}\left(\frac{\partial^2\phi}{\partial z \partial x} - \frac{\partial^2\phi}{\partial x \partial z}\right) + \hat{\mathbf{k}}\left(\frac{\partial^2\phi}{\partial x \partial y} - \frac{\partial^2\phi}{\partial y \partial x}\right)$$

As $\dfrac{\partial^2\phi}{\partial y \partial z} = \dfrac{\partial^2\phi}{\partial z \partial y}$ etc, hence **curl grad ϕ = 0.**

(ii) div curl $\vec{A} = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = \vec{\nabla} \cdot \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix}$

$$= \vec{\nabla} \cdot \left[\hat{\mathbf{i}}\left\{\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right\} + \hat{\mathbf{j}}\left\{\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right\} + \hat{\mathbf{k}}\left\{\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right\}\right]$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_z}{\partial y}\right)$$

$$= \frac{\partial^2 A_z}{\partial x \partial y} - \frac{\partial^2 A_y}{\partial x \partial z} + \frac{\partial^2 A_x}{\partial y \partial z} - \frac{\partial^2 A_z}{\partial y \partial x} + \frac{\partial^2 A_y}{\partial z \partial x} - \frac{\partial^2 A_z}{\partial z \partial y}$$

If $\vec{A}$ is a perfect differential, then $\dfrac{\partial^2 A_x}{\partial x \partial y} = \dfrac{\partial^2 A_x}{\partial y \partial x}$ etc and therefore **div curl $\vec{\mathbf{A}}$ = 0.**

(iii) curl $(\phi \vec{A}) = \vec{\nabla} \times (\phi \vec{A}) = \vec{\nabla} \times (\phi A_x \hat{\mathbf{i}} + \phi A_y \hat{\mathbf{j}} + \phi A_z \hat{\mathbf{k}})$

$$= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \phi A_x & \phi A_y & \phi A_z \end{vmatrix}$$

$$= \hat{\mathbf{i}}\left[\frac{\partial}{\partial y}(\phi A_z) - \frac{\partial}{\partial z}(\phi A_y)\right] + \hat{\mathbf{j}}\left[\frac{\partial}{\partial z}(\phi A_x) - \frac{\partial}{\partial x}(\phi A_z)\right] + \hat{\mathbf{k}}\left[\frac{\partial}{\partial x}(\phi A_y) - \frac{\partial}{\partial y}(\phi A_x)\right]$$

$$= \hat{\mathbf{i}}\left[\phi\frac{\partial A_z}{\partial y} + A_z\frac{\partial \phi}{\partial y} - \phi\frac{\partial A_y}{\partial z} - A_y\frac{\partial \phi}{\partial z}\right] + \hat{\mathbf{j}}\left[\phi\frac{\partial A_x}{\partial z} + A_x\frac{\partial \phi}{\partial z} - \phi\frac{\partial A_z}{\partial x} - A_z\frac{\partial \phi}{\partial x}\right]$$

$$+ \hat{\mathbf{k}}\left[\phi\frac{\partial A_y}{\partial x} + A_y\frac{\partial \phi}{\partial x} - \phi\frac{\partial A_x}{\partial y} - A_x\frac{\partial \phi}{\partial x}\right]$$

$$= \phi\left[\hat{\mathbf{i}}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{\mathbf{j}}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{\mathbf{k}}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\right]$$

$$+\left[\hat{\mathbf{i}}\left(A_z\frac{\partial \phi}{\partial y} - A_y\frac{\partial \phi}{\partial z}\right) + \hat{\mathbf{j}}\left(A_x\frac{\partial \phi}{\partial z} - A_z\frac{\partial \phi}{\partial x}\right) + \hat{\mathbf{k}}\left(A_y\frac{\partial \phi}{\partial x} - A_x\frac{\partial \phi}{\partial y}\right)\right]$$

$$= \phi\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} + \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \\ A_x & A_y & A_z \end{vmatrix} = \phi\ \textbf{curl}\ \vec{\mathbf{A}} + (\textbf{grad}\ \phi) \times \vec{\mathbf{A}}.$$

Ex. 4. *Prove that* div $(\vec{A}\times\vec{B}) = \vec{B}\cdot \text{curl}\,\vec{A} - \vec{A}\cdot \text{curl}\,\vec{B}$. **(Jabalpur 2003; Ujjain 1998, 96; Sagar 97; Rewa 2004, 92; Raipur 97; Bhopal 92; Indore 90)**

Sol. Let $\vec{A}\times\vec{B} = \vec{C}$, then in components

$$\vec{C} = C_x\hat{i} + C_y\hat{j} + C_z\hat{k} = \vec{A}\times\vec{B}$$

$$= (A_yB_z - A_zB_y)\hat{i} + (A_zB_x - A_xB_z)\hat{j} + (A_xB_y - A_yB_x)\hat{k}$$

Now, div $(\vec{A}\times\vec{B})$ = div $\vec{C} = \vec{\nabla}\cdot\vec{C}$

$$= \frac{\partial C_x}{\partial x} + \frac{\partial C_y}{\partial y} + \frac{\partial C_z}{\partial z}$$

$$= \frac{\partial}{\partial x}(A_yB_z - A_zB_y) + \frac{\partial}{\partial y}(A_zB_x - A_xB_z) + \frac{\partial}{\partial z}(A_xB_y - A_yB_x)$$

$$= B_x\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + B_y\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + B_z\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

$$-A_x\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right) + A_y\left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right) + A_z\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)$$

$$= (B_x\,\hat{i} + B_y\,\hat{j} + B_z\,\hat{k})\cdot\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ A_x & A_y & A_z\end{vmatrix}$$

$$-(A_x\,\hat{i} + A_y\,\hat{j} + A_z\,\hat{k})\cdot\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ B_x & B_y & B_z\end{vmatrix}$$

$$= \vec{\mathbf{B}}\cdot \textbf{curl}\,\vec{\mathbf{A}} - \vec{\mathbf{A}}\cdot\textbf{curl}\,\vec{\mathbf{B}}.$$

Ex. 5. *If* $\vec{a}$ *is a constant vector and* $\vec{r}$ *is a position vector, then prove that* curl $(\vec{a}\times\vec{r}) = 2\vec{a}$. **(Indore 2004, 2000; Gwalior 2000; Sagar 1999; Raipur 92)**

Sol. Here, $\vec{a}\times\vec{r} = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ a_x & a_y & a_z\\ x & y & z\end{vmatrix} = (a_yz - a_zy)\,\hat{i} + (a_zx - a_xz)\,\hat{j} + (a_xy - a_yx)\,\hat{k}$

Hence, curl $(\vec{a}\times\vec{r}) = \vec{\nabla}\times(\vec{a}\times\vec{r}) = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ a_yz - a_zy & a_zx - a_xz & a_xy - a_yx\end{vmatrix}$

$$= \hat{i}\left[\frac{\partial}{\partial y}(a_xy - a_yx) - \frac{\partial}{\partial z}(a_zx - a_xz) + \ldots\right] = \hat{i}(a_x + a_x) + \hat{j}(a_y + a_y) + \hat{k}(a_z + a_z)$$

$$= 2(a_x\hat{i} + a_y\hat{j} + a_z\hat{k}) = \mathbf{2}\,\vec{\mathbf{a}}.$$

Ex. 6. *If* $\vec{A}$ *and* $\vec{B}$ *are irrotational vectors, then prove that* $\vec{A}\times\vec{B}$ *is a solenoidal vector.* **(Indore 2000)**

Sol. Given : $\vec{\nabla}\times\vec{A} = 0$ and $\vec{\nabla}\times\vec{B} = 0$

We know that $\vec{\nabla}\cdot(\vec{A}\times\vec{B}) = \vec{B}\cdot\text{curl}\,\vec{A} - \vec{A}\cdot\text{curl}\,\vec{B}$.

$$\therefore \quad \vec{\nabla}\cdot(\vec{A}\times\vec{B}) = \vec{B}\cdot(\vec{\nabla}\times\vec{A}) - \vec{A}\cdot(\vec{\nabla}\times\vec{B}) = 0$$

Thus, $\vec{A}\times\vec{B}$ is solenoidal vector.

Ex. 7. *If* $\vec{F} = xy^2\hat{\mathbf{i}} + 2x^2yz\hat{\mathbf{j}} - 3yz^2\hat{\mathbf{k}}$, *find* div $\vec{F}$, curl $\vec{F}$ *at the point* (1, –1, 1).

Sol. (a) $$\text{div } \vec{F} = \vec{\nabla}\cdot\vec{F} = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}\right)\cdot(xy^2\hat{\mathbf{i}} + 2x^2yz\hat{\mathbf{j}} - 3yz^2\hat{\mathbf{k}})$$

$$= \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(2x^2yz) + \frac{\partial}{\partial z}(-3yz^2) = y^2 + 2x^2z - 6yz.$$

At the point (1, –1, 1), div $\vec{F}$ = 1 + 2 + 6 = **9.**

(b) $$\text{curl } \vec{F} = \vec{\nabla}\times\vec{F} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy^2 & 2x^2yz & -3yz^2 \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y}(-3yz^2) - \frac{\partial}{\partial z}(2x^2yz)\right]\hat{\mathbf{i}} + \left[\frac{\partial}{\partial z}(xy^2) - \frac{\partial}{\partial x}(-3yz^2)\right]\hat{\mathbf{j}}$$
$$+ \left[\frac{\partial}{\partial x}(2x^2yz) - \frac{\partial}{\partial y}(xy^2)\right]\hat{\mathbf{k}}$$

$$= -(3z^2 + 2x^2y)\ \hat{\mathbf{i}} + (4xyz - 2xy)\ \hat{\mathbf{k}}$$

At the point (1, –1, 1), curl $\vec{F} = -\hat{\mathbf{i}} - 2\hat{\mathbf{k}}$.

Ex. 8. *Prove that* : $\vec{F} = a(x\hat{\mathbf{i}} + y\hat{\mathbf{j}})$ *is a conservative force field.* **(Bhopal 1997)**

Sol. Here, for a conservative force field, curl $\vec{F} = 0$

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ ax & ay & 0 \end{vmatrix}$$

$$= \hat{\mathbf{i}}\left[\frac{\partial(0)}{\partial y} - \frac{\partial(ay)}{\partial z}\right] - \hat{\mathbf{j}}\left[\frac{\partial(0)}{\partial x} - \frac{\partial(ax)}{\partial z}\right] + \hat{\mathbf{k}}\left[\frac{\partial(ay)}{\partial x} - \frac{\partial(ax)}{\partial y}\right]$$

$$= \hat{\mathbf{i}}|0-0| - \hat{\mathbf{j}}|0-0| + \hat{\mathbf{k}}|0-0| = 0$$

Therefore, $\vec{F} = a(x\hat{\mathbf{i}} + y\hat{\mathbf{j}})$ **is a conservative force field.**

Ex. 9. *If* $\vec{A} = -y\hat{\mathbf{i}} + x\hat{\mathbf{j}}$, *then calculate* (a) *curl* $\vec{A}$ (b) *the line integral* $\int\vec{A}\cdot d\vec{l}$ *for the closed curve* $x^2 + y^2 = r^2$, $z = 0$. *Hence verify Stoke's theorem.*

Sol. (a) Here, $A_x = -y$, $A_y = x$, $A_z = 0$; $\therefore \dfrac{\partial A_x}{\partial y} = -1$, $\dfrac{\partial A_y}{\partial x} = 1$ and other derivatives are zero.

$$\text{Hence} \quad \text{curl } \vec{A} = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\mathbf{k}} = (1+1)\hat{\mathbf{k}} = \mathbf{2\hat{k}}.$$

(b) $$\int\vec{A}\cdot d\vec{l} = \int(-y\hat{\mathbf{i}} + x\hat{\mathbf{j}})\cdot(dx\hat{\mathbf{i}} + dy\hat{\mathbf{j}}) = \int -y\,dx + \int x\,dy.$$

For the circle $x^2 + y^2 = r^2$, $x = r\cos\theta$, $y = r\sin\theta$;

$$\therefore \quad dx = -r\sin\theta\, d\theta,\ dy = r\cos\theta\, d\theta.$$

Hence, $\int \vec{A}\cdot d\vec{l} = \int_0^{2\pi} r^2(\cos^2\theta + \sin^2\theta)d\theta = \int_0^{2\pi} r^2 d\theta = 2\pi r^2$

Now, according to Stoke's theorem, $\int \vec{A}\cdot d\vec{l} = \iint_s \text{curl}\,\vec{A}\cdot d\vec{s}$

Here, $\int_S \text{curl}\,\vec{A}\cdot d\vec{S} = \iint 2\hat{k}\cdot(dS)\hat{k} = \iint_s 2dS = 2S = 2\pi r^2.$

Therefore, $\int \vec{A}\cdot d\vec{l} = \iint_S \text{curl}\,\vec{A}\cdot d\vec{S}$

Thus, **Stoke's theorem is verified.**

1·18. Use of Multiple Del Operators

(1) div grad $\phi = \vec{\nabla}\cdot\vec{\nabla}\phi = \nabla^2\phi$ ...(1)

(Rewa 2000)

Proof : $\vec{\nabla}\cdot\vec{\nabla}\phi = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\cdot\left(\hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}\right) = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = \nabla^2\phi$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called **Laplacian operator.**

(2) curl curl $\vec{A}$ = grad div $\vec{A} - \nabla^2\vec{A}$ ...(2)

(Bhopal 1999; Indore 99; Raipur 98, 90; Gwalior 93; Ujjain 97)

Proof : Since $\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}\cdot\vec{C}) - (\vec{A}\cdot\vec{B})\vec{C}$

$\therefore$ curl curl $\vec{A} = \vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - (\vec{\nabla}\cdot\vec{\nabla})\vec{A}$ = grad div $\vec{A} - \nabla^2\vec{A}$

(3) curl grad $\phi = 0$ ...(3)

which has been proved already.

In the application of del operator, in addition to the above identities, we give below some identities which have been proved already in solved examples.

(4) grad $(\phi\psi) = \phi$ grad $\psi + \psi$ grad ϕ ...(4)

(5) div $(\phi\vec{A})$ = grad $\phi\cdot\vec{A} + \phi$ div $\vec{A}$ ...(5)

(6) div $(\vec{A}\times\vec{B}) = \vec{B}\cdot$ curl $\vec{A} - \vec{A}\,.$ curl $\vec{B}$ ...(6)

(7) div (curl $\vec{A}$) = 0. ...(7)

EXERCISE 1.5

1. If $\vec{A}$ is a irrotational vector, prove that $\vec{A}\times\vec{r}$ is a solenoidal vector.
2. If $\vec{F} = (x+y+1)\hat{i} + \hat{j} - (x+y)\hat{k}$, prove that $\vec{F}\cdot\text{curl}\,\vec{F} = 0$
3. If $\vec{F} = x^2y\hat{i} + xz\hat{j} + 2yz\hat{k}$, find div curl $\vec{F}$.
   [Ans. 0]
4. If $\vec{B}$ = curl $\vec{A}$, prove that div $\vec{B} = 0$. **(Bhopal 1997; Sagar 98)**
5. (i) Prove that curl $(\vec{r}/r^3) = 0$.
   (ii) Prove that for any value of n, $\vec{F} = Kr^n\hat{r}$ is irrotational field. **(Bhopal 2000)**
6. Find div and curl for $\vec{F} = (x^2 - z^2)\hat{i} + y^2\hat{j} + (2x+z)z\hat{k}$.
   [Ans. $4x + 2y + 2z$; $-4z\hat{j}$]

7. Is the following field conservative : $\vec{F} = 5x\hat{i} + 16y\hat{j} + 7z\hat{k}$? **(Indore 1994)**

[**Ans.** Yes, because $\vec{\nabla} \times \vec{F} = 0$.]

8. Prove that $\vec{F} = (x + y)\hat{i} + (x + z)\hat{j} + (y - z)\hat{k}$ is a conservative force field. **(Raipur 1993; Sagar 97)**

9. The velocity of body $\vec{v} = 2x\hat{i} + 2y\hat{j}$, prove that the motion is irrotational. **(Ujjain 1994; Raipur 91; Bilaspur 89)**

10. Prove that $\vec{\nabla} \times \vec{v} = 2\vec{\omega}$, where $\vec{v}$ and $\vec{\omega}$ are the linear and angular velocities respectively in circular motion. **(Ujjain 1999)**

11. For the rotational motion of a rigid body, prove that, the curl of linear velocity $\vec{v}$ of a particle of the body is equal to two times the angular velocity $\vec{\omega}$. **(Ujjain 1993; Rewa 92; Gorakhpur 80)**

12. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, show that

 (i) curl $\vec{r} = 0$. **(Agra 1982; Bundelkhand 94; Purvanchal 93; Purvanchal 1995)**

 (ii) curl $(\vec{r}/r^3) = 0$. **(Agra 1976; Avadh 82; Purvanchal 95)**

13. Verify Stoke's theorem when $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ and surface S is the part of the sphere $x^2 + y^2 + z^2 = 1$ above XY-plane.

14. Verify Stoke's theorem for the function $\vec{F} = x^2\hat{i} + xy\hat{j}$ integrated round the square in the z = 0 plane whose sides are along the lines $x = 0$, $x = a$, $y = a$.

QUESTIONS

Long Answer Questions

1. What is a unit vector ? Show that a unit vector in three dimensions can be expressed as

$$\hat{A} = \hat{i}\cos(\vec{A}, x) + \hat{j}\cos(\vec{A}, y) + \hat{k}\cos(\vec{A}, z).$$ **(Agra 1992)**

2. If vector $\vec{A}$ makes angles α, β and γ respectively with the X, Y- and Z-axes of a cartesian coordinate system, prove that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$. **(Agra 1980)**

3. (a) Define scalar and vector products of two vectors. Give physical examples.

 (b) Prove that : $$\left|\vec{A} \times \vec{B}\right| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$ **(Rewa 1978)**

4. If $\hat{i}$, $\hat{j}$, $\hat{k}$ are unit vectors along x, y and z axes respectively, show that

$$\hat{i}\cdot\hat{i} = 1;\ \hat{i}\cdot\hat{j} = 0;\ \hat{i} \times \hat{i} = 0 \text{ and } \hat{i} \times \hat{j} = \hat{k}$$ **(Meerut 1970)**

5. Show that the component of $\vec{A}$ along $\vec{B}$ may be expressed as $(\vec{A}\cdot\vec{B})\vec{B}/B^2$.

6. What is a vector area ? Show that the area of a parallelogram of sides $\vec{A}$ and $\vec{B}$ will be $\vec{A} \times \vec{B}$.

7. If $\vec{A} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $\vec{B} = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$, find the condition for $\vec{A}$ and $\vec{B}$ to be parallel.

[**Ans.** $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$]

8. (a) Suppose that $\vec{A}, \vec{B}$ and $\vec{C}$ are any three non-coplanar vectors. They are not necessarily at right angles. Show that $\vec{A}\cdot(\vec{B}\times\vec{C}) = \vec{B}\cdot(\vec{C}\times\vec{A})$ **(Meerut 1971)**

   (b) Show that $\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}\,(\vec{A}\cdot\vec{C}) - \vec{C}\,(\vec{A}\cdot\vec{B})$ **(Gwalior 2004; Agra 1985, 83)**

9. If $\vec{A}$, $\vec{B}$ and $\vec{C}$ are three non-coplanar vectors, show that $\vec{A}\cdot(\vec{B}\times\vec{C}) = \vec{B}\cdot(\vec{C}\times\vec{A}) = \vec{C}\cdot(\vec{A}\times\vec{B})$ and it is equal to the volume of that parallelopiped whose edges are along the vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$. **(Gwalior 2004)**

10. Explain the meaning of scalar field and vector field. Give examples of each field. How are they graphically represented ? **(Ujjain 1994; Indore 93; Jabalpur 93)**

11. What is meant by gradient of a scalar field ? Derive its expression in term of the operator $\vec{\nabla}$. **(Indore 2004)**

12. What is meant by gradient of a scalar function ? Prove that $\vec{\nabla}\phi = (\partial\phi/\partial n)\hat{n}$ and $\vec{\nabla}\phi$ is always normal to the surface ϕ = constant. Give two examples of gradient of a scalar field. **(Rewa 2001; Gwalior 2003; Bhopal 2004; Bilaspur 98)**

13. What do you understand by the vector integration ? Explain the meanings of (i) line integral. (ii) surface integral, and (iii) volume integral. **(Sagar 1997; Jabalpur 90)**

14. Explain the meaning of divergence of a vector field and give its physical significance. **(Raipur 2003; Ujjain 2003; Sagar 2003; Rewa 1999; Indore 99)**

15. Define divergence of a vector field. Obtain its value in cartesian coordinates and prove that. div $\vec{A} = \vec{\nabla}\cdot\vec{A}$ where $\vec{\nabla}$ is a vector operator. **(Ujjain 2004; Bhopal 2001; Raipur 1996)**

16. State and prove Gauss's divergence theorem. **(Raipur 2003; Vikarm 2004; Indore 2004; Agra 2004; Jabalpur 2003; Bhopal 2003;Ujjain 1997)**

17. State and prove that two forms of Green's theorem. **(Agra 2004; Indore 2003; Rewa 2003; Gwalior 2003; Ujjain 2003)**

18. Explain the meaning of curl (or rotation) of a vector field and state its physical significance. Give an example of non-curl field. **(Ujjain 2003; Jabalpur 1998; Indore 99)**

19. Define curl of a vector field and obtain its value in cartesian coordinates. Prove that curl $\vec{A} = \vec{\nabla}\times\vec{A}$ where $\vec{\nabla}$ is a vector operator. **(Bhopal 2000; Bilaspur 2003; Indore 2003; Gwalior 2001)**

20. Define and explain the physical significance of (a) Divergence, (b) Curl. **(Agra 2006)**

21. State and prove Stoke's theorem related to curl of a vector field. **(Bhopal 2004; Jabalpur 2004; Ujjain 2003; Rewa 2000; Bilaspur 200)**

22. If $\vec{A} = \theta\,\vec{\nabla}\phi$ where θ and ϕ are the scalar functions of position. show that

$$\iiint_V (\theta\,\nabla^2\phi + \vec{\nabla}\theta\cdot\vec{\nabla}\phi)\,dV = \iint_S (\theta\,\vec{\nabla}^2\phi)\cdot d\vec{S}.$$ **(Avadh 1996)**

Short Answer Questions

1. Is a physical quantity, having magnitude and direction, necessarily a vector ? Give examples. [**Ans.** No; electric current; finite angle]
2. Define : (i) Unit vector, (ii) Zero vector, (iii) Co-planer vectors, (iv) Axial vector, (v) Polar vector.
3. What is a position vector ?
4. Can the resultant be zero in case of three coplanar vectors ? [**Ans.** Yes]

5. Can the resultant be zero in case of three non-coplanar vectors ?
   [**Ans.** No]
6. Show that $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$.
7. Let $\vec{A} = 4\hat{i} + 3\hat{j}$. Write vector $\vec{B}$ such that $\vec{A} \neq \vec{B}$, but $A = B$.
8. Find the condition to show that vector $\vec{A} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $\vec{B} = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$ are parallel to each other.
   [**Ans.** $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$]
9. Prove that $\vec{A}\cdot(\vec{A}\times\vec{B}) = 0$
10. Explain scalar and vector fields with examples. **(Bhopal 2001)**
11. What do you understand by gradient of a scalar function ?
12. What is the meaning of potential and temperature gradients ?
13. What is represented by the gradient of a electrostatic potential ? Explain.
14. Give physical interpretation of : (a) Gradient of a scalar field; (b) Divergence and curl of a vector field. **(Jabalpur 1998)**
15. If $\vec{F} = 3x\,\hat{\mathbf{i}} + 4y\,\hat{\mathbf{j}} + 2z\,\hat{\mathbf{k}}$, find div $\vec{F}$.
    [**Ans.** 9]
16. Find the value of div $\vec{E}$, if $\vec{E} = -$ grad ϕ. **(Bilaspur 1999)**
17. Divergence of vector is scalar and the gradient of a scalar is vector. Explain them with examples. **(Ujjain 1997)**
18. What is the meaning of $\vec{\nabla}\cdot\vec{A} = 0$, where $\vec{A}$ is a vector. **(Jabalpur 1997)**
    [**Ans.** The meaning of $\vec{\nabla}\cdot\vec{A} = 0$, is the flux lines of vector $\vec{A}$ are closed]
19. Show that $\iint_s \vec{r}\cdot d\vec{s} = 3V$.
20. Define curl of a vector. Give its physical significance. **(Bhopal 1998)**
21. What is the divergence of a vector - scalar or vector ?
22. What is a solenoidal vector ? Give example.
23. Prove that uniform electric field is irrotational. **(Gwalior 1997)**
24. Prove that curl of a conservative field is zero.
25. Give an example of irrotational field. **(Rewa 2000)**
26. Prove that curl grad $\phi = 0$ or $\vec{\nabla}\times\vec{\nabla}\phi = 0$. **(Raipur 1999)**
27. Exaplain the meaning of (i) Solenoidal field and (ii) Lamellar field. **(Bilaspur 1999; Ujjain 95)**
28. If curl $\vec{A} = 0$ and div $\vec{A} = 0$, state the nature of the vector field. **(Bilaspur 1999)**
    [**Ans.** For curl $\vec{A} = 0$, the field is irrotational and $\vec{A}$ = grad ϕ. For div $\vec{A} = 0$, the field is solenoidal.
    Hence, div $\vec{A} = \vec{\nabla}\cdot\vec{\nabla}\phi = \nabla^2\phi = 0$, which is Laplace equation.]
29. Show that the vector $\vec{A} = \hat{i}yz + \hat{j}xz + \hat{k}xy$ is solenoidal non-rotational. **(Jabalpur 2004)**
30. curl $(\vec{A}\times\vec{B}) = \vec{A}$ div $\vec{B} - \vec{B}$ div $\vec{A}$. **(Jabalpur 2004)**
    [**Hint :** $\vec{\nabla}\times(\vec{A}\times\vec{B}) = \vec{A}\times(\vec{\nabla}\cdot\vec{B}) - \vec{B}\times(\vec{\nabla}\cdot\vec{A})$]

Objective Type Questions

1. If vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} - 3\hat{k}$ and $3\hat{i} + a\hat{j} + 5\hat{k}$ are coplanar, the value of a will be :
(a) 3 (b) 2 (c) -4 (d) None of these. **(Agra 2004)**
2. The magnitude of the resultant of two forces of equal magnitude (F and F) is equal to the magnitude of one force. The angle between them is :
(a) 0° (b) 45° (c) 60° (d) 120°.
3. If $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, then angle between $\vec{A}$ and $\vec{B}$ vectors is :
(a) 0° (b) 90° (c) 60° (d) 30°.
4. If $\vec{A} = 5\hat{i} - 7\hat{j} + 3\hat{k}$ and $\vec{B} = -4\hat{i} + 7\hat{j} - c\hat{k}$ are perpendicular to each other, then the value of c is :
(a) -23 (b) 3 (c) 4 (d) None of these.
5. If $\vec{A}$ and $\vec{B}$ are non-zero vectors and $\vec{A} \times \vec{B} = 0$, then :
(a) angle between $\vec{A}$ and $\vec{B}$ is zero (b) Both $\vec{A}$ and $\vec{B}$ are null-vectors
(c) $\vec{A}$ or $\vec{B}$ is null vector (d) $\vec{A}$ is normal to $\vec{B}$.
6. If ϕ is scalar field, then grad ϕ is :
(a) scalar (b) vector (c) zero (d) either vector or scalar.
7. The divergence of position vector $\vec{r}$ is :
(a) 1 (b) 0 (c) 3 (d) 2.
(Rewa 2003; Bundelkhand 2004)
8. If ϕ is gravitational potential at a point, then $\vec{F}$ will be the intensity of the gravitation field at that place (point) where : **(Gwalior 1991)**
(a) $\vec{F}$ = curl ϕ (b) $\vec{F} = \vec{\nabla}\phi$ (c) $\vec{F} = -\vec{\nabla}\phi$ (d) $\vec{F}$ = div ϕ.
9. The divergence of a vector is :
(a) positive or negative (b) positive or zero
(c) always positive (d) positive or negative or zero.
10. The divergence of $\vec{r}$ is :
(a) $x + y + z$ (b) r (c) 3 (d) $\hat{i} + \hat{j} + \hat{k}$
11. The line intergral of a vector between two points in a conservative vector field depends upon :
(a) the displacement between the points (b) positions of the points
(c) both (a) and (b) are corret (d) both (a) and (b) are incorret.
12. The value of $\vec{\nabla}(1/r)$ is :
(a) $-\dfrac{\vec{r}}{r^3}$ (b) $\vec{r}$ (c) 0 (d) r^2.
13. The value of grad $(\vec{r} \cdot \vec{r})$ is :
(a) $\vec{r}$ (b) $2\vec{r}$ (c) $\vec{r}/2$ (d) $2(x + y + z)$. **(Rewa 1999)**
14. If $\vec{A}$ = grad ϕ, then $\vec{A}$ is :
(a) rotational vector (b) irrotational vector
(c) solenoidal vector (d) non-solenoidal. **(Rewa 1996)**

15. For a vector field $\vec{A}$:

(a) curl $\vec{A} = \vec{\nabla}\vec{A}$ (b) curl $\vec{A} = \vec{\nabla}\cdot\vec{A}$ (c) curl $\vec{A} = \vec{\nabla}\times\vec{A}$ (d) curl $\vec{A} = 0$. **(Rewa 2004)**

16. Curl of vector function at a point is :

(a) a vector (b) a scalar

(c) an exponential function (d) depends on the position vector of the point.

17. If $\oint \vec{A}\cdot d\vec{l} = 0$ for a closed curve, then :

(a) $\vec{\nabla}\times\vec{A} = 0$ (b) $\vec{\nabla}\cdot\vec{A} = 0$ (c) $\vec{A} = 0$ (d) $\int \vec{A}\cdot d\vec{S} = 0$.

18. If $\vec{F} = (x^2 - z^2)\hat{i} + 2\hat{j} + 2xz\hat{k}$, then div $\vec{F}$ is

(a) $2x$ (b) 0 (c) $4x$ (d) $2x + 2z$.

19. Curl of gradient V i.e., $\vec{\nabla}\times(\vec{\nabla}V)$ is :

(a) -1 (b) 1 (c) 0 (d) ∞. **(Agra 2006)**

20. If $\vec{A}$ is vertical upward and $\vec{B}$ is along north, then the direction of $\vec{A}\times\vec{B}$ is :

(a) west (b) east (c) south (d) vertically downwards.

21. Surface area $\vec{S} = 5\hat{k}$ lies in :

(a) x-y plane (b) y-z plane (c) z-x plane (d) along z-axis.

(Purvanchal 2001)

22. The incorrect equation is :

(a) curl $\vec{r} = 0$ (b) curl grad $\phi = 0$

(c) div curl $\vec{E} = 0$ (d) div $\vec{r} = 0$.

23. The value of $\iint_s \vec{r}\cdot d\vec{s}$ is :

(a) V (b) $2V$ (c) $3V$ (d) 0. **(Agra 2005)**

24. The function $\vec{E} = (3x - 4y + az)\,\hat{i} + (cx + 5y - 2z)\,\hat{j} + (x - by + 7z)\,\hat{k}$ is irrotational, if :

(a) $a = -1,\ b = 2,\ c = -4$ (b) $a = 1,\ b = 2,\ c = -4$

(c) $a = 1,\ b = -1,\ c = -4$ (d) $a = 1,\ b = -2,\ c = -4$ **(Agra 2005)**

25. Divergence theorem relates surface intergration with volume intergration as :

(a) $\frac{d}{dV}\int_V \vec{\nabla}\cdot\vec{J}\,dV = \int_S \vec{J}\cdot d\vec{S}$ (b) $\int_V (\vec{\nabla}\cdot\vec{J})dV = \vec{J}.d\vec{S}$

(c) $\int_J \vec{\nabla}\cdot\vec{J}\,dV = \oiint_S \vec{J}\cdot d\vec{S}$ (d) $\int_V (\vec{\nabla}\cdot\vec{J})dV = \oint_S \vec{J}\cdot d\vec{S}$ **(Agra 2006)**

ANSWERS

1. (c),	**2.** (d)	**3.** (b),	**4.** (a),	**5.** (a),	**6.** (b),	**7.** (c),	**8.** (c),	**9.** (d),
10. (c),	**11.** (b),	**12.** (a),	**13.** (b),	**14.** (b),	**15.** (c),	**16.** (a),	**17.** (a),	**18.** (c),
19. (c),	**20.** (a),	**21.** (a),	**22.** (d),	**23.** (c),	**24.** (b),	**25.** (d)		

2

Differentiability of Functions of Two and Three Variables

2·1. Differentiation and Differential Coefficient

The object of differential calculus is to find the procedure in which how one quantity changes on changing another related quantity. Those quantities whose values remain unchanged during a mathematical operation are called **constants** and those quantities whose values change in a mathematical operation are called **variables.** In general, the constants are represented by the begining letters of English alphabets $a, b, c, ...$ and the variables by the ending letters u, v, w, x, y, z.

Let y be a quantity which depends on x. This quantity y is said to be the **function** of x and is written as

$$y = f(x) \qquad ...(1)$$

If we give any value to x, then we obtain a specific value of y. Now, if we give different values to x, we will get different values of y. Thus, on changing x, the value of y changes. Hence, we call x as **independent variable** and y as **dependent variable.** For example, we consider the equation of a straight line :

$$y = 2x + 3$$

Here if we have $x = 0, 1, 2, ...$ etc, then we obtain $y = 3, 5, 7, ...$ etc. Clearly x is a independent variable and y is the dependent variable.

Let $y = f(x)$ be a function of x. If we change x by small increase δx, the corresponding change in y, say δy, occurs. The quantity $\delta y/\delta x$ tells us the average rate of change of y with respect to x. As δx approaches to zero, then the **limit of** $\delta y/\delta x$ is called the **derivative** or **differential coefficient** of y with respect x and it is represented by dy/dx *i.e.*,

$$\frac{dy}{dx} = \lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x} \qquad ...(2)$$

This differential coefficient is represented by y' or y_1, generally.

2·2. Functions of Two or Three Variables

If the length of a rectangle is x and breadth is y, then area A is given by

$$A = xy \qquad ...(1)$$

Hence, the area of a rectangle is the function of two variables x and y.

Similarly, the volume of a rectangular parallelopiped with length x, breadth y and height z is represented as

$$V = xyz \qquad ...(2)$$

Thus the volume (V) of the rectangular parallelopiped is the function of three variables x, y and z.

In general, **any variable u is said to be the function of x and y variables, when for given values of x and y, u acquires a definite value.** We write a function u of two variables x and y as

$$u = f(x, y) \qquad ...(3)$$

Here, x and y are *independent variables* and u is the *dependent variable*. This u is also written as $u(x, y)$.

Similarly **any variable u will be a function of three variables x, y and z, when we obtain a definite value of u for given values of x, y and z.**

Thus,

$$u = f(x, y, z) \text{ or } u = u(x, y, z) \qquad ...(4)$$

is the function of three variables x, y and z.

2·3. Partial Derivatives and Partial Differential Coefficients

Let $u = f(x, y)$ is the function of two variables x and y. If we assume y as constant and change only the variable x, then the derivative of u, so obtained, is called the **partial derivative** or **partial differential coefficient** of u **with respect to** x and it is represented by $\partial u/\partial x$ or $\partial f/\partial x$. Thus

$$\frac{\partial u}{\partial x} \text{ or } \frac{\partial f}{\partial x} = \lim_{\delta x \to 0} \frac{f(x+\delta x, y) - f(x, y)}{\delta x} \qquad \text{...(1)}$$

Similarly, if we assume x as constant and change y, then the **derivative** of u **with respect** to y will be

$$\frac{\partial u}{\partial y} \text{ or } \frac{\partial f}{\partial y} = \lim_{\delta y \to 0} \frac{f(x, y+\delta y) - f(x, y)}{\delta y} \qquad \text{...(2)}$$

Sometimes we represent $\partial f/\partial x$ and $\partial f/\partial y$ as f_x and f_y.

Second Order Partial Derivatives :

Partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ can be differentiated partially with respect to x and y. Thus, the partial derivatives of $\partial f/\partial x$ with respect to x and y will be represented respectively as

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} \text{ and } \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} \qquad \text{..(3)}$$

They are also written as f_{xx} and f_{xy}.

Similarly, the partial derivatives of $\partial f / \partial y$ with respect to x and y can be written respectively as

$$\frac{\partial^2 f}{\partial x \partial y} \text{ and } \frac{\partial^2 f}{\partial y^2} \qquad \text{...(4)}$$

which can also be written as f_{yx} and f_{yy}.

Clearly,
$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} \qquad \text{...(5)}$$

Higher Order of Partial Derivatives :

Higher order partial derivatives of $u = f(x, y)$ are written as $\frac{\partial^3 f}{\partial x^3}$, $\frac{\partial^3 f}{\partial x^2 \partial y}$, $\frac{\partial^3 f}{\partial x \partial y^2}$, $\frac{\partial^3 f}{\partial y^3}$, $\frac{\partial^3 f}{\partial y^2 \partial x}$, $\frac{\partial^3 f}{\partial y \partial x^2}$, $\frac{\partial^4 f}{\partial x^4}$,.... etc. For example, suppose

$$u = u(x, y) = x^3 y \qquad \text{...(6)}$$

Then, $\frac{\partial u}{\partial x} = 3x^2 y$, $\frac{\partial u}{\partial y} = x^3$, $\frac{\partial^2 u}{\partial x^2} = 6xy$, $\frac{\partial^2 u}{\partial x \partial y} = 3x^2$, $\frac{\partial^2 u}{\partial y^2} = 0$, $\frac{\partial^3 u}{\partial x^3} = 6y$, $\frac{\partial^3 u}{\partial x^2 \partial y} = 6x$, ... etc.

Partial Derivatives of Function of Three Variables :

If u is function of three variables x, y, z *i.e.*, $u = f(x, y, z)$, then the partial derivatives of u with respect to x, y, z are written respectively as $\partial u / \partial x$, $\partial u / \partial y$ and $\partial u / \partial z$. In order to find the partial derivative of u with respect to x, we take y and z as constants and determine the ordinary differential of u with respect to x (similar to the function of one variable) . Thus

$$\frac{\partial u}{\partial x} \text{ or } \frac{\partial f}{\partial x} = \lim_{\delta x \to 0} \frac{f(x+\delta x, y, z) - f(x, y, z)}{\delta x} \qquad \text{...(7)}$$

Similarly, we can write the partial derivatives of u *i.e.*, $\partial u/\partial y$ and $\partial u/\partial z$. Higher order partial derivatives of $u = f(x, y, z)$, can be obtained as discussed above for $u = f(x, y)$.

Important Note on Partial Derivatives : Generally, the meaning of the partial derivative $\partial u / \partial x$ is $(\partial u / \partial x)_{y = \text{constant}}$ for $u = u(x, y)$ and that of $\partial u / \partial y = (\partial u / \partial y)_{x = \text{constant}}$.

Similarly for $u = u(x, y, z)$, the meaning of $\partial u / \partial x$ is $(\partial u / \partial x)_{y,\, z = \text{constant}}$, which is also written as $(\partial u / \partial x)_{y, z}$.

In thermodynamics, the partial derivatives are written as $(\partial u / \partial x)_y$ which means that the value of $\partial u / \partial x$ is obtained by taking y as constant. For example, in case of an ideal gas, pressure (P), volume (V) and temperature (T) are related by the following gas equation :

$$PV = RT \quad \text{or} \quad V = \frac{RT}{P} \quad \text{or} \quad V = f(T, P) \qquad ...(8)$$

The derivative of volume V with respect to T at constant P is written as $(\partial V / \partial T)_P$ and here,

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{R}{P} \qquad ...(9)$$

Similarly, in thermodynamics, other derivatives $\left(\frac{\partial T}{\partial P}\right)_V$, $\left(\frac{\partial T}{\partial V}\right)_S$, $\left(\frac{\partial P}{\partial K}\right)_T$, $\left(\frac{\partial P}{\partial T}\right)_V$ etc are used.

The writing of the derivatives in the above way will be clear by the following example. Suppose

$$z = x^2 - y^2 \qquad ...(10)$$

In plane polar coordinates (r, θ),

$$x = r\cos\theta,\ y = r\sin\theta \text{ and } x^2 + y^2 = r^2 \qquad ...(11)$$

We can write z in other different ways and then obtain $\partial z / \partial r$:

$$z = r^2\cos^2\theta - r^2\sin^2\theta, \quad \left(\frac{\partial z}{\partial r}\right)_\theta = 2r(\cos^2\theta - \sin^2\theta) \qquad ...(12)$$

$$z = x^2 - (r^2 - x^2) = 2x^2 - r^2, \quad \left(\frac{\partial z}{\partial r}\right)_x = -2r \qquad ...(13)$$

$$z = r^2 - y^2 - y^2 = r^2 - 2y^2, \quad \left(\frac{\partial z}{\partial r}\right)_y = 2r \qquad ...(14)$$

The values of the three expressions of $(\partial z / \partial r)$ are different and in fact, they are the derivatives of three different functions. Hence, in order to distinguish them, the other variable (keeping as constant) is written as **subscript.**

2·4. Geometrical Interpretation of Partial Derivatives of Functions of Two Variables

In two dimensions or in $x - y$ coordinates, we consider the following equation :

$$y = f(x) \qquad ...(1)$$

where y is a function of variable x. If we give different values to x, then we obtain different values of y and hence a curve $y = f(x)$ in two dimensions. **Examples :** (i) The equation

$$y = mx + c \qquad ...(2)$$

represents a **straight line.** The equation

$$x^2 + y^2 = a^2 \quad \text{or} \quad y = \pm\sqrt{a^2 - x^2} \qquad ...(3)$$

gives a **circle.**

(ii) If we consider the derivative dy/dx of y in eq. (1) with respect to x, then dy/dx can be thought as the **slope** or **gradient** of the curve $y = f(x)$.

(iii) The quantity dy/dx tells us **the rate of change** y **with respect** to x. Such rates occur frequently in physics. If $x = t$, then the rate of change of a quantity (y) occurs with time (t) variable. Its examples are **velocity** $(d\vec{r}/dt)$, **acceleration** $(d\vec{v}/dt)$, **rate of cooling** $(d\theta/dt)$ of a hot body etc.

(iv) The derivative dy/dx is also used to find the **maximum** and **minimum** of a curve $y = f(x)$. Such applications occur also, when we consider a function of two or three variables.

Let us consider a function z of x and y variables *i.e.*,

$$z = f(x, y) \qquad ...(4)$$

We interpret this equation geometrically similar to $y = f(x)$. If x, y, z are cartesian coordinates, then for each x and y eq. (4) gives us a value of z and a point (x, y, z) in three dimensions is obtained. All the points satisfying this equation ordinarily form a surface in three dimensional space. For eample, the equation may be of the form :

$$x^2 + y^2 + z^2 = 1 \quad \text{or} \quad z = \sqrt{1 - x^2 - y^2} \qquad ...(5)$$

If we give different values to x and y, we obtain a surface ABC in three dimension in between the X, Y, Z axes. Now suppose x is a constant, this means that x = constant is a plane which intersects the surface ABC in the curve DE [**fig. 2.1**]. We may need the slope, maximum and minimum points of such a curve. Since z is a function of y on this curve (x = constant), the **partial derivative** $\partial z/\partial y$ **gives** the **slope of this curve** DE. Similarly, we can keep y = constant and find the partial derivative $\partial z/\partial x$, which may be interpreted as the slope of corresponding curve. This discussion gives the geometrical interpretation of a function $z = f(x, y)$ of two variables.

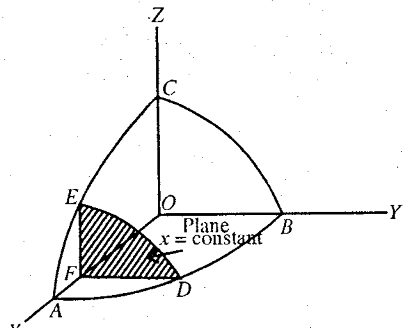


Fig. 2.1.

Ex. 1. *Calculate the first and second order partial differentials of* $f(x, y) = x^3 + y^3 + 3axy$. **(Raipur 2003)**

Sol. Here, $f(x, y) = x^3 + y^3 + 3axy$; $\therefore \dfrac{\partial f}{\partial x} = \mathbf{3x^2 + 3ay}$, $\dfrac{\partial f}{\partial y} = \mathbf{3y^2 + 3ax}$

Second order partial differentials are :

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}(3x^2 + 3ay) = \mathbf{3a}, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}(3y^2 + 3ax) = \mathbf{3a}.$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(3x^2 + 3ay) = \mathbf{6x} \text{ and } \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}(3y^2 + 3ax) = \mathbf{6y}.$$

Ex. 2. *If* $F(x, y) = xy + \sin(x + y)$, *then find the value of*

$$\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial^2 F}{\partial x^2}, \frac{\partial^2 F}{\partial y^2}, \frac{\partial^2 F}{\partial x \partial y} \text{ and } \frac{\partial^2 F}{\partial y \partial x}$$

Sol. Here, $F(x, y) = xy + \sin(x + y)$; $\therefore \dfrac{\partial F}{\partial x} = \mathbf{y + \cos(x + y)}$, $\dfrac{\partial F}{\partial y} = \mathbf{x + \cos(x + y)}$

Now,
$$\frac{\partial^2 F}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial F}{\partial x}\right) = \frac{\partial}{\partial x}[y + \cos(x + y)] = \mathbf{-\sin(x + y)};$$

$$\frac{\partial^2 F}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial F}{\partial y}\right) = \frac{\partial}{\partial y}[x + \cos(x + y)] = \mathbf{-\sin(x + y)};$$

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial F}{\partial y}\right) = \frac{\partial}{\partial x}[x + \cos(x + y)] = \mathbf{1 - \sin(x + y)}$$

and
$$\frac{\partial^2 F}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial F}{\partial x}\right) = \frac{\partial}{\partial y}[y + \cos(x + y)] = \mathbf{1 - \sin(x + y)}.$$

Ex. 3. *If* $z = x^2 + 2y^2$, $x = r\cos\theta$, $y = r\sin\theta$, *find the following :*

$$\text{(i)} \left(\frac{\partial z}{\partial x}\right)_y \quad \text{(ii)} \left(\frac{\partial z}{\partial x}\right)_r \quad \text{and} \quad \text{(iii)} \left(\frac{\partial z}{\partial x}\right)_\theta$$

Sol. (i) $z = x^2 + 2y^2$, $\left(\dfrac{\partial z}{\partial x}\right)_y = \mathbf{2x}$.

(ii) $x = r\cos\theta$ and $y = r\sin\theta$

Hence, $x^2 + y^2 = r^2\cos^2\theta + r^2\sin^2\theta$ or $x^2 + y^2 = r^2$

Now, $z = x^2 + 2y^2 = x^2 + 2(r^2 - x^2) = 2r^2 - x^2$; $\therefore$ $\left(\dfrac{\partial z}{\partial x}\right)_r = \mathbf{-2x}.$

(iii) $x = r\cos\theta$ and $y = r\sin\theta$, $\therefore$ $\dfrac{y}{x} = \tan\theta$ or $y = x\tan\theta$

Hence, $z = x^2 + 2y^2 = x^2 + 2x^2\tan^2\theta = x^2(1 + 2\tan^2\theta)$; $\therefore$ $\left(\dfrac{\partial z}{\partial x}\right)_\theta = \mathbf{2x(1 + 2\tan^2\theta)}.$

Ex. 4. *If* $V = (x^2 + y^2 + z^2)^{-1/2}$, *show that*

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0.$$

Sol. Here, $V = (x^2 + y^2 + z^2)^{-1/2}$, $\therefore \dfrac{\partial V}{\partial x} = -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}.2x = -x(x^2 + y^2 + z^2)^{-3/2}$

and
$$\begin{aligned}
\frac{\partial^2 V}{\partial x^2} &= -\left[(x^2+y^2+z^2)^{-3/2} + x\left(-\frac{3}{2}\right)(x^2+y^2+z^2)^{-5/2}.2x\right] \\
&= -[(x^2 + y^2 + z^2)^{-3/2} - 3x^2(x^2 + y^2 + z^2)^{-5/2}] \\
&= -(x^2 + y^2 + z^2)^{-3/2}\,[1 - 3x^2(x^2 + y^2 + z^2)^{-1}] \\
&= -(x^2 + y^2 + z^2)^{-3/2}\left[1 - \frac{3x^2}{x^2+y^2+z^2}\right] \\
&= -(x^2 + y^2 + z^2)^{-5/2}\,[y^2 + z^2 - 2x^2] = \frac{2x^2-y^2-z^2}{(x^2+y^2+z^2)^{5/2}} \qquad \text{...(i)}
\end{aligned}$$

Similarly, $$\frac{\partial^2 V}{\partial y^2} = \frac{2y^2-z^2-x^2}{(x^2+y^2+z^2)^{5/2}} \qquad \text{...(ii)}$$

and $$\frac{\partial^2 V}{\partial z^2} = \frac{2z^2-x^2-y^2}{(x^2+y^2+z^2)^{5/2}} \qquad \text{...(iii)}$$

Adding (i), (ii) and (iii), $\dfrac{\partial^2 V}{\partial x^2} + \dfrac{\partial^2 V}{\partial y^2} + \dfrac{\partial^2 V}{\partial z^2} = 0.$

Ex. 5. *If* $u = f(y/x)$, *prove that*

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0.$$

Sol. Here, $u = f\left(\dfrac{y}{x}\right)$

$\therefore$ $$\frac{\partial u}{\partial x} = f'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right) \text{ or } x\frac{\partial u}{\partial x} = -\frac{y}{x}f'\left(\frac{y}{x}\right)$$

Similarly $$\frac{\partial u}{\partial y} = f'\left(\frac{y}{x}\right).\frac{1}{x} \text{ or } y\frac{\partial u}{\partial y} = \frac{y}{x}f'\left(\frac{y}{x}\right).$$

Hence, $$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0.$$

Ex. 6. *If* $u = f(r)$, *where* $r^2 = x^2 + y^2$, *prove that*

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{f'(r)}{r}.$$

Sol. Here, $r^2 = x^2 + y^2$

Differentiating this equation with respect to x

$$2r\frac{\partial r}{\partial x} = 2x \text{ or } \frac{\partial r}{\partial x} = \frac{2x}{2r} = \frac{x}{r} \qquad \text{...(i)}$$

Similarly, $\dfrac{\partial r}{\partial y} = \dfrac{y}{r}$ ..(ii)

Now, $u = f(r), \therefore \dfrac{\partial u}{\partial x} = \dfrac{\partial f(r)}{\partial r}\dfrac{\partial r}{\partial x} = f'(r)\dfrac{x}{r}$ [from eq. (i)]

Hence, $$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{f'(r)x}{r}\right] = \frac{r\left[f'(r)\cdot 1 + xf''(r)\cdot\left(\frac{\partial r}{\partial x}\right)\right] - xf'(r)\left(\frac{\partial r}{\partial x}\right)}{r^2}$$

i.e. $$\frac{\partial^2 u}{\partial x^2} = \frac{1}{r^2}\left[rf'(r) + x^2 f''(r) - \frac{x^2}{r}f'(r)\right]$$ [from eq. (i)] ...(iii)

Similarly $$\frac{\partial^2 u}{\partial y^2} = \frac{1}{r^2}\left[rf'(r) + y^2 f''(r) - \frac{y^2}{r}f'(r)\right]$$ ...(iv)

Adding equations (iii) and (iv), we obtain

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{1}{r^2}\left[2rf'(r) + (x^2+y^2)f''(r) - \frac{(x^2+y^2)}{r}f'(r)\right] = \frac{1}{r^2}\left[2rf'(r) + r^2 f''(r) - rf'(r)\right]$$

$[\because x^2 + y^2 = r^2]$

i.e., $$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{1}{r}f'(r) + f''(r).$$

EXERCISE 2·1

1. If $V = \dfrac{1}{\sqrt{x^2+y^2+z^2}}$, then show that $x\dfrac{\partial V}{\partial x} + y\dfrac{\partial V}{\partial y} + z\dfrac{\partial V}{\partial z} = -V$.

2. If $u = \dfrac{x^2}{(x^2+y^2)}$, find $\dfrac{\partial u}{\partial x}$ and $\dfrac{\partial u}{\partial y}$.

   $\left[\textbf{Ans.}\ \dfrac{2xy^2}{(x^2+y^2)^2}, -\dfrac{2x^2y}{(x^2+y^2)^2}.\right]$

3. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, prove that $\left(\dfrac{\partial}{\partial x} + \dfrac{\partial}{\partial y} + \dfrac{\partial}{\partial z}\right)^2 u = -\dfrac{9}{(x+y+z)^2}$.

4. If $u = x^2 + y^2 + z^2$, then show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = 2u$.

5. If $z = x^2 + 2y^2$, $x = r\cos\theta$, $y = r\sin\theta$, find the following partial derivatives :

   (i) $\left(\dfrac{\partial z}{\partial x}\right)_y$, (ii) $\left(\dfrac{\partial z}{\partial x}\right)_\theta$, (iii) $\left(\dfrac{\partial z}{\partial y}\right)_r$,

   (iv) $\left(\dfrac{\partial z}{\partial \theta}\right)_x$, (v) $\left(\dfrac{\partial z}{\partial \theta}\right)_r$, (vi) $\left(\dfrac{\partial z}{\partial r}\right)_x$,

   (vii) $\dfrac{\partial^2 z}{\partial r\partial y}$, (viii) $\dfrac{\partial^2 z}{\partial y\partial \theta}$, (ix) $\dfrac{\partial^2 z}{\partial r\partial \theta}$.

   [**Ans.** (i) $2x$, (ii) $2x(1 + \tan^2\theta)$, (iii) $2y$, (iv) $4r^2\tan\theta$, (v) $r^2\sin 2\theta$, (vi) $4r$, (vii) 0, (viii) $-4x\,\mathrm{cosec}^2\theta$, (ix) $2r\sin 2\theta$.]

6. Find $\partial u/\partial x$ and $\partial u/\partial y$, if (a) $u = \log_e(x^2 + y^2)$ and (b) $u = e^{my}\cos mx$.

   $\left[\textbf{Ans.}\ (a)\ \dfrac{\partial u}{\partial x} = \dfrac{2x}{x^2+y^2}, \dfrac{\partial u}{\partial y} = \dfrac{2y}{x^2+y^2},\ (b)\ \dfrac{\partial u}{\partial x} = -e^{my}m\sin mx, \dfrac{\partial u}{\partial y} = me^{my}\cos mx.\right]$

7. Prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$, if (a) $u = x^y + y^x$, (b) $u = x \sin y + t \cos x$,

(c) $u = \log_e\left(\frac{x^2+y^2}{xy}\right)$, (d) $u = \log_e \tan\frac{y}{x}$.

8. Prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ if (a) $u = e^{my} \cos mx$, (b) $u = \tan^{-1}\left(\frac{y}{x}\right)$, (c) $u = \log_e (x^2 + y^2)$.

9. $u = x^2(y - z) + y^2(z - x) + z^2(x - y)$, prove that $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z} = 0$.

10. If $r = (x^2 + y^2 + z^2)^{1/2}$, show that $\frac{\partial^2 r}{\partial x^2}+\frac{\partial^2 r}{\partial y^2}+\frac{\partial^2 r}{\partial z^2} = \frac{2}{r}$. **(Indore 2003)**

2·5. Total Differential of a Function of Two or Three Variables

In case of partial differentiation of a function of two or three variables, only one variable changes. However, when we do total differentiation, the changes or increments are given to all the variables. Suppose that

$$u = f(x, y) \quad ...(1)$$

is the function of x and y variables.

If the infinitesimal change in the free variable x is $\delta x = dx$ and the infinitesimal change in the variable y be $\delta y = dy$, then the total differential of u is defined as

$$du = \frac{\partial u}{\partial x}dx+\frac{\partial u}{\partial y}dy \quad ...(2a)$$

or

$$du = \frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy \quad ...(2b)$$

Similarly, if $u = f(x, y, z)$ is the function of three variables x, y, z, then the total differential of u is given by

$$du = \frac{\partial u}{\partial x}dx+\frac{\partial u}{\partial y}dy+\frac{\partial u}{\partial z}dz \quad ...(3)$$

If $u = f(x_1, x_2 ...., x_n)$ is the function of several variables $x_1, x_2, ... x_n$, then the total differential of u is

$$du = \frac{\partial u}{\partial x_1}dx_1+\frac{\partial u}{\partial x_2}dx_2+.....+\frac{\partial u}{\partial x_n}dx_n \quad ...(4)$$

For example, the pressure (P) of a gas is the function of variables volume V and temperature T *i.e.*,

$$P = f(V, T) \quad ...(5)$$

The total differential of P is given by

$$dP = \left(\frac{\partial P}{\partial V}\right)_T dV+\left(\frac{\partial P}{\partial T}\right)_V dT \quad ..(6)$$

We write the total differential in the form (6) in thermodynamics.

Scalar quantities, such as temperature, electric potenital etc, are represented by scalar function ϕ (x, y, z) in space and we say that a scalar field is present in that space. The total differential $d\phi$ of this field ϕ (x, y, z) is obtained as

$$d\phi = \frac{\partial \phi}{\partial x}dx+\frac{\partial \phi}{\partial y}dy+\frac{\partial \phi}{\partial z}dz \quad ...(7)$$

2·6. Total Differential Coefficient of a Function of Two or Three Variables

Let u be a function of two variables *i.e.*,

$$u = f(x, y) \quad ...(1)$$

If x and y are functions of t *i.e.*,

$$x = x(t) \text{ and } y = y(t) \qquad ...(2)$$

$$\text{or } x = \phi(t) \text{ and } y = \psi(t)$$

Then the **total differential coefficient** of u is defined as

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} \qquad ...(3)$$

Similarly, if $u = f(x_1, x_2, ..., x_n)$ and $x_i = x_i(t)$, then the total differential coefficient will be given by

$$\frac{du}{dt} = \frac{\partial u}{\partial x_1}\frac{dx_1}{dt} + \frac{\partial u}{\partial x_2}\frac{dx_2}{dt} + ..., + \frac{\partial u}{\partial x_n}\frac{dx_n}{dt} \qquad ...(4)$$

2·7. First Differential Coefficient of an Implicit Function

When $z = f(x, y)$, where $y = \psi(x)$, then

$$\frac{dz}{dx} = \frac{\partial z}{\partial x}.\frac{dx}{dx} + \frac{\partial z}{\partial y}.\frac{dy}{dx} \text{ or } \frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}.\frac{dy}{dx} \qquad ...(1)$$

If an implicit relation $z = f(x, y) = c$ is given between x and y, where c is a constant and y is a function of x. Then

$$\frac{dz}{dx} = 0 \text{ or } \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}.\frac{dy}{dx} = 0$$

Hence

$$\frac{dy}{dx} = -\frac{\partial z/\partial x}{\partial z/\partial y} \text{ or } \frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y} \qquad ...(2)$$

2·8. Higher Order Derivatives

Let the quantity y be function of the variable x *i.e.*,

$$y = f(x) \qquad ...(1)$$

Its derivative dy/dx may be the function of x and this derivative can be differentiated again. The derivative of dy/dx with respect x is called the **second order derivative** of y with respect to x. Similarly, the derivative of second order derivative with respect to x is called the **third order derivative** of y with respect to x. Similarly, we can get successive derivatives or differential coefficients, which are represented as

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, ..., \frac{dy^n}{dx^n}, \qquad ...(2)$$

where dy^n/dx^n is the n[th] **derivative of** y **with respect to** x.

They are also represented as

$$y', y'', y''', ..., y^n, ... \text{ or } y_1, y_2, y_3, ..., y_n, ... \text{ or}$$

$$f'(x), f''(x), f'''(x), ..., f^n(x), ... \text{ or } Df(x), D^2f(x), ..., D^n f(x), ... \qquad ...(3)$$

Some formulae for n^{th} derivative

(1) If $y = (ax + b)^p$, then

$$y' = p(ax + b)^{p-1} a, \; y'' = p(p-1)(ax + b)^{p-2} a^2, ... \; .... \; .... \; ....$$

$$y^n = p(p-1)(p-2) ..... (p-n+1)(ax + b)^{p-n} a^n \qquad ...(4)$$

(2) If $y = e^{ax}$, then $y' = ae^{ax}$, $y'' = a^2e^{ax}$ etc and $y^n = a^n e^{ax}$ ...(5)

(3) If $y = \sin(ax + b)$, then

$$y' = a\cos(ax + b) = a\sin\left(ax + b + \frac{\pi}{2}\right), \; y'' = a^2\cos\left(ax + b + \frac{\pi}{2}\right) = a^2\sin\left(ax + b + \frac{2\pi}{2}\right),$$

Hence, $y^n = a^n\sin\left(ax + b + \frac{n\pi}{2}\right)$ ...(6)

Ex.1. *If* $(\tan x)^y + (y)^{\cot x} = a$, *find* $\frac{dy}{dx}$

Sol. Here,
$$f(x, y) = (\tan x)^y + (y)^{\cot x} = a,$$

$\therefore$
$$\frac{\partial f}{\partial x} = y(\tan x)^{y-1} \sec^2 x + y^{\cot x} \cdot \log y(-\operatorname{cosec}^2 x) \text{ and}$$
$$\frac{\partial f}{\partial y} = (\tan x)^y \cdot \log \tan x + (\cot x) \cdot y^{\cot x - 1}$$

Hence,
$$\frac{dy}{dx} = -\frac{\partial f / \partial x}{\partial f / \partial y} = -\frac{y(\tan x)^{y-1} \sec^2 x - y^{\cot x} \cdot \log y \cdot \operatorname{cosec}^2 x}{(\tan x)^y \log \tan x + (\cot x) y^{\cot x - 1}}$$

Ex. 2. *If* $z = xy$ *and* $x = 2t^2$, $y = \sin t$, *find* $\frac{dz}{dt}$.

Sol.
$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

Here, $z = xy$, $x = 2t^2$ and $y = \sin t$

Hence,
$$\frac{\partial z}{\partial x} = y, \frac{\partial z}{\partial y} = x, \frac{dx}{dt} = 4t \text{ and } \frac{dy}{dt} = \cos t$$

$\therefore$
$$\frac{dz}{dt} = y.4t + x.\cos t = 4t \sin t + 2t^2 \cos t.$$

Ex. 3. *If* $z = \sin (x^2 + y^2)$, *where* $a^2x^2 + b^2y^2 = c^2$, *find* dz/dx.

Sol.
$$\frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}\frac{dy}{dx} \qquad \text{...(i)}$$

Now, $z = \sin (x^2 + y^2)$; $\therefore$ $\frac{\partial z}{\partial x} = 2x \cos (x^2 + y^2)$ and $\frac{\partial z}{\partial y} = 2y \cos (x^2 + y^2)$

But, $a^2x^2 + b^2y^2 = c^2$, $\therefore$ $2a^2x + 2b^2y\left(\frac{dy}{dx}\right) = 0$ or $\frac{dy}{dx} = -\frac{a^2x}{b^2y}$

Now, substituting for $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ and $\frac{dy}{dx}$ in eq. (i), we get

$$\frac{dz}{dx} = 2x \cos (x^2 + y^2) - [2y \cos (x^2 + y^2)]\left[\frac{a^2x}{b^2y}\right] = 2x \cos (x^2 + y^2) \left\{- \frac{a^2}{b^2}\right..$$

Ex. 4. *Find the different derivatives of the following function*
$$f(x) = 2x^3 + x^2 + 4x + 1.$$

Sol.
$$f'(x) = 6x^2 + 2x + 4, \; f''(x) = 12x + 2 \text{ and } f'''(x) = 12$$

Other higher order derivatives are zero.

Ex. 5. *If* $y = \frac{ax + b}{cx + b}$, *show that* $2y' y''' = 3y''^2$.

Sol.
$$y = \frac{ax + b}{cx + b} = \frac{a}{c} + \frac{bc - ad}{c(cx + d)}$$

$\therefore$
$$y' = -\frac{bc - ad}{(cx + d)^2}, y'' = \frac{2c(bc - ad)}{(cx + d)^3} \text{ and } y''' = -\frac{6c^2(bc - ad)}{(cx + d)^4}$$

Therefore,
$$2y'y''' = \frac{6c^2(bc - ad)^2}{(cx + d)^6} = 3y''^2$$

Ex. 6. *If $x = f(t)$, $y = g(t)$, prove that* $\dfrac{d^2y}{dx^2} = \dfrac{x'y'' - y'x''}{x'^3}$, *where* $x' = \dfrac{dx}{dt}$, $y'' = \dfrac{d^2y}{dt^2}$ etc.

Sol. Here,
$$x' = \frac{dx}{dt} = f'(t) \text{ and } y' = \frac{dy}{dt} = g'(t)$$

$\therefore$
$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{y'}{x'}$$

Hence,
$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{y'}{x'}\right) = \frac{d}{dt}\left(\frac{y'}{x'}\right)\frac{dt}{dx} = \frac{x'\dfrac{dy'}{dt} - y'\dfrac{dx'}{dt}}{(x')^2}\cdot\frac{1}{x'} = \frac{x'y'' - y'x''}{x'^3}.$$

EXERCISE 2·2

1. If $z = x \log xy$, where $x^3 + y^3 + 3xy = 1$, find the value of $\dfrac{dz}{dx}$.

   $\left[\textbf{Ans. } 1 + \log xy - \dfrac{x}{y}\left(\dfrac{x^2 + y}{x + y^2}\right).\right]$

2. If $\sqrt{1 - x^2} + \sqrt{1 - y^2} = a(x - y)$, prove that $\dfrac{dy}{dx} = \dfrac{\sqrt{1 - y^2}}{\sqrt{1 - x^2}}$.

3. If $z = u^2 + v^2$, $u = at^2$, $v = 2at$, find $\dfrac{dz}{dt}$.

   [**Ans.** $4a\,(at^3 + 2at)$.]

4. If $y = ae^{-kt}\cos(pt + \phi)$, prove that $\dfrac{d^2y}{dt^2} + 2k\dfrac{dy}{dt} + n^2y = 0$, where $n^2 = p^2 + k^2$.

5. If $x = 2\cos\phi - \cos 2\phi$ and $y = 2\sin\phi - \sin 2\phi$, find $\dfrac{d^2y}{dx^2}$ at $\phi = \dfrac{\pi}{2}$.

   $\left[\textbf{Ans. } -\dfrac{3}{2}.\right]$

6. Find the second order derivative of $y = x^4e^{3x}$.

   [**Ans.** $x^4e^{3x}(25x^2 + 40x + 12)$.]

7. If $y = x\log\dfrac{x}{a + bx}$, show that $x^3y'' = (y - xy')^2$.

2·9. Applications of the Differentiability of Functions

(1) To find the rate of change of a physical quantity with respect to another : Consider a physical quantity represented by

$$y = f(x) \qquad ...(1)$$

This quantity is a function of variable x. If we change x by δx, then the corresponding change in y is δy (say). Now, the quantity $\delta y/\delta x$ tells us the **average rate of change of the physical qauntity y with respect to the variable x.**

For example, if the position vector of a particle changes from $\vec{r}$ to $\vec{r} + \delta\vec{r}$ in the time interval δt, then the ratio $\delta\vec{r}/\delta t$ gives the average velocity in this interval *i.e.*,

$$\vec{v}_{av} = \frac{\delta\vec{r}}{\delta t} \qquad ...(2)$$

If we want to find the rate (dy/dx) of change of y with respect to x, then taking the limit $\delta x \to 0$,

$$\frac{dy}{dx} = \lim_{\delta x \to 0} \frac{\delta y}{\delta x}. \qquad ...(3)$$

This quantity dy/dx tells us the **rate of change of y with respect x.**

The rate of change of the position vector $\vec{r}$ of a particle with respect to time t is called the velocity of the particle. This is obtained from (2) in the limit $\delta t \to 0$ *i.e.,*

$$\therefore \qquad \vec{v} = \frac{d\vec{r}}{dt} = \lim_{\delta t \to 0} \frac{\delta \vec{r}}{\delta t}. \qquad ...(4)$$

Further, **rate of change of velocity of a particle is called its acceleration** *i.e.,*

$$\vec{a} = \frac{d\vec{v}}{dt} = \lim_{\delta t \to 0} \frac{\delta \vec{v}}{\delta t} \qquad ...(5)$$

The acceleration can also be written as

$$a = \frac{dv}{dt} = \frac{dv}{ds} \cdot \frac{ds}{dt} = v\frac{dv}{ds} \qquad ...(6)$$

where ds is the distance moved in infinitesimal time dt and v is the magnitude of the velocity.

Now let x and y quantities are the functions of time t *i.e.,*

$$x = f(t) \text{ and } y = g(t) \qquad ...(7)$$

In such a case the rate of change of y with respect to x is given by

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\text{Rate of change of } y \text{ with respect to } t}{\text{Rate of change of } x \text{ with respect to } t} = \frac{g'(t)}{f'(t)} \qquad ...(8)$$

(2) To determine the slope of a tangent at a point (x, y) **of the curve** $y = f(x)$ **:**

The slope of a tangent to the curve $y = f(x)$ at a point (x, y) is given by

$$\frac{dy}{dx} = \tan \phi$$

If AB be a tangent at the point $P(x, y)$ of the curve $y = f(x)$, then **[fig. 2.2]**

$$\tan \phi = \lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \frac{dy}{dx} \qquad ...(9)$$

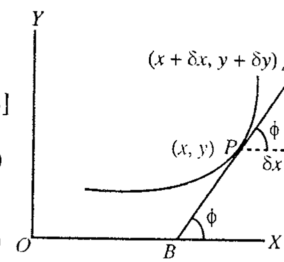


Fig. 2.2.

If the equation of the curve is represented as

$$f(x, y) = 0 \qquad ...(10)$$

then $$\frac{df}{dx} = 0 = \frac{\partial f}{\partial x}\frac{dx}{dx} + \frac{\partial f}{\partial y}\frac{dy}{dx} \text{ or } \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx} = 0$$

Hence $$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y} \qquad ...(11)$$

Example : *A curve is represented by* $x = a(t + \sin t)$ *and* $y = a(1 - \cos t)$. *Find* dy/dx *at a point* (x, y) *of the curve.*

Sol. $$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a \sin t}{a(1 + \cos t)} = \frac{2a \sin\frac{t}{2}\cos\frac{t}{2}}{2a\cos^2\frac{t}{2}} = \mathbf{\tan\frac{t}{2}}.$$

(3) Estimation of Error : Let a quantity be given in the form

$$u = xy$$

which depends on the variables x and y.

If the errors (small) in the measurement of x and y are δx and δy respectively, then the error δu in the measurement of u will be given by

$$u \pm \delta u = (x \pm \delta x)(y \pm \delta y)$$

or $$u \pm \delta u = xy \pm x\delta y \pm y\delta x \qquad (\delta x\delta y \text{ is negligible})$$

Therefore, $$\delta u = \pm\, x\delta y \pm y\delta x \text{ or } \frac{\delta u}{u} = \pm\frac{\delta y}{y} \pm \frac{\delta x}{x} \qquad ...(12)$$

Hence, the maximum relative error δu in u is given by

$$\left|\frac{\delta u}{u}\right| = \left|\frac{\delta x}{x}\right| + \left|\frac{\delta y}{y}\right| \qquad ...(13)$$

In order to find the maximum relative error by differential calculus, we write

$$\ln u = \ln xy \quad \text{or} \quad \ln u = \ln x + \ln y$$

On differentiation, we get

$$\frac{du}{u} = \frac{dx}{x} + \frac{dy}{y}$$

Maximum relative error $$\left|\frac{du}{u}\right| = \left|\frac{dx}{x}\right| + \left|\frac{dy}{y}\right|$$

Example : *The resistance R of wire of length l and radius r is given by*

$$R = K\, l/r^2$$

If the relative errors in the measement of the length and radius are 5% and 10% respectively, find the maximum relative error in R.

Sol. $$R = \frac{Kl}{r^2}, \ \therefore \ \ln R = \ln K + \ln l + 2 \ln r$$

On differentiation, $$\frac{dR}{R} = \frac{dl}{l} + \frac{2dr}{R}$$

Relative maximum error $$\left|\frac{dR}{R}\right| = \left|\frac{dl}{l}\right| + 2\left|\frac{dr}{r}\right| = \frac{5}{100} + \frac{2\times 10}{100} = \frac{25}{100} = \mathbf{25\%}.$$

(4) To find the maxima or minima of a curve : If we want to know the maxima or minima of a curve $y = f(x)$, then we differentiate y with respect to x and put it equal to zero *i.e.*,

$$\frac{dy}{dx} = \frac{df(x)}{dx} = 0 \qquad ...(14)$$

Let this condition be satisfied for the condition $x = a$. Then at $x = a$, there may be maxima or minima. If we find d^2f/dx^2 at $x = a$ and its value is negative or positive, then correspondingly there will be maxima or minima at the point $x = a$ of the curve.

In case $d^2y/dx^2 = 0$ at $x = a$, we find d^3y/dx^3 at $x = a$. If this quantity is not equal to zero, the function will have neither maxima or minima at $x = a$. But if $d^3y/dx^3 = 0$ at $x = a$, we should find out d^4y/dx^4 at $x = a$ and if it is negative, the function will have a maximum at $x = a$ and if it is positive, the function will have a minimum. If it is zero, we should find out d^5y/dx^5 and so on.

For a **function of two variables**, say $z = f(x, y)$, we find

$$\frac{\partial z}{\partial x} = 0 \text{ and } \frac{\partial z}{\partial y} = 0$$

Let these equations give $x = a$ and $y = b$. Then

(1) There will be a minimum at the point (a, b), if at $x = a$ and $x = b$, we have

$$\frac{\partial^2 f}{\partial x^2} > 0, \frac{\partial^2 f}{\partial y^2} > 0 \text{ and } \frac{\partial^2 f}{\partial x^2}\cdot\frac{\partial^2 f}{\partial y^2} > \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2;$$

(2) For maximum at the point (a, b), we have

$$\frac{\partial^2 f}{\partial x^2} < 0, \frac{\partial^2 f}{\partial y^2} < 0 \text{ and } \frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} > \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2.$$

(3) At the point (a, b), there will be neither a maximum nor a minimum, if

$$\frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} < \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2.$$

Note that this includes $\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} < 0$ *i.e.*, $\frac{\partial^2 f}{\partial x^2}$ and $\frac{\partial^2 f}{\partial y^2}$ are of opposite sign.

Ex. 1. *If $y = x^5 - 5x^4 + 5x^3 - 1$, calculate the maximum and minimum values of y.*

(Raipur 2003)

Sol. Here, $y = x^5 - 5x^4 + 5x^3 - 1$; $\therefore \frac{dy}{dx} = 5x^4 - 20x^3 + 15x^2$

For maximum and minimum values of y, the value of dy/dx should be equal to zero *i.e.*,
$5x^4 - 20x^3 + 15x^2 = 0$ or $5x^2(x^2 - 4x + 3) = 0$ or $5x^2(x - 3)(x - 1) = 0$.
$\therefore x = 1, x = 3$ and $x = 0$

Also, $$\frac{d^2y}{dx^2} = \frac{d}{dx}(5x^4 - 20x^3 + 15x^2) = 20x^3 - 60x^2 + 30x$$

At $x = 1$, $\frac{d^2y}{dx^2} = 20 - 60 + 30 = -10$, hence at $x = 1$, the value of y will be maximum. The maximum value of y is

$$y_{max} = 1 - 5 + 5 - 1 = \mathbf{0}$$

At $x = 3$, $\frac{d^2y}{dx^2} = 20 \times (3)^3 - 60 \times (3)^2 + 30 \times 3 = 90$, hence at $x = 3$, the value of y will be minimum. The minimum value of y is

$$y_{min} = (3)^5 - 5 \times (3)^4 + 5 \times (3)^3 - 1 = \mathbf{-28}.$$

At $x = 0$, $\frac{d^2y}{dx^2} = 0$. Hence we cannot say whether the function has a maximum or minimum at $x = 0$.

Therefore, we find, $$\frac{d^3y}{dx^3} = 60x^2 - 120x + 30$$

At $x = 0$, $d^3y/dx^3 = 30$. This is not zero and hence **the function has neither a maximum nor a minimum at $x = 0$.**

Ex. 2. *Find the maximum and minimum points of the following function :*

$$f(x, y) = x^2 + y^2 + 2x - 4y + 10$$

Sol. $\frac{df}{dx} = 2x + 2 = 0$ or $x + 1 = 0$ or $x = -1$ and $\frac{df}{dy} = 2y - 4 = 0$ or $y = 2$

Now, $\frac{\partial^2 f}{\partial x^2} = +2$, $\frac{\partial^2 f}{\partial y^2} = +2$ and $\frac{\partial^2 f}{\partial x \partial y} = 0$

Hence **at point (–1, 2), there will be minimum.**

Ex. 3. *Find the maximum and minimum points of the folllowing function.*

$$f(x, y) = x^2 - y^2 + 2x - 4y + 5$$

Sol. Here, $\frac{\partial f}{\partial x} = 2x + 2 = 0$ or $x = -1$ and $\frac{\partial f}{\partial y} = -2y - 4$ or $y = -2$

Now, $$\frac{\partial^2 f}{\partial x^2} = 2,\ \frac{\partial^2 f}{\partial y^2} = -2 \text{ and } \frac{\partial^2 f}{\partial x \partial y} = 0$$

Hence, **at the point (–1, –2), there will be neither maximum nor minimum.**

Ex. 4. *The volume of tent V is given. If has ends but no floor. It is to be made using least possible cloth. Find the angle* θ **[fig. 2.3]** *and height of the tent.*

Sol. Here l is the length and 2a is the width of the base of the tent **[fig. 2.3]**.

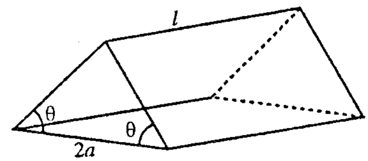


Fig. 2.3.

$$\text{Volume, } V = \frac{1}{2}\cdot 2a\cdot l\cdot a\tan\theta = a^2 l\tan\theta \qquad \text{...(i)}$$

$$\text{Area, } A = 2a^2\tan\theta + \frac{2la}{\cos\theta} \qquad \text{...(ii)}$$

As V is constant and $l = V/(a^2\tan\theta)$ from (i), hence

$$A = 2a^2\tan\theta + \frac{2a}{\cos\theta}\cdot\frac{V}{a^2\tan\theta} \text{ or } A = 2a^2\tan\theta + \frac{2V}{a}\text{cosec}\,\theta$$

The area A is the function of two independent variables θ and a *i.e.*, $A = A(\theta, a)$. Hence for minimum area A, we have

$$\frac{\partial A}{\partial a} = 4a\tan\theta - \frac{2V\text{cosec}\,\theta}{a^2} = 0 \qquad \text{...(iii)}$$

and $$\frac{\partial A}{\partial \theta} = 2a\sec^2\theta - \frac{2V}{a}\text{cosec}\,\theta\cot\theta = 0 \qquad \text{...(iv)}$$

Hence $a^3 = \frac{V\text{cosec}\,\theta}{2\tan\theta} = \frac{V\text{cosec}\,\theta\,\cot\theta}{\sec^2\theta}$ or $\frac{\cos\theta}{2\sin^2\theta} = \frac{\cos\theta\,\cos^2\theta}{\sin^2\theta}$ or $\cos^2\theta = \frac{1}{2}$ or $\cos\theta = \frac{1}{\sqrt{2}}$.

Here neither $\sin\theta = 0$ nor $\cos\theta = 0$, because for these values the volume of the tent will be zero.

Hence $$\cos\theta = 1/\sqrt{2} \text{ or } \theta = \mathbf{45°}.$$

This is the desired angle.

Now, for $\tan\theta = 1$, $V = a^2 l$ and from equation (iii), $\frac{\partial A}{\partial a} = 0$, $2a = l\sqrt{2}$ or $a = l/\sqrt{2}$. In such a condition the height of the tent will be

$$h = a\tan\theta = a = \frac{l}{\sqrt{2}}$$

Ex. 5. *The displacement of a particle moving away from origin at any instant t is* $s = a\sin t + b\cos 2t$, where a and b are constants. Calculate the velocity and acceleration of the particle.

Sol. $$\text{Velocity, } v = \frac{ds}{dt} = \frac{d}{dt}(a\sin t + b\cos 2t) = \boldsymbol{a\cos t - 2b\sin 2t}.$$

$\therefore$ $$\text{Acceleration, } a = \frac{dv}{dt} = \frac{d}{dt}(a\cos t - 2b\sin 2t) = \boldsymbol{-a\sin t - 4b\cos 2t}.$$

Ex. 6. *A particle in XY-plane moves according to the following equation :*

$$x = a\sin\omega t \text{ and } y = a(1 - \cos\omega t)$$

where a and ω *are positive constants.*

(a) *Find the distance moved by the particle in* t_0 *time.*

(b) *What is the angle between the velocity and acceleration of the particle ?*

Sol. $x = a \sin \omega t$ and $y = a(1 - \cos \omega t)$

Position vector of the particle, $\vec{r} = x\hat{i} + y\hat{j}$ or $\vec{r} = a \sin \omega t\,\hat{i} + a(1 - \cos \omega t)\,\hat{j}$

Hence, velocity of the particle, $\vec{v} = \dfrac{d\vec{r}}{dt} = \omega a \cos \omega t\,\hat{i} + \omega a \sin \omega t\,\hat{j}$

Acceleration of the particle, $\vec{w} = \dfrac{d\vec{v}}{dt} = -\,\omega^2 a \sin \omega t\,\hat{i} + \omega^2 a \cos \omega t\,\hat{j}$

(a) Speed of the particle $v = \sqrt{\omega^2 a^2(\cos^2 \omega t + \sin^2 \omega t)} = \omega a$

Distance travelled by the particle in time t_0, $s = \int ds = \int \frac{ds}{dt}dt = \int_0^{t_0} v dt = \int_0^{t_0} \omega a dt = \omega a t_0$

(b) $\vec{v} \cdot \vec{w} = vw \cos \alpha$, where α is the angle between $\vec{v}$ and $\vec{w}$.

$$\therefore \quad \cos \alpha = \frac{\vec{v} \cdot \vec{w}}{vw} = \frac{-\omega^3 a^2 \sin \omega t \cos \omega t + \omega^3 a^2 \sin \omega t \cos \omega t}{vw} = 0; \text{ i.e., } \alpha = \frac{\pi}{2}.$$

Ex. 7. *Calculate the slope at the point (2, 3) of curve* $y = ax^3 + b$.

(Ujjain 2003; Gwalior 2003)

Sol. $y = ax^3 + b$ or $y = \sqrt{ax^3 + b}$, $\therefore$ Slope $\dfrac{dy}{dx} = \dfrac{3ax^2}{2\sqrt{ax^3 + b}} = \dfrac{3ax^2}{2y}$

Hence, $\dfrac{dy}{dx}$ at $(2, 3) = \dfrac{3a \times 2^2}{2 \times 3} = \mathbf{2a}$.

EXERCISE 2·3

1. Prove that the function $f(x) = \sin x(1 + \cos x)$ will have a maximum at $x = \pi/3$.
2. Find the maxima and minima for the function $f(x, y) = x^3y^2(1 - x - y)$.

   [**Ans.** Maximum at the point $\left(\frac{1}{2}, \frac{1}{3}\right)$ and then $f(x, y) = 1/432$.]
3. Discuss the maximum or minimum for the function $f(x, y) = x^2y^2 - 5x^2 - 8xy - 5y^2$.
   [**Ans.** Maximum at $x = y = 0$.]
4. A body is projected with velocity u at an angle θ from the horizontal and the formula for the range is $R = \dfrac{u^2 \sin^2\theta}{g}$. For the maximum value of the range find the value of θ for given initial velocity u. What is the maximum value of R ?

   [**Ans.** $\theta = \pi/4$; $R_{max} = u^2/g$.]
5. The van der-Wall's equation of state for a real gas is $\left(P + \dfrac{a}{V^2}\right)(V - b) = RT$, where P is pressure, V is volume, T is temperature and a, b, R are constants. Find the rate of variation of volume with pressure.
   [**Ans.** $(V - b)\,V^3/(aV - PV^3 - 2ab)$.]
6. If the velocity of a particle is represented as $v = k\sqrt{x}$, where k is a positive constant. If at $t = 0$, $x = 0$, how the velocity and acceleration will depend on time ?

   [**Ans.** $v = \frac{1}{2}k^2 t$; $a = \frac{1}{2}k^2$.]

7. A particle moves in XY-plane according to the law $x = at$ and $y = at(1 - bt)$, where a and b are positive constants and t is the time. Find (a) the equation of the path of the particle, (b) the speed v and acceleration w as the function of time t. (c) the time t_0 at which the angle between velocity vector and acceleration vector is $\pi/4$.

[**Ans.** (a) $y = x - (b/a)x^2$, (b) $v = a\sqrt{1+(1-2bt)^2}$; $w = 2ba$ = constant, (c) $t_0 = 1/b$.]

8. Resistance is given by the formula : $R = V/I$. If $V = 100 \pm 5$ volt and $I = 10 \pm 0.2$ ampere, find the maximum error in R.

[**Ans.** 7%.]

9. A physical quantity is related to four physical quantities a, b, c and d as $Q = \dfrac{a^3b^2}{d\sqrt{c}}$. The errors in the measurements of a, b, c and d are 1%, 3%, 4% and 2% respectively what is the maximum percentage error in Q ?

[**Ans.** 13%.]

10. Prove that the area of a rectangle, drawn inside a circle, will be maximum when it is square.

QUESTIONS

Long Answer Questions

1. What do you understand by the function of two or more variables ? Find the expressions for the partial differential coefficients of first and second order for the function of two or three variables.
2. If $z = z(x, y)$ and $x = x(t)$, $y = y(t)$, find the total differential coefficient of the function z with respect to t.
3. Explain the geometrical meaning of the partial derivatives of function of two variables.

(Rewa 2004)

Short Answer Questions

1. What do you understand by a function of two variables ? Give one example.
2. If $u = f(x, y)$ and $x = \phi(t)$, $y = \psi(t)$, find the total differential coefficient of u with respect to t.

$$\left[\textbf{Ans.}\;\; du = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}\right]$$

3. If $P = f(V, T)$, find the total differential of P.

$$\left[\textbf{Ans.}\;\; dP = \frac{\partial P}{\partial V}dV + \frac{\partial P}{\partial T}dT\right]$$

4. What are the second order partial differential coefficients of a function $u = f(x, y)$?
5. What is the total differential of function of two variables ? Explain.

Objective Type Questions

1. If $u = x^2y$, then the value of $\partial^2u/\partial x^2$ is :

(a) x^2y (b) $2xy$ (c) $2y$ (d) x^2.

2. If $u = x/y$, the value of $\partial u/\partial y$ is :

(a) $\dfrac{1}{y}$ (b) $\dfrac{x}{y}$ (c) $-\dfrac{x}{y^2}$ (d) $\dfrac{1}{y} - \dfrac{x}{y^2}$.

3. If $x = r\cos\theta$ and $y = r\sin\theta$, then :

(a) $\left(\dfrac{\partial r}{\partial x}\right)_\theta = \sec\theta$ (b) $\left(\dfrac{\partial r}{\partial x}\right)_\theta = \cos\theta$ (c) $\left(\dfrac{\partial r}{\partial x}\right)_y = \sec\theta$ (d) $\left(\dfrac{\partial r}{\partial x}\right)_y = \dfrac{y}{r}$

4. If $\frac{PV}{T} = R$, then :

(a) $\left(\frac{\partial P}{\partial T}\right)_V = \frac{P}{T}$ (b) $\left(\frac{\partial P}{\partial T}\right)_V = -\frac{R}{V^2}$ (c) $\left(\frac{\partial P}{\partial T}\right)_V = R$ (d) $\left(\frac{\partial V}{\partial T}\right)_P = R$.

5. If $f(x, y) = x^3 + y^3 - 3axy$, then $\frac{\partial f}{\partial x}$ is :

(a) $3x^2 + 3ay$ (b) $3x^2 - 3ay$ (c) $3y^2 - 3ax$ (d) $3x^2$.

6. If $f(x, y) = x \cos y + y \cos x$, the correct answer is

(a) $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ (b) $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ (c) $\frac{\partial^2 f}{\partial x \partial y} \neq \frac{\partial^2 f}{\partial y \partial x}$ (d) none of these.

7. If $u = x^3 + y^3 - 3xy^2$, then the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is :

(a) $2u$ (b) u (c) $3u$ (d) 0

8. If $u = \tan^{-1}\frac{x}{y}$, then the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is :

(a) 0 (b) u (c) $2u$ (d) $\frac{y}{x}$.

9. For the gas equation $PV = RT$, the correct relation is :

(a) $\left(\frac{\partial P}{\partial V}\right)_T = R$ (b) $\left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{P}$ (c) $\left(\frac{\partial P}{\partial V}\right)_T = \frac{-P}{V}$ (d) $\left(\frac{\partial P}{\partial T}\right)_P = \frac{R}{V}$.

10. The acceleration of a moving particle is given by the equation $a = 4t^3 + 3t^2 + 1\ ms^{-2}$. The displacement of the particle at $t = 1$ second will be :

(a) $\frac{20}{19}$ m (b) 8 m (c) 3 m (d) $\frac{19}{20}$ m. **(Bhopal 2004)**

ANSWERS

1. (c)	**2.** (c)	**3.** (a)	**4.** (a)	**5.** (b)	**6.** (a)
7. (c)	**8.** (a)	**9.** (c)	**10.** (d)		

3

Integration of Functions of Two and Three Variables

3·1. Integration

Integration is the inverse operation of differentiation. If $dy/dx = f(x)$, its integration with respect to the variable x is given by

$$y = \int f(x)dx + C \qquad ...(1)$$

where $f(x)$ is a function of the variable x and C is a constant, called as *constant of integration.*

In eq. (1), the function $f(x)$ is called **integrand** and the quantity $\int f(x)dx$ is called **integral of** $f(x)$ **with respect to the variable** x. The integral represented by eq. (1) is called **indefinite integral.**

Now if we differentiate $y = \int f(x)dx$ with respect to x, then

$$\frac{dy}{dx} = f(x) \qquad ...(2)$$

Hence, any integral $\int f(x)dx$ is in the form of antiderivative.

In fact, *integration is an operation of summation or addition.* If we think about $\Sigma f(x)\delta x$, then in the condition of δx approaching zero, we have

$$\sum_{\delta x \to 0} f(x)dx = \int f(x)dx \qquad ...(3)$$

In this situation, the summation sign (Σ) changes to integration sign ($\int$).

Now, if the integration is taken in between the two limits $x = a$ and $x = b$, then such an integral is called **definite integral**. It is represented as follows :

$$\int_a^b f(x)dx = F(b) - F(a) \qquad ...(4)$$

Examples : (1) Determination of area under a curve : We will find the **area under a curve** $y = f(x)$. In **fig. 3.1**, the curve $y = f(x)$, *i.e.*, y has been drawn as ordinate as a function of the variable x on abscissa. If we divide the area between the curve $y = f(x)$ and X-axis from $x = a$ to $x = b$ by different rectangles or strips and consider a strip of infinitesimal width δx and height y, then the area of this strip will be

$$\delta A = y\,\delta x = f(x)\,\delta x \qquad ...(5)$$

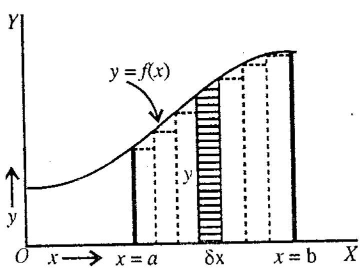


Fig. 3.1 : Area under a curve.

where y or $f(x)$ has been taken constant in the infinitesimal width δx. Hence, the area (A) between the curve $y = f(x)$ and X-axis from $x = a$ to $x = b$ will be equal to the sum of areas of all the similar strips *i.e.*,

$$A = \sum_{x=a}^{x=b} y\delta x = \sum_{x=a}^{x=b} f(x)dx \qquad ...(6)$$

where Σ represents the sign of summation.

Now, if we increase the number of strips and reduce δx, then in the limit $\delta x \to 0$, the sum of the areas of all the strips will give the area in between $y = f(x)$ curve and X-axis from $x = a$ to $x = b$ *i.e.*,

$$A = \sum_{x=a}^{x=b} \lim_{\delta x \to 0} f(x)\,\delta x = \int_a^b f(x)dx \qquad ...(7)$$

This is the equation used to find the area in between a curve $y = f(x)$ and X-axis for definite limits.

(2) **Work :** If $f(x)$ represents a force, acting along X-axis with variable magnitude depending on x, then the area in between the curve $y = f(x)$ and X-axis for the displacement $x = a$ to $x = b$ will give us the work done *i.e.*,

$$W = \int_a^b f(x)\,dx \qquad ...(8)$$

3·2. Double Integral

Let a function $f(x, y)$ be given in X-Y plane at all the points of a region R. If $\delta A = \delta x\,\delta y$ is an infinitesimal area around a point $P(x, y)$ in the region R, then

$$\lim_{\delta x \to 0} \lim_{\delta y \to 0} \Sigma\Sigma f(x, y)\delta x\,\delta y = \iint_R f(x,y)\,dx\,dy \qquad ...(1)$$

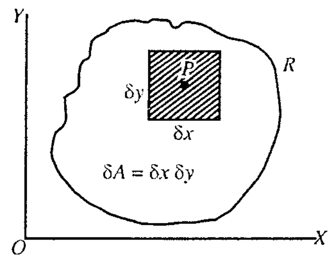


Fig. 3.2.

is called the **double integral** of the function $f(x, y)$ on the region R.

If the region R is bounded by the limits $x = x_1$, $x = x_2$, $y = y_1$, and $y = y_2$, then

$$\iint_R f(x,y)\,dx\,dy = \int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x,y)\,dx\,dy \qquad ...(2)$$

Now, suppose

$$I = \int_{y_1}^{y_2}\left[\int_{x_1}^{x_2} f(x,y)\,dx\right]dy \qquad ...(3)$$

where in general x_1 and x_2 may be the functions of the variable y. Double integral represented by equation (3) has been shown in the form of two single integrals. In this first we solve the integral in between the square brackets. Let its value be $F(y)$. Now the integral $\int_{y_1}^{y_2} F(y)\,dy$ is calculated. The integral represented by equation (3) is called **repeated integral**, because after performing one integration, the other integration is done.

Evaluation of Double Integral : (i) Let the double integral be as follows :

$$I = \int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x,y)\,dx\,dy \qquad ...(4)$$

We want to determine the value of this double integral. In order to solve it, we adopt the following procedure :

(a) If x_1, x_2, y_1 and y_2 are constants, then the order of integration is arbitrary *i.e.*,

$$I = \int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x,y)\,dx\,dy = \int_{x_1}^{x_2}\int_{y_1}^{y_2} f(x,y)\,dy\,dx \qquad ...(5)$$

(b) If x_1 and x_2 are the functions of y *i.e.*, $x_1 = \psi_1(y)$ and $x_2 = \psi_2(y)$ and y_1 and y_2 are constants, then

$$I = \int_{y_1}^{y_2}\int_{x_1=\psi_1(y)}^{x_2=\psi_2(y)} f(x, y)\;dx\;dy \qquad ...(6)$$

(c) If y_1 and y_2 are the functions of x *i.e.*, $y_1 = \psi_1(x)$ and $y_2 = \psi_2(x)$ and x_1 and x_2 are constants, then

$$I = \int_{x_1}^{x_2}\int_{y_1=\psi_1(x)}^{y_2=\psi_2(x)} f(x, y)\;dy\;dx \qquad ....(7)$$

It is clear from the above relations that *first the integration is done with respect to that variable whose limits are variable and then the integration is done with respect to the variable whose limits are constants.*

(ii) If $f(x, y) = 1$, then the double integral $\iint_A dx\,dy$ tells us the area of the region *i.e.*,

$$\int_{y_1}^{y_2}\int_{x_1}^{x_2} dy\,dx = A \text{ (Area)} \qquad ...(8)$$

(iii) In order to determine the value of the double integral

$$\int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x, y)\,dx\,dy = \int_{y_1}^{y_2}\left[\int_{x_1}^{x_2} f(x, y)\,dx\right]dy,$$

assuming y as constant we integrate $f(x, y)$ with respect to x. In this way, either we obtain a constant or a function of y, say $F(y)$ and then we integrate $F(y)$ with respect to y. Thus we obtain the value of the integral I.

3·3. Use of Plane Polar Coordinates in Solving Double Integrals—Area of a Surface

In order to solve several problems in physics, it is convenient to use polar coordinate system or spherical coordinate system instead of cartesian coordinate system. In two dimensions, there is the following relation between the cartesian coordinates (x, y) and plane polar coordinates (r, θ) of a point P :

$$x = r\cos\theta \text{ and } y = r\sin\theta \qquad \text{...(1)}$$

or $\quad r^2 = x^2 + y^2$ and $\tan\theta = y/x$ or $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1}(y/x) \quad$ ...(2)

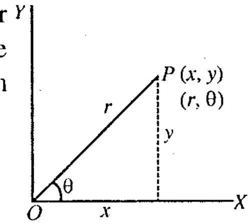


Fig. 3.3.

Hence, the double integral may be written as

$$\int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x, y)\,dx\,dy = \int_{\theta_1}^{\theta_2}\int_{r_1}^{r_2} f(r, \theta)\, r d\theta\, dr \qquad \text{...(3)}$$

where the limits for r and θ can be obtained by using the transformation equations (2).

Length of a curve and Area of a region : Let us consider a point Q close to the point $P(r, \theta)$, its polar coordinates will be $(r + dr, \theta + d\theta)$. The distance $PQ = ds$ will be the **line element [fig. 3.4]** :

$$ds^2 = dr^2 + r^2 d\theta^2 \text{ or } ds = \sqrt{dr^2 + r^2 d\theta^2} \qquad \text{....(4)}$$

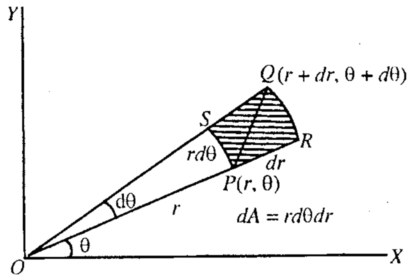


Fig. 3.4 : Area element in polar coordinates.

If in the r-θ plane, P and Q points are situated close to each other on a curved line AB **[fig. 3.5]**, then we may obtain the line element PQ as follows :

We differentiate eq. (1) and obtain

$dx = -r\sin\theta\, d\theta + dr\cos\theta$ and $dy = r\cos\theta\, d\theta + dr\sin\theta$

Now, $ds^2 = dx^2 + dy^2 = (-r\sin\theta\, d\theta + dr\cos\theta)^2 + (r\cos\theta\, d\theta + dr\sin\theta)^2$

$= dr^2 + r^2\, d\theta^2$

$$\therefore \quad ds = dx\sqrt{1 + (dy/dx)^2} = dr\sqrt{1 + r^2(d\theta/dr)^2} \qquad \text{...(5)}$$

We can obtain the length s of the curve AB **[fig. 3.5]** by calculating the integral :

$$s = \int_A^B dx\sqrt{1 + (dy/dx)^2} \text{ or } s = \int_A^B dr\sqrt{1 + r^2(d\theta/dr)^2} \qquad \text{...(6)}$$

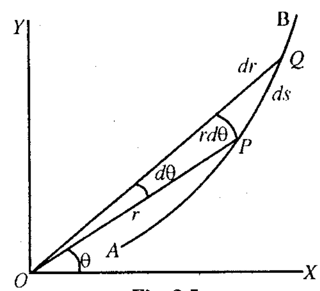


Fig. 3.5.

If we draw a straight line θ = constant and a circle of radius r = constant, and also a straight line $\theta + d\theta$ = constant and a circle of radius $r + dr$ = constant, then the area element $PSQR = dA$ **[fig. 3.4]** will be given by

$$dA = r\, d\theta\, dr \text{ or } dA = r\, dr\, d\theta \qquad \text{...(7)}$$

Now, the area of the region will be obtained by integrating eq. (7) for proper limits :

$$A = \iint dA = \int_{\theta_1}^{\theta_2}\int_{r_1}^{r_2} r\, dr\, d\theta \qquad \text{...(8)}$$

This is the equation (3) with $f(r, \theta) = 1$.

For example, the area of a circle or surface area of a disc of radius R can be obtained by using eq. (8). In this case the area element $dA = r\, d\theta\, dr = r\, dr\, d\theta$ is shown in **fig. 3.6** and the total area of the circle is obtained as

$$A = \int_0^{2\pi}\int_0^R r\, dr\, d\theta = \int_0^{2\pi}\left[\int_0^R r\, dr\right]d\theta = \int_0^{2\pi}\left[\frac{r^2}{2}\right]_0^R d\theta$$

$$= \int_0^{2\pi}\frac{R^2}{2}d\theta = \frac{R^2}{2}\int_0^{2\pi}d\theta = \frac{R^2}{2}[\theta]_0^{2\pi} = \pi R^2$$

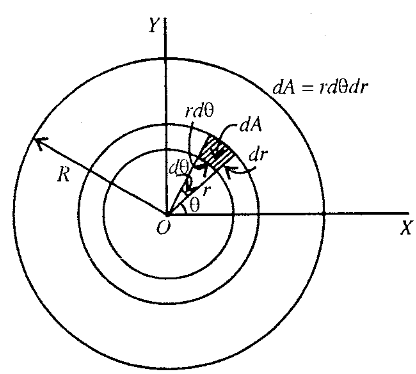


Fig. 3.6 : Calculation of area of a circle.

3·4. Triple Integral

Let us consider a function $f(x, y, z)$ in a finite region in three diemensional space. If $\delta V = \delta x\, \delta y\, \delta z$ is the infinitesimal volume element enclosing a point $P(x, y, z)$, then

$$\underset{\delta x \to 0}{\lim \Sigma}\ \underset{\delta y \to 0}{\lim \Sigma}\ \underset{\delta z \to 0}{\lim \Sigma}\ f(x,y,z)\delta x\, \delta y\, \delta z = \iiint_R f(x,y,z)\, dx\, dy\, dz \quad ...(1)$$

is called the **triple integral** of $f(x, y, z)$ in the region R.

If the region R is bound by the surfaces $x = x_1$, $x = x_2$, $y = y_1$, $y = y_2$ and $z = z_1$, $z = z_2$, then

$$\iiint_V f(x,y,z)\, dx\, dy\, dz = \int_{x_1}^{x_2}\int_{y_1}^{y_2}\int_{z_1}^{z_2} f(x,y,z)\, dx\, dy\, dz \quad ...(2)$$

Evaluation of triple integral : (i) If $x_1, x_2, y_1, y_2, z_1, z_2$ are constants, then the order of integration does not affect the evaluation of the integral. In such a case,

$$\int_{z_1}^{z_2}\int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x,y,z)\, dx\, dy\, dz = \int_{z_1}^{z_2}\int_{x_1}^{x_2}\int_{y_1}^{y_2} f(x,y,z)\, dx\, dy\, dz = \int_{x_1}^{x_2}\int_{y_1}^{y_2}\int_{z_1}^{z_2} f(x,y,z)\, dx\, dy\, dz \quad ...(3)$$

(ii) If $z_1 = \psi_1(x, y)$, $z_2 = \psi_2(x, y)$, $y_1 = \phi_1(x)$, $y_2 = \phi_2(x)$ and x_1, x_2 are constants, then

$$\int_{x_1}^{x_2}\int_{y_1}^{y_2}\int_{z_1}^{z_2} f(x,y,z)\, dx\, dy\, dz = \int_{x_1}^{x_2}\left[\int_{y_1=\phi_1(x)}^{y_2=\phi_2(x)}\left\{\int_{z_1=\psi_1(x,y)}^{z_2=\psi_2(x,y)} f(x,y,z)\, dz\right\}dy\right]dx \quad ...(4)$$

(iii) If $f(x, y, z) = 1$, then the triple integral gives us the volume *i.e.*,

$$\int_{z_1}^{z_2}\int_{y_1}^{y_2}\int_{x_1}^{x_2} dx\, dy\, dz = V \text{ (volume)} \quad ...(5)$$

These double and triple integrals are called **multiple integrals**.

3·5. Use of Spherical and Cylindrical Coordinates in Solving Triple Integrals

(1) Spherical coordinates : It is convenient to use spherical coordinates (r, θ, ϕ) for systems which possess spherical symmetry; for example in the study of the motion of an electron around a nucleus, the motion of a satellite around the earth etc.

In spherical coordinates the position of a particle P is represented by spherical coordinates (r, θ, ϕ), where r is called radial distance, θ polar angle and ϕ azimuthal angle. The polar angle θ is measured from Z-axis downwards and the azimuthal angle ϕ gives the position of that plane whose rotation is measured from X-axis [**fig. 3.7**]. Remember that the distance r can have values from 0 to ∞, angle θ from 0 to π and angle ϕ from 0 to 2π.

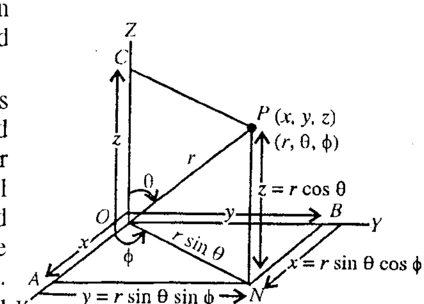


Fig. 3.7 : Cartesian and spherical coordinates.

The spherical coordinates (r, θ, ϕ) of a point P are related to its cartesian coordinates by the following equations :

$$x = OA = ON \cos\phi = r \sin\theta \cos\phi \qquad [\because ON = OP \sin\theta = r \sin\theta]$$

$$y = OB = ON \sin\phi = r \sin\theta \sin\phi,\ z = OC = OP \cos\theta = r\cos\theta$$

i.e., $$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta \quad ...(1)$$

Now, $$x^2 + y^2 + z^2 = r^2\sin^2\theta(\sin^2\phi + \cos^2\phi) + r^2\cos^2\theta = r^2\sin^2\theta + r^2\cos^2\theta = r^2$$

Hence, $$r = \sqrt{x^2+y^2+z^2}$$

and also $$\tan\theta = \frac{PC}{OC} = \frac{ON}{z} = \frac{\sqrt{x^2+y^2}}{z} \text{ and } \tan\phi = \frac{AN}{OA} = \frac{y}{x}$$

i.e., $$r = \sqrt{x^2+y^2+z^2},\ \theta = \tan^{-1}\frac{\sqrt{x^2+y^2}}{z} \text{ and } \phi = \tan^{-1}\frac{y}{x} \quad ...(2)$$

These are the reverse relations.

In this coordinate system a triple integral is written as :

$$\iiint_V f(x, y, z)\, dx\, dy\, dz = \iiint_V f(r,\theta,\phi)r^2 dr \sin\theta\, d\theta\, d\phi = \int_{\phi_1}^{\phi_2}\left[\int_{\theta_1}^{\theta_2}\left[\int_{r_1}^{r_2} f(r,\theta,\phi)r^2 dr\right]\sin\theta\, d\theta\right]d\phi$$

where the limits of integration for r, θ, ϕ can be obtained by the transformation equations (2). Here in spherical coordinate system, $dV = r^2\, dr \sin\theta\, d\theta\, d\phi$ is the volume element.

The sides of the volume element dV will be dr, $rd\theta$ and $r \sin d\phi$ and its value will be **[fig. 3.8]** :

$$dV = dr\; r\, d\theta\; r\sin\theta\, d\phi = r^2\, dr \sin\theta\, d\phi \qquad ...(3)$$

The line element in the spherical coordinate system is given by

$$ds^2 = dr^2 + r^2\, d\theta^2 + r^2 \sin^2\theta\, d\phi^2$$

or
$$ds = \sqrt{dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta\, d\phi^2} \qquad ...(4)$$

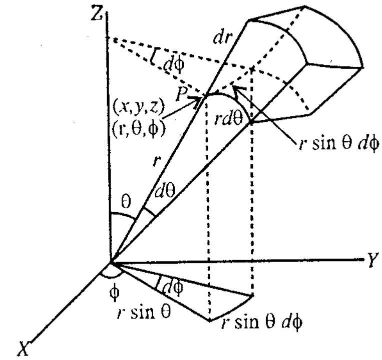


Fig. 3.8 : Volume elementary in spherical coordinates.

In several problems, involving spherical symmetry, we use spherical coordinate system directly. For example, here we deduce an expression for the volume of a sphere of radius R **[fig. 3.9]**.

$$V = \iiint dV = \iiint r^2 dr \sin\theta\, d\theta\, d\phi$$

$$= \int_0^{2\pi}\left[\int_0^{\pi}\left\{\int_0^{R} r^2 dr\right\}\sin\theta\, d\theta\right]d\phi$$

$$= \int_0^{2\pi}\left[\int_0^{\pi}\frac{R^3}{3}\sin\theta\, d\theta\right]d\phi \quad = \int_0^{2\pi}\frac{R^3}{3}[-\cos\theta\,]_0^{\pi}\, d\phi$$

$$= \int_0^{2\pi}\frac{R^3}{3}\cdot 2\,d\phi = \frac{2R^3}{3}[\phi]_0^{2\pi} = \frac{4}{3}\pi R^3 \qquad ...(5)$$

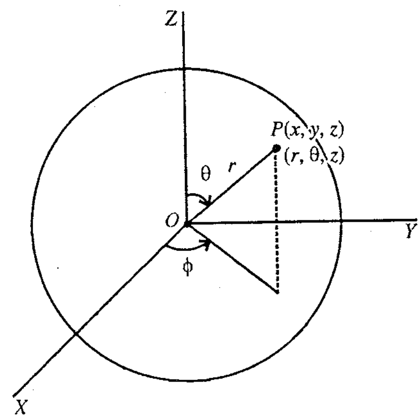


Fig. 3.9 : Calculation of the volume of sphere.

(2) Cylindrical coordinates : Consider the position of a particle P, located at a distance r from the origin O. This position P can be expressed in rectangular (x, y, z) or cylindrical (ρ, ϕ, z) coordinates as shown in **fig. 3.10.** The two systems of coordinates are related as

$$x = \rho\cos\phi, \qquad y = \rho\sin\phi, \qquad z = z \; ...(6)$$

Also $\rho = \sqrt{x^2 + y^2}$, $\tan\phi = \dfrac{y}{x}$ or $\phi = \tan^{-1}\dfrac{y}{x}$, $z = z$ ...(7)

Observe that in cylindrical coordinates ρ has replaced r and ϕ has replaced θ of plane polar coordinates.

In the cylindrical coordinate system the triple integral is written as

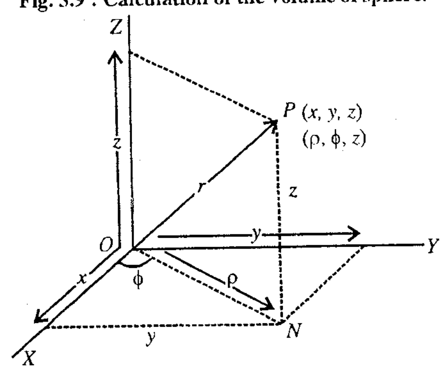


Fig. 3.10 : Cartesian and cylindrical coordinates.

$$\iiint_V f(x,y,z)\,dx\,dy\,dz = \iiint_V f(\rho,\phi,z)\,\rho\, d\rho\, d\phi\, dz$$

$$= \int_{z_1}^{z_2}\left[\int_{\phi_1}^{\phi_2}\left\{\int_{\rho_1}^{\rho_2} f(\rho,\phi,z)\,\rho\, d\rho\right\}d\phi\right]dz \; ...(8)$$

where the limits of integration can be obtained by transformation equations (7).

In the cylindrical coordinate system, the sides of the volume element are $d\rho$, $\rho\, d\phi$ and dz **[fig. 3.11]** and its value is

$$dV = d\rho\; \rho\, d\phi\; dz \qquad ...(9)$$

The line element ds in this system is given by

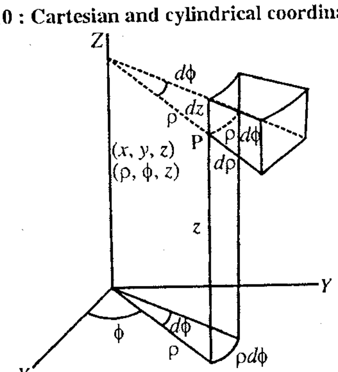


Fig. 3.11 : Volume element in cylindrical coordinates.

$$ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2$$

or $$ds = \sqrt{d\rho^2 + \rho^2 d\phi^2 + dz^2} \quad \text{...(10)}$$

Ex. 1. *Find the line integral* $\int_C xy^3\, ds$, *where C is the segment of a straight line* $y = 2x$ *in x-y plane from point A* $(-1, -2, 0)$ *to point B* $(1, 2, 0)$. **(Sagar 2003)**

Sol. The line element ds is

$$ds = \sqrt{(dx)^2 + (dy)^2} = dx\sqrt{1 + (dy/dx)^2}$$

Here $y = 2x$; $dy/dx = 2$ and $ds = dx\sqrt{1 + (2)^2} = \sqrt{5}\, dx$

Hence $$\int_C xy^3 ds = \int_{-1}^{1} x(2x)^3\sqrt{5}\, dx = 8\sqrt{5}\left(\frac{x^5}{5}\right)_{-1}^{1} = \frac{8}{\sqrt{5}}[(1)^5 - (-1)^5] = \frac{16}{\sqrt{5}} = \mathbf{7.16.}$$

Ex. 2. *Find the value of the integral* $I = \int_0^a \int_0^b xy\, dx\, dy$ **(Agra 2004)**

Sol. $$I = \int_0^a \int_0^b (y\, dy)x\, dx = \int_0^a [y^2/2]_0^b x\, dx = \int_0^a (b^2/2)x\, dx = \frac{b^2}{2}\left[\frac{x^2}{2}\right]_0^a = \frac{\mathbf{a^2 b^2}}{\mathbf{4}}.$$

Ex. 3. *Find the value of the double integral* : $\int_{y=1}^{2} \int_{x=0}^{3y} y\, dy\, dx$.

Sol. $$\int_{y=1}^{2} \int_{x=0}^{3y} y\, dy\, dx = \int_1^2 \left\{y\int_0^{3y} dx\right\} dy = \int_1^2 3y^2 dy = (y^3)_1^2 = \mathbf{7.}$$

Ex. 4. *Find the value of* : $\int_0^2 \int_1^3 xy(1 + x + y)dx\, dy$.

Sol. $$\int_0^2 \int_1^3 xy(1 + x + y)dx\, dy = \int_0^2 \left[\int_1^3 xy(1 + x + y)dx\right] dy = \int_0^2 \left[\int_1^3 (xy + x^2 y + xy^2)dx\right] dy$$

$$= \int_0^2 \left[\frac{x^2 y}{2} + \frac{x^3 y}{3} + \frac{x^2 y^2}{2}\right]_1^3 dy = \int_0^2 \left[\left(\frac{9y}{2} + \frac{27y}{3} + \frac{9y^2}{2}\right) - \left(\frac{y}{2} + \frac{y}{3} + \frac{y^2}{2}\right)\right] dy$$

$$= \int_0^2 \left[4y + \frac{26y}{3} + 4y^2\right] dy = \left[\frac{4y^2}{2} + \frac{26y^2}{3.2} + \frac{4y^3}{3}\right]_0^2 = \left[2y^2 + \frac{13}{3}y^2 + \frac{4y^3}{3}\right]_0^2$$

$$= \left[\frac{19y^2}{3} + \frac{4y^3}{3}\right]_0^2 = \frac{76}{3} + \frac{32}{3} = \frac{108}{3} = \mathbf{36.}$$

Ex. 5. *Find the value of* : $\int_0^1 \int_0^{x^2} e^{y/x} dy\, dx$.

Sol. $$\int_0^1 \int_0^{x^2} e^{y/x} dy\, dx = \int_0^1 \left[\int_0^{x^2} e^{y/x} dy\right] dx = \int_0^1 \left[\frac{e^{y/x}}{1/x}\right]_0^{x^2} dx = \int_0^1 [xe^{y/x}]_0^{x^2} dx$$

$$= \int_0^1 [xe^{x^2/x} - xe^{0/x}]\, dx = \int_0^1 (xe^x - x)dx = \int_0^1 \underset{\text{I}}{x}\underset{\text{II}}{(e^x - 1)}dx$$

(Integrating by parts)

$$= [x(e^x - x)]_0^1 - \int_0^1 1(e^x - x)dx = e - 1 - \left[e^x - \frac{x^2}{2}\right]_0^1$$

$$= e - 1 - (e - \tfrac{1}{2} - 1) = \tfrac{1}{2}.$$

Ex. 6. *Find the value of the integral* $I = \int_0^3\int_0^2\int_0^1 (x+y+z)\,dx\,dy\,dz$

Sol. $I = \int_0^3\int_0^2\int_0^1 (x+y+z)\,dx\,dy\,dz = \int_0^3\int_0^2\left[\int_0^1 (x+y+z)\,dz\right]dx\,dy = \int_0^3\int_0^2\left[xz + yz + \frac{z^2}{2}\right]_0^1 dx\,dy$

$= \int_0^3\int_0^2\left(x + y + \frac{1}{2}\right)dx\,dy = \int_0^3\int_0^2\left[\left(x + y + \frac{1}{2}\right)dy\right]dx = \int_0^3\left[xy + \frac{y^2}{2} + \frac{1}{2}y\right]_0^2 dx$

$= \int_0^3 (2x+3)\,dx = [x^2 + 3x]_0^3 = 9 + 9 = $ **18.**

Ex. 7. *Calculate* $I = \int_0^a\int_0^x\int_0^{x+y} e^{x+y+z}\,dx\,dy\,dz$ **(Jabalpur 2003)**

Sol. $I = \int_0^a\int_0^x\left\{\int_0^{x+y} e^{x+y+z}\,dz\right\}dx\,dy = \int_0^a\int_0^x (e^{x+y+z})_0^{x+y}\,dx\,dy$

$= \int_0^a\int_0^x (e^{2(x+y)} - e^{x+y})\,dx\,dy = \int_0^a\left\{\int_0^x (e^{2(x+y)} - e^{(x+y)})\,dy\right\}dx = \int_0^a\left\{\frac{1}{2}e^{2(x+y)} - e^{(x+y)}\right\}_0^x dx$

$= \int_0^a\left[\frac{1}{2}(e^{4x} - e^{2x}) - (e^{2x} - e^x)\right]dx = \int_0^a\left(\frac{1}{2}e^{4x} - \frac{3}{2}e^{2x} + e^x\right)dx = \left(\frac{1}{2}\frac{e^{4x}}{4} - \frac{3}{2}\frac{e^{2x}}{2} + e^x\right)_0^a$

$= \frac{1}{8}(e^{4a} - e^0) - \frac{3}{4}(e^{2a} - e^0) + (e^a - e^0) = \frac{1}{8}e^{4a} - \frac{3}{4}e^{2a} + e^a - \frac{3}{8}$

$= \boldsymbol{\frac{1}{8}(e^{4a} - 6e^{2a} + 8e^a - 3)}.$

Ex. 8. *Find the value of* $I = \int_0^\pi\int_{r=0}^{a\cos\theta} r\,dr\,d\theta$ **(Ujjain 2004)**

Sol. $I = \int_0^\pi\left\{\int_0^{a\cos\theta} r\,dr\right\}d\theta = \int_0^\pi\left[\frac{r^2}{2}\right]_0^{a\cos\theta} d\theta = \int_0^\pi \frac{1}{2}a^2\cos^2\theta\,d\theta$

$= \frac{a^2}{4}\int_0^\pi 2\cos^2\theta\,d\theta = \frac{a^2}{4}\int_0^\pi (1+\cos 2\theta)\,d\theta = \frac{a^2}{4}\left[\theta + \frac{\sin 2\theta}{2}\right]_0^\pi = \boldsymbol{\frac{a^2\pi}{4}}.$

Ex. 9. *Solve* : $I = \int_0^\pi\int_0^{a(1+\cos\theta)} r^3\sin\theta\,\cos\theta\,d\theta\,dr.$

Sol. $I = \int_0^\pi \sin\theta\,\cos\theta\left\{\int_0^{a(1+\cos\theta)} r^3 dr\right\}d\theta = \int_0^\pi \sin\theta\,\cos\theta\left[\frac{r^4}{4}\right]_0^{a(1+\cos\theta)} d\theta$

$= \frac{a^4}{4}\int_0^\pi (1+\cos\theta)^4\sin\theta\,\cos\theta\,d\theta$

Let $1 + \cos\theta = t$, then $-\sin\theta\,d\theta = dt$ and hence

$I = \frac{a^4}{4}\int_2^0 t^4(t-1)(-dt) = \frac{a^4}{4}\int_0^2 (t^5 - t^4)\,dt = \frac{a^4}{4}\left[\frac{t^6}{6} - \frac{t^5}{5}\right]_0^2 = \boldsymbol{\frac{16}{15}a^4}.$

Ex. 10. *Calculate* $I = \int_0^{2\pi}\int_0^\pi\int_0^a r^2\sin\theta\,d\theta\,dr\,d\phi$. **(Jabalpur 2004, 03)**

Sol. $I = \int_0^{2\pi}\left\{\int_0^\pi\left(\int_0^a r^2 dr\right)\sin\theta\,d\theta\right\}d\phi = \int_0^{2\pi}\left(\int_0^\pi \frac{a^3}{3}\sin\theta\,d\theta\right)d\phi = \int_0^{2\pi}\left(\frac{a^3}{3}(-\cos\theta)\right)_0^\pi d\phi$

$= \int_0^{2\pi}\frac{2a^3}{3}d\phi = \frac{2a^3}{3}(\phi)_0^{2\pi} = \boldsymbol{\frac{4}{3}\pi a^3}.$

Ex. 11. *Find the value of* $I = \int_0^{\pi/2} d\theta\int_0^{a\sin\theta} dr\int_0^{\frac{a^2-r^2}{a}} r\,dz.$

Sol. $I = \int_0^{\pi/2} d\theta\int_0^{a\sin\theta} dr\int_0^{\frac{a^2-r^2}{a}} r\,dz = \int_0^{\pi/2} d\theta\int_0^{a\sin\theta} dr\,(rz)_0^{\frac{a^2-r^2}{a}} = \int_0^{\pi/2} d\theta\int_0^{a\sin\theta}\frac{r(a^2-r^2)}{a}dr$

$$= \frac{1}{a}\int_0^{\pi/2} d\theta \int_0^{a\sin\theta} (ra^2 - r^3)dr = \frac{1}{a}\int_0^{\pi/2} d\theta \left[\frac{r^2a^2}{2} - \frac{r^4}{4}\right]_0^{a\sin\theta}$$

$$= \frac{1}{a}\int_0^{\pi/2}\left[\frac{a^4\sin^2\theta}{2} - \frac{a^4\sin^4\theta}{4}\right]d\theta = \frac{a^3}{4}\int_0^{\pi/2}(2\sin^2\theta - \sin^4\theta)\,d\theta$$

$$= \frac{a^3}{4}\left[\frac{\pi}{2} - \frac{3\pi}{16}\right] = \frac{5a^3\pi}{64}.$$

EXERCISE 3·1

Find the value of integrals :

1. $\int_{x=0}^{1}\int_{y=2}^{4} 3x\,dy\,dx$ **[Ans.** 3.**]**
2. $\int_{y=0}^{2}\int_{x=2y}^{4} dx\,dy$ **[Ans.** 4.**]**
3. $\int_0^2\int_0^2 (x+y)dx\,dy$ **[Ans.** 8.**]**
4. $\int_0^a\int_0^b (x^2+y^2)dx\,dy$ **[Ans.** $\frac{ab}{3}(a^2+b^2)$.**]**
5. $\int_0^1\int_0^1 \frac{dx\,dy}{\sqrt{1-x^2}\sqrt{1-y^2}}$ **[Ans.** $\frac{\pi^2}{4}$.**]**
6. $\int_1^2\int_0^x \frac{dx\,dy}{x^2+y^2}$ **[Ans.** $\frac{\pi}{4}\log 2$.**]**
7. $\int_1^a\int_1^b \frac{dx\,dy}{xy}$ **[Ans.** $\log a \,.\, \log b$.**]**
8. $\int_0^{\pi/2}\int_{\pi/2}^{\pi} \cos(x+y)dy\,dx.$ **[Ans.** -2.**]**
9. $\int_0^{\pi/2}\int_{\pi/2}^{\pi} \sin(x+y)dy\,dx.$ **[Ans.** 0.**]**
10. $\int_1^2\int_3^4 (xy+e^y)dy\,dx.$ **[Ans.** $\frac{21}{4}+e^4-e^3$.**]**
11. $\int_{-1}^{2}\int_{x^2}^{x+2} dy\,dx.$ **[Ans.** $\frac{9}{2}$.**]**
12. $\int_0^1\int_y^{y^2+1} x^2y\,dx\,dy.$ **[Ans.** $\frac{67}{120}$.**]**
13. $\int_0^1\int_x^{\sqrt{x}} (x^2+y^2)\,dy\,dx.$ **[Ans.** $\frac{8}{35}$.**]**
14. $\int_0^1\int_0^2\int_1^2 x^2yz\,dz\,dy\,dx.$ **[Ans.** 1.**]**
15. $\int_0^a\int_0^{\sqrt{a^2-x^2}}\int_y^{2y} \sqrt{a^2-y^2}\,dx\,dy\,dz.$ **[Ans.** $\frac{a^4}{4}$.**]** **(Jabalpur 2004)**
16. $\int_0^1\int_0^1\int_0^1 e^{x+y+z}dx\,dy\,dz.$ **[Ans.** $(e-1)^3$.**]**
17. $\int_{-c}^{c}\int_{-b}^{b}\int_{-a}^{a} (x^2+y^2+z^2)dz\,dy\,dx.$ $\left[\textbf{Ans. } \frac{8}{3}abc(a^2+b^2+c^2)\right]$
18. $\int_0^{\pi/2}\int_0^{4\sin\theta} r\,dr\,d\theta.$ **[Ans.** 2π.**]**
19. $\int_0^{2\pi}\int_0^{\pi}\int_0^{a} r^2\sin\theta\,dr\,d\theta\,d\phi.$ **[Ans.** $\frac{4}{3}\pi a^3$.**]**
20. $\int_0^{\pi/2}\int_1^{a\cos\theta} r\sin\theta\,d\theta\,dr.$ **[Ans.** $\frac{a^2}{6}$.**]**

3·6. Applications of Multiple Integrals

Double integrals have various geometrical and physical applications.

(1) To determine the area in a plane by double integral : The area of a region R in XY plane is determined by

$$A = \iint dx\,dy \qquad ...(1)$$

In polar coordinates,

$$A = \iint r\,d\theta\,dr \qquad ...(2)$$

(2) To determine the volume by multiple integral : (i) The volume between the surface $z = f(x, y)$ and XY-plane can be determined by the double integral :

$$V = \iint_A f(x, y)\,dx\,dy \qquad ...(3)$$

where the area A is in the XY-plane. This has been explained in the next article.

(ii) If in a triple integral, we put $f(x, y, z) = 1$, then we obtain the volume for the given limits *i.e.*,

$$V = \iiint_V dx\, dy\, dz \qquad ...(4)$$

(3) To determine the mass by multiple integrals : Suupose in XY-plane the mass per unit area of a plate is $\sigma = f(x, y)$, then the mass of the plate will be given by

$$M = \iint_A \sigma\, dx\, dy = \iint_A f(x, y)\, dx\, dy \qquad ...(5)$$

For a solid, if the density at a point (x, y, z) is $\rho = f(x, y, z)$, then the mass of the solid is given by

$$M = \iiint_V \rho\, dx\, dy\, dz = \iint_V f(x, y, z)\, dx\, dy\, dz \qquad ...(6)$$

(4) To determine the centre of mass : In XY-plane, the centre of mass (X, Y) of a plate, having mass per unit area $\sigma = f(x, y)$, is given by

$$X = \frac{1}{M}\iint_A x\, f(x, y,)\, dx\, dy, \quad Y = \frac{1}{M}\iint_R y\, f(x, y)\, dx\, dy \qquad ...(7)$$

where M is the mass of the plate.

For a solid of mass M and density $\rho = f(x, y, z)$, the centre of mass is given by

$$X = \frac{1}{M}\iiint_V x\, f(x, y, z)\, dx\, dy\, dz, \; Y = \frac{1}{M}\iiint_V y\, f(x, y, z)\, dx\, dy\, dz \text{ and } Z = \frac{1}{M}\iiint_V z\, f(x, y, z)\, dx\, dy\, dz \quad ...(8)$$

3·7. Geometrical Interpretation of Integration of the Function of two Variables (To find the volume a solid)

Let $u = f(x, y)$ be a function of two variables x and y. We take $z = u$, then $z = f(x, y)$. We want to find out the volume of a cylinder in **fig. 3.12** between the surfaces $z = f(x, y)$ and XY-plane. We cut the XY-plane into little rectangles $\delta A = \delta x\, \delta y$ [**fig. 3.12**]. Above each $dxdy$ we take a rod or column of height z, which reaches upto the plane $z = f(x, y)$. The sum of the volumes all these rectangular section rods will give us the volume of the cylinder in the limits $\delta x \to 0$ and $\delta y \to 0$ *i.e.*,

$$V = \sum_{\substack{\delta x \to 0 \\ \delta y \to 0}} z\, \delta x\, \delta y = \iint_A f(x, y)\, dx\, dy \qquad ...(1)$$

This discussion gives us the geometrical interpretation of a double integral.

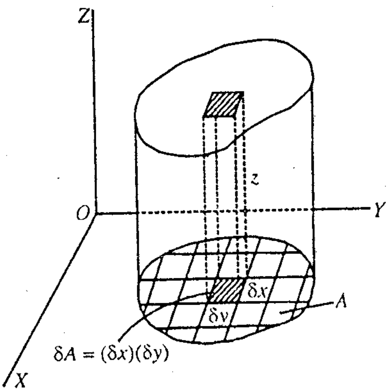


Fig. 3.12 : Calculation of volume.

Example : *Find the volume of the solid below the plane $z = 1 + y$ bounded by the coordinate planes ($x = 0$ and $y = 0$) and the vertical plane $2x + y = z$* [**fig. 3.13**].

Sol. The volume of the solid is given by

$$V = \iint_A z\, dx\, dy \quad \text{or} \quad V = \iint f(x, y)\, dx\, dy \qquad ...(2)$$

Here, $z = 1 + y$, hence

$$V = \iint_A (1 + y)\, dx\, dy \qquad ...(3)$$

where A is the area of the triangle AOB in XOY plane. We adopt the following procedure to determine the volume of the cylinder :

(1) In the area of the triangle AOB, we take an elementary area $\delta A = \delta x\, \delta y$ and consider a column of height $z = 1 + y$. Its volume will be

$$z\, \delta x\, \delta y = (1 + y)\, \delta x\, \delta y \qquad ...(4)$$

(2) Next we find the volume of the slab $PQRS$, standing on the strip PS *i.e.*,

$$dV = \delta x \int_P^S (1 + y)\, dy = \delta x \int_0^{2-2x} (1 + y)\, dy$$

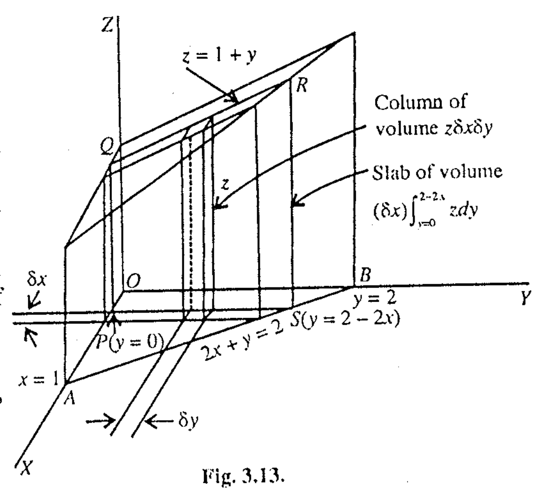


Fig. 3.13.

$$= \delta x\left[y + \frac{y^2}{2}\right]_0^{2-2x} = \delta x\,(4 - 6x + 2x^2) \qquad ...(5)$$

where for constant x, y coordinates of P and S are $y_1 = 0$ and $y_2 = 2 - 2x$ respectively.

(3) Now if we add the volumes of all the slabs from $x = 0$ to $x = 1$, we get the volume of the solid *i.e.*,

$$V = \int_0^1 dx \int_0^{2-2x} (1 + y)\,dy = \int_0^1 (4 - 6x + 2x^2)\,dx = \frac{5}{3}$$

where the limits of x are from $x_1 = 0$ to $x_2 = 1$, because for $y = 0$, from $2x + y = 2$, we have $x = 1$.

In short,

$$V = \int_0^1\int_0^{2-2x} (1 + y)\,dx\,dy = \int_0^1\left[\int_0^{2-2x}(1 + y)\,dy\right]dx = \int_0^1\left[y + \frac{y^2}{2}\right]_0^{2-2x} dx.$$

$$= \int_0^1\left[(2 - 2x) + \frac{(2-2x)^2}{2}\right]dx = \int_0^1 (4 - 6x + 2x^2)\,dx = \left[4x + \frac{6x^2}{2} + \frac{2x^3}{3}\right]_0^1$$

$$= 4 - 3 + \frac{2}{3} = \frac{5}{3}$$

The above integral is an example of repeated integral.

Determination of the volume of above solid by triple integral.

$$V = \iiint_V dx\,dy\,dz = \int_{x=0}^1\int_{y=0}^{2-2x}\left[\int_{z=0}^{1+y} dz\right]dx\,dy = \int_{x=0}^1\int_{y=0}^{2-2x}(1 + y)\,dy\,dz = \frac{5}{3}$$

where first we obtained the volume of a column of base $\delta x \delta y$ or $dxdy$ with the limits of z from $z = 0$ to $z = 1 + y$. Next we have done the integration as given above.

Ex. 1. *Determine the mass of a rectangular plate bounded by $x = 0$, $x = 2$, $y = 0$ and $y = 1$. The density or mass per unit area of the plate is $\sigma = f(x, y) = xy$.*

Sol. The mass of an infinitesimal rectangle $dx\,dy$ is

$$dM = \sigma\,dx\,dy = xy\,dx\,dy$$

$$\therefore \quad \text{Mass of the plate, } M = \iint_A xy\,dx\,dy = \int_{x=0}^2\int_{y=0}^1 xy\,dx\,dy$$

$$= \int_{x=0}^2 x^2 dx \int_{y=0}^1 y\,dy = \left[\frac{x^2}{2}\right]_0^2\left[\frac{y^2}{2}\right]_0^1 = \mathbf{1.}$$

Ex. 2. *A curve $y = x^2$ is given from $x = 0$ to $x = 1$. Calculate the area between X-axis and this curve. If this area is taken in the form of a plate, then find its mass for $\rho = f(x, y) = xy$.*

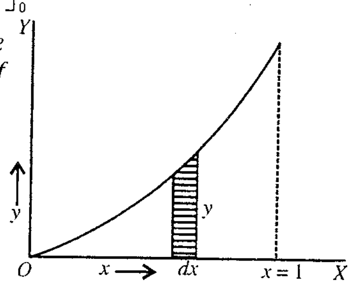


Fig. 3.14.

Sol. Area, $A = \int_0^1 y\,dx = \int_0^1 x^2 dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$

This area can be obtained as double integral

$$A = \iint_A dA = \iint dx\,dy = \int_{x=0}^1\int_{y=0}^{x^2} dy\,dx = \int_0^1 x^2 dx = \frac{1}{3}$$

$$\text{Mass, } M = \iint \rho\,dx\,dy = \int_{x=0}^1\int_{y=0}^{x^2} xy\,dy\,dx = \int_0^1 x\,dx\left[\frac{y^2}{2}\right]_0^{x^2} = \int_0^1 \frac{x^5}{2}\,dx = \frac{\mathbf{1}}{\mathbf{12}}.$$

Ex. 3. *Find the area between the parabola $y = 4x - x^2$ and the straight line $y = x$.*

Sol. Here $y = 4x - x^2$ and $y = x$ and hence $4x - x^2 = x$ or $3x - x^2 = 0$ or $x(3 - x) = 0$ *i.e.*, $x_1 = 0$ and $x_2 = 3$. This meains that the parabola $y = 4x - x^2$ and the straight line intersect at $x_1 = 0$ and $x_2 = 3$ and (from $y = x$) $y_1 = 0$ and $y_2 = 3$.

In order to find to the desired area, first we determine the area of the shaded strip of area $dx\int_{y_1=x}^{y_2=4x-x^2} dy$ and then integrate it for x from $x_1 = 0$ to $x_2 = 3$. Thus

$$A = \int_0^3\left[\int_{y_1=x}^{y_2=4x-x^2} dy\right] dx = \int_0^3 [y]_x^{4x-x^2}\, dx$$

$$= \int_0^3 (3x - x^2)\,dx = \left[\frac{3}{2}x^2 - \frac{1}{3}x^3\right]_0^3 = \frac{27}{2} - 9 = \mathbf{4.5.}$$

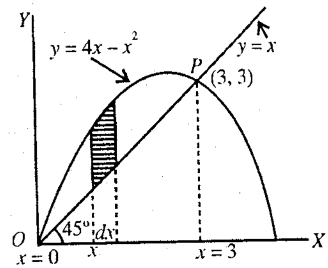


Fig. 3.15.

EXERCISE 3·2

1. Find the value of the double integral given by

$$I = \iint_A (2x - 3y)\, dx\, dy$$

where A is the area of a triangle with vertices (0, 0), (2, 1) and (2, 0) **(fig. 3.16)**.

[**Hint.** $I = \int_{x=0}^{2}\int_{y=0}^{x/2} (2x - 3y)\, dy\, dx$.]

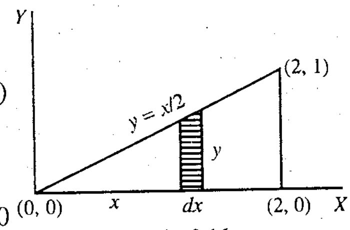


Fig. 3.16.

2. If the area between parabola $y = x^2$ and the straight line $2x - y + 8 = 0$ is A, then find the integral

$$I = \iint_A x\, dx\, dy$$

[**Ans.** 36.]

3. Find the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$.
[**Ans.** $7.5a^2$.]

4. Find the mass of a plate in the form of quadrant of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $\rho = kxy$.

[**Ans.** $\frac{1}{8}ka^2b^2$.]

3·8. Jacobian

If x and y are functions of two independent variables u and v *i.e.*, $x = x(u, v)$ and $y = y(u, v)$, then

$$J = J\left(\frac{x, y}{u, v}\right) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{vmatrix} \qquad ...(1)$$

is called the **Jacobian** of x, y with respect to u, v. If (x, y) and (u, v) represent coordinates of a point in two coordinate systems, then J is called the **Jacobian of transformation** from the coordinates (x, y) to (u, v).

Now, if in three dimensions, x, y and z are the functions of three independent variables u, v and w *i.e.*, $x = x(u, v, w)$, $y = y(u, v, w)$ and $z = z(u, v, w)$, then the **Jacobian** of x, y, z with respect to u, v, w is

$$J = J\left(\frac{x, y, z}{u, v, w}\right) = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \partial x/\partial u & \partial x/\partial v & \partial x/\partial w \\ \partial y/\partial u & \partial y/\partial v & \partial y/\partial w \\ \partial z/\partial u & \partial z/\partial v & \partial z/\partial w \end{vmatrix} \qquad ...(2)$$

Similarly, if $x_1, x_2, x_3, ..., x_n$ are the functions of n independent variables $u_1, u_2, u_3, ..., u_n$, then the Jacobian of $x_1, x_2, ...x_n$ with respect to the variables $u_1, u_2, ..., u_n$ is

$$J = J\left(\frac{x_1, x_2, \ldots, x_n}{u_1, u_2, \ldots, u_n}\right) = \frac{\partial(x_1, x_2, \ldots, x_n)}{\partial(u_1, u_2, \ldots, u_n)} = \begin{vmatrix} \partial x_1/\partial u_1 & \partial x_1/\partial u_2 & \ldots & \partial x_1/\partial u_n \\ \partial x_2/\partial u_1 & \partial x_2/\partial u_2 & \ldots & \partial x_2/\partial u_n \\ \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots \\ \partial x_n/\partial u_1 & \partial x_n/\partial u_2 & \ldots & \partial x_n/\partial u_n \end{vmatrix} \quad \ldots(3)$$

Some Theorems on Jacobian :

Here we have giving the statements of some theorems related to Jacobian :

(1) If x and y are the functions of two independent variables, then

$$\frac{\partial(x, y)}{\partial(u, v)} \frac{\partial(u, v)}{\partial(x, y)} = 1 \quad \ldots(4)$$

Now, if $u = u\,(s, t)$ and $v = v\,(s, t)$, then

$$\frac{\partial(x, y)}{\partial(u, v)} \frac{\partial(u, v)}{\partial(s, t)} = \frac{\partial(x, y)}{\partial(s, t)} \quad \ldots(5)$$

(2) For implicit functions, suppose we are given following relations between x, y, z and u, v, w. $f_1(x, y, z, u, v, w) = 0$, $f_2(x, y, z, u, v, w) = 0$, $f_3(x, y, z, u, v, w) = 0$, then

$$J = (-1)^3 \frac{\dfrac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)}}{\dfrac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)}} \quad \ldots(6)$$

3·9. Change of Variables of Double and Triple Integrals—Applications of Jacobian

Sometimes it is convenient to evaluate multiple integrals by changing the variables. The process of changing the variables in a multiple integral with the help of Jacobian is known as the **transformation of multiple integrals.**

Let the double integral be

$$I = \iint f(x, y)\, dx\, dy \quad \ldots(1)$$

If x and y are the functions of two new variables u and v *i.e.*, $x = x\,(u, v)$ and $y = y\,(u, v)$, then the double integral (1) will be transformed into u, v variables with the help of Jacobian as

$$I = \iint f[x(u, v), y(u, v)]\, |J|\, du\, dv \quad \ldots(2)$$

where $J = \dfrac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{vmatrix}$ is the **Jacobian** of x, y with respect to u, v.

Similarly, the triple integral

$$I = \iiint f(x, y, z)\, dx\, dy\, dz \quad \ldots(3)$$

is transformed from x, y, z variables to u, v, w variables as

$$I = \iiint f(u, v, w)\, |J|\, du\, dv\, dw \quad \ldots(4)$$

where $x = x\,(u, v, w)$, $y = y\,(u, v, w)$ and $z = z\,(u, v, w)$.

Here,
$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \partial x/\partial u & \partial x/\partial v & \partial x/\partial w \\ \partial y/\partial u & \partial y/\partial v & \partial y/\partial w \\ \partial z/\partial u & \partial z/\partial v & \partial z/\partial w \end{vmatrix}$$

is the **Jacobian** of transformation from x, y, z system to u, v, w system.

Examples : (1) To find the area element in polar coordinates (r, θ) **from cartesian coodinates** (x, y) :

The cartesian coordinates (x, y) and polar coordinates (r, θ) of a point in a plane are related as

$$x = r\cos\theta \text{ and } y = r\sin\theta \quad \ldots(5)$$

We know how to find the area element $dA = r\,dr\,d\theta$ in polar coordinates by **geometrial method.** In the **mathematical method,** we obtain the area element in polar coordinates from cartesian coordinates as

$$dA = |J|\,dr\,d\theta \qquad ...(6)$$

where $|J|$ is the absolute value of the Jacobian. Now,

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \partial x/\partial r & \partial x/\partial\theta \\ \partial y/\partial r & \partial y/\partial\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= r\cos^2\theta + r\sin^2\theta = r(\cos^2\theta + \sin^2\theta) = r \qquad ...(7)$$

Hence, from eq. (6), we have

$$dA = r\,dr\,d\theta \qquad ...(8)$$

This can be used to find the area of a circle of radius R as

$$A = \iint dx\,dy = \iint |J|\,dr\,d\theta = \int_0^{2\pi}\int_0^R r\,dr\,d\theta = \int_0^{2\pi}\frac{R^2}{2}d\theta = \pi R^2 \qquad ...(9)$$

If we want to find area of a circle by using cartesian coordinates, it will be a little difficult process.

(2) To transform a multiple integral from cartesian coordinates (x, y, z) **to spherical coodinates** (r, θ, ϕ) :

The relations between cartesian coordinates and spherical coordinates are given by

$$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta \qquad ...(10)$$

The transformation of multiple integral from (x, y, z) coordinates to (r, θ, ϕ) coordinates is

$$\iiint f(x, y, z)\,dx\,dy\,dz = \iiint f(r, \theta, \phi)|J|\,dr\,d\theta\,d\phi \qquad ...(11)$$

$$\text{Here } J = \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \begin{vmatrix} \partial x/\partial r & \partial x/\partial\theta & \partial x/\partial\phi \\ \partial y/\partial r & \partial y/\partial\theta & \partial y/\partial\phi \\ \partial z/\partial r & \partial z/\partial\theta & \partial z/\partial\phi \end{vmatrix} = \begin{vmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix} = r^2\sin\theta \qquad ...(12)$$

$$\text{Hence,}\quad \iiint f(x, y, z)\,dx\,dy\,dz = \iiint f(r, \theta, \phi)r^2\sin\theta\,dr\,d\theta\,d\phi \qquad ...(13)$$

(3) To transform a multiple integral from cartesian coordinates (x, y, z) **to cylindrical coordinates** (ρ, ϕ, z) : The transformation equations in two coordinate systems are

$$x = \rho\cos\phi,\ y = \rho\sin\phi,\ z = z \qquad ...(14)$$

The transformation of multiple integral from (x, y, z) coordinates to (ρ, ϕ, z) coordinates is given by

$$\iiint f(x, y, z)\,dx\,dy\,dz = \iiint f(\rho, \phi, z)|J|\,d\rho\,d\phi\,dz \qquad ...(15)$$

$$\text{Now,}\quad J = \begin{vmatrix} \partial x/\partial\rho & \partial x/\partial\phi & \partial x/\partial z \\ \partial y/\partial\rho & \partial y/\partial\phi & \partial y/\partial z \\ \partial z/\partial\rho & \partial z/\partial\phi & \partial z/\partial z \end{vmatrix} = \begin{vmatrix} \cos\phi & -\rho\sin\phi & 0 \\ \sin\phi & \rho\cos\phi & 0 \\ 0 & 0 & 1 \end{vmatrix} = \rho(\cos^2\phi + \sin^2\phi) = \rho$$

$$\text{Hence,}\quad \iiint f(x, y, z)\,dx\,dy\,dz = \iiint f(\rho, \phi, z)\rho\,d\rho\,d\phi\,dz \qquad ...(16)$$

(4) To find the volume element from cartesian coordinates (x, y, z) **to spherical coordinates :** The volume element dV is given by

$$dV = dx\,dy\,dz = |J|\,dr\,d\theta\,d\phi \qquad ...(17)$$

$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \begin{vmatrix} \partial x/\partial r & \partial x/\partial\theta & \partial x/\partial\phi \\ \partial y/\partial r & \partial y/\partial\theta & \partial y/\partial\phi \\ \partial z/\partial r & \partial z/\partial\theta & \partial z/\partial\phi \end{vmatrix} = \begin{vmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix} = r^2\sin\theta \qquad ...(18)$$

$$\text{Hence,}\quad dV = r^2\sin\theta\,dr\,d\theta\,d\phi$$

Similarly, we can find the volume element in cylindrical coordinates as

$$dV = |J|\,d\rho\,d\phi\,dz = r\,d\rho\,d\phi\,dz \qquad ...(19)$$

Ex. 1. *Find the value of the following integral :*

$$I = \int_0^\infty \int_\delta^\infty e^{-(x^2+y^2)} dx\, dy$$

by changing to polar coordinates (r, θ).

Sol. The relations between cartesian coordinates (x, y) to polar coordinates (r, θ) are :

$$x = r \cos \theta \text{ and } y = r \sin \theta$$

Hence $$x^2 + y^2 = r^2$$

When the limits of integration will be $x = 0$ to $x = \infty$ and $y = 0$ to $y = \infty$, then limits of r and θ will be $r = 0$ to $r = \infty$ and $\theta = 0$ to $\theta = \pi/2$ respectively.

Here, $$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \partial x/\partial r & \partial x/\partial\theta \\ \partial y/\partial r & \partial y/\partial\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & -r \sin\theta \\ \sin\theta & r \cos\theta \end{vmatrix} = r$$

Hence $$\int_0^\infty \int_\delta^\infty e^{-(x^2+y^2)} dx\, dy = \int_0^{\pi/2} \int_0^\infty e^{-r^2} r\, dr\, d\theta = \int_0^{\pi/2} \int_0^\infty \frac{1}{2} e^{-r^2}\, 2r\, dr\, d\theta$$

$$= \int_0^{\pi/2} \int_0^\infty \frac{1}{2} e^{-u} du\, d\theta \text{ , where } r^2 = u \text{ and } 2r\, dr = du$$

$$= \int_0^{\pi/2} \left[-\frac{1}{2} e^{-u} \right]_0^\infty d\theta = \frac{1}{2} \int_0^{\pi/2} (0 - 1)\, d\theta = \frac{1}{2} [\theta]_0^{\pi/2} = \frac{\pi}{4}.$$

EXERCISE 3·3

1. If $x = r \cos \theta$, $y = r \sin \theta$, find (a) $\dfrac{\partial(x, y)}{\partial(r, \theta)}$ and (b) $\dfrac{\partial(r, \theta)}{\partial(x, y)}$.
   [**Ans.** (a) r, (b) $1/r$.]
2. If $u = ax + by$ and $v = cx + dy$, then find Jacobian $J = \dfrac{\partial(u, v)}{\partial(x, y)}$.
   [**Ans.** $J = ad - bc$.]
3. If $x = u\,(1 + v)$ and $y = v\,(1 + u)$, then find Jacobian $J = \dfrac{\partial(x, y)}{\partial(u, v)}$.
   [**Ans.** $J = u + v + 1$.]
4. If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$, then find the Jacobian $J = \dfrac{\partial(x, y, z)}{\partial(r, \theta, \phi)}$.
   [**Ans.** $J = r^2 \sin \theta$.]
5. Find the value of the following integral by changing into polar coordinates : $\displaystyle\int_0^a \int_y^a \frac{x\, dx\, dy}{x^2 + y^2}$.
   [**Ans.** $\pi a/4$.]
6. If $y_1 = \cos x_1$, $y_2 = \sin x_1 \cos x_2$ and $y_3 = \sin x_1 \sin x_2 \cos x_3$, then show that :
   $$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \sin^3 x_1 \sin^2 x_2 \sin x_3.$$
7. Prove that $\dfrac{\partial(u, v)}{\partial(x, y)} \dfrac{\partial(x, y)}{\partial(u, v)} = 1$.

QUESTIONS

Long Answer Questions

1. Explain the meaning of double and triple integrals and hence describe the method of finding them in cartesian and polar coordinates. **(Ujjain 2004; Indore 03)**

2. What are multiple integrals ? The quantities x, y, z are given in terms of three independent variables u, v, w. How will you transform the integral :

$$I = \iiint f(x, y, z)\, dx\, dy\, dz$$

in the u, v, w system with the help of Jacobian ?

3. How can the variable in integration be changed ? Explain. **(Rewa 2004, 03)**

4. What is Jacobian ? If the functions relating u and x are as follows :
$u_1 = f_1(x_1)$, $u_2 = f_2(x_1, x_2)$, $u_3 = f_3(x_1, x_2, x_3)$ ......, $u_n = f_n(x_1, x_2, ..., x_n)$, then prove that

$$\frac{\partial(u_1, u_2, ..., u_n)}{\partial(x_1, x_2, ..., x_n)} = \frac{\partial u_1}{\partial x_1} \cdot \frac{\partial u_2}{\partial x_2} \cdot \frac{\partial u_3}{\partial x_3} \cdots\cdot \frac{\partial u_n}{\partial x_n}.$$

5. Define Jacobian. The relations for the coordinates of a point between the cartesian and spherical coordinate systems are given as follows :

$$x = r \sin\theta \cos\phi,\ y = r \sin\theta \sin\phi,\ z = r \cos\theta.$$

Obtain the expression for volume element in spherical coordinates with the help of Jacobian.

Short Answer Questions

1. What are double integrals ? Explain by giving an example.
2. What is triple integral ? **(Rewa 2003)**
3. What are multiple integrals ?
4. How do you evaluate a double integral as repeated integrals ?
5. Explain the evaluation of a triple integral as repeated integrals.
6. Find the value of $\int_0^{\pi}\int_0^{a\sin\theta} r\, dr\, d\theta$.

   [**Ans.** $\frac{1}{4}\pi a^2$]
7. What is Jacobian ?
8. What is the advantage of the change of variables in multiple integrals ?
9. Calculate the value of $\int_0^{\pi/2}\int_0^{a\cos\theta} r \sin\theta\, dr d\theta$.

   [**Ans.** $a^2/6$.]
10. Find the area of a circle of radius R in polar coordinates by using the relations $x = r\cos\theta$, $y = r\sin\theta$ and Jacobian.
11. If $x = x(u, v)$, $y = y(u, v)$, write the Jacobian of x, y with respect to u, v.
12. If (x, y) and (u, v) represent the coordinates of a point in two different systems, then write the Jacobian of u, v with respect to x, y.

$$\left[\textbf{Ans. } J = \begin{vmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{vmatrix}.\right]$$

Objective Type Questions

1. The value of the integral $\int_0^1\int_0^1 x\, dx\, dy$ is :

   (a) 1 (b) $\frac{1}{2}$ (c) 0 (d) $\frac{1}{4}$

2. The value of $\int_0^a\int_0^b xy\, dx\, dy$ is :

   (a) ab (b) a^2b^2 (c) $\frac{1}{4}ab$ (d) $\frac{1}{4}a^2b^2$ **(Agra 2004)**

3. The value of $\int_0^3\int_0^2\int_0^1 x\, dx\, dy\, dz$ will be :

   (a) 9 (b) 2 (c) 0.5 (d) none of these

4. The value of the integral $\int_1^2\int_0^{3/2}(x/y^2)\,dy\,dx$ is :

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{3}{4}$ (d) $\frac{5}{4}$

5. Evaluation of $\int_0^2\int_0^3(x+y)\,dy\,dx$ gives :

(a) 3 (b) 5 (c) 7 (d) 9

6. $\int_0^1\int_0^1(x^2+y^2)\,dx\,dy =$

(a) 1 (b) 0 (c) $\frac{1}{3}$ (d) $\frac{2}{3}$

7. $\int_0^{\pi/2}\int_{\pi/2}^{\pi}\sin(x+y)\,dy\,dx =$

(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) -1

8. The value of $\int_0^{\pi/2}\int_{\pi/2}^{\pi}\cos(x+y)\,dy\,dx$ is :

(a) -2 (b) -1 (c) 0 (d) 1

9. The value of the integral $\int_0^{\pi}\int_0^{R} r\,d\theta\,dr$ is :

(a) πR^2 (b) $\frac{1}{2}\pi R^2$ (c) $2\pi R^2$ (d) $2\pi R$

10. The calculation of the integral $\int_0^{2\pi}\int_0^{\pi/2}\int_0^{R} r^2\sin\theta\,d\theta\,dr\,d\phi$ gives :

(a) $\frac{4}{3}\pi R^3$ (b) $\frac{2}{3}\pi R^3$ (c) πR^2 (d) $4\pi R^2$

11. In spherical coordinates system, the volume element (dV) is :

(a) $dr\,d\theta\,d\phi$ (b) $d\rho\,d\phi\,dz$ (c) $r^2\sin\theta\,dr\,d\theta\,d\phi$ (d) $r^2\,dr\,d\theta\,d\phi$

12. In cylindrical coordinates, the volume element (dV) is :

(a) $r^2\sin\theta\,dr\,d\theta\,d\phi$ (b) $d\rho\,d\phi\,dz$ (c) $\phi\,d\rho\,d\phi\,dz$ (d) $\rho\,d\rho\,d\phi\,dz$

13. The value of the integral $\int_0^3\int_0^2\int_0^1(x+y+z)\,dx\,dy\,dz$ is :

(a) $\frac{15}{8}$ (b) $\frac{3}{8}$ (c) $\frac{8}{3}$ (d) $\frac{8}{15}$

14. $\int_0^1\int_0^x xe^y\,dx\,dy =$

(a) $\frac{1}{2}e$ (b) $1-e$ (c) $e-1$ (d) $\frac{1}{2}e-1$

15. If $x = r\cos\theta$ and $y = r\sin\theta$, the value of $\dfrac{\partial(x, y)}{\partial(r, \theta)}$ is :

(a) r (b) θ (c) $r\theta$ (d) r/θ

ANSWERS

1. (b), **2.** (d) **3.** (a), **4.** (c), **5.** (a), **6.** (d), **7.** (a), **8.** (a)
9. (b), **10.** (c), **11.** (c), **12.** (d), **13.** (a), **14.** (d), **15.** (a)

4

Electrostatics—Electric Field and Potential

4·1. Coulomb's Law

There are two types of electric charges : positive and negative. Like charges repel each other, while unlike charges have mutual attraction. According to Coulomb's law, *the magnitude of the electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.* In mathematical form, the magnitude F of the force that one charge q_1 exerts on another charge q_2 at a distance r apart is given by

$$F = k\frac{q_1 q_2}{r^2} \qquad ...(1)$$

where k is a constant. For reasons in S.I. system , k is written as $1/4\,\pi\varepsilon_0$, where ε_0 is a constant, called the permittivity of free space, if we suppose the charges are situated in **vacuum.** Thus

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2} \qquad ...(2)$$

In this expression, force F is measured in newtons, charge q in coulombs and distance r in meters. The value of ε_0 is 8.85×10^{-12} C^2/N-m^2 and that of $1/4\pi\varepsilon_0$ as 8.99×10^9 N-m^2/C^2 or 9×10^9 N-m^2/C^2.

In order to express the Coulomb's law in vector form, let a point charge q_2 be situated at a vector distance $\vec{r}$ from a point charge q_1 at the origin O. The force $\vec{F}$ experienced by the charge q_2 **[fig. 4.1]** is given by

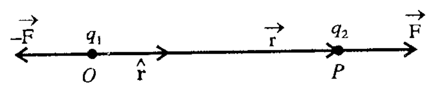


Fig. 4.1 : Coulomb's law.

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{\mathbf{r}} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^3}\vec{r} \qquad ...(3)$$

where $\hat{\mathbf{r}} = \vec{r}/r$ is unit vector along $\vec{r}$. **Fig. 4.1** shows that if $\vec{F}$ acts upon q_2, then $-\vec{F}$ will act on q_1 and the magnitude of each force is $F = q_1 q_2/4\pi\varepsilon_0 r^2$.

Unit of charge : In S.I. system the unit of charge is **coulomb** and is denoted by **C.**

A coulomb is defined as the amount of charge which flows through a cross-section of a wire in 1 *second if there is a steady current of* 1 *ampere in the wire i.e.,*

1 coulomb (C) = 1 ampere-second (A-s)

This definition is obtained on the basis of S.I. unit of current that is **ampere.**

Another alternative definition of coulomb may be given using Coulomb's law. According to this law [eq. (2)],

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2} \text{ or } F = 9 \times 10^9\frac{q_1 q_2}{r^2}$$

If $q_1 = q_2 = 1$ C and $r = 1$ m, then $\qquad F = 9 \times 10^9$ N.

Hence, *Coulomb is defined as that amount of charge which when placed at a distance of* 1 *meter from an equal and similar charge in vacuum, repels it with a force of* 9×10^9 *newton.*

The **most fundamental unit of charge** is the magnitude of charge on an electron or proton, which is denoted by e. The most precise value of e is

$$e = 1.60217733\ (49) \times 10^{-19}\ \text{C}$$

The charge on an electron is $-e$ and that on a proton is $+e$.

This fundamental unit of charge e is very small. 1 coulomb represents the negative of the total charge of about 6×10^{18} electrons *i.e.*, 6×10^{18} electrons pass through a cross-section of wire per second due to a current of 1 ampere.

Typical values of charge range from about 10^{-9} C to 10^{-6} C. Hence the microcoulomb (μC) and nanocoulomb (nC) units are used in measurements *i.e.*,

$$1 \text{ microcoulomb} = 1\ \mu\text{C} = 10^{-6}\ \text{C}$$

$$1 \text{ nanocoulomb} = 1\ \text{nC} = 10^{-9}\ \text{C}$$

Scope and Importance of Coulomb's Law

Coulomb's law holds good only for the point charges but it can be extended to cover distributions of charges. Coulomb's law holds good over a wide range from very small distances of the order of 10^{-15} m to large distances of the order of several kilometers.

If this law is incorporated with quantum theory, it satisfactorily explains the forces which bind (i) the electrons with the nucleus in an atom, (ii) two or more atoms to form a molecule, and (iii) large number of atoms or molecules to form liquids and solids.

The nucleus of an atom contains positively charged protons and uncharged neutrons and its size is of the order of 10^{-15} m. According to Coulomb's law, the protons of the nucleus should repel each other and fly apart. However the atomic nucleus of a nonradioactive element is **stable**. Hence *Coulomb's law cannot account the stability of an atomic nucleus**. In fact, in the nucleus, some nonelectrical very short ranged attractive forces are operative between the nuclear particles in addition to electrical forces and these forces are called **nuclear forces.** The nuclear forces are weaker than the Coulomb force, when the separation between the particles is more than approximately 10^{-14} m. But for the separations of the order of 10^{-15} m or less, the attractive nuclear forces are much stronger than the Coulomb repulsion forces and hence are able to make the nucleus stable.

In case of a big nucleus, *e.g.*, that of uranium containing 92 protons, the electrical forces become comparable to nuclear forces and hence it is easy to break such nucleus by nuclear fission.

Quantization of Charge

Experimentally it is found that the charge on an object is always an integral multiple of charge e ($= 1.6 \times 10^{-19}$ C), charge of an electron *i.e.*, $q = ne$, where $n = \pm 1, \pm 2, \pm 3, \ldots\ldots$ This property of charge is called **quantization of charge**. *e is the smallest unit of charge* and is called *elementary charge.*

The **charge is additive** *i.e.*, the total charge on a body is the algerbraic sum of the charges at different parts of the body.

Conservation of Charge

The total amount of charge in an isolated system is always constant. Of course, the charge may be transferred from one part of the system to another, but the net charge will remain the same. Thus, the charge can neither be created nor it can be destroyed. This is called **law of conservation of charge.**

Law of conservation of charge is supported by following examples :

(1) When we rub a glass rod with silk, a positive charge is obtained on the glass rod and at the same time an equal negative charge is acquired by the silk. Thus, the net charge on the glass rod-silk system is zero before and after rubbing.

(2) Experimentally a γ-ray photon may produce a pair of electron with charge $-e$ and positron with charge $+e$. Here also the net charge is zero before and after the process.

Similarly, an electron ($-e$) and a positron ($+e$), when come together, may annihilate each other and produce a γ-ray photon. Again the net charge is zero both before and after the process.

(3) In a number of nuclear reactions, we find that the total charge before and after the reaction remains the same. For example,

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* Between the protons and neutrons of the nucleus one may think of gravitational attractive forces. But these gravitational attractive forces, when calculated, are negligible in comparison to coulumb repulsion between protons.

(i) A neutron may change into a proton and simultaneously emitting an electron

$$_0n^1 \rightarrow {}_1p^1 + {}_{-1}e^0$$

so that the total charge is zero before and after the reaction.

(ii) In a radioactive decay, uranium nucleus with 92 protons, *i.e.*, + 92 e charge, changes into a thorium nucleus with 90 protons (*i.e.*, + 90 e charge) by emitting an α particle with 2 protons (*i.e.*, +2e charge) :

$$_{92}U^{238} \rightarrow {}_{90}Th^{234} + {}_2He^4$$

Thus there is + 92e charge before and after the decay.

4·2. Coulomb's Law and Newton's Law of Gravitation

According to Coulomb's law, the electric force between two point charges q_1 and q_2 is given by

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}\hat{\mathbf{r}} \qquad ...(1)$$

Newton's law of gravitation gives the force of attraction between two point masses m_1 and m_2 *i.e.*,

$$\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}} \qquad ...(2)$$

Thus the forces given by the two basic laws are two different types of inverse-squre-law force (*i.e.*, $\vec{F} = -\frac{K}{r^2}\hat{\mathbf{r}}$). We also see that charge q plays the same role in electrostatics as mass m in gravitation. However the magnitude of electrostatic force acting between two charged particles is very strong when compared to the gravitational force between them. For example, we consider the electrostatic force and gravitational force between a proton and an electron at a distance r situated in air or vacuum. The charges and masses of the two particles are as follows :

For proton, $q_1 = +\,1.6 \times 10^{-19}$ C and $m_1 = 1.67 \times 10^{-27}$ kg

For electron, $q_2 = -\,1.6 \times 10^{-19}$ C and $m_2 = 9.1 \times 10^{-31}$ kg

The gravitational force between two masses is always attractive and hence it is attractive for proton and electron.

Between proton and electron due to positive and negative charges respectively, the electrostatic force is also attractive type. Here, we consider the magnitudes of the two types of force.

Electrostatic force,
$$F_E = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2} = \frac{9\times10^9\times(1.6\times10^{-19})\times(1.6\times10^{-19})}{r^2}$$
$$= \frac{2.304\times10^{-28}}{r^2}\text{ newton}$$

Gravitational force,
$$F_G = \frac{Gm_1m_2}{r^2} = \frac{6.67\times10^{-11}\times(1.67\times10^{-27})\times(9.1\times10^{-31})}{r^2}$$
$$= \frac{1.014\times10^{-67}}{r^2}\text{ newton}$$

$$\therefore \quad \frac{F_E}{F_G} = \frac{2.304\times10^{-28}}{1.014\times10^{-67}} = 2.27\times10^{39}.$$

Thus we see that the electrostatic force between a proton and an electron is approximately 10^{39} times stronger than the gravitational force between them at a particular distance. The electrostatic forces between two protons and two electrons are respectively 10^{36} times and 10^{43} times the gravitational forces between them.

Similarities between electrostatic and gravitational forces : (1) Both the forces are central forces. (2) Inverse square law is obeyed by electrostatic as well as gravitational forces.

(3) Both the forces are conservative in nature.

(4) Both the forces act even in vacuum.

Dissimilarities between electrostatic and gravitational forces : (1) The electrostatic force can be attractive or repulsive type, when the two charges are unlike or similar respectively. The gravitational force is always attractive in nature.

(2) The electrostatic force is relatively too strong in comparison to gravitational force.

(3) The electrostatic force between two charges at a distance decreases in the presence of dielectric medium when compared to the electrostatic force between them in vacuum. However the gravitational force between two masses is not influenced by the medium.

4·3. Principle of Superposition

When more than two charges are present in a system, Coulomb's law holds good for every pair of charges. Thus *the total force on any charge is equal to the vector sum of all the forces, exerted on it by other charges of the system.* This is called **principle of superposition** in electrostatics.

Thus, if there is a system of three charges q_1, q_2 and q_3, then the net force on q_3 due to q_1 and q_2 is given by [**fig. 4.2**]

$$\vec{F}_3 = \vec{F}_{31} + \vec{F}_{32} = \frac{1}{4\pi\varepsilon_0}\frac{q_3 q_1}{r_{31}^2}\hat{r}_{31} + \frac{1}{4\pi\varepsilon_0}\frac{q_3 q_2}{r_{32}^2}\hat{r}_{32}$$

where $\vec{F}_{31}$ denotes the force exerted on 3rd charge by the 1st charge and $\hat{r}_{31} (= \vec{r}_{31}/r_{31})$ is the unit vector directed from 1st to the 3rd charge. Similar definitions hold, in general, for $\vec{F}_{ij}$ and $\vec{r}_{ij}$.

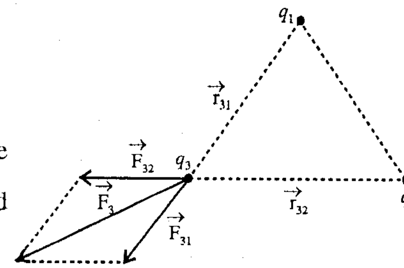


Fig. 4.2 : Principle of superposition.

Thus in view of principle of superposition, the *force which two charges exert on each other is not changed by the presence of a third charge.*

4·4. Electric Field and Intensity of Electric Field

Electric field : When a charged particle experiences an electric force, we say that it is present in an **electric field.** A charge produces an electric field in the space around it and this field exerts a force on any charge (except the source charge itself) placed in it. In fact the electric field has its own existence and it is present even if no additional charge is present to experience it.

Intensity of electric field : If a charge q at a point in an electric field experiences an electric force $\vec{F}$, we define the intensity of the electric field $\vec{E}$ at the given point by the relation

$$\vec{F} = q\vec{E} \text{ or } \vec{E} = \frac{\vec{F}}{q} \qquad ...(1)$$

Thus *the intensity of electric field* $\vec{E}$ *is defined as the force per unit positive charge at the position of charged particle.* We often call *the intensity of electric field as only* **electric field.**

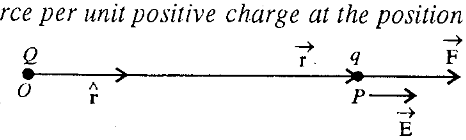


Fig. 4.3 : Electric field intensity $\vec{E}$ at a point P due to a point charge Q.

The charge q, used in eq. (1) to define the intensity of electric field, is called **test charge**. This test charge should be very small so that it does not disturb other charges.

In **fig. 4.3**, if Q is the point charge at the origin O, then according to Coulomb's law the force on a charge q at a distance $\vec{r}$ is given by

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{Qq}{r^2}\hat{r}, \qquad \text{where } \hat{r} = \frac{\vec{r}}{r} \qquad ...(2)$$

Hence from (1) and (2), we have

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{\mathbf{r}} \qquad ...(3)$$

This is *the expression for intensity electric field due to a point charge at a distance* $\vec{r}$ *from it.*

S.I. unit of electric field is newton/coulomb (N/C) or volt/meter (V/m). The dimensional formula for electric field is $[MLT^{-3}A^{-1}]$.

The electric field can be mapped by electric lines of force. A number of lines per unit area, taken perpendicularly at a point is considered as a measure of the electric field at that point.

4·5. Electric Field due to a System of Charges at a Point

The electric field due to a point charge q at a distance $\vec{r}$ apart in vacuum or air is given by

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\vec{r} \qquad ...(1)$$

where $\hat{\mathbf{r}} = \vec{r}/r$ is the unit vector along $\vec{r}$.

Consider a system of two point charges q_1 and q_2. Let a point P be situated at vector distances $\vec{r}_1$ and $\vec{r}_2$ from the charges q_1 and q_2 respectively [**fig. 4.4**].

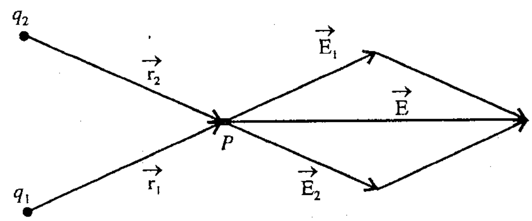


Fig. 4.4 : Electric field at a point P due to a system of two charges.

The electric fields due to charges q_1 and q_2 at P are

$$\vec{E}_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1^2}\vec{r}_1 \text{ and } \vec{E}_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{4\pi r_2^2}\hat{\mathbf{r}}_2$$

respectively. Now resultant electric field $\vec{E}$ at P due to the two charges is obtained by the principle of superposition *i.e.*,

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1^2}\hat{\mathbf{r}}_1 + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2^2}\hat{\mathbf{r}}_2 = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{2}\frac{q_i}{r_i^2}\hat{\mathbf{r}}_i \qquad ...(2)$$

Hence the electric field at a point P due to a system of charges q_1, q_2 ......, q_n is given by

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + ... + \vec{E}_n = \sum_{i=1}^{n}\vec{E}_i = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i^2}\hat{\mathbf{r}}_i \qquad ...(3)$$

where $\vec{E}_i$ is the electric field at the point P situated at a vector distance $\vec{r}_i$ from the charge q_i of the system.

4·6. Electric Field due to a Continuous Charge Distribution

We want to calculate the electric field at a point P due to a continuous distribution of charge (say a charged body). Consider an infinitesimally small element dq of the charged region. The electric field at the point P at a distance $\vec{r}$ from the charge dq is given by [**fig. 4.5**],

$$d\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}\hat{\mathbf{r}}, \text{ where } \hat{\mathbf{r}} = \frac{\vec{r}}{r} \qquad ...(4)$$

The charge distribution of the body may be considered to be composed of large number of such infinitesimally small elements. According to the principle of superposition, the resultant electric field will be the sum of electric fields due to all the elements of the charged body *i.e.*,

$$\vec{E} = \Sigma d\vec{E} = \frac{1}{4\pi\varepsilon_0}\Sigma\frac{dq}{r^2}\hat{\mathbf{r}} \qquad ...(5)$$

For the continuous distribution of the charge on the body, the sum may be changed to an integral

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r^2}\hat{r} \qquad ...(6)$$

where the integral is done over the entire charge distribution.

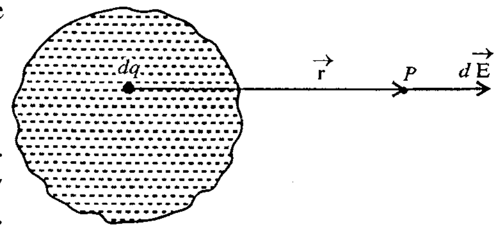


Fig. 4.5 : Electric field due to a continuous charge distribution.

The region over which the charge is continuously distributed may be in the form of a line, an area or a volume.

(1) Electric field at a point due to a linear charge distribution : When the charge is distributed on a length, such as a long wire, the amount of charge per unit length of the wire is called **linear charge density.** It is represented by λ and its unit is coulomb/metre.

Now, if dq is the charge on an infinitesimally small element dl of the wire, then $dq = \lambda dl$ and the electric field due to this charge at a point P [**fig. 4.6**] is given by

$$d\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dl}{r^2}\hat{r}$$

Hence the electric field at P due to the entire length of the wire is obtained to be,

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\lambda dl}{r^2}\hat{r} \qquad ...(7)$$

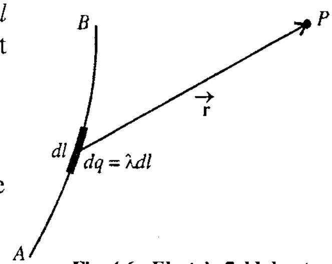


Fig. 4.6 : Electric field due to a linear charge distribution.

where the integral is to be taken over the total length of the wire.

(2) Electric field at a point due to a surface charge distribution : When the charge is spread over a surface, the charge per unit area of the surface is called **surface charge density.** It is denoted by σ and its unit is coulomb/m^2.

Consider an infinitesimally small area dS of a charged surface. The charge on the dS area will be σdS [**fig. 4.7**]. Hence the electric field $\vec{E}$ due to the charged surface is given by

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\iint_S \frac{\sigma dS}{r^2}\hat{r} \qquad ...(8)$$

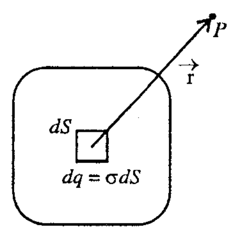


Fig. 4.7 : Electric field due to a charged surface.

(3) Electric field due to a volume charge distribution : When the charge is distributed continuously in the volume of a body, the amount of charge per unit volume is called the **volume charge density.** It is represented by ρ and its unit is coulomb/m^3.

Consider an infinitesimally small volume element dV of the charged body around a point, where the charge density is ρ. The charge dq in the volume element dV will be $dq = \rho dV$.

Hence the electric field at a point P due to the entire charged volume distribution is obtained to be [**fig. 4.8**],

$$\vec{E} = \iiint_V \frac{\rho dV}{r^2}\hat{r} \qquad ...(9)$$

where the integral extends over the entire charged region.

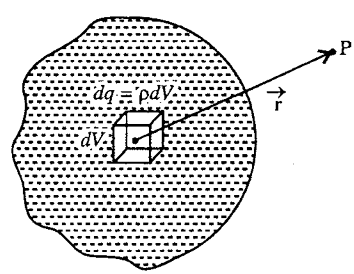


Fig. 4.8 : Electric field due to a volume charge distribution.

4·7. Electric Field due to an Infinitely Long Uniformly Charged Straight Wire (or Rod)

Consider an uniformly positively charged straight wire with a linear charge density λ. Let us determine the electric field at a point P at a distance r normal to the wire. Draw a perpendicular PO from P on the wire so that the distance $PO = r$ and O is the foot of the perpendicular. Consider a small element of length dx of the wire at a distance x from O [**fig. 4.9**]. The charge on the element is λdx.

The intensity of electric field at the point P due to this element at A is

$$dE_1 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\lambda dx}{AP^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dx}{r^2+x^2} \text{ along } PC$$

where $AP = (r^2 + x^2)^{1/2}$.

Similarly, the intensity of electric field due to a charge λdx on a small element dx at B, situated at a distance x from O on the other side of the wire, is

$$dE_2 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\lambda dx}{r^2+x^2} \text{ along } PD.$$

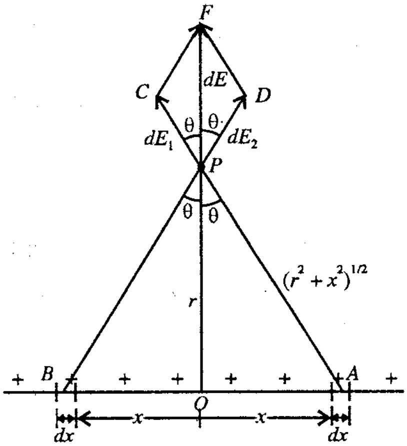


Fig. 4.9 : Electric field due to uniformly charged straight wire.

We see that dE_1 and dE_2 are equal in magnitude. Now resolve dE_1 and dE_2 in components along OP and perpendicular to OP. The perpendicular components of dE_1 and dE_2 cancel each other because they are in opposite directions. However their components along PF (or OP) are added and the resultant electric field at P is obtained to be

$$dE = dE_1 \cos\theta + dE_2 \cos\theta$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{\lambda dx \cdot 2\cos\theta}{r^2+x^2} \text{ along } OP \qquad \left[\because dE_1 = dE_2 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\lambda dx}{r^2+x^2}\right]$$

or

$$dE = \frac{1}{2\pi\varepsilon_0}\frac{\lambda dx \cdot r}{(r^2+x^2)(r^2+x^2)^{1/2}} \qquad \left[\because \cos\theta = \frac{r}{(r^2+x^2)^{1/2}}\right]$$

$$= \frac{1}{2\pi\varepsilon_0}\frac{\lambda r dx}{(r^2+x^2)^{3/2}} \text{ along } OP \qquad \text{...(1)}$$

Hence the total electric field due to the wire of infinite length can be obtained by integrating the above expression from $x = 0$ to $x = \infty$ *i.e.*,

$$E = \int_0^\infty \frac{1}{2\pi\varepsilon_0}\frac{\lambda r dx}{(r^2+x^2)^{3/2}} \text{ along } OP.$$

From the **fig. 4.9**, we see

$$\tan\theta = \frac{x}{r} \text{ or } x = r\tan\theta.$$

Hence $\quad r^2 + x^2 = r^2(1+\tan^2\theta) = r^2\sec^2\theta$ and $dx = r\sec^2\theta\, d\theta$

Therefore, $$E = \int_0^{\pi/2}\frac{1}{2\pi\varepsilon_0}\frac{\lambda r^2\sec^2\theta\, d\theta}{r^3\sec^3\theta} = \frac{\lambda}{2\pi\varepsilon_0 r}\int_0^{\pi/2}\cos\theta\, d\theta$$

where at $x = 0$, $\theta = 0$ and at $x = \infty$, $\theta = \pi/2$.

$$\therefore \quad E = \frac{\lambda}{2\pi\varepsilon_0 r}[\sin\theta]_0^{\pi/2} \text{ or } E = \frac{\lambda}{2\pi\varepsilon_0 r} \text{ along } \vec{OP} \qquad \text{...(2)}$$

Thus *the electric field due to a uniformly positively charged wire of infinite length is inversely proportional to the distance from the wire and is directed perpendicular to the length of the wire (or radially outward).*

Special case :

Electric field due to a uniformly charged straight wire (or rod) of finite length : If the length of the wire is finite, say L, we take O as the midpoint of the wire and determine the electric field at a point P distant r, situated on the perpendicular bisector OP. Consider two symmetrical elements dx of the wire at a distance x from the midpoint O of the wire. Due to symmetry, the X-components of the fields dE_1 and dE_2 due to the two elements dx at A and B will cancel each other and the Y-components $dE_1 \cos\theta_1$ and $dE_2 \cos\theta_2$ will be added. Thus the intensity of the electric field at P will be given by **[fig. 4.9]**,

$$dE = 2dE_1 \cos\theta = \frac{1}{2\pi\varepsilon_0}\frac{r\lambda dx}{(r^2+x^2)^{3/2}} \text{ along } OP \qquad ...(3)$$

because $dE_1 = dE_2 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\lambda dx}{r^2+x^2}$ and λdx is the charge on the element dx.

Hence the electric field at P will be obtained by integrating (3) for the limits $x = 0$ to $x = L/2$ *i.e*,

$$E = \int_0^{L/2} \frac{1}{2\pi\varepsilon_0}\frac{\lambda r\, dx}{(r^2+x^2)^{3/2}} \text{ along } OP$$

$$= \frac{\lambda}{2\pi\varepsilon_0 r}\int_0^{\phi} \cos\theta\, d\theta \text{ along } OP \qquad ...(4)$$

where ϕ is the angle subtended by one half of the rod ($L/2$) at P **[fig. 4.10]**.

$$\therefore \quad E = \frac{\lambda}{2\pi\varepsilon_0 r}\sin\phi \text{ or } E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2\lambda\sin\phi}{r} \qquad ...(5)$$

But $\sin\phi = \frac{L/2}{(r^2+L^2/4)^{1/2}} = \frac{L}{(L^2+4r^2)^{1/2}}$.

Thus $\quad E = \frac{\lambda}{2\pi\varepsilon_0 r}\cdot\frac{L}{(L^2+4r^2)^{1/2}}$ along OP $\qquad ...(6)$

When $L \to \infty$,

$$E = \lim_{L\to\infty}\frac{\lambda}{2\pi\varepsilon_0 r}\cdot\frac{1}{(1+4r^2/L^2)^{1/2}} = \frac{\lambda}{2\pi\varepsilon_0 r} \qquad ...(7)$$

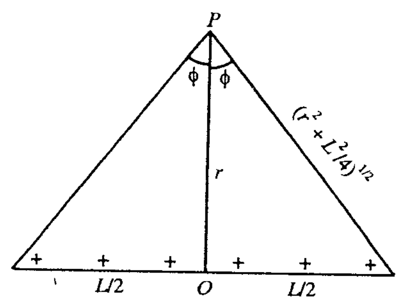


Fig. 4.10 : Electric field due to a uniformly charged straight wire.

This is the same expression (2) calculated for infinitely long wire.

4·8. Electric Field due to a Uniformly Charged Ring at an Axial Point

Consider a ring of radius R with centre O. Let a charge q be uniformly distributed over the ring. The linear charge density or charge per unit length on the ring is

$$\lambda = \frac{q}{2\pi R} \qquad ...(1)$$

Let us calculate the electric field due to this ring at a point P on its axis. We consider a very small element dl at a point A on the circumference of the ring. The charge on the dl element is $dq = \lambda dl$. The electric field at the point P due to this small charge is given by **[fig. 4.11]**,

$$dE_1 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\lambda\, dl}{r^2+R^2} \text{ along } \vec{PC}$$

In the same way, the electric field at the point P due to the charge on the element of length dl at a point B diametrically opposite to the point A is

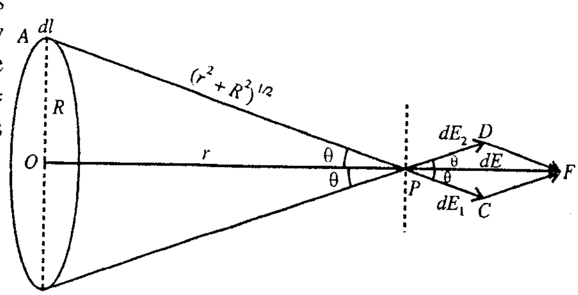


Fig. 4.11 : Electric field due to a uniformly charged ring at an axial point.

given by

$$dE_2 = \frac{\lambda}{4\pi\varepsilon_0}\cdot\frac{\lambda\, dl}{r^2 + R^2} \text{ along } \vec{PD}$$

We find $dE_1 = dE_2$ in magnitude. Let us resolve dE_1 and dE_2 into axial and perpendicular components. We see that the perpendicular components $dE_1 \sin\theta$ and $dE_2 \sin\theta$ cancel each other and axial components $dE_1 \cos\theta$ and $dE_2 \cos\theta$ are added.

Hence the resultant electric field due to the two small elements dl at A and B at P is given by

$$dE = dE_1 \cos\theta + dE_2 \cos\theta = 2\, dE_1 \cos\theta$$

$$= \frac{2}{4\pi\varepsilon_0}\cdot\frac{\lambda\, dl}{(r^2 + R^2)}\cdot\frac{r}{(r^2 + R^2)^{1/2}} = \frac{\lambda r\, dl}{2\pi\varepsilon_0 (r^2 + R^2)^{3/2}} (\text{along } \vec{OP}) \quad ...(2)$$

where $\cos\theta = r/(r^2 + R^2)^{1/2}$.

Thus the total elelctric field due to the entire ring is given by

$$E = \int_0^{\pi R} \frac{\lambda r dl}{2\pi\varepsilon_0 (r^2 + R^2)^{3/2}} = \frac{\lambda r}{2\pi\varepsilon_0 (r^2 + R^2)^{3/2}} \int_0^{\pi R} dl = \frac{\lambda r \times \pi R}{2\pi\varepsilon_0 (r^2 + R^2)^{3/2}}$$

where the integral has been taken only for half length of the ring.

But $\lambda = q/2\pi R$, hence

$$E = \frac{qr}{4\pi\varepsilon_0 (r^2 + R^2)^{3/2}} \text{ along } \vec{OP} \quad ...(3)$$

Thus the electric field due to a uniformly positively charged ring at a point on its axis is along the axis of the ring in the outward direction.

Distance of the point of maximum electric field intensity : Differentiating (3) with respect to r, we obtain

$$\frac{dE}{dr} = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(r^2 + R^2)^{3/2}} - \frac{3r^2}{(r^2 + R^2)^{5/2}}\right]$$

For maximum intensity $dE/dr = 0$ *i.e.*,

$$\frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(r^2 + R^2)^{3/2}} - \frac{3r^2}{(r^2 + R^2)^{5/2}}\right] = 0 \text{ or } 1 - \frac{3r^2}{r^2 + R^2} = 0 \text{ or } 3r^2 = r^2 + R^2 \text{ or } 2r^2 = R^2$$

i.e.,

$$r = \pm R/\sqrt{2} \quad ...(4)$$

It can be shown that at $r = \pm R/\sqrt{2}$, d^2E/dr^2 is negative and hence the electric field is maximum at $r = \pm R/\sqrt{2}$. The value of the maximum field is

$$E_{max} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2q}{3\sqrt{3}R^2}. \quad ...(5)$$

4·9. Electric Field due to a Uniformly Charged Disc along its Axis

Consider a charge q is distributed on the surface of a circular disc of radius R. Hence surface charge density, $\sigma = q/\pi R^2$. Now we consider a ring of radius x and thickness dx [**fig. 4.12**].

Surface area of ring $= 2\pi x\, dx$

Charge on the ring $dq = 2\pi x\, dx\, \sigma$

The electric field at a point P along the axis of a ring at a distance r is given by

$$dE = \frac{1}{4\pi\varepsilon_0}\cdot\frac{dq\, r}{(r^2 + x^2)^{3/2}} \text{ along } \vec{OP}$$

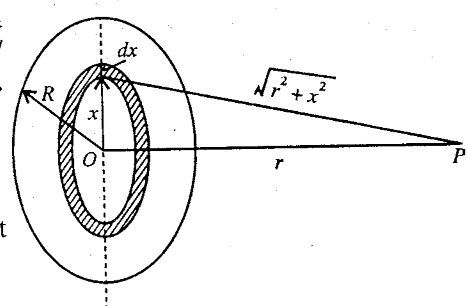


Fig. 4.12 : Electric field due to a uniformly charged disc along its axis.

Hence the total electric field due to the charged disc is

$$E = \frac{1}{4\pi\varepsilon_0}\int_0^R \frac{2\pi x\,dx\,r\sigma}{(r^2+x^2)^{3/2}} = \frac{r\sigma}{2\varepsilon_0}\int_0^R \frac{x\,dx}{(r^2+x^2)^{3/2}} \qquad ...(1)$$

Let $r^2 + x^2 = t^2$; $\therefore$ $x\,dx = t\,dt$

$\therefore$

$$E = \frac{r\sigma}{2\varepsilon_0}\int_r^{\sqrt{r^2+R^2}} \frac{dt}{t^2} = \frac{r\sigma}{2\varepsilon_0}\left[-\frac{1}{t}\right]_r^{\sqrt{r^2+R^2}} = \frac{r\sigma}{2\varepsilon_0}\left[\frac{1}{r} - \frac{1}{\sqrt{r^2+R^2}}\right]$$

or

$$E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{r}{\sqrt{r^2+R^2}}\right) = \frac{q}{2\pi\varepsilon_0 R^2}\left(1 - \frac{r}{\sqrt{r^2+R^2}}\right). \qquad ...(2)$$

Ex. 1. *Two small metallic spheres, each of mass 5 gm, are suspended from the same point by silk threads, each of length 30 cm. The spheres are then charged with equal quantities of positive electricity so that when equilibrium is obtained, the threads make an angle 60° with each other. Calculate the charge on each sphere.*

Sol. Let A and B be the two small metallic spheres **[fig. 4.13]** and q be the charge on each of them.

Since the length of each silk thread = 30 cm = 0.3 m and $\angle AOB = 60°$, it is clear from the figure that $AB = 2 \times 0.3 \sin 30° = 0.3$ m.

Evidently, the sphere A (or B) is in equilibrium under three forces, *viz;*

(i) force of repulsion $F = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{(0.3)^2}$

(ii) force due to gravity, $mg = 5 \times 10^{-3} \times 9.8$ N

(iii) tension of the thread, T.

So that taking moments about O, we have

$$F \times ON = mg \times AN.$$

$$8.99 \times 10^9 \times \frac{q^2}{(0.3)^2} \times 0.3\cos 30° = 5 \times 10^{-3} \times 9.8 \times 0.3 \sin 30°$$

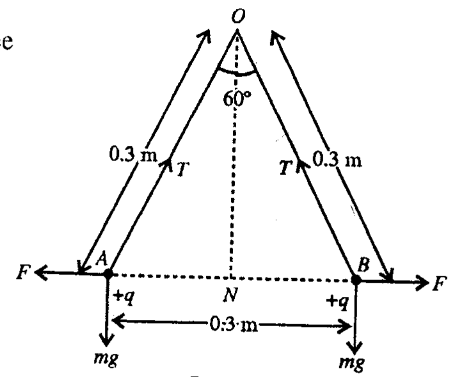


Fig. 4.13.

or

$$q^2 = \frac{5 \times (0.3)^2 \times 9.8 \times 10^{-3}}{\sqrt{3} \times 8.99 \times 10^9} = 28.32 \times 10^{-14}$$

$\therefore$ $q = 5.32 \times 10^{-7}$ coulomb = **53.2 μC.**

The charge on each sphere is thus **53.2 μC.**

Ex. 2. *The distance between the electron and the proton in hydrogen atom is* 5.3×10^{-11} *m. Calculate the electric force acting between them. Compare this force with the gravitational force.* ($G = 6.7 \times 10^{-11}$ $N\,m^2kg^{-2}$, *mass of electron* $= 9.1 \times 10^{-31}$ *kg, mass of proton* $= 1.7 \times 10^{-27}$ *kg, charge on electron,* $e = 1.6 \times 10^{-19}$ *C;* $1/4\pi\varepsilon_0 = 9 \times 10^9$ $N\text{-}m^2/C^2$).

Sol. According to Coulomb's law, the electric force of attraction between two charges q_1 and q_2 at a distance r apart is

$$F_E = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}.$$

Here, $\frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m² C⁻², charge on the electron, $q_1 = 1.6 \times 10^{-19}$ C (negative), charge on the proton, $q_2 = 1.6 \times 10^{-19}$ C (positive) and $r = 5.3 \times 10^{-11}$ m. Hence

$$F_E = (9.0 \times 10^9) \times \frac{(1.6 \times 10^{-19}) \times (1.6 \times 10^{-19})}{(5.3 \times 10^{-11})^2} = \mathbf{8.2 \times 10^{-8}\ N.}$$

According to Newton's law of universal gravitation, the gravitational force between the electron and the proton is given by

$$F_G = G\frac{m_1 m_2}{r^2}.$$

Here, mass of electron, $m_1 = 9{\cdot}1 \times 10^{-31}$ kg; mass of proton $m_2 = 1.7 \times 10^{-27}$ kg; $r = 5.3 \times 10^{-11}$ m and $G = 6.7 \times 10^{-11}$ N m^2 kg^{-2}. Hence

$$F_G = (6.7 \times 10^{-11}) \times \frac{(9.1 \times 10^{-31}) \times (1.7 \times 10^{-27})}{(5.3 \times 10^{-11})^2} = 3.7 \times 10^{-47}\text{ N}.$$

$$\therefore \quad \frac{F_E}{F_G} = \frac{8.2 \times 10^{-8}}{3.7 \times 10^{-47}} = \mathbf{2.2 \times 10^{39}}.$$

Ex. 3. *Two point charges* $+3q$ and $-q$ *are placed in air at a separation of* 0.2 *m. At what point on the line joining the two charges should a third charge Q be placed so that it will be in equilibrium ?*

Sol. As shown in **fig. 4.14**, the resultant force on the charge Q at P due to the two charges $+3q$ and $-q$ will be zero, only when Q is placed to the right of both along the line joining them. Let Q be placed at P at a distance x from charge $-q$.

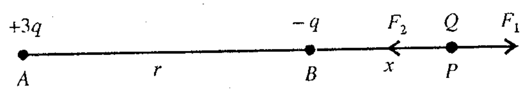


Fig. 4.14.

Force on Q due to $+ 3q$ charge,

$$F_1 = \frac{1}{4\pi\varepsilon_0}\frac{3qQ}{(r+x)^2} \text{ towards right}$$

where r is the distance between $+ 3q$ and $- q$ charges.

Force on Q due to $- q$ charge, $F_2 = \dfrac{1}{4\pi\varepsilon_0}\dfrac{qQ}{x^2}$ towards left

Force on Q will be zero, when F_1 and F_2 are equal in magnitude *i.e.*,

$$F_1 = F_2 \text{ or } \frac{1}{4\pi\varepsilon_0}\frac{3qQ}{(r+x)^2} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{x^2}$$

or
$$\frac{3}{(r+x)^2} = \frac{1}{x^2} \text{ or } \sqrt{3}x = r + x \text{ or } x = \frac{r}{\sqrt{3}-1} = \frac{0.2}{0.732} = \mathbf{0.273\ m.}$$

Ex. 4. *A, B, C, and D are the four corners of a square of side a. Equal charges of value q are placed at the four corners. Find out the force experienced by the charge placed at D.*

Sol. As shown in **fig. 4.15,** the sides of the squre are

$AB = BC = CD = DA = a$, and diagonal is $BD = a\sqrt{2}$.

Due to charge q at the corner A, thee force on the charge q placed at D is given by

$$F_1 = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2} \text{ along } \vec{AD}$$

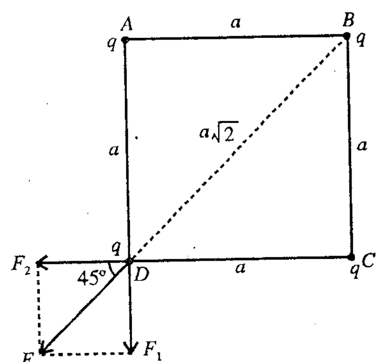


Fig. 4.15.

The force due to the charge q at the corner C on q at D is

$$F_2 = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2} \text{ along } \vec{CD}$$

The force due to the charge q at the conrner B on q at D is

$$F_3 = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{2a^2} \text{ along } \vec{BD}$$

Hence the resultant force on the charge q placed at D is given by

$$= F_3 + F_1 \cos 45^\circ + F_2 \cos 45^\circ \text{ along } \vec{BD}$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{q^2}{2a^2} + \frac{q^2}{a^2}\cos 45^\circ + \frac{q^2}{a^2}\cos 45^\circ\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}\left[\frac{1}{2}+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right] = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}\left(\frac{1}{2}+\sqrt{2}\right) = \frac{\mathbf{1.914}\,q^2}{4\pi\varepsilon_0\,a^2} \textbf{ along } \vec{BD}.$$

Ex. 5. *A charge q is to be divided on two bodies. What should be the values of the charges on the bodies so that the electric force between the two bodies can be maximum at a particular distance ?*

Sol. Let the charge on one body be q'. Hence the charge on the other body will be $q - q'$.

Force between the two bodies, $$F = \frac{1}{4\pi\varepsilon_0}\frac{q'(q-q')}{d^2}$$

For F to be maximum at a particular distance, $\frac{dF}{dq'} = 0$ or $q - 2q' = 0$ or $q' = \frac{q}{2}$

Hence, *the charge is to be divided equally on the two bodies to obtain maximum electric force.*

Ex. 6. *Volume charge density of a spherical charge distribution is given by the following equation :*

$$\rho = \rho_0\left(1-\frac{r}{a}\right) \quad \textit{when} \quad 0 < r < a$$

and $$\rho = 0 \textit{ when } r = a$$

Find the total charge in the spherical region.

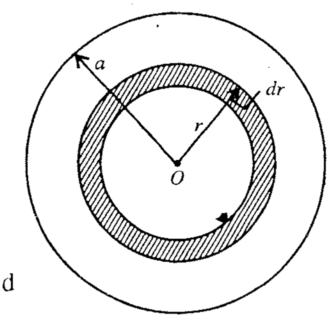


Fig. 4.16.

Sol. Total charge, $$q = \int_0^a \rho\, dV = \int_0^a \rho_0\left(1-\frac{r}{a}\right)4\pi r^2 dr$$

where $dV = 4\pi r^2 dr$ is the volume of the spherical shell of radius r and thickness dr [**fig. 4.16**].

$$\therefore \quad q = 4\pi\rho_0\int_0^a r^2 dr - \frac{4\pi\rho_0}{a}\int_0^a r^3 dr$$

$$= 4\pi\rho_0\frac{a^3}{3} - \frac{4\pi\rho_0}{a}\frac{a^4}{4} = 4\pi\rho_0 a^3\left[\frac{1}{3}-\frac{1}{4}\right] = \frac{\pi\rho_0 a^3}{3}.$$

Ex. 7. *Charge 2e is placed at the point A* (1, 2, 3) *cm. Find the electric field intensity at the point B* (4, 5, 10) *cm.*

Sol. $$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}} = \frac{1}{4\pi\varepsilon_0}\frac{2e}{r^2}\hat{\mathbf{r}}$$

Here $$\vec{r} = \vec{r}_B = \vec{r}_A = (4\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 10\hat{\mathbf{k}}) - (\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}})\text{ cm}$$

$$= 3\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 7\hat{\mathbf{k}}\text{ cm} = 10^{-2}(3\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 7\hat{\mathbf{k}})\text{ m}$$

$$\therefore \quad r = 10^{-2}\sqrt{9+9+49} = 10^{-2}\sqrt{67}\text{ m and } \hat{\mathbf{r}} = \frac{\vec{r}}{r} = \frac{3}{\sqrt{67}}\hat{\mathbf{i}} + \frac{3}{\sqrt{67}}\hat{\mathbf{j}} + \frac{7}{\sqrt{67}}\hat{\mathbf{k}}$$

Now, $$E = 9\times10^9\frac{2e}{(67)^{3/2}\times10^{-4}}(3\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 7\hat{\mathbf{k}}) = 5.25\times10^{-8}(3\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 7\hat{\mathbf{k}})\textbf{ volt/m.}$$

Ex. 8. *Adjoining figure shows a particular charge distribution $\frac{1}{2}q$, $-q$, $\frac{1}{2}q$. Show that for $r >> a$ the electric field at P is given by $3Q/4\pi\varepsilon_0 r^4$ where $Q = qa^2$.*

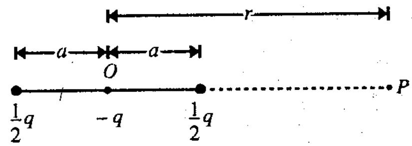


Fig. 4.17.

Sol. Electric field at P due to $q/2$ charge, situated left to $-q$,

$$\vec{E}_1 = \frac{1}{4\pi\varepsilon_0}\frac{q\hat{\mathbf{r}}}{2(r+a)^2},$$

where $\hat{\mathbf{r}}$ is a unit vector along OP **[fig. 4.17]**.

Electric field at P due to $q/2$ charge, right to q,

$$\vec{E}_2 = \frac{1}{4\pi\varepsilon_0}\frac{q\hat{\mathbf{r}}}{2(r-a)^2}$$

Electric field at P due to $-q$ charge, $\vec{E}_3 = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}$

$\therefore$ Total electric field at P is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{2(r+a)^2} + \frac{q}{2(r-a)^2} - \frac{q}{r^2}\right]\hat{\mathbf{r}}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{2r^2}\left[\left(1+\frac{a}{r}\right)^{-2} + \left(1-\frac{a}{r}\right)^{-2} - 2\right]\hat{\mathbf{r}}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{2r^2}\left[\left(1 - \frac{2a}{r} + \frac{(-2)(-3)}{2!}\frac{a^2}{r^2}\right) + \left(1 + \frac{2a}{r} + \frac{(-2)(-3)}{2!}\frac{a^2}{r^2}\right)\right]\hat{\mathbf{r}}$$

[Expanding by bionomial series in the condition $(a/r) << 1$ or $a << r$]

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{2r^2}\left[\frac{3a^2}{r^2} + \frac{3a^2}{r^2}\right]\hat{\mathbf{r}} = \frac{1}{4\pi\varepsilon_0}\frac{3qa^2}{r^4}\hat{\mathbf{r}} = \frac{1}{4\pi\varepsilon_0}\frac{3Q}{r^4}\hat{\mathbf{r}} \qquad [\because qa^2 = Q]$$

Thus the electric field at P is $\frac{1}{4\pi\varepsilon_0}\frac{3Q}{r^4}$ directed away from P.

Note :- The charge distribution in this problem is similar to an electric quadrupole and the electric field due to it has been found in the end on position.

Ex. 9. *Charges 5×10^{-7} C and -2.5×10^{-7} C are held at the two corners A and B of an equilateral triangle of 5 cm side* **[fig. 4.18]**. *Find the intensity of the electric field at the third corner C of the triangle due to the two charges.*

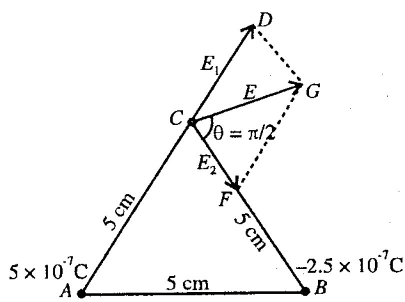


Fig. 4.18.

Sol. Intensity of electric field at C due to charge at A

$$E_1 = \frac{1}{4\pi\varepsilon_0} \times \frac{5 \times 10^{-7}}{(0.05)^2} = \frac{9 \times 10^9 \times 5 \times 10^{-7}}{(0.05)^2}$$

$= 1.8 \times 10^6$ N/C along $\vec{CD}$

Intensity of electric field at C due to charge at B

$$E_2 = \frac{1}{4\pi\varepsilon_0} \times \frac{2.5 \times 10^{-7}}{(0.05)^2} = 0.9 \times 10^6 \text{ N/C along } \vec{CF}$$

The triangle ABC is an equilateral triangle.

Hence the angle between CD and CF is $2\pi/3$.

$\therefore$ Resultant electric field at C is given by

$$E^2 = E_1^2 + E_2^2 + 2E_1E_2 \cos 2\pi/3$$
$$= (1.8 \times 10^6)^2 + (0.9 \times 10^{-6})^2 - 2 \times 1.8 \times 10^6 \times 0.9 \times 10^6 \times \tfrac{1}{2}$$
$$= (3.24 + 0.81 - 1.62)\, 10^{12} = 2.43 \times 10^{12}$$
$$\therefore \quad E = \mathbf{1.56 \times 10^6\ N/C}$$

The angle made by this resultant with CB is given by

$$\tan\theta = \frac{1.8 \times 10^6 \sin 120^\circ}{0.9 \times 10^6 + 1.8 \times 10^6 \cos 120^\circ} = \infty, \therefore \theta = \pi/2.$$

Ex. 10. *Charge is uniformly distributed over the circumference of a circle of radius R. Calculate the electric field at the centre of the ring charge if a part, corresponding to an angle 2α radians at its centre, has been removed. Assume the linear density of charge to be λ* **[Fig. 4.19].**

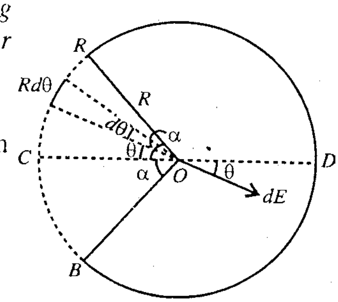


Fig. 4.19.

Sol. Charge per unit length of the ring = λ

Consider an element of length $Rd\theta$ with charge $dq = \lambda R d\theta$ at an angle θ with the radius OC **[fig. 4.19]**.

Due to this element the electric field at the centre O is

$$dE = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\lambda R d\theta}{R^2} = \frac{1}{4\pi\varepsilon_0 R} d\theta \quad \text{at an angle } \theta \text{ with } OD. \quad ...(i)$$

If we consider a similar element on the ring diametrically opposite to it, the electric field due to this element will have the same magnitude as dE but opposite to the direction in (i). Hence they will cancel each other. If we consider the entire ring similarly, we find electric field due to the complete ring at the centre O,

$$E = 0 \quad ...(ii).$$

Let us consider ACB portion of the ring, subtending an angle 2α at O. Now, let us resolve the electric field dE at O along OD and perpendicular to OD. The components are $dE \cos\theta$ along OD and $dE \sin\theta$, perpendicular to OD (down). Now if we consider a similar element $Rd\theta$ below to OC at an angle θ, the electric field at O due to this element will be $dE \cos\theta$ along OD and $dE \sin\theta$ perpendicular to OD (up).

Hence the perpendicular components will cancel and the components $dE \cos\theta$ along OD will add. Now, if we consider the ACB portion of the ring, the perpendicular components due to all its elements will add up to zero, the total component along OD *i.e.*, the electric field due to the ACB portion of the ring at its centre O will be obtained by integrating $dE \cos\theta$ *i.e.*,

$$E = \frac{\lambda}{4\pi\varepsilon_0 R} \int_{-R}^{R} d\theta \cos\theta = \frac{2\lambda \sin\alpha}{4\pi\varepsilon_0 R} \text{ along } \vec{OD}$$

But the electric field due to the entire ring at O is zero. If we remove the part ACB, then the electric field at O due to ADB portion of the ring will be

$$E' = -\frac{2\lambda \sin\alpha}{4\pi\varepsilon_0 R} \text{ opposite to } \vec{OC}.$$

Thus the electric field at the centre O uniformly charged ring with linear charge density λ due to ADB portion of the ring has the magnitude $2\lambda \sin\alpha/4\pi\varepsilon_0 R$ along $\vec{OC}$.

Ex. 11. *A charge q is uniformly distributed on a hemisphere. The radius of hemisphere is R. Find the intensity of electric field at the centre of hemisphere.* **(Agra 1971)**

Sol. Let, **fig. 4.20** represent a hemisphere of radius R. Surface density of charge on its surface is $\sigma = \dfrac{q}{2\pi R^2}$. Let us calculate the intensity of electric field at its centre O. Consider a small ring $ABDC$ obtained by two parallel planes AB and CD intersecting the hemisphere which subtends an angle $d\theta$ at is centre O **[fig. 4.20]**. Let G be the centre of the ring. Then $\angle GOA = \theta$.

Radius of the ring $GA = R \sin\theta$, thickness of ring $= Rd\theta$.

$\therefore$ Area of ring $= 2\pi R \sin\theta \times Rd\theta = 2\pi R^2 \sin\theta\, d\theta$.

Charge on the ring, $dq = 2\pi R^2 \sin\theta\, d\theta\sigma$

Intensity of electric field at centre O due to ring,

$$dE = \frac{1}{4\pi\varepsilon_0} \times \frac{2\pi R^2 \sin\theta\ d\theta\sigma}{R^2} \cos\theta$$

$$= \frac{\sigma}{2\varepsilon_0} \sin\theta \cos\theta\, d\theta \quad \text{along } \vec{GO}$$

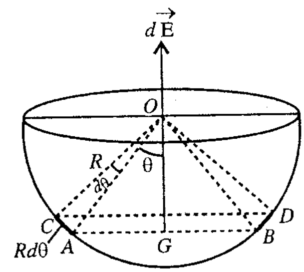


Fig. 4.20 : Electric field at the centre of a uniformly charged hemisphere.

Hence the resultant intensity of electric field at the point O due to the entire hemisphere is

$$E = \int_0^{\pi/2} \frac{\sigma}{2\varepsilon_0} \sin\theta \cos\theta\, d\theta = \frac{\sigma}{4\varepsilon_0} \int_0^{\pi/2} \sin 2\theta\, d\theta$$

$$= \frac{\sigma}{4\varepsilon_0} \left(\frac{-\cos 2\theta}{2} \right)_0^{\pi/2} = \frac{\sigma}{4\varepsilon_0} \left(\frac{1+1}{2} \right) = \frac{\sigma}{4\varepsilon_0} = \frac{q}{8\varepsilon_0 \pi R^2} \text{ along } \vec{GO}.$$

Ex. 12. *Two positive point charges, each* $+q$, *are placed on Y-axis at* $(0, -a)$ *and* $(0, a)$ *respectively. Find :* (i) *the intensity of electric field at any point* $(x, 0)$ *on X-axis,* (ii) *the value of x where the intensity of electric field is maximum,* (iii) *the intensity of electric field at the origin* $(0,0)$, (iv) *the type of motion of an electron if it is placed at a point* $(x, 0)$ *for* $a >> x$. (v) *Next if one positive charge on Y-axis is replaced by an equal negative charge, then show that the intensity of electric field is maximum at origin. If it is simple harmonic, find the period.*

Sol. Fig. 4.21 represents the location of charges.

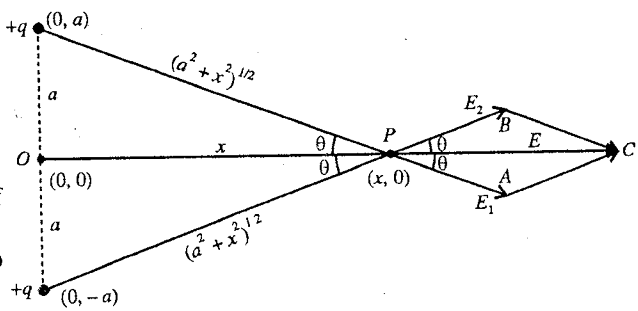


Fig. 4.21.

(i) Intensity of electric field at P due to upper $+q$ charge

$$E_1 = \frac{1}{4\pi\varepsilon_0} \frac{q}{a^2 + x^2} \quad \text{along } \vec{PA}$$

Intensity of electric field at P due to lower $+q$ charge

$$E_2 = \frac{1}{4\pi\varepsilon_0} \frac{q}{a^2 + x^2} \quad \text{along } \vec{PB}$$

As E_1 and E_2 have equal magnitudes, their perpendicular components $E_1 \sin\theta$ and $E_2 \sin\theta$ will cancel out and components along X-axis will add up. Thus the internsity of electric field at P *is*

$$E = E_1 \cos\theta + E_2 \cos\theta = 2E_1 \cos\theta$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{2qx}{(a^2 + x^2)^{3/2}} \quad \text{along } X\text{-axis} \qquad \left[\because \cos\theta = \frac{x}{(a^2 + x^2)^{1/2}} \right]$$

(ii) For maximum intensity of electric field,

$$\frac{dE}{dx} = 0 \text{ i.e., } \frac{1}{(a^2 + x^2)^{3/2}} - \frac{3}{2} \frac{x(2x)}{(a^2 + x^2)^{5/2}} = 0$$

or $$a^2 + x^2 - 3x^2 = 0 \text{ or } 2x^2 = a^2 \text{ or } x = \pm a/\sqrt{2}.$$

If we calculate d^2E/dx^2 at $x = a/\sqrt{2}$ or $x = -a/\sqrt{2}$, we find its value to be negative. Hence the intensity of electric field is maximum at $x = a/\sqrt{2}$ and $x = -a/\sqrt{2}$.

(iii) At the origin $x = 0$, $E = 0$ from the expresion of (i) *i.e.,* the intensity of electric field is zero at the origin.

(iv) If an electron of charge $-e$ and mass m is placed at $(x, 0)$, then the force acting on the electron is

$$F = -eE = -\frac{1}{4\pi\varepsilon_0} \cdot \frac{2eqx}{(a^2 + x^2)^{3/2}} \text{ towards } O.$$

For $a >> x$, $$F = -\frac{2eqx}{4\pi\varepsilon_0 a^3}.$$

According to Newton's second law, the above equation can be written as

$$m\frac{d^2x}{dt^2} = -\frac{2eq}{4\pi\varepsilon_0 a^3}x \text{ or } \frac{d^2x}{dt^2} + \omega^2 x = 0$$

where $\omega^2 = 2eq/4\pi\varepsilon_0 a^3 m$. This equation represents a simple harmonic motion, whose time period is given by

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{4\pi\varepsilon_0 a^3 m}{2eq}}.$$

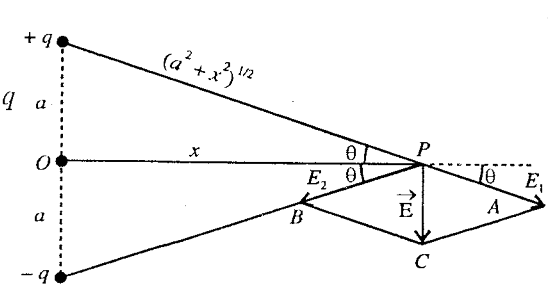


Fig. 4.22.

(v) Let the lower $+q$ charge be replaced by $-q$ **[fig. 4.22]**.

Intensity of the field at P due to $+q$,

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{a^2 + x^2} \text{ along } \vec{PA}$$

Intensity of the field at P due to $-q$,

$$E_2 = \frac{1}{4\pi\varepsilon_0}\frac{q}{a^2 + x^2} \text{ along } \vec{PB}$$

If we resolve E_1 and E_2 along X-axis and Y-axis, we find that the X-components are cancelled out and Y-components are added up along $\vec{PC}$.

Thus the resultant intensity of the electric field at P is

$$E = E_1 \sin\theta + E_2 \sin\theta = 2E_1 \sin\theta = \frac{1}{4\pi\varepsilon_0}\frac{2qa}{(a^2 + x^2)^{3/2}} \text{ along } \vec{PC}, \left[\because \sin\theta = \frac{a}{(a^2 + x^2)^{1/2}}\right]$$

At the origin ($x = 0$), it is obvious that the intensity of the electric field is maximum.

EXERCISE 4·1

1. Two charges of same magnitude exert a force of 2 N on each other. The force between them becomes 18 N on decreasing the distance by 0.5 m. (a) What was the initial distance between the two charges ? (b) What is the magnitude of each charge ?
[**Ans.** (a) 0.75 m or 0.375 m; (b) For $r = 0.75$ m, $q = \pm 11.18\ \mu$C and for $r = 0.375$m, $q = \pm 5.59\ \mu$C.]
2. Three small balls, each of mass 10^{-2} kg are suspended separately from a common point by silk threads, each 1 metre long. The balls are identically charged and hanged at the corners of an equilateral triangle of side 0.1m. Find the charge on each ball
[**Ans.** 60 nanocoulomb.]
3. Two similar balls each of mass m and charged by an equal charge q, are suspended from a point by the two different silk threads each of length l. If the distance between the balls in equilibrium is d, show that $d = \left(\dfrac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{1/3}$.
4. Three point charges + 2, + 2 and − 2 nanocoulombs are placed at the vertices of an equilateral triangle of side 10 cm. Find the force on one of the + 2 nanocoulombs charges.
[**Ans.** 3.6 μC at 60° from the side (+ 2, − 2).]

5. Two charges 8 μC and 2 μC respectively are held fixed with a separation of 40 cm. Where should a third charge 2 μC be placed so that it does not experience any force ?
[**Ans.** 40 cm.]
6. Three charges, each equal to q, are placed at the three corners of a square of side a. Determine the electric field at the fourth corner.
[**Ans.** $\frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{a^2}\left(\frac{1}{2} + \sqrt{2}\right)$.]
7. Infinite charges, each q coulomb, are placed on X-axis at distances $x = 1, 2, 4, 8$ .... metre respectively from the origin. Find the electric field at $x = 0$.
[**Ans.** $q/3\pi\varepsilon_0$.]
[**Hint :** $E = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{(1)^2} + \frac{q}{(2)^2} + \frac{q}{(4)^2} + \frac{q}{(8)^2} + ...\right] = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + ...\right)$
$= \frac{q}{4\pi\varepsilon_0} \times \frac{1}{1-(1/4)} = \frac{q}{3\pi\varepsilon_0}$.]
8. Two similar helium filled balloons, tied to a 5 g weight, float in equilibrium. The length of each thread is 1.0 metre and the balloons are 0.60 metre apart. Find the charge on each of them ?
[**Ans.** 5.55×10^{-7} coul.]
9. Two identical spheres, having unequal, opposite charges are placed at a distance of 0.90 m apart. After touching the two spheres mutually, they are again placed at the same distance apart. Now they repel each other with a force of 0.025 N. Find the final charge on each of them.
[**Ans.** $\pm 1.5 \times 10^{-4}$ C.]
10. Two equal point-charges $q = \pm\sqrt{2}$ μC are placed at each of the two opposite corners of a sqaure, and equal point-charges q' at each of the other two corners. What must be the value of q' so that the resultant force on q is zero ?
[**Ans.** – 0.5 C.]
11. A point charge of Q coulomb is placed at each of the corners of a cube of side a metre. Calculate the magnitude of the force on any of the charges. What is the direction of this force relative to the cube edges ?
[**Ans.** $29.6 \times 10^9\ Q^2/a^2$ N; Along the outgoing extended diagonal through that charge.]
12. Two equal positive charges are kept at a distance $2a$ from each other. A small test charge is moved on a plane perpendicular to the line joining the two charges and passing midway between them. Find the radius of the circle of symmetry at which the force on the test charge is maximum. What is the direction of this force if the test charge is positive.
[**Ans.** $a\sqrt{2}$; In the plane away from the line joining the two charges.]
13. A charge q is placed at each of the eight corners of a cube of side a. Find the resultant force acting on any of the charge.
[**Ans.** $\frac{0.261q^2}{\varepsilon_0 a^2}$, directed outwards along the diagonal passing through that charge.]
14. Determine in volt per meter the intensity of electric field which can just balance the weight of (i) an electron, (ii) an oil drop of mass 0.001 milligram containing one extra electron.
[**Ans.** 5.57 volt/m; 6×10^{10} volt/m.]
[**Hint :** $eE = mg$.]
15. An electron is separated from a proton by a distance of 0.53 Å. Find the electric field at the location of the electron.
[**Ans.** 5.13×10^{11} N/C.]
16. Charges of $+ 1.2 \times 10^{-8}$ C and $- 1.6 \times 10^{-8}$ C are placed at two points A and B respectively, distant 5 cm from each other. Calculate the electric field at a point C distant 3 cm from A and 4 cm from B.
[**Ans.** 1.5×10^5 N/C at $\theta = \tan^{-1}(3/4)$ from $\overrightarrow{AC}$.]

17. Two point-charges of 5×10^{-19} C and 20×10^{-19} C are separated by a distance 2 meter. At which point on the line joining them, the electric field is zero ? If a charge q is placed at this point then what will be the force acting on it ?
[**Ans.** 2/3 m from small charge along the line in between the two charges; 0]

18. Charges of +144 and + 81 nC are separated by 0.42 metre. Find the electric field at a point midway between the charges. Also find a point where the electric field is zero. How many such points are there ?
[**Ans.** 1.3×10^4 N/C. towards + 81 nC.; at 0.24 m from + 144 nC. towards + 81 nC.; only one.]

19. A pith ball coated with a layer of tin whose mass is m kg, is suspended in a uniform electric field E N/C by means of a silk thread of length l metre. If a charge q coulomb is given to the ball, it displaces by a distance d metre. Show that $E = mgd/(q\sqrt{l^2 - d^2})$.

20. Two particles A and B having charges q and $2q$ are placed on a smooth horizontal table with a separation d. A third particle is to be clamped in such a way that the particles A and B remain at rest on the table due to electric forces. Find the charge on C and where it is to be clamped ?
[**Ans.** $-0.344q$ between q and $2q$ at a distance of 0.41 d from q.]

21. A rod of length 10 cm carries a charge 50×10^{-6} C distributed uniformly along its length. Determine the magnitude of the electric field at a point 10 cm from both ends of the rod.
[**Ans.** 5.2×10^7 N/C.]

22. A rod of length l has a total charge q distributed uniformly along its length. It is bent in the shape of a semicircle. Find the magnitude of the electric field at the centre of the semicircle.
[**Ans.** $q/2\varepsilon_0 l^2$.]

23. A uniformly charged rod of linear charge density λ has been bent in the form of an arc of a circle of radius R. This arc subtends an angle 2α at the centre. Calculate the electric field at the centre of the circle.
[**Ans.** $2\lambda \sin \alpha/4\pi\varepsilon_0 R$ along the symmetrical radius.]

24. A thin glass rod is bent in the form of a semicircle of radius r and its diameter is kept in vertical position. In the upper half part of it, charge $+q$ is uniformly distributed and in the lower half part charge $-q$ is uniformly distributed. Find the electric field intensity at the centre of the semicircle.

[**Ans.** $\frac{1}{4\pi\varepsilon_0} \cdot \frac{4q}{\pi r^2}$ along vertical diameter downwards.]

25. A charge $+q$ is uniformly distributed over a ring of radius a. Calculate the electric field strength at a point situated on the axis of the ring charge. Also, show that an electron, constrained to move along the axis of this charge, can perform small oscillations about its centre. Calculate the frequency of oscllations.

[**Ans.** Simple harmonic oscillations for a >> x; $\nu = \frac{1}{2\pi}\sqrt{\frac{eq}{4\pi\varepsilon_0 m a^3}}$.]

26. A circular surface of radius R has a charge q distributed uniformly over it. Calculate the electric field at a point on its axis at a distance d from the centre.

[**Ans.** $\frac{1}{4\pi\varepsilon_0} \cdot \frac{2q}{R^2}\left(1 - \frac{d}{\sqrt{a^2 + d^2}}\right)$ along the axis away from the centre.]

27. A charge q_1 is uniformly distributed on $\frac{1}{6}$ th of circumference of an insulating ring of radius R and a charge q_2 on the remaining $\frac{5}{6}$ th of its circumference. Show that the intensity of electric field at the centre of ring is $\frac{1}{4\pi\varepsilon_0} \cdot \frac{3(q_2 - 5q_1)}{5\pi R^2}$.

4·10. Electric Dipole

A combination of two equal and opposite charges separated by a small distance constitute an **electric dipole.**

A dipole is characterised by its **dipole moment** *whose magnitude is equal to the product of one of its charges and the separation between the charges.* **Fig. 4.23** shows an electric dipole which consists of two charges $-q$ and $+q$ placed at A and B points respectively at a separation of d. The dipole moment $\vec{p}$ is given by

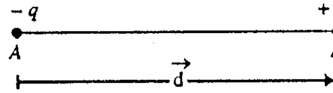


Fig. 4.23 : Electric dipole.

$$\vec{p} = q\vec{d} \qquad ...(1)$$

where $\vec{d}$ is the vector distance, joining the negative charge $-q$ to the positive charge $+q$. The line along the direction of the dipole moment ($\vec{p}$) is called the **axis of the dipole.** A molecule consisting of a positive and negative ion is an example of an electric dipole *e.g.*, HCl molecule, CO molecule etc.

If we take $d = 2a$, the dipole moment in magnitude is given by

$$p = qd = q \times 2a \qquad ...(2)$$

Its S.I. unit is coulomb-metre (C-m).

4·11. Electric Field due to a Dipole

Now, we shall determine the electric field at a point due to a dipole in three positions :

(i) When the point lies along the axis of the dipole; this is also called *end on position* or *longitudinal position.*

(ii) When the point lies on the perpendicular bisector of the dipole axis; this is also called *transverse position* or *broad side on position.*

(iii) When the point lies in any position.

(i) When the point lies along the axis of the dipole (i.e., **end on position**) : Consider an electric dipole AB, consisting $-q$ and $+q$ charges at a separation of $2a$. We want to determine the electric field due to the dipole at a point P at a distance r from the midpoint O of the dipole along its axis [**fig. 4.24**]. The distance of the point P from $-q$ charge is $AP = r + a$ and from $+q$ charge is $BP = r - a$.

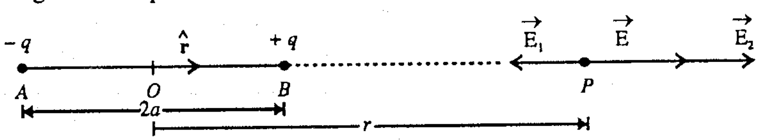


Fig. 4.24 : Electric field due to a dipole at a point along its axis.

The electric field at the point P due to the charge $-q$ (at A) is

$$\vec{E}_1 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{-q}{(r+a)^2}\hat{r} = -\frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{(r+a)^2}\hat{r}$$

and due to the charge $+q$ (at B) is

$$\vec{E}_2 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{(r-a)^2}\hat{r}$$

According to the principle of superposition, the resultant electric field at P is given by

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = \frac{q}{4\pi\varepsilon_0}\hat{r}\left[-\frac{1}{(r+a)^2} + \frac{1}{(r-a)^2}\right]$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{4ar}{(r^2-a^2)^2}\hat{r} = \frac{1}{4\pi\varepsilon_0}\frac{2\vec{p}\,r}{(r^2-a^2)^2} \qquad ...(1)$$

where $\vec{p} = 2aq\hat{r}$.

If P is at a large distance from the dipole *i.e.*, $r >> a$, than $r^2 - a^2 \simeq r^2$ and hence

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{2\vec{p}}{r^3} \qquad ...(2)$$

(ii) When the point lies on the perpendicular bisector of the dipole axis (*i.e.*, **broadside on position)** : Let the distance between the centre O of the dipole and the point P in this case be r.

Here as shown in **fig. 4.25** the distances

$$AP = BP = (r^2 + a^2)^{1/2}.$$

The electric field due to $-q$ at P is along PC and its magnitude is

$$E_1 = \frac{1}{4\pi\varepsilon_0}.\frac{q}{(r^2 + a^2)}$$

The electric field due to $+q$ at P is along PD and the magnitude is

$$E_2 = \frac{1}{4\pi\varepsilon_0}.\frac{q}{(r^2 + a^2)}$$

Hence $\vec{E}_1$ and $\vec{E}_2$ have equal magnitudes, but they have different directions. As shown in **fig. 4.25**, obviously the angle between $\vec{E}_1$ and $\vec{E}_2$ is 2θ. The resultant field $\vec{E}$ is given by law of addition of parallelogram of vectors *i.e.*,

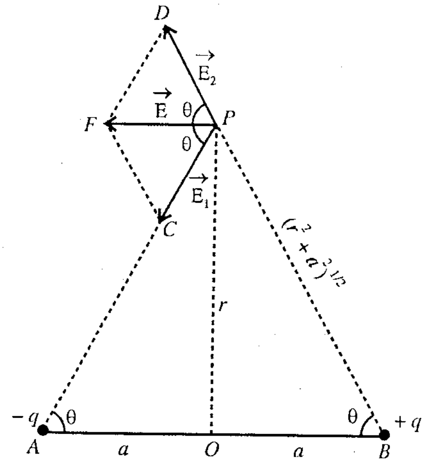


Fig. 4.25 : Electric field at a point on the perpendicular bisector of the dipole axis.

$$E^2 = E_1^2 + E_2^2 + 2E_1E_2 \cos 2\theta$$

or $$E^2 = \frac{q^2}{(4\pi\varepsilon_0)^2}\left[\frac{1}{(r^2 + a^2)^2} + \frac{1}{(r^2 + a^2)^2} + \frac{2\cos 2\theta}{(r^2 + a^2)^2}\right]$$

$$= \frac{q^2}{(4\pi\varepsilon_0)^2}\frac{2}{(r^2 + a^2)^2}(1 + \cos 2\theta) = \frac{4q^2\cos^2\theta}{(4\pi\varepsilon_0)^2(r^2 + a^2)^2}$$

Thus, $$E = \frac{2q\cos\theta}{4\pi\varepsilon_0(r^2 + a^2)} = \frac{2qa}{4\pi\varepsilon_0(r^2 + a^2)^{3/2}} \qquad \left[\because \cos\theta = \frac{a}{(r^2 + a^2)^{1/2}}\right]$$

As $\vec{E}_1$ and $\vec{E}_2$ are equal in magnitude, hence their resultant $\vec{E}$ will be along $\vec{PF}$ inclined with the same angle θ with $\vec{E}_1$ and $\vec{E}_2$. But $\vec{PF}$ is opposite to $\vec{p}$ $(p = 2qa)$, hence the resultant electric field at P is

$$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\frac{\vec{p}}{(r^2 + a^2)^{3/2}} \qquad ...(3)$$

For $r >> a$, $$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\frac{\vec{p}}{r^3} \qquad ...(4)$$

A comparison of eqs. (2) and (4) shows that *in magnitude the electric field due to a dipole at a far point in the end on position is double to that in the broad side on position.*

(iii) When the point lies in any position : Consider a point P with polar coordinates (r, θ), where $r = OP$ [**fig. 4.26**]. In order to determine the electric field at the point P, we resolve the dipole moment $\vec{p}$ into two components $p \cos\theta$ along OP and $p \sin\theta$ perpendicular to OP. Hence for dipole moment $p \cos\theta$, the point P will be in the end on position and for dipole moment $p \sin\theta$, the point P will be in the broad side on position. Thus as shown in **fig. 4.26**.

$$E_r = \frac{1}{4\pi\varepsilon_0}\frac{2p\cos\theta}{r^3}$$

and $$E_\theta = \frac{1}{4\pi\varepsilon_0}\frac{p\sin\theta}{r^3}$$

The resultant electric field at P is

$$E = \sqrt{E_r^2 + E_\theta^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{r^3}\sqrt{4\cos^2\theta + \sin^2\theta}$$

$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{r^3}\sqrt{3\cos^2\theta + 1} \qquad ...(5)$$

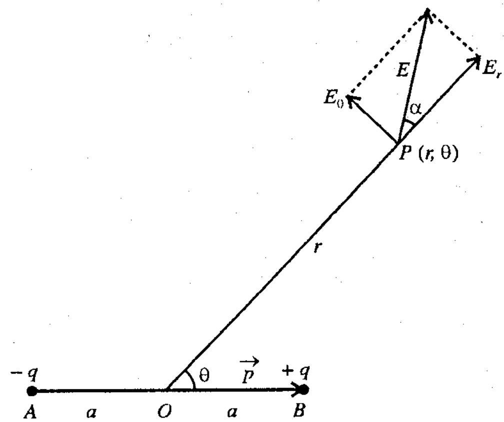


Fig. 4.26 : Electric field due to a dipole at any point.

If the resultant field makes an angle α with the radial direction OP, then

$$\tan\alpha = \frac{E_\theta}{E_r} = \frac{1}{2}\tan\theta$$

or $$\alpha = \tan^{-1}\left(\frac{1}{2}\tan\theta\right) \qquad ...(6)$$

From eq. (5) we can obtain the expressions for electric field due to a dipole in the end on and broadside on positions corresponding to $\theta = 0$ and $\theta = \pi/2$ respectively. Thus,

(i) for $\theta = 0$, $\quad E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2p}{r^3}$ (end on position)

(ii) for $\theta = \pi/2$, $\quad E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{r^3}$ (broad side on position)

4·12. Torque on an Electric Dipole in a Uniform Electric Field

Let an electric dipole AB be placed in a uniform electric field $\vec{E}$. It consists of two charges $-q$ and $+q$ placed at A and B respectively at a vector distance $\vec{AB} = \vec{d}$. Suppose O is the midpoint of AB and the axis of the dipole AB makes an angle θ with the electric field $\vec{E}$ at an instant **[fig. 4.27]**.

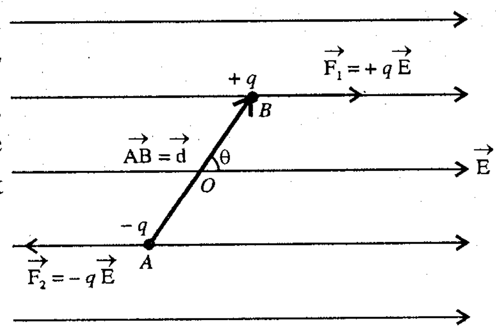


Fig. 4.27 : Torque on an electric dipole.

Force on charge $+q$, $\quad \vec{F}_1 = q\vec{E}$

Force on charge $-q$, $\quad \vec{F}_2 = -q\vec{E}$

Torque of $\vec{F}_1$ about O, $\quad \vec{\tau}_1 = \vec{OB}\times\vec{F}_1$

$$= q(\vec{OB}\times\vec{E}) \qquad \left[\because \text{Torque } \vec{\tau} = \vec{r}\times\vec{F}\right]$$

Torque of $\vec{F}_2$ about O, $\quad \vec{\tau}_2 = \vec{OA}\times\vec{F}_2 = -q(\vec{OA}\times\vec{E}) = q(\vec{AO}\times\vec{E})$

Hence the resultant torque acting on the dipole is

$$\vec{\tau} = \vec{\tau}_1 + \vec{\tau}_2 = q(\vec{OB}\times\vec{E}) + q(\vec{AO}\times\vec{E}) = q\left[(\vec{AO}+\vec{OB})\times\vec{E}\right] = q(\vec{AB}\times\vec{E}) \qquad ...(1)$$

But the dipole moment is $\vec{p} = q\vec{d} = q\vec{AB}$,

$\therefore$ Torque, $\vec{\tau} = \vec{p} \times \vec{E}$ ...(2)

Its magnitude $\tau = p\,E \sin\theta$ ...(3)

Thus the torque, acting on a dipole of dipole moment $\vec{p}$, in a uniform electric field $\vec{E}$ is a vector quantity whose magnitude is $pE \sin\theta$ and direction is perpendicular to the plane containing the dipole axis ($\vec{p}$) and electric field ($\vec{E}$). If $\vec{p}$ and $\vec{E}$ are in the plane of the paper, then direction of the torque $\vec{\tau}$ is perpendicular to the plane of the paper downward.

Now, if the dipole is placed perpendicular to the direction of the electric field $\vec{E}$, then $\theta = \pi/2$ and hence the value of the torque is

$$\tau = pE \sin\theta = pE \sin \pi/2 = pE \quad ...(4)$$

Obviously this is the maximum value of the torque *i.e.*, $\tau_{max} = pE$.

Now, if $E = 1$, then

$$\tau_{max} = p \quad ...(5)$$

Thus the **dipole moment** can be defined as *the maximum value of the torque, acting on the dipole when it is placed perpendicular to the direction of the uniform electric field of unit intensity.*

4·13. Work Done in Rotating an Electric Dipole in Uniform Electric Field

Let an electric dipole be rotated in a uniform electric field $\vec{E}$ through an angle θ to the position $A'B'$ from its equilibrium position AB. In this position, the torque acting on the dipole is $\vec{\tau} = \vec{p} \times \vec{E}$ or $\tau = pE \sin\theta$ in the clockwise direction [**fig. 4.28**]. Obviously the work has been done by us in rotating the dipole through an angle θ from the equilibrium position. If the dipole is further rotated through an angle $d\theta$, then the additional work done by us

$$dW = \text{Torque} \times \text{angular displacement}$$
$$= pE \sin\theta \, d\theta \quad ...(1)^*$$

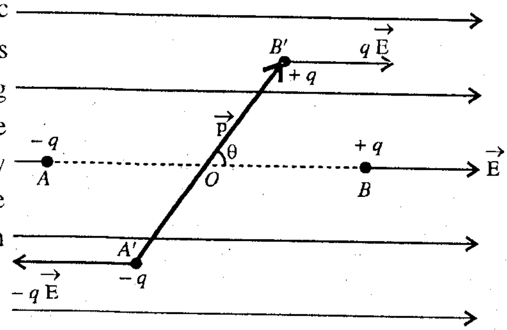


Fig. 4.28 : Calculation of work done in rotating an electric dipole in a uniform electric field.

Hence the work done in rotating the dipole from $\theta = 0$ to $\theta = \theta$ is

$$W = \int_0^\theta pE \sin\theta \, d\theta = pE\,(-\cos\theta)_0^\theta \text{ or } W = pE\,(1 - \cos\theta) \quad ...(2)$$

This is the expression for the work done in rotating an electric dipole by an external agent (or us) in a uniform electric field through an angle θ from the direction of the field or equilibrium position.

Let us find the amount of work done if the dipole is rotated through $\theta = 90°$ and $\theta = 180°$.

When $\theta = 90°$, $W = pE$ ...(3)

When $\theta = 180°$, $W = 2pE$ ...(4)

4·14. Potential Energy of Electric Dipole

Usually the potential energy of an electric dipole is assumed to be **zero**, when $\theta = \pi/2$. The potential energy in the θ position of the dipole is equal to the work done in rotating the dipole from $\theta = \pi/2$ (zero potential energy position) to this (θ) position *i.e.*,

$$U = -\int \tau \, d\theta = \int_{\pi/2}^{\theta} pE \sin\theta \, d\theta = pE\,(-\cos\theta)_{\pi/2}^{\theta} = -pE\cos\theta = -\vec{p}\cdot\vec{E} \quad ...(1)$$

* $dW = -\tau\, d\theta$, where τ is torque due to conservative force field and is equal to $-pE \sin\theta$, if we consider positive for θ increasing angle. Hence $dW = pE \sin\theta\, d\theta$.

We find from eq. (1) that

(i) When the dipole is along the direction of the electric field (*i.e.*, $\theta = 0°$),

$$U_{\theta = 0°} = -pE \qquad ...(2)$$

In fact this is the minimum potential energy and the electric dipole in this position is in **stable equilibrium**.

(ii) When the dipole is perpendicular to the field (*i.e.*, $\theta = \pi/2$),

$$U_{\theta = \pi/2} = -pE \cos \pi/2 = 0 \qquad ...(3)$$

which is the **zero potential energy position.**

(iii) When the dipole is opposite to the direction of the electric field (*i.e.*, $\theta = \pi$),

$$U_{\theta = \pi} = -pE \cos \pi = pE \qquad ...(4)$$

This is the maximum potential energy of the dipole and in this position it is said to be in **unstable equilibrium position.**

4·15. Dipole in a Non-uniform Electric Field

Consider a small dipole in a non-uniform electric field *e.g.*, the electric field due to a point charge. In such a case, the forces acting on the two charges of the dipole are not equal in magnitude and also their directions are not exactly opposite to each other [**fig. 4.29**].

For a small dipole, if we take the electric field $\vec{E}$ at the centre of the dipole, the torque is still given by

$$\vec{\tau} = \vec{p} \times \vec{E} \qquad ...(1)$$

as in case of uniform field. However in addition to the torque, the dipole experiences a net translatory force because the forces acting on the two charges are not equal in magnitude.

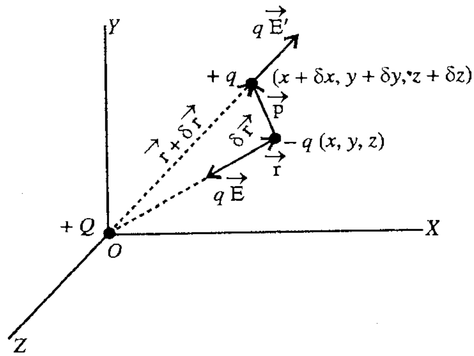


Fig. 4.29 : Dipole is a nonuniform electric field.

Calculation of the Net Translatory Force on the Dipole

Let $\vec{E}$ be the electric field at the site (x, y, z) of the charge $-q$ of the dipole and $\vec{E}'$ at the position $(x + \delta x, y + \delta y, z + \delta z)$ of the charge $+q$ of the dipole. $\vec{E}$ and $\vec{E}'$ can be expressed in components form as

$$\vec{E} = E_x\hat{\mathbf{i}} + E_y\hat{\mathbf{j}} + E_z\hat{\mathbf{k}} \quad \text{and} \quad \vec{E}' = E'_x\hat{\mathbf{i}} + E'_y\hat{\mathbf{j}} + E'_z\hat{\mathbf{k}} \qquad ...(2)$$

As the variation of the electric field is continuous from the position of $-q$ to the position of $+q$, hence

$$E'_x = E_x + \delta E_x = E_x + \frac{\partial E_x}{\partial x}\delta x + \frac{\partial E_x}{\partial y}\delta y + \frac{\partial E_x}{\partial z}\partial z \qquad ...(3)$$

Hence the net force on the dipole in X-direction is

$$F_x = qE'_x - qE_x = \frac{\partial E_x}{\partial x}q\delta x + \frac{\partial E_x}{\partial y}q\delta y + \frac{\partial E_x}{\partial z}q\delta z \qquad ...(4)$$

But dipole moment

$$\vec{p} = q\delta\vec{r} = p_x\hat{\mathbf{i}} + p_y\hat{\mathbf{j}} + p_z\hat{\mathbf{k}} = q\delta x\,\hat{\mathbf{i}} + q\delta y\,\hat{\mathbf{j}} + q\delta z\,\hat{\mathbf{k}}$$

i.e.,

$$p_x = q\delta x,\ p_y = q\delta y \ \text{ and } p_z = q\delta z \qquad ...(5)$$

$$\therefore \quad F_x = p_x\frac{\partial E_x}{\partial x} + p_y\frac{\partial E_x}{\partial y} + p_z\frac{\partial E_x}{\partial z} = \left(\vec{p}\cdot\vec{\nabla}\right)E_x \qquad ...(6)$$

Similarly, we can write the y and z components of the net force on the dipole as

$$F_y = \left(\vec{p} \cdot \vec{\nabla}\right) E_y \text{ and } F_z = \left(\vec{p} \cdot \vec{\nabla}\right) E_z \qquad ...(7)$$

Hence the net transtatory force acting on a small dipole in a nonuniform field $\vec{E}$ is given by

$$\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k} = \left(\vec{p} \cdot \vec{\nabla}\right) E_x\hat{i} + \left(\vec{p} \cdot \vec{\nabla}\right) E_y\hat{j} + \left(\vec{p} \cdot \vec{\nabla}\right) E_z\hat{k}$$

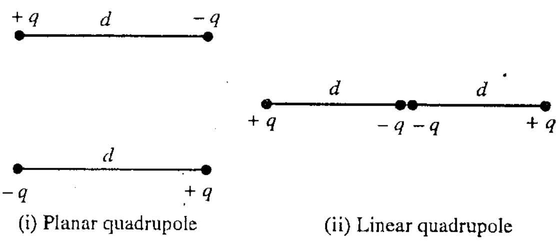
$$= \left(\vec{p} \cdot \vec{\nabla}\right)(E_x\hat{i} + E_y\hat{j} + E_z\hat{k})$$

i.e.,

$$\vec{F} = \left(\vec{p} \cdot \vec{\nabla}\right)\vec{E} \qquad ...(8)$$

4·16. Electric Field due to an Electric Quadrupole in End on Position*

A quadrupole is an arrangement of two parallel dipoles, having dipole moments equal in magnitude but opposite in direction. **Fig. 4.30** shows two quadrupole arrangements : (i) planar quadrupole and (ii) linear quadrupole.



(i) Planar quadrupole (ii) Linear quadrupole

Fig. 4.30 : Quadrupoles.

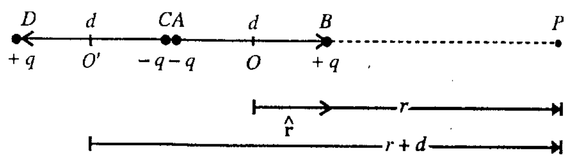


Fig. 4.31 : Electric field due to a quadrupole in end on position.

In order to calculate the electric field, we shall consider the case of linear quadrupole and determine the electric field in an end on position at an axial point *P*. This linear quadrupole is composed of two dipoles *AB* and *CD* [**fig. 4.31**] whose negative charges $-q$ and $-q$ are at the same place. The distance of *P* from the centre *O* of *AB* dipole is *r* and from the centre *O'* of *CD* dipole is $r + d$.

The electric field $\vec{E}_1$ due to *AB* dipole at *P* is

$$\vec{E}_1 = \frac{1}{4\pi\varepsilon_0} \frac{2\vec{p}r}{(r^2 - d^2/4)^2}$$

and the electric field $\vec{E}_2$ due to *CD* dipole at *P* is

$$\vec{E}_2 = -\frac{1}{4\pi\varepsilon_0} \cdot \frac{2\vec{p}(r+d)}{\left[(r+d)^2 - d^2/4\right]^2}$$

where $\vec{p} = q\vec{d} = qd\,\hat{r}$.

Hence the resultant electric field due to the two dipoles or quadrupole is obtained by using the principle of superposition *i.e.,*

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = \frac{1}{4\pi\varepsilon_0}\left[\frac{2\vec{p}r}{(r^2 - d^2/4)^2} - \frac{2\vec{p}(r+d)}{\left[(r+d)^2 - d^2/4\right]^2}\right] \qquad ...(1)$$

At large *r i.e.,* $r >> d$, then

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\left[\frac{2\vec{p}}{r^3} - \frac{2\vec{p}}{(r+d)^3}\right] = \frac{1}{4\pi\varepsilon_0} \cdot \frac{2\vec{p}}{r^3}\left[1 - \left(1 + \frac{d}{r}\right)^{-3}\right]$$

* For electric field and potential due to a quadrupole at a general point see Art. 4.26.

or
$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{2qd\hat{\mathbf{r}}}{r^3}\left[1 - \left(1 - \frac{3d}{r}\right)\right] = \frac{1}{4\pi\varepsilon_0} \cdot \frac{6qd^2\hat{\mathbf{r}}}{r^4} \quad ...(2)$$

The quantity $2qd^2\hat{\mathbf{r}} = \vec{Q}_d$ is called **electric quadrupole moment.** Hence

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \frac{3\vec{Q}_d}{r^4} \quad ...(3)$$

Thus, *the electric field due to a linear quadrupole varies as inversely as the fourth power of the distance.* Hence the electric field due to a linear quadrupole decreases more rapidly (as $1/r^4$) than that due to a dipole where the field decreases as $1/r^3$.

Ex. 1. *Two charges + 2.0 μC and – 2.0 μC are situated at a separation of 2.5 mm and form an electric dipole. Calculate :* (i) *dipole moment,* (ii) *electric field at distance 0.30 m in end on position,* (iii) *electric field at distance 0.30 m in broad side on position.*

Sol. Here, one charge of dipole $q = 2.0\ \mu C = 2.0 \times 10^{-6}$ C, separation between the two charges $d = 2.5 \times 10^{-3}$ m, $r = 0.30$ m.

(i) Dipole moment $p = qd = 2.0 \times 10^{-6} \times 2.5 \times 10^{-3} = \mathbf{5.0 \times 10^{-9}}$ **C-m.**

(ii) Electric field in end on position, $E = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3} = \frac{9 \times 10^9 \times 2 \times 5.0 \times 10^{-9}}{(0.3)^3} = \mathbf{3.3 \times 10^3}$ **N/C.**

(iii) Electric field in board side on position

$$E = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3} = \frac{9 \times 10^9 \times 5.0 \times 10^{-9}}{(0.3)^3} = \mathbf{1.67 \times 10^3\ N/C.}$$

Ex. 2. *A sample of* HCl *gas is placed in an elctric field of* 2.5×10^4 *N/C. The dipole momnet of* HCl *molecule is* 3.4×10^{-30} *C-m. Calculate the maximum torque that can act on a molecule.*

Sol. Torque $= pE \sin\theta$

Its maximum value will be when $\sin\theta = 1$ or $\theta = \pi/2$ *i.e.,*

$$\text{Maximum torque} = pE = 3.4 \times 10^{-30} \times 2.5 \times 10^4 = \mathbf{8.5 \times 10^{-26}\ N\text{-}m.}$$

Ex. 3. *Two point-charges of* $+3.2 \times 10^{-19}$ *C and* -3.2×10^{-19} *C are situated at a distance of* 2.4×10^{-10} *m from each other and constitute an electric dipole. This dipole is placed in a uniform electric field of* 4.0×10^5 *V m*$^{-1}$*. Calculate :* (i) *electric dipole moment,* (ii) *potential energy of the dipole in equilibrium position,* (iii) *work done in rotating the dipole through* 180° *from the equilibrium position* ($1/4\pi\varepsilon_0 = 9.0 \times 10^9\ N\ m^2\ C^{-2}$).

Sol. (i) Electric dipole moment, $p = qd = 3.2 \times 10^{-19} \times 2.4 \times 10^{-10} = \mathbf{7.68 \times 10^{-29}}$ **C-m.**

(ii) Potential energy of the dipole in the elctric field $\vec{E}$ is

$$U = \vec{p} \cdot \vec{E} = -pE\cos\theta.$$

where θ is the angle between the axis of the dipole and the field. In equilibrium θ = 0°, and hence

$$U_0 = -pE$$

Here, $p = 7.68 \times 10^{-29}$ C-m and $E = 4.0 \times 10^5$ V-m^{-1}.

$$\therefore \quad U_0 = -7.68 \times 10^{-29} \times 4.0 \times 10^5 = \mathbf{-3.07 \times 10^{-23}\ J.}$$

(iii) The work done in rotating the dipole from the direction of the electric field E is given by

$$W = pE\,(1 - \cos\theta)$$

For θ = 180° or cos θ = – 1,

$$W = 2pE = 2 \times 7.68 \times 10^{-29} \times 4.0 \times 10^5 = \mathbf{6.14 \times 10^{-23}\ J.}$$

EXERCISE 4·2

1. Two equal and opposite charges of magnitude 2 μC are located at a separation of 3.0 cm. Calculate the dipole moment. **(Ujjain 2003)**
[**Ans.** 6×10^{-8} C-m.]

2. Calculate the electric field at a point at distance 1 cm from an electric dipole of dipole moment $\frac{5}{3} \times 10^{-27}$ cm when the point is (i) on the axis of dipole. (ii) on the perpendicular bisector of dipole. **(Ujjain 2004)**
[**Ans.** (i) 3×10^{-11} N/C; (ii) 1.5×10^{-11} N/C.]

3. Two particles with charges -2×10^{-6} C and 2×10^{-6} C are placed at a separation of 1 cm. (i) What is the electric dipole moment of the pair ? (ii) Find the electric field on the axis of the dipole 1 m away from the centre of the dipole. (iii) Find the electric field at a point on the perpendicular bisector of the dipole and 1 m away from the centre.
[**Ans.** (i) 2×10^{-8} C-m; (ii) 360 N/C; (iii) 180 N/C.]

4. The dipole moment vector $\vec{p}$ of a short electric dipole point is the positive x-direction. Taking the origin at the centre of the dipole, show that the cartesian components of the electric field at a point P (x, y) are given by

$$E_x = \frac{1}{4\pi\varepsilon_0}\frac{p(2y^2 - x^2)}{(x^2+y^2)^{5/2}} \quad \text{and} \quad E_y = \frac{1}{4\pi\varepsilon_0}\frac{3pxy}{(x^2+y^2)^{5/2}}.$$

[**Hint :** $E_x = E_r \cos\theta - E_\theta \sin\theta$; $E_y = E_r \sin\theta - E_\theta \cos\theta$.

Here $$E_r = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2p\cos\theta}{r^3} \quad \text{and } E_\theta = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p\sin\theta}{r^3}$$

$$\therefore \quad E_x = \frac{1}{4\pi\varepsilon_0}\left(\frac{p\cos^2\theta}{r^3} - \frac{p\sin^2\theta}{r^3}\right) \quad \text{and} \quad E_y = \frac{1}{4\pi\varepsilon_0}\left(\frac{2p\cos\theta\sin\theta}{r^3} + \frac{p\sin\theta\cos\theta}{r^3}\right).$$

But $r\cos\theta = x$, $r\sin\theta = y$ and $r^2 = x^2 + y^2$, hence

$$E_x = \frac{p}{4\pi\varepsilon_0}\left(\frac{2r^2\cos^2\theta}{r^5} - \frac{r^2\sin^2\theta}{r^5}\right) \quad \text{and} \quad E_y = \frac{p}{4\pi\varepsilon_0}\left(\frac{3(r\cos\theta)(r\sin\theta)}{r^5}\right)$$

or $$E_x = \frac{p}{4\pi\varepsilon_0}\frac{(2x^2 - y^2)}{(x^2+y^2)^{5/2}} \quad \text{and } E_y = \frac{p}{4\pi\varepsilon_0}\frac{3xy}{(x^2+y^2)^{5/2}}.]$$

4·17. Work Done on a Charge in an Electrostatic Field Expressed as a Line Integral

Let a charge Q be placed at O. The inensity of the electric field due to this point charge at a distance $\vec{r}$ is given by [**fig. 4.32**].

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{r} \qquad ...(1)$$

where $\hat{r} = \vec{r}/r$.

The direction of $\vec{E}$ is along $\vec{r}$.

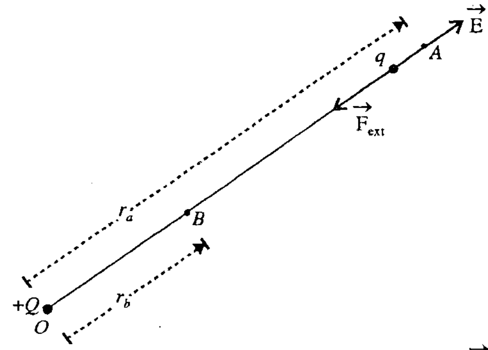


Fig. 4.32 : Work done on a charge in an electric field $\vec{E}$.

If a charge q is placed in the electric field $\vec{E}$, then the force acting on the charge is given by

$$\vec{F} = q\vec{E} \qquad ...(2)$$

If we move this charge q against the force $\vec{F}$ through a distance $d\vec{r}$, then the amount of work done on the charge will be given by

$$dW = -\vec{F}\cdot d\vec{r} = -q\vec{E}\cdot d\vec{r} \qquad ...(3)$$

If we move this charge from a point A to another point B, against the electrostatic force, then the total amount of work done on the charge will be obtained by integrating (2) for the given limits *i.e.*,

$$W = -\int_A^B \vec{F}\cdot d\vec{r} = -q\int_A^B \vec{E}\cdot d\vec{r} \qquad ...(4)$$

In order to move the charge q against the force $\vec{F}$ due to another charge Q, an external force $\vec{F}_{ext}$ has to be applied. This force $\vec{F}_{ext}$ is equal in magnitude but opposite to $\vec{F}$ *i.e.*, $\vec{F}_{ext} = -\vec{F}$.

If r_a and r_b the distances of A and B points from O, then the amount of work done on the charge is

$$W = -\int_A^B \frac{1}{4\pi\varepsilon_0}\frac{Qq}{r^2}\hat{r}\cdot d\vec{r} = -\int_{r_a}^{r_b}\frac{1}{4\pi\varepsilon_0}\frac{Qq}{r^2}dr = \frac{Qq}{4\pi\varepsilon_0}\left[\frac{1}{r}\right]_{r_a}^{r_b} = \frac{Qq}{4\pi\varepsilon_0}\left[\frac{1}{r_b}-\frac{1}{r_a}\right] \qquad ...(5)$$

Here, we have moved the charge q radially. Instead, we may move the charge along a curve between A and B point **[fig. 4.33]**, then the amount of work done on the charge q can be expressed as

$$W = -q\int_A^B \vec{E}\cdot d\vec{l} \qquad ...(6)$$

where $d\vec{l}$ is the small displacement along the curve.

Hence the work done per unit charge is given by

$$W' = \frac{W}{q} = -\int_A^B \vec{E}\cdot d\vec{l} \qquad ...(7)$$

The integral $\int_A^B \vec{E}\cdot d\vec{l}$ in eq. (5) is called the **line integral of the electric field.**

4·18. Conservative Nature of Electrostatic Field

We consider first the electric field due to a point charge Q. The electric field $\vec{E}$ at a distance $\vec{r}$ from the charge Q is given by

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{r} \qquad ...(1)$$

Consider two points A and B at radial distances r_a and r_b from the charge Q. Now we carry a unit positive charge from A to B by different possible paths. We will find that *the work done on the unit positive charge is the same for all possible paths* from A and B *i.e.*, the line integral of the electric field $\int_A^B \vec{E}\cdot d\vec{l}$ is independent of the path between A and B. In fact this is the property of a conservative force field. Hence **electrostatic field is conservative in nature.**

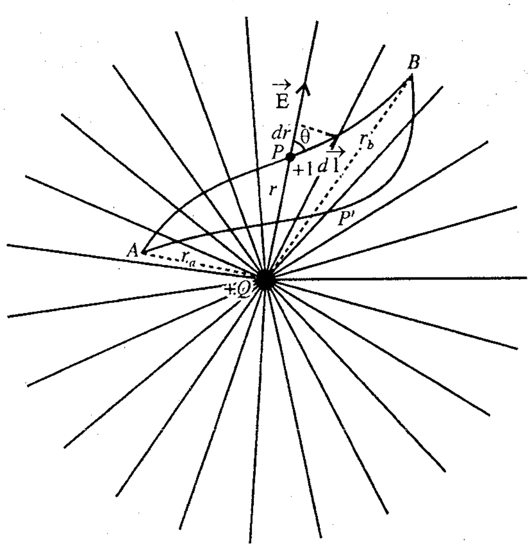


Fig. 4.33 : Work done along different paths between the points *A* and *B* in an electric field due to a point charge.

Proof : Suppose we carry a unit positive charge from A to B along APB path **[fig. 4.33]**. The line integral of the electric field can be written as

$$\int_A^B \vec{E}\cdot d\vec{l} = \int_A^B E\cos\theta\, dl = \int_A^B \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\cos\theta\, dl \qquad ...(2)$$

But from **fig. 4.33**, we see that $\cos\theta\, dl$ or $dl\cos\theta = dr$. This means that the work done during a small displacement $d\vec{l}$ depends only on its radial component dr. Thus the value of the line integral is

$$\int_A^B \vec{E}\cdot d\vec{l} = \int_A^B \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\,dr = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_a}-\frac{1}{r_b}\right] \qquad \text{...(3)}$$

Thus the value of the line integral of the electric field depends only on the initial and final positions r_a and r_b and the value will be the same, if we move the unit charge along another path, say $AP'B$ [**fig. 4.33**].

In other words, the work done in carrying unit positive charge from one point to another in an electric field $\vec{E}$, is given by

$$W' = \frac{W}{q} = -\int_A^B \vec{E}\cdot d\vec{l} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_b}-\frac{1}{r_a}\right] \qquad \text{...(4)}$$

is independent of the path followed between the two points. Hence **the electrostatic field is a conservative force field.**

Now, if a charge q is carried from point A to point B, along any curve between the two points, the work done on the charge will be given by

$$W = -q\int_A^B \vec{E}\cdot d\vec{l} = \frac{Qq}{4\pi\varepsilon_0}\left[\frac{1}{r_b}-\frac{1}{r_a}\right] \qquad \text{...(5)}$$

This is the same expression for the work done on a charge q in an electric field (obtained in last section), if the charge is moved radially from r_a distance to r_b distance.

4·19. Electric Potential V

The electric potential difference $(V_2 - V_1)$ *between two points in an electric field* $\vec{E}$ *is defined as the amount of work done in moving a unit positive charge from first point to second point.* Thus,

$$V_2 - V_1 = -\int_1^2 \vec{E}\cdot d\vec{r} \qquad \text{...(1)}$$

The potential is usually taken zero at infinite distance. Now if we consider the first point to be at infinity and take $V_1 = 0$, then the potential V at second point at $\vec{r}$ is given by

$$V = -\int_\infty^{\vec{r}} \vec{E}\cdot d\vec{r} \qquad \text{...(2)}$$

Thus *the potential at a point in an electric field is defined as the amount of work done in moving a unit positive charge from infinity to that point.* As the electric potential is work done per unit charge, it is a scalar quantity.

Now, if a charge q is moved between the two points, then the amount of work done gives, the potential energy difference between them *i.e.*,

$$U_2 - U_1 = q\,(V_2 - V_1) \qquad \text{...(3)}$$

Electric potential at a point due to a point charge : Consider a point charge $+Q$ placed at a point O. We want to determine the potential V at a point P distant r from O. From the definition of the electric potential at a point P, we write

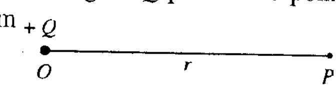


Fig. 4.34 : Electric potential due to a point charge + Q.

$$V = -\int_\infty^{d} \vec{E}\cdot d\vec{r} \qquad \text{...(4)}$$

But $\vec{E}$ due to the point charge Q at $\vec{r}$ is given by

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\,\hat{\mathbf{r}} \qquad \text{...(5)}$$

Hence, $$V = -\int_\infty^{r} \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\,\hat{\mathbf{r}}\cdot d\vec{r} = -\frac{Q}{4\pi\varepsilon_0}\int_\infty^{r}\frac{dr}{r^2} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r}\right]_\infty^{r} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r} \qquad \text{...(6)}$$

The electric potential at a point due to a system of point charges Q_1, Q_2..... can be obtained by finding the potential due to the individual charges using eq. (6) and then adding them. Thus,

$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q_1}{r_1} + \frac{1}{4\pi\varepsilon_0}\frac{Q_2}{r_2} + \ldots\ldots = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{Q_i}{r_i} \qquad \ldots(7)$$

where r_1, r_2 ...... are the distances of the point from the point charges Q_1, Q_2, ... respectively.

S.I. unit of electric potential is volt (= joule/coulomb) and the dimensional formula for it is $ML^2T^{-3}A^{-1}$.

4·20. Relation between $\vec{E}$ and V at a Point $\vec{E} = -\vec{\nabla} V$

The potential at a point $\vec{r}$ is given by

$$V = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} \qquad \ldots(1)$$

We express $\vec{E}$ and $d\vec{r}$ in rectangular cartesian coordinates,

$$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k} \text{ and } d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\therefore \qquad V = \int (E_x\hat{i} + E_y\hat{j} + E_z\hat{k})\cdot(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

or

$$V = -\int E_x\,dx - \int E_y\,dy - \int E_z\,dz \qquad \ldots(2)$$

Differentiating this equation partially with respect to x, y and z, we get

$$\frac{\partial V}{\partial x} = -E_x,\ \frac{\partial V}{\partial y} = -E_y \text{ and } \frac{\partial V}{\partial z} = -E_z$$

Hence,

$$E_x = -\frac{\partial V}{\partial x},\ E_y = -\frac{\partial V}{\partial y} \text{ and } E_z = -\frac{\partial V}{\partial z}$$

Therefore,

$$\vec{E} = \hat{i}E_x + \hat{j}E_y + \hat{k}E_z = -\hat{i}\frac{\partial V}{\partial x} - \hat{j}\frac{\partial V}{\partial y} - \hat{k}\frac{\partial V}{\partial z}$$

i.e.,

$$\vec{E} = -\left(\hat{i}\frac{\partial V}{\partial x} + \hat{j}\frac{\partial V}{\partial y} + \hat{k}\frac{\partial V}{\partial z}\right) = -\left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)V$$

or

$$\vec{E} = -\text{grad } V = -\vec{\nabla}V \qquad \ldots(3)$$

Thus, *the intensity of electric field at a point is equal to the negative gradient of electric potential.*

Now, if along $\vec{r}$, taken perpendicular to any constant potential surface, the maximum rate of change of potential is dV/dr, then

$$\vec{E} = -\vec{\nabla}V = -\frac{dV}{dr}\hat{r} \text{ or } E = -\frac{dV}{dr} \qquad \ldots(4)$$

where $\hat{r}$ is unit vector along $\vec{r}$.

If the plates of a parallel plate condensers are at V_1 and V_2 potentials at a separation d, then

$$E = -\frac{dV}{dr} = -\frac{V_2 - V_1}{d} = \frac{V_1 - V_2}{d} \qquad \ldots(5)$$

Curl of $\vec{E}$ (conservative force field) is zero : We know that the electric field $\vec{E}$ at a point can be expressed as the negative gradient of the electric potential V *i.e.,*

$$\vec{E} = -\vec{\nabla}V$$

In rectangular component form $\vec{E}$ is expressed as

$$\vec{E} = E_x\hat{\mathbf{i}} + E_y\hat{\mathbf{j}} + E_z\hat{\mathbf{k}}$$

Now, $$\text{curl } \vec{E} = \vec{\nabla}\times\vec{E} = -\vec{\nabla}\times\vec{\nabla}V$$

$$= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = -\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial V}{\partial x} & \frac{\partial V}{\partial y} & \frac{\partial V}{\partial z} \end{vmatrix}$$

$$= \hat{\mathbf{i}}\left(\frac{\partial^2 V}{\partial y\partial z} - \frac{\partial^2 V}{\partial z\partial y}\right) + \hat{\mathbf{j}}\left(\frac{\partial^2 V}{\partial z\partial x} - \frac{\partial^2 V}{\partial x\partial z}\right) + \hat{\mathbf{k}}\left(\frac{\partial^2 V}{\partial x\partial y} - \frac{\partial^2 V}{\partial y\partial x}\right)$$

But V is taken as a perfect differential,

$$\therefore \qquad \frac{\partial^2 V}{\partial y\partial z} = \frac{\partial^2 V}{\partial z\partial y}, \ \frac{\partial^2 V}{\partial z\partial x} = \frac{\partial^2 V}{\partial x\partial z} \text{ and } \frac{\partial^2 V}{\partial x\partial y} = \frac{\partial^2 V}{\partial y\partial x}$$

Hence $$\text{Curl } \vec{E} = 0.$$

Thus, **the curl of the electric field** (conservative force field) **vanishes.** Thus electric field is an **irrotational vector.**

4·21. Gain in Kinetic Energy by a Charged Particle in an Electric Field

If a positive charge q is accelerated from rest by the electric field $\vec{E}$, then the work done on the charged particle moving from one point to another, appears in the form of change in its kinetic energy (*work-energy theorem*) *i.e.*,

$$\int_1^2 \vec{F}\cdot d\vec{r} = \frac{1}{2}mv^2 \qquad ...(1)$$

where v is the speed acquired by the particle in the electric field.

But, $$\int_1^2 \vec{F}\cdot d\vec{r} = \int_1^2 q\vec{E}\cdot d\vec{r} = q\int_1^2 \vec{E}\cdot d\vec{r}$$

By definition of potential difference between the two points

$$V_1 - V_2 = \int_1^2 \vec{E}\cdot d\vec{r} = V \text{ (say)}; \ \therefore \ \int_1^2 \vec{F}\cdot d\vec{r} = qV \qquad ...(2)$$

Thus from eqs. (6) and (7), we have

$$\frac{1}{2}mv^2 = qV \qquad ...(3)$$

Gain in kinetic energy = Charge × Potenital difference

Thus *a positively charged particle gains kinetic energy in moving from a state of higher potential to a state of lower potential and loses kinetic energy in moving from a state of lower potential to that of higher potential.* The reverse case occurs for a negatively charged particle.

The energy of a charged particle is usually measured in electron volts. *When an electron moves through a potential difference of* 1 *volt, then the energy gained by it is* **one electron volt** *i.e.,*

1 eV = electronic charge × 1 volt potential difference

or **1 eV = 1.6 × 10^{-19} × 1 = 1.6 × 10^{-19} joule** ...(4)

The other multiples of electron volt, namely million electron volt (MeV) and billion electron volt (BeV), are also used.

$$\mathbf{1\ MeV = 10^6\ eV\ ;\ 1\ BeV = 10^9\ eV.}$$

For constant $\vec{E}$,
$$\int_1^2 \vec{F} \cdot d\vec{r} = q\vec{E} \cdot \int_1^2 d\vec{r} = q\vec{E} \cdot (\vec{r}_2 - \vec{r}_1) = q\vec{E} \cdot \vec{d}$$

where $\vec{r}_2 - \vec{r}_1 = \vec{d}$ is the displacement of the particle. If $\vec{E}$ and $\vec{d}$ are along the same direction, then $q\vec{E} \cdot \vec{d} = qEd$ and hence

$$\int_1^2 \vec{F} \cdot d\vec{r} = qEd \quad ...(5)$$

Using eq. (2), we have
$$qEd = qV \text{ or } Ed = V \text{ or } V = \frac{E}{d} \quad ...(6)$$

Useful data : Charge on electron or proton = 1.6×10^{-19} coulomb; Mass of electron = 0.91×10^{-30} Kg; Mass of proton = 1.67×10^{24} kg.

4·22. Equipotential Surface

An equipotential surface is the surface, at all the points of which the potential is constant. The potential due to a point charge at a distance r is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$$

and it is constant on a spherical surface of radius r. This is the equipotential surface [**fig. 4.35**]. The intensity of the field $\vec{E}$ at $\vec{r}$ is given by

$$\vec{E} = -\Delta V = -\frac{dV}{dr}\hat{\mathbf{r}}$$

where $\hat{\mathbf{r}} = \vec{r}/r$ is along OA [**fig. 4.35**].

Now if a unit charge moves a small distance $\vec{AB} = d\vec{r}$ on the surface, then

$$\vec{E} \cdot d\vec{r} = -\frac{dV}{dr}\hat{\mathbf{r}} \cdot d\vec{r} = 0$$

because vector $\hat{\mathbf{r}}$ is perpendicular to the vector $d\vec{r}$ at the surface. This means that for equipotential surface, the intensity of the field $\vec{E}$ is perpendicular to the surface at any point A. If a unit charge is moved from one point to the other on an equipotential surface, the amount of work done is equal to zero (because $-\int_A^B \vec{E} \cdot d\vec{r} = V_B - V_A = 0$).

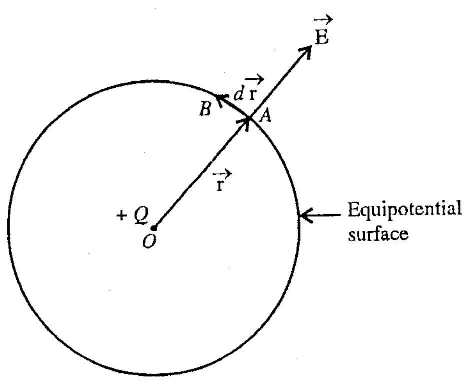


Fig. 4.35 : Equipotential surface.

If we have a charge q' at A, then force on the charge is given by.

$$\vec{F} = -\vec{\nabla}U = -\vec{\nabla}(qV) = -q\vec{\nabla}V = q\vec{E}.$$

Thus the force is also directed perpendicular to the equipotential surface at A. Hence, if any charge is moved on an equipotential surface, no work will be done.

In **fig. 4.35**, the equipotential surface due to a point charge is spherical. The equipotential surface may be plane (*e.g.*, plane, in case of a parallel surface between the plates of a parallel plate condenser) or any shape, provided at all of its points the potential is constant.

4·23. Electric Potential due to a Uniformly Charged Rod at a Point on Perpendicular Bisector

Consider a charge q uniformly distributed over a rod AB of length l with centre O. Charge per unit length on rod will be $\lambda = q/l$.

We want to find the electric potential at a point P at a distance r, situated on the perpendicualr bisector OP of the rod [**fig. 4.36**].

Consider a small element dx of the rod at a distance x from O. The charge on the element dx is $dq = \lambda\, dx$. The electric potential at P due to this element will be

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dx}{(r^2+x^2)^{1/2}}. \qquad ...(1)$$

Hence the potential at the point P due to the entire rod is obtained to be

$$V = \int_{-l/2}^{l/2}\frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dx}{(r^2+x^2)^{1/2}}$$

where the limits of integration are from $x = -l/2$ (at A) to $x = l/2$ (at B).

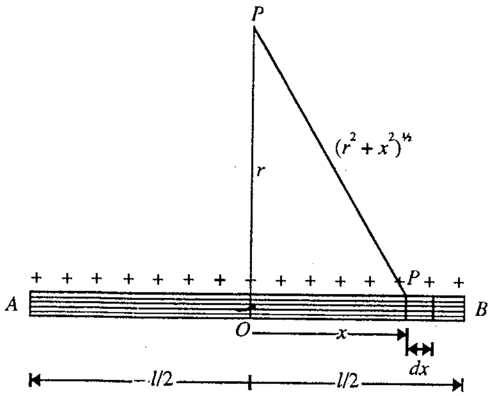


Fig. 4.36 : Electric potential due to a uniformly charged rod at a point on perpendicular bisector.

Thus, $$V = \frac{2\lambda}{4\pi\varepsilon_0}\int_0^{l/2}\frac{dx}{(r^2+x^2)^{1/2}} = \frac{\lambda}{2\pi\varepsilon_0}\left[\log_e\left(x+\sqrt{r^2+x^2}\right)\right]_0^{l/2}$$

or $$V = \frac{\lambda}{2\pi\varepsilon_0}\log_e\frac{(l/2)+\sqrt{r^2+l^2/4}}{r} \text{ or } V = \frac{q}{2\pi\varepsilon_0 l}\log_e\frac{(l/2)+\sqrt{r^2+l^2/4}}{r}. \qquad ...(2)$$

4·24. Electric Potential and Field due to Uniformly Charged Disc along its Axis

Consider a point P on the axis of a uniformly charged circular disc at a distance r from its centre O. Let the radius of the disc be R and its charge per unit area be σ.

Imagine the disc to be divided into a large number of concentric rings with O as centre, and consider one such ring of radius x and width dx. Join P to the extremity A of radius OA of the ring. Let $AP = p$ [**fig. 4.37**].

Now, area of the ring $= 2\pi x\, dx$,

and mass of the ring $= 2\pi x{\cdot}dx{\cdot}\sigma$

Each element of the ring is at the same distance p from P.

Therfore, the potential at P due to this ring, is given by

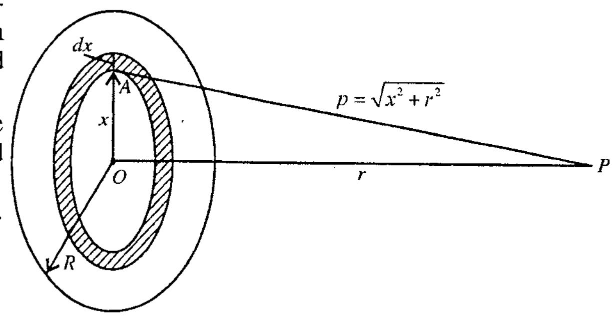


Fig. 4.37 : Electric potential and field due to a uniformly charged disc at an axial point.

$$dV = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\text{charge on the ring}}{p} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2\pi x\cdot dx\cdot\sigma}{p}$$

$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{2\pi x\cdot dx\,.\sigma}{\sqrt{x^2+r^2}}. \qquad \left[\because p^2 = x^2+r^2\right] \qquad ...(1)$$

The electric potential at the point P due to the whole disc is obtained by integrating it between the limits $x = 0$ and $x = R$. Hence

$$V = \int_0^R\frac{1}{4\pi\varepsilon_0}\cdot\frac{2\pi x\cdot dx\cdot\sigma}{\sqrt{x^2+r^2}} = \frac{\sigma}{2\varepsilon_0}\int_0^R\frac{x\,dx}{\sqrt{x^2+r^2}} = \frac{\sigma}{2\varepsilon_0}\left[\sqrt{x^2+r^2}\right]_0^R = \frac{\sigma}{2\varepsilon_0}\left[\sqrt{R^2+r^2}-r\right] \qquad ...(2)$$

The electric field at the point P will be given by

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left[\frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+r^2}-r\right)\right] = \frac{\sigma}{2\varepsilon_0}\left(1-\frac{r}{\sqrt{R^2+r^2}}\right). \qquad ...(3)$$

4·25. Electric Potential and Field due to a Dipole

Consider a dipole AB with charges $-q$ and $+q$ at a distance d apart. The midpoint of the dipole is at O. We want to calculate the electric potential at a point P, distant r from O. As d is very small compared to r, hence we can write distance AP and BP [fig. 4.38] as

$$AP = A'P = PO + OA' = r + \frac{d}{2}\cos\theta$$

and
$$BP = B'P = PO + OB' = r - \frac{d}{2}\cos\theta$$

where AA' and BB' are the perpendiculars drawn from A and B respectively on the line PO.

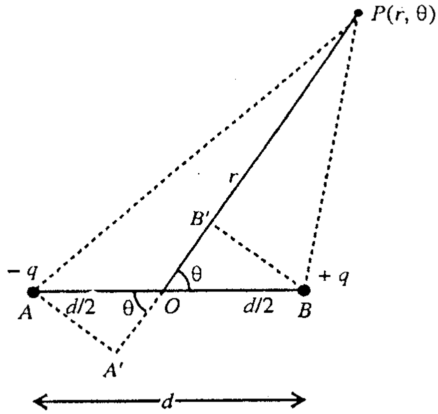


Fig. 4.38 : Electric potential due to a dipole at P (r, θ).

The potential at P due to $-q$ and $+q$ charges is given by

$$V = \frac{1}{4\pi\varepsilon_0}\frac{-q}{AP} + \frac{1}{4\pi\varepsilon_0}\frac{q}{BP}$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{r - \frac{d}{2}\cos\theta} - \frac{q}{r + \frac{d}{2}\cos\theta}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{qd\cos\theta}{r^2 - \frac{d^2}{4}\cos^2\theta} \quad ...(1)$$

For $r >> d$, $r^2 - \frac{d^2}{4}\cos^2\theta = r^2$ and hence

$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{qd\cos\theta}{r^2} \text{ or } V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p\cos\theta}{r^2} \quad ...(2)$$

Thus, *the electric potential due to a dipole at a point is inversely proportional to the square of its distance from the dipole.*

Special Cases :

(1) Electric potential due to a dipole in the end on (axial) position [fig. 4.39] : The potential at the axial point P is obtained from eq. (1) for $\theta = 0$ as

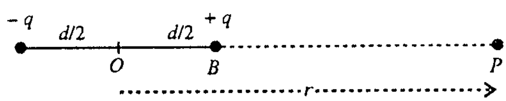


Fig. 4.39 : Electric potential due to a dipole in end on position.

$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{qd}{r^2 - d^2/4} \quad ...(3)$$

For $r >> d$,
$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{qd}{r^2} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{r^2} \quad ...(4)$$

(2) Electric potential due to a dipole in the broadside on (transverse) position : The potential at P in the transverse position is obtained from eq. (1) for $\theta = \pi/2$ [Fig. 4.40]. Thus,

$$V = 0. \quad ...(5)$$

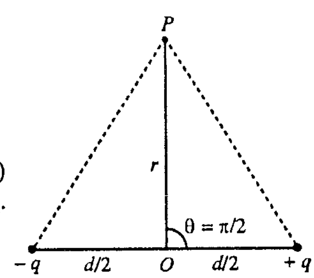


Fig. 4.40 : Electric field due to a dipole in broadside on position.

Calculation of Electric Field due to a Dipole from Electric Potential :

The electric potential at a point P (r, θ) is given by

$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p\cos\theta}{r^2} \text{ or } V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\vec{p}\cdot\vec{r}}{r^3} \quad ...(6)$$

The electric field $\vec{E}$ can be expressed as

$$\vec{E} = -\text{grad } V = -\vec{\nabla} V \qquad ...(7)$$

But in polar coordinates, the $\vec{\nabla}$ operator in two dimensions can be written as

$$\vec{\nabla} = \hat{r}\frac{\partial}{\partial r} + \hat{\theta}\frac{\partial}{r\partial\theta} \qquad ...(8)$$

where $\hat{r}$ and $\hat{\theta}$ are the unit vectors along $\vec{r}(=\vec{OP})$ and perpendicular to $\vec{r}$. Hence

$$\vec{E} = -\left(\hat{r}\frac{\partial V}{\partial r} + \hat{\theta}\frac{\partial V}{r\partial\theta}\right) = \hat{r}E_r + \hat{\theta}E_\theta \qquad ...(9)$$

where $E_r = -\frac{\partial V}{\partial r}$ and $E_\theta = -\frac{1}{r}\frac{\partial V}{\partial\theta}$ are the radial and transverse components of the field $\vec{E}$ respectively.

$\therefore$ Radial component of electric field

$$E_r = -\frac{\partial V}{\partial r} = \frac{1}{4\pi\varepsilon_0}\frac{\partial}{\partial r}\left(\frac{p\cos\theta}{r^2}\right) = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2p\cos\theta}{r^3} \qquad ...(10)$$

Transverse component of electric field

$$E_\theta = -\frac{\partial V}{r\partial\theta} = -\frac{\partial V}{r\partial\theta}\left(\frac{1}{4\pi\varepsilon_0}\cdot\frac{p\cos\theta}{r^2}\right) = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p\sin\theta}{r^3} \qquad ...(11)$$

$$\therefore \qquad \vec{E} = \frac{1}{4\pi\varepsilon_0}\left[\left(\frac{2p\cos\theta}{r^3}\right)\hat{r} + \left(\frac{p\sin\theta}{r^3}\right)\hat{\theta}\right]$$

The resultant electric field at P **[Fig. 4.41]** is

$$E = \sqrt{E_r^2 + E_\theta^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\sqrt{\left(\frac{2p\cos\theta}{r^3}\right)^2 + \left(\frac{p\sin\theta}{r^3}\right)^2}$$

or

$$E = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}\sqrt{3\cos^2\theta + 1} \qquad ...(12)$$

Its direction will be at an angle α from radial direction OP will be given by

$$\tan\alpha = \frac{E_\theta}{E_r} = \frac{p\sin\theta/r^3}{2p\cos\theta/r^3} = \frac{1}{2}\tan\theta$$

or

$$\alpha = \tan^{-1}\left(\frac{1}{2}\tan\theta\right) \qquad ...(13)$$

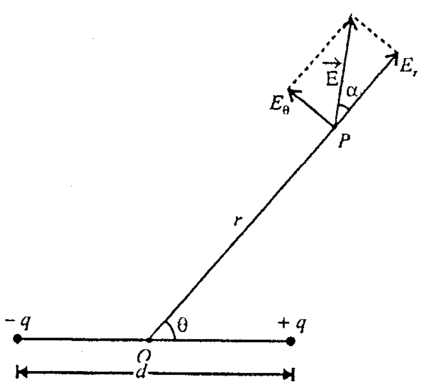


Fig. 4.41 : Electric field due to a dipole at (r, θ).

Thus, from eq. (12) *we see that the electric field due to a dipole at a point is inversely proportional to the cube of its distance from the dipole.*

Special Cases : (1) In the end on position $(\theta = 0)$:

$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2p}{r^3}, \ \alpha = 0 \qquad ...(14)$$

(2) In the broad side on position $(\theta = \pi/2)$:

$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{r^3}, \ \alpha = \pi/2 \qquad ...(15)$$

The field is antiparallel to the dipole axis.

4·26. Electric Potential and Field Due to a Quadrupole

Consider a linear quadrupole and take the origin O of the coordinaters at the centre of the quadrupole.

The electric potential at P due to the quadrupole is given by [**fig. 4.42**],

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{r_1} - \frac{2q}{r} + \frac{q}{r_2}\right]$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r_1} + \frac{1}{r_2} - \frac{2}{r}\right] \qquad ...(1)$$

Using cosine law in ΔOPA, $r_1^2 = r^2 + d^2 - 2rd\cos\theta$. and hence

$$\frac{1}{r_1} = \frac{1}{\left[r^2 + d^2 - 2rd\cos\theta\right]^{1/2}}$$

$$= \frac{1}{r}\frac{1}{\left[1 + \frac{d^2}{r^2} - \frac{2d}{r}\cos\theta\right]^{1/2}}$$

$$= \frac{1}{r}\left[1 + \left(\frac{d^2}{r^2} - \frac{2d}{r}\cos\theta\right)\right]^{-1/2}$$

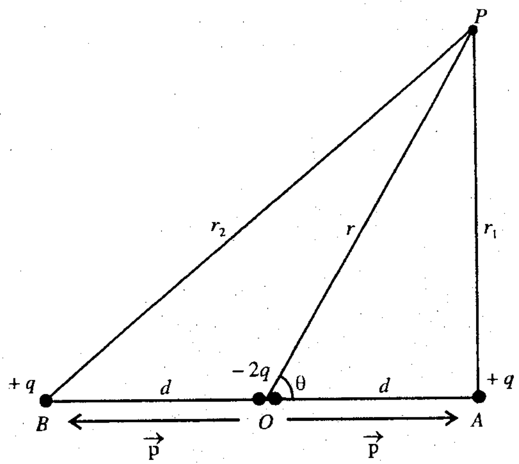


Fig. 4.42 : Electric potential and field at a point due to a quadrupole.

Using Bionomial expansion, we can write

$$\frac{1}{r_1} = \frac{1}{r}\left[1 - \frac{1}{2}\left(\frac{d^2}{r^2} - \frac{2d}{r}\cos\theta\right) + \frac{3}{8}\left(\frac{d^2}{r^2} - \frac{2d}{r}\cos\theta\right)^2 + \ldots\ldots\ldots\right]$$

As $r >> d$, hence retaining the terms upto the order of d^2/r^2 we have

$$\frac{1}{r_1} = \frac{1}{r}\left[1 - \frac{1}{2}\left(\frac{d^2}{r^2} - \frac{2d}{r}\cos\theta\right) + \frac{3}{8}\frac{4d^2}{r^2}\cos^2\theta + \ldots\right]$$

or

$$\frac{1}{r_1} = \frac{1}{r}\left[1 + \frac{d^2}{2r^2}\left(3\cos^2\theta - 1\right) - \frac{d}{r}\cos\theta\right] \qquad ...(2)$$

Similarly,

$$\frac{1}{r_2} = \frac{1}{r}\left[1 + \frac{d^2}{2r^2}\left(3\cos^2\theta - 1\right) + \frac{d}{r}\cos\theta\right] \qquad ...(3)$$

Substituting the values of $1/r_1$ and $1/r_2$ from eqs. (2) and (3) in (1), we obtain

$$V = \frac{q}{4\pi\varepsilon_0}\frac{1}{r}\left[1 + \frac{d^2}{2r^2}\left(3\cos^2\theta - 1\right) - \frac{d}{r}\cos\theta + 1 + \frac{d^2}{2r^2}\left(3\cos^2\theta - 1\right) + \frac{d}{r}\cos\theta\right]$$

or

$$V = \frac{qd^2}{4\pi\varepsilon_0 r^3}(3\cos^2\theta - 1) \qquad ...(4)$$

But $Q_d = 2qd^2$ is called **quadrupole moment,** hence

$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q_d}{2r^3}\left(3\cos^2\theta - 1\right) \qquad ...(5)$$

Thus *the electric potential due to a linear quadrupole is inversely proportional to the cube of the distance.*

Special Cases :

(1) Potential in the end on position : In this position, $\theta = 0$ and the point P lies on the axis. Hence

$$V = \frac{1}{4\pi\varepsilon_0}\frac{2qd^2}{r^3} = \frac{1}{4\pi\varepsilon_0}\frac{Q_d}{r^3} \qquad ...(6)$$

(2) Potential in the broad side on position : In this position $\theta = \pi/2$ and the point lies on the line perpendicular to the axis of the quadrupole and passing through the centre. Hence

$$V = -\frac{1}{4\pi\varepsilon_0}\frac{qd^2}{r^3} = -\frac{1}{4\pi\varepsilon_0}\frac{Q_d}{2r^3} \quad ...(7)$$

Calculation of Electric Field due to Quadrupole from Electric Potential

$$\vec{E} = -\vec{\nabla}V = -\left(\hat{\mathbf{r}}\frac{\partial V}{\partial r} + \hat{\boldsymbol{\theta}}\frac{\partial V}{r\partial\theta}\right) = \hat{\mathbf{r}}E_r + \hat{\boldsymbol{\theta}}\frac{\partial V}{\partial\theta} \quad ...(8)$$

∴ Radial component of electric field is

$$E_r = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r}\left[\frac{1}{4\pi\varepsilon_0}\frac{qd^2}{r^3}(3\cos^2\theta - 1)\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{3qd^2}{r^4}(3\cos^2\theta - 1) \quad ...(9)$$

Transverse component of electric field is

$$E_\theta = -\frac{1}{r}\frac{\partial V}{\partial\theta} = -\frac{1}{r}\frac{\partial}{\partial\theta}\left[\frac{1}{4\pi\varepsilon_0}\frac{qd^2}{r^3}(3\cos^2\theta - 1)\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{3qd^2}{r^4}\sin 2\theta \quad ...(10)$$

Hence
$$\vec{E} = \hat{\mathbf{r}}E_r + \hat{\boldsymbol{\theta}}E_\theta = \frac{1}{4\pi\varepsilon_0}\cdot\frac{3qd^2}{r^4}\left[(3\cos^2\theta - 1)\hat{\mathbf{r}} + \sin 2\theta\,\hat{\boldsymbol{\theta}}\right] \quad ...(11)$$

Its magnitude,
$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{3qd^2}{r^4}\left[5\cos^4\theta - 2\cos^2\theta + 1\right]^{1/2} \quad ...(12)$$

Thus, *the electric field of a linear quadrupole is inversely proportional to the fourth power of distance.*

Special Cases :

(1) Electric field on the axis of the dipole ($\theta = 0$) :

$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{6qd^2}{r^4} = \frac{1}{4\pi\varepsilon_0}\frac{3Q_d}{r^4} \quad ...(13)$$

(2) Electric field on the equatorial line of the dipole ($\theta = \pi/2$) :

$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{3qd^2}{r^4} = \frac{3}{8\pi\varepsilon_0}\frac{Q_d}{r^4} \quad ...(14)$$

4·27. Electrostatic Potential Energy of a System of Charges

The potential energy of a system of two charges at a distance apart is usually defined as the amount of work done by an external agent to bring the charges at that distance starting from infinite separation. Thus, if there are two charges q_1 and q_2 at a distance r_{12} apart, then the *electrostatic potential energy of the system* is

$$U_{12} = -\int_\infty^{r_{12}}\vec{F}\cdot d\vec{r} = -\frac{1}{4\pi\varepsilon_0}\int_\infty^{r_{12}}\frac{q_1q_2}{r^2}dr = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r_{12}} \quad ...(1)$$

where $\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r_{12}}\hat{\mathbf{r}}$ is the Coulomb (conservative) force.

Now, suppose that we bring a third charge q_3 from infinity at a position P, whose distance from the charge q_1 is r_{31} and from charge q_2, r_{32} [**fig. 4.43**]. The amount of work done in this process *i.e.*, the increase in the potential energy of the system is

$$= -\int_{\infty}^{P} \vec{F}_3 \cdot d\vec{r}\text{ , where } \vec{F}_3 \text{ is the Coulomb force on } q_3.$$

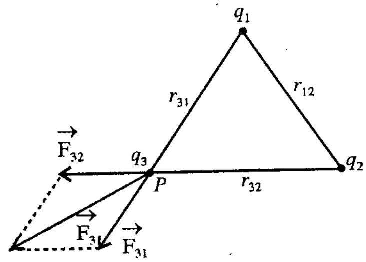


Fig. 4.43 : Calculation of potential energy for a system of three charges.

As the electrical interactions are additive,

$$\vec{F}_3 = \vec{F}_{31} + \vec{F}_{32} \qquad ...(2)$$

$$\therefore \quad -\int_{\infty}^{P} \vec{F}_3 \cdot d\vec{r} = -\int \vec{F}_{31} \cdot d\vec{r} - \int \vec{F}_{32} \cdot d\vec{r}$$

$$= -\frac{1}{4\pi\varepsilon_0}\int_{\infty}^{r_{31}} \frac{q_1 q_2}{r^2}\,dr - \frac{1}{4\pi\varepsilon_0}\int_{\infty}^{r_{31}} \frac{q_2 q_3}{r}\,dr$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_3}{r_{31}} + \frac{1}{4\pi\varepsilon_0}\frac{q_2 q_3}{r_{32}} = U_{13} + U_{23} \qquad ...(3)$$

Thus, the amount of work done in bringing q_3 to P is the sum of the work needed when q_1 is present alone and that needed when q_2 is present alone.

The total work done in assembling this arrangement of three charges *i.e.*, the potential energy of the system of three charges is

$$U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}} + \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\varepsilon_0}\frac{q_2 q_3}{r_{23}} \qquad ...(4)$$

or

$$U = U_{12} + U_{13} + U_{23} \qquad ...(5)$$

The potential energy belongs to the configuration as a whole. It is not meaningful to assign a certain fraction of it to one of the charges of the system.

Now, if we have a system of n charges, the potential energy of the system will be equal to the sum of the potential energy of all pairs of charges *i.e.*,

$$U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}} + \frac{1}{4\pi\varepsilon_0}\frac{q_2 q_3}{r_{23}} + ... + \frac{1}{4\pi\varepsilon_0}\frac{q_i q_j}{r_{ij}} + ... \text{ or } U = \sum_{\substack{\text{all pairs}\\ i \neq j}} \frac{q_i q_j}{r_{ij}} \qquad ...(6)$$

In expression (6) the summation is to be done for all pairs of particles i and j, excluding the case $i = j$, which is not a pair at all. The expression (6) can also be written in the following two forms :

$$U = \frac{1}{4\pi\varepsilon_0}\sum_{i>j}^{i=n}\sum_{j=1}^{j=n-1}\frac{q_i q_j}{r_{ij}} \qquad ...(7)$$

or

$$U = \frac{1}{2}\frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{i=n}\sum_{\substack{j=1\\ \neq i}}^{j=n}\frac{q_i q_j}{r_{ij}} \qquad ...(8)$$

In expression (7), first summation is for all values of i which are greater than j so that the same pair has not been counted twice (verify this for a system of three charges). In expression (8), each energy actually counted twice and the factor $\frac{1}{2}$ corrects for the over counting.

For example, in case of a system of three charges,

$$U = \frac{1}{2}\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1q_2}{r_{12}} + \frac{q_2q_1}{r_{21}} + \frac{q_1q_3}{r_{13}} + \frac{q_3q_1}{r_{31}} + \frac{q_2q_3}{r_{23}} + \frac{q_3q_2}{r_{32}}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1q_2}{r_{12}} + \frac{q_1q_3}{r_{13}} + \frac{q_2q_3}{r_{23}}\right] \qquad \left[\because \frac{q_iq_j}{r_{ij}} = \frac{q_jq_i}{r_{ji}}\right]$$

Now, suppose all the n charges are equal *i.e.*, $q_1 = q_2 = q_3 = \ldots = q_{n-1} = q_n = q$, and they are at equal distances from each other *i.e.*, $r_{12} = r_{23} = r_{31} = \ldots.. r$, then from eq. (8) we obtain

$$U = \frac{1}{2}\frac{1}{4\pi\varepsilon_0}\frac{n(n-1)q^2}{r} \qquad \ldots(9)$$

In eq. (8) the sum $\sum_{j=1, j\neq i}^{n} \frac{1}{4\pi\varepsilon_0}\frac{q_j}{r_{ij}} = V_i$ represents the potential at the position of i^{th} charge due to all the j^{th} charges ($j \neq i$). Hence we can express the **electrostatic potential energy** of n charges as

$$U = \frac{1}{2}\sum_{i=1}^{n} q_iV_i \qquad \ldots(10)$$

This potential energy represents the amount of work needed to put together the configuration of point charges.

4·28. Potential Energy of a Continuous Charge Distribution

If we have a continuous charge distribution, the total charge in a volume element dv will be

$$dq = \rho\, dv \qquad \ldots(1)$$

where ρ is the charge density at a point in dv. If V is the electric potential at the point (in dv), then the potential energy of the continuous charge distribution is [from eq. (10), Art. 4.27]

$$U = \frac{1}{2}\int_v \rho V\, dv \qquad \ldots(2)^*$$

Using Gauss's law,

$$\vec{\nabla}\cdot\vec{E} = \rho/\varepsilon_0 \text{ or } \rho = \varepsilon_0\,\vec{\nabla}\cdot\vec{E} \qquad \ldots(3)$$

$$\therefore \quad U = \frac{1}{2}\int_v \varepsilon_0(\vec{\nabla}\cdot\vec{E})V\, dv \qquad \ldots(4)$$

We know that

$$\vec{\nabla}\cdot(\vec{E}V) = V(\vec{\nabla}\cdot\vec{E}) + \vec{E}\cdot(\vec{\nabla}V);\ \therefore\ V(\vec{\nabla}\cdot\vec{E}) = \vec{\nabla}\cdot(\vec{E}V) + E^2 \qquad \ldots(5)$$

where $\vec{E}\cdot\vec{\nabla}V = \vec{E}\cdot(-\vec{E}) = -\vec{E}\cdot\vec{E} = -E^2$

Hence eq. (4) takes the form

$$U = \frac{\varepsilon_0}{2}\int_V [\vec{\nabla}\cdot(\vec{E}V)]\, dv + \frac{\varepsilon_0}{2}\int_V E^2\, dv \qquad \ldots(6)$$

According to Gauss's divergence theorem

$$\int_v (\vec{\nabla}\cdot\vec{E}V)\, dv = \int_S V\vec{E}\cdot d\vec{S}$$

$$\therefore \quad U = \frac{\varepsilon_0}{2}\int_S V\vec{E}\cdot d\vec{S} + \frac{\varepsilon_0}{2}\int_v E^2\, dV \qquad \ldots(7)$$

At large distances from the charge, E varies as $1/r^2$ and V as $1/r$, while surface area increases as r^2. Hence the surface integral decreases as $1/r$. Thus for very large distance r, the surface integral vanishes. Hence

$$U = \frac{1}{2}\int_v \varepsilon_0 E^2\, dv \qquad \ldots(8)$$

* The corresponding integrals for line and surface charges would be $\int \lambda V\, dl$ and $\int \sigma V\, dS$ respectively.

Thus the energy density u, where the electric field is E, is

$$u = \frac{1}{2}\varepsilon_0 E^2 \qquad ...(9)$$

This *energy stored per unit volume of the electric field.*

4·29. Electrostatic (Self) Energy

The potential energy of a charged body itself is called its electrostatic (self) energy. In other words, *the self energy of a charge distribution is defined as the amount of work done in assembling the constituent charges from infinite separations.*

Electrostatic energy of a uniformly charged non-conducting sphere : Consider a charge Q to be uniformly distributed over the volume of a shpere of radius R. Hence, the volume charge density is given by

$$\rho = \frac{Q}{(4/3)\pi R^3} \qquad ...(1)$$

Now, consider a spherical core of radius r, then the charge on the core will be given by

$$q = \frac{4}{3}\pi r^3 \rho$$

Potential at the surface of the spherical core of radius r is given by

$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{r} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\frac{4}{3}\pi r^3\rho}{r} = \frac{r^2\rho}{3\varepsilon_0}.$$

Next, if a thin layer of charge thickness dr is made to surround the core of radius r, then the charge in the spherical layer is

$$dq = 4\pi r^2\, dr\, \rho$$

In surrounding the core with charge q by a charge $dq = 4\pi r^2 dr\rho$, the amount of work done or the charge in the potential energy will be given by

$$dU = V\,dq = \frac{r^2\rho}{3\varepsilon_0}\cdot 4\pi r^2 dr\ \rho = \frac{4}{3\varepsilon_0}\pi\rho^2 r^4 dr \qquad ...(2)$$

Hence we obtain the electrostatic potential energy of the total sphere by integrating eq. (2) from $r = 0$ to $r = R$ *i.e.*,

$$U = \frac{4\pi\rho^2}{3\varepsilon_0}\int_0^R r^4 dr = \frac{4\pi\rho^2 R^5}{3\varepsilon_0 \times 5} = \frac{1}{4\pi\varepsilon_0}\times\frac{3}{5}\times\left(\frac{4}{3}\pi R^3\rho\right)^2 / R = \frac{1}{4\pi\varepsilon_0}\times\frac{3}{5}\frac{Q^2}{R} \qquad ...(3)$$

because from (1), $Q = \frac{4}{3}\pi R^3\rho$.

Thus, *the energy required to assemble a shpere of charge is directly proportional to the square of total charge and inversely proportional to the radius of the sphere.*

Electrostatic energy of a charged conducting sphere or uniformly charged spherical shell : In case of a conducting sphere, the charge resides over its surface uniformly and the system acts as a charged spherical shell. If a charge q is distributed over the spherical surface of radius R, then the potential at the surface is $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{R}$. Now, if the charge is increased to a value Q on the sphere starting from $q = 0$, then the amount of work done or electrostatic energy of the conducting sphere is

$$U_S = \int_0^Q V dq = \int_0^Q \frac{q\,dq}{4\pi\varepsilon_0 R} = \frac{1}{2}\cdot\frac{1}{4\pi\varepsilon_0}\cdot\frac{Q^2}{R} \qquad ...(4)$$

Ex. 1. *An electric field 10 N/C exists along the X-axis in space. Calculate the potential difference $V_B - V_A$ where the points A and B have the coordinates (4m, 2m) and (6m, 5m). Now, if a charge $-10^{-4}C$ is moved from A to B, find the change in potential energy.*

Sol.

$$V_B - V_A = -\int_A^B \vec{E}\cdot d\vec{r} \qquad ...(i)$$

Here, $\vec{E} = 10\,\hat{i}$ N/C *i.e.*, the electric field is constant. Thus from eq. (i)

$$V_B - V_A = -\vec{E}\cdot\int_A^B d\vec{r} = -\vec{E}\cdot(\vec{r}_B - \vec{r}_A)$$

Here, $\vec{r}_B = 6\hat{i} + 5\hat{j}$ m and $\vec{r}_A = 4\hat{i} + 2\hat{j}$ m ; $\therefore$ $\vec{r}_B - \vec{r}_A = 2\hat{i} + 3\hat{j}$ m

Hence $\quad V_B - V_A = -10\hat{i}\cdot(2\hat{i} + 3\hat{j}) = \mathbf{-20\ volt}$

Change in potential energy $= U_B - U_A = q(V_B - V_A) = -10^{-4}(-20) = \mathbf{2 \times 10^{-3}\ J.}$

Ex. 2. *The electric potential exists in space is*

$$V = 20(xy + yz + zx) \text{ volt.}$$

Find the expression for the electric filed. What is the magnitude of the electric field at (1 *m*, 1 *m*, 0).

Sol. $\quad V = 20(xy + yz + zx)$ volt

Now, $\quad \vec{E} = -\vec{\nabla}V = -\dfrac{\partial V}{\partial x}\hat{i} - \dfrac{\partial V}{\partial y}\hat{j} - \dfrac{\partial V}{\partial z}\hat{k}$

Here, $\dfrac{\partial V}{\partial x} = 20(y + z)$; $\dfrac{\partial V}{\partial y} = 20(x + z)$; $\dfrac{\partial V}{\partial z} = 20(y + x)$

$\therefore \quad \vec{E} = -20(y + z)\hat{i} - 20(x + z)\hat{j} - 20(y + x)\hat{k}$ **volt/m**

Also, $\quad \vec{E}$ at $(1\,m, 1\,m, 0) = -20\hat{i} - 20\hat{j} - 40\hat{k} = \mathbf{-20(\hat{i} + \hat{j} + 2\hat{k})\ volt/m.}$

Ex. 3. *Calculate the potential at the centre of a square of side* $\sqrt{2}$ *meter which carries at its four corners charges* $q_1 = 2 \times 10^{-6}$ *C*, $q_2 = -3 \times 10^{-6}$ *C*, $q_3 = -4 \times 10^{-6}$ *C* and $q_4 = 7 \times 10^{-6}$ *C*.

Sol. The distance of the centre from each corner of the square of side $\sqrt{2}$ m is $r = 1$ m. The potential at the centre due to the charges q_1, q_2, q_3 and q_4 is given by [fig. **4.44**],

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r} + \frac{q_2}{r} + \frac{q_3}{r} + \frac{q_4}{r}\right)$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{1}{r}(q_1 + q_2 + q_3 + q_4)$$

$$= (9 \times 10^9)\left(\frac{1}{1}\right)\{(2 - 3 - 4 + 7) \times 10^{-6}\}$$

$$= \mathbf{1.8 \times 10^4\ volt.}$$

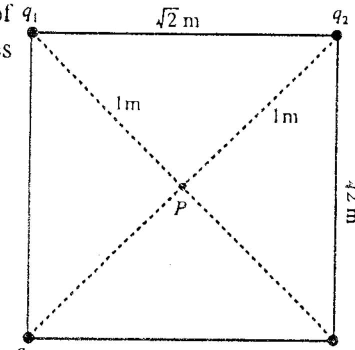


Fig. 4.44.

Ex. 4. *The radius of nucleus of silver (atomic number Z = 47) is* 3.4×10^{-14} *m. Find the electric potential on the surface of nucleus.*

Sol. Here, charge on nucleus of silver $q = Ze = 47 \times 1.6 \times 10^{-19}$ C $= 7.52 \times 10^{-18}$ C.

Distance of surface from the centre of nucleus = radius of nucleus $= 3.4 \times 10^{-14}$ m.

Hence, the potential at the surface of nucleus is given by

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} = 9 \times 10^9 \times \frac{7.52 \times 10^{-18}}{3.4 \times 10^{-14}}$$

$$= \mathbf{1.99 \times 10^6\ volt.}$$

Ex. 5. *A cube of side a has a charge q at each of its vertices. Calculate the electric potential and field due to this arrangement of charges at the centre of the cube.*

Sol. As shown in **fig 4.45**, the centre O of the cube is the point of intersection of the diagonals AC' and BD' and is equidistant from all vertices.

Here, $AC' = \sqrt{a^2 + a^2 + a^2} = \sqrt{3}a$ and $AO = \sqrt{3}a/2$

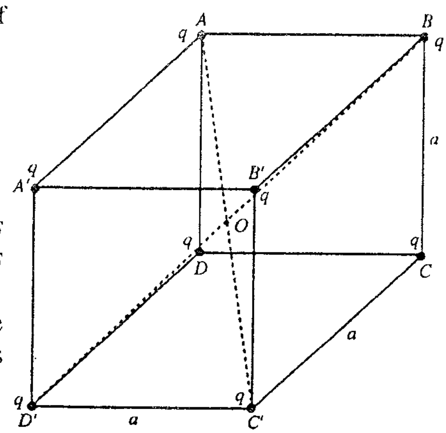


Fig. 4.45.

Thus the distance of a vertex from the centre of the cube is $r = \frac{\sqrt{3}}{2}a$.

As the potential at the centre due to a charge q at a vertex is $q/4\pi\varepsilon_0 r$, the potential due to the 8 charges, each of magnitude q, is given by

$$V = \frac{8q}{4\pi\varepsilon_0 r} = \frac{4q}{\sqrt{3}\pi\varepsilon_0 a}.$$

We know that the electric fields are added vectorially. It is seen from the symmetry of the eight charges with respect to the centre O of the cube, the fields at the centre due to two opposite charges are cancelled out in pairs. This means that the resultant electric field is **zero** at the centre of the cube.

Ex. 6. *Charges +2 μC and – 4 μC are placed at the corners A and C of rectangle of side AB = 20 cm and BC = 10 cm* [**fig. 4.46**]. *Find the electric potential at B and D corners. Calculate the work done in taking a charge 1μC from B to D.*

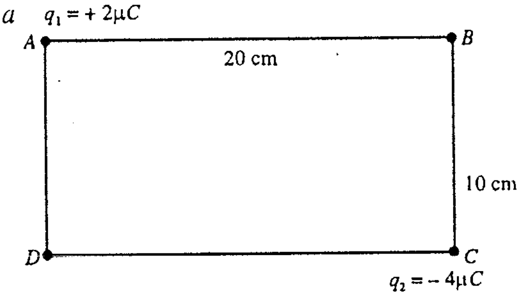


Fig. 4.46.

Sol. Potential at $B, V_B = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{AB} + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{BC}$

$$= \frac{9 \times 10^9 \times 2 \times 10^{-6}}{0.2} + \frac{9 \times 10^9 \times -4 \times 10^{-6}}{0.1}$$

$$= 9 \times 10^9 \times 10^{-5} - 9 \times 10^9 \times 4 \times 10^{-5}$$

$$= \mathbf{-2.7 \times 10^5\ volt.}$$

Potential at D, $V_D = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{AD} + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{CD}$

$$= \frac{9 \times 10^9 \times 2 \times 10^{-6}}{0.1} + \frac{9 \times 10^9 \times -4 \times 10^{-6}}{0.2}$$

$$= 9 \times 10^9 \times 2 \times 10^{-5} - 9 \times 10^9 \times 2 \times 10^{-5} = \mathbf{0}.$$

Work done in carrying a charge $q = 1\ \mu C = 10^{-6}$ C from B to D is

$$W = q\,(V_D - V_B) = 10^{-6} \times 2.7 \times 10^5 = \mathbf{0.27\ J.}$$

Ex. 7. *A square of side* 1 *m has a point charge* $q_1 = 0.1$ *nanocoulomb at the upper left corner, a point charge* $q_2 = -1$ *nanocoulomb at the lower left corner and a line distribution of charge* $\lambda = 1$ *nano-coulomb per metre along the right edge* [**fig. 4.47**]. *Find the potential at P the centre of the square.*

$$\left[\textit{Given}: I = \int_0^a \frac{dy}{\sqrt{a^2 + y^2}} = 0.88.\right]$$

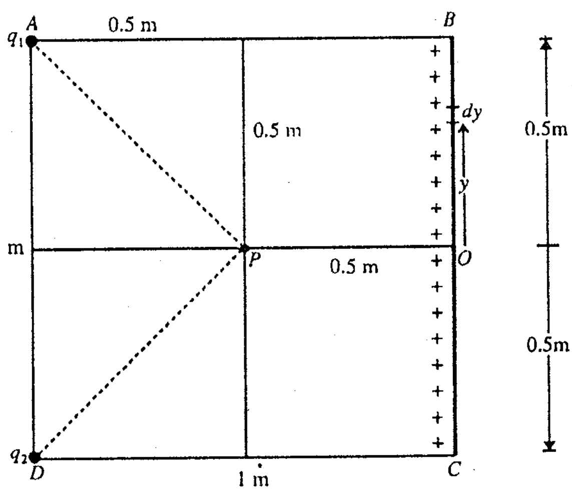


Fig. 4.47.

Sol. Distances $AP = PD = \sqrt{(0.5)^2 + (0.5)^2}$

$$= 0.5\sqrt{2} = 0.707 \text{ m}$$

Potential at O due to $q_1 = 10^{-10}$ C and $q_2 = 10^{-9}$ C is

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{AP} + \frac{q_2}{PD}\right)$$

$$= 9 \times 10^9\left(\frac{10^{-10}}{0.707} - \frac{10^{-9}}{0.707}\right)$$

$$= \mathbf{-11.46\ volt.}$$

The charge is uniformly distributed on the side BC with centre O having a linear charge density $\lambda = 10^{-9}$ C/m. Considering a small line element dy with $dq = \lambda dy$ at a distance y from O, the potential at P due to the linear charge BC will be given by

$$V_2 = \frac{1}{4\pi\varepsilon_0}\int_{-0.5m}^{0.5m}\frac{10^{-9}dy}{\sqrt{(0.5)^2+y^2}}$$

$$= 2\times10^9\times10^{-9}\int_0^{0.5m}\frac{dy}{\sqrt{(0.5)^2+y^2}}$$

But $\displaystyle\int_0^a\frac{dy}{\sqrt{a^2+y^2}} = 0.88,$*

$\therefore$ $V_2 = 2\times9\times10^9\times10^{-9}\times0.88 =$ **15.84 volt.**

Hence total potential at the centre (P) $= -11.46 + 15.84 =$ **4.38 volt.**

Ex. 8. *Consider an electric field* $\vec{E} = 2x\,\hat{i} + 4y\,\hat{j}$ *N/C in a region of space. If the coordinates of points A, B and C are* (0, 0), (x, 0) *and* (x, y) *in XY plane* [**fig. 4.48**], *find the line integral of the electric field E for the paths AB, BC and AC. Show that the electric field is conservative. Show that curl* $\vec{E} = 0$.

Sol. Here $\vec{E} = 2x\,\hat{i} + 4y\,\hat{j}$ N/C

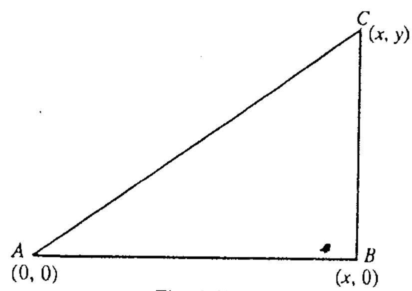


Fig. 4.48.

Line integral of $\vec{E}$ along $AB = \int_A^B \vec{E}\cdot d\vec{r}$

$$= \int_{(0,0)}^{(x,0)}(E_x dx + E_y dy) = \int_0^x E_x dx$$

$$= \int_0^x 2x\,dx = \left[x^2\right]_0^x = x^2$$

Line integral of $\vec{E}$ along $BC = \int_B^C \vec{E}\cdot d\vec{r}$

$$= \int_{(x,0)}^{(x,y)}(E_x dx + E_y dy) = \int_0^y 4y\,dy = 2y^2$$

Line integral of $\vec{E}$ along $AC = \int_A^C \vec{E}\cdot d\vec{r} = \int_{(0,0)}^{(x,y)}(E_x dx + E_y dy) = \int_0^x E_x dx + \int_0^y E_y dy$

$$= \int_0^x 2x\,dx + \int_0^y 4y\,dy = x^2 + 2y^2$$

Also, line integral along ABC path $= x^2 + 2y^2$.

Hence the line integral between the points A and C in the field along the two paths AC and ABC is the same. In other words, the line integral between two points in the field $\vec{E}$ is independent of the path. Hence *the field* $\vec{E}$ *is conservative.*

Now, curl $\vec{E} = \vec{\nabla}\times\vec{E} = \vec{\nabla}\times(2x\,\hat{i} + 4y\,\hat{j})$

---

* $I = \displaystyle\int_0^a\frac{dy}{\sqrt{a^2+y^2}}$ · Let $y = a\tan\theta$; $\therefore$ $dy = a\sec^2\theta\,d\theta$

$\therefore$ $I = \displaystyle\int_0^{\pi/4}\sec\theta\,d\theta = \left[\ln\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right]_0^{\pi/4} = 0.88.$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2x & 4y & 0 \end{vmatrix} = 0$$

Remerber that if curl $\vec{E} = 0$, then the field $\vec{E}$ is conservative.

Ex. 9. *The potential function is given by*

$$V = \frac{100}{x^2 + y^2}, \text{ where } V \text{ is in volts and } x,\ y \text{ in metre.}$$

Calculate the following :

(i) *An expression for the gradient of the potential.*

(ii) *The value of the gradient at the point* (1, 1) *m.*

(iii) *Intensity of electric field at the point* (1, 1) *m.*

Sol. (i) $$\text{grad } V = \vec{\nabla} V = \hat{i}\frac{\partial V}{\partial x} + \hat{j}\frac{\partial V}{\partial y} + \hat{k}\frac{\partial V}{\partial z}$$

Here, $$V = \frac{100}{x^2 + y^2}$$

$$\therefore \quad \frac{\partial V}{\partial x} = -\frac{100 \times 2x}{(x^2 + y^2)^2} = -\frac{200x}{(x^2 + y^2)^2};\ \frac{\partial V}{\partial y} = -\frac{200y}{(x^2 + y^2)^2} \text{ and } \frac{\partial V}{\partial z} = 0.$$

Hence $$\text{grad } V = -\frac{200}{(x^2 + y^2)^2}(x\hat{i} + y\hat{j})$$

(ii) At the point (1, 1) m, $x = 1$ m, $y = 1$ m,

$$\text{grad } V = -\frac{200}{(1^2 + 1^2)^2}(\hat{i} + \hat{j}) = -50(\hat{i} + \hat{j}), \text{ volt/m}$$

(iii) Intensity of electric field at (1, 1) m

$$\vec{E} = -\text{ grad } V = 50(\hat{i} + \hat{j}) \textbf{ volt/m.}$$

Ex. 10. *The electric potential at a point in XY plane is*

$$\phi = 5x(x^2 + y^2)^{-1/2} + y(x^2 + y^2)^{1/2}.$$

Find the X and Y components of the electric field intensity at such a point. **(Kanpur 2005)**

Sol. Here, $$\phi = 5x(x^2 + y^2)^{-1/2} + y(x^2 + y^2)^{1/2}$$

$$\therefore \quad \frac{\partial \phi}{\partial x} = 5(x^2 + y^2)^{-1/2} + 5x\left(-\frac{1}{2}\right)(x^2 + y^2)^{-3/2} \cdot 2x + y\frac{1}{2}(x^2 + y^2)^{-1/2} \cdot 2x$$

$$= 5(x^2 + y^2)^{-1/2} - 5x^2(x^2 + y^2)^{-3/2} + xy(x^2 + y^2)^{-1/2}$$

$$= \frac{5(x^2 + y^2) - 5x^2 + xy(x^2 + y^2)}{(x^2 + y^2)^{3/2}} = \frac{y}{(x^2 + y^2)^{3/2}}\left[5y + x(x^2 + y^2)\right]$$

and $$\frac{\partial \phi}{\partial y} = 5x\left(-\frac{1}{2}\right)(x^2 + y^2)^{-3/2} 2y + (x^2 + y^2)^{1/2} + y\left(\frac{1}{2}\right)(x^2 + y^2)^{-1/2} 2y$$

$$= \frac{-5xy}{(x^2 + y^2)^{3/2}} + (x^2 + y^2)^{1/2} + \frac{y^2}{(x^2 + y^2)^{1/2}}$$

$$= \frac{-5xy+(x^2+y^2)^2+y^2(x^2+y^2)}{(x^2+y^2)^{3/2}} = \frac{x^4+2y^4+3x^2y^2-5xy}{(x^2+y^2)^{3/2}}.$$

Now, electric field (in XY plane), $\vec{E} = -\vec{\nabla}\phi = -\frac{\partial\phi}{\partial x}\hat{i} - \frac{\partial\phi}{\partial y}\hat{j}$

But, $\vec{E} = E_x\hat{\mathbf{i}} + E_y\hat{\mathbf{j}}$

$$\therefore \quad E_x = -\frac{\partial\phi}{\partial x} = -\frac{y}{(x^2+y^2)^{3/2}}(x^3 + xy^2 + 5y)$$

and

$$E_y = -\frac{\partial\phi}{\partial y} = -\frac{x^4 + 2y^4 + 3x^2y^2 - 5xy}{(x^2+y^2)^{3/2}}.$$

Ex. 11. *Electric potential of a system at any point in a field is given by*

$$V = (20 + 6x^2 - 5xy + 4y^2 + 3z^2).$$

Calculate the force acting on the charge 2×10^{-15} coulomb located at a point (2, 0, –3) in the field region in terms of unit vectors $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$. **(Lucknow 2005)**

Sol. $V = (20 + 6x^2 - 5xy + 4y^2 + 3z^2)$

$$\therefore \quad \frac{\partial V}{\partial x} = 12x - 5y, \quad \frac{\partial V}{\partial y} = -5x + 8y, \quad \frac{\partial V}{\partial z} = 6z$$

$$\therefore \quad \vec{E} = -\vec{\nabla}V = -\left[\hat{\mathbf{i}}\frac{\partial V}{\partial x} + \hat{\mathbf{j}}\frac{\partial V}{\partial y} + \hat{\mathbf{k}}\frac{\partial V}{\partial z}\right]$$

$$= -\left[(12x - 5y)\hat{\mathbf{i}} + (-5x + 8y)\hat{\mathbf{j}} + 6z\hat{\mathbf{k}}\right]$$

$\therefore$ At (2, 0, –3) point, $\vec{E} = -\left[24\hat{\mathbf{i}} - 10\hat{\mathbf{j}} - 18\hat{\mathbf{k}}\right]$

$$\therefore \quad \text{Force } \vec{F} = q\vec{E} = 2 \times 10^{-15}\left[-24\hat{\mathbf{i}} + 10\hat{\mathbf{j}} + 18\hat{\mathbf{k}}\right]$$

$$= \mathbf{4 \times 10^{-15}\left[-12\hat{i} + 5\hat{j} + 9\hat{k}\right] N.}$$

Ex. 12. *Three charges + 2q, + 2q and + q coulomb are fixed at the corners A, B, C respectively of an isosceles triangle of side a metre. Calculate the electrostatic potential energy of the system.*

Sol. Here, three charges $+2q$, $+2q$ and $+q$ coulomb are fixed respectively at the corners A, B and C of a triangle ABC **[fig. 4.49]**. The potential energy of the system will be the sum of potential energies of all pairs of charges.

Potential energy due to AB pair of charges $+ 2q$ and $+ 2q$,

$$U_1 = \frac{1}{4\pi\varepsilon_0}\frac{(+2q)\times(+2q)}{a} = \frac{1}{4\pi\varepsilon_0}\times\frac{4q^2}{a} \text{ joule}$$

Potential energy due to BC pair of charges $+2q$ and $+ q$,

$$U_2 = \frac{1}{4\pi\varepsilon_0}\frac{(+2q)\times(+q)}{a} = \frac{1}{4\pi\varepsilon_0}\times\frac{2q^2}{a} \text{ joule}$$

Potential energy due to CA pair of charges $+ q$ and $+ 2q$,

$$U_3 = \frac{1}{4\pi\varepsilon_0}\frac{(+q)\times(+2q)}{a} = \frac{1}{4\pi\varepsilon_0}\times\frac{2q^2}{a} \text{ joule}$$

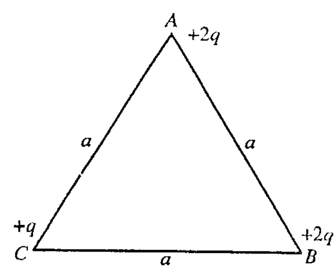


Fig. 4.49.

$\therefore$ Total potential energy $U = U_1 + U_2 + U_3$

$$= \frac{1}{4\pi\varepsilon_0}\left[\left(\frac{+4q^2}{a}\right)+\left(\frac{+2q^2}{a}\right)+\left(\frac{+2q^2}{a}\right)\right] = \frac{1}{4\pi\varepsilon_0}\cdot\frac{8q^2}{a}\text{ J}.$$

Ex. 13. *Eight negative charges, each of magnitude q, are kept at the corners of a cube of side a. Calculate the potential energy of the system if a positive charge 2q is placed at the centre of the cube.* **(Agra 1993, 83)**

Sol. The potential energy is contributed by the different pairs of charges, as shown in **fig. 4.50.**

(i) Potential energy corresponding to eight pairs each with charges $-q$ and $+2q$ separated by $a\sqrt{3}/2$ is $-(1/4\pi\varepsilon_0)\ (2q^2 \times 2/a\sqrt{3}) \times 8$.

(ii) Potential energy for 12 pairs each with charges $-q$ and $-q$ separated by a is $(1/4\pi\varepsilon_0)\ (q^2/a) \times 12$.

(iii) Potential energy for 12 pairs of charges $-q$ and $-q$ separated by $a\sqrt{2}$ is $(1/4\pi\varepsilon_0)(q^2/a\sqrt{2}) \times 12$.

(iv) Potential energy for four pairs of charges $-q$ and $-q$ separated by $a\sqrt{3}$ is $(1/4\pi\varepsilon_0)(q^2/a\sqrt{3}) \times 4$.

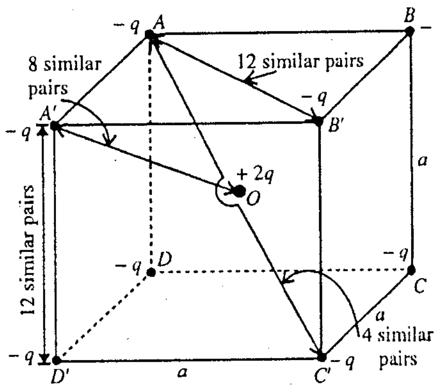


Fig. 4.50.

Hence the total potential energy for the given arrangement is given by

$$U = \frac{1}{4\pi\varepsilon_0}\left[12\frac{q^2}{a} + 12\frac{q^2}{a\sqrt{2}} + 4\frac{q^2}{a\sqrt{3}} - 32\frac{q^2}{a\sqrt{3}}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{4.32q^2}{a}.$$

EXERCISE 4·3

1. An electric field 30 N/C exists along the X-axis in the space. Calculate the potential difference $V_B - V_A$ between the points A (0, 0) and B (4m, 2m). Now if a charge -10^{-3}C is moved from A to B, find the change in potential energy.
[**Ans.** −120 volt; 0.12 J.]

2. A point charge 4 nanocoulomb is situated at the origin. Calculate the electric potential difference between the point A (1, 5, 2) metre and point B (0, 2, 6) metre.
[**Ans.** − 0.88 volt.]

3. In the nucleus of U^{238}, two protons are at a distance of 6.0×10^{-15} m. Calculate their mutual electrostatic potential energy.
[**Ans.** 3.84×10^{-14} J.]

4. Two point-charges of 3 μC and − 6μC are 0.6 m apart. Find the point at which the electric potential is zero.
[**Ans.** 0.2 m from 3 μC in between the two charges.]

5. Calculate the electric potentials at the corners A and C of the rectangle shown in **fig. 4.51** ? What will be the work done in taking a charge 2.0×10^{-6} C from A to C ?
[**Ans.** $V_A = -7.8 \times 10^5$ volt; $V_C = 3.6 \times 10^5$ volt; 1.68 J.]

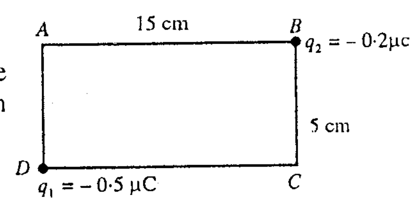


Fig. 4.51.

6. The electric potential at a point (x, y) is given by

$$V = 5x\,(x^2 + y^2)^{1/2} + y\,(x^2 + y^2)^{-1/2}.$$

Find an expression for the electric field $\vec{E}$ and calculate it at a point (1, 1).

[**Ans.** $\vec{E} = -\dfrac{1}{(x^2 + y^2)^{3/2}}\left[(10x^4 + 15x^2y^2 + 5y^4 - xy)\hat{\mathbf{i}} + (5x^3y + 5xy^3 + x^2)\hat{\mathbf{j}}\right]$;

$\vec{E} = -\dfrac{1}{2\sqrt{2}}(29\hat{\mathbf{i}} + 11\hat{\mathbf{j}})$.]

7. A point charge of 1 μC is situated at the origin of coordinates. Calculate the potential difference between the points A and B when their coordinates are :

(i) $(a, 0, 0)$ and $(0, a/\sqrt{2}, -a/\sqrt{2})$; (ii) (0, 1, 5) and (2, 3, 8) respectively.

[**Ans.** (i) zero; (ii) 738 volt.]

8. Two point charges + 0.12 μC and – 0.06 μC respectively are placed at a distance 3.0 m. Find : (i) electric potential at the mid point between them; (ii) work done in bringing a charge + 1.0 μC from infinity to the mid point between the two charges.

[**Ans.** (i) 360 volt; (ii) 3.6×10^{-5} joule.]

9. Charges –3.0 μC and +1.5 μC are placed at the opposite corners A and C of a diagonal of square $ABCD$ of side 10 cm. Find the electric potential at the corners B and D of other diagonal of square and how much work is needed to move a charge 2 μC from the corner B to the other corner D ?

[**Ans.** $V_B = -1.35 \times 10^5$ V; $V_D = -1.35 \times 10^5$ V; zero.]

10. Three charges $q_1 = 10^{-7}$ C, $q_2 = -4 \times 10^{-7}$ C and $q_3 = 2 \times 10^{-7}$ C are placed respectively at the corners A, B and C of an equilateral triangle of side 0.1 m. Find the electrostatic potential energy of the system

[**Ans.** -9×10^{-3} J.]

11. How much work has to be done in assembling three charges 20 μC, 30 μC and 40 μC at the vertices of an equilateral triangle ?

[**Ans.** 234 J.]

12. A ring of radius R is uniformly charged by a charge q. Show that the electrostatic potential at distance r from the centre of ring on its axis is $V = q/[4\pi\varepsilon_0\,(r^2 + R^2)^{1/2}]$.

13. Find the energy per ion in an infinite linear array of alternate positive and negative ions, each of charge q. The separation between successive ions is a.

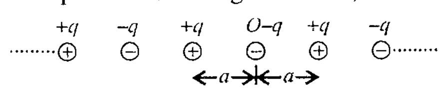


Fig. 4.52.

[**Ans.** $-(1/2\pi\varepsilon_0)\,(q^2/a)\log_e 2$.]

[**Hint :** Potental at O,

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{2q}{a} - \frac{2q}{2a} + \frac{2q}{3a} - \frac{2q}{4a} + \dots\dots\dots\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{2q}{a}\left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots\dots\right] = \frac{q}{2\pi\varepsilon_0 a}\cdot\log_e 2$$

∴ Potential energy of $-q$ charge (per –ve ion) at $O = -qV = -\dfrac{q^2}{2\pi\varepsilon_0 a}\cdot\log_e 2$

Similarly potential energy per +ve ion (+ q charge) $= \dfrac{q^2}{2\pi\varepsilon_0 a}\cdot\log_e 2$.]

14. n positive charges, each of value q are arranged symmetrically on the circumference of a circle of radius a. If an equal negative charge be kept at the centre, calculate the potential energy of the arrangement for n=3 and 4.

[**Ans.** $-(1/4\pi\varepsilon_0)\,1.268\,(q^2/a)$; $-(1/4\pi\varepsilon_0)\,0.172\,(q^2/a)$.]

15. It is known that the electrostatic field is conservative in nature. Which of the following field may respresent electrostatic field :

(i) $\vec{F} = C(x\hat{i} + y\hat{j})$; (ii) $\vec{F} = C(xy^2\hat{i} + y^3\hat{j})$; (iii) $\vec{F} = C(x^{3/4}y^{7/4}\hat{i} + x^{7/4}y^{3/4}\hat{j})$

[**Ans.** (i) and (iii).]

16. If the potential in a region is $V = x(y^2 - 4x^2)$, find the corresponding electric field $\vec{E}$.

[**Ans.** $\vec{E} = (12x^2 - y^2)\hat{i} - 2xy\hat{j}$.]

17. The potential in the space near the origin in X-Y plane is $V = Ax^2 - By^2$ volt, where A and B are positive constants. Find : (i) the position of zero potential and (ii) intensity of electric field.

[**Ans.** (i) (0, 0); (ii) $-2Ax\hat{i} + 2By\hat{j}$.]

QUESTIONS

Long Answer Questions

1. State Coulomb's law in vector form. Define unit charge. Discuss the scope and limitations of Coulomb's law.

2. What is the law of conservation of charge? Explain it with an example. **(Bilaspur 2003)**

3. Explain quantization of charge and define elementary charge.

4. State and discuss the principle of superposition.

5. What do you understand by the intensity of electric field ? Deduce an expression for the intensity of electric field at a point due to a point charge and due to a continuous distributed charge. **(Indore 2004)**

6. Find the electric field due to an infinite line charge of uniform density λ by direct integration.

7. Charge is distributed uniformly along an infinite straight line with constant linear density λ. Find the electric field $\vec{E}$ at the general point at distance r from line. **(Agra 2006)**

8. A thin rod of length l is placed along X-axis whose one end is at origin. Total charge q is uniformly distributed on the rod. Calculate the intensity of electric field at point $(x, 0)$.

[**Ans.** $\dfrac{q}{4\pi\varepsilon_0 x(x-l)}$.]

9. A charge q is uniformly distributed on a circular surface of radius a. Deduce an expression for the intensity of electric field at a distance d from the centre on its axis. **(Raipur 2003)**

10. Charge is uniformly distributed on the inner surface of an insulating hemisphere. If the radius of the sphere is R and surface density of charge is σ, show that the intensity of electric field at the centre of hemisphere will be $\sigma/4\varepsilon_0$ outwards along the axis of hemisphere.

11. What is meant by an electric dipole ? Obtain an expression for the intensity of electric field at a point in end on position due to an electric dipole. **(Bhopal 2004; Jabalpur 03, Raipur 03; Rewa 04; Ujjain 04)**

12. What is menat by dipole moment ? Obtain an expression for the intensity of electric field at a point in broad side on position due to an electric dipole. **(Rewa 2003; Ujjain 03)**

13. Find the electric field E on the equatorial line of a dipole with units and direction. From this find the potential at a far away point. **(Agra 2005)**

14. Establish the necessary formula for the torque acting on an electric dipole in a uniform electric field. Use it to define the dipole moment. **(Bhopal 2003; Sagar 03)**

15. An electric dipole is turned by an angle θ in a uniform electric field. Calculate the amount of work done in this process. **(Bhopal 2003; Ujjain 03)**

16. What is meant by the potential energy of a dipole ? Obtain expression for the potential energy of a dipole in a uniform electric field, when the dipole axis makes an angle θ with the direction of the field. **(Sagar 2003)**

17. What is electric potential. Deduce an expression for the electric potential due to a point charge.
18. Show that the curl for a conservative force field is zero ?
19. Find an expression for electric potential and field due to a uniformly charged disc along its axis.
20. Establish the formula for the force acting on an electric dipole placed in a non-uniform electric field. **(Kanpur 2005)**
21. Show that the electric potential due to a linear charge density λ of length l at distance r on its perpendicular bisector is $V = \dfrac{\lambda}{2\pi\varepsilon_0}\log_e\left[\dfrac{l+\sqrt{l^2+4r^2}}{2r}\right]$.
22. What is meant by an electric quadrupole ? Deduce an expression for the intensity of electric field at any point in end on position due to an electric quadrupole. **(Bhopal 2003)**
23. Show that the electric potential at a point P (r, θ) due to an electric dipole is directly proportional to $1/r^2$ and intensity of electric field is directly proportional to $1/r^3$. **(Bilaspur 2003)**
24. Find the expressions for electric potential at a point (i) in the end on position, (ii) in the broad side on position, due to an electric dipole.
25. Prove that the electric potential at any point due to an electric quadrupole is inversely proportional to the cube of the distance of that point and the intensity of electric field is inversely proportional to the fourth power of the distance.
26. What is meant by the electrostatic potential energy ? Deduce an expression for the electrostatic potential energy of a system of n charges.
27. Show that the electric energy per unit volume stored in an electric field E is $\varepsilon_0 E^2/2$.

Short Answer Questions

1. "Coulomb's law obeys Newton's third law of motion". Comment.
2. Define coulomb.
3. Which is more stronger, electrostatic force between an electron and a proton, or gravitational force between them ? Explain.
   [Ans. Electric force.]
4. Discuss the scope and importance of Coulomb's law.
5. What do you understand by quantization of charge ?
6. What is principle of superposition ? Discuss.
7. Does the Coulomb's law hold good for atomic and nuclear distances.
   [Ans. Yes.]
8. What is elementary charge ? How many electrons will have a charge of 1 coulomb ?
9. Is the coulomb force between two charges changed if other charges are brought nearby ?
   [Ans. No.]
10. Find the unit of ε_0 with the help of Coulomb's law.
    [Ans. $C^2/N\text{-}m^2$.]
11. A conductor has 1.60×10^{-18} C positive charge. What excess or deficit of electrons it has from its normal state ?
    [Ans. Deficit of 10 electrons.]
12. Let a charge Q be divided in two parts q_1 and q_2. What are the values of q_1 and q_2 so that the repulsive force between them at a distance d is maximum ?
    [Ans. $q_1 = q_2 = Q/2$.]
13. The weight of an electron is balanced by producing an electric field between two parallel horizontal plates. What is the direction of electric field ?
    [Ans. Downwards.]

14. Calculate the electric field intensity due to charge on an electron at distance 1.0 m in vacuum. **(Rewa 2003)**

    [Ans. 1.44×10^{-9} N C^{-1}.]

15. At the four corners of a square of side l metre, equal charges q are placed. Another charge Q is placed at its centre. Obtain relationship between q and Q for the system to be in equilibrium.

    [Ans. $Q = -\frac{q}{2}\left(\sqrt{2} + \frac{1}{2}\right)$.]

16. A point charge is taken from a point A to a point B in an electric field. Does the work done by the electric field depend on the path followed by the charge ?

    [Ans. No.]

17. What is meant by an electric dipole ? Give two examples. **(Rewa 2004)**

    [Ans. HCl and CO molecules.]

18. Compare the way E varies with r for (a) point charge; (b) a dipole and (c) a quadrupole.

    [Ans. (a) $E \propto 1/r^2$; (b) $E \propto 1/r^3$; (c) $E \propto 1/r^4$.]

19. Write the unit and dimensional formula of electric dipole moment. **(Rewa 2004)**

    [Ans. C-m; $[M^0L^1T^1A^1]$.]

20. An electric dipole is placed in a nonuniform electric field. Is there a net force on it ? Explain.

    [Ans. Yes.]

21. What is line integral of electric field ?

22. If the electric potential is constant in a region, what inference can be drawn about the electric field ? **(Lucknow 2005)**

    [Ans. $\vec{E} = 0$.]

23. Find an expression for electric potential and intensity of electric field due to a linear electric quadrupole at end on position. **(Kanpur 2005)**

24. Show that the electric field intensity at a point in an electric field is equal to the negative gradient of potential at that point. **(Agra 2004)**

25. Show that $\vec{E} = -$ grad V. **(Kanpur 05; Indore 2004; Lucknow 05)**

26. From the relation $\vec{E} = -\vec{\nabla}V$, prove that the electrostatic field is an irrotational field. **(Lucknow 2005)**

    [**Hint** : Curl $\vec{E} = 0$ for irrotational field.]

27. A dipole is constituted by two charges -3.2×10^{-19}C and $+3.2 \times 10^{-19}$C at a separation of 2.4 Å. Find the intensity of electric field at a point at distance 0.30 m from the dipole when the point lies in (i) end on position, (ii) broad side on position, of the dipole.

    [Ans. (i) 5.12×10^{-17} N/C, (ii) 2.56×10^{-17} N/C.]

28. A circular ring of radius 0.1 metre is uniformly charged with a total charge 1 μC. Calculate the electric field (i) at centre of the ring, (ii) at distance 0.3 m from the centre of the ring on its axis. Where will the electric field be maximum on the axis of the ring ?

    [Ans. (i) Zero; (ii) 3×10^5 N/C, 7.07 cm.]

29. Would electrons move from higher potential to lower potential or from lower potential to higher potential ?

    [Ans. From lower potential to higher potential.]

30. Can two equipotential surface intersect ?

    [Ans. No.]

31. Two point charges $-q$ and $+q$ are placed at a distance apart. What are the points at which the resultant field is parallel to the line joining the two charges ?

[**Ans.** The resultant field will be parallel on (i) any point on the line joining the charges and (ii) perpendicular bisector of the line joining the charges.]

32. If the electric field intensity $\vec{E}$ is zero at a point, will the electric potential be necessarily zero at that point ?

[**Ans.** No; For example, the electric field is zero at a point, exactly midway between $+q$ and $+q$ charges at a distance d apart, but the potential at that point is $\frac{1}{4\pi\varepsilon_0} \cdot \frac{2q}{d/2}$.]

33. Can electric potential be zero at a point where the electric field is not zero ?

[**Ans.** Yes; For example, at a point on the bisector plane of a dipole axis.]

Objective Type Questions

1. The number of electrons in 1 coulomb of charge are :
(a) 6.24×10^{18} (b) 6.24×10^{17} (c) 1.6×10^{-19} (d) 1

2. The vector form of Coulomb's electrostatic force in air, in S.I. system is :
(a) $\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}\hat{r}$ (b) $\vec{F} = \frac{q_1q_2}{r^2}\hat{r}$
(c) $\vec{F} = \frac{1}{4\pi K}\frac{q_1q_2}{r^2}\hat{r}$ (d) $\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}\vec{r}$ **(Bhopal 2004)**

3. F_G and F_e represent gravitational and electrostatic force respectively between one electron and one proton at a distance of 1 Å. The ratio of F_G and F_e is of the order of :
(a) 10^{39} (b) 10^{-39} (c) 1 (d) 10^4

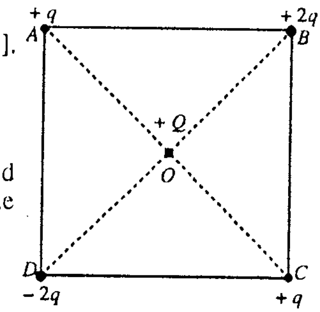


Fig. 4.53.

4. Four charges are arranged at the corners of a square $ABCD$ [**fig. 4.53**]. The force on the $+Q$ charge kept at the centre O is :
(a) zero (b) along AC
(c) along BD (d) $\perp$ to AB

5. Two equal negative charges $-q$ are fixed at the points $(0, a)$ and $(0, -a)$ on the Y-axis. A positive charge $+Q$ is released from rest at the point $(2a, 0)$ on the X-axis. The charge Q will :
(a) execute SHM about the origin.
(b) move to the origin and remain at rest.
(c) move to ∞.
(d) execute oscillatory motion but not SHM.

6. A point charge q is rotated along a circle in the electric field of another point charge $+q_0$. The work done by the electric field on the rotating charge in one complete rotation is :
(a) positive (b) negative
(c) may be positive or negative (d) zero

7. Which one of the following is not a unit of electric field :
(a) NC^{-1} (b) $V\ m^{-1}$ (c) JC^{-1} (d) $JC^{-1}m^{-1}$

8. The electrostatic potential inside a charged hollow sphere :
(a) is zero (b) is constant
(c) increases with distance from centre (d) decreases with distance from centre.

9. Tick the incorrect statement(s) :
(a) S.I. unit of electric potential is J/C.
(b) Electric field vector is irrotational.
(c) Electric field vector is equal to the negative gradient of electric potential
(d) Electric flux is a vector quantity.

10. Which statement is incorrect for an equipotential surface ?
(a) The potential difference between any two points on the surface is zero.
(b) The electric field is always perpendicular to the surface.
(c) Equipotential surfaces are always spherical.
(d) No work is done in moving a charge along the surface.

11. A non-conducting ring of radius 0.5 m carries a total charge of 1.11×10^{-10}C distributed uniformly on its circumference producing an electric field $\vec{E}$ everywhere in space. The value of the line integral $\int_{l=\infty}^{l=0} -\vec{E} \cdot d\vec{l}$ ($l = 0$ being the centre of the ring) in volt is :
(a) + 2 (b) – 1 (c) – 2 (d) zero.
[(**Given :** $1/4\pi\varepsilon_0 = 9.0 \times 10^9$ N m^2 C^{-2}).] **(IIT 1997)**

[**Hint :** $\int_{\infty}^{0} -\vec{E} \cdot d\vec{l}$ = the potential difference between $l = 0$ (centre of the ring) and $l = \infty$

$$= \int_{l=\infty}^{l=0} -\vec{E} \cdot d\vec{l} = V_0 - V_\infty = \frac{1}{4\pi\varepsilon_0}\frac{q}{R} - 0 = (9.0\times10^9)\times\frac{1.11\times10^{-10}}{0.5} - 0 = -2\text{V.}]$$

12. The electric field intensity at a point in broad side on position of a dipole is in direction :
(a) opposite to the direction of dipole moment
(b) parallel to the direction of dipole moment
(c) perpendicular to the direction of dipole moment
(d) making an angle 45° with the dipole moment.

13. An electric dipole moment $\vec{p}$ placed in a uniform electric field $\vec{E}$ has minimum (maximum negative) potential energy when the angle between $\vec{p}$ and $\vec{E}$ is :
(a) $\pi/2$ (b) zero (c) π (d) $3\pi/2$

14. The electric field at a point at a distance r from an electric dipole is proportional to :
(a) $1/r$ (b) $1/r^2$ (c) $1/r^3$ (d) r^2

15. The net force experienced by a dipole of dipole moment $\vec{p}$ in a uniform electric field $\vec{E}$ is :
(a) $\vec{p} \cdot \vec{E}$ (b) $\vec{p} \times \vec{E}$ (c) zero (d) infinity **(Agra 2006)**

16. A given charge is situated at a certain distance from an electric dipole in the end on position, experiences a force F. If the distance of the charge is doubled, the force acting on the charge will be :
(a) $2F$ (b) $F/2$ (c) $F/4$ (d) $F/8$

17. The electric field due to an electric quadrupole varies as :
(a) $1/r$ (b) $1/r^2$ (c) $1/r^3$ (d) $1/r^4$

18. The work done in turning a dipole of dipole moment p by 180° from the direction of electric field E is :
(a) rE (b) $-pE$ (c) zero (d) $2pE$

19. An electric dipole is placed in an electric field produced by a point charge :
(a) The net electric force on the dipole may be zero.
(b) The net electric force on the dipole must be zero.
(c) The torque on the dipole due to the electric field must be zero.
(d) The torque on the dipole due to the electric field may be zero.

20. For the conservative electric field $\vec{E}$:
(a) curl $\vec{E} = 0$ (b) div $\vec{E} = 0$ (c) $\int_a^b \vec{E} \cdot d\vec{l} = 0$ (d) grad div $\vec{E} = 0$

21. The relationship between the electric field $\vec{E}$ and potential difference V is :

(a) $\vec{E} = -\vec{\nabla}V$ (b) $\vec{E} = \vec{\nabla} \cdot V$ (c) $\vec{E} = \vec{\nabla} \times V$ (d) $\vec{E} = \int V\,dr.$ **(Rewa 2004)**

22. When the separation between two charges is increased, the electric potential energy of the system :

(a) decreases (b) increases

(c) remains the same (d) may increase or decrease

23. The electric potential is given as a function of distance x (m) by $V = 5x^2 + 10x - 9$ volt. Value of the electric field at x = 1m is :

(a) 20 V/m (b) 6 V/m (c) 11 V/m (d) – 23 V/m

24. Potential energy stored in a system of n point charge is given by :

(a) $\frac{1}{2}\sum_{i=1}^{n} q_i V_i$ (b) $\frac{1}{2}\sum_{i=1}^{n} q_i$ (c) $\frac{1}{2}\sum_{i=1}^{n} q_i V_i^2$ (d) $\frac{1}{2}\sum_{i=1}^{n} q_i \sqrt{V_i}$ **(Agra 2006)**

ANSWERS

1. (a) **2.** (a) **3.** (b) **4.** (c) **5.** (d) **6.** (d) **7.** (c) **8.** (b) **9.** (d) **10.** (c) **11.** (a)
12. (a) **13.** (b) **14.** (c) **15.** (c) **16.** (d) **17.** (d) **18.** (d) **19.** (d) **20.** (a) **21.** (a) **22.** (d)
23. (a) **24.** (a)

5
Gauss's Law and Capacitors

5·1. Electric Flux

Consider an arbitrary surface S, situated in an electric field. The electric field may not be uniform. Consider an infinitesimal area dS of the surface around a point P, where the intensity of electric field is $\vec{E}$. Vector area corresponding to the area element dS is $d\vec{S}$. The angle between the area element $d\vec{S}$ and electric field $\vec{E}$ at P is θ. The *scalar product* $\vec{E} \cdot d\vec{S}$ *defines the electric flux through the area element* $d\vec{S}$. Hence the **electric flux** through *the* entire surface S **[fig 5.1]** is given by

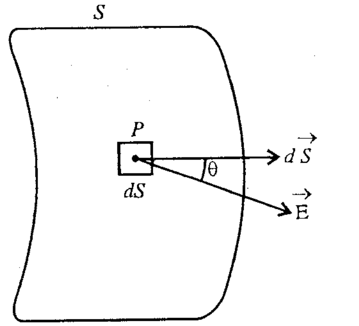


Fig. 5.1. Electric flux through a surface.

$$\phi = \int_S \vec{E} \cdot d\vec{S} \qquad ...(1)$$

We see that this electric flux is the surface integral of the electric filed $\vec{E}$ over the surface S. Thus **the electric flux linked with a surface in an electric field may be defined as the surface integral of the electric field over that surface.**

If the surface S is closed, enclosing a volume (like the surface of a balloon), total flux through the surface is given by

$$\phi = \oint_S \vec{E} \cdot d\vec{S} \qquad ...(2)$$

In case of a plane area S with its normal making an angle θ with a **uniform electric field** $\vec{E}$, the electric flux through this plane surface **[fig. 5.2]** is given by

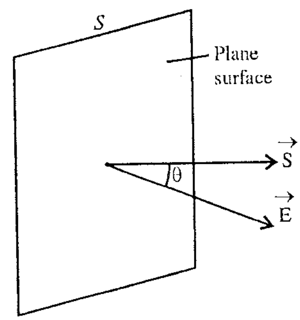


Fig. 5.2. Electric flux through a plane surface.

$$\phi = \int_S \vec{E} \cdot d\vec{S} = \vec{E} \cdot \int d\vec{S} = \vec{E} \cdot \vec{S} = ES\cos\theta \qquad ...(3)$$

If the uniform electric field $\vec{E}$ is perpendicular to the surface S, *i.e.*, $\vec{E}$ and $\vec{S}$ are in the same direction ($\theta = 0$), then

$$\phi = ES \qquad ...(4)$$

This is to be remembered that an electric field $\vec{E}$ represents the number of electric lines of force per unit area at a point (the area being taken perpendicular to the lines of force). Hence the electric flux $\int_S \vec{E} \cdot d\vec{S}$ linked with a surface is equal to the total number of electric lines of force passing normally through that surface in an electric field.

Unit of electric flux is N-m^2-C^{-1} or volt-m and its dimensional formula is $[ML^3T^{-3}A^{-1}]$. Electric flux is a *scalar quantity*.

5·2. Gauss's Law

According to Gauss's law, **the total electric flux through a closed surface is equal to $1/\varepsilon_0$ 1times the net charge enclosed by the surface** *i.e.*,

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \qquad ...(1)$$

where q is the net charge enclosed by the surface. Such a surface is called **Gaussian surface.** Eq. (1) is the **integral form** of Gauss's law.

Further, if the closed surface does not enclose any charge and there may be any charge outside it, then the electric flux through the surface is zero *i.e.*,

$$\oint_S \vec{E} \cdot d\vec{S} = 0 \qquad ...(2)$$

Proof : We deduce Gauss's law by using Coulomb's law. Consider a point charge $+q$ placed at a point O inside a closed surface S **[fig. 5.3]**. Take a point P on the surface and consider a small area dS on the surface around P. Suppose $OP = r$. The area element dS may be represented by a vector $d\vec{S}$, being drawn outward along the normal to the element. The electric field at P due to $+q$ charge at O is

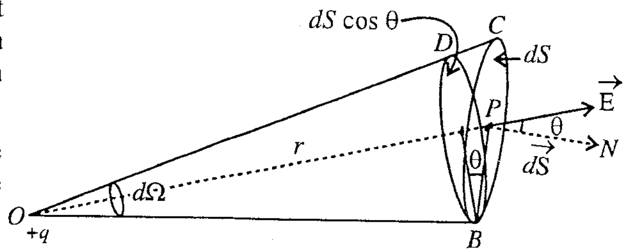


Fig. 5.3.

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}} \qquad ...(3)$$

where $\hat{\mathbf{r}} = \vec{r}/r$ is the unit vector along OP and it makes an angle θ with the vector $d\vec{S}$. The electric flux $d\phi$ through the area dS is

$$d\phi = \vec{E} \cdot d\vec{S} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}} \cdot d\vec{S} = \frac{q}{4\pi\varepsilon_0}\frac{dS\cos\theta}{r^2}$$

But $\dfrac{dS\cos\theta}{r^2} = d\Omega$ is the solid angle $d\Omega$ subtended by the area dS at the point O. Hence

$$d\phi = \frac{q}{4\pi\varepsilon_0}d\Omega \qquad ...(4)$$

Therefore the total electric flux through the entire closed surface due to the internal charge q is

$$\phi = \oint_S \vec{E} \cdot d\vec{S} = \oint_S \frac{q}{4\pi\varepsilon_0}d\Omega = \frac{q}{4\pi\varepsilon_0}\oint_S d\Omega$$

But $\oint_S d\Omega = 4\pi$ is the total solid angle subtended by the entire closed surface S at the point O. Hence,

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \qquad ...(5)$$

This is the **Gauss's law.**

Gauss's law for a set of discrete charges : If there are several discrete charges $+q_1, +q_2, -q_3, +q_4, -q_5,$.. etc., the same argument is true for all other charges anywhere inside the closed surface. Hence, the total flux through the surface S is

$$\phi = \oint_S \vec{E} \cdot d\vec{S} = \oint_S \vec{E}_1 \cdot d\vec{S} + \oint_S \vec{E}_2 \cdot d\vec{S} + ...$$

where the electric field at P is obtained by the principle of superposition due to all the charges inside the closed surface. Now,

$$\oint_S \vec{E}_1 \cdot d\vec{S} = \frac{q_1}{\varepsilon_0}, \oint_S \vec{E}_2 \cdot d\vec{S} = \frac{q_2}{\varepsilon_0}, ...$$

and hence

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q_1}{\varepsilon_0} + \frac{q_2}{\varepsilon_0} - \frac{q_3}{\varepsilon_0} + ...$$

or
$$\oint_S \vec{E}\cdot d\vec{S} = \frac{\Sigma q_i}{\varepsilon_0} = \frac{q}{\varepsilon_0} \qquad ...(6)$$

where $\Sigma q_i = q$ is the algebraic sum of all the charges or net charge inside the closed surfaces.

Eq. (6) represents the **Gauss's law for a number of charges enclosed by a closed surface.**

Gauss's law for continuous distribution of charge : If the charge is distributed over a volume enclosed inside a closed surface, then we can represent the enclosed charge by the volume integral *i.e.*,

$$q = \Sigma q_i = \int_V \rho dV \qquad ...(7)$$

where ρ is the charge density ($= dq/dV$) at a point inside the charge distribution and dV is the volume around the point. Hence for continuous distribution of charge, Gauss's law takes the form :

$$\oint_S \vec{E}\cdot d\vec{S} = \frac{1}{\varepsilon_0}\int_V \rho dV \qquad ...(8)$$

Gauss's law is a *fundamental theorem of electrostatics.*

Electric flux due to a charge $+q$ situated outside the closed surface : Let a charge $+q$ be placed at the point O outside the closed surface S. As shown in **fig. 5.4**, the surfaces dS_1 and dS_2 or cones made by them are subtending the same solid angle $d\Omega$ at O. The flux going outward through the surface dS_2 is $\frac{q}{4\pi\varepsilon_0}d\Omega$ and entering the surface dS_1, is $-\frac{q}{4\pi\varepsilon_0}d\Omega$. Hence the total flux through the two small surfaces dS_1 and dS_2 of the surface S is zero. Now, in the same way, if we divide the entire closed surface into a large number of such small elements, the total flux linked with all such elements will be zero. Hence for a charge situated outside a closed surface, the total electric flux linked with the surface is

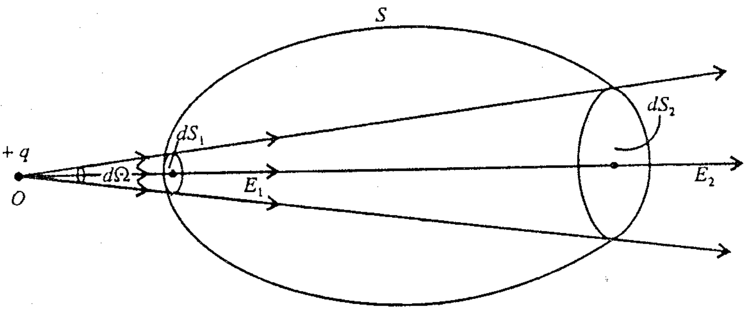


Fig. 5.4.

$$\phi = \oint_S \vec{E}\cdot d\vec{S} = 0 \qquad ...(9)$$

This means that *if the closed surface does not enclose any charge, the electric flux linked with the surface is zero.*

5·3. Differential Form of Gauss's Law

When a charge is distributed over a volume V, the charge density (ρ) at a point P inside the volume is given by

$$\rho = \frac{dq}{dV}$$

Hence
$$dq = \rho dV$$

is the charge in the volume dV around the point P.

Thus the charge enclosed by the surface S enclosing the volume V is given by

$$q = \int_V \rho dV \qquad ...(1)$$

According to Gauss's law (in integral form)

$$\oint_S \vec{E}\cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } \oint_S \vec{E}\cdot d\vec{S} = \frac{1}{\varepsilon_0}\int_V \rho dV \qquad ...(2)$$

According to Gauss's theorem in vectors,

$$\int_S \vec{E} \cdot d\vec{S} = \int_V (\vec{\nabla} \cdot \vec{E}) dV \quad ...(3)$$

Hence from eqs. (2) and (3)

$$\int_V (\vec{\nabla} \cdot \vec{E}) dV = \frac{1}{\varepsilon_0} \int_V \rho\, dV \text{ or } \int_V (\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\varepsilon_0}) dV = 0 \quad ...(4)$$

Eq. (4) is true for any arbitrary volume, hence

$$\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\varepsilon_0} = 0$$

Thus,

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \text{ or } \operatorname{div} \vec{E} = \frac{\rho}{\varepsilon_0} \quad ...(5)$$

This is the **differential form of Gauss's law.** Eq. (5) is also called **poisson's equation.**

The electric displacement vector $\vec{D}$ in vacuum is given by

$$\vec{D} = \varepsilon_0 \vec{E}$$

Hence Gauss's law is also expressed as

$$\vec{\nabla} \cdot \vec{D} = \rho \text{ or } \operatorname{div} \vec{D} = \rho \quad ...(6)$$

This form of Gauss's law is fundamentally important in electromagnetic theory.

5·4. Applications of Gauss's Law

We can use Gauss's law to determine the electric field at a point due to a charge or charge distribution. In order to find the field, we construct such a closed surface near the charge or the charge distribution so that the integral $\oint_S \vec{E} \cdot d\vec{S}$ is easily evaluated. This surface is called **Gaussian surface.** The Gauss's law $\oint \vec{E} \cdot d\vec{S} = q/\varepsilon_0$ can readily be applied for uniform symmetrical charge distribution and symmetrical Gaussian surface.

If we know the intensity of the electric field $\vec{E}$ due to a charge distribution, we can evaluate the electric potential at a point $\vec{r}$ by using the relation

$$V = -\int_\infty^r \vec{E} \cdot d\vec{r} \quad ...(1)$$

In case of radial symmetry,

$$V = -\int_\infty^r E\, dr \text{ or } E = -\frac{dV}{dr} \quad ...(2)$$

5·5. Deduction of Coulomb's Law from Gauss's Law–Electric Field due to a Point Charge

Consider a point charge $+q$ at O. First we find the electric field at a point P, distant r from O. Draw a spherical Gaussian surface of radius r around the charge $+q$ [fig. 5.5]. The electric field is radial by symmetry *i.e.*, perpendicular to the spherical surface. Hence,

$$\oint_S \vec{E} \cdot d\vec{S} = \oint_S E\, dS = E \oint_S dS = E \times 4\pi r^2 \quad ...(1)$$

But according to Gauss's law,

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \quad ...(2)$$

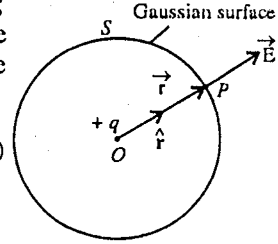


Fig. 5.5 : Electric field due to a point charge.

Hence from eqs. (1) and (2),

$$E \times 4\pi r^2 = \frac{q}{\varepsilon_0} \text{ or } E = \frac{q}{4\pi\varepsilon_0 r^2}$$

As the electric field is radial,

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}} \qquad ...(3)$$

This is the **electric field due to a point charge** q at a distance r.

Now, we put a point charge q' at the point P, where we have calculated the electric filed $\vec{E}$ [eq. (3)]. The force, acting on the charge q', is

$$\vec{F} = q'\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{qq'}{r^2}\hat{\mathbf{r}} \qquad ...(4)$$

This is the **Coulomb's law.** In fact, *Gauss's law is one of the four basic (Maxwell's) equations of electromagnetic theory and it is equivalent to Coulomb's law.*

Electric potenial due to a point charge q :

$$\text{Electric potential, } V = -\int_\infty^r E\,dr = -\int_\infty^r \frac{q}{4\pi\varepsilon_0 r^2}\,dr = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \qquad ...(5)$$

5·6. Electric Field (and Electric Potential) due to a Uniformly Charged (Nonconducting) Sphere

Let a charge q be uniformly distributed in a spherical volume of radius R. We want to find the electric field at a point P which is at a distance r from the centre of the sphere.

Case (i), **Electric field at a point outside the sphere :** For a point P outside the sphere, we have $r > R$. We construct a Gaussian spherical surface of radius r, passing through the point P **[fig. 5.6]**. Due to symmetry, the electric field at P and everywhere on the Gaussian surface will be perpendicular to the surface. Let the magnitude of this field be E. As the electric field $\vec{E}$ is normal to any surface element $d\vec{S}$, hence $\vec{E}\cdot d\vec{S} = E\,dS$. Thus the electric flux through the entire surface of the sphere is

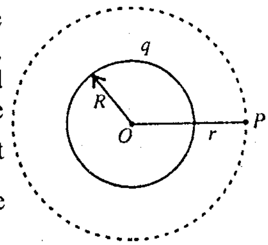


Fig. 5.6 : Electric field at a point outside a uniformly charged sphere.

$$\oint_S \vec{E}\cdot d\vec{S} = \oint_S E\,dS = E\oint_S dS = E \times 4\pi r^2$$

According to Gauss's law, this electric flux should be equal to the charge (q) contained inside the gaussian surface divided by ε_0. Hence

$$E \times 4\pi r^2 = \frac{q}{\varepsilon_0} \text{ or } E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \qquad ...(1)$$

Thus *the electric field due to a uniformly charged sphere at a point outside it is the same as if the charge on the sphere is concentrated at the centre.*

Case (ii), **Electric field at a point on the surface of sphere :** Here $r = R$, and therefore

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^2}. \qquad ...(2)$$

Case (iii), **Electric field at a point inside the sphere :** Now we find the electric field E at a point P which is inside the sphere at a distance r from its centre. We draw a spherical Gaussian surface of radius r passing through P with O, the centre of the sphere **[fig. 5.7]**. By symmetry, the electric field is radial at all the points of the sphere and has a constant magnitude E. The electric flux through the spherical surface of radius r is

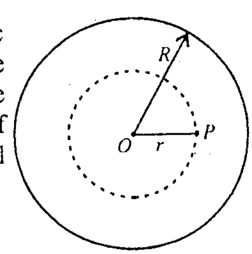


Fig. 5.7 : Electric field at a point inside a uniformly charged sphere.

$$\oint \vec{E}\cdot d\vec{S} = \oint E\,dS = E\oint dS = E \times 4\pi r^2$$

The charge enclosed by the Gaussian spherical surface of radius r is

$$q' = \text{Volume enclosed by it} \times \text{Charge density}$$

$$= \frac{4}{3}\pi r^3 \times \frac{q}{\frac{4}{3}\pi R^3} = \frac{qr^3}{R^3}$$

where q is the charge, uniformly distributed over the volume of the sphere of radius R, so that the charge density (ρ) is $q\Big/\left(\frac{4}{3}\pi R^2\right)$.

According to Gauss's law

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } E \times 4\pi r^2 = \frac{qr^3}{\varepsilon_0 R^3}$$

Hence $$E = \frac{qr}{4\pi\varepsilon_0 R^3} \qquad ...(3)$$

or $$E \propto r \qquad ...(4)$$

Thus *the electric field due to a uniformly charged sphere at an internal point is proportional to its distance from the centre.*

Fig. 5.8 shows the variation of electric field E as a function of radial distance r from the centre of uniformly charged sphere.

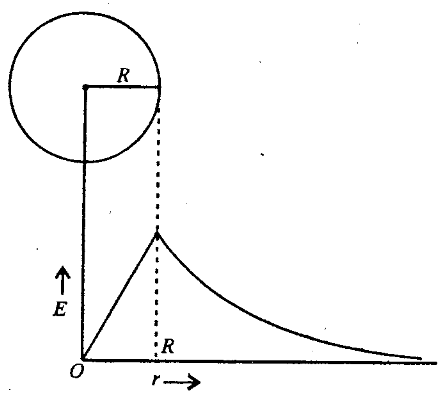


Fig. 5.8 : Variation of electric field E with distance r due to a uniformly charged sphere.

Electric potential due to a uniformly (nonconducting) Charged sphere

Case **(i), Point outside the shere ($r > R$) :** Here, $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$,

$$\therefore V = -\int_\infty^r E\,dr = -\int_\infty^r \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}dr = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \qquad ...(5)$$

Case **(ii), Point on the surface ($r = R$) :** At $r = R$,

$$V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{R}. \qquad ...(6)$$

Case **(iii), Point inside the sphere ($r < R$) :**

$$V = -\int_\infty^r E\,dr = -\int_\infty^R E\,dr - \int_R^r E\,dr$$

where $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$ from ∞ to R and $E = \frac{1}{4\pi\varepsilon_0}\frac{qr}{R^3}$ from R to r.

$$\therefore \quad V = -\int_\infty^R \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}dr - \int_R^r \frac{1}{4\pi\varepsilon_0}\frac{qr}{R^3}dr$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r}\right]_\infty^R - \frac{1}{4\pi\varepsilon_0 R^3}\left[\frac{r^2}{2}\right]_R^r$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{R} - \frac{r^2 - R^2}{2R^3}\right] = \frac{q}{4\pi\varepsilon_0}\frac{(3R^2 - r^2)}{2R^3} \qquad ...(7)$$

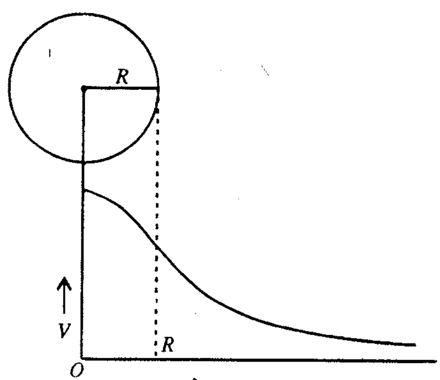


Fig. 5.9 : Variation of electric potential V with distance r due to a uniformly charged sphere.

Fig. 5.9 shows the variation of electric potential V as a function of radial distance r from the centre of the sphere.

5·7. Electric Field (and Electric Potential) due to Uniformly Charged Spherical Shell or Conducting Sphere

In case of a shpere of conducting material, its total charge resides on its outer surface and there is no charge inside the sphere. Hence *a uniformly charged conducting sphere may be considered as uniformly* *charged spherical shell.* A spherical shell is a hollow sphere of very small thickness. Let q be the charge uniformly distributed over the spherical shell of radius R.

Case (i), **Electric field at an external point :** Let the point P be situated at a distance r from the centre O of the spherical shell ($r > R$). Draw a spherical Gaussian surface of radius r with centre O [fig. 5.10]. By symmetry, the electric field will be radial and perpendicular to the Gaussian surface. Hence applying Gauss's law,

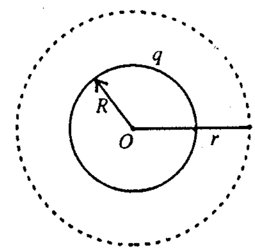


Fig. 5.10 : Electric field outside a uniformly charged spherical shell or conducting sphere.

$$\oint \vec{E}\cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } \oint E\, dS = \frac{q}{\varepsilon_0}$$

or
$$E\oint dS = \frac{q}{\varepsilon_0} \text{ or } E \times 4\pi r^2 = \frac{q}{\varepsilon_0}$$

i.e.,
$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}. \qquad \text{...(1)}$$

Thus *the electric field due to a uniformaly charged spherical shell (or conducting sphere) at a point outside it is the same as if the total charge is concentrated at the centre.*

Case (ii), **Electric field at a point on the surface of the sherical shell :** Here $r = R$, and therefore

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^2} \qquad \text{...(2)}$$

Case (iii), **Electric field at a point inside the spherical shell (or conducting sphere) :** Consider a point P which is inside the spherical shell at a distance r from its centre. Now draw a shperical Gaussian surface of radius r with centre O, which passes through P [fig. 5.11]. Thus *Gaussian surface does not enclose any charge.* Now, applying Gauss's law,

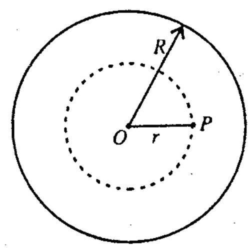


Fig. 5.11 : Electric field inside a uniformly charged spherical shell.

$$\oint \vec{E}\cdot d\vec{S} = 0 \text{ or } \oint E\, dS = 0$$

or
$$E\oint dS = 0 \text{ or } E \times 4\pi r^2 = 0$$

i.e.,
$$E = 0 \qquad \text{...(3)}$$

Thus *the electric field at any point inside a uniformly charged spherical shell or conducting sphere is zero.*

The variation of electric field E with distance r from the centre of the uniformly charged sphere is shown in **fig.5.12.**

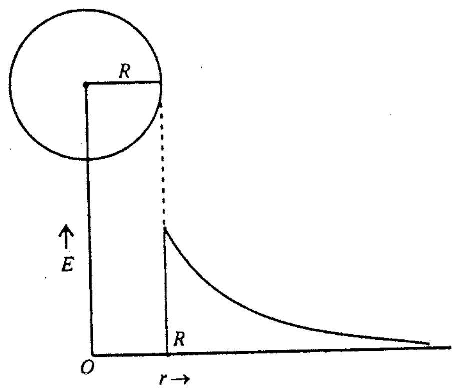


Fig. 5.12 : Variation of electric field E with distance r due to a uniformly charged spherical shell or conduction sphere.

Electric potential due to a uniformly charged spherical shell or conducting sphere :

Case (i), **Point outside ($r > R$) :** Here, $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$,

$$\therefore V = -\int_\infty^r E\, dr = -\int_\infty^r \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\, dr = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \qquad \text{...(4)}$$

Case (ii), **Point on the surface ($r = R$) :** At $r = R$,

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{R} \qquad \text{...(5)}$$

Case (iii), **Point inside ($r < R$) :**

$$V = -\int_\infty^r E\, dr = -\int_\infty^R E\, dr - \int_R^r E\, dr$$

Here $\quad E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$ from ∞ to R and $E = 0$ from R to r.

$$\therefore \qquad V = -\int_{\infty}^{R} \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r^2}\, dr - \int_{R}^{r} 0\, dr$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{q}{R} \qquad \text{...(6)}$$

Thus *the electric potential at any point inside a uniformly charged spherical shell or conducting sphere is constant and equal to the potential on its surface.*

Fig. 5.13 shows the variation of electric potential V with distance r from the centre of the uniformly charged spherical shell (or conducting sphere).

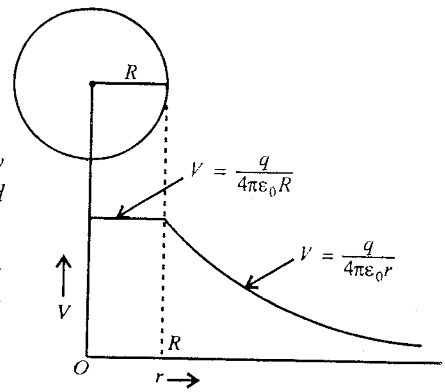


Fig. 5.13 : Variation of electric potential V with distance r due to a uniformly charged spherical shell.

5·8. Electric Field and Potential due to a Non-uniform Spherical Charge Distribution

We consider a charge distribution in a spherical volume of radius a with the charge density given by

$$\left.\begin{aligned} \rho &= \rho_0\left(1 - \frac{r}{a}\right) \text{ for } 0 < r \le a \\ \text{and} \quad \rho &= 0 \qquad\qquad \text{for } r > 0 \end{aligned}\right\} \qquad \text{...(1)}$$

where ρ_0 is constant and r is the distance from the centre of spherical charge distribution.

We plan to calculate the electric field and electric potential for (i) $r > a$, (ii) $r = a$ and (iii) $r < a$. We also want to find the distance r from the centre at which the electric field is maximum.

Consider a thin spherical shell of radius x and thickness dx. The volume of this shell is $dV = 4\pi x^2\, dx$.

Charge on the shell $dq = 4\pi x^2\, dx\, \rho = 4\pi x^2\, dx\, \rho_0\, (1 - x/a)$ ...(2)

Total charge in the spherical charge distribution of radius a is

$$q = \int_V \rho\, dV = \int_0^a 4\pi\rho_0 x^2\left(1 - \frac{x}{a}\right) dx = 4\pi\rho_0\left[\frac{x^3}{3} - \frac{x^4}{4a}\right]_0^a dx = 4\pi\rho_0 \times \frac{a^3}{12} = \frac{\pi\rho_0 a^3}{3} \qquad \text{...(3)}$$

***Case* (i) :** When $r > a$, the point lies outside the charge distribution **[fig. 5.14]**. Applying Gauss's law,

$$\int_S \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } E \times 4\pi r^2 = \frac{\pi\rho_0 a^3}{3\varepsilon_0}$$

$$\therefore \qquad E = \frac{\rho_0 a^3}{12\varepsilon_0 r^2} \qquad \text{...(4)}$$

The electric potential at a distance r is

$$V = -\int_{\infty}^{r} E\, dr = -\int_{\infty}^{r} \frac{\rho_0 a^3}{12\varepsilon_0 r^2}\, dr$$

$$= \frac{\rho_0 a^3}{12\varepsilon_0}\left[\frac{1}{r}\right]_{\infty}^{r} = \frac{\rho_0 a^3}{12\varepsilon_0 r} \qquad \text{...(5)}$$

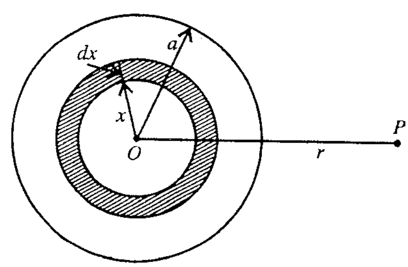


Fig. 5.14.

***Case* (ii) :** When $r = a$, then

$$E = \frac{\rho_0 a^3}{12\varepsilon_0 a^2} = \frac{\rho_0 a}{12\varepsilon_0} \qquad \text{...(6)}$$

and

$$V = \frac{\rho_0 a^3}{12\varepsilon_0 a} = \frac{\rho_0 a^2}{12\varepsilon_0} \qquad \text{...(7)}$$

***Case* (iii) :** When $r < a$, the point lies on the surface of sphere of radius r **[fig. 5.15]**. Here, the total charge enclosed inside the Gaussian surface, *i.e.*, the sphere of radius r is

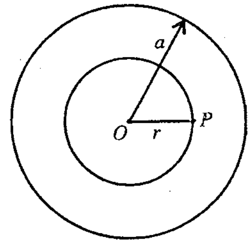


Fig. 5.15.

$$q' = \int_V \rho\, dV = \int_0^r 4\pi\rho_0 x^2 \left(1 - \frac{x}{a}\right) dx$$

$$= 4\pi\rho_0 \left[\frac{x^3}{3} - \frac{x^4}{4a}\right]_0^r = 4\pi\rho_0 \left[\frac{r^3}{3} - \frac{r^4}{4a}\right] \quad ...(8)$$

Applying Gauss's law,

$$E \times 4\pi r^2 = \frac{q'}{\varepsilon_0} \text{ or } E = \frac{q'}{4\pi\varepsilon_0 r^2} = \frac{4\pi\rho_0}{4\pi\varepsilon_0 r^2}\left[\frac{r^3}{3} - \frac{r^4}{4a}\right] = \frac{\rho_0}{\varepsilon_0}\left[\frac{r}{3} - \frac{r^2}{4a}\right] \quad ...(9)$$

$\therefore$ Electric potential, $V = -\int_\infty^r E\, dr = -\int_\infty^a E\, dr - \int_a^r E\, dr$

$$= -\int_\infty^a \frac{\rho_0 a^3}{12\varepsilon_0 r^2} dr - \int_a^r \frac{\rho_0}{\varepsilon_0}\left(\frac{r}{3} - \frac{r^2}{4a}\right) dr$$

$$= \frac{\rho_0 a^2}{12\varepsilon_0}\left[\frac{1}{r}\right]_\infty^a - \frac{\rho_0}{\varepsilon_0}\left[\frac{r^2}{6} - \frac{r^3}{12a}\right]_r^a$$

$$= \frac{\rho_0 a^2}{12\varepsilon_0} + \frac{\rho_0}{\varepsilon_0}\left[\frac{a^2 - r^2}{6} - \frac{(a^3 - r^3)}{12a}\right]$$

i.e.,

$$V = \frac{\rho_0 a^2}{6\varepsilon_0}\left[1 - \frac{r^2}{a^2} + \frac{r^3}{2a^3}\right] \quad ...(10)$$

Maximum value of the electric field will be somewhere in between $0 < r \le a$. For maximum value of E,

$$\frac{dE}{dr} = 0 \text{ i.e., } \frac{d}{dr}\left[\frac{\rho_0}{\varepsilon_0}\left(\frac{r}{3} - \frac{r^2}{4a}\right)\right] = 0$$

or

$$\frac{1}{3} - \frac{2r}{4a} = 0 \text{ or } r = \frac{2}{3}a \quad ...(11)$$

$$\therefore \quad E_{max} = \left[\frac{\rho_0}{\varepsilon_0}\left(\frac{r}{3} - \frac{r^2}{4a}\right)\right]_{r=\frac{2}{3}a} = \frac{\rho_0}{\varepsilon_0}\left(\frac{2}{9}a - \frac{1}{9}a\right) = \frac{\rho_0 a}{9\varepsilon_0} \quad ...(12)$$

5·9. Electric Field due to a Linear Charge (or Uniformly Charged Wire) of Infinite Length

Consider *a* long straight uniformly positively charged wire of infinite length. Let the charge per unit length of the wire be λ. From the symmetry, the electric field due to the wire must be radially outward. Let us find the electric field E due to the wire at a point P which is at a distance r (from the wire). Now, we construct a Gaussian surface in the form of a circular hollow cylinder of radius r and length l with the wire as its axis **[fig. 5.16]**. This cylindrical surface passes through the point P. The electric field E is constant over the cylindrical surface and perpendicular to it. The flux of the electric field E through the cylindrical surface is

$$\oint \vec{E} \cdot d\vec{S} = \oint E\, dS = E \oint dS = E \times 2\pi r l.$$

The *flux through the end faces of the cylinder is* **zero** because E is radial and tangentail to an end face. Hence the total outward flux linked with the cylindrical surface is $E \times 2\pi r l$.

Charge enclosed by the Gaussian surface = λl.

Hence by applying Gauss's law, we have

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } E \times 2\pi r l = \frac{\lambda l}{\varepsilon_0}$$

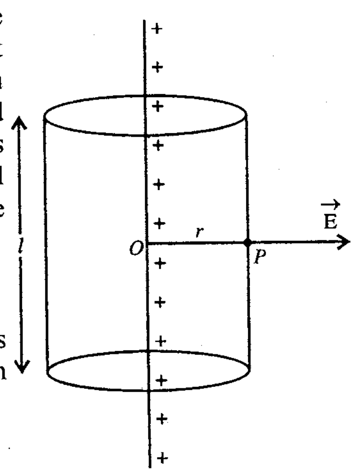


Fig. 5.16 : Electric field due to a linear charge.

i.e.,
$$E = \frac{\lambda}{2\pi\varepsilon_0 r} \qquad ...(8)$$

This is the expression for electric field at a distance r from a uniformly charged wire of infinite length. The direction of the field is away from the wire for positive charge and towards the line for negative charge.

5·10. Electric Field due to a Uniformly Charged Cylindrical Conductor (or Hollow Cylinder) of Infinite Length

Consider a cylindrical conductor of infinite length and radius R. It is uniformly charged with charge per unit length (linear charge density) λ. As the cylinder is conducting, the charge resides on the outer surface of the cylinder. Hence it is equivalent to a uniformly charged hollow cylinder of radius R. Now consider a point P at a distance r from the axis of the cylinder at which we want to calculate the electric field **[fig. 5.17]**. The point P may lie (i) outside, (ii) on the surface, and (iii) inside the cylinder.

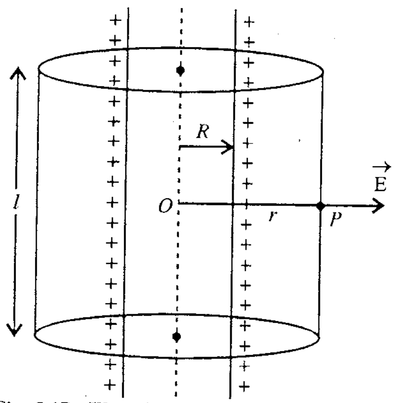


Fig. 5.17 : Electric field due to a uniformly charged cylindrical conductors.

Consider a Gaussian cylindrical surface of radius r and length l. The Gaussian surface passes through the point P. Due to the infinitely long uniformly charged hollow cylinder, the electric field at any point of the curved Gaussian surface is the same and radial *i.e.,* it is acting perpendicular to the surface in the outward direction. The electric flux through the Gaussian curved surface is

$$\oint \vec{E} \cdot d\vec{S} = \oint E\, dS = E \oint dS = E \times 2\pi r l$$

As the electric field is radial, the *flux through the upper and lower faces of the Gaussian cylindrical surface will be zero.* Now, let us consider different cases :

Case **(i), Point P lies outside the cylinder** (*i.e.,* $r > R$) : Applying Gauss's law, we have

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } E \times 2\pi r l = \frac{\lambda l}{\varepsilon_0}$$

because $\lambda l = q$ is the charge on the length l of the hollow cylinder and it is the charge enclosed by the Gaussian surface.

Hence
$$E = \frac{\lambda}{2\pi\varepsilon_0 r} \qquad ...(1)$$

Case **(ii), Point P lies on the surface of the cylinder** (*i.e.,* $r = R$) : In such a case from eq. (1), we have

$$E = \frac{\lambda}{2\pi\varepsilon_0 R} \qquad ...(2)$$

Case **(iii), Point P lies inside the cylinder** (*i.e.,* $r < R$) : As the cylinder is conducting, the total charge lies on its cylindrical surface *i.e.*, the charged conducting cylinder acts like a charged hollow cylinder. Hence there is no charge inside the region within the cylinder of radius R. Therefore the electric flux through the Gaussian surface of radius r ($< R$) is zero *i.e.,*

$$E \times 2\pi r l = 0 \text{ or } E = 0 \qquad ...(3)$$

Electric field due to a solid (nonconducting) cylinder with uniform volume charge distribution

(i) Point outside ($r > R$) : In this case, obviously

$$E = \frac{\lambda}{2\pi\varepsilon_0 r} \qquad ...(4)$$

(ii) Point inside ($r < R$) : Let ρ be the volume charge density. Applying Gauss's law for a Gaussian cylindrical surface of radius r ($< R$),

$$\oint \vec{E}\cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } E\times 2\pi r l = \frac{\pi r^2 l \rho}{\varepsilon_0} \text{ or } E = \frac{r\rho}{2\varepsilon_0} \quad ...(5)$$

But, $$\rho = \lambda/\pi R^2, \quad \therefore \quad E = \frac{r\lambda}{2\pi\varepsilon_0 R^2} \quad ...(6)$$

5·11. Gaussian Pill Box and Electric Field due to an Infinite Plane Sheet of Charge

Consider a uniformly charged infinite (nonconducting) plane sheet of charge with charge per unit area (surface charge density) σ. We want to determine the electric field at a nearby point P in front of the sheet **[fig. 5.18]**.

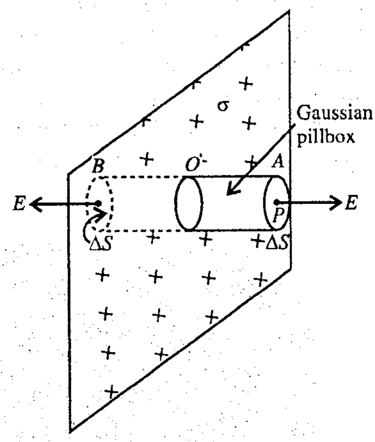


Fig. 5.18 : Electric field due to an infinite plane sheet of charge.

Construction of Gaussian pill box : Draw a small plane circular surface A passing through the point P and parallel to the charge sheet. Draw a cylinder with this surface as a cross-section and extend it to the other side of the plane charge sheet. Now, close the cylinder on the other side by a plane circular surface B such that A and B are equidistant from the sheet. The cylinder goes through the sheet at O. *This cylindrical box forms a Gaussian surface and is called* **Gaussian pill box**. Let the area of cross-section of the cylindrical box be ΔS.

Calculation of the electric field : By symmetry, the electric field at all the points of the area ΔS at A has the same magnitude. The direction of the electric field is along the positive normal to ΔS at A (for positively charged sheet). The electric flux through ΔS at A is

$$\vec{E}\cdot\Delta\vec{S} = E\Delta S \quad ...(1)$$

As the two sides of the charge sheet are equivalent and A and B surfaces are equidistant from the sheet, the electric field at any point of ΔS at B is also equal to E and is along the outward normal. Hence the electric flux through ΔS at B is

$$\vec{E}\cdot\Delta\vec{S} = E\Delta S \quad ...(2)$$

Thus the total flux through the plane surfaces of the cylindrical body is $2E\Delta S$.

The electric field E is parallel to the curved surface of the cylindrical box and hence the electric flux across the curved surface is zero.

Thus, the total electric flux through the Gaussian cylinder or pill box is

$$\oint \vec{E}\cdot d\vec{S} = 2E\Delta S \quad ...(3)$$

Since the area of the charged sheet enclosed in the cylindrical box is ΔS, the charge enclosed in the box is $\sigma\Delta S$. Applying Gauss's law, we have

$$\oint \vec{E}\cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } 2E\Delta S = \frac{\sigma\Delta S}{\varepsilon_0}$$

i.e., $$E = \frac{\sigma}{2\varepsilon_0} \quad ...(4)$$

Thus *the electric field due to an infinite uniformly charged sheet is uniform and is independent of the distance from the sheet.* The electric field at any point in front of the positively charged sheet is directed away from it and the field is directed towards the negatively charged sheet.

5·12. Electric Field due to a Uniformly Charged Infinite Conducting Plane

Consider a positively charged infinite conducting plane surface with surface charge density σ. The entire charge is distributed only on the right plane surface of the conductor and there is no charge inside the

If E_1 and E_2 be the electric fields at any point due to positively charged plane sheets S_1 and S_2 respectively, then at a point P inside the sheets the total electric field is given by **[fig. 5.21]**,

$$E = E_1 - E_2 = \frac{\sigma_1}{2\varepsilon_0} - \frac{\sigma_2}{2\varepsilon_0} \qquad ...(2)$$

When $\sigma_1 = \sigma_2$, $E = 0$. ...(3)

Thus *the electric field due to two infinite uniformly charged plates sheets of charges with positive surface charge density at a point between the plates is zero.* Similar is the case for two negatively charged sheets.

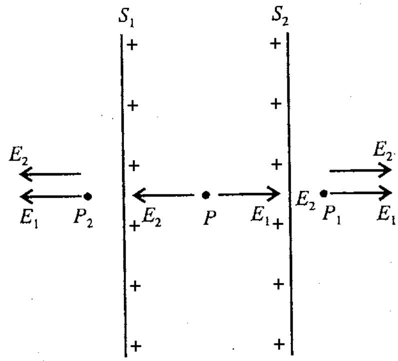


Fig. 5.21 : Electric field due to two uniformly positively charged plane sheets.

Case **(ii), One sheet is positively charged and other negatively charged :** Let the sheets S_1 and S_2 be charged with surface charge densities σ_1 and σ_2 respectively. At a point P inside the plates, the electric field is given by **[fig. 5.22]**,

$$E = E_1 + E_2 = \frac{\sigma_1}{2\varepsilon_0} + \frac{\sigma_2}{2\varepsilon_0} \qquad ...(4)$$

As E_1 and E_2 both are directed to the right, the resultant field E at P is also directed towards right. If σ_1 ans σ_2 are equal and $\sigma_1 = \sigma_2 = \sigma$ (say), then

$$E = \frac{\sigma}{\varepsilon_0} \qquad ...(5)$$

This resultant electric field is the same at any point between the plates. Thus *the electric field between the two plane sheets of opposite charges is uniform between the plates (except near the edges) and is directed towards the negative sheet.*

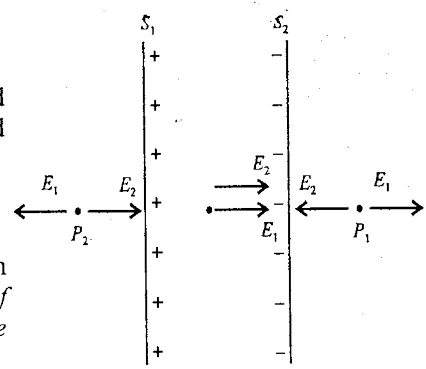


Fig. 5.22 : Electric field due to two uniformly charged sheets—one positively and other negatively charged.

The electric field at a point P_1 or P_2 **(fig. 5.22)** is given by

$$E = E_1 - E_2 = \frac{\sigma_1}{2\varepsilon_0} - \frac{\sigma_2}{2\varepsilon_0} \qquad ...(6)$$

which is directed away from the positively charged sheet for $\sigma_1 > \sigma_2$. If $\sigma_1 = \sigma_2$, then

$$E = 0 \qquad ...(7)$$

In such a case, the electric field is zero outside the two sheets.

Thus we find that *the resultant electric field due to two infinite uniformly oppositely charged plane sheets with equal surface charge densities is uniform in the region between them and zero elsewhere.* We can follow this principle to produce a uniform field in a limited region of space, provided the sheets are much larger in linear dimensions than the separation between them.

5·14. Electric Field inside a Conductor (Conductor in an Electric Field)

In a solid conductor, we have free electrons, moving in random directions, and an arrangement of positively charged ions. These electrons are almost uniformly distributed in the conductor. Any small volume, of the conductor contains a large number of electrons and ions but the negative charge of the elctrons is equal to the positive charge of the ions with almost uniform distribution. Thus the net charge in the volume is zero and hence there is no net electric field inside the conductor. Also if we consider a Gaussian surface of this small volume inside the conductor, the electric flux or electric field will be zero because the net enclosed charge within the surface is zero.

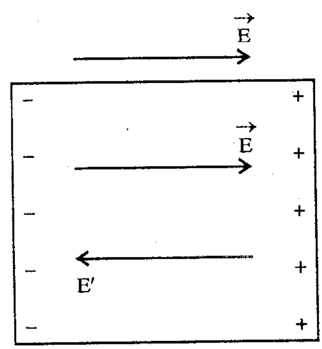


Fig. 5.23 : Electric field inside a conductor is zero.

Now, suppose the conductor is placed, in an uniform electric field $\vec{E}$ **[fig. 5.23]**. This field exerts force on the few electrons of the conductor from right to left and consequently the electrons move to the left, resulting increase of electrons on the left face and decrease of electrons on the right face. Thus the left face of the conductor becomes negatively charged and right face positively charged. Hence an electric field E' is created from right to left opposite to the field E. The free electrons will contiune to drift and deposit on the left face till the electric field E' becomes equal and opposite to the applied field E. Thus the net electric field at any point inside the conductor becomes zero. In such a situation, the free electrons inside the conductor do not experience any net force and the process of further drifting of the electrons stops. In the steady, sate*, *there are positive charges on the right face and negative charges in the same amount on the left face and the electric field inside the conductor is zero.* Further, *whatever be the shape of a conductor, if it is placed in an electric field, there is no electric field inside the conductor.*

5·15. Screening of Electric Field by a Conductor

Cavity in a conductor : Within a conductor of any shape, the electric field is zero at every point. Further any extra charge given to the conductor, it resides entirely on its surface **[fig. 5.24(a)]**. Let us consider a cavity inside conductor **[fig. 5.24(b)]**. If there is no charge within the cavity, a Gaussian surface can be constructed within the material of the conductor to show that there is no net charge on the surface of the cavity and the electric field inside the cavity is zero. Now, even if any charge is given to the conductor, this will reside on its outer surface.

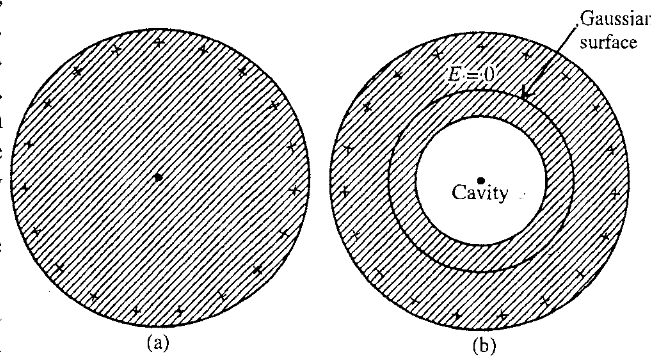


Fig. 5.24.

Suppose we place a small body with a charge $+q$ inside the cavity within a uncharged conductor. The conductor is insulated from the charge $+q$. If we draw a Gaussian surface in the conductor **[fig. 5.25]**, the electric field $E = 0$ on the Gaussian surface and hence according Gauss's law, the total charge inside this surface is zero. Hence there must be a charge $-q$ distributed on the surface of the cavity, appeared due to the charge $+q$ inside the cavity. Further the total charge on the conductor must be zero, hence a charge $+q$ appears on its outer surface. In case the conductor has originally a charge q_c, then the total charge on the outer surface must be $q + q_c$ after inserting the charge $+q$ into the cavity.

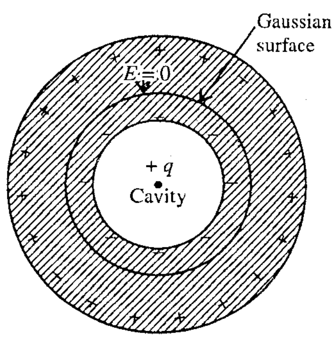


Fig. 5.25.

Screening of electric field or Electrostatic sheilding : Suppose that a conductor with cavity is placed in an external electric field. The external electric field redistributes the free electrons in the conductor so that a net positive charge is developed in some regions on the outer surface of the conductor and a net equal negative charge in other regions **(fig. 5.26)**. The charge distribution causes an additional electric field such that the total field at every point inside the conductor is zero in view of the Gauss's law. Therefore, if we place a charge or sensitive equipment inside the cavity, it is shielded from the external field. This is called **electrostatic screening or scruitning of the electric field by a conductor.**

Thus during a thunder storm, the safest place will be a metal closure or metal car. If the lightening falls on the car, the charge remains on the metal skin of the vehicle and the man inside the car is shielded by the metal enclosure.

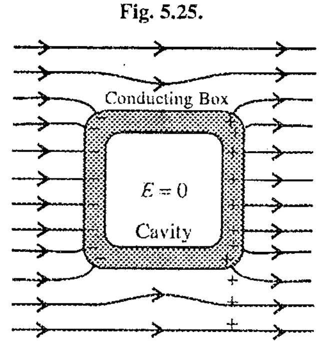


Fig. 5.26 : Screening of the electric field by a conductor – Conducting box (Faraday cage) immersed in an electric field.

A metal enclosure, used to protect the sensitive instruments from external stray electric fields is sometimes called **Faraday Cage.** It is not

* The steady state is reached in general in a time less than a millisecond.

required that the enclosure should be a solid conductor. A wire mesh can shield the charges and equipments, placed inside the cage, from external strong electric fields.

Ex. 1. *A sphere of radius* 10 *m encloses a charge of* 10 *coulombs. Find the total electric flux.* **(Agra 2006)**

Sol. Electric flux, $\phi = \oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} = \frac{10}{8.85 \times 10^{-12}} = \mathbf{1.13 \times 10^{12}\ NC^{-1}m^2}$

Ex. 2. 2*C electrostatic charge is placed at the centre of a cube of side* 1 *m. Find the flux density coming out of any one face of the cube.*

Sol. As all the 6 faces of the cube are symmetrically situated about the point-charge at the *centre*, the flux through each face will be the same. Let ϕ be the flux through a face of the cube. Hence the total flux through the entire surface of the cube will be 6ϕ. Now, by Gauus's law, we have

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } 6\phi = \frac{q}{\varepsilon_0} \text{ or } \phi = \frac{q}{6\varepsilon_0}$$

Here, $q = 2C$, $\therefore$
$$\phi = \frac{2}{6 \times 8.85 \times 10^{-12}} = \mathbf{3.77 \times 10^{10}\ NC^{-1}m^2}.$$

Since the area of each of the cube is 1 m^2, the flux density (flux per unit area) through each face is $\mathbf{3.77 \times 10^{10}\ NC^{-1}}$.

Ex. 3. *A thin spherical shell of metal has a radius of* 0.25 *meter and carries a charge of* 0.1 μ*C. Calculate the electric field intensity at a point* (i) *inside the shell,* (ii) *just outside the shell and* (iii) 3.0 *meter from the centre of the shell.*

Sol. (i) The electric field intensity at a point inside a charged shell is **zero.**

(ii) For an external point the spherical shell behaves as if its entire charge is concentrated at its centre. Hence, if R be the radius of the shell, the intensity at a point just outside the shell is

$$E = \frac{1}{4\pi\varepsilon_0} \frac{q}{R^2}.$$

Here, $q = 0.1\ \mu C = 0.1 \times 10^{-6}$ C, $R = 0.25$ m and $1/4\pi\varepsilon_0 = 9 \times 10^9$ Nm^{2}C^{-2}

$$\therefore \quad E = (9.0 \times 10^9) \times \frac{0.1 \times 10^{-6}}{(0.25)^2} = \mathbf{1.44 \times 10^4\ NC^{-1}}.$$

(iii) The electric field intensity at an external point at a distance r from the centre of the shell is given by

$$E = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2}.$$

Here, $r = 3.0$ m, $\therefore$
$$E = (9.0 \times 10^9) \times \frac{0.1 \times 10^{-6}}{(3.0)^2} = \mathbf{100\ NC^{-1}}.$$

Ex. 4. *A hollow metallic sphere of radius* 0.1 *m is uniformly charged with a charge* 10^{-8} *coulomb. Calculate the electric potential and electric field intensity inside the sphere. What will be the potential on the surface of sphere ?* **(Jabalpur 2004)**

Sol. Here, $q = 10^{-8}$ C and $R = 0.1$ m.

We know that no charge resides in the hollow space inside a conducting sphere and all the charge resides on the surface of sphere. Hence the electric potential at any point inside the sphere is the same as on its surface, *i.e.*,

Potential inside the sphere, $V = \frac{1}{4\pi\varepsilon_0} \frac{q}{R} = 9.0 \times 10^9 \times \frac{10^{-8}}{0.1} = \mathbf{900\ volt.}$

Electric field inside the sphere, $E = 0$

On the surface of sphere, the electric potential will be same as inside the sphere, *i.e.*, potential on the surface of sphere = **900 volt.**

Ex. 5. *A charge of* 2×10^{-8} *C is distributed uniformly on the surface of sphere of radius* 1 *cm. It is covered by a hollow conducting sphere of radius* 4 *cm.*

(a) *Calculate the electric field at a point* 2 *cm away from the centre of the sphere.*

(b) *A charge of* 5×10^{-8} *C is placed on the hollow sphere. Calculate the charge on the outer surface of the hollow sphere.*

Sol. (a) Draw a Gaussian spherical surface of radius $r = 2$ cm through P at which we want to find the electric field. Charge on the shpere of radius 1 cm is 2×10^{-8}C. Hence, according to Gauss's law

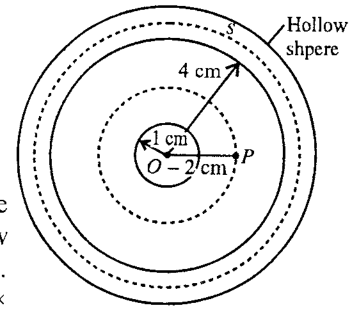


Fig. 5.27.

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \text{ or } E \cdot 4\pi r^2 = \frac{q}{\varepsilon_0} \therefore E = \frac{q}{4\pi\varepsilon_0 r^2}$$

$$= \frac{9.0 \times 10^9 \times 2 \times 10^{-8}}{(2 \times 10^{-2})^2} = \mathbf{4.5 \times 10^5\ NC^{-1}}$$

(b) Let us draw a spherical Gaussian surface S in the material of the hollow sphere. Since the electric field in the conducting material of the hollow sphere is zero, hence the electric flux through the Gaussian surface is zero. This implies that the charge on the inner surface of hollow sphere is -2×10^{-8}C. As 5×10^{-8}C charge is given to the hollow sphere, hence the charge on its outer surface is $2 \times 10^{-8}C + 5 \times 10^{-8}C = \mathbf{7 \times 10^{-8}C}$.

Ex. 6. 8 *identical drops of mercury are charged to the same potential of* 20 *volt. What will be the potential if all the charged drops are made to combine to form one large drop. Assume the drops to be spherical.*

Sol. If r be the radius of each small drop and q be the charge on each drop, then the potential at the surface of a drop is given by

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} = \mathbf{20\ volt.} \qquad \text{...(i)}$$

When all the 8 drops combine, the volume of the large drop will be the same as the volume of the 8 drops. Thus, if the radius of the large drop is R, then we have

$$\frac{4}{3}\pi R^3 = 8\left(\frac{4}{3}\pi r^3\right) \text{ or } R = 2r.$$

The charge on the large drop is $Q = 8q$. Therefore, the electric potential of the large drop will be

$$V' = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R} = \frac{1}{4\pi\varepsilon_0}\frac{8q}{2r} = \frac{1}{4\pi\varepsilon_0}\frac{4q}{r} = 4V = 4 \times 20 = \mathbf{80\ volt.}$$

Ex. 7. *A particle of mass* 10^{-5} *gm is kept at a small distance over a large horizontal sheet of charge density* 4×10^{-6} C/m^2 **[fig. 5.28]**. *Calculate the amount of charge given to this particle so that if released, it does not fall down.*

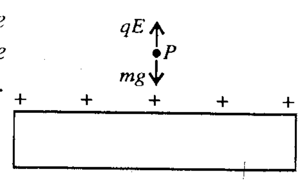


Fig. 5.28.

Sol. The electric field in front of the sheet is given by

$$E = \frac{\sigma}{2\varepsilon_0} = \frac{4 \times 10^{-6}}{2 \times 8.85 \times 10^{-12}} = 2.26 \times 10^5\ NC^{-1}.$$

Let a positive charge q be given to the particle P.

Upward force on the ponticle = qE

Downward force due to gravity = mg

The particle P will not fall down, if the upward force balances the weight of the particle *i.e.*,

$$qE = mg \text{ or } q = \frac{mg}{E} = \frac{10^{-5} \times 10^{-3} \times 9.8}{2.26 \times 10^{6}} = 4.44 \times 10^{-13} \text{ C.}$$

Ex. 8. *In fig 5.29 two parallel plane non-conducting sheets A and B, have surface density of charges $+\sigma$ and $-\sigma$ C/m^2 respectively. Calculate the intensity of electric field at points P_1, P_2 and P_3.*

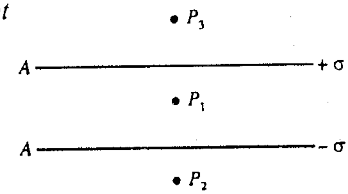


Fig. 5.29.

Sol. At point P_1, electric field due to sheet A is [**fig. 5.29**]

$$\vec{E}_1 = \frac{\sigma}{2\varepsilon_0} \text{ (from } A \text{ to } B)$$

where we have taken +ve downward

Electric field due to sheet B, $\vec{E}_2 = \frac{\sigma}{2\varepsilon_0}$ (from A to B)

$\therefore$ Resultant electric field at P_1, $\vec{E} = \vec{E}_1 + \vec{E}_2 = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0}$

$$= \frac{\sigma}{\varepsilon_0} \text{ (from } A \text{ to } B) \quad \text{...(i)}$$

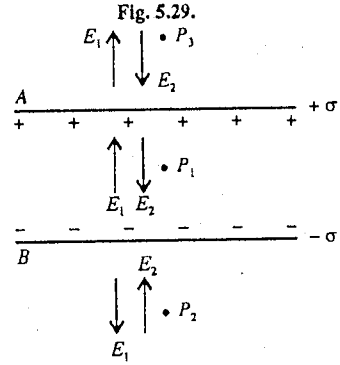


Fig. 5.30.

At point P_2, electric field due to sheet A is

$$\vec{E}_1 = \frac{\sigma}{2\varepsilon_0} \text{ (away from the sheet } A)$$

Electric field due to sheet B, $\vec{E}_2 = -\frac{\sigma}{2\varepsilon_0}$ (towards the sheet B)

$\therefore$ Resultant electric field at P_2, $\vec{E} = \vec{E}_1 + \vec{E}_2 = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0$ ...(ii)

At point P_3, electric field due to sheet A

$$\vec{E}_1 = -\frac{\sigma}{2\varepsilon_0} \text{ (away from the sheet } A)$$

Electric field due to sheet B, $\vec{E}_2 = \frac{\sigma}{2\varepsilon_0}$ (towards the sheet B)

$\therefore$ Resultant electric field at P_3, $\vec{E} = -\frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = 0.$ ...(iii)

Ex. 9. *A dielectric cylinder of radius 'a' is infinitely long. It is non-uniformly charged such that volume charge density varies directly as the distance from the axis of the cylinder. Calculate the intensity at distance r from the axis such that* (i) $r > a$, (ii) $r = a$ *and* (iii) $r < a$. *The volume density of charge at the axis is zero and on the surface is* ρ_S. **(Agra 1970S)**

Sol. Here, the volume charge density ρ varies directly as the distance x from the axis of cylinder, *i.e.*,

$$\rho \propto x \text{ or } \rho = kx \text{ (} k \text{ is a constant)} \quad \text{...(i)}$$

According to the problem at $x = a$, $\rho = \rho_S$.

$$\therefore \quad \rho_S = ka \text{ or } k = \rho_S/a. \quad \text{...(ii)}$$

Now consider a coaxial thin cylindrical shell of length l and radius x and thickness dx. The volume dV of the shell is

$$dV = \pi\{(x + dx)^2 - x^2\}\, l = 2\pi x dx l \quad \text{...(iii)}$$

(i) For $r > a$, the point lies outside the cylinder. We can calculate the intensty E at a distanc r from the axis of cylinder by Gauss's law *i.e.*,

$$\int_S \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} = \frac{1}{\varepsilon_0}\int_V \rho\, dV$$

or $$(2\pi r l)E = \frac{1}{\varepsilon_0}\int_V \rho\, dV = \frac{1}{\varepsilon_0}\int_0^a kx(2\pi\, x dx)l$$

$$\therefore \quad E = \frac{k}{\varepsilon_0 r}\int_0^a x^2 dx = \frac{ka^3}{3\varepsilon_0 r} = \frac{1}{3\varepsilon_0 r}\left(\frac{\rho_S}{a}\right)a^3 = \frac{\rho_S a^2}{3\varepsilon_0 r}. \quad \text{...(iv)}$$

(ii) For $r = a$, the point lies at the surface of the cylinder. Hence,

$$E = \frac{\rho_S a^2}{3\varepsilon_0 a} = \frac{\rho_S a}{3\varepsilon_0} \quad \text{...(v)}$$

(iii) For $r < a$, the point lies inside the surface of the cylinder. Here, there will not be any contribution towards electric flux of the charge lying outside the Gaussian cylindrical surface of radius r. Hence

$$2\pi r l\, E = \frac{1}{\varepsilon_0}\int_0^r kx(2\pi\, x\, dx)\, l \text{ or } E = \left(\frac{1}{3\varepsilon_0}\right)kr^2 = \left(\frac{\rho_S}{3\varepsilon_0 a}\right)r^2. \quad \text{...(vi)}$$

In all cases, the direction of field is away from the charge distribution and is normal to the axis of the cylinder.

Ex. 10. *Two conducting plates A and B are placed parallel to each other. If A is given a charge q_1 and B a charge q_2, find the distribution of charges on the four surfaces of the plates.*

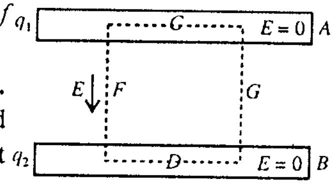


Fig. 5.31.

Sol. Let a cylindrical Gaussian surface be drawn as shown in **fig. 5.31.** C and D faces of the Gaussian surface are inside the conducting plates and hence the electric field $E = 0$ there, resulting electric flux to be zero throught the C and D faces. The electric flux through F and G surfaces of the Gaussian surface are also zero, because the electric field is parallel to F and G faces. Thus the total electric flux through the Gaussian surface is zero.

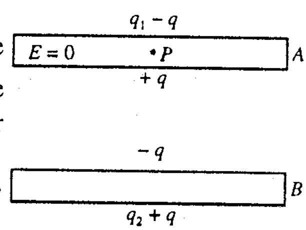


Fig. 5.32.

According to Gauss's law, $(\oint \vec{E}\cdot d\vec{S} = q/\varepsilon_0)$, hence the net charge inside the Gaussian surface should be zero. This means that if $+q$ is the charge on the inner surface of plate A, then $-q$ should be the charge on the inner surface of plate B. The resulting charge distribution is shown in **fig. 5.32.**

The electric field due to the charges at the four faces of the A and B plates at P is zero *i.e.,*

E_1 (due to $q_1 - q$) + E_2 (due to $+q$) + E_3 (due to $-q$) + E_4 (due to $q_2 + q$) = 0

If A is the area of plate, then using the formula $E = \sigma/2\varepsilon_0$ for electric field,

$$\frac{q_1 - q}{2A\varepsilon_0}\text{(downward)} + \frac{q}{2A\varepsilon_0}\text{(upward)} + \frac{q}{2A\varepsilon_0}\text{(downward)} + \frac{q_2 + q}{2A\varepsilon_0}\text{(upward)} = 0$$

Taking direction of the electric field to be positive downward,

$$\frac{q_1 - q}{2A\varepsilon_0} - \frac{q}{2A\varepsilon_0} + \frac{q}{2A\varepsilon_0} - \frac{q_2 + q}{2A\varepsilon_0} = 0$$

or $$q_1 - q - q_2 - q = 0 \text{ or } q = \frac{q_1 - q_2}{2} = 0$$

Thus the charges on the four surfaces of the plates are

(i) $q_1 - q = q_1 - \dfrac{(q_1 - q_2)}{2} = \dfrac{(q_1 + q_2)}{2}$ on upper face of A

(ii) $+q = \dfrac{q_1 - q_2}{2}$ on lower face of A

(iii) $-q = -\dfrac{q_1 - q_2}{2}$ on upper face of B

(iv) $q_2 + q = q_2 + \dfrac{q_1 - q_2}{2} = \dfrac{q_1 + q_2}{2}$ on lower face of B.

EXERCISE 5·1

1. A point charge Q coulomb is kept at the centre of cube of each side a metre. Calculate the electric flux through one face of the cube. **(Rewa 2003)**
[**Ans.** $Q/6\varepsilon_0$.]
2. Calculate the electric potential on the surface of the gold nucleus having a radius of 6.6×10^{-15} m. The atomic number of gold is 79.
[**Ans.** 1.72×10^7 V.]
[**Hint :** Charge on the gold nucleus = $79e$]
3. Eight charged drops of water, each of radius 1 mm and having a charge of 10^{-10} C, combine to form a larger drop. Determine the potential of the bigger drop.
[**Ans.** 3600 V.]
4. An infinite line of charge produces a field of 9×10^{-4} NC^{-1} at a distance of 2.0 m. Find the linear charge density.
[**Ans.** 10 μC/m.]
5. A particle with charge -2μC is placed near a non-conducting plate having a surface charged density 4μC/m^2. Calculate the force of attraction between the particle and the plate.
[**Ans.** 0.45 N.]
6. Two concentric metallic spherical shells of radii R_1 and R_2 ($R_1 < R_2$) are charged uniformly with charges q_1 and q_2 respectively. Calculate the intensity of electric field at a distance r from the centre when (i) $r < R_1$. (ii) $R_1 < r < R_2$, (iii) $r > R_2$.
[**Ans.** (i) $E = 0$; (ii) $E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1}{r^2}$; (iii) $E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 + q_2}{r^2}$.]
7. In a non-conducting sphere of radius 10 cm, a charge of 1.0 μC is uniformly distributed in the entire volume. Calculate the electric field due to it : (i) at a distance of 20 cm from the centre of the sphere, (ii) at the surface of the sphere and (iii) inside the sphere at a distance of 4.0 cm from the centre.
[**Ans.** (i) 2.25×10^5 NC^{-1}; (ii) 9.0×10^5 NC^{-1}; (iii) 3.6×10^5 NC^{-1}.]
8. A charge 4×10^{-9} coulomb is uniformly distributed on a circular disc of radius 30 cm. Calculate the intensity of electric field and electric potential at a distance 40 cm from its centre on the axis.
[**Ans.** 80 V, 160 V/m.]
9. The charge per unit length distributed on a conducting cylinder of infinite length and 0.01 m radius is 0.1 C/m. Calculate the electric field at a point situated at distance (i) 1 m and (ii) 0.5 cm, from the axis of cylinder.
[**Ans.** (i) 1.8×10^9 N/C; (ii) Zero.]
10. Electrical charge of uniform density ρ per unit volume is distributed within a spherical shell of inner and outer radii a and b respectively. Estimate the electric field at a point (P) distant r from the centre such that (i) $0 < r < a$, (ii) $a < r < b$ and (iii) $b < r$.
$\left[\textbf{Ans.}\ \text{(i) } 0;\ \text{(ii) } \dfrac{\rho}{3\varepsilon_0}\left(r - \dfrac{a^3}{r^2}\right);\ \text{(iii) } \dfrac{\rho}{3\varepsilon_0 r^2}\left(b^3 - a^3\right).\right]$
11. Two large conducting plates are placed parallel to each other and they have equal and opposite charges

with surface density σ **[fig. 5.33]**. Find the electric field at P_1, P_2 and P_3.
[**Ans.** 0, σ/ε_0, 0.]

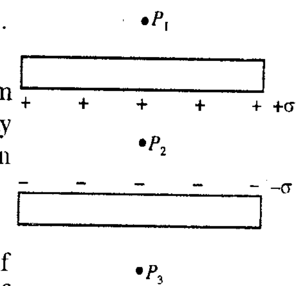


Fig. 5.33.

12. Two large metal plates of area 1.0 m^2 face each other are separated by 5 cm and carry equal and opposite charges on their inner surfaces. If the intensity of the electric field between the plates is 400 N/C, calculate the charge on the plates.
[**Ans.** 3.54 nC.]

13. An electric dipole consists of charges $\pm 10^{-8}$ C. separated by a distance of 2 mm. It is placed with its axis perpendicular to a very long line charge of linear charge density 4×10^{-4} C/m such that the negative charge is closer at a distance of 2 cm from the line charge. Calculate the force on the dipole.
[**Ans.** 0.3 N towards the line charge.]

14. A bob of mass 10^{-3} gm uniformly charged with a charge 4.0×10^{-8}C is suspended from a metallic sheet by means of a silk thread. When the sheet is uniformly charged, in equilibrium the thread makes an angle of 30° with the vertical sheet. Find the surface density of charge on the sheet.
[**Ans.** 1.2×10^{-9} Cm^{-2}.]

15. Three similar large conducting plates A, B and C are kept parallel to each other **[fig. 5.34]**. A is given a charge q and C a charge $-2q$ and B remains neutral. Determine the charge appearing on the outer surface of plate C.
[**Ans.** $-q/2$.]

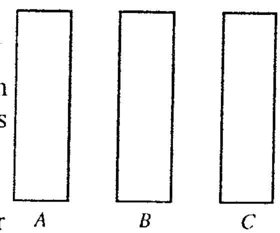


Fig. 5.34

16. Electric charge is uniformly distributed in an infinite cylindrical shell of inner and outer radii r_1 and r_2 respectively. If ρ be the charge per unit volume, determine the electric field at a point P distant r from the axis of the charge distribution when (i) $0 < r < r_1$, (ii) $r_1 < r < r_2$, and (iii) $r_2 < r$. Plot roughly the nature of variation of E with r.

$$\left[\textbf{Ans.}\ \text{(i) } 0;\ \text{(ii) } \frac{\rho}{2\varepsilon_0}\left(r - \frac{r_1^2}{r}\right);\ \text{(iii) } \frac{\rho}{2\varepsilon_0 r}\left(r_2^2 - r_1^2\right).\right]$$

17. A spherically symmetric charge distribution of radius 'a' is characterized by the charge density function $\rho(r) = \rho_0(1 - r^3/a^3)^{1/2}$. There is no charge for $r > a$. Find the electric field at a point (i) inside and (ii) outside the charge distribution.

$$\left[\textbf{Ans.}\ \text{(i) } \frac{2a^3\rho_0}{9\varepsilon_0 r^2}\left\{1 - \left(1 - \frac{r^3}{a^3}\right)^{3/2}\right\};\ \text{(ii) } \frac{2a^3\rho_0}{9\varepsilon_0 r^2}.\right]$$

5·16 . Force per Unit Area of a Charged Conductor

We know that in a charged conductor charge resides on its surface. Consider a very small area element of a positively charged conductor with surface charge density σ. Now, consider a point P just outside the area element dS **[fig. 5.35]**. The electric field at the point P is given by

$$E = \frac{\sigma}{\varepsilon_0} \qquad \text{...(1)}$$

which acts along the outward normal to the element dS.

This electric field at P is in fact the sum of two fields :

(1) E_1 due to the charge on the element dS, and (2) E_2 due to the charge on the rest of the surface of the conductor.

Thus, $$E = E_1 + E_2 = \frac{\sigma}{\varepsilon_0} \qquad \text{...(2)}$$

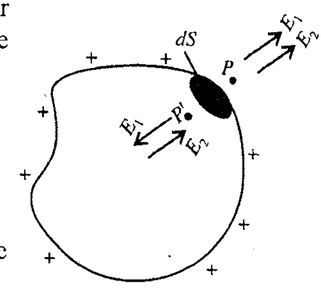


Fig. 5.35 : Mechanical force on the charged surface of a conductor.

Next consider a point P' just inside the surface element dS. The electric field at P' may again be taken as composed of two parts :

(1) As P' is on the opposite side of the element dS relative to P, the electric field at P' is $-E_1$ due to the charge on the element dS.

(2) The field at P' due to the charge on the rest of the surface is the same as that at P *i.e.*, E_2 because P' is very close to P.

Hence the electric field at P' is $-E_1 + E_2$, which should be zero because the electric field at a point inside a conductor is zero *i.e.*,

$$-E_1 + E_2 = 0 \text{ or } E_1 = E_2 \quad ...(3)$$

Hence from eq. (2)

$$E_2 = \frac{\sigma}{2\varepsilon_0} \quad ...(4)$$

This is the intensity of the electric field on the infinitesimally small area element dS due to the charge on the rest of the surface. Thus $\sigma/2\varepsilon_0$ is the outward force per unit charge (intensity) at the element dS. As the charge on the area element dS *is* σdS, the outward force acting on the charge of dS area is given by

$$F = \frac{\sigma}{2\varepsilon_0} \times \sigma\, dS = \frac{\sigma^2}{2\varepsilon_0} dS \quad ...(5)$$

Hence the foce per unit area or pressure (P) on the charged purface of the conductor is

$$\frac{F}{dS} = \frac{\sigma^2}{2\varepsilon_0} \text{ or } P = \frac{\sigma^2}{2\varepsilon_0} \quad ...(6)$$

This is an outward force arising due to mutual repulsion between charges.

From eq. (1) $\sigma = \varepsilon_0$ E, hence

$$P = \frac{(\varepsilon_0 E)^2}{2\varepsilon_0} \text{ or } P = \frac{1}{2}\varepsilon_0 E^2 \quad ...(7)$$

This normal force per unit area experienced by the surface of a charged conductor is also called **electrostatic pressure.**

Expansion of a Soap Bubble due to Charging

When we gradually charge a soap bubble, it is found to expand.

If T is the surface tension of soap solution, the excess of pressure inside a soap bubble of radius r is given by

$$p = \frac{4T}{r}$$

Let a charge q be given to the bubble, producing a surface charge density σ. Hence the additional outward pressure due to the charge will be [using eq. (6)] $\sigma^2/2\varepsilon_0$. Because of this additional pressure, the bubble expands and therefore, the *excess of air pressure* inside the bubble decreases, say, to p'. Hence now, the total excess of pressure is $p' + \sigma^2/2\varepsilon_0$. Let r' be the new radius of the expanded bubble. The value of surface charge density of the bubble is $\sigma = q/4\pi r'^2$. In equilibrium, this new excess of pressure is balanced by surface tension *i.e.*,

$$p' + \frac{\sigma^2}{2\varepsilon_0} = \frac{4T}{r'} \text{ or } p' = \frac{4T}{r'} - \frac{q^2}{32\pi^2\varepsilon_0 r'^4}$$

i.e.,

$$p' = \frac{4}{r'}\left(T - \frac{q^2}{128\pi^2\varepsilon_0 r'^3}\right) \quad ...(8)$$

This is the excess of air pressure inside the soap bubble after charging it with a charge q. From eq. (8), we see that the charging of the bubble effectively reduces the value of surface tension and hence the bubble increases.

As a special case, when the pressure inside the charged bubble is equal to the outside pressure, we have $p' = 0$. In such a case, from eq. (8)

$$T - \frac{q^2}{128\pi^2\varepsilon_0 r'^3} = 0, \quad \therefore \; q = 8\pi r' (2\varepsilon_0 T r')^{1/2} \qquad ...(9)$$

In such a case, radius of the bubble is given by

$$r' = \left(\frac{q^2}{128\pi^2\varepsilon_0 T}\right)^{1/3} = \frac{1}{4}\left(\frac{q^2}{2\pi^2\varepsilon_0 T}\right)^{1/3} \qquad ...(10)$$

Increase in radius of bubble due to charging : Let excess of pressure inside a soap bubble of radius r be p $(= 4T/r)$. If P_a is the atmospheric pressure, then the pressure inside the bubble is $P = P_a + p$. If V is the volume of the bubble, then according to Boyle's law,

$$PV = K, \text{ constant or } P \times \frac{4}{3}\pi r^3 = K \text{ or } Pr^3 = K' \qquad ...(11)$$

If due to change of pressure δP, the change in radius of the bubble is δr, then from eq. (11), we have

$$\delta P r^3 + 3Pr^2 \delta r = 0 \qquad \text{or} \qquad \delta r = -\left(\frac{r}{3P}\right)\delta P \qquad ...(12)$$

The decrease in pressure $(-\delta P)$ inside the bubble has been done by additional outward pressure $\sigma^2/2\varepsilon_0$ due to charging. This means that $-\delta P = \sigma^2/2\varepsilon_0$. Hence from eq. (12) the increase in radius of the bubble is given by

$$\delta r = \left(\frac{r}{3P}\right)\frac{\sigma^2}{2\varepsilon_0} = \left(\frac{r}{3P}\right)\frac{1}{2\varepsilon_0}\left(\frac{q}{4\pi r^2}\right)^2 \qquad \left[\because \sigma = \frac{q}{4\pi r^2}\right]$$

or

$$\delta r = \frac{q^2}{96\varepsilon_0 \pi^2 r^3 P} \qquad .(3)$$

5·17. Electrostatic Field Energy

According to Faraday, the electrostatic energy due to a charged conductor is contained in the medium in the electric field surrounding the conductor.

Consider an electrostatic field around a charged conductor. We know that the charged surface of the conductor experiences a force $\varepsilon_0 E^2/2$ per unit area which is along the outward normal at each point of the surface.

Let the surface be displaced through an infinitesimally small distance dl against the normal force. If the area of the surface be S, then the amount of work done

$$= \text{Froce} \times \text{distance}$$

$$= \frac{\varepsilon_0 E^2}{2} S \times dl$$

This work done is stored up as electrostatic field energy in the medium of volume $S \times dl$. Hence the electrostatic field energy stored per unit volume of the medium is

$$u = \frac{\varepsilon_0 E^2}{2} \qquad ...(1)$$

If the charged conductor is placed in a dielectric medium of dielectric constant K, then the stored energy per unit volume is

$$u = \frac{1}{2}\varepsilon_0 K E^2 \qquad ...(2)$$

5·18. Point Charge in front of a Grounded Infinite Conductor (Method of Electrical Images)

Let us consider a point charge $+q$ placed at a point P at a distance d from an infinite plane earthed conductor **[fig. 5.36]**. As the conductor is earthed, the potential of its plane face is zero. The total induced charge on the conductor due to the charge q at P will be $-q$. However this induced charge will be distributed non-uniformly on the conductor. Now if we want to find the electric field at any point, we should find the effect of non-uniform charge distribution on the conductor in addition to the effect due to the actual charge q at P. Lord Kelvin suggested a method to find the electric effect at any point due to the charge induced on the plane surface and the method itself is known as the **method of electrical images.**

An elelctrical image *is a point charge situated towards one side of the conducting surface which produces the same electrical field on the other side of the surface which is produced by the induced charge on that surface.*

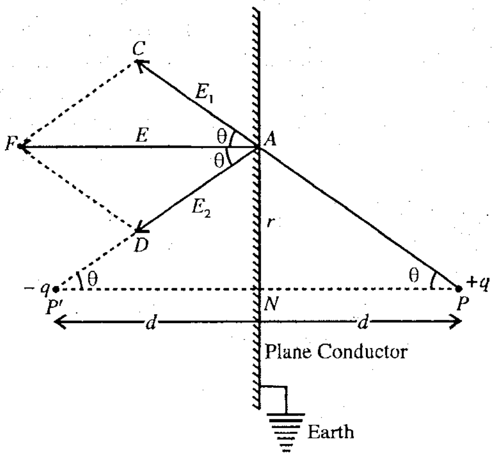


Fig. 5.36 : Method of electrical images.

Hence to find the electric field at a point due to the induced charge on the earthed conductor, it is replaced by a charge $-q$ situated on the other side of the conductor at P', which is at the same distance d from the conductor (as the charge $+q$ from it **[fig. 5.36]**. This $-q$ charge at P' is the **electrical image** of the charge $+q$ at P.

Now we want to determine the electric potential, electric field and the surface density of the induced charge on the conductor.

Electric potential at a point on the conductor : As shown in **fig. 5.36**, the electric potential at a point P on the conductor due to $+q$ and $-q$ charges is

$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{PA} - \frac{1}{4\pi\varepsilon_0}\frac{1}{P'A} = 0 \qquad [\because PA = P'A]$$

Electric field at a point on the conductor : As shown in **fig. 5.36**, the electric field at a point A due to the charge $+q$ at P is

$$E_1 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{PA^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2+d^2} \text{ along } \overrightarrow{AC}$$

and due to the charge $-q$ at P' is

$$E_2 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{P'A^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2+d^2} \text{ along } \overrightarrow{AD}$$

where $PA^2 = P'A^2 = r^2 + d^2$ and $r = NA$ (N is the foot point of the perpendicular PN on the conductor from P).

As E_1 and E_2 have the same magnitude, hence their perpendicular components (to AF) $E_1 \sin\theta$ and $E_2 \sin\theta$ will cancel each other and the components along AF will add up and give the resultant electric field at A *i.e.*,

$$
\begin{aligned}
E &= E_1 \cos\theta + E_2 \cos\theta \qquad \textbf{[fig. 5.36]}\\
&= \frac{1}{4\pi\varepsilon_0}\cdot\frac{2q}{r^2+d^2}\cos\theta \text{ along } \overrightarrow{AF}\\
&= \frac{1}{4\pi\varepsilon_0}\cdot\frac{2q}{r^2+d^2}\cdot\frac{d}{(r^2+d^2)^{1/2}} \qquad \left[\because \cos\theta = \frac{PN}{PA} = \frac{d}{(r^2+d^2)^{1/2}}\right]\\
&= \frac{qd}{2\pi\varepsilon_0(r^2+d^2)^{3/2}} \qquad \text{...(1)}
\end{aligned}
$$

This electric field is directed along AF or directed normally towards the conductror.

Surface induced charge density on the conductor : As the electric field at A is directed normally towards the conductor, the surface density of induced charge σ must be negative at A. As the electric field just outside a conductor is σ/ε_0 along the normal, hence

$$\frac{\sigma}{\varepsilon_0} = -\frac{qd}{2\pi\varepsilon_0(r^2+d^2)^{3/2}} \text{ or } \sigma = -\frac{qd}{2\pi(r^2+d^2)^{3/2}} \qquad ...(2)$$

Total induced charge on the conducting plane : Equation (2) can be used to determine the total induced charge on the conducting plane. We consider a ring area between the radii r and $r + dr$ on the plane conductor around N **[fig. 5.37]**.

$$\text{Area of the ring} = 2\pi r dr$$

$$\text{Induced charge density (at } r) \ \sigma = -\frac{qd}{2\pi(r^2+d^2)^{3/2}}$$

$$\text{Induced charge on the ring area } dq = -\frac{qd \times 2\pi r\, dr}{2\pi(r^2+d^2)^{3/2}}$$

Hence the induced charge on the conducting plane is

$$q_i = \int_0^\infty -\frac{qd \times 2\pi r\, dr}{2\pi(r^2+d^2)^{3/2}} = -qd\int_0^\infty \frac{rdr}{(r^2+d^2)^{3/2}}$$

Let $r = d \tan\theta$. Hence $dr = d\sec^2\theta\, d\theta$. Therefore,

$$q_i = -qd\int_0^{\pi/2} \frac{d\tan\theta\, d\sec^2\theta\, d\theta}{(d^2+d^2\tan^2\theta)^{3/2}}$$

$$= -q\int_0^{\pi/2} \frac{\tan\theta\sec^2\theta\, d\theta}{\sec^3\theta} = -q\int_0^{\pi/2} \sin\theta\, d\theta$$

$$= -q[-\cos\theta]_0^{\pi/2} = -q \qquad ...(3)$$

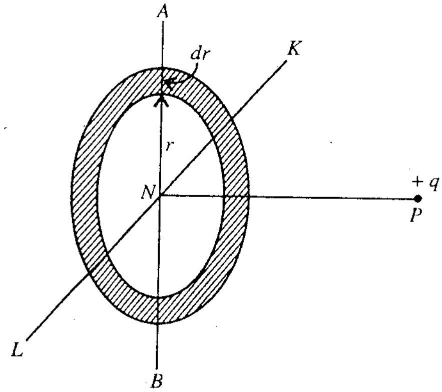


Fig. 5.37 : Calculation of total induced charge on the conducting plane.

In fact we expected the total induced charge on the plane conductor to be $-q$ and as such eq. (3) is a further check to see the correctness of our calculation.

Force of attraction between the charge q and the conductor : We can calculate the force of attraction F between the charge $+q$ and the conductor, if we take in place of the induced charge in the conductor the image charge $-q$ at P'. Thus

$$F = -\frac{q^2}{4\pi\varepsilon_0(2d)^2} = -\frac{q^2}{16\pi\varepsilon_0 d^2} \qquad ...(4)$$

Work done in removing the charge q to infinity : The work done W in removing the charge from the distance d to infinity can be calculated by taking X-axis along NP direction with N as origin *i.e.*,

$$W = -\int_d^\infty F dx = \int_d^\infty \frac{q^2}{16\pi\varepsilon_0 x^2}dx = \frac{q^2}{16\pi\varepsilon_0 d} \qquad ...(5)$$

where $F = -\frac{q^2}{16\pi\varepsilon_0 x^2}$ is the force when charge $+q$ is at a distance x from N.

Ex. 1. *A sphere of diameter* 2 *cm is charged to a potential of* 100 *volt. Calculate the outward force per unit area.*

Sol. Potential at the surface of sphere, $V = \frac{q}{4\pi\varepsilon_0 r}$

Electric field at the surface of the sphere, $E = \frac{q}{4\pi\varepsilon_0 r^2} = \frac{V}{r}$

Outward force per unit area, $F = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0\frac{V^2}{r^2} = \frac{1}{2}\times\frac{8.85\times10^{-12}\times100^2}{(10^{-2})^2} = \mathbf{4.42 \times 10^{-4}\ N/m^2}$.

Ex. 2. *2 gm of a metal is beaten into a thin leaf of area 1 sq.m. A small piece is cut off from this and placed upon a conductor. Calculate the charge density required by the conductor so that the piece of metal is just lifted up.*

Sol. Let σ is the charge density on the surface of the conductor. The upward force per sq.m. due to charge is given by

$$F = \frac{\sigma^2}{2\varepsilon_0} \text{ N/m}^2$$

The downward force on the gold due to its gravity on 1 sq. m. of metal leaf is

$$mg = 2 \times 10^{-3} \times 9.8 \text{ N/m}^2$$

For just lifting up the metal piece,

$$F = mg \text{ or } \frac{\sigma^2}{2\varepsilon_0} = 2\times10^{-3}\times9.8$$

or $$\sigma^2 = 2\times8.5\times10^{-12}\times2\times10^{-3}\times9.8$$

whence $$\sigma = \mathbf{4.1 \times 10^{-10} \text{ C/m}^2}.$$

Ex. 3. *To what potential must a soap bubble be raised so that the pressure inside the bubble equals to the pressure outside. The radius of the bubble in this condition is 1 cm. ($T = 25 \times 10^{-3}$ N/m).*

Sol. We know that the excess of pressure in a charged bubble is given by

$$p = \frac{4T}{r} - \frac{\sigma^2}{2\varepsilon_0} = \frac{4T}{r} - \frac{q^2}{32\pi^2\varepsilon_0 r^4} \qquad \left[\because \sigma = \frac{q}{4\pi r^2}\right] \qquad ...(i)$$

The excess of pressure p will be zero, when the pressure inside the bubble equals the outside pressure

i.e.,

$$\frac{4T}{r} - \frac{q^2}{32\pi^2\varepsilon_0 r^4} = 0 \qquad ...(ii)$$

The potential for a spherical bubble of radius r with charge q is

$$V = \frac{q}{4\pi\varepsilon_0 r}; \quad \therefore q = 4\pi\varepsilon_0 rV \qquad ...(iii)$$

Substituting for q form eq. (iii) into eq. (ii), we get

$$\frac{4T}{r} - \frac{(4\pi\varepsilon_0 rV)^2}{32\pi^2\varepsilon_0 r^4} = 0 \text{ or } \frac{4T}{r} - \frac{\varepsilon_0 V^2}{2r^2} = 0$$

$$\therefore \quad V = \sqrt{\frac{8Tr}{\varepsilon_0}} = \sqrt{\frac{8\times25\times10^{-3}\times10^{-2}}{8.85\times10^{-12}}} = \mathbf{1.5\times10^4 \text{ volt.}}$$

Ex. 4. *A soap bubble of radius 1 cm is given a charge 2 nC. Calculate the change in radius of the bubble due to charging. (Atmospheric pressure = 10^5 N/m^2)*

Sol. Due to charging, the increase in radius is given by

$$\delta r = \frac{q^2}{96\varepsilon_0\pi^2 r^3 P}$$

$$= \frac{(2\times10^{-9})^2}{96\times8.85\times10^{-12}\times(3.14)^2\times(0.01)^3\times10^5}$$

$$= \mathbf{4.76 \times 10^{-9} \text{ m.}}$$

where we have used the value of the atmospheric pressure for total pressure inside the bubble because

$$P = P_a + \frac{4T}{r} \simeq P_a.$$

Ex. 5. *Prove that the charge required to increase the radius of a soap bubble from radius R to $2R$ is given by*

$$q = 8\pi R^2[\varepsilon_0(7P_a + 12T/R)]^{1/2}$$

where P_a is the atmospheric pressure.

Sol. Initially, volume of the bubble, $V_1 = \frac{4}{3}\pi R^3$

Pressure inside the bubble, $\quad P_1 = P_a + \frac{4T}{R}$

Finally, due to charging the bubble by the charge, say q, the radius of the bubble becomes $2R$.

Now, volume of the bubble, $\quad V_2 = \frac{4}{3}\pi(2R)^3 = \frac{32}{3}\pi R^3$

Pressure inside the bubble, $\quad P_2 = P_a + \frac{4T}{2R} - \frac{\sigma^2}{2\varepsilon_0} = P_a + \frac{2T}{R} - \frac{q^2}{512\pi^2\varepsilon_0 R^4} \quad \left[\because \sigma = \frac{q}{4\pi(2R)^2}\right]$

According to Boyle's law,

$$P_1V_1 = P_2V_2$$

i.e.,
$$\left(P_a + \frac{4T}{R}\right)\times\frac{4}{3}\pi R^3 = \left(P_a + \frac{2T}{R} - \frac{q^2}{512\pi^2\varepsilon_0 R^4}\right)\times\frac{32\pi R^3}{3}$$

or
$$P_a + \frac{4T}{R} = \left(P_a + \frac{2T}{R} - \frac{q^2}{512\pi^2\varepsilon_0 R^4}\right)\times 8$$

or
$$7P_a + \frac{12T}{R} - \frac{q^2}{64\pi^2\varepsilon_0 R^4} = 0 \text{ or } \frac{q^2}{64\pi^2\varepsilon_0 R^4} = 7P_a + \frac{12T}{R}$$

whence
$$q = 8\pi R^2\left[\varepsilon_0\left(7P_a + 12T/R\right)\right]^{1/2}.$$

Ex. 6. *Charge 10^{-15}C is placed at a distance of 20Å from a plane conductor. Calculate the force acting on the charge and the work required to take the charge from its present position to infinity.*

Sol. The **method of electrical image** is used for calculation.

$$\text{Force on the charge, } F = -\frac{q^2}{16\pi\varepsilon_0 d^2} = -\frac{(10^{-15})^2\times 9\times 10^9}{4\times(20\times 10^{-10})^2}$$

$$= \mathbf{5.6\times 10^{-2}\ N\ (attractive\ force)}$$

$$\text{Work done, } W = \frac{q^2}{16\pi\varepsilon_0 d} = \frac{(10^{-15})^2\times 9\times 10^9}{4\times 20\times 10^{-10}} = \mathbf{1.1\times 10^{-12}\ J.}$$

Ex. 7. *Two charges q and q are placed at a distance d from a conducting plane at zero potential and at a distance $3d/2$ from each other. Find the resultant force and the angle made by it on a charge q with the normal to the surface.*

Sol. Let CD be the conducting plane at zero potential [fig. 5·38]. Two charges q and q are placed at A and B respectively so that their corresponding images are having charges $-q$ (at A') and $-q$ (at B').

The charge q at A experiences following forces :

(i) Due to charge q at B,

$$F_B = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q^2}{(3d/2)^2} = \frac{q^2}{9\pi\varepsilon_0 d^2} \text{ along } \overrightarrow{BA}$$

(ii) Due to charge $-q$ at A', $$F_{A'} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q^2}{(2d)^2} = \frac{q^2}{16\pi\varepsilon_0 d^2} \text{ along } \overrightarrow{AA'}$$

(iii) Due to charge $-q$ at B', $$F_{B'} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q^2}{(4d^2+9d^2/4)} = \frac{q^2}{25\pi\varepsilon_0 d^2} \text{ along } \overrightarrow{AB'}$$

Force parallel to the conducting plane (along $\overrightarrow{BA}$),

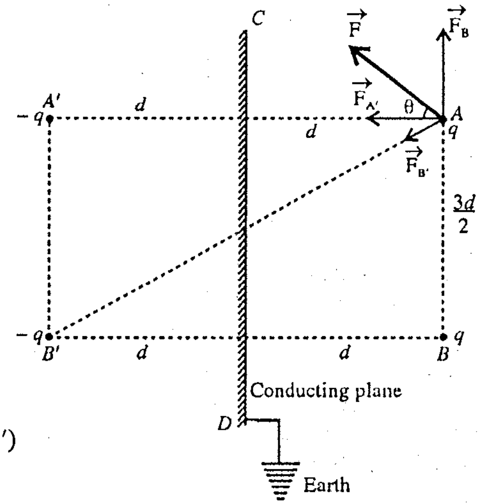


Fig. 5.38.

$$F_1 = \frac{q^2}{9\pi\varepsilon_0 d^2} - \frac{q^2}{25\pi\varepsilon_0 d^2}\cos B'AB$$

$$= \frac{q^2}{9\pi\varepsilon_0 d^2} - \frac{q^2}{25\pi\varepsilon_0 d^2}\cdot\frac{3}{5}$$

$$\left[\because \cos B'AB = \frac{3d/2}{\sqrt{(4d^2+9d^2/4)}} = \frac{3}{5}\right]$$

$$= \frac{q^2}{\pi\varepsilon_0 d^2}\left[\frac{1}{9} - \frac{3}{125}\right] = \frac{98}{112.5}\,\frac{q^2}{\pi\varepsilon_0 d^2}$$

Force perpendicular to the conducting plane (allong AA')

$$F_2 = \frac{q^2}{16\pi\varepsilon_0 d^2} + \frac{q^2}{25\pi\varepsilon_0 d^2}\sin B'AB$$

$$= \frac{q^2}{16\pi\varepsilon_0 d^2} + \frac{q^2}{25\pi\varepsilon_0 d^2}\cdot\frac{4}{5}$$

$$= \frac{q^2}{\pi\varepsilon_0 d^2}\left[\frac{1}{16} + \frac{4}{125}\right] = \frac{189}{2000}\cdot\frac{q^2}{\pi\varepsilon_0 d^2}$$

$\therefore$ Resultant force, $F = \sqrt{F_1^2 + F_2^2}$

$$= \frac{q^2}{\pi\varepsilon_0 d^2}\sqrt{\left(\frac{98}{1125}\right)^2 + \left(\frac{189}{2000}\right)^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{0.51q^2}{d^2}$$

Direction, $\theta = \tan^{-1}\frac{F_1}{F_2} = \tan^{-1}(0.92) = \mathbf{42.6°}$ **with** AA'.

EXERCISE 5·2

1. A soap bubble of radius 0.1 m is given a charge of 7.5×10^{-19} C. Calculate the change in radius of the bubble. (Atmospheric pressure = 10^5 N/m^2) **(Lucknow 2005)**
[**Ans.** 6.7×10^{-30} m.]
2. The charge on a soap bubble is q. If its internal pressure is equal to its external pressure, find the radius of bubble. (Surface tension of soap solution = T.)
$$\left[\textbf{Ans. } \frac{1}{4}\left(\frac{1}{2\pi^2\varepsilon_0 T}\right)^{1/3}.\right]$$

3. A sphere of radius 1 cm is charged at a potential of 1000 volt. Find the outward force per unit area.
[**Ans.** 0.44 Nm^{-2}.]
4. If the radius of a charged spherical soap bubble is r and its internal pressure is equal to the atmospheric pressure, show that the potential of the bubble will be $\sqrt{8Tr/\varepsilon_0}$, where T is the surface tension of the soap solution.
5. An electron is placed at a distance 10 Å from a plane conductor. Find the force acting on the electron, and the work required to take the electron from its present position to infinity. **(Rohilkhand 1993)**
[**Ans.** 5.76×10^{-11} N (attractive); 5.76×10^{-20} J.]
6. A point charge of 30 μC is placed at a distance of 3 cm from a plane earthed conductor. Calculate :
(i) the froce with which it is attracted towards conductor.
(ii) work done in taking the charge to infinity.
(iii) induced charge density at a distance of 4 cm on the conductor from point charge.
[**Ans.** (i) 2250 N; (ii) 67.5 J; (iii) 1.146×10^{-3} C/m^2.]
7. A point charge 2 μC is placed at a distance 40 cm above an infinite linear conductor. Find the force of attraction on charge and surface density of induced charge at the point on the conductor, at distance 50 cm from the point charge.
[**Ans.** 0.05625 N; -1.02×10^{-6} C/m^2.]
8. Two equal point charges of opposite sign are placeed at A and B at $2a$ distance apart in front of an infinite earthed plane conductor. The distance of each charge from the plane conductor is a. Show that the force needed to maintain either charge in its position is inclined to the plane at an angle of $\pi/4$ and is less than that would be if the plane were absent in the ratio $(2\sqrt{2}-1)/2$.

5·19. Capacitor

An arrangement of two conductors placed close to each other is called **capacitor**. A positive charge is given to one of the two conductors and an equal negative charge is given to the other conductor. We call the conductors with the positive and negative charges as the positive and negative plates of the capacitors respectively. *The charge on the positive plate is called the* **charge on the capacitor.**

We represent a capacitor as shown in **fig. 5.39.** One of the plate of the capacitor has charge $+Q$ and the other $-Q$ and we say Q is the charge on the capacitor. It is to be mentioned that the term charge on a capacitor does not mean the total charge on the capacitor because the total charge on the capacitor is $Q - Q = 0$.

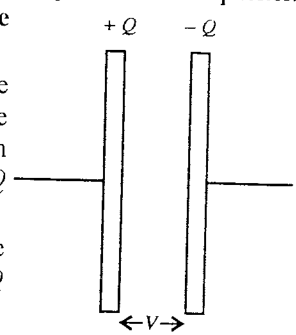


Fig. 5.39 : Capacitor.

Due to $+Q$ and $-Q$ charges on the two plates of the capacitor, let V be the difference of potential between its plates. Then it is found that the charge Q on the capacitor is proportional to the potential difference V, *i.e.*,

$$Q \propto V \text{ or } Q = CV \qquad ...(1)$$

where C is the constant of proportionality and is called the **capacitance** of the capacitor. The value of C depends on the shape, size, spacing and the medium between the conductor.

SI unit of capacitance is **coulomb/volt** and is written as farad (F). It is a large unit and in practice smaller units **microfarad** (μF) and **pekofarad** (pF) are used.

$$1\ \mu F = 10^{-6}\ F \text{ and } 1\ pF = 10^{-12}\ F$$

Dimensions of capacitance are $[M^{-1}L^{-2}T^4A^2]$.

In order to give equal and opposite charges on the two plates of the capacitor, they are connected to the terminals of a battery.

5·20. Capacitance of a Parallel Plate Capacitor

A parallel plate capacitor consists of two large plane metallic plates, placed parallel to each other with a small distance between them. The space between the plates contains air or vacuum or some dielectric material. Here first we consider vacuum in the space between the plates.

Let a charge $+Q$ be given to one plate and $-Q$ to the other plate. The separation between the plates is d. The charges will appear on the facing surfaces of the capacitor **[fig. 5.40]**. Surface charge density (σ) on any of these surfaces will be given by

$$\sigma = \frac{Q}{A} \quad ...(1)$$

where A is the area of a plate.

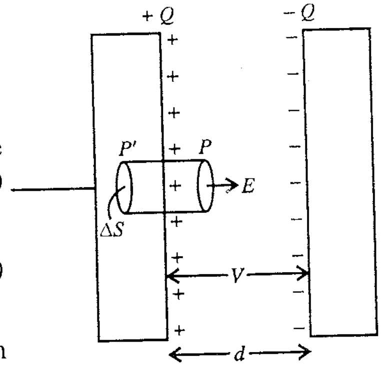


Fig. 5.40 : Parallel plate capacitor.

The plates are large when compared to the separation between them. In such a case, the electric field between the two plates of the capacitor is uniform and perpendicular to the plates except near the edge. This uniform electric field can be calculated by using Gauss's law.

We draw a Gaussian pill box PP' (cylinder) of cross-sectional area ΔS as shown in **fig. 5.40.** P face of the cylinder is in the space between the plates and P' face is inside the conducting plate.

The electric flux through ΔS at P' and curved portion of the cylinder PP' inside the plate is zero because the electric field inside the conductor is zero. The electric flux through the curved part of the Gaussian cylinder (surface) is also zero because the direction of the electric field E is parallel to this surface. The electric flux through the surface ΔS at P is

$$\vec{E} \cdot \vec{\Delta S} = E\,\Delta S \quad ...(2)$$

This is the total flux through the Gaussian cylindrical surface. The charge enclosed by the Gaussian surface is

$$\Delta Q = \sigma \Delta S = \frac{Q}{A}\Delta S \quad ...(3)$$

Using Gauss's law, $\oint \vec{E} \cdot \vec{dS} = \frac{Q}{\varepsilon_0}$, we have

$$E\,\Delta S = \frac{Q}{\varepsilon_0 A}\Delta S \text{ or } E = \frac{Q}{\varepsilon_0 A} \quad ...(4)$$

Let the potential difference between the plates be V. It will be related to the electric field E as

$$E = \frac{V}{d} \text{ or } V = Ed \quad ...(5)$$

Putting the value of E from eq. (4), we obtain

$$V = \frac{Q}{\varepsilon_0 A}d \text{ or } Q = \frac{\varepsilon_0 A}{d}V \quad ...(6)$$

Hence using the relation $Q = CV$, the **capacitance of the parallel plate condenser** is given by

$$C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d} \quad ...(7)$$

where $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m for free space (or air).

If the space between the plates of the dielectric is filled with a dielectric of dielectric constant K, then $E = \sigma/K\varepsilon_0$ and hence

$$C = \frac{K\varepsilon_0 A}{d} \quad ...(8)$$

Thus in order to increase the capacity of parallel plate capacitor, (i) area of the plates (A) should be large, (ii) the separation between the plates (d) should be small and (iii) the space between the plates is to be filled with a dielectric of high dielectric constant (K).

5·21. Capacitance of a Spherical Capacitor

A spherical capacitor (or condenser) consists of a solid or hollow spherical conductor surrounded by another concentric hollow spherical conductor. Let the radii of the inner and outer spheres be R_1 and R_2 respectively. Also let the charges $+Q$ and $-Q$ be given to the inner and outer spheres.

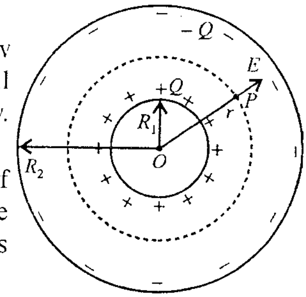


Fig. 5.41 : Spherical capacitor.

The electric field at any point P between the two spherical surfaces of the spheres is radially outward and its magnitude depends only on the distance r of the point from the centre O. We draw a Gaussian spherical surface of radius r with centre O through P **[fig. 5.41]**.

The flux through the Gaussian surface is

$$\oint \vec{E} \cdot \vec{dS} = \oint E dS = E \oint dS = E \times 4\pi r^2$$

The charge enclosed by the Gaussian surface is $+Q$. Hence, using Gauss's law $\oint \vec{E} \cdot \vec{dS} = q/\varepsilon_0$, we have

$$E \times 4\pi r^2 = \frac{Q}{\varepsilon_0} \text{ or } E = \frac{Q}{4\pi\varepsilon_0 r^2} \qquad ...(1)$$

The potential difference $V_2 - V_1 = V$ between the positive and negative conductors is given by

$$V = V_2 - V_1 = -\int_{R_2}^{R_1} \vec{E} \cdot d\vec{r} = -\int_{R_2}^{R_1} \frac{Q}{4\pi\varepsilon_0 r^2} dr$$

$$= \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r}\right]_{R_2}^{R_1} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = \frac{Q(R_2 - R_1)}{4\pi\varepsilon_0 R_1 R_2} \qquad ...(2)$$

Hence the capacitance of the spherical capacitor is given by

$$C = \frac{Q}{V} = \frac{4\pi\varepsilon_0 R_1 R_2}{R_2 - R_1} \qquad ...(3)$$

If a dielectric of dielectric constant K is put in between the two spheres, then the expression of capacity will be modified as

$$C = \frac{4\pi\varepsilon_0 K R_1 R_2}{R_2 - R_1} \qquad ...(4)$$

Hence the capacity of the condenser can be increased by putting a dielectric material in between the two spheres.

Capacity of an Isolated Conducting Sphere

If we consider the outer spherical surface at infinity *i.e.*, $R_2 = \infty$, we obtain an isolated single sphere of radius $R_1 = R$ (say). Then from eq. (3), we have

$$C = \frac{4\pi\varepsilon_0 R_1}{1 - R_1/R_2} = 4\pi\varepsilon_0 R \qquad [\text{for } R_2 \to \infty] \qquad ...(5)$$

This is the expression for the capacity of a spherical conductor of radius R.

5·22. Force between the Plates of a Parallel Plate Capacitor

The electric field due to the positive plate only is

$$E_+ = \frac{\sigma}{2\varepsilon_0} = \frac{Q}{2\varepsilon_0 A}.$$

where $\sigma = Q/A$.

The negative charge $-Q$ of the negative plate finds itself in the electric field of the positive plate [fig. 5.42]. Hence the force acting on the negative plate or charge $-Q$ is

$$F = -QE_+ = -Q\left(\frac{Q}{2\varepsilon_0 A}\right) = -\frac{Q^2}{2\varepsilon_0 A}$$

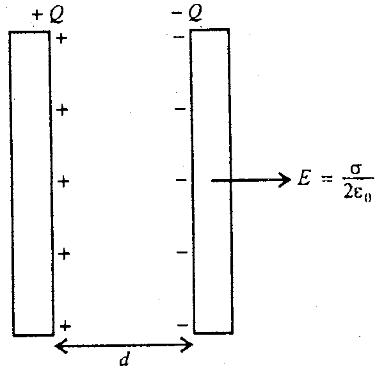


Fig. 5.42 : Calculation of force between the plates of a parallel plate capacitor.

Hence the positive plate is attracting the negative plate by a force of magnitude $Q^2/2\varepsilon_0 A$ and also by this force the negative plate is attracting the positive plate. Thus the force of **attraction between the positive and negative plates of a parallel plate capacitor** is given by

$$F = \frac{Q^2}{2\varepsilon_0 A} = \frac{\sigma^2 A}{2\varepsilon_0} \qquad [\because Q = \sigma A]$$

$$\therefore \qquad \text{Force per unit area} = \frac{F}{A} = \frac{\sigma^2}{2\varepsilon_0}$$

If E is the resultant intensity of the electric field between the positive and negative plates, then we know that $E = \sigma/\varepsilon_0$ and hence

$$\text{Force per unit area} = \frac{1}{2}\varepsilon_0 E^2$$

5·23. Potential Energy Stored in a Capacitor

Consider a parallel plate capacitor. Let initially the positive plate with charge $+Q$ and negative plate with charge $-Q$ be very close to each other. If we hold the positive plate and slowly pull away the negative plate, then one has to apply a force equal to $F = Q^2/2A\varepsilon_0$ and the work will be done in pulling apart the plates. The amount of work done if the plates are moved apart by a distance d is

$$W = Fd = \frac{Q^2 d}{2A\varepsilon_0} = \frac{1}{2}\frac{Q^2}{C}$$

This work done is stored in the electric field created in the space between the plates as the **electrostatic potential energy** U *i.e.*,

$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}QV = \frac{1}{2}CV^2 \qquad [\because Q = CV]$$

The electrostatic potential energy per unit volume of the electric field is given by

$$u = \frac{U}{Ad} = \frac{1}{2}\frac{Q^2}{CAd} = \frac{(\varepsilon_0 EA)^2}{2(\varepsilon_0 A/d)Ad} = \frac{1}{2}\varepsilon_0 E^2$$

where Ad is the volume between the plates, $Q = \varepsilon_0 EA$ [$\because E = \sigma/\varepsilon_0 = Q/A\varepsilon_0$] and $C = \varepsilon_0 A/d$.

We see that the region, having the electric field E has the energy $\frac{1}{2}\varepsilon_0 E^2$ per unit volume. This is true for any electric field E which is inside a parallel capacitor or elsewhere.

Ex. 1. *A parallel-plate capacitor has plates of area* 100 *cm*^{2} *and separation between the plates* 1.00 *mm. What potential difference will be developed if a charge of* 2.00 *nC is given to the capacitor ? If the plate separation is now increased to* 2.00 *mm, calculate new potential difference ?*

Sol. The capacitance of the capacitor is $C = \frac{\varepsilon_0 A}{d}$

$$= 8.85 \times 10^{-12} \times \frac{10 \times 10^{-4}}{1 \times 10^{-3}} = 8.85 \times 10^{-10}\ \text{F}$$

The potential difference between the plates

$$V = \frac{Q}{C}$$

$$= \frac{2 \times 10^{-9}}{0.885 \times 10^{-10}} = \mathbf{22.56\ volt.}$$

When the separation is increased from 1 mm to 2 mm, obviously the capacitance will become one half of the previous value *i.e.*,

$$\text{New capacitance, } C' = \frac{0.885 \times 10^{-10}}{2} \text{ F}$$

If the charge remains the same *i.e.*, $Q = 2 \times 10^{-9}$C, then the new potential difference will be

$$V = \frac{2 \times 10^{-9} \times 2}{0.885 \times 10^{-10}} = \mathbf{45.12\ volt.}$$

Ex. 2. *The area of each plate of a parallel plate capacitor is* 100 cm^2 *and the intensity of electric field between the two plates is* 100 NC^{-1}. *What is the charge on each plate ?* ($\varepsilon_0 = 8.85 \times 10^{-12}\ C^2 N^{-1} m^{-2}$.)

Sol. The capacitance of a parallel plate air capacitor

$$C = \frac{\varepsilon_0 A}{d}$$

If the potential difference between the plates is V, then the charge on each plate is

$$Q = CV = \varepsilon_0 A \times \frac{V}{d}.$$

If the electric field between the plates is E, then $E = V/d$. Hence

$$Q = \varepsilon_0 AE = 8.85 \times 10^{-12} \times 100 \times 10^{-4} \times 100 = \mathbf{8.85 \times 10^{-12} C.}$$

Ex. 3. *The area of each plate of a parallel plate condenser is* 200 cm^2 *and the plates are at a separation of* 1 *mm. The medium between the plates is of dielectric constant* 6. *Calculate the capacity of condenser. What will be the radius of a conducting sphere of same capacity ?* $\varepsilon_0 = 8.86 \times 10^{-12}\ F m^{-1}$.

Sol. Here, $A = 200 \text{ cm}^2 = 200 \times 10^{-4} \text{ m}^2 = 2 \times 10^{-2} \text{ m}^2$, $d = 1 \text{ mm} = 10^{-3}$ m, $K = 6$.

$$\text{Capacity of parallel plate condenser, } C = \frac{\varepsilon_0 KA}{d}$$

$$= \frac{8.85 \times 10^{-12} \times 6 \times 2 \times 10^{-2}}{10^{-3}} = 10.62 \times 10^{-10} \text{ F}$$

If R be the radius of conducting sphere, then its capacity

$$C = 4\pi\varepsilon_0 R.$$

Hence for $C = 10.62 \times 10^{-10}$ F, we have

$$10.62 \times 10^{-10} = \frac{R}{9 \times 10^9}$$

or

$$R = 9 \times 10.62 \times 10^{-1} = \mathbf{9.558\ m.}$$

Ex. 4. *The potential difference between the plates of a parallel plate condenser is* 100 *volt. The area of each plate is* 100 cm^2 *and separation between them is* 1 *mm. If the medium in between the plates is air, calculate the charge on condenser. If there be a medium of dielectric constant* 5 *in between the plates, what potential difference is needed for the same charge ?*

Sol. Here, $V = 100$ volt, $d = 1 \text{ mm} = 10^{-3}$ m, $A = 100 \text{ cm}^2 = 100 \times 10^{-4} \text{ m}^2 = 10^{-2} \text{ m}^2$.

If there is air in between the plates, capacity of condenser is given by

$$C = \frac{\varepsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 10^{-2}}{10^{-3}} = 8.85 \times 10^{-11} \text{ F}$$

$\therefore$ Charge on condenser, $Q = CV = 8.85 \times 10^{-11} \times 100 = \mathbf{8.85 \times 10^{-9}\ C}$

When there is a medium of dielectric constant $K = 5$ between the plates, the capacity of condenser will become K times, *i.e.*,

$$C' = KC = 5 \times (8.85 \times 10^{-11}) = 4.425 \times 10^{-10} \text{ F}$$

The new potential difference between the plates for the charge $Q = 8.85 \times 10^{-9}$ C is

$$V' = \frac{Q}{C'} = \frac{8.85 \times 10^{-9}}{4.425 \times 10^{-10}} = \mathbf{20 \text{ volt.}}$$

Ex. 5. *Capacitors and battery are connected as shown in* **fig. 5.43.** (a) *Find the capacitance between the points A and B.* (b) *Find the charges on the three capacitors.* (c) *Taking the potential at the point B to be zero, find the potential at the point D.*

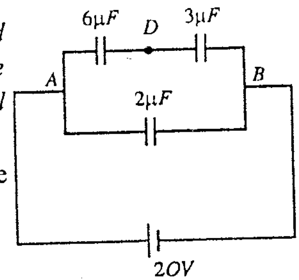


Fig. 5.43

Sol. (a) The capacitors of capacity 6 μF and 3 μF are joined in series. The equivalent capacitance will be given by

$$\frac{1}{C} = \frac{1}{6\mu\text{F}} + \frac{1}{3\mu\text{F}} = \frac{1}{2\mu\text{F}} \text{ or } C = 2\ \mu\text{F}$$

This combination of the two capacitors is connected in parallel with the 2 μF capacitor. Thus, the equivalent capacitance between A and B is

$$2\ \mu\text{F} + 2\ \mu\text{F} = \mathbf{4\ \mu F.}$$

(b) The charge supplied by the battery is

$$Q = CV = 4\ \mu\text{F} \times 24\ \text{V} = \mathbf{80\ \mu C}$$

The potential difference across the 2 μF capacitors is 20 V. The charge on this capacitor is, therefore, 2 μF × 20 V = 40 μC.

The charge on the 6 μF and 3 μF capacitor is the same and given by 80 μC – 40 μC = **40 μC.**

(c) The potential difference across the 3 μF capacitor is

$$\frac{40\mu\text{C}}{3\mu\text{F}} = \mathbf{13.3 \text{ volt.}}$$

Since the potential at the point B is taken to be zero, the potential at the point D is **13.3 volt.**

Ex. 6. *An uncharged cpacitor is connected to a battery. Show that half the energy supplied by the battery is lost as heat during charging the capacitor.*

Sol. Let the capacitance of the capacitor be C and the emf of the battery be V. The charge on the capacitor is $Q = CV$. Hence the work done by the battery is

$$W = QV$$

This energy is supplied by the battery.

The energy stored in the capacitor is given by

$$U = \frac{1}{2}CV^2 = \frac{1}{2}QV.$$

Hence, the remaining energy $QV - \frac{1}{2}QV = \frac{1}{2}QV$ is lost as heat, which is one-half the energy supplied by the battery.

Ex. 7. *Calculate the capacitance of spherical capacitor consisting of two concentric spheres of radii* 0.50 *m and* 0.60 *m. The dielectric constant of the material filled in the space between the sphere is* 3.

Sol. The capacitance of spherical capacitor is given by

$$C = 4\pi\varepsilon_0 K\left(\frac{ab}{b-a}\right).$$

where a and b are the radii of the spheres. Substituting the given values, we have

$$C = \left(\frac{3}{9 \times 10^9}\right)\left(\frac{0.50 \times 0.60}{0.60 - 0.50}\right) = \mathbf{10^{-9}\ F.}$$

Ex. 8. *Find the capacity of earth assuming its radius to be 6400 km. The intenstiy of electric field near the earth surface is 130 Vm^{-1}. What amount of charge will produce this electric field on earth ? Calculate the electric potential on the earth surface due to this charge ?*

Sol. Here, radius of earth, $R = 6400$ km $= 6.4 \times 10^6$ m, $E = 130$ V/m.

$$\text{Capacity of the earth, } C = 4\pi\varepsilon_0 R = \frac{6.4 \times 10^6}{9 \times 10^9} = \mathbf{7.1 \times 10^{-4}\,F}$$

Let the charge on earth be q. Then the electric field on earth surface will be given by

$$E = \frac{Q}{4\pi\varepsilon_0 R^2}$$

Hence,

$$q = 4\pi\varepsilon_0 R^2 E$$

$$= \frac{1}{9 \times 10^9} \times (6.4 \times 10^6)^2 \times 130 = \mathbf{5.6 \times 10^5\ C.}$$

Therefore the potential on earth surface due to charge q is

$$V = \frac{Q}{4\pi\varepsilon_0 R}$$

$$= \frac{9 \times 10^9 \times 5.6 \times 10^5}{6.4 \times 10^6} = \mathbf{7.9 \times 10^8\ volt.}$$

Ex. 9. *A parallel plate capacitor with the plate area 100 cm^2 and the separation between the plates 1.0 cm is connected across a battery of emf 24 volts. Find the force of attraction between the plates.*

Sol.

$$F = \frac{Q^2}{2\varepsilon_0 A} = \frac{C^2V^2}{2\varepsilon_0 A} = \left(\frac{\varepsilon_0 A}{d}\right)^2 \frac{V^2}{2\varepsilon_0 A} = \frac{\varepsilon_0 A V^2}{2d^2}$$

$$= \frac{8.85 \times 10^{-12} \times 100 \times 10^{-4} \times 20^2}{2 \times (10^{-2})^2} = \mathbf{1.77 \times 10^{-7}\ N.}$$

EXERCISE 5·3

1. How much area of a paper wil be required to construct a parallel plate capactor of 0.004 μF, if the dielectric constant of the paper be 2.5 and its thickness be 0.025 mm ?
   [Ans. 45 cm^2.]
2. A parallel plate capacitor having plate area 25 cm^2 and separation 1.00 mm is connected to a battery of 12 V. Find the charge flown through the battery. Calculate work done by the battery during the process.
   [Ans. 2.66×10^{-10} C; 3.2×10^{-9} J.]
3. An air capacitor has a capacitance of 2.0 μF and the capacity becomes 12 μF when same material is filled in the space between the plates. Calculate the dielectric constant of the material.
   [Ans. 6.]
4. A parallel plate air capacitor has plate area 100 cm^2 and separation 5.0 mm. A potential difference of 300 V is acquired between its plates by a battery. After disconnecting the battery, the space between the plates is filled by ebonite a dielectric of dielectric constant 2.6. Calculate (i) new potential difference between the plates, (ii) initial and final capacitances, (iii) initial and final surface densities of charge on the plates.
   [Ans. (i) 115 V; (ii) 1.77×10^{-11} F, 4.6×10^{-11} F; (iii) in each case 5.31×10^{-7} Cm^{-2}.]

5. In the given network [**fig. 5.44**], calculate (i) equivalent capacitance between points A and B. (ii) If a voltage of 150 volt is applied between A and B, calculate the p.d. between the plates of 6 μF capacitance.
   [**Ans.** (i) 2 μF; (ii) 50 V.]

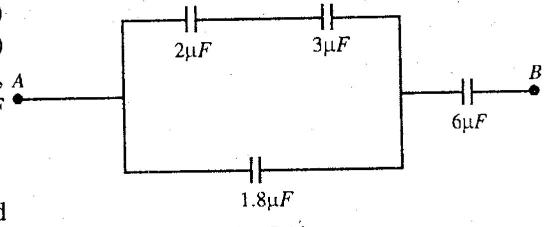


Fig. 5.44

6. A parallel plate capacitor of plate area 0.2 m^2 and spacing 10^{-2} m is charged to 10^3 V and is then disconnected from the battery. How much work is required if the plates are pulled apart to double the plate spacing ? Calculate the final voltage on the capacitor. ($\varepsilon_0^{-1} = 36\,\pi \times 10^9$ V-m/A-s)
   [**Ans.** 8.85×10^{-5} J; 2000 volt.]
7. A 2 μF parallel plate capacitor with a dielectric slab ($K = 5$) between the plates is charged to 100 volt and then isolated. What will be the potential difference if the dielectric be removed ? How much work would be done in removing the dielectric ?
   [**Ans.** 500 volt; 0.2 J.]
8. The radii of inner and outer spheres of a spherical condenser are 10 cm and 12 cm respectively. If a charge 60 nano-coulomb is imparted to the inner sphere and there is a medium of dielectric constant 5 in between the two spheres, calculate the potential difference between the spheres and capacity of the condenser.
   [**Ans.** 180 volt; 3.33×10^{-10} F.]
9. Two parallel plates each of area 100 cm^2 are kept at 2 mm distance apart at a pontential difference of 1000 V. Calculate the force of attraction between them.
   [**Ans.** 0.011 N.]

QUESTIONS

Long Answer Questions

1. What do you understand by electric flux ? Under what conditions is it maximum and minimum ? State its units and dimensions. **(Gwalior 1994)**
2. State and prove Gauss's law in electrostatics. **(Bhopal 2004, 03; Bilaspur 03; Rewa 04; Indore 03; Japalpur 04)**
3. State Gauss's law in electrostatics. Deduce Coulomb's law from Gauss's law.
4. Deduce Gauss's law in differential form *i.e.*
$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}.$$
5. Prove that if a closed surface does not enclose any charge, the electric flux linked with the surface is zero.
6. Show that the total electric flux associated with a closed surface is equal to $1/\varepsilon_0$ times the total charge present inside that closed surface, where ε_0 is the permittivity of free space.
7. State Gauss's law in electrostatics. Derive expressions for the electric field due to a uniformly charged spherical shell at a point (i) inside, (ii) outside the shell.
8. Deduce expressions for electric field due to a uniformly charged conducting sphere by using Gauss's law at a point (i) outside and (ii) inside the sphere.
9. State Gauss's law. Use it to find the electric field due to a uniformly charged, non-conducting solid sphere at points (i) outside; (ii) inside the sphere. How the above results would change, if the sphere were metallic ?

10. Deduce expressions for the intensity of electric field due to a uniformly charged spherical shell at a point situated (i) outside, (ii) on the surface and (iii) inside it. Hence draw a graph to show the variation of electric field intensity with distance from the centre of shell.
(Gwalior 2003; Sagar 03; Raipur 03; Indore 1994)
11. Find the intensity of electric field due to a uniformly charged non-conducting solid sphere at a point (i) outside, (ii) on the surface and (iii) inside it. Show that for the points outside the sphere, it behaves as if the total charge given to the sphere is concentrated at its centre. **(Bhopal 2003; Raipur 1996)**
12. Find the electric potential at any point situated outside the uniformly charged spherical shell and show that the electric potential at each point inside it is equal to that on its surface.
13. Apply Gauss's law to prove that the intensity of electric field at a point inside the uniformly charged solid sphere is directly proportional to the distance of that point from the centre of sphere.
14. If a spherical charge distribution is given as

$$\rho = \rho_0\left(1 - \frac{r}{a}\right) \qquad \text{when } r \le a$$

and
$$\rho = 0 \qquad \text{when } r > a,$$

then using Gauss's theorem show that (i) the electric field at centre will be zero, (ii) potential difference between the centre and surface will be $\frac{\rho_0 a^2}{12\varepsilon_0}$. **(Gwalior 2004)**
15. Deduce using Gauss's law, an expression for the electric field of an infinitely long line of charged wire having a linear charge density of λ coulomb/meter.
16. The charge per unit length distributed uniformly over a conducting cylinder of radius R and infinite length is λ. Using Gauss's theorem, calculate the electric field at a point situated at distance r from the axis of cylinder when (i) $r > R$, (ii) $x = R$, and (iii) $r < R$. **(Bilaspur 1996; Bhopal 94)**
17. Use Gauss's law to find the electric field intensity at a point near an infinite plane non-conducting plate of charge of uniform charge density σC/m^2.
18. Prove that the electric field just outside a charged conductor is σ/ε_0, where σ is surface charge density on the conductor and ε_0 is permittivity of free space.
19. Positive charge is uniformly spread on the Y-Z surface of an infinite plane sheet of charge. Obtain intensity of electric field at points on the X-axis.
[**Ans.** $E = \sigma/2\varepsilon_0$, away from the sheet.]
20. Derive expressions for the electric field in between and outside of two parallel, oppositely charged, non-conducting plates with the same charge density.
21. Deduce expression for the electric field in between and outside of two parallel non-conducting sheets which are uniformly charged positively.
22. Derive an expression for the force acting on unit area the surface of a charged conductor situated in an electrical field. **(Agra 2004)**
23. Deduce an expression for the mechanical force directed outwards on the surface of a charged conductor.
24. What do you understand by an electrical image ? Derive expressions, for (i) potential and (ii) intensity of electric field on the surface of an earthed conductor due to a point charge placed near it.
25. What is an electric image ? Show that the total induced charge on a grounded infinite conductor due to a charge q placed near it is $-q$.
26. If a charge is placed near a grounded plane conductor, deduce expressions for (i) force between the charge and conductor and (ii) amount of work done to take the charge to infinity.
27. What do you understand by the capacitance of a conductor ? Obtain an expression for the capacity of a parallel plate condenser.
28. Deduce the formula for the capacity of a spherical condenser.

29. Deduce an expression for the force between the plates of a parallel plate condenser. Show that the force per unit area is obtained to be $\sigma^2/2\varepsilon_0$ or $\varepsilon_0 E^2/2$.
30. Derive an expression for the total electrostatic energy stored in a parallel plate capacitor of capacitance C. **(Agra 2006)**
31. Show that the electrostatic potential energy per unit volume of the electric field can be expressed as $\frac{1}{2}\varepsilon_0 E^2$.

Short Answer Questions

1. What is electric flux ? Show that the electric flux through a surface parallel to the field is zero.
2. State Gauss's theorem of electrostatics for discrete and continuous charges. **(Agra 2005)**
3. What is a Gaussian surface ?
   [**Ans.** An imaginary surface which is drawn to enclose a charge or charge distribution is called Gaussian surface.]
4. What is the use of Gaussian surface ?
   [**Ans.** A Gaussian surface is used to calculate the electric field due to a charge system where use of Coulomb's law is difficult.]
5. Prove the differential form of Gauss's law. **(Agra 2004)**
6. How are Gauss's and Coulomb's laws related ? **(Kanpur 2005)**
7. How does the intensity of electric field vary inside a uniformly charged conducting sphere ? Explain.
8. A non-conducting solid sphere is uniformly charged in volume. How does the intensity of electric field inside the sphere vary with the distance from its centre ?
   [**Ans.** The intensity of electric field inside a uniformly charged non-conducting solid sphere varies directly with the distance of point from its centre.]
9. A hollow metallic sphere of radius R meter is given a charge of q coulomb. Obtain (i) potential at its surface, (ii) directions of electric lines of force on its surface, (iii) intensity of electric field at its centre.
   [**Ans.** (i) $\frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{R}$; (ii) perpendicular to the surface; (iii) zero.]
10. Use Gauss's law to find out electric intensity due to an infinite sheet of charge having surface charge density σ. **(Agra 2006)**
11. Two infinite parallel plates have uniform charge densities $+\sigma$ and $-\sigma$. What is the intensity of electric field at a point (i) in between the plates, (ii) outside ?
    [**Ans.** (i) σ/ε_0; (ii) zero.]
12. Find the electric field $\vec{E}$ between two plane sheets of surface charge densities σ_1 and σ_2 respectively in vacuum. **(Agra 2005)**
13. Apply Gauss's law to show that electric field does not exist inside a charged conductor.
    [**Ans.** According to Gauss's law, $\oint \vec{E} \cdot d\vec{S} = q/\varepsilon_0$. As the charge does not reside inside a 'conductor' ($q = 0$), the electric field, $\vec{E} = 0$.]
14. A large hollow metallic sphere A is charged positively to a potential of 200 volt, and a small sphere B to a potential of 100 volt. Now B is placed inside A and they are connected by a wire. In which direction will the charge flow ? Explain.
    [**Ans.** The (positive) charge will flow from B to A until the whole charge of B flows to A, because there cannot be any electric field inside a hollow charged conductor.]
15. There are two copper spheres A and B of same radius, out of which A is hollow and B is solid. Both are charged to the same potential. Which sphere will have more charge ? Explain.
    [**Ans.** The two spheres will have same charge.]

16. A solid metallic sphere of radius R carries a charge Q. What is the value of surface charge density ? **(Lucknow 2005)**

 [**Ans.** $\sigma = Q/4\pi R^2$.]

17. A cylindrical Gaussian surface has its axis along an infinite line of uniformly distributed positive charge, λ per unit length [**fig. 5.45**]. Find the following :

 (i) For which surface is the electric flux zero.

 (ii) Over which surface E is constant ?

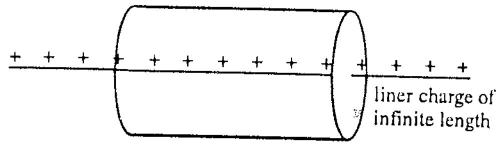


Fig. 5.45.

 [**Ans.** (i) The field at a point distant r from the line charge is $E = \lambda/2\pi\varepsilon_0 r$, directed radially outward. Therefore, the flux $\int \vec{E} \cdot d\vec{S}$ is zero for the plane face.

 (ii) E is constant over the curved surface of the cylinder.]

18. An isolated conductor of any shape has a net charge of + 20 μC. Inside the conductor is a cavity within which is a point-charge of + 20 μC. What is the charge on the cavity wall ? What is the charge on the outer surface of the conductor ?

 [**Ans.** – 20 μC, + 22 μC.]

19. Electric field $\vec{E}$ has the same value for all points in front of an infinite uniformly charged sheet. Is this statement reasonable ? This can be thought that the electric field should be stronger near the sheet because the charges are close.

 [**Ans.** Yes; because the electric field is normal to the uniformly charged surface and it is uniform. At very large distances, the field will deviate towards the edges.]

20. Show that the electric field near a charged conducting sheet is twice as great as that near a non-conducting sheet with the same surface charge density. **(Luknow 2005)**

21. Using Gauss's law find an expression for the force per unit area acting on the surface of a charged conductor. **(Kanpur 2005)**

22. If we charge a soap bubble, its radius increases. Explain.

23. Two large metal plates of area 1 m^2 face each other at a separation of 5 cm. They carry equal and opposite charges on their inner surfaces. If the electric field between the plates is 100 N/C, calculate the charge on the plates.

 [**Ans.** 8.85×10^{-10}C.]

 [**Hint :** $E = \dfrac{\sigma}{\varepsilon_0} = \dfrac{q}{A\varepsilon_0}$.]

24. State how the following quantities depend upon distance r : (i) electric field due to a point-charge, (ii) potential due to a point-charge, (iii) potential at a distance r from the centre of a charged metallic sphere (r < radius of the sphere). **(IIT 1980)**

 [**Ans.** (i) electric field $E \propto 1/r^2$; (ii) potential $\propto 1/r$; (iii) potential will not depend upon r.]

25. What is a condenser ? What is meant by its capacity ? Explain the principle of a condenser. **(Rewa 2004)**

26. The capacitance of a spherical conductor is 2 picofarad. Determine its radius.

 [**Ans.** 0.018 m.]

27. A 6 μF capacitor is raised in potential difference from 10 to 20 volt. Find the increase in its electric potential energy.

 [**Ans.** 9×10^{-4}J.]

28. A sensitive instrument which is to be saved from external electric fields in its environment is covered by a metallic enclosure. Why ?

[**Ans.** We know that there is no electric field inside a charged conductor. This implies that the external electric field cannot enter the inner empty space of any hollow conductor. Hence conductors can act as electrostatic shields. If we want to save an electrical instrument from external electric fields, the instrument is covered by a metallic enclosure. The external fields will not enter inside. Thus the instrument inside the metllalic surface will remain safe. We call this as **electrostatic shielding.**]

29. Obtain an expression for energy stored in a static electric field. **(Agra 2006)**

30. Show that the energy density associated with a charge distribution is $\frac{1}{2}\varepsilon_0 E^2$. **(Agra 2005)**

Objective Type Questions

1. A rectangular frame of area 20 m^2 is placed in a uniform electric field of 20 NC^{-1}, with normal drawn on the surface of the frame making 60° angle with the direction of field. The electric flux through the frame is :
   (a) 400 Vm (b) 200 Vm
   (c) $100\sqrt{3}$ Vm (d) $200\sqrt{3}$ Vm
2. If a surface is parallel to the electric field, the electric flux linked with it is :
   (a) zero (b) infinite
   (c) between 0 and ∞ (d) maximum **(Bhopal 2004)**
3. At the centre of a cubical box of side 1 m, there is a charge $+Q$. The total flux coming out of a wall of the cube will be :
   (a) Q/ε_0 (b) $Q/3\varepsilon_0$
   (c) $Q/4\varepsilon_0$ (d) $Q/6\varepsilon_0$ **(Agra 2004)**
4. If the net electric flux through a surface is zero, then :
   (a) There are no charges inside the surface.
   (b) The electric field is zero everywhere on the surface.
   (c) The net charge inside the surface is zero.
   (d) The number of electric field lines entering the surface equals the number leaving the surface.
5. Given $E = 3 \times 10^3\ \hat{i}\ NC^{-1}$, the flux through a square of side 10 cm, when normal to the plane makes an angle 30° with the X-axis (in $NC^{-1}m^2$) :
   (a) 15 (b) $30\sqrt{3}/2$
   (c) 3×10^3 (d) 30 **(Agra 2005)**
6. Select the correct statement :
   (a) Gauss's law is valid only for charges placed in vacuum.
   (b) Gauss's law is valid only for symmetrical charge distribution.
   (c) The electric flux through a closed surface due to all the charges is equal to the electric flux due to the charges inside the surface.
   (d) The electric field obtained by using Gauss's law is the field due to the charges enclosed by a closed surface.
7. In a conductor the static electric field is :
   (a) unity (b) zero
   (c) infinity (d) always parallel to the surface **(Agra 2006)**
8. The electrostatic potential at a point inside a uniformly positively charged non-conducting sphere :
   (a) is constant (b) is zero
   (c) increases with distance from centre (d) decreases with distance from centre.

9. The intensity of electric field E due to infinitely long charge line dependes on distance r as :
(a) $E \propto 1/r^2$ (b) $E \propto 1/r$
(c) $E \propto r$ (d) $E \propto r^2$

10. A solid conducting sphere of radius a meter carries a charge of $+2q$ coulomb. It is surrounded by a concentric conducting spherical shell of larger radius b meter and carrying a charge of $-q$ coulomb. The electric field at a distance of r meter ($a < r < b$) from the centre is :
(a) $\frac{1}{4\pi\varepsilon_0}\frac{q}{r}$ (b) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{r}$
(c) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{r^2}$ (d) $\frac{1}{4\pi\varepsilon_0}\frac{3q}{r^2}$

11. The quadrupole moment of a charge distribution is :
(a) qa^2 (b) $3\,qa^2$
(c) $2\,qa^2$ (d) $4\,qa^2$
(where $2a$ is the length of the linear quadrupole.) **(Agra 2005)**

12. The potential at the centre of a square of side 1 m, when equal charges q each are placed at the corners, is :
(a) $q/2\sqrt{2}\pi\varepsilon_0$ (b) $q\sqrt{2}/\pi\varepsilon_0$
(c) $q/4\pi\varepsilon_0$ (d) $q/4\sqrt{2}\pi\varepsilon_0$ **(Agra 2005)**

13. A hollow sphere and a solid sphere, both of same radii and made of same metal are charged to the same potential. Which of the statement is true ?
(a) The charge will be more on the hollow sphere.
(b) The charge will be more on the solid sphere.
(c) Both spheres will have equal charge.
(d) Only solid sphere has the charge.

14. A hollow sphere of charge does not produce an electric field at any point :
(a) inside (b) outside
(c) beyond 2 meter (d) beyond 20 meter.

15. The electric field at a point inside a unformly charged spherical conductor :
(a) is constant but not zero (b) is zero
(c) depends on the distance from centre (d) depends on the charge on the conductor

16. The electric charge of a conducting sphere :
(a) resides on its surface (b) is concentrated at its centre
(c) is uniformly distributed throughout (d) none of these

17. A hollow metallic sphere of radius 5 cm is charged such that the potential on its surface is 10 V. The potential at the centre of the sphere is :
(a) 5 V (b) 10 V
(c) 15 V (d) zero **(IIT 1983)**

18. Select the correct alternative(s) : A non-conducting solid sphere of radius R is uniformly charged. The magnitude of the electric field due to the sphere at a distance r from its centre :
(a) increases as r increases, for $r < R$ (b) decreases as r increases, for $0 < r < \infty$
(c) decreases as r increases, for $R < r < \infty$ (d) is discontinuous at $r = R$. **(IIT 1998)**

19. The electric field at a distance r near a plane sheet of charge with surface charge density σ is :
(a) σ/ε_0 (b) $\sigma/2\varepsilon_0$
(c) $\sigma r/\varepsilon_0$ (d) $\sigma/r\varepsilon_0$

20. Two parallel plates of large area are charged uniformly. The surface densities of charge on them are $+\sigma$ and $-\sigma$ [**fig. 5·46**]. The electric field will be zero :

(a) only at the point P_1 (b) only at the point P_2

(c) only at the point P_3 (d) both at points P_1 and P_3

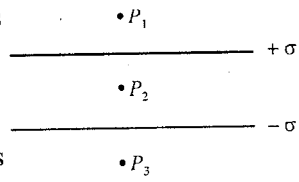


Fig. 5.46.

21. Two parallel plates of large area are charged uniformly. The surface densities of charge on them are $+\sigma$ and $-\sigma$. The electric field at the point P_2 (**fig. 5.46**) is :

(a) zero (b) σ/ε_0

(c) $\sigma/2\varepsilon_0$ (d) $2\sigma/\varepsilon_0$

22. Mark the correct answer(s) : An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the centre of the cavity. The points A and B are on the cavity surface, as shown in the figure. Then :

(a) electric field near A in the cavity = electric field near B in the cavity

(b) charge density at A = charge density at B

(c) potential at A = potential at B

(d) total electric flux through the surface of the cavity is q/ε_0

(IIT 1999)

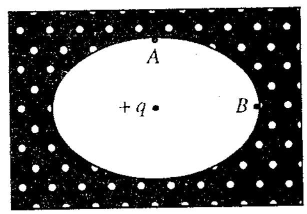


Fig. 5.47.

23. There is no net charge an a metallic particle. It is placed near a metallic plate of finite dimensions carrying a positive charge. The electric a force on the particle will be :

(a) zero (b) towards the plate

(c) away from the plate (d) parallel to the plate

24. In case of a charged bubble the mechanical force due to the charge is balanced by :

(a) viscous force (b) gravitational force

(c) surface tenssion (d) none of these

25. When a soap bubble is charged, then :

(a) it contracts (b) it expands

(c) it does not undergo any change in size (d) none of these

26. A parallel plate condenser with separation d between its plates is given a charge Q. Now the distance is increased to $2d$. The potential difference between its plates will :

(a) be 2 times (b) be one-half

(c) remain same (d) be zero

27. The capacitance of an isolated conducting sphere of radius R is :

(a) zero (b) $4\pi\, \varepsilon_0 R$

(c) $4\pi\, \varepsilon_0/R$ (d) $4\pi\varepsilon_0 R^2$ **(Agra 2006)**

28. On inserting a plate of dielectric between the plates of a parallel plate condenser, the stored energy is increased 3 times. The dielectric constant of the material is :

(a) 1/3 (b) 3

(c) 1/9 (d) 9

ANSWERS

1. (b) **2.** (b) **3.** (d) **4.** (c, d) **5.** (b) **6.** (c) **7.** (b) **8.** (d) **9.** (b) **10.** (c) **11.** (c) **12.** (b) **13.** (c) **14.** (a) **15.** (b) **16.** (a) **17.** (b) **18.** (a) **19.** (b) **20.** (d) **21.** (b) **22.** (c, d) **23.** (b) **24.** (c) **25.** (b) **26.** (b) **27.** (b) **28.** (b)

●●

6

Electric Field in Dielectrics

6·1. Dielectrics

Basically any substance is composed of atoms and molecules. An atom or molecule is electrically neutral. An atom consists of a positively charged nucleus and negatively charged electrons moving around it in different orbits. The electrons in the orbits close to the nucleus (called as core electrons) are rigidly bound to the nucleus and outer orbit electrons, called as valence electrons, are lightly bound to the nucleus. In some substances, these valence electrons are detached from the atom and freely move within the substance. We call these electrons as free electrons. There are also some materials which do not possess any free electrons and all the electrons of an atom are tightly bound to the nucleus. We may broadly categorize the materials into two classes :

(1) **Conductors :** These susbstances have good number of conduction or free electrons which roam around inside the material. When a potential difference or electric field is applied to these materials, an electric current flows through them. Examples of such susbtances are the metals like Cu, Ag, Au, Fe etc.

(2) **Insulators or Dielectrics :** These substances have effectively no free electrons for conduction. When a potential difference or electric field is applied to a dielectric, no current flows through it, but this application of the electric field changes the charge distribution of the constituting atoms or moelcules and hence results the change in the characteristic behaviour of the dielectric. Examples of dielectrics are glass, plastic, rubber, quartz, water, oil etc.

One may have susbstances between the two extremes *i.e.*, conductors and dielectrics, known as **semiconductors,** which are insulators at 0 K and become conductors at higher temperatures. Here we shall not go into the details of conductors and semiconductors, but in the present chapter our main interest is to study the dielectrics and the effect of an external electric field on these materials.

6·2. Induced Dipoles–Atomic Polarizability

An atom being as a whole electrically neutral, consists of a positively charged nucleus with charge $+q$ and a negatively charged electron cloud with $-q$ charge surrounding it. We may consider the electron charge distribution to be spherical in shape and nucleus as a point. In absence of the electric field, the centre of negative charge distribution coincides with the centre of nucleus [**fig. 6.1**]. When such a natural atom is placed in an electric field $\vec{E}_0$, the two regions of charge within the atom are influenced by the applied electric field. The positively charged nucleus is pushed in the direction of the field $\vec{E}_0$ and the negatively charged electron distribution in the opposite direction. Equilibrium is established, when the nucleus N is at a distance, say d, apart from the centre of the negative charge O. In equilibrium, the net electric field on the nucleus N must be zero *i.e.*, the applied electric field E_0 to the right of the nucleus is equal to E_e to the left of the nucleus, where E_e results from the negative charge distribution so that $E_e = E_0$. Thus, when an external field is applied on a neutral atom, in equilibrium one may consider $+q$ and $-q$ charges at a distance d apart, forming an **induced dipole** [**fig. 6.2**]. Its dipole moment is given by

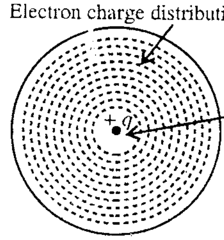


Fig. 6.1 : Neutral atom (primitive model).

$$\vec{p} = q\vec{d} \qquad ...(1)$$

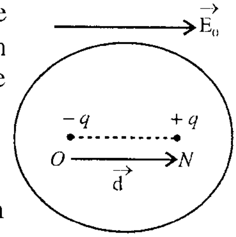


Fig. 6.2 : Induced dipole with dipole moment $\vec{p} = q\vec{d}$.

which has the direction of $\vec{E}_0$. As the positive and negative charges of an atom are get displaced along the direction of the electric field, the atom is said to be **polarized.** Further, this dipole moment is found approximately proportional to the applied field *i.e.*,

$$\vec{p} \propto \vec{E}_0 \quad \text{or} \quad \vec{p} = \alpha \vec{E}_0 \qquad \text{...(2)}$$

This constant of proportionality is called **atomic polarizability**. Generally the dipole moment disappears on the removal of the electric field.

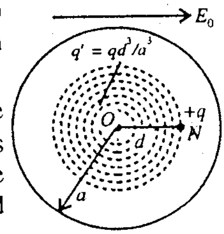


Fig. 6.3 : Calculation of atomic polarizability.

Calculation of atomic polarizability : Consider an atom (say hydrogen atom) with positively charged nucleus ($+q$) and a negatively charged spherical cloud (with charge $-q$) around the nucleus.

In the presence of the applied electric field E_0, the nucleus is shifted to the right and the electron cloud to the left. We assume that the electron cloud retains its spherical shape on shifting and in equilibrium the centre of negative charge ($-q$) is at O and that of nucleus at N ($+q$) [**fig. 6.3**]. We know that the electric field E_e at a distance d from the centre of a uniformly charged sphere is

$$E_e = \frac{1}{4\pi\varepsilon_0}\frac{qd}{a^3} \qquad \text{...(3)*}$$

where a is the radius of the charged sphere. But dipole moment $p = qd$ and $E = E_0$,

$$\therefore \quad E_0 = \frac{1}{4\pi\varepsilon_0}\frac{p}{a^3} \text{ or } p = (4\pi\varepsilon_0 a^3)E_0 \qquad \text{...(4)}$$

But $p = \alpha E_0$, hence

$$\alpha = 4\pi\varepsilon_0 a^3 \qquad \text{...(5)}$$

This is the *expression for atomic polarizability of an atom.*

In case of hydrogen atom, its radius is the Bohr radius $a = 0.53$ Å and hence one can calculate its polarizability using eq. (5)

6·3. Polar and Non-polar Molecules and Dipole Moment

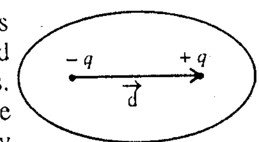


Fig. 6.4 : Dipole with moment $\vec{p} = q\,\vec{d}$.

A molecule is composed of atoms. In a molecule the positive charge is concentrated in the nuclei of its atoms, but the negative charge is distributed outside the constituent nuclei. Polyatomic dielectric materials are made of molecules. The centre of negative charge distribution in a molecule may or may not coincide with the centre of positive charge distribution. Accordingly, there are principally two types of molecules : (1) Polar molecules and (2) Nonpolar molecules.

(1) Polar molecules : If the centre of the negative charge distribution in a molecule does not coincide with the centre of the positive charge distribution, *each molecule has a permanent dipole moment,* given by

$$\vec{p} = q\vec{d}$$

where $\vec{d}$ is the vector from the centre of negative charge to the centre of positive charge [**fig. 6.4**].

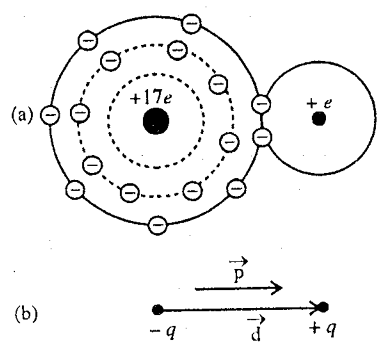


Fig. 6.5 : HCl molecule, having permanent dipole moment $\vec{p} = q\,\vec{d}$.

Such a molecule is known as **polar molecule** and the related material as **polar material.** Examples of polar molecules are HCl, H_2O, CO, NH_3, N_2O, CH_3OH etc. A polar molecule in an electric field has permanent as well as induced dipole moments. The later is produced by a small shift of positive and negative charges in the presence of the electric field.

Examples of polar molecules : HCl molecule is one example of polar molecules. This has a permanent dipale moment. In case of a HCl molecule, one electron of H-atom and one electron of Cl-atom *i.e.*, two electrons are shared mutually by the two atoms to obtain

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* Charge inside a sphere of radius d is $q' = \dfrac{q}{(4/3)\pi a^3} \times \dfrac{4}{3}\pi d^3 = \dfrac{qd^3}{a^3}$.

Electric field due to this charge q' at N *i.e.*, $E_e = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q'}{d^2} = \dfrac{1}{4\pi\varepsilon_0}\cdot\dfrac{qd^3}{a^3 d^2} = \dfrac{qd}{4\pi\varepsilon_0 a^3}$.

closed shell stable electronic configuration [**fig. 6.5(a)**]. This results in the formation of covalent bond. Due to this bond, an unsymmetric charge distribution is produced : a net positive charge towards H-atom and a negative charge towards Cl-atom. This molecule is equivalent to a dipole [**fig. 6.5(b)**] with dipole moment $\vec{p} = q\vec{d}$.

(2) Water molecule H_2O is another example of polar molecule. This has also built-in permanent dipole moment. In a water molecule, the electrons tend to cluster around the oxygen atom and the molecule is bent at 105° so that a negative charge ($-q$) at the vertex and a positive charge ($+q$) at the opposite end. Consequently H_2O molecule is equivalent to a dipole, having dipole moment $\vec{p}$ [**fig. 6.6**]. The dipole moment of water molecule is 6.1×10^{-30} C-m.

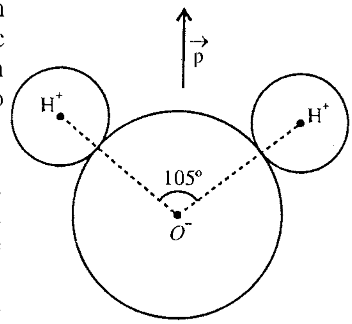

Fig. 6.6 : Water molecule and its dipole moment $\vec{p}$.

Dipole moment in a polar material : In absence of an external electric field, different molecules of a polar material have different directions of the dipole moment due to random thermal agitation and the net dipole moment of the material is zero [**fig. 6.7(a)**]. When an external electric field is applied on such a material, the individual small dipoles experience torque due to the electric field and try to align along the direction of the field. However, thermal agitation tries to randomise the orientation of the dipoles. Even then a large number of tiny dipoles are aligned along the direction of the field [**fig. 6.7(b)**] and we obtain a net dipole moment in any volume of the material.

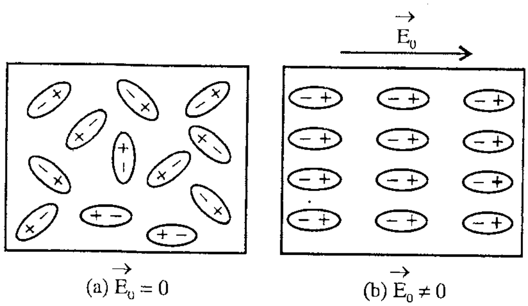

Fig. 6.7 : (a) Randomly oriented dipoles in a dielectric in absence of external electric field, (b) Alignment of dipoles in the presence of the electric field $\vec{E}_0$.

(2) **Nonpolar molecules :** If the centre of the negative charge distribution in a molecule (or atom) coincides with the positive charge distribution, then it does not possess any permanent dipole moment. Such molecules are known as **nonpolar molecules** and the related material as **nonpolar material.** When such a material is placed in an electric field, the electron charge distribution is slightly shifted opposite to the electric field [**fig. 6.8**], so that the centre of negative charge distribution and the centre of positive distribution are at a small distance apart. Thus dipole moment in each molecule of the material is **induced** and as a result, we obtain a *net dipole moment in any volume of the nonpolar material.* Examples of nonpolar molecules are H_2, N_2 O_2, CO_2, CH_4 etc.

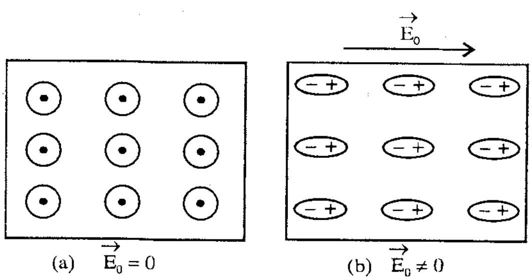

Fig. 6.8 : Polarization of nonpolar dielectric.

We see that in both polar and nonpolar materials, the same basic result is obtained in the presence of an electric rield : *a large number of small dipoles pointing along the direction of the filed, producing a net dipole moment in any volume of the material.* We say that the volume of the dielectric material is **polarized.**

From now ahead, we shall not worry how this polarization has been produced except in the discussion of molecular polarizability. It is also possible that the polarization persists after the removal of the field. The polarized material also produces an electric field.

6·4. Dielectric Polarization and Polarization Vector ($\vec{P}$)

When a dielectric material is placed in an electric field, dipole moment appears in any volume of the material. This is known as **dielectric polarization.** The dipole moment per unit volume of a dielectric material is called **polarization vector** $\vec{P}$ *i.e.,*

$$\text{Polarization vector } \vec{P} = \frac{\text{Dipole moment}}{\text{Volume of dielectric}}$$

In order to understand dielectric polarization, consider a rectangular slab of a dielectric material. In absence of an external electric field, the net dipole moment of the slab is zero. When an external electric field $\vec{E}_0$ is applied, either the dipoles are induced in the nonpolar dielectric or dipoles are aligned in the polar dielectric along the direction of the field $\vec{E}_0$ [**fig. 6.9(a)**]. This results in net dipole moment in the volume of the slab. We see that the interior of the dielectric slab is charge free because inside the dielectric dipolar charges cancel the effect of each other. However, *the left surface of the slab gets negative charge and the right surface gets positive charge* [**fig. 6.9(b)**]. The charge, produced on the surface of a dielectric on the application of an electric field in called **induced charge.** This induced charge produces a field $\vec{E}_P$ in the direction opposite to the applied electric field $\vec{E}_0$. Since the induced charge appears due to a shift in the electrons bound to the nuclei, we call this charge as **bound charge.** If Q_p be the induced charge appearing on the surface of the slab of area A, then the surface charge density is given by

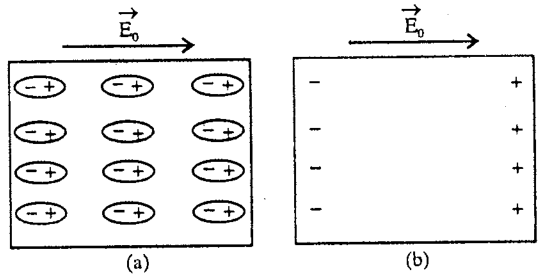

Fig. 6.9 : (a) Dielectric polarization in the presence of external electric field $\vec{E}_0$, (b) Induced charges on left and right surfaces of the dielectric slab.

$$\sigma_p = \frac{Q_p}{A} \qquad \text{...(1)}$$

The polarization P is related to the surface charge density σ_p. Let the length of the rectangular slab be l and its area of cross-section A [**fig. 6.10**]. If Q_p be the magnitude of charge on a surface, then

$$\text{dipole moment of the slab} = Q_p l = \sigma_p Al \qquad \text{...(2)}$$

where $\sigma_p = Q_p/A$ is the surface charge density. The polarization P is given by

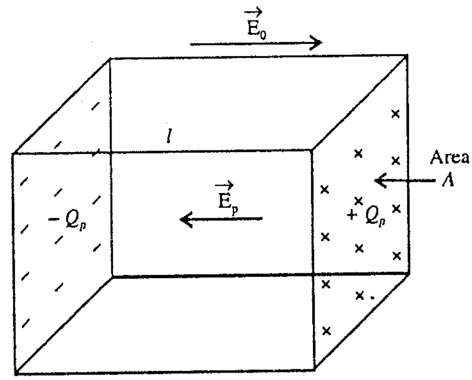

Fig. 6.10 : Induced charges on the surfaces of the dielectric slab due to polarization.

$$P = \frac{\text{Dipole moment}}{\text{Volume}} = \frac{\sigma_p Al}{Al} = \sigma_p \qquad \text{...(3)}$$

Thus, *the dielectric polarization in magnitude is equal to the surface density of induced charge, when the surface of the dielectric slab is perpendicular to the direction of the field.* Though this result has been deduced for rectangular slab, yet it is true in general. The unit of polarization is C/m^2.

6·5. Dielectric Constant

We consider a rectangular slab of a dielectric material. When an external electric $\vec{E}_0$ is applied on a dielectric material, positive and negative induced charges developed due to polarization on the surfaces of the dielectric produce an extra field $\vec{E}_P$ whose direction is opposite to the direction of $\vec{E}_0$. This field $\vec{E}_P$ due to polarization is called the **depolarizing field.** For a homogeneous and isotropic dielectric, the polarization is uniform and the resultant field ($\vec{E}$) inside the dielectric is given by [**fig. 6.11**].

$$\vec{E} = \vec{E}_0 + \vec{E}_P \qquad \text{...(1)}$$

where the resultant field $\vec{E}$ is in the direction of the applied field $\vec{E}_0$. Thus, we see that the electric field $\vec{E}$ inside the

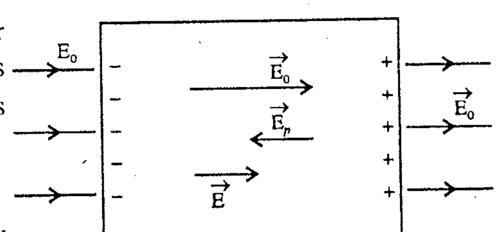

Fig. 6.11 : Resultant field $\vec{E} = \vec{E}_0 + \vec{E}_p$.

dielectric is, of course, in the direction of the field $\vec{E}_0$, but its magnitude of E is less than the magnitude E_0 of the external field. For a particular dielectric material, the ratio of the applied field E_0 and the resultant field E inside the dielectric is constant K, whose value is greater than 1. This K is called **dielectric constant** of the material. Thus

$$K = \frac{E_0}{E} \text{ or } \vec{E} = \frac{\vec{E}_0}{K} \quad \text{...(2)}$$

This dielectric constant is also called **relative permittivity** ε_r of the material *i.e.*,

$$K = \varepsilon_r = \frac{\varepsilon}{\varepsilon_0} \quad \text{...(3)}$$

where ε is the permittivity of the dielectric and ε_0 that of vacuum.

In case of **vacuum**, there is no polarization, hence $E = E_0$ and $K = \varepsilon_r = 1$.

Dielectric Breakdown and Dielectric Strength : In a dielectric material, if a very high electric field is produced, the outer electrons of the atoms may get detached from them. The dielectric then behaves like a conductor and this phenomenon is called **dielectric breakdown.** The minimum electric field at which the beakdown occurs is known as the **dielectric strength** of the material.

6·6. Parallel Plate Capacitor Filled with a Dielectric

We consider a parallel plate capacitor with plate area A and distance d between the plates. Now, we insert a dielectric slab of dielectric constant K in the space between the plates. The slab almost fills the entire space between the plates. Now, we give a $+Q$ charge to one plate and $-Q$ charge to the other plate of the capacitor.

Suppose the induced charges due to the polarization of the dielectric are $+Q_p$ and $-Q_p$ on the two faces of the slab close to the plates [**fig. 6.12**].

The surface charge density on the plates is $\sigma = Q/A$. Then the electric field between the plates due to the charges $+Q$ and $-Q$ on the capacitor plates is given by

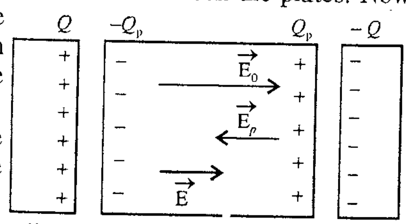

Fig. 6.12 : Dielectric between the plates of parallel plate capacitor.

$$E_0 = \frac{\sigma}{\varepsilon_0} = \frac{Q}{A\varepsilon_0} \quad \text{...(1)}$$

According to the definition of the dielectric constant, the resultant field inside the slab is given by

$$E = \frac{E_0}{K} = \frac{Q}{\varepsilon_0 AK} \quad \text{...(2)}$$

Hence, the corresponding potential difference V between the plates is

$$V = Ed \qquad [\because E = V/d] \quad \text{...(3)}$$

Using the value of E from (2), we obtain

$$V = \frac{Qd}{\varepsilon_0 AK} \text{ or } Q = \frac{\varepsilon_0 AK}{d}V \quad \text{...(4)}$$

Let C be the capacitance of the parallel plate capacitor with the dielectric in between its plates. Then

$$Q = CV \quad \text{...(5)}$$

Comparing eqs. (4) and (5), we obtain

$$C = \frac{\varepsilon_0 AK}{d} = KC_0 \quad \text{...(6)}$$

where $C_0 = \varepsilon_0 A/d$ is the capacitance without the dielectric. *Thus we see that the capacitance of a parallel plate capacitor is increased by K times, when the space between the plates is filled with a dielectric of dielectric constant K.*

Also from (6), we may define the dielectric constant as

$$K = \frac{C}{C_0} = \frac{\text{Capacitance of parallel plate capacitor filled with dielectric}}{\text{Capacitance of the capacitor without dielectric}} \quad ...(7)$$

Now, if the dielectric is removed from the space between the capacitor plates, the potential difference, say V_0, will increase because for fixed amount of charge Q,

$$Q = CV \text{ and } Q = C_0V_0 \text{ or } C_0V_0 = CV$$

$$\therefore \quad V_0 = \frac{C}{C_0}V = KV \text{ or } V_0 > V \qquad [\because K > 1] \quad ...(8)$$

Magnitude of Induced Charge : The electric field at a point between the plates due to $+Q, -Q$ charges on the plates is

$$E_0 = \frac{Q}{A\varepsilon_0}$$

The electric field E_p due to $Q_p, -Q_p$ charges is directed opposite to E_0 and has the magnitude, given by

$$E_p = \frac{\sigma_p}{\varepsilon_0} = \frac{Q_p}{A\varepsilon_0}$$

$$\therefore \quad \text{Resultant field, } E = E_0 - E_p = \frac{Q - Q_p}{A\varepsilon_0} \quad ...(9)$$

From eqs. (2) and (9), we get

$$\frac{Q}{\varepsilon_0 AK} = \frac{Q - Q_p}{A\varepsilon_0} \text{ or } \frac{Q}{K} = Q - Q_p \text{ or } Q_p = Q\left(1 - \frac{1}{K}\right) \quad ...(10)$$

Thus the *induced surface charge Q_p is always less than the free charge Q* and is zero when $K = 1$ or dielectric is not present.

Parallel Plate Capacitor Partly Filled with Dielectric : Consider a parallel plate capacitor, having plate area A and plate separation d. The space between the plates of the capacitor is filled up to a thickness t ($< d$) with a slab of dielectric constants K **[fig. 6.13]**.

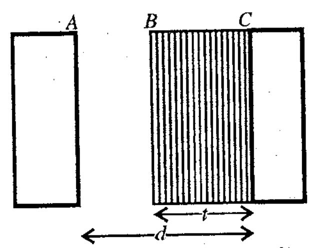

Fig. 6.13 : Parallel plate capacitor partly filled with dielectric.

We may consider the given arrangement equivalent to a series combination of two parallel plate condensers one between A and B and other between B and C. Here, B represents the left surface of the dielectric and as the potential of this surface is constant, we may think a thin metal plate being placed there.

The capacitance of the capacitor between A and B is

$$C_1 = \frac{\varepsilon_0 A}{d - t} \quad ...(11)$$

Also the capacitance between B and C is given by

$$C_2 = \frac{\varepsilon_0 KA}{t} \quad ...(12)$$

The equivalent capacitance for the series combination is

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d - t}{\varepsilon_0 A} + \frac{t}{\varepsilon_0 KA} = \frac{1}{\varepsilon_0 A}\left[d - t + \frac{t}{K}\right]$$

$$\therefore \quad C = \frac{\varepsilon_0 A}{d - t + t/K} \quad ...(13)$$

Special Cases : (i) If the slab of dielectric constant K fills complety the space between the plates, then $t = d$ and hence from (13)

$$C = \frac{\varepsilon_0 KA}{d} \qquad ...(14)$$

(ii) If $t = 0$ *i.e.*, the dielectric slab is not present between the plates of the capacitor, so that in the entire space, there is the vacuum or air, then the capacitance will be

$$C_0 = \frac{\varepsilon_0 A}{d} \qquad ...(15)$$

(iii) In case, there is a metal slab, for which $K = \infty$, of thickness t between the plates, then the capacitance is

$$C = \frac{\varepsilon_0 A}{d - t} \qquad ...(16)$$

(iv) If the metal slab fills the entire space so that $t = d$, then the capacitance will be infinite.

(v) Further if we have several slabs of thicknesses $t_1, t_2, t_3, ...$ of dielectric constants $K_1, K_2, K_3, ...$ respectively, between the condenser plates, then from (13) the capacitance of the condenser will be

$$C = \frac{\varepsilon_0 A}{d - (t_1 + t_2 + t_3 + ...) + \left(\frac{t_1}{K_1} + \frac{t_2}{K_2} + \frac{t_3}{K_3} + ...\right)} \qquad ...(17)$$

Also if the entire space between the plates is filled with dielectric slabs, then $d = t_1 + t_2 + t_3 + ...$ and hence

$$C = \frac{\varepsilon_0 A}{\frac{t_1}{K_1} + \frac{t_2}{K_2} + \frac{t_3}{K_3} + ...} \qquad ...(18)$$

6·7. Displacement Vector $\vec{D}$

When an external electric field $\vec{E}_0$ is applied on a dielectric, an electric field $\vec{E}_p$ due to polarization developed in the opposite direction of the applied field. The resultant field $\vec{E}$ in the dielectric is given by [fig. 6.11].

$$\vec{E} = \vec{E}_0 + \vec{E}_p \qquad ...(1)$$

We know that the electric field due to the polarization $\vec{E}_p$ is

$$\vec{E}_p = -\frac{\vec{P}}{\varepsilon_0} \qquad \left[\because E_p = \frac{\sigma_p}{\varepsilon_0} = \frac{P}{\varepsilon_0}\right] \qquad ...(2)$$

where $\vec{P}$ is polarization vector with magnitude P (which is equal to the surface induced charge density σ_p). Negative sign shows that the direction of $\vec{E}_p$ is opposite to $\vec{E}_0$.

From eqs. (1) and (2), we obtain

$$\vec{E} = \vec{E}_0 - \frac{\vec{P}}{\varepsilon_0} \text{ or } \varepsilon_0\vec{E} + \vec{P} = \varepsilon_0\vec{E}_0 \qquad ...(3)$$

But, $\qquad \vec{E} = \vec{E}_0/K \text{ or } \vec{E}_0 = K\vec{E}$

Hence From (3)

$$\varepsilon_0\vec{E} + \vec{P} = \varepsilon_0 K\vec{E} \text{ or } \vec{P} = \varepsilon_0(K-1)\vec{E} \qquad ...(4)$$

which gives the relation between $\vec{P}$ and $\vec{E}$.

We can write eq. (3) as

$$\varepsilon_0\vec{E} + \vec{P} = \varepsilon_0\vec{E}_0 = \varepsilon_0 K\vec{E} \qquad ...(5)$$

The quantity $\varepsilon_0 \vec{E} + \vec{P}$ is called the displacement vector $\vec{D}$ *i.e.*,

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} \qquad ...(6)$$

This gives the **relation between $\vec{D}$, $\vec{E}$ and $\vec{P}$.**

Also by using eq. (5), we get $\vec{D} = \varepsilon_0 K \vec{E} = \epsilon_0 \vec{E}$ $\quad [\because \varepsilon_0 K = \varepsilon_0 \varepsilon_r = \varepsilon]$ ...(7)

which gives a **relation between $\vec{D}$ and $\vec{E}$**. Here, ε is the *permittivity of the dielectric medium.*

In a homogeneous and isotropic dielectric, polarization vector $\vec{P}$ is directly proportional to the resultant field $\vec{E}$ in the dielectric *i.e.*,

$$\vec{P} \propto \vec{E} \quad \text{or} \quad \vec{P} = \varepsilon_0 \chi \vec{E} \qquad ...(8)$$

where χ is called the **electrical susceptibility** of the dielectric material. It is a scalar dimensionless quantity.

Comparing (4) and (8), we obtain

$$\chi = K - 1 \text{ or } K = 1 + \chi \qquad ...(9)$$

This is the **relation between the dielectric constant and electrical susceptibility of the dielectric material.**

6·8. Gauuss's Law in Dielectrics (Discussion for $\vec{E}$ and $\vec{D}$ Vectors)

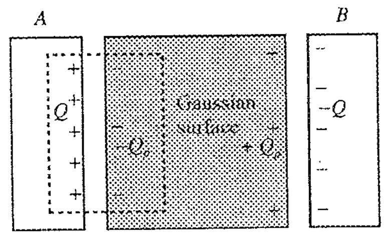

Fig. 6.14 : Gauss's law in a dielectric.

We consider a parallel plate capacitor with charge Q, $-Q$ on the plates. The space between the plates is filled with a dielectric material of dielectric constant K.

Now, we take a Gaussian surface [**fig. 6.14**], which encloses the charge $Q - Q_p$, where Q_p, $-Q_p$ are the induced charge developed on the surfaces of the dielectric. Using Gauss's law,

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q - Q_p}{\varepsilon_0} \qquad ...(1)$$

Thus in case of a dielectric, the total electric flux linked with a closed surface depends on free charges (Q) as well as induced charges (Q_p) due to polarization.

But, $Q_p = Q\,(1 - 1/K)$, hence $Q - Q_p = Q - Q\,(1 - 1/K) = Q/K$. Thus

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{free}}{\varepsilon_0 K} \quad \text{or} \quad \oint \varepsilon_0 K \vec{E} \cdot d\vec{S} = Q_{free} \qquad ...(2)$$

where Q_{free} is written for Q, which emphasizes the fact that it is the **free charge** given to the plates.

But we know that $\varepsilon_0 K \vec{E} = \vec{D}$, hence from (2)

$$\oint \vec{D} \cdot d\vec{S} = Q_{free} \qquad ...(3)$$

Thus, *the total displacement flux linked with a closed surface is equal to the free charge inside the surface.* This is the **Gauss's law in a dielectric.**

We observe from eq. (3) that the flux of the displacement $\vec{D}$ does not involve any induced charge due to polarization and it depends only on free charges Q_{free}. However, the flux of the electric vector $\vec{E}$ depends on free charges (Q_{free}) and polarization charges (Q_p). Thus we find that there is no effect on displacement flux whether the medium is present or not, while electric flux depends on the medium.

It is to be remembered that

$$\vec{D} = \varepsilon \vec{E} = \varepsilon_0 K \vec{E} = \varepsilon_0 \vec{E}_0 \qquad ...(4)$$

Thus, the vector $\vec{D}$ is similar to vector $\vec{E}$ and for homogeneous isotropic medium, $\vec{D}$ is in the direction of $\vec{E}$. However in case of anisotropic medium $\vec{D}$ and $\vec{E}$ may have different directions and ε is a *tensor*.

If there is no polarization *i.e.*, there is vacuum or air in the space between the parallel plate capacitor, then $\vec{D} = \varepsilon_0 \vec{E}$ and then eq. (3) takes the form

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{free}}{\varepsilon_0} \quad ...(5)$$

This is the usual form of the Gauss's law and Q_{free} is the total charge inside the Gaussian surface. According to Gauss's theorem,

$$\oint_S \vec{D} \cdot d\vec{S} = \int_V (\vec{\nabla} \cdot \vec{D}) dV \quad ...(6)$$

But from (3)

$$\oint_S \vec{D} \cdot d\vec{S} = Q_{free} = \int_V \rho \, dV \quad ...(7)$$

where $\rho = dQ/dV$ is the (free) charge density so that $Q = \int_V \rho \, dV$.

From (6) and (7),

$$\int_V (\vec{\nabla} \cdot \vec{D}) dV = \int_V \rho \, dV \quad ...(8)$$

This is correct for any arbitrary volume, hence

$$\vec{\nabla} \cdot \vec{D} = \rho \quad \text{or} \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad ...(9)$$

Thus, *at any point in a medium the divergence of electric displacement is equal to the free charge density at that point.* Eq. (9) represents the **differential form of Gauss's law for a dielectric medium.** This is also one of the Maxwell's well known four equations.

6·9. Electric Field due to a Point Charge Q in an Dielectric

Let a point charge Q be placed at O, inside an infinite dielectric medium. We want to calculate the electric field at a distance r from the point charge Q. We draw a spherical surface through P with the centre at O [**fig. 6.15**]. According to Gauss's law in dielectric,

$$\oint \vec{D} \cdot d\vec{S} = Q \quad \text{or} \quad \oint \varepsilon_0 K \vec{E} \cdot d\vec{S} = Q$$

or

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q}{\varepsilon_0 K} \quad \text{or} \quad E \cdot 4\pi r^2 = \frac{Q}{\varepsilon_0 K} \quad \text{or} \quad E = \frac{Q}{4\pi\varepsilon_0 K r^2} \quad \text{or} \quad \vec{E} = \frac{Q\hat{r}}{4\pi\varepsilon_0 K r^2} \quad ...(1)$$

The direction of the field is radially away from the charge Q. It may be noted that Q is the total free charge within the Gaussian surface.

This electric field $\vec{E}$ having magnitude $Q/4\pi\varepsilon_0 K r^2$ is due to the free charge Q and polarization charges induced in the dielectric medium.

If we consider Q charge to be positive, then due to outward radial field negative charges shift inwardly. Hence, an induced charge $Q_p = -Q\,(1 - 1/K)$ is produced on the surface of the spherical cavity in the dielectric in which Q charge is placed. Consequently, effective charge is $Q - Q_p = Q - Q\,(1 - 1/K) = Q/K$ due to which the electric field at r is $Q/4\pi\varepsilon_0 K r^2$.

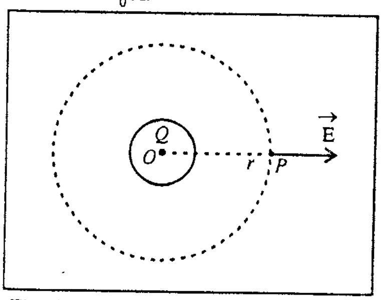

Fig. 6.15 : Electric field due to a point charge in a dielectric.

6·10. Energy in the Electric Field in a Dielectric

We consider a parallel plate capacitor filled with a dielectric of dielectric constant K. The energy stored in the capacitor is

$$U = \frac{1}{2}CV^2 = \frac{1}{2}KC_0V^2 \quad ...(1)$$

where $C = KC_0 = \varepsilon_0 KA/d$.

The energy density u in the volume of the dielectric between the plates is given by

$$u = \frac{U}{Ad} = \frac{\frac{1}{2}(\varepsilon_0 KA/d)V^2}{Ad} = \frac{1}{2}\varepsilon_0 K\left(\frac{V}{d}\right)^2 = \frac{1}{2}\varepsilon_0 KE^2 = \frac{1}{2}\varepsilon E^2 \quad ...(2)$$

where $E = V/d$ and d is the distance between the plates of the capacitor.

From (2), we note that the energy density in dielectric is K times of the energy density in vacuum for the same electric field.

6·11. Molecular Interpretation of Clausius-Mossotti Equation (Macroscopic and Microscopic Electric Fields)

When an external electric field $\vec{E}_0$ is applied on a dielectric slab, due to polarization an electric field $\vec{E}_p = -\vec{P}/\varepsilon_0$ is developed in the opposite direction of the applied field, where $\vec{P}$ is the polarization (or dipole moment per unit volume). The resultant field in the dielectric is given by

$$\vec{E} = \vec{E}_0 + \vec{E}_p \text{ or } \vec{E} = \vec{E}_0 - \frac{\vec{P}}{\varepsilon_0} \quad ...(1)$$

This electric field is **macroscopic.**

Local Field : The electric field that acts at the site of a molecule of the dielectric material is called the **local field** and is denoted by $\vec{E}_{Loc}$. This is a **microscopic field** significantly different from the macroscopic electric field $\vec{E}$. The induced dipole moment $\vec{p}$ in the molecule is directly proportional to the local electric field $\vec{E}_{Loc}$ *i.e.*,

$$\vec{p} = \alpha E_{Loc} \quad ...(2)$$

where α is called **molecular polarizability.**

If the number of molecules per unit volume of the dielectric substance is n, then the polarization $\vec{P}$, being dipole moment per unit volume is given by

$$\vec{P} = n\vec{p} = n\,\alpha\,\vec{E}_{Loc} \quad ...(3)$$

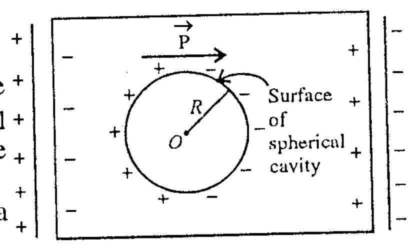

Fig. 6.16 : Spherical cavity in a dielectric.

We now develop an expression for the local field $\vec{E}_{Loc}$ at the site of a molecule inside the dielectric. In order to calculate the local field, Lorentz suggested that the dielectric around a particular molecule should be divided into two parts :

(1) A spherical region, centred on the molecule, having a radius R, of many molecular spacings.

(2) The region outside the sphere of radius R.

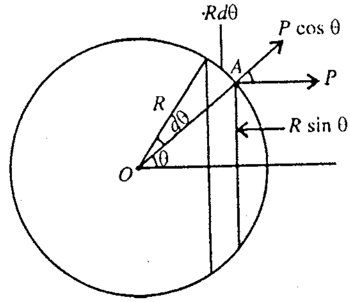

Fig. 6.17 : Calculation of field at O due to polarization of the dielectric.

For an isotropic dielectric, the field at the site of the molecule due to all the molecular dipoles inside the spherical region can be shown equal to zero. Removing this spherical region, a spherical cavity of radius R is produced in the dielectric [fig. 6.16]. The region beyond the spherical cavity is sufficiently large enough so that it may be treated as continuum with a uniform polarization $\vec{P}$. Corresponding to this polarization there is the charge density on the surface of the sphere of radius R. If θ be the angle, referred to the polarization direction, then the surface charge density at A is $\sigma_p = P\cos\theta$. Now, the electric field at O, the site of the molecule under consideration, due to a ring shaped element between the angles θ and $\theta + d\theta$, is

$$dE' = \frac{1}{4\pi\varepsilon_0}\frac{\sigma_p dS}{R^2}\cos\theta = \frac{1}{4\pi\varepsilon_0}\frac{P\cos\theta\cdot 2\pi R\sin\theta\, Rd\theta}{R^2}\cos\theta = \frac{P}{2\varepsilon_0}\cos^2\theta\,\sin\theta\, d\theta \quad ...(4)$$

along the direction of $\vec{P}$ **[fig. 6.17]**. Here, the area of the ring is $dS = 2\pi R\sin\theta\, Rd\theta$.

Hence the total electric field E' at the centre O of the spherical cavity is

$$E' = \frac{P}{2\varepsilon_0}\int_0^{\pi}\cos^2\theta\,\sin\theta\, d\theta = \frac{P}{3\varepsilon_0} \quad ...(5)$$

This is called the Lorentz field along the direction of polarization $\vec{P}$. Thus

$$\vec{E}' = \frac{\vec{P}}{3\varepsilon_0} \quad ...(6)$$

In addition to this field, electric field $\vec{E}_0$ due to free charges on the condenser plates and $E_p = -\vec{P}/\varepsilon_0$ due to induced charges on the surfaces of the dielectric are also acting. Hence, the local field at the site of a molecule inside the dielectric is given by

$$\vec{E}_{Loc} = \vec{E}_0 - \frac{\vec{P}}{\varepsilon_0} + \frac{\vec{P}}{3\varepsilon_0} = \vec{E} + \frac{\vec{P}}{\varepsilon_0} - \frac{\vec{P}}{\varepsilon_0} + \frac{\vec{P}}{3\varepsilon_0} = \vec{E} + \frac{\vec{P}}{3\varepsilon_0} \qquad \left[\because \vec{E}_0 = \vec{E} + \frac{\vec{P}}{\varepsilon_0}\right] \quad ...(7)$$

This relation for local field is valid for a *homogeneous* and *isotropic dielectric.*

Now, from eqs. (3) and (7), we obtain

$$\vec{P} = n\alpha\left(\vec{E} + \frac{\vec{P}}{3\varepsilon_0}\right) \text{ or } \vec{P}\left(1 - \frac{n\alpha}{3\varepsilon_0}\right) = n\alpha\vec{E}$$

or
$$\vec{P} = \frac{n\alpha}{1 - (n\alpha/3\varepsilon_0)}\vec{E} \quad ...(8)$$

We know that
$$\vec{P} = \varepsilon_0\chi\vec{E} \quad ...(9)$$

where χ is the electrical susceptibility of the dielectric material. From eqs. (8) and (9)

$$\chi = \frac{n\alpha/\varepsilon_0}{1 - (n\alpha/3\varepsilon_0)} \quad ...(10)$$

This is the relation between the electrical susceptibility and molecular polarization of a dielectric.

Also, $\vec{P} = \varepsilon_0(K-1)\vec{E}$, hence from eqs. (8),

$$\varepsilon_0(K-1) = \frac{n\alpha}{1 - (n\alpha/3\varepsilon_0)} \text{ or } \varepsilon_0(K-1) - \frac{1}{3}n\alpha(K-1) = n\alpha$$

or
$$n\alpha\left(1 + \frac{K-1}{3}\right) = \varepsilon_0(K-1) \text{ or } n\alpha(K+2) = 3\varepsilon_0(K-1)$$

Thus,
$$\frac{n\alpha}{3\varepsilon_0} = \frac{K-1}{K+2} \quad ...(11)$$

This relation between the polarizability (α) and dielectric constant (K) is known as **Clausius-Mossotti relation.**

But $K = \varepsilon_r = \varepsilon/\varepsilon_0$, hence eq. (11) can be written in another form

$$\frac{n\alpha}{3\varepsilon_0} = \frac{\varepsilon_r - 1}{\varepsilon_r + 2} \text{ or } \frac{n\alpha}{3} = \frac{\varepsilon - \varepsilon_0}{\varepsilon + 2\varepsilon_0} \quad ...(12)$$

If μ be the refractive index of the dielectric medium, then $\mu = \sqrt{K}$ or $K = \mu^2$. Hence from eq. (11), we get a relation between the polarizability α and the refractive index μ of the dielectric material as

$$\frac{n\alpha}{3\varepsilon_0} = \frac{\mu^2 - 1}{\mu^2 + 2} \quad ...(13)$$

6·12. Molecular Polarizability—Electronic, Ionic and Orientational Polrizablities

When the molecules of a dielectric are acted by an electric field, the polarization can arise in two ways :

(i) The electric field distorts the charge distribution of a molecule so that the centre of negative charge distribution and the centre of positive charge distribution are at a small distance apart. This results in an induced dipole moment in each molecule of the dielectric.

(ii) The electric field exerts torques on the randomly oriented permanent dipoles in case of polar molecules of a dielectric. This results in the alignment of the polar molecules in the direction of the field and hence producing polarization.

The electric field $\vec{E}$ acting on a single molecule of the dielectric is contributed by the external field as well as by the polarized molecules of the dielectric except the molecule under consideration. The dipole moment $\vec{p}$ of each molecule is proportional to the field $\vec{E}$ *i.e.*,

$$\vec{p} = \alpha \vec{E} \quad ...(1)$$

This constant of proportionality α called **molecular polarizability**. The molecular polarizability may consist of one or more of the following three contributions :

(1) **Electronic Polarizability** (α_e) : This part arises due to the displacement of the nuclei relative to the electron cloud in the atoms of the molecule in the presence of the electric field. In this process, the molecule develops dipole moment and the process is known as **electronic polarization.** The contribution due to electronic polarization is called **electronic polarizability** and is denoted by α_e.

(2) **Ionic Polarizability** (α_i) : Some materials have ionic molecules *i.e.*, the molecules are composed of positive and negative ions with ionic bond. In the presence of the electric field, the positive ion is displaced with respect to the negative ion in the direction of the field and this results in the induced dipole moment in the molecule.

This process is called **ionic polarization.** Note that this process is different to electronic polarization in the sense that in the electronic polarization the positive nucleus is displaced with respect to the electron cloud. The contribution due to the ionic polarization is called **ionic polarizability** and is denoted by α_i.

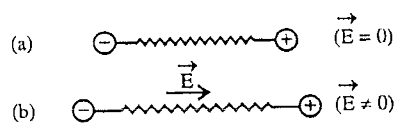

Fig. 6.18 : Induced dipole moment in an ionic molecule.

(3) **Orientational or Dipolar Polarizability** (α_0) : When the dielectric has polar molecles with permanent dipole moments, randomly oriented, they tend to align due to torques in the direction of the applied field. This develops a net dipole moment and the process is known as **orientational** or **dipolar polarization.** The contribution due to dipolar polarization is known as **orientational** or **dipolar polarizability** and is written as α_0. This polarizability is temperature dependent and an expression for it at absolute temperature T, can be obtained as

$$\alpha_0 = \frac{p_0}{3 - kT} \quad ...(2)$$

where p_0 is the dipole moment of the polar molecule and k is Boltzman constant.

Thus the polarization may be considered as the sum of three contributions :

$$\vec{P} = \vec{P}_e + \vec{P}_i + \vec{P}_0 \quad ...(3)$$

The molecular polarizability due to these polarization can be written as

$$\alpha = \alpha_e + \alpha_i + \alpha_0 \text{ or } \alpha = \alpha_d + \alpha_0 \quad ...(4)$$

where $\alpha_d = \alpha_e + \alpha_i$ is called the **deformation polarizability** caused by the deformation of the molecules in the presence of the electric field. Non-polar molecules may possess only deformation polarizability ($\alpha_d = \alpha_e + \alpha_i$). However polar molecules may have both deformation as well as orientational polarizabilities.

6·13. Boundary Conditions for $\vec{E}$ and $\vec{D}$ Vectors Between Two Homogeneous Dielectrics

Sometimes we need to know the way the electrostatic vectors $\vec{E}$ and $\vec{D}$ behave at the boundary or interface between two dielectric media of permittivities ε_1 and ε_2 respectively. Simple relations exist between the tangential and normal components of two field vectors which are called the **boundary conditions** on the field vectors.

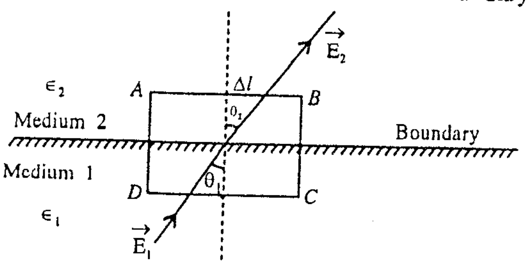

Fig. 6.19 : Continuity of the tangential component of E.

Let us consider first the **electric vector** at the interface between two media 1 and 2. We know that the line integral $\oint \vec{E} \cdot d\vec{l}$ for a closed path is zero for electrostatic field. So, considering path $ABCD$ we get

$$\oint \vec{E} \cdot d\vec{l} = \int_A^B \vec{E} \cdot d\vec{l} + \int_B^C \vec{E} \cdot d\vec{l} + \int_C^D \vec{E} \cdot d\vec{l} + \int_D^A \vec{E} \cdot d\vec{l} = 0 \qquad ...(1)$$

The contribution of the integral along the paths DA and BC can be made vanishingly small as these paths can be taken of infinitesimally small lengths. If $E_{1t} = E_1 \sin\theta_1$ and $E_{2t} = E_2 \sin\theta_2$ are the tangential components of electric field vectors $\vec{E}_1$ and $\vec{E}_2$ in mediums 1 and 2, then

$$E_{2t}\,\Delta l - E_{1t}\,\Delta l = 0 \qquad ...(2)$$

where Δl is distance AB or CD.

Hence

$$E_{1t} = E_{2t} \quad \text{or} \quad E_1 \sin\theta_1 = E_2 \sin\theta_2 \qquad ...(3)$$

Thus, *the component of $\vec{E}$ tangential to the boundary of the two dielectrics is the same on each side of the boundary. In other words, the tangential component across such a boundary is continuous.*

Let us assume that there are no free charges on the boundary surface and consider a small cylindrical pill box shaped Gaussian surface. Since there are no free charges on the surface and the cylinder is of vanishingly small height and the total charge enclosed by the Gaussian surface is zero, hence

$$\oint \vec{D} \cdot d\vec{S} = 0$$

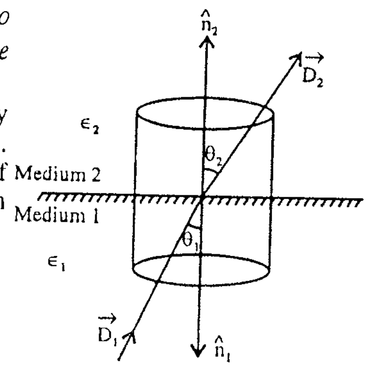

Fig. 6.20 : Continuity of the normal component of D.

If the perpendicular components of the displacement vectros $\vec{D}_1$ and $\vec{D}_2$ in two dielectrics are $D_{1n} = D_1 \cos\theta_1$ and $D_{2n} = D_2 \cos\theta_2$ then

$$\vec{D}_1 \cdot \hat{n}_1\,\Delta S + \vec{D}_2 \cdot \hat{n}_2\,\Delta S = 0$$

or

$$D_{1n}\,\Delta S - D_{2n}\,\Delta S = 0 \qquad [\because \hat{n}_2 = -\hat{n}_1]$$

where ΔS is the area of cross-section of the cylinderical box. Thus

$$D_{1n} = D_{2n} \quad \text{or} \quad D_1 \cos\theta_1 = D_2 \cos\theta_2 \qquad ...(4)$$

Thus *in the absence of any free charges on the interface*, the *normal component of $\vec{D}$ is the same on each side of the boundary*. In other words, *the normal component of $\vec{D}$ is continuous across a charge-free boundary between two dielectrics*. From relation $\vec{D} = \varepsilon\,\vec{E}$, we can also write

$$\varepsilon_1 E_{1n} = \varepsilon_2 E_{2n} \quad \text{or} \quad \varepsilon_1 E_1 \cos\theta_1 = \varepsilon_2 E_2 \cos\theta_2 \qquad ...(5)$$

Equations (3), (4) and (5) are the boundary conditions on electrostatic field vectors $\vec{E}$ and $\vec{D}$ at the interface between the two homogeneous dielectrics.

The above 'boundary conditions' can be used to establish a law of refraction for ideal, homogeneous and isotropic dielectrics. For such dielectrics, $\vec{P}$ and hence $\vec{D}$ is paralle to $\vec{E}$. So that, we have from **fig. 6.21,**

$$E_1 \sin\phi_1 = E_2 \sin\phi_2 \qquad [\text{From } E_{1t} = E_{2t}] \qquad ...(6)$$

and
$$D_1 \cos\phi_1 = D_2 \cos\phi_2 \qquad [\text{From } D_{1n} = D_{2n}] \qquad ...(7)$$

For linear dielectrics, $\vec{D}_1 = \varepsilon_1 \vec{E}_1$ and $\vec{D}_2 = \varepsilon_2 \vec{E}_2$, Hence eq. (7) is

$$\varepsilon_1 E_1 \cos\phi_1 = \varepsilon_2 E_2 \cos\phi_2 \qquad ...(8)$$

Dividing (6) by (8)

$$\frac{E_1 \sin\phi_1}{\varepsilon_1 E_1 \cos\phi_1} = \frac{E_2 \sin\phi_2}{\varepsilon_2 E_2 \cos\phi_2} \text{ or } \frac{\tan\phi_1}{\tan\phi_2} = \frac{\varepsilon_1}{\varepsilon_2} = \frac{\varepsilon_0 K_1}{\varepsilon_0 K_2} = \frac{K_1}{K_2} \qquad ...(9)$$

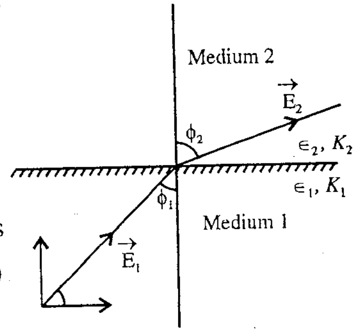

Fig. 6.21.

Above result is similar to the **Snell's law** for refraction of light rays and is known as the **law of refraction** for lines of force at the boundary of two dielectrics.

Ex. 1. *Two parallel rectangular plates each of area* 6.45×10^{-4} m^2 *are placed at a separation of* 2×10^{-3} *m and a potential difference of* 10 V *is applied between them. Now if a slab of dielectric constant* 6.0 *is introduced between the plates, then claculate :* (i) *capacity,* (ii) *charge on each plate,* (iii) *displacement vector* $\vec{D}$ *and* (iv) *polarisation vector* $\vec{P}$. **(Sagar 2003)**

Sol. Here, $A = 6.45 \times 10^{-4}$ m², $d = 2 \times 10^{-3}$ m, $V = 10$ volt and $K = 6.0$

We assume that the potential difference V between the plates remains the same after introducing the dielectric slab between the plates.

(i) Capacity of condenser, $C = \dfrac{\varepsilon_0 KA}{d} = \dfrac{(8.85 \times 10^{-12}) \times 6 \times 6.45 \times 10^{-4}}{2 \times 10^{-3}} = \mathbf{1.71 \times 10^{-11}}$ **farad**

(ii) Charge on each plate, $Q = CV = (1.71 \times 10^{-11}) \times 10 = \mathbf{1.71 \times 10^{-10}}$ **C.**

(iii) Displacement vector, $D = \varepsilon_0 KE = \varepsilon_0 K \dfrac{V}{d} = \dfrac{(8.85 \times 10^{-12}) \times 6 \times 10}{2 \times 10^{-3}} = \mathbf{2.66 \times 10^{-7}}$ **C/m²**

(iv) Polarisation vector, $P = D - \varepsilon_0 E = D - \varepsilon_0 \dfrac{V}{d}$

$$= (2.66 \times 10^{-7}) - \left(\frac{8.85 \times 10^{-12} \times 10}{2 \times 10^{-3}}\right) = \mathbf{2.22 \times 10^{-7}\ C/m^2}$$

The directions of $\vec{D}$ and $\vec{P}$ are along $\vec{E}$ (positive plate to negative plate).

Ex. 2. *The dielectric constant of gaseous argon is* 1.00054 *at NTP. If an electric field of* 10 *KV/m is applied, then calculate the induced dipole moment in argon atom.*

Sol. One mole of a gas at NTP has volume 22.4 litre = 22.4×10^{-3} m³

$\therefore$ Number of atoms per unit volume, $n = \dfrac{\text{Avogadro number}}{22.4 \times 10^{-3}} = \dfrac{6.02 \times 10^{23}}{22.4 \times 10^{-3}} = \dfrac{6.02 \times 10^{26}}{22.4}$

Now, dipole moment per unit volume, $P = \varepsilon_0 (K - 1) E$

$\therefore$ Induced dipole moment per atom, $p = \dfrac{\varepsilon_0 (K-1)E}{n}$

Here, $K - 1 = 0.00054$ and $E = 10^4$ volt/m

$$\therefore \qquad p = \frac{0.00054 \times 8.85 \times 10^{-12} \times 10^4 \times 22.4}{6.02 \times 10^{26}} = \mathbf{1.78 \times 10^{-36}\ C/m}$$

Ex. 3. *The permittivity of a dielectric susbtance is* 1.46×10^{-10} $C^2/N \times m^2$. *Calculate its dielectric constant and electrical susceptibility.* **(Ujjain 2003; Gwalior 2003; Jabalpur 2003; Bilaspur 1999)**

Sol. Dielectric constant, $K = \dfrac{\varepsilon}{\varepsilon_0} = \dfrac{1.46 \times 10^{-10}}{8.86 \times 10^{-12}} = \mathbf{16.48}$

Electric susceptibility, $\chi = K - 1 = 16.48 - 1 = \mathbf{15.48}$

Ex. 4. *A homogeneous and isotropic slab of dielectric constant* $K = 5$ *and volume* $0.5\ cm^3$ *is kept normal to an electric field due to which the magnitude of displacement vector* $\vec{D}$ *is* $5 \times 10^{-12}\ C/m^2$. *Calculate* : (i) *the magnitude of electric field* $\vec{E}$, (ii) *the magnitude of electrical polarisation* $\vec{P}$ *and* (iii) *total induced dipole moment inside the slab.* **(Indore 2004)**

Sol. (i) From the relation $D = \varepsilon_0 KE$, we have

$$E = \frac{D}{\varepsilon_0 K} = \frac{5 \times 10^{-12}}{8.86 \times 10^{-12} \times 5} = \mathbf{0.113\ N/C}$$

(ii) From the relation $D = \varepsilon_0 E + P$, we obtain

$$P = D - \varepsilon_0 K = D - \frac{D}{K} = D\left(1 - \frac{1}{K}\right) = 5 \times 10^{-12} \times \left(1 - \frac{1}{5}\right) = 5 \times 10^{-12} \times 0.8 = \mathbf{4 \times 10^{-12}\ C/m^2}$$

(iii) Toral induced dipole moment inside the slab = $P \times$ Volume of slab

$= 4 \times 10^{-12} \times (0.5 \times 10^{-6}) = \mathbf{2 \times 10^{-18}\ C\text{-}m}$

Ex. 5. *Calculate the dipole moment of the molecules of carbon tetra chloride, if relative permittivity,* $K = 2.24$, *density* $= 1.60\ gm/cm^3$ *molecular weight* $= 156$ *and electric field* $= 10^7$ volt/metre.

Sol. If n be the number of molecules per unit volume, then

$$n = \frac{\text{Avogadro's number}}{\text{Molecular weight}} \times \text{Density} = \frac{6.02 \times 10^{23}}{156} \times 1.60$$

$= 6.17 \times 10^{21}$ molecules/cm^3 $= 6.17 \times 10^{27}$ molecules/m^3

So, dipole moment of one molecule

$$p = \frac{P}{n} = \frac{\varepsilon_0 (K-1)E}{n} = \frac{8.85 \times 10^{-12} \times 1.24 \times 10^7}{6.17 \times 10^{27}} = \mathbf{1.77 \times 10^{-32}\ C\text{-}m.}$$

Ex. 6. *Show that the local field on a dielectric molecule is* $\vec{E}_{Local} = \dfrac{(K+2)}{3}\vec{E}$.

Sol. Local electric field on a molecule in a dielectric is given by

$$\vec{E}_{Local} = \vec{E} + \frac{\vec{P}}{3\varepsilon_0}$$

We knwo that $\vec{P} = \varepsilon_0 (K-1)\vec{E}$,

$$\vec{E}_{Local} = \vec{E} + \frac{1}{3\varepsilon_0}[(K-1)]\vec{E} = \left(1 + \frac{K-1}{3}\right)\vec{E} = \left(\frac{3 + K - 1}{3}\right)\vec{E} \text{ or } \vec{E}_{total} = \frac{K+2}{3}\vec{E}$$

EXERCISE 6·1

1. The electric susceptibility of a material is 35.4×10^{-12} coulomb per newton-metre^2. Calculate the dielectric constant and permittivity of that material. **(Indore 1998)**
   [**Ans.** $K = 5$; 4.43×10^{-11} F/m.]
2. Two parallel plates each of area $10^{-2}\ m^2$ are charged with equal and opposite charges 1.0×10^{-7} C. If the space between the plates is filled with a dielectric and the intensity of electric field within the dielectric is 3.3×10^5 V/m, calculate : (i) dielectric constant, (ii) surface density of charge on the plate.
   [**Ans.** (i) 3.42; (ii) $10^{-5}\ C/m^2$.]

3. A homogeneous and isotropic dielectric slab of dielectric constant 6.0 is kept normal to an electric field of intensity 0.1 N/C. Calculate tha values of displacement D and polarization P within the slab.
[Ans. $D = 5.3 \times 10^{-12}$ C/m^2; $P = 4.43 \times 10^{-12}$ C/m^2.]
4. The dielectric constant of helium gas at 0°C and 1 atmospheric pressure is 1.000074. If it is kept in an electric field of intensity 100 V/m, calculate the dipole moment induced in each helium molecule and molecular polarizability.
[Ans. 2.44×10^{-39} C-m; 2.44×10^{-41} C^2-m/N.]
5. The electrical susceptibility of a substance is 4.0. Calculate the dielectric constant and the absolute pemittivity of the substance. **(Indore 2003)**
[Ans. 5.0; 4.43×10^{-11} C^2/N-m^2.]
6. The dielectric constant of glass is 4.2 Calculate its permittivity and susceptibility. **(Indore 1991)**
[Ans. 4.43×10^{-11} Fd/m; 3.2.]
7. A parallel plate air capacitor having area of each plate 100 cm^2 and separation between its plates 1 cm, is charged to a potential of 100 V. Now a dielectric slab of dielectric constant 7.0 and thickness 0.5 cm is inserted between the plates of capacitor. Calculate the values of $\vec{E}, \vec{D}$ and $\vec{P}$ in (i) dielectric, (ii) air.
[Ans. (i) $E = 1.43 \times 10^3$ V/m, $D = 8.86 \times 10^{-8}$ C/m^2, $P = 7.59 \times 10^{-6}$ C/m^2;
(ii) $E = 1.0 \times 10^4$ V/m, $D = 8.86 \times 10^{-8}$ C/m^2, $P = 0$.]

QUESTIONS

Long Answer Questions

1. What do you mean by a dielectric substance ? How does it differ from a conductor ? **(Indore 1999; Raipur 92; Ujjain 93)**
2. Obtain an expression for the atomic polarisability of hydrogen atom. Assume that according to Bohr's model, its negative charge is uniformly distributed in a spherical volume of radius a around the nucleus. **(Gwalior 2004)**
3. Differentiate between the polar and non-polar molecules. Explain the mechanism of polarisation in them. **(Raipur 2003)**
4. What do you understand by dielectric polrisation. Explain the electric field $\vec{E}$, electric polarisation $\vec{P}$ and electric displacement $\vec{D}$ in the dielectric material and establish a relation between them. **(Indore 2004; Ujjain 2003; Gwalior 2003; Bhopal 1996; Jabalpur 96)**
5. Prove that : (i) $\vec{D} = \varepsilon \vec{E}$, (ii) $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$ **(Kanpur 2005; Jabalpur 2003; Sagar 1996; Indore 95)**
6. Explain the dielectric polarisation and define polarisation vector $\vec{P}$. Prove that the electric field due to polarisation in any dielectric medium is $-\vec{P}/\varepsilon_0$. **(Sagar 1997; Gwalior 89)**
7. Prove that when a dielectric medium is kept in an external field, then induced dipole moment is equal to the induced surface charge density. **(Bhopal 2000, 1997)**
8. Find an expression for the capacity of a parallel plate capacitor, being filled with a dielectric between the plates of the capacitor.
9. Find an expression for the capacitance of a parallel plate capacitor of separation d, if it is filled with a dielectric of thickness t and dielectric constant K in the middle. What happens when $t = 0$, $t = d$. Define permittivity through K. **(Agra 2005)**
10. Deduce an expression for the displacement vector $\vec{D}$ in a parallel plate condenser with dielectric. **(Bundelkhand 2004)**

11. Explain the concept of electrical susceptibility with the help of a capacitor filled with a dielectric medium and establish the relationship between dielectric constant and electrical susceptibility. **(Jabalpur 2004)**
12. Discuss the Gauss's law in the presence of dielectrics.
13. Explain the macroscopic and microscopic (local) electric fields in a dielectric medium. **(Ujjain 2003; Jabalpur 1999; Sagar 99; Bilaspur 97; Bhopal 97)**
14. What is local field ? Deduce an expression for the local field at the site of an atomic dipole in a dielectric.
15. State Clausius-Mossotti equation and derive it. **(Ujjain 2004; Indore 2003; Raipur 2003; Bundelkhand 2004)**
16. Show that the molecular polarisability $\alpha = \frac{3\varepsilon_0}{n}\left(\frac{K-1}{K+2}\right)$, where symbols have their usual meanings. **(Gwalior 2004)**
17. Discuss the boundary conditions for electric vector $\vec{E}$ and displacement vector $\vec{D}$ at the interface of two dielectrics.
18. Show that at the interface of two homogeneous dielectrics, the tangential components of electric field and normal components of displacement vector are continuous.
19. Write short notes on the following :
    (i) Dielectric polarisation **(Raipur 2003)**
    (ii) Displacement vector **(Bilaspur 1993)**
    (iii) Dielectric constant **(Raipur 2003)**
    (iv) Electrical susceptibility.

Short Answer Questions

1. What are dielectric materials ?
2. What do you understand by atomic polarisability ?
3. What are polar and non-polar molecules. Give two examples of each.
4. Discuss polarisation and polarisation vector.
5. Show that in the presence of an external electric field, the electric field due to polarisation is $\vec{E}_p = -\vec{P}/\varepsilon_0$, where $\vec{P}$ is polarisation vector.
6. Show that for given charge on the plates of parallel plate condenser, when the dielectric is introduced between the plates, the potential difference between the plates is reduced.
7. State Gauss's law in the presence of a dielectric.
8. What is dielectric breakdown ?
9. What do you understand by the dielectric strength of a dielectric material ?
10. If Q is the charge on the plate of a parallel plate capacitor and K is the dielectric constant of the material between the plates, then show that the induced charge on the surface of the dielectric material is given by $Q_p = Q\,(1 - 1/K)$.
11. Deduce the relation : $\vec{P} = \varepsilon_0(K - 1)\vec{E}$.
12. Deduce the relation between $\vec{D}$, $\vec{E}$ and $\vec{P}$.
13. Show that : $\vec{D} = \varepsilon\,\vec{E}$.
14. What is electrical susceptibility show that $\chi = K - 1$.
15. What do you understand by local field ?
16. List and explain the boundary conditions of $\vec{E}$ and $\vec{D}$ at the boundary of two homogeneous dielectrics.
17. The dielectric constant of a substance is 4. What is its electrical susceptibility ? **(Agra 2005)**
    [Ans. $\chi = 3$.]

18. The presence of a dielectirc to increase the capacitance. Explain. **(Lucknow 2005)**
19. Explain the term polarisation in dielectrics. **(Lucknow 2005)**

Objective Type Questions

1. Which one of the following is correct relation between $\vec{P}$ and $\vec{E}$?

   (a) $\vec{P} = \chi\varepsilon_0\vec{E}$ (b) $\vec{P} = \varepsilon_0 K\vec{E}$ (c) $\vec{P} = \varepsilon_0\vec{E}$ (d) $\vec{P} = (\chi - 1)\vec{E}$

2. The polar molecule is :

   (a) H_2 (b) N_2 (c) H_2O (d) O_2 **(Bundelkhand 2004)**

3. The relation netween $\vec{D}, \vec{E}$ and $\vec{P}$ is :

   (a) $\vec{D} = \varepsilon\vec{E}$ (b) $\vec{D} = \vec{E} + \frac{\vec{P}}{K}$

   (c) $\vec{D} = \varepsilon_0(K - 1)\vec{E} + \vec{P}$ (d) $\vec{D} = \varepsilon_0\vec{E} + \vec{P}$ **(Rewa 1995)**

4. The correct relation in vacuum is :

   (a) $\vec{D} = 0$ (b) $\vec{E} = 0$ (c) $\vec{D} = \varepsilon_0\vec{E} - 1$ (d) $\vec{P} = 0$ **(Ujjain 1995)**

5. The relationship between the diplacement vector $\vec{D}$, electric field $\vec{E}$ and dielectric constant K is :

   (a) $\vec{D} = \vec{E}/K$ (b) $\vec{D} = \varepsilon_0 K\vec{E}$ (c) $\vec{D} = \frac{1}{2}K\vec{E}$ (d) $\vec{D} = K/\vec{E}$ **(Bundelkhand 2004)**

6. Inside a dielectric medium, if the resultant electric field due to external field and polarisation is $\vec{E}$, the local field at a point is :

   (a) $\vec{E}_{Local} = \vec{E} + \vec{P}$ (b) $\vec{E}_{Local} = \vec{E} + \vec{P}/3\varepsilon_0$

   (c) $\vec{E}_{Local} = \vec{P}$ (d) $\vec{E}_{Local} = \vec{P}/2\varepsilon_0$ **(Agra 2004; Rewa 1995)**

7. A parallel plate capacitor has charges $+q$ and $-q$ on its plates, the induced charges on the surface of a dielectric substance between the plates are $-q'$ and $+q'$. Then :

   (a) $q < q'$ (b) $q = q'$ (c) $q > q'$ (d) $q' = 0$

8. When a slab of dielectric material is inserted between the plates of a parallel plate capacitor, the energy stored is increased five times. The dielectric constant of the material is :

   (a) 25 (b) 5 (c) 1/5 (d) 1/25

9. The dielectric constant of a dielectric material is 3. Its electrical susceptibility is :

   (a) 3 (b) 2 (c) 4 (d) 1/3

10. The unit of displacement vector $\vec{D}$ is same as that of :

    (a) $\vec{E}$ (b) $\vec{P}$ (c) charge (d) $\vec{E}$/charge

11. The differential form of Gauss's law for a dielectric is :

    (a) $\vec{\nabla}\cdot\vec{D} = \rho$ (b) $\vec{\nabla}\cdot\vec{E} = \rho$ (c) $\vec{\nabla}\cdot\vec{E} = \rho/\epsilon$ (d) $\vec{\nabla}\cdot\vec{D} = 0$

    where ρ is the density of free charges.

ANSWERS

1. (a) **2.** (c) **3.** (a),(d) **4.** (b) **5.** (b) **6.** (b) **7.** (c) **8.** (b) **9.** (b) **10.** (b)
11. (a), (c)

7

Electric Currents—Steady and Varying Currents

7·1. Steady Electric Current

When there is a flow of charge from one side of an area to the other, it is said that an electric current is passing through the area. If a charge Δq passes through an area in time Δt, then the **average electric current** through the area is defined as

$$i_{av} = \frac{\Delta q}{\Delta t} \qquad ...(1)$$

In the limit $\Delta t \to 0$, we obtain the electric current as

$$i = \lim_{\Delta t \to 0} \frac{\Delta q}{\Delta t} = \frac{dq}{dt} \qquad ...(2)$$

Thus, *an electric current is defined as the rate of transfer of charge through an area.* Here, generally we will call electric current simply as current. If the moving charges are positive, the current is said to be in the direction of motion, while for negative moving charges it is said to be in the opposite direction of motion.

The S.I. unit of electric current is coulomb/second (C/s) or ampere (A). Current is a scalar quantity.

If the value of the current does not change with time or the current remains constant through an area, then the current is called **steady current.**

7·2. Current Density

Current density at a point is defined as the current flowing through unit area of a surface, taken normal to the direction of flow of charge. This is a **vector quantity** and written as $\vec{J}$. If we consider an area $d\vec{S}$ around a point P of surface in a conductor, then the current through this $d\vec{S}$ area is given by

$$di = \vec{J} \cdot d\vec{S} = J \cos \theta \, dS \qquad ...(1)$$

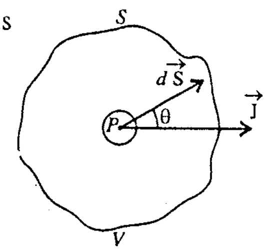

Fig. 7.1.

where θ is the angle between $\vec{J}$ and vector area $d\vec{S}$ **[fig. 7.1]**.

Hence, the current flowing through the total surface S is

$$i = \int_S \vec{J} \cdot d\vec{S} \qquad ...(2)$$

Thus, current flowing through a surface is equal to the flux of current density linked with the surface. If the current i is uniformly distributed over the surface of a plane area S and is in the direction perpendicular to it, then the current density is constant over the area S and $\vec{J}$ and $\vec{S}$ are in the same direction so that $\vec{J} \cdot \vec{S} = JS$. Thus

$$i = \int_S \vec{J} \cdot d\vec{S} = \vec{J} \cdot \int_S d\vec{S} = \vec{J} \cdot \vec{S} = JS \qquad ...(3)$$

i.e.,
$$J = \frac{i}{S}$$

S.I. unit of current density is ampere/metre^2 (A/m^2).

7·3. Nonsteady Currents and Equation of Continuity for Charge and Current

If the current through an area in a conductor does not remain constant with time, the current is said to be **time dependent** *or* **time-varying current.**

The current (steady or nonsteady) through a surface S is expressed as

$$i = \int_S \vec{J} \cdot d\vec{S} \qquad ...(1)$$

where $\vec{J}$ is the current density at a point in an infinitesimal small area dS.

Consider an arbitrary volume V bounded by a closed surface S **[fig. 7.1]**. Hence, eq. (1) will represent the charge flowing out per second normally through the surfaces S. Therefore, according to the law of conservation of charge, the charge must decrease with the same rate inside the volume V enclosed by the surface S.

This means that

$$\oint_S \vec{J} \cdot d\vec{S} = -\frac{dq}{dt} \qquad ...(2)$$

where q is the charge at any time t inside the surface S in volume V.

If ρ be the charge density at a point inside the volume V and dq be the charge in the volume dV around that point, then

$$\rho = \frac{dq}{dV} \quad \text{or} \quad dq = \rho dV$$

Hence the charge in the volume V enclosed by the surface S is

$$q = \int_V \rho dV \qquad ...(3)$$

Thus, eq. (2) assumes the form

$$\oint_S \vec{J} \cdot d\vec{S} = -\frac{d}{dt}\int_V \rho dV \quad \text{or} \quad \oint_S \vec{J} \cdot d\vec{S} = -\int_V \frac{\partial \rho}{\partial t} dV \qquad ...(4)$$

According to Gauss's divergence theorem

$$\oint_S \vec{J} \cdot d\vec{S} = \int_V (\text{div } \vec{J})\, dV \qquad ...(5)$$

From eqs. (4) and (5), we have

$$\int_V (\text{div } \vec{J})\, dV = -\int_V \frac{\partial \rho}{\partial t} dV \quad \text{or} \quad \int_V \left(\text{div } \vec{J} + \frac{\partial \rho}{\partial t} \right) dV = 0 \qquad ...(6)$$

Eq. (6) is correct for any arbitrary volume. Hence,

$$\text{div } \vec{J} + \frac{\partial \rho}{\partial t} = 0 \qquad ...(7)$$

This is known as **equation of continuity for charge and current.** Eq. (7) has been obtained for time-dependent charge density or current distribution.

Nonsteady currents or time varying currents flow in the conducting wires when we charge or discharge a condenser. During charging, the charge density at a point in the conducting wire does not remain constant. However in case of **steady currents**, the charge density at point in the conductor remains constant with time *i.e.*, ρ = constant in time or $\partial\rho/\partial t = 0$. Hence for **steady currents**, equation of continuity (7) takes the form

$$\text{div } \vec{J} = 0 \qquad ...(8)$$

7·4. Kirchoff's Laws

Simple electrical circuits may be analyzed by using Ohm's law. Many electrical circuits comprise a number of sources of e.m.f. and resistors connected in a more or less complicated manner. Such electrical circuits are called **netweoks.** All networks can not be reduced to simple series-parallel combination arrangements. When we are dealing with complicated electrical circuits, we may find the combined resistances or the current in any part of the circuit by using **Kirchoff's laws.** These laws are the following :

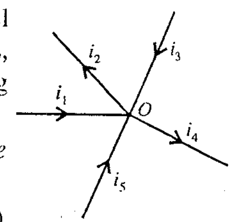

Fig. 7.2.

1. Kirchoff's First Law—The Junction Law : *The algebraic sum of all the currents flowing towards a point is zero. This means that*

$$\Sigma i = 0 \qquad ...(1)$$

Applying this law to the currents meet at the junction point O [**fig. 7.2**], we obtain

$$i_1 - i_2 + i_3 - i_4 + i_5 = 0 \qquad ...(2)$$

where positive sign has been given to the currents flowing towards the point O and negtive sign to those flowing away from the point O. From eq. (2)

$$i_1 + i_3 + i_5 = i_2 + i_4 \qquad ...(3)$$

Hence, *the sum of all the currents entering a junction point is equal to the sum of all the currents leaving it.* This is an alternative statement of the Kirchoff's first law.

This law is a consequence of the fact that a junction point can not accumulate the charge or supply the charge so that the charge flowing towards the point should be equal to the charge going away from the point in the same time. This is in accordance with the law of conservation of charge.

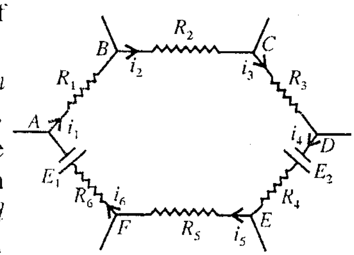

Fig. 7.3 : Kirchoff's second law.

2. Kirchoff's Second Law—The Loop Law : *The algebraic sum of all the potential differences along a closed loop in a circuit is zero.*

In order to use this rule, we start from a point on the loop and move along the loop in clockwise or anticlockwise direction and finally reach the same point again. We consider *any potential drop to be positive and any potential rise to be negative.* In **fig. 7.3** we start from the point A move along the loop $ABCDEFA$ and finially reach the point A. According to the Kirchoff's second law,

$$i_1R_1 + i_2R_2 + i_3R_3 - E_2 + i_4R_4 + i_5R_5 + i_6R_6 + E_1 = 0 \qquad ...(4)$$

$$i_1R_1 + i_2R_2 + i_3R_3 + i_4R_4 + i_5R_5 = -E_1 + E_2 \text{ or } \Sigma iR = \Sigma E \qquad ...(5)$$

In other words, Kirchoff's second law can be stated as "*In a close circuit the algebraic sum of the products of the current flowing in different parts and their corresponding resistances is equal to the algebraic sum of different electromotive forces present in the circuit.*" In such a case, the product of current and the corresponding resistance is taken positive in the direction of flow of the current and negative opposite to the direction of flow of the current. Further inside the cell, the e.m.f. is taken positive from the negative electrode to positive electrode and negative from the positive electrode to negative electrode.

The second law is a consequence of the fact that the electrostatic force is a conservative force and the work done by it in any closed path is zero.

We also call Kirchoff's first law as **Current Law** and second law as **Voltage Law.**

Application of Kirchoff's Laws : Derivation of the Condition of Balance of Wheatstone's Bridge

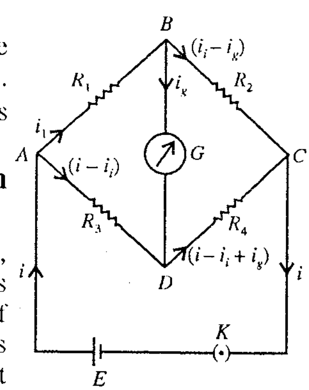

Fig. 7.4.

Fig. 7.4 shows the Wheatstone's bridge, having four resistances R_1, R_2, R_3 and R_4 in its four arms. A battery of e.m.f. E is connected between the points A and C and a galvanometer of resistance G between the points B and D. If i current flows from the battery and i_1 in the resistance R_1, then using Kirchoff's first law to the juction A, the current in R_3 is $(i - i_1)$. Next if i_g be the current in the galvanometer G, then, applying same law to the junctions B and D, the current in R_2 is $(i_1 - i_g)$ and that in R_4 is $(i - i_1 + i_g)$.

Now, applying Kirchoff's second law to the loop $ABDA$, we obtain

$$i_1R_1 + i_gG - (i - i_1)R_3 = 0 \text{ or } i_1(R_1 + R_3) + i_gG - i R_3 = 0. \quad ...(1)$$

Similarly, for the loop $BCDB$, we get

$$(i_1 - i_g)R_2 - (i - i_1 + i_g)R_4 - i_gG = 0$$

or

$$i_1(R_2 + R_4) - i_g(R_2 + R_4 + G) - i R_4 = 0 \quad ...(2)$$

When the bridge is balanced, no current flows through the galvanometer *i.e.*, $i_g = 0$. Eqs. (1) and (2) then become

$$i_1 (R_1 + R_3) = i R_3 \quad ...(3)$$

and

$$i_1 (R_2 + R_4) = i R_4. \quad ...(4)$$

Dividing eq. (3) by eq. (4), we get

$$\frac{R_1 + R_3}{R_2 + R_4} = \frac{R_3}{R_4} \text{ or } R_1R_4 + R_3R_4 = R_2R_3 + R_3R_4 \text{ or } R_1R_4 = R_2R_3$$

Thus

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad ...(5)$$

This is the *condition of balance of Wheatstone's bridge.*

Ex. 1. *Determine the equivalent resistance between the points a and b of the circuit shown in* **fig. 7.5.**

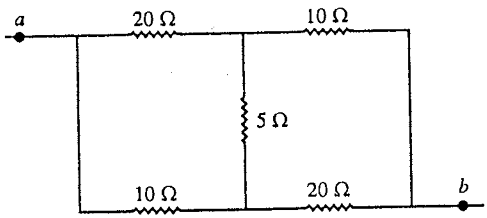

Fig. 7.5.

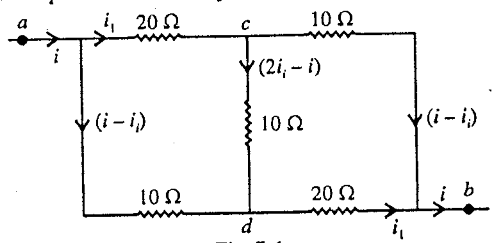

Fig. 7.6.

Sol. Let a current i enter at the point a. A part i_1 of the current i goes through the resistor 20 Ω and remaining current $i - i_1$ through the 10 Ω resistor. As current i comes out from the point b, by symmetry it will be composed of i_1 from 20 Ω and $i - i_1$ from 10 Ω [**fig. 7.6**]. Using Kirchoff's first law, the current through the middle 10 Ω is

$$i_1 - (i - i_1) = 2\, i_1 - i \text{ [fig. 7.6].}$$

Now, for path acb

$$V_a - V_b = (V_a - V_c) + (V_c - V_b) = 20\, i_1 + 10\,(i - i_1) = 10\, i + 10\, i_1 \quad ...(i)$$

Also, for path acdb

$$V_a - V_b = (V_a - V_c) + (V_c - V_d) + (V_d - V_b) = 20\, i_1 + 10\,(2\, i_1 - i) + 20\, i_1 = -\,10\, i + 60\, i_1 \quad ...(ii)$$

Multiply eq. (i) by 6 and then suptract eq. (ii)

$$5\,(V_a - V_b) = 70\, i \text{ or } \frac{V_a - V_b}{i} = 14\ \Omega$$

Hence, the equivalent resistance between the points a and b is **14 Ω.**

Ex. 2. *Calculate, in the given electrical circuit* [**fig. 7.7**] *current in each resistance. The internal resistances of the cells are negligible.*

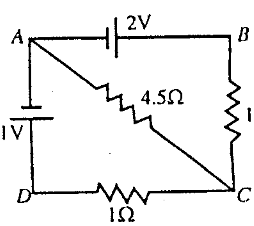

Fig. 7.7.

Sol. *If* i_1 and i_2 be the currents flowing from the 1 V and 2 V cells respectively [**fig. 7.8**]., then by Kirchoff's first law at the junction C, the current in the 4.5 Ω resistance will be $i_1 + i_2$.

Applying Kirchoff's second law for the loop $ADCA$, we have

$$i_1 \times 1 + (i_1 + i_2) \times 4.5 = 1$$

or $$5.5\, i_1 + 4.5\, i_2 = 1. \qquad ...(i)$$

Similarly, for the loop $ABCA$, we have

$$i_2 \times 1 + (i_1 + i_2) \times 4.5 = 2$$

or $$4.5\, i_1 + 5.5\, i_2 = 2. \qquad ...(ii)$$

Solving eqs. (i) and (ii), we obtain

$$i_1 = -\mathbf{0.35\ A} \text{ and } i_2 = \mathbf{0.65\ A.}$$

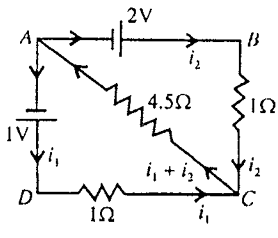

Fig. 7.8.

Hence, the current in DC arm with 1 Ω resistance is – 0.35 A *i.e.*, current 0.35 A is flowing along the path CDA. The current in BC arm with 1 Ω resistance is 0.45 A.

The current in the 4.5 Ω resistance is

$$i_1 + i_2 = -0.35 + 0.65 = \mathbf{0.30\ A.}$$

Ex. 3. *In the circuit shown in* **fig. 7.9,** *E_1, E_2, E_3 and E_4 are cells of e.m.f.* 2, 1, 3 *and* 1 *V respectvely. The resistances* 2, 1, 3 *and* 1 Ω *are their respective internal resistances. Find* (a) *the potential difference between B and D and* (b) *the potential differences across the terminals of each of the cells E_3 and E_4.*

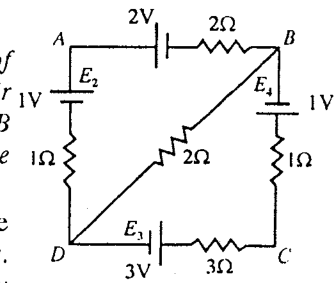

Fig. 7.9.

Sol. Let a current i_1 flow in the branch BAD and a current i_2 in the branch DCB. The current along DB will be $i_1 - i_2$ from Kirchoff's first law. The circuit with the currents is shown in **fig 7.10.** Applying the loop law to the circuit $BADB$, we get

$$2i_1 - 2\text{ V} + 1\text{ V} + 1i_1 + 2\,(i_1 + i_2) = 0$$

or $$5i_1 - 2i_2 = \text{V}. \qquad ...(i)$$

Similarly for the loop $DCBD$, we get

$$-3\text{ V} + 3i_2 + 1i_2 + 1\text{ V} - 2\,(i_1 - i_2) = 0$$

or $$-i_1 + 3i_2 = \text{V} \qquad ...(ii)$$

Solving eqs. (i) and (ii),

$$i_1 = \frac{5}{13}\text{ A}, i_2 = \frac{6}{13}\text{ A} \quad \therefore\ i_1 - i_2 = -\frac{1}{13}\text{ A.}$$

where, the current in BD arm is from B to D.

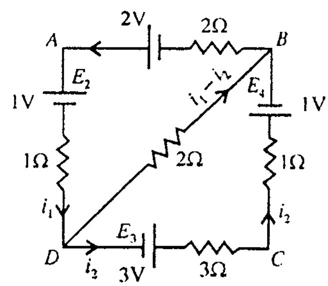

Fig. 7.10.

(a) $V_B - V_D = 2 \times \frac{1}{13} = \frac{2}{13}$ Volt

(b) The potential difference across the terminals (C and D) of cell E_3 is

$$V_C - V_D = -3\, i_2 + 3\text{V} = \left(3\text{V} - \frac{18}{13}\text{ V}\right) = \frac{21}{13}\text{ V.}$$

The potential difference across the terminals (C and B) of cell E_4 is

$$V_C - V_B = 1 \times i_2 + 1 = 1 \times \frac{6}{13} + 1 = 1\frac{6}{13}\text{ volt.}$$

Ex. 4. *Calculate the current passing through the resistor* 3 Ω *in the circuit shown in* **fig. 7.11.** *Also find the power supplied by both voltage sources.*

Sol. If i_1 is the current in AB branch, i_2 in the BC branch and i_3 in the DGF branch, then in other branches of the circuit the currents will be as shown in fig 7.12. Applying Kirchoff's second law to I, II and III loops,

$$2i_1 + 3(i_1 - i_2) + 5(i_1 - i_3) = 20 \qquad ...(i)$$
$$2i_2 + 5(i_2 - i_3) - 3(i_1 - i_2) = 10 \qquad ...(ii)$$
$$5i_3 - 5(i_1 - i_3) - 5(i_2 - i_3) = 0 \qquad ...(iii)$$

or $$10i_1 - 3i_2 - 5i_3 = 20 \qquad ...(iv)$$
$$-3i_1 + 10i_2 - 5i_3 = 10 \qquad ...(v)$$

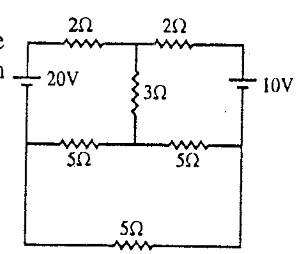

Fig. 7.11.

$$-i_1 - i_2 + 3i_3 = 0 \qquad \text{...(vi)}$$

Subtracting eq. (v) from eq. (vi), we get

$$13i_1 - 13i_2 = 10$$

or

$$i_1 - i_2 = \frac{10}{13} = 0.77 \text{ A} \qquad \text{...(vii)}$$

Thus the current through 3 Ω is **0.77 A.**

From (vi) $i_1 + i_2 = 3i_3$ or $i_3 = \dfrac{i_1 + i_2}{3}$ ...(viii)

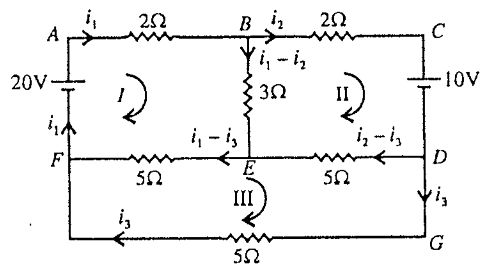

Fig. 7.12.

Putting this value of i_3 in eq. (iv)

$$10i_1 - 3i_2 - \frac{5}{3}(i_1 + i_2) = 20 \quad \text{or} \quad 25i_1 - 14i_2 = 60 \qquad \text{...(ix)}$$

Solving (vii) and (ix), we get $i_1 = 4.48$ A and $i_2 = 3.71$ A.

Power supplied by 20V source $= 20i_1 = 20 \times 4.48 =$ **89.6 watt.**

Power supplied by 10V source $= 10i_2 = 10 \times 3.71 =$ **371 watt.**

Ex. 5. *Twelve equal wires each of resistance R are joined to form a skeleton cube. The current enters at one corner and leaves at diagonally opposite corner. Find the total resistance between these corners.*

Sol. Let $ABCDEFGH$ be a skeleton cube, formed by twelve equal wires AB, BC, CD....etc., each of resistance R [**fig 7.13**].

The current enters at the corner A and leaves at the diagonally opposite corner G.

Let us, for convenience, take the current to be $6i$.

As the resistance of each wire is the same, the total current $6i$, divides at the point A into three equal parts, one along AB, the other along AD and the third, along AE so that the current in each of these wires is $2i$. At the points B, D and E the current again divides into two equal parts, each of magnitude i. At the points C, F and H these currents recombine to give currents $2i$ along CG, FG and HG, respectively. At the point G these three currents re-unite so that the current leaving at G is $6i$.

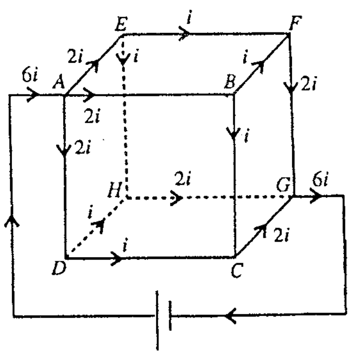

Fig. 7.13.

If E be the potential difference across A and G, then taking any one of the paths between A and G, say $ABFG$, we have

$$2i \times R + i \times R + 2i \times R = E \quad \text{or} \quad 5iR = E, \qquad \text{...(i)}$$

where R is the resistance of each wire.

Let R' be the total resistance between the corners A and G. Then, from **Ohm's law**, we have

$$6i \times R' = E \qquad \text{...(ii)}$$

From equations (i) and (ii), we have

$$6iR' = 5iR \quad \text{or} \quad r = \frac{5}{6}R.$$

Thus, the total resistance encountered by the current is **5/6th** of the resistance of each wire.

Ex. 6. *Twelve equal resistances each* 1 *ohm form a cube. A* 2 *volt accumulator of internal resistance* 0.09 *ohm is put directly across one arm of the cube. Calculate the distribution of current in the network.*

Sol. Let an accumulator of e.m.f. E volts and internal resistance r be put directly across one arm AB of the cube [**fig. 7.14**]. Let the resistance of each arm of the cube be R.

The current enters at A, and a part i_1 flows along AB and the rest is equally divided along AD and AE (by symmetry). Let it be i_2. At D a part i_3 flows along DC, while the remaining part $(i_2 - i_3)$, flows along DF. Since E is situated in the similar position to D, a current i_3 flows along EH, while a current $(i_2 - i_3)$ flows along EF. The current along FG will thus be $2(i_2 - i_3)$. At G it is equally divided and thus the current along GH and GC are each $(i_2 - i_3)$. The current $(i_2 - i_3)$ along GH unites with current i_3 coming along EH to give current i_2 along HB.

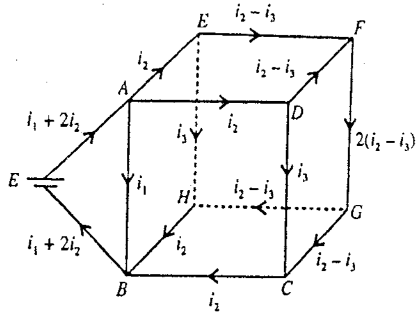

Fig. 7.14.

Similarly, the current $(i_2 - i_3)$ along GC combines with the current i_3 coming along DC to give current i_2 along CB.

Applying *Kirchoff's second law* to the circuit containing the cell and the arm AB, we have

$$r(i_1 + 2i_2) + i_1 R = E \quad \text{...(i)}$$

where r is the internal resistance of the accumulator.

Similarly, for the circuit $AEHBA$,

$$i_2R + i_3R + i_2R - i_1R = 0 \quad \text{or} \quad 2i_2 + i_3 - i_1 = 0 \quad \text{...(ii)}$$

For circuit $DFGCD$,

$$(i_2 - i_3)R + 2(i_2 - i_3)R + (i_2 - i_3)R - i_3R = 0$$

or

$$4i_2 - 5i_3 = 0 \quad \text{or} \quad i_2 = (5/4)i_3 \quad \text{...(iii)}$$

Substituting for i_2 in eq. (ii), we have

$$2 \cdot \frac{5}{4} i_3 + i_3 - i_1 = 0 \quad \text{or} \quad i_1 = \frac{7}{2} i_3 \quad \text{...(iv)}$$

Substituting for i_2 and i_3 in eq. (i), we get

$$r\left(\frac{7i_3}{2} + \frac{5i_3}{2}\right) + R \cdot \frac{7i_3}{2} = E \text{ or } 6i_3 r + 3.5 i_3 R = E$$

Here, $r = 0.09$ ohms ; $R = 1$ ohm and $E = 2$ volt.

$$\therefore \quad 0.09\,(6i_3) + 3.5i_3 = 2$$

or,

$$4.04\, i_3 = 2, \text{ whence } i_3 = 2/4.04 = 0.495 \text{ amp.}$$

$$\therefore \quad i_1 = 7 \times 0.495/2 = 1.732 \text{ amp.} \quad \text{and} \quad i_2 = 5 \times 0.495/4 = 0.619 \text{ amp.}$$

Thus, the currents in various branches are as indicated below :

Current in $AB = i_1 =$ **1.732 amp.**; Current in AE, HB or $CB = i_2 =$ **0.619 amp.**; Current in EH or $DC = i_3 =$ **0.495 amp.**; Current in EF, FD, GH or $GC = i_2 - i_3 =$ **0.124 amp** and Current in $FG = 2(i_2 - i_3) =$ **0.248 amp.**

EXERCISE 7·1

1. Find the currents i_1 and i_2 and voltage V across 24 Ω in the circuit, shown in **fig. 7.15**. The *emf* and the internal resistance of the upper cell are 11 V and 2 Ω, and those of the lower cell are 9V and 1 Ω.

   [**Ans.** $i_1 = (59/74)$ A; $i_2 = -(30/74)$ A; $V = 9.4$ volt.]

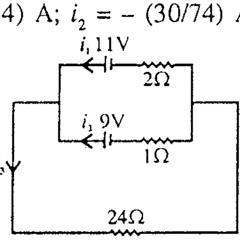

Fig. 7.15.

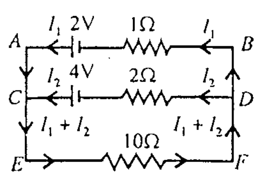

Fig. 7.16.

2. Calculate the values of current passing through each resistance of circuit shown in **fig. 7.16**.

   [**Ans.** $I_1 = -0.5$ A, $I_2 = 0.75$ A, $I_1 + I_2 = 0.25$ A.]

3. In the circuit shown in **fig. 7.17**, determine the value of current in 5 Ω resistance and equivalent resistance between points A and B.

   [Ans. 0.1 A; 2.47 Ω.]

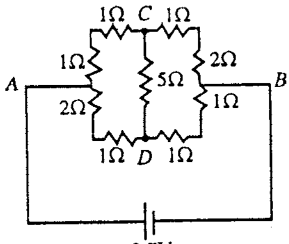

Fig. 7.17.

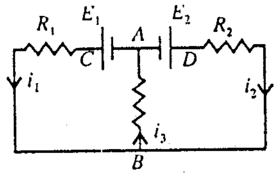

Fig. 7.18.

4. Calculate the currents flowing through the three resistors R_1, R_2 and R_3 in the circuit of **fig. 7.18.**

   [**Ans.** $i_1 = \dfrac{E_1(R_1 + R_3) - E_2R_3}{R}$, $i_2 = \dfrac{E_2(R_1 + R_3) - E_1R_3}{R}$,

   $i_3 = \dfrac{E_1R_2 + E_2R_1}{R}$, where $R = R_1R_2 + R_2R_3 + R_3R_1$.]

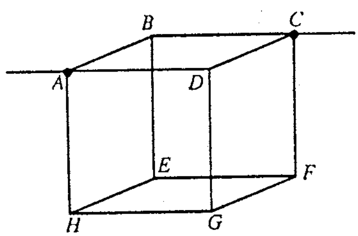

Fig. 7.19.

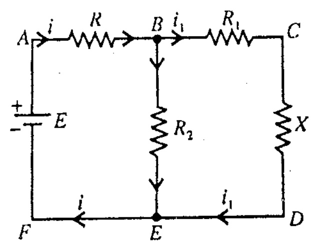

Fig. 7.20.

5. Twelve wires, each having resistance r, are joined to form a cube as shown in **fig. 7.19**. Find the equivalent resistance between the ends of a face diagonal such as A and C.

   [Ans. $\frac{3}{4}r$.]

6. Find the value of resistance X of the network shown in **fig. 7.20** in terms of R_1 and R_2, if X be the total resistance between the terminals A and F.

   $\left[\textbf{Ans. } \sqrt{R_1^2 + 2R_1R_2}.\right]$

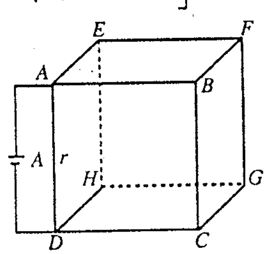

Fig. 7.21.

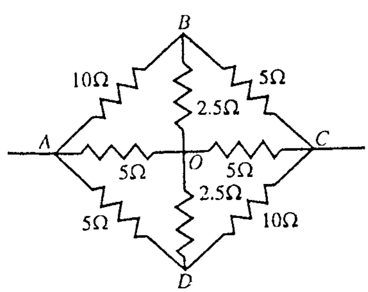

Fig. 7.22.

7. All the edges of a skeleton cube are made up of wires of 2 ohm resistance each. A battery of e.m.f. E and internal resistance r is connected parallel to an edge of the cube [**fig. 7.21**]. Show that the value of current drawn from the battery is $12E/(7R + 12r)$.

8. Calculate the equivalent resistance of the network shown in **fig. 7.22** between the point A and C.

   [Ans. 4.1 Ω.]

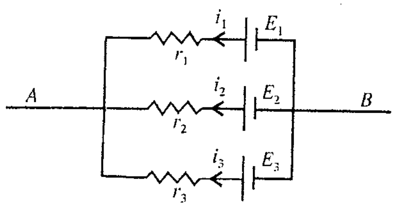

Fig. 7.23.

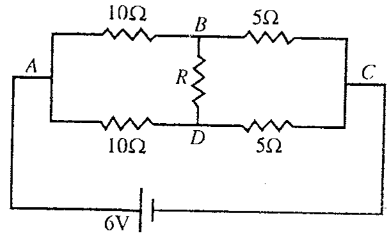

Fig. 7.24.

9. In the circuit shown in **fig. 7.23**, $E_1 = 3$ V, $E_2 = 2$ V, $E_3 = 1$ V and $r_1 = r_2 = r_3 = 1\Omega$. Calculate the potential difference between the points A and B and the current through each branch of the circuit.
[**Ans.** 2 volt; $i_1 = 1$ A, $i_2 = 0$, $i_3 = -1$ A.]

10. Find the value of R in **fig. 7.24** for which the current in it is zero.
[**Ans.** 0.4 A.]

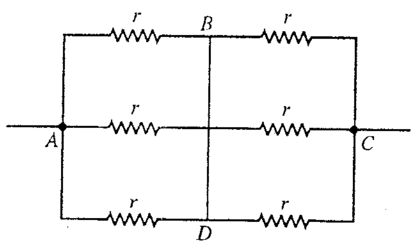

Fig. 7.25.

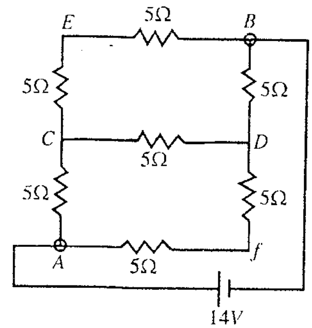

Fig. 7.26.

11. Find the equivalent resistance between the points A and C of the network shown in **fig. 7.25.** The value of each resistance is r.

12. In the circuit shown in **fig. 7.26**, determine the current in the resistance CD and equivalent resistance between the points A and B. The internal resistance of cell is negligible.
[**Ans.** 7 Ω; 0.4 A.]

7·5. Varying Currents-Transients

When we connect an electrical circuit with a direct current (d.c.) source, a current starts to flow in the circuit. The value of current depends on the applied e.m.f. and different elements (as resistance R, inductance L, capacitance C) present in the circuit. If the value of current flowing in the circuit remains constant *i.e.*, it does not vary with time, then the current is said to be **steady current**. But when the value of current flowing in the circuit does not remain constant and varies with time, it is called **varying current**. For example, if only an ohmic resistance is connected in the circuit, then current instantaneously achieves its maximum value i_0 ($= E/R$) corresponding to e.m.f. of applied source and as the source is withdrawn, the current becomes zero. Clearly, the current flowing in a pure ohmic circuit is a steady current not a varying current. When an inductor L or capacitor C or both are connected alongwith an ohmic resistance, then the current does not gain its maximum value instantaneously by connecting the combination to a d.c. source and similarly the current does not become zero instantaneously in the circuit by withdrawing the source. In such circuits the current gradually increases from zero to its maximum value (i_0) in some time or decreases from its maximum value to the lowest (zero) value in some time. Thus, when in the circuit an inductance or capacitance or both alongwith resistance are connected or disconnected suddenly to a source of e.m.f., a varying current flows in the circuit for some time. *Such time-dependent* *currents which flow in the circuit for some time when the e.m.f. source is connected or disconnected from the circuit are known as* **transient currents**. In fact when the circuit transits from one steady state to another steady state, the process of transition is called **transient process** or simply **transient**.

Any electric circuit may have three elements—resistance R, inductance L and capacitance C and all these three elements have distinct effects on the flow of current.

When a source of constant e.m.f. E is connected to a resistance R, the current i in the circuit follows the Ohms law similar to a steady current *i.e.*,

$$E = iR$$

If the source is withdrawn suddenly, then for $E = 0$, the current becomes zero (*i.e.*, $i = 0$) instantaneously. The cases when the circuit consists of inductance L or capacitance C or both in addition to resistance will be discussed in the next few articles.

7·6. Rise and Decay of Current in *LR* Circuit

(a) Rise of current in *LR* circuit : Consider a circuit containing an inductance L, a resistance R and a source of e.m.f. E in series through a switch S [**fig. 7.27**]. At $t = 0$, the circuit is closed by connecting S to A. As the current increases in the circuit, a self-induced e.m.f. is produced in the inductor L which opposes the growth of the current in the circuit. When the current is growing, let i be the current at any instant t. The opposing induced e.m.f. is $-L\,di/dt$. Hence, at the instant t, the effective e.m.f. driving the current in the circuit is $E - di/dt$ which must be equal to Ri according Ohm's law *i.e.*,

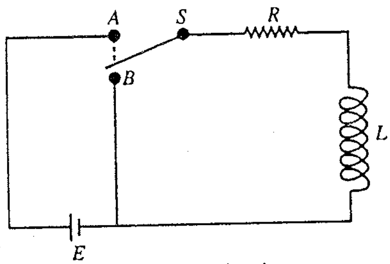

Fig. 7.27 : *LR* circuit.

$$E - L\frac{di}{dt} = Ri \quad \text{or} \quad L\frac{di}{dt} = E - Ri \quad \text{or} \quad \frac{di}{E - Ri} = \frac{dt}{L} \qquad ...(1)$$

Integrating (1) for the limits at $t = 0$, $i = 0$ and at time t, the current being i,

$$\int_0^i \frac{di}{E - Ri} = \int_0^t \frac{dt}{L} \quad \text{or} \quad -\frac{1}{R}\log_e \frac{E - Ri}{E} = \frac{t}{L} \quad \text{or} \quad \frac{E - Ri}{E} = e^{-(R/L)t} \quad \text{or } E - Ri = Ee^{-(R/L)t}$$

Thus,

$$i = \frac{E}{R}\left[1 - e^{-(R/L)t}\right] \qquad ...(2)$$

In eq. (2), the constant L/R has dimensions of time and is called the **time constant** τ of the *LR* circuit *i.e.*, $L/R = \tau$. Writing $E/R = i_0$, eq. (2) takes the form

$$i = i_0(1 - e^{-t/\tau}) \qquad ...(3)$$

This $i_0 = E/R$ is the final steady value of the current when $t \to \infty$. According to eq. (3), the current in the circuit rises according to an exponential law. A graph between current and time is shown in **fig. 7.28**. The current gradually rises from $t = 0$ and attains the maximum value i_0 in a long time.

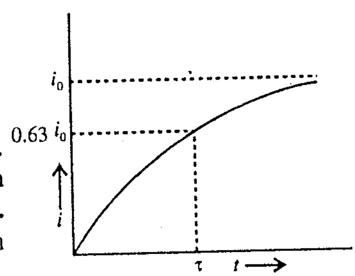

Fig. 7.28 : Rise of current in *LR* circuit.

If we put $t = \tau$, the current is

$$i = i_0\left(1 - \frac{1}{e}\right) = 0.632\, i_0 \qquad [\because e = 2.718] \qquad ...(3)$$

Thus the *time constant* ($\tau = L/R$) *of an LR circuit is the time-interval in which the current rises from zero to* 0.632 *times its maximum value*. The rate of growth of the current is

$$\frac{di}{dt} = \frac{i_0 e^{-t/\tau}}{\tau} = \frac{i_0 - i}{\tau} \qquad ...(4)$$

Thus if the time constant is small, more rapidly the current will approach to its maximum value. In principle, current takes infinite time to attain its maximum value. In practice however the current acquires almost its maximum value in few time constants.

(b) Decay of current in LR circuit : In fig. **7.27**, at $t = 0$ the switch is connected to B and hence the applied e.m.f. E becomes zero in the circuit. The equivalent circuit is redrawn [**fig. 7.29**]. As the source of e.m.f. E is disconnected, the current in the circuit decreases . If i be the current at any instant t, then the rate of decrease of current induces the e.m.f. $-L\, di/dt$. This e.m.f. opposes the fall of the current. As this is the only e.m.f. in the circuit across R,

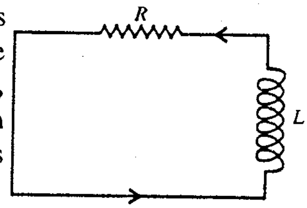

Fig. 7.29.

$$-L\frac{di}{dt} = Ri \quad \text{or} \quad \frac{di}{i} = -\frac{R}{L}\,dt \qquad ...(5)$$

Integrating for the limits at $t = 0$, $i = i_0$ and at time t, current $= i$,

$$\int_0^i \frac{di}{i} = \int_0^t -\frac{R}{L}\,dt \quad \text{or} \quad \log_e \frac{i}{i_0} = -\frac{R}{L}t \quad \text{or } i = i_0 e^{-(R/L)t}$$

Thus,

$$i = i_0\, e^{-t/\tau} \qquad ...(6)$$

where $L/R = \tau$ is the **time constant** of the LR circuit.

Eq. (6) shows that the current i gradually decreases as time t increases [**fig. 7.30**]. At $t = \tau$,

$$i = \frac{i_0}{e} = 0.37\, i_0 \qquad ...(7)$$

Thus, *we can also define the time constant of an LR circuit as the time interval in which the current falls from the maximum value to its $1/e$ value. For small time constant, the current will decay rapidly.*

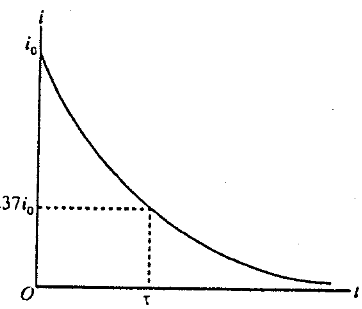

Fig. 7.30 : Decay of current in LR circuit.

Energy in LR circuit

If **during the growth of the current** in a LR circuit i is the current at any instant t, then the instantaneous *rate of energy delivered by the battery* is

$$E\,i \qquad ...(8)$$

where E is the e.m.f. of the battery assumed to be constant.

The magnetic energy stored in the inductance is $U = \frac{1}{2}Li^2$. Hence, the rate of energy stored in the magnetic field is

$$\frac{dU}{dt} = \frac{d}{dt}\left(\frac{1}{2}Li^2\right) = Li\frac{di}{dt}$$

But $$\frac{di}{dt} = \frac{d}{dt}\left[i_0(1 - e^{-(R/L)t})\right] = i_0\frac{R}{L}e^{-(R/L)t} = \frac{R}{L}(i_0 - i)$$

because $i = i_0\left[1 - e^{-(R/L)t}\right]$, or $i_0 e^{-(R/L)t} = i_0 - i$.

$$\therefore \quad \frac{dU}{dt} = Li\frac{R}{L}(i_0 - i) = iR(i_0 - i) \qquad ...(9)$$

The rate of energy (heat) developed in the resistance R is

$$i^2R \qquad ...(10)$$

Adding (9) and (10), we get

$$iR(i_0 - i) + i^2R = iRi_0 = Ei \qquad [\because i_0 = E/R]$$

Thus, *during the growth of current in a circuit containing inductance and resistance, at any instant the rate of energy delivered by the battery is the sum of the rate of energy stored in the magnetic field and the rate of energy used in the production of heat.*

During **decay of the current,** *the magnetic energy stored in the inductance* $\left(\frac{1}{2} Li_0^2\right)$ *is dissipated as thermal energy in the resistor,* bacause thermal energy developed

$$= \int_0^\infty i^2 R\, dt = \int_0^\infty [i_0^2 e^{-(R/L)t}]^2 R\, dt = \frac{1}{2} Li_0^2 \qquad ...(11)$$

Larger Self-induced e.m.f. in a Circuit at Break than at Make

When the current in a circuit is switched off by breaking the circuit, a very high resistance of the air gap is introduced in the circuit. This results in the value of time constant (L/R) to be much less at break than at make. Note that at the make the time constant is due to only the resistance present in the circuit. Hence the rate of fall of current (di/dt) at break is much greater than the rate of growth of the current at make. Therefore the induced e.m.f. ($-L\, di/dt$) at break is much greater than at the make. Often the e.m.f. at break is so high which breaks down the insulation of the air between the switch contacts and produces a **spark** in the gap. This is the cause that sometime when we switch off the circuit with glowing electric bulb, it fuses.

Ex. 1. *A coil of resistance of* 5 *ohm and an inductance of* 4 *henry is connected with a battery of e.m.f.* 10 *volt and negligible internal resistance. How much time will current take to become* 1 *ampere ?*

Sol. In the process of current-growth, the current i at any instant t is given by following relation :

$$i = i_0 (1 - e^{-Rt/L}) = \frac{E}{R}(1 - e^{-Rt/L})$$

Here, E = 10 volt, R = 50 ohm and i = 1 ampere.

$$\therefore \quad 1 = \frac{10}{5}(1 - e^{-Rt/L}) \text{ or } 1 - e^{-Rt/L} = \frac{1}{2} \text{ or } e^{-Rt/L} = 1 - \frac{1}{2} = \frac{1}{2} \text{ or } e^{Rt/L} = 2$$

Therefore, $\frac{Rt}{L} = \log_e 2 = 2.3026 \log_{10} 2$ or $t = \frac{L}{R} \times 2.3026 \times 0.3010$

or $\qquad t = \frac{4}{5} \times 2.3026 \times 0.3010 = \mathbf{0.554}$ **second.**

Ex. 2. *A coil of inductance* 3.0 *henry and a resistance of* 6.0 *ohm are connected in series with a battery of e.m.f.* 12.0 *volt and negligible internal resistance. Calculate* : (i) *time constant of circuit,* (ii) *maximum or steady current established in circuit,* (iii) *the stored magnetic energy after establishment of steady current and* (iv) *the current in the circuit after* 0.2 *second of connecting the battery.*

Sol. Here, R = 6.0 ohm, L = 3.0 henry and E = 12.0 volt.

(i) Time constant of L-R circuit, L/R = 3.0/6.0 = **0.5 second.**

(ii) Steady current in circuit, $i_0 = E/R$ = 12.0/6.0 = **2 ampere.**

(iii) Stored magnetic energy after establishment of steady current,

$$U_B = \frac{1}{2} \times Li_0^2 = \frac{1}{2} \times 3.0 \times (2)^2 = \mathbf{6.0 \text{ joule}}$$

(iv) The current in circuit after 0.2 second of connecting the battery,

$$i = i_0 (1 - e^{-Rt/L}) = 2.0 \times (1 - e^{-6.0 \times 0.2/3.0}) = 2.0\,(1 - e^{-0.4}) = 2.0 \times 0.33 = \mathbf{0.66 \text{ ampere.}}$$

Ex. 3. *With reason state which of the following sets is suitable for obtaining rapid growth and decay of current in L-R circuit :*

$$\left.\begin{array}{l} L = 1\, mH \\ R = 1000\, ohm \end{array}\right\} \qquad \left.\begin{array}{l} L = 10\, mH \\ R = 10\, ohm \end{array}\right\}$$

(Meerut 2002 S)

Sol. Time-constant for the first set $= \frac{L}{R} = \frac{1 \times 10^{-3} \text{ henry}}{1000 \text{ ohm}} = 10^{-6}$ sec.

Time constant for the second set $= \dfrac{L}{R} = \dfrac{10 \times 10^{-3} \text{ henry}}{10 \text{ ohm}} = 10^{-3}$ sec.

As the time-constant is less for the **first set**, it is suitable for obtaining rapid growth and decay of current in L-R circuit.

Ex. 4. *A coil of resistance* 10 Ω *and self-inductance of* 0.4 *H is connected to a* 60 *V source of constant e.m.f. and negligible internal resistance. Calculate the induced e.m.f. across the coil at the instant the current in the coil has risen to* 25% *of its steady value.* **(Meerut 1997)**

Sol. The current growth in an L-R circuit is given by

$$i = i_0 \left[1 - e^{-(R/L)t}\right].$$

$$\therefore \quad \frac{di}{dt} = i_0 \frac{R}{L} e^{-(R/L)t} = i_0 \frac{R}{L}\left(1 - \frac{i}{i_0}\right) = \frac{R}{L}(i_0 - i)$$

The e.m.f. induced across the coil $= |e| = L\dfrac{di}{dt} = R\,(i_0 - i)$.

At the instant the current has risen to 25% of its steady value i_0 *i.e.*, when $i = 0.25\, i_0$, we have

$$|e| = R\,(i_0 - 0.25\, i_0) = R \times 0.75\, i_0 = 0.75\, E = 0.75 \times 60 = \mathbf{45\ V}.$$

Ex. 5. *A* 12 *volt battery connected to a* 6 Ω, 10 H *coil through a switch drives a constant current in the circuit. The switch is suddenly opened. Assuming that it took* 2 *ms to open the switch, calculate the average e.m.f. induced across the coil.*

Sol. The steady-state current is $i_0 = \dfrac{12 \text{ V}}{6\,\Omega} = 2$ A. The final current is zero. Thus,

$$\frac{di}{dt} = -\frac{2 \text{ A}}{2 \text{ ms}} = -10^3 \text{ A/sec.}$$

The induced e.m.f. is $\quad E = -L\dfrac{di}{dt} = -(10) \times (-10^3) = \mathbf{10000\ V.}$

Note that such a high e.m.f. may cause sparks across the open switch.

Ex. 6. *An L-R circuit having inductance* $L = 4.0$ *H and* $R = 1.0\ \Omega$ *and applied d-c e.m.f. of* 6.0 *V is switched on at* $t = 0$. *Find the power dissipated as joule heat in the circuit at* $t = 4.0$ *sec.*

(Meerut 1995)

Sol. At time t the current in an L-R circuit given by

$$i = i_0 \left[1 - e^{-(R/L)t}\right].$$

where $i_0\ (= E/R)$ is the final steady current.

Here, $\quad i_0 = \dfrac{E}{R} = \dfrac{6.0 \text{ V}}{1.0\,\Omega} = 6.0$ A.

Hence, the current in the circuit at $t = 4.0$ sec is given by

$$i = 6.0\left[1 - e^{-(1.0/4.0) \times 4.0}\right] = 6.0\left[1 - e^{-1.0}\right] = 6.0\left[1 - \frac{1}{2.72}\right] = 3.8 \text{ A.}$$

Rate of joule heating in the circuit at $t = 4.0$ sec is

$$i^2 R = (3.8)^2 \times 1 = \mathbf{14.4\ watt.}$$

Ex. 7. *A resistance R and an inductance L are connected to a battery of e.m.f. E. When will the potential difference across inductance be equal to that across resistance ?*

Sol. The instantaneous current in the L-R circuit in the presence of a battery of e.m.f. E. is given by

$$i = i_0 \left[1 - e^{-(R/L)t}\right], \qquad \left[\because\ i_0 = \frac{E}{R}\right]$$

Hence, the instantaneous potential difference across the resistance is

$$Ri = Ri_0 \left[1 - e^{-(R/L)t}\right] \qquad \text{...(i)}$$

and that across the inductance is

$$L\frac{di}{dt} = R\, i_0 e^{-(R/L)t} \qquad ...(ii)$$

Equating (i) and (ii), we get

$$1 - e^{-(R/L)t} = e^{-(R/L)t} \text{ or } 1 = 2e^{-(R/L)t} \text{ or } e^{(R/L)t} = 2;$$

$$\therefore \qquad t = \frac{L}{R}\log_e 2$$

Hence at $t = (L/R)\log_e 2$, the potential difference across the inductance L will be equal to that across the resistance R.

Ex. 8. *A coil of inductance of* 2 *henry a resistance of* 4 *ohm is connected with a battery of e.m.f.* 8 *volt and negligible internal resistance. Calculate :*

(i) *Initial rate of current growth in the circuit.*

(ii) *The rate of current growth at that instant when the current in circuit is* 1 *ampere.*

(iii) *Instantaneous value of current after* 0.2 *second of connecting the circuit.*

(iv) *Final steady state current.*

Sol. Here, $L = 2$ henry, $R = 4$ ohm and $E = 8$ volt.

(i) In the process of current growth in L-R circuit, the value of current i at any instant t is given by

$$i = i_0\left(1 - e^{-Rt/L}\right), \text{ where } i_0 = E/R.$$

So, rate of current growth

$$\frac{di}{dt} = i_0\left(-e^{-Rt/L}\right)\left(-\frac{R}{L}\right) = \frac{E - Ri}{L} \qquad \left[\because i_0 e^{-Rt/L} = (i_0 - i) = \frac{E}{R} - i\right]$$

Initially at instant $t = 0$, $i = 0$, so, initial rate of current growth is

$$\left(\frac{di}{dt}\right)_{t=0} = \frac{E}{L} = \frac{8}{2} = \mathbf{4.0\ A/sec.}$$

(ii) When $i = 1$ ampere, rate of current growth

$$\frac{di}{dt} = \frac{E - Ri}{L} = \frac{8 - 4 \times 1}{2} = \mathbf{2.0\ A/sec.}$$

(iii) Instantaneous value of current at $t = 0.2$ second

$$i = i_0\left(1 - e^{-Rt/L}\right) = \frac{E}{R}\left(1 - e^{-Rt/L}\right) = \frac{8}{4}\left[1 - e^{-4\times 0.2/2}\right] = 2 \times (1 - e^{-0.4})$$

$$= 2 \times [1 - (2.72)^{-0.4}] = 2\,[1 - 0.67] = 2 \times 0.33 = \mathbf{0.66\ A.}$$

(iv) Final steady state current, $i_0 = E/R = 8/4 = \mathbf{2\ A.}$

Ex. 9. *A* 2.0 *henry inductor is connected in series with a* 10 *ohm resistor, an e.m.f. of* 100 *volt being suddenly applied to the combination. At* 0.1 *second after the contact is made, what is* (a) *the current* (b) *rate at which energy is being stored in the magnetic field,* (c) *the rate at which heat is appearing in the resistor and* (d) *the rate at which energy is being delivered by the battery ?*

Sol. (a) Final steady current i_0 is given by

$$i_0 = \frac{E}{R} = \frac{100}{10} = 10\text{ A.} \qquad ...(i)$$

$$\text{Instantaneous current, } i = i_0\,[1 - e^{-(R/L)t}]. \qquad ...(ii)$$

At $t = 0$ sec, we have $L = 2.0$ henry and $R = 10$ ohm. Hence

$$i = 10\,[1 - e^{-(10/2.0)\,0.1}] = 10\,[1 - e^{-0.5}] = \mathbf{3.9\ A.} \qquad ...(iii)$$

(b) The instantaneous energy in the magnetic field associated with the inductor carrying an instantaneous current i is

$$U = \frac{1}{2}Li^2$$

$\therefore$ Rate of storing energy, $\dfrac{dU}{dt} = Li\dfrac{di}{dt}$ ...(iii)

From (ii), $\dfrac{di}{dt} = i_0 \dfrac{R}{L} e^{-(R/L)t} = i_0 \dfrac{R}{L}\left(1 - \dfrac{i}{i_0}\right) = \dfrac{R}{L}(i_0 - i).$

$\therefore$ $\dfrac{dU}{dt} = Li\dfrac{R}{L}(i_0 - i) = iR(i_0 - i) = 3.9 \times 10 \times (10 - 3.9) =$ **238 watt.**

(c) Rate of heat appearing in the resistor $= i^2R = (3.9)^2 \times 10 =$ **152 watt.**

(d) Rate at which energy is being delivered by the battery $= E\,i = 100 \times 3.9 =$ **390 watt.**

EXERCISE 7·2

1. A solenoid has the inductance of 50 henry and resistance of 30 ohm. If it is connected with a battery of 100 volt, then calculate the time in which current becomes half of the maximum steady value. [$\log_{10} 2 = 0.3010$]
[**Ans.** 11.55 second.]

2. A coil of inductance 10 henry and resistance 15 ohm is connected with a battery of 110 volt. After how much time the current becomes half of the maximum value. **(Kanpur 2005)**
[**Ans.** 0.46 sec.]

3. A resistance of 100 Ω, an inductance of 20 mH and a battery of e.m.f. 10 V are connected in series. Find the time constant of the circuit, maximum current in the circuit and time required to attain 99% of its maximum value. **(Meerut 2000 S)**
[**Ans.** 2×10^{-4} sec, 0.1 A, 9.2×10^{-4} sec.]

4. A coil having a resistance of 15 ohm and an inductance of 10 henry is connected to a 90 volt supply. Determine the value of the current (i) after 0.67 sec and (ii) after 2 sec ($e = 2.718$).
[**Ans.** 3.8 A; 5.7 A.] **(Meerut 1998)**

5. A solenoid of inductance 50 mH and resistantce 10 Ω is connected to a battery of 6 V. Calculate the time elapsed before the current acquires half of its steady value.
[**Ans.** 3.5×10^{-3} sec.]

6. A solenoid of inductance 53 mH and resistance 0.37 ohm is connected to a battery. How long will it take to acquire the current equal to half the maximum steady value ? **(Sagar 2003)**
[**Ans.** 0.1 sec.]

7. A coil of inductance 5.0 H and resistance 1.0 ohm is connected with a battery of e.m.f. 2 volt. Calculate : (i) the time constant of circuit, (ii) maximum steady current, (iii) current in the circuit after 5.0 sec from joining the battery, (iv) rate of growth of current after 5.0 sec from joining the battery.
[**Ans.** (i) 5 sec, (ii) 2.0 A, (iii) 1.264 A, (iv) 0.147 A sec^{-1}.]

8. An *L-R* circuit with a time constant of 50 ms is connected to an ideal battery of e.m.f. *E*. Find the time elapsed before (i) the current reaches half its maximum value, (ii) the power dissipated in heat reaches half its maximum value and (iii) the magnetic field energy stored in the ciruit reaches half its maximum value.
[**Ans.** (i) 0.035 sec, (ii) 0.061 sec, (iii) 0.061 sec.]

9. A solenoid of inductance 2 henry and resistance 10 ohm is suddenly connected with a battery of e.m.f. 100 volt and negligible resistance.
(i) What is the time constant of the circuit ?
(ii) What is the value of steady current established in circuit ?
(iii) What is the value of stored magnetic energy if circuit has some current ?
[**Ans.** (i) 0.2 second; (ii) 10 ampere; (iii) 100 joule.]

10. The time constant of any inductive coil is 2.0×10^{-3} second. The time constant reduces to 0.5×10^{-3} second when it is connected with a resistance of 10 ohm in series. Find the inductance and resistance of the coil

[Ans. 30 Ω; 60 mH.]

[Hint : Let L be the inductance and R the resistance of the coil. $\therefore$ $2 \times 10^{-3} = L/R$ and $0.5 \times 10^{-3} = L/(R + 10)$; hence obtain L and R.]

11. A direct voltage of 120 volt is applied to a circuit containing a coil of resistance 5 ohm and inductance 10 henry and a separate resistor of 5 ohm. Find the current 1 second after switching and potential difference across the inductance.

[Ans. 7.58 A; 82 volt.]

12. A coil of inductance 1.0 H and resistance 100 Ω is connected to a battery of e.m.f. 12 V. Calculate the energy stored in the magnetic field associated with the coil at an instant 10 ms after the circuit is switched on.

[Ans. 2.8×10^{-3} J.]

13. A coil has the inductance 1 henry and resistance 0.2 ohm. If an e.m.f. of 3.0 volt is applied, then what is the value of stored magnetic energy after establishing the steady current in the circuit ?

[Ans. 112.5 joule.]

14. A solenoid of inductance 4.0 H and resistance 10 Ω is connected to a 4.0 V battery at $t = 0$. Calculate : (i) the time constant, (ii) the time elapsed before the current reaches 0.63 of its steady-state value, (iii) the power delivered by the battery at this instant and (iv) the power dissipated in joule heating at this instant.

[Ans. (i) 0.4 sec; (ii) 0.4 sec; (iii) 1 watt; (iv) 0.64 watt.]

7·7. Charging and Discharging of a Condenser through Resistance (Rise and Decay of Current in *CR* Circuit)

(a) **Charging of a condenser** : Consider a circuit consisting of a capacitor having capacitace C in series with a resistance R and a source of e.m.f. or battery E [**fig. 7.31**]. When the circuit is closed by connecting the switch S to A, charge or current flows from the battery and the capacitor C is charged. The charging current stops when the potential difference between the plates of the capacitor becomes equal to the e.m.f. E of the battery. In charging of the capacitor, there is the flow of the **varying current** which is maximum in the beginning and reduces to zero when the capacitor is fully charged.

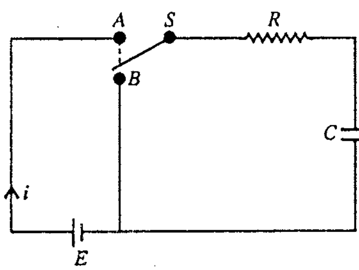

Fig. 7.31 : *CR* circuit.

Let the battery be connected at time $t = 0$. Suppose the charge on the capacitor is q and the current in the circuit is i at time t. The potential difference between the plates of the capacitor at time t is q/C and the potential drop on the resistor is Ri. Obviously potential difference q/C acts opposite to the battery e.m.f. E. Thus at time t the effective e.m.f. in the circuit is $E - (q/C)$ which must be equal to Ri according to Ohm's law. Thus

$$E - \frac{q}{C} = Ri \qquad ...(1)$$

But $i = dq/dt$, hence

$$E - \frac{q}{C} = R\frac{dq}{dt} \text{ or } \frac{dq}{dt} = \frac{EC - q}{RC} \text{ or } \frac{dq}{EC - q} = \frac{dt}{RC}$$

Integrating for the limits at $t = 0$, $q = 0$ and at time t, charge $= q$,

$$\int_0^q \frac{dq}{EC - q} = \int_0^t \frac{dt}{CR} \text{ or } -\log_e \frac{EC - q}{EC} = \frac{t}{CR} \text{ or } 1 - \frac{q}{EC} = e^{-t/CR}$$

Thus, $$q = EC(1 - e^{-t/CR}) \qquad ...(2)$$

This gives the charge on the capacitor C at any time t. As t increases, q also increases. The maximum charge is obtained on the plates when $t = \infty$ and its value is $q_0 = EC$.
This is the final steady charge on the capacitor. Hence

$$q = q_0 (1 - e^{-t/CR}) \qquad ...(3)$$

This equation represents the charge on the capacitor at any instant t. The growth of the charge on the capacitor with time is shown in **fig 7.32.**

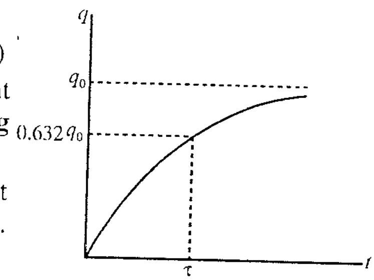

Fig. 7.32 : Charging of a condenser in CR circuit

Time constant $\tau = CR$: In eq. (3) the dimension of the constant CR is that of time. Hence we call CR as time constant τ of the CR circuit. Thus

$$q = q_0(1 - e^{-t/\tau}) \qquad ...(4)$$

If $t = \tau$, $q = q_0(1 - e^{-1}) = 0.632\, q_0$, $\quad [e = 2.718] \qquad ...(5)$

Thus *the time constant of CR circuit is that time in which the charge rises from zero to 0.632 times of its final steady value.*

Rate of growth of charge from eq. (3) is given by

$$\frac{dq}{dt} = \frac{q_0}{CR} e^{-t/CR} = \frac{q_0}{CR}\left(1 - \frac{q}{q_0}\right) = \frac{q_0 - q}{CR} \qquad ...(6)$$

Thus, smaller the value of the time constant CR, more rapidly the charge increases on the capacitor

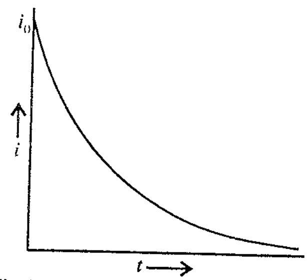

Fig. 7.33 : Exponential decrease of current as time increases.

Current during charging : At time t, the current in the circuit is given by

$$i = \frac{dq}{dt} = \frac{q_0}{CR} e^{-t/CR} \text{ or } i = i_0\, e^{-t/CR} \qquad ...(7)$$

where $i_0 = q_0/CR = E/R$ is the current at the start ($t = 0$) of charging and this is the maximum value of the current. We see that during charging of the capacitor, the current starts with its maximum value and falls of exponentially [**fig. 7.33**].

(b) **Discharging of a condenser :** After charging the capacitor, suppose the switch S is connected to B [**fig. 7.31**] at $t = 0$. The equivalent of the circuit is shown in **fig 7.34**. There is a flow of charge (or current) from positive to negative plate through the resistance R and the capacitor discharges gradually.

Let q be the charge left on the capacitor at any instant t and i be the instantaneous current in the circuit. The potential differenc q/C across the capacitor at this instant drives the current through the resistance R. Hence

$$\frac{q}{C} = Ri \qquad ...(8)$$

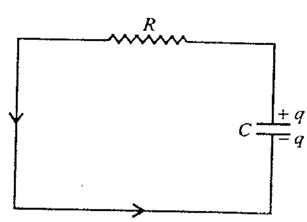

Fig. 7.34 : Circuit for discharging of a condenser.

But $i = -dq/dt$, because the charge decreases as time increases

$$-R\frac{dq}{dt} = \frac{q}{C} \text{ or } \frac{dq}{q} = -\frac{dt}{CR}$$

Integrating it for the limits at $t = 0$, $q = q_0$ and at time t, charge $= q$,

$$\int_{q_0}^{q} \frac{dq}{q} = -\int_0^t \frac{dt}{CR} \text{ or } \log_e \frac{q}{q_0} = -\frac{t}{CR}$$

Thus,

$$q = q_0\, e^{-t/CR} \qquad ...(9)$$

In principle the capacitor discharges completely at $t = \infty$. A graph between the charge q and time t is shown in **fig 7.35.**

In eq. (9) CR is the **time constant.** At $t = CR$, from eq. (9), we obtain

$$q = q_0\, e^{-1} = q_0/e \qquad \text{...(10)}$$

Thus *time constant of a CR circuit is that time in which the maximum charge falls to its 1/e (or 0.37) value.*

$$\text{Rate of discharge} = \frac{dq}{dt} = -\frac{q_0}{CR}e^{-t/CR} = -\frac{q}{CR} \qquad \text{...(11)}$$

Thus, *smaller the value of the time constant CR, the quicker is the discharge of the capacitor.*

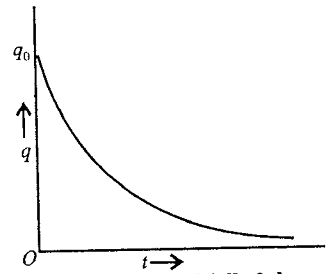

Fig. 7.35 : Exponential fall of charge on the condenser with time during discharging.

Current during discharging : From eq. (9)

$$i = \frac{dq}{dt} = -\frac{q_0}{CR}e^{-t/CR} = -i_0\, e^{-t/CR} \qquad \text{...(12)}$$

where $i = -i_0$ is the current at the start ($t = 0$) of discharging. Thus the direction of discharging current is opposite to the charging current. We see from eq. (12) that the discharge current in magnitude starts with its maximum value $i_0 = q_0/CR$ and falls off exponentially **[fig. 7.36]**.

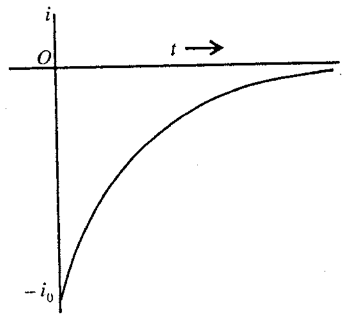

Fig. 7.36 : Current during discharging of the capacitor

Energy in CR Circuit

During charging the condenser at any instant t the rate of energy given by the battery is

$$Ei = E\, i_0 e^{-t/CR} = \frac{E^2}{R}e^{-t/CR} \qquad \text{...(13)}$$

Rate of Joule heat production in the resistance R is

$$i^2R = \left(i_0 e^{-t/CR}\right)^2 R = \frac{E^2}{R}e^{-2t/CR} \qquad \text{...(14)}$$

Rate of stored electrostatic energy $\left(U = \frac{1}{2}CV^2 = q^2/2C\right)$ in the capacitor is

$$\frac{dU}{dt} = \frac{d}{dt}\left(\frac{q^2}{2C}\right) = \frac{q}{C}\frac{dq}{dt} = \frac{q_0}{C}(1 - e^{-t/CR})\,\frac{E}{R}e^{-t/CR} = \frac{E^2}{R}e^{-t/CR}(1 - e^{-t/CR}) \quad [\because q_0 = EC] \qquad \text{...(15)}$$

The sum of (14) and (15) is

$$\frac{E^2}{R}e^{-2t/CR} + \frac{E^2}{R}e^{-t/CR}[1 - e^{-t/CR}] = \frac{E^2}{R}e^{-t/CR} \qquad \text{...(16)}$$

From eq. (13) and eq. (16), we see that *at any instant the rate of energy supplied by the battery is equal to the sum of* (i) *the rate of Joule heat production in the resistor and* (ii) *the rate of stored electrostatic energy in the capcacitor.*

Maximum stored energy in the capacitor $= \frac{1}{2}CE^2$ for $V = E$ finally.

During discharging *the total energy of the capacitor* $\left(\frac{1}{2}CE^2\right)$ *is developed as Joule heat in the resistor* (R). This can be shown as follows :

Rate of Joule heat developed in the circuit $= i^2R$

$\therefore$ When the condenser is discharged completely in time 0 to ∞, total heat (H) produced in the resistance is

$$H = \int_0^\infty i^2R\,dt = \int_0^\infty \frac{E^2}{R}e^{-2t/CR} = \frac{E^2}{R}\left[\frac{e^{-2t/CR}}{-2/CR}\right]_0^\infty$$

$$= -\frac{1}{2}CE^2(e^{-\infty} - e^0) = -\frac{1}{2}CE^2(0 - 1) = \frac{1}{2}CE^2 \qquad \text{...(17)}$$

Measurement of High Resistance by the Method of Leakage :

If a condenser of capacity C be first charged to charge q_0 and then it is discharged for time t through a high resistance R, the charge q during discharging at time t left on the condenser is given by

$$q = q_0\, e^{-t/CR} \text{ or } \frac{q_0}{q} = e^{t/CR} \qquad ...(1)$$

Hence,
$$\log_e \frac{q_0}{q} = \frac{t}{CR} \quad \text{or} \quad R = \frac{t}{C \log_e (q_0/q)} \qquad ...(2)$$

The connections are made as shown in **fig. 7.37.** Specially a ballistic galvanometer (B.G.) is connected in the circuit. The throw (θ) of the ballistic galvanometer is proportional to the charge (q) passed through it [*i.e.*, $q = k\theta\ (1 + \lambda/2)$], where λ is logarithmic decrement].

When the key K is pressed, the condenser is charged to a potential V_0, say corresponding to a charge q_0 ($= CV_0$). It is then discharged through ballistic galvanometer B.G., and the first throw θ_0 of the galvanometer is noted.

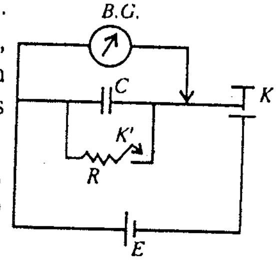

Fig. 7.37 : High resistance by the method of leakage.

The condenser again charged to the same potential V_0 as before but, now the key K is released and K' is pressed so that the charge is allowed to leak for a time t through the high resistance R until the charge on the condenser falls to q with potential V ($q = CV$). It is then instantaneously discharged through the ballistic galvanometer and first throw θ_1 in it is observed. The calculations are then made as follows :

We have
$$q_0 = CV_0 = K\theta_0\ (1 + \lambda/2)$$
and
$$q = CV = K\theta_1\ (1 + \lambda/2)$$

$$\therefore \qquad \frac{q_0}{q} = \frac{\theta_0}{\theta_1} = \frac{V_0}{V} \qquad ...(3)$$

If C be the capacity of the condenser, then the high resistance R is obtained by using eq. (2) *i.e.*,

$$R = \frac{t}{C \log_e (q_0/q_1)} = \frac{t}{C \log_e (\theta_0/\theta_1)} = \frac{t}{C \log_e (V_0/V)} \qquad ...(4)$$

Ex. 1. *A condenser of capacity* 0.5 μF *is discharged through a resistance of* 10 *mega-ohm. How much time will it take to discharge to half of its charge ?* ($\log_e 2 = 0.6931$)

(Ujjain 2003; Gwalior 03; Meerut 1990 S)

Sol. Here, $q = \frac{1}{2} q_0$, $C = 0.5\ \mu\text{F} = 0.5 \times 10^{-6}$ F, and $R = 10$ mega-ohm $= 10 \times 10^6$ ohm.

During discharge, charge at an instant t is $q = q_0\, e^{-t/CR}$. Therefore

$$\frac{q_0}{2} = q_0\, e^{-t/CR} \text{ or } \frac{1}{2} = e^{-t/CR} \text{ or } e^{t/CR} = 2 \text{ or } \frac{t}{CR} = \log_e 2$$

$$\therefore \qquad t = CR \log_e 2 = (0.5 \times 10^{-6}) \times (10 \times 10^6) \times 0.6931 = \mathbf{3.47\ sec.}$$

Ex. 2. *A* 4.0 μF *capacitor is connected to* 240 *volts D.C. source through a resistor of* 0.25 *mega-ohm. Find the charge on the capacitor, potential and the current in the circuit after* 1.0 *second.*

Sol. The charge on the capacitor t second after switching on the D. C. source is given by

$$q = q_0\, (1 - e^{-t/CR}) \qquad ...(i)$$

where q_0 is the final steady charge.

Here, $q_0 = CE = 4.0\ \mu\text{F} \times 240$ volt $= 960\ \mu\text{C}$ (microcoulomb), $t = 1$ sec, $C = 4.0 \times 10^{-6}$ farad and $R = 0.25 \times 10^6$ ohm, hence $CR = 1$ sec. Substituting in (i), we get

$$q = (960\ \mu\text{C})\ (1 - e^{-1}) = (960\ \mu\text{C})\ (1 - 1/2.718) = \mathbf{606.8\ \mu C}$$

$$\text{Potential, } V = \frac{q}{C} = \frac{606.8\ \mu\text{C}}{4.0\ \mu\text{F}} = \mathbf{151.7\ volt.}$$

$$\text{Current, } i = \frac{dq}{dt} = \frac{q_0}{CR} e^{-t/CR} = \frac{960 \mu\text{C}}{1\ \text{sec}} e^{-1} = 960 \mu\text{A} \times \frac{1}{2.718} = \mathbf{353.2\ \mu A}$$

Ex. 3. *A condenser is being charged from a D. C. source through a resistance of two mega-ohms. If it takes 0.5 sec for the charge to reach three quarter of its final value, what is the capacity of condenser ?* **(Rajasthan)**

Sol. We have the relation, $q = q_0(1 - e^{-t/CR})$

Here $q = \frac{3}{4}q_0$; $t = 0.5$ sec; and R = mega-ohms = 2×10^6 ohms.

$$\therefore \quad \frac{3}{4}q_0 = q_0\left(1 - e^{-\frac{0.5}{2\times10^6\times C}}\right) \text{ or } e^{-\frac{0.5}{2\times10^6\times C}} = 1 - \frac{3}{4} = \frac{1}{4} \text{ or, } e^{\frac{0.5}{2\times10^6\times C}} = 4$$

$$\therefore \quad \frac{0.5}{2\times10^6\times C} = \log_e 4 = 2.3026 \log_{10} 4,$$

whence,
$$C = \frac{0.5}{2\times10^6\times 2.32026\times0.6021} = \mathbf{0.1803\ \mu F.}$$

Ex. 4. *A capacitor charged to 50 volt is discharged by connecting the two plates at $t = 0$. If the potential difference across the plates drops to 1.0 V at $t = 10$ ms, calculate the potential difference at $t = 20$ ms.*

Sol. The potential difference at time t is given by

$$V = q/C = (q_0/C)e^{-t/RC} \text{ or } V = V_0e^{-t/RC}$$

Here, $V = 1$ volt, $V_0 = 50$ volt and $t = 10 \times 10^{-3} = \mathbf{0.01}$ **sec.**

$$\therefore \quad 1 = 50\, e^{-0.01/RC}$$

or,
$$e^{-0.01/RC} = 1/50.$$

The potential difference at $t = 20 \times 10^{-3} = 0.02$ sec. is

$$V = V_0e^{-t/RC} = 50\, e^{-0.02/RC} = 50\left(e^{-0.01/RC}\right)^2 = 50 \times \left(\frac{1}{50}\right)^2 = \frac{1}{50} = \mathbf{0.02\ volt.}$$

Ex. 5. *A condenser of capacity 1.0 μF is charged by a battery of e.m.f. 4 volt through a resistance 3 mega-ohm. Calaculate after 1 sec,* (i) *the rate of growth of charge on condenser,* (ii) *the rate of energy supplied by battery,* (iii) *the rate of production of heat in resistance and* (iv) *the rate of energy stored in condenser* ($e^{-1/3} = 0.72$).

Sol. Here, $C = 1.0\ \mu F = 1.0 \times 10^{-6}$ F, $E = 4$ volt, $t = 1$sec and $R = 3$ mega-ohm $= 3 \times 10^6$ ohm.

(i) Rate of grwoth of charge on condenser $= \frac{dq}{dt} = \frac{d}{dt}\left[q_0\left(1 - e^{-t/CR}\right)\right] = \frac{E}{R}e^{-t/CR}$

$$\therefore \quad \frac{dq}{dt} = \frac{4}{3\times10^6}e^{-1/(1\times10^{-6}\times3\times10^6)} = \frac{4}{3\times10^6}e^{-1/3} = \frac{4}{3\times10^6}\times0.72 = 9.6\times10^{-7} \text{ C/sec.}$$

(ii) Rate of energy supplied by the battery $= Ei = \frac{E^2}{R}e^{-t/CR}$

$$= \frac{(4)^2}{3\times10^6}e^{-1/3} = \frac{16}{3\times10^6}\times0.72$$

$$= \mathbf{3.84\times10^{-6}\ watt.}$$

(iii) Rate of production of heat in resistance $= i^2R = \frac{E^2}{R}e^{-2t/CR}$

$$= \frac{(4)^2}{3\times10^6}e^{-2/3} = \frac{16}{3\times10^6}\times(0.72)^2$$

$$= \mathbf{2.76\times10^{-6}\ watt.}$$

(iv) Rate of energy stored on the condenser

$$\frac{dU}{dt} = \frac{d}{dt}\left(\frac{q^2}{2C}\right) = \frac{q}{C}\frac{dq}{dt} = \frac{E^2}{R}e^{-t/CR}\left(1 - e^{-t/CR}\right) = \frac{(4)^2}{3 \times 10^6}e^{-1/3}\left(1 - e^{-1/3}\right) = \mathbf{1.08 \times 10^{-6}\ watt.}$$

Ex. 6. *The electric field between the plates of a parallel-plate capacitor of capacitance 1.0 μF drops to one-third of its initial value in 4.4 μs when the plates are connected by a thin wire. Find the resistance of the wire.*

Sol. The electric field between the plates is given by

$$E = \frac{q}{A\varepsilon_0} = \frac{q_0}{A\varepsilon_0}e^{-t/RC} \quad \text{or} \quad E = E_0 e^{-t/RC}.$$

Here, $E = \frac{1}{3}E_0$ at $t = 4.4$ μs. $= 4.4 \times 10^{-6}$ sec.

$$\therefore \quad \frac{1}{3} = e^{-\frac{4.4 \times 10^{-6}}{RC}} \quad \text{or} \quad \frac{4.4 \times 10^{-6}}{RC} = \log_e 3 \doteq 1.1$$

$$\therefore \quad R = \frac{4.4 \times 10^{-6}}{1.1 \times 1 \times 10^{-6}} = \mathbf{4\ ohm.}$$

Ex. 7. *A resistance R and a 2 μF capacitor in series are connected to a 200 volt direct supply. Across the capacitor is a neon lamp that ignites at 120 volt. Calculate the value of R to make the lamp to ignite 5 second after the switch is made on.* (*Given* : $\log_e 2 = 0.6931$, $\log_e 5 = 1.6094$)

(Meerut 1995 S)

Sol. The resistance R must be such that the potential difference across the capacitor should rise to 120 volt in 5 second after the switch is made on. The lamp would then ignite.

The charge on the capacitor at time t after the switch is on, is given by

$$q = q_0(1 - e^{-t/CR}) \qquad \text{...(i)}$$

where q_0 is the final steady charge.

In a capacitor, $q \propto V$. Hence we may write eq. (i) as

$$V = V_0(1 - e^{-t/CR}).$$

Here $V = 120$ volt, $V_0 = 200$ volt, $t = 5$ sec. and $C = 2 \times 10^{-6}$ farad.

$$\therefore \quad 120 = 200\{1 - e^{-5/(2\times10^{-6}R)}\}$$

or

$$e^{-5/(2\times10^{-6}R)} = 1 - \frac{120}{200} = 1 - \frac{3}{5} = \frac{2}{5} \quad \text{or} \quad e^{5/(2\times10^{-6}R)} = \frac{5}{2}$$

$$\therefore \quad \frac{5}{2 \times 10^{-6}R} = \log_e 5 - \log_e 2 = 1.6094 - 0.6931 = 0.9163$$

whence

$$R = \frac{5}{2 \times 10^{-6} \times 0.9163} = 2.73 \times 10^6\ \text{ohm} = \mathbf{2.73\ M\Omega}$$

Ex. 8. *How many time constants will be taken by a condenser to gain 99% of its steady state charge in a RC circuit ?*

Sol. Let q be the charge on a condenser at any instant and q_0 be its steady charge, then

$$q = q_0(1 - e^{-t/CR})$$

Here $q/q_0 = 99\% = 0.99$

Putting the value, we have

$$0.99 = 1 - e^{-t/\lambda} \quad \text{or} \quad e^{-t/\lambda} = 1 - 0.99 = 0.01$$

$$\therefore \quad e^{t/\lambda} = \frac{1}{0.01} = 100 \quad \text{or} \quad t/\lambda = \log_e 100 = 2.3026 \times \log_{10} 100$$

or

$$t/\lambda = 2.3026 \times 2, \ \therefore\ t = \mathbf{4.6\ \lambda}.$$

Ex. 9. *A condenser of capacity 2 μF is charged through a 5 MΩ resistance to any potential. After how much time the condenser will acquire half of maximum energy ?*

Sol. Here, $C = 2\ \mu F = 2 \times 10^{-6}$ farad, and $R = 5\ M\Omega = 5 \times 10^{6}$ ohm

So, time constant, $CR = (2 \times 10^{-6}) \times (5 \times 10^{6}) = 10$ second.

If q_0 be the maximum (final) charge on condenser then the maximum energy stored in condenser will be $U_0 = \frac{1}{2}\frac{q_0^2}{C}$. Similarly, if q be the charge on condenser at any instant t, then the energy of capacitor will be $U = \frac{1}{2}\frac{q^2}{C}$. Hence

$$\frac{U}{U_0} = \frac{q^2/2C}{q_0^2/2C} = \left(\frac{q}{q_0}\right)^2 \qquad \text{...(i)}$$

The charge q on condenser at any instant t during charging is given by following relation :

$$q = q_0(1 - e^{-t/CR}) \quad \text{or} \quad q/q_0 = (1 - e^{-t/CR}) \qquad \text{...(ii)}$$

From equations (i) and (ii)

$$\frac{U}{U_0} = \left(\frac{q}{q_0}\right)^2 = (1 - e^{-t/CR})^2 \text{ or } \frac{1}{2} = (1 - e^{-t/CR})^2 \text{ or } 1 - e^{-t/CR} = \frac{1}{\sqrt{2}} = 0.707$$

$\therefore$ $$e^{-t/CR} = 1 - 0.707 = 0.293 \quad \text{or} \quad e^{t/CR} = 1/0.293 = 3.413$$

or $$t/CR = \log_e 3.413 = 2.3026 \times \log_{10} 3.413 = 2.3026 \times 0.5332$$

whence $$t = CR \times 2.3026 \times 0.5332 = 10 \times 2.3026 \times 0.5332 = \mathbf{12.28\ sec.}$$

EXERCISE 7·3

1. A condenser of 0.5 μF is charged through a resistance of 10 mega-ohm at the potential difference of 100 volt. Calculate the time constant and maximum stored charge.
[**Ans.** 5 second, 5×10^{-6} coulomb.]

2. On the series combination of a resistance R and a condenser C, a steady e.m.f. is applied. Find the time in which the condenser will acquire 90% of total charge.
[**Ans.** 2.3026 CR.]

3. A condenser is charged with a D.C. source and a resistance of 2 mega-ohm is conneted in the circuit. If this condenser takes 0.5 second to acquire $\frac{3}{4}$ part of its maximum charge, then what is the capacity of the condenser ?
[**Ans.** 0.184 μF.]

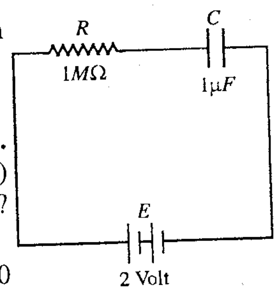

Fig. 7.38.

4. Obtain the expression for charge q and time t for the circuit given in **fig. 7.38.** How much time the condenser will take to acquire the 63% (approx) value of its maximum charge ? What is the value of maximum charge ?
[**Ans.** 1 second, 2×10^{-6} coulomb.]

5. A condenser of capacity 0.05 μF is discharged through a resistance of 10 mega-ohm. Find the required time to leak half of the charge in the condenser. ($\log_e 2 = 0.3010$)
[**Ans.** 3.46 sec.]

6. A condenser of capacity 0.1 μF is first charged and then discharged through a resistance of 10 MΩ. In how much time the potential difference of condenser falls to half its initial potential difference ?
[**Ans.** 0.693 sec.]

7. A condenser of capacity C is charged through a resistance R. In how many time constants. the energy of condensers will become half of its total energy ? **(Bundelkhand 2004)**
[**Ans.** 1.23 time constants.]

8. A condenser is charged through a resistance of 3 MΩ with a battery to a potential. If it takes 0.6 second to acquire $\frac{2}{3}$ of its final potential, then calculate the capacity of condenser.
[**Ans.** 0.2 μF.]

9. How many time constants will elapse before the charge falls to 1% of its maximum value in a discharging RC circuit ?
[**Ans.** 6.9.]

10. A capacitor is charged to a certain potential through a resistance of 3 mega-ohm. If it reaches 3/4 of its final potential in 0.5 second, calculate its capacitance. **(Meerut 1988 S)**
[**Ans.** 0.12 μF.]

11. A 2 μF capacitor is discharged through a high resistance. The time taken for half the charge on the capacitor to leak was found to be 20 sec. Compute the value of the resistance.
[**Ans.** 14.5 mega-ohm (MΩ).] **(Meerut 1988 S)**

12. Show that in the discharge of a capacitor through a resistance the charge falls to $1/e$ (= 0.368) of its initial value after a time equal to one time-constant of the circuit.

13. A capacitor of 0.5 μF is first charged and then discharged through a resistance of 10 MΩ. Calculate the time in which the charge on the capacitor will become 36.8% of its initial value.
[**Ans.** 5 sec.]

14. A condenser of capacity 2 μF is connected in series with a resistance of 1 mega-ohm and this combination is connected with a d.c. source of 300 volt. Find
(i) Initial charging current, (ii) Final charge, (iii) Time taken by charge to acquire 99% of its final value and (iv) The potential difference at the ends of the condenser and resistance after time interval equal to one time constant.
[**Ans.** (i) 3×10^{-4}A, (ii) 6×10^{-4}C, (iii) 9.21 sec., (iv) 3.79 C.]

15. A condenser of capacity 2 μF is charged through a resistance of 4 MΩ. In how much time, the condenser will acquire half of its maximum energy ?
[**Ans.** 5.5 second.]

16. In fig 7.39 the switch S is closed at $t = 0$. Find the values of i, $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at an instant t. **(Jabalpur 2004)**
[**Ans.** $i = 0.1\, e^{-10^4 t}$, $\frac{di}{dt} = -1000\, e^{-10^4 t}$, $\frac{d^2i}{dt^2} = 10^7 e^{-10^4 t}$.]

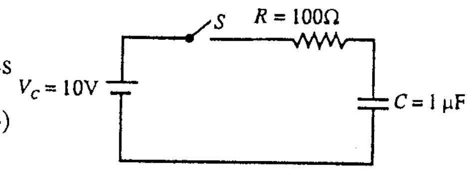

Fig. 7.39.

17. A 5.0 μF capacitor with a charge of 20 μC is discharged through a wire of resistance 5.0 Ω. Calculate the heat dissipated in the wire between 25 μs to 50 μs after the connections are made.
[**Ans.** 4.7 μJ.]

7·8. Discharge of a Capacitor through Inductance

Consider the circuit containing a condenser with capacity C, a coil of inductance L of negligible resistance and a source of e.m.f. E with a switch S. Let the capacitor be charged to a charge q_0 by connecting the switch S to A [fig. 7.40]. Now by throwing S to B, the condenser is discharged through the coil L. At any time t during the discharge, let q be the charge left on the capacitor and i be the current through the inductance. Then the potential difference across the capacitor is q/C and the induced e.m.f. in the inductance is $-L\,(di/dt)$ which acts opposite to that across the capacitor. Hence the effective e.m.f. in the circuit is $q/C - L\, di/dt$. Since there is no resistance in the circuit, this effective e.m.f. must be equal to zero *i.e.*,

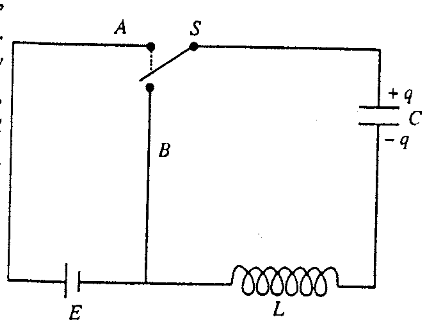

Fig. 7.40 : *LC* **circuit.**

$$\frac{q}{C} - L\frac{di}{dt} = 0 \qquad ...(1)$$

But $i = -dq/dt$ (minus sign indicates that q is decreasing with time). Hence

$$\frac{q}{C} - L\frac{d}{dt}\left(-\frac{dq}{dt}\right) = 0 \text{ or } L\frac{d^2q}{dt^2} + \frac{q}{C} = 0 \text{ or } \frac{d^2q}{dt^2} + \frac{q}{LC} = 0$$

Thus,
$$\frac{d^2q}{dt^2} + \omega_0^2 q = 0 \qquad ...(2)$$

where $\omega_0^2 = 1/LC$. This is a linear differential equation of second order as in the case of harmonic oscillator. Let the solution of eq. (2) be

$$q = Ae^{\alpha t} \qquad ...(3)$$

$$\therefore \quad \frac{dq}{dt} = A\alpha e^{\alpha t} \text{ and } \frac{d^2q}{dt^2} = A\alpha^2 e^{\alpha t}$$

Substituting for $\frac{d^2q}{dt^2}$ and q in eq. (2), we obtain

$$A\alpha^2 e^{\alpha t} + \omega_0^2 A e^{\alpha t} = 0 \text{ or } Ae^{\alpha t}(\alpha^2 + \omega_0^2) = 0 \text{ or } \alpha^2 + \omega_0^2 = 0;$$

$$\therefore \quad \alpha = \pm j\omega_0 \qquad ...(4)$$

where $j = \sqrt{-1}$. Thus, the eq. (5) has two solutions given by

$$q = Ae^{+j\omega_0 t} \text{ and } q = Ae^{-j\omega_0 t} \qquad ...(5)$$

Hence the general solution is

$$q = A_1 e^{+j\omega_0 t} + A_2 e^{-j\omega_0 t} \qquad ...(6)$$

where A_1 and A_2 are new arbitrary constants.

At $t = 0$, $q = q_0$, hence eq. (6) gives

$$q_0 = A_1 + A_2 \qquad ...(7)$$

Differentiating eq. (6), we get

$$i = \frac{dq}{dt} = A_1 e^{j\omega_0 t}(j\omega_0) + A_1 e^{-j\omega_0 t}(-j\omega_0).$$

At $t = 0$, $i = 0$ and therefore

$$0 = A_1 j\omega_0 - A_2 j\omega_0 \text{ or } A_1 = A_2 \qquad ...(8)$$

Therefore from eq. (7), $\quad A_1 = A_2 = q_0/2.$

Hence eq. (6) assumes the form

$$q = \frac{q_0}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t}) \text{ or } q = q_0 \cos\omega_0 t \qquad [\because \omega_0 = 1/\sqrt{LC}] \qquad ...(9)$$

Thus the discharge of the capacitor through an inductance is oscillatory and the charge executes simple harmonic motion.

The period (T) and frequency (ν) of the charge oscillations are given by

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{LC} \text{ and } \nu = \frac{1}{T} = \frac{1}{2\pi\sqrt{LC}} \qquad ...(10)$$

Thus after charging the condenser, if it is discharged through an inductance, then the charge oscillations in the LC circuit continue indefinitely, provided the resistance of the circuit is zero. Practically there is some resistance in the circuit and due to damping, the oscillation die out after some time. We consider next the charging and discharging of a condenser through an inductance L and resistance R.

7·9. *LCR* Circuit—Charging of a Capacitor through *LR* Circuit

We consider a circuit consisting of a capacitor C, an inductance L and a resistance R, joined in series. Now, a constant e.m.f. E is applied by connecting a switch S to A [**fig. 7.41**]. This results in a momentary current flow in the circuit and the capacitor gets charged.

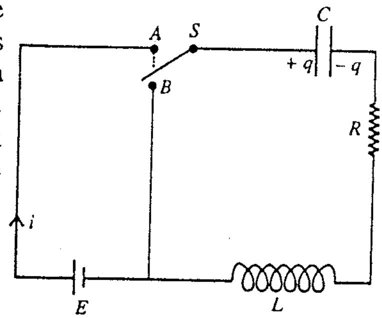

Fig. 7.41 : Charging of condenser through a *LR* circuit.

At instant t let q be the charge on the capacitor and i the current during charging. At this instant the potential difference across the capacitor is q/C and the self-induced e.m.f. in the inductance is $L\,(di/dt)$, both being opposite to the e.m.f. E. Therefore the effective e.m.f. in the circuit is $E - \frac{q}{C} - L\frac{di}{dt}$ which, by Ohm's law must be equal to Ri, *i.e.*,

$$E - \frac{q}{C} - L\frac{di}{dt} = Ri$$

or

$$L\frac{di}{dt} + Ri + \frac{q}{C} = E.$$

But $i = \frac{dq}{dt}$ and $\frac{di}{dt} = \frac{d^2q}{dt^2}$, hence

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = E$$

or

$$\frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} + \frac{q}{LC} = \frac{E}{L}. \qquad ...(1)$$

Let $\frac{R}{L} = 2k$ and $\frac{1}{LC} = \omega_0^2$. Therefore, from eq. (1), we get

$$\frac{d^2q}{dt^2} + 2k\frac{dq}{dt} + \omega_0^2 q = \frac{E}{L}$$

or

$$\frac{d^2q}{dt^2} + 2k\frac{dq}{dt} + \omega_0^2\left(q - \frac{E}{\omega_0^2 L}\right) = 0 \qquad ...(2)$$

We want to solve this equation for the instantaneous charge q on the capacitor. We introduce a variable x, where $x = q - E/\omega_0^2 L$;

$$\therefore \qquad \frac{dx}{dt} = \frac{dq}{dt} \text{ and } \frac{d^2x}{dt^2} = \frac{d^2q}{dt^2}$$

Now, eq. (2) takes the form

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + \omega_0^2 x = 0 \qquad ...(3)$$

Let the solution of this equation be

$$x = Ae^{\alpha t} \qquad ...(4)$$

$$\therefore \qquad \frac{dx}{dt} = A\alpha e^{\alpha t} \text{ and } \frac{d^2x}{dt^2} = A\alpha^2 e^{\alpha t}$$

Substituting in eq. (3), we get

$$A\alpha^2 e^{\alpha t} + 2kA\alpha e^{\alpha t} + \omega_0^2 A e^{\alpha t} = 0$$

or

$$\alpha^2 + 2k\alpha + \omega_0^2 = 0 \text{ or } \alpha = -k \pm \sqrt{k^2 - \omega_0^2}.$$

Hence the general solution of eq. (3) is

$$x = A_1 e^{(-k+\sqrt{k^2-\omega_0^2})t} + A_2 e^{(-k-\sqrt{k^2-\omega_0^2})t}$$

Hence $q = \dfrac{E}{\omega_0^2 L} + x$ or $q = \dfrac{E}{\omega_0^2 L} + A_1 e^{(-k+\sqrt{k^2-\omega_0^2})t} + A_2 e^{(-k-\sqrt{k^2-\omega_0^2})t}$ $\left[\because x = q - \dfrac{E}{\omega_0^2 L}\right]$

Now, $\dfrac{E}{\omega_0^2 L} = \dfrac{E}{(1/LC)\times L} = CE = q_0$ where q_0 is the *final steady charge on the capacitor.*

$$\therefore \quad q = q_0 + A_1 e^{(-k+\sqrt{k^2-\omega_0^2})t} + A_2 e^{(-k-\sqrt{k^2-\omega_0^2})t} \qquad ...(5)$$

At $t = 0$, charge $q = 0$ on the capacitor. Therefore from eq. (5), we get

$$0 = q_0 + A_1 + A_2 \text{ or } A_1 + A_2 = -q_0. \qquad ...(6)$$

Differentiating eq. (5), we obtain

$$i = \frac{dq}{dt} = A_1[-k + \sqrt{k^2-\omega_0^2}]e^{(-k+\sqrt{k^2-\omega_0^2})t} + A_2[-k - \sqrt{k^2-\omega_0^2}]e^{(-k+\sqrt{k^2-\omega_0^2})t}$$

But $i = 0$ at $t = 0$, hence

$$\begin{aligned} 0 &= A_1[-k + \sqrt{k^2-\omega_0^2}] + A_2[-k - \sqrt{k^2-\omega_0^2}] \\ &= -k(A_1 + A_2) + \sqrt{k^2-\omega_0^2}(A_1 - A_2) \\ &= kq_0 + \sqrt{k^2-\omega_0^2}(A_1 - A_2) \qquad [\because A_1 + A_2 = -q_0] \end{aligned}$$

$$\text{Therefore, } A_1 - A_2 = -\frac{kq_0}{\sqrt{k^2-\omega_0^2}}. \qquad ...(7)$$

From eq. (6) and (7), we obatin

$$A_1 = -\frac{q_0}{2}\left(1 + \frac{k}{\sqrt{k^2-\omega_0^2}}\right) \text{ and } A_2 = -\frac{q_0}{2}\left(1 - \frac{k}{\sqrt{k^2-\omega_0^2}}\right).$$

Substituting these values of A_1 and A_2 in eq. (5), we get

$$q = q_0 - \frac{q_0}{2}\left(1 + \frac{k}{\sqrt{k^2-\omega_0^2}}\right)e^{(-k+\sqrt{k^2-\omega_0^2})t} - \frac{q_0}{2}\left(1 - \frac{k}{\sqrt{k^2-\omega_0^2}}\right)e^{(-k-\sqrt{k^2-\omega_0^2})t} \qquad ...(8)$$

This is the solution of eq. (2). There are three cases :

(i) $k^2 > \omega_0^2$, (ii) $k^2 = \omega_0^2$ and (iii) $k^2 < \omega_0^2$.

Case I. When $k^2 > \omega_0^2$, *i.e.*, $\dfrac{R^2}{4L^2} > \dfrac{1}{LC}$, both the exponential terms in eq. (8) are real. In this case the charge gradually acquires its final value q_0 as shown by curve I in **fig. 7.43.**

Case II. When $k^2 = \omega_0^2$, *i.e.*, $\dfrac{R^2}{4L^2} = \dfrac{1}{LC}$. In this case eq. (8) breaks down because the two coefficients of the exponential terms become infinite. For such a case, we return to eq. (5) and put $\sqrt{k^2-\omega_0^2} = h$, where h is a very small positive quantity. Hence

$$\begin{aligned} q &= q_0 + A_1 e^{(-k+h)t} + A_2 e^{(-k-h)t} \\ &= q_0 + e^{-kt}(A_1 e^{ht} + A_2 e^{-ht}) \end{aligned}$$

For small ht, writing for exponential terms

$$\begin{aligned} q &= q_0 + e^{-kt}\{A_1(1 + ht + ...) + A_2(1 - ht + ...)\} \\ &= q_0 + e^{-kt}\{(A_1 + A_2) + h(A_1 - A_2)t\} \end{aligned}$$

$$\text{Thus,} \quad q = q_0 + e^{-kt}(P + Qt). \qquad ...(9)$$

where $P = (A_1 + A_2)$ and $Q = h\,(A_1 - A_2)$.

At $t = 0$, $q = 0$, hence from eq. (9), $P = -q_0$ ...(10)

Differentiating eq. (9), we get

$$i = \frac{dq}{dt} = -ke^{-kt}(P + Qt) + e^{-kt}Q.$$

Using the condition $i = 0$ at $t = 0$, we get

$$0 = -kP + Q = kq_0 + Q.$$

Hence $$Q = -kq_0 \qquad ...(11)$$

Putting these values of P and Q in eq. (9), we obtain

$$q = q_0 + e^{-kt}(-q_0 - kq_0 t)$$

or $$q = q_0 - q_0(1 + kt)e^{-kt} \qquad ...(12)$$

The growth of charge on the capacitor with time is shown by curve II in **fig. 7.43.**

Case III. For $k^2 < \omega_0^2$ *i.e.*, $\frac{R^2}{4L^2} < \frac{1}{LC}$. In this case $\sqrt{k^2 - \omega_0^2}$ is imaginary.

We put $\sqrt{k^2 - \omega_0^2} = j\sqrt{\omega_0^2 - k^2} = j\omega$, where $\omega = \sqrt{\omega_0^2 - k^2}$ and $j = \sqrt{-1}$. Hence eq. (8) is

$$\begin{aligned} q &= q_0 - \frac{q_0}{2}\left(1 + \frac{k}{j\omega}\right)e^{(-k+j\omega)t} - \frac{q_0}{2}\left(1 - \frac{k}{j\omega}\right)e^{(-k-j\omega)t} \\ &= q_0 - \frac{q_0}{2}e^{-kt}\left(e^{j\omega t} + \frac{k}{j\omega}e^{j\omega t} + e^{-j\omega t} - \frac{k}{j\omega}e^{-j\omega t}\right) \\ &= q_0 - q_0 e^{-kt}\left(\frac{e^{j\omega t} + e^{-j\omega t}}{2} + \frac{k}{\omega}\,\frac{e^{j\omega t} - e^{-j\omega t}}{2j}\right) \\ &= q_0 - q_0 e^{-kt}\left(\cos\omega t + \frac{k}{\omega}\sin\omega t\right) \qquad ...(13) \end{aligned}$$

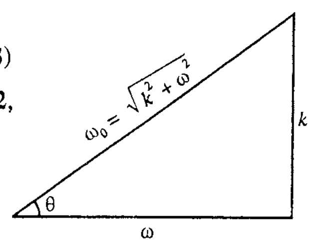

Fig. 7.42.

If we put $k/\omega = \tan\theta$ (where θ is a new constant), then from **fig. 7.42,** $\sin\theta = k/\omega_0$ and $\cos\theta = \omega/\omega_0$. Therefore,

$$\begin{aligned} q &= q_0 - q_0 e^{-kt}(\cos\omega t + \tan\theta\ \sin\omega t) \\ &= q_0 - q_0\frac{e^{-kt}}{\cos\theta}(\cos\theta\ \cos\omega t + \sin\theta\ \sin\omega t) \\ &= q_0 - q_0\frac{\omega_0 e^{-kt}}{\omega}\cos(\omega t - \theta), \qquad ...(14) \end{aligned}$$

where $\omega = \sqrt{\omega_0^2 - k^2}$. Thus, we see that for $\omega_0^2 > k^2$, **the charge is damped oscillatory.** After some oscillations it acquires the final steady value q_0, as shown by curve III in **fig. 7.43.** In this case the frequency of damped oscillations is given by

$$\nu = \frac{1}{2\pi}\sqrt{\omega_0^2 - k^2} = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \qquad ...(15)$$

Period of oscillations, $$T = \frac{1}{\nu} = \frac{2\pi}{\sqrt{(1/LC) - (R^2/4L^2)}} \qquad ...(16)$$

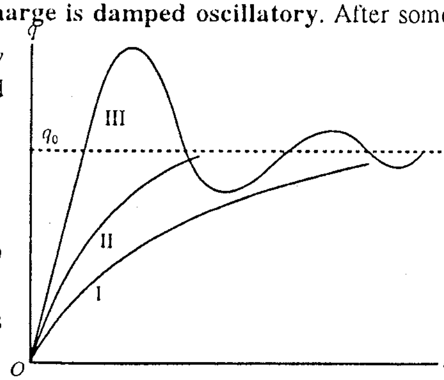

Fig. 7.43 : Charging of the condenser in ***LCR*** **circuit.**

Instantaneous current in the circuit when the charge is oscillatory, can be obtained by differentiating eq. (14) *i.e.*,

$$i = \frac{dq}{dt} = q_0\omega_0 e^{-kt}\sin(\omega t - \theta) + q_0\frac{\omega_0 k e^{-kt}}{\omega}\cos(\omega t - \theta)$$

$$= q_0 \frac{\omega_0 e^{-kt}}{\omega} [\omega \sin(\omega t - \theta) + k \cos(\omega t - \theta)]$$

$$= q_0 \frac{\omega_0^2 e^{-kt}}{\omega} \left[\frac{\omega}{\omega_0} \sin(\omega t - \theta) + \frac{k}{\omega_0} \cos(\omega t - \theta) \right]$$

$$= q_0 \frac{\omega_0^2 e^{-kt}}{\omega} [\cos\theta \sin(\omega t - \theta) + \sin\theta \cos(\omega t - \theta)]$$

$$\therefore \quad i = q_0 \frac{\omega_0^2 e^{-kt}}{\omega} \sin \omega t \qquad ...(17)$$

Thus when $\omega_0^2 > k^2$, the current in the circuit is also oscillatory.

7·10. *LCR* Circuit—Discharging of a Condenser through *LR* Circuit

We consider a circuit consisting of a capacitor C, an inductance L and resistance R, all connected in series. We connect a battery of e.m.f. E and switch S in the circuit **[fig. 7.41]**. When S is connected to A the capacitor C gets a steady charge q_0. Next the switch S is connected to B so that the capacitor discharges through the inductance L and resistance R and a current i flows through the circuit. Equivalent circuit is shown in **fig. 7.44.** During the discharge, at an instant t, if q is the charge left on the capacitor and i the current in the circuit, then the induced e.m.f. $-L\, di/dt$ set up in the inductance acts opposite to the potential difference q/C across the capacitor. Hence effective e.m.f. in the circuit is $q/C - L\, di/dt$ which drives the current i through the resistance R. Thus,

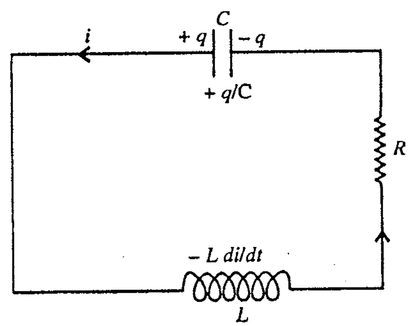

Fig. 7.44 : Discharge fo condenser through *LR* circuit.

$$\frac{q}{C} - L\frac{di}{dt} = Ri$$

As the charge is decreasing, $i = -dq/dt$. Hence

$$\frac{q}{C} - L\frac{d}{dt}\left(-\frac{dq}{dt}\right) = R\left(-\frac{dq}{dt}\right) \text{ or } L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0$$

or

$$\frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} + \frac{q}{LC} = 0 \qquad ...(1)$$

We put $R/L = 2k$ and $1/LC = \omega_0^2$ so that eq. (1) takes the form

$$\frac{d^2q}{dt^2} + 2k\frac{dq}{dt} + \omega_0^2 q = 0. \qquad ...(2)$$

Let the solution of this equation be, $q = Ae^{\alpha t}$. Hence

$$\frac{dq}{dt} = A\alpha e^{\alpha t} \text{ and } \frac{d^2q}{dt^2} = A\alpha^2 e^{\alpha t}$$

Substituting in eq. (2), we obtain

$$A\alpha^2 e^{\alpha t} + 2kA\alpha e^{\alpha t} + \omega_0^2 A e^{\alpha t} = 0 \text{ or } \alpha^2 + 2k\alpha + \omega_0^2 = 0$$

$$\therefore \quad \alpha = -k \pm \sqrt{k^2 - \omega_0^2} \qquad ...(3)$$

Hence the general solution of eq. (2) is

$$q = A_1 e^{(-k+\sqrt{k^2-\omega_0^2})t} + A_2 e^{(-k-\sqrt{k^2-\omega_0^2})t} \qquad ...(4)$$

where A_1 and A_2 are constants to be determined by the initial conditions.

As at $t = 0$, $q = q_0$. Therefore from eq. (4), we get

$$q_0 = A_1 + A_2. \quad ...(5)$$

Differeniating eq. (4) with respect to t, we get

$$i = \frac{dq}{dt} = A_1[-k + \sqrt{k^2 - \omega_0^2}]e^{(-k+\sqrt{k^2-\omega_0^2})t} + A_2[-k - \sqrt{k^2 - \omega_0^2}]e^{(-k-\sqrt{k^2-\omega_0^2})t} \quad ...(6)$$

But at $t = 0$, $i = 0$. Hence from eq. (6)

$$0 = A_1[-k + \sqrt{k^2 - \omega_0^2}] + A_2[-k - \sqrt{k^2 - \omega_0^2}]$$

or $$-k(A_1 + A_2) + \sqrt{k^2 - \omega_0^2}(A_1 - A_2) = 0$$

or $$-kq_0 + \sqrt{k^2 - \omega_0^2}(A_1 - A_2) = 0 \qquad [\because A_1 + A_2 = q_0]$$

$\therefore$ $$A_1 - A_2 = \frac{kq_0}{\sqrt{k^2 - \omega_0^2}} \quad ...(7)$$

Using eqs. (6) and (7), we get

$$A_1 = \frac{q_0}{2}\left(1 + \frac{k}{\sqrt{k^2 - \omega_0^2}}\right) \text{ and } A_2 = \frac{q_0}{2}\left(1 - \frac{k}{\sqrt{k^2 - \omega_0^2}}\right)$$

Substituting these values in eq. (4), we get

$$q = \frac{q_0}{2}\left(1 + \frac{k}{\sqrt{k^2 - \omega_0^2}}\right)e^{(-k+\sqrt{k^2-\omega_0^2})t} + \frac{q_0}{2}\left(1 - \frac{k}{\sqrt{k^2 - \omega_0^2}}\right)e^{(-k-\sqrt{k^2-\omega_0^2})t} \quad ...(8)$$

Now, we consider three cases : (i) $k^2 > \omega_0^2$, (ii) $k^2 = \omega_0^2$ and $k^2 < \omega_0^2$.

Case I. For $k^2 > \omega_0^2$, the quantity $\sqrt{k^2 - \omega_0^2}$ is real and positive. Hence as t approaches ∞, the exponential terms with negative powers in eq. (8) approach to zero, *i.e.*, the charge q decreases exponentially with time. This is the case of **over-damped, non-oscillatory** or **dead-beat discharge** and is shown **in fig. 7.46** by curve I.

Case II. When $k^2 = \omega_0^2$, eq. (8) breaks down as the coefficients of the exponential terms become infinite. We return to eq. (4) and use $\sqrt{k^2 - \omega_0^2} = h$, where h is very small positive quantity. Therefore,

$$q = A_1e^{(-k+h)t} + A_2e^{(-k-h)t} = e^{-kt}(A_1e^{ht} + A_2e^{-ht}) = e^{-kt}[A_1(1 + ht + ...) + A_2(1 - ht + ...)]$$
$$= e^{-kt}[(A_1 + A_2) + h(A_1 - A_2)t]$$

Let us put $(A_1 + A_2) = P$ and $h(A_1 - A_2) = Q$. Then

Thus, $$q = e^{-kt}(P + Qt) \quad ...(9)$$

Now, at $t = 0$, $q = q_0$ so that from eq. (9), we have $q_0 = P$ or $P = q_0$.

Differentiating eq. (9), we get

$$i = \frac{dq}{dt} = -ke^{-kt}(P + Qt) + e^{-kt}Q.$$

As at $t = 0$, $i = 0$, hence

$$0 = -kP + Q \text{ or } = -kq_0 + Q = 0 \qquad [\because P = q_0]$$

$\therefore$ $Q = kq_0$.

Putting these values of P and Q in eq. (9), we obtain

$$q = e^{-kt}(q_0 + kq_0t) \text{ or } q = q_0(1 + kt)e^{-kt} \quad ...(10)$$

According to this equation as t increases, first term of right hand side increases but its second term decreases exponentially. The discharge in this case is just non-oscillatory and is said to **be critically damped** and is shown in **fig. 7.46** by the curve II.

Case III. For $k^2 < \omega_0^2$, $\sqrt{k^2 - \omega_0^2}$ is an imaginary quantity. We put $\sqrt{k^2 - \omega_0^2} = j\sqrt{\omega_0^2 - k^2} = j\omega$ where $\omega = \sqrt{\omega_0^2 - k^2}$ and $j = \sqrt{-1}$. Then from eq. (8), we have

$$q = \frac{q_0}{2}\left(1 + \frac{k}{j\omega}\right)e^{(-k+j\omega)t} + \frac{q_0}{2}\left(1 - \frac{k}{j\omega}\right)e^{(-k-j\omega)t} = \frac{q_0}{2}e^{-kt}\left(e^{j\omega t} + \frac{k}{j\omega}e^{j\omega t} + e^{-j\omega t} - \frac{k}{j\omega}e^{-j\omega t}\right)$$

$$= q_0 e^{-kt}\left(\frac{e^{j\omega t} + e^{-j\omega t}}{2} + \frac{k}{\omega}\frac{e^{j\omega t} - e^{-j\omega t}}{2j}\right)$$

Thus, $q = q_0 e^{-kt}\left(\cos \omega t + \frac{k}{\omega}\sin \omega t\right)$ ...(11)

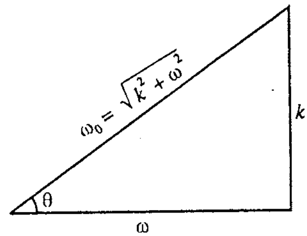

Fig. 7.45.

We take $k/\omega = \tan\theta$, (where θ is a constant). Then from **fig. 7.45**, sin $\theta = \sin k/\omega_0$ and $\cos\theta = \omega/\omega_0$. Therefore eq. (11) can be written as

$$q = q_0 e^{-kt}(\cos\omega t + \tan\theta \sin\omega t)$$

$$= q_0 \frac{e^{-kt}}{\cos\theta}(\cos\theta \cos\omega t + \sin\theta \sin\omega t)$$

i.e., $$q = q_0 \frac{\omega_0 e^{-kt}}{\omega}\cos(\omega t - \theta),$$ ...(12)

For small damping $\omega = \sqrt{\omega_0^2 - k^2} \approx \omega_0$,

$$\therefore\ q = q_0 e^{-kt}\cos(\omega t - \theta)$$ ...(13)

where $\omega = \sqrt{\omega_0^2 - k^2}$. Hence the discharge is **damped oscillatory**, as shown in **fig. 7.46** by curve III. The condition for oscillatory discharge is

$$k^2 < \omega_0^2 \text{ or } \frac{R^2}{4L^2} < \frac{1}{LC} \text{ or } R < 2\sqrt{\frac{L}{C}}$$ ...(14)

The frequency oscillation is, ν

$$= \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\omega_0^2 - k^2} = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$ ...(15)

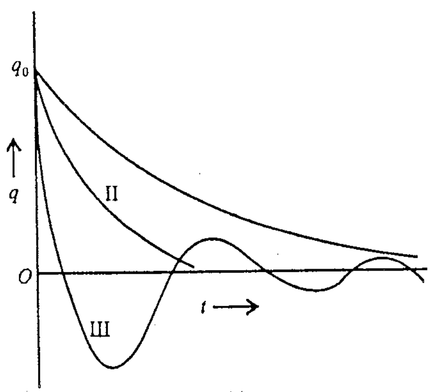

Fig. 7.46.

Period of oscillation, $$T = \frac{1}{\nu} = \frac{2\pi}{\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}}$$ ...(16)

Discharge in *LCR* Circuit with Negligible Resistance ($R = 0$)

If in *LCR* circuit, the damping is zero (*i.e.*, $k = 0$) or resistance is negligible ($R = 0$), then from eq. (12) instantaneous charge in the circuit is given by

$$q = q_0 \cos\omega_0 t$$ ...(17)

where angular frequency of oscillation $\omega_0 = 1/\sqrt{LC}$.

Hence in absence of damping ($R = 0$), the frequency and period of oscillation of the charge is given by

$$\nu_0 = \frac{1}{2\pi\sqrt{LC}} \text{ and } T_0 = \frac{1}{\nu_0} = 2\pi\sqrt{LC}$$ ...(18)

This is the expression for the natural frequency of an *LC* circuit (in absence of damping). Thus $\nu < \nu_0$ *i.e.*, the frequency in the presence of damping is less than that in absence of damping. Reverse case is for the period of oscillations *i.e.*, $T > T_0$.

The current in the circuit is given by

$$i = -\frac{dq}{dt} = q_0\omega_0 \sin \omega_0 t = \frac{q_0}{\sqrt{LC}} \sin \frac{t}{\sqrt{LC}} \qquad ...(19)$$

Importance in Wireless Telegraphy :

If in an *LCR* circuit the resistance *R* is less than $2\sqrt{L/C}$ [eq. (14)], the discharge of a capacitor is oscillatory. During the discharge, the energy of the charged capacitor is stored in the magnetic field of the inductance *L*, then again back in the electric field between the plates of capacitor *C* and the process continues. If the fields change rapidly, some energy from the circuit comes out in the form of electromagnetic waves, moving in the space with speed of light. Such electromagnetic waves make the base for wireless telegraphy.

7·11. Quality Factor *Q*

The relative damping of any oscillatory circuit is represented by a dimensionless quantity *Q*, which is called the **quality factor.** If the damping is low, then the value of *Q* is high. *The quality factor is defined as 2π times the ratio of energy stored at any instant within the circuit to the loss in energy in one period of oscillation i.e.,*

$$q = 2\pi \times \frac{\text{Energy stored in circuit}}{\text{Loss in energy in one period}}$$

When the condenser has the maximum charge during discharging, all the energy is stored in the form of electrical energy of the condenser. The maximum value of charge at any instant *i.e.*, amplitude of oscillation is given by (for small damping) [from eq. (13), Art. 7.10] :

$$q_m = q_0 e^{-kt} \qquad ...(1)$$

because for small damping $\omega = \sqrt{\omega_0^2 - k^2} \approx \omega_0.$

Hence, the energy stored in condenser at any instant *t* is,

$$U = \frac{q_m^2}{2C} = \frac{1}{2C} q_0^2 e^{-(R/L)t} = U_0 e^{-2kt} \qquad ...(2)$$

where we have put $U_0 = q_0^2/2C$.

Energy stored in the condenser after one complete cycle,

$$U' = U_0 e^{-2k(t+T)} \qquad ...(3)$$

Thus, the loss of energy in one complete cycle,

$$\Delta U = U - U' = U_0\left[e^{-2kt} - e^{-2k(t+T)}\right] = U_0 e^{-2kt}(1 - e^{-2kT})$$

If the damping is low, *i.e.*, the value of *k* is small, then

$$\Delta U = U_0 e^{-2kt}[1 - (1 - 2kT + ...)] = U_0 e^{-2kt} \times 2kT \qquad ...(4)$$

$$\text{Quality factor, } Q = 2\pi \frac{U}{\Delta U} = 2\pi \times \frac{U_0 e^{-2kt}}{U_0 e^{-2kt}(2kT)}$$

or

$$Q = \frac{\pi}{kT} = \frac{\pi}{\frac{R}{2L} \times \frac{2\pi}{\omega}} = \frac{\omega L}{R} \qquad ....(5)$$

For low value of damping or resistance, $\omega \approx \omega_0$ and then

$$Q = \frac{\omega_0 L}{R}. \qquad ...(6)$$

Ex. 1. *Find whether the discharge of a condenser through the following circuit is oscillatory :* $C = 0.1$ *micro-farad,* $L = 10$ *mlli-henries,* $R = 200$ *ohms ? If the discharge is oscillatory find its frequency.* **(Meerut 1978; Agra)**

Sol. As we know, the discharge is oscillatory, when $(R^2/4L^2) < (1/LC)$

Here, $R = 200$ ohms, $L = 10 \times 10^{-3}$ henries and $C = 0.1 \times 10^{-6}$ farad. So that

$$\frac{R^2}{4L^2} = \frac{200 \times 200}{4 \times (10 \times 10^{-3})^2} = 10^8 \text{ and } \frac{1}{LC} = \frac{1}{10 \times 10^{-3} \times 0.1 \times 10^{-6}} = 10^9$$

Hence, $\frac{1}{LC} > \frac{R^2}{4L^2}$. *i.e.*, the discharge of the condenser is oscillatory in character and its frequency is given by

$$\omega = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \frac{1}{2\pi}\sqrt{10^9 - 10^8} = \frac{10^4}{2\pi}\sqrt{10-1} = \frac{10^4 \times 3 \times 7}{2 \times 22} = \mathbf{4772.7\ Hz.}$$

Ex. 2. *Calculate the frequency of oscillations produced in a circuit composed of an inductance of* 0.5 *henry, a capacity of* 0.1 *micro-farad and a resistance of* 4000 *ohms. What will be the frequency, if the resistance is reduced to a negligible value ?* **(Poona 1964)**

Sol. The frequency of an oscillatory discharge, as we know, is given by

$$\omega = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

Here, $R = 4000$ ohm; $L = 0.5$ henry and $C = 0.1 \times 10^{-6}$ farad.

$$\therefore \quad \frac{1}{LC} = \frac{1}{0.5 \times 0.1 \times 10^{-6}} = 2 \times 10^7 \text{ and } \frac{R^2}{4L^2} = \frac{4000 \times 4000}{4 \times 0.5 \times 0.5} = 16 \times 10^6$$

Hence, $$\omega = \frac{1}{2\pi}\sqrt{2 \times 10^7 - 16 \times 10^6} = \frac{10^3 \times 2 \times 7}{2 \times 22} = \mathbf{318.2\ Hz.}$$

If $R = 0$, $$\omega = \frac{1}{2\pi}\sqrt{2 \times 10^7} = \frac{10^3\sqrt{20}}{2 \times 3.142} = \mathbf{711.7\ Hz.}$$

Ex. 3. *In an oscillatory circuit, L* = 0.2 *henry, C* = 0.0012 *microfarad. What is the maximum value of the resistance so that the circuit may oscillate.* **(Meerut 1982; Punjab; Delhi B Sc. (Hons.))**

Sol. For the discharge to be oscillatory, the condition, as we know, is that $\frac{R^2}{4L^2} < \frac{1}{LC}$.

Thus, the maximum value of R, for which the circuit may be oscillated, is just less than that given by

$$\frac{R^2}{4L^2} = \frac{1}{LC} \quad \text{or} \quad R = \sqrt{\frac{4L}{C}}$$

$$\therefore \quad R = \sqrt{\frac{4 \times 0.2}{0.0012 \times 10^{-6}}} = \mathbf{2.58 \times 10^4\ ohms.}$$

Ex. 4. *In an LCR circuit, C* = $2\mu F$, *L* = $2H$, *R* = $1\ \Omega$. *Calculate* : (i) *the quality factor of the circuit,* (ii) *the frequency of electrical oscillations,* (iii) *the time in which the amplitude of oscillations of discharge falls to* 1%. **(Ujjain 2004)**

Sol. (i) Quality factor, $$Q = \frac{\omega L}{R}$$

Here, $$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \sqrt{\frac{1}{2 \times 2 \times 10^{-6}} - \frac{(1)^2}{4 \times (2)^2}} = \frac{10^3}{2} = 500$$

$$\therefore \quad Q = \frac{\omega_0 L}{R} = \frac{500 \times 2}{1} = \mathbf{1000}$$

(ii) Frequency of electrical oscillations $\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \times 500 = \mathbf{79.6\ Hz.}$

(iii) The charge at any instant t on the condenser during its discharge is $q = q_0 e^{-kt}$, where $k = R/2L$

At $t = 0$, amplitude of charge $= q_0$

At instant t, amplitude of charge $q = \frac{1}{100} \times q_0$

$$\therefore \frac{1}{100} q_0 = q_0 e^{-kt} \text{ or } e^{-kt} = \frac{1}{100} \text{ or } e^{kt} = 100$$

or $$kt = \log_e 100 \text{ or } t = \frac{\log_e 100}{k} = \frac{2.3026 \times \log_{10} 100}{R/2L} = 2.3026 \times \frac{2L}{R} \times 2.000$$

$$\therefore \quad t = \frac{2.3026 \times 2 \times 2 \times 2.000}{1} = \mathbf{18.42 \text{ sec.}}$$

Ex. 5. *A condenser of capacity* 0.05 μF *is charged through a battery of potential difference* 200 *volt, connecting a coil of resistance* 100 Ω *and inductance* 10 *mH. Calculate the final charge on condenser. Show that during charging, the current in the circuit will be damped oscillatory. Calculate the time period of oscillatory current.*

Sol. Here, $C = 0.05\ \mu F = 0.05 \times 10^{-6}$ F, $R = 100\ \Omega$, $L = 10$ mH $= 10 \times 10^{-3}$ H and V = 200 volt.

Final charge on condenser $q_0 = CV = 0.05 \times 10^{-6} \times 200 = 10^{-5}$ coulomb $= \mathbf{10\ \mu C}$

Here, $$\frac{1}{LC} = \frac{1}{10 \times 10^{-3} \times 0.05 \times 10^{-6}} = 2 \times 10^9 \text{ sec}^{-1}$$

and $$\frac{R^2}{4L^2} = \frac{(100)^2}{4 \times (10 \times 10^{-3})^2} = 2.5 \times 10^7 \text{ sec}^{-1}$$

As $\frac{R^2}{4L^3} << \frac{1}{LC}$, during charging, the current in the circuit is oscillatory. Its time period is given by

$$T = \frac{2\pi}{\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}} = \frac{2 \times 3.14}{\sqrt{2 \times 10^9 - 2.5 \times 10^7}} = \frac{6.28}{\sqrt{(20 - 0.25) \times 10^8}} = \mathbf{1.413 \times 10^{-4} \text{ sec.}}$$

Ex. 6. *In a series LCR circuit, L = 10 mH, C = 1.0 μF and R = 0.1 ohm. How much time does the discharge oscillation take to decay half of its amplitude ? Find the angular frequency and Q-factor of the circuit.*

Sol. At any instant t, for oscillatory discharge

$$q = q_0 \frac{\omega_0 e^{-kt}}{\omega} \cos(\omega t - \theta)$$

where $\omega = \sqrt{\omega_0^2 - k^2}$, $\omega_0 = 1/\sqrt{LC}$ and $k = R/2L$.

The amplitude of this discharge is $q_0 \frac{\omega_0 e^{-kt}}{\omega}$.

Here, $$k = \frac{R}{2L} = \frac{0.1 \text{ ohm}}{2(10 \times 10^{-3} \text{ henry})} = 5 \text{ sec}^{-1}$$

If a be the amplitude at time $t = 0$ and $a/2$ at t, then we have

$$a = q_0 \frac{\omega_0}{\omega} \quad ...(i) \qquad \text{and} \quad \frac{a}{2} = q_0 \frac{\omega_0}{\omega} e^{-kt} \quad ...(ii)$$

Dividing eq. (ii) by eq. (i),

$$e^{-kt} = \frac{1}{2} \text{ or } e^{kt} = 2 \text{ or } kt = \log_e 2 = 0.69 \text{ or } t = \frac{0.69}{k} = \frac{0.69}{5} = \mathbf{0.138 \text{ sec.}}$$

$$\text{Angular frequency, } \omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \sqrt{\frac{1}{10 \times 10^{-3} \times 10^{-6}} - \frac{1}{4 \times (10 \times 10^{-3})^2}}$$

$$= \sqrt{10^8 - 2.5 \times 10^{-5}} = \mathbf{10^4\ Hz.}$$

Q-factor of the circuit, $Q = \dfrac{\omega L}{R} = \dfrac{10^4 \times 10 \times 10^{-3}}{1} = \mathbf{10^3}.$

Ex. 7. *A condenser of capacity* $(20/\pi)$ μF *discharged with a coil of inductance* $(50/\pi)$ *mH. Calculate the frequency of discharged oscillations.* **(Gwalior 2004)**

Sol. Here, $C = \dfrac{20}{\pi}\,\mu\text{F} = \dfrac{20}{\pi} \times 10^{-6}\,\text{F}$, $L = \dfrac{50}{\pi}\,\text{mH} = \dfrac{50}{\pi} \times 10^{-3}\,\text{H}$

Frequency of discharge oscillations in the circuit is given by

$$\nu = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{\left(\frac{50}{\pi} \times 10^{-3}\right) \times \left(\frac{20}{\pi} \times 10^{-6}\right)}} = \frac{1}{2\sqrt{10^{-6}}} = \frac{10^3}{2} = \mathbf{500\ Hz.}$$

Ex. 8. *If a condenser of* 20 *micro-farad capacity be charged to a potential difference of* 15000 *volts and then suddenly short-circuited by a coil of negligible resistance and of inductance* 0.002 *henry, calculate the frequency and maximum amplitude of the current oscillations.* **(Gorakhpur; Delhi B.Sc. (Hons))**

Sol. If the resistance in the circuit is zero, the charge at any instant t is given by

$$q = q_0 \cos (t/\sqrt{LC})$$

Hence, the resulting oscillatory current is given by

$$i = -\frac{dq}{dt} = \frac{q_0}{\sqrt{LC}} \sin \frac{t}{\sqrt{LC}}$$

The amplitude of this oscillatory current is $q_0/\sqrt{LC}$ and its frequency $1/2\pi\sqrt{LC}$.

Here $C = 20$ μfd $= 20 \times 10^{-6}$ F; $L = 0.002$ H and potential difference $V = 15000$ V.

$\therefore\ q_0 = C \times V = 20 \times 10^{-6} \times 15000 = 0.3$ C

Hence, current amplitude $= \dfrac{q_0}{\sqrt{LC}} = \dfrac{0.3}{\sqrt{0.002 \times 20 \times 10^{-6}}} = \mathbf{1500\ A.}$

and, frequency $= \dfrac{1}{2\pi\sqrt{LC}} = \dfrac{1}{2\pi\sqrt{0.002 \times 20 \times 10^{-6}}} = \mathbf{795\ Hz.}$

Ex. 9. *A* $\sqrt{3}\mu F$ *capacitor is charged to* 2 *volt and then connected to a pure inductor of inductance* 1 *mH. How much charge is present on the capacitor when one-third of the total energy is in the electric field and the rest is in the magnetic field ?*

Sol. Total energy on the charged capacitor is $U = \frac{1}{2}CV^2 = \frac{1}{2}\frac{q^2}{C}$

When the capacitor is connected to the inductor, one-third of the total energy *i.e.*, $U/3$ is present in the electric field between the plates of the capacitor.

If q be the *charge on the capacitor corresponding the energy* $U/3$, then

$$\frac{U}{3} = \frac{1}{2}\frac{q^2}{C} \text{ or } q^2 = \frac{2CU}{3} = \frac{2}{3}C\left(\frac{1}{2}CV^2\right) = \frac{1}{3}C^2V^2 \text{ or } q = \frac{CV}{\sqrt{3}}$$

Therefore, for the given data, $q = \dfrac{\sqrt{3} \times 10^{-6} \times 2}{\sqrt{3}} = 2 \times 10^{-6}\ \text{C} = \mathbf{2\ \mu C}.$

Ex. 10. *A condenser of an oscillatory circuit of low resistance is enclosed in a container. When the container is, evacuated the frequency of the circuit is* 100 *kilo-cycles and when filled with carbon dioxide, the frequency changes by* 50 *cycles. Calculate the dielectric constant of the gas.* **(Agra)**

Sol. Let C_0 be the capacity of the condenser in vacuum. When filled with gas of dielectric constant K, the capacity becomes $C = KC_0$.

In the first case, $10^5 = \frac{1}{2\pi\sqrt{LC_0}}$ and in second case, $99950 = \frac{1}{2\pi\sqrt{LKC_0}}$

$\therefore \quad \frac{10^5}{99950} = \sqrt{K}$, whence, $K = \left(\frac{10^5}{99950}\right)^2$ = **1.001.**

EXERCISE 7·4

1. In a LCR circuit, a capacity 25 μF, inductance 1.0 H and resistance 2.0 Ω are connected. Calculate the oscillatory frequency and quality factor Q of the circuit. **(Bilaspur 2003)**
   [**Ans.** 31.8 Hz, 100.]
2. A condenser of capacity 0.25 μF, is charged upto a certain voltage and then discharged through a circuit of inductance 1/10 henry and resistance 100 ohms. Find the frequency of the discharge.
   [**Ans.** 1000 Hz.] **(Allahabad)**
3. Deduce the condition under which the discharge of a condenser is oscillatory in character. Examine the discharge to be oscillatory if R = 400 ohms, L = 0.1 henry and C = 1 μF. **(Agra)**
   [**Ans.** Yes.]
4. Examine if the discharge will be oscillatory, when C = 2 μF, L = 0.15 henry, and R = 150 ohms. If so, find the frequency. **(Sagar; Agra)**
   [**Ans.** Yes; 279.5 Hz.]
5. A condenser of capacity 1 μF, an inductance 0.2 henry and a resistance of 800 ohms are joined in series. Is the circuit oscillatory ? **(Delhi)**
   [**Ans.** Yes.]
6. A capacitor of 2 μF, an inductor of 0.15 H and a resistor of 15 Ω are joined in series. Is the circuit oscillatory ? If yes, than calculate its frequency. **(Meerut 1990)**
   [**Ans.** Yes; 291 Hz.]
7. A condenser of capacity 0.25 μF is charged upto a certain voltage and then dischaged through a circuit of inductance 0.1 henry and resistance 100 ohm. Show that the discharge is oscillatory. Also calculate the frequency of the discharge. **(Meerut 1983)**
   [**Ans.** 1000 Hz.]
8. There is a coil of self-inductance 5 mH and resistance 0.5 ohm. Calculate the capacitance required to generate oscillations of frequency 10^3 Hz.
   [**Ans.** 5 μF.]
9. In an oscillatory electric circuit L = 0.3H, C = 30 μF. For what maximum value of resistance, the circuit is oscillatory ? **(Rewa 2004, 03)**
   [**Ans.** 200 ohm.]
10. A condenser of capacity 1 micro-farad is connected to an inductance of 1 milli-henry. Calculate the natural frequency of the circuit. **(Delhi)**
    [**Ans.** 1891 Hz.]
11. Determine the frequency of a circuit of negligible resistance when C = 0.001 μfd and L = 0.003 henry. **(Allahabad)**
    [**Ans.** 9.194×10^4 Hz.]
12. A condenser of capacity 1.5 μF is charged to a certain voltage and then discharged through a circuit containing an inductance of 0.015 henry and a resistance of 1000 ohms. Is the discharge oscillatory ?
    [**Ans.** No.]
13. A condenser of capacity 20 μF is first charged to a potential difference of 15000 volt and then is short circuited through a coil of negligible resistance and inductance 0.002 henry. Find the frequency of oscillations and maximum amplitude of the resultant current.
    [**Ans.** 796.2 Hz; 1500 A.]

14. If a capacitor of 2.0 μF is charged to 20 volt and then suddenly short-circuited by a coil of negligible resistance and of inductance 8.0 μH, calculate the maximum amplitude and frequency of the resulting current oscillations. **(Meerut 1980)**
[**Ans.** 10 A.]
15. A condenser of capacity 1.0 μF is charged to a potential difference of 40 volt and then it is discharged through a coil of negligible resistance and inductance 10 mH. Find : (i) maximum current in the inductance, (ii) the remaining charge on condenser and the time in which half of the total energy is stored in electric field and half in the magnetic field.
[**Ans.** (i) 0.4 A; (ii) 0.283 μC, 78.5 μs.]

QUESTIONS

Long Answer Questions

1. What are steady and non-steady currents. Define electric current and current density. How are they related ?
2. Establish the equation of continuity $\operatorname{div} J + \partial\rho/\partial t = 0$ for time varying or non-steady current. **(Kanpur 2005; Gwalior 03; Jabalpur 03; Raipur 03)**
3. Derive the equation of continuity for steady current. **(Agra 2004)**

or

Show that for steady current the equation of continuity is $\operatorname{div} \vec{J} = 0$.

4. Derive equation of continuity for charge and current. Hence show that for a non-steady current distribution, divergence of current density is equal to the rate of decrease in charged density. **(Indore 2003)**
5. State and explain Kirchoff's laws of electrical network. **(Bhopal 2004)**
6. Establish equation for the growth of current in a circuit containing a resistance and inductance and solve it. Explain the time constant of circuit and find its value. **(Jabalpur 2004; Ujjain 03; Sagar 03; Raipur 03; Rewa 04)**
7. Derive expressions for the growth and decay of current in a circuit containing an inductance, a resistance and a source of constant e.m.f. in series. Explain the significance of time constant and show that the rate of growth of current is complementary of the rate of decay of current in the circuit. What is the source of decaying current ? **(Meerut 1998, 97)**
8. Analyse the rise and decay of current in *C-R* circuit. **(Agra 2004)**
9. Discuss the charging and discharging of a capacitor through a resistance. What is time-constant ? **(Meerut 2002 S, 1999)**
10. Show graphically the variation of current in a *R-C* circuit while charging and discharging of the capacitor and compare the behaviour in the two cases. **(Meerut 1990, 88)**
11. Obtain an expression for growth of charge in a *LCR* circuit. Under what condition the charging will be oscillatory and then what will be the frequency of oscillation ? **(Meerut 2003)**
12. Discuss the nature of discharge of a condenser in a circuit containing an inductance and a resistance. Obtain condition required for the discharge to be oscillatory. Establish the expression for the frequency and quality factor of oscillatory discharge. **(Rewa 2003; Bhopal 03)**

Short Answer Questions

1. Define current and current density. Why the former is scalar and the later is vector ?
2. What are steady and non-steady currents ?
3. What are varying currents ?
4. What are transient currents ?
5. What is the equation of continuity for charge and current ?
6. Discuss the equation of continuity for steady currents ?
7. What is meant by time constant in a circuit containing a resistance and an inductance ? Write its value. **(Ujjain 1993; Rewa 95)**

8. Explain why a larger e.m.f. is induced in a circuit at break than at make. **(Indore 1995; Rewa 95)**
9. Define time constant for *CR* circuit.
10. A coil of inductance 2 mH and 1 ohm resistance are connected in series with a battery of e.m.f. 2 volt. Find the value of current after 5 seconds. **(Ujjain 2003)**
[**Ans.** Nearly 2 A.]
11. A coil of inductance 10 H is connected in series with a resistance of 10 ohm and a battery of e.m.f. 100 volt. Find the time constant of the circuit. What is the current in the circuit after 1.0 s ? What will be the maximum steady current in the circuit ? **(Raipur 1994; Bhopal 93)**
[**Ans.** Time constant, $L/R = 1$ s, current in circuit after 1.0 s will be 0.632 i_0 where the maximum steady current $i_0 = E/R = 100/10 = 10$ A.]
12. Define time constant of a circuit containing a resistance and a condenser. Write its expression. If $C = 1.0$ μF and $R = 1.0$ mega-ohm, find the value of time constant. **(Jabalpur 1995; Bhopal 93; Rewa 94)**
[**Ans.** Time constant = *CR*; time constant = 1.0 s]
13. What is the effect of introducing a dielectric between the plates of the condenser in *LCR* circuit on the frequency of discharge oscillations ?
[**Ans.** Frequency will decrease.]
14. Write the condition for discharge of a condenser in *LCR* circuit to be oscillatory. How is the frequency of the oscillations affected on increasing the resistance in the circuit ?
[**Ans.** Condition $R < 2\sqrt{L/C}$; Increase of *R* decreases the frequency and the discharge is non-oscillatory for $R \geq 2\sqrt{L/C}$.]
15. What do you mean by the quality factor of a *LCR* circuit ? **(Gwalior 1991)**
16. Discuss the importance of oscillatory discharge in *LCR* circuit in wireless telegraphy.
17. In an *LCR* circuit, if the resistance is reduced to zero, whether the frequency of oscillatory discharge will increase or decrease ?
[**Ans.** Increase.]
18. Find an expresion for the impedance of *LCR* circuit. **(Agra 2006)**

Objective Type Questions

1. Choose the correct statment :
(a) Current is a vector, while current density is a scalar.
(b) Current is a scalar, while current density is a vector.
(c) Both current and current density are vectors.
(d) Both current and current density are scalars.
2. The current *i* flowing through a conductor of cross-section area *A* is related to current density *J* as :
(a) $J \propto i$ (b) $J \propto 1/i$ (c) $J = i$ (d) None of these **(Bundelkhand 2004)**
3. The equation of continuity for steady curren is :
(a) $\text{div}\vec{J} = 0$ (b) $\text{div}\vec{J} = \partial\rho/\partial t$ (c) $\text{div}\vec{J} = -\partial\rho/\partial t$ (d) $\text{div}\vec{J} = \rho$ **(Rewa 2003)**
4. The equation of continuity for charge-current is :
(a) $\text{div}\vec{J} = 0$ (b) $\text{div}\vec{J} = \rho$ (c) $\text{div}\vec{J} = \partial\rho/\partial t$ (d) $\text{div}\vec{J} = -\partial\rho/\partial t$
5. Kirchoff's first law is based on :
(a) conservation of mass (b) conservation of charge
(c) conservation of energy (d) conservation of momentum.

6. Due to flow of current i in a conductor of self inductance L, the e.m.f. produced at any instant is :
(a) $L\frac{di}{dt}$ (b) $\frac{1}{L}\frac{di}{dt}$ (c) $-L\frac{di}{dt}$ (d) $-L^2\frac{di}{dt}$ **(Bundelkhand 2004)**

7. The time constant in discharge of a condenser through a resistance is the time in which charge q on condenser is related with the maximum charge q_0 as :
(a) $q = 0.368\ q_0$ (b) $q = 0.03689\ q_0$
(c) $q = 3.68\ q_0$ (d) None of these **(Gwalior 1991)**

8. A condenser of capacity 1 μF is discharged through a resistance 2 mega-ohm. The time constant will be :
(a) 0.5 s (b) 1 s (c) 2 s (d) 3 s **(Agra 2004; Bhopal 1993)**

9. The time constant of L-R circuit is :
(a) L/R (b) R/L (c) $1/RL$ (d) RL. **(Bhopal 1994; Gwalior 90)**

10. In a circuit containing resistance $R = 10\ \Omega$ and inductance, $L = 1.0$ mH, the time taken to acquire 63% of maximum current is :
(a) 10^{-4} s (b) 10^{-8} s (c) 10 s (d) 100 s **(Rewa 1992)**

11. In LCR circuit, if the resistance is reduced to zero, the period of oscillatory discharge :
(a) will increase (b) will decrease
(c) may increase or decrease (d) will be infinite

12. Time constant of an CR circuit is :
(a) C/R (b) R/C (c) CR (d) $\sqrt{CR}$

13. The period of oscillation of the charge in LC circuit is :
(a) $\frac{1}{2\pi\sqrt{LC}}$ (b) $2\pi\sqrt{LC}$ (c) $\frac{\sqrt{LC}}{2\pi}$ (d) $\frac{2\pi}{\sqrt{LC}}$

14. The condition for oscillatory discharge is :
(a) $R < 2\sqrt{L/C}$ (b) $R > 2\sqrt{L/C}$ (c) $\frac{R^2}{4L^2} > \frac{1}{LC}$ (d) $\frac{R^2}{4L^2} > LC$

15. A condenser of 0.1 μF capacity is first charged and then discharged through a resistance of 10 mega-ohms. The potential will take time to fall to half to original value is :
(a) ln 2 (b) 10 ln 2 (c) ln 1/2 (d) 1 **(Agra 2005)**

ANSWERS

1. (b)	2. (a)	3. (a)	4. (d)	5. (b)	6. (c)	7. (a)	8. (c)
9. (a)	10. (a)	11. (b)	12. (c)	13. (b)	14. (a)	15. (a)	

8

Alternating Currents

8·1. Alternating Current (A.C.)

An alternating current (A.C.) is the current which changes periodically in magnitude and direction. The simplest form of an alternating current which changes with time simple harmonically or sinusoidally is represented as

$$i = i_0 \sin \omega t \qquad ...(1)$$

where i is the instantaneous current at any time t. The maximum value of the current i_0 is called the **peak value** or **amplitude**. It increases with time from zero to peak value, then decreases to zero and reverses in direction, increascing to peak value in this direction and then decreasing to zero. This variation of current completes **one cycle [fig. 8.1]**. Thus in one complete cycle, during one-half of the cycle the current flows in one direction and in the other half cycle it flows in the reverse direction.

The time interval in which an A.C. goes through one cycle of changes is called its **time period**. It is given by

$$T = \frac{2\pi}{\omega} \qquad ...(2)$$

The number of cycles completed by A.C. per second is called **frequency** (ν) and is given by

$$\nu = \frac{1}{T} = \frac{\omega}{2\pi} \qquad ...(3)$$

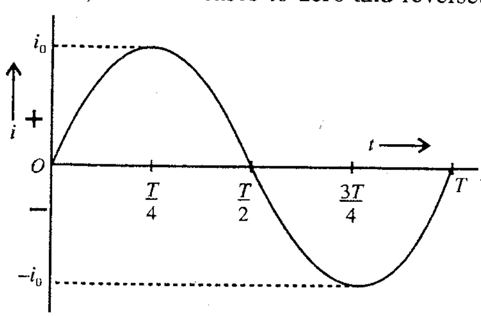

Fig. 8.1 : Alternating current $i = i_0 \sin \omega t$.

where $\omega = 2\pi\nu$ is called **angular frequency**. Sometimes we will write the word frequency for ω, but it is meant angular frequency.

Mean value of A.C. : The average current or mean value of an alternating current for a time interval 0 to t is given by

$$i_{av} = \bar{i} = \frac{\int_0^t i\,dt}{\int_0^t dt} = \frac{1}{t}\int_0^t i\,dt \qquad ...(4)$$

The average value of the current for a cycle is given by

$$\bar{i} = \frac{1}{T}\int_0^T i\,dt = \frac{1}{T}\int_T^{T=2\pi/\omega} i_0 \sin \omega t\,dt = 0 \qquad ...(5)$$

Hence the average value of A.C. for a cycle is zero. Therefore, for an A.C. we define the average value for one-half cycle *i.e.*,

$$i_{av} = \frac{1}{T/2}\int_0^{T/2} i\,dt = \frac{\omega}{\pi}\int_0^{\pi/\omega} i_0 \sin \omega t\,dt$$

$$= \frac{\omega i_0}{\pi\omega}\left[-\cos\omega t\right]_0^{\pi/\omega} = \frac{2}{\pi} i_0 \qquad ...(6)$$

Thus *the average current for one-half cycle is $2/\pi$ times the peak value.*

Root-mean-square value of A.C. : *The root-mean-square or r.m.s. value of an alternating current is defined as the square root of the average of i^2 for a complete cycle, where i is the instantaneous alternating current.* Thus

$$i_{rms} = \sqrt{\overline{i^2}} \text{ or } i_{rms}^2 = \overline{i^2}$$

$$\therefore \quad i_{rms}^2 = \frac{1}{T}\int_0^T i^2 dt = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} i_0^2 \sin^2 \omega t\,dt = \frac{\omega}{2\pi} i_0^2 \int_0^{2\pi/\omega} \frac{[1 - \cos 2\omega t]}{2} dt$$

$$= \frac{\omega}{2\pi}\frac{i_0^2}{2}\left[t - \frac{\sin 2\omega t}{2\omega}\right]_0^{2\pi/\omega} = \frac{\omega}{2\pi}\cdot\frac{i_0^2}{2}\times\frac{2\pi}{\omega} = \frac{i_0^2}{2}$$

Hence, $$i_{rms} = \frac{i_0}{\sqrt{2}} = 0.707\ i_0 \quad ...(7)$$

Thus, *the root mean square value of an alternating current is* $1/\sqrt{2}$ *or* 0.707 *times the peak value.*

Similar to alternating current, an alternating voltage is given by

$$E = E_0 \sin \omega t \quad ...(8)$$

The r.m.s. value of the alternating voltage is given by

$$E_{rms} = \frac{E_0}{\sqrt{2}} = 0.707\ E_0 \quad ...(9)$$

When we say that the household supply is 220 V AC, we understand that the r.m.s. value of supply voltage is 220 V and the corresponding peak value would be $220\sqrt{2}$ V = 311 volt.

The significance of r.m.s. values of the alternating current and voltage can be shown by considering a resistance R in which an alternative current $i = i_0 \sin \omega t$ is passing. The instantaneous rate of heating the resistance is i^2R.

Hence the average rate of heating during a cycle

$$= \frac{1}{T}\int_0^T i^2R\,dt = \frac{R}{T}\int_0^T i^2dt = \frac{i_0^2}{2}R = (i_{rms})^2 R \quad \left[\because \frac{1}{T}\int_0^T i^2dt = \frac{i_0^2}{2}\right] \quad ...(10)$$

If a direct current of value $i_{dc} = i_{rms}$ is passed through the resistance R, then it will produce the rate of heating $i_{dc}^2R = i_{rms}^2R$ *i.e.*, the average rate of heating during a cycle.

Thus, *the r.m.s. value of an alternating current can also be defined as that direct current which produces the same rate of heating in a resistance R as the alternating current.* Hence, the r.m.s value of alternating current is also called as the **effective** or the **virtual value of the current** *i.e.*,

$$i_{effective} = i_{virtual} = \frac{i_0}{\sqrt{2}} = i_{rms}. \quad ...(11)$$

Similarly, *the r.m.s value of an alternating voltage can be defined as that direct voltage which produces the same rate of heating in a resistance R as the alternating voltage.* The r.m.s. value of alternating voltage is also called as the **effective** or the **virtual value of the voltage** *i.e.*,

$$E_{effective} = E_{virtual} = \frac{E_0}{\sqrt{2}} = E_{rms}. \quad ...(12)$$

A.C. ammeter (or voltmeter) measures the r.m.s. or virtual value of the current (or votage). In a hot wire type ammeter or voltmeter, based on heating effect of the current, the deflection ($\theta \propto i^2$ or $\theta \propto E^2$) gives the average value of the square of the current or voltage. Hence in the A.C. meters the scale is so graduated that its reading gives directly the r.m.s. values. This is why the marks on the scale are not equidistant, the marks on the scale in the beginning are closer but gradually the distance between them increases.

8·2. Simple A.C. Circuits

Phase lag and phase lead in A.C. circuits : When an alternating voltage $E = E_0 \sin \omega t$ is applied in a circuit, the corresponding alternating current may not necessarily be in phase with the voltage. The actual difference in phase between the voltage and current depends upon the type of the circuit. If in an A.C. circuit, the maximum current occurs after the maximum voltage, the current is said to **lag** in phase behind the voltage. However, if in an A.C. circuit, the maximum current occurs before the maximum voltage, then the current is said to **lead** in phase the voltage.

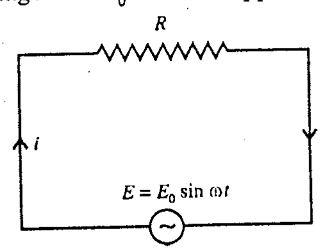

Fig. 8.2 : Pure resistance in A.C. circuit.

Now, we consider following three simple circuits to understand the concept of phase lag and lead between the A.C. voltage and current.

(1) **A.C. circuit containing only a resistor :** Consider a circuit containing an A.C. source of e.m.f., given by

$$E = E_0 \sin \omega t \qquad \text{...(1)}$$

and a resistor of resistance R [fig. **8.2**]. This is known as a purely resistive circuit. If i be the current at any instant t due to the applied e.m.f., then using Kirchoff's loop law, we have

$$E = Ri \quad \text{or} \quad E_0 \sin \omega t = Ri$$

or
$$i = \frac{E_0}{R} \sin \omega t \quad \text{or} \quad i = i_0 \sin \omega t \qquad \text{...(2)}$$

where $i_0 (= E_0/R)$ is the maximum value of the current.

Thus from eqs. (1) and (2), we see that *in a pure resistance, the current is always in phase with the applied e.m.f.* This is represented in fig. **8.3** graphically and vectorially.

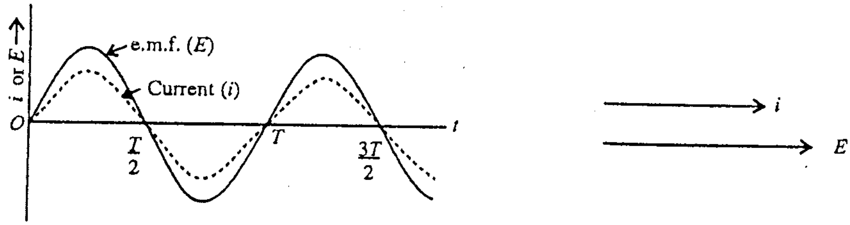

Fig. 8.3 : Current i and e.m.f. E in phase for pure R circuit.

(2) A.C. circuit containing only an inductance : Consider an alternating e.m.f. $E = E_0 \sin \omega t$ be applied across an inductance L [fig. **8.4**]. The changing current i produces an opposite e.m.f. $- L\, di/dt$ in the coil so that from Kirchoff's loop law,

$$E - L\frac{di}{dt} = 0 \quad \text{or} \quad \frac{di}{dt} = \frac{E_0 \sin \omega t}{L}$$

Integrating it, we get

$$i = -\frac{E_0}{\omega L} \cos \omega t \quad \text{or} \quad i = i_0 \sin(\omega t - \pi/2) \qquad \text{...(3)}$$

where $i_0 = E_0/\omega L$ is the maximum value of current. Here the constant of integration has been taken to be zero as the average of i for one time period must be zero.

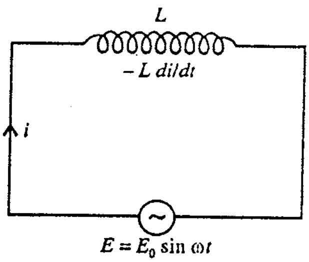

Fig. 8.4 : Pure inductance in an A.C. circuit.

From eq. (3), we see that *in a pure inductance the current lags behind*

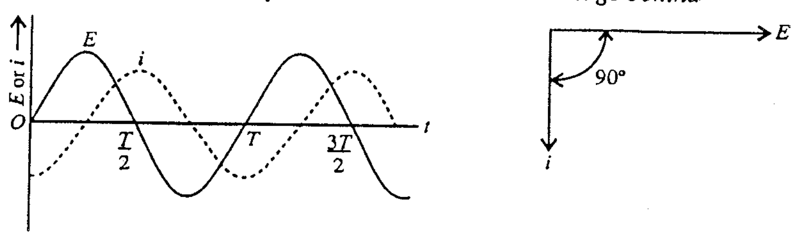

Fig. 8.5 : Current i lags in phase behind E by $\pi/2$ for pure L circuit.

the e.m.f. in phase by $\pi/2$ [fig. **8.5**]. From the expression $i_0 = E_0/\omega L$, we see that ωL is similar to resistance and is called **reactance** of the coil or **inductive reactance**. This is denoted by X_L and its unit is ohm. Thus

$$X_L = \omega L = 2\pi \nu L \qquad \text{...(4)}$$

where ν is the frequency of A.C. This inductive reactance increases with the frequency of A.C. The reciprocal of reactance is called **susceptance.**

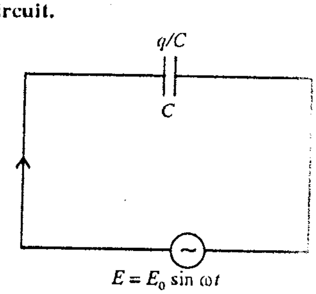

Fig. 8.6 : Pure capacitance in an A.C. circuit.

(3) A.C. circuit containing only a capacitance : Consider an alternating e.m.f. $E = E_0 \sin \omega t$ be applied across the plates of a capacitor of capacitance C. Suppose at any instant t, the charge on the capacitor is q and the current in the circuit is i. The instantaneous potential difference across the capacitor is q/C. Applying Kirchoff's loop law,

$$E_0 \sin \omega t = \frac{q}{C} \text{ or } q = CE_0 \sin \omega t$$

$$\therefore \quad i = \frac{dq}{dt} = CE_0\, \omega \cos \omega t = \frac{E_0}{1/\omega C} \sin\left(\omega t + \frac{\pi}{2}\right) \text{ or } i = i_0 \sin\left(\omega t + \frac{\pi}{2}\right) \quad ...(5)$$

where $i_0 = E_0/(1/\omega C)$ is the maximum value of the current. From eq. (5), we note that *in a pure capacitor the current leads the e.m.f. in phase by* $\pi/2$ [**fig. 8.7**]. Here $1/\omega C$ represents analogous to resistance and is called **reactance of the capacitor** or **capacitative reactance**. This is denoted by X_C and its unit is ohm. Thus

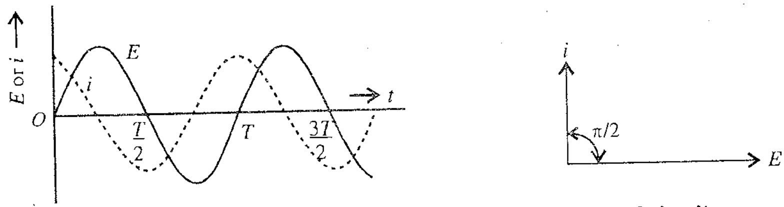

Fig. 8.7 : Current leads in phase the e.m.f. E **by** $\pi/2$ **for pure** C **circuit.**

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C} \quad ...(6)$$

We find from eq. (6) that the capacitative reactance decreases with increase of the frequency of the applied e.m.f.

8·3. Complex Number Representation

The A.C. circuit analysis becomes simple and convenient by the complex number representation. A complex number (z) can be written as

$$z = x + jy \qquad \left[j = \sqrt{-1}\right] \quad ...(1)$$

where x is the real part and y is the imaginary part.

The complex number z can be represented by a vector $z = OP$ in a complex XY-plane. The real part of the complex number is represented on X-axis and imaginary part on Y-axis [**fig. 8.8**]. Therefore, in complex XY-plane, X-axis is called the real axis and Y-axis, the imaginary axis. In plane polar coordinates, if (r, θ) be the coordinates of the point P, then

$x = r \cos \theta$ and $y = r \sin \theta$.

$$\therefore \quad z = r \cos \theta + jr \sin \theta = r (\cos \theta + j \sin \theta) = re^{j\theta} \quad ...(2)$$

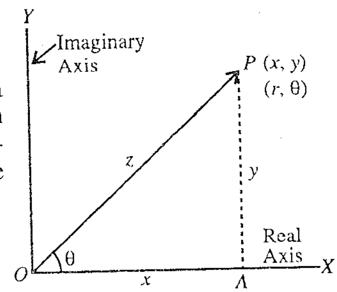

Fig. 8.8 : Complex number as coplanar vector.

where $r = \sqrt{x^2 + y^2}$ is called the **modulus** or **magnitude** of the complex number z and $\theta = \tan^{-1}(y/x)$ is called the angle of **argument** or **phase angle** of z *i.e.*, $|z| = r$ and arg $z = \theta$.

***j*-operator** : The imaginary quantity $\sqrt{-1}$ is represented by j and in A.C. circuits we call it ***j*-operator** *i.e.*, $j = \sqrt{-1}$. *When j operates on any vector in the complex XY-plane, it rotates the vector by 90° or (π/2) in the anticlockwise direction.* If $A = OP$ be a vector along X-axis, then jA will be vector OQ along Y-axis. If we apply to $OQ = jA$ the operator j, then

$$j(jA) = j^2A = -A$$

which is the vector OR of the same magnitude as that of A but in opposite direction. Thus the application of j^2 to a vector A rotates it through 180°. Similarly the application of j^3 to A will rotate it through 270°, because

$$j^3A = -jA = \text{vector } OS$$

This is equivalent to the multiplication of A by $-j$ in the clockwise direction by 90°.

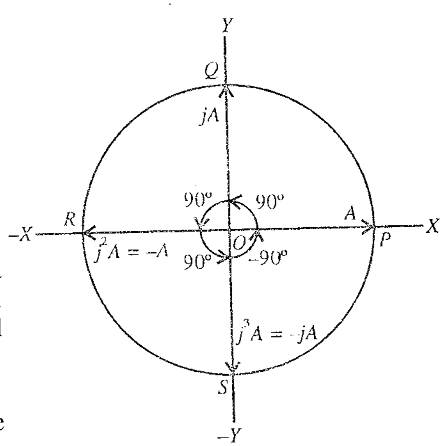

Fig. 8.9 : *j*-operator.

Phasor : If θ changes with time t and $\theta = \omega t$ (ω representing angular velocity), then $z = re^{j\omega t}$. The phase of this vector z changes with time, hence such a vector is called **phasor** or **rotating vector**. This is represented as [**fig. 8.10**].

$$z = re^{j\theta} \text{ or } z = r\angle\theta \text{ i.e., } z = re^{j\omega t} \text{ or } z = r\angle\omega t \quad ...(3)$$

If $z_1 = r_1e^{j\theta_1}$ and $z_2 = r_2e^{j\theta_2}$, then

$$z_1 + z_2 = r_1e^{j\theta_1} + r_2e^{j\theta_2}$$
$$= (r_1\cos\theta_1 + r_2\cos\theta_2) + j(r_1\sin\theta_1 + r_2\sin\theta_2) \quad ...(4)$$
$$z_1 - z_2 = r_1e^{j\theta_1} - r_2e^{j\theta_2}$$
$$= (r_1\cos\theta_1 - r_2\cos\theta_2) + j(r_1\sin\theta_1 - r_2\sin\theta_2) \quad ...(5)$$
$$z_1 z_2 = r_1r_2e^{j(\theta_1+\theta_2)} = r_1r_2\angle(\theta_1 + \theta_2) \quad ...(6)$$
$$\frac{z_1}{z_2} = \frac{r_1e^{j\theta_1}}{r_2e^{j\theta_2}} = \frac{r_1}{r_2}e^{j(\theta_1-\theta_2)} = \frac{r_1}{r_2}\angle\theta_1 - \theta_2 \quad ...(7)$$

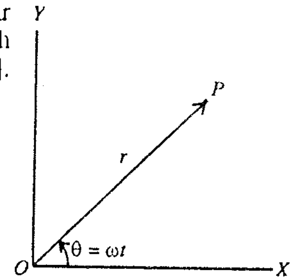

Fig. 8.10 : Phasor $z = re^{j\omega t}$. The vector $z = OP$ changes in direction with time, but its magnitude r remains constant.

Complex Representation of Alternating Voltage and Current

The A.C. circuit analysis can be simplified by expressing sinusoidal voltages and currents in complex number representation in place of vector representation. These complex numbers, used in A.C. circuits, can be added and subtracted like coplanar vectors.

If $E = E_0 \sin \omega t$ is the appplied voltage in an A.C. circuit, then this may be considered as the imaginary part of the complex A.C. voltage, given by

$$E = E_0e^{j\omega t} \quad ...(8)$$

where E_0 is the peak value of A.C. voltage. As in A.C. circuit, there may be a phase difference between the current and the voltage, the complex alternating current can be represented by

$$i = i_0e^{j(\omega t \pm \theta)} \quad ...(9)$$

where i_0 is the peak value of alternating current. Here positive sign means that the current is leading in phase with respect to voltage, while negative sign represents the lag in phase. The actual current flowing in the circuit is $i = i_0 \sin(\omega t \pm \theta)$, being the real part of $i = i_0\, e^{j(\omega t \pm \theta)}$.

8·4. Impedance and Admittance of an A.C. Circuit

In an A.C. circuit, *the ratio of complex voltage E and complex current i defines the* **impedance** *of the circuit* and it is represented by Z. Thus

$$Z = \frac{E}{i} = \frac{E_0e^{j\omega t}}{i_0e^{j(\omega t+\theta)}} = \frac{E_0e^{-j\theta}}{i_0} = Z_0e^{-j\theta} \quad ...(1)$$

where Z_0 is the magnitude of the impedance and $-\theta$ is the phase angle.

From (1), we see that the impedance in general a complex quantity and can be represented as

$$Z = R + jX = Z_0e^{-j\theta} \quad ...(2)$$

where $Z_0 = |Z| = \sqrt{R^2 + X^2}$ and $\theta = \tan^{-1}(-X/R)$. Here R, the **resistance** and X, the **reactance** are the resistive and reactive components of the impedance Z. The unit of R, X or Z is **ohm.**

Reactance X : The imaginary part of the complex impedance Z is called the **reactance.** In an A.C. circuit, the effective resistance in the flow of alternating current due to inductance or capacitance or both is called **reactance.** If in the circuit resistance R is not present, the values of the reactance X and impedance Z are equal. The reactance due to inductance L is called **inductive reactance** and that due to capacitance C is called **capacitive reactance.** If the frequency of the applied alternating voltage is ω, then the *inductive reactance* is $X_L = \omega L$ and *capacitive reactance* is $X_C = 1/\omega C$.

Admittance Y : The reciprocal of impdance Z is called **admittance** Y *i.e.,*

$$Y = \frac{1}{Z} = \frac{i_0e^{j(\omega t+\theta)}}{E_0e^{j\omega t}} = Y_0e^{j\omega\theta} \quad ...(3)$$

where Y_0 is the magnitude of the admittance and θ is its phase angle. In general, admittance is a complex quantity, given by

$$Y = G + jB,\ Y_0 = \sqrt{G^2 + B^2} \text{ and } \theta = \tan^{-1}(B/G) \quad ...(4)$$

where the real part G of Y is called **conductance** and the imaginary part B as **susceptance**. The unit of Y, G or B is **ohm**$^{-1}$ or **mho.**

According to the definition

$$Y = \frac{1}{Z} \text{ or } G + jB = \frac{1}{R + jX}$$

or
$$G + jB = \frac{R - jX}{(R + jX)(R - jX)} = \frac{R - jX}{R^2 + X^2} \quad ...(5)$$

$\therefore$
$$G = \frac{R}{R^2 + X^2} = \frac{R}{Z_0^2} \text{ and } B = -\frac{X}{R^2 + X^2} = -\frac{X}{Z_0^2} \quad ...(6)$$

8·5. A.C. Circuit Containing Pure Resistance

Consider an alternating voltage

$$E = E_0 \sin \omega t \quad ...(1)$$

applied across a resistance R. This applied voltage is the imaginary part of the complex voltage, given by

$$E = E_0 e^{j\omega t} \quad ...(2)$$

Using Kirchoff's law at an instant t,

$$E - Ri = 0 \text{ or } i = \frac{E}{R} = \frac{E_0 e^{j\omega t}}{R} \text{ or } i = i_0 e^{j\omega t} \quad ...(3)$$

where $i_0 = E_0/R$ is the peak value of the current. Hence the current flowing in the circuit is

$$i = i_0 \sin \omega t \quad ...(4)$$

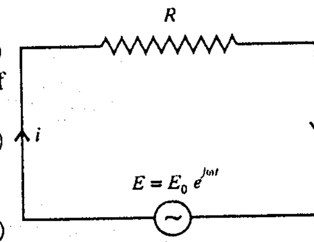

Fig. 8.11 : Circuit containing pure resistance.

From eqs (1) and (4), we observe that the current through the resistance is in phase with the voltage. Time variations of the current and voltage are shown in **fig. 8.12(a)**. The complex number representation of current and voltage is given in **fig. 8.12(b).**

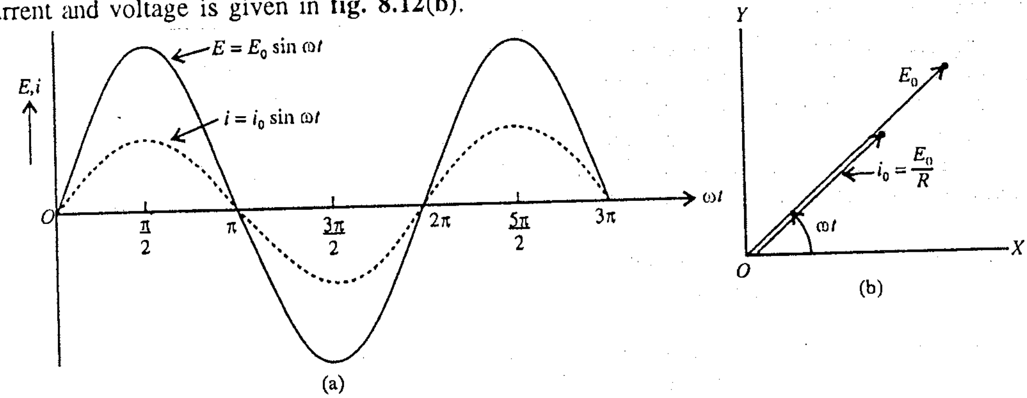

Fig. 8.12 : (a) Time variations of voltage and current for pure resistance, (b) Phasor diagram (current in phase with voltage)

The ratio of complex E and i defines the impedance Z *i.e.*,

$$Z = \frac{E}{i} = R \quad ...(5)$$

Here the resistance R is the impedance (Z) of the circuit.

8·6. A.C. Circuit Containing Inductance Only

Let an alternating voltage

$$E = E_0 \sin \omega t \quad ...(1)$$

be applied across an inductor of inductance L [**fig. 8.13**]. This voltage corresponds to the complex voltage, given by

$$E = E_0 e^{j\omega t} \quad ...(2)$$

Let at any instant t, the current in the circuit be i. If di/dt be the rate of change of current at the instant, then the induced e.m.f. in the inductance L is $- L\,di/dt$ opposing the applied alternating voltage E. Applying Kirchoff's law to the A.C. circuit, we have

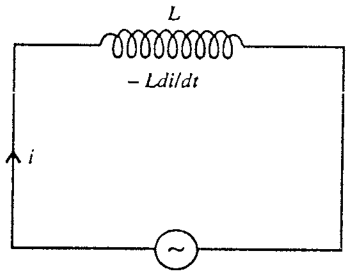

Fig. 8.13 : Circuit containing pure inductance.

$$E - L\frac{di}{dt} = 0 \text{ or } \frac{di}{dt} = \frac{E_0 e^{j\omega t}}{L} \quad ...(3)$$

Integrating (3), we obtain

$$i = \frac{E_0}{j\omega L} e^{j\omega t} = \frac{E}{j\omega L} \quad ...(4)$$

But, $\quad \frac{1}{j} = \frac{-1}{-j} = -\frac{j^2}{j} = -j = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} = e^{-j\pi/2},$

hence $\quad i = \frac{E_0}{\omega L} e^{j(\omega t - \pi/2)} = i_0 e^{j(\omega t - \pi/2)} \quad ...(5)$

where $i_0 = E_0/\omega L$ is the peak value of the current. The current flowing in the circuit is

$$i = i_0 \sin(\omega t - \pi/2) \quad ...(6)$$

From eqs. (1) and (6), we see that the current in the circuit is sinusoidal with frequency $\omega/2\pi$ similar to the applied voltage. Further we see that **the phase of the current is lagging behind the phase of the applied voltage by $\pi/2$.**

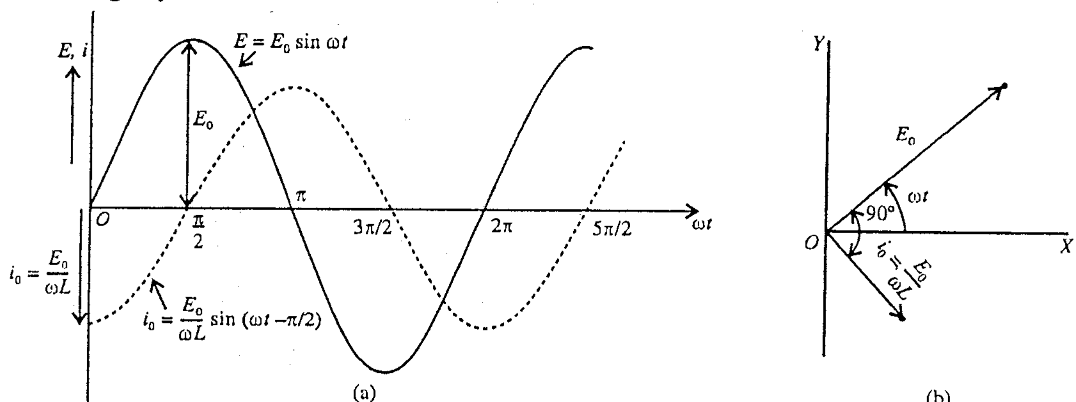

Fig. 8.14 : (a) Time variations of voltage and current for pure inductance, (b) Phasor diagram (current lags voltage in phase by $\pi/2$.

Fig. 8.14(a) gives the time variations of the current and voltage in the circuit, showing phase lag of current by $\pi/2$ with respect to applied voltage. Further the phase diagram is shown in [**fig. 8.14(b)**]. The complex current $i_0 \angle(\omega t - \pi/2)$ is lagging in phase by $\pi/2$ from complex voltage $E_0 \angle\omega t$.

The ratio of E and i gives the **impedance** (Z) of the circuit *i.e.*, from (4)

$$Z = \frac{E}{i} = j\omega L \quad ...(7)$$

Thus the impedance of A.C. circuit containing an inductance L only, is complex *i.e.*, $Z = j\omega L$. The quantity $\omega L = X_L$ is called the **inductive reactance.**

Since $i_0 = E_0/\omega L = E_0/X_L$, the inductive reactance is given by

$$X_L = \frac{E_0}{i_0} = \omega L \quad ...(8)$$

Thus, *the inductive reactance is obtained by dividing the peak value of the voltage by peak value of the current.* Further $X_L = \omega\, L = 2\pi\nu L$ shows that the inductive reactance is directly proportional to the frequency of alternating voltage. The effect of inductive reactance $X_L = \omega L$ in an A.C. circuit is the same as the resistance in D.C. circuit to reduce the current in it (as $i_0 = E_0/X_L$), but **without loss of power.**

8·7. A. C. Circuit Containing Capacitance Only

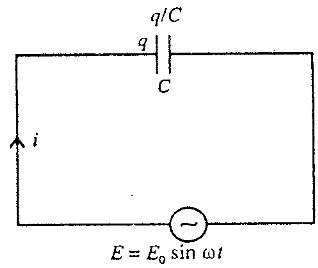

Fig. 8.15 : Circuit containing pure capacitance.

Consider an alternating e.m.f.

$$E = E_0 \sin \omega t \qquad ...(1)$$

applied to a capacitor of capacity C. Corresponding complex voltage is give by

$$E = E_0 e^{j\omega t} \qquad ...(2)$$

If at any instant t, the charge on the capacitor is q, then using Kirchoff's loop law,

$$E - \frac{q}{C} = 0 \text{ or } q = CE_0 e^{j\omega t} \qquad ...(3)$$

Hence the instantaneous current is

$$i = \frac{dq}{dt} = j\omega CE_0 e^{j\omega t} = j\omega CE \text{ or } i = \frac{E}{(1/j\omega C)} \qquad ...(4)$$

But $\qquad j = \cos \pi/2 + j \sin \pi/2 = e^{j\pi/2}$

$$\therefore \qquad i = \frac{E_0}{(1/\omega C)} e^{j(\omega t + \pi/2)} \text{ or } i = i_0 e^{j(\omega t + \pi/2)} \qquad ...(5)$$

where $i_0 = E_0/(1/\omega C)$ is the peak value of the current. The current flowing in the circuit is given by

$$i = i_0 \sin (\omega t + \pi/2) \qquad ...(6)$$

Thus we see from eqs. (1) and (6) that the current is leading in phase to the applied voltage by $\pi/2$. **Fig. 8.16(a)** shows the time variations of the voltage and current in the circuit. The phasor diagram is shown in **fig. 8.16(b)** The complex current $i_0 \angle(\omega t + \pi/2)$ is leading in phase by $\pi/2$ with respect to the complex voltage $E_0 \angle \omega t$.

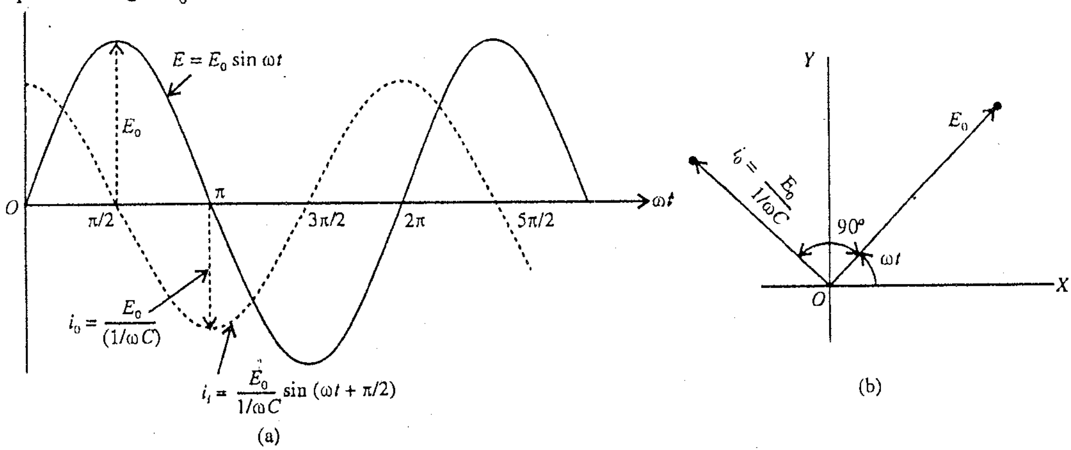

Fig 8.16 : (a) Time variations of current and voltage for pure capacitor, (b) Phasor diagram.

The impedance Z of the circuit is [from eq. (4)]

$$Z = \frac{E}{i} = \frac{1}{j\omega C} \qquad ...(7)$$

Thus the impedance of the circuit, in which an alternating voltage is applied to a pure capacitance C, is a complex quantity *i.e.*, $Z = 1/j\omega C$. The quantity $1/\omega C = X_C$ is called the **capacitative reactance.** As $i_0 = E_0/(1/\omega C) = E_0/X_C$, the capacitative reactance is given by

$$X_C = \frac{E_0}{i_0} = \frac{1}{\omega C}$$

Thus, the *capacitative reactance is inversely proportional to the frequency* ($\nu = \omega/2\pi$) *of the applied alternating voltage.* Hence condenser offers negligible reactance at high frequency and it works as **by-pass capacitor.** However the condenser offers very high reactance at low frequencies and acts as **blocking condenser.**

8·8. A.C. Networks

Any circuit can be made from three circuit elements, namely, the resistance, the inductance and the capacitance. If the A.C. circuit contains R, the impedance Z is equal to R and corresponding to the inductance L and capacitance C, the impedances are $j\omega L$ and $1/(j\omega C)$ respectively. When these elements or combination of elements are connected in series in an A.C. circuit, their impedances add directly. Let E be the applied alternating voltage to a circuit containing Z_1 and Z_2 impedances. If i is the instantaneous current in the circuit, then the voltages across Z_1 is $E_1 = iZ_1$ and that across Z_2 is $E_2 = iZ_2$. Hence

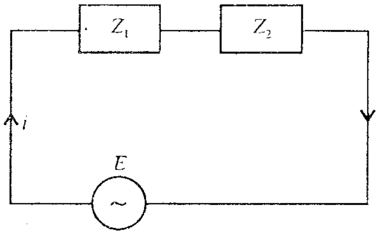

Fig. 8.17 : Impedances in series.

$$E = E_1 + E_2 = iZ_1 + iZ_2 = i(Z_1 + Z_2) \quad ...(1)$$

Therefore, the current in the circuit is given by

$$i = \frac{E}{Z_1 + Z_2} \quad ...(2)$$

Thus, when two or more impedances are connected in series in an A.C. circuit, their equivalent impedance Z is obtained by adding two impedances *i.e.*, $Z = Z_1 + Z_2$.

Now, if the impedances Z_1 and Z_2 are in parallel in an A.C. circuit, then

$$E = i_1 Z_1 = i_2 Z_2 \quad ...(3)$$

where i_1 is the current in Z_1 and i_2 in Z_2. If the equivalent impedance of the A.C. circuit is Z, then

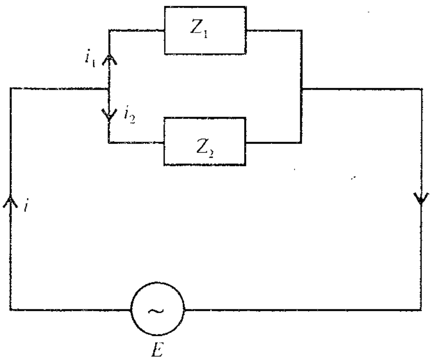

Fig. 8.18 : Impedances in parallel.

$$E = iZ \quad ...(4)$$

But $\quad i = i_1 + i_2 \quad ...(5)$

Using eqs. (3) and (4), we obtain

$$\frac{E}{Z} = \frac{E}{Z_1} + \frac{E}{Z_2} \text{ or } \frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2} \text{ or } Z = \frac{Z_1 Z_2}{Z_1 + Z_2} \quad ...(6)$$

8·9. Inductance and Resistance in Series

Let an alternating e.m.f.

$$E = E_0 \sin \omega t \quad ...(1)$$

applied to a circuit, containing an inductance L and resistance R in series [**fig. 8.19**]. This e.m.f. is an imaginary part of the complex alternating e.m.f., given by

$$E = E_0 e^{j\omega t} \quad ...(2)$$

The current at any instant t is given by

$$i = \frac{\text{Applied alternating voltage}}{\text{Impedance of the circuit}} = \frac{E}{Z} \quad ...(3)$$

Here, vector impedance $Z = R + j\omega L \quad ...(4)$

where R is the impedance due to resistance and $j\omega L$ due to inductance L.

This complex impedance can be written as

$$Z = Z_0 e^{j\theta} \quad ...(5)$$

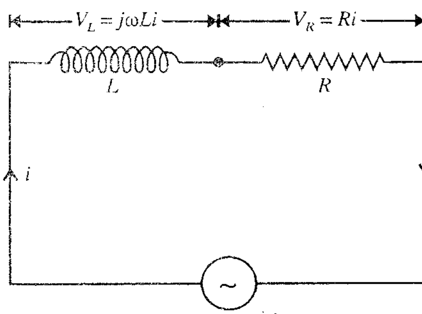

Fig. 8.19 : L and R in series in A.C. circuit.

where $Z_0 = |Z| = \sqrt{R^2 + \omega^2 L^2} = \sqrt{R^2 + X_L^2}$ (X_L = inductive reactance) and $\theta = \tan^{-1}(\omega L/R)$. This $Z_0 = \sqrt{R^2 + \omega^2 L^2}$ is the magnitude of the vector impedance of the LR circuit and θ is the phase angle. Hence,

$$i = \frac{E}{Z} = \frac{E}{R + j\omega L} = \frac{E_0 e^{j\omega t}}{e^{j\theta}} = \frac{E_0 e^{j(\omega t - \theta)}}{\sqrt{R^2 + \omega^2 L^2}} = i_0 e^{j(\omega t - \theta)} \quad ...(6)$$

where $$i_0 = \frac{E_0}{Z_0} = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}} \qquad ...(7)$$

is the peak value of the current in *LR* circuit.

The actual current in the circuit is

$$i = i_0 \sin(\omega t - \theta) \qquad ...(8)$$

From eqs. (1) and (8), we see that the current in *LR* circuit is alternating and sinusoidal with angular frequency ω similar to the applied alternating voltage. Further we find that the current i in the circuit lags the applied e.m.f. by phase angle

$$\theta = \tan^{-1}(\omega L/R).$$

The phasor diagram in view of eqs. (2) and (6) is represented in the complex plane in **fig. 8.20**. This figure also indicates that the current $i_0 \angle(\omega t - \theta)$ lags behind the applied e.m.f. $E_0 \angle \omega t$ in phase by an angle $\theta = \tan^{-1}(\omega L/R)$.

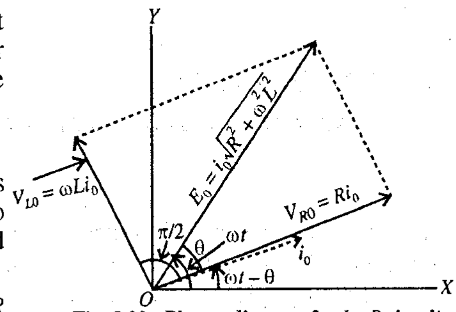

Fig. 8.20 : Phasor diagram for *L* - *R* circuit.

If V_L and V_R are the vector voltages across L and R respectively, then from **fig. 8.19**,

$$E = V_L + V_R = j\omega Li + Ri \qquad ...(9)$$

where $V_L = j\omega Li = e^{j\pi/2}\omega L i_0 e^{j(\omega t - \theta)} = \omega L i_0 e^{j[\pi/2 + (\omega t - \theta)]} = V_{L0} e^{j[\pi/2 + (\omega t - \theta)]} = V_{L0}\angle[\pi/2 + (\omega t - \theta)]$ ...(10)

and $$V_R = Ri = Ri_0 e^{j(\omega t - \theta)} = V_{R0} e^{j(\omega t - \theta)} = V_{R0}\angle(\omega t - \theta) \qquad ...(11)$$

Here, $$V_{L0} = \omega L i_0 \text{ and } V_{R0} = Ri_0 \qquad ...(12)$$

Thus the voltage $V_L = V_{L0} \angle [\pi/2 + (\omega t - \theta)]$ across the inductance leading in phase the current $i = i_0 \angle (\omega t - \theta)$ by $\pi/2$ and the voltage $V_R = V_{R0} \angle (\omega t - \theta)$ is in phase with the current $i = i_0 \angle (\omega t - \theta)$ [**fig. 8.20**].

8·10. Capacitance and Resistance in Series

Let an alternating e.m.f.

$$E = E_0 \sin \omega t \qquad ...(1)$$

be applied to a circuit containing a capacitance C and resistance R in series [**fig. 8.21**]. The corresponding complex e.m.f. is

$$E = E_0 e^{j\omega t} \qquad ...(2)$$

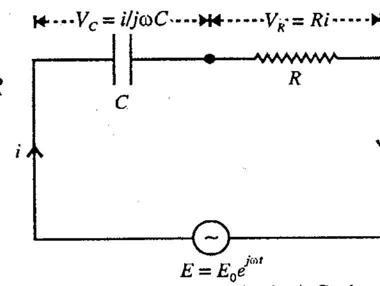

Fig. 8.21 : *C* and *R* in series in A.C. circuit.

At any instant, the current in the circuit is given by

$$i = \frac{E}{Z} = \frac{E_0 e^{j(\omega t - \theta)}}{R + 1/j\omega C} \qquad ...(3)$$

where $$Z = R + \frac{1}{j\omega C} = R - \frac{j}{\omega C} \qquad ...(4)$$

is the total vector impedance of the circuit. This can be written as

$$Z = Z_0 e^{-j\theta} \qquad ...(5)$$

where $Z_0 = |Z| = \sqrt{R^2 + (1/\omega^2 C^2)} = \sqrt{R^2 + X_C^2}$ ($X_C = 1/\omega C$ = capacitative reactance) and $\theta = \tan^{-1}(X_C/R)$ $= \tan^{-1}(1/\omega CR)$.

Hence, from eq. (3) and (5)

$$i = \frac{E_0 e^{j\omega t}}{Z_0 e^{-j\theta}} = \frac{E_0}{Z_0} e^{j(\omega t + \theta)} = i_0 e^{j(\omega t + \theta)} \qquad ...(6)$$

where $$i_0 = \frac{E_0}{Z_0} = \frac{E_0}{\sqrt{R^2 + (1/\omega C)^2}} \qquad ...(7)$$

is the peak value of the current.

The actual current in the circuit is

$$i = i_0 \sin(\omega t + \theta) \qquad ...(8)$$

Thus, when an alternating e.m.f. $E = E_0 \sin \omega t$ is applied to a C-R circuit, the current i in the circuit leads in phase the applied voltage by $\theta = \tan^{-1}(1/\omega CR)$.

The phasor diagram corresponding to eqs. (2) and (6) is shown in complex plane [**fig. 8.22**]. Here, the current $i_0 \angle (\omega t + \theta)$ leads the applied e.m.f. $E_0 \angle \omega t$ in phase by an angle $\theta = \tan^{-1}(1/\omega CR)$.

If V_C and V_R are the vector voltage across C and R respectively, then from eq. (3)

$$E = Ri + \frac{i}{j\omega C} = Ri - \frac{j}{\omega C}i = V_R + V_C \qquad ...(9)$$

where $V_R = Ri = Ri_0 e^{j(\omega t + \theta)} = V_{R0}e^{j(\omega t + \theta)} = V_{R0} \angle (\omega t + \theta)$ ...(10)

and $$V_C = -\frac{j}{\omega C}i = -\frac{j}{\omega C}i_0 e^{j(\omega t+\theta)} = \frac{i_0}{\omega C}e^{-j\pi/2}e^{j(\omega t+\theta)}$$

$$= V_{C0}e^{j[(\omega t+\theta)-\pi/2]} = V_{C0} \angle [(\omega t + \theta) - \pi/2] \qquad ...(11)$$

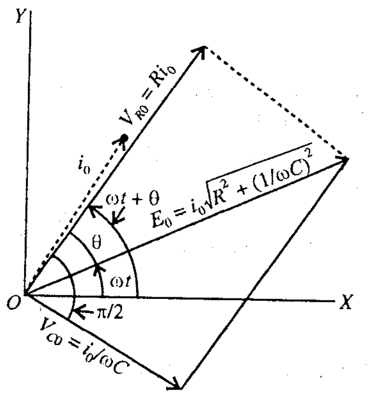

Fig. 8.22 : ***C*** **and** ***R*** **in series in A.C. circuit.**

where $V_{R0} = Ri_0$ and $V_{C0} = i_0/\omega C$. ...(12)

Thus the voltage $V_C = V_{C0} \angle [(\omega t + \theta) - \pi/2]$ is lagging in phase the current $i = i_0 \angle (\omega t + \theta)$ by $\pi/2$ and the voltage $V_R = V_{R0} \angle (\omega t + \theta)$ is in phase with the current $i = i_0 \angle (\omega t + \theta)$ [**fig. 8.22**].

8·11. A.C. Circuit Containing Inductance, Capacitance and Resistance in Series—Series LCR Circuit

Let us consider an alternating e.m.f. of peak value E_0 and frequency $\omega/2\pi$, given by

$$E = E_0 \sin \omega t \qquad ...(1)$$

This e.m.f. is applied to a circuit, containing an inductance L, a capacitance C and a resistance R in series [**fig. 8.23**]. The complex e.m.f. corresponding to (1) is represented as

$$E = E_0 e^{j\omega t} \qquad ...(2)$$

The complex impedances of the circuit elements L, C and R are $j\omega L$, $1/j\omega C$ and R respectively and hence the total impedance of the circuit is given by

$$Z = R + j\omega L + \frac{1}{j\omega C} \text{ or } Z = R + j\left(\omega L - \frac{1}{\omega C}\right) \qquad ...(3)$$

The current in the circuit is obtained to be

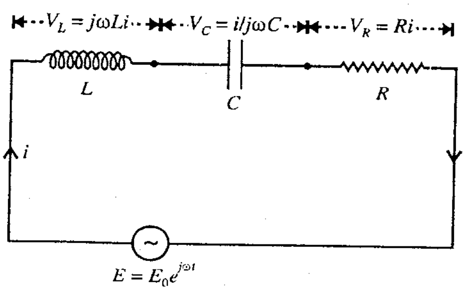

Fig. 8.23 : ***LCR*** **alternating current circuit.**

$$i = \frac{E}{Z} = \frac{E_0 e^{j\omega t}}{R + j(\omega L - 1/\omega C)} \qquad ...(4)$$

But,

$$Z = R + j(\omega L - 1/\omega C) = Z_0 e^{j\theta} \qquad ...(5)$$

where $Z_0 = |Z| = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$ is the magnitude of the impedance and $\theta = \tan^{-1}[(\omega L - 1/\omega C)/R]$ is the phase angle. Thus from eqs. (4) and (5), we get

$$i = \frac{E_0 e^{j\omega t}}{Z_0 e^{j\theta}} = \frac{E_0}{Z_0}e^{j(\omega t-\theta)} = i_0 e^{j(\omega t - \theta)} \qquad ...(6)$$

where

$$i_0 = \frac{E_0}{Z_0} = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} \qquad ...(7)$$

is the peak value of the current in the LCR circuit.

The actual current in the circuit is

$$i = i_0 \sin(\omega t - \theta) \qquad ...(8)$$

From eqs. (1) and (8), we see that there is a phase difference θ between the current and the applied e.m.f. and the value of θ depends on the relative values of the inductive reactance $X_L = \omega L$ and capacitive reactance $X_C = 1/\omega C$. These reactances are frequency dependent, on increasing the frequency $(\omega/2\pi)$, inductive reactance increases, while capacitive reactance decreases.

The impedance of the circuit will be **inductive** for $\omega L > (1/\omega C)$, **capacitive** for $\omega L < (1/\omega C)$ and ohmic for $\omega L = 1/\omega C$.

From eq. (4), we obtain

$$E = Zi = Ri + j\omega Li - ji/\omega C = V_R + V_L + V_C \qquad ...(9)$$

where V_R, V_L and V_C are the vector voltages across the resistance R, inductance L and capacitance C respectively. They can be represented as

$$V_R = Ri = Ri_0 e^{j(\omega t - \theta)} = V_{R0} e^{j(\omega t - \theta)} = V_{R_0} \angle (\omega t - \theta) \qquad ...(10)$$

$$V_L = j\omega Li = \omega Li_0 e^{j[\pi/2 + (\omega t - \theta)]} = V_{L0} \angle [\pi/2 + (\omega t - \theta)] \qquad ...(11)$$

and

$$V_C = -\frac{ji}{\omega C} = e^{-j\pi/2} \frac{i_0}{\omega C} e^{j(\omega t - \theta)} = \frac{i_0}{\omega C} e^{j[(\omega t - \theta) - \pi/2]}$$

$$= V_{C0} \angle [(\omega t - \theta) - \pi/2] \qquad ...(12)$$

Here, $V_{R0} = Ri_0$, $V_{L0} = \omega L i_0$ and $V_{C0} = i_0/\omega C$ ...(13)

From the expression for $\theta = \tan^{-1}[(\omega L - 1/\omega C)/R]$, ...(14)

we find that the current may lag, lead or in same phase relative to the voltage for $\omega L \gtreqless (1/\omega C)$ respectively. We discuss below these three cases :

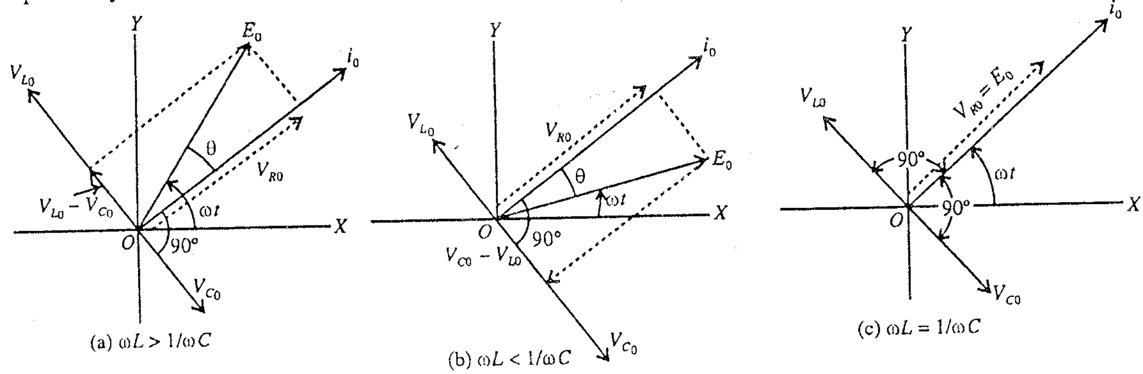

Fig. 8.24 : Phasor diagram for *LCR* circuit.

Case I : When $\omega L > (1/\omega C)$, $V_{L0} > V_{C0}$. Here θ is positive and the impedance is inductive. In this case, the current lags in phase the applied e.m.f. by phase angle θ **[fig. 8.24(a)]**.

Case II : When $\omega L < (1/\omega C)$, $V_{L0} < V_{C0}$. Here θ is negative and the impedance is capacitive. In this case the current leads in phase the applied e.m.f. by phase angle θ **[fig. 8.24(b)]**.

Case III Resonance : When $\omega L = (1/\omega C)$, $V_{L0} > V_{C0}$. However due to the opposite phase of V_L and V_C, they cancel each other **[fig. 8.24(c)]**. In consequence, in such a case, $E = V_R$ and the value of impedance Z_0 is R and it is minimum **[fig. 8.24(c)]**. Here, the value of θ is equal to zero *i.e.*, $\theta = 0$ and hence for $\omega L = 1/\omega C$, **the current *i* is in phase with the applied e.m.f.** *i.e.*,

$$i = i_0 e^{j\omega t} \qquad ...(15)$$

where $i_0 = \dfrac{E_0}{Z_0} = \dfrac{E_0}{R}$ and therefore **the peak value of the current is maximum.** This is the case of **resonance** and the *LCR* circuit is said to be in **series resonance.** The phasor diagram for the case of resonance is shown in fig. 8.24(c). The condition for resonance is

$$\omega L = \frac{1}{\omega C} \text{ or } \omega = \frac{1}{\sqrt{LC}} = \omega_r(\text{say}) \quad \text{...(16)}$$

This relation gives the angular frequency of the applied e.m.f. at which the phenomenon of resonance will occur. If ν_r is the frequency at resonance, then

$$\nu_r = \frac{\omega_r}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \quad \text{...(17)}$$

This frequency of the applied e.m.f. ($\nu_r = 1/2\pi\sqrt{LC}$) *for which the current in LCR circuit is maximum is called the* **resonant frequency**. Further, *the phenomenon due to which the current in a given LCR circuit attains a maximum value is called* **resonance**.

In fact, we know that the **natural frequency** of LCR circuit (for low value of the resistance) or undamped frequency (ν_0) is given by

$$\nu_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{...(18)}$$

This is the frequency (ν_r) of applied e.m.f. at resonance.

Thus, *when the natural frequency of the LCR circuit is equal to the frequency of the applied e.m.f., the phenomenon of series resonance takes place and the frequency, given by* eq. (18) *is the resonant frequency.*

8·12. Variation of Current Amplitude and Phase with the Frequency of Applied e.m.f.

In a series LCR circuit, when an e.m.f. $E = E_0 e^{j\omega t}$ of angular frequency is applied, then the current in the circuit is given by

$$i = i_0 e^{j(\omega t - \theta)} \quad \text{...(1)}$$

where

$$i_0 = \frac{E_0}{Z_0} = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} = \frac{E_0}{\sqrt{R^2 + (X_L - X_C)^2}} \quad \text{...(2)}$$

is the current amplitude and $\theta = \tan^{-1}\frac{\omega L - 1/\omega C}{R} = \tan^{-1}\frac{X_L - X_C}{R}$ ...(3)

Now, if the values of the circuit elements L, C and R are kept constant, then on increasing the frequency (ω) of the applied alternating e.m.f. gradually, we discuss below the variation of current amplitude and phase in three conditions :

(1) **Case I :** When the frequency of the applied e.m.f. is low *i.e.*, $\omega << \omega_0$ ($\omega_0 = 1/\sqrt{LC}$) = undamped frequency of the circuit), the value of X_L ($= \omega L$) is smaller than X_C ($= 1/\omega C$). On increasing the frequency X_L increases, but X_C decreases. Hence on increasing the frequency the value of $|X_L - X_C|$ decreases and consequently from eq. (2) the current amplitude increases. The phase difference θ between the current and applied e.m.f. is $-\pi/2$ at $\omega = 0$ and changes from $-\pi/2$ towards 0 on increasing ω.

(2) **Case II :** When $\omega = \omega_0$, $X_L = X_C$, (because $\omega = 1/\sqrt{LC}$ or $\omega^2 = 1/LC$ or $\omega L = 1/\omega C$). In this

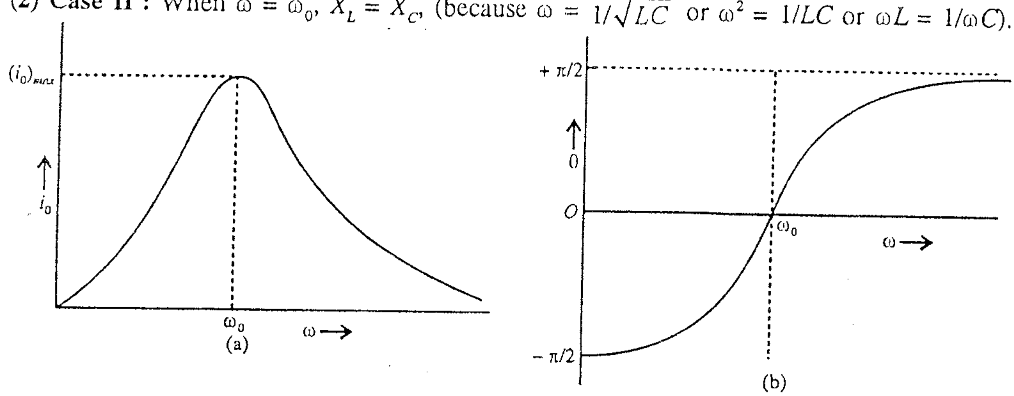

Fig. 8.25 : (a) Variation of current amplitude (peak value) with frequency, (b) Variation of the phase angle θ with frequency.

condition the value of the impedance $Z_0 = R$ which is minimum (Z_{min}) and hence the current amplitude is maximum, given by [from eq. (2)],

$$(i_0)_{max} = \frac{E_0}{R} \qquad ...(4)$$

This is the case of **current resonance.**

From eq. (3) for $X_L = X_C$, the value of $\theta = 0$ *i.e.*, in this condition the current and applied e.m.f. are in the same phase.

(3) **Case III :** When the frequency of the applied e.m.f. is high *i.e.*, $\omega > \omega_0$, X_L becomes greater than X_C and hance on increasing ω, the value of $X_L - X_C$ increases. Consequently from eq. (2) the current amplitude i_0 dereases from its maximum value. Further when $\omega \to \infty$, the current amplitude $i_0 \to 0$. In view of eq. (3), the phase difference θ between the current and applied e.m.f. changes from $-\pi/2$ to $+\pi/2$ on increasing the value of ω and becomes $\theta = \pi/2$ for $\omega = \infty$.

Fig. 8.25(a) shows the variation of current amplitude i_0 and **fig. 8.25(b)** shows the variation of phase difference θ between current and applied e.m.f. with frequency ω. From eq. (1) we see that the current lags or leads the applied e.m.f. in phase for θ to be positive or negative respectively. Hence, when ω changes from 0 to ω_0, θ changes from $-\pi/2$ to 0 and hence for this interval of frequency current leads the e.m.f. in phase. At $\omega = \omega_0$, $\theta = 0$ *i.e.*, both the current and e.m.f. are in the same phase. Further, when ω becomes greater than ω_0, θ changes 0 to positive value and as $\omega \to \infty$, $\theta \to +\pi/2$. Thus for $\omega > \omega_0$, the current lags the applied e.m.f. in phase.

8·13. Sharpness of Resonance

In *LCR* circuit, the current amplitude is maximum at a particular frequency of the applied e.m.f. and this frequency is known as the *resonant frequency*. The maximum of the current amplitude has greater value for smaller the resistance (R) of the circuit [**fig. 8.26**].

Sharpness of resonace depends on the rate at which the current amplitude decreases on either side of resonant frequency, with change in frequency of applied e.m.f. If the frequency of applied e.m.f. is slightly changed from its resonant value and the current amplitude falls rapidly, the resonance is said to be **sharp**. However if on changing the frequency of applied e.m.f., the current decreases slowly, the resonance is said to be **flat**. For low resistance, the resonance is sharp, while for high resistance the resonance is flat.

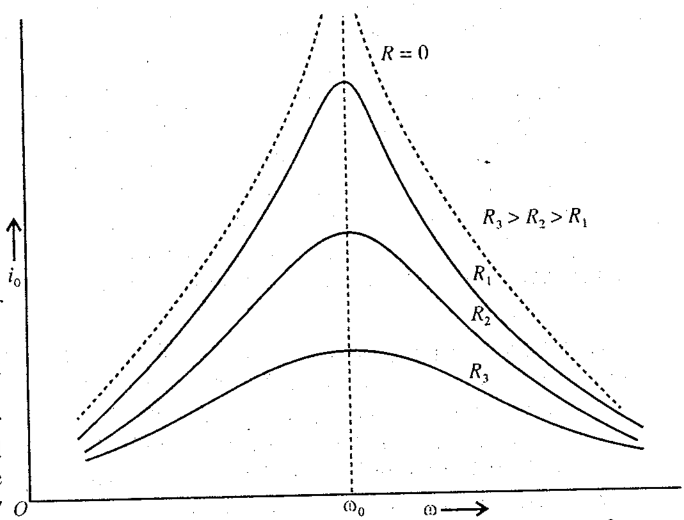

Fig. 8.26 : Variation of current amplitude with frequency for different values of resistance (R).

In **fig. 8.26**, we have drawn curves for the variation of current amplitude with angular frequency ω of the applied e.m.f. for different values of resistances $R = 0$, $R_1 < R_2 < R_3$. At resonance ($\omega = \omega_0$), the maximum current amplitude is greater for low value of the resistance (R_1) when compared to that for high value of the resistance (R_3). For R_1, the resosnace may be said to be **sharp**, and for R_3, **flat.** If $R = 0$ the maximum value of i_0 becomes infinite at $\omega = \omega_0$. However there is always some resistance in the circuit and the maximum current amplitude remains finite.

If for a given frequency the value of the current amplitude is maximum and it is very small or negligible for other frequencies, the circuit is said to be **frequency selector**. Frequency selectivity for low resistance (sharp resonance) circuit is more, when compared to that for high resistance (flat resonance) circuit.

8·13·1. Half-Power Frequencies and Band-Width

The half-power frequencies are the frequencies at which the power dissipation in the circuit drops to one half to one-half of its resonance value.

The power dissipation at resonant frequency

$$= (i_0)^2_{max} R = \left(\frac{E_0}{R}\right)^2 R \qquad \left[\because (i_0)_{\max} = \frac{E_0}{R}\right] \quad ...(1)$$

If the power dissipation be one-half of the resonant value, when the current amplitude is i_0, then the power dissipation $= i_0^2 R$.

Hence $\quad \frac{1}{2}(i_0)^2_{max} R = i_0^2 R$

or
$$i_0 = \frac{(i_0)_{max}}{\sqrt{2}} = \frac{1}{\sqrt{2}}\frac{E_0}{R} \qquad ...(2)$$

Thus *at the half-power frequencies the current amplitude becomes* $1/\sqrt{2}$ *times the current amplitude at resonance.*

From **fig. 8.27**, we note that the half-power dissipation occurs at the current $i_0 = (i_0)_{max}/\sqrt{2}$ at the two frequencies ω_1 and ω_2; ω_1 is the *lower half-power frequency and* ω_2 *the upper half-power frequency*. If $Z = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$ is the impedance corresponding to the current i_0, of eq. (2) then

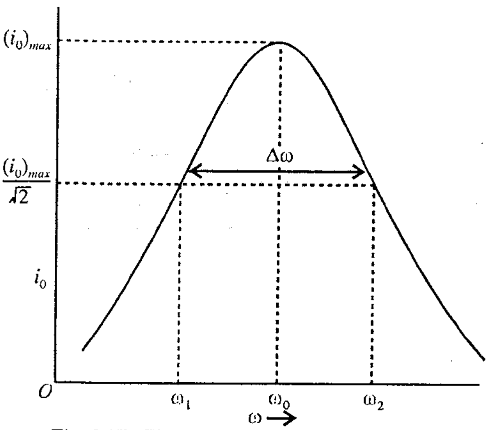

Fig. 8.27 : Half-power frequencies (ω_1 and ω_2) and band-with ($\Delta\omega = \omega_2 - \omega_1$).

$$i_0 = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} = \frac{1}{\sqrt{2}}\frac{E_0}{R} \text{ or } R^2 + (\omega L - 1/\omega C)^2 = 2R^2$$

or
$$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2 \text{ or } \omega L - \frac{1}{\omega C} = \pm R \qquad ...(3)$$

Eq. (3) has two solutions. One for lower half-power frequency ω_1 and the other for upper half-power frequency ω_2. Thus

$$\omega_1 L - \frac{1}{\omega_1 C} = -R \text{ and } \omega_2 L - \frac{1}{\omega_2 C} = +R \qquad ...(4)$$

Adding these two equations, we obtain

$$(\omega_1 + \omega_2)L - \frac{1}{C}\left(\frac{1}{\omega_1} + \frac{1}{\omega_2}\right) = 0 \text{ or } (\omega_1 + \omega_2)L - \frac{(\omega_1 + \omega_2)}{\omega_1\omega_2}\frac{1}{C} = 0 \text{ or } \omega_1\omega_2 = \frac{1}{LC} \qquad ...(5)$$

But $\omega_r = \omega_0 = 1/\sqrt{LC}$ is the resonant frequency, hence

$$\omega_r^2 = \omega_1\omega_2 \text{ or } \omega_r = \sqrt{\omega_1\omega_2} \text{ or } \nu_r = \sqrt{\nu_1\nu_2} \qquad ...(6)$$

Thus, *the resonant frequency* ω_r or ν_r *is the geometric mean of the lower and upper half-power frequencies.*

In eq. (1), if we subtract from one equation to the other, we get

$$(\omega_2 - \omega_1)L + \frac{1}{C}\left(\frac{1}{\omega_1} - \frac{1}{\omega_2}\right) = 2R \text{ or } (\omega_2 - \omega_1)L + \frac{(\omega_2 - \omega_1)}{\omega_1\omega_2 C} = 2R$$

or
$$(\omega_2 - \omega_1)L + \frac{(\omega_2 - \omega_1)}{(1/LC)}\cdot\frac{1}{C} = 2R \text{ or } 2(\omega_2 - \omega_1)L = 2R$$

Thus
$$\omega_2 - \omega_1 = R/L \text{ or } \nu_2 - \nu_1 = R/2\pi L \qquad ...(7)$$

We call the frequency difference $\omega_2 - \omega_1 = \Delta\omega$ as the **band-width.**

Hence
$$\text{band-width } \Delta\omega = R/L \text{ or } \Delta\nu = R/2\pi L \qquad ...(8)$$

8·13·2. Quality Factor Q

If $\omega_r = \omega_0 = 1/\sqrt{LC}$ is the resonant frequency, then the quality factor for LCR circuit at the resosnace is given by

$$Q = \frac{\omega_0 L}{R} = \frac{\omega_r L}{R} \quad ...(9)$$

But at resosnace $\omega_0 L = 1/\omega_0 C$, hence

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR} \quad ...(10)$$

Hence, *for an LCR circuit quality factor is the ratio of inductive reactance (or capacitative reactance) at the resonant frequency and the resistance of the circuit.*

Now, $$Q = \frac{\omega_0 L}{R} = \frac{\omega_0}{\Delta\omega} \text{ or } \Delta\omega = \frac{\omega_0}{Q_0} \quad ...(11)$$

Thus *the band-with is inversely proportional to the quality factor. Larger the value of quality factor, smaller is the band-width.* Since resonance is sharper for small value of band-width, the *larger the quality factor Q for a series LCR circuit, sharper will be the resonance.*

8·13·3. Selectivity—Acceptor Circuit

If the applied alternating voltage has a number of frequency components, then for a series *LCR* circuit, maximum current amplitude will occur for only that frequency component which has the frequency,

$$\omega_r = \frac{1}{\sqrt{LC}} \text{ or } \nu_r = \frac{1}{2\pi\sqrt{LC}} \quad ...(12)$$

Thus *out of number of available frequencies the LCR circuit selects one frequency for which the current is maximum and for other frequencies the current is relatively very small, in other words the LCR circuit shows selectivity.* The resonant circuit is therefore known as the **acceptor circuit** as it practically accepts one frequency component of the applied voltage and rejects other. We note that sharper the resonance, *i.e.*, higher the Q value, the more highly selective will be the acceptor circuit.

8·13·4. Voltage Magnification

Quality factor Q is also a measure of vlotage magnification in the series *LCR* circuit. This is shown below :

We know that at resonance the potential difference across inductance and capacitance are equal and 180° out of phase and hence cancel. Therefore, the only potential difference at resonance is across the resistance R.

The current is maximum at resonance and is given by,

$$(i_0)_{max} = \frac{E_0}{R} \quad ...(13)$$

i.e., potential difference across resistance $R = (i_0)_{max} R = E_0$

We see that, at resonance the potential difference across the resistance is equal to the applied e.m.f. The voltage magnification in *LCR* circuit is given by

$$\text{Voltage magnification, } M = \frac{\text{P.D. across the inductance (or capacitance) at resonance}}{\text{Applied e.m.f.}} \quad ...(14)$$

Thus, $$M = \frac{(i_0)_{max}\omega_r L}{E_0} = \frac{E_0}{R}\frac{\omega_r L}{E_0} = \frac{\omega_r L}{R} = Q = \frac{1}{R}\sqrt{\frac{L}{C}} \quad \left[\because \omega_r = 1/\sqrt{LC}\right] \quad ...(15)$$

Thus, *the voltage magnification of LCR circuit is equal to its quality factor Q at resonance.*

8·14. Power Consumed by an A.C. Circuit

If an e.m.f. $E = E_0 \sin \omega t$ is applied to an A.C. circuit, then the current flowing in the circuit is represented as

$$i = i_0 \sin (\omega t - \theta) \quad ...(1)$$

where θ is the phase difference between the e.m.f. and the current.

The instantaneous power consumed by the circuit is

$$P = Ei = E_0 i_0 \sin \omega t \sin (\omega t - \theta) = \frac{E_0 i_0}{2}[\cos\theta - \cos (2\omega t - \theta)] \quad ...(2)$$

Hence the **average power** consumed by the circuit over a cycle is given by

$$P_{av} = \frac{1}{T}\int_0^T Ei\,dt = \frac{1}{T}\int_0^T \frac{E_0 i_0}{2}\,[\cos\theta - \cos(2\omega t - \theta)]\,dt$$

$$= \frac{1}{T}\int_0^T \frac{1}{2}E_0 i_0 \cos\phi\,dt - \frac{1}{T}\int_0^{T=2\pi/\omega} \frac{1}{2}E_0 i_0 \cos(2\omega t - \theta)dt = \frac{1}{2}E_0 i_0 \cos\theta$$

Thus, $$P_{av} = E_{rms}\, i_{rms} \cos\theta \qquad ...(3)$$

where E_{rms} $(= E_0/\sqrt{2})$ is the rms value of the applied e.m.f. and i_{rms} $(= i_0/\sqrt{2})$ is the rms value of the current. If the current is in ampere and e.m.f. is in volts, then the unit of power is **watt**.

The product $E_{rms}\, i_{rms}$ is the apparent power, because E_{rms} and i_{rms} are also called as virtual e.m.f. and virtual current respectively. The quantity $\cos\theta$ is called **power factor**, because the multiplication of this factor to the apparent power gives the average power consumed in a cycle. Thus

Average power (P_{av}) = Apparent power × Power factor

$$\therefore \quad \text{Power factor} = \frac{\text{Average power}}{\text{Apparent power}} = \cos\theta \qquad ...(4)$$

Thus, in any A.C. circuit, *the ratio of the actual average power and apparent power is called power factor* $(\cos\theta)$.

As the value of phase difference depends on the nature of the circuit, the power factor also depends on the circuit elements. In case of series LCR circuit,

$$\tan\theta = \frac{\omega L - 1/\omega C}{R},$$

$$\therefore \text{Power factor, } \cos\theta = \frac{R}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} = \frac{R}{Z_0} \qquad ...(5)$$

Since $Z_0 > R$, $\cos\theta < 1$ *i.e.*, the power factor of LCR circuit is always less than 1. Average power for LCR circuit is given by

$$P_{av} = E_{rms}\, i_{rms} \cos\theta = i^2_{rms} R \qquad ...(6)$$

because $$E_{rms} = Z_0 i_{rms} \text{ and } \cos\theta = R/Z_0$$

Power Factor in Different A.C. Circuits

(1) **A.C. circuit with pure resistance** R **:** We knwo that if an A.C. circuit contains a pure resistance R, the current i is in phase with the applied *e.m.f.* E. This means that the phase difference $\theta = 0$ and hence

$$\text{Power factor, } \cos\theta = \cos 0 = 1 \qquad ...(7)$$

$$\therefore \quad \text{Average power, } P_{av} = E_{rms} i_{rms} \cos 0^\circ = E_{rms} i_{rms} = i^2_{rms} R \qquad ...(8)$$

(2) **A.C. circuit with pure inductance :** In this case, the current lags behind in phase the applied e.m.f. by $\theta = \pi/2$. Hence

$$\text{Power factor, } \cos\theta = \cos\pi/2 = 0 \qquad ...(9)$$

$$\therefore \quad \text{Average power, } P_{av} = E_{rms} i_{rms} \cos\pi/2 = 0 \qquad ...(10)$$

Thus we see from eq. (10) that in the presence of pure inductance only in the A.C. circuit, the average power consumed by the circuit in a cycle is zero. In such a circuit, the current is called **wattless current** or **idle current**.

(3) **A.C. circuit with pure capacitance :** When the circuit contains pure capacitance, the current leads in phase the applied e.m.f. by $\theta = \pi/2$. Hence

$$\text{Power factor, } \cos\theta = \cos\pi/2 = 0 \qquad ....(11)$$

$$\therefore \quad \text{Average power, } P_{av} = E_{rms} i_{rms} \cos\pi/2 = 0 \qquad ...(12)$$

Hence, in pure capacitative circuit, the average power consumed is also zero and the current in the circuit is **watteless** or **idle**.

(4) **A.C. circuit containing inductance** L **and resistance** R **:** In this case, the current lags behind in phase the applied e.m.f by θ, given by

$$\tan\theta = \frac{\omega L}{R}, \ \therefore \text{Power factor, } \cos\theta = \frac{R}{\sqrt{R^2 + \omega^2 L^2}} \qquad ...(13)$$

Hence, $$P_{av} = E_{rms} i_{rms} \cos\theta = i_{rms}^2 R \qquad ...(14)$$

because $$E_{rms} = i_{rms} Z_0 = i_{rms}\sqrt{R^2+\omega^2L^2} \text{ and } \cos\theta = R/\sqrt{R^2+\omega^2L^2}.$$

Since for any value of R, $\cos\theta < 1$ *i.e.*, the power factor of LR circuit is less than 1.

(5) **A.C. circuit containing capacitance C and resistance R** : In case of CR alternating current circuit, the current leads in phase the applied e.m.f. by θ, given by

$$\tan\theta = \frac{1/\omega C}{R}; \therefore \text{ Power factor, } \cos\theta = \frac{R}{\sqrt{R^2+(1/\omega C)^2}} \qquad ...(15)$$

Hence, $$P_{av} = E_{rms} i_{rms} \cos\theta = i_{rms}^2 R \qquad ...(16)$$

The power factor of CR circuit is also less than 1.

Alternatively, we can obtain the expression for the power factor $\cos\theta$ from eq. (5), as discussed below :

(1) For an A.C. circuit with **pure resistance** R, $L = 0$ and $C = \infty$, and we have from (5) the power factor, given by

$$\cos\theta = \frac{R}{\sqrt{R^2+(\omega L - 1/\omega C)^2}} = \frac{R}{\sqrt{R^2+0}} = \frac{R}{R} = 1 \text{ and } \theta = 0.$$

(2) When the circuit contains **pure inductance** L with $R = 0$ and $C = \infty$, the power factor is

$$\cos\theta = \frac{0}{\sqrt{R^2+\omega^2L^2}} = 0 \text{ and } \theta = \frac{\pi}{2}.$$

(3) For an A.C. circuit with **pure capacitance,** $L = 0$, $R = 0$, we have

$$\cos\theta = \frac{0}{\sqrt{R^2+(1/\omega C)^2}} = 0 \text{ and } \theta = -\frac{\pi}{2}$$

because $$\tan\theta = \frac{\omega L - 1/\omega C}{R} = \frac{0 - 1/\omega C}{0} = -\infty.$$

(4) For an alternating current LR circuit with $C = \infty$,

$$\cos\theta = \frac{R}{\sqrt{R^2+\omega^2L^2}}.$$

(5) In case of CR alternating current circuit ($L = 0$),

$$\cos\theta = \frac{R}{\sqrt{R^2+(1/\omega C)^2}}.$$

8·15. Choke Coil

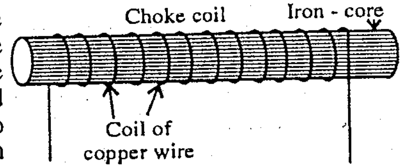

Fig. 8.28 : Choke coil.

When an alternating e.m.f. is applied to a pure inductor, the current lags behind the applied e.m.f. in phase by $\pi/2$. In case of pure capacitor, the current leads the applied e.m.f. in phase by $\pi/2$. Hence as the power factor $\cos\theta$ is equal to zero, the average power dissipated in both cases is zero *i.e.*, there is no loss of power in the circuit. Also in both cases, the current is said to be **wattless.** Therefore, we can reduce the current in an A.C. circuit ($i_0 = E_0/Z_0$) by including a capacitor or inductor or both without loss of energy. *If we use an inductor to reduce the current in an A.C. circuit without (or negligible) loss of energy, this inductor is said to be* **choke** *or* **choking coil.** A choke coil is made by a thick copper wire which is wound closely with a large number of turns over a soft iron laminated core [fig. **8.28**]. As the choke coil has negligible resistance, the power loss $i_{rms}^2 R$ is negligible.

In case R is the resistance of the choke coil, then the average power dissipated per cycle is given by

$$P_{av} = E_{rms} i_{rms} \cos\theta = i_{rms}^2 R$$

where the power factor $\cos\theta = \dfrac{R}{\sqrt{R^2+\omega^2L^2}}$.

For a choke coil, the value of L is large but R is very small so that $\cos\theta$ is approximately zero. Hence the power dissipated by the choke coil is almost negligible.

If we connect a resistance in place of inductance in the circuit to reduce the current, there will be average power loss $i_{rms}^2 R$. Hence *a choke coil is preferred over a resistance to reduce the alternating current in order to avoid the power loss.*

Note : The given values of alternating currents and voltages represent the rms or virtual values, unless indicated otherwise.

Ex. 1. *If the A.C. main supply is given to be 220 volt, calculate average e.m.f. during a positive half-cycle ?*

Sol. The given value of the e.m.f. of the A.C. supply is the rms value *i.e.*, $E_{rms} = 220$ volt.

But
$$E_{rms} = \frac{E_0}{\sqrt{2}}, \text{ where } E_0 = \text{peak value of e.m.f.}$$

$$\therefore \quad E_0 = \sqrt{2} \times E_{rms} = 1.414 \times 220 = 311 \text{ volt.}$$

The average e.m.f. during half-cycle is given by

$$E_{av} = \frac{2}{\pi} E_0 = \frac{2 \times 311}{3.14} = \mathbf{198\ volt.}$$

Ex. 2. *A coil has an inductance of* $0.4/\pi$ *henry and is joined in series with a resistance of* 30 *ohms. Calculate the current flowing in the circuit when connected to A.C. mains of* 200 *volts and frequency* 50 *Hz.* **(Agra 1966)**

Sol.
$$\text{Current } i_{rms} = \frac{e.m.f.}{impedance} = \frac{E_{rms}}{Z}.$$

Here, $E_{rms} = 200$ volts and impedance $= \sqrt{R^2 + \omega^2 L^2} = \sqrt{(30)^2 + (2\pi \times 50 \times 0.4/\pi)^2}$

$$= \sqrt{(30)^2 + (40)^2} = 50 \text{ ohms}$$

$$\therefore \quad i_{rms} = 200/50 = \mathbf{4\ amp.}$$

Ex. 3. *A lamp in which 10 A current can flow at 15 V, is connected with an alternating source of potential 220 V. The frequency of source is 50 cycle/sec. Calculate the inductance of choke coil required to ignite the bulb.* **(Ujjain 1992)**

Sol. The lamp requires 10 A current at 15 V. Therefore resistance of lamp $R = V/i = 15/10 = 1.5\ \Omega$.

In order to use the lamp with 220 V A.C. source, let a choke coil of inductance L be connected in series with the lamp so that 10 A current flows through it. Hence the inpedance of circuit is

$$Z = \frac{V}{i} = \frac{220}{10} = 22 \text{ ohm}$$

But
$$Z = \sqrt{R^2 + \omega^2 L^2} \text{ or } 22 = \sqrt{(1.5)^2 + (2\pi \times 50)^2 L^2}$$

$$\therefore \quad (2\pi \times 50)^2 L^2 = 484 - 2.25 = 481.75 \text{ or } L = \frac{\sqrt{481.75}}{2\pi \times 50} = \mathbf{0.07\ H.}$$

Ex. 4. *A 50 W, 100 V lamp is to be used with 200 V, 50 Hz A.C. source. Calculate the capacity of the condenser to be connected in series with the lamp.* **(Indore 2004; Sagar 1992)**

Sol. Power, $P = Vi, \quad \therefore \ i = \dfrac{P}{V}$

Here, $P = 50$ watt, $V = 100$ volt, hence current through the lamp is $i = \dfrac{50}{100} = 0.5$ A

Resistance of the lamp, $R = \dfrac{V}{i} = \dfrac{100}{0.5} = 200\ \Omega$.

If C be the capacity of the condenser connected in series with the lamp to use it with A.C. so that 0.5 A current flows in the circuit, then the impedance of circuit is

$$Z = \frac{V}{i} = \frac{200}{0.5} = 400 \text{ ohm}$$

But $\quad Z = \sqrt{R^2 + (1/\omega C)^2}$, where $\omega = 2\pi\nu = 2\pi \times 50 = 314$ rad/sec.

$\therefore \quad 400 = \sqrt{(200)^2 + (1/314C)^2}$ or $(1/314C)^2 = (400)^2 - (200)^2$

or $\quad = (1/314C) = \sqrt{12} \times 10^2$

$$\therefore \quad C = \frac{1}{314 \times \sqrt{12} \times 10^2} = 9.2 \times 10^{-6}\text{F} = \mathbf{9.2\ \mu F.}$$

Ex. 5. *A 8 μF capacitor is placed across the 200 volt rms with 50 Hz supply. Calculate the rms current which flows in the circuit. What is the peak value of the voltage across the capacitor ?*

Sol. Here, capacitance, $C = 8\ \mu\text{F} = 8 \times 10^{-6}\text{F}$, e.m.f., $E_{rms} = 200$ volt. and frequency, $\nu = 50$ Hz.

$$\therefore \quad \text{Reactance, } X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C} = \frac{1}{2 \times 3.14 \times 50 \times (8 \times 10^{-6})} = 400 \text{ ohm (nearly)}$$

$$\therefore \quad \text{Current, } i_{rms} = \frac{E_{rms}}{X_C} = \frac{200}{400} = \mathbf{0.5\ amp.}$$

Now, $\quad E_{rms} = \dfrac{E_0}{\sqrt{2}}$, hence

$$E_0 = E_{rms}\sqrt{2} = 200 \times \sqrt{2} = 200 \times 1.414 = \mathbf{283\ volt.}$$

Ex. 6. *A resistance and an inductacne are connected in series across an alternating voltage $V = 28.3 \sin 314\, t$ volt. The current is found to be $4 \sin (314\, t - \pi/4)$ amp. Find the values of the inductance and the resistance.*

Sol. Here, $E = 28.3 \sin 314t$ volt and $i = 4 \sin (314t - \pi/4)$ amp.

Hence the impedance of the circuit is given by

$$Z = \sqrt{R^2 + (\omega L)^2} = \frac{E_0}{i_0} = \frac{28.3}{4} = 7.07 \text{ ohm.} \qquad ...(i)$$

The current lags the voltage in phase by angle $\pi/4$ *i.e.*, 45°, so that $\tan\theta = 1$. Therefore,

$$\tan\theta = \frac{\omega L}{R} = 1. \text{ or } \omega L = R. \qquad ...(ii)$$

Hence from eq. (i), we get

$$\sqrt{R^2 + R^2} = 7.07 \text{ or } \sqrt{2}R = 7.07 \text{ or } R = \frac{7.07}{\sqrt{2}} = \frac{7.07}{1.414} = \mathbf{5\ ohm.}$$

Here $\omega = 314$. Therefore, from eq. (ii) the inductance is

$$L = \frac{R}{\omega} = \frac{5}{314} = \mathbf{0.016\ henry.}$$

Ex. 7. *An alternating voltage of 110 V, 50 cycles is applied to a circuit which contains an inductance of 0.02 henry and a resistance of 10 ohms in series. Find the current and the phase lag.*

Sol. $\quad$ Current, $i = \dfrac{E}{\sqrt{R^2 + \omega^2 L^2}} = \dfrac{110}{\sqrt{10^2 + (2 \times 3.1416 \times 50 \times 0.02)^2}} = \mathbf{10.5\ amp.}$

$$\text{Phase lag, } \theta = \tan^{-1}\frac{\omega L}{R} = \tan^{-1}\frac{2 \times \pi \times 50 \times 0.02}{10} = \tan^{-1}\frac{\pi}{5} = \mathbf{32°8'.}$$

Ex. 8. *An alternating voltage of 100 virtual volts is applied to a circuit of resistance 0.5 ohm and inductance 0.01 henry, the frequency being 50 cycles/sec. What is the current and lag in time between voltage and current ?*

Sol. $\quad$ Virtual current $= \dfrac{\text{Virtual e.m.f.}}{\text{Impedance}} = \dfrac{E_{rms}}{Z_0}$

Here, $E_{rms} = 100$ volts and $Z_0 = \sqrt{R^2 + \omega^2 L^2} = \sqrt{(0.5)^2 + (2\pi \times 50)^2 (0.01)^2} = 3.2\ \Omega.$

$\therefore$ Virtual current $= 100/3.2 =$ **31.25 amp.** (nearly)

Also $\tan\theta = \dfrac{\omega L}{R} = \dfrac{2\pi \times 50 \times 0.01}{0.5} = 6.28 \therefore \theta = 81° = \dfrac{81}{360} \times 2\pi$ radian

Here period, $T = \dfrac{1}{50}$ sec.

Phase angle 2π corresponds to $\dfrac{1}{50}$ sec. (period).

Hence phase angle $\dfrac{81}{360} \times 2\pi$ corresponds to $\dfrac{81}{360} \times 2\pi \times \dfrac{1}{50} \times \dfrac{1}{2\pi} = \mathbf{4.5 \times 10^{-3}}$ **sec.**

This is the lag in time between voltage and current.

Ex. 9. *A coil of inductance of* 0.04 *henry is joined in series with a wire of zero inductance and resistance* 15 *ohm. An e.m.f. of* 120 *volt and* 50 *Hz is applied. Find* (i) *the current,* (ii) *the potential difference across the resistance and that across the inductance,* (iii) *angle of lag and,* (iv) *time lag.*

Sol. Here, $E_{rms} = 120$ volt, $\nu = 50$ Hz, $L = 0.04$ H, $R = 15\ \Omega$.

$\therefore$ Inductive reactance, $X_L = \omega L = 2\pi\nu L = 2 \times 3.14 \times 50 \times 0.04 = 12.56$ ohm.

and Impedance, $Z = \sqrt{R^2 + X_L^2} = \sqrt{(15)^2 + (12.56)^2} = \sqrt{225 + 158} = 19.6\ \Omega.$

(i) Current, $i_{rms} = \dfrac{E_{rms}}{Z} = \dfrac{120}{19.6} = \mathbf{6.12\ A.}$

(ii) P.D. across resistance $=$ Current $\times$ Resistance $= i_{rms} \times R = 6.12 \times 15 =$ **91.84 volt.**

P.D. across inductance $=$ Current $\times$ Reactance $= i_{rms} \times X_L = 6.12 \times 12.56 =$ **76.87 volt.**

(iii) Lag in phase angle is given by

$$\tan\theta = \frac{X_L}{R} = \frac{12.56}{15} = 0.84; \therefore \phi = \mathbf{40°}$$

(iv) Time lag $= \dfrac{\phi}{360°} \times$ Period $= \dfrac{40°}{360°} \times \dfrac{1}{50}$ sec $= \dfrac{1}{450} =$ **0.0022 sec.**

Ex. 10. *In a seies LCR circuit containing* $L = 1$ mH, $C = 0.1$ μF and $R = 10\ \Omega$, *an alternating e.m.f. of* 100 *volt is applied. Calculate :* (i) *frequency of resonance,* (ii) *frequency interval between the half power points and* (iii) *power factor at resonance. Show that at resonance, voltage magnification = quality factor.*

(Gwalior 2003; Ujjain 1990)

Sol. Here, $L = 1$ mH $= 10^{-3}$ H, $C = 0.1$ μF $= 10^{-7}$F, $R = 10\ \Omega$.

(i) Resonant frequency, $\nu_0 = \dfrac{1}{2\pi\sqrt{LC}} = \dfrac{1}{2 \times 3.14 \times \sqrt{10^{-3} \times 10^{-7}}} = \dfrac{10^5}{6.28} = \mathbf{1.59 \times 10^4\ Hz}$

(ii) Frequency interval between the half-power points

$$\nu_2 - \nu_1 = \frac{R}{2\pi L} = \frac{10}{2 \times 3.14 \times 10^{-3}} = \mathbf{1.59 \times 10^3\ Hz.}$$

(iii) Power factor, $\cos\theta = \mathbf{1}$ [$\because$ at resonance, current is in phase with the e.m.f. *i.e.*, $\phi = 0$]

At resonance,

$$\text{Voltage magnification} = \frac{\text{Potential difference across the inductance}}{\text{Applied e.m.f.}}$$

$$= \frac{i\omega_0 L}{E} = \frac{E}{R} \cdot \frac{\omega_0 L}{E} = \frac{\omega_0 L}{R} \qquad \left[\because \text{ at resonance } i = \frac{E}{R}\right]$$

$$= \frac{2\pi \times 1.59 \times 10^3 \times 10^{-3}}{10} = \mathbf{10}$$

But quality factor, $Q = \dfrac{\omega_0 L}{R}$ and hence $Q = 10$.

Thus at **resonance,** **voltage magnification = quality factor.**

Ex. 11. *A 20 volt, 5 watt lamp is to be used with A.C. source of 200 volt, 50 hertz. Calculate the inductance of the choke coil required to be put in series to run the lamp. How much pure resistance should be used in place of the choke coil, so that the lamp may run on its rated voltage ?* **(Meerut 2000)**

Sol. The current taken by the lamp, $i = \dfrac{\text{Power}}{\text{Voltage}} = \dfrac{5 \text{ watt}}{20 \text{ volt}} = 0.25$ A

Its resistance, $R = \dfrac{20 \text{ volt}}{0.25 \text{ amp}} = 80$ ohm.

If we want to run the lamp at 200 volt 50 Hz A.C. mains, a choke coil is to be placed in series with it so that its effective resistance or impedance is increased and the current through it does not exceed 0.25 A. If L be the inductance of the required choke, then its impedance is

$$Z = \sqrt{R^2 + \omega^2 L^2} = \sqrt{(80)^2 + (2 \times 3.14 \times 50)^2 L^2} = \sqrt{(80)^2 + (314L)^2}$$

Now, Current $= \dfrac{\text{Voltage}}{\text{Impedance}}$, *i.e.*, $0.25 = \dfrac{200}{\sqrt{(80)^2 + (314L)^2}}$ or $(80)^2 + (314\,L)^2 = (800)^2$

whence $L =$ **2.53 henry.**

If R' be the pure resistance of the circuit for which the current in the lamp would be 0.25 amp, then

$$i = \frac{E}{R'} \text{ or } 0.25 = \frac{200}{R'}, \therefore R' = 800 \text{ ohm.}$$

As the resistance R of the lamp itself is 80 ohm, the additional resistance required in the circuit is $R' - R = 800 - 80 =$ **720 ohm.**

Ex. 12. *A circuit contains a resistance of 4 ohm and an inductance of 0.68 henry, and an alternating e.m.f. of 500 volt at a frequency of 120 cycles/sec is applied to it. Find the value of the effective current in the circuit and the power factor.* **(Agra; Lucknow)**

Sol. Effective current $= \dfrac{\text{Effective e.m.f.}}{\text{Impedance}}$

Here, Effective e.m.f. = 500 volts,

$$\text{Impedance} = \sqrt{R^2 + \omega^2 L^2} = \sqrt{4^2 + (2\pi \times 120 \times 0.68)^2}$$
$$= \sqrt{16 + 262600} = 512.4 \text{ ohms}$$

$\therefore$ Effective current $= \dfrac{500}{512.4} =$ **0.976 amp.**

and, power factor, $\cos\theta = \dfrac{R}{\sqrt{R^2 + \omega^2 L^2}} = \dfrac{4}{512.4} = \mathbf{\dfrac{10}{1285}}$.

Ex. 13. *An E.M.F. of 100 volt R.M.S. at frequency of 50 cycles per second is applied to a circuit containing an inductance of 80 mH and a resistance of 20 ohm in series. Calculate* (a) *the magnitude and phase angle of the current,* (b) *the rate at which the energy is dissipated in the circuit.* **(Allahabad; Lucknow)**

Sol. R.M.S. current $= \dfrac{\text{R.M.S. E.M.F.}}{\sqrt{R^2 + \omega^2 L^2}}$

$$= \frac{100}{\sqrt{20^2 + (2 \times 3.1416 \times 50 \times 80 \times 10^{-3})^2}} = \frac{100}{\sqrt{20^2 + (25.13)^2}}$$

$$= \frac{100}{32.12} = \textbf{3.114 amp.}$$

Now, $\tan\theta = \omega L/R = 25.13/20 = 1.256 \quad \therefore \theta = \mathbf{\tan^{-1}(1.256)}$

Also, $\cos\theta = \dfrac{20}{\sqrt{20^2 + (25.13)^2}} = \dfrac{20}{31.12}$

$$\therefore \quad \text{Power} = E \times i \times \cos\theta$$

$$= 100 \times 3.114 \times \frac{20}{31.12} = \textbf{200 watt.}$$

Ex. 14. *An electric lamp which runs at 40 volts D.C. and consumes 10 amp. current is connected to A.C. mains at 100 volt 50 and Hz. Calculate the inductance of the choke.* **(Agra; Rajasthan)**

Sol. Clearly, resistance of the lamp $= \dfrac{40 \text{ volt}}{10 \text{ amp}} = 4$ ohm

Let L be the inductance of the required choke.

Then, $\quad \text{impedance} = \sqrt{R^2 + \omega^2 L^2} = \sqrt{4^2 + (2\pi \times 50)^2 L^2}$

Also, $\quad \text{impedance} = 100/10 = 10$ ohm

$$\therefore \quad \sqrt{4^2 + (2\pi \times 50)^2 \times L^2} = 10 \text{ or, } 16 + (100)^2\pi^2 L^2 = 100$$

$$\therefore \quad L = \frac{\sqrt{100-16}}{100\pi} = \frac{\sqrt{84}}{100 \times 3.142} = \textbf{0.029 henry.}$$

Ex. 15. *An alternating E.M.F. of 220 volt and 50 Hz is applied to a condenser in series with 20 votls, 5 watt lamp. Find the capacity required to run the lamp.*

(Poona; Agra; Allahabad)

Sol. Power of the lamp $= V \times i = 5$ watts.

$\therefore$ Current through the lamp $= i = 5/V = 5/20 = 0.25$ amp.

Resistance of the lamp, $R = V/i = 20/0.25 = 80$ ohm

Total impedance of the circuit, $Z = \dfrac{\text{Applied E.M.F.}}{\text{Current}} = \dfrac{200}{0.25} = 800$ ohm

But, $Z^2 = R^2 + (1/\omega C)^2$. Here, $R = 80$ ohm, $\omega = 2\pi \times 50$ and $C = ?$

$$\therefore (800)^2 = (80)^2 + \frac{1}{(2\pi \times 50 \times C)^2} \text{ or } \frac{1}{100\pi C} = \sqrt{800^2 - 80^2} = 80\sqrt{99}$$

$$\therefore \quad C = \frac{1}{8000 \times 3.142\sqrt{99}} = \textbf{4 μF (approx.)}$$

Ex. 16. *A 60 cps A.C. circuit has $R = 2\ \Omega$ and $L = 10.5$ mH. Find the value of the condenser placed in series that will make the power factor unity.* **(Agra 2005)**

Sol. Power factor, $\cos\theta = \dfrac{R}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$

For $\cos\theta = 1$,

$$\frac{R}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} = 1 \quad \text{or} \quad R^2 = R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2$$

or
$$\left(\omega L - \frac{1}{\omega C}\right)^2 = 0 \quad \text{or} \quad \omega L = \frac{1}{\omega C}$$

Hence
$$C = \frac{1}{L\omega^2} = \frac{1}{10.5 \times 10^{-3} \times (2\pi \times 60)^2} = \mathbf{6.7 \times 10^{-4}\ F.}$$

Thus $C = 6.7 \times 10^{-4}$ F capacity is to be placed in series to make the power factor unity.

Ex 17. *Find the capacity of the condenser which must be placed in series with a resistance of 5 ohm and inductance of 200 milli-henry to bring the current in phase with the voltage if the frequency be 60 c.p.s What current will flow if 100 volt be impresed on the circuit ?* **(Delhi, B.Sc. Hons.)**

Sol. For the current and voltage to be in phase, $\theta = 0$.

$$\therefore \quad \tan\theta = \frac{\omega L - (1/\omega C)}{R} = 0 \text{ or } \omega L - \frac{1}{\omega C} = 0$$

$$\therefore \quad C = \frac{1}{L\omega^2} = \frac{1}{200 \times 10^{-3} \times (2 \times 3.14 \times 60)^2} = \mathbf{35.2 \times 10^{-6} F = 35.2 \mu F}$$

$\therefore$ Current when 100 volt are impressed upon the circuit = 100/5 = **20 amp.**

Ex. 18. 2000-*ohm resistor and a* 1 μF *capacitor are connected in series across a* 120 *volt* (*rms*), 60 *cps line.* (a) *What is the impedance ?* (b) *What is the rms value of the current ?* (c) *What is the power dissipated in the circuit ?* (d) *What will be the readings of an A.C. voltmeter connected across the resistor and across the capacitor ?* **(Meerut 1983)**

Sol. Here, $E_{rms} = 120$ volt, $C = 10^{-6}$ F, $R = 2000\ \Omega$ and $\omega = 2\pi\nu = 120\pi$.

The capacitative reactance, $X_C = \dfrac{1}{\omega C} = \dfrac{1}{120 \times 3.14 \times 10^{-6}} = 2654$ ohm.

(a) Impedance, $Z = \sqrt{R^2 + X_C^2} = \sqrt{(2000)^2 + (2654)^2} = \mathbf{3323\ ohm.}$

(b) Current, $i_{rms} = \dfrac{E_{rms}}{Z} = \dfrac{120}{3323} = \mathbf{0.036\ amp.}$

(c) For a circuit containing capacitance and resistance in series, we have

$$\tan\theta = \frac{X_C}{R} = \frac{1/\omega C}{R}$$

$$\therefore \quad \text{power factor, } \cos\theta = \frac{R}{\sqrt{R^2 + 1/\omega^2 C^2}} = \frac{R}{Z} = \frac{2000}{3323} = 0.60.$$

and power dissipated $= E_{rms} \times i_{rms} \times \cos\theta = 120 \times 0.036 \times 0.60 = \mathbf{2.60\ watt.}$

(d) P.D. (*rms*) across the resistor $= i_{rms} \times R = 0.036 \times 2000 = \mathbf{72\ volt.}$

and P.D. (*rms*) across the capacitor $= i_{rms} \times X_C = 0.036 \times 2654 = \mathbf{95.5\ volt.}$

Ex. 19. *A coil of resistance R and self-inductance L is connected to a 60 Hz generator of e.m.f. 100 V (rms). The current through the coil is 0.2 A (rms) and the average power absorbed is 12 W. Find R and L. Show that the rms current in the coil is maximum when 6.6 μF capacitor is connected in series.*

Sol. Here, $\nu = 60$ Hz, $E_{rms} = 100$ V, $i_{rms} = 0.2$ A, $P_{av} = 12$ W

Therefore, $\omega = 2\pi\nu = 120\pi$, $E_0 = 100\sqrt{2}$ V and $i_0 = 0.2\sqrt{2}$ A.

Peak current through an *L-R* circuit, $i_0 = \dfrac{E_0}{\sqrt{R^2 + \omega^2 L^2}}$

$$\therefore \quad i_0^2 = \frac{E_0^2}{R^2 + \omega^2 L^2} = (0.2\sqrt{2})^2 \text{ or } \frac{E_0^2}{R^2 + \omega^2 L^2} = 0.08 \qquad \text{...(i)}$$

Average power absorbed in *L-R* circuit, $P_{av} = \dfrac{E_0^2 R}{2(R^2 + \omega^2 L^2)}$.

$$\text{But, } P_{av} = 12 \text{ W}, \qquad \therefore \quad \frac{E_0^2 R}{2(R^2 + \omega^2 L^2)} = 12. \qquad \text{...(ii)}$$

Dividing eq. (ii) by eq. (i), we obtain $R = \mathbf{300\ \Omega}$.

Putting the values of E_0, R and ω in eq. (i), we have

$$\frac{(100\sqrt{2})^2}{(300)^2 + (120\pi L)^2} = 0.08 \text{ or } (300)^2 + (120\pi L)^2 = \frac{(100\sqrt{2})^2}{0.08} = 250000 = (500)^2$$

or $$(120\pi L)^2 = (500)^2 - (300)^2 = (400)^2 \text{ or } 120\pi L = 400 \ \therefore\ L = \frac{400}{120 \times 3.14} = \mathbf{1.06\ H.}$$

The current will be maximum at resonance when a capacitor C is joined in series so that at $\omega L = 1/\omega C$ *i.e.*, when

$$C = \frac{1}{\omega^2 L} = \frac{1}{(120\pi)^2 \times 1.06} = 6.6 \times 10^{-6}\ \text{F} = 6.6\ \mu\text{F}.$$

Thus current is maximum for C = **6.6 μF**.

EXERCISE 8·1

1. A resistance of 4 Ω and a reactance of 3 Ω are connected in series in an A.C. circuit. What is the impedance of the circuit ? **(Meerut 2000 S)**
[**Ans.** 5 Ω.]

2. An electric lamp which runs at 100 volt D.C. and 10 amp is connected to 200 volt 50 cycles/sec A.C. mains. Calculate the inductance of the choke needed to be used in series with the lamp.
[**Ans.** 0.054 H.]

3. A 50 cycle A.C. of amplitude 5.0 amp. is flowing through a coil of resistance 10 ohm and inductance 50 millihenry. Calculate the rms value of the potential difference across the coil and also its phase with respect to the current.
[**Ans.** 65.8 volt; 57.5°.]

4. When an e.m.f. $E = 200 \sin 377t$ volt is applied across an inductance L having a resistance of 1.0 Ω, the maximum current is found to be 10 amp. Calculate L.
[**Ans.** 53 H.]

5. A filament lamp which requires 10 amp. at 100 volt is connected to A.C. mains of 150 volt and 50 cycles with a choke coil in series. Calculate the self-inductance and the drop of voltage across the choke. Neglect the resistance of choke. **(Magadh)**
[**Ans.** 0.35 H; 110 volts.]

6. A 60 volt 10 watt lamp is to be run on 100 volt 60 cycles mains. (a) Calculate the inductance of a choke coil that would be required in the circuit. (b) How much pure resistance would be necessary in the circuit to achieve the same result ? **(Rajasthan)**
[**Ans.** (a) 1.274 H; (b) 600 Ω.]

7. A steady current of 20 amperes flows through a coil of inductance 0.03 henry when connected to a 200 volt D.C. supply. Calculate the current, power absorbed and power factor of the coil when connected to 200 volt 50 cycles per second A.C. supply. **(Ujjain)**
[**Ans.** 14.56 A; 2119 watt; 0.728.]

8. An alternating voltage of 220 r.m.s. volts at a frequency of 40 cycles per second is applied to a circuit containing a pure inductance of 0.01 henry and a pure resistance of 6 ohms in series. Calculate (a) the current; (b) the potential drop across (i) resistance, (ii) inductance; (c) lag in time between the voltage and current. **(Patna)**
[**Ans.** (a) 32.83 A; (b) (i) 202.98 volt; (ii) 96.83 volt; (c) 1.6×10^{-3} sec.]

9. An alternating supply of 100 volts at a frequency of 100 cycles per second is connected to a condenser of capacity 0.1 μF and a resistance of 500 ohms in series. Calculate the power dissipated and the power factor.
[**Ans.** 1.8 watt; 0.3.]

10. Across a 100 volt, 25 cycles line, there are connected in series a resistance of 200 ohm, an inductance having a reactance of 100 ohm and a resistance of 40 ohm and a capacity of 25 micro-farad. Calculate (a) the impedance of the circuit, (b) the current, (c) the angle by which the current leads or lags the applied E.M.F. **(Lucknow)**
[**Ans.** (a) 285.5 Ω; (b) 0.38 amp.; (c) leads by $\theta = 32°48'$.]

11. An arc lamp consumes 10 amp current at 40 voltage. Calculate the power factor when it is connected with a suitable choke coil required to run the arc lamp on A.C. mains of 200 volt rms and 50 Hz. Calculate also the energy wasted per second if it were to run with a resistance on 200 volt D.C.
[**Ans.** 0.2; 1600 watt.]

12. A coil of inductance 0.05 H and resistance 12.4 Ω is connected to an A.C. source of 200 V and 50 Hz. Find (i) power factor, (ii) angle of lag, (iii) power consumed and (iv) rate of production of heat in the coil. **(Meerut 1997 S)**
[**Ans.** (i) 0.62; (ii) 51.7°; (iii) 1240 W; (iv) 1240 W.]
13. Find the capacitance of a capacitor to run a 20 volt, 5 watt lamp when connected in series to an alternating e.m.f. of 200 volt and 50 cycles per second. **(Jabalpur 1996; Sagar 92)**
[**Ans.** 4 μF.]
14. In an A.C. circuit an inductance of 1 henry, a capacitance of 0.2 microfarad and a resistance of 3000 ohm are connected in series to a 10 volt A.C. source of $1000/2\pi$ cycles per second. Find the peak current in the circuit and the power consumed. **(Meerut 1992, 89)**
[**Ans.** $2\sqrt{2}$; 12 mW.]
15. Determine the value of an inductance which should be connected in series with a capacitor of 5 microfarad, a resistance of 100 ohm and an A.C. source of 50 Hz so that the power factor of the circuit is unity.
[**Ans.** 2 H (nearly).]
16. A coil has inductance L and resistance 1 Ω. When an E.M.F. $E = 200 \sin 377t$ volt is applied, then the peak value of the current is found 10 A. Calculate the value of inductance L. **(Agra 2003)**
[**Ans.** 0.053 H.]
17. A coil of resistance 50 ohm and inductance 5 henry is connected in series with a condenser of 2.5 μF and A.C. supply of 200 volt and 50 Hz. Calculate : (i) the impedance in the circuit, (ii) the phase difference between current and voltage, (iii) potential differences across inductor and (iv) capacitor.
[**Ans.** (i) 300 W; (ii) 80.4°; (iii) 1047 volt; (iv) 849 volt.] **(Meerut 1994, 90)**
18. Find the capacity of the condenser which must be placed in series with a resistance of 5 ohm and inductance of 200 millihenry to bring the current in phase with the voltage, if the frequency be 60 cycles/sec. **(Meerut 1983 S)**
[**Ans.** 35 μF.]
19. In a series L-C-R circuit we have $E = 100$ volt, $f = 60$ Hz, $R = 1.0$ ohm, $L = 1.0$ henry and $C = 7.04$ μF. Prove that the voltage magnification is equal to the quality factor Q.
20. What is band-width of resonanace frequency 1.0 mega-hertz if the quality factor of circuit is 1.26×10^3 ? **(Bhopal 1990; Bilaspur 90)**
[**Ans.** 193.7 hertz.]
21. A condenser of capacity 1.5 μF and an inductance 1.0 mH are connected in series in an A.C. circuit. Calculate the natural frequency of the circuit in kilo-hertz. **(Rewa 1993; Indore 89)**
[**Ans.** 4.11 kilo-hertz.]
22. In a series resonant circuit $L = 1$ mH, $C = 10$ μF and $R = 10$ ohm. Calculate the resonant frequency and quality factor of the circuit. **(Bhopal 04; Jabalpur 2003; Gwalior 03)**
[**Ans.** 1.59×10^3 hertz, 1.0]

8·16. Parallel Resonant Circuit–Rejector Circuit

Let us consider an inductance L and resistance R in series which are in parallel with a capacitor of capacitance C. An alternating e.m.f.

$$E = E_0 e^{j\omega t} \quad ...(1)$$

is applied across this parallel circuit. [**fig. 8.29**].

If the current through the capacitance is i_C and that through the inductance L and resistance R is i_L, then the total current through the circuit is

$$i = i_C + i_L \quad ...(2)$$

The current i_C through the capacitance C leads in phase the e.m.f. E by $\pi/2$ and i_L through LR circuit lags in phase by an angle θ ($\theta < \pi/2$). Thus the phase difference between i_C and i_L is less than π. However, if R = 0, the phase difference between i_C and i_L will be π i.e., the currents in the two branches of the parallel circuit differ in phase by π or they are out of phase with each other.

Note that in the parallel circuit, the voltage across C is the same as voltage across L and R and both are in phase at every instant.

If Z is the impedance of total parallel circuit and Z_1 and Z_2 are the impedance of the C and LR branches, then

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2} \qquad ...(3)$$

Here $$Z_1 = \frac{1}{j\omega C} \text{ and } Z_2 = R + j\omega L \qquad ...(4)$$

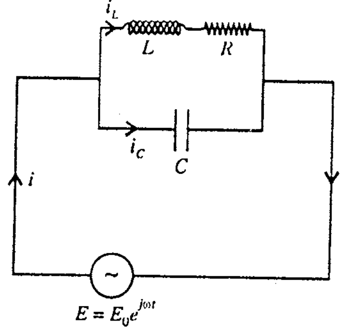

Fig. 8.29 : Parallel resonant circuit.

$\therefore$ Admittance, $$Y = \frac{1}{Z} = j\omega C + \frac{1}{R + j\omega L}$$

$$= j\omega C + \frac{R - j\omega L}{(R + j\omega L)(R - j\omega L)}$$

$$= j\omega C + \frac{R - j\omega L}{R^2 + \omega^2 L^2}$$

$$= \frac{R}{R^2 + \omega^2 L^2} + \frac{j\omega (CR^2 + C\omega^2 L^2 - L)}{R^2 + \omega^2 L^2} \qquad ...(5)$$

Hence the current i is given by

$$i = \frac{E}{Z} = EY = E\left[\frac{R}{R^2 + \omega^2 L^2} + \frac{j\omega (CR^2 + C\omega^2 L^2 - L)}{R^2 + \omega^2 L^2}\right] \qquad ...(6)$$

The resonance will occur, when the current is in phase with the applied e.m.f. In such a case, the impedance of the circuit is purely resistive and the **power factor is unity**. For current and applied e.m.f. to be in phase, the imaginary part in (6) must vanish *i.e.*, for resonance

$$\omega(CR^2 + C\omega^2 L^2 - L) = 0 \text{ or } CR^2 + C\omega^2 L^2 - L = 0 \quad [\because \omega \neq 0] \text{ or } \omega^2 L^2 = \frac{L}{C} - R^2,$$

$$\therefore \qquad \omega^2 = \frac{1}{LC} - \frac{R^2}{L^2} \text{ or } \omega = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} = \omega_r \qquad ...(7)$$

The frequency for which the power factor is unity is called **resonant frequency**. Hence ω_r, given by eq. (7), is the **resonant frequency of the parallel circuit.**

Clearly, parallel resonance will occur, when the resonant frequency ω_r is real *i.e.*,

$$\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} > 0 \text{ or } \frac{1}{LC} > \frac{R^2}{L^2} \text{ or } \frac{L}{C} > R^2 \text{ or } \sqrt{\frac{L}{C}} > R$$

Thus the necessary condition for parallel resonance is

$$R < \sqrt{\frac{L}{C}} \qquad ...(8)$$

For the values of L, C, R and ω, which satisfy eq. (7), the admittance Y is real and its value is minimum or impedance is maximum. Hence **at resonance, the circuit has maximum impedance and the circuit current i is minimum.** The variations of Z and i with frequency ω are shown in **fig. 8.30**. Hence *for the resonant frequency ω_r, the circuit has maximum impedance and minimum current and therefore does not allow this frequency from the source to pass through itself.* Due to this reason, the circuit works as **rejector** for this frequency (ω_r). However in series resonance, the impedance is minimum ($Z_0 = R$) and the current is maximum ($i_0 = E_0/R$). Hence the parallel resonance is also called **antiresonance.**

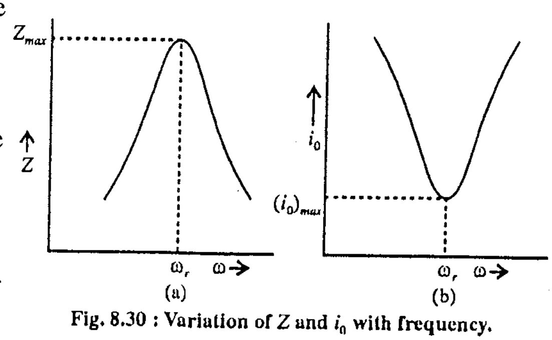

Fig. 8.30 : Variation of Z and i_0 with frequency.

The frequency given by eq. (7) at parallel resonance reduces to the frequency at series resonance $\omega_r = 1/\sqrt{LC}$, if R is very small so that R^2/L^2 is negligible relative to $1/LC$.

Dynamic resistance : The impedance of parallel circuit at resonance is called the **dynamic resistance** or **parallel resistance.** The current from supply in parallel resonance is given by [from eq. (6)]

$$i = \frac{E}{Z} = EY = E\frac{R}{R^2 + \omega_r^2 L^2} \quad ...(9)$$

Hence, the **impedance of parallel resonant circuit or dynamic resistance** is given by

$$Z = \frac{1}{Y} = \frac{R^2 + \omega_r^2 L^2}{R} = \frac{R^2 + L^2\left(\frac{1}{LC} - \frac{R^2}{L^2}\right)}{R} = \frac{L}{CR} \quad ...(10)$$

Current magnification : In parallel resonance, the current obtained from the supply is called **make up current** and the current through the capacitor or inductor is called **forced oscillatory current**.

Peak value of make up current, $i_0 = \frac{E_0}{L/CR} = \frac{CR}{L}E_0$ ...(11)

Peak value of the oscillatory current through capacitor,

$$i_{10} = \frac{E_0}{|Z_1|} = \frac{E_0}{1/\omega_r C} = \omega_r C E_0 \quad ...(12)$$

$$\text{Current magnification} = \frac{\text{Peak value of oscillatory current}}{\text{Peak value of make up current}} = \frac{\omega_r C E_0}{CRE_0/L} = \frac{\omega_r L}{R} = Q_0 \quad ...(13)$$

Thus, *the current magnification in parallel resonance is equal to the quality factor of the circuit.* Thus the rejector circuit at resonance shows a current magnification of Q_0-factor ($= \omega_r L/R$) similar to the voltage magnification in series resonance.

The current through the parallel resonant circuit is minimum and is zero for $R = 0$ and hence such a circuit works as choke for a particular frequency (ω_r). Due to the working of the circuit as rejector of particular frequency, it is used in filter circuits.

Comparision of Series and Parallel Resonant Circuits

Series Resonant Circuit	Parallel Resonant Circuit
1. In a series resonant circuit, L, C and R are connected in series with the alternating e.m.f.	1. In a parallel resonant circuit, L and R are connected in parallel with the capacitance C and then the system is connected to alternating e.m.f.
2. Resonant frequency is given by $\nu_r = \frac{1}{2\pi\sqrt{LC}}$.	2. Here, $\nu_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$.
3. At resonance, the impedance $Z = R$ (ohmic) which is minimum, when compared to that at all other frequencies.	3. At resonance, the impedance $Z = L/CR$ (ohmic), which is maximum when compared to that at all other frequencies.
4. At resonance, the power factor is unity.	4. At resonance, the power factor is also unity.
5. The current amplitude is maximum at resonance.	5. The current amplitude is minimum at resonance.
6. This circuit is called **acceptor circuit** as it accepts a particular frequency and rejects others.	6. This circuit is called rejector circuit, because it rejects one particular frequency and accepts others.
7. At resonance, the circuit shows **voltage magnification** which is equal to the quality factor $Q = \omega_r L/R$.	7. At resonance, the circuit exhibits current magnification, equal to quality factor $Q = \omega_r L/R$
8. At resonance, the potential differences across L and C are equal, but in opposite phase.	8. Here, the currents in L and C are in opposite phase.

Ex. 1. *A coil of inductance* 10 *mH and resistance* 10 Ω *is connected in parallel with a condenser of capacity* 0.1 μ*F. Find the impedance of the circuit at resonance.* **(Meerut 1989)**

Sol. Impedance of a parallel resonance L-C-R circuit is given by

$$Z = \frac{L}{CR}$$

Here, $L = 10$ mH $= 10 \times 10^{-3}$ henry, $C = 0.1$ μF $= 0.1 \times 10^{-6}$F and $R = 10$ ohm.

$$\therefore \qquad Z = \frac{10 \times 10^{-3}}{0.1 \times 10^{-6} \times 10} = \mathbf{10^4 \text{ ohm.}}$$

Ex. 2. *For the given circuit* [**fig. 8.31**], *find the resonance frequency and the impedance at resonance.* **(Meerut 1997)**

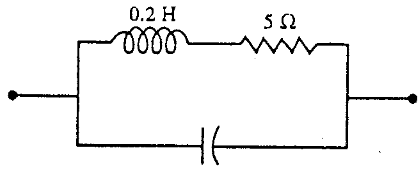

Fig. 8.31.

Sol. The given circuit is a parallel circuit in which the frequency at resonance is given by

$$\nu = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

Here, $\frac{R^2}{L^2} << \frac{1}{LC}$. Therefore,

$$\nu_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2 \times 3.14 \times \sqrt{0.2\text{H} \times (30 \times 10^{-6}\text{F})}} = \mathbf{65 \text{ Hz.}}$$

The impedance of resonant parallel circuit

$$Z = \frac{L}{RC} = \frac{0.2\text{ H}}{5\Omega \times (30 \times 10^{-6}\text{F})} = \mathbf{1.33 \times 10^3\ \Omega.}$$

EXERCISE 8·2

1. Calculate the impedance of an inductance of 2.0 H in parallel with a capacitor of 100 μF at resonance. The resistance of the inductance is 10 ohm. **(Meerut 2001)**
   [**Ans.** 2000 ohm.]
2. A condenser of capacity 100 pF is connected in parallel with a coil of inductance 20 μH and resistance 10 ohm, so as to obtain a parallel resonant circuit. Calculate the resonant frequency and impedance of the circuit.
   [**Ans.** 3.56 mega-hertz; 2×10^4 ohm.]
3. A coil of inductance 2 mH and resistance 15 Ω is connected in parallel with a capacitance of 0.001 μF. Calculate (a) the frequency at which the current from an a.c. source to this circuit is minimum, (b) the peak value of this make-up current if the peak value of supply voltage is 2 volt.
   [**Ans.** 1.126×10^5 Hz; 15 μA.]

   [**Hint** : Peak value of make up current at resonance $= \frac{E_0}{L/CR} = \frac{CRE_0}{L}$.]

8·17. *Y* and Δ Networks

Three elements or impedances Z_1, Z_2 and Z_3 can be connected in *Y* and Δ networks as suggested by their shapes.

***Y* Network** : In *Y* network, two impedanc Z_1 and Z_2 are connected in series arm and Z_3 impedance in the third arm.

Δ Network : In this network, Z_1' impedance is between Z_2' and Z_3' impedance, as shown in **fig. 8.33.**

We can transform *Y* network into Δ network or Δ network to *Y* network. The formulae for conversion are deduced below.

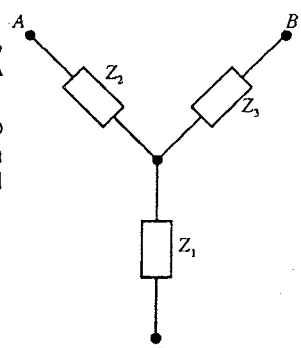

Fig. 8.32 : *Y*-Network.

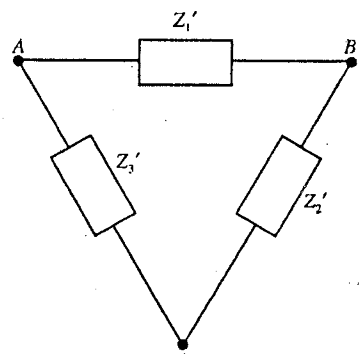

Fig. 8.33 : Δ-Network.

Conversion of Δ network into *Y* networks : In fig. 8.32, the impedance between *A* and *B* in *Y* network is

$$Z_{AB} = Z_2 + Z_3 \qquad ...(1)$$

In fig 8.33, the impedance Z_{AB} between *A* and *B* is Δ network is given by

$$\frac{1}{Z_{AB}} = \frac{1}{Z_1'} + \frac{1}{Z_2' + Z_3'} \text{ or } Z_{AB} = \frac{Z_1'(Z_2' + Z_3')}{Z_1' + Z_2' + Z_3'} \qquad ...(2)$$

Equating the impedances between *A* and *B* of the two networks from eqs. (1) and (2), we obtain

$$Z_2 + Z_3 = \frac{Z_1'(Z_2' + Z_3')}{Z_1' + Z_2' + Z_3'} \qquad ...(3)$$

In the same way, equating the impedances of two networks between *B* and *C*, we get

$$Z_3 + Z_1 = \frac{Z_2'(Z_3' + Z_1')}{Z_1' + Z_2' + Z_3'} \qquad ...(4)$$

and equating the impedances of two networks between *A* and *C*, we obtain

$$Z_2 + Z_1 = \frac{Z_3'(Z_1' + Z_2')}{Z_1' + Z_2' + Z_3'} \qquad ...(5)$$

Adding eqs. (4) and (5) and then subtracting eq. (3), we have

$$2Z_1 = \frac{Z_2'(Z_3' + Z_1') + Z_3'(Z_1' + Z_2') - Z_1'(Z_2' + Z_3')}{Z_1' + Z_2' + Z_3'}$$

$$= \frac{2Z_2'Z_3'}{Z}, \text{ where } Z = Z_1' + Z_2' + Z_3'$$

$$\therefore \quad Z_1 = \frac{Z_2'Z_3'}{Z} \qquad ...(6)$$

Similarly adding eqs. (3) and (5) and subtracting eq. (4), we get

$$Z_2 = \frac{Z_1'Z_3'}{Z} \qquad ...(7)$$

and adding eqs. (3) and (4) and subtracting eq. (5), we obtain

$$Z_3 = \frac{Z_1'Z_2'}{Z} \qquad ...(8)$$

Thus, if we know the impedances Z_1', Z_2' and Z_3' of Δ network, the impedances Z_1, Z_2 and Z_3 of *Y* network can be calculated with the help of eqs. (6), (7) and (8).

Conversion *Y* network into Δ network : From eqs. (6), (7) and (8), we obtain

$$Z_1Z_2 = \frac{Z_2'Z_3'}{Z} \times \frac{Z_1'Z_3'}{Z} = \frac{Z_1'Z_2'Z_3'^2}{Z^2} \qquad ...(9)$$

$$Z_1Z_3 = \frac{Z_2'Z_3'}{Z} \times \frac{Z_1'Z_2'}{Z} = \frac{Z_1'Z_2'^2Z_3'}{Z^2} \qquad ...(10)$$

and

$$Z_2Z_3 = \frac{Z_1'Z_3'}{Z} \times \frac{Z_1'Z_2'}{Z} = \frac{Z_1'^2Z_2'Z_3'}{Z^2} \qquad ...(11)$$

Adding eqs. (9), (10) and (11), we get

$$Z_1Z_2 + Z_1Z_3 + Z_2Z_3 = \frac{Z_1'Z_2'Z_3'}{Z^2}(Z_1' + Z_2' + Z_3') = \frac{Z_1'Z_2'Z_3'}{Z} \qquad [\because Z = Z_1' + Z_2' + Z_3']$$

But from eq. (6), $\dfrac{Z_2'Z_3'}{Z} = Z_1$,

Hence $\quad Z_1Z_2 + Z_1Z_3 + Z_2Z_3 = Z_1'Z_1$

$$\therefore \quad Z_1' = \frac{Z_1Z_2 + Z_1Z_3 + Z_2Z_3}{Z_1} \qquad ...(12)$$

Similarly we can get

$$Z_2' = \frac{Z_1Z_2 + Z_1Z_3 + Z_2Z_3}{Z_2} \quad ...(13)$$

$$Z_3' = \frac{Z_1Z_2 + Z_1Z_3 + Z_2Z_3}{Z_3} \quad ...(14)$$

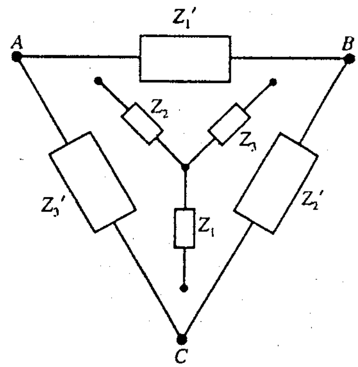

Fig. 8.34.

Thus if we know the impedances Z_1, Z_2 and Z_3 of Y network the impedances Z_1', Z_2' and Z_3' of Δ network can be calculated by using eqs. (12), (13) and (14).

It is to be noted that the impedance Z_1 is opposite to closed side Z_1', Z_2 to Z_2' and Z_3 to Z_3'. Further each impedance in an open arm has two adjacent impedances : Z_1 has Z_2' and Z_3'; Z_2 has Z_1' and Z_3' and Z_3 has Z_1' and Z_2' as adjacent impedances. In order to remember the formulae, place Y network inside the Δ network [fig. **8.34**].

Ex. 1. *Construct Y network equivalent to Δ network shown by* fig. **8.35.** **(Gwalior 2004)**

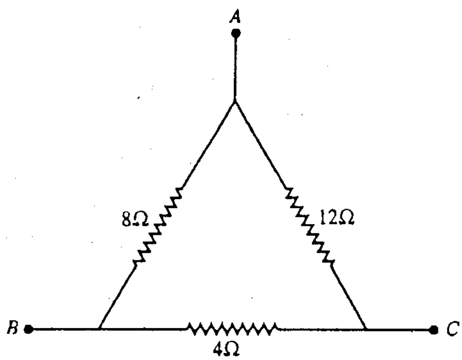

Fig. 8.35.

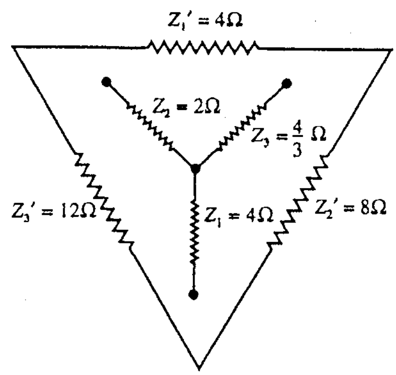

Fig. 8.36.

Sol. The given Δ network can be drawn as in fig. **8.36**.

Hence $\qquad Z = Z_1' + Z_2' + Z_3' = 4 + 8 + 12 = 24\ \Omega.$

Therefore, the impedances Z_1, Z_2 and Z_3 of Y network are given by

$$Z_1 = \frac{Z_2'Z_3'}{Z} = \frac{8 \times 12}{24} = 4\ \Omega$$

$$Z_2 = \frac{Z_1'Z_3'}{Z} = \frac{4 \times 12}{24} = 2\ \Omega$$

$$Z_3 = \frac{Z_1'Z_2'}{Z} = \frac{4 \times 8}{24} = \frac{4}{3}\ \Omega$$

Ex. 2. *Convert the given network* [fig. **8.37**] *to equivalent Δ network.*

Sol.

$$R_1' = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1} = \frac{5\times3+3\times4+4\times5}{5} = \frac{47}{5}\ \Omega$$

$$R_2' = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2} = \frac{5\times3+3\times4+4\times5}{3} = \frac{47}{3}\ \Omega$$

$$R_3' = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3} = \frac{5\times3+3\times4+4\times5}{4} = \frac{47}{4}\ \Omega$$

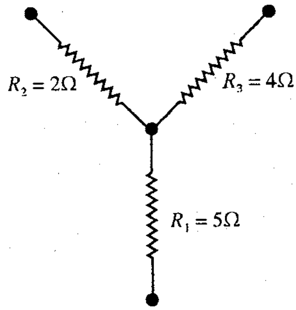

Fig. 8.37.

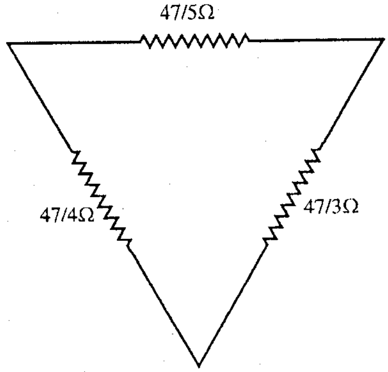

Fig. 8.38.

The equivalent Δ network is shown in **fig. 8.38.**

EXERCISE 8·3

1. Calculate the resistances of Y network equivalent to Δ network as shown in **fig. 8.39.**

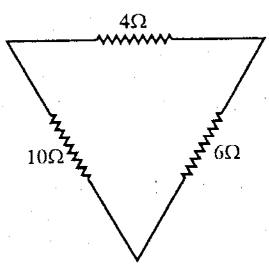

Fig. 8.39.

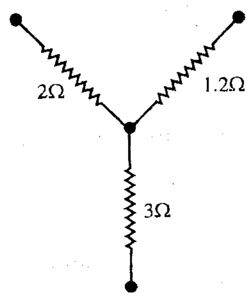

Fig. 8.40.

[**Ans. Fig. 8.40** shows Y network equivalent to the given network.]

2. Calculate Δ network equivalent to Y network as shown in [**fig. 8.41**].

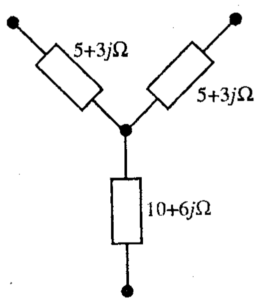

Fig. 8.41.

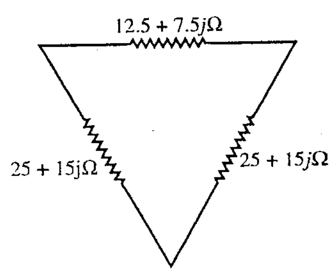

Fig. 8.42.

[**Ans. Fig 8.42** shows the equivalent Δ network.]

8·18. Transmision of Electric Power—Use of Y and Δ Connections

In a simple A.C. generator a coil rotates in a magnetic field and a sinusoidal e.m.f. $E = E_0 \sin \omega t$ is produced at the two terminals. This kind of generator is called *single phase generator*. **Polyphase A.C. generators** are used for power generation because (1) in these generators, the total power does not fluctuate, while it fluctuates in a single phase generator; (2) these generators give more output power than single phase generator of same size and (3) these generators are superior power for transmission and distribution purposes.

In a three phase generator, there are three coils equally spaced around a circle and inclined at 120° to one another. They are rotated at constant angular speed in a uniform magnetic field, so that they produce e.m.f.'s 120° out of phase with each other. The e.m.f.'s, so generated, are given by

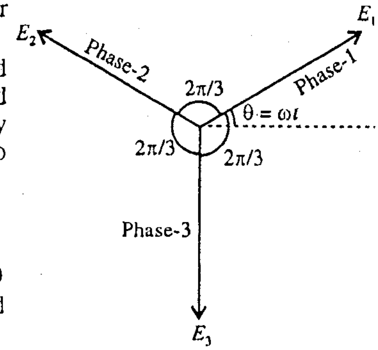

Fig. 8.43.

$$\left.\begin{aligned} E_1 &= E_0 \sin \omega t, \\ E_2 &= E_0 \sin (\omega t - 2\pi/3) \\ \text{and} \qquad E_3 &= E_0 \sin (\omega t - 4\pi/3) \end{aligned}\right\} \qquad ...(1)$$

where the e.m.f. of second coil will lag behind that of A by 1/3 period (or $2\pi/3$) and that of third coil by 2/3 period or $4\pi/3$ [**fig. 8.43**].

In a three phase generator, there will be six end terminals from the three coils of the generator and hence for power transmission six heavy wires will be required. In order to reduce the cost of material of wires, the end terminals of the wires are so joined that the number of heavy wires to carry the currents of the three phases is reduced. There are following two ways to connect the three coils of the generator; so that the distribution of current is brought about with a minimum number of conducting wires.

(1) Y (or Star) connection, (2) Δ (or Delta) connection.

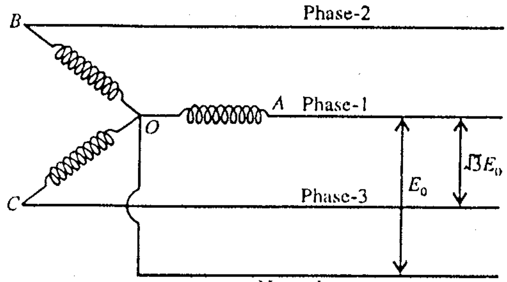

Fig. 8.44 : Y or Star connection.

(1) **Y (or Star) connection :** In this system, the three coils (or windings) of the generator are joined at one end, called as **common point** O, and the opposite ends of the three coils are used for the output terminals A, B and C. In the Y connection any pair of terminals is across two coils in series. A wire, known as the **neutral wire** is connected to the common point O and three heavy wires, connected to the end terminals A, B and C of the three coils, are called **phase wires or lines,** which carry the currents of phase 1, phase 2 and phase 3 [**fig. 8.44**]. Thus Y connection is **three phase-four wires system.** We can obtain a phase current by connecting a load (*e.g.*, lamp, fan etc.) between one of the phase wires and the neutral. The voltage across one of the phase wires and the neutral wire is called **phase voltage.** Let these voltages be denoted by E_1, E_2 and E_3. For identical coils, the amplitude of each phase voltage is the same. If E_0 is the amplitude of the phase voltage, the voltage difference between two phase wires (lines) is

$$E_{12} = E_1 - E_2 = E_{23} = E_2 - E_3 = E_{31} = E_3 - E_1$$

$$E_{12} = E_1 - E_2 = E_0 \sin \omega t - E_0 \sin\left(\omega t - \frac{2\pi}{3}\right)$$

$$= 2E_0 \cos\left(\omega t - \frac{\pi}{3}\right) \sin\frac{\pi}{3} = \sqrt{3}E_0 \cos\left(\omega t - \frac{\pi}{3}\right) = E_{23} = E_{31} \qquad ...(2)$$

Thus the voltage amplitude is $\sqrt{3}E_0$. Generally for our house supply we get single phase A.C., where one line wire and a neutral wire is connected to the supply. The house supply voltage or phase voltage is 230 volts. The voltage between any two phase wires called the **line voltage,** is $\sqrt{3} \times 230$ = $1.73 \times 230 = 400$ volts. For running heavy machinary in factories, three phase A.C. supply at 400 volts is used.

We call the total load to be balanced when all the three phase currents have same amplitude and differing in phase from one another by $2\pi/3$. In such a case, the sum of three currents, returning through the neutral wire, is zero *i.e.*, $i_1 + i_2 + i_3 = 0$ *i.e.*, the total current at star point O is zero*. As no current passes through the neutral wire, the line current is equal to phase current.

---

* Total current at O, $\quad i = i_1 + i_2 + i_3 = i_0 \sin \omega t + i_0 \sin\left(\omega t - \frac{2\pi}{3}\right) + i_0 \sin\left(\omega t - \frac{4\pi}{3}\right)$

$$= i_0 \sin \omega t + 2i_0 \sin (\omega t - \pi) \cos \frac{\pi}{3} = i_0 \sin \omega t - 2i_0 \sin \omega t \times \frac{1}{2} = 0$$

(2) Δ (Delta) connection : In this type connection, three coils of the generator are joined in series **[fig. 8.45]**. Therefore, the line voltage *i.e.*, the difference of voltage between two phase wires is the same as that (phase voltage) across each coil *i.e.*, line voltage is equal to the phase voltage. The line current is given by

$$i_L = i_1 - i_2$$
$$= i_0 \sin \omega t - i_0 \sin (\omega t - 2\pi/3)$$
$$= 2i_0 \cos (\omega t - \pi/3) \sin \pi/3 = \sqrt{3} i_0 \cos(\omega t - \pi/3) \quad ...(3)$$

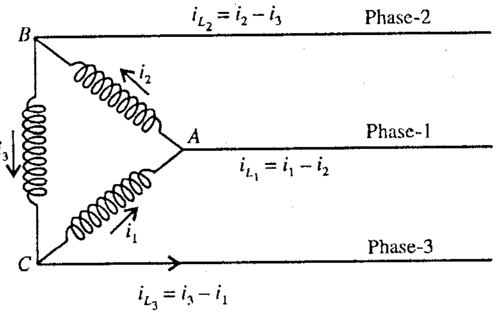

Fig. 8.45 : Δ-connection.

Hence the line current amplitude is $\sqrt{3} i_0$ *i.e.*, $\sqrt{3}$ times of phase current amplitude. Thus *the power consumption is the same in both cases.*

Thus we see that by the use of Y or Δ connections, the number of current carrying wires is reduced resulting in saving a lot of material of wires specially for long distances. Further, for the transmission of power for long distances, transformers are used to reduce the power losses.

Generally, A.C. generating stations are far away from users in the cities or towns. At the generating station, first the A.C. is stepped up by step up transformer in voltage and then at very high voltage, it is transmitted to long distances for use. At very high voltage, the current is very small and hence the power loss $i^2 R$ in transmission in the wires is very much reduced. In a town or city, where the electricity is to be used, the voltage is stepped down by step down transformer to the desired value *e.g.* 230 volts.

Comparison of Y and Δ Connections

Quantity	Y connection	Δ connection
1. System	3 Phase, 4 wire system	3 phase, 3 wire system
2. Line current	Phase current	$\sqrt{3}$ × phase current
3. Line voltage	$\sqrt{3}$ × phase voltage	Phase voltage
4. Power consumption	P	P (same)
5. Cost	At the same cost, the e.m.f. produced per phase is $\sqrt{3}$ times more in the form of line voltage.	To obtain the same voltage for each phase, the cost of copper increases and size of generator is bigger.

QUESTIONS

Long Answer Questions

1. What is a complex number ? Why are the complex numbers used in A.C. circuit analysis ?
2. What are peak value, mean value and rms value of an alternating current ? How are these related ? What is the virtual value of the alternating current ? **(Meerut 2003)**
3. (a) Show that the current in an A.C. circuit containing pure inductance lags in phase the e.m.f. by $\pi/2$.

   (b) Show that the current in an A.C. circuit containing pure capacitance leads in phase the e.m.f. by $\pi/2$.
4. A circuit contains an inductance and resistance in series. An altenating e.m.f. is applied to it. Find the effective value of current in circuit and also the phase difference between the current and potential difference. Explain the meaning of the terms reactance of different components of circuit and the impedance of circuit. **(Ujjain 1992)**
5. An alternating e.m.f. is applied to a series circuit containing a condenser and a resistance. Find the effective value of current and the phase difference between the current and potential difference. Explain the terms reactance and impedance. **(Jabalpur 2003, 1992; Meerut 2000 S)**
6. An alternating e.m.f. $E_0 \sin \omega t$ is applied to a circuit containing an inductance L and resistance R in series. Calculate the current at an instant. Hence explain the terms 'impedance' and 'reactance'

of the circuit. Also find the voltage across the resistance and across the inductance. Use it to obtain the impedance of the circuit. **(Meerut 2002)**

7. An alternating e.m.f. $V = V_0 \sin \omega t$ is applied to a series combination of resistance R, inductance L and capcacitance C. Find : (i) current in circuit. (ii) phase difference between the current and potential difference, (iii) potentail difference across the resistance, inductance and capacitance, (iv) average power consumed in the circuit, (v) condition of resonance and (vi) the resonant frequency. **(Rewa 2004; Bhopal 03; Bilaspur 03; Gwalior 1993; Indore 95)**
8. An alternating e.m.f. $E = E_0 \sin \omega t$ is applied to a circuit containing resistance R, inductance L and capacitance C in series. Find expression for the current at any instant, and calculate the phase angle and impedance of the circuit. Discuss the condition of resonance. **(Meerut 2002 S, 2000)**
9. What is a series resonant circuit ? How does the current depend on the frequency of applied potential difference ? Draw diagram to represent it. Obtain expression for the resonant frequency. Establish relationship between the band-width and the quality factor. How does the sharpness of resonance depend upon the quality factor of the circuit ? **(Gwalior 2004; Bundelkhand 04; Bhopal 1995)**
10. Discuss the varitaion of phase angle (ϕ) with driving frequency in a series LCR circuit. **(Agra 2006)**
11. Deduce an expression for current in a series alternating LCR circuit by the method of complex numbers. **(Bundelkhand 2004)**
12. Define quality factor of a series LCR circuit and show that the quality factor of this circuit is equal to the voltage magnification. **(Rewa 1995)**
13. Deduce an expression for average power in a complete cycle in an alternating current circuit. What do you understand by the wattless current ? Explain with example. **(Indore 2004; Ujjain 03)**
14. A condenser of capacity C and an inductance L are connected in parallel with an alternating e.m.f. Obtain expressions for the resonant frequency, impedance and current magnification of the circuit. **(Sagar 2003)**
15. What is a parallel resonant circuit ? Obtain expressions for the resonant frequency, impedance and current magnification for the circuit. Why is this circuit called the rejector circuit ? **(Ujjain 2004; Indore 03; Raipur 1992)**
16. Explain quality factor and sharpness of resonance for a LCR circuit. **(Agra 2004)**
17. What is meant by sharpness of resonance ? Explain by sketching the frequency response curve of LCR circuit. How is sharpness of resonance related to the Q of the circuit ? **(Meerut 2005)**
18. What is meant by resonance in an electric circuit ? Distinguish between the series and parallel resonant circuits. **(Jabalpur 2004; Gwalior 1995; Bhopal 90)**
19. Compare between the series and resonant circuits. **(Indore 2004, 03)**
20. What is meant by resonance in parallel circuit ? Obtain the condition of resonance in such a circuit. Why this circuit is known as rejector circuit ? **(Meerut 2005)**
21. What is meant by Y and Δ networks ? How will you obtain Y network from a Δ network and Δ network from a Y network ? **(Gwalior 2004; Ujjain 2004)**
22. What is the use of Y and Δ connections in three phase power transmission.

Short Answer Questions

1. What do you mean by rms value of current ?
2. Why rms value of alternating current is called virtual or effective value of current ?
3. What do you undestand by phase lag and phase lead in an A.C. circuit.
4. Explain the difference in behaviour of a ohmic resistance and a reactance in an A.C. circuit. **(Ujjain 1996, 92)**

   [**Ans.** Reactance $X_L = \omega L$ or $X_C = 1/\omega C$ depend on the A.C. frequency, while resistance is independent of the frequency. Further when an alternating e.m.f. is applied to an A.C. circuit, the current is in phase with the e.m.f. in a resistance but it has phase difference with the e.m.f. in a reactance.]
5. How are the inductive reactance X_L and capacitative reactance X_C affected, if the frequency of A.C. source is (i) doubled, (ii) made zero ? **(Gwalior 1992)**

[**Ans.** (i) When the frequency is doubled $X_L = \omega L$ is doubled, but $X_C = 1/\omega C$ is halved; (ii) for $\omega = 0$, $X_L = 0$ and $X_C = \infty$.]

6. Draw phasor diagrams for alternating current *LR* and *CR* circuits.
7. In an alternating *LC* circuit, the potential difference across the inductance *L* is $V_L = 50$ volt and the potential difference across the condenser *C* is $V_C = 30$ volt. What is the potential difference applied ? **(Rewa 1995)**

   [**Hint :** In *LC* circuit, applied e.m.f. = $V_L \sim V_C$]
8. A bulb and a condenser are connected in series with an alternating source. What happens when the frequency of A.C. source is increased ?

   [**Ans.** On increasing the frequency, the reactance of the capacitative circuit will decrease and hence due to the increase in the current, the bulb will glow brighter.]
9. Explain the behaviour of a condenser in an A.C. and D.C. circuit. **(Rewa 1994)**

   [**Ans.** A.C. can pass through the condenser, but D.C. will not pass through it.]
10. Show that power factor of purely inductive or capacitative A.C. circuit is zero. **(Agra 2006)**
11. What is wattless current in an A.C. circuit ? **(Kumaun 1992)**
12. Show that for *LCR* circuit the band-width is more for more value of the resistance.
13. Show that the voltage appearing across *L* or *C* at resonance in a series *LCR* circuit is *Q* times the applied voltage. **(Meerut 1997)**
14. What do you understand by voltage magnification ?
15. What is current magnification ?
16. Write the expression for resonant frequency in series and parallel circuits.

$$\left[\textbf{Ans.}\ \nu_s = \frac{1}{2\pi\sqrt{LC}}; \nu_p = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}.\right]$$

17. What are acceptor and rejector circuits ?
18. Distinguigh between series and parallel resonance.
19. Show that in a resonant circuit the band-width is inversely proportional to the *Q* of the circuit. **(Kanpur 2005)**
20. What is the necessary condition for the state of resonance in the parallel circuit when impedance has some resistance. **(Lucknow 2005)**

    [**Ans.** $R < \sqrt{L/C}$]
21. Show that in series resonance *LCR* circuit, sharpness of resonance is equal to the quality factor. **(Lucknow 2005)**
22. How is electric power transmitted from a power station to houses ? **(Agra 2004)**

Objective Type Questions

1. An A.C. voltmeter measures 20 volt in a circuit. The peak value of the voltage is :

   (a) $20/\sqrt{2}$ volt (b) $20\sqrt{2}$ volt (c) 400 volt (d) 10 volt
2. The r.m.s. value alternating current and its peak value I_0 related as :

   (a) $I_{rms} = 2I_0$ (b) $I_0 = I_{rms}/\sqrt{2}$ (c) $I_{rms} = I_0/\sqrt{2}$ (d) $I_0 = 2I_{rms}$ **(Bhopal 1992)**
3. The phase difference between the applied alternating e.m.f. and the current in a pure inductance is :

   (a) 0 (b) $\pi/2$ (c) π (d) 2π
4. The phase difference between the voltage and the current in an A.C. circuit containg resistance is :

   (a) 0 (b) $\pi/2$ (c) π (d) $3\pi/2$
5. In a purely capacitative A.C. circuit, the current :

   (a) in phase with the e.m.f. (b) lags behind the e.m.f. by π

   (c) lags behind the e.m.f. by $\pi/2$ (d) leads the e.m.f. by $\pi/2$

6. An alternating e.m.f. $E = 200\sqrt{2} \sin(100t)$ is applied to 1 μF capacitor through an ammeter. The ammeter will read :
(a) 10 mA (b) 20 mA (c) 40 mA (d) 80 mA

7. An A.C. having the peak value 1.4 A is used to heat a wire. A D.C. producing the same rate of heating will be :
(a) 2.0 A (b) 1.41 A (c) 0.707 A (d) 1 A

8. A hot wire ammeter can measure :
(a) D.C. only (b) A.C. only (c) Both A.C. and D.C. (d) temperature

9. A series circuit cotains two pure components. The current flowing in circuit and the applied e.m.f. are respectively,

$$I = 13.43 \sin(500t - 53.4°) \text{ amp and } E = 150 \sin(500t + 10°) \text{ volt}$$

The components of circuit are :
(a) resistance and condenser (b) resistance and inductance
(c) inductance and condenser (d) only two resistances **(Gwalior 1989)**

10. If Z is the impedance of LCR circuit, the power factor is :
(a) $\omega L/Z$ (b) $1/\omega CR$ (c) R/Z (d) L/C **(Bhopal 1996)**

11. In an A.C. circuit, the voltages across R, L and C are 40 volt, 90 volt and 60 volt respectively. The applied e.m.f. is :
(a) 50 V (b) 70 V (c) 100 V (d) 150 V

12. In a circuit with alternating source of 200 V and 50 Hz, the phase difference between the e.m.f. and current is $\pi/6$. The power factor of the circuit will be :
(a) 1/2 (b) $1/\sqrt{2}$ (c) $\sqrt{3}/2$ (d) $\sqrt{2}$ **(Bhopal 1994)**

13. A coil of inductance 10 H and resistance 10 Ω is connected to an alternating e.m.f. of 100 V and frequency 50 Hz. The alternating current will be
(a) 0.03 A, (b) 1 A, (c) 5 A, (d) 10 A
and the power factor of circuit will be :
(a) 1 (b) 0.5 (c) 0.003 (d) zero **(Bhopal 1993)**

14. The value of power factor is :
(a) 1 in pure inductance (b) 1 in pure capacitance
(c) 1 in a special case in LCR circuit, (d) never 1. **(Bhopal 1994)**

15. In a LCR circuit with an A.C. source, inductive reactance is X_L an capacitive reactance is X_C. Resonance is obtained when :
(a) $X_L = X_C$ (b) $X_L + X_C = 0$ (c) $X_C = 1/X_L$ (d) none of these **(Gwalior 1991)**

16. In an alternating resonant LCR circuit, the resonant frequency depends on :
(a) R (b) L and C (c) R and L (d) C and R **(Rewa 1992, 94)**

17. In an alternating current circuit, at resonance the phase difference between the applied e.m.f. current is :
(a) + 90° (b) 0° (c) – 90° (d) 180° **(Rewa 1993)**

18. A resistance R, inductance L and capacitance C are connected in series to an oscillator of frequency f. If f_r is the resonant frequency, the current will lag-behind the voltage when :
(a) $f < f_r$ (b) $f > f_r$ (c) $f = 0$ (d) $f = f_r$

19. Power in A.C. circuit when voltage and current are in phase, is
(a) Ei (b) $Ei \sin\theta$ (c) $Ei \cos\theta$ (d) zero

20. At resonance, the impedance of A.C. circuit is :
(a) Purely capacitative (b) Purely inductive
(c) Purely resistive (d) None of these

21. In a series *LCR* A.C. circuit, when $\omega L = 1/\omega C$:
(a) current is maximum (b) current is minimum
(c) the phase difference between the current and potential difference is 90°
(d) the phase difference between the current and potential difference is 180° **(Bhopal 1990)**

22. In parallel resonant circuit, at resonance the impedance is :
(a) Minimum (b) Maximum
(c) Equal to difference of inductive and capacitative impedances
(d) Neither maximum nor minimum.

23. Quality factor Q of a *LCR* circuit at resonance is :
(a) $Q = \omega_0 L/R$ (b) $Q = \omega_0 LR$ (c) $Q = R/\omega_0 L$ (d) $Q = 1/\omega_0 LR$

24. The impedances of the acceptor and rejector circuit in the resonance conditions are respectively :
(a) R, R (b) R, L/CR (c) L/CR, R (d) $1/R$, C/LR **(Agra 2004)**

25. In a parallel resonant circuit, at resonance :
(a) current is maximum (b) impedance is maximum
(c) impedance is minimum (d) impedance is zero **(Bhopal 1992)**

26. The phase difference between the wattless current and the applied e.m.f. is :
(a) zero (b) $\pm \pi/4$ (c) $\pm \pi/3$ (d) $\pm \pi/2$ **(Rewa 1995)**

27. An alternating current can induce voltage because it has magnetic field which is :
(a) time varying (b) constant
(c) stronger than d.c. field (d) of closed loop nature **(Agra 2006)**

ANSWERS

1. (b)	**2.** (c)	**3.** (b)	**4.** (a)	**5.** (d)	**6.** (b)	**7.** (a)	**8.** (c)
9. (b)	**10.** (c)	**11.** (a)	**12.** (c)	**13.** (a),(c)	**14.** (c)	**15.** (a)	**16.** (b)
17. (b)	**18.** (b)	**19.** (a)	**20.** (c)	**21.** (a)	**22.** (b)	**23.** (a)	**24.** (b)
25. (b)	**26.** (d)	**27.** (a)					

9

Magnetostatics

9·1. Introduction

The word magnetism comes from the name of place Magnesia (now Manisa, in Western Turkey) where certain stones were found about 2500 years ago which have the property of attracting small pieces of iron. These stones are in fact examples of natural *permanent magnets*. When a magnet is freely suspended by a string, it turns and acquires rest position in north-south direction. The end of the magnet pointing towards north were named as the **north pole** or **N-pole** and the other end as **south pole** or **S-pole**. Opposite poles attract each other and like poles repel each other. Earlier it was considered that a magnet was a magnetic dipole with N- and S- poles similar to an electric dipole with positive and negative charges at a distance. This is why certain magnetic phenomena are studied under the heading 'Magnetostatics' similar to the study of static charges in 'Electrostatics'.

The concept of magnetic poles may seem similar to that of electric charges, and north and south poles may appear analogous to positive and negative charges. However, in case of an electric dipole, *it is possible to isolate a single positive or negative charge, but there is no experimental evidence that a single magnetic pole exists.* In fact, the magnetic poles always appear in pairs. If a bar magnet is broken into two pieces, we will not get north pole and south pole separately, but get each broken piece having two opposite poles (or get two complete smaller bar magnets). Thus we conclude that (i) *an isolated magnetic monopole does not exist* and (ii) *magnetic poles always appear as dipoles.*

In 1820, Oersted discovered that when a compass needle is brought near a current carrying wire, it suffers deflection. This means that a current carrying conductor produces magnetic field in the space around. The charges in motion constitute an electric current and hence a **magnetic field is set up by the moving charges.** Remember that the charges at rest produce only electric field (not magnetic field) and moving charges produce magnetic field in addition to electric field. If a current flows through a conducting wire, there is only the magnetic field around the wire because in any volume of the wire on an average, the positive charge is neutralized by the negative charge of moving electrons. Now, if a charge moves (or a current carrying wire exists) in the magnetic field, in general it experiences a (magnetic) force.

Thus we conclude the following :

(1) *A moving charge or a current produces a magnetic field in the surrounding space* (in addition to the electric field).

(2) *The magnetic field exerts a force on any other moving charge or current, present in the field.*

At present, magnetism is regarded as a part of the effects produced by moving electric charges or electric currents and all electrical and magnetic phenomena are studied under the same title, **Electromagnetism** or **Electrodynamics**.

In this chapter, first we will study the effect of magnetic field on moving charge or current carrying conductor and next we shall discuss the production of magnetic field by moving charge or current carrying conductor.

9·2. Force on a Moving Charge in a Magnetic Field – Lorentz Force

If a charged particle moves in a magnetic field, it experiences a force. This *magnetic force* is known as **Lorentz force**. The direction of this force is perpendicular to the direction of the magnetic field as well as the direction of the velocity. Consider a particle with charge $+q$, moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$, then the **Lorentz force** on the particle is given by

$$\vec{F} = q(\vec{v} \times \vec{B}) \quad \text{...(1a)}$$

In magnitude, $$F = qvB \sin\theta \quad \text{...(1b)}$$

where θ is the angle between the velocity of the particle $\vec{v}$ and magnetic field $\vec{B}$ [fig. 9.1]. The direction of the Lorentz force is perpendicular to the plane containing the vectors $\vec{v}$ and $\vec{B}$.

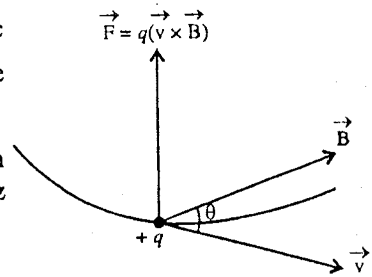

Fig. 9.1 : Lorentz force.

Eq. (1) was not deduced theoretically. It has been established on the basis of experimental observations. This equation is called **Lorentz force equation.**

From eq. (1), we observe the following :

(a) The force $\vec{F} = 0$ (or minimum) for (i) $\vec{v} = 0$, (ii) $\vec{B} = 0$, or (iii) $\vec{v} \times \vec{B} = 0$ *i.e.*, $\vec{v}$ is parallel to $\vec{B}$.

(b) When $\vec{v}$ is perpendicular to $\vec{B}$, (*i.e.*, $\theta = 90°$ or $\sin\theta = 1$), the force $\vec{F}$ has the **maximum value,** given by $F = qvB$. As this force is perpendicular to the velocity of the particle, the particle will move on a circular path of radius r in a transverse magnetic field and the necessary centripetal force for circular motion is provided by the magnetic force [fig. 9.2] *i.e.*,

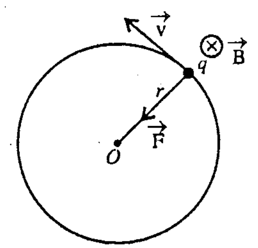

Fig. 9.2 : Motion of a charged particle in a uniform transverse magnetic field.

$$\frac{mv^2}{r} = qvB \quad \text{...(2)}$$

$\therefore$ Radius of circular path, $$r = \frac{mv}{qB} \quad \text{...(3)}$$

This is called **gyroradius** or cyclotron radius.

Thus the radius of curvature of the path of the charged particle is proportional to its momentum (mv).

Angular velocity in circular motion, $$\omega = \frac{v}{r} = \frac{qB}{m} \quad \text{...(4)}$$

Period of rotation, $$T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB} \quad \text{...(5)}$$

Frequency of rotation, $$\nu = \frac{\omega}{2\pi} = \frac{qB}{2\pi m} \quad \text{...(6)}$$

Thus the period and frequency are independent of the speed of the particle.

(c) If the charged particle moves with velocity $\vec{v}$ at an angle with the magnetic field $\vec{B}$, the velocity will have two components $\vec{v}_1$, along the field and $\vec{v}_2$ perpendicular to the field. The magnetic force on the particle is

$$\vec{F} = q(\vec{v} \times \vec{B}) = q[(\vec{v}_1 + \vec{v}_2) \times \vec{B}] = q(\vec{v}_2 \times \vec{B}) \qquad [\because \vec{v}_1 \parallel \vec{B} \text{ and } \vec{v}_1 \times \vec{B} = 0]$$

The force is perpendicular to $\vec{v}_2$ and hence changes its direction and not magnitude and $\vec{v}_1$ remains constant along $\vec{B}$. Due to transverse velocity, the charged particle will perform circular motion and due to component $\vec{v}_1$, it will move with constant velocity along $\vec{B}$. Therefore the path of the particle will be **helix** with uniform pitch in uniform magnetic field [fig. 9.3].

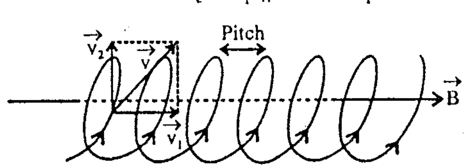

Fig. 9.3 : Helical path of a charged particle in a magnetic field.

Pitch of the helix = $v_1 T$

Here, radius of helical path, $r = \dfrac{mv_2}{qB}$; Angular frequency, $\omega = \dfrac{qB}{m}$;

$$\text{Period, } T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}\text{; Frequency, } \nu = \frac{1}{T} = \frac{qB}{2\pi m}.$$

$$\therefore \qquad \text{Pitch} = v_1 T = \frac{2\pi m}{qB} m_1$$

(d) **Magnetic or Lorentz force does not do any work on the charged particle** because

$$W = \int \vec{F} \cdot d\vec{r} = \int q(\vec{v} \times \vec{B}) \cdot \frac{d\vec{r}}{dt} dt = q\int (\vec{v} \times \vec{B}) \cdot \vec{v}\, dt = 0 \qquad ...(7)$$

where for a scalar triple product

$$(\vec{v} \times \vec{B}) \cdot \vec{v} = \vec{v} \cdot (\vec{v} \times \vec{B}) = (\vec{v} \times \vec{v}) \cdot \vec{B} = 0$$

This means that the *kinetic energy or speed of a charged particle remains constant, when moving in a magnetic field.*

In case a **charge q moves in a region where both the electric field $\vec{E}$ and magnetic field $\vec{B}$ are present, then the Lorentz force equation takes the form,**

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B}) \qquad ...(8)$$

9·3. Definition and Unit of Magnetic Field $\vec{B}$

The magnetic field $\vec{B}$ is defined by the Lorentz relation

$$\vec{F} = q(\vec{v} \times \vec{B}) \qquad ...(1)$$

Thus by measuring the magnetic force $\vec{F}$ acting on a charge q, moving with velocity $\vec{v}$, we can obtain magnetic field $\vec{B}$. As the value of the force is $F = qvB \sin\theta$, the **magnitude of the magnetic field** $\vec{B}$ can be determined from the relation

$$B = \frac{F}{qv\sin\theta} \qquad ...(2)$$

where θ is the angle between $\vec{v}$ and $\vec{B}$. From eq. (1), when $\vec{v}$ is parallel to $\vec{B}$, then $\vec{F} = q(\vec{v} \times \vec{B}) = 0$. Thus, *when a charge moving in a magnetic field does not experience any magnetic force, the direction of the motion of the charge defines the* **direction of** $\vec{B}$ *at that instant.* Thus to find the direction of $\vec{B}$, we vary the direction of $\vec{v}$ through a point, say P, keeping the magnitude of $\vec{v}$ as constant, then when $\vec{v}$ is parallel to $\vec{B}$, the force on the charged particle will be found zero. Now, this direction of the velocity of the the particle gives the direction of the magnetic field.

Magnetic field vector $\vec{B}$ is called **magnetic induction.**

Unit of magnetic field : From relation (2), we have $B = F/qv \sin\theta$. As $\sin\theta$ has no unit, the unit of B is the same as that of F/qv. Hence,

$$\text{Unit of } B = \frac{1 \text{ newton}}{1 \text{ coulomb} \times 1 \text{ metre/second}} = \frac{1 \text{ newton}}{1 \text{ ampere-metre}} = 1 \text{ NA}^{-1}\text{m}^{-1} \qquad ...(3)$$

because 1 coulomb/sec = 1 ampere.

Thus, S.I. unit of magnetic field B is $\mathbf{N\,A^{-1}m^{-1}}$ and it is known as **tesla** (T) *i.e.*,

$$1 \text{ N A}^{-1} \text{ m}^{-1} = 1 \text{ T} \qquad ...(4)$$

This unit of B can be expressed in terms of fundamental units of S.I. system as $\mathbf{kg\ s^{-2}\ A^{-1}}$ and its dimensional formula is $[MT^{-2}A^{-1}]$.

Another S.I. unit of B is **1 weber/metre²** (Wb/m²) and therefore

$$1\ \mathrm{N\,A^{-1}\,m^{-1}} = 1\ \mathrm{T} = 1\ \mathrm{Wbm^{-2}} \qquad ...(5)$$

There is also another unit in common use is **gauss** (or **oersted**). It is related to tesla as

$$10^4\ \text{gauss} = 1\ \text{tesla or } 1\ \text{gauss} = 10^{-4}\ \text{tesla} \qquad ...(6)$$

If q is measured in e.s.u., $\vec{v}$ in cm/sec and $\vec{B}$ in gauss, q will be replaced by q/c and the force will be in dynes. In such a case, Lorentz equation (1) takes the form :

$$\vec{F} = \frac{q}{c}(\vec{v} \times \vec{B}) \qquad ...(7)$$

9·4. Magnetic Field Lines

We can represent a magnetic field by **magnetic field lines***, which can be traced with the help of a compass needle. The tangent to a magnetic field line at a point gives the direction of $\vec{B}$ at that point. The magnetic field lines are drawn so that the number of lines per unit cross-sectional area at a point is proportional to the magnitude of magnetic field vector $\vec{B}$ at that point. The lines, where they are close together, represent strong magnetic field and where they are far apart, represent weak magnetic field.

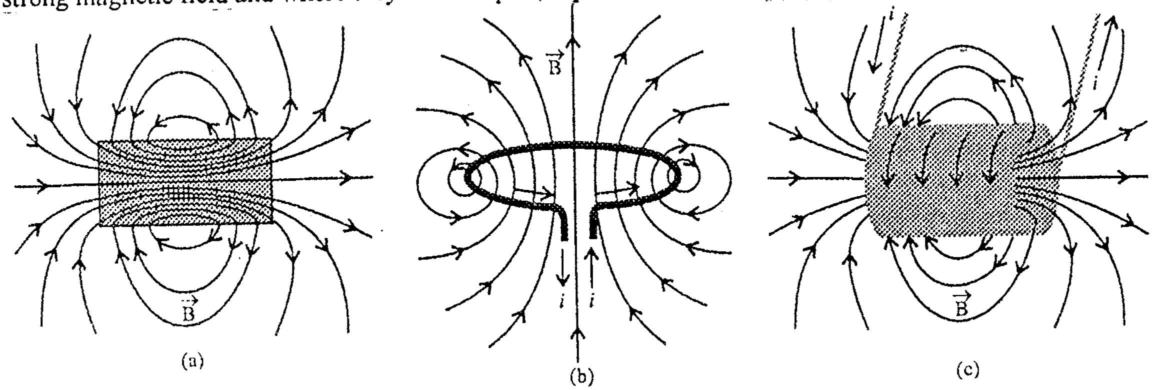

Fig. 9.4 : Magnetic field lines due to (a) bar magnet, (b) a circular current carrying loop and (c) a solenoid.

In a magnetic field, the direction of the field vector $\vec{B}$ at every point is unique and hence two magnetic field lines never intersect.

Magnetic field lines due to a bar magnet, a circular current carrying loop and a solenoid are shown in fig. 9.4 (a), (b) and (c).

9·5. Magnetic Flux and Gauss's Law in Magnetostatics

Consider a surface S situated in a magnetic field. Let $d\vec{S}$ be an elementary area around a point P [fig. 9.5]. If the magnetic field at P is $\vec{B}$, then the magnetic flux $d\phi$ through the area $d\vec{S}$ is defined by the scalar product $\vec{B}\cdot d\vec{S}$ *i.e.*,

$$d\phi = \vec{B}\cdot d\vec{S} \qquad ...(1)$$

* We find in literature the name 'magnetic lines of force' for 'magnetic field lines'. This seems not to be correct name for the magnetic field lines. In case of electric field lines, the name 'electric lines of forces' is all right because they point in the direction of force on the test charge in the field. However a magnetic field line does not point in the direction of force on a charge. In eq. $\vec{F} = q(\vec{v} \times \vec{B})$, the force on a moving charged particle is always perpendicular to the magnetic field and therefore, the force is perpendicular to the magnetic field line which passes through the position of the charged particle. As the direction of the force depends on the velocity of the particle and sign of its charge, just looking at magnetic field lines one can not know the direction of the force on the moving charged particle.

The total magnetic flux through the entire surface S is given by

$$\phi = \int_S \vec{B} \cdot d\vec{S} \qquad \text{...(2)}$$

If the surface is closed one, then

$$\phi = \oint \vec{B} \cdot d\vec{S} \qquad \text{...(3)}$$

If $\vec{B}$ is constant over the surface, then

$$\phi = \vec{B} \cdot \vec{S} = BS \cos \theta \qquad \text{...(4)}$$

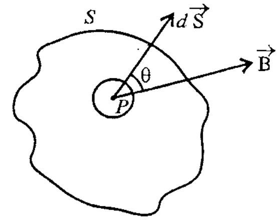

Fig. 9.5 : Magnetic flux through a surface.

If $\vec{B}$ and $\vec{S}$ have the same direction *i.e.*, the magnetic field is perpendicular to a plane area S, **[fig. 9.6]**, then

$$\phi = \vec{B} \cdot \vec{S} = BS \qquad \text{...(5)}$$

S.I. unit of magnetic flux is **weber**.

Now, $$B = \frac{\phi}{S} = \frac{\text{Magnetic flux}}{\text{Area}} \qquad \text{...(6)}$$

Thus the magnetic field is magnetic flux per unit area. This is why **magnetic field** or **magnetic induction is called magnetic flux density** and its unit is weber/metre^2 (Wb/m^2).

Magnetic flux may also be expressed in terms of magnetic field lines. As the number of lines per unit cross-sectional area at a point gives the magnitude of the magnetic field B at that point, the quantity $\int_S \vec{B} \cdot d\vec{S}$ represents the total number of field lines through the surface S.

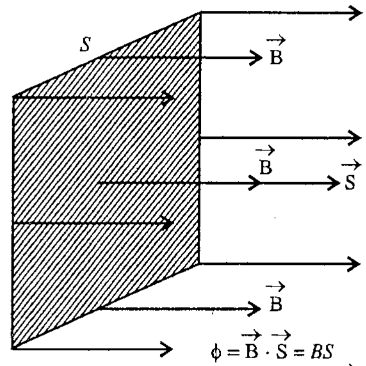

Fig. 9.6 : Magnetic flux of a uniform field $\vec{B}$ through a plane surface of vector area $\vec{S}$.

Gauss's law in magnetism : The magnetic field lines are always in the form of continuous closed curves around the current which produces that field. This is also true for magnetised material because the magnetisation of a material is equivalent to surface currents around the material. Hence in any situation, if we consider a closed surface, then the number of field lines entering the surface is the same as that leaving it. In other words, *the total magnetic flux through a closed surface is zero.*

This is called **Gauss's law in magnetism.** Mathematically we can express the Gauss's law as

$$\oint \vec{B} \cdot d\vec{S} = 0 \qquad \text{...(7)}$$

This is the **integral form of Gauss's law in magnetism**.

Now, according to Gauss's divergence theorem

$$\int_V \text{div}\, \vec{B}\, dV = \int_S \vec{B} \cdot d\vec{S} \qquad \text{...(8)}$$

where V is the volume enclosed by the surface S.

From eqs. (7) and (8), we have

$$\int_V \text{div}\, \vec{B}\, dV = 0 \qquad \text{...(9)}$$

This is true for any arbitrary volume. Hence

$$\text{div}\, \vec{B} = 0 \text{ or } \vec{\nabla} \cdot \vec{B} = 0 \qquad \text{...(10)}$$

This is the **differential form of Gauss's law in magnetism**.

9·6. Force on a Straight Conductor Carrying Current in a Uniform Magnetic Field

Consider a straight conducting wire of length l, carrying a current i. When the wire is placed in a magnetic field $\vec{B}$, we find a force acting on the wire, given by

$$\vec{F} = i\vec{l} \times \vec{B} \qquad \text{...(1a)}$$

In magnitude, $$F = ilB \sin\theta \qquad \text{...(1b)}$$

where $\vec{l}$ vector, having the magnitude l, is along the direction of the current and θ is the angle between $\vec{l}$ and $\vec{B}$ [fig. 9.7].

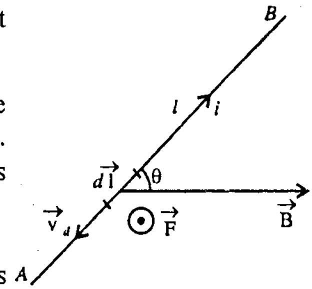

Fig. 9.7 : Force on a current carrying straight conductor.

Proof : In the conducting wire, the current i is flowing due to free electrons. These electrons drift with a v_d opposite to the direction of the current.

We know that the relation between the current i and the drift speed v_d is given by

$$i = nev_dA \qquad \text{...(2)}$$

where A is the cross-sectional area of the wire and n is the number of electrons per unit volume. Each electron experiences on an average a magnetic force (or Lorentz force), given by

$$\vec{F} = -e\vec{v}_d \times \vec{B} \qquad [\because \text{Lorentz force, } \vec{F} = q(\vec{v} \times \vec{B})] \qquad \text{...(3)}$$

Let us consider a small element dl of the wire. The number of free electrons in the dl length of the wire are $nAdl$. Hence, the magnetic force on the dl length of the wire is

$$d\vec{F} = (nAdl)(-e\vec{v}_d \times \vec{B}) = (nev_dA)(d\vec{l} \times \vec{B}) \qquad \text{...(4)}$$

where $d\vec{l}$ is along the direction of the current i or $-\vec{v}_d$ *i.e.*, $-dl\,\vec{v}_d = d\vec{l}\,v_d$.

But, $i = nev_dA$, hence

$$d\vec{F} = i\,d\vec{l} \times \vec{B} \qquad \text{...(5)}$$

We call $i\,d\vec{l}$ as the **current element.**

Thus, the magnetic force, acting on the wire of length l, is given by

$$\vec{F} = i\,\vec{l} \times \vec{B} \qquad \text{...(6)}$$

which has the magnitude $F = ilB \sin\theta$ and direction is perpendicular to the plane containing the vectors $\vec{l}$ and $\vec{B}$.

If the magnetic field is acting in a direction perpendicular to the length or current flow, then

$$F = ilB \qquad [\because \sin\theta = \sin\pi/2 = 1] \qquad \text{...(7)}$$

In this condition the magnetic force, $F = ilB$, is **maximum.**

If the magnetic field is acting along the direction of the current flow, then $\theta = 0$ or $\sin\theta = 0$, and hence the magnetic force is zero or minimum *i.e.*,

$$F = 0 \qquad \text{...(8)}$$

Fleming's Left Hand Rule :

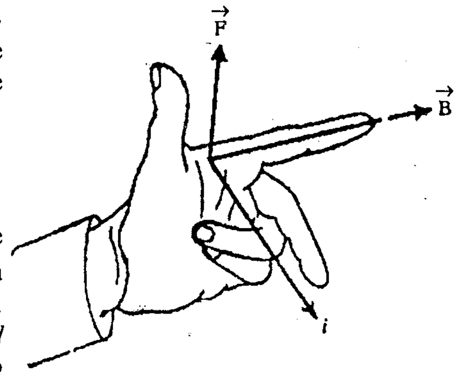

Fig. 9.8 : Fleming's left hand rule.

If the magnetic field $\vec{B}$ is acting perpendicular to the direction of the current i, then the direction of the force acting on the wire can be easily remembered by Fleming's left hand rule. According to this rule, *if we stretch the thumb, fore finger and the middle finger of the left hand perpendicualr to each other so that the fore finger points in the direction of the magnetic field*

$\vec{B}$ and the middle finger points in the direction of the current i (positive charge flow), then the thumb will point in the direction of the force $\vec{F}$ on the conductor [fig 9.8]. In fact, this rule is contained obviously in the expression (6) for magnetic force $\vec{F}$ *i.e.*, if $\vec{B}$ is perpendicular to the current flow direction (*i.e.*, that of $\vec{I}$), then $\vec{F}$ is perpendicular to the plane contain $\vec{I}$ and $\vec{B}$.

9·7. Torque on a Current Loop in a Magnetic Field (Magnetic Dipole Moment)

A coil or a loop carrying a current in a magnetic field experiences a torque. This is the principle on which the working of direct current measuring instruments, *e.g.*, galvanometers, ammeters and voltmeters, depends.

Consider a rectangular loop (or coil) $PQRS$ of sides $PQ = RS = l$ and $QR = SP = b$, carrying a current i. It is freely suspuspended in a uniform magnetic field $\vec{B}$. Initially the direction of $\vec{B}$ is in the plane of the coil *i.e.*, $\vec{B}$ is perpendicular to the sides PQ and RS and is along the sides QR and PS [fig. 9.9]. As angle θ between the side PQ and $\vec{B}$ is 90°, hence

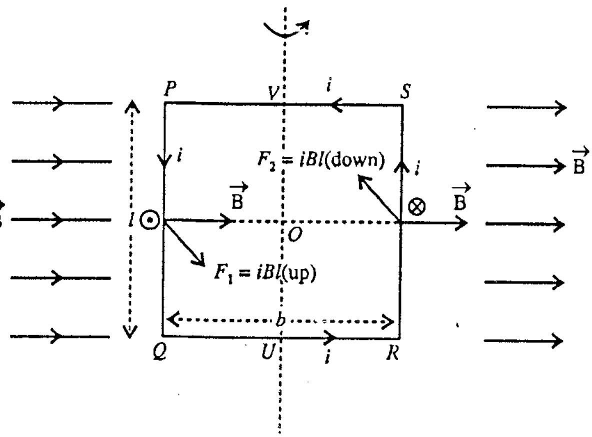

Fig. 9.9 : Torque on a current loop in a magnetic field.

Force on side PQ,

$$F_1 = ilB \qquad [\because F = ilB \sin\theta] \qquad ...(1)$$

The direction of F_1 is perpendicular to the plane of the paper *i.e.*, up towards the reader.

Similarly, **force on side RS,** $\quad F_2 = ilB \qquad ...(2)$

The direction of F_2 is perpendicular to the plane of the paper but down away from the reader, because the direction of current i in RS arm is opposite to that in PQ arm.

The forces F_1 and F_2 are equal in magnitude and opposite in direction at a perpendicualr distance b and hence constitute a **torque,** given by [**fig. 9.10(a)**],

$$\tau = ilBb = iAB \qquad ...(3)$$

where $lb = A$, area of the rectangular loop.

As the current in QR and SP arms of the rectangular loop is parallel or antiparallel to the direction of the magnetic field $\vec{B}$, θ = 0 or π *i.e.*, sin θ = 0 for both sides and the magnetic force ($ilB \sin\theta = 0$) is zero for QR and SP sides of the loop.

Thus eq. (3) represents the total torque on the loop $PQRS$ in the magnetic field $\vec{B}$. Here, the direction $\vec{B}$ is in the plane of the loop or $\vec{B}$ is perpendicular to the normal ON (or vector area $\vec{A}$) of the loop [**fig. 9.10(b)**].

Next, if the loop is rotated about the axis UV so that the normal ON to the loop makes an angle ϕ with the direction of $\vec{B}$, then as shown in **fig. 9.10(b)**, the torque in this position is given by

$$\tau = ilB\, b \sin\phi = i\, AB \sin\phi \qquad ...(4)$$

whose direction is towards the axis of rotation UV.

In the rotated position of the loop, the magnetic forces on QR and SP sides of the loop are equal and opposite and act along the same line. Hence these forces do not produce any torque and eq. (4) gives the net torque on the loop.

Let us consider the vector area $\vec{A}$ of the loop. The magnitude of the area vector $\vec{A}$ is the area ($A = lb$) enclosed by the loop and its direction is towards the side from which the current looks anticlockwise. Here, the direction of $\vec{A}$ is up or towards the reader **[fig. 9.9]*** and in **fig. 9.10(a)** along ON perpendicular to $\vec{B}$. In **fig 9.10(b)**, the direction of $\vec{A}$ is along ON' (the normal to the plane of the coil in the other position). The definition of vector area $\vec{A}$ is valid for a closed loop of any shape, not necessarily rectangular.

Thus, the torque $\vec{\tau}$ can can be represented vectorially as

$$\vec{\tau} = i\vec{A} \times \vec{B} \qquad ...(5)$$

If there are n turns in the loop (or coil) each turn experiences a torque. Hence, the total torque on the loop of n turns is

$$\vec{\tau} = ni\vec{A} \times \vec{B} \qquad ...(6)$$

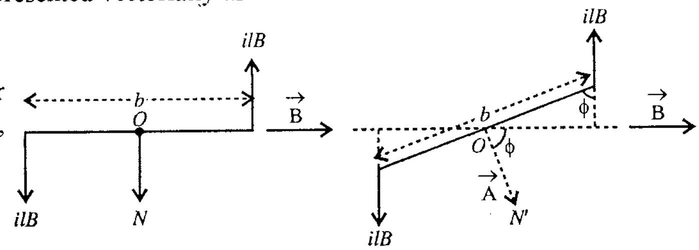

Fig. 9.10 : The figure is drawn in the plane perpendicular to the plane of the loop.

whose magnitude is $niAB \sin\phi$ and direction is along the axis of rotation. Eq. (6) has been obtained for torque by considering a rectangular loop. However it is valid for a plane loop of any shape *e.g.*, circular loop.

The torque tries to rotate the loop to bring its plane normal to the magnetic field or vector area parallel to the magnetic field. When the plane of the loop becomes perpendicular to the direction of the field, then the torque on the loop is equal to zero.

In galvanometers, the poles of the magnet are made concave and a soft iron core is put inside the coil so that (the iron core does not touch the coil) the magnetic field due to the magnet becomes radial **[fig. 9.11]**. In such a case, the plane of the coil remains parallel to the direction of the field and the vector area of the coil is normal to the direction of the field (ϕ = 90°) resulting the deflecting couple on the coil to be maximum.

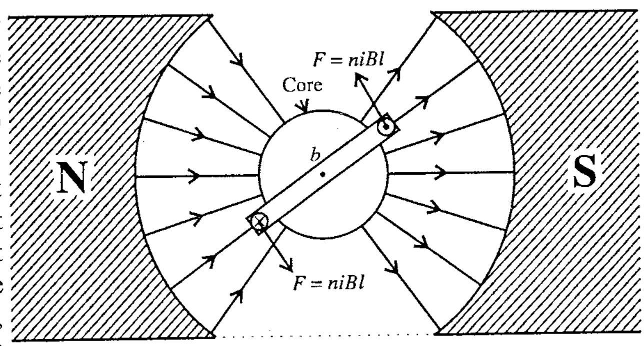

Fig. 9.11 : Torque on a loop in a radial magnetic field; the figure is drawn in a plane perpendicular to the plane of the loop.

Magnetic dipole moment : Torque on a loop (or coil) of n turns in a magnetic field $\vec{B}$ is given by [from eq. (6)],

$$\vec{\tau} = ni\vec{A} \times \vec{B} \qquad ...(7)$$

where $\vec{A}$ is the vector area of the coil.

In eq. (7), the quantity $ni\vec{A}$ is called the **magnetic dipole moment** or simply the **magnetic moment** of the current loop and is denoted by $\vec{m}$ *i.e.*,

$$\vec{m} = ni\vec{A} \qquad ...(8)$$

whose direction is the same as that of $\vec{A}$ given by right hand thumb rule.

Thus, eq. (7) takes the form

* The direction of the area vector can be obtained by **right hand thumb rule** for currents. According to this rule, if one curls his fingers of the right hand along the current, the stretched thumb gives the direction of the area vector $\vec{A}$.

$$\vec{\tau} = \vec{m} \times \vec{B} \quad ...(9a)$$

Its magnitude, $\tau = mB \sin \phi$ ...(9b)

A current loop or any body that experiences a magnetic torque given by eq. (7), is called a magnetic dipole.

The name comes from a similar expression for torque $\vec{\tau}$ on an electric dipole of dipole moment $\vec{p} = q\vec{d}$ in an electric field $\vec{E}$ *i.e.*,

$$\vec{\tau} = \vec{p} \times \vec{E} \quad ...(10)$$

with magnitude $\tau = pE \sin \phi$, where ϕ is angle between $\vec{p}$ and $\vec{E}$.

Expression (8) for dipole magnetic moment is valid for a plane loop of any shape with vector area $\vec{A}$ and turns n. This is because any plane loop C, carrying a current i may be considered to be divided into a large number of small adjacent rectangular loops **[fig. 9.12]**. We take the same current i to flow clockwise in each small loop. Let us look at one small loop **[fig. 9.12]**. The magnetic effect due to the current in any of its sides is annulled by the equal and opposite current in its adjacent loop. This neutralization of current happens to be for each small loop inside. The magnetic effect of the current remains only in those sides of the loops which coincide with the boundary of the big loop C. Thus, the total magnetic effect of all loops is the same as that due to the big loop C, carrying current i. The sum of all the small vector areas (having the same direction as that of $\vec{A}$ in view of right hand thumb rule) is equal to the vector area $\vec{A}$ of the loop. Hence, the vector area $\vec{A}$ of a plane loop of any shape, carrying a current i, gives rise the dipole magnetic moment $\vec{m} = ni\vec{A}$.

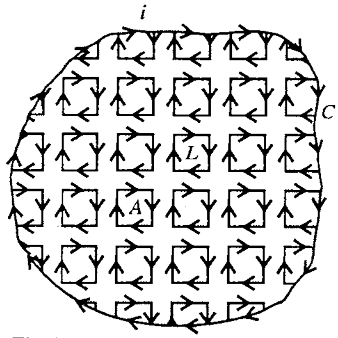

Fig. 9.12 : Current carrying plane loop of any shape.

9·8. Angular Momentum and Gyromagnetic Ratio

In an atom, electrons move around the positively charged nucleus in different orbits. We know that a current carrying loop has a magnetic dipole moment and hence an electron due to its orbital motion will have magnetic dipole moment, because a moving electron (charge) is equivalent to a current.

Consider an electron with charge $-e$ moving in a circular orbit of radius r around a nucleus O with uniform angular velocity ω or linear speed v **[fig. 9.13]**. This moving charge constitutes a current. The current is the flow of charge per second through a point. As the electron is making $n = \omega/2\pi$ rounds per second, hence the charge passing through the point per second or the current in the orbital loop is

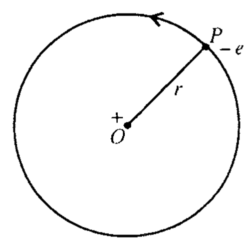

Fig. 9.13 : Circulating electron about a positively charged nucleus.

$$i = en = e\frac{\omega}{2\pi} \quad ...(1)$$

This current due to the orbital motion of the electron gives rise to a magnetic dipole moment, given by

$$\vec{m}_l = i\vec{A} \quad ...(2)$$

In magnitude, $m_l = i \times \pi r^2 = \frac{1}{2} e\omega r^2$ ...(3)

where $A = \pi r^2$ is the area of orbital loop.

As the current in the loop is in opposite direction to the motion of the electron, the direction of the area $\vec{A}$ and hence that of $\vec{m}$ is directed into the plane of the page and perpendicular to it.

If m_e is the mass of the moving electron, the angular momentum $\vec{L}$ of the electron is given by

$$\vec{L} = m_e \vec{r} \times \vec{v} \qquad ...(4)$$

In magnitude,

$$L = m_e rv = m_e \omega r^2 \qquad [\because v = r\omega] \qquad ...(5)$$

The direction of the angular momentum is along the cross product $\vec{r} \times \vec{v}$ *i.e.*, up perpendicular to the page. Thus the direction of angular momentum $\vec{L}$ is opposite to the direction of magnetic moment $\vec{m}$.

From eqs. (3) and (5)

$$\frac{m_l}{L} = \frac{\frac{1}{2} e\omega r^2}{m_e \omega r^2} = \frac{e}{2m_e} \qquad ...(6)$$

The ratio of the magnetic dipole moment to its angular momentum is called **gyromagnetic ratio.** Thus the gyromagnetic ratio for orbitting electron is $e/2m_e$.

Eq. (6) can be written as

$$m_l = \left(\frac{e}{2m_e}\right) L \qquad ...(7)$$

In vector form,

$$\vec{m}_l = -\frac{e}{2m_e}\vec{L} \qquad ...(8)$$

because $\vec{m}_l$ is opposite to $\vec{L}$.

Eq. (8) represents *the relation between the magnetic dipole moment and angular momentum of an electron in orbital motion.*

Remember that if a **positive charge** $+q$ is rotating in a circle of radius r, then the magnetic moment $m_l = \frac{1}{2} q\omega r^2$ and the angular momentum $L = m_q \omega r^2$ will be in the same direction. Hence

$$\frac{m_l}{L} = \frac{q}{2m_q} \text{ or } m_l = \frac{q}{2m_q} L \qquad ...(9)$$

In vector form,

$$\vec{m}_e = \left(\frac{q}{2m_q}\right)\vec{L} \qquad ...(10)$$

According to quantum mechanics, the angular momentum L is an integral multiple of $\hbar = h/2\pi$ *i.e.*,

$$L = l\hbar = l\frac{h}{2\pi} \qquad ...(11)$$

where $l = 0, 1, 2, ...$ is called the **orbital quantum number** and h is Planck's constant.

Hence, form eq. (7), the magnitude of orbital magnetic moment of the electron is

$$m_l = \frac{e}{2m_e}\frac{lh}{2\pi} \text{ or } m_l = l\left(\frac{eh}{4\pi m_e}\right) \qquad ...(12)$$

This quantity $eh/4\pi m_e$ is called **Bohr magneton** and is denoted by μ_B *i.e.*,

$$1\ \mu_B = \frac{eh}{4\pi m_e} \qquad ...(13)$$

Its value *i.e.*, 1 μ_B (Bohr magneton) $= \dfrac{1.6 \times 10^{-19} \times 6.62 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31}} = \mathbf{9.3 \times 10^{-24}\ A\text{-}m^2}$ ...(14)

Bohr magneton is a convenient unit of atomic magnetic moment.

Spin magnetic moment : In an atom, an electron in orbital motion also performs spin motion. Due to its spin motion, the electron also possesses magnetic moment and angular momentum. In fact, the spin magnetic moment and spin angular momentum are its intrinsic properties and permanently associated with an electron whether it is in orbital motion or not.

The spin magnetic dipole moment $\vec{m}_s$ is of an electron is related to its spin angular momentum $\vec{S}$ as

$$\vec{m}_S = -\frac{e}{m_e}\vec{S} \qquad \text{...(15a)}$$

In magnitude,
$$m_S = \frac{e}{m_e} S \qquad \text{...(15b)}$$

According to quantum meihanics,
$$S = \hbar s = \frac{h}{2\pi} s \qquad \text{...(16)}$$

where $s = \pm\frac{1}{2}$ is called spin quantum number.

For spin motion the gyromagnetic ratio m_S/S is e/m_e. However for orbital motion the gyromagnetic ratio $m/L = e/2m_e$. Thus *the value of gyromagnetic ratio for spin motion is two times that of orbital motion.*

Atomic magnetic dipole moment : In an atom, usually there are several electrons. Because of orbital and spin motions of an electron, it posseses corresponding angular momentum and magnetic dipole moment. We have seen that the direction of the magnetic moment is opposite to that of angular momentum for a moving electron. Due to the motion of all the electrons in an atom, the total magnetic dipole moment ($\vec{m}$) and total angular momentum ($\vec{J}$) can be written as

$$\vec{m} = -g\left(\frac{e}{2m_e}\right)\vec{J} \qquad \text{...(17)}$$

where g is called **Lande g-factor.**

This factor is characteristic of the state of atom. For pure orbital motion of an electron $g = 1$ and pure spin motion $g = 2$. Its value is in between 1 and 2 for complicated system *e.g.*, an atom.

When the total magnetic dipole moment of an atom is not zero, we may consider an atom as a **magnetic dipole** having permanent dipole moment. We shall discuss later materials made of such atoms.

Ex. 1. *A proton of velocity* $(3\hat{i} + 2\hat{j})\,10^6$ *m/sec enters a magnetic field* $2000\hat{j} + 3000\hat{k}$ *gauss. Calculate the accleration. Also deduce the simultaneous electric field which will keep the particle undeflected. (Mass of proton* $= 1.7 \times 10^{-27}$ *kg)* **(Agra 1970)**

Sol.
$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$\therefore \quad \text{Acceleration} = \frac{\vec{F}}{m_p} = \frac{q}{m_p}(\vec{v} \times \vec{B})$$

$$= \frac{1.6 \times 10^{-19}}{1.7 \times 10^{-27}}[(3\hat{i} + 2\hat{j})10^6 \times (2000\hat{j} + 3000\hat{k})10^{-4}]$$

$$= \mathbf{2.8 \times 10^{13}(2\hat{i} - 3\hat{j} + 2\hat{k})\ m/sec^2.}$$

Let $\vec{E}$ be the electric field, required to balance the force due to magnetic field. So that

$$\vec{F} = q(\vec{v} \times \vec{B}) + q\vec{E} = 0$$

or
$$\vec{E} = -(\vec{v} \times \vec{B}) = \mathbf{-3 \times 10^5(2\hat{i} - 3\hat{j} + 2\hat{k})\ N/C.}$$

Ex. 2. *A proton enters a magnetic field of intensity* 1.5 *Wb* m^{-2} *with a velocity* 2.0×10^7 *m* s^{-1} *in a direction at an angle* 30° *with the field. Calculate the force acting on the proton. The charge on proton is* 1.6×10^{-19} *coulomb.* **(Gwalior 2003)**

Sol. Here, $q = 1.6 \times 10^{-19}$ coulomb, $B = 1.5$ Wb m^{-2}, $v = 2.0 \times 10^{7}$ ms^{-1}, $\theta = 30°$, $\sin 30° = 0.5$

Force on proton, $F = qvB \sin\theta = (1.6 \times 10^{-19}) \times (2 \times 10^{7}) \times 1.5 \times 0.5.$ = $\mathbf{2.4 \times 10^{-12}}$ **N.**

Ex. 3. *A 5 MeV proton enters vertically downward in a horizontal field of magnetic induction* $\vec{B}$ *of magnitude* 1.5 *tesla. Calculate the acceleration of the proton and describe the resultant trajectory of the proton. (The mass of proton is* 1.7×10^{-27} kg) **(Meerut 1997)**

Sol. The kinetic energy of the proton,

$$K = \frac{1}{2} mv^2 = 5 \text{ MeV} = 5 \times 10^6 \text{ eV} = (5 \times 10^6) \times (1.6 \times 10^{-19}\text{ J}) = 8 \times 10^{-13}\text{ J}.$$

$$\therefore \qquad \text{Speed, } v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2 \times (8 \times 10^{-13})}{1.7 \times 10^{-27}}} = 3.07 \times 10^7 \text{ m/s}.$$

Now, the magnitude of the magnetic force acting on the proton is

$$F = qvB \sin\theta = (1.6 \times 10^{-19})(3.07 \times 10^7)(1.5) \sin 90° = 7.37 \times 10^{-12}\text{ N}.$$

Hence, the acceleration of the proton is given by

$$a = \frac{F}{m} = \frac{7.37 \times 10^{-12}\text{ N}}{1.7 \times 10^{-27}\text{ kg}} = \mathbf{4.33 \times 10^{15}\ m/s^2}.$$

The trajectory of the proton is circle, because it enters the magnetic field normally.

Ex. 4. *When a proton is released from rest in a room, it starts to move with an initial acceleration* α_0 *towards west. When it is projected towards north with a speed* v_0, *it moves with an initial acceleration* $3\alpha_0$ *towards west. Determine the electric field and the minimum possible magnetic field in the room.*

Sol. Obviously, both electric and magnetic fields, say E and B, are present. Initially, when proton is at rest, only electric force eE (e = charge on proton) will act on it. If m is the mass of the proton, then according to the problem,

$$eE = m\alpha_0 \text{ or } E = \frac{m\alpha_0}{e} \qquad \text{...(i)}$$

As acceleration α_0 of the proton is towards west, hence **electric field** $E = m\alpha_0/e$ **is acting on it towards west.**

When the proton is projected towards north with a speed v_0, it moves with $3\alpha_0$ acceleration towards west. Hence the magnetic force $e(\vec{v}_0 \times \vec{B})$ acts on the proton and produces an initial acceleration $2\alpha_0$ towards west. Clearly the magnetic field B must be in the vertical plane so that

$$e\,|\vec{v}_0 \times \vec{B}| = 2m\alpha_0$$

or

$$ev_0 B \sin\alpha = 2m\alpha_0 \text{ or } B = \frac{2m\alpha_0}{ev_0 \sin\alpha}$$

The magnetic field will be minimum for $\sin\alpha = 1$ or $\alpha = 90°$. Hence the minimum magnetic field is $2m\alpha_0/ev_0$.

Ex. 5. *A vertical conductor of length* 1 *m carries a current* 2 *A in downward direction. The horizontal component of earth's magnetic field is* 3×10^{-5} *T. Calculate the force (magnitude and direction) on the conductor.* **(Sagar 2003)**

Sol. We know that the force on a current carrying conductor placed in a magnetic field in direction perpendicular to the field is given by

$$F = iBl$$

Here, $i = 2$ A, $B = 3 \times 10^{-5}$ T, $l = 1$ m

$$\therefore \qquad F = 2 \times (3 \times 10^{-5}) \times 1 = \mathbf{6 \times 10^{-5}\ N}.$$

Fleming's left hand rule gives the direction of this force **towards west.**

Ex. 6. *A wire is bent in the form of an equilateral triangle PQR of side* 10 *cm and carries a current of* 0.5 *A. It is placed in a magnetic filed B of magnitude* 1.0 *T directed perpendicularly to the plane of the loop. Find the forces on the three sides of the triangle.*

Sol. Fig 9.14 shows the direction of the current in the triangle of wire in a transverse magnetic field $\vec{B}$, directed into the plane of the paper. The force on PQ side is

$$F_1 = ilB = 5 \times 0.1 \times 1 = 0.5 \text{ N}$$

which is directed, according to Fleming's left hand rule, perpendicular to PQ and towards inside in the plane of triangle.

As the triangle is equilateral, forces F_2 and F_3 have the same magnitude *i.e.*, 0.5 N and directed inwardly [fig. 9.14].

Obviously the resultant force of the three forces F_1, F_2 and F_3 is **zero**.

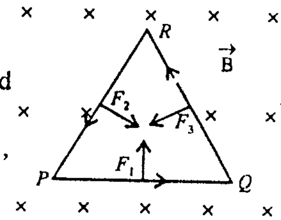

Fig. 9.14.

Ex. 7. *A wire of length l metre carries a current i ampere along the X-axis. A magnetic field exists which is given as* $\vec{B} = B_0(\hat{i} + \hat{j} + \hat{k})$ *T. Determine the magnitude of the magnetic force acting on the wire.*

Sol. Here $\vec{l} = l\hat{i}$ metre and $\vec{B} = B_0(\hat{i} + \hat{j} + \hat{k})$ tesla

$$\therefore \qquad \text{Force, } \vec{F} = i(\vec{l} \times \vec{B}) = i\,[l\hat{i} \times B_0(\hat{i} + \hat{j} + \hat{k})] = ilB_0(\hat{k} - \hat{j})$$

$$\therefore \qquad |\vec{F}| = \sqrt{2}\,ilB_0.$$

Ex. 8. *A horizontal wire of length* 5 *cm and carrying a current of* 2 *A is placed at the centre of a long solenoid at right angles to its axis. Find magnetic induction at the centre of solenoid if force on the wire is* 10^{-4} *N. What is the force on the wire if it lies parallel to the axis of solenoid ?* **(Meerut 1989)**

Sol. We know that the magnetic induction $\vec{B}$ at the centre of the solenoid is along the axis of the solenoid. The current-carrying wire of length 5 cm and with current 2 A is placed at the centre at right angles to the axis of the solenoid *i.e.*, perpendicular to $\vec{B}$. Hence the value of the force on the wire $\vec{F} = i\,\vec{l} \times \vec{B}$ is given by

$$F = ilB$$

Here $F = 10^{-4}$ N and hence

$$B = \frac{F}{il} = \frac{10^{-4}}{(2)(0.05)} = 10^{-3} \text{ T}.$$

If the wire lies parallel to the axis of the solenoid, then it will be parallel to $\vec{B}$. Hence the force is given by

$$\vec{F} = i\,\vec{l} \times \vec{B} \quad \text{or} \quad F = ilB\sin\theta,$$

where θ is the angle between $\vec{l}$ and $\vec{B}$. But here $\theta = 0$, therefore

$$F = 0.$$

Ex. 9. *A rectangular coil PQRS is placed in a uniform magnetic field* $\vec{B}$ *with magnitude* 2.00×10^{-2} $NA^{-1}m^{-1}$, *as shown in* **fig 9.15**. *Compute the forces on the four sides of the coil and the initial torque on the coil when it carries a current of* 2.0 *A.*

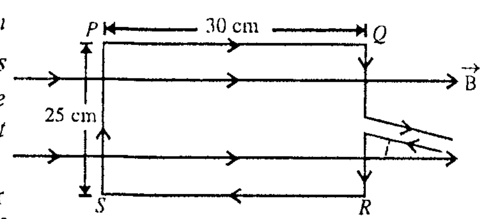

Fig. 9.15.

Sol. We know that straight current-carrying conductor of length l placed in a magnetic field B experiences a force of magnitude given by

$$F = ilB \sin\theta \qquad ...(i)$$

where θ is the angle between the straight wire and the field. Here, the sides PQ and RS of the coil are parallel to the field *i.e.*, $\theta = 0$ or $\sin\theta = 0$. Hence the force on each of these two sides is **zero**.

The sides PS and QR make an angle of 90° with the field and therefore the force on each of these sides will be given by

$$F = ilB \qquad ...(ii)$$

Here, $i = 2.0$ A, $l = 25$ cm $= 0.25$ m and $B = 2.00 \times 10^{-2}$ N A^{-1} m^{-1}.

$$\therefore \quad F = 2.00 \times 0.25 \times (2.00 \times 10^{-2}) = \mathbf{1.0 \times 10^{-2}\ N.}$$

By applying the Fleming's left hand rule, we see that the forces acting on the sides PS and QR are equal, parallel and opposite. Hence they constitute a couple and the torque is given by

$$\tau = F \times PQ = (1.0 \times 10^{-2}\ \text{N}) \times (0.30\ \text{m}) = \mathbf{3.0 \times 10^{-3}\ Nm.}$$

Ex. 10. *Two long metal rails placed horizontally and parallel to each other at a separation* l **[fig. 9.16]**. *A uniform magnetic field B is acting in the vertically downward direction. A straight wire of mass m can slide on the rails. The rails are connected to a constant current source S which drives a current i in the circuit. The coefficient of friction between the rails and the wire is* μ. (a) *Calculate the minimum value of* μ *which can just stop the wire from sliding on the rails ?* (b) *Discuss the motion of the wire if the value of* μ *is half the value calculated in* (a).

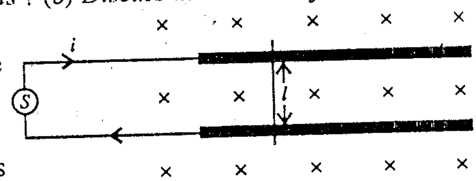

Fig. 9.16.

Sol. (a) The force on the straight wire of length l in the magnetic field B is

$$F = ilB$$

which acts towards right on the wire **[fig. 9.16]**. If the wire does not slide on the rails, the force of friction by the rails should be equal to F. If μ_0 be the value of **minimum coefficient of friction** which can just stop sliding, this force is also equal to $\mu_0 mg$ *i.e.*,

$$\mu_0 mg = ilB \text{ or } \mu_0 = \frac{ilB}{mg}.$$

(b) When the friction coefficient is $\mu = \dfrac{\mu_0}{2} = \dfrac{ilB}{2mg}$, the wire will slide towards right. The frictional force by the rails is

$$f = \mu mg = ilB/2,$$

which is acting towards left. However, the force due to the current in the wire is $F = ilB$ towards right. Hence the resultant force is $ilB - \dfrac{ilB}{2} = \dfrac{ilB}{2}$ towards right. The acceleration of the wire be $a = \dfrac{ilB}{2m}$ **and the wire will slide with this acceleration towards right.**

Ex. 11. *Calculate the maximum torque on a coil 4 cm × 15 cm of 600 turns, when carrying a current of* 10^{-5} *A in a field where the flux density B is 0.10 Wb/m*^{2}.

Sol. The torque on a coil with N turns, carrying a current i in a magnetic field B, is given by

$$\tau = N\, i\, A\, B \sin\theta,$$

where A is the area of the coil and θ is the angle between the normal to the plane of the coil and the field.

The value of the torque will be a maximum when $\theta = 90°$ so that $\sin\theta = 1$ *i.e.*,

$$\tau_{max} = N\,i\,A\,B = 600 \times 10^{-5} \times 60 \times 10^{-4} \times 0.10 = \mathbf{3.6 \times 10^{-6}\ N\text{-}m.}$$

Ex.12. *A circular coil of radius 2.0 cm with 100 turns carrying a current of 5.0 A is rotated in a magnetic field of strength 0.20 T.* (a) *Calculate the maximum torque that acts on the coil .* (b) *In a particular position of the coil, the torque acting on it is half of this maximum. What is the angle between the magnetic field and the plane of the coil ?*

Sol. Here, $i = 5$ A, $B = 0.2$ T, $n = 100$, $R = 2 \times 10^{-2}$ m.

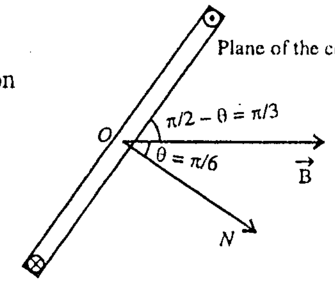

Fig. 9.17.

Area of the coil, $A = \pi R^2 = \pi(2 \times 10^{-2})^2$.

$$\text{Torque, } \tau = n\, i\, A\, B \sin\theta$$

where θ is the angle between the normal to the plane of the coil and direction of $\vec{B}$.

(a) Maximum torque will be for $\theta = 90°$ or $\sin\theta = 1$ *i.e.*,

$$\tau_{max} = niAB = 100 \times 5 \times \pi \times (2 \times 10^{-2})^2 \times 0.2$$
$$= \mathbf{0.126\ N\text{-}m}.$$

(b) $\frac{\tau_{max}}{2} = n\,i\,AB\sin\theta$ or $\frac{0.126}{2} = 0.126\sin\theta$ or $\sin\theta = \frac{1}{2}$ or $\theta = \frac{\pi}{6}$.

Hence, the angle between the magnetic field and plane of the coil [**fig. 9.17**],

$$= \frac{\pi}{2} - \theta = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}.$$

Ex. 13. *A loop has* 100 *turns and carries a current* 2.0 *A. If the radius of loop is* 10 *cm, calculate the magnetic moment of its equivalent magnetic dipole.*

Sol. Here, $n = 100$, $i = 2.0$ A, $r = 10$ cm $= 0.1$ m

$\therefore$ Area of loop, $A = \pi r^2 = 3.14 \times (0.1)^2$

Magnetic moment, $M = n\,i\,A = 100 \times 2 \times 3.14 \times (0.1)^2 = \mathbf{6.28\ A\text{-}m^2}$.

Ex. 14. *A conducting circular loop of radius a carries a clockwise current of i amp. The loop lies in X-Y plane in a uniform magnetic field of B weber m^{-2} acting parallel to X-axis. Calculate the torque (about Y-axis) experienced by the coil.*

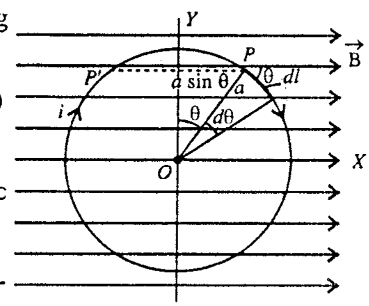

Fig. 9.18.

Sol. The magnetic force, acting on an element of length dl carrying current i is given by

$$d\vec{F} = i\,d\vec{l} \times \vec{B} \text{ or } dF = idl\,B\sin\theta \quad \text{...(i)}$$

where θ is the angle between $d\vec{l}$ and $\vec{B}$.

The loop lies in the X-Y plane (plane of the paper) and the magnetic field $\vec{B}$ is along X-axis [**fig. 9.18**].

The direction of this force $d\vec{F}$ at a point P, distant $a\sin\theta$ from Y-axis, is up perpendicular to the plane of the paper. The direction of force at a point P', distant $a\sin\theta$ on the other side from Y-axis, is down and opposite. Hence these forces constitute couple whose torque about Y-axis is given by

$$d\tau = dF \times 2a\sin\theta \quad \text{...(ii)}$$

Therefore, the total torque on the loop about Y-axis is

$$\tau = \int_0^\pi 2dF \times 2a\sin\theta \quad \text{...(iii)}$$

As $dl = a\,d\theta$, hence using eq. (i), we obtain

$$\tau = \int_0^\pi 2ia\,d\theta\,B\sin\theta \times 2a\sin\theta$$
$$= 2i\,Ba^2\int_0^\pi \sin^2\theta\,d\theta = i\,B\,\pi a^2 = i\,AB$$

where $A = \pi r^2$ area of the loop.

If there are n turns in the loop, then total torque on the loop

$$\tau = n\,i\,A\,B$$

This can be written in the vector form

$$\vec{\tau} = ni\,\vec{A} \times \vec{B} \text{ or } \vec{\tau} = \vec{m} \times \vec{B} \quad \text{..(iv)}$$

where $\vec{m} = ni\vec{A}$ is the magnetic moment of the loop carrying current i along the direction of vector area $\vec{A}$. The direction of $\vec{A}$ or $\vec{m}$ is down into the page according to right hand thumb rule. Further the direction of the torque $\vec{\tau}$ or $\vec{m} \times \vec{B}$ is along negative Y-axis in the present case.

Ex. 15. *A charge q moves with a constant speed v along a circle of radius r.* (a) *What is the equivalent current through a point on its path ?* (b) *Find the magnetic moment of the circulating charge.* (c) *What is the ratio of the magnetic moment to the angular momentum of the charge.*

Sol. (a) Frequency of revolution, $n = \dfrac{v}{2\pi r}$

The charge q passes through a point on the circle $n = v/2\pi r$ times per second. Hence the charge crossing the point per second or the current is given by

$$i = qn = q\frac{v}{2\pi r} \quad \text{...(i)}$$

(b) Magnetic moment,
$$m = iA = \frac{vq}{2\pi r}\cdot \pi r^2 = \frac{1}{2}qvr \quad \text{...(ii)}$$

(c) Angular momentum, $L = \mu v r$

where μ is the mass of the charged particle.

Both m and L have same direction for positive q but opposite to each other for negative q. Now,

$$\frac{m}{L} = \frac{\frac{1}{2}qvr}{\mu vr} = \frac{q}{2\mu} \quad \text{...(iii)}$$

Ex. 16. *Suppose a nonconducting ring of radius r and mass μ which has a total charge Q distributed uniformly on it* **[fig. 9.19]**. *The ring is rotated about its axis with an angular speed ω.* (a) *What is the equivalent electric current in the ring ?* (b) *What is the magnetic moment m of the ring ?* (c) *Show that $m = \dfrac{q}{2\mu}L$, where L is the angular momentum of the ring about its axis of rotation.*

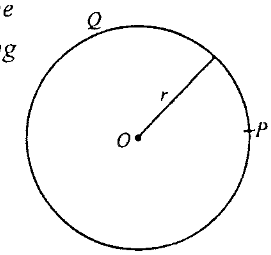

Fig. 9.19.

Sol. (a) Charge per unit length $= \dfrac{Q}{2\pi r}$

Linear speed of any particle of the ring $v = r\omega$

Equivalent current, i = charge passing through a point P on the circumference of the ring per second

= Charge on length v

$$= \frac{Q}{2\pi r}\times v = \frac{Q}{2\pi r}\times r\omega = \frac{Q\omega}{2\pi} \quad \text{...(i)}$$

(b) Magnetic moment,
$$m = iA = \frac{Q\omega}{2\pi}\cdot \pi r^2 = \frac{Q\omega r^2}{2} \quad \text{...(ii)}$$

(c) Angular momentum, $L = I\omega = \mu r^2\omega$ ...(iii)

From eq. (iii) $\omega r^2 = L/\mu$. Hence from eq. (ii)

$$m = \frac{Q}{2}\cdot\frac{L}{\mu} \text{ or } m = \frac{Q}{2\mu}L \quad \text{...(iv)}$$

Ex. 17. *An electron is moving in a circular path of radius 0.5×10^{-10} m, with an angular velocity 4×10^{16} rad/sec. Find the maximum torque an the electron if the magnetic induction B is 0.4×10^{-3} T.*

(Lucknow 2004)

Sol. Equivalent current of the circulating electron

$$i = \text{charge} \times \text{frequency of rotation} = e\frac{\omega}{2\pi} \qquad \text{...(i)}$$

$$\text{Torque, } \tau = mB \sin\theta \qquad \text{...(ii)}$$

$$\text{Maximum torque, } \tau_{max} = mB \text{ (for } \theta = \pi/2) \qquad \text{...(iii)}$$

Now, $$\text{Magnetic moment, } m = iA = \frac{e\omega}{2\pi}\cdot \pi r^2 = \frac{e\omega r^2}{2} \qquad \text{...(iv)}$$

$$\therefore \quad \tau_{max} = mB = \frac{e\omega r^2}{2}\cdot B \qquad \text{...(v)}$$

Here, $e = 1.6 \times 10^{-19}$ C, $\omega = 4 \times 10^{16}$ rad/sec, $r = 0.5 \times 10^{-10}$ m and $B = 0.4 \times 10^{-3}$ T

$$\therefore \quad \tau_{max} = \frac{1.6 \times 10^{-19} \times 4 \times 10^{16} \times (0.5 \times 10^{-10})^2 \times 0.4 \times 10^{-3}}{2}$$

$$= 3.2 \times 0.25 \times 0.4 \times 10^{-26} = \mathbf{3.2 \times 10^{-27}\ N\text{-}m.}$$

Ex. 18. *A charge q is uniformly distributed in the volume of a sphere of radius r. If the sphere is rotating with a uniform angular velocity ω about its diameter, find the magnetic moment of charge distribution.*

(Gwalior 2004)

Sol. Angular momentum of rotating sphere (about its diameter) $= I\omega = \frac{2}{5}\mu r^2\omega$

where $I = \frac{2}{5}\mu r^2$ is the moment of inertia of the sphere of mass μ and radius r about a diameter.

$$\text{Magenetic moment, } m = \frac{q}{2\mu} \times \text{Angular momentum} = \frac{q}{2\mu} \times \frac{2}{5}\mu r^2\omega = \frac{1}{5}qr^2\omega.$$

EXERCISE 9·1

1. Find the force on an electron, when it moves with a velocity 10^7 m/sec making an angle 45° with a magnetic field of strength 10 gauss.
   [**Ans.** 1.13×10^{-15} N.]

2. Calculate the magnitude and direction of the force acting on a proton moving with a velocity $(2\hat{i} + \hat{j})10^6$ m/sec in a magnetic field $2000\,(\hat{i} + 2\hat{j} - \hat{k})$ gauss.
   $\left[\text{Ans. } 3.2 \times 10^{-14}\sqrt{14}\text{ N; Direction cosines}\left(-\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right).\right]$

3. A proton moving with a speed 3.4×10^7 ms^{-1} enters in a magnetic field from direction perpendicular to the field. If intensity of magnetic field is 2.0 Wb m^{-2}, calculate the force exerted on proton.
   [**Ans.** 1.088×10^{-11} N.] **(Jabalpur 2003)**

4. An electron is moving with a velocity $(2\hat{i} + 3\hat{j})$ m/s in an electric field $E = 3\hat{i} + 6\hat{j} + 2\hat{k}$ V/m and magnetic field $2\hat{j} + 3\hat{k}$ T. Calculate the force acting on the electron. **(Bundelkhand 2004)**
   [**Ans.** $9.6 \times 10^{-19}\ (2\hat{i} + \hat{k})$ N.]

5. A parallel-plate capacitor has a separation of 4 mm and a potential difference of 200 V between its plates. The capacitor is placed in a uniform magnetic field $\vec{B}$. An electron, projected vertically upwards parallel to the plates with a velocity of 10^6 m s^{-1}, passes between the plates undeflected [**fig. 9.20**]. Find the magnitude and direction of the magnetic field $\vec{B}$ between the plates. **(I.I.T. 1981)**
   [**Ans.** .0.05 Wb/m^2.]

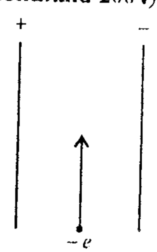

Fig. 9.20.

6. A charge of 2.0 μC moves with a speed of 2.0×10^6 m/s along the positive X-axis. A magnetic field $\vec{B}$ of strength $(0.20\hat{j} + 0.40\hat{k})$ T exists in space. Calculate the magnetic force acting on the charge.
[Ans. $0.8(-2\hat{j} + \hat{k})$ N.]
7. A 10 volt electron is circulating in a plane at right angles to a uniform field of magnetic induction of 1.0×10^{-4} Wb/m². Compute the radius of the circle. ($m = 9.1 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C, 1 eV = 1.6×10^{-19}J). **(Meerut 1989 S)**
[Ans. 10.6 cm.]
8. A proton enters a magnetic field of intensity 2000 gauss directed along X-axis. Calculate the force acting on the proton if (i) initial velocity of proton is zero, (ii) if the initial velocity of proton is 2×10^7 ms^{-1} along Y-axis, (iii) if the initial velocity of proton is 2×0^7 ms^{-1} along X-axis.
[Ans. (i) 0; (ii) 6.4×10^{-13} N along Z-axis; (iii) 0.]
9. Protons having kinetic energy E emerge from an accelerator as a narrow beam. The beam is bent by a perpendicular magnetic field so that it just misses a plane target kept at a distance d in front of the accelerator. Find the magnetic field.
[Ans. $\sqrt{2m_pE}/ed$, where e is charge and m the mass of proton.]
[Hint : $E = \frac{1}{2}mv^2, \frac{mv^2}{d} = evB$ in transverse B.]
10. A straight, horizontal wire of mass 20 mg and length 1.0 m carries a current of 2.0 A. What minimum magnetic field B should be applied in the region so that the magnetic force on the wire may balance its weight ?
[Ans. 10^{-4} T.]
11. Calculate the magnetic force on a wire one meter long and carrying a current of 10 ampere when placed in a uniform field of magnetic induction 1.5 weber/meter² making an angle of 30° with the direction of the field.
[Ans. 7.5 N.]
12. A rectangular coil of breadth x is suspended from the insulated hanger of a spring balance as shown in [fig. 9.21]. A current i flows in the anticlockwise direction in the loop. A magnetic field B exists in the lower region. Find the change in the tension of the spring if the current in the loop is reversed.
[Ans. $2iBx$.]

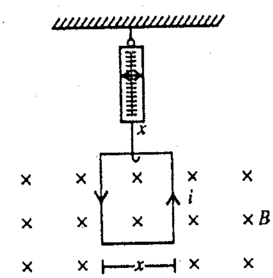

Fig. 9.21.

13. A 30 cm long wire, having mass 20 gm is hanged by two flexible wires in a magnetic field of 0.40 T. The field is perpendicular to the plane of paper, directed downwards [fig. 9.22]. Find the magnitude and direction of the current required to be flown to neutralize the tension in the hanging wires.
[Ans. 1.64 A from left to right.]

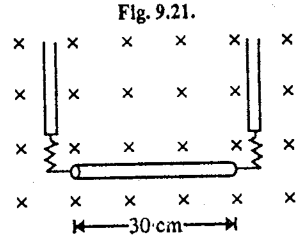

Fig. 9.22.

14. A wire is bent in the form of $pqrs$ and is placed in a magentic field $\vec{B}$ directed perpendicular to the plane of paper outwards [fig. 9.23]. If the wire carries a current i ampere, find the total force acting on the wire. Given that $pq = rs = a$ metre and radius of part $or = R$ metre.
[Ans. $2iB(a + R)$.]

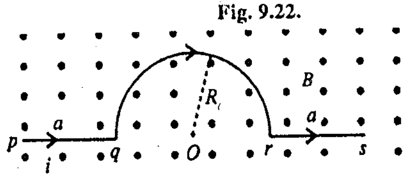

Fig. 9.23.

15. A stright wire AB of mass 10 g is at rest on two parallel metal rails. The separation between the rails is 4.9 cm. A magnetic field of 0.80 tesla is applied perpendicular to the plane of the rails, directed downwards [fig. 9.24]. The resistance of the circuit is slowly decreased. When the resistance decreases to below 20 ohm, the wire AB begins to slide on the rails. Find the coefficient of friction between the wire and the rails.
[**Ans.** 0.12.]

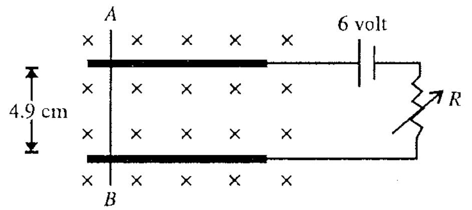

Fig. 9.24.

16. A metal wire of mass m slides without friction on two horizontal rails spaced at a distance d apart as shown in the figure. The rails are situated in a uniform magnetic field B, directed vertically upward and a battery is sending a current i through them [fig. 9.25]. Find the velocity of the wire as a function of time, assuming it to be at rest initially. **(Roorkee 1983)**
[**Ans.** $v = iBdt/m$ away from the battery.]

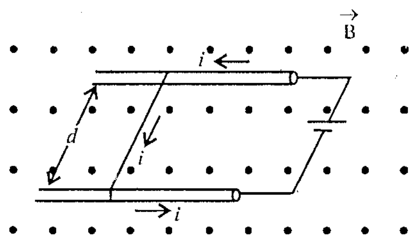

Fig. 9.25.

17. A rectangular coil of 200 turns has length 5 cm and width 4 cm. It is placed with its plane parallel to a uniform magnetic field and a current of 2 A flows through the coil. Calculate the magnitude of the magnetic field B, if the torque acting on the coil is 0.2 N-m.
[**Ans.** 1 T.]

18. A rectangular loop of side 10 cm and 20 cm carries a current of 5.0 A. A uniform magnetic field of magnitude 0.10 T exists parallel to the longer side of the loop. Calculate the force and the torque acting on the loop.
[**Ans.** zero; 0.01 N parallel to shorter side.]

19. The magnetic field existing in a region is given by

$$\vec{B} = B_0\left(1 + \frac{x}{a}\right)\hat{k}.$$

A square loop of edge a and carrying a current i, is placed with its edges parallel to the X-Y axes. Determine the magnitude of the net magnetic force experienced by the loop.
[**Ans.** iB_0a.]

20. Suppose a nonconducting plate of radius R and mass M, on which a charge Q is distributed uniformly, is rotated about its axis with an angular speed ω. Calculate (i) the equivalent current and (ii) magnetic moment of rotating plate. Show that the magnetic moment m and the angular momentum L of the plate are related as $m = \dfrac{Q}{2M}L$.
[**Ans.** (i) $Q\omega/4\pi$; (ii) $m = Q\omega R^2/4$.]

[**Hint :** $L = I\omega = \frac{1}{2}MR^2\omega$. Current due to a ring of width dr [fig. 9.26] is given by

$$di = (\text{charge on the ring})\,\frac{\omega}{2\pi} = \frac{Q}{2\pi}\cdot 2\pi r\,dr\,\frac{\omega}{2\pi} = \frac{Q\omega}{\pi R^2}r\,dr$$

$\therefore$ Equivalent current, $i = \dfrac{Q\omega}{2\pi R^2}\displaystyle\int_0^R r\,dr = \dfrac{Q\omega}{4\pi}$.

$\therefore$ Magnetic moment, $m = iA = \dfrac{Q\omega}{4\pi}\cdot\pi R^2 = \dfrac{Q\omega R^2}{4}$; $\therefore\ \dfrac{m}{L} = \dfrac{Q}{2M}$.]

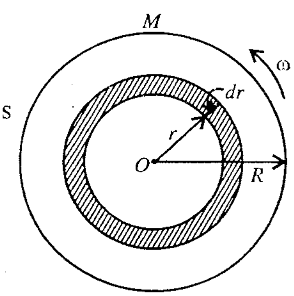

Fig. 9.26.

21. In a solid sphere of radius R and mass M, the charge Q is distributed uniformly over its volume. If the sphere is rotated about a diameter with an angular speed ω, find the equivalent current, angular

momentum and magnetic moment for the rotating sphere. Show that the ratio of magnetic moment and angular momentum is $Q/2M$.

[**Ans.** $i = Q\omega/4\pi$, $L = \frac{2}{5} mR^2\omega$, $m = Q\omega R^2/5$.]

[**Hint** : Magnetic moment, $m = \frac{Q}{2M} L$.]

22. A current of 10 A is flowing through a coil of radius 0.5 m and 10 turns. Calculate the magnetic moment of the coil. **(Meerut 2003)**

[**Ans.** 25π A-m^2.]

9·9. Biot-Savart's Law

When an electric current flows through a conductor, it produces a magnetic field around the conductor. The magnitude and direction of this field at a point due to this current can be obtained by a law, formulated on the basis of experiments by Biot and Savart. This law is known as **Biot-Savart law**.

Consider a conducting wire, carrying a current i. If $d\vec{l}$ be a vector length element in the direction of the current i, then according to Biot-Savart law the magnetic field $d\vec{B}$ at a point P at a vector distance $\vec{r}$ from the element is given by **[fig. 9.27]**.

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{i\, d\vec{l} \times \vec{r}}{r^3} \qquad ...(1)$$

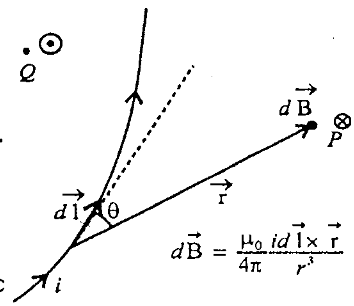

Fig. 9.27 : Biot-Savart's law.

where μ_0 is the permeability of the vacuum and its value is $4\pi \times 10^{-7}$ T-m/A. The magnitude of the field is

$$dB = \frac{\mu_0}{4\pi}\frac{i\, dl \sin\theta}{r^2} \qquad ...(2)$$

where θ is the angle between $d\vec{l}$ and $\vec{r}$. The direction of the magnetic field $d\vec{B}$ is perpendicular to the plane containing the element $d\vec{l}$ and vector $\vec{r}$ along the direction of the cross-product $d\vec{l} \times \vec{r}$.

Remember that the direction of cross-product is given by right hand screw rule according to which if the screw is rotated from $d\vec{l}$ towards $\vec{r}$, the screw advances in the direction of cross-product. In **fig. 9.27**, if $d\vec{l}$ and $\vec{r}$ are in the plane of the paper, then the direction of the field $d\vec{B}$ is perpendicular to the plane of the paper and directed into it.

Usually the direction of the magnetic field going into the plane (of the paper) is denoted by encircled cross $\otimes$ and the direction coming out of the plane (of the paper) by an encircled dot $\odot$. In **fig. 9.27**, the direction of the field at the point P, represented by $\otimes$, goes into the plane of the paper and that $\odot$ at Q comes out of this plane.

The resultant magnetic field due to a current carrying length of the wire can be found by integrating eq. (1) *i.e.*,

$$d\vec{B} = \int d\vec{B} = \frac{\mu_0}{4\pi}\int \frac{i\, d\vec{l} \times \vec{r}}{r^3}$$

between suitable limits.

9·10. Magnetic Field due to Current Carrying Straight Wire

Consider a straight conducting straight wire carrying a current i. We want to find the magnetic field at a point P distance d from the wire. The distance d is the length of the perpendicular PO (= d) drawn from the point P on the wire and O is the foot of the perpendicular **[fig. 9.28]**.

Field due to a finite length of the wire : Consider CD portion of the current carrying straight wire. Let dl be a length element of the wire at a distance l from the point O. The distance of the point P from the element is r. If θ be the angle between $\vec{dl}$ and $\vec{r}$, the magnetic field dB at P due to the current element idl is given by

$$dB = \frac{\mu_0}{4\pi}\frac{i\,dl\sin\theta}{r^2} \qquad ...(1)$$

The direction of the field is along the cross-product $\vec{dl} \times \vec{r}$ *i.e.*, perpendicular to the plane of the paper and directed into it. If we consider various elements of the wire, then the direction of the field at P due to all these elements is the same. Thus the magnetic field due to the CD portion of the wire can be obtained by integrating eq. (1) *i.e.*,

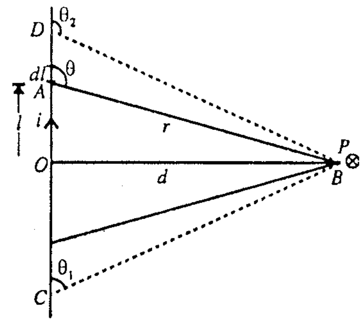

Fig. 9.28 : Magnetic field due to a current carrying straight conductor.

$$B = \frac{\mu_0}{4\pi}\int_{\theta_1}^{\theta_2}\frac{i\,dl\sin\theta}{r^2} \qquad ...(2)$$

where θ_1 and θ_2 are the values of θ at the point C and D of the wire.

From fig. 9.28, we have $\angle OAP = \pi - \theta$ and $\tan OAP = \tan(\pi - \theta) = \frac{OP}{OA} = \frac{d}{l}$ or $-\tan\theta = \frac{d}{l}$

$$\therefore \qquad l = -d\cot\theta$$

Differentiating l with θ, we obtain

$$dl = d\,\text{cosec}^2\theta\; d\theta \qquad ...(3)$$

Also $\quad \sin OAP = \sin(\pi - \theta) = \frac{d}{r}$ or $\sin\theta = \frac{d}{r}$ or $r = d\,\text{cosec}\,\theta \qquad ...(4)$

Substituting the values of dl and r from (3) and (4) in eq. (2), we get

$$B = \frac{\mu_0 i}{4\pi}\int_{\theta_1}^{\theta_2}\frac{d\,\text{cosec}^2\theta\,\sin\theta\,d\theta}{d^2\,\text{cosec}^2\theta} = \frac{\mu_0 i}{4\pi d}\int_{\theta_1}^{\theta_2}\sin\theta\,d\theta = \frac{\mu_0 i}{4\pi d}[-\cos\theta]_{\theta_1}^{\theta_2}$$

i.e.,
$$B = \frac{\mu_0 i}{4\pi d}(\cos\theta_1 - \cos\theta_2) \qquad ...(5)$$

This is the expression for the magnetic field due to a finite length of the current carrying straight wire with the value θ_1 and θ_2 of angle θ at the lower and upper ends respectively.

Field due to a long straight wire (Infinite length) : If the straight wire is very long or of infinite length, in fig.9.28, $\theta_1 = 0$ and $\theta_2 = \pi$. Hence the magnetic field at a distance d from such a wire is [using eq. (5)]

$$B = \frac{\mu_0 i}{4\pi d}(\cos 0 - \cos\pi) = \frac{\mu_0 i}{2\pi d} \qquad ...(6)$$

Field at a point on a perpendicular bisector of finite length : Let CD be a wire of finite length L and O be its mid point. The point P lies on the perpendicular bisector OP. CO or OD portion of the wire subtends angle α at the point P [fig. 9.29].

Here, $\cos\theta_1 = \frac{L/2}{CP} = \frac{L/2}{\sqrt{(L/2)^2 + d^2}} = \frac{L}{\sqrt{L^2 + 4d^2}}$

and $\quad \cos\theta_2 = \cos(\pi - \angle PDO) = -\cos PDO$

$$= -\frac{L/2}{\sqrt{(L/2)^2 + d^2}} = -\frac{L}{\sqrt{L^2 + 4d^2}}$$

Hence from eq. (5)

$$B = \frac{\mu_0 i}{4\pi d}\left[\frac{L}{\sqrt{L^2 + 4d^2}} + \frac{L}{\sqrt{L^2 + 4d^2}}\right] = \frac{\mu_0 i L}{2\pi d\sqrt{L^2 + 4d^2}} \qquad ...(7)$$

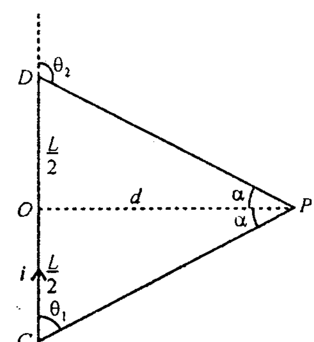

Fig. 9.29.

Also $\cos\theta_1 = \cos\left(\frac{\pi}{2} - \alpha\right) = \sin\alpha$

and $\cos\theta_2 = \cos(\pi - \angle ODP) = \cos\left[\pi - \left(\frac{\pi}{2} - \alpha\right)\right] = \cos\left(\frac{\pi}{2} + \alpha\right) = -\sin\alpha$

Hence, from eq. (5)

$$B = \frac{\mu_0 i}{4\pi d}[\sin\alpha + \sin\alpha] = \frac{\mu_0 i \sin\alpha}{2\pi d} \qquad ...(8)$$

where 2α is the angle subtended by the straight wire at a point P lying on perpendicular bisector.

In order to find the direction of the magnetic field at a point P due to a long straight conductor, we may apply **right hand thumb rule for linear currents (fig. 9.30)**. According to this rule *if we grasp the wire with right hand such that the thumb points in the direction of the current and the fingers are curled around the wire through the point P, then the direction of the fingers at P gives the direction of the magnetic field at that point.*

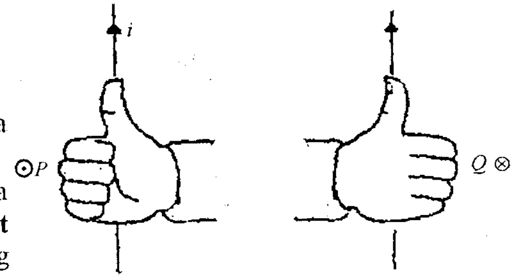

Fig. 9.30.

Magnetic field lines : The magnetic field lines due to a long straight conductor are concentric circles around the wire with their centres situated on the wire. The field lines are denser near the wire and rarer as the distance increases **[fig. 9.31(a)]**.

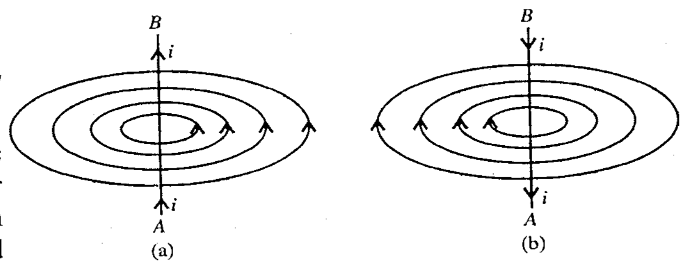

Fig. 9.31 : Magnetic field lines due to a straight wire (a) Current from A to B (b) Current from B to A.

On reversing the direction of the current, the direction of field lines is also reversed **[fig. 9.31(b)]**.

9·11. Force between Two Parallel Currents—Definition of Ampere

Consider two long straight parallel wires PQ and RS, placed at a distance d apart in vacuum or air. Let the wire PQ and RS carry the current i_1 and i_2 respectively **[fig. 9.32]**. The magnetic field at any point of the RS wire due to the current i_1 in PQ is

$$B = \frac{\mu_0 i_1}{2\pi d} \qquad ...(1)$$

The direction of this field is perpendicular to the plane of the diagram and directed into it. If we consider a small element dl of the wire RS, then the magnetic force acting on the dl element due to this field is

$$d\vec{F} = i_2\, d\vec{l} \times \vec{B} \qquad ...(2)$$

The magnitude of the force on dl length of the wire is given by

$$dF = i_2\, dl\, B \sin 90^\circ = i_2\, dl\left(\frac{\mu_0 i_1}{2\pi d}\right) = \frac{\mu_0}{2\pi}\frac{i_1 i_2}{d}\, dl \qquad ...(3)$$

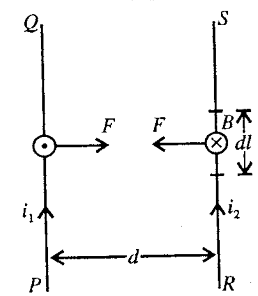

Fig. 9.32 : Force of attraction between two parallel currents.

The direction of this force $d\vec{F}$ is along $d\vec{l} \times \vec{B}$ or towards the wire PQ. Hence the force per unit length of the wire RS due to the wire PQ is

$$F = \frac{dF}{dl} = \frac{\mu_0}{2\pi}\frac{i_1 i_2}{d} \qquad ...(4)$$

which acts towards the wire PQ.

Similarly if we consider a length element dl' in the wire PQ and determine the magnetic force per unit length of the wire PQ due to the wire RS, it is again found to be F, given by eq. (4), acting towards the wire RS.

Thus we find that if two parallel straight wires at a distance d apart carry the currents i_1 and i_2 in the same direction, then they attract each other and the force of attraction per unit length is given by

$$F = \frac{\mu_0}{2\pi}\frac{i_1 i_2}{d} \qquad ...(5)$$

If the two parallel wires at a distance d apart carry current i_1 and i_2 in opposite directions [fig. 9.33], the wires repel each other with a force per unit length given by eq. (5).

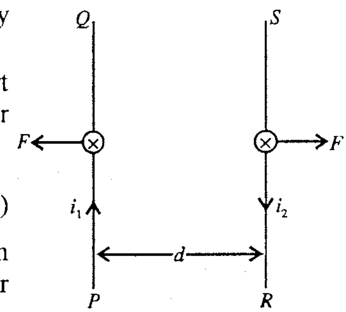

Fig. 9.33 : Force of repulsion between two antiparallel currents.

Thus we conclude that *the two parallel conductors attract each other, if the currents in them are in the same direction and repel each other, if the currents in them are in opposite directions.*

If the currents in the two wires are the same *i.e.*, $i_1 = i_2 = i$ (say), then the force per unit length of each of the two wires will be given by

$$F = \frac{\mu_0}{2\pi}\frac{i^2}{d} \text{ newton/metre} \qquad ...(6)$$

Definition of ampere : If we consider two parallel wires separated by 1 m and carrying a current 1 A in each wire, then $d = 1$ m and $i_1 = i_2 = i = 1$ A. In such a case, from eq. (6) the force per unit length of a wire is given by

$$F = \frac{4\pi \times 10^{-7}}{2\pi} \times \frac{1^2}{1} = 2 \times 10^{-7} \text{ N/m} \qquad ...(7)$$

This is used to define the S.I. unit of current ampere. Thus, *if two long parallel wires, placed at 1 m distance apart in vacuum, carry equal currents and there is a force of* 2×10^{-7} *newton per metre between them, the current in each wire is said to be* 1 *ampere.*

9·12. Magnetic Field at the Centre of a Current Carrying Circular Coil (or Loop)

Consider a circular coil (or loop) of radius R, carrying a current i. The coil lies in the plane of the paper and the current is flowing in the clockwise direction. We want to determine the magnetic field at the centre O of the coil due to this current. If we take a small element $d\vec{l}$ of the coil [fig. 9.34], the magnetic field $d\vec{B}$ at the centre O of the coil due to the current element $i\,d\vec{l}$ is given by

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{i\,d\vec{l} \times \vec{r}}{r^3} \qquad ...(1)$$

where $\vec{r}$ is the vector distance from the element to the centre O.

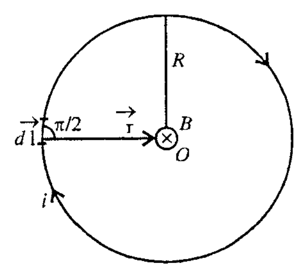

Fig. 9.34 : Magnetic field at the centre of a current carrying circular coil.

The direction of this field $d\vec{B}$ is perpendicular to the plane of the paper and directed into it ⊗. The magnitude of this magnetic field is

$$dB = \frac{\mu_0}{4\pi}\frac{i\,dl}{R^2} \qquad ...(2)$$

because the angle between $d\vec{l}$ and $\vec{r}$ is $\pi/2$ so that $|d\vec{l} \times \vec{r}| = dl\,R \sin\frac{\pi}{2} = dl\,R$ with $|\vec{r}| = R$, radius of the coil.

We see that the field due to all such elements of the coil is in the same direction. Hence the resultant field B at O is also in the same direction *i.e.*, it is perpendicular to the plane of the paper and directed into it. The magnitude of the resultant field can be obtained by integrating eq. (2) *i.e.*,

$$B = \frac{\mu_0}{4\pi}\int\frac{i\,dl}{R^2} = \frac{\mu_0 i}{4\pi R^2}\int dl = \frac{\mu_0 i}{R^2}\times 2\pi R = \frac{\mu_0 i}{2R} \quad ...(3)$$

The direction of the magnetic field B at the centre is along the axis of the coil downward.

The direction of the magnetic field due to a current carrying circular coil at its centre may be found by using the **right hand thumb rule for loop currents.** If the fingers of the right hand are curled along the current, the stretched thumb will point in the direction of the magnetic field* **[fig. 9.35]**.

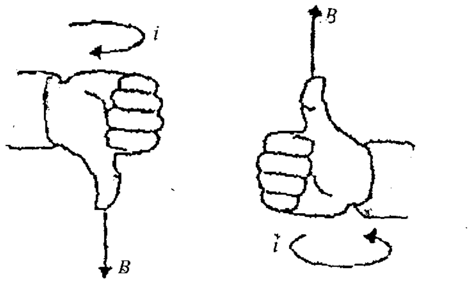

Fig. 9.35 : Right hand thumb rule for loop current..

If there are n turns of wire in the coil, then the magnetic field at the centre coil will be

$$B = \frac{\mu_0 ni}{2R} \quad ...(4)$$

Thus the magnetic field at the centre of the coil can be increased (i) by increasing the number of turns n in the coil, (ii) by increasing the current i in the coil and (iii) by decreasing the radius R of the coil.

Magnetic field lines due to a current carrying circular coil are shown in **fig 9.36**. We see that the field lines are concentric circles near the wire and they are approximately parallel to each other at the centre of the coil. Therefore, *the magnetic field at the centre of a current carrying circular coil is nearly uniform and perpendicular to the plane of the coil.*

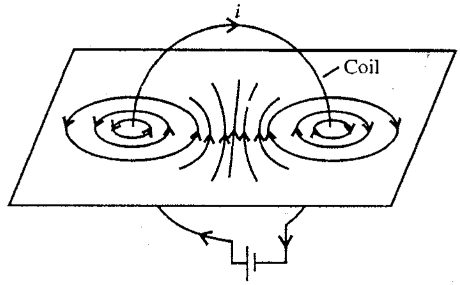

Fig. 9.36 : Magnetic field lines due to circular current; the field lines are seen to be uniform at the centre of the coil.

Magnetic field at the centre of a current carrying circular arc PQ **of radius** R **and subtending an angle** α **at the centre** O **[fig. 9.37]** can be obtained by considering the field dB at O due to an element dl of the coil, given by eq. (2)

$$dB = \frac{\mu_0}{4\pi}\frac{i\,dl}{R^2}$$

and integrating it for proper limits P to Q *i.e.*,

$$B = \int dB = \frac{\mu_0 i}{4\pi R}\int dl = \frac{\mu_0 i}{4\pi R}(\alpha R) = \frac{\mu_0 i\alpha}{4\pi R}$$

$$\left[\because \int dl = \text{lenght of arc } PQ = \alpha R\right]$$

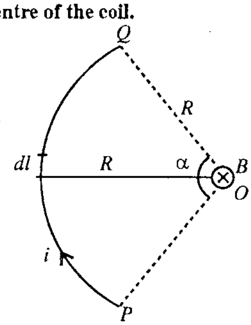

Fig. 9.37 : Magnetic field at the centre of a current carrying circular arc.

9·13. Magnetic Field due to a Charge in Uniform Circular Motion

Let a charge q be revolving uniformaly, around O an a circular path of radius R with speed v **[fig. 9.38]**. The period of revolution is

$$T = \frac{\text{Circumference of the circular path}}{\text{Speed}} = \frac{2\pi R}{v} \quad ...(1)$$

$$\text{Frequency of revolution, } n = \frac{1}{T} = \frac{v}{2\pi R} \quad ...(2)$$

Equivalent current due to flow of charge revolving on a circle with frequency n is given by

$$i = nq = \frac{qv}{2\pi R} \quad ...(3)$$

Hence the magnetic field at the centre of the circular path is

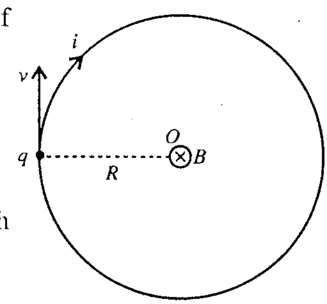

Fig. 9.38 : Magnetic field due to a charge q in uniform circular motion.

* Another way to set the direction is to look into the loop along its axis. If the current in the coil is anticlockwise, the direction of the magnetic field is up along the axis or towards the reader. If the current in the coil is clockwise, the direction of the magnetic field is down along the axis or away from the reader.

$$B = \frac{\mu_0 i}{2R} = \frac{\mu_0 nq}{2R} = \frac{\mu_0 qv}{4\pi R^2} \qquad ...(4)$$

9·14. Magnetic Field at an Axial Point of a Current Carrying Circular Coil

Let us consider a circular coil (or loop) of radius R carrying a current i. We want to find the magnetic field B at a point P on the axis of the coil at a distance x from its centre O **[fig. 9.39]**. The coil is shown perpendicular to the plane of the paper and its axis OP lies in the plane of the paper. At the point C, the current comes out of the plane and at the point D, it goes into the plane. Consider a small element $d\vec{l}$ of the wire at C. If $\vec{r}$ be the vector distance of the point P from the point C, then the magnetic field at P due to the current element $i\, d\vec{l}$ is given by

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{i\, d\vec{l} \times \vec{r}}{r^3} \qquad ...(1)$$

We see that $d\vec{l}$ is perpendicular to the plane of the paper and $\vec{r}$ is in the plane of the paper. Hence the direction of $d\vec{l} \times \vec{r}$, or $d\vec{B}$ is perpendicular to $\vec{r}$ but in the plane of the paper.

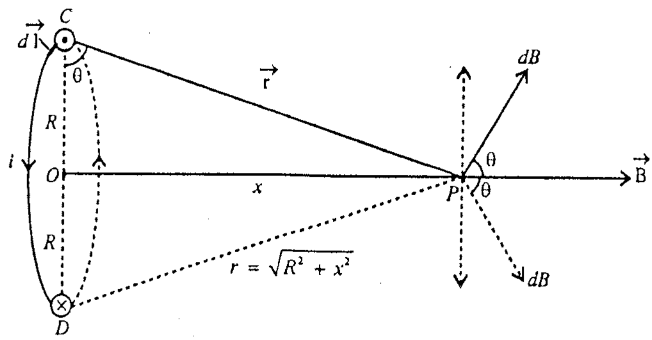

Fig. 9.39 : Magnetic field at an axial point of a current carrying circular coil.

From eq. (1), the magnitude of the field $d\vec{B}$ at P is

$$dB = \frac{\mu_0 i dl}{4\pi r^2} = \frac{\mu_0 i dl}{4\pi (R^2 + x^2)} \qquad [\because r = \sqrt{R^2 + x^2}] \qquad ...(2)$$

If we resolve $d\vec{B}$ along the axis of the coil and perpendicular to the axis, we find the components $dB \cos\theta$ and $dB \sin\theta$, where θ is the angle between $d\vec{B}$ and the axis of the coil and it is obviously equal to $\angle PCO$ as $\angle\theta = \frac{\pi}{2} - \angle CPO = \angle PCO$.

Now, if we consider a diametrically opposite current element idl at D, the magnetic field at P due to this element will have the same magnitude dB with direction along the dotted arrow, making the angle θ with the axis of the coil. This dB can also be resolved as $dB \cos\theta$ along the axis of the coil and $dB \sin\theta$ perpendicular to the axis. The perpendicular components ($dB \sin\theta$) due to the two fields (dB each) due to the element at C and D cancel each other and the axial components $dB \cos\theta$ are added. Let us divide the coil in such pairs of diametrically opposite elements, we find that the resultant magnetic field B at P is along the axis of the coil and can be obtained by integrating $dB \cos\theta$ *i.e.*,

$$B = \int dB \cos\theta = \int \frac{\mu_0 i dl}{4\pi (R^2 + x^2)} \cdot \frac{R}{(R^2 + x^2)^{1/2}} \qquad \left[\because \cos\theta = \frac{R}{(R^2 + x^2)^{1/2}}\right]$$

$$= \frac{\mu_0 iR}{4\pi (R^2 + x^2)^{3/2}} \int dl = \frac{\mu_0 iR \times 2\pi R}{4\pi (R^2 + x^2)^{3/2}} = \frac{\mu_0 iR^2}{2(R^2 + x^2)^{3/2}} \qquad ...(3)$$

If there are n turns in the coil, then each turn will equally contribute to the magnetic field and the *total magnetic field due to the coil* at P is obtained to be

$$B = \frac{\mu_0 niR^2}{2(R^2 + x^2)^{3/2}} \qquad ...(4)$$

which has the direction along the axis of the coil *i.e.*, along $\vec{OP}$. Right hand thumb rule for loop current may be used to find the direction of the field.

Special Cases :

(1) **Magnetic field at the centre of the circular coil :** When $x = 0$, we obtain from eq. (4) the expression for magnetic field at the centre of the coil *i.e.,*

$$B = \frac{\mu_0 ni}{2R} \quad ...(5)$$

This is the **maximum value of the magnetic field** due to a current carrying circular coil on its axis.

(2) **Magnetic field at a point far away from the centre of the coil :** When $x >> R$, from eq. (4) we obtain

$$B = \frac{\mu_0 niR^2}{2x^3} = \frac{2\mu_0 ni(\pi R^2)}{4\pi x^3} = \frac{2\mu_0 (ni\,A)}{4\pi x^3} \quad ...(6)$$

where $A = \pi R^2$ is the area of the coil and the quantity $niA = m$ represents the **magnetic dipole moment** of the current carrying coil. According to right hand thumb rule, the direction of the area vector $\vec{A}$ is along $\vec{B}$ and hence the magnetic moment $\vec{m}$ $(= ni\,\vec{A})$ has also the direction of $\vec{B}$. Thus, the magnetic field due to the current carrying coil at an axial point at a distance x is given by

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{2\vec{m}}{x^3} \quad ...(7)$$

We know that the electric field due to an electric dipole at a point distant x on its axis is given by

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{2\vec{p}}{x^3} \quad ...(8)$$

Thus one can see the similarity of the expressions (7) and (8) for magnetic and electric dipole moments.

The magnetic field lines due to a current carrying coil is shown in **fig. 9.40** and they are similar to the magnetic field lines due to a bar magnet or magnetic dipole.

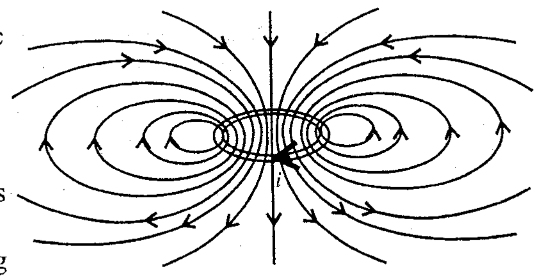

Fig. 9.40 : Magnetic field lines due to a current carrying circular coil.

9·15. Rate of Variation of Magnetic Field on the Axis of Current Carrying Circular Coil and Helmholtz Coils

The expression for magnetic field at a coil distant x on the axis of a current carrying circular coil of radius R is given by

$$B = \frac{\mu_0 niR^2}{2(R^2 + x^2)^{3/2}} \quad ..(1)$$

At $x = 0$ *i.e.,* centre of the circular coil, the value of the magnetic field is

$$B = \frac{\mu_0 ni}{2R} \quad ...(2)$$

This is the **maximum value of the magnetic field due to a circular coil on its axis.** If we move on either side of the centre O of the coil, the value of x increases and hence from eq. (1), the value of magnetic field decreases [**fig. 9.41**]. At $x = \infty$, the value of B becomes zero.

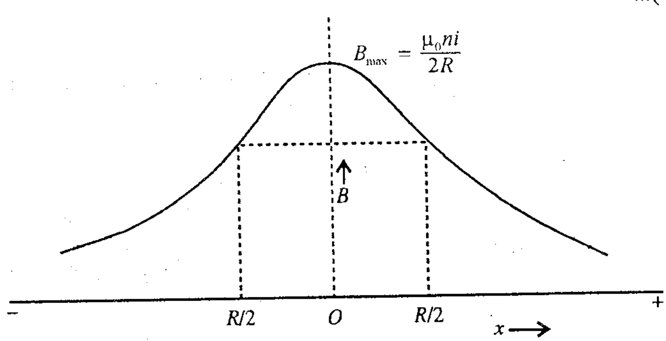

Fig. 9.41 : Variation of magnetic field on the axis of a current carrying coil.

The rate of variation of magnetic field with distance x along the axis of the coil is obtained by differentiating B of eq. (1) with respect to x *i.e.,*

$$\frac{dB}{dx} = \frac{\mu_0 niR^2}{2\,(R^2 + x^2)^{5/2}}\left(-\frac{3}{2}\right)2x = -\frac{3\,\mu_0 niR^2 x}{2\,(R^2 + x^2)^{5/2}} \qquad ...(3)$$

Also,

$$\frac{d^2B}{dx^2} = -\frac{3\mu_0 niR^2}{2}\left[\frac{1}{(R^2 + x^2)^{5/2}} - \frac{5}{2}\,\frac{2x^2}{(R^2 + x^2)^{7/2}}\right]$$

$$= -\frac{3}{2}\,\frac{\mu_0 niR^2}{2(R^2 + x^2)^{7/2}}(R^2 + x^2 - 5x^2) = -\frac{3}{2}\,\mu_0 niR^2(R^2 - 4x^2)$$

If $\frac{d^2B}{dx^2} = 0$, then $R^2 - 4x^2 = 0$ or $x = \pm\frac{R}{2}$ ...(4)

This means that dB/dx is constant at $x = + R/2$ and $x = - R/2$.

Thus we find that *near the points distant R/2 from the centre of the coil on both sides, the rate of change of magnetic field from a current carrying coil becomes constant or the field decreases uniformly with increasing distance.* This result is applied in **Helmholtz galvanometer** in which two identical coils are mounted on a common axis at a distance apart equal to their common radius (R) [fig. 9.42]. Around the point on the axis mid-way between two coils, the magnetic field B is obtained very nearly uniform over an appreciable region and due to two coils it is given by using eq. (1) *i.e.*,

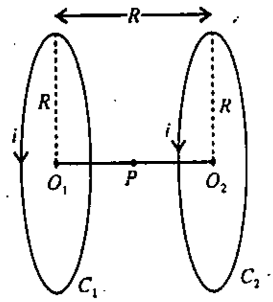

Fig. 9.42 : Identical coils C_1 and C_2 carry the same current in the same direction in Helmholtz galvanometer; magnetic field is nearly uniform around P.

$$B = 2 \times \frac{\mu_0 niR^2}{2\left(R^2 + \frac{R^2}{4}\right)^{3/2}} = \frac{8\mu_0 ni}{5\sqrt{5}\,R} \qquad ...(5)$$

The variations of the magnetic field with distance due to two coils of the Helmholtz galvanometer with their resultant are shown in fig. 9.43. We see that the resultant magnetic field is nearly uniform in the region between the two coils.

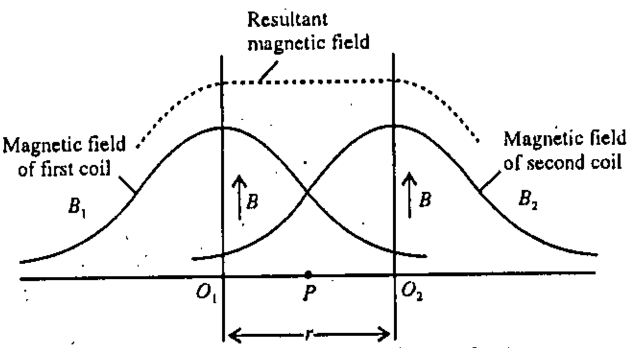

Fig. 9.43 : Variation of magnetic field with distance due to two coils of Helmholtz galvanometer.

9·16. Magnetic Field due to a Current Carrying Solenoid

If a coil of conducting wire is made by winding closely a large number of turns in the form of helix, then this arrangement is called **solenoid.** The wire is coated with an insulating material so that the adjacent turns, which touch physically each other, are electrically insulated [fig. 9.44]. A solenoid is made by winding the conducting wire on the length of a hollow cylinder of nonconducting material *e.g.*, card board, china clay etc.

Consider a solenoid of radius R, carrying a current i. We consider the winding of the solenoid to be uniform so that there are n turns per unit length. Let us calculate the magnetic field at a point P on the axis of the solenoid due to the current i flowing in it.

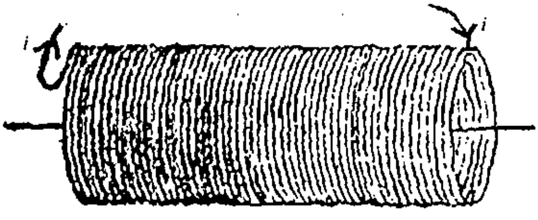

Fig. 9.44 : Solenoid.

We consider an infinitesimally small length dx of the solenoid. The number of turns in this dx length are $n\,dx$. Let O be the centre of the coil of length dx and x be the distance of P from O **[fig. 9.45]**. According to Biot-Savart law, the magnetic field at P due to this coil of length dx is

$$dB = \frac{\mu_0 (ndx) iR^2}{2(R^2 + x^2)^{3/2}} \quad ...(1)$$

Let the angle APX be θ and the length dx subtend angle $d\theta$ at P.

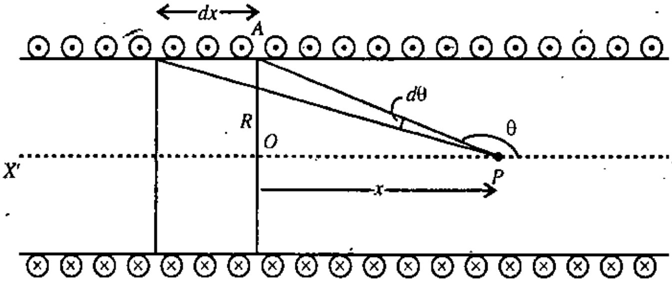

Fig. 9.45 : Calculation of magnetic field at a point p on the axis of a solenoid.

Now, $\tan APO = \frac{R}{x}$

or $\tan(\pi - \theta) = \frac{R}{x}$

or $-\tan\theta = \frac{R}{x}$

or $x = -R\cot\theta, \therefore dx = R\,\text{cosec}^2\theta\, d\theta$

and $(R^2 + x^2)^{3/2} = R^3 (1 + \cot^2\theta)^{3/2} = R^3\,\text{cosec}^3\theta$

$$\therefore \quad dB = \frac{\mu_0 niR^2 (R\,\text{cosec}^2\theta\, d\theta)}{2R^3\,\text{cosec}^3\theta} = \frac{\mu_0 ni}{2}\sin\theta\, d\theta \quad ...(2)$$

Total field B at P due to the entire length of the solenoid is obtained by integrating eq. (2) for θ from θ_1 (one end) to θ_2 (other end) **[fig. 9.46]**.

$$B = \frac{1}{2}\int_{\theta_1}^{\theta_2} \mu_0 ni \sin\theta\, d\theta = \frac{1}{2}\mu_0 ni \left[-\cos\theta\right]_{\theta_1}^{\theta_2}$$

or $$B = \frac{1}{2}\mu_0 ni (\cos\theta_1 - \cos\theta_2) \quad ...(3)$$

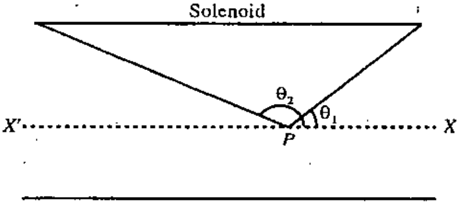

Fig. 9.46.

The direction of the magnetic field at P will be given by right hand thumb rule *i.e.*, if the fingers of the right hand are curled around the solenoid in the direction of the current, the thumb will point the direction of the magnetic field. In the present case, the direction of B is along positive X-axis.

Special Cases :

(1) Magnetic field due to a very long solenoid (infinite length) : In case of very long solenoid and the point P is well inside around the middle of the solenoid, $\theta_1 = 0$ and $\theta_2 = \pi$. Hence from eq. (3), the expression for magnetic field B is obtained to be

$$B = \mu_0 ni \quad ...(4)$$

because $\cos\theta_1 - \cos\theta_2 = \cos 0 - \cos\pi = 1 + 1 = 2$.

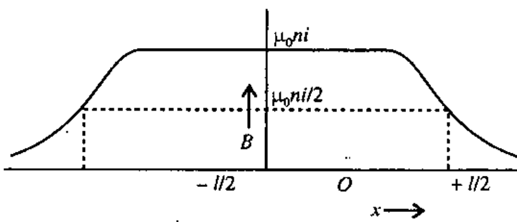

Fig. 9.47 : Variation of magnetic field on the axis of current carrying solenoid.

(2) Magnetic field at an end of a long solenoid : When the point P is at an end on the axis of the solenoid, $\theta_1 = \pi/2$ and $\theta_2 = \pi$. Hence from eq. (3),

$$B = \frac{1}{2}\mu_0 ni \quad ...(5)$$

because $\cos\theta_1 - \cos\theta_2 = \cos\frac{\pi}{2} - \cos\pi = 0 + 1 = 1$.

Thus the magnetic field at an end is one half that at the middle of a long solenoid.

The variation of magnetic field on the axis of current carrying solenoid is shown in **fig.9.47.** If the solenoid is long enough, the magnetic field is uniform at all points inside the solenoid except at its ends.

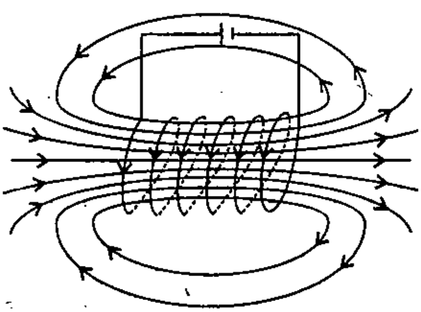

Fig. 9.48 : Magnetic field lines due to a current carrying solenoid.

Magnetic field lines due to a solenoid are shown in **fig.9.48.** The field lines inside the solenoid are nearly parallel to its axis and therefore the magnetic field inside the solenoid is uniform. The magnetic field lines are similar to a bar magnet. If we look towards the face of the solenoid and find the current to be anticlockwise, this face of the solenoid behaves like the north pole of a bar magnet. The face of the solenoid will behave as south pole for clockwise current (fig. 9.49).

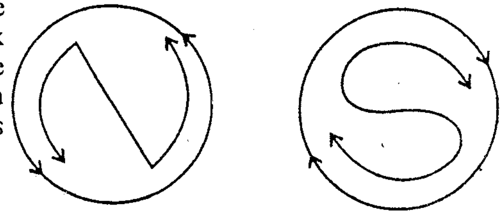

Fig. 9.49 : Representation of North and South poles at the ends of solenoid.

9·17. Ampere's Law

According to this law, *the line integral of magnetic field $\vec{B}$ along a closed curve C is equal to μ_0 times the total current i flowing through the area bounded by the curve C i.e.,*

$$\oint_C \vec{B}\cdot d\vec{l} = \mu_0 i \qquad \text{(1)}$$

where μ_0 is the permeability of free space (or air). The direction of i in eq. (1) is given by right hand thumb rule *i.e.,* if you curl your fingers in the direction of the line integral around the curve C, the thumb points the direction of the current. Eq. (1) repsesents the **integral form of Ampere's law.**

If the total current through the area bounded by the curve C is zero, then the line intigral of magnetic field $\vec{B}$ along the curve C is equal to zero *i.e.,*

$$\oint_C \vec{B}\cdot d\vec{l} = 0 \qquad \text{...(2)}$$

Proof : Consider a long straight wire carrying a current i directed upward in the plane of the paper [**fig. 9.50**]. Consider a point O on the wire and a circular path of radius R around O in a plane perpendicular to the plane of the paper and straight wire. According to Biot-Savart law, the magnetic field due to the currents i at a point P on the circular path is given by

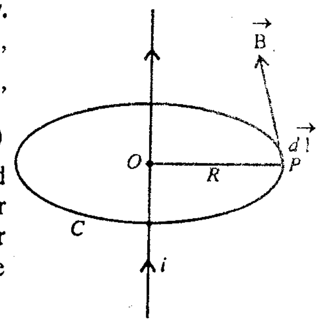

Fig. 9.50.

$$B = \frac{\mu_0 i}{2\pi R} \qquad \text{...(3)}$$

whose direction is along the tangent to the circle at P. The magnetic field $\vec{B}$, given by eq. (3) is constant in magnitude at all points of the circular path. If we move an infinitesimally small distance $d\vec{l}$ along $\vec{B}$, then the line integral of the magnetic field is given by

$$\oint_C \vec{B}\cdot d\vec{l} = \oint_C B\,dl = B\oint_C dl = \frac{\mu_0 i}{2\pi R}\times 2\pi R = \mu_0 i \qquad \text{...(4)}$$

where we have put $\oint dl = 2\pi R$ as the length of the circular path.

Thus we find that the line integral $\oint_C \vec{B}\cdot d\vec{l}$ is equal to the current through the area bounded by the circular curve C. This proves the Ampere's law. This law holds good for an algebraic sum of currents through the surface area bounded by a closed curve C.

Ampere's law and Biot-Savart's law both are equivalent, but under certain symmetrical conditions, Ampere's law is more useful. Here, we have proved Ampere's law by using Biot-savart's law.

This is to be noted that eq. (1), the mathematical form of *Ampere's law is valid only for stationary currents i.e.,* the currents which do not change with time.

9·18. Differential Form of Ampere's Law : $\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$

The differential form of Ampere's law is

$$\vec{\nabla}\times\vec{B} = \mu_0\vec{J} \text{ or curl } \vec{B} = \mu_0\vec{J} \qquad \text{...(1)}$$

where $\vec{B}$ is te magnetic field at a point and $\vec{J}$ is the current density at the same point.

The Ampere's law in differential form can be stated as : *The curl of magnetic field $\vec{B}$ at a point is equal to μ_0 times the current density at the point.*

Proof : Acceding to Ampere's law, the line integral of magnetic field $\vec{B}$ along a closed curve C in the field is equal to μ_0 times the net current i through the area bounded by the curve *i.e.*,

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 i \qquad ...(1)$$

This is the **integral from of Ampere's law.**

Let a surface S be bound by a curve C. Consider a small area $d\vec{S}$ around a point P on this surface. If $\vec{J}$ is the current density at that point, then the quantity $\vec{J} \cdot d\vec{S}$ represents the current through $d\vec{S}$ [**fig. 9.51**]. Hence the total current through the surface S is given by

$$i = \int_S \vec{J} \cdot d\vec{S}$$

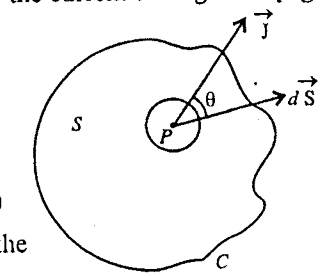

Fig. 9.51.

Hence the Ampere's law can be put in another from :

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S \vec{J} \cdot d\vec{S} \qquad ...(2)$$

According to Stoke's theorem, we can express the line integral of the magnetic field as

$$\oint \vec{B} \cdot d\vec{l} = \int_S \text{curl}\, \vec{B} \cdot d\vec{S} \qquad ...(3)$$

From eqs. (2) and (3), we have

$$\int_S \text{curl}\, \vec{B} \cdot d\vec{S} = \mu_0 \int_S \vec{J} \cdot d\vec{S} \text{ or } \int_S (\text{curl}\, \vec{B} - \mu_0 \vec{J}) \cdot d\vec{S} = 0$$

This equation is true for any surface. Hence

$$\text{curl}\, \vec{B} - \mu_0 \vec{J} = 0$$

i.e.,

$$\text{curl}\, \vec{B} = \mu_0 \vec{J} \text{ or } \vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \qquad ...(4)$$

This is the **differential form of Ampere's law** and is the general relation between the magnetic field $\vec{B}$ and the current density $\vec{J}$, causing that field. Eq. (4) *is correct only for stationary* (*not time-varying*) *currents.*

It is to be mentioned that eq. (4) is not enough to determine the magnetic field $\vec{B}$ uniquely at a point from the given value of the current density $\vec{J}$ at that point, because many different vector fields can have the same curl. Hence another condition is required and that condition is

$$\text{div}\, \vec{B} = 0 \qquad ...(5)$$

This equation together with eq. (4) uniquely determines $\vec{B}$ for known $\vec{J}$. This equation is established in the next section.

9·19. Divergence of $\vec{B}$ is Equal to Zero i.e., $\vec{\nabla} \cdot \vec{B} = 0$

Magnetic field lines always form closed curves. The lines of magnetic field do not originate at any definite point *i.e.*, magnetic field lines do not have sources or sinks. It is to be mentioned that in case of electric field, the lines emerge from a positive charge and end on the negative charge.

If we consider a closed surface, enclosing a volume in space, in a magnetic field, we find that as many lines enter into it as are coming out of it. Hence the total magnetic flux over any closed surface S in a magnetic field $\vec{B}$ is always equal to zero *i.e.*,

$$\oint_S \vec{B} \cdot d\vec{S} = 0 \qquad ...(1)$$

According to Gauss's divergence theorem,

$$\oint_S \vec{B} \cdot d\vec{S} = \int_V (\text{div}\,\vec{B})\, dV \qquad ...(2)$$

Hence from eqs. (1) and (2), we have

$$\int_V (\text{div}\,\vec{B})\, dV = 0 \qquad ...(3)$$

This is true for any arbitrary volume. Hence

$$\text{div}\,\vec{B} = 0 \text{ or } \vec{\nabla} \cdot \vec{B} = 0 \qquad ...(4)$$

Eq. (4) is a basic equation of electromagnetism and is one of the Maxwell's equations of electromagnetic theory. The significance of equation (4) is that there exists no isolated magnetic poles as sources or sinks for the magnetic field lines which in fa :t are closed curves.

Proof of div B = 0 by using Biot-Savart's Law

For volume current, the Biot-Savart law is

$$\vec{B} = \frac{\mu_0}{4\pi} \int_V \vec{J}(\vec{r}') \times \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}\, dV \qquad ...(5)$$

This formula gives the magnetic field $\vec{B}$ at the position vector $\vec{r}$ relative to an origin O [fig. 9.52]. There is the current distribution in the volume and $J(\vec{r}')$ is the current density at the point $\vec{r}'$. $dV = dx'\, dy'\, dz'$ is the volume element around the point $\vec{r}'$.

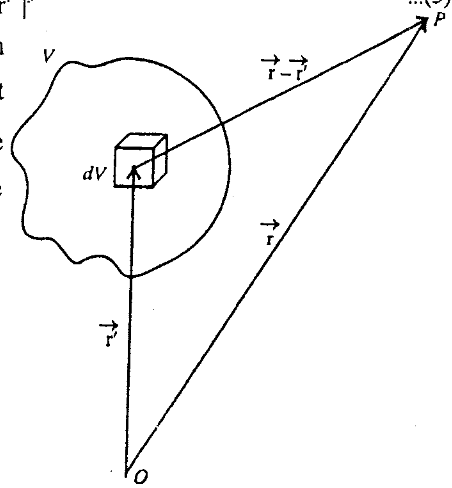

Fig. 9.52.

The integration is to be taken over the primed coordinates and we take divergence with respect to unprimed coordinates.

Taking divergence of eq. (5)

$$\vec{\nabla} \cdot \vec{B} = \frac{\mu_0}{4\pi} \int_V \vec{\nabla} \cdot \left[\vec{J}(\vec{r}') \times \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right] dV \qquad ...(6)$$

From vector identity,

$$\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})$$

$$\therefore \vec{\nabla} \cdot \left[\vec{J}(\vec{r}') \times \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right] = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \cdot [\vec{\nabla} \times \vec{J}(\vec{r}')] - \vec{J}(\vec{r}') \cdot \left[\vec{\nabla} \times \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right] \qquad ...(7)$$

As $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$ and $\vec{J}(\vec{r}')$ does not depend on unprimed variables (x, y, z), hence

$$\vec{\nabla} \times \vec{J}(\vec{r}') = 0. \qquad ...(8)$$

Also, we know $\qquad \vec{\nabla}\left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$

$$\therefore \qquad \vec{\nabla} \times \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} = \vec{\nabla} \times \vec{\nabla}\left(\frac{1}{|\vec{r} - \vec{r}'|}\right) \qquad ...(9)$$

Here $\frac{1}{|\vec{r}-\vec{r}'|}$ is a scalar function, say ϕ, and we know that curl grad $\phi = 0$ or $\vec{\nabla}\times\vec{\nabla}\phi = 0$.

$$\therefore \qquad \vec{\nabla}\times\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^3} = 0 \qquad ...(10)$$

In view of eqs. (8) and (10), we obtain from eq. (7)

$$\vec{\nabla}\cdot\left[\vec{J}(\vec{r}')\times\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^3}\right] = 0 \qquad ...(11)$$

Hence eq. (6) becomes

$$\vec{\nabla}\cdot\vec{B} = 0 \quad \text{or div } \vec{B} = 0 \qquad ...(12)$$

9·20. Applications of Ampere's Law

(1) **Magnetic field due to a straight long wire :** Consider a long straight wire carrying a current i. We want to determine the magnetic field due to this wire at a point P distance d **[fig. 9.53(a)]**. The magnetic field lines are concentric circles, centred on the wire. In **fig. 9.53(b)** we represent the current i coming out of the page by a central dot. We draw a circular path of radius d around the wire. The magnetic field is along the tangent at the point and due to symmetry it has the same value at any point of the circular path.

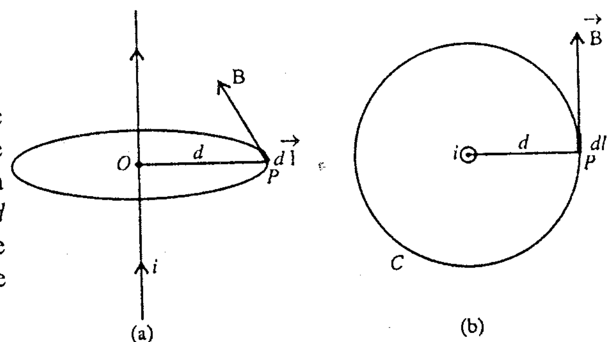

Fig. 9.53 : Magnetic field due to a straight long wire.

As the value of $\vec{B}\cdot d\vec{l} = B\,dl$, applying Ampere's law,

$$\oint_C \vec{B}\cdot d\vec{l} = \mu_0 i \text{ or } B\oint_C dl = \mu_0 i \text{ or } B\times 2\pi d = \mu_0 i$$

$$\therefore \qquad B = \frac{\mu_0 i}{2\pi d} \qquad ...(1)$$

(2) **Magnetic field due to a long cylindrical wire of radius R :** Consider a long cylindrical solid wire of radius R, carrying a current i. The current is distributed uniformly on the circular cross-section of the wire. We want to find the magnetic field at a point P, distant r from the point O on the axis of the cylindrical wire, when (a) $r < R$ (P inside the wire); (b) $r = R$ (P on the surface of the wire) and (c) $r > R$ (P outside the wire). Consider a circular path of radius r about the point O.

By symmetry, the magnetic field $\vec{B}$ at a point P on the circular path is along the tangent and has the same magnitude at each point of the circle. Applying Ampere's law,

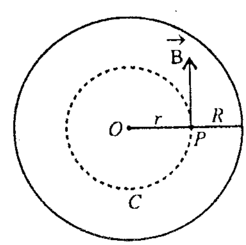

Fig. 9.54 : Magnetic field at a point P inside the cylindrical solid wire.

$$\oint_C \vec{B}\cdot d\vec{l} = \mu_0 \times \text{Current flowing through the area bounded by the curve } C \qquad ...(2)$$

(a) When $r < R$ **[fig. 9.54]**, the point P is inside the wire and the current through the area πr^2 bounded by the curve C is

$$i' = \frac{i}{\pi R^2} \times \pi r^2 = \frac{ir^2}{R^2} \quad ...(3)$$

where the total current i is uniformly distributed on the cross-sectional area πR^2 of the wire. Hence $i/\pi R^2$ is the current per unit area and the current through πr^2 area is $\pi r^2(i/\pi R^2)$ or ir^2/R^2. Therefore,

$$\oint_C B\, dl = \mu_0 \frac{ir^2}{R^2} \text{ or } B \times 2\pi r = \mu_0 \frac{ir^2}{R^2}$$

$$\therefore \quad B = \frac{\mu_0 i r}{2\pi R^2} \quad ...(4)$$

Thus *the magnetic field at a point inside a cylindrial long wire is proportional to its distance from the axis i.e.,* $B \propto r$. The magnetic field is zero at the centre (O) of the wire.

(b) When $r = R$, the point P exists on the surface of the wire and the value of magnetic field B is

$$B = \frac{\mu_0 i R}{2\pi R^2} = \frac{\mu_0 i}{2\pi R} \quad ...(5)$$

(c) When $r > R$, the point P lies outside of the wire. In this case the Ampere's law gives [fig. 9.55].

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 i \text{ or } \oint_C B\, dl = \mu_0 i \text{ or } B \times 2\pi r = \mu_0 i$$

$$\therefore \quad B = \frac{\mu_0 i}{2\pi r} \quad ...(6)$$

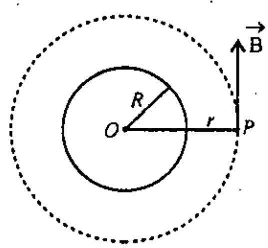

Fig. 9.55 : Magnetic field outside the cylindrical wire.

Thus, when the point P lies outside the cylindrical wire, the magnetic field B is inversely proportional to its distance from the axis of the wire.

The variation of magnetic field with distance (r) from the axis of a cylindrical wire of radius R is shown in fig. 9.56. Note *that the magnetic field is maximum at a point* ($r = R$) *just at the surface of the cylinder.*

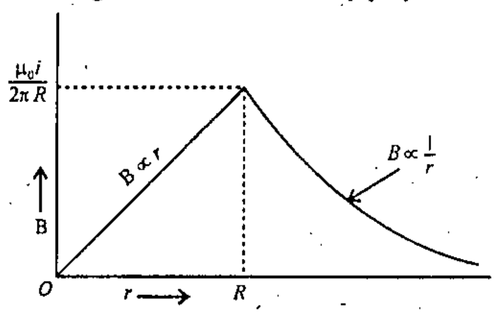

Fig. 9.56 : Variation of magnetic field with distance from the axis of a cylindrical wire.

(3) **Magnetic field due to a long hollow current carrying cylindrical conductor :** Consider R_1 and R_2 to be the inner and outer radii of a long hollow cylindrical conductor. It carries a current i, uniformly distributed over its cross-section. We want to determine the magnetic field at a point P distant r from its axis.

(i) **Point P lies inside the hollow space** ($0 \le r < R_1$) : **Fig. 9.57(a)** represents the cross-section of the cylindrical conductor. The axis of the

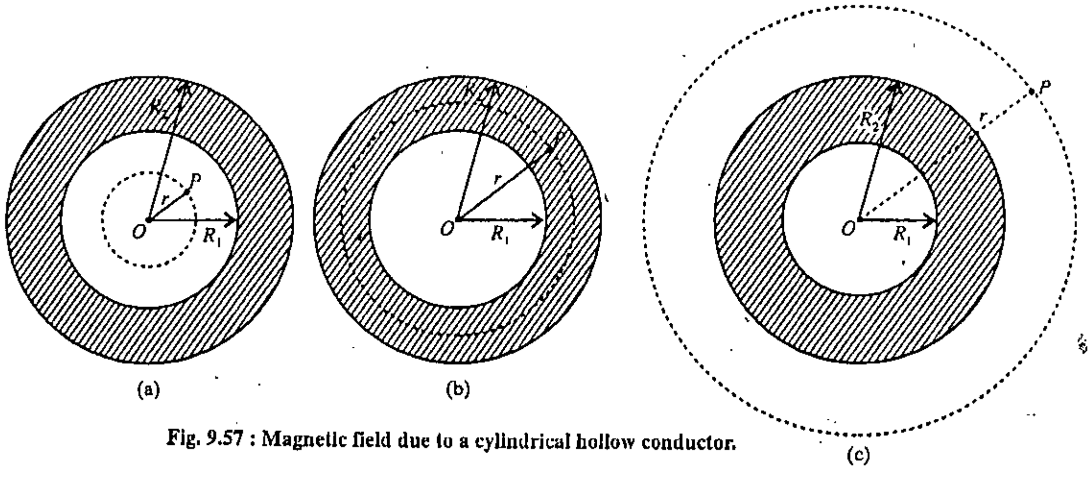

Fig. 9.57 : Magnetic field due to a cylindrical hollow conductor.

conductor passes through O and R_1 and R_2 are its inner and outer radii respectively. Draw a circle of radius r around O, passing through the point P. Due to symmetry, the magnetic field $\vec{B}$ at any point of the circle is tangential and has the same magnitude at its each point. According to Ampere's law,

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 i' \quad ...(7)$$

Since there is no current inside the circle of radius r *i.e.*, $i' = 0$ and the line integral of constant $\vec{B}$ along the circumference of the circle is $B \times 2\pi r$, eq. (6) gives

$$B \times 2\pi r = 0 \text{ or } B = 0 \quad ...(8)$$

Thus, *the magnetic field in the hollow space of the hollow cylindrical current carrying conductor is zero.*

(ii) Point P lies within the material of hollow conductor *i.e.*, $R_1 \le r < R_2$. Draw a circle of radius r, passing through the point P, lying in the material of the conductor in between its inner and outer radii R_1 and R_2 **[fig. 9.57(b)]**. Since the current i is uniformly distributed over the cross-section of the conductor, hence the current per unit area of the cross-section is $i/[\pi(R_2^2 - R_1^2)]$ and the current through the area in between the circles of radii R_1 and r is

$$i = \frac{i}{\pi(R_2^2 - R_1^2)} \cdot \pi(r^2 - R_1^2) = \frac{i(r^2 - R_1^2)}{R_2^2 - R_1^2}$$

Applying Ampere's law,

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 i \text{ or } B \times 2\pi r = \frac{\mu_0 i(r^2 - R_1^2)}{R_2^2 - R_1^2}$$

$$\therefore \quad B = \frac{\mu_0 i}{2\pi r}\left(\frac{r^2 - R_1^2}{R_2^2 - R_1^2}\right) \quad ...(9)$$

(iii) Point P lies outside the conductor *i.e.*, $R_2 \le r < \infty$: Draw a circle of radius r, passing through the point P **[fig. 9.57(c)]**. Using Ampere's law,

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 i \text{ or } B \times 2\pi r = \mu_0 i$$

$$\therefore \quad B = \frac{\mu_0 i}{2\pi r} \quad ...(10)$$

Thus, *the magnetic field at a point outside a hollow current carrying conductor is inversely proportional to its distance.*

(4) Magnetic field due to a current carrying solenoid : A solenoid is a long wire wound closely in the form of helix. Let it carry a current i. The magnetic field due to a single turn at points close to it, is similar to that of a straight conductor. The resultant field is the vector sum of the fields produced by all the turns. The fields at an outside point due to the neighbouring turns oppose each other, while at an inside point are in the same direction. Thus the magnetic field inside a solenoid is uniform everywhere and approximately zero outside it.

We apply Ampere's law

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \times \text{Total current inside the boundary } C \quad ...(11)$$

to the rectangular path $PQRS$ shown in **fig. 9.58**.

This integral for the closed path $PQRS$ is the sum of four integrals, one for each side of the rectangle *i.e.*,

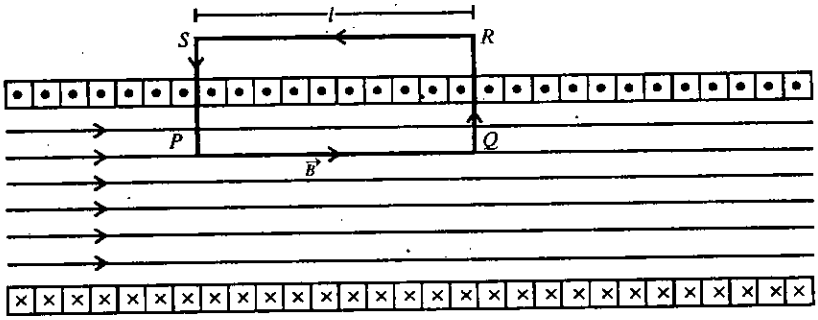

Fig. 9.58 : Magnetic field due to a current carrying solenoid

$$\oint_{PQRS} \vec{B} \cdot d\vec{l} = \int_P^Q \vec{B} \cdot d\vec{l} + \int_Q^R \vec{B} \cdot d\vec{l} + \int_R^S \vec{B} \cdot d\vec{l} + \int_S^P \vec{B} \cdot d\vec{l} \quad ...(12)$$

For the path PQ, $\vec{B}$ is along $d\vec{l}$ and the value of the integral is

$$\int_P^Q \vec{B} \cdot d\vec{l} = \int_P^Q B\,dl = Bl$$

where l is the length of the side PQ.

As the paths QR and SP are perpendicular to $\vec{B}$ and hence

$$\int_Q^R \vec{B} \cdot d\vec{l} = \int_S^P \vec{B} \cdot d\vec{l} = 0$$

Further the field is zero outside the solenoid, hence

$$\int_R^S \vec{B} \cdot d\vec{l} = 0$$

Thus, the value of the line integral along $PQRS$ path is

$$\oint_{PQRS} \vec{B} \cdot d\vec{l} = Bl \quad ...(13)$$

If there are n turns per unit length of the solenoid, then there are nl turns inside the rectangle, carrying a current nli (each turn carries a current i). Hence from eqs. (11) and (13), we have

$$Bl = \mu_0 nil \text{ or } B = \mu_0 ni \quad ...(14)$$

This relation, of course deduced for a very long solenoid, is true for an actual solenoid (whose length is much greater than its diameter) at internal points near the centre of the solenoid. In eq. (14), the value of B does not depend on the radius and the length of the solenoid and it is constant over the cross-section of the solenoid. This relation (14) was earlier derived by using Biot-Savart law.

(5) Magnetic field due to a toroid : If we wind a conducting wire closely on a non-conducting ring, we obtain a toroid. In other words, a toroid is obtained by bending a solenoid in a circular shape with its ends joined together. Let i be the current in the toroid. We want to find the magnetic field at a point P inside the toroid [**fig. 9.59**]. The distance of P from the centre of the toroid is r.

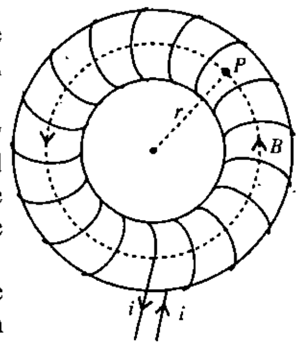

Fig. 9.59 : Magnetic field due to a toroid.

We draw a circular path (shown dotted) through the point P inside the toroid. Due to symmetry, the magnetic field will have the same magnitude at each point of the circle and its direction at a point will be along the tangent to the circle. Hence

$$\oint_C \vec{B} \cdot d\vec{l} = \oint B\,dl = B \oint dl = B \times 2\pi r = 2\pi r B \quad ...(15)$$

If N is the number of turns in the toroid, then Ni will be the total current through the area of the circular boundary.

Applying Ampere's law,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 Ni \text{ or } 2\pi r B = \mu_0 Ni$$

$$\therefore \qquad B = \frac{\mu_0 Ni}{2\pi r} \quad ...(16)$$

We see that the magnetic field is different at different points of the toroid. If the internal and external radii of the toroid are r_1 and r_2, then the corresponding magnetic fields will be

$$B_1 = \frac{\mu_0 Ni}{2\pi r_1} \text{ and } B_2 = \frac{\mu_0 Ni}{2\pi r_2}$$

Thus in contrast to the solenoid, the magnetic field B is not constant over the cross-section of the solenoid.

Ex. 1. *10 ampere current is flowing in a long straight wire. What will be the magnetic field at a distance 10 cm from the wire ?*

Sol. Here $i = 10$ A, $d = 10$ cm $= 0.1$ m

Magnetic field at a disatance $r = 0.1$ m due to a straight current carrying wire is given by

$$B = \frac{\mu_0}{4\pi} \times \frac{2i}{d} = 10^{-7} \times \frac{2 \times 10}{0.1} = 2 \times 10^{-5} \text{ Wb/m}^2.$$

Ex. 2. *A long straight vertical wire carries a current of 20 ampere directed upward. Calculate the position of neutral point if the horizontal component of earth's magnetic field is 2×10^{-5} Wb/m^2.*

Sol. Let d metre be the distance of the neutral point from the wire. The magnetic field at this point due to the long vertical wire is given by

$$B = \frac{\mu_0 i}{2\pi d} = \frac{4\pi \times 10^{-7} \times 20}{2 \times \pi \times d} = \frac{4 \times 10^{-6}}{d}$$

According to right hand screw rule, this field is horizontal and, at the neutral point, it is balanced by the horizontal component of earth's magnetic field. Thus,

$$\frac{4 \times 10^{-6}}{d} = 2 \times 10^{-5}, \therefore\ d = \frac{4 \times 10^{-6}}{2 \times 10^{-5}} = 0.2 \text{ m}.$$

Ex. 3. *A metal wire of diameter 0.8 mm carries a current of 10 A. Find the maximum magnitude of the magnetic field $\vec{B}$ due to this current.*

Sol. The maximum value of the magnetic field occurs at a point just on the surface and its value is given by

$$B = \frac{\mu_0 i}{2\pi R} = \frac{2 \times 10^{-7} \times 10}{0.4 \times 10^{-3}} = 5 \times 10^{-3} \text{ T}.$$

Ex. 4. *A long straight wire carrying a current of 2.0 A is placed horizontally in a uniform magnetic field $B = 2.0 \times 10^{-5}$ T pointing vertically upward* [fig. 9.60]. *Find the magnitude of the resultant magnetic field at the points P_1 and P_2, both situated at a distance of 2.0 cm from the wire in the same horizontal plane.*

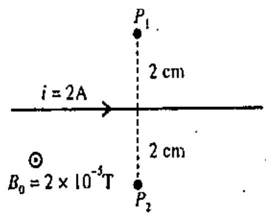

Fig. 9.60.

Sol.
$$B = \frac{\mu_0 i}{2\pi d} = \frac{2 \times 10^{-7} \times 2}{2 \times 10^{-2}} = 2 \times 10^{-5} \text{ T}.$$

Let the plane of paper represent horizontal plane. Magnetic field $B_0 = 2 \times 10^{-5}$ T is acting vertically upward.

At P_1, the magnetic field B is up (along vertical).

Hence the resultant field at $P_1 = 2 \times 10^{-5}\text{ T} + 2 \times 10^{-5}\text{ T} = 4 \times 10^{-5}$ T

At P_2, the magnetic field B is acting downwards.

$\therefore$ Resultant field at $P_2 = -2 \times 10^{-5}\text{ T} + 2 \times 10^{-5}\text{ T} = 0$

Ex. 5. *A current of 20 A is flowing east to west in a long wire kept in the east-west direction. Find magnetic field in a horizontal plane at a distance of* (i) 10 *cm north,* (ii) 20 *cm south from the wire; and in a vertical plane at a distance of* (iii) 40 *cm below;* (iv) 50 *cm up.*

Sol. The magnetic field at a distance of d from a long wire carrying a current of i is given by

$$B = \frac{\mu_0 i}{2\pi d}$$

(i) The magnetic field in a horizontal plane at a distance of 10 cm or 0.1 m north from the wire is

$$B_1 = \frac{2 \times 10^{-7} \times 20}{0.10} = 4 \times 10^{-5} \text{ T}.$$

The direction of the field at the point will be **downward in a vertical plane.**

(ii) The magnetic field at a distance of 20 cm or 0.2 m south of the wire is

$$B_2 = \frac{2 \times 10^{-7} \times 20}{0.20} = 2 \times 10^{-5}\ \text{T}.$$

The direction of the field is **upward in a vertical plane.**

(iii) The magnetic field at 40 cm or 0.4 m from the wire downward in the vertical plane is

$$B_3 = \frac{2 \times 10^{-7} \times 10}{0.40} = 5 \times 10^{-6}\ \text{T}.$$

The direction of the field is **in a horizontal plane pointing south.**

(iv) The field at 50 cm or 0.5 m above the wire in the vertical plane is

$$B_4 = \frac{2 \times 10^{-7} \times 10}{0.50} = 4 \times 10^{-6}\ \text{T}.$$

The field is **in a horizontal plane pointing north.**

Ex. 6. *Two parallel infinitely long wires carry same current i. The separation between the wires is d. Calculate the magnetic field at a point exactly at the middle between them when the direction of current in them is* (a) *same*; (b) *opposite.*

Sol. Consider two wires W_1 and W_2, each carrying a current and lying in the plane of paper. We want to calculate the field at a point P just in the middle of the two wires *i.e.*, at a distance $d/2$ from each wire (d = separation between the two wires). We represent the fields at P due to W_1 and W_2 wires by B_1 and B_2 respectively.

The values of the fields at P are

$$|\vec{B}_1| = \frac{\mu_0 i}{2\pi (d/2)} = \left(\frac{\mu_0}{2\pi}\right)\frac{2i}{d} = |\vec{B}_2|$$

(a) When the currents i are flowing in the same direction in the two wires [fig. 9.61(a)], B_1 is down and B_2 is up and they cancel each other.

Hence the magnetic field at P is **zero.**

(b) When the currents in W_1 and W_2 are flowing in the opposite directions [fig. 9.61(b)], both B_1 and B_2 are down and hence the field at P is

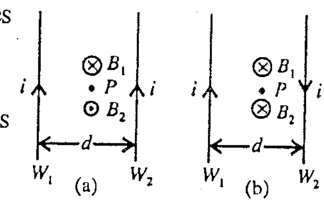

Fig. 9.61.

$$B = |\vec{B}_1| + |\vec{B}_2|$$

$$= \left(\frac{\mu_0}{2\pi}\right)\frac{2i}{d} + \frac{\mu_0}{2\pi}\frac{2i}{d} = \left(\frac{\mu_0}{2\pi}\right)\frac{4i}{d}\ \textbf{downward}$$

Ex. 7. *Two parallel wires PQ and RS are* 20 *cm apart. The currents flowing in the wires are as shown in* **fig. 9.62**. *Calculate the force acting over a length of* 3 *m of the wires. What is the nature of this force ?*

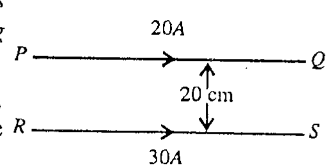

Fig. 9.62.

Sol. When currents flow in two long, parallel wires in the same direction, the wires exert a force of attraction on each other. The magnitude of this force acting per meter length of the wires is given by

$$F = \frac{\mu_0}{2\pi}\frac{i_1 i_2}{d}$$

Here $i_1 = 20$ A, $i_2 = 30$ A and $d = 20$ cm $= 0.2$ m

$$\therefore \quad F = (2 \times 10^{-7})\frac{20 \times 30}{0.2} = 6 \times 10^{-4}\ \text{N m}^{-1}.$$

$\therefore$ Force on 3 meter length of the wires = 3 m × (6×10^{-4} N m^{-1}) = $\mathbf{1.8 \times 10^{-3}}$ **N.**

The force is **attractive type.**

Ex. 8. *A current of* 20 *A flows through each of the two parallel long wires which are* 4 *cm apart. Compute the force exerted per unit length of each wire* ($\mu_0 = 4\pi \times 10^{-7}$ *weber/ampere-meter*).**(Meerut 2001)**

Sol. The force per unit length between two parallel long wires carrying currents i_1 and i_2 is given by

$$F = \frac{\mu_0}{2\pi}\frac{i_1 i_2}{d}$$

where d is the distance between the conductors.

If $i_1 = i_2 = i$, then the force per unit length of each wire is

$$F = \frac{\mu_0 i^2}{2\pi d}.$$

Here $i = 20$ amp and $d = 4 \times 10^{-2}$ m.

$$\therefore \qquad F = \frac{4\pi \times 10^{-7}}{2\pi}\left(\frac{20\times 20}{4\times 10^{-2}}\right) = \mathbf{2 \times 10^{-3}\ N.}$$

There is the force of attraction, if the currents in the wires are in the same direction, but of repulsion if the currents are in opposite directions.

Ex. 9. *A long straight wire carries a current of* 10 *amperes. An electron is travelling at the speed of* 10^7 *m/sec, is* 2.0 *cm from the wire. Calculate the force acting on the electron if its motion is directed* (a) *towards the wire,* (b) *parallel to the wire and,* (c) *at right angles to the direction given in* (a) *and* (b).

Sol. The magnetic field B due to a current carrying long wire is

$$B = \frac{\mu_0 i}{2\pi d} = \frac{4\pi\times 10^{-7}\times 10}{2\pi\times 0.02} = 10^{-4}\ \mathrm{T}.$$

Force on the moving electron, $\vec{F} = q(\vec{v}\times\vec{B})$

(a) When electron travels towards the wire, electron velocity v is perpendicular to the magnetic field B due to wire. Then,

$$F = qvB = 1.6 \times 10^{-19} \times 10^7 \times 10^{-4} = \mathbf{1.6 \times 10^{-16}\ N.}$$

The force F is parallel to the current.

(b) When electron travels parallel to the wire, its velocity v will again be perpendicular to the magnetic field B and hence the magnitude will be $\mathbf{1.6 \times 10^{-16}}$ **N** radially away.

(c) When the electron is moving at right angles to the directions given by (a) and (b), then v and B will be parallel or anti-parallel. In such a case,

$$\vec{F} = q(\vec{v}\times\vec{B}) = 0.$$

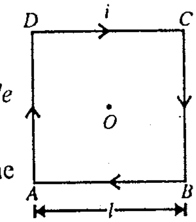

Fig. 9.63.

Ex. 10. *Find out the magnetic field at the centre O of a square frame ABCD of side l meter and carrying current i ampere, as shown in* fig. 9.63.

Sol. Due to the current i in each arm of the frame $ABCD$, the magnitude of the magnetic field $\vec{B}$ at the centre O of the frame [fig. 9.64] is

$$B = \frac{\mu_0}{4\pi}\frac{i}{l/2}\ (\cos 45^\circ - \cos 135^\circ),\ \text{where } \frac{\mu_0}{4\pi} = 10^{-7}\ \mathrm{N A^{-2}}$$

$$= \frac{\mu_0}{4\pi}\frac{2\sqrt{2}i}{l} = 2\sqrt{2}\times 10^{-7}\frac{i}{l}\ \mathrm{N A^{-1} m^{-1}}.$$

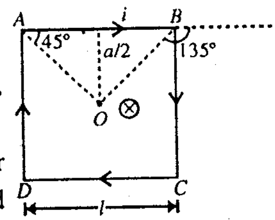

Fig. 9.64.

The magnetic field $\vec{B}$ will be perpendicular to the plane of the paper, directed downwards.

The magnetic fields at the centre O of the frame due to currents in the four arms are the same in magnitude and direction. Thus, the resultant magnetic field at the centre O due to all the four current-carrying arms is

$$4\times\left(2\sqrt{2}\times10^{-7}\frac{i}{l}\right) = 8\sqrt{2}\times10^{-7}\frac{i}{l}\text{N A}^{-1}\text{m}^{-1}.$$

which is perpendicular to the plane of the paper and directed into it.

Ex. 11. *A wire is bent in the shape of regular hexagon of each side a and a current i flows in it. Calculate the intensity of magnetic field at the centre of hexagon.*

Sol. Consider a current i flowing in the hexagon of side a [fig. 9.65]. Field at the centre O of the hexagon due to AB part is ($d = a \sin \pi/3 = a\sqrt{3}/2$)

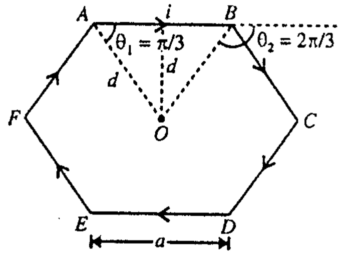

Fig. 9.65.

$$B_1 = \frac{\mu_0 i}{4\pi d}(\cos\theta_1 - \cos\theta_2)$$

$$= \frac{\mu_0 i}{4\pi d}(\cos \pi/3 - \cos 2\pi/3)$$

$$= \frac{\mu_0}{4\pi}\frac{2i}{a\sqrt{3}}\left(\frac{1}{2}+\frac{1}{2}\right) = \frac{\mu_0}{4\pi}\cdot\frac{2i}{a\sqrt{3}}$$

The direction of the field at O is down (into the page). Obviously the field due to each of the arm is the same and in the same direction. Hence the magnetic field due to the current in hexagon is

$$B = 6B_1 = 6\cdot\frac{\mu_0}{4\pi}\cdot\frac{2i}{a\sqrt{3}} = \frac{\sqrt{3}\mu_0 i}{\pi a}.$$

Ex. 12. *A square loop ABCD carrying a current of* 6.0 *A is placed near a long wire carrying* 10 *A as shown in* fig 9.66. (a) *Show that the magnetic force acting on the part AB is equal and opposite to that on the part CD.* (b) *Find the magnetic force on the square loop.*

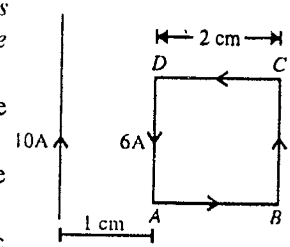

Fig. 9.66.

Sol. Consider $i_1 = 10$ A current flowing in wire W and $i_2 = 6$ A in the loop $ABCD$.

(a) Let us evaluate the magnetic force on AB wire carrying current i_2 due to current i_1.

Consider a point P on AB wire at a distance x [fig. 9.67]. The magnetic field at P due to i_1 current is

$$B = \frac{\mu_0 i_1}{2\pi x}$$

Hence force acting on current element $i_2\, dx$ is

$$dF = Bi_2\, dx = \frac{\mu_0\, i_1\, i_2\, dx}{2\pi\, x}$$

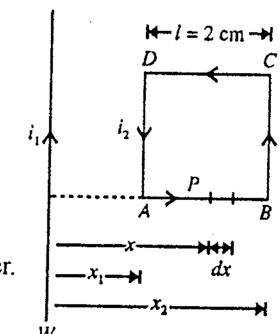

Fig. 9.67.

$\therefore$ Force on AB part, $F_{AB} = \int_A^B \frac{\mu_0\, i_1\, i_2}{2\pi\, x}dx = \frac{\mu_0 i_1\, i_2}{2\pi}\log\frac{x_2}{x_1}$

The direction of this force is up perpendicular to AB in the plane of the paper.

Similarly on CD part, $F_{CD} = \frac{\mu_0\, i_1\, i_2}{2\pi}\log\frac{x_2}{x_1}$

The direction of this force is down perpendicular to CD in the plane of the paper. Thus F_{AB} and F_{CD} are equal in magnitude but in opposite directions.

Hence the magnetic force on part AB is equal and opposite to that on the part CD.

(b) Forces on AB and CD parts of the loop cancel each other.

Force due to i_1 on DA part,

$$F_1 = B_1\, i_2\, l = \frac{\mu_0\, i_1}{2\pi\, x_1}i_2 l \quad \text{towards wire } W$$

Force due to i_1 on BC part.

$$F_2 = B_2 i_2 l = \frac{\mu_0 i}{2\pi x_2} i_2 l \text{ away from wire } W$$

Hence, the resultant force on the loop

$$= F_1 - F_2 = \frac{\mu_0 i_1 i_2 l}{2\pi x}\left[\frac{1}{x_1} - \frac{1}{x_2}\right]$$

$$= \frac{4\pi \times 10^{-7} \times 10 \times 6 \times 0.02}{2\pi}\left[\frac{1}{1 \times 10^{-2}} - \frac{1}{3 \times 10^{-2}}\right]$$

$$= \frac{2 \times 10^{-7} \times 10 \times 6 \times .02}{10^{-2}} \times \frac{2}{3}$$

$= \mathbf{1.6 \times 10^{-5}}$ **N towards wire** W.

Ex. 13. *The diameter of a circular coil is 0.2 m. It has 1000 turns of wire and carries a current of 0.1 A. Calculate the intensity of magnetic field at the centre of coil.* **(Rewa 2004; Jabalpur 2003)**

Sol. Here, $r = 0.2/2$ m $= 0.1$ m, $n = 1000$, $i = 0.1$ A.

The intensity of magnetic field at the centre of coil is given by

$$B = \frac{\mu_0 n i}{2R} = \frac{4\pi \times 10^{-7} \times 1000 \times 0.1}{2 \times 0.1} = \mathbf{6.28 \times 10^{-4}\ T.}$$

Ex. 14. *The magnetic field B due to a current carrying circular coil of radius 5 cm at its centre is 1 gauss. Find the magnetic field due to this coil at a point on the axis at a distance of 12 cm from the centre.*

Sol. Magnetic field at the centre of coil,

$$B_1 = \frac{\mu_0 i}{2R} \quad \text{...(i)}$$

and at an axial point,

$$B_2 = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}} \quad \text{...(ii)}$$

$$\therefore \quad \frac{B_2}{B_1} = \frac{R^3}{(R^2 + x^2)^{3/2}} \text{ or } B_2 = \frac{R^3}{(R^2 + x^2)^{3/2}} \times B_1$$

Here, $R = 5 \times 10^{-2}$ m, $x = 12 \times 10^{-2}$ m and $B_1 = 10^{-4}$ T.

$\therefore$ Field at the axial point is

$$B_2 = \frac{(5 \times 10^{-2}) \times 10^{-4}}{[(5 \times 10^{-2})^2 + (12 \times 10^{-2})^2]^{3/2}} = \frac{125 \times 10^{-4}}{[25 + 144]^{3/2}}$$

$$= \frac{125 \times 10^{-4}}{[13]^3} = \mathbf{5.7 \times 10^{-6}\ T.}$$

The direction of the field is along the axis. If the current in the coil is anticlockwise (as seen from the point) the magnetic field will be directed away from the coil and for clockwise current, it is towards the coil.

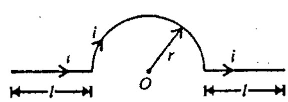

Fig. 9.68.

Ex. 15. *A long wire having a semi-circular loop of radius r carries a current i, as shown in* **fig. 9.68**. *Find the magnetic field at the centre O of the loop due to the entire wire.*

Sol. The magnetic field at a point due to a current-element $i\, d\vec{l}$ is given by

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{i\, d\vec{l} \times \vec{r}}{r^3} \quad \text{...(i)}$$

where $\vec{r}$ is the radius vector from the element to the point.

From the fig. 9.68, we see that the angle between a current-element of the straight parts of the wire and the radius vector from the element to the point O is zero (or 180°) and hence $i\,d\vec{l} \times \vec{r}$ is zero. Thus the magnetic field at O due to both the straight parts of the wire is zero.

The angle between a current-element $i\,d\vec{l}$ of the semi-circular portion of the wire and the radius vector from the element of the point O is 90°. Hence, the magnitude of the magnetic field at O due to this element is

$$dB = \frac{\mu_0}{4\pi}\frac{i\,(dl)(r)\sin\pi/2}{r^3} = \frac{\mu_0}{4\pi}\frac{i\,dl}{r^2} \qquad \text{...(ii)}$$

Therefore, the magnetic field due to the entire semi-circular loop is obtained to be

$$B = \frac{\mu_0}{4\pi}\frac{i}{r^2}\int dl = \frac{\mu_0}{4\pi}\frac{i}{r^2}(\pi r) = \frac{\mu_0 i}{4\pi} \qquad \text{...(iii)}$$

As the contribution of the straight parts of the wire are zero, hence the magnetic field due to the entire wire is

$$B = \frac{\mu_0 i}{4\pi} \qquad \text{...(iv)}$$

Ex. 16. (a) *In a hydrogen atom the electron circulates around the nucleus in a path of radius* 5.1×10^{-11} *m at a frequency of* 6.8×10^{15} *rev/sec. Calculate the intensity of magnetic field at the centre of the orbit. Also find the equivalent dipole moment.* **(Lucknow 2005; Agra 1993; Meerut 96)**

(b) *Calculate the current in a loop of radius* 4π *cm to produce the same magnitude of magnetic field at its centre as in part* (a). **(Meerut 1996)**

Sol. The circulating electron is equivalent of a circular current-loop. The current is the rate at which charge passes any point of the orbit. If n is the frequency or number of revolutions per second, then the equivalent current is

$$i = ne = 6.8 \times 10^{15} \times 1.6 \times 10^{-19} = 1.1 \times 10^{-3}\ \text{A}$$

(a) The magnetic field on the centre of a circular coil of radius R and carrying a current i is given by

$$B = \frac{\mu_0 i}{2R}.$$

Here, $i = 1.1 \times 10^{-3}$ A and $R = 5.1 \times 10^{-11}$ m.

$$\therefore \qquad B = \frac{4\pi \times 10^{-7} \times i}{2(4\pi \times 10^{-2})} = \mathbf{13.5\ T.}$$

Equivalent dipole moment $\quad m = iA = 1.1 \times 10^{-3} \times 3.14 \times (5.1 \times 10^{-11})^2 = \mathbf{9 \times 10^{-2}\ A\text{-}m^2.}$

(b) The magnetic field at the centre of a circular loop of radius 4π cm, carrying a current i, is

$$B = \frac{\mu_0 i}{2R} = \frac{4\pi \times 10^{-7} \times i}{2(4\pi \times 10^{-2}\ \text{m})} = 5 \times 10^{-8}\, i$$

But $B = 13.5$ T, $\qquad \therefore\ i = \dfrac{13.5}{5 \times 10^{-8}} = \mathbf{2.7 \times 10^8\ A.}$

Ex. 17. *A plastic disc of radius R has a charge Q uniformly distributed over its surface. If the disc is rotated at an angular frequency ω about its axis, find the magnetic field at the centre of the disc.*

Sol. The charge Q is distributed over the surface of the disc of radius R. Hence the charge per unit area $\sigma = Q/\pi R^2$.

We divide the disc into a large number of concentric rings with radii increasing from 0 to R, and consider one such ring of radius r and width dr [fig. 9.69]. The surface area of this ring is $2\pi r\,dr$ and the charge on it is $2\pi r\,dr\,\sigma$.

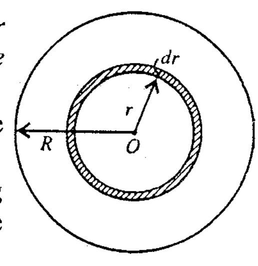

Fig. 9.69.

As the disc rotates about its centre O with angular velocity ω, this circulating ring is equivalent to a current-loop. The charge on this ring $2\pi r\, dr\, \sigma$, passes any point of the loop with frequency $n = \omega/2\pi$. Therefore, the charge flowing per second through the point *i.e.*, the current is

$$i = 2\pi r\, dr\, \sigma \cdot \frac{\omega}{2\pi} = r\, dr\, \sigma\, \omega. \qquad \text{...(i)}$$

The magnetic field at the centre of circular coil of radius r and carrying a current i is given by

$$dB = \frac{\mu_0 i}{2r}.$$

Hence the magnetic field at the centre O of circulating ring is

$$dB = \frac{\mu_0\, dr\, \sigma\, \omega}{2}.$$

Thus, the field due to the entire disc is

$$B = \frac{\mu_0 \sigma \omega}{2} \int_0^R dr = \frac{\mu_0 \sigma \omega R}{2} = \frac{\mu_0 \omega R}{2} \times \frac{Q}{\pi R^2} = \frac{\mu_0 Q \omega}{2\pi R}.$$

Ex. 18. *A cell is connected across two points A and B of a closed circular wire* **[fig. 9.70]**. *Prove that the magnetic field at its centre O will be zero.* **(Meerut 1988)**

Sol. Suppose the lengths of the two parts ACB and ADB of the wire be x_1 and x_2, and the resistance per unit length of the wire be α. Then the resistance of these portions will be $R_1 = x_1 \alpha$ and $R_2 = x_2 \alpha$ respectively.

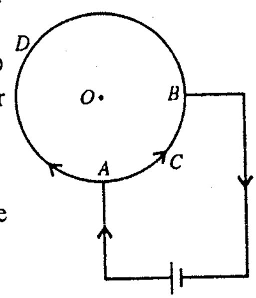

Fig. 9.70.

Let i_1 and i_2 be the currents in ACB and ADB portions respectively. These two portions ACB and ADB are in parallel. Hence the potential difference across their ends will be the same *i.e.*,

$$i_1 R_1 = i_2 R_2 \text{ or } i_1 x_1 \alpha = i_2 x_2 \alpha \text{ or } i_1 x_1 = i_2 x_2.$$

According to Biot-Savart's law, the magnetic fields at the centre O due to the circular loops ACB and ABD, are

$$B_1 = \frac{\mu_0 i_1 x_1}{4\pi r^2} \text{ and } B_2 = \frac{\mu_0 i_2 x_2}{4\pi r^2}.$$

As $i_1 x_1 = i_2 x_2$, hence, $B_1 = B_2$.

According to the right hand thumb rule, the directions of B_1 and B_2 are consequently in opposite directions at O. Consequently the resultant field at O is zero.

Ex. 19. *A current of* 200 *A is flowing through a long straight copper tube of internal and external radii* 1 *cm and* 2 *cm respectively. Calculate the magnetic field at a distance* 1.5 *cm from its axis.* **(Jabalpur 2004)**

Sol. Consider a long straight copper tube. The section of the tube perpendicular to its length has the internal radius R_1 = 1 cm = 10^{-2} m and external radius R_2 = 2 cm = 2×10^{-2} m. Consider a circular path of radius r around the axis of the tube [fig. 9.70]. Due to the symmetry, the magnetic field $\vec{B}$ due to the current in the tube at any point over this path has the same value and is tangential to it everywhere.

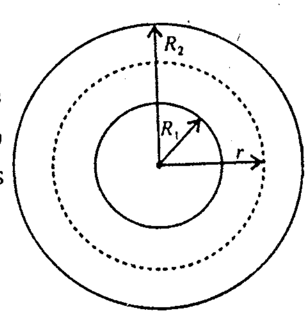

Fig. 9.71.

Current in the tube, i = 200 A

Current per unit area of the cross-section of the tube $= \dfrac{i}{\pi (R_2^2 - R_1^2)}$

$$= \frac{200}{\pi[(2\times10^{-2})^2-(1\times10^{-2})^2]}$$

$$= \frac{2\times10^6}{3\pi}\ \text{A/m}^2$$

$\therefore$ Current in the section between radii, $R_1 = 10^{-2}$ m to r,

$$i' = \pi(r^2 - R_1^2)\times\frac{2\times10^6}{3\pi}$$

$$= \pi[(1.5\times10^{-2})^2-(1\times10^{-2})^2]\times\frac{2\times10^6}{3\pi}$$

$$= \frac{2}{3}\times1.25\times10^2 = \mathbf{83.33\ A.}$$

Applying Ampere's law,

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 i' \text{ or } B\times2\pi r = \mu_0\times83.33$$

$$\therefore \qquad B = \frac{\mu_0\times83.33}{2\pi r} = \frac{4\pi\times10^{-7}\times83.33}{2\pi\times1.5\times10^{-2}} = \mathbf{1.1\times10^{-3}\ Wb/m^2.}$$

Ex. 20. *In 40 cm length of a long solenoid there are 500 turns. If it carries a current of 1.0 A, calculate the intensity of magnetic field inside the solenoid on its axis.* **(Ujjain 2003; Indore 03)**

Sol. The intensity of magnetic field inside a solenoid carrying a current i at a point on its axis is given by

$$B = \mu_0 n i$$

where $n = N/l$ is the number of turns per unit length. Here, $n = 500/0.4 = 1250$ turns per unit length. Therefore,

$$B = 4\pi\times10^{-7}\times1250\times1$$

$$= 3.14\times5\times10^{-4} = \mathbf{1.57\times10^{-3}\ Wb/m^2.}$$

Ex. 21. *A solenoid of length 20 cm and radius 2 cm is closely wounded with 200 turns. Calculate the magnetic field intensity at the centre of an end of the solenoid. The current in the winding is 5 ampere.* **(Meerut 2002)**

Sol. Magnetic field intensity at the centre of an end of a long solenoid (length much greater than its radius) is given by

$$B = \frac{1}{2}\mu_0 n i$$

Here $n = \frac{N}{l} = \frac{200}{0.2} = 1000$ turns per unit length and $i = 5$ A.

$$\therefore \qquad B = \frac{1}{2}\times4\pi\times10^{-7}\times1000\times5 = \mathbf{3.14\times10^{-3}\ Wb/m^2.}$$

Ex. 22. *A solenoid 1 m long and of mean diameter 20 cm has 1000 turns of wire. It carries 12 A current. Calculate the magnetic field at a point on the axis of solenoid when the point is (i) at centre, (ii) at end, (iii) at distance 20 cm outside the end of solenoid.*

Sol. Here, $i = 12$ A, $n = 1000$ turns per meter, $R = 0.1$ m and $l = 1$ m.

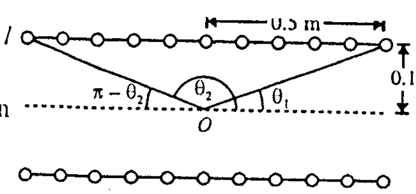

Fig. 9.72.

Magnetic field at any point on the axis of the solenoid is given by

$$B = \frac{\mu_0 n i}{2}(\cos\theta_1 - \cos\theta_2)$$

(i) At the centre of the solenoid [fig. 9.72] :

Now, $\cos\theta_1 = \dfrac{0.5}{\sqrt{(0.5)^2 + (0.1)^2}} = \dfrac{5}{\sqrt{26}}$

and $\cos(\pi - \theta_2) = \dfrac{5}{\sqrt{26}}$ or $\cos\theta_2 = -\dfrac{5}{\sqrt{26}}$

$$\therefore \quad B = \frac{4\pi \times 10^{-7} \times 1000 \times 12}{2}\left(\frac{5}{\sqrt{26}} + \frac{5}{\sqrt{26}}\right) = 0.0148 \text{ T}.$$

(ii) At one end of the solenoid [fig. 9.73] :

For the point P at one end, $\theta_1 = \pi/2$

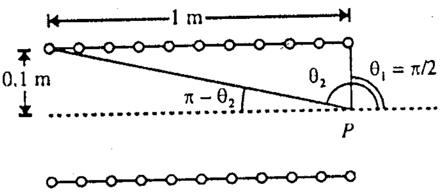

Fig. 9.73.

and $\cos(\pi - \theta_2) = \dfrac{0.1}{\sqrt{1^2 + (0.1)^2}} = \dfrac{1}{\sqrt{1.01}}$

or $\cos\theta_2 = -\dfrac{1}{\sqrt{1.01}}$

Hence $B = \dfrac{4\pi \times 10^{-7} \times 1000 \times 12}{2}\left(0 + \dfrac{1}{\sqrt{1.01}}\right)$

$= 7.5 \times 10^{-3}$ T.

(iii) At distance 20 cm or 0.2 m outside from one end [fig. 9.74] :

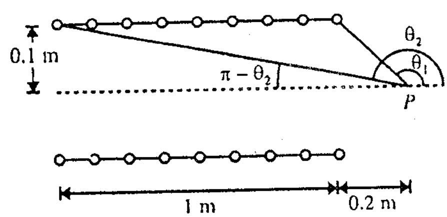

Fig. 9.74.

Here, $\cos(\pi - \theta_1) = \dfrac{0.2}{\sqrt{(0.2)^2 + (0.1)^2}} = 0.894$

$\therefore \quad \cos\theta_1 = -0.894.$

and $\cos(\pi - \theta_2) = \dfrac{0.12}{\sqrt{(0.12)^2 + (0.1)^2}} = 0.996$

$\therefore \quad \cos\theta_2 = -0.996.$

$$\therefore \quad B = \frac{4\pi \times 10^{-7} \times 1000 \times 12}{2}(-0.894 + 0.996)$$

$= 2 \times 3.14 \times 12 \times 10^{-4} \times 0.102 = 7.69 \times 10^{-4}$ T.

EXERCISE 9·2

1. A long straight wire carries a currents of 6 ampere. Calculate the intensity of magnetic field at a distance of 48 cm from the wire.
   [**Ans.** 2.6 μT.]
2. A copper wier 0.2 cm in diameter carries a current of 50 ampere. Find the magnetic field induction B at the surface of the wire.
   [**Ans.** 0.01 T.]
3. A long, vertical wire carrying a current of 20 A in the upward direction is placed in a region where a horizontal magnetic field of magnitude 2.0×10^{-3} T exists from south to north. Find the point where the resultant magnetic field is zero.
   [**Ans.** 0.2 mm.]
4. Two parallel wires are separated by distance d carrying same current i in opposite directions. Find the magnetic induction for a point between the wires at a distance x from one wire.
   $\left[\textbf{Ans. } \dfrac{\mu_0 i}{2\pi}\left[\dfrac{1}{x} + \dfrac{1}{dx}\right].\right]$

5. Two long wires W_1 and W_2 carrying equal currents of 20.0 A, are placed parallel to each other with a separation of 8.00 cm between them as shown in fig. 9.75. Find the magnetic field B at the points P_1, P_2 and P_3.

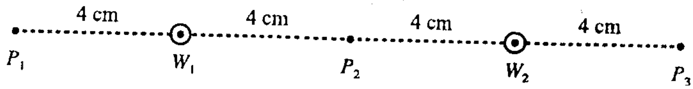

Fig. 9.75.

[**Ans.** 1.33×10^{-4} T perpendicular to the line P_1 P_3 downwards in the plane of the paper; zero; 1.33×10^{-4} T perpendicular to the line P_1 P_3 upwards in the plane of the paper.]

6. Two straight wires A and B of lengths 10 m and 16 m and carrying currents of 4.0 and 5.0 A respectively in opposite directions, lie parallel to each other 4.0 cm apart. Compute the force on a 10 cm long section of the wire B near its centre.

[**Ans.** 10^{-5} N away from the wire A.]

7. **Fig. 9.76**, shows two current-carrying parallel wires W_1 and W_2. The currents in W_1 and W_2 are 10 A and 15 A in up and down directions respectively. Find the magnitudes and directions of the magnetic fields at the points P, Q and R.

[**Ans.** 10^{-5} T; 5 mT; 2.35×10^{-5} T.]

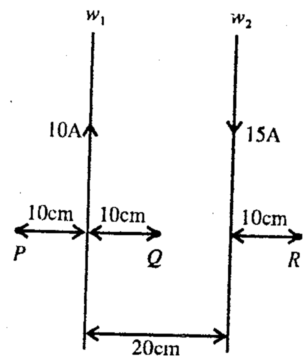

Fig. 9.76.

8. **Fig. 9.77** shows two parallel wires separated by a distance of 4.0 cm and carrying equal currents of 10 A along opposite directions. Find the magnitude of the magnetic field B at the points P_1, P_2 and P_3.

[**Ans.** 0.27 mT; 0.2 mT; 0.1 mT.]

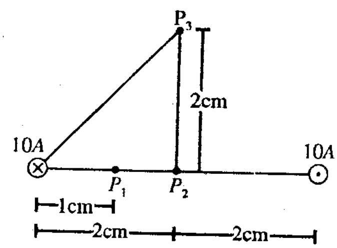

Fig. 9.77.

9. Two long wires at a distance d apart carry equal antiparallel currents i as shown in **fig. 9.78**. Show that the magnetic field at a point P which is equidistant from the wires is given by

$$B = \frac{2\mu_0 id}{\pi(4R^2 + d^2)}.$$ **(Lucknow 2005)**

10. A square loop of wire of side a carries a current i. Show that the value of magnetic field B at the centre is given by

$$B = \frac{2\sqrt{2}\,\mu_0 i}{\pi a}.$$

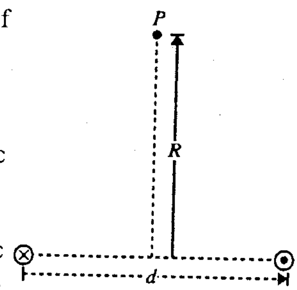

Fig. 9.78.

11. A current 1.00 A flows in a square loop of side 1 m . Find the magnetic field B at the centre of the square loop. **(Meerut 2005)**

[**Ans.** 1.1 μT.]

12. **Fig. 9.79** shows a square loop of side a of metallic wire. Find the magnetic field at the centre of the square if a current i flows as shown in **fig.9.79**.

[**Ans.** Zero.]

13. Calculate the distance between two parallel wires if they carry currents of 100 A and 20 A respectively and repel each other with a force of 0.08 N/m.

[**Ans.** 5 mm.]

14. A long straight wire carries a current of 50 A. An electron moving at 10^7 m/s, is 5.0 cm from the wire. Find the force acting on the electron if its velocity is directed (a) toward the wire, (b) parallel to the wire and (c) perpendicualr to the direction defined by (a) and (b). $\mu_0/4\pi = 10^{-7}$ N/A^2. **(Meerut 1997)**

[**Ans.** (a) 3.2×10^{-16} N parallel to the current; (b) 3.2×10^{-16} N radially away ($\vec{v}$ of electron parallel to current); (c) zero.]

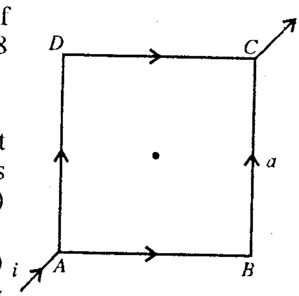

Fig. 9.79.

15. A current of 1.0 A is flowing in the sides of an equilateral triangle of side 4.5×10^{-2} m. Find the magnetic field at the centroid of the triangle. (Permeability constant $\mu_0 = 4\pi \times 10^{-7}$ V-s/A-m). **(Roorkee 1991)**
[**Ans.** 40 μT.]

16. Find the magnetic field at the point P by the curved wire as shown in **fig. 9.80**. The curved portion is a semicircle and the straight wires are long enough.
[**Ans.** $\frac{\mu_0 i}{2d}\left(1+\frac{2}{\pi}\right)$ down into the plane of the page.]

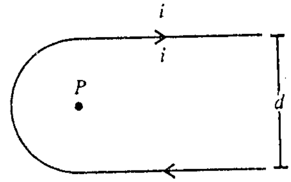

Fig. 9.80.

17. Find the resultant force on the current-carrying rectangular loop $ABCD$ [fig. 9.81] due to the current-carrying long wire. Take $i_1 = 30$ A, $i_2 = 20$ A, $l = 30$ cm; $a = 1.0$ cm and $b = 8.0$ cm.
[**Ans.** 3.2×10^{-3} N towards wire.]

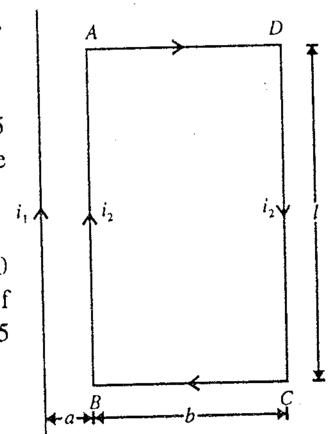

Fig. 9.81.

18. A wire shaped as a regular hexagon of side 2 cm carries a current of 5 amperes. Calculate the strength of magnetic field at the centre of the circle just enclosing the hexagon.
[**Ans.** 1.73×10^{-4} T.]

19. A circular coil of copper wire has 100 turns and its mean diameter is 30 cm. If 10 A current flows through the coil, then calculate the intensity of magnetic field and its direction at (i) the centre of coil, (ii) at distance 25 cm from the centre on the axis of coil.
[**Ans.** (i) 4.2×10^{-3} T; (ii) 5.7×10^{-4} T along the axis.]

20. Two circular turns, each of radius a carrying equal currents i in the same direction are placed a distance x apart with their planes parallel to each other. Calculate the value of magnetic field at the centre of either turn.
$$\left[\textbf{Ans. } \frac{\mu_0 i a^2}{2}\left\{\frac{1}{a^3}+\frac{1}{(a^2+x^2)^{3/2}}\right\}\right]$$

21. In **fig. 9.82**, a long wire is bent in the form of a circle. Calculate the magnetic field at the centre O of the circular part.
[**Ans.** $\frac{\mu_0 i}{2R}\left(1+\frac{1}{\pi}\right)$, perpendicualr to the plane of paper upwards.]

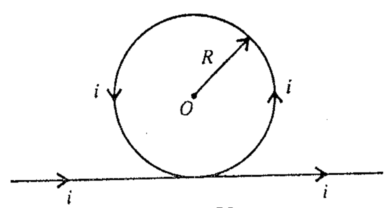

Fig. 9.82.

22. A current-carrying circular coil of 100 turns and radius 5.0 cm produces a magnetic field of 3.0×10^{-5} T at its centre. Find the value of the current.
[**Ans.** 24 mA.]

23. Find the magnetic field B due to a semicircular wire of radius 10.0 cm carrying a current of 10 A at its centre of curvature.
[**Ans.** 3.2×10^{-5} T.]

24. A piece of wire carrying a current of 3.00 A is bent in the form of a circular arc of radius 10.0 cm, and it subtends an angle of 120° at the centre. Find the magnetic field B due to this piece of wire at the centre.
[**Ans.** 6.3×10^{-4} T.]

25. A conducting circular loop of radius R is connected to two long, straight wires. The straight wires carry a current i [**fig. 9.83**]. Find the magnetic field B at the centre O of the loop.
[**Ans.** Zero.]

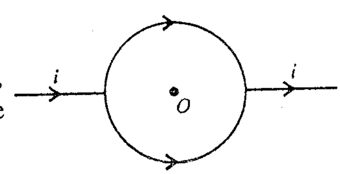

Fig. 9.83.

26. A current i_1 flows round a circular loop of wire of radius a and another current i_2 flows in a long straight wire placed at a distance x from the centre of the circular loop. If the circular loop and the straight wire lie in the same plane, calculate the force between them.

$$\left[\text{Ans. } \mu_0 i_1 i_2 \left\{\frac{b}{\sqrt{x^2-a^2}}-1\right\}\text{N.}\right]$$

27. The magnetic field B inside a long solenoid, carrying a current of 5.00 A, is 3.14×10^{-4} T. Find the number of turns per unit length of the solenoid.
[**Ans.** 50 turns/m.]

28. A solenoid of length 50 cm, radius 8.5 cm and number of turns 1990, carries a current 8.0 A. Calculate the magnetic field at mid-point inside the solenoid on its axis. **(Ujjain 2004; Rewa 2003)**
[**Ans.** 0.04 N A^{-1} m^{-1}.]

29. A solenoid of length 30 cm is wound uniformly with 3,000 turns of wire. It carries a current of 10 A. What is the value of (i) $\vec{B}$ on the axis of the solenoid at the middle, (ii) $\vec{B}$ on the axis at an end ?
[**Ans.** (i) 0.125 Wb/m^2; (ii) 0.0628 Wb/m^2.] **(Meerut 1983 S)**

30. There are 1600 turns of mean radius 10 cm distributed uniformly over a 20 cm long solenoid. A current of 1 ampere is made to flow in it. Calculate the magnetic field on its axis (i) 10 cm inside from one end, (ii) at one end.
[**Ans.** (i) 7.1×10^{-3} Wb/m^2; (ii) 4.5×10^{-3} Wb/m^2.]

31. A solid wire of radius 10 cm carries a current of 10.0 A distributed uniformly over its cross-section. Find the magnetic field B at a point at a distance (a) 2 cm; (b) 10 cm and (c) 20 cm away from the axis. Draw a graph of B versus x for $0 < x < 20$ cm.
[**Ans.** (a) 4 μT; (b) 20 μT; (c) 10 μT.]

32. A solenoid is 1.0 m long and 3.0 cm in mean diameter. It has five layers of windings of 850 turns each and carries a current of 5.0 A. What is the value of the magnetic field induction $\vec{B}$ at its centre ? What is the magnetic flux ϕ for a cross-section of the solenoid at its centre ?
[**Ans.** $B = 0.027$ Wb/m^2; $\phi = 1.9 \times 10^{-5}$ Wb.]
[**Hint.** $B = \mu_0 ni$, $n = 5 \times 850$ turns/m; Flux $\phi = BA$.]

33. A long co-axial cable consists of a solid conductor of radius R on the axis of a hollow cylindrical conductor of inner and outer radii R_1 and R_2 respectively. Section of the cable perpendicular to its length is shown in fig. 9.84. The inner conductor carries a steady current i in one direction and the outer conductor a steady current (known as the return current) of the same magnitude in opposite direction. Find the magnitude of magnetic field B at a distance x from the axis of the cable when (a) $x < R$, (b) $R < x < R_1$, (c) $R_1 < x < R_2$, (d) $x > R_2$.

[**Ans.** (a) $\dfrac{\mu_0 i x}{2\pi R^2}$; (b) $\dfrac{\mu_0 i}{2\pi x}$; (c) $\dfrac{\mu_0 i}{2\pi x}\left(\dfrac{R_2^2-x^2}{R_2^2-R_1^2}\right)$; (d) 0.]

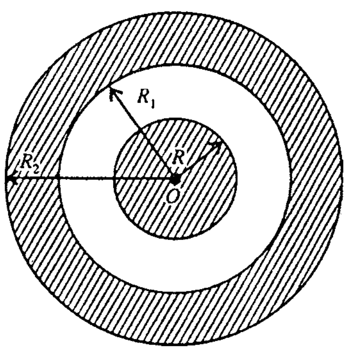

Fig. 9.84.

34. A long solenoid consists of 500 turns of wire uniformly wound on a wooden cylinder. The length of the windings is 100 cm and the mean diameter is 20 cm. When a current of 24 amperes flows in the windings, calculate the magnetic intensity on the axis (i) at the centre, (ii) at one end and (iii) 20 cm beyond one end of the solenoid. **(Agra 1970)**

[**Ans.** (i) $\approx 1.48 \times 10^{-2}$ weber m^{-2}; (ii) $\approx 7.5 \times 10^{-3}$ weber m^{-2}; (iii) 7.7×10^{-4} weber m^{-2}.]

9·21. Magnetic Dipole and its Potential Energy in a Magnetic Field

The torque on a loop (or coil) of n turns, carrying a current i,in a magnetic field $\vec{B}$ is given by

$$\vec{\tau} = \vec{m} \times \vec{B} \qquad ...(1a)$$

Its magnitude,
$$\tau = mB \sin\theta \qquad ...(1b)$$

The quantity $\vec{m}$ in eqs. (1) is called the **magnetic moment** of the current loop and the expression for it is given by

$$\vec{m} = ni\vec{A} \qquad ...(2)$$

where $\vec{m} = ni\,\vec{A}$ and $\vec{A}$ is the vector area of the loop. The direction of the area vector is given by right hand thumb rule for loop current. According to the right hand thumb rule, if one curls his fingers of the right hand along the current, the stretched thumb gives the direction of the area vector.

A current loop or any body that experiences a magnetic torque, given by eq. (1), is called a magnetic dipole.

The name comes from a similar expression for torque $\vec{\tau}$ acting on an electric dipole placed in an electric field $\vec{E}$ *i.e.*,

$$\vec{\tau} = \vec{p} \times \vec{E} \qquad ...(3)$$

where $\vec{p} = q\vec{d}$ is the electric dipole moment with $\vec{d}$ is the vector distance from $-q$ charge to $+q$ charge in the electric dipole.

A number of physical systems have magnetic dipole moments : bar magnets, current loops, the earth, atoms, nuclei, elementary particles etc.

We know that when a short bar magnet, having magnetic field moment $\vec{m}$, is placed in a magnetic $\vec{B}$, a torque $\vec{\tau}$ acts on the magnet, given by

$$\vec{\tau} = \vec{m} \times \vec{B} \qquad ...(4a)$$

Its magnitude,
$$\tau = mB\sin\theta \qquad ...(4b)$$

where θ is the angle between $\vec{m}$ and $\vec{B}$. The direction of $\vec{m}$ is from S-pole to N-pole **[fig. 9.85]**.

Thus the expressions for torque on a current loop and torque on a small bar magnet are identical (with $\vec{m} = ni\,\vec{A}$ for current loop). Further the magnetic field lines of a small bar magnet and current loop are similar. Hence we can consider a small bar magnet and a current carrying loop both as magnetic dipoles.

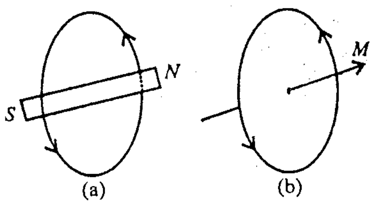

Fig. 9.85 : Magnetic dipole.

Potential energy of a magnetic dipole is an external magnetic field : When the orientation of a magnetic dipole (magnet, current loop etc.) is changed from θ to $\theta + d\theta$ in a magnetic field, the work done on it, given by $\tau\, d\theta$, gives the change in potential energy *i.e.*, $dU = \tau\, d\theta = mB\sin\theta\, d\theta$. Hence the work done in turning the dipole from an orientation θ_0 to θ,

$$W = \int_{\theta_0}^{\theta} mB\sin\theta\, d\theta = mB\left[-\cos\theta\right]_{\theta_0}^{\theta} = mB(\cos\theta_0 - \cos\theta) \qquad ...(5)$$

This work done on the dipole is stored as potential energy in the dipole in the field. If at $\theta_0 = 90°$, the potential energy is assumed to be zero, Then the potential energy in the orientation θ is given by

$$U = -mB\cos\theta = -\vec{m}\cdot\vec{B} \qquad ...(6)$$

If the magnetic dipole lies along the direction of the external magnetic field $\vec{B}$, then for $\theta = 0$ its potential energy is given by

$$U_0 = -\vec{m} \cdot \vec{B} = -mB\cos\theta = -mB \tag{7}$$

This is the value of the minimum potential energy of the dipole in the external magnetic field and represents the position of **stable equilibrium** of the magnetic dipole.

Work done in rotation of the magnetic dipole from equilibrium position (or direction of the external magnetic field) : If the magnetic dipole of magnetic moment $\vec{m}$ is rotated through angle θ from its equilibrium position ($\theta = 0$), the amount of work done on the dipole will be

$$W = \int_0^\theta mB\sin\theta\, d\theta = mB\left[-\cos\theta\right]_0^\theta = mB(1-\cos\theta) \qquad ...(8)$$

Now, if the dipole is rotated through 180° from the position of stable equilibrium, the amount of work done by the external agent will be

$$W = \int_0^\pi mB\sin\theta\, d\theta = mB\left[-\cos\theta\right]_0^\pi = 2mB \qquad ...(9)$$

This corresponds to the maximum potential energy of the dipole [$U_{max} = mB$ for $\theta = \pi$ in eq. (6)] in the external magnetic field and represents the **unstable equilibrium position** of the dipole.

9·22. Magnetic Field due to a Magnetic Dipole

A current carrying loop or coil is considered as a magnetic dipole. In order to find the magnetic field due to a magnetic dipole, we may use a plane current carrying loop. For convenience, we take a rectangular loop of length l and width b, carrying a current i. The dipole moment of the rectangular loop is given by

$$m = niA = nilb$$

where n are the turns in the loop and $A = lb$ is the area of the loop. The direction of the magnetic moment will be given by right hand thumb rule.

We plan to calculate first the manetic field at a point P due to this rectangular loop or magnetic dipole in the axial (or end on) position and in broad side on position. In the axial position the point is situated on the axis of the coil and in the broad side on position the point P lies perpendicular to the axis in the plane of the coil. Finally we calculate the magnetic field due to a magnetic dipole at an arbitrary point.

(1) Magnetic field in the end on position : Let us consider a rectangular loop $pqrs$. Its length $pq = rs = l$ and width $qr = sp = b$. It carries a current i in the anticlockwise direction seen from the side of P [fig. 9.86].

The point P is situated at a distance x from the centre O of the coil. Let the distance x be much more when compared to the lengths of the sides of the loop. Therefore we may consider the sides of the rectangular loop as straight length elements.

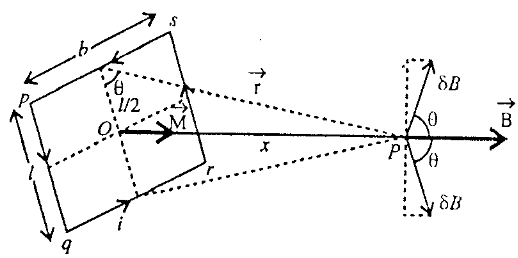

Fig. 9.86 : Magnetic field due to a magnetic dipole at an axial point.

Let the magnetic field at P due to the current in sp side be δB. It is given by

$$\delta B = \frac{\mu_0}{4\pi}\frac{ib\sin\pi/2}{r^2} \qquad \left[\because \vec{B} = \frac{\mu_0}{4\pi}\frac{i\,d\vec{l}\times\vec{r}}{r^3}\right]$$

$$= \frac{\mu_0}{4\pi}\frac{ib}{(x^2 + l^2/4)} \qquad \left[\because r^2 = x^2 + (l/2)^2\right]$$

The direction of this field is perpendicular to $\vec{r}$. This field δB has two components : (i) $\delta B\cos\theta$ along OP, and (ii) $\delta B\sin\theta$ perpendicular to OP. The perpendicular component will be cancelled by an equal and opposite component of the magnetic field δB due qr side of length b. However the axial components will be added up.

Hence the field at P due to sp and qr sides is

$$B_1 = 2\delta B \cos\theta = 2\frac{\mu_0}{4\pi}\cdot\frac{ib}{(x^2+l^2/4)}\cdot\frac{l/2}{(x^2+l^2/4)^{1/2}}$$

$$\left[\because \cos\theta = \frac{l/2}{r} = \frac{l/2}{(x^2+l^2/4)^{1/2}}\right]$$

$$= \frac{\mu_0}{4\pi}\cdot\frac{ibl}{(x^2+l^2/4)^{3/2}} \text{ along } \overrightarrow{OP}$$

For $x >> l$ or b,

$$B_1 = \frac{\mu_0}{4\pi}\cdot\frac{iA}{x^3} \text{ along } \overrightarrow{OP} \qquad ...(1)$$

where $A = bl$ is the area of the loop. Similarly, the field B_2 at P due to pq and rs sides is given by

$$B_2 = \frac{\mu_0}{4\pi}\cdot\frac{iA}{x^3} \text{ along } \overrightarrow{OP} \qquad ...(2)$$

Hence the total magnetic field at P due to the current loop is

$$B = B_1 + B_2 = \frac{\mu_0}{4\pi}\cdot\frac{iA}{x^3} + \frac{\mu_0}{4\pi}\cdot\frac{iA}{x^3} = \frac{\mu_0}{4\pi}\cdot\frac{2iA}{x^3} \qquad ...(3)$$

If there are n turns in the loop, then the magnetic field at the point P is given by

$$B = \frac{\mu_0}{4\pi}\cdot\frac{2niA}{x^3} \text{ along } \overrightarrow{OP} \qquad ...(4)$$

But $niA = m$ is the magnetic moment of the current loop. Hence the magnetic field at an axial point due to a magnetic dipole is given by

$$B = \frac{\mu_0}{4\pi}\cdot\frac{2m}{x^3} \text{ along } \overrightarrow{OP} \qquad ...(5a)$$

The magnetic moment $\vec{m} = ni\vec{A}$ is also along $\overrightarrow{OP}$ in view of the right hand thumb rule. Hence in vector form,

$$\vec{B} = \frac{\mu_0}{4\pi}\cdot\frac{2\vec{m}}{x^3} \qquad ...(5b)$$

(2) Magnetic field in the broad side on position : Let the point P be situated at a distance x from the centre O of the rectangular loop in its plane [fig. 9.87]. The direction of the magnetic moment $\vec{m} = ni\vec{A}$ will be up perpendicular to the plane of the loop and the point P is in the broad side on position with respect to the current loop or magnetic dipole.

We consider the distance x to be much more than the lengths of sides l and b *i.e.*, $x >> l$ or b. Hence we can take any current carrying side of the loop as a current element.

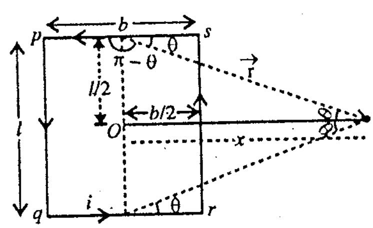

Fig. 9.87 : Magnetic field due to a magnetic dipole in the broad side on position.

The magnetic field at P due to the current in rs side is

$$B_1 = \frac{\mu_0}{4\pi}\cdot\frac{il\sin\pi/2}{(x-b/2)^2} = \frac{\mu_0}{4\pi}\cdot\frac{il}{(x-b/2)^2} \qquad ...(6)$$

whose direction is perpendicular to the plane of the page and into the page.

The magnetic field at P due to the current in pq side in obviously

$$B_2 = \frac{\mu_0}{4\pi}\cdot\frac{il}{(x+b/2)^2} \qquad ...(7)$$

whose direction is perpendicular to the plane of the page in the upward direction. Hence B_2 is in the opposite direction to B_1. We take the direction of B_1 to be positive, hence the magnetic field at P due to pq side bears a *negative sign.*

The field at P due to the current in sp side

$$B_3 = \frac{\mu_0}{4\pi} \cdot \frac{ib \sin(\pi - \theta)}{r^2} = \frac{\mu_0}{4\pi} \cdot \frac{ib \sin\theta}{r^2} \qquad ...(8)$$

But from fig. 9.87, $\sin\theta = \frac{l/2}{r} = \frac{l/2}{(x^2 + l^2/4)^{1/2}}$

$$\therefore \qquad B_3 = \frac{\mu_0}{4\pi} \cdot \frac{ibl/2}{(x^2 + l^2/4)^{3/2}}$$

which acts perpendicular to the plane of the page in the upward direction. Hence it is in the opposite direction to B_1 and bears *negative sign.*

Similarly the field at P due to current in qr side is

$$B_4 = \frac{\mu_0}{4\pi} \cdot \frac{ibl/2}{(x^2 + l^2/4)^{3/2}} \qquad ...(9)$$

in the upward direction and hence bears a *negative sign.*

Thus, the total magnetic field at P due to the current in the rectangular loop is

$$B = B_1 - B_2 - B_3 - B_4 \text{ in the downward direction}$$

$$= \frac{\mu_0 i}{4\pi}\left[\frac{l}{(x - b/2)^2} - \frac{l}{(x + b/2)^2} - \frac{bl/2}{(x^2 + l^2/4)^{3/2}} - \frac{bl/2}{(x^2 + l^2/4)^{3/2}}\right]$$

$$= \frac{\mu_0 il}{4\pi}\left[\frac{2xb}{(x^2 - b^2/4)^2} - \frac{b}{(x^2 + l^2/4)^{3/2}}\right]$$

For $x >> l$ or b,

$$B = \frac{\mu_0 il}{4\pi}\left[\frac{2b}{x^3} - \frac{b}{x^3}\right] = \frac{\mu_0}{4\pi} \cdot \frac{ilb}{x^3} = \frac{\mu_0}{4\pi}\frac{iA}{x^3} \qquad ...(10)$$

where $lb = A$ is the area of the rectangular loop. If there are n turns in the loop, then the magnetic field due to the current carrying loop or the magnetic dipole in the broad side on position is given by

$$B = \frac{\mu_0}{4\pi}\frac{niA}{x^3} = \frac{\mu_0}{4\pi} \cdot \frac{m}{x^3} \qquad ...(11a)$$

whose direction is into the page. As the direction of m is up, hence in vector form,

$$\vec{B} = -\frac{\mu_0}{4\pi} \cdot \frac{\vec{m}}{x^3} \qquad ...(11b)$$

Thus from eqs. (5a) and (11a), *we see that for the same distance (x)* **the magnitude of magnetic field due to a magnetic dipole in the end on position is two times the magnitude of the magnetic field in the broad side on position.**

(3) **Magnetic field at any arbitrary point :** Let us find the magnetic field due to a current loop or magnetic dipole at an arbitrary point $P(r, \theta)$, with respect to the origin O, the centre of the loop. Let the magnetic moment $\vec{m} = ni\vec{A}$ of the current loop be along X-axis. Now we resolve the magnetic moment $\vec{m}$ (i) $m\cos\theta$ along $\vec{OP}$ and (ii) $m\sin\theta$ perpendicular to $\vec{OP}$ [fig. 9.88]. Thus, the point P lies for $m\cos\theta$ in the end on position and for $m\sin\theta$ in broadside on position. The expressions for the magnetic field in the two positions are given by

$$B_r = \frac{\mu_0}{4\pi} \cdot \frac{2m\cos\theta}{r^3} \text{ and } B_\theta = \frac{\mu_0}{4\pi} \cdot \frac{2m\sin\theta}{r^3}$$

where B_r is the radial magnetic field at P along $OP(= r)$ and B_θ is the transverse magnetic field at P perpendicular to OP. Note that the direction of B_θ is opposite to that of $M \sin \theta$.

The resultant magnetic field at P due to the magnetic dipole is

$$B = \sqrt{B_r^2 + B_\theta^2}$$

$$= \frac{\mu_0 m}{4\pi r^3} \sqrt{4 \cos^2 \theta + \sin^2 \theta}$$

or $$B = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3} \sqrt{3 \cos^2 \theta + 1} \quad ...(12)$$

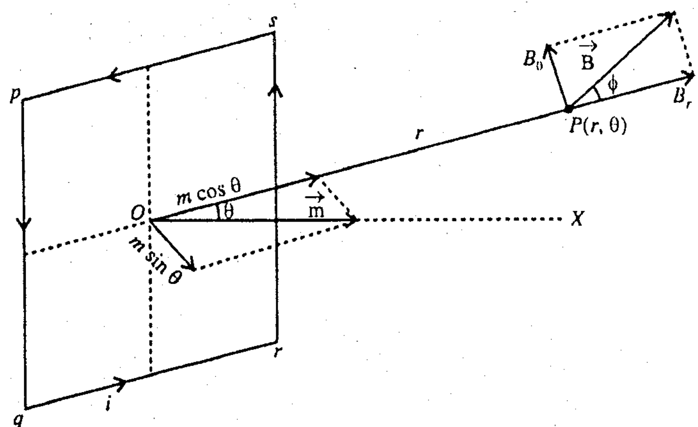

Fig. 9.88 : Magnetic field due to a magnetic dipole at a point $P(r, \theta)$.

If the resultant magnetic field makes an angle ϕ with the radial direction OP, then

$$\tan \phi = \frac{B_\theta}{B_r} = \frac{\sin \theta}{2 \cos \theta} = \frac{1}{2} \tan \theta \text{ or } \phi = \tan^{-1}\left(\frac{1}{2} \tan \theta\right) \quad ...(13)$$

Thus, we see that the expressions for the magnetic field due to a magnetic dipole are similar to those for the electric field due to an electric dipole, when the fields are considered at large distances.

9·23. Magnetic Field due to a Magnetic Dipole in the Form of Current Carrying Solenoid at an Axial Point

Let us consider a current carrying solenoid of N turns with radius R and length l. If it carries a current i, then it acts as a magnetic dipole. We want to calculate the magnetic field due to this dipole at a point P situated on its axis. We know that the magnetic field at an axial point P is given by

$$B = \frac{\mu_0 N i}{2l} (\cos \theta_1 - \cos \theta_2) \quad ...(1)$$

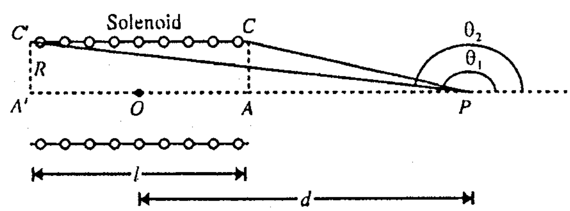

Fig. 9.89.

Let the point P lie on the axis of the solenoid at a distance d from its centre. From **fig. 9.89**, we have

$$\cos CPA = \cos (\pi - \theta_1) = \frac{d - l/2}{\left[R^2 + (d - l/2)^2\right]^{1/2}}$$

$$= \frac{1}{\left[1 + \left(\frac{R}{d - l/2}\right)^2\right]^{1/2}} = \left[1 + \left(\frac{R}{d - l/2}\right)^2\right]^{-1/2}$$

$$= 1 - \frac{R^2}{2(d - l/2)^2} \quad \text{[using Bionomial expansion for small } R \text{ and large } d\text{]}$$

Thus, $$\cos \theta_1 = -\left[1 - \frac{R^2}{2(d - l/2)^2}\right]$$

Similarly, $$\cos \theta_2 = -\left[1 - \frac{R^2}{2(d + l/2)^2}\right]$$

$\therefore$ $$\cos \theta_1 - \cos \theta_2 = \frac{R^2}{2(d - l/2)^2} - \frac{R^2}{2(d + l/2)^2} = \frac{ldR^2}{(d^2 - l^2/4)^2}$$

Therefore,
$$B = \frac{\mu_0 Ni}{2l} \times \frac{ldR^2}{(d^2 - l^2/4)^2}$$
$$= \frac{\mu_0 Ni(\pi R^2)d}{2\pi(d^2 - l^2/4)^2} = \frac{\mu_0}{4\pi} \cdot \frac{2md}{(d^2 - l^2/4)^{1/2}} \quad ...(2)$$

where $m = N\,i\,(\pi R^2)$ is the magnetic moment of the dipole. If the length of the solenoid is small when compared to distance d, then

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2m}{d^3}. \quad ...(3)$$

9·24. Magnetization Vector

In a magnetic material, the magnetic moments of the constituent atoms are randomly oriented and the net magnetic moment is usually zero. However, when the material is kept in an external magnetic field, torques act on the atomic dipoles and these torques try to align them parallel to the magnetic field. The alignment of the atomic dipoles is only partial as the thermal motion of the atoms tries to randomize these dipoles. The degree of alignment becomes more if the strength of the applied field is increased and also if the temperature is reduced. For sufficiently large magnetic fields, the alignment of the atomic dipoles is approximately perfect and we call the material to be magnetically saturated.

If the atomic dipoles in a material are aligned, partialty or fully, we have a net magnetic moment parallel to the applied magnetic field and we say that the material is magnetically polarised or magnetised. **The magnetic moment per unit volume is called the magnetization vector and is denoted by** $\vec{M}$. Thus,

$$\vec{M} = \frac{\text{Magnetic moment}}{\text{Volume}} \quad ...(1)$$

The magnetization vector plays the role analogous to the polarization $\vec{P}$ in electrostatics. It is also termed as **intensity of magnetization** or simply **magnetization.**

The unit of magnetic moment is ampere-metre^2 (A-m^2), hence from relation (1) unit of magnetization is *ampere/metre (A/m).*

The magnetization of a material is said to be uniform if all the atomic magnetic dipoles are uniformly distributed throughout the volume of the material and are oriented in the same direction as the magnetising field. Hence the magnetization $\vec{M}$ has the same value throughout a uniformly magnetized material.

For continuous magnetization, the **magnetization** $\vec{M}$ **at a point is defined as**

$$\vec{M} = \text{Lim}_{\Delta V \to 0} \frac{\Sigma \vec{m}}{\Delta V} \quad ...(2)$$

where $\Sigma \vec{m}$ is the net dipole moment of the atomic dipoles in a small volume ΔV and ΔV is allowed to shrink to zero around the point.

9·25. Magnetization Current

Consider a cylindrical block of uniformly magnetized material of length l with plane faces. The magnetization vector $\vec{M}$ is perpendicular to the upper plane along Z-axis [**fig. 9.90(a)**]. Now let us consider a plane sheet of thickness dz [**fig. 9.90(b)**] and divide into a large number of small peices, each of thickness dz [**fig. 9.90(c)**]. If dS is the cross-sectional area of one such piece, then the volume of the piece is $dS\,dz$ and its magnetic moment is

$$m = M\,dS\,dz \quad ...(1)$$

The magnetic field due to this small piece of magnetic material at distant points (greater than 10^{-9}m) is the same as due to any magnetic dipole of dipiole moment $\vec{m}$. *This dipole moment is to be produced by a loop current i around the boundary of the piece* **[fig. 9.90(c)]** *i.e.,*

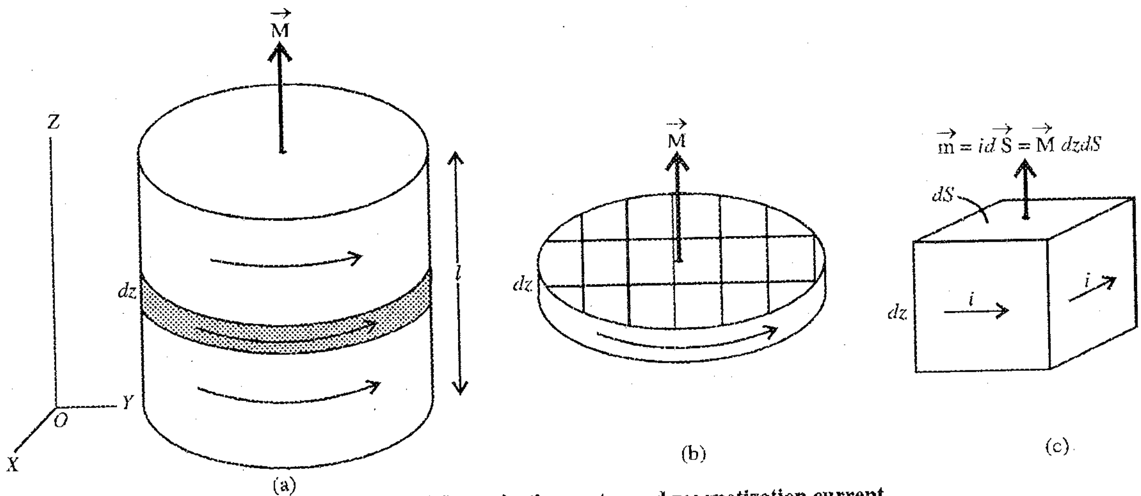

Fig. 9.90 : Magnetization vector and magnetization current.

$$m = i\, dS = M\, dS\, dz \qquad ...(2)$$

whence

$$i = M\, dz \qquad ...(3)$$

Now, we replace each piece of the sheet of **fig 9.90(b)** by its equivalent current loop, shown in **fig. 9.91(a)**. The current i is the same along the boundary of every piece and inside the sheet for two adjacent pieces, the boundary currents are opposite and cancel mutually. Hence we are left with a current $i = M\, dz$ along the external boundary of the magnetized sheet **[fig. 9.91(b)]**.

If we add a large number of such sheets one over the other, we get the cylindrical block of length l and hence the current i_s along the surface of the block is obtained by using eq. (3) *i.e.,*

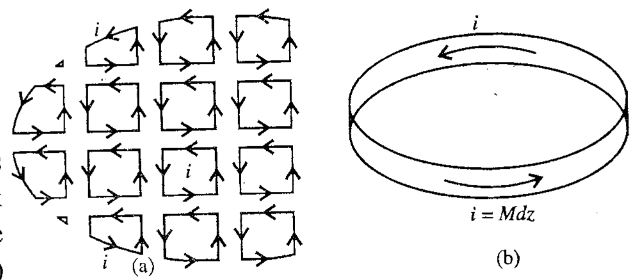

Fig. 9.91 : Magnetization current.

$$i_s = \Sigma M\, dz = M\, \Sigma\, dz = Ml \qquad ...(4)$$

This current circulates along the surface of the block and is called **magnetization current.**

From eq. (3) or (4), surface current per unit length of the magnetized block is given by

$$j_s = \frac{i}{dz} = \frac{i_s}{l} = M \qquad ...(5)$$

Thus *the surface current flowing per unit length at the surface of magnetized block is numerically equal to the magnetization (M).* This surface current per unit length is also called **surface magnetization current density** or **linear current density.** It can be represented vectorially.

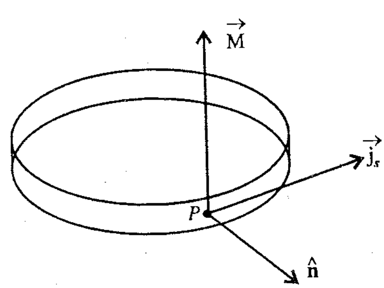

Fig. 9.92 : Surface magnetization current density $\vec{j}_s = \vec{M} \times \hat{n}$.

$$\vec{j}_s = \vec{M} \times \hat{n} \qquad ...(6)$$

where $\hat{n}$ is the out drawn normal to the surface **[fig. 9.92]**.

9·26. Interpretation of a Bar Magnet as a Surface Distribution of Solenoidal Current

Let us consider a cylindrical magnet of length l. For simplicity, this magnet may be considered as a cylindrical block with uniform magnetization M parallel to its axis. Such a uniform magnetization implies uniform circular current on its surface. The surface current per unit length (j_s) is equal to the magnetization M *i.e.*,

$$j_s = M \qquad ...(1)$$

Due to such a distribution of circular currents on the surface, the cylindrical magnet is equivalent to a closely wound current carrying solenoid and hence it produces a magnetic field similar to a solenoid **[fig. 9.93]**.

From the sense of the surface current, clockwise or anticlockwise, an end of the magnet behaves as south pole or north pole respectively.

The magnetic field due to a solenoid carrying a current i and having N turns and length l, is given by

$$B = \mu_0 \frac{N}{l} i = \mu_0 j^s \qquad ...(2)$$

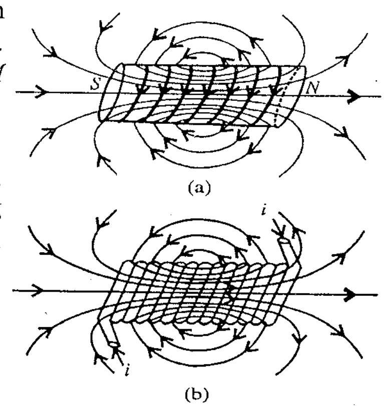

Fig. 9.93 : (a) Cylindrical bar magnet and magnetic field lines; (b) Magnetic field lines due to a current carrying solenoid.

where $j^s = Ni/l$ is the solenoidal current per unit length. Thus the magnetic field inside the cylindrical bar magnet will be $\mu_0 j_s$, where j_s is the surface magnetization current density.

9·27. Non-uniform Magnetization–Relation $\vec{J}_m = \text{curl}\,\vec{M}$

We consider a magnetized material with magnetization parallel to Z-axis. The magnetization is considered to be non-uniform and assumed to increase along positive Y-axis. Let us consider some small rectangular blocks of the material of dimensions δx, δy and δz along the three axes, which lie along Y-axis **[fig. 9.94(a)]**. A block is considered to be so small that the magnetization through each of them is taken to be constant. As magnetization M is increasing along Y-axis, each successive rectangular block has its magnetization larger than the preceding one by

$$\delta M_z = \left(\frac{\partial M_z}{\partial y}\right)\delta y \qquad ...(1)$$

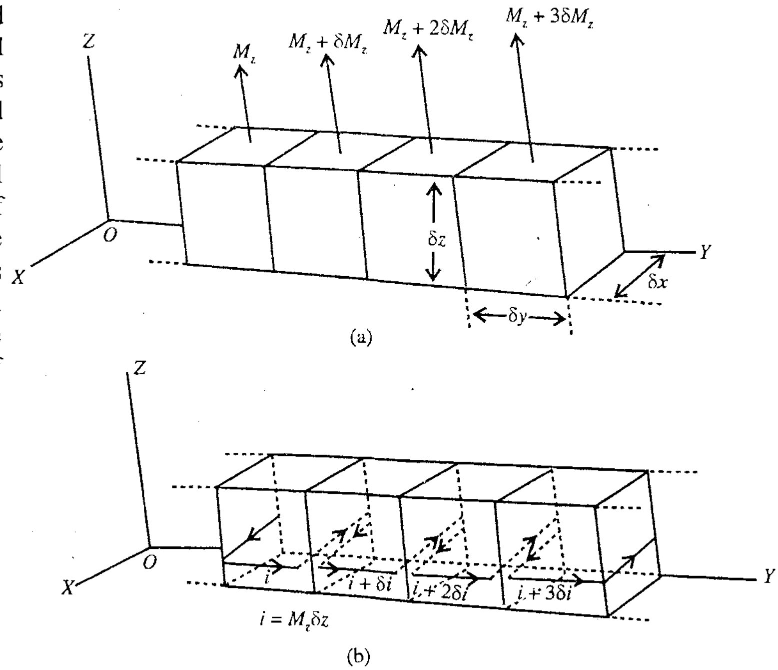

Fig. 9.94 : Non-unifrom magnetization.

If we replace the magnetization of each small block by its equivalent surface loop current **[fig. 9.94(b)]**, we find that the current in any loop is greater than the current to its left loop by

$$\delta i = (\delta M_z)\,\delta z \qquad [\because i = M_z \delta z]$$

or
$$\delta i = \left(\frac{\partial M_z}{\partial y}\right)\delta y\,\delta z \qquad ...(2)$$

Thus, in the magnetized material at each interface, after cancellation of internal currents, a net current of magnitude $\delta i = (\delta M_z/\delta y)\,\delta y\,\delta z$ along X-axis is left. This δi current along X-axis is flowing across the cross-section $(\delta y\,\delta z)$ of each block. Hence the current density along X-axis is [from eq. (2)].

$$J_x = \frac{\delta i}{\delta y\,\delta z} = \frac{\partial M_z}{\partial y} \qquad ...(3)$$

Let the magnetization also possess a Y-component, increasing along Z-axis. Therefore there is a gradient of magnetization along Z-axis, which gives to a contribution to the current density along the negative X-axis, given by

$$J_x = -\frac{\partial M_y}{\partial z} \qquad ...(4)$$

Hence the X-component of the current density $\vec{J}_m$ at any point of the magnetized material is

$$J_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z} = (\text{curl}\,\vec{M})_x = (\vec{\nabla}\times\vec{M})_x \qquad ...(5)$$

Similarly for the general direction of non-uniform direction $\vec{M}$, the Y- and Z- components of current density $\vec{J}_m$ at a point in the material are given by

$$J_y = \frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x} = (\text{curl}\,\vec{M})_y = (\vec{\nabla}\times\vec{M})_y \qquad ...(6)$$

and
$$J_z = \frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y} = (\text{curl}\,\vec{M})_z = (\vec{\nabla}\times\vec{M})_z \qquad ...(7)$$

Hence the current density at a point due to the non-uniform magnetization of the material is

$$\vec{J}_m = J_x\,\hat{i} + J_y\,\hat{j} + J_z\,\hat{k}$$

$$= \left(\frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial x}\right)\hat{i} + \left(\frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x}\right)\hat{j} + \left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y}\right)\hat{k}$$

i.e.,
$$\vec{J}_m = \text{curl}\,\vec{M} \qquad ...(8)$$

This equation is the distribution of current density at different points inside the material of atomic origin in a non-uniformly magnetized material. At the surface, the magnetization of the material results in surface magnetization current density $\vec{J}_S = \vec{M}\times\hat{\mathbf{n}}$.

9·28. Magnetizing Field $\vec{H}$

Free currents : *Ordinary conduction current which flows through the wires and other macroscopic paths are called* **free currents.** We can switch on and off such currents and measure with an ammeter.

Bound currents : The currents which are responsible for magnetic moments of atomic or molecular dipoles. Magnetigation current density $\vec{J}_m$ inside the magnetized material and surface magnetization current density $\vec{J}_S$ at its surface are due to bound currents. The magnetic field due to magnetization of the material is the field produced by these bound currents.

Thus the magnetic field $\vec{B}$ inside a magnetic material is not only due to free current density $\vec{J}_f$ but also due to the bound or equivalent magnetization current density $\vec{J}_m$. In other words, $\vec{B}$ is due to the sum $\vec{J}_f + \vec{J}_m$ of two current densities in which $\vec{J}_f$ may be due to the current in a wire and $\vec{J}_m$ due to the magnetization.

Thus, the differential form of Ampere's law can be written as

$$\text{curl}\,\vec{B} = \mu_0(\vec{J}_f + \vec{J}_m) \quad ...(1)$$

But

$$\vec{J}_m = \text{curl}\,\vec{M} \quad ...(2)$$

where $\vec{M}$ is magnetization vector.

Hence

$$\text{curl}\,\vec{B} = \mu_0\,\vec{J}_f + \mu_0\,\text{curl}\,\vec{M} \text{ or curl } \frac{\vec{B}}{\mu_0} = \vec{J}_f + \text{curl}\,\vec{M}$$

or

$$\text{curl}\left(\frac{\vec{B}}{\mu_0} - \vec{M}\right) = \vec{J}_f \quad ...(3)$$

The vector quantity in the bracket is replaced by a new vector $\vec{H}$ *i.e.*,

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} \quad ...(4)$$

This vector $\vec{H}$ is called **magnetizing field intensity** or **magnetic field strength.**

The unit of $\vec{H}$ is the same as that of $\vec{M}$ *i.e.*, ampere/metre (A/m).

In terms of $\vec{H}$, the Ampere's law becomes [from eqs. (3) and (4)]

$$\text{curl}\,\vec{H} = \vec{J}_f \text{ or } \vec{\nabla} \times \vec{H} = \vec{J}_f \quad ...(5)$$

Its intergral from is

$$\oint_C \vec{H} \cdot d\vec{l} = i_f \quad ...(6)^*$$

where i_f is the free current passing the area enclosed by the curve (Amperian) C along which the line integral of $\vec{H}$ is calculated.

The magnetizing field vector $\vec{H}$ possesses great practical significance. In applying eq. (6), we can control the free current i_f easily. However the bound current (J_m) due to magnetization in eq. (1) [or $\oint \vec{B} \cdot d\vec{l} = \mu_0 i\ (i = i_f + i_m)$] can not be turned on and off independently similar to free current (i_f) in a wire. Further, when symmetry permits, we can calculate H immediately from eq. (6) by using Ampere's law methods.

We see that magnetizing field vector $\vec{H}$ is related to free currents [eqs. (5) and (6)] similarly as $\vec{B}$ related to the total current (free + bound) [eqs. (1) and $\oint \vec{B} \cdot d\vec{l} = \mu_0 i = \mu_0(i_f + i_m)$]. **Difference between $\vec{B}$ and $\vec{H}$** is that always $\text{div}\,\vec{B} = 0$, but $\text{div}\,\vec{H}$ may not be zero, because from eq. (4) for $\text{div}\,\vec{B} = 0$, $\text{div}\,\vec{H} = -\,\text{div}\,\vec{M}$, which is clearly not zero.

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* $\vec{\nabla} \times \vec{H} = \vec{J}_f$ or $\int_s (\vec{\nabla} \times \vec{H}) \cdot d\vec{S} = \int_s \vec{J}_f \cdot d\vec{S} = i_f$

According to stokes theorem $\int_s (\vec{\nabla} \times \vec{H}) \cdot d\vec{S} = \oint_c \vec{H} \cdot d\vec{l}$

Therefore, $\oint_c \vec{H} \cdot d\vec{l} = i_f$

9·29. Magnetic Permeability and Susceptibility

For most of the materials (paramagnetic and diamagnetic), the magnetization $\vec{M}$ is found proportional to the magnetizing field $\vec{H}$ *i.e.*,

$$\vec{M} \propto \vec{H} \text{ or } \vec{M} = \chi_m \vec{H} \quad ...(1)$$

where the constant of proportionality χ_m is called **magnetic susceptibility** of the material. Magnetic susceptibility χ_m is a dimensionless quantity. It varies from one substance to another. χ_m is positive for paramagnetic substances and negative for diamagnetic substances.

A material that obeys the relation $\vec{M} = \chi_m \vec{H}$ *is called a* **linear medium.**

We know that the magnetic field $\vec{B}$ is related to the magnetization $\vec{M}$ and magnetizing field $\vec{H}$ as

$$\vec{B} = \mu_0(\vec{H} + \vec{M}) \quad ...(2)$$

Hence for a linear medium, using relation (1), eq. (2) becomes

$$\vec{B} = \mu_0(1 + \chi_m)\vec{H} \quad ...(3)$$

Thus we find that for a linear medium, the magnetic field $\vec{B}$ is found to be proportional to the magnetizing field $\vec{H}$:

$$\vec{B} \propto \vec{H} \text{ or } \vec{B} = \mu \vec{H} \quad ...(4)$$

where μ is called the **magnetic permeability** of the material. Thus *the magnetic permeability of a medium is the ratio (B/H) of the magnetic field and the magnetizing field i.e.,*

$$\mu = \frac{B}{H} \quad ...(5)$$

Comparing eqs. (3) and (4), we obtain

$$\mu = \mu_0(1 + \chi_m) \quad ...(6)$$

This is the *relation between the magnetic permeability* (μ) *and magnetic susceptibility* χ_m *of a linear medium.*

In vacuum, where there is no matter to magnetize, the magnetic susceptibility χ_m is zero and hence from eq. (6)

$$\mu = \mu_0 \quad ...(7)$$

This is why μ_0 is called the **permeability of free space.**

Thus for vacuum (or air) $\quad \mu_0 = \frac{B}{H}$ or $B = \mu_0 H \quad ...(8)$

The ratio of the permeability of a medium to that of free space is called the **relative permeability** (μ_r) of the medium *i.e.*,

$$\frac{\mu}{\mu_0} = \mu_r \text{ or } \mu = \mu_0 \mu_r \quad ...(9)$$

Hence eq. (6) assumes the form

$$\mu_r = 1 + \chi_m \quad ...(10)$$

or

$$\chi_m = \mu_r - 1 \quad ...(11)$$

Hence the relations (4) and (1) take the forms :

$$\vec{B} = \mu_0 \mu_r \vec{H} \text{ and } \vec{M} = (\mu_r - 1)\vec{H} \quad ...(12)$$

For vaçuum $\vec{M} = 0$, hence

$$\mu_r - 1 = 0 \text{ or } \mu_r = 1 \quad ...(13)$$

Volume bound current density or magnetization current density $\vec{J}_m$ for a homogeneous linear medium is given by

$$\vec{J}_m = \text{curl}\,\vec{M} = \vec{\nabla}\times\vec{M} = \vec{\nabla}\times(\chi_m\vec{H}) = \chi_m(\vec{\nabla}\times\vec{H}) = \chi_m\vec{J}_f \qquad ...(14)$$

because $\vec{M} = \chi_m\vec{H}$ and $\vec{\nabla}\times\vec{H} = \vec{J}_f$. Thus **for a linear medium, the bound current density is proportional to the free current density.**

Unit of magnetic permeability μ : As $\mu = B/H$, hence

$$\text{unit of } \mu = \frac{\text{unit of } B}{\text{unit of } H} = \frac{\text{weber/m}^2}{\text{amper/m}} = \frac{\text{weber}}{\text{ampere}\times\text{meter}} = \frac{\textbf{henry}}{\textbf{metre}} \qquad ...(15)$$

because weber/ampere = henry.

Unit of μ_0 is the same as that of μ *i.e.*, henry/metre.

Classification of Magnetic Materials

All materials can be classified into three groups on the basis of the value of magnetic susceptibility and permeability.

(1) **Paramagnetic materials** : For such materials the magnetic susceptibility (χ_m) is small but positive and the relative permeability (μ_r) is slightly more than 1. Examples of paramagnetic materials are Pt, Al, Cr, CaO, CuO etc.

(2) **Diamagnetic materials** : The magnetic susceptibility for diamagnetic materials is small and negative and the relative permeability is slightly less than 1. Examples of these materials are Bi, Cu, Ag, Ge, Si etc.

(3) **Ferromagnetic materials** : For such materials, the magnetic susceptibility is large and positive of the order of several thousands. The relative permeability of these materials is also large of the order of several thousands. Examples of ferromagnetic materials are Fe, Ni, Co etc.

Ex. 1. *The diameter of a* 40 *turns circular coil is* 8.0 *cm and it carries a current of* 3.0 *A.* (i) *Find the magnetic moment of the coil.* (ii) *If the coil is suspended in a uniform magnetic field of* 0.50 *Wb* m^{-2} *so that its plane is parallel to the direction of the field, then find the torque acting on it.* (iii) *If the coil is suspended freely in the magnetic field and then turned through* 180°, *what will be the work done ?*

Sol. (i) The magnitude of the magnetic moment (m) of a current-carrying coil is given by

$$m = NiA$$

Here $N = 40$, $i = 3.0$ A and $A = \pi r^2 = 3.14 \times (4.0 \times 10^{-2}\text{m})^2 = 5.0 \times 10^{-3}\ \text{m}^{-2}$.

$$\therefore \qquad m = 40 \times 3.0 \times (5.0 \times 10^{-3}) = \mathbf{0.60\ Am^2}.$$

(ii) When a current-carrying coil is suspended in a uniform magnetic field $\vec{B}$ so that the normal to the plane of the coil (axis of the coil) makes an angle θ with the direction of $\vec{B}$, then the magnitude of the torque acting on it is given by

$$\tau = mB\sin\theta.$$

When the plane of the coil is parallel to the magnetic frield $\vec{B}$, *i.e.*, its axis is perpendicular to the field $\vec{B}$ ($\theta = 90°$ or $\sin\theta = 1$), then the torque is maximum. Hence

$$\tau_{max} = mB = 0.60\ \text{A m}^2 \times 0.50\ \text{N A}^{-1}\text{m}^{-1} = \mathbf{0.3\ N\ m}.$$

(iii) The work done in rotating a current-coil suspended freely in a magnetic field $\vec{B}$ through an angle θ from the equilibrium position is given by

$$W = mB(1 - \cos\theta).$$

If the coil is turned through 180°, then the work done

$$W = mB(1 - \cos 180°) = 2mB = 2 \times 0.30\ \text{J} = \mathbf{0.60\ J}.$$

Ex. 2. *A circular coil of* 100 *turns and having an effective radius of* 0.05 *m carries a current of* 0.1 *A. How much work is required to turn it in an external magnetic field of* 1.5 *Wb m*$^{-2}$ *through* 180° *about an axis perpendicular to the magnetic field. The plane of the coil is initially perpendicular to the magnetic field.* **(Roorkee 1986)**

Sol. Magnetic moment of the coil, $m = NiA = 100 \times 0.1 \times \pi \times (0.05)^2 = 7.85 \times 10^{-2}$ A-m^2

The work done in rotating a current-carrying coil through 180° in a magnetic field of magnitude B is given by

$$W = mB(1 - \cos\theta) = mB((1 - \cos 180^\circ) = 2mB = 2 \times 7.85 \times 10^{-2} \times 1.5 = \mathbf{0.2355\ J.}$$

Ex. 3. *A current of* 10 *A is flowing in a coil of radius* 10 *cm having* 20 *turns. Calculate the dipole moment and the magnetic field due to the dipole at a distance of* 20 *cm in end on position.*

Sol. Dipole moment of the magnetic dipole

$$m = NiA = 20 \times 10 \times \pi \times (0.1)^2 = \mathbf{6.28\ A\text{-}m^2}$$

Magnetic field due to the magnetic dipole in the end on position is given by

$$B = \frac{\mu_0}{4\pi} \times \frac{2m}{x^3} = \frac{10^{-7} \times 2 \times 6.28}{(0.2)^3} = \mathbf{1.57 \times 10^{-4}\ T.}$$

Ex. 4. *The magnetic moment of a magnetic dipole is* 2.0 *Am*^{2}. *Calculate the intensity of magnetic field at a distance* 20 *cm in broad side on position.* **(Ujjain 2003)**

Sol. Intensity of magnetic field due to a magnetic dipole of dipole moment m in the broad side on position is given by

$$B = \frac{\mu_0}{4\pi} \times \frac{m}{x^3} = \frac{10^{-7} \times 2}{(0.2)^3} = \mathbf{2.5 \times 10^{-5}\ Wb/m^2.}$$

Ex. 5. *Compare the intensities of magnetic field due to a magnetic dipole in its end on position at distances* 10 *cm and* 20 *cm from its centre.* **(Bilaspur 2003)**

Sol. Intensity of magnetic field in end on position of magnetic dipole is

$$B = \frac{\mu_0}{4\pi} \times \frac{2m}{x^3}$$

When $x = 10$ cm $= 0.1$ m, $B_1 = \dfrac{\mu_0}{4\pi} \times \dfrac{2m}{(0.1)^3}$ ...(i)

When $x = 20$ cm $= 0.2$ m, $B_2 = \dfrac{\mu_0}{4\pi} \times \dfrac{2m}{(0.2)^3}$ ...(ii)

Dividing eq. (i) by eq. (ii)

$$\frac{B_1}{B_2} = \frac{(0.2)^3}{(0.1)^3} = \mathbf{8.}$$

Ex 6. *The horizontal component of flux density of earth's magnetic field is* 3.4×10^{-5} *Wb/m*^{2}. *Find the horizontal component of earth's magnetizing field.* **(Gwalior 2004)**

Sol. The relation between magnetic field B (flux density) and the magnetising field H is

$$B = \mu_0 H$$

$\therefore$ Magnetising field, $H = \dfrac{B}{\mu_0} = \dfrac{3.4 \times 10^{-5}}{4\pi \times 10^{-7}} = \mathbf{27.1\ A/m.}$

Ex. 7. *A solenoid of length* 0.5 *m has* 2000 *turns and carries a current of* 2 *A. Calculate the magnetising field H and magnetic flux density B at the centre of the solenoid.*

Sol. The magnetic field or flux density B at the centre of the solenoid is given by

$$B = \mu_0 ni$$

Here, n = number of turns per unit length $= \dfrac{2000}{0.5} = 4000$ turns/m

$$\therefore \quad B = 4\pi \times 10^{-7} \times 4000 \times 2 = \mathbf{1.005 \times 10^{-2}\ Wb/m^2.}$$

$\therefore$ Magnetizing field, $H = \dfrac{B}{\mu_0} = ni = 4000 \times 2 = \mathbf{8000\ A/m.}$

Ex. 8. *The magnetic moment of a piece of magnetized substance is* 0.9 *A* m^2. *The mass of piece is* 0.24 *kg and density of its substance is* 8×10^3 *kg/m*^{3}. *Find the intensity of magnetization.* **(Jabalpur 2004)**

Sol. Intensity of magnetization,

$$M = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{m}{V}$$

Here, $m = 0.9$ Am^2 and $$V = \frac{\text{Mass}}{\text{Density}} = \frac{0.24}{8 \times 10^3} = 3 \times 10^{-5} \text{ m}^3$$

$\therefore$ $$M = \frac{0.9}{3 \times 10^{-5}} = \mathbf{3 \times 10^4 \text{ A/m.}}$$

Ex. 9. *The magnetic susceptibility of a material is* 948×10^{-11}. *Calculate its permeability.* **(Delhi 1995)**

Sol. $$\text{Permeability, } \mu = \mu_0(1 + \chi_m) = 4\pi \times 10^{-7}(1 + 948 \times 10^{-11})$$
$$= \mathbf{12.56 \times 10^{-7} \text{ henry/m.}}$$

Ex. 10. *An iron rod of length* 50 *cm and area of cross-section* 2 *mm*^{2} *is kept inside a long solenoid along its axis. The number of turns in the solenoid are* 25 *per cm and it carries a current* 2 *A. The intensity of magnetization produced in the rod is* 8×10^{-5} *A/m. Calculate :* (i) *the magnetic moment produced in the rod,* (ii) *the magnetic susceptibility of iron,* (iii) *magnetic field B and magnetization field H within the rod and* (iv) *relative permeability of iron.* **(Jabalpur 1990)**

Sol. (i) Intensity of magnetization, $M = \dfrac{\text{Magnetic moment, } m}{\text{Volume, } V}$

$\therefore$ Magnetic moment of rod, $m = M \times V$

Here, $M = 8 \times 10^5$ A/m and $V = lA = 0.5 \times 2 \times (10^{-3})^2 = 10^{-6}$ m^3

$\therefore$ $$m = (8 \times 10^5) \times 10^{-6} = \mathbf{0.8 \text{ A m}^2}$$

(ii) Magnetizing field due to solenoid

$$H = ni = \frac{25}{10^{-2}} \times 2 = 5000 \text{ A/m}$$

$\therefore$ Magnetic susceptibility of iron, $\chi_m = \dfrac{M}{H} = \dfrac{8 \times 10^5}{5000} = \mathbf{160.}$

(iii) Magnetic field within the rod, $B = \mu_0(H + M) = 4\pi \times 10^{-7} \times (5000 + 8 \times 10^5) = \mathbf{1.01 \text{ Wb/m}}$

Magnetization field, $H = \mathbf{5000 \text{ A/m}}$

(iv) Relative permeability of iron, $\mu_r = \chi_m + 1 = 160 + 1 = \mathbf{161.}$

Ex. 11. *A rod of iron of length* 20 *cm and diameter* 1 *cm is kept inside a solenoid along its axis, having* 3 *turns per cm. Relative permeability of iron is* 1000 *and the solenoid carries current* 0.5 *A. Calculate the magnetic moment produced in the rod.* **(Indore 1993)**

Sol. Magnetizing field due to solenoid,

$$H = n\,i$$

Here, $n = 3/10^{-2} = 300$ turns/m and $i = 0.5$ A

$\therefore$ $$H = 300 \times 0.5 = \mathbf{150 \text{ A/m}}$$

Magnetic field inside the rod, $B = \mu H = \mu_0 \mu_r H = (4\pi \times 10^{-7}) \times 1000 \times 150 = \mathbf{0.1884 \text{ Wb/m}^2.}$

Intensity of magnetization of the rod,

$$M = \frac{B}{\mu_0} - H = \frac{0.1884}{4\pi \times 10^{-7}} - 150 = \mathbf{1.4985 \times 10^5 \text{ A/m}}$$

$\therefore$ Magnetic moment of the rod, $m = M \times$ Volume of the rod

Here, Volume of the rod $= \pi r^2 l = 3.14 \times (0.5 \times 10^{-2}) \times 0.2 = 1.56 \times 10^{-5}$ m^3

$\therefore$ $$m = (1.4985 \times 10^5) \times 1.56 \times 10^{-5} = \mathbf{2.34 \text{ A m}^2.}$$

Ex. 12. *A toroid has a mean radius R equal to 20/π cm, and a total of 400 turns of wire carrying a current of 1 A. An aluminium ring at temperature 280 K inside the toroid provides the core. If the magnetization M is 4.8×10^{-2} A/m, find the susceptibility of aluminium.*

Sol. The magnetic intensity H in the core is

$$H = \frac{Ni}{2\pi R} = \frac{400 \times 1}{2\pi \times \frac{20}{\pi} \times 10^{-2}} = \mathbf{1000\ A/m.}$$

The susceptibility is

$$\chi_m = \frac{M}{H} = \frac{4.8 \times 10^{-2}\ \text{A/m}}{1000\ \text{A/m}} = \mathbf{4.8 \times 10^{-5}}.$$

Ex. 13. *The intensity of magnetization of a plane magnetized disc of radius 1 cm and thickness 1.2 cm is 100 A/m along its axis. Calculate* : (i) *bound current* i_b; (ii) *magnetic field B and magnetizing field H at the centre of disc*; (iii) *magnetic moment of disc*; (iv) *magnetic susceptibility and relative permeability of the material of disc.*

Sol. (i) Magnetization current, $i_b = M \times \text{thickness} = 100 \times (1.2 \times 10^{-2}) = \mathbf{1.2\ A.}$

(ii) The magnetic field at the centre of disc is given by (due to magnetization current at the boundary)

$$B = \frac{\mu_0 i_b}{2r} = \frac{(4\pi \times 10^{-7}) \times 1.2}{2 \times (10^{-2})} = \mathbf{7.54 \times 10^{-5}\ Wb/m^2}$$

∴ Magnetizing field, $H = \frac{B}{\mu_0} - M = \frac{7.54 \times 10^{-5}}{4\pi \times 10^{-7}} - 100 = 60 - 100 = \mathbf{-40\ A/m.}$

(iii) Magnetic moment of disc = Intensity of magnetization × Volume of disc

$= 100 \times [\pi (0.01)^2 \times 0.012]$

$= \mathbf{3.77 \times 10^{-4}\ A\ m^2}.$

(iv) Magnetic susceptibility, $\chi_m = \frac{M}{H} = \frac{100}{-40} = \mathbf{-2.5}$

Relative permeability, $\mu = \chi_m + 1 = -2.5 + 1 = \mathbf{-1.5}.$

EXERCISE 9·3

1. A coil of area 5 cm² and having 100 turns is placed in a uniform magnetic field, 1.5 N $A^{-1}m^{-1}$. When a current of 0.2 A is passed through the coil, find the magnetic dipole moment of the coil and the maximum torque produced.
[**Ans.** 1.0×10^{-2} A m²; 1.5×10^{-2} N m.]
2. A current of 2*A* flows in a circular coil of 100 turns and 4 cm radius. Find : (i) magnetic moment of the coil, (ii) the torque acting on the coil when it is suspended in a uniform magnetic field of 0.1 Wb m^{-2} parallel to the field, (iii) work done in rotating the coil through 180°, when freely suspended in the magnetic field.
[**Ans.** (i) 1.04 A m²; (ii) 0.10^4 N m; (iii) 0.208 J.]
3. The magnetic dipole moment of earth is 6.4×10^{21} A m². If this magnetic moment is to be considered due to a current-loop wound around the magnetic equator of earth, then calculate the current. (Radius of earth = 6.4×10^6 m).
[**Ans.** 5.0×10^7 A.]
4. The magnetic moment of a magnetic dipole is 2.5 A m². Find the intensity of magnetic field at a distance of 30 cm in the (i) end on position and (ii) broad side on position.
[**Ans.** (i) 18.52 μT. (ii) 9.26 μT.]
5. The dipole moment of a short bar-magnet is 1.25 A m². Find the magnetic field on its axis at a distance of 50 cm from the centre of the magnet.
[**Ans.** 2 μT.]

6. A long straight wire carries a current 7.5 A. Calculate the magnetizing field H and the magnetic flux density B at a point at distance 10.5 cm from the axis of wire.
[**Ans.** $H = 11.37$ A/m, $B = 1.4 \times 10^{-5}$ T.]
7. A solenoid of length 0.48 m has 4000 turns and carries current 3.2 A. Find the magnetizing field H and magnetic flux density B at the centre of solenoid.
[**Ans.** $H = 26.66 \times 10^3$ A/m, $B = 3.36 \times 10^{-2}$ Wb/m^2.]
8. A toroid has 600 turns in its mean perimeter 0.5 m, each turn carrying current 0.15 A. Calculate (i) the values of H and B with air core; (ii) the values of B and H with iron core (relative magnetic permeability = 5000)
[**Ans.** (i) $H = 180$ A/m, $B = 2.26 \times 10^{-4}$ T; (ii) $H = 180$ A/m, $B = 1.13$ T.]
[**Hint :** $B = \dfrac{\mu_0 N i}{2\pi r}$.]
9. A solenoid having 4000 turns/m has an aluminium core and carries a current of 2.0 A Calculate the magnetization M developed in the core and the magnetic field B at the centre. The susceptibility χ_m of aluminium is 2.3×10^{-5}.
[**Ans.** $M = 0.18$ A/m; $B = 10^{-3}$ T.]
[**Hint :** $M = \chi H$, $H = ni$, $B = \mu_0 (H + M)$]
10. The magnetic field B and the magnetizing field H in a material are found to be 1.6 T and 1000 A/m respectively. Find the relative permeability μ_r and the susceptibility χ of the material.
[**Ans.** 1.3×10^3; 1.3×10^3.]
11. An iron ring of mean length 0.3 m and area of cross-section 10^{-4} m^2 is wound with 300 closely space turns of wire. When a current 0.032 amp flows in the winding, a flux 2×10^{-6} weber is produced. Find (i) flux density in the ring; (ii) the magnetic intensity *i.e.*, $\vec{M}$; (iii) the permeability; (iv) the relative permeability and (v) the magnetic susceptibility of the material of the ring.
[**Ans.** (i) 2×10^{-2} Wb/m^2; (ii) 32 amp./m; (iii) 6250×10^{-7} henry/m; (iv) 498 and (v) 6240×10^{-7} henry/m .]

QUESTIONS

Long Answer Questions

1. What is meant by the Lorentz force ? Explain the conditions for it to be minimum and maximum ? **(Bhopal 2004; Sagar 2003)**
2. Define magnetic field $\vec{B}$ on the basis of the force on a charge moving in a magnetic field. Write its unit. **(Meerut 1997)**
3. Write expression for the Lorentz force acting on a moving charge in a magnetic field. State Fleming's left hand rule to determine the direction of this force. **(Bundelkhand 2004)**
4. On the basis of the Lorentz force acting on a moving charge, find an expression for the force on a current carrying conductor placed in a uniform magnetic field. Find the unit of the magnetic field on this basis.
5. Define magnetic field on the basis of force acting on a current carrying conductor in a magnetic field. Write its unit and dimensional formula. State Fleming's left hand rule. **(Bhopal 2003)**
6. Establish the expression for the torque acting on a current carrying coil, freely suspended in a uniform magnetic field. Under what condition this torque will be (i) maximum; (ii) minimum ? **(Sagar 2003)**
7. What is gyromagnetic ratio ? Show that the ratio of magnetic moment to its angular momentum due to rotation of a uniformly charged body (mass m, charge q) is equal to $q/2m$. **(Gwalior 2004, 03; Bhopal 03; Ujjain 03; Jabalpur 03; Raipur 03)**
8. Explain, giving diagram, Biot and Savart's law. Give the expression for the magnetic field produced. Discuss analogies and differences between Coulomb's law and Biot-Savart's law.

9. Write Biot-Savart's law and use it to establish expression for the intensity of magnetic field produced at a point near a long straight current carrying conductor. **(Jabalpur 2004; Gwalior 03; Rewa 03)**
10. Using Biot-Savart's law find the magnetic field due to a circular current carrying coil along the axis perpendicular to its plane. (i) What is the field at the centre of the coil ? (ii) What is magnetic moment of the coil ? **(Agra 2002)**
11. Write Biot-Savart's law. Use it to obtain the expression for the magnetic field at the centre of a current carrying circular coil. **(Raipur 2004, 03; Indore 03)**
12. What is Biot-Savart's law ? Find an expression for the magnetic field on the axis of a circular coil due to flow of current in it. **(Bundelkhand 2004; Bhopal 03; Vikram 03; Sagar 03)**
13. Show that two long parallel conductors carrying currents in the same direction attract each other. Find the magnitude of this force. Define ampere on this basis. **(Lucknow 2004)**
14. Deduce formula for the force acting between two parallel current carrying conductors. When will this force be attractive and when will it be repulsive ? **(Ujjain 2003; Indore 04)**
15. Find the expression for magnetic field of induction $\vec{B}$ due to a circular coil of radius a, carrying a current i at a point on the axis in terms of the magnetic dipole moment. Find $\vec{B}$ when the point P is (i) at the centre of the coil, (ii) far away from the coil. **(Agra 2005)**
16. Derive expression for the magnetic field produced at a point on the axis of a current carrying solenoid inside it. **(Ujjain 2004; Raipur 03)**
17. State Ampere's circuital law and prove it. **(Ujjain 2003,04; Jabalpur 2004; Raipur 03)**
18. Express Ampere's circuital law. Find also its integral and differential forms. **(Agra 2002)**
19. State Ampere's law in circuital form and use it to find an expression for the magnetic field at the axis of a long solenoid. **(Agra 2004, 02)**
20. A long, straight and solid metallic rod of radius R carries current i. Assuming the current distribution to be uniform, obtain expressions for the intensity of magnetic field at a point inside and outside the wire. **(Jabalpur 2004)**
21. What is Ampere's circuital law ? Derive it in the form :

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

where $\vec{B}$ is the magnetic field and $\vec{J}$ is the current density ? What is the other equation which together with this equation uniquely determines $\vec{B}$, if $\vec{J}$ is known ? What is the significance of the equation ? **(Lucknow 2005, 04; Meerut 1994)**

[Hint : Other equation is div $\vec{B} = 0$.]

22. Prove that

(i) div $\vec{B} = 0$ **(Ujjain 2004; Rewa 03; Bilaspur 03; Indore 04)**

(ii) curl $\vec{B} = \mu_0 \vec{J}$ **(Ujjain 2004; Bilaspur 03; Indore 04)**

23. What do you mean by a magnetic dipole ? Explain the meaning of magnetic moment of dipole and hence find the magnetic moment of magnetic dipole equivalent to a current loop.
24. What is magnetic dipole ? Find an expression for the magnetic field due to a magnetic dipole at a point situated on its axis.
25. Find an expression for magnetic field due to a magnetic dipole at an arbitrary point.
26. Show that the magnetic field due to a short magnetic dipole at a distant point (r, θ) is given by

$$B = \frac{\mu_0}{4\pi} \cdot \frac{M}{r^3} \sqrt{3\cos^2\theta + 1}$$

27. For the same distance of a point, the magnitude of magnetic field due to a short magnetic dipole in the end on position is two times the field in the broad side on position.

28. Show that the magnetic field due to a magnetic dipole in the form of a current carrying solenoid at an axial point distant x from its centre is given by

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2mx}{(x^2 - l^2/4)^2}$$

where $m = NiA$ is the magnetic moment of the dipole.

29. (a) What is magnetization vector ? Define the magnetization at a point.
    (b) What is magnetization current ? Show that for a cylindrical block having uniform magnetization along its axis, the magnetization current is equal to the product of magnetization and the length of the block.
30. What is surface magnetization current density ? Show that it is numerically equal to the magnitude of the magnetization vector.
31. (a) Interpret a bar magnet as a surface distribution of solenoidal currents.
    (b) Prove that $\vec{J}_m = \text{curl}\,\vec{M}$
    where the terms have usual meaning. **(Gwalior 2004; Bilaspur 1996)**
32. Define magnetization $\vec{M}$ and field $\vec{H}$ and establish the relation
    $$\vec{B} = \mu_0(\vec{H} + \vec{M})$$ **(Agra 2003, 02)**
33. Explain the meaning of the terms $\vec{B}, \vec{H}$ and $\vec{M}$ in a magnetized medium and establish a relationship between them. **(Gwalior 2004; Jabalpur 03; Indore 03)**
34. What are free and bound currents ? Show that the curl of a magnetizing field vector is equal to the free current density.
35. Prove that the divergence of magnetic field $\vec{B}$ is zero everywhere, whereas it is not essential that the divergence of magnetising field $\vec{H}$ be also zero. **(Bhopal 1996)**
36. (a) What are magnetic permeability and susceptibility ? Discuss their units. What is relative permeability ?
    (b) Show that the magnetic susceptibility (χ_m) is related to the relative permeability (μ_r) as
    $$\chi_m = \mu_r - 1.$$ **(Indore 1993; Ujjain 1995)**
37. Explain the terms magnetic permeability and magnetic susceptibility of a magnetised medum and obtain relationship between them. **(Bundelkhand 2004; Bhopal 1995; Sagar 95)**
38. (a) Show that $\oint \vec{H} \cdot d\vec{l} = i_f$, where i_f is free current.
    (b) Show that for a linear medium the bound current density is proportional to the free current density.
39. Write short notes on the following :
    (i) Magnetization vector
    (ii) Magnetization current **(Bilaspur 2003)**
    (iii) Free and bound currents **(Bilaspur 2003)**
    (iv) Magnetization field
    (v) Difference between $\vec{B}$ and $\vec{H}$.

Short Answer Questions

1. Discuss the motion of a charged particle in a magnetic field. **(Meerut 2002)**
2. A moving charged particle enters in a uniform magnetic field in a perpendicular direction. Prove that the work done by the force acting on the particle due to magnetic field is zero.

3. An electron, moving in a magnetic field, is not experiencing any force. When is it possible ?
   [**Ans.** When the electron moves along the direction of magnetic field.]
4. A proton having the charge e is (i) at rest in a magnetic field, (ii) moving with velocity v in direction of magnetic field B, (iii) moving with velocity v perpendicular to the magnetic field B. In each case, find the force acting on the particle.
   [**Ans.** (i) Zero; (ii) Zero; (iii) evB.]
5. A charge q moves in a magnetic field $\vec{B}$ with a velocity $\vec{v}$ making an angle of 30° with $\vec{B}$. What will be the magnitude of the force acting on the charge ?
   [**Ans.** $\frac{1}{2}qvB$.]
6. An electron beam is moving towards the east. If the beam passes through a uniform magnetic field acting vertically donwards, in what direction will it deflect ?
   [**Ans.** Towards south.]
7. The speed of a charged particle moving in a magnetic field does not change. Why ?
8. An electron and a proton moving with the same speed, enter a uniform magnetic field perpendicularly. (i) Which particle will have a larger radius of its circular path ? (ii) Why it will be large ? (iii) How many times larger ? (Proton is 1840 times heavier than electron.)
   [**Ans.** (i) Proton; (ii) $r = mv/qB$. bacause the mass of proton is larger; (iii) 1840 times.]
9. A conductor, carrying a current i ampere, is placed in a uniform magnetic field of intensity B Wb m^{-2}. Find the force on the conductor if (i) direction of current in conductor is parallel to the magnetic field; (ii) direction of current in conductor is perpendicular to the magnetic field
   [**Ans.** (i) Zero; (ii) ilB newton.]
10. A long, straight current carrying conductor PQ is situated in the plane of the paper. Near to it another small movable conductor RS is situated in the plane of paper perpendicular to PQ. Discuss the effect on the position of RS on passing a current through it (fig. 9.95).
   [**Ans.** Current carrying conductor RS is in a non-uniform magnetic field, acting ⊥ to the plane of the paper downwards, due to the current i_1. Force on the conductor RS is perpendicular upwards in the plane of the paper but decreasing from R to S. Hence, the conductor RS will move upward and rotate in clockwise direction. When it will become parallel to i_1, then the currents i_1 and i_2 will be antiparallel and hence the RS conductor will move away due to repulsion and remain parallel to PQ.]

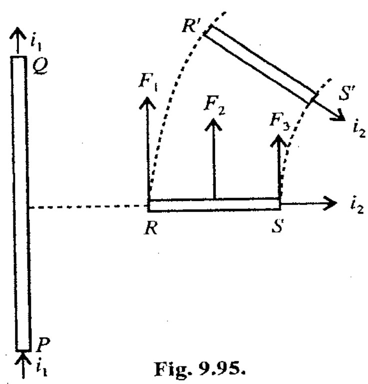

Fig. 9.95.

11. What is a magnetic dipole ? What is the torque acting on a magnetic dipole of dipole moment $\vec{m}$ placed in a uniform magnetic field $\vec{B}$. **(Lucknow 2004)**
12. What do you understand by the magnetic dipole moment of a current loop ?
13. Show that a current loop is equivalent to a magnetic dipole of magnetic moment $\vec{M} = ni\vec{A}$, where the symbols have their usual meanings. **(Raipur 2003)**
14. A current of 10 μA is established in a circular loop of radius 5 cm. Calculate the magnetic dipole moment of the current loop.
   [**Ans.** 7.85×10^{-8} A-m^2.]
15. Show that due to orbital motion of electron, magnetic moment is an integer multiple of $eh/4\pi m$, where the symbols have their usual meanings. **(Gwalior 2004)**
16. What is gyromagnetic ratio ? Write its value for the orbital motion of electron. **(Gwalior 2003)**

17. A circular loop of area 2 cm^2, carrying a current of 2 A, is placed in a magnetic field of 0.1 T perpendicular to the plane of the loop. Find the torque on the loop due to the magnetic field.
[Ans. Zero.]

18. (a) Give the statement of Biot-Savart's law and mention its use as well. **(Agra 2003)**
(b) State and explain Biot-Savart's law. **(Agra 2006)**

19. A straight wire carrying a current i_1 is situated along the axis of a circular conductor of radius r carrying a current i_2 [fig. 9.96]. Find the force produced due to the interaction between the two currents.
[Ans. Zero.]

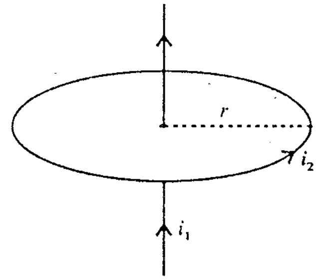

Fig. 9.96.

20. Find out the distance between two parallel wires, when currents of 100 A and 20 A flow in them and they attract each other with a force of 0.08 N m^{-1}.
[Ans. 0.5 cm.]

21. Find the force between two parallel wires carrying current in the same direction. **(Meerut 2002)**

22. Find out the magnetic field at the centre O due to flow of electric current i in the two wires as shown in fig. 9.97.
[Ans. Zero.]

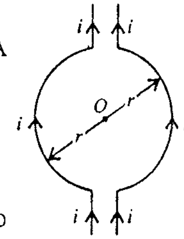

Fig. 9.97.

23. A charge q makes n revolutions per second in a circular path of radius r. Establish an expression for the magnetic field produced at the centre of path.

24. The electron in hydrogen atom is revolving in a circular path of 0.5×10^{-10} m, with a constant speed of 2.0×10^6 m s^{-1}. Calculate the magnetic field at the centre of the circular path.
[Ans. 12.8 $NA^{-1}m^{-1}$.]

25. (a) Why two coils are used in Helmholtz golavanometer ? **(Lucknow 2005)**
(b) Why do we need Helmholtz coils ? Expalin. **(Agra 2003)**

26. A current i flows in a circular sector of radius R which subtends an angle θ at its centre. Find the magnetic field at the centre of sector.

[Ans. $\frac{\mu_0}{4\pi} \cdot \frac{i\theta}{r} \cdot$]

27. A long, straight wire is turned into a loop of radius r as shown in fig. 9.98. If a current of i is passed through the loop, find the magnetic field at the centre O of the loop.

[Ans. $\frac{\mu_0 i}{2r}\left(1 - \frac{1}{\pi}\right)$ downward into the page.]

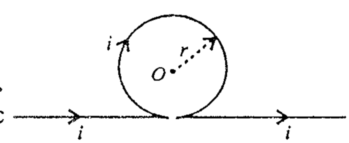

Fig. 9.98.

28. (a) State and explain Ampere's law. **(Agra 2006)**
(b) Prove that $\quad \text{curl } \vec{B} = \mu_0 \vec{J}$. **(Agra 2004)**

29. Use Ampere's law to find the magnetic field of a solenoid carrying a current i and having n turns per unit length. **(Agra 2006)**

30. A current is set up in a long copper pipe. Is there a magnetic field (i) inside and (ii) outside pipe ?
[**Ans.** (i) There is not magnetic field inside the pipe, because any closed path in the cross-section of the tube will not enclose any current so that $\oint \vec{B} \cdot d\vec{l} = 0$ and it imphies $\vec{B} = 0$ inside.
(ii) There is magnetic field outside the pipe.]

31. A long thin cylindrical conductor carries a current I. Using Ampere's law find $\vec{B}$ at a point lying (i) inside the conductor, (ii) outside the conductor. **(Lucknow 2004)**
32. Prove that $\vec{\nabla} \cdot \vec{B} = 0$.
33. Give the physical significance of div $\vec{B} = 0$.
34. Deduce the differential form of Ampere's law.
35. What is magnetization vector ? Write its unit.
36. Explain magnetization current.
37. How surface current in a magnetic material is related to magnetization.
38. What are free and bound currents ?
39. Explain the difference between magnetic induction $\vec{B}$ and magnetizing field $\vec{H}$. **(Lucknow 2005)**
40. In case of a magnetic material, deduce the relation between $\vec{B}, \vec{H}$ and $\vec{M}$.
41. Discuss the equivalence of a bar magnet and a solenoid.
42. Deduce the relation between current density in a magnetic material due to non-uniform magnetization and the magnetization vector.
43. Deduce the Ampere's law in differential form in terms of the magnetizing field.
44. Show that the magnetic permeability (μ) and magnetic susceptibility (χ_m) are related as
$$\mu = \mu_0 (1 + \chi_m).$$

Objective Type Questions

1. A moving charge produces :
(a) magnetic field only (b) electric field only
(c) both magnetic and electric fields (d) None of these
2. A proton moving with a constant velocity passes through a region without any change in its velocity. If E and B denote electric and magnetic fields respectively, this region may have :
(a) $E = 0, B = 0$ (b) $E = 0, B \neq 0$
(c) $E \neq 0, B = 0$ (d) $E \neq 0, B \neq 0$. **(IIT 1985)**
3. A positive charge moving up enters a magnetic field in the south direction. The direction of the force on the charge is towards :
(a) East (b) West (c) North (d) South. **(Agra 2005)**
4. A positively charged particle ($+q$) enters a magnetic field $\vec{B}$ with a uniform velocity $\vec{v}$, making an angle θ with the field. The path of the particle in the field will be :
(a) circular (b) parabolic (c) helical (d) straight line.
5. A charged particle moves in a region having a uniform magnetic field and parallel uniform electric field. At some instant the velocity of the particle is perpendicular to the field direction, the path of the particle will be a :
(a) straight line (b) circle
(c) helix with uniform pitch (d) a helix with non-uniform pitch
(Agra 2003)
6. When a charged particle moves in a uniform magnetic field, its kinetic energy :
(a) remains constant (b) increases
(c) decreases (d) may increase or decrease.
7. A charged particle is released from rest in a region of steady and uniform electric and magnetic fields which are parallel to each other. The particle will move in a :
(a) straight (b) circle (c) helix (d) cycloid. **(IIT 1999)**

8. Select the correct statement :
(a) Two parallel current carrying wires attract each other
(b) Two parallel current carrying wires repel each other
(c) Two antiparallel current carrying wires attract each other
(d) Two antiparallel current carrying wires repel each other.

9. Magnetic field at a distance of 2 m at right angles to a long wire of radius R and carrying a current I is
(a) $\frac{\mu_0 I}{4\pi R}$ (b) $\frac{\mu_0 I}{2\pi R}$ (c) $\frac{4\pi I}{2\mu_0}$ (d) $\frac{\mu_0 I}{4\pi}$ **(Agra 2004)**

10. The force acting on a current carrying conductor kept in a magnetic field is :
(a) $F = IBl\sin\theta$ (b) $F = InBl\sin\theta$ (c) $F = InBl$ (d) $F = IBl$ **(Bhopal 2004)**

11. A vertical wire carries a current in the downward direction. An electron beam sent horizontally towards the wire will be directed :
(a) towards left (b) towards right (c) upwards (d) downwards

12. A wire carries an electric current from west to east and is placed in a magnetic field directed along north. The force acting on the wire will be directed towards :
(a) vertically downwards (b) vertically upwards
(c) east (d) west

13. An electric conductor of length l is placed in a magnetic field B. If a current i flows in the conductor, the force on conductor will be proportional to :
(a) $\frac{1}{l}$ (b) $\frac{1}{B}$ (c) $\frac{1}{i}$ (d) i **(Rewa 2003)**

14. A long straight conductor carries a current along Z-axis. We can find two points in the XY-plane for which :
(a) the directions of the magnetic fields are the same
(b) the magnitudes of the magnetic fields are the same
(c) the magnetic fields are equal
(d) the magnetic field at one point is opposite to that at the other point.

15. A circular loop has a radius 5 cm and it is carrying a current of 0.1 A. Its magnetic moment is :
(a) 1.32×10^{-4} Am^2 (b) 2.65×10^{-4} Am^2
(c) 5.25×10^{-4} Am^2 (d) 7.85×10^{-4} Am^2 **(Agra 2004)**

16. A coducting circular loop of radius r carries a constant current i. It is placed in a uniform magnetic field $\vec{B}$ such that $\vec{B}$ is perpendicular to the plane of the loop. The magnitude of the magnetic force acting on the loop is :
(a) irB (b) $2\pi irB$ (c) zero (d) πirB. **(IIT 1983)**

17. A particle carrying a charge 1000 times the charge on an electron is revolving once per second on a circular path of radius 0.8 m. The magnetic field produced at the centre will be :
(a) $10^{-7}\ \mu_0$ (b) $10^{-16}\ \mu_0$ (c) $10^{-6}\ \mu_0$ (d) None of these

18. A rectangular loop carrying a current i is situated near a long, straight wire placed parallel to one of the sides of the loop and in the plane of the loop, as shown. If a steady current I is passed in the wire **[fig. 9.99]**, the loop will :
(a) rotate about an axis parallel to the wire
(b) move away from the wire
(c) move towards the wire
(d) remain stationary. **(IIT 1985)**

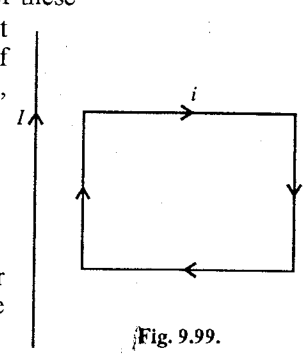

Fig. 9.99.

19. In hydrogen stom, the electron revolves around the nucleus in a circular orbit of radius 5.4×10^{-11} m with a frequency 9.8×10^{15} Hz. The magnitude of the magnetic field produced at the centre of the orbit is :

(a) 44.6 Wb m^{-2} (b) 18.2 Wb m^{-2}
(c) 25.3 Wb m^{-2} (d) 70.7 Wb m^{-2}

20. A wire of length l carrying a current i is bent into a circular loop of one turn. The magnetic field produced at the centre of the loop is B. When this wire is bent into a loop having 2 turns, then the magnetic field produced at the centre of the loop will be given by :
(a) B (b) $2B$ (c) $B/2$ (d) $4B$

21. Fig. 9.100 shows a square loop $PQRS$ with side l. The resistance of the length PQR is R and that of PSR is $2R$. The magnetic field at the centre of the loop is :
(a) 0 (b) $\frac{\sqrt{2}\mu_0 i}{3\pi l}$ (c) $\frac{\mu_0 i}{2l}$ (d) $\frac{\mu_0 i}{2\pi l}$

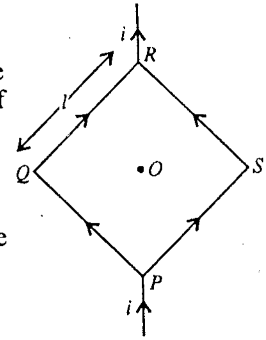

Fig. 9.100.

22. The magnetic field inside a toroid having N turns and current i at a distance r from its centre is :
(a) $B = \mu_0 Ni$ (b) $\frac{\mu_0}{2} Ni$ (c) $\frac{\mu_0 Ni}{2\pi r}$ (d) $\frac{\mu_0 Ni}{2r}$

23. A circular coil of radius a carries a current and the magnetic field at its centre is B. At what distance from the centre on the axis of coil, the magnetic field will be $B/8$:
(a) a (b) $2a$ (c) $\sqrt{3}a$ (d) $8a$

24. The magnetic field inside a long solenoid carrying current in the middle is :
(a) constant throughout its cross-section
(b) maximum at the centre of cross-section and decreases towards the periphery
(c) proportional to the length of the solenoid
(d) inversely proportional to the current flowing in it.

25. The magnetic field inside a long solenoid with 5000 turns per metre carrying a current of 5 A is :
(a) 2.5×10^4 T (b) 2.5×10^{-4} T (c) 3.14×10^{-2} T (d) $25\pi \times 10^{-4}$ T

26. The magnetic field due to a current carrying long solenoid :
(a) is zero inside (b) is zero outside
(c) varies along its length (d) None of these.

27. Divergence of a magnetic field vector is :
(a) 4π (b) 2π (c) 0 (d) none of these **(Agra 2004)**

28. A long straight wire of radius R, carries a current distributed uniformly over its cross-section. The magnitude of the magnetic field is :
(a) maximum at its axis
(b) maximum at distance $R/2$ from its axis
(c) maximum at the surface of the wire
(d) uniform at every point of the cross-section. **(Agra 2003)**

29. The energy density of a magnetic field is :
(a) $\mu H/2$ (b) $\mu H^2/2$ (c) μ/H (d) μH **(Agra 2006)**

30. A current flows along the length of a infinitely long, straight, thin-walled pipe. Then :
(a) the magnetic field at all points inside the pipe is same, but not zero
(b) the magnetic field at any point inside the pipe is zero
(c) the magnetic field is zero only on the axis of the pipe
(d) the magnetic field is different at different points inside the pipe. **(IIT 1993)**

31. In a straight coaxial cable, the central conductor and outer conductor carry equal currents in the opposite directions. The magnetic field will be zero :
(a) inside the inner conductor (b) inside the outer conductor
(c) outside the cable (d) in between the two condutors.

32. An electric current is flowing in a hollow copper tube along its length and it is distributed uniformly over its surface. The magnetic field :

(a) is zero inside the tube

(b) is zero just outside the surface

(c) increases linearly from the axis to the surface

(d) is directly proportional to the radial distance fròm the axis outside the surface.

33. A current of 10 A is flowing through a coil of radius 0.5 m and 10 turns. The magnetic moment of the equivalent dipole is :

(a) 1 Am^2 (b) 7.85 Am^2 (c) 78.5 Am^2 (d) 0.785 Am^2

34. In hydrogen atom, the electron is making 6.6×10^{15} rev/sec, around the nucleus in an orbit of radius 0.528 Å. The magnetic moment (A m^2) will be :

(a) 1×10^{-14} (b) 1×10^{-23} (c) 1×10^{-15} (d) 1×10^{-27}

35. Two of the magnetic vectors $\vec{B}, \vec{H}$ and $\vec{M}$ (magnetization) have the same unit :

(a) $\vec{B}$ and $\vec{H}$ (b) $\vec{B}$ and $\vec{M}$ (c) $\vec{H}$ and $\vec{M}$ (d) None of these

36. SI unit of magnetizing field $\vec{H}$ is :

(a) A/m (b) weber (c) Wb/m^2 (d) $Wb\text{-}m^2$

37. SI unit of magnetic field $\vec{B}$ is :

(a) A/m (b) weber (c) Wb/m^2 (d) $Wb\text{-}m^2$

38. SI unit of intensity of magnetization :

(a) A/m (b) weber (c) Wb/m^2 (d) $Wb\text{-}m^2$

39. The correct relationship between $\vec{B}, \vec{H}$ and $\vec{M}$ is :

(a) $\vec{H} = \frac{\vec{B}}{\mu_0} + \vec{M}$ (b) $\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$ (c) $\vec{B} = \frac{\vec{H}}{\mu_0} - \vec{M}$ (d) $\vec{H} = \mu_0(\vec{B} + \vec{M})$

40. The corrent relation is :

(a) $\text{div}\,\vec{B} = -\varepsilon_0\mu_0$ (b) $\text{div}\,\vec{B} = 0$ (c) $\text{curl}\,\vec{B} = 0$ (d) $\text{div}\,\vec{B} = \text{div}\,\vec{H}$

41. The correct relationship between $\vec{B}, \vec{H}$ and $\vec{M}$ is :

(a) $\vec{B} = \mu_0(\vec{H} + \vec{M})$ (b) $\vec{B} = \mu_0(\vec{H} - \vec{M})$

(c) $\vec{H} = \mu_0(\vec{B} - \vec{M})$ (d) $\vec{B} = \mu_0\,\vec{H} + \vec{M}$ **(Gwalior 1990, 89; Ujjain 88)**

42. The correct relationship between the intensity of magnetization M and the magnetizing field H is :

(a) $M = \frac{1}{\chi} H$ (b) $M = \chi H$

(c) $M = \frac{\chi}{H}$ (d) $M = \chi H^2$ **(Bundelkhand 2004)**

43. The permeability μ, relative permeability μ_r and the magnetic susceptibility are related as :

(a) $\mu = 1 + \chi_m$ (b) $\mu_r = \mu_0 (1 + \chi_m)$

(c) $\chi_m = \mu_r - 1$ (d) $\chi_m = \mu/\mu_0$

44. Outside a magnetized rod :

(a) $\vec{B} = 0$ (b) $\vec{H} = 0$ (c) $\vec{M} = 0$ (d) $\vec{H} = \vec{B}$

45. The correct relation is :

(a) $\vec{\nabla} \times \vec{B} = \vec{J}_f$ (b) $\vec{\nabla} \times \vec{H} = \vec{J}_f$

(c) $\vec{\nabla} \times \vec{H} = \vec{J}_f + \vec{J}_m$ (d) $\vec{\nabla} \cdot \vec{H} = \vec{J}_f$

where $\vec{B}$ = magnetic field, $\vec{H}$ = magnetizing field, $\vec{J}_f$ = free current density and $\vec{J}_m$ = magnetization current density.

46. The vector form of Biot-Savart's law for a current carrying element is :

(a) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{idl \sin\theta}{r^2}$ (b) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\vec{l} \times \hat{r}}{r^2}$

(c) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\vec{l} \times \hat{r}}{r^3}$ (d) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\vec{l} \times \vec{r}}{r^2}$ **(Bhopal 2004; Bundelkhand 04)**

ANSWERS

1. (c)	**2.** (a),(b),(d)	**3.** (a)	**4.** (d)	**5.** (d)	**6.** (a)	**7.** (a)
8. (a),(d)	**9.** (d)	**10.** (a)	**11.** (d)	**12.** (b)	**13.** (d)	**14.** (a),(b),(d)
15. (d)	**16.** (c)	**17.** (b)	**18.** (c)	**19.** (b)	**20.** (d)	**21.** (b)
22. (c)	**23.** (c)	**24.** (a)	**25.** (c)	**26.** (b)	**27.** (c)	**28.** (c)
29. (b)	**30.** (b)	**31.** (c)	**32.** (a)	**33.** (c)	**34.** (b)	**35.** (c)
36. (a)	**37.** (c)	**38.** (a)	**39.** (b)	**40.** (b)	**41.** (a)	**42.** (b)
43. (c)	**44.** (c)	**45.** (b)	**46.** (b)			

10

Electromagnetic Induction

10·1. Magnetic Flux

If we consider a plane surface perpendicular to a magnetic field, then the product of the magnetic field and the area of that surface is called the **magnetic flux.** Suppose a plane surface of area S is placed perpendicular to a magnetic field $\vec{B}$, then the magnetic flux (ϕ) through that surface will be given by [fig. 10.1(a)].

$$\phi = BS \qquad ...(1)$$

If the normal to the plane surface makes an angle θ with the uniform magnetic field $\vec{B}$, then the component of magnetic field perpendicular to that surface will be $B \cos \theta$ and in this position, the value of the magnetic flux becomes [fig. 10.1(b)]

$$\phi = (B \cos \theta)\, S = BS \cos \theta = \vec{B} \cdot \vec{S} \qquad ...(2)$$

Here $\vec{S}$ is the vector area of the surface.

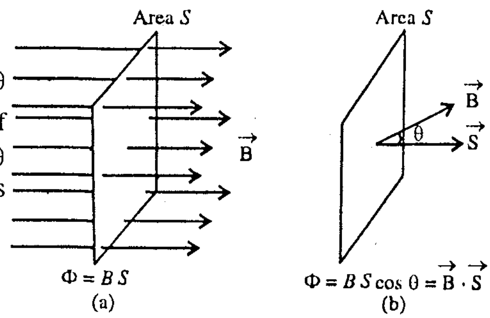

Fig. 10.1 : Magnetic flux.

If the field $\vec{B}$ is non-uniform and surface is not in a plane, then we use an infinitesimal area $d\vec{S}$ of that surface to calculate the magnetic flux. If a magnetic field $\vec{B}$ is associated with area $d\vec{S}$, then the value of magentic flux through this area is $d\phi = \vec{B} \cdot d\vec{S}$ and the total magnetic flux through the entire surface can be found by integration *i.e.*,

$$\phi = \int_S \vec{B} \cdot d\vec{S} \qquad ...(3)$$

SI unit of magnetic flux is **weber**, written as Wb and its dimensional formula is $[ML^2T^{-2}A^{-1}]$. As $\phi = BS$, the magnetic field B is also represented in unit 'weber/meter^2' or Wb/m^2. Therefore we also call magnetic field B as **magnetic flux density.**

We can represent the magnetic flux by magnetic lines of force. The uniform magnetic field $\vec{B}$ can be represented by B lines of force passing through unit area of surface normal to the field. The magnetic flux ϕ represents BS magnetic lines of force passing through area A normal to the field.

10·2. Electromagnetic Induction—Induced E.M.F. and Induced Current

First of all in 1831, Faraday performed a number of experiments and discovered that if a closed circuit moved across a magnetic field, a current flowed in the circuit. The current is also produced, if the closed circuit is kept stationary and the magnetic field, say due to a magnet, is moved. In fact, a current is always produced whenever there is a relative motion between the closed circuit and magnetic field. *The phenomenon in which an electric current is produced in a closed circuit due to the relative motion between a magnetic field and the circuit itself is called electromagnetic induction and the current so produced, is known as* **induced current.** The electromotive force due to which the current is induced, is known as **induced electromotive force (or induced e.m.f.).** As shown in **fig 10.2** if a magnet approaches a closed circuit consisting of a coil and a galvanometer, the galvanometer indicates a deflection *i.e.*, an electric current is induced in the closed circuit and the direction of electric current depends on the direction of motion of the magnet.

When the north pole N of a bar magnet NS is brought near the coil, an instantaneous deflection is produced in the galvanometer due to the flow of current. The direction of this current is such that the face of the coil near the magnet behaves like a north pole [**fig. 10.2**]. If the north pole N of the magnet is taken away from the coil, a deflection again produces in the galvanometer but now it is in opposite direction. Hence the current produced in the coil is in opposite direction and the front face of the coil works as a south pole S. Similarly, an instantaneous deflection is produced in bringing the south pole S of the magnet towards or away from the coil. The induced current exists only so long as there is a relative motion between the coil and the magnet. The flow of current comes to zero as we stop the relative motion of both coil and magnet. The current so produced in the coil is the **induced current.** Larger the relative speed between the magnet and the coil, more is the deflection in the galvanometer, *i.e.*, more strong current is produced in the coil. On increasing the number of turns in the coil, the current produced in the circuit increases. A deflection is also found in the galvanometer if the magnet is kept stationary and the coil is taken towards or away from the magnet. It is concluded that the current in a coil is produced due to the relative motion between the coil and the magnet. Faraday also observed that when two coils are kept close to each other and a current is switched on in one coil, a momentary deflection is produced in the galvanometer in the circuit of the other coil [**fig. 10.3**]. This indicates that an induced current is produced in the second coil. Now when the circuit is broken by pulling out the key from the first coil circuit, a momentary current is again induced in the secondary coil. But the directions of the induced currents in the two cases are opposite to each other.

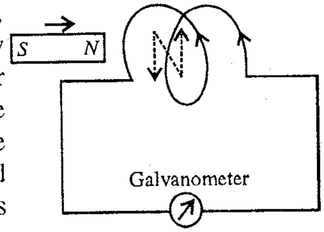

Fig. 10.2 : Production of induced current due to the motion of a magnet towards closed circuit.

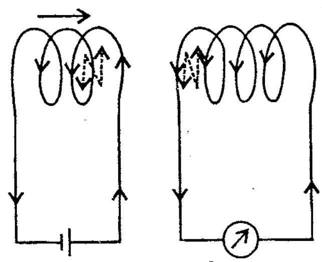

Fig. 10.3 : Production of induced current due to the relative motion between a current carrying coil and a closed circuit with galvanometer.

10·3. Faraday's Law of Electromagnetic Induction

On the basis of his experiments, Faraday established the following law of electromagnetic induction :

Whenever the magnetic flux linked with a conductor changes, an e.m.f. is induced in the conductor. The e.m.f., so induced, is equal to the negative rate of change of magnetic flux.

If the rate of change of magnetic flux linked with any conductor is $d\phi/dt$, then according to Faraday's law, the induced e.m.f. in the conductor will be given by

$$e = -d\phi/dt \qquad ...(1)$$

If the change in flux in time Δt is $\Delta\phi = \phi_2 - \phi_1$, then

$$e = -\frac{\phi_2 - \phi_1}{\Delta t} = -\frac{\Delta\phi}{\Delta t} \qquad ...(2)$$

In the limit $\Delta t \to 0$,

$$e = \lim_{\Delta t \to 0} -\frac{\Delta\phi}{\Delta t} = -\frac{d\phi}{dt} \qquad ...(3)$$

If the conductor is in the form of a coil, and the coil has N turns of wire, then

$$e = -N\frac{d\phi}{dt} \text{ or } e = -N\frac{\Delta\phi}{\Delta t} \qquad ...(4)$$

If the conductor is in the form of a closed circuit, then a current will be produced in the circuit due to induced electromotive force. If R be the resistance in the circuit, then the current i is given by

$$i = \frac{e}{R} = -\frac{1}{R}\frac{d\phi}{dt} \qquad ...(5)$$

Hence, the electrical charge induced in the circuit in time Δt is

$$q = i\Delta t = \frac{1}{R}\frac{\Delta\phi}{\Delta t}\Delta t = \frac{1}{R}\Delta\phi \qquad ...(6)$$

The magnetic flux (ϕ) linked with a coil can be changed by several ways. If a coil of surface area S is placed in a uniform field B with its plane normal to the field and if the coil is suddenly withdrawn from the field, then $\phi_1 = BS$ and $\phi_2 = 0$ and hence $\Delta\phi = \phi_2 - \phi_1 = -BA$. If the coil, placed normally in the magnetic field B, is rotated by angle $\pi/2$ (or 90°) about an axis perpendicular to B and the surface area of coil is S, then the change in the magnetic flux linked with the coil becomes $\Delta\phi = \phi_2 - \phi_1 = 0 - BS = -BS$ and if the coil is rotated by angle π, then the change in the magnetic flux becomes $\Delta\phi = -BS - BS = -2BS$. Thus, due to the continuous rotation of the coil in the magnetic field, the magnetic flux linked with the coil continuously changes, which sets up a continuous flow of current in the coil because of the e.m.f. induced in it. On the other hand if the coil is kept at rest and the magnetic field changes, it also produces a change in flux.

10·4. Lenz's Law and Fleming's Right Hand Rule

The direction of induced e.m.f. and hence the direction of induced current can be obtained by Lenz's law. *According to this law, the direction of the induced current (or induced e.m.f.) in a circuit is such that it always opposes the cause that produces it.* In Faraday's law of electromagnetic induction, the negative sign indicates this fact. The Lenz's law is based on conservation of energy.

For example, if we study the relative motion of a magnet and a coil, then according to Lenz's law, when the north pole of a magnet is brought towards the coil, the direction of induced current in the coil is anticlockwise as seen from the side of the magnet *i.e.*, the face of the coil near the N-pole of the magnet becomes a north pole which opposes the motion of the magnet towards the coil. Similarly, when the magnet recedes from the coil, S-pole is created on the face of the coil which attracts the N-pole of the magnet and thus opposes the receding.

The Lenz's law is applicable for the induced currents *i.e.*, it tells us the direction of the induced current in any closed conduction circuit. If the circuit is open, then we think that what will happen if the circuit were closed and thus, we can find the direction of the induced electromotive force.

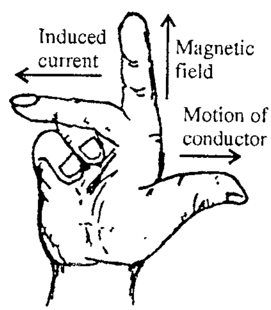

Fig. 10.4 : Fleming's right hand rule

Fleming's right and rule : If a conductor is movinŗ normally in a magnetic field, then the direction of the induced current can be found by **Fleming's right hand rule.** According to this law, *if we stretch the thumb, middle finger and the forefinger of our right hand mutually perpendicular to each other and if the forefinger indicates the direction of the magnetic field and the thumb indicates the direction of motion of the conductor, then the middle finger will indicate the direction of induced current.* **[fig. 10.4].**

10·5. Origin of Induced Electromotive Force

An electromotive force is induced due to the change in the magnetic flux linked with any conductor or coil. The magnetic flux ($\phi = \int_S \mathbf{B}\cdot d\mathbf{S}$) linked with any conductor can be changed by following methods :

(i) Keep the magnetic field constant with time and move the coil (or conductor).

(ii) Keep the coil at rest and change the magnetic field with time.

(iii) By the combined effect of (i) and (ii) *i.e.*, simultaneous motion of the coil (partly or whole) and change the magnetic field.

The mechanisms of electromotive force, produced in the two processes (i) and (ii), are different. We will stduy both of these mechanisms under the titles **motional e.m.f.** and **induced electric field.**

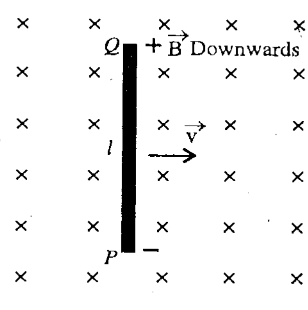

Fig. 10.5 : Motional e.m.f.

(1) Motional e.m.f. : Figure 10.5 shows a conducting rod of length l

moving in a uniform magnetic field normally with a velocity $\vec{v}$ towards right in the plane of the paper. The free electrons of the rod are also moving with the same velocity $\vec{v}$ and also perform the random motion in the rod. The average magnetic force is equal to zero due to the random velocity. Thus the magnetic field applies an average force $\vec{F}_m = q\vec{v} \times \vec{B}$ on each free electron, where q (=1.6×10^{-19} coulomb) is the charge on an electron. The force is along the QP direction, due to which the electrons move towards P. Therefore, negative charge is accumulated at P and the positive charge at Q. An electrostatic field $\vec{E}$ is produced from Q to P. This field exerts a force $\vec{F}_e = q\vec{E}$ on each free electron. The accumulation of charge continues till the total force on an electron will become zero *i.e.*, $\vec{F}_e + \vec{F}_m = 0$. The total force $\vec{F}_e + \vec{F}_m = q\vec{E} + q(\vec{v} \times \vec{B})$, acting on any charge q is called the **Lorentz force.** When the Lorentz force on an electron is zero, then

$$q\vec{E} + q(\vec{v} \times \vec{B}) = 0 \text{ or } qE - qvB = 0 \qquad [\because |\vec{v} \times \vec{B}| = vB \sin 90^\circ = vB]$$

i.e.,

$$E = vB \qquad ...(1)$$

Now, the resultant force on the free electrons of rod PQ becomes zero *i.e.*, no force acts on the free electrons of the rod. Hence, the potential difference between the ends Q and P is given by

$$V = El = vBl \qquad ...(2)$$

Thus, this is the magnetic force on the moving free electrons maintains the potential difference $V = vBl$ constant and hence it produces an electromotive force

$$e = vBl \qquad ...(3)$$

This electromotive force is produced due to the motion of a conductor, so it is called the **motional electromotive force** (or **motional e.m.f.**). If the ends P and Q of the rod are connected with an external resistance R, [**fig. 10.6(a)**] then an electric field is produced across the resistance due to the potential difference and the current starts to flow in the circuit. The electrons start to flow from P to Q in outer circuit and this tends to neutralize the charges collected at P and Q. The magnetic force qvB on the free electrons in QP rod accelerates the electrons towards P from Q to maintain the potential difference between both ends and hence the current.

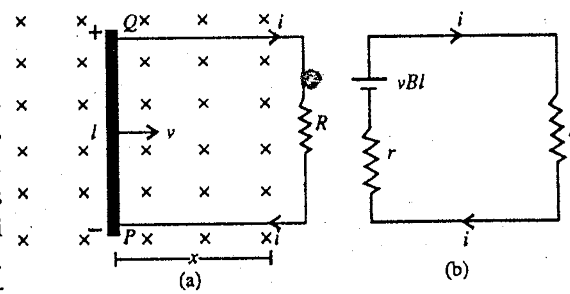

Fig. 10.6 : Flow of in induced current in the resistive circuit

Therefore, we can replace the rod QP moving in the magnetic field by a battery of electromotive force vBl in which positive end may be considered at Q and negative end at P. The resistance r of the rod QP may be considered as the internal resistance of the bettery. An equivalant circuit of this system is shown in fig **10.6(b)**. The induced current i in the clockwise direction is given by

$$i = \frac{vBl}{r + R} \qquad ...(4)$$

The induced electromotive force or induced current in the circuit of **fig. 10.6(b)** can be calculated by Faraday's law. If the length of the circuit in the magnetic field at time t is x, then the magnetic flux through the area bounded by the loop is given by

$$\phi = Blx \qquad ...(5)$$

Hence, the value of induced electromotive force is

$$e = \left|\frac{d\phi}{dt}\right| = Bl\frac{dx}{dt} = vBl \qquad ...(6)$$

It causes a current $i = \dfrac{vBl}{r + R}$ in the circuit, whose direction can be found clockwise using the Lenz's law.

(2) **Induced electric field and Induced electromotive force :** Let us assume that a conducting coil with single turn is kept in a magnetic field $\vec{B}$. Till time $t = 0$, the magnetic field remains constant and then it starts to vary with time. An induced current starts at time $t = 0$.

The free electrons of the conductor are at rest till $t = 0$. We are not interested in the random motion of the electrons. The magnetic field do not produce any force on the electrons at rest. Hence no current will be induced due to the magnetic force. The electrons can be accelerated only by an electric field. Hence, an electric field is produced as the magnetic field starts to vary at time $t = 0$. This electric field, is not produced by the charged particles according to Coulomb's law. The electric field produced by the varying magnetic field is of *non-electrostatic nature* and is a *non-conservative field*. We can not define any potential corresponding to this field. We call this field an **induced electric field.** The lines of force of induced electric field are in the form of closed curves. These lines have no starting and end points.

If $\vec{E}$ is an induced electric field, the force on a charge q placed in it will be $q\vec{E}$. The work done per unit charge on displacing it by $d\vec{r}$ is $\vec{E} \cdot d\vec{r}$, so the induced electromotive force produced in the coil is

$$e = \oint \vec{E} \cdot d\vec{r}$$

The presence of a conducting coil is not necessary to have an induced electromotive force. The electric field will be induced in any conductor as long as the magnetic field varies. If the conductor is in the form of a coil, the free electrons will drift and in consequence an induced electric current will flow.

10·6. Integral and Differenital Forms of Faraday's Law

(i) **Faraday's law in the integral form :** According to Faraday's law of electromagnetic induction, the induced electromotive force in any circuit is equal to the negative time rate of change of magnetic flux linked with that circuit *i.e.*, in mathematical form

$$e = -\frac{d\phi}{dt}, \text{ where } \phi = \int_s \vec{B} \cdot d\vec{S} \text{ is the magnetic flux} \quad ...(1)$$

If $\vec{E}$ be the induced electric field in the circuit, then the value of the induced e.m.f. is given by

$$e = \oint \vec{E} \cdot d\vec{r} \quad ...(2)$$

Hence, from (1) and (2) $$\oint \vec{E} \cdot d\vec{r} = -\frac{d}{dt}\left(\int_s \vec{B} \cdot d\vec{S}\right) \quad ...(3)$$

where electromagnetic force acts in the circuit which encloses area S. Equation (3) is known as the **Faraday's law in the integral form.**

(ii) **Faraday's law in the differential form :** According to Stoke's theorem,

$$\oint \vec{E} \cdot d\vec{r} = \int_s (\text{curl } \vec{E}) \cdot d\vec{S} \quad ...(4)$$

But from Faraday's law, $$\oint \vec{E} \cdot d\vec{r} = -\frac{d}{dt}\left(\int_s \vec{B} \cdot d\vec{S}\right) \quad ...(5)$$

Hence, $$\int_s \text{curl } \vec{E} \cdot d\vec{S} = -\frac{d}{dt}\left(\int_s \vec{B} \cdot d\vec{S}\right) \quad ...(6)$$

If $\vec{B}$ is a time varying function, then.

$$\int_s \text{curl } \vec{E} \cdot d\vec{S} = -\int_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad ...(7)$$

As above equation is valid for every surface, so the integrands of both sides of equation (7) must be equal, *i.e.*,

$$\text{curl } \vec{E} = -\frac{\partial \vec{B}}{\partial t} \qquad ...(8)$$

The above equation (8) is the **differential form of Faraday's law.** It should be noted that both the forms of Faraday's law are equivalent to each other.

10·7. Self-Induction

When there is a change in the current in a coil (or circuit), an e.m.f. is induced in that circuit itself which opposes the actual current due to which this e.m.f. is produced. This phenomenon is called **self-induction.** This name self-induction comes from the fact that the e.m.f. is induced in the coil by the change of current in the same coil.

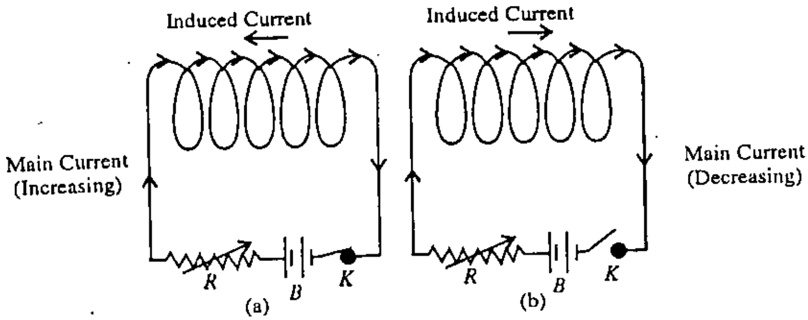

Fig. 10.7.

When a current starts to flow in a coil or loop, then the current increases with time. This current produces a magnetic field and hence there is the flux through the area bounded by the coil. As the current changes with time, the flux through the coil also changes with time, resulting in an induced e.m.f. in the coil. The current due to this induced e.m.f. opposes the actual current [**fig. 10.7(a)**]. Hence, the electric current in the coil takes more time to achieve its peak value. Further as the circuit is broken (or current flow is stopped), the magnetic flux passing through it decreases, due to which induced current flows in the coil in the direction of actual (main) current so as to oppose the decrease in the current in the coil [**fig. 10.7(b)**]. Due to the induced current, a **spark** is produced generally at that place where the circuit is broken. Some momentary current continues to flow in the same direction due to the sparking. This momentary current is called the **extra current** of self induction.

Self-inductance or Coefficient of self-induction : Due to a current in a coil, the magnetic field produced at a point is proportional to the current flowing in the coil. Hence, the magnetic flux (ϕ) passing through the area bounded by the current carrying coil, is proportional to current (i), *i.e.,*

$$\phi \propto i \text{ or } \phi = Li \qquad ...(1)$$

where L is called the **coefficient of self-induction** or **self-inductance**

If $i = 1$ ampere, then from equation (1), $L = \phi$, *i.e., the self-inductance of a coil is numerically equal to the magnetic flux linked with the coil when* 1 *ampere current flows in the coil.*

If the current flowing in the coil changes, the induced electromotive force (e.m.f.) produced due to the change in magnetic flux linked with the coil, is given by Faraday's law as

$$e = -\frac{d\phi}{dt}$$

Using equation (1),
$$e = -L\frac{di}{dt} \qquad ...(2)$$

Here, negative sign indicates that the induced e.m.f. opposes the change in current.

If $di/dt = 1$ ampere/second, then from equation (2), $L = |e|$, *i.e., the self-inductance of a coil may also be defined as the induced e.m.f. produced around the coil per unit rate of change of current with time flowing through it.*

The S.I. unit of self-inductance is **henry,** which is equal to 1 **weber/ampere** or 1 **volt-second/ampere** and may be defined as : *The self inductance of a circuit is said to be* 1 *henry if an induced e.m.f. of* 1 *volt is produced in it when the current flowing through it changes at the rate of* 1 *ampere per second.*

If we have a coil or solenoid with N turns, then the flux through each turn is $\int \vec{B} \cdot d\vec{S}$, where $\vec{B}$

is the magnetic field. If the flux is varying, an induced e.m.f. is produced in each turn and the total induced e.m.f. between the ends of the coil is the sum of these e.m.f. *i.e.*,

$$e = -N\frac{d\phi}{dt} = -N\frac{d}{dt}\int_s \vec{B}\cdot d\vec{S} \quad ...(3)$$

The value of self-inductance L can be found by equating equation (2) and (3)

The self-inductance of a coil depends upon the area of cross-section, shape, number of turns and the geometry of the winding *i.e.*, on the geometry of the coil. Any conductor, whether straight or bent, possesses self-inductance. A straight conductor possesses far less inductance than when it is bound in the form of a coil or solenoid. Further the self-inductance of a coil with iron core has an inductance μ (permeability) times that without an iron core.

A coil or solenoid made from thick wire has negligible resistance but sufficient inductance. Such an element is called as an ideal **inductor** and is repsesented as ⎍ꝏꝏꝏ⎍.

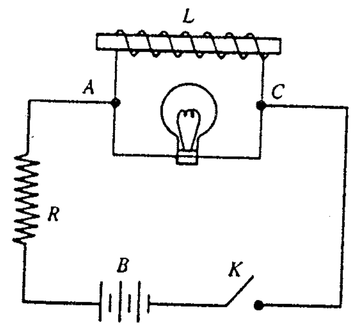

Fig. 10.8 : Demonstration of self-induction.

Practical Demonstration of Self-Induction : We take a coil L, obtained by winding a thin insulated copper wire on a soft iron rod. This is connected to a rheostat R, battery B and a tapping key K in series. An electric bulb E is connected in parallel to the coil between the points A and C [**fig. 10.8**].

Now, we press the key and find that the bulb E glows. Next if the key K is suddenly released, the bulb glows more brightly for a while and then it stops glowing. The reason of this more bright glow is as follows : As the circuit is broken, the current in the coil falls soon to zero, resulting in the fall (change) of magnetic flux in the coil. Hence a momentary and strong e.m.f. is induced in the coil due to self-induction, due to which for a moment a strong induced current flows in the bulb in the same direction in which the actual current was flowing. This is why the electric bulb glows more brightly due to self-induction.

Examples of the effect of self-induction : (1) *The coils in a standard resistance box are made non-inductive by winding the wire on doubling it* [**fig. 10.9**]. In such a case, in half part of the wire the current flows in one direction and in the other half part of the wire, the current flows in the opposite direction. Hence the magnetic fields due to the two parts of the wire cancel each other so that the magnetic flux linked with the coil is zero. In consequence, the effect of self-induction in the coil (resistance wire) is zero *i.e.*, due to self-induction the current in the circuit will neither decrease or increase.

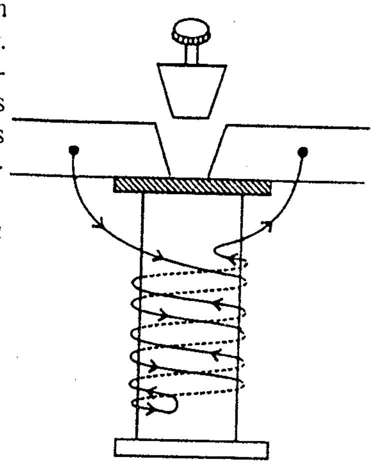

Fig. 10.9 : Doubling the resistance wire inside a standard resistance box.

(2) *In order to guard against the effect of self-inductance in a post office box or wheatstone bridge, the battery key is pressed before the galvanometer key* so that the current becomes first steady. If the galvanometer key were closed before the battery key the changing current may produce deflection in the galvanometer when a balance was actually obtained in the galvanometer.

10·8. Calculation of Self-Inductance in Some Simple Cases

(1) **Self-inductance of a plane coil :** Consider a plane circular coil of radius r with N turns. If it carries a current i, then the magnetic field at the centre of the coil is given by

$$B = \frac{\mu_0 Ni}{2r} \quad ...(1)$$

If we take the magnetic field to be uniform over the area of the coil, the magnetic flux linked with one turn of the coil is

$$\phi_1 = BA \qquad ...(2)$$

where $A = \pi r^2$ is the area of the plane of the coil. Hence the flux linked with N turns of the coil

$$\phi = N\phi_1 = NBA = N \cdot \frac{\mu_0 Ni}{2r} \cdot \pi r^2 = \frac{1}{2}\mu_0 \pi N^2 ri \qquad ...(3)$$

If L is the inductance of the coil, then

$$\phi = Li \qquad ...(4)$$

Comparing (3) and (4), we obtain

$$L = \frac{1}{2}\mu_0 \pi N^2 r \qquad ...(5)$$

(2) Self-inductance of a long current carrying solenoid : Consider a long solenoid of radius r with n turns per unit length and carrying an electric current i [**fig. 10.10**].

The magnetic field B produced inside the solenoid is given by

$$B = \mu_0 ni \qquad ...(1)$$

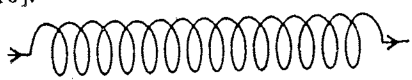

Fig. 10.10.

where μ_0 is the permeability of vacuum or air.

The flux (ϕ) passing through each turn of the solenoid is

$$\phi = BA = (\mu_0 ni)\, \pi r^2$$

$$\therefore \quad \text{Induced e.m.f. in each turn} = -\frac{d\phi}{dt} = -\mu_0 \pi n r^2 \frac{di}{dt} \qquad ...(2)$$

The number of turns in length l of solenoid will be nl. Hence, the induced electromotive force (e.m.f.) between the ends of the solenoid is

$$e = -(nl)\,(\mu_0 n\pi r^2)\, di/dt \qquad ...(3)$$

Comparing equation (3) with equation $e = -L\, di/dt$, we get

$$L = \mu_0 n^2 \pi r^2 l \text{ or } L = \mu_0 n^2 Al \qquad ...(4)$$

where $A = \pi r^2$ and is the cross-sectional area of the solenoid. It can be observed that the self-inductance of the solenoid depends on the geometrical factors *i.e.*, shape, size, medium, number of turns etc of the solenoid.

10·9. Magnetic Energy Stored in an Inductor

When we charge a capacitor, electric field builds up between its plates and energy is stored. In a similar way, when an inductor carries a current, a magnetic field builds up in it and magnetic energy is strored in it.

Let us consider an electric circuit shown in **fig. 10.11**. The circuit consists of an an inductance L, resistance R, a battery with an e.m.f. e and a key. As the key is closed, the current in the circuit increases and the magnetic field in the inductor also increases. Some part of the work done by the battery during this process is stored in the inductor as magnetic energy and remaining part appears in the form of heat in the resistance R. After sufficient time, the current as well as magnetic field become constant and further work done by the battery appears in the form of heat only. If i be the current in the circuit at time t, then

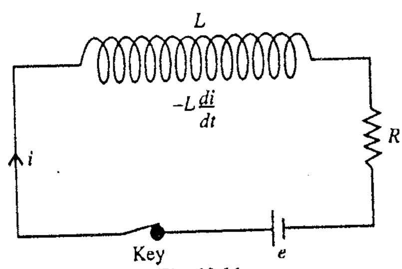

Fig. 10.11.

$$e - L\frac{di}{dt} = iR \qquad ...(1)$$

Multiplying equation (1) by idt and on integrating, we get

$$\int_0^t eidt = \int_0^t i^2 Rdt + \int_0^i Li\, di \qquad ...(2)$$

where the value of current becomes i starting from 0, in time $t = 0$ to $t = t$.

Hence,
$$\int_0^t eidt = \int_0^t i^2 Rdt + \frac{1}{2}Li^2 \qquad ...(3)$$

Now, idt ($= dq$) is the charge passing through the circuit during time dt, thus $eidt$ is the work done by the battery during this time interval (dt). So, the left hand side of equation (3) is the total work done by the battery during time $t = 0$ to $t = t$. Similarly, the first term of the right hand side of equation (3) represents the heat produced in the resistance R in time t. Thus, when current increases from 0 to i, the expression $\frac{1}{2}Li^2$ gives the magnetic energy stored in the inductor. Hence, the expression for the magnetic energy (U) stored in the inductor carrying current i will be

$$U = \frac{1}{2}Li^2 \qquad ...(4)$$

Energy stored in a constant magnetic field : Consider a solenoid of radius r and length l, having n turns per unit length. If it carries a current i, the magnetic field (B) inside the solenoid is given by

$$B = \mu_0 ni \qquad ...(5)$$

We assume that the magnetic field B inside the solenoid is uniform and constant and zero outside. The self-inductance (L) of this solenoid is

$$L = \mu_0 n^2 \pi r^2 l \qquad ...(6)$$

Hence, the magnetic energy stored in the solenoid is given by

$$U = \frac{1}{2}Li^2 = \frac{1}{2}\mu_0 n^2 \pi r^2 l i^2 \text{ or } U = \frac{1}{2\mu_0}(\mu_0 ni)^2 V = \frac{B^2}{2\mu_0}V \qquad ...(7)$$

where $V = \pi r^2 l$ is the volume enclosed by the solenoid.

Therefore, the energy stored per unit volume in the magnetic field *i.e.*, energy density u is obtained to be

$$u = \frac{U}{V} \text{ or } u = \frac{B^2}{2\mu_0} \qquad ...(8)$$

10·10. Mutual Induction

Whenever the magnetic field associated with any coil changes due to the change in current flowing through it, then a current is produced due to induction in the second coil placed close to it. This phenomenon is known as **mutual induction.** The coil in which the current flows is called **primary** and another coil in which the current is induced is called **secondary.**

Fig **10.12** shows two coils primary P and secondary S close to each other. The value of flux at any instant linked with the coils S depends upon its position and the current flowing (say i_1) in the coil P. If the relative position of the coil S is constant, then a momentary deflection is obtained in the galvanometer as the key of primary coil P is pressed. It is because on pressing the key, the current builds up in the coil P and produces a magnetic field around it. This magnetic flux starts to pass through the coil S, so that the magnetic flux changes through it and an electromotive force is induced in the coil S. Consequently an induced current flows in the secondary coil S which causes a momentary deflection in the galvanometer. According to Lenz's law, the direction of induced current in the coil S is opposite to the direction of current flowing in the coil P [fig. **10.12**].

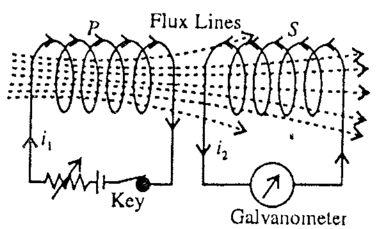

Fig. 10.12 : Mutual Induction.

As the key connected with coil P is released, again a momentary deflection in opposite direction is obtained in the galvanometer. The reason is that the flux lines decrease to zero and the direction of current in S is the same as that of the current flowing in the coil P. The cause of this flow of the induced current in the secondary coil S due to the change in the current in the primary coil P is mutual induction. Let us considet that i ampere current is flowing through the coil P, due to which the magnetic flux linked with each turn of the secondary coil is ϕ. If the secondary coil has N_2 turns, then $N_2\phi_2$ is the total magnetic flux linked with the secondary coil; the value of this flux will be proportional to the current i_1 flowing through the primary coil P, *i.e.*,

$$N_2\phi_2 \propto i_1, \text{ or } N_2\phi_2 = Mi_1 \qquad ...(1)$$

where the constant M is called the **mutual inductance** or **coefficient of mutual induction** of the two coils.

If e_2 is the induced electromotive force produced in the secondary coil due to the flow of current in the primary coil, then according to Faraday's law,

$$e_2 = -\frac{d}{dt}(N_2\phi_2) \quad ...(2)$$

where $\frac{d}{dt}(N_2\phi_2)$ is the rate of change of flux in the secondary coil due to the change of current in the primary coil.

But, $$N_2\phi_2 = Mi_1,$$

hence $$e_2 = -\frac{d}{dt}(Mi_1) \text{ or } e_2 = -M\frac{di_1}{dt} \quad ...(3)$$

If $di_1/dt = 1$ amp/sec, then $M = |e_2|$. Hence *the mutual inductance between two coils is numerically equal to the e.m.f. induced in the secondary coil when the rate of change of current in the primary coil is unity.* The S.I. unit of mutual inductance is **henry.**

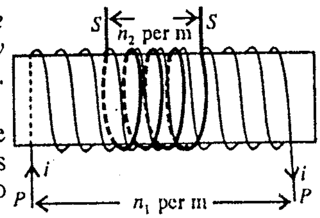

Fig. 10.13.

Mutual inductance between two coaxial long solenoids : Let PP be a solenoid of raduis r, having n_1 turns per metre. Another small coil SS is wound in the middle of the solenoid PP and the raduis of this coil is also r and it has n_2 turns per metre [**fig. 10.13**].

Now assume that i ampere current flows through the primary coil PP, then the magnetic field (B) inside the solenoid is

$$B = \mu_0 n_1 i \quad ...(4)$$

Hence, the flux linked with each turn of the secondary coil SS may be written as

$$B\pi r^2 = \mu_0 n_1 i \cdot \pi r^2$$

If the length of the secondary coil is l, then $n_2 l$ will be the number of turns in the secondary and the total flux linked with the secondary coil will be given by

$$\phi = \mu_0 n_1 i\pi r^2 n_2 l = (\mu_0 n_1 n_2 \pi r^2 l)\, i \quad ...(5)$$

But the total flux through the secondary coil is proportional to the current i in the primary *i.e.,*

$$\phi = Mi \quad ...(6)$$

Comparing (5) and (6), we have

$$M = \mu_0 n_1 n_2 \pi r^2 l \quad ...(7)$$

Thus, the mutual inductance between two coils depends on the geometerical factors as shape, size, number of turns in coils, and medium inside the coil.

10·11. Relation between the Mutual Inductance between Two Coils and their Self Inductances ($M = \sqrt{L_1 L_2}$)

Suppose that the number of turns per unit length in the primary coil is n_1, its length is l and area of cross-section is A. When a current i is passed through it, the magnetic flux linked with it is given by

$$\phi_1 = \mu_0 n_1 i\, (n_1 A) \quad ...(1)$$

where $(n_1 A)$ is the effective area of the coil.

$\therefore$ Self-inductance of the primary coil is

$$L_1 = \frac{\phi_1}{i} = \mu_0 n_1^2 A \quad ...(2)$$

Similarly, if n_2 is the number of turns per unit length of the secondary coil of the same cross-section, then self-inductance of the secondary coil (L_2) is

$$L_2 = \mu_0 n_2^2 A \quad ...(3)$$

Now by passing current i in the primary coil, the magnetic flux linked with the secondary coil S is

$$\phi_2 = \mu_0 n_1 i \times (n_2 A) \quad ...(4)$$

Hence, mutual inductance between the two coils is

$$M = \frac{\phi_2}{i} = \mu_0 n_1 n_2 A \quad ...(5)$$

$\therefore$ From equations (2) and (3)

$$L_1L_2 = \mu_0 n_1^2 A \cdot \mu_0 n_2^2 A = \mu_0^2 n_1^2 n_2^2 A^2 \quad ...(6)$$

From eq. (5),

$$M^2 = \mu_0 n_1^2 n_2^2 A^2,$$

Hence

$$L_1L_2 = M^2 \text{ i.e., } M = \sqrt{L_1L_2} \quad ...(7)$$

This is the desired relation between the mutual inductance for two coils and their self-inductances.

In the above deduction, we have assumed the ideal coupling between the coils in which the magnetic field produced due to the flow of current in the primary coil is completely linked with the secondary coil. However in general the magnetic field produced by the primary coil is not totally linked with the secondary coil. In such a case we can write eq. (7) as

$$M = k\sqrt{L_1L_2} \quad ...(8)$$

where k is called the coupling coefficient of the coils and for ideal coupling its value is unity *i.e.*, $k = 1$.

Comparison between the self-induction and mutual induction :

(i) The phenomenon of the production of induced e.m.f. in a coil on account of change of current flowing through the same coil known as **self-induction** whereas the production of induced e.m.f. in the secondary coil on account of change of current flowing through the primary coil is knwon as **mutual induction.**

(ii) The induced current has a direct effect on the main current of the coil in case of self-induction but it has no direct effect on the main current of the primary coil in case of mutual induction because the induced current flows in the secondary coil.

(iii) In case of self induction, we have only one coil but the phenomenon of mutual induction needs two coils.

10·12. Inductors in Series and Parallel

We consider two coils of self-inductances L_1 and L_2, separated by large distance so that there is no flux linkage between them *i.e.*, the mutual inductance between these coils is negligible.

(a) Series arrangement : If the two coils of self-inductances L_1 and L_2 are connected in series and a current i is passed through them, then the flux linked with the two coils is the sum of the fluxes linked with the individual coils *i.e.*,

$$\phi = \phi_1 + \phi_2 \quad ...(1)$$

But,

$$\phi_1 = L_1 i \text{ and } \phi_2 = L_2 i$$

Hence

$$\phi = L_1 i + L_2 i = (L_1 + L_2)\, i \quad ...(2)$$

If L be the *equivalent inductance of the two coils in series*, then

$$\phi = Li \quad ...(3)$$

From (2) and (3), we obtain

$$L = L_1 + L_2 \quad ...(4)$$

(b) Parallel arrangement : When two coils L_1 and L_2 are connected in parallel between two points and a current (i) is divided between them. Then

$$i = i_1 + i_2 \quad ...(5)$$

where i_1 is the current flowing in L_1 and i_2 in L_2.

Hence

$$\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt} \quad ...(6)$$

However in the parallel arrangement the induced e.m.f. at the ends of the two coils will be the same. This means that

$$e = -L_1\frac{di_1}{dt} = -L_2\frac{di_2}{dt} \quad ...(7)$$

If the *equivalent self-inductance of the parallel arrangement of the inductance is L*, then

$$e = -L\frac{di}{dt} = -L\frac{d}{dt}(i_1 + i_2) = -L\frac{di_1}{dt} - L\frac{di_2}{dt} \text{ or } \frac{e}{L} = -\frac{di_1}{dt} - \frac{di_2}{dt} \quad ...(8)$$

But, from (7) $-di_1/dt = e/L_1$ and $-di_2/dt = e/L_2$, hence

$$\frac{e}{L} = \frac{e}{L_1} + \frac{e}{L_2} \text{ or } \frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} \text{ or } L = \frac{L_1 L_2}{L_1 + L_2} \qquad ...(9)$$

10·13. Transformer

Using the principle of mutual induction, a high AC voltage can be obtained from a low-voltage AC source or a low AC voltage from a high-voltage AC source. This device is called **transformer**. There are two types of transformers.

(1) **Step-up transformer** : This transformer changes low voltage AC to high voltage AC.

(2) **Step-down transformer** : This transformer changes high voltage AC to low voltage AC.

In the above mentioned changes of voltages by transformers, the frequency of the alternating current (AC) remains unaltered. Remember that a transformer can be used for an alternating current (AC) not for direct current (DC).

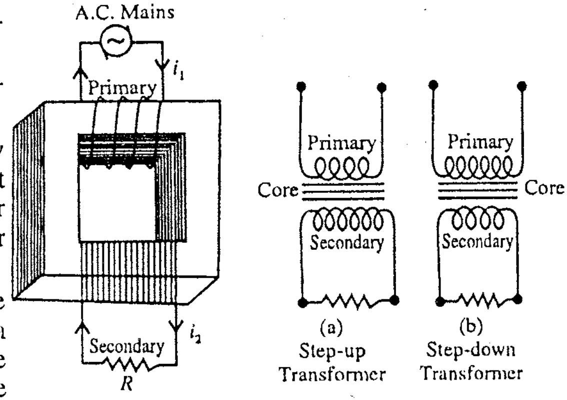

Fig. 10.14 : Transformer.

Construction : The design of a simple transformer is shown in **fig. 10.14**. It consists of a thick rectangular laminated iron core, which is made by placing soft iron laminated strips one above the other. Due to lamination, the soft iron strips remain sepearated from each other so that the eddy currents in the core are reduced resulting in small loss of electrical energy in the transformer. On this core, two coils of copper wire are wound which are kept sepearated from each other and iron core. One of the coils is called **primary** and the other as **secondary.** In the step-up transformer, the number of turns in the primary coil is less than the number of turns in the secondary coil, while in the step-down transformer, the number of turns in the primary coil is more than the number of turns in the secondary coil [**fig. 10.14**].

Principle of working : When the main source of the alternating e.m.f. is connected with the primary coil, an induced e.m.f. is produced between the ends of the secondary coil, which is used to derive a current in any circuit.

Suppose the number of turns in the primary and secondary coils are N_1 and N_2 respectively. If an alternating e.m.f. e_1 is applied between the ends of the primary coil, a current is produced in the primary circuit and a current i_2 in the secondary circuit. The currents in the coils produce a magnetization in the core of soft iron and there is corresponding magnetic field, say B, inside the core. Here we assume that the magnetic field is uniform inside the core and its flux (BA) remains the same in each turn of the both primary and secondary coils. Let ϕ be the flux in each turn. Then e.m.f. induced in the primary coil is $-N_1 d\phi/dt$ and e.m.f. induced in the secondary coil is $e_2 = -N_2 d\phi/dt$.

If the resistance of the primary coil is negligible, then applying the Kirchoff's law to the primary circuit, we have

$$e_1 - N_1 \frac{d\phi}{dt} = 0 \text{ or } e_1 = N_1 \frac{d\phi}{dt} \qquad ...(1)$$

and

$$e_2 = -N_2 \frac{d\phi}{dt} \qquad ...(2)$$

From eqs. (1) and (2), we obtain

$$\frac{e_2}{e_1} = -\frac{N_2}{N_1} \text{ or } e_2 = -\frac{N_2}{N_1} e_1 \qquad ...(3)$$

Here, negative sign shows e_2 has a phase difference of 180° with e_1. In magnitude,

$$\frac{e_2}{e_1} = \frac{N_2}{N_1} = r \qquad ...(4)$$

where r is called the **transformation ratio.** The value of r is more than 1 in case of step-up transformer and is less than 1 for step-down transformer.

Power Transfer : First we consider the situation in which the secondary coil is not connected with any external circuit. Thus the secondary circuit is incomplete and there is no flow of current through it. Let i_0 be the current in the primary coil in this case. We consider the resistance of the primary coil to be zero, so it is a purely an inductive circuit. The current has a phase difference of $\pi/2$ with the applied e.m.f. e_1 and hence the power supplied by the alternating current source will be zero. The power in the secondary circuit will also be zeo as there is no current in this circuit.

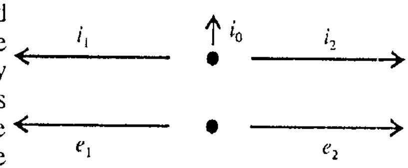

Fig. 10.15.

Now if the terminals of the secondary coil are connected to a resistance R, an alternating current i_2 starts to flow through it. Hence there will be extra e.m.f.'s in both the primary and secondary coils due to i_2. But, the resultant induced e.m.f., in the primary coil should be equal and opposite to the e.m.f. e_1 [eq. (1)]. Hence, there is an extra current i_1 in the primary circuit which will cancel the induced e.m.f. due to the current i_2. Thus, the current in the primary coil is $i_0 + i_1$ and in the secondary coil i_2. The e.m.f. in the secondary coil remains e_2 as given by equation (2).

As the induced electromotive forces due to i_1 and i_2 in the primary coil cancel each other, hence the two alternating currents should be at 180° phase difference. Current i_2 is in phase with e.m.f. e_2 (purely resistive circuit) and e_1 is 180° out of phase with the e.m.f. e_2 according to equation (3). Thus i_1 is in phase with e_1 **[fig. 10.15]**.

Hence, the component i_0 of the current $i_0 + i_1$ in the primary circuit is at a phase difference of $\pi/2$ from the applied e.m.f. e_1 and second component i_1 is in phase with e.m.f. e_1. The power supplied by the alternating current source will be $e_1 i_1$ and the power taken by the secondary circuit will be $e_2 i_2$. If there is no energy loss, then

$$e_1 i_1 = e_2 i_2 \qquad ...(5)$$

Using equation (3)

$$i_2 = -\frac{N_1}{N_2} i_1 \qquad ...(6)$$

Here, negative sign shows that i_2 has a phase difference of π with e_1.

Generally, the extra current i_1 is more than the initial current i_0 in the primary coil. It can be shown easily by connecting a bulb in series in the primary circuit. The bulb glows more brightly only when the secondary circuit is completed with respect to the open circuit. If i_0 is negligible, then eq. (6) relates the net currents in the two circuits.

Step-up and Step-down Transformers : From eqs. (3), (5) and (6) (in magnitude)

$$\frac{e_2}{e_1} = \frac{N_2}{N_1} = \frac{i_1}{i_2} \qquad ...(7)$$

When $N_2 > N_1$, the e.m.f. e_2 in the secondary circuit is more than the primary e.m.f. e_1. This is the case of **step-up transformer.** In such a case, the secondary current is less than the primary current. Therefore, the primary coil is made of a thick wire to sustain high current.

When $N_2 < N_1$, the e.m.f. in the secondary is less in magnitude than the primary e.m.f. This is the case of **step-down transformer.** In this case, the secondary current is more than the primary current. This is why the wire of secondary coil is made thick to carry high current.

Energy Losses in a Transformer

Practically, the energy obtained from the secondary coil is little less than the energy supplied to the primary coil, hence there is some energy loss in the transformer. There are following four forms of the energy loss in a transformer.

(a) **Iron loss :** Some eddy currents are produced in the transformer due to which the energy is wasted in the form of heat. This is called the **iron loss.** The core is taken laminated to reduce the eddy currents and hence the iron loss.

(b) **Copper loss :** Some part of the energy supplied to the primary coil is wasted in the form of heat in its wires. This is known as **copper loss.** To reduce this loss, the wires of the primary coil in a step-up transformer and the wires of the secondary coil in step-down transformer are made thick.

(c) **Hysterisis loss :** Due to the flow of AC, the iron core is magnetized and demagnetized again and again and hence some part of the energy is wasted in this process. This is known as **Hysterisis loss.** The core is made of soft iron to reduce this loss.

(d) **Loss due to leakage of magnetic flux :** As the total magnetic flux of the primary coil is not linked with the secondary coil, so there is some loss of energy in the form of heat due to the leakage of magnetic flux. The soft iron-core is taken in the form of a closed frame to reduce this loss.

Thus we see that heat is produced due to the loss of energy in a transformer. In order to absorb this heat, oil or cold air is passed around the transformer.

Efficiency of Transformer

There is some loss of energy in a normal transformer due to the resistance, hysteresis and eddy currents etc. The efficiency of a tranformer is defined as *the ratio of energy obtained from the secondary coil to the energy supplied to the primary coil i.e.,*

$$\eta = \frac{\text{Output power}}{\text{Input power}} = \frac{e_2 i_2}{e_1 i_1}$$

The efficiency of an ideal tranformer is 1 (or 100%) but in practice, due to loss of energy, it is always less than 1 or less than 100%.

Transmission of Power : The voltage of an alternating current may be step-down or step-up with the help of a transformer. This fact is used in power transmission at the generating stations. These generating stations usually are very far from the place where the power is used. Sometimes, very long wires (upto hundreds of kilometeres) are used to transmit this power, which causes the loss of power in the form of heat (i^2Rt) due to the resistance of the carrying wires. A generating station can supply fixed power according to its capacity. If we supply this power at high voltage, then the loss of power in transmission becomes less due to less current. Therefore the voltage is stepped-up with the help of step-up transformer at the generating station and then it is supplied to transmission lines. To use this transmitted power at any place, the voltage is stepped-down to appropriate value with the help of a step-down transformer.

Uses of transformer : Following are the uses of transformers :

(i) Transformers are used in electrical appliances which operate at a voltage other than the supplied voltage by mains.

(ii) Step-up transformers are used in power transmission at generating stations and step-down transformers are used at power sub-stations to step down the voltage before its distribution.

(iii) Step-up transformers are also used in T.V., in wireless sets and X-ray tubes to provide a high alternating voltage.

(iv) Step-down transformers are used in electric bells, in radio-sets for the valve heaters with night lamps (to reduce 220 V to 6 V).

(v) For welding, a strong alternating current is required at a low voltage, for which a step down transformer is used.

Ex. 1. *A* 1 *metre long wire is moving with velocity* 10 *cm/sec. normal to the magnetic field of intensity* 1000 *gauss. Calculate the value of induced electromotive force.* [10^4 *gauss* = 1 *tesla* (*S.I. unit*)].

Sol.
$$e = -\frac{d\phi}{dt} = -\frac{d}{dt}(BA)$$

If the magnetic field is uniform and the area A is varying, then

$$e = -B\frac{dA}{dt}$$

Here, $B = 10^3$ gauss = 0.1 tesla, length of wire, l = 1 metre and velocity of wire v = 0.1 m/sec.

Now, $dA/dt = lv = 1 \times 0.1 = 0.1$ m/sec.

Therefore, $|e| = B\ dA/dt = 0.1 \times 0.1 =$ **0.01 volt.**

Ex. 2. *A coil of area of cross section* 100 *cm*^{2}*, having* 50 *turns of wire is placed in a magnetic field of intensity* 2×10^{-2} *T with its plane normal to the field. If the coil is withdrawn from the field in* 0.1 *second, calculate the e.m.f. induced in the coil.* **(Raipur 2003)**

Sol. Here, $A = 100\ cm^2 = 10^{-2}\ m^2$, $n = 50$, $B = 2 \times 10^{-2}$ T

$$\text{Initial flux, } \phi_1 = BnA = 2 \times 10^{-2} \times 50 \times 10^{-2} = 10^{-2}\ \text{Wb}$$

$$\text{Final flux, } \phi_2 = 0;\ \text{Time } \Delta t = 0.1\ \text{sec.}$$

$$\therefore \quad \text{Induced e.m.f. } e = -\frac{\Delta\phi}{\Delta t} = -\frac{(\phi_2 - \phi_1)}{t} = -\frac{(0 - 10^{-2})}{0.1} = \mathbf{0.1\ volt.}$$

Ex. 3. *A circular coil having 10000 turns is of area 0.02 metre. Keeping its plane normal to a magnetic field of intensity 5.0×10^{-5} Wb/m², it is rotated by 90° in 0.2 second. Calculate the maximum magnetic flux linked with the coil and average electromotive force induced due to rotation of coil.*

Sol. The magnetic flux passing through each turn of the coil of area A situated normal to magnetic field B is given by

$$\phi = BA$$

Here, $B = 5.0 \times 10^{-5}\ \text{Wb/m}^2$ and $A = 0.02\ m^2$

$$\therefore \quad \phi = (5.0 \times 10^{-5}) \times (0.02) = 10^{-6}\ \text{weber.}$$

The magnetic flux passing through coil is reduced to zero on rotaing it by 90°. Hence the change in magnetic flux

$$\Delta\phi = \phi_2 - \phi_1 = 0 - 10^{-6} = -10^{-6}\ \text{weber}$$

This flux change takes place in time $\Delta t = 0.2$ second.

Therefore, the induced in the coil is

$$e = -N\Delta\phi/\Delta t = 10000 \times 10^{-6}/0.2 = \mathbf{0.05\ volt.}$$

Ex. 4. *A closed coil is formed by a rectangular frame of area 4.0 cm² with 500 turns on it and the resistance of coil is 50 ohm. Keeping its plane normal to a magnetic field of intensity 0.2 Wb/m², it is rotated by 180°. Calculate the charge flowing through it. Will this answer depend on the speed of rotation of the coil ?*

Sol. The change in magnetic flux $\Delta\phi = \phi_2 - \phi_1$

If the area of the coil is A, then

$$\phi_1 = \vec{B} \cdot \vec{A} = BA\cos\theta = BA \quad [\because \theta = 0] \text{ and } \phi_2 = -BA \qquad [\because \theta = 180°]$$

$$\therefore \quad \Delta\phi = \phi_2 - \phi_1 = -BA - BA = -2BA$$

$$\text{Hence,} \quad e = -N\frac{\Delta\phi}{\Delta t} = \frac{-N(-2BA)}{\Delta t} = \frac{2NBA}{\Delta t}$$

Here N is the number of turns in the coil. If current i flows due to above e.m.f., then

$$i = \frac{e}{R} = \frac{2NBA}{R\Delta t}$$

If q be the change flowing in time Δt, then

$$q = i\Delta t = 2NBA/R$$

Here, $N = 500$, $B = 0.2\ \text{Wb/m}^2$, $A = 4 \times 10^{-4}\ \text{metre}^2$ and $R = 50$ ohm,

$$\therefore \quad q = (2 \times 500 \times 0.2 \times 4 \times 10^{-4})/50 = \mathbf{1.6 \times 10^{-3}\ coulomb.}$$

If the coil is rotated rapidly by 180°, the change in magnetic flux will remain the same, but due to increase in rate of change of magnetic flux, the e.m.f. induced and so the induced current will increase. However due to the decrease in time Δt, the charge $i\Delta t = q = 2NBA/R$ flowing through the coil will remian the same. Thus the answer will not depend on the speed of rotation of the coil.

Ex. 5. *Calculate the e.m.f. induced between the ends of 1.75 m long axle of a railway carriage travelling north-south on level ground with a uniform speed of 50 km/hour. The vertical component of earth's magnetic field V is given to be 0.5 gauss.*

Sol. The horizontal axle of the carriage is perpendicular to the vertical component of the earth's magnetic field V and it moves perpendicular to its length as well as to v. The induced e.m.f. can therefore be computed as follows :

$$e = lvB = 1.75 \times (50 \times 10^3) \times 60 \times 60 \times 5 \times 10^{-5} = \mathbf{1.2 \times 10^{-3}\ volt.}$$

The horizontal component 'H' of the earth's magnetic field induces no e.m.f. in the axle rod since it is collinear with its velocity and there is no rate of change of flux.

Ex. 6. *A coil is placed in a magnetic field B such that the direction of magnetic field is normal to the plane of the coil. The magnetic flux ϕ linked with the coil depends on the time t as follows.*

$\phi = (3t^2 - 10t + 11)$ *milliweber. Calculate the e.m.f. induced in the coil at t = 3 second.*

Sol. Here, $\phi = (3t^2 - 10t + 11)$ milliweber

$\therefore$ Induced e.m.f., $e = -\dfrac{d\phi}{dt} = -\dfrac{d}{dt}(3t^2 - 10t + 11)$ *i.e.*, $e = -(6t - 10)$ milli volt

At t = 3 second, $e = -[(6 \times 3) - 10] = -$ **8mV**

Ex. 7. *A copper rod of length L is rotating about one of its ends with an angular velocity ω perpendicular to a magnetic field of intensity B. Prove that the electromotive force induced between the ends of the rod is given by* $|e| = \frac{1}{2}\omega BL^2$.

Sol. As the rod of length L is rotating about one of its ends with an angular velocity ω perpendicular to a magnetic field B, the rod completes $\omega/2\pi$ rotations per second. As the area covered by the rod in one rotation is πL^2, it covers area $(\omega/2\pi)\,\pi L^2$ or $\omega L^2/2$ in one second (*i.e.*, in $\omega/2\pi$ rotations). Thus

$$\frac{dA}{dt} = \frac{\omega L^2}{2}$$

As the field B is constant, the e.m.f. induced due to the rotation of the rod is

$$|e| = \frac{d\phi}{dt} = \frac{d}{dt}(BA) = \frac{BdA}{dt} = \frac{B\omega L^2}{2}.$$

Ex. 8. *The coefficient of self-induction of a coil is 0.4 mH. The current flowing through it changes by 250 milliampere in 0.1 second. Calculate the induced electromotive force.*

Sol. Induced e.m.f., $e = -L\,\Delta i/\Delta t$

Here, $L = 0.4$ mH $= 0.4 \times 10^{-3}$ henry, $\Delta i = 250$ mA $= 0.250$ ampere and $\Delta t = 0.1$ second.

$\therefore$ $e = -(0.4 \times 10^{-3}) \times 0.25/0.1 = -1 \times 10^{-3}$ volt $=$ **– 1 millivolt.**

Ex. 9. *The mutual inductance between the two coils C_1 and C_2 placed closed to each other is 20 mH. If the current in the coil C_1 changes from 5A to 2A in 0.01 second, calculate the induced e.m.f. and induced charge in the coil C_2 assuming the resistance of coil C_2 to be 10 ohm.*

Sol. Induced electromotive force (e.m.f.) in coil C_2,

$$e_2 = -M\Delta i_1/\Delta t = -(20 \times 10^{-3}) \times (2 - 5)/0.01 = \mathbf{6\ volt.}$$

Induced charge in coil C_2,

$$q_2 = -M\Delta i_1/R = -(20 \times 10^{-3}) \times (2 - 5)/10 = \mathbf{6 \times 10^{-3}\ coulomb.}$$

Ex. 10. *If the current 3.0 A flowing in the primary coil is made zero in 0.1 sec, the e.m.f. induced in the secondary coil is 1.5 volt. Calculate the mutual inductance between the coils.*

(Ujjain 2003; Jabalpur 03)

Sol. Here, $e = 1.5$ volt, $\Delta i = 0 - 3.0 = -3.0$ A and $\Delta t = 0.1$ sec.

If M is the mutual inductance between the coils and the current in the primary coil changes by Δi in time Δt, then the e.m.f. induced in the secondary coil is

$$|e| = M\,\Delta i/\Delta t$$

$\therefore$ $1.5 = M \times (3.0)/0.1$ or $M = 1.5 \times 0.1/3.0 =$ **0.05 henry.**

Ex. 11. *The ratio of number of turns in the primary and secondary coils of a step-up tranformer is 1 : 20.* (i) *If it is connected with the main line of 200 volt, what voltage will be obtained from it ?* (ii) *If the current obtained in the secondary coil is 2A, find the current flowing in the primary coil.*

(Ujjain 2004)

Sol. Here, $N_1/N_2 = 1/20$, $V_1 = 200$ volt and $i_2 = 2$A

(i) $$\frac{e_2}{e_1} = \frac{N_2}{N_1} \text{ (numerically)}$$

$\therefore$ $$e_2 = e_1\left(\frac{N_2}{N_1}\right) = 200 \times \left(\frac{20}{1}\right) = \mathbf{4000\ volt.}$$

(ii) Also, $\frac{i_2}{i_1} = \frac{N_1}{N_2}$

$\therefore$ $i_1 = (N_2/N_1)\, i_2 = 20 \times 2 = \mathbf{40\ A.}$

Ex. 12. *If in a transformer, the current in the primary coil reduces to zero from 0.5 ampere in 1 millisecond, then an e.m.f. of 500 volt is induced in the secondary coil. Calculate the mutual inductance coefficient.*

Sol. Induced e.m.f., $e = -M\Delta i_1/\Delta t$; $\therefore M = -e/\Delta i_1/\Delta t$

Here, $e = 500$ volt and $\Delta t = 0.001$ second

$\therefore$ $M = -\frac{500}{(0-0.5)/0.001} = -\frac{500 \times 0.001}{0.5} = -1$ henry or $|M| = $ **1 henry.**

EXERCISE 10·1

1. Calculate the potential difference induced between the ends of a horizontal wire of length 1 metre after 2 seconds when it starts to fall from rest. (The horizontal component of the earth's magnetic field, $H = 0.2$ gauss)
[**Ans.** 3.924×10^{-4} volt.] (Remember : 10^4 gauss = 1 tesla)
2. A circular loop is kept in an uniform magnetic field of strength $B = 0.02$ tesla normally. If the radius of the loop is shrinking at the rate of 1 mm/second any how, then calculate the induced electromotive force at that instant when the radius of loop is 2 cm.
[**Hint :** $\phi = \pi r^2 B$, for uniform B, $d\phi/dt = 2\pi rB dr/dt$]
[**Ans.** 2.5×10^{-6} volt.]
3. A coil having 25 turns is of area 1.6 cm^2. It is kept in a magnetic field of 1.8 Wb/m^2 in 0.3 second such that its plane remains perpendicular to the flux-lines of the field. What is the induced e.m.f. in the coil ? If the resistance of the coil is 10 ohm, then how much charge will flow through it ?
[**Ans.** 2.4×10^{-2} volt., 7.2×10^{-4} coulomb.]
4. A conducting rod of length 15 cm can rotate about a peg attached at its end. If it is rotating at a rate of $33\frac{1}{3}$ revolutions per minute in a horizontal plane and the vertical component of earth's magnetic field is 0.01 weber/metre^2, calculate the potential difference induced across the ends of rod. **(Sagar 2003)**
[**Ans.** 3.925×10^{-4} volt.]
5. A disc of gramophone is made of brass metal and is of diameter 30 cm. It is rotating in a horizontal plane at the rate of $33\frac{1}{3}$ turns per minute. If the vertical component of the earth's magnetic field is 0.01 Wb/m^2, then calculate the induced e.m.f. between the centre and circumference of the disc.
[**Ans.** 3.9×10^{-4} volt.]
6. The length of the wing of an aeroplane is 20 metre. It is moving with a speed 250 metre/second in south direction parallel to the earth's surface. If the horizontal component of earth's magnetic field is 2×10^{-5} Wb/m^2 and the angle of dip is 60°, calculate the e.m.f. induced between the ends of its wing.
[**Ans.** 0.1732 volt.]
7. A ring of radius 10 cm and resistance 2 ohm, is kept normal to a uniform magnetic field. If the magentic field decreases at a rate of 0.1 tesla per second, then calculate the value of induced e.m.f. and induced current in the ring.
[**Ans.** 3.14 millivolt, 1.57 milliampere.]
8. A 1.2 m wide railway line lies in magnetic meridian. The vertical component of earth's magnetic field is 0.5 oersted. If a train moves with a speed 60 km/hour on the railway line, find the e.m.f. induced. [**Hint :** 1 oersted = 1 gauss = 10^{-4} tesla (S.I. unit)] **(Bundelkhand 2004)**
[**Ans.** 1 mV.]

9. The two rails of a railway track insulated from each other and the ground, are connected to a millivoltmeter. What is the reading of the millivoltmeter when a train travels at a speed of 180 km/hour along the track, given that the horizontal component of earth's magnetic field is 0.2×10^{-4} weber/m^2, and the rails are separated by 1 m ?
[**Ans.** 1 millivolt.]

10. A uniform magnetic field of 40×10^{-4} Wb/m^2 is applied perpendicular to the plane of coil of radius 0.02 m and of 500 turns. If the magnetic field is reversed in 0.01 second, calculate the average induced e.m.f. in coil.
[**Ans.** 5.03 volt.]

11. A fan blade of length $2a$ rotates with frequency f cycle per second perpendicular to a magnetic field B. Find the potential difference between the centre and end of the blade.
[**Ans.** $e = -\pi B a^2 f$.]

12. A rod MN of length l moves with a uniform velocity v parallel to a long straight wire carrying a current i as shown in **fig. 10.16**. What is the e.m.f. induced across the two ends of this rod.

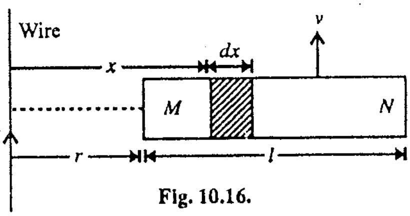

Fig. 10.16.

$$\left[\textbf{Ans. } e = \frac{\mu_0 i v}{2\pi}\ln\left(\frac{r+l}{r}\right)\right]$$

$$\left[\textbf{Hint : } de = \frac{\mu_0 i}{2\pi x} v\, dx, \text{ and } e = \int_r^{r+l} de\right]$$

13. A coil of wire of certain radius has 600 turns and a self inductance of 108 mH. What will be self-inductance of a second similar coil with 500 turns ?
[**Ans.** 75 mH.]
[**Hint :** $L \propto n^2$.]

14. Calculate the self-inductance of a wire having 2000 turns if a current of 5 ampere produces a magnetic flux of 0.5 weber through the coil.
[**Ans.** 0.5 henry.]

15. Two parallel wires AL and BM are l distance apart and are connected through resistance R. They are kept in a magnetic field which is perpendicular to the plane containing the wire. Another wire CD is connected perpendicular to both wires and it is moving with velocity v [**fig. 10.17.**]. Calculate the work done per second to move the wire CD if the resistance of all wires is zero.

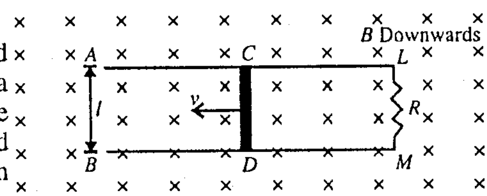

Fig. 10.17.

[**Ans.** $W = B^2 v^2 l^2 / R$.]
[**Hint :** $i = \dfrac{e}{R} = \dfrac{Bvl}{R}$ and $F = ilB$, $\therefore W = F \times v$.]

16. A coil is made by winding the wire on a soft iron rod of length 0.1 m and radius 0.01 m. If the relative permeability of soft iron is 1200, and the self-inductance of coil is 0.25 H. Calculate the number of turns in the coil.
[**Ans.** 230.]

$$\left[\textbf{Hint : } L = \frac{\mu_0 \mu_r N^2 A}{l}.\right]$$

17. In a coil of self inductance 5 mH, the current rises from 0 to 1 A in 0.1 second at a uniform rate. Calculate the magnitude of e.m.f. induced.
[**Ans.** 0.05 V.]

18. The ratio of number of turns in the primary and secondary coils of a step-up transformer is 1 : 10.

If it is connected to a main line of 220 volt carrying current 2 ampere, find the e.m.f. and current across the secondary coil.
[**Ans.** 2200 volt, 0.2 ampere.]

19. A transformer is used to glow a 100 W-240 V bulb at 240 V A.C. If current in the primary coil is 0.5 A, calculate the efficiency of the transformer.
[**Ans.** 0.83.]
[**Hint :** Input power = $e_1 i_1 = 240 \times 0.5 = 120$ watt; output power = 100 watt; $\eta = \frac{\text{Output power}}{\text{Input power}}$ $= \frac{100}{120} = 0.83$.]

20. 2.0 kW power is transmitted by a line of resistance 10 ohm at 40 kV. What will be the power loss ?
[**Ans.** 0.025 watt.]
[**Hint :** Power loss = $i^2 R$, where $i = \frac{P}{V} = \frac{2 \times 10^3}{40 \times 10^3} = 0.05$ A and $R = 10$ ohm.]

21. A step-down transformer changes the transmitted line voltage 2200 volt to 220 volt. The number of turns in the primary coil is 5000. Its efficiency is 90% and the output power is 10 kilo-watt. Calculate : (i) the number of turns in the secondary coil, (ii) input power.
[**Ans.** (i) 500, (ii) 11.1 kilo-watt.]

QUESTIONS

Long Answer Questions

1. What do you understand by electromagnetic induction ? Discuss Faradays's law of electro-magnetic induction and find an expression for induced e.m.f. **(Rewa 2004; Bilaspur 03)**
2. Discuss Lenz's law ? How would you determine the direction of induced current using Lenz's law.
3. What do you understand by the induced electric field ? How does it differ from the static electric field and how is it related to the induced electromotive force. ?
4. Discuss Faraday's law of electromagnetic induction. Obtain the differential and integral forms of Faraday's law. **(Indore 2004; Jabalpur 04; Agra 04; Sagar 03)**
5. Explain the meaning of self-induction. What is meant by self-inductance of a coil. State its unit and dimensional formula. **(Indore 2004)**
6. What is self-induction ? Define the coefficient of self-induction. Obtain an expression for the coefficient of self-induction of a long solenoid.
7. (a) Show that the energy spent in establising a current from zero to I in a coil of self-inductance L is $\frac{1}{2} LI^2$. **(Raipur 2003)**
(b) Deduce an expression for the energy stored is a static magnetic field B. **(Jabalpur 2003)**
8. What is mutual induction ? Define the coefficient of mutual induction. Obtain and expression for the coefficient of mutual induction of two coaxial solenoids.
9. What is a transformer ? How many types of tranformers you know ? Discuss its construction and working. How is it used in power transmission from one place to far situated another place ?
10. Draw labelled diagrams for a step-up transformer and a step-down transformer and differentiate between them. **(Jabalpur 2003)**
11. On what principle does a transformer work ? How are the transformer used in transmission of electrical energy to the long distances ? **(Raipur 2003)**
12. Explain the construction and working of transformers. What are the causes of loss of energy in a transformer ? How are they minimised ?
(Indore 2003; Rewa 04; Bholpal 04; Bilaspur 03; Ujjain 04)
13. Derive an expression for mutual inductance of two solenoids. **(Agra 2006)**
14. (a) The self-inductances of two inductive coils are L_1 and L_2. Show that when they are joined in series, the equivalent inductance is $L_1 + L_2$ and when they are joined in parallel, the equivalent inductance is $\frac{L_1 L_2}{(L_1 + L_2)}$. **(Ujjain 2003)**

(b) The self-inductance of two coils P and S are L_1 and L_2. The coupling between them is ideal. Show that the mutual inductance between these coils is $M = \sqrt{L_1 L_2}$. **(Bundelkhand 2004)**

Short Answer Questions

1. What is the magnetic flux ? State its unit and dimensional formula. **(Rewa 2003)**
2. (a) State Lenz's Law. **(Agra 2006)**
   (b) What is the use of Lenz's law of electromagnetic induction ?
3. Obtain the integral and differential forms of Faraday's law. **(Agra 2004)**
4. Obtain the differential form of Faraday's law : curl $\vec{E} = -\partial B/\partial t$. **(Agra 2005)**
5. Is the induced electric field conservative or non-conservative ?
6. How does the self-inductance of any coil depend on number of turns in it ?
7. Explain the meaning of mutual induction with an example.
8. What is an inductor ?
9. Differentiate between the self-induction and mutual induction.
10. The iron core of a transformer is made laminated. Why ?
11. On what factors does the induced current in any coil depend ?
12. The plane of the coil is parallel to the magnetic frield B in which it is situated. What is the magnetic flux linked with the coil of area A ?
    [Ans. Zero.]
13. The value of magnetic flux passing through a ring increases from ϕ_1 to ϕ_2 at a uniform rate. If the resistance of ring is R, then find (i) induced e.m.f., (ii) induced current, (iii) total induced charge on the ring.
    [Ans. (i) $\frac{\phi_1 - \phi_2}{t}$, (ii) $\frac{\phi_1 - \phi_2}{tR}$, (iii) $\frac{\phi_1 - \phi_2}{R}$.]
14. Discuss the principle on which a transformer work.
15. Why is a transformer used only with an a.c. source not with a d.c. source ?
16. Why is the core of a transformer made of soft iron ?
17. Deduce an expression for the energy stored in a static magnetic field B.
18. Show that the mutual inductance between two coils is $M = \sqrt{L_1 L_2}$, where L_1 and L_2 are the self-inductance of both coils.
19. Discuss the causes of loss of energy in a transformer. How are they minimised ?
20. Write short note on Faraday's law of electromagnetic induction. **(Agra 2006)**

Objective Type Questions

1. An area $A = 0.5\ \text{m}^2$ is placed in a uniform magnetic field $B = 2\ \text{Wb/m}^2$. The angle between the perpendicular on the area and the field direction is 60°. The magnetic flux linked with the area A is
   (a) 2 Wb (b) 1 Wb (c) 0.5 Wb (d) $\sqrt{3}$ Wb
2. Faraday's law relates electric field $\vec{E}$ and magnetic field $\vec{B}$ as
   (a) $\oint_C \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{S}$ (b) $\oint_C \vec{E} \cdot d\vec{r} = -\int_S \vec{B} \cdot d\vec{S}$
   (c) $\oint_C \vec{E} \cdot d\vec{r} = \frac{d}{dt} \int_S \vec{B} \cdot d\vec{S}$ (d) $\oint_C \vec{E} \cdot d\vec{r} = \int_S \vec{B} \cdot d\vec{S}$ **(Agra 2006)**
3. Consider the following two statements :
   (i) An e.m.f. can be induced by changing the magnetic field.
   (ii) An e.m.f. can be induced by moving a conductor in a magnetic field.
   (a) (i) is true (ii) is false (b) (i) is false (ii) is true
   (c) Both (i) and (ii) are true (d) Both (i) and (ii) are false

4. A copper ring is held horizontal and a bar magnet is dropped through the ring with its length along the axis of the ring. The acceleration of the falling magent is :
(a) equal to g (b) more than g (c) less than g (d) none of the these.
5. There are 500 turns in a coil of area 100 cm^2. A magnetic field 0.1 Wb/m^2 is perpendicular to the plane of the coil. If the magnetic field is reduced to zero in 0.1 sec, the induced e.m.f. in the coil is
(a) 1 volt (b) 5 volt (c) 50 volt (d) zero
6. The e.m.f. is induced only when :
(a) there is no change in magnetic flux
(b) the resistance of coil is low
(c) inductance of coil is large
(d) the magnetic flux linked with the closed coil changes. **(Bhopal 2004)**
7. A conducting wire is moving in a magnetic field B towards right in the plane of the paper. The direction of induced current is shown in **fig. 10.18** The direction of magnetic field B with respect to the plane of the paper is :

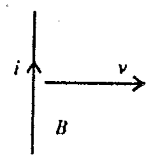

Fig. 10.18.

(a) up (b) down
(c) in the direction of i (d) in the direction of v
8. The south pole of a magnet is brought near a metallic ring as shown in fig. **10.19.** The direction of the induced current in the ring as seen from the side of the magnet is :

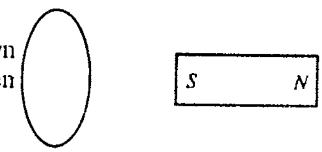

Fig. 10.19.

(a) clcokwise
(b) anticlockwise
(c) first clockwise then anticlockwise
(d) first anticlockwise then clockwise
9. The area of each turn of a circular coil of 500 turns is 0.1 m^2. It is placed normal to the magnetic field of intensity 0.2 T. The coil is rotated about its diameter, normal to the magnetic field by 180° in 0.1 *sec*. If the coil is connected to a galvanometer and the total resistance of circuit is 40 ohm, the charge passed in the circuit will be:
(a) 0.5 Coul. (b) 0.4 Coul. (c) 2 Coul.
(d) 4 Coul.
10. A conducting square loop of each side l and resistance R moves with a velocity v with plane in a direction perpendicular to its one side. A constant magnetic field B acts everywhere perpendicular to the plane of the loop and downwards. The current induced in the loop is :

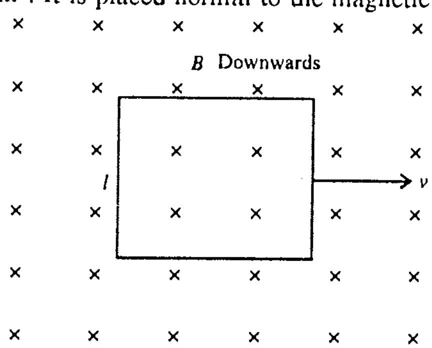

Fig. 10.20.

(a) Blv/R anticlockwise (b) zero
(c) $2Blv/R$ clockwise (d) $BlvR$
11. A rod of length l rotates with uniform angular velocity ω about its perpendicular bisector. There is a uniform magnetic field B along the axis of rotation. The potential difference between the centre of the rod and an end is :

(a) $\frac{1}{8}\omega Bl^2$ (b) $\frac{1}{2}\omega Bl^2$ (c) $B\omega l^2$ (d) zero
12. A conducting loop is placed in a uniform magnetic field with its plane perpendicular to the field. An e.m.f. will be induced in the loop, if
(a) it is rotated about a diameter (b) it is rotated about its axis
(c) it is translated (d) it is deformed

13. There is one meter long vertical aerial in a car, moving east-west at 100 km/hr. The horizontal earth's magnetic field is 0.18×10^{-4} T. The e.m.f. induced in the aerial is :
(a) 0.5 mV (b) 0.25 mV (c) 0.75 mV (d) 1 mV
14. When the current in a coil changes from 8 A to 2 A in 0.03 sec, the e.m.f. induced in the coil is 2 volt. The self-inductance of the coil is :
(a) 1 mH (b) 5 mH (c) 10 mH (d) 20 mH
15. S.I. unit of inductance can be written as :
(a) Wb/A (b) ohm-sec (c) volt-sec/A (d) Wb
16. An e.m.f. of 100 mV is induced in a coil when current in a nearby placed another coil changes from 0 to 10 A in 0.1 sec. The coefficient of mutual induction between the two coils will be :
(a) 10 mH (b) 1 mH (c) 100 mH (d) 0.1 mH
17. If the number of turns and length both, of a solenoid are doubled, its self-inductance will become :
(a) double (b) four times (c) eight times (d) unchanged **(Agra 2004)**
18. The quantity which remains unchanged in a transformer is :
(a) voltage (b) frequency (c) current (d) none of the above.
19. A transformer has N_1 turns in its primary coil and N_2 turns in its secondary coil. The transformer ratio will be :
(a) N_1/N_2 (b) N_2/N_1 (c) $N_2 \times N_1$ (d) $N_2 - N_1$ **(Bundelkhand 2004)**
20. A horse-shoe magnet is held in a vertical position with its north pole on the upper side. A sheet of copper is pushed into the gap of the magnet. On viewing from above, the eddy currents in the sheet would flow :
(a) north direction (b) anticlockwise direction
(c) south direction (d) clockwise direction
21. In a transformer of negligible energy loss, the number of turns in the primary coil are 200 and in the secondary coil are 1000. The readings of secondary coil are 200 volt and 8 ampere. The readings of primary coil will be :
(a) 100 volt, 16 ampere (b) 80 volt, 20 ampere
(c) 160 volt, 10 ampere (d) 40 volt, 40 ampere.

ANSWERS

1. (c)	**2.** (a)	**3.** (c)	**4.** (c)	**5.** (b)	**6.** (d)	**7.** (b)	**8.** (a)
9. (a)	**10.** (b)	**11.** (a)	**12.** (a), (d)	**13.** (a)	**14.** (a)	**15.** (a)	**16.** (a)
17. (a)	**18.** (b)	**19.** (b)	**20.** (b)	**21.** (d)			

11

Maxwell's Equations and Electromagnetic Waves

11·1. Introduction

Both the electric and magnetic fields are associated with the charges in motion and have space and time variations. This phenomenon is called **electromagnetism** and under this topic, we shall study electromagnetic waves. This study inolves time-dependent electromagnetic fields, whose nature can be described by a set of equations, known as **Maxwell's equations.** Before discussing about these equations, first we will describe the **equations of continuity** for charge and current and on its basis, the concept of Maxwell's displacement current will be given because this discussion presents one of the Maxwell's equations.

11·2. Equation of Continuity

The rate of flow of charge is known as current, *i.e.*,

$$i = \frac{dq}{dt} \qquad ...(1)$$

If ρ be the charge density at any point of a conductor and dq be the charge in volume dV around that point then

$$\rho = \frac{dq}{dV} \text{ or } dq = \rho dV \text{ or } q = \int_V \rho \, dV \qquad ...(2)$$

where volume V contains charge q and surface S is surrounding it. Remember that the charge density may be different at different points inside the volume V. Hence from (1) and (2), we obtain

$$i = \frac{d}{dt}\int_V \rho \, dV \qquad ...(3)$$

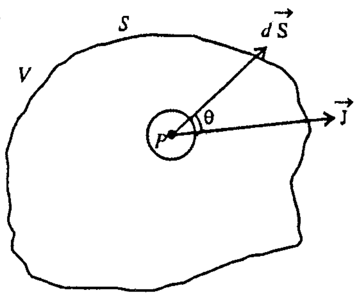

Fig. 11.1 : Equation of continuity.

The current density ($\vec{J}$) can be defined as the charge passing per unit time through unit area of a surface, being held perpendicular to the direction of flow. If we consider an area $d\vec{S}$ around a point, where the current density is $\vec{J}$, then the total current passing through the surface S is

$$i = \int_S \vec{J} \cdot d\vec{S} \qquad ...(4)$$

According to the law of conservation of charge, if an amount of current flowing outward of a surface S, enclosing volume V, then by the same rate the charge must decrease inside that volume V *i.e.*,

$$\int_S \vec{J} \cdot d\vec{S} = -\frac{dq}{dt} = -\frac{d}{dt}\int_S \rho \, dV$$

As charge density ρ varies with time,

$$\int_S \vec{J} \cdot d\vec{S} = -\int_S \frac{d\rho}{dt} \, dV \qquad ...(5)$$

From Gauss's divergence theorem,

$$\int_S \vec{J} \cdot d\vec{S} = \int_S (\text{div } \vec{J}) \, dV \qquad ...(6)$$

From equations (3) and (4),

$$\int_V (\text{div}\,\vec{J})\,dV = -\int_V \frac{\partial \rho}{\partial t}\,dV \quad i.e., \quad \int_V \left(\text{div}\,\vec{J} + \frac{\partial \rho}{\partial t}\right) dV = 0 \qquad ...(7)$$

As equation (7) is valid for any arbitrary volume, the integrand must be equal to zero *i.e.*,

$$\text{div}\,\vec{J} + \frac{\partial \rho}{\partial t} = 0 \qquad ...(8)$$

This equation is called the **equation of continuity** and it is a mathematical expression for conservation of charge.

In case of steady currents, charge density at any point within the region remains constant *i.e.*, $\partial\rho/\partial t = 0$. Hence, from equation (8) for steady currents

$$\text{div}\,\vec{J} = 0 \qquad ...(9)$$

11·3. Maxwell's Displacement Current

According to Ampere's circuital law, in the magnetic field produced by the flow of current, the line integral of magnetic vector $\vec{B}$ (*i.e.*, $\oint \vec{B} \cdot \vec{dl}$) along any closed plane curve is μ_0 times the total current passing through the area closed by that curve, where μ_0 is the permeability of vacuum, *i.e.*,

$$\oint_C \vec{B} \cdot \vec{dl} = \mu_0 i = \mu_0 \int_S \vec{J} \cdot d\vec{S} \qquad ...(1)$$

This is known as the **integral form of Ampere's law.** Here $\vec{J}$ is the **current density.**

From Stoke's theorem.

$$\oint_C \vec{B} \cdot \vec{dl} = \int_S \text{curl}\,\vec{B} \cdot d\vec{S}$$

Hence, $$\int_S \text{curl}\,\vec{B} \cdot d\vec{S} = \mu_0 \int_C \vec{J} \cdot d\vec{S}$$

Whence, $$\text{curl}\,\vec{B} = \mu_0 \vec{J} \qquad ...(2)$$

Eq. (2) is the differential form of **Ampere's law.**

Taking divergence of both sides of eq. (2), we get

$$\text{div curl}\,\vec{B} = \mu_0\,\text{div}\,\vec{J}$$

But, $$\text{div curl}\,\vec{B} = 0, \quad \therefore \quad \text{div}\,\vec{J} = 0 \qquad ...(3)$$

However in general div $\vec{J} \neq 0$. Of course, div $\vec{J} = 0$, for steady currents, but if we go beyond magnetostatics Ampere's law can not be correct.

According to the equation of continuity

$$\text{div}\,\vec{J} + \frac{\partial \rho}{\partial t} = 0 \quad \text{or} \quad \text{div}\,\vec{J} = -\frac{\partial \rho}{\partial t} \qquad ...(4)$$

Thus when the charge density ρ is a function of time, div $\vec{J} \neq 0$. Also, according to Gauss's law in electrostatics,

$$\text{div}\,\vec{E} = \frac{\rho}{\varepsilon_0} \quad \text{or} \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \qquad ...(5)$$

and hence from eq. (4), we have

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t} = -\frac{\partial(\varepsilon_0 \vec{\nabla} \cdot \vec{E})}{\partial t} = -\vec{\nabla} \cdot \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \qquad ...(6)$$

Thus div $\vec{J} \neq 0$ *for time-varying electric field as well as for time-varying charge density.* Time-varying electric field actually occurs during charging or discharging of a condenser. Hence Ampere's law is not correct for time-varying electric field (or charge density). Therefore *Ampere's law needs modification for time-varying electric fields.*

First of all the modification in Ampere's law was done by Maxwell. He suggested that the total current density is not only due to conduction charges but also due to time-varying electric field. If the quantity $\varepsilon_0\, \partial \vec{E}/\partial t$ is added to $\vec{J}$ in Ampere's law, then the Ampere's law becomes

$$\text{curl } \vec{B} = \mu_0 \left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \quad \text{...(7)}$$

So that
$$\text{div curl } \vec{B} = \mu_0 (\vec{\nabla} \cdot \vec{J} + \varepsilon_0 \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{E}) \quad \text{...(8)}$$

But div curl $\vec{B}$ = 0, hence right hand side of eq. (7) gives

$$\vec{\nabla} \cdot \vec{J} + \varepsilon_0 \frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{E}) = 0 \text{ or div } \vec{J} + \frac{\partial \rho}{\partial t} = 0 \left[\because \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \text{ (Gauss's law)} \right] \quad \text{...(9)}$$

Eq. (7) is called the **modified form of Ampere's law.** Thus if we take the Ampere's law in the form of eq. (7), the equation of continuity is satisfied and hence **the modified Ampere's law is correct for time-varying electric field or charge density.** However when $\vec{E}$ is constant, we still have, curl $\vec{B} = \mu_0 \vec{J}$ which is Ampere's law valid for steady currents.

The term $\varepsilon_0 \partial \vec{E}/\partial t$ was added to conduction charge density $\vec{J}$ by Maxwell and Maxwell called it **displacement current** $\vec{J_d}$ *i.e.*,

$$\vec{J_d} = \varepsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{\partial \vec{D}}{\partial t} \qquad \left[\because \vec{D} = \varepsilon_0 \vec{E} \right] \quad \text{...(10)}$$

Since $\vec{J_d}$ arises due to the variation of electric displacement $\vec{D}$ with time, this is why Maxwell called it displacement. Further the quantity $\varepsilon_0 \partial \vec{E}/\partial t = \vec{J_d}$ has the dimensions of current. It is to be very clearly noted that the displacement current (even in vacuum) is not like conduction current in terms of the motion of charged particles but it is the time-variation of the electric field *i.e.*, $\vec{J_d} = \varepsilon_0 \partial \vec{E}/\partial t$.

The concept of displacement current was a great theoretical discovery of Maxwell. The concept makes not only the new Ampere's law consistent with the equation of continuity (or law of conservation of charge), but it also led to a new induction effect : **Time-varying electric field induces magnetic field** ($\because$ curl $\vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \partial \vec{E}/\partial t$) and even for conduction charge density $\vec{J} = 0$, curl $\vec{B} = \mu_0 \varepsilon_0 \partial \vec{E}/\partial t$) similar to a time-varying magnetic field producing an electric field (Faraday's law of electromagnetic induction). This new induction effect for time-varying field is found experimentally correct because when we charge a parallel plate condenser by a battery and place a compass needle between the plates, the needle, in general, deflects. This shows the presence of magnetic field in the region between the plates.

Example of Displacement Current and Its Physical Significance

Consider a parallel plate condenser connected to a battery. When the conduction current starts from the battery, it gradually charges the condenser plates. After some time, the voltage across the capacitor becomes equal to the battery voltage and conduction current stops. During the process of charging the capacitor, the conduction current* does not flow across the gap between the plates. However according to Maxwell, the changing electric field between the plates is equivalent to the conduction current. In fact the displacement current in the gap produced by the changing electric field is found exactly equal to the conduction current and hence we can think of continuity of the current in the circuit.

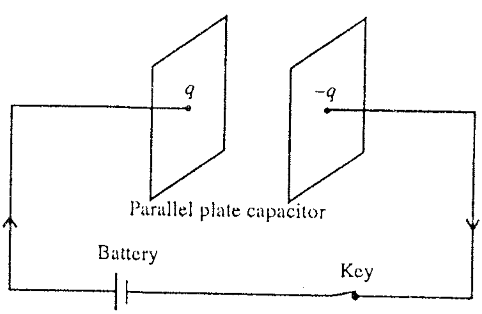

Fig. 11.2 : Time-varying electric field and displacement current.

* Conduction current is produced by the rate of flow of charges.

If q be the instantaneous charge on one of the plates of the capacitor, then the electric field in between the plates of the capacitor is given by

$$E = \frac{q}{\varepsilon_0 A}$$

Hence
$$\frac{dE}{dt} = \frac{1}{\varepsilon_0 A}\frac{dq}{dt}$$

But dq/dt is the current flowing in the circuit, say $dq/dt = i$. Hence

$$\frac{dE}{dt} = \frac{i}{\varepsilon_0 A} \quad \text{or} \quad \varepsilon_0 \frac{dE}{dt} = \frac{i}{A}$$

As $\varepsilon_0 dE/dt = J_d$, displacement current and $i/A = J$ the conduction current density,

$$\therefore \qquad J_d = J$$

Thus we find that **the displacement current in the gap of the capacitor plates is found to be equal to the conduction current in the circuit wires.** It is to be noted that the displacement current in vacuum or dielectric is not as the conduction current constituted by the motion of charged particle. In fact it is the current in the sense, that this displacement current or time-varying electric field in the gap has an associated magnetic field similar to an electric current.

Characteristics and Significance of Displacement Current

(1) The concept of displacement current modifies the Ampere's law for time-varying electric fields (or charge density) so that this law becomes consistent with the equation of continuity.

(2) Displacement current is considered as a current in the sense that it produces a magnetic field similar to a conduction current due to motion of charges. In case of displacement current, there is not the motion of charges because it exists even in vacuum.

(3) The displacement current makes the current to be continuous with the conduction current in the gap of a parallel plate capacitor.

(4) The concept of displacement current makes a symmetry in the magnetic and electric field effects. Just as a time-varying magnetic field produces an electric field, a time-varying electric field produces a magnetic field.

(5) Magnetic fields are entirely produced in vacuum by time-varying electric fields or displacement currents and this is responsible for the propagation of electromagetic waves in vacuum.

11·4. Maxwell's Equations

In 1862, Maxwell represented the basic laws of electric and magnetic fields in the form of four equations, which are known as **Maxwell's equations.** These equations can be derived from the four fundamental laws, related to electric and magnetic fields. These laws are Gauss's law in electrostatics, Gauss's law in magnetism, Faraday's law of electromagnetic induction and Ampere's (modified) circuital law. The Maxwell's equations for electromagnetic fields lead to the propogation of electromagnetic waves in free space or medium.

The divergence and curl relations of electric and magnetic fields are called Maxwell's equations. Following are the four equations of Maxwell :

(1) div $\vec{D} = \rho$ or $\vec{\nabla} \cdot \vec{D} = \rho$

(2) div $\vec{B} = 0$ or $\vec{\nabla} \cdot \vec{B} = 0$

(3) curl $\vec{E} = -\frac{\partial \vec{B}}{\partial t}$ or $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

(4) curl $\vec{B} = \mu_0\left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right)$ or $\vec{\nabla} \times \vec{B} = \mu_0\left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right)$

If we take $\vec{D} = \varepsilon_0 \vec{E}$ and $\vec{B} = \mu_0 \vec{H}$, then the Maxwell's equations are written as

(1) div $\vec{E} = \frac{\rho}{\varepsilon_0}$; (2) div $\vec{H} = 0$; (3) curl $\vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$; (4) curl $\vec{H} = \vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$

where $\vec{D}$ = electric displacement; $\vec{E}$ = electric field intensity; $\vec{B}$ = magnetic induction, $\vec{H}$ = magnetic field indensity; ρ = charge density, μ_0 = permeability of vacuum and ε_0 = permittivity of vacuum.

If μ be the permeability and ε be the permittivity of any medium, then μ_0 and ε_0 of the Maxwell's equations are replaced by μ and ε respectively and the Maxwell's equations assume the following form :

(1) div $\vec{D} = \rho$ or div $\vec{E} = \rho/\varepsilon$;

(2) div $\vec{B} = 0$

(3) curl $\vec{E} = -\frac{\partial \vec{B}}{\partial t}$

and (4) curl $\vec{B} = \mu\left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right)$ or curl $\vec{B} = \mu\left(\vec{J} + \varepsilon\frac{\partial \vec{E}}{\partial t}\right)$

where $\vec{D} = \varepsilon\vec{E}$, $\vec{B} = \mu\vec{H}$ and $\vec{J} = \sigma\vec{E}$, where σ be the conductivity of the medium.

In the present study, we assume that the medium is *homogeneous, isotropic* and *source free.* The homogeneous medium means that the quantities ε, μ and σ are constant throughout the medium. The isotropic medium is the medium for which ε is a scalar constant so that both $\vec{D}$ and $\vec{E}$ are in the same direction *i.e.*, in an isotropic medium, its properties are the same in all directions.

Derivation of Maxwell's Equations

(1) Maxwell's First Equation : div $\vec{D} = \rho$ or div $\vec{E} = \frac{\rho}{\varepsilon_0}$.

Let us consider a surface S bounding a volume V in the vacuum. According to Gauss's law, the flux of electric field $\oint_S \vec{E} \cdot d\vec{S}$ over any closed surface is equal to $1/\varepsilon_0$ times the total charge q enclosed within the surface, *i.e.*,

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\varepsilon_0} \quad ...(1)$$

If ρ is the charge density of the continuously distributed charge at a point, then

$$q = \int_V \rho \, dV \quad ..(2)$$

Hence,
$$\oint_S \vec{E} \cdot d\vec{S} = \frac{1}{\varepsilon_0}\int_V \rho \, dV \quad ...(3)$$

From Gauss's divergence theorem,

$$\oint_S \vec{E} \cdot d\vec{S} = \int_V (\text{div}\,\vec{E}) \, dV \quad ...(4)$$

From equations (3) and (4), we obtain

$$\int_V (\text{div}\,\vec{E}) \, dV = \frac{1}{\varepsilon_0}\int_V \rho \, dV \text{ or } \int_V \left(\text{div}\,\vec{E} - \frac{\rho}{\varepsilon_0}\right) dV = 0 \quad ...(5)$$

Since this equation is true for any volume,

$$\text{div}\,\vec{E} - \rho/\varepsilon_0 = 0 \text{ or div}\,\vec{E} = \rho/\varepsilon_0 \quad ...(6)$$

or
$$\varepsilon_0 \,\text{div}\,\vec{E} = \rho \text{ or div}(\varepsilon_0 \vec{E}) = \rho \text{ or div}\,\vec{D} = \rho \quad ...(7)$$

where $\vec{D} = \varepsilon_0 \vec{E}$ is the electric displacement. Eq. (6) or eq. (7) is the **Maxwell's first equation.**

(2) Maxwell's Second Equation (div $\vec{B}$ = 0).

As the magnetic field lines are either closed curves or go off to infinity, the number of magnetic field lines entering any arbitrary closed surface is exactly the same as leaving it. This implies that the flux of magnetic induction across any closed surface is always zero (Gauss's law in magnetism), *i.e.*,

$$\oint_S \vec{B} \cdot d\vec{S} = 0 \quad ...(8)$$

Using Gauss's divergence theorem

$$\oint_S \vec{B} \cdot d\vec{S} = \int_V (\text{div}\, \vec{B})\, dV \quad ...(9)$$

and eq. (8), we have

$$\int_V (\text{div}\, \vec{B})\, dV = 0 \quad ...(10)$$

The integrand in (10) should vanish for any arbitrary volume *i.e.*,

$$\text{div}\, \vec{B} = 0 \quad ...(11)$$

This is **Maxwell's second equation**

(3) Maxwell's Third Equation $\left(\text{curl}\, \vec{E} = -\dfrac{\partial \vec{B}}{\partial t}\right)$.

According to Faraday's law of electromagnetic induction, the induced e.m.f. $\left(e = \oint_C \vec{E} \cdot d\vec{l}\right)$ in a closed loop is equal to negative rate of change magnetic flux passing through the surface S bounded by that loop, *i.e.*,

$$e = -\frac{d\phi}{dt} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{S} \quad ...(12)$$

But $$e = \oint_C \vec{E} \cdot d\vec{l},$$

$$\therefore \quad \oint_C \vec{E} \cdot d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad ...(13)$$

According to Stoke's theorem,

$$\oint_C \vec{E} \cdot d\vec{l} = \int_S \text{curl}\, \vec{E} \cdot d\vec{S} \quad ...(14)$$

From equations (13) and (14), we obtain

$$\int_S \text{curl}\, \vec{E} \cdot d\vec{S} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad ...(15)$$

This equation is true for any surface whether small or large situated in the field. Hence the two vectors in the integrands must be equal at every point, *i.e.*,

$$\text{curl}\, \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad ...(16)$$

This is the **Maxwell's third equation.**

(4) Maxwell's Fourth Equation : curl $\vec{B} = \mu_0(\vec{J} + \partial \vec{D}/\partial t)$ or curl $\vec{B} = \mu_0(\vec{J} + \varepsilon_0 \partial \vec{E}/\partial t)$.

According to Ampere's law,

$$\oint_C \vec{E} \cdot d\vec{l} = \mu_0 i = \mu_0 \int_S \vec{J} \cdot d\vec{S}$$

From Stoke's theorem,

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S \text{curl}\, \vec{B} \cdot d\vec{S}$$

Hence,

$$\int_s \text{curl}\,\vec{B}\cdot d\vec{S} = \mu_0 \int_s \vec{J}\cdot d\vec{S} \qquad ...(17)$$

This is true for any surface. Hence from (17)

$$\text{curl}\ \vec{B} = \mu_0 \vec{J} \qquad ...(18)$$

It is true for steady currents. For changing electric fields,

$$\text{div}\ \vec{J} = -\frac{\partial \rho}{\partial t} \qquad \text{[From equation of continuity]} \qquad ...(19)$$

Taking the divergence of both sides of eq. (18), we get

$$\text{div curl}\ \vec{B} = \mu_0\, \text{div}\, \vec{J} \qquad ...(20)$$

But $\qquad \text{div curl}\ \vec{B} = \vec{\nabla}\cdot\vec{\nabla}\times\vec{B} = 0$

Hence $\qquad \text{div}\ \vec{J} = 0$ or $\partial\rho/\partial t = 0$ or ρ = constant $\qquad ...(21)$

Thus eq. (18) derived from Ampere's law is not consistent with the equation of continuity because this gives the steady state condition in which the charge density is not changing. This is correct for steady closed currents and therefore for the changing electric fields, the current density should be modified. Therefore, Maxwell realised that eq. (21) for the definition of total current density is incomplete and suggested to add **displacement current** or **displacement current density** $\vec{J}_d$ to $\vec{J}$. Hence

$$\text{curl}\ \vec{B} = \mu_0(\vec{J} + \vec{J}_D) \qquad ...(22)$$

Taking divergence, $\text{div curl}\ \vec{B} = \mu_0\, \text{div}\ (\vec{J} + \vec{J}_d)$ or $0 = \text{div}\ (\vec{J} + \vec{J}_d)$

Thus, $\qquad \text{div}\, \vec{J}_d = -\,\text{div}\ \vec{J} = \partial\rho/\partial t \qquad ...(23)$

But, $\qquad \vec{\nabla}\cdot\vec{D} = \rho,$

where $\vec{D}$ is electric displacement.

Hence from eq. (23) $\qquad \vec{\nabla}\cdot\vec{J}_d = \dfrac{\partial}{\partial t}(\vec{\nabla}\cdot\vec{D}) = \vec{\nabla}\cdot\dfrac{\partial \vec{D}}{\partial t} \qquad ...(24)$

Eq. (24) implies that $\qquad \vec{J}_d = \dfrac{\partial \vec{D}}{\partial t} \qquad ...(25)$

Thus, eq. (22) takes the following form :

$$\text{curl}\ \vec{B} = \mu_0\left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \qquad ...(26)$$

This is **Maxwell's fourth equation.**

As $\vec{D} = \varepsilon_0 \vec{E}$, $\text{curl}\ \vec{B} = \mu_0\left(\vec{J} + \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t}\right) \qquad ...(27)$

11·5. Maxwell's Equations in Integral Form

The integral forms of the Maxwell's equations tell us the physical significance of these equations and are obtained as follows :

(1) *The total flux of electric displacement* $\left(\oint_s \vec{D}\cdot d\vec{S}\right)$ *through the surface enclosing a volume (V) is equal to the total charge (q) within that volume, i.e.,*

$$\oint_s \vec{D} \cdot d\vec{S} = q \quad ...(1)$$

Corresponding to this equation, Maxwell's first equation is,

$$\vec{\nabla} \cdot \vec{D} = \rho \quad ...(2)$$

Proof : Integrating equation (2) over a volume V, we get

$$\int_V (\vec{\nabla} \cdot \vec{D})\, dV = \oint_V \rho\, dV = q \quad ...(3)$$

But from Gauss's theorem,

$$\int_V (\vec{\nabla} \cdot \vec{D})\, dV = \oint_s \vec{D} \cdot d\vec{S} \quad ...(4)$$

From eqs. (3) and (4), we obtain

$$\oint_s \vec{D} \cdot d\vec{S} = q \quad ...(5)$$

which is the **integral form of Maxwell's first equation.**

(2) *The total outward flux* $\oint \vec{B} \cdot d\vec{S}$ *of magnetic induction* $\vec{B}$ *through any closed surface* $\vec{S}$ *is equal to zero, i.e.,*

$$\oint_s \vec{B} \cdot d\vec{S} = 0 \quad ...(6)$$

It corresponds to Maxwell's second equation

$$\vec{\nabla} \cdot \vec{B} = 0 \quad ...(7)$$

Proof : Integrating equation (7) over a volume V, $\oint_V (\vec{\nabla} \cdot \vec{B})\, dV = 0$.

Now, using Gauss's theorem,

$$\int_V (\vec{\nabla} \cdot \vec{B})\, dV = \oint_s \vec{B} \cdot d\vec{S},$$

we obtain

$$\oint_s \vec{B} \cdot d\vec{S} = 0, \quad ...(8)$$

This is the **integral form of Maxwell's second equation.**

(3) *The electromotive force (e.m.f.) around a closed path is equal to the time derivative of the magnetic flux passing through any surface bounded by the path, i.e.,*

$$\oint_C \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t}\int_s \vec{B} \cdot d\vec{S} \quad ...(9)$$

This corresponds to Maxwell's third equation

$$\text{curl } \vec{E} = -\partial \vec{B}/\partial t \quad ...(10)$$

Proof : Integrating equation (10) over a surface S (whose bounding curve is C)

$$\int_s (\text{curl } \vec{E}) \cdot d\vec{S} = -\int_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} = -\frac{\partial}{\partial t}\int_s \vec{B} \cdot d\vec{S} \quad ...(11)$$

Using Stoke's theorem to convert the surface integral into line integral, we obtain

$$\int_s (\text{curl } \vec{E}) \cdot d\vec{S} = \oint_c \vec{E} \cdot d\vec{r} \quad ...(12)$$

The right hand side of the above equation denotes the electromotive force around the closed path. Hence from eqns. (11) and (12)

$$\oint_c \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t}\int_s \vec{B} \cdot d\vec{S} \quad ...(13)$$

which is the **integral form of Maxwell's third equation.**

(4) *The magnetomotive force around any closed path is μ_0 times the conduction current* $\int_s \vec{J} \cdot d\vec{S}$ *plus the time derivative of the electric displacement* $\int_s \vec{D} \cdot d\vec{S}$ *through any surfaces bounded by the path, i.e.,*

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S \left(\vec{J} + \partial \vec{D}/\partial t\right) \cdot d\vec{S} \quad \text{...(14)}$$

It corresponds to Maxwell's fourth equation

$$\text{curl } \vec{B} = \mu_0 \left(\vec{J} + \partial \vec{D}/\partial t\right) \quad \text{...(15)}$$

Proof : Integrating equation (15) for surface S,

$$\int_S \text{curl } \vec{B} \cdot d\vec{S} = \mu_0 \int_S \left(\vec{J} + \partial \vec{D}/\partial t\right) \cdot d\vec{S}$$

Using Stoke's theorem,

$$\int_S \text{curl } \vec{B} \cdot d\vec{S} = \oint_C \vec{B} \cdot d\vec{l}$$

Hence,
$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S \left(\vec{J} + \partial \vec{D}/\partial t\right) \cdot d\vec{S} \quad \text{...(16)}$$

This is the **integral form of Maxwell's fourth equation.**

Ex. 1 *Using Maxwell's equations establish the equation of continuity for the current charge.*

(Agra 2006; Jabalpur 2004)

Sol. From Maxwell's fourth equation.

$$\text{curl } \vec{B} = \mu_0(\vec{J} + \partial \vec{D}/\partial t) \quad \text{...(i)}$$

Taking divergence of both sides of eq. (i), we obtain

$$\vec{\nabla} \cdot \text{curl } \vec{B} = \mu_0 \left(\vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t}\right) \quad \text{...(ii)}$$

But, $\vec{\nabla} \cdot \text{curl } \vec{B} = \vec{\nabla} \cdot \vec{\nabla} \times \vec{B} = 0$ and $\vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t} = \frac{\partial}{\partial t}(\vec{\nabla} \cdot \vec{D}) = \frac{\partial \rho}{\partial t}$ $\quad [\because \vec{\nabla} \cdot \vec{D} = \rho]$

Hence, $0 = \mu_0 \left(\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t}\right)$ or $\text{div } \vec{J} + \frac{\partial \rho}{\partial t} = 0$ ...(iii)

This is the equation of continuity for the current charge.

Ex. 2. *Using Maxwell's equation* div $\vec{E} = \rho/\varepsilon_0$, *establish the Coulomb's law for electrostatics.*

(Raipur 1999; Gwalior 93)

Sol. Integrating equation div $\vec{E} = \rho/\varepsilon_0$, over a volume V with surface S,

$$\int_S \text{div } \vec{E}\, dV = \int_S \frac{\rho}{\varepsilon_0} dV$$

But from Gauss's divergence theorem,

$$\int_S \text{div } \vec{E}\, dV = \int_S \vec{E} \cdot d\vec{S}$$

Hence,
$$\int_S \vec{E} \cdot d\vec{S} = \int_V \frac{\rho}{\varepsilon_0} dV = \frac{q}{\varepsilon_0} \qquad \left[\text{As } \int_V \rho\, dV = q\right] \quad \text{...(i)}$$

If we draw a spherical surface of radius r around charge q, then

$$\int_S \vec{E} \cdot d\vec{S} = \text{E} \cdot 4\pi r^2 \quad \text{...(ii)}$$

From eqns. (i) and (ii),

$$E \cdot 4\pi r^2 = \frac{q}{\varepsilon_0} \text{ or } E = \frac{q}{4\pi\varepsilon_0 r^2} \quad \text{...(iii)}$$

Force (F) on any charge q' in this electric field E, will be

$$F = q'E = \frac{qq'}{4\pi\varepsilon_0 r^2} \quad \text{...(iv)}$$

It is called the **Coulomb's law.**

Ex. 3. *When a dielectric medium is kept in a variable electric field, show that the value of the displacement current is equal to the conduction current.*

Sol. Displacement current density, $\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$

So, displacement current is given by

$$i_d = \int_S \vec{J}_d \cdot d\vec{S} = \varepsilon_0 \int_S \frac{\partial \vec{E}}{\partial t} \cdot d\vec{S} = \varepsilon_0 \frac{\partial}{\partial t}\int_S \vec{E} \cdot d\vec{S} = \varepsilon_0 \frac{\partial}{\partial t}\left(\frac{q}{\varepsilon_0}\right) \quad \text{[From Gauss's law]}$$

Hence,
$$i_d = \varepsilon_0 \frac{\partial}{\partial t}\left(\frac{q}{\varepsilon_0}\right) = \frac{\partial q}{\partial t} = \frac{dq}{dt} = i$$

i.e., Displacement current = Conduction current.

Ex. 4. *Using Maxwell's equation* curl $\vec{E} = -\partial\vec{B}/\partial t$, *prove that* : div $\vec{B} = 0$. **(Rewa 1997)**

Sol. Here, curl $\vec{E} = -\partial\vec{B}/\partial t$

Taking divergence of both sides

$$\text{div curl } \vec{E} = -\text{div}\left(\frac{\partial \vec{B}}{\partial t}\right) = -\frac{\partial}{\partial t}(\text{div } \vec{B})$$

But,
$$\text{div curl } \vec{E} = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{E}) = 0,$$

Hence,
$$\frac{\partial}{\partial t}(\text{div } \vec{B}) = 0 \text{ or div } \vec{B} = 0.$$

where the constant of integration is zero because only then the total magnetic flux through a closed surface will be zero.

Ex. 5. *Using Maxwell's equation,* curl $\vec{B} = \mu_0(\vec{J} + \partial\vec{D}/\partial t)$, *prove that*

div $\vec{D} = \rho$ **(Raipur 2000; Rewa 2000; Indore 1999)**

Sol. Here curl $\vec{B} = \mu_0(\vec{J} + \partial\vec{D}/\partial t)$ ...(i)

Taking divergence on both sides

$$\frac{1}{\mu_0}\text{div curl } \vec{B} = \text{div } \vec{J} + \frac{\partial}{\partial t}(\text{div } \vec{D})$$

But,
$$\text{div curl } \vec{B} = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) = 0, \quad \therefore \text{ div } \vec{J} + \frac{\partial}{\partial t}(\text{div } \vec{D}) = 0 \quad \text{...(ii)}$$

Using equation of continuity, div $\vec{J} = -\partial\rho/\partial t$, we have

$$\frac{\partial}{\partial t}(\text{div } \vec{D}) = \frac{\partial \rho}{\partial t} \text{ or div } \vec{D} = \rho.$$

Here integration constant will be zero because only then this equation will satisfy the Gauss's law,

$$\oint_S \vec{D} \cdot d\vec{S} = q.$$

Ex. 6. *Show that* $\frac{\partial \rho}{\partial t} + \frac{\sigma}{\varepsilon}\rho = 0$ *with the help of equation of continuity and Gauss's law, for a conducting medium obeying Ohm's law,* $\vec{J} = \sigma\vec{E}$.

Sol. Let us assume a homogeneous conducting medium of permittivity ε and conductivity σ.

From Maxwell's equation, $\vec{\nabla} \cdot \vec{E} = \rho/\varepsilon$ ...(i)

Within the conductor, $\vec{J} = \sigma \vec{E}$ ...(ii)

Using the relation (ii), equation of continuity div $\vec{J} + \partial\rho/\partial t = 0$ can be written as

$$\text{div } (\sigma \vec{E}) + \frac{\partial \rho}{\partial t} = 0 \quad \text{or} \quad \sigma \vec{\nabla} \cdot \vec{E} + \partial\rho / \partial t = 0$$

Using $\vec{\nabla} \cdot \vec{E} = \rho/\varepsilon$ from (i), we get

$$\frac{\sigma\rho}{\varepsilon} + \frac{\partial \rho}{\partial t} = 0$$

Ex. 7. *Using Ex. 6, show that if initially there is charge density* ρ_0 *at any point inside the conducting medium, the charge density at any time t can be expressed as*

$$\rho = \rho_0 \, e^{-(\sigma/\varepsilon).t}$$

where ε/σ *is known as relaxation time.*

Sol. From Ex. 6, we know

$$\frac{\partial \rho}{\partial t} + \frac{\sigma\rho}{\varepsilon} = 0 \quad \text{or} \quad \frac{\sigma}{\varepsilon} = \frac{1}{\rho}\frac{d\rho}{dt} \qquad \text{...(i)}$$

Now, at $t = 0$, $\rho = \rho_0$ is the charge density and at $t = t$, $\rho = \rho$ is the charge density. Hence, integrating eq. (i) with the limits $t \to 0$ to $t \to t$, we obtain

$$\frac{\sigma}{\varepsilon}\int_0^t dt = -\int_{\rho_0}^{\rho} \frac{d\rho}{\rho} \quad \text{or} \quad \frac{\sigma}{\varepsilon} \cdot t = -\log\frac{\rho}{\rho_0} \quad \text{or} \quad \rho = \rho_0 e^{-(\sigma/\varepsilon)\cdot t}$$

Ex. 8. *A conductor is held in a varying electric field* $\vec{E} = \vec{E}_0 \sin \omega t$. *Prove that if the frequency is less than* 10^5 Hz, *then the value of the displacement current density is negligible, when compared to the value of conduction current density. (For conductor,* $\sigma \cong 10^8$ ohm^{-1}-m^{-1}*)*

Sol. Displacement current vector $\vec{D} = \varepsilon_0 \vec{E} = \varepsilon_0\omega \vec{E}_0 \sin \omega t$.

Hence, Displacement current,. $\vec{J}_d = \partial \vec{D}/\partial t = \varepsilon_0 \vec{E}_0 \cos \omega t$ or $|\vec{J}_d| = \varepsilon_0 \omega E_0$

Conduction current density, $\vec{J} = \sigma \vec{E} = \sigma \vec{E}_0 \sin \omega t$ or $|\vec{J}| = \sigma E_0$

$$\frac{J_d}{J} = \frac{\varepsilon_0\omega E_0}{\sigma E_0} = \frac{\varepsilon_0 \omega}{\sigma} = \frac{\varepsilon_0}{\sigma} \cdot 2\pi\nu = 2\pi \times \nu \frac{\varepsilon_0}{\sigma} \qquad [\because \omega = 2\pi\nu]$$

Therefore,
$$\frac{J_d}{J} = \frac{2 \times 3.14 \times \nu \times 8.85 \times 10^{-12}}{10^8} = 10^{-19}\nu$$

If $\nu \cong 10^5$ Hz, $J_d/J \cong 10^{-14}$

Thus the displacement current density is negligible when compared to conduction current density for $\nu \le 10^5$ Hz.

EXERCISE 11·1

1. Establish the equation of continiuty for current charge using Maxwell's equations.
   (Bilaspur 2000; Rewa 1999)
2. Using Maxwell's equations div $\vec{D} = \rho$ and curl $\vec{B} = \mu_0(\vec{J} + \partial \vec{D}/\partial t)$, show that
   (i) $\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}\hat{r}$, (Coulomb's law), (ii) div $\vec{J} + \frac{\partial \rho}{\partial t} = 0$ (Equation of continuity)
3. The charge on a paralled plate capacitor is given by $q = q_0 \sin 2\pi nt$. Find the value of displacement current.
   (Raipur 1996)

[**Ans.** $2\pi n q_0 \cos 2\pi n t$.]

[**Hint** : Displacement current $= J_d A = \frac{\partial D}{\partial t} A = \varepsilon_0 \frac{\partial E}{\partial t} A$, where $E = \frac{\sigma}{\varepsilon_0} = \frac{q}{A\varepsilon_0}$.]

4. A conductor is held in an electric field, given by $\vec{E} = \vec{E}_0 \sin \omega t$. What will be the value of conduction and displacement current densities ? **(Gwalior 1998)**
[**Ans.** $J_d = \varepsilon_0 \omega E$, $J = \sigma E_0$.]
5. A capacitor consists of two plates each of radius 10 cm and the plates are at 3.19 cm apart. The capacitor is now charged by a source, giving a current of 0.2 ampere.
(a) What is the rate of potential charge between plates ?
(b) What is the value of displacement current between plates ? **(Sagar 1998)**
[**Ans.** (a) 2.26×10^{10} volt/sec. (b) 0.2 ampere.]

11·6. Electromagnetic waves

An electromagnetic wave is composed of the oscillations of electric and magnetic fields in mutually perpendicular planes and the direction of propagation of the wave is perpendicular to the oscillating electric and magnetic fields. Thus the electromagnetic waves are transverse in character. These waves can propagate in free space or vacuum as well as in material media. Electromagnetic waves transport energy from the source to some place in space. Light and heat waves are electromagnetic waves and they reach the earth from the sun practically through vacuum for most of the distance.

The equation of a plane harmonic electromagnetic wave propagating along Z-direction in vacuum may be represented as

$$E = E_0 \sin \omega (t - z/c) \qquad \text{...(1)}$$

where E is the sinusoidally varying electric field at the position x in space at time t. The constant c is the speed of light in vacuum. The electric field is in the XY-plane, that is, perpendicular to the direction of propagation.

When an electromagnetic wave propagates, there is also a sinusoidally varying magnetic field associated with the changing electric field. This magnetic field is perpendicular to the direction of propagation as well as to the electric field E. The magnetic field associated with plane electromagnetic wave is represented as

$$B = B_0 \sin \omega (t - z/c) \qquad \text{...(2)}$$

In fact such a composition of mutually perpendicular electric and magnetic fields, propagating in vacuum (or medium) is referred as an **electromagnetic wave**. The theory of electromagnetic waves was mainly developed by Maxwell around 1864. We discuss this theory below.

11·7. Wave Equations Satisfied by Electric Field $\vec{E}$ and Magnetic Field $\vec{B}$ in Free Space or Vacuum

The Maxwell's equations in free space (or vacuum) having no charges and currents (*i.e.*, $\rho = 0$ and $\vec{J} = 0$) are as follows :

$$\text{(i) } \vec{\nabla} \cdot \vec{E} = 0, \quad \text{(ii) } \vec{\nabla} \cdot \vec{B} = 0, \quad \text{(iii) } \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \text{(iv) } \vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \qquad \text{...(1)}$$

Taking the curl of Maxwell's third equation,

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\vec{\nabla} \times \partial \vec{B}/\partial t \qquad \text{...(2)}$$

Left hand side of this equation can be written as by using vector algebra :

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - (\vec{\nabla} \cdot \vec{\nabla})\vec{E} = -\vec{\nabla}^2 \vec{E} \qquad [\because \vec{\nabla} \cdot \vec{E} = 0]$$

Right hand side of eq. (2) is

$$-\vec{\nabla} \times \frac{\partial \vec{B}}{\partial t} = -\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B}) = -\frac{\partial}{\partial t}\left(\mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right) = -\mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \qquad \left[\because \vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \partial \vec{E}/\partial t\right]$$

Hence eq. (2) takes the form

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad ...(3)$$

This is the **wave equation satisfied by the electric fields vector** $\vec{E}$. Similarly taking the curl of eq. (iv) and using eqs. (ii) and (iii), we obtain

$$\nabla^2 \vec{B} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} \quad ...(4)$$

This is the **wave euqation satisfied by the magnetic field vector** $\vec{B}$. Eqs. (3) and (4) are in three dimensions for $\vec{E}$ and $\vec{B}$ vectors. The standard wave equation is

$$\nabla^2 \psi = v^2 \frac{\partial^2 \psi}{\partial t^2} \quad ...(5)$$

which represents waves travelling with a velocity v.

Comparing eq. (3) or (4) with (5), we obtain an expression for the velocity of electromagnetic waves in free space (or vacuum) as

$$v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \quad ...(6)$$

We know that for vacuum $\mu_0 = 4\pi \times 10^{-7}$ N/A^2 is the permeability and $\varepsilon_0 = 8.854 \times 10^{-12} C^2$/N-m^2 is the permittivity. Hence

$$v = \frac{1}{\sqrt{(4\pi \times 10^{-7}) \times (8.854 \times 10^{-12})}} = 3 \times 10^8 \text{ m/sec.} \quad ...(7)$$

This is precisely the speed of light c in vacuum and hence the *speed of electromagnetic waves in vacuum is equal to the speed of light in vacuum,* given by

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \text{ m/sec.} \quad ...(8)$$

Thus, according to Maxwell's equations, empty or vacuum supports the propagation of electromagnetic waves with speed of light c. Maxwell concluded from this that propagation of light is a special case of the propagation of electromagnetic waves. Thus **according to Maxwell's electromagnetic theory, the nature of light is electromagnetic in character.** Remember that the mechanical waves *e.g.*, sound waves require material medium for their propagation, while electromagnetic waves can travel even in vacuum.

It is to be noted that the unit of $1/\sqrt{\mu_0 \varepsilon_0}$ is the same as that of velocity *i.e.*,

$$\left[\frac{1}{\sqrt{\mu_0 \varepsilon_0}}\right] = \left[\frac{\text{Newton}}{\text{Amp.}^2} \times \frac{\text{Coul.}^2}{\text{Newton-m}^2}\right]^{-1/2} = \frac{\text{Amp-m}}{\text{Coul}}$$

$$= \frac{(\text{Coul/sec})\text{m}}{\text{Coul}} = \frac{\text{m}}{\text{sec}} = \text{unit of velocity}$$

11·8. Propagation of Electromagnetic Waves in a Dielectric Medium

In a dielectric medium, the charge density and the current density are zero *i.e.*, $\rho = 0$ and $\vec{J} = 0$. Further we consider homogeneous and isotropic medium so that $\vec{D} = \varepsilon \vec{E}$, and $\vec{B} = \mu \vec{H}$, where ε is the permittivity and μ is the permeability of the medium. Maxwell's equations in an isotropic and homogeneous dielectric medium are :

$$\text{(i) } \vec{\nabla} \cdot \vec{E} = 0, \quad \text{(ii) } \vec{\nabla} \cdot \vec{B} = 0, \quad \text{(iii) } \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \text{ and } \quad \text{(iv) } \vec{\nabla} \times \vec{B} = \mu\varepsilon \frac{\partial \vec{E}}{\partial t} \quad ...(1)$$

Taking curl of Maxwell's third equation

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = -\vec{\nabla}\times\partial\vec{B}/\partial t \qquad ...(2)$$

But, from vector calculus,

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = \vec{\nabla}\times(\vec{\nabla}\cdot\vec{E}) - (\vec{\nabla}\cdot\vec{\nabla})\vec{E} = -\nabla^2\vec{E} \qquad [\because \vec{\nabla}\cdot\vec{E}=0]$$

Also,

$$\vec{\nabla}\times\frac{\partial\vec{B}}{\partial t} = \frac{\partial}{\partial t}(\vec{\nabla}\times\vec{B}) = \frac{\partial}{\partial t}\left(\mu\varepsilon\frac{\partial\vec{E}}{\partial t}\right) = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}$$

Hence eq. (2) takes the form

$$\nabla^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad ...(3)$$

Similarly taking the curl of eq. (iv) and using eqs. (ii) and (iii), we obtaion

$$\nabla^2\vec{B} = \mu\varepsilon\frac{\partial^2\vec{B}}{\partial t^2} \qquad ...(4)$$

Equations (3) and (4) are the wave equations in a dielectric medium. The standard wave equation is

$$\nabla^2\psi = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2} \qquad ...(5)$$

where v is the speed of the waves. Comparing eqs. (3) and (4) with eq. (5), we obtain the speed (v) of electromagnetic waves in a dielectric medium *i.e.*,

$$v = \frac{1}{\sqrt{\mu\varepsilon}} \qquad ...(6)$$

But, $\mu_r = \mu/\mu_0$ and $\varepsilon_r = \varepsilon/\varepsilon_0$ or $\mu = \mu_0\mu_r$ and $\varepsilon = \varepsilon_0\varepsilon_r$, where μ_r and ε_r are the relative permeability and relative permittivity of the dielectric medium respectively.

Hence,

$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_0\varepsilon_0\mu_r\varepsilon_r}} = \frac{c}{\sqrt{\mu_r\varepsilon_r}} \qquad \left[\because c = \frac{1}{\sqrt{\varepsilon_0\mu_0}}\right] \qquad ...(7)$$

Obviously the *speed of propagation of an electromagnetic wave in a dielectric medium is less than the speed of light* (c).

Refractive index n of the dielectric medium :

$$n = \frac{\text{Speed of wave in vacuum}}{\text{Speed of wave in medium}} = \frac{c}{v} = \sqrt{\mu_r\varepsilon_r} \qquad ...(8)$$

For a nonmagnetic medium $\mu_r = 1$ and then,

$$n = \sqrt{\varepsilon_r} \text{ or } n^2 = \varepsilon_r \qquad ...(9)$$

Hence the square of refractive index of a *dielectric medium is equal to its relative permittivity.*

We can write the wave equations for electric field vector $\vec{E}$ and magnetic field vector $\vec{B}$ in a dielectric medium as

$$\nabla^2\vec{E} = \frac{1}{v^2}\frac{\partial^2\vec{E}}{\partial t^2} \qquad ...(10)$$

and

$$\nabla^2\vec{B} = \frac{1}{v^2}\frac{\partial^2\vec{B}}{\partial t^2} \qquad ..(11)$$

where $v = 1/\sqrt{\mu\varepsilon}$.

Below we will solve the wave equations for electric field and magnetic field vectors $\vec{E}$ and $\vec{B}$ in vacuum and discuss the propagation of electrimagnetic waves. The same procedure can be followed for the solution of the wave equations and travelling electromagnetic waves in an isotropic dielectric medium. Only by replacing μ, ε and v for this case in place of μ_0, ε_0 and c for vacuum.

11·9. Plane Electromagnetic Waves in Vacuum

Maxwell's equations lead to the following equations for electric field vector $\vec{E}$ and magnetic field vector $\vec{B}$ in vacuum :

$$\nabla^2 \vec{E} = \frac{1}{c^2}\frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{...(1)}$$

and

$$\nabla^2 \vec{B} = \frac{1}{c^2}\frac{\partial^2 \vec{B}}{\partial t^2} \quad \text{...(2)}$$

We will see that $\vec{E}$ and $\vec{B}$ are related with each other and the varying electric and magnetic fields ($\vec{E}$ and $\vec{B}$) of eqs. (1) and (2) travel in the form of a wave. This wave is called **electromagnetic wave,** moving with speed of light $c = 1/\sqrt{\mu_0\varepsilon_0}$. In fact, an electromagnetic wave is composed of two coupled waves—(i) electric wave and (ii) magnetic wave. In general, solutions of eqs. (1) and (2) may be any kind of electromagnetic wave *e.g.*, plane, spherical or cylindrical, but here we plan to study plane electromagnetic waves in vacuum.

A wave which travels in a particular direction with plane wave front is called **plane wave.** In a plane electromagnetic wave, at all the points on the plane taken perpendicular to the direction of its propagation, the electric field and magnetic field vectors ($\vec{E}$ and $\vec{B}$) are in the same phase and this surface taken perpendicular to the direction of propogation is the **plane wave-front.**

Let a plane electromagnetic wave be travelling along *Z*-axis. The electric and magnetic field vectors $\vec{E}$ and $\vec{B}$ related to this wave will be the functions of distance z covered along *Z*-axis and time t *i.e.*,

$$\vec{E} = \vec{E}(z, t) \quad \text{...(3)}$$

and

$$\vec{B} = \vec{B}(z, t) \quad \text{...(4)}$$

Hence, the eqs. (1) and (2) can be written in the following form for the wave propagating along *Z*-axis :

$$\frac{\partial^2 \vec{E}}{\partial z^2} = \frac{1}{c^2}\frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{...(5)}$$

and

$$\frac{\partial^2 \vec{B}}{\partial z^2} = \frac{1}{c^2}\frac{\partial^2 \vec{B}}{\partial t^2} \quad \text{...(6)}$$

The general solution of these wave equations of plane wave propagating along *Z*-axis is as follows–

$$\vec{E}(z, t) = \vec{E}(t - z/c) + \vec{E}(t + z/c) \quad \text{...(7)}$$

and

$$\vec{B}(z, t) = \vec{B}(t - z/c) + \vec{B}(t + z/c) \quad \text{...(8)}$$

where the wave of first term of right hand side is propagating along positive *Z*-axis while the wave of second term is propagating along negative *Z*-axis. Here we are dealing only with plane waves propagating along positive *Z*-axis, hence the terms $\vec{E}(t + z/c)$ and $\vec{B}(t + Z/c)$ respectively in eqs. (7) and (8) are not present. Hence,

$$\vec{E}(z, t) = \vec{E}(t - z/c) \quad \text{...(9)}$$

and

$$\vec{B}(z, t) = \vec{B}(t - z/c) \quad \text{...(10)}$$

The plane harmonic electromagnetic wave propagating along positive Z-axis will have the following form :

$$\vec{E}(z, t) = \vec{E}_0 \sin \omega (t - z/c)$$

or
$$\vec{E}(z, t) = \vec{E}_0 \sin(\omega t - kz) \quad ...(11)$$

and
$$\vec{B}(z, t) = \vec{B}_0 \sin(\omega t - kz) \quad ...(12)$$

where $\frac{\omega}{c} = \frac{2\pi\nu}{\nu\lambda} = \frac{2\pi}{\lambda} = k$, is the propagation constant; and ω, ν and λ are angular frequency, frequency and wavelength of the wave respectively.

If we put the expressions of $\vec{E}$ and $\vec{B}$ from eqs. (11) and (12) in eqs. (5) and (6) respectively, we will see that these equations satisfy wave equations (5) and (6).

The plane harmonic electromagnetic waves will have following form in any arbitrary direction of three dimensional space :

$$\vec{E}(\vec{r}, t) = \vec{E}_0 \sin\left(\omega t - \vec{k} \cdot \vec{r}\right) \quad ...(13)$$

and
$$\vec{B}(\vec{r}, t) = \vec{B}_0 \sin\left(\omega t - \vec{k} \cdot \vec{r}\right) \quad ...(14)$$

which are solutions of eqs. (1) and (2) and satisfy both of them.

11·10. Characteristics of Plane Electromagnetic Waves in Vacuum

(1) **Plane electromagnetic waves are transverse :** A plane electromagnetic wave moves along a particular direction. Let this wave be travelling along Z-axis, so that $\vec{E}$ and $\vec{B}$ will vary along Z-axis. In such a case,

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial y} = 0 \text{ and } \frac{\partial}{\partial z} \neq 0 \quad ...(1)$$

and we may write
$$\vec{E} = \vec{E}(z, t) \text{ and } \vec{B} = \vec{B}(z, t) \quad ...(2)$$

Maxwell's equations will have the following form in vacuum :

(i) $\vec{\nabla} \cdot \vec{D} = 0$, (ii) $\vec{\nabla} \cdot \vec{B} = 0$, (iii) $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$, (iv) $\vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$

[As in vacuum, $\rho = 0$ and $\vec{J} = 0$].

From Maxwell's first equation,

$$\vec{\nabla} \cdot \vec{E} = 0 \qquad [\text{As } \vec{D} = \varepsilon_0 \vec{E}]$$

where
$$\vec{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z} \text{ and } \vec{E} = \hat{i}\,E_x + \hat{j}\,E_y + \hat{k}\,E_z$$

i.e.,
$$\vec{\nabla} \cdot \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0 \quad ...(3)$$

But according to equation (1), the values of $\partial E_x/\partial x$ and $\partial E_y/\partial y$ are zero, hence

$$\frac{\partial E_z}{\partial z} = 0 \quad \text{or} \quad E_z = \text{constant} \quad ...(4)$$

Similarly
$$\frac{\partial B_z}{\partial z} = 0 \quad \text{or} \quad B_z = \text{constant} \quad ...(5)$$

where $\vec{B} = \hat{i}\,B_x + \hat{j}\,B_y + \hat{k}\,B_z$.

We see that in (4) and (5) E_z and B_z are *constants in space*. From Maxwell's third equation.

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \text{ or } \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ E_x & E_y & E_z \end{vmatrix} = -\hat{i}\frac{\partial B_x}{\partial t} - \hat{j}\frac{\partial B_y}{\partial t} - \hat{k}\frac{\partial B_z}{\partial t}$$

The z-component of $\vec{\nabla}\times\vec{E}$ vector wil be

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -\frac{\partial B_z}{\partial t} \qquad ...(6)$$

But according to equation (1) $\frac{\partial E_y}{\partial x} = 0 = \frac{\partial E_x}{\partial y}$

Hence, $\frac{\partial B_z}{\partial t} = 0$ or B_z = constant in time ...(7)

Similary, form Maxwell's fourth equation,

$$E_z = \text{constant in time} \qquad ...(8)$$

Thus, we conclude that E_z and B_z *remain constants with respect to space and time.* Here, we are not interested in their constant values because no wave is set up by them (as an electromagnetic wave requires the variations in the fields). Therefore, take their values as zero *i.e.*,

$$E_z = B_z = 0 \qquad ...(9)$$

This implies that *electromagnetic waves do not possess longitudinal components.*

Thus, $$\vec{E} = \hat{i}\,E_x + \hat{j}\,E_y \qquad ...(10)$$

and $$\vec{B} = \hat{i}\,B_x + \hat{j}\,B_y \qquad ...(11)$$

As the wave is propagating along Z-axis and both $\vec{E}$ and $\vec{B}$ have no z-components and both vectors $\vec{E}$ and $\vec{B}$ are in the XY-plane perpendicular to the direction of propagation (Z-axis). Hence, *Maxwell's electromagnetic waves are transverse in nature.*

(2) Electric field $\vec{E}$ and magnetic field $\vec{B}$ are mutually perpendicular in plane electromagnetic waves : Remembering the condition (1), X and Y components of Maxwell's third and fourth equations are :

$$-\frac{\partial E_y}{\partial z} = -\frac{\partial B_x}{\partial t} \text{ or } \frac{\partial E_y}{\partial z} = \frac{\partial B_x}{\partial t} \qquad ...(12)$$

$$\frac{\partial E_x}{\partial z} = -\frac{\partial B_y}{\partial t} \qquad ...(13)$$

$$-\frac{\partial B_y}{\partial z} = \mu_0\varepsilon_0\frac{\partial E_x}{\partial t} \text{ or } \frac{\partial B_y}{\partial z} = -\mu_0\varepsilon_0\frac{\partial E_x}{\partial t} \qquad ...(14)$$

and $$\frac{\partial B_x}{\partial z} = \mu_0\varepsilon_0\frac{\partial E_y}{\partial t} \qquad ...(15)$$

We see that in an electromagnetic wave, propagating along Z-axis, x-component of $\vec{B}$ is related to y-component of $\vec{E}$ and y-component of $\vec{B}$ to x-component of $\vec{E}$. Hence, we can assume a plane electromagaanetic wave equivalent to the superposition of two independent plane electromagnetic waves (E_x, B_y) and (E_y, B_x). For a general plane electromagnetic wave propagating along Z-axis :

$$\vec{E}(z,t) = \vec{E}(t - z/c) \text{ and } \vec{B}(z,t) = \vec{B}(t - z/c) \qquad ...(16)$$

From which $$\frac{\partial E_y}{\partial z} = \frac{\partial E_y}{\partial(t - z/c)}\cdot\frac{\partial(t - z/c)}{\partial z} = -\frac{1}{c}\frac{\partial E_y}{\partial(t - z/c)}$$

and $$\frac{\partial E_y}{\partial t} = \frac{\partial E_y}{\partial(t - z/c)}\cdot\frac{\partial(t - z/c)}{\partial t} = \frac{\partial E_y}{\partial(t - z/c)}$$

Hence, $$\frac{\partial E_y}{\partial z} = -\frac{1}{c}\frac{\partial E_y}{\partial t} \quad ...(17)$$

Putting this value of $\partial E_z/\partial z$ in equation (12), we find

$$-\frac{1}{c}\frac{\partial E_y}{\partial t} = \frac{\partial B_x}{\partial t} \text{ or } \frac{1}{c}\frac{\partial E_y}{\partial t} = -\frac{\partial B_x}{\partial t} \quad ...(18)$$

Integrating eq. (18) with respect to t,

$$\frac{1}{c}E_y = -B_x + C \quad ...(19)$$

where C is an integration constant. As the electric and magnetic fields associated with an electromagnetic wave are varying, so their constant part (C) may be taken zero. Hence,

$$\frac{1}{c}E_y = -B_x \text{ or } E_y = -c\,B_x \quad ...(20)$$

Similarly, $$\frac{\partial E_x}{\partial z} = -\frac{1}{c}\frac{\partial E_x}{\partial t} \quad ...(21)$$

Using equation (13), $$-\frac{1}{c}\frac{\partial E_x}{\partial t} = -\frac{\partial B_y}{\partial t} \text{ or } \frac{1}{c}\frac{\partial E_x}{\partial t} = \frac{\partial B_y}{\partial t} \quad ...(22)$$

On integrating, $$\frac{1}{c}E_x = B_y \text{ or } E_x = c\,B_y \quad ...(23)$$

Now, $$\vec{E}\cdot\vec{B} = (E_x\,\hat{i} + E_y\,\hat{j})\cdot(B_x\,\hat{i} + B_y\,\hat{j}) = E_x B_x + E_y B_y$$

Using eqs. (20) and (23), we obtain

$$\vec{E}\cdot\vec{B} = E_x\left(-\frac{E_y}{c}\right) + E_y\left(\frac{1}{c}E_x\right) = \frac{-E_xE_y}{c} + \frac{E_yE_x}{c} = 0 \quad ...(24)$$

Hence, *Electric field $\vec{E}$ and magnetic field $\vec{B}$ are mutually perpendicular in a plane electromagnetic wave.*

If we assume that $\vec{E}$-vector of the electromagnetic wave propagating along Z-axis, is along X-axis, *i.e.*, $\vec{E} = \hat{i}\,E_x$, then $\vec{B}$-vector will be along Y-axis *i.e.*, $\vec{B} = \hat{j}\,B_y$.

If it is a plane harmonic electromagnetic wave, then

$$E_x = E_0 \sin(\omega t - kz) \quad ...(25)$$

and $$B_y = B_0 \sin(\omega t - kz) \quad ...(26)$$

This plane harmonic electromagnetic wave is represented in **fig 11.3.** A plane electromagnetic wave which is propagating along Z-direction and in which the electric field vector $\vec{E}(= \hat{i}\,E_x)$ is always along X-axis (and associated magnetic vector $\vec{B}(= \hat{j}\,B_y$ is along Y-axis), is called plane polarised *plane electromagnetic wave in XZ plane* [fig. **11.3**].

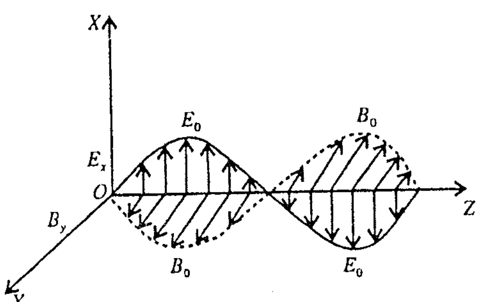

Fig. 11.3 : Propagation of an electromagnetic waves.

(3) *Both electric field $\vec{E}$ and magnetic field $\vec{B}$ are perpendicular to the direction of propagation in a plane electromagnetic wave* : From equation (11),

$$\vec{B} = \hat{i}\,B_x + \hat{j}\,B_y \quad ...(27)$$

Putting the values of B_x and B_y in equation (27) from euqations (20) and (23),

$$\vec{B} = -\hat{i}\frac{E_y}{c} + \hat{j}\frac{E_x}{c}$$

Putting $\hat{i} = \hat{j} \times \hat{k}$ or $-\hat{i} = \hat{k} \times \hat{j}$ and $\hat{j} = \hat{k} \times \hat{i}$ in the above equation, we obtain

$$\vec{B} = \frac{1}{c}\left(\hat{k} \times \hat{j}\, E_y + \hat{k} \times \hat{i}\, E_x\right) = \frac{1}{c}\hat{k} \times \left(E_x\, \hat{i} + E_y\, \hat{j}\right) \text{ or } \vec{B} = \frac{1}{c}\hat{k} \times \vec{E} \quad [\because \vec{E} = E_x\, \hat{i} + E_y\, \hat{j}] \quad ...(28)$$

Similarly, equation $\vec{E} = E_x\, \hat{i} + E_y\, \hat{j}$ can be obtained in the following form using equations (20) and (23),

$$\vec{E} = c(\vec{B} \times \hat{k}) \quad ...(29)$$

Now,
$$\vec{B} \cdot \hat{k} = \frac{1}{c}(\hat{k} \times \vec{E}) \cdot \hat{k} = \frac{1}{c}\hat{k} \cdot (\hat{k} \times \vec{E})$$

$$= \frac{1}{c}(\hat{k} \times \hat{k}) \cdot \vec{E} = 0 \quad ...(30)$$

because
$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A} \text{ and } \vec{A} \cdot (\vec{A} \times \vec{B})$$

$$= (\vec{A} \times \vec{A}) \cdot \vec{B} = 0 [\because \vec{A} \times \vec{A} = 0]$$

Similarly
$$\vec{E} \cdot \hat{k} = 0 \quad ...(31)$$

As Z-axis or $\hat{k}$ is the direction of propagation, hence *both $\vec{E}$ and $\vec{B}$ are perpendicular to the direction of propagation and also are mutually perpendicular.*

(4) Both electric field $\vec{E}$ and magnetic field $\vec{B}$ are in the same phase in a plane electromagnetic wave : Following are the equations of a plane electromagnetic wave propagating along Z-axis :

$$\vec{E} = \vec{E}_0 \sin(\omega t - kz) \text{ and } \vec{B} = \vec{B}_0 \sin(\omega t - kz) \quad ...(32)$$

Hence, both $\vec{E}$ and $\vec{B}$ are in the same phase.

(5) *The direction of vector $\vec{E} \times \vec{B}$ is the direction of wave propagation* : From eqs. (10) and (11),

$$\vec{E} = \hat{i}\, E_x + \hat{j}\, E_y \text{ and } \vec{B} = \hat{i}\, B_x + \hat{j}\, B_y$$

Hence,
$$\vec{E} \times \vec{B} = (\hat{i}\, E_x + \hat{j}\, E_y) \times (\hat{i}\, B_x + \hat{j}\, B_y) = \hat{k}(E_x B_y - E_y B_x)$$

Putting the values of B_x and B_y from equations (20) and (23), we obtain

$$\vec{E} \times \vec{B} = \hat{k}\left[E_x\left(\frac{E_x}{c}\right) - E_y\left(\frac{-E_y}{c}\right)\right] = \hat{k}\left[\frac{E_x^2}{c} + \frac{E_y^2}{c}\right] = \hat{k}\frac{E^2}{c} [\because E_x^2 + E_y^2 = E^2] \quad ...(33)$$

It is clear that the direction of $\vec{E} \times \vec{B}$ is along $\hat{k}$ or Z-axis *i.e.*, the direction of vector $\vec{E} \times \vec{B}$ is along the direction of propagation of a plane electromagnetic wave.

(6) Relation between electric field vector $\vec{E}$ and magnetic field vector $\vec{B}$: The electric and magnetic field vectors of a plane harmonic electromagnetic wave propagating along Z-axis are given by

$$\vec{E} = \vec{E}_0 \sin(\omega t - kz) \text{ and } \vec{B} = \vec{B}_0 \sin(\omega t - kz) \quad ...(34)$$

where
$$\vec{E} = E_x\, \hat{i} + E_y\, \hat{j} \text{ and } \vec{B} = B_x\, \hat{i} + B_y\, \hat{j}$$

But for a plane electromagnetic wave propagating along Z-axis, from eq. (28), we have

$$\vec{B} = \frac{1}{c}(\hat{k} \times \vec{E}) \text{ or } |\vec{B}| = \frac{1}{c}|\hat{k} \times \vec{E}| \text{ or } B = \frac{E}{c} \quad ...(35)$$

[As $\hat{k}$ and $\vec{E}$ are mutually perpendicular, $|\hat{k} \times \vec{E}| = E \sin \pi/2 = E$]

Thus, $$\frac{E}{B} = c = \frac{1}{\sqrt{\varepsilon_0\mu_0}} \quad ...(36)$$

Hence, *the ratio of the values of electric field vector and magnetic field vector of any plane electromagnetic wave in vacuum is equal to the velocity of light* ($c = 1/\sqrt{\varepsilon_0\mu_0}$).

If $\vec{B} = \mu_0\vec{H}$, then from equation (36),

$$\frac{E}{B} = \frac{E}{\mu_0 H} = \frac{1}{\sqrt{\varepsilon_0\mu_0}} \text{ or } \frac{E}{H} = \sqrt{\frac{\mu_0}{\varepsilon_0}} \quad ...(37)$$

This quantity is called the **characteristic impedence of the vacuum.** On putting the values of ε_0 and μ_0, the value of characteristic impedence of vacuum comes out to be **376.7 ohm or 377 ohm (approx).**

For any plane harmonic electromagnetic wave, from equation (34)

$$\frac{|\vec{E}|}{|\vec{B}|} = \frac{|\vec{E}_0|}{|\vec{B}_0|} \text{ or } \frac{E}{B} = \frac{E_0}{B_0}$$

Hence, $$\frac{E_0}{B_0} = c = \frac{1}{\sqrt{\varepsilon_0\mu_0}} \text{ and } \frac{E_0}{H_0} = \sqrt{\frac{\mu_0}{\varepsilon_0}} \quad ...(38)$$

Hence, *the ratio of the extreme values of electric and magnetic field vectors of a plane harmonic electromagnetic wave in value is equal to c* ($= 1/\sqrt{\varepsilon_0\mu_0}$) *and the ratio of the extreme values of electric field vector and intensity of magnetic field is equal to* $\sqrt{\mu_0/\varepsilon_0}$ *i.e., characteristic impedence* (**= 376.7 ohm**).

11·11. Energy Density in Electromagnetic Field – Poynting Vector

Energy is transported by electromagnetic waves from source to some place in space. Here for plane electromagnetic waves, we will establish relation between *energy density, rate of transfer of energy* and *the electromagnetic field vectors* ($\vec{E}$ and $\vec{B}$). This relation is called **Poynting vector**. The direction of an electromagnetic wave at a point is that in which the energy is transferred.

Definition of Poynting Vector : If electric field vector $\vec{E}$ and magnetic field vector $\vec{B}$ are associated with an electromagnetic wave, then the Poynting vector is defined by

$$\vec{P} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) \text{ or } \vec{P} = \vec{E} \times \vec{H} \quad ...(1)$$

The energy per unit time (or power) transported by the field through unit area, taken perpendicular to the direction of propagation of an electromagnetic wave, is called Poynting vector $\vec{P}$. First of all, J. H. Poynting discussed the characteristics of this vector $\vec{P}$ and this is why the vector represented by eq. (1) is called Poynting vector. Below, we will show that the Poynting vector represents the flow of the power per unit area for an electromagnetic wave. Alongwith we will obtain an expression for energy density in an electromagnetic field and relate it with the Poynting vector. *Poynting theorem* will also be discussed.

Poynting Vector and Energy Density : From Maxwell's third and fourth equations in vacuum

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad ...(2)$$

and $$\vec{\nabla} \times \vec{B} = \mu_0\varepsilon_0 \frac{\partial \vec{E}}{\partial t} \quad ...(3)$$

Taking the dot product of eq. (2) with $\vec{B}$,

$$\vec{B} \cdot (\vec{\nabla} \times \vec{E}) = -\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} = -\frac{1}{2}\frac{\partial}{\partial t}B^2 \quad ...(4)$$

because $B^2 = \vec{B}\cdot\vec{B}$ and $\dfrac{\partial B^2}{\partial t} = \dfrac{\partial(\vec{B}\cdot\vec{B})}{\partial t} = \vec{B}\cdot\dfrac{\partial\vec{B}}{\partial t} + \dfrac{\partial\vec{B}}{\partial t}\cdot\vec{B} = 2\vec{B}\cdot\dfrac{\partial\vec{B}}{\partial t}$

Also taking the scalar product of $\vec{E}$ with eq. (3), we obtain

$$\vec{E}\cdot(\vec{\nabla}\times\vec{B}) = \mu_0\varepsilon_0\vec{E}\cdot\frac{\partial\vec{E}}{\partial t} = \frac{\mu_0\varepsilon_0}{2}\frac{\partial}{\partial t}E^2 \qquad ...(5)$$

Subtracting eq. (4) from eq. (5), we get

$$\vec{E}\cdot(\vec{\nabla}\times\vec{B}) - \vec{B}\cdot(\vec{\nabla}\times\vec{E}) = \frac{\mu_0\varepsilon_0}{2}\frac{\partial}{\partial t}E^2 + \frac{1}{2}\frac{\partial}{\partial t}B^2 \qquad ...(6)$$

From vector identity

$$\vec{\nabla}\cdot(\vec{A}\times\vec{B}) = \vec{B}\cdot(\vec{\nabla}\times\vec{A}) - \vec{A}\cdot(\vec{\nabla}\times\vec{B})$$

Eq. (6) can be written as

$$\vec{\nabla}\cdot(\vec{B}\times\vec{E}) = \frac{\mu_0\partial}{\partial t}\left(\frac{\varepsilon_0 E^2}{2} + \frac{B^2}{2\mu_0}\right) \text{ or } \vec{\nabla}\cdot\left(\frac{\vec{E}\times\vec{B}}{\mu_0}\right) = -\frac{\partial}{\partial t}\left(\frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0}\right) \qquad ...(7)$$

i.e.,

$$\vec{\nabla}\cdot\vec{P} + \frac{\partial u}{\partial t} = 0 \qquad ...(8)$$

where,

$$\vec{P} = \frac{1}{\mu_0}(\vec{E}\times\vec{B}) = \vec{E}\times\frac{\vec{B}}{\mu_0} = \vec{E}\times\vec{H} \qquad ...(9)$$

and

$$u = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0} \text{ or } u = \frac{1}{2}(\vec{B}\cdot\vec{H} + \vec{E}\cdot\vec{D}) \qquad ...(10)$$

We know that $\frac{1}{2}\varepsilon_0 E^2$ represents the energy per unit volume of the electrostatic field and $\frac{1}{2}\frac{B^2}{\mu_0}$ the energy per unit volume of the magnetostatic field in vacuum. Let us assume that these expressions are valid when magnetic and electric fields are varying rapidly. Thus, the **total electromagnetic energy** (u) **per unit volume** or the **energy density** in vacuum will be

$$u = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0} \qquad ...(11)$$

Hence, the expression given by equation (11) represents the energy density in an electromagnetic field in vacuum.

Now, from (11) we can obtain the total electromagnetic energy in any given volume V in vacuum as

$$U = \int_V u\,dV = \int_V\left(\frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0}\right)dV \qquad ...(12)$$

Integrating eq. (7) for volume V surrounded by surface S, we obtain

$$\int_V \vec{\nabla}\cdot\frac{1}{\mu_0}(\vec{E}\times\vec{B})dV = -\frac{\partial}{\partial t}\int_V\left(\frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0}\right)dV \qquad ...(13)$$

Using Gauss's divergence theorem.

$$\int_V \vec{\nabla}\cdot\frac{1}{\mu_0}(\vec{E}\times\vec{B})dV = \frac{1}{\mu_0}\int_S(\vec{E}\times\vec{B})\cdot d\vec{S} \qquad ...(14)$$

Hence,

$$\int_S\frac{1}{\mu_0}(\vec{E}\times\vec{B})\cdot d\vec{S} = -\frac{\partial}{\partial t}\int_V\left(\frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0}\right)dV \qquad ...(15)$$

Using eq. (12)

$$\int_S \frac{1}{\mu_0}(\vec{E} \times \vec{B}) \cdot d\vec{S} = -\frac{\partial U}{\partial t} \qquad ...(16)$$

The quantity $-\frac{\partial U}{\partial t}$ represents the rate of decrease of energy in any given volume V. The rate of decrease of energy should be equal to the rate of energy flow outwards that volume. Hence, the left hand side of equation (16) represents the total energy flow per unit time outwards the surface S surrounding the volume V. Consequently,

$$\vec{P} = \frac{1}{\mu_0}(\vec{E} \times \vec{B})$$

represents the energy flow per unit area per unit time outwards the surface.

Thus, the power transmitted through any arbitrary surface will be

$$-\frac{\partial U}{\partial t} = \int_S \vec{P} \cdot d\vec{S} \qquad ...(17)$$

Eq. (17) [or eq. (8)] represents the **Poynting theorem.** This theorem tells us an important characteristic of the Poynting vector $\vec{P}$ *i.e.* the Poynting vector

$$\vec{P} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) \text{ or } (\vec{E} \times \vec{H}) \text{ watt/m}^2 \qquad ...(18)$$

represents the power transmitted through unit area normal to the direction of propagation of an electromagnetic wave.

Unit and Dimensions of $\vec{P}$:

$$\vec{P} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) \text{ or } [P] = \frac{[E][B]}{[\mu_0]}$$

Here, $$[E] = \frac{[F]}{[q]} = \frac{[MLT^{-2}]}{[AT]} = [MLA^{-1}\,T^{-3}] \qquad [\because F = qE]$$

$$\frac{[B]}{[\mu_0]} = \frac{[\mu_0][AL]}{[\mu_0][L^2]} = [AL^{-1}] \qquad \left[\because B = \frac{\mu_0}{4\pi}\frac{i\,dl\,\sin\theta}{r^2}\right]$$

Thus, dimensions of $\vec{P} = [MLA^{-1}\,T^{-3}]\,[AL^{-1}] = [MT^{-3}]$

Since Poynting vector $\vec{P}$ represetns energy flow per unit area per unit time,

$$\text{Unit of } \vec{P} = \frac{[\text{energy}]}{[\text{area}] \times [\text{time}]} = \frac{\text{joule}}{\text{metre}^2 \times \text{second}} = \textbf{watt/metre}^2$$

Energy density for an electromagnetic wave : If $\vec{E}$ and $\vec{B}$ respectively are the varying electric field and magnetic field vectors at any instant for any electromagnetic wave, then the corresponding energy densities in vacuum will be $\frac{1}{2}\varepsilon_0 E^2$ and $\frac{1}{2}\frac{B^2}{\mu_0}$. Hence, the value of *instantaneous energy density* at any point associated with an electromagnetic wave is

$$u = u_e + u_m = \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{B^2}{\mu_0}\right) \qquad ...(19)$$

where ε_0 is the pemittivity and μ_0 is the permeability of vacuum.

We know that for an electromagnetic wave

$$B = E/c \qquad ...(20)$$

$$\therefore \quad u_m = \frac{B^2}{2\mu_0} = \frac{1}{2}\frac{E^2}{\mu_0 c^2} = \frac{1}{2}\frac{E^2\mu_0\varepsilon_0}{\mu_0} = \frac{1}{2}\varepsilon_0 E^2 = u_e \qquad ...(21)$$

Hence, $$u_e = u_m = \frac{1}{2}\varepsilon_0 E^2 \qquad ...(22)$$

Thus *in a plane electromagnetic wave instantaneous electric energy density and magnetic energy density are equal.* Consequently the instantaneous energy density at any point associated with any electromagnetic wave will be given by

$$u = \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{E^2}{c^2\mu_0}\right) = \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{E^2}{\mu_0(1/\mu_0\varepsilon_0)}\right) \text{ or } u = \varepsilon_0 E^2 \qquad ...(23)$$

The expression for instantaneous energy density for a plane harmonic electromagnetic wave $\vec{E} = \vec{E}_0 \sin(\omega t - kz)$ will be of the form

$$u = \vec{\varepsilon}_0 E_0^2 \sin^2(\omega t - kz) \text{ joule/metre}^3 \qquad ...(24)$$

It is clear from eq. (24) that the value of energy density at any point is oscillatory. The value of average energy density (u_{av}) for a time period in plane harmonic electromagnetic wave can be obtained as

$$\bar{u} = \frac{1}{T}\int_0^T \varepsilon_0 E_0^2 \sin^2(\omega t - kz)dt = \frac{1}{T}\int_0^T \frac{\varepsilon_0 E_0^2}{2}(1 - \cos 2(\omega t - kz))dt = \frac{1}{2}\varepsilon_0 E_0^2 \qquad ...(25)$$

because $$\frac{1}{T}\int_0^T \cos 2(\omega t - kz)\, dt = \frac{1}{T}\left[\frac{\sin 2(\omega t - kz)}{2\omega}\right]_0^{T=2\pi/\omega} = 0$$

Similarly $$\bar{u}_e = \frac{1}{T}\int_0^T \frac{1}{2}\varepsilon_0 E_0^2 \sin^2(\omega t - kz)dt = \frac{1}{4}\varepsilon_0 E^2 = \overline{u_m} \qquad ...(26)$$

11·12. Momentum in an Electromagnetic Wave

The energy is transported with the electromagnetic waves and hence the momentum must be associated with the travelling electromagnetic waves. The momentum $\vec{p}$ of a particle moving with velocity $\vec{v}$ is given by

$$\vec{p} = m\vec{v} \qquad ...(1)$$

But from Einstein's mass-energy relation, the energy (U) is given by

$$U = mc^2 \text{ or } m = U/c^2$$

$$\therefore \quad \vec{p} = \frac{U}{c^2}\vec{v} \qquad ...(2)$$

This is the relation between momentum and energy of a moving particle.

The energy density in vacuum for an electromagnetic wave is

$$u = \varepsilon_0 E^2 \qquad ...(3)$$

Hence, we can represent the momentum density $\vec{p}$ *i.e.*, momentum per unit volume associated with an electromagnetic wave similar to eq. (2),

$$\vec{p} = \frac{u}{c^2}\vec{v} \qquad ...(4)$$

As the electromagnetic waves are propagating along Z-axis, so $\vec{v} = c\,\hat{k}$, hence

$$\vec{p} = \frac{u}{c^2}c\,\hat{k} \text{ or } \vec{p} = \frac{u}{c}\hat{k} \qquad ...(5)$$

But Poynting vector, $\vec{P} = \dfrac{\vec{E} \times \vec{B}}{\mu_0}$ and $\vec{E} \times \vec{B} = \dfrac{E^2}{c}\hat{k}$ [eq. (33) Art. 11.10]

$\therefore$ $\vec{P} = \dfrac{E^2}{c}\dfrac{\hat{k}}{\mu_0}$

But, $u = \varepsilon_0 E^2$ or $E^2 = u/\varepsilon_0$,

hence $\vec{P} = \dfrac{u}{\varepsilon_0 c \mu_0}\hat{k} = uc\hat{k}$ $\left[\because \dfrac{1}{\varepsilon_0\mu_0} = c^2\right]$ ...(6)

Now, form eq. (5) and (6),

$$\vec{p} = \frac{u}{c}\hat{k} = \frac{\vec{P}}{c^2} = \frac{\vec{E} \times \vec{B}}{\mu_0 c^2} \quad ...(7a)$$

or

$$\vec{p} = \frac{\varepsilon_0 E^2}{c}\hat{k} = \varepsilon_0(\vec{E} \times \vec{B}) \quad \left[\because C = 1/\sqrt{\mu_0\varepsilon_0}\right] \quad ...(7b)$$

Eqs. (7) represent the **momentum for unit volume of any electromagnetic wave.**

Magnitude of the momentum, $p = u/c$ or $u = pc$ ...(8)

i.e., **Energy density = Momentum of wave × Velocity of wave**

11·13. Relation between Radiation Pressure and Energy Density

We know that momentum transfer is associated with the electromagnetic wave propagation. Hence when these waves strike any surface, there is the change in their momentum and the same amount of momentum is received by the surface. The momentum transfer per unit time to the surface by the electromagnetic wave is equivalent to the force (rate of change of momentum) according to Newton's second law. This force acting on unit area of the surface due to the electromagnetic wave produces the **radiation pressure** on the surface.

Consider a plane electromagnetic wave normally incident on a surface situated in vacuum. Due to this wave, the momentum density $p = \varepsilon_0 E^2/c$, is normal to the surface. Hence, the amount of electromagnetic radiations incident on unit area of the surface in 1 second will be the same amount of radiations, contained volume $c \times 1 = c$ of a cylinder of length c and unit area (say circular). Hence the momentum of volume c will be pc, as p is the momentum density. Thus the radiation pressure (P_r) on the surface will be

$$P_r = pc \quad ...(1)$$

because this is the momentum gained per second per unit area or force per unit area or pressure on the surface while total amount of incident rediation is completely absorbed by the surface

We know that

$$p = u/c,$$

hence $P_r = u$ ...(2)

Thus **in case of normal incidence, the pressure due to electromagnetic radiations on any ideal absorbing surface, is equal to the energy density.**

If the surface is a ideal reflector, then the change in momentum of electromagnetic waves will be doubled resulting in the radiation pressure on the surface be two times *i.e.,*

$$P_r = 2pc = 2u \quad ...(3)$$

Ex. 1. *The electric field of a plane electromagnetic wave in vacuum is represented by following equation :*

$$E_x = \frac{1}{\sqrt{2}}\cos\left[4\pi \times 10^7 (t - z/c)\right],\ E_y = 0,\ E_z = 0.$$

Calculate the following :

(i) *Wavelength, nature of polarization and direction of propagation.*
(ii) *Magnetic field of wave.*
(iii) *Flux or Energy flow per unit area per second.*

Sol. Here, $E_x = \frac{1}{\sqrt{2}} \cos [4\pi \times 10^7(t - z/c)]$, $E_y = 0$ and $E_z = 0$.

(i) Standard cosine wave along Z-direction is given by

$$E_x = E_0 \cos [\omega t - kz]$$

Hence, $E_0 = 1/\sqrt{2}$ volt/metre, $\omega = 4\pi \times 10^7$ radian/second and $k = (4\pi \times 10^7)/c$ per metre.

Now, $$k = \frac{2\pi}{\lambda};\ \therefore\ \lambda = \frac{2\pi}{k} = \frac{2\pi \times 3 \times 10^8}{4\pi \times 10^7} = \textbf{15 metre.}$$

As the electric field $\vec{E} = E_x\,\hat{i}$ in this wave vibrates along X-direction, the wave is **plane polarised** in X-direction (X-Z plane) and **propagates along Z-axis.**

(ii) $$\vec{B} = \frac{1}{c}\,\hat{k} \times \vec{E} = \frac{1}{c}\,\hat{k} \times E_x\,\hat{i} = \frac{E_x}{c}\,\hat{j}.$$

$$\therefore \quad \vec{B} = \frac{1}{\sqrt{2} \times 3 \times 10^8} \cos\left[4\pi \times 10^7(t - z/c)\right] \ Wb/\text{m}^2 \text{ along } Y\text{-direction}$$

(iii) $$\text{Flux} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) = \frac{1}{\mu_0}[E_x\,\hat{i} \times B_y\,\hat{j}] = \frac{E_x B_y}{\mu_0}\hat{k}$$

i.e., $$\text{Value of flux} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2} \times 3 \times 10^8} \times \frac{1}{4\pi \times 10^{-7}} = \mathbf{13.32 \times 10^{-4}\ \textbf{watt/metre}^2}.$$

Ex. 2. *If the radius of the sun is* 7×10^8 *metre and radiated energy is* 38×10^{28} *watt, then find the value of energy propagation Poynting vector on the surface of the sun.*

(Rewa 2004; Gwalior 03; Ujjain 03, 2000; Bhopal 1999; Indore 99; Bilaspur 98)

Sol. Value of Poynting vector = Energy flow per unit time per unit area

$$= \frac{\text{Energy radiated per second by sun}}{\text{Area of the surface of the sun}}$$

$$= \frac{38 \times 10^{28}\,\text{watt}}{4\pi \times (7 \times 10^8)^2\,\text{metre}} = \mathbf{6.17 \times 10^{10}\ \textbf{watt/metre}^2}$$

Ex. 3. *The components of the electric field in free space are* $E_x = E_y = 0$ *and* $E_z = E_0 \cos (2\pi x/\lambda) \cos \omega t$. *Find the magnetic induction* $\vec{B}$. **(Bhopal 1991)**

Sol. $$\vec{E} = E_z \cdot \hat{k}$$

Clearly, $$\vec{B} = \frac{\hat{i} \times \vec{E}}{c} = \frac{\hat{i} \times E_z\,\hat{k}}{c} = \frac{-E_0}{c} \cos\left(\frac{2\pi x}{\lambda}\right) \cos \omega t\ \hat{j}.$$

Ex. 4. *With the help of Maxwell's electromagnetic theory, prove that the momentum of a photon associated with a wave of wavelength* λ *is* h/λ, *where* h *is Planck's constant.*

(Gwalior 2004; Indore 1993; Rewa 94)

Sol. We know that : Radiation Pressure = Energy density

If there are n photons per unit volume, then

energy density = $nh\nu$

where $h\nu$ is the energy of one photon

Radiation pressure = P = Number of photons incident on unit area × Momentum

Let p (say) be the momentum of a photon. Then

$$P = nc \times p = ncp$$

where nc is the number of photons falling on unit area of the surface per second.

Hence, $$ncp = nh\nu \quad i.e., \; p = \frac{h\nu}{c} = \frac{h}{\lambda} \qquad [\because c = \nu\lambda]$$

Ex. 5. *The energy flux of a laser beam is* 10 *watt/m*^{2}. *It falls on an ideal reflecting mirror for* 1 *hour. Calculate the momentum transferred and the force on the mirror.* **(Indore 2004)**

Sol. As the energy flux is 10 watt/metre^2 or 10 joule/second-metre^2, the energy incident on unit area of the mirror in one second = 10 joule.

This energy will be in a cylinder of length c and surface of unit area *i.e.*, volume $c \times 1 = c$ contains 10 joule energy.

$\therefore$ Energy density, $$u = \frac{10}{c} \text{ joule/metre}^3$$

So, the momentum of unit volume towards the mirror $= p = \dfrac{u}{c} = \dfrac{10}{c^2}$

Now, the momentum falling on unit area of surface in one second will be of volume $c \times 1 = c$, *i.e.*, total momentum $pc = 10/c$ will fall on mirror in one second. As the mirror is ideal reflector, so the momentum transferred to the unit area of the surface in one second will be $2 \times 10/c$ *i.e.*,

$$\text{Force on unit area} = \frac{2 \times 10}{c} = \frac{2 \times 10}{3 \times 10^8} = \mathbf{6.67 \times 10^{-8}\ Newton.}$$

Now, momentum transferred to the unit area of surface in one hour is

$$= \frac{2 \times 10}{c} \times 60 \times 60 = 6.67 \times 10^{-8} \times 3600$$

$$= \mathbf{2.4 \times 10^{-4}\ kg\text{-}m/second.}$$

Ex. 6. *A plane electromagnetic wave is travelling along – X-axis. Its frequency is* 100 *MHz and electric field is normal to Z-axis. What will be the expressions of electric field vector* $\vec{E}$ *and magnetic field vector* $\vec{B}$ *of the wave ?*

Sol. The wave is travelling along X-axis, therefore electric vector will be normal to X-axis. As the direction of $\vec{E}$ is given to be normal to Z-axis, therefore $\vec{E}$ will be along Y-axis. Thus

$$\vec{E} = E_0\, \hat{j} \sin(\omega t + kx)$$

Here, $\nu = 100$ MHz $= 10^8$ Hz, $\omega = 2\pi\nu = 2\pi \times 10^8$ rad/s and $k = \dfrac{2\pi}{\lambda} = \dfrac{2\pi\nu}{c} = \dfrac{2\pi}{3}\ \text{m}^{-1}$

$$\therefore \quad \vec{E} = E_0\, \hat{j} \sin 2\pi (10^8 t + x/3)$$

$$\text{Magnetic field vector } \vec{B} = \frac{1}{c}\hat{i} \times \vec{E} = \frac{1}{c}(\hat{i} \times \hat{j})\, E_0 \sin 2\pi(10^8 t + x/3)$$

$$= \hat{k}\frac{E_0}{c} \sin 2\pi(10^8 t + x/3) = B_0\, \hat{k} \sin 2\pi(10^8 t + x/3).$$

Ex. 7. *If the magnitude of H in a plane electromagnetic wave is* 2 *amp/m. Find the value of E for the wave in free space.*

Sol. $$E = \sqrt{\frac{\mu_0}{\varepsilon_0}}\, H = 376.7 \times 2 \text{ ohm-amp/m.} = \mathbf{753.4\ volt/m}$$

EXERCISE 11·2

1. Energy radiated per second by a radio transmitter is 100 W. Calculate the value of Poynting vector for a distance of 45 km. **(Rewa 2000; Raipur 1997)**
[Ans. 3.93×10^{-9} W/m^2.]

[**Hint** : $P = \dfrac{100\,\text{W}}{4\pi \times (45 \times 10^3)^2\,\text{m}^2} = 3.93 \times 10^{-9}$ W/m^2.]

2. The energy radiated by a radio transmitter is 50 watt. Calculate the value of Poynting vector at a distance of 15 km. **(Raipur 1996)**
[Ans. 6.9×10^{-8} watt/metre^2.]

3. The energy flux of a laser beam is 10 watt/cm^2. It falls on 1 cm^2 of the surface of an ideal reflecting mirror for 1 hour. Calculate the momentum transferred and the force exerted on the mirror. **(Indore 2004; Rewa 1997)**
[Ans. 2.4×10^{-4} kg-m/second, 6.7×10^{-8} newton.]

4. If the values of permittivity ε_0 and permeability μ_0 of the vacuum are 8.83×10^{-12} C^2/N-m^2 and 12.56×10^{-7} N/A^2 respectively, then calculate the velocity of electromagnetic waves in vacuum.
[Ans. 3×10^8 m/s.] **(Sagar 2000)**

5. In a plane electromagnetic wave, the electric field oscillates sinusoidally with an amplitude 6 V m^{-1} and frequency 1.5×10^{12} Hz. Calculate : (i) the wavelength of the wave, (ii) the amplitude of oscillations of the magnetic field.
[Ans. (i) 2×10^{-4} m. (ii) 2×10^{-8} tesla.]

6. The components of electric field of an electromagnetic wave in vacuum are given by
$$E_x = 0.\ E_y = 10 \cos (2\pi \times 10^{10} t - 2\pi x/5),\ E_z = 0,$$
where E is in V/m, t in second and x in metre.
Calculate : (i) frequency ν, (ii) wavelength λ, (iii) direction of wave propagation, (iv) direction of magnetic field.
[Ans. (i) 10^{10} Hz, (ii) 5 m, (iii) X-axis, (iv) Z-axis.]

7. The electric field in an electromagnetic wave is given by $E = (50\ N/c) \sin (\omega t - kx)$.
Calculate the energy contained in a cylinder of cross-section 10 cm^2 and length 50 cm along the X-axis. Find the flux of the wave.
[Ans. 5.5×10^{-12} J; 3.3 W/m^2.]

QUESTIONS

Long Answer Questions

1. Establish the equation of continuity for the charge and current density. How will it change for the steady current ? **(Bhopal 2000, 1998; Indore 98; Ujjain 99)**

2. What do you understand by current and current density ? How are both related ? Prove that for the steady current, the equation of continuity is : $\text{div}\ \vec{J} = 0$.
(Sagar 1999, 97; Gwalior 98; Raipur 98; Bilaspur 99; Indore 97; Jabalpur 98)

3. Prove the equation of continuity $\text{div}\,\vec{J} + \dfrac{\partial \rho}{\partial t} = 0$ for the time varying current distribution.
(Jabalpur 1998; Indore 97; Sagar 97)

4. Prove that the Ampere's law, curl $\vec{B} = \mu_0 \vec{J}$ is valid only for the steady currents. Explain its inconsistency in the time varying electric field. How was it modified by Maxwell ?
(Ujjain 2003; Raipur 03)

5. Explain the concept of displacement current density in the time varying electric and magnetic fields. Prove that the displacement current density $\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$.
(Agra 2004; Gwalior 03; Indore 1998; Rewa 96)

6. Prove theat Ampere's law curl $\vec{B} = \mu_0 \vec{J}$ is valid only for the steady currents. Discuss the inconsistency of this relation in time varying electric field. How did Maxwell modify it ? Obtain its modified form. **(Bilaspur 1997; Gwalior 97; Bhopal 98; Ujjain 97)**

7. Explain the concept of displacement current density and prove that for the magnetic field $\vec{B}$ produced in a time varying electric field $\vec{E}$, curl $\vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$, where the symbols have their usual meanings. **(Rewa 2004; Bhopal 03)**

8. Show that curl $\vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$, where the symbols have their usual meanings.
(Bhopal 2003; Indore 03; Jabalpur 1995)

9. Why did it become necessary to modify Ampere's law ? What is its modified form ? Explain its importance. **(Jabalpur 1999; Bilaspur 99)**

10. What is Ampere's law ? Explain how this law is inconsistent with the time varying fields ? Hence discuss the concept of displacement current. **(Gwalior 1999, 96; Sagar 98; Indore 99)**

11. How does the displacement current density explain the discharge of a capacitor through a resistor ?

12. Write down the Maxwell's equations in an isotropic medium and deduce them.
(Bundelkhand 2004; Sagar 03)

13. Express the Maxwell's equations in their differential and integral forms and use them to obtain the equation of continuity. **(Jabalpur 2004)**

14. What are the fundamental laws of electricity and magnetism ? Deduce Maxwell's equations.
(Raipur 1997, 96; Indore 1994)

15. Obtain Maxwell's electromagnetic equations. On which laws, these depend ? Discuss.
(Ujjain 1999; Bhopal 2000, 1997)

16. Write down Maxwell's equations. Use them to establish the wave equation of electromagnetic wave. Prove that the velocity of electromagnetic wave is $v_0 = \sqrt{\frac{1}{\mu_0 \varepsilon_0}}$ and discuss the nature of wave,
(Jabalpur 2003; Indore 1999, 95; Gwalior 96; Ujjain 99; Bilaspur 99)

17. Deduce wave equations for $\vec{E}$ and $\vec{B}$ in electromagnetic waves in vacuum and show that in vacuum, the speed of propagation of wave is $c = \sqrt{\frac{1}{\mu_0 \varepsilon_0}}$. **(Agra 2004; Indore 03)**

18. Obtain the wave equations for $\vec{E}$ and $\vec{B}$ in electromagnetic waves in a dielectric medium and prove that the speed of waves in a dielectric medium is $v = \sqrt{\frac{1}{\mu \varepsilon}}$. **(Gwalior 2004)**

19. Establish the following relations :

(i) $\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$ (ii) $\nabla^2 \vec{B} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$

and show that the speed of electromagnetic waves in vacuum is $c = 1/\sqrt{\mu_0 \varepsilon_0}$.
(Bilaspur 1999; Ujjain 99; Gwalior 97)

20. Establish the wave equations for E.M. waves from Maxwells equations and find the velócity of E.M. waves in free space. **(Agra 2006)**

21. Establish the wave equations for $\vec{E}$ and $\vec{H}$ in electromagnetic waves and prove that the velocity of wave propagation in vacuum is $c = (\mu_0 \varepsilon_0)^{-1/2}$, where $c = 3 \times 10^8$ metre/second. **(Raipur 1999, 98; Indore 99; Jabalpur 98; Sagar 98)**

22. Establish an expression for the rate of flow energy flux in an electromagnetic field. In this reference define Poynting vector and obtain its dimensions. **(Ujjain 2003)**

23. What is Poynting's vector ? Show that the direction of energy flow is normal to the plane of $\vec{E}$ and $\vec{B}$. **(Bundelkhand 2004)**

24. Define Poynting vector and write down its expression. Prove that $\vec{P} = \frac{1}{\mu_0}(\vec{E} \times \vec{B})$, where the symbols have their usual meanings. Discuss its physical importance. **(Bilaspur 2003; Bhopal 04)**

25. Explain Poynting vector and obtain the expression for energy density of electromagetic field,

$$W = \left(\frac{\vec{B} \cdot \vec{H} + \vec{E} \cdot \vec{D}}{2}\right).$$

26. Obtain the expression for the energy density in electromagnetic field. **(Bhopal 1996)**

27. Discuss how are the elecromagnetic waves propagated in a dielectric medium ? **(Bhopal 2000; Bilaspur 1997; Jabalpur 97)**

28. Establish the expression for the rate of flow of energy flux in electromagnetic field. Define Poynting vector and find its dimensions. **(Jabalpur 1997; Ujjain 98)**

29. Write short notes on the following :
    (i) Equation of continuity
    (ii) Displacement current
    (iii) Poynting theorem **(Agra 2006)**

Short Answer Questions

1. Define current and current density. **(Gwalior 1999)**
2. What is equation of continuity of charge and current ? **(Jabalpur 1999; Sagar 98)**
3. From curl $\vec{E} = -\frac{\partial \vec{B}}{\partial t}$ deduce div $\vec{B} = 0$. **(Rewa 1997)**
4. Prove that curl $\vec{E} = -\frac{\partial \vec{B}}{\partial t}$. **(Gwalior 2003)**
5. Prove that $\vec{\nabla} \cdot \vec{D} = \rho$. **(Bhopal 2003)**
6. Prove that $\vec{\nabla} \times \vec{E} = -\frac{\partial H}{\partial t}$.
7. Explain the inconsistency of Ampere's law in time varying fields.
8. Discuss maxwell's correction to Ampere's circuital law for time varying currents. **(Agra 2006)**
9. What is the modified form of Ampere's law ? **(Ujjain 1999)**
10. What is the number of equations relating electric field, magnetic field and their effects ? What are the names of these equations ? Write down all equations. **(Raipur 1992; Ujjain 91)**
11. Derive the wave equation for E.M. waves from Maxwell's equations. **(Agra 2006)**

12. Prove that $\vec{E}, \vec{H}$ and $\hat{k}$ are mutually perpendicular. **(Indore 1996; Bhopal 98; Gwalior 98)**
13. Prove that the electric field vector and magnetic field vector are mutually perpendicular in any electromagnetic wave. **(Sagar 1994)**
14. Prove that $\vec{E}$ and $\vec{H}$ are in same phase in plane electromagnetic wave.
15. What do you understand by displacement current density ? **(Bhopal 1997)**
16. Write down the fundamental laws of electricity and magnetism. **(Raipur 1996)**
17. State Poynting's theorem. **(Agra 2006; Sagar 1999)**
18. Explain the transverse nature of electromagnetic waves in dielectric medium. **(Bhopal 1996)**
19. Explain the flow of energy in electromagnetic field. **(Rewa 1992)**
20. Write down the expression for velocity of electromagnetic waves in vacuum in terms of permittivity and permeability. **(Ujjain 2000)**

    [**Ans.** $c = 1/\sqrt{\mu_0\varepsilon_0}$]
21. Write a short note on Poynting vector. **(Raipur 1999; Jabalpur 97)**

Objective Type Questions

1. Electromagnetic waves are :

   (a) α-rays (b) β-rays (c) γ-rays (d) X-rays
2. The speed of electromagnetic wave in vacuum is :

   (a) $c = \sqrt{\mu_0/\varepsilon_0}$ (b) $c = \sqrt{\mu_0\varepsilon_0}$ (c) $1/\sqrt{\mu_0\varepsilon_0}$ (d) $c = \sqrt{\varepsilon_0/\mu_0}$ **(Agra 2006)**
3. In an electromagnetic wave, the amplitudes of oscillating electric field $\vec{E}$ and magnetic field $\vec{B}$ are related as :

   (a) $B = \mu_0\varepsilon_0 E$ (b) $E = B$ (c) $B = \sqrt{\mu_0\varepsilon_0}\,E$ (d) $E = \sqrt{\mu_0\varepsilon_0}\,B$ **(Agra 2004)**
4. Select the equation which is not Maxwell's equation :

   (a) $\vec{\nabla}\cdot\vec{B} = 0$ (b) $\vec{\nabla}\cdot\vec{D} = \rho$ (c) $\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$ (d) $\vec{\nabla}\cdot\vec{E} = -\frac{\partial\vec{B}}{\partial t}$ **(Agra 2006)**
5. The value of displacement current for steady current is :

   (a) zero (b) $\varepsilon_0\frac{\partial E}{\partial t}$ (c) $-\varepsilon_0\frac{\partial E}{\partial t}$ (d) $\frac{\partial\vec{D}}{\partial t}$ **(Rewa 1994)**
6. Equation of continuity for varying charge distribution is :

   (a) $\text{div}\vec{J} = -\frac{\partial\rho}{\partial t}$ (b) $\text{div}\vec{J} = 0$ (c) $\text{div}\vec{B} = 0$ (d) $\text{div}\vec{E} = 0$ **(Bhopal 1991)**
7. Equation of continuity for steady current is :

   (a) $\text{div}\vec{J} = 0$ (b) $\text{div}\vec{J} = \frac{\partial\rho}{\partial t}$ (c) $\text{div}\vec{J} = \frac{-\partial\rho}{\partial t}$ (d) $\text{div}\vec{J} = \rho$

   (Ujjain 1995; Gwalior 90)
8. According to modified Ampere's law :

   (a) curl $\vec{E} = \mu_0(\vec{J}_{free} + \vec{J}_d)$ (b) curl $\vec{B} = \mu_0(\vec{J}_{free} + \vec{J}_d)$

   (c) curl $\vec{E} = \mu_0(\vec{J}_{free} - \vec{J}_d)$ (d) curl $\vec{B} = \mu_0(\vec{J}_{free} - \vec{J}_d)$
9. Ampere's law for time varying current is :

   (a) $\oint\vec{B}\cdot\vec{dl} = \mu_0 I$ (b) curl $\vec{B} = \mu_0\vec{J}$

(c) $\text{curl}\,\vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$ (d) $\oint \vec{B}\cdot \vec{dl} = \mu_0 \int_S [\vec{J} + \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t}].d\vec{a}$

(Ujjain 1991; Rewa 94)

10. Ampere's law tells us :
(a) Propagation of magnetic field (b) Divergence of magnetic field
(c) Flux of magnetic field (d) Flux of current density **(Rewa 1998)**

11. Maxwell's curl equation for static magnetic field is given by :
(a) $\vec{\nabla}\cdot\vec{B} = 0$ (b) $\vec{\nabla}\cdot\vec{B} = \mu_0 \vec{J}$ (c) $\vec{\nabla}\cdot\vec{B} = \mu_0 \vec{I}$ (d) $\vec{\nabla}\times\vec{E} = 0$ **(Agra 2006)**

12. The unit of Poynting vector is :
(a) watt (b) watt/metre^2 (c) watt-metre (d) Newton

13. If $\vec{J}$ is the current density and ρ is the charge density, then $\vec{\nabla}\cdot\vec{J} = -\partial\rho/\partial t$ is called :
(a) Gauss's law (b) Ampere's law
(c) Equation of continuity (d) Coulomb's law **(Rewa 1996)**

14. Propagation vector $\vec{S}$ of an electromagnetic wave is :
(a) parallel to $\vec{H}$ but perpendicular to $\vec{E}$. (b) parallel to $\vec{E}$ but perpendicular to $\vec{H}$.
(c) perpendicular to both $\vec{E}$ and $\vec{H}$. (d) parallel to both $\vec{E}$ and $\vec{H}$. **(Gwalior 1990)**

15. The velocity of electromagnetic waves is :
(a) maximum in vacuum (b) zero in vacuum
(c) maximum in dielectric medium (d) same in any medium

16. The speed of an electromagnetic wave is c in vacuum and v in a dielectric medium. if $n = c/v$ (refractive index), then correct relation is :
(a) $n = \sqrt{\mu_r \varepsilon_r}$ (b) $n = \sqrt{\mu_r/\varepsilon_r}$ (c) $n = \sqrt{\mu_r \mu_0/\varepsilon_0}$ (d) $c = \sqrt{\mu_0 \varepsilon_0}$

17. The direction of propagation of electromagnetic waves :
(a) $\vec{E}$ (b) $\vec{B}$ (c) $\vec{E}\times\vec{B}$ (d) $\vec{B}\times\vec{E}$ **(Agra 2006)**

18. Intrinsic or characteristic impedance of free space has a value of :
(a) zero (b) π ohm (c) 73 ohm (d) 377 ohm **(Agra 2006)**

ANSWERS

1. (c), (d)	**2.** (c)	**3.** (c)	**4.** (d)	**5.** (a)	**6.** (a)	**7.** (a)
8. (b)	**9.** (d)	**10.** (d)	**11.** (d)	**12.** (b)	**13.** (c)	**14.** (c)
15. (a)	**16.** (a)	**17.** (c)	**18.** (d)			

12

Reflection and Refraction of Electromagnetic Waves

12·1. Boundary Conditions

Let us consider plane electromagnetic waves be incident at the interface of two linear dielectric media* *e.g.*, light passing from air to water, glass to air, oil to water etc. It is assumed that there are no free charges or currents in the materials. The interface or the boundary between two dielectric media is considered to be plane and the two media are having different values of permittivity (ε) and permeability (μ) *i.e.*, ε_1, μ_1, for medium 1 and ε_2, μ_2 for media, 2. When a plane electromagnetic wave is incident on the boundary of two media from medium 1, in general we get a reflected wave in midium 1 and a refracted or transmitted wave in medium 2. We plan to find the expression for the reflected and refracted waves in terms of the incident wave. We will also determine the ratio of the amplitudes (or the energies) of the reflected and refracted waves with that of incident wave. In order to do this analysis, we need to know certain conditions, called **boundary conditions**, to be satisfied between the field vectors $\vec{B}$, $\vec{E}$ and $\vec{D}$ on the two sides of the interface. These boundary conditions can be deduced from Maxwell's equations and are given below :

(1) The normal component of the electric displacement vector is continuous across the boundary (in absence of surface charges) *i.e.*,

$$D_{1n} = D_{2n} \text{ or } \varepsilon_1 E_{1n} = \varepsilon_2 E_{2n} \quad \text{...(1)}$$

(2) The normal component of the magnetic field is continuous across the boundary *i.e.*,

$$B_{1n} = B_{2n} \quad \text{...(2)}$$

(3) The tangential component of the electric field across the interface of two media is continuous *i.e.*,

$$E_{1t} = E_{2t} \quad \text{...(3)}$$

(4) The tangential component of the magnetic field intensity is continuous across the interface of two media *i.e.*,

$$H_{1t} = H_{2t} \text{ or } \frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2} \quad \text{...(4)}$$

Deduction of Boundary Conditions from Maxwell's Equations

For a charge and current free dielectric, Maxwell's equations in integral form are :

(i) $\oint_S \vec{D}\cdot d\vec{S} = 0$ or $\varepsilon \oint_S \vec{E}\cdot d\vec{S} = 0$ (over a closed suface)

(ii) $\oint_S \vec{D}\cdot d\vec{S} = 0$ (over a closed surface)

(iii) $\oint_C \vec{E}\cdot d\vec{l} = -\frac{d}{dt}\int_S \vec{B}\cdot d\vec{S}$

---

*By a linear medium, we mean that $\vec{D}$ and $\vec{B}$ are proportional to $\vec{E}$ and $\vec{H}$ respectively *i.e.*, $\vec{D} = \varepsilon \vec{E}$ and $\vec{B} = \mu \vec{H}$, where ε and μ are independent of position and direction. This implies that the media are homogeneous, isotropic and linear.

(iv) $\frac{1}{\mu}\oint_C \vec{B}\cdot d\vec{l} = \frac{d}{dt}\int_S \vec{D}\cdot d\vec{S}$ or $\frac{1}{\mu}\oint_C \vec{B}\cdot d\vec{l} = \varepsilon\frac{d}{dt}\int_S \vec{E}\cdot d\vec{S}$

We consider medium 1 and medium 2 of respective permitivities ε_1 and ε_2 with plane boundary XX'. Now consider a very small area ΔS on the boundary. Let $\vec{D}_1$ and $\vec{D}_2$ are electric displacement vectors in the two media on the two sides of ΔS, making angles θ_1 and θ_2 with the normals to ΔS [fig. 12.1] Consider a tiny Gaussian cylinder of very small height over ΔS, extending a little bit on either side of the boundary. Now, we apply Maxwell's equation (i) to this cylinder *i.e.*,

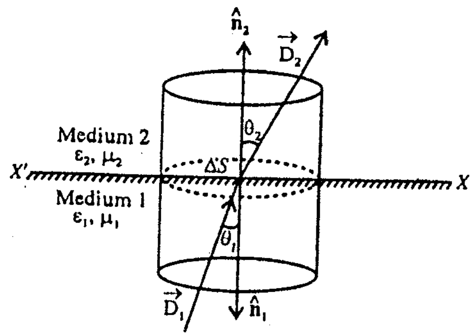

Fig. 12.1.

$$\oint_S \vec{D}\cdot d\vec{S} = 0 \qquad ...(5)$$

We consider the cylinder of being of vanishingly small height so that the flux through the curved surface is negligible. Hence there is only the flux $\vec{D}_1\cdot\Delta\vec{S} = \vec{D}_1\cdot\hat{n}_1\Delta S$ and $\vec{D}_2\cdot\Delta\vec{S} = \vec{D}_2\cdot\hat{n}_2\Delta S$ through the plane ends. Thus from eq. (5)

$$\vec{D}_1\cdot\hat{n}_1\Delta S + \vec{D}_2\cdot\hat{n}_2\Delta S = 0$$

But unit normal vector $\hat{n}_2 = -\hat{n}_1$, hence

$$(\vec{D}_1 - \vec{D}_2)\cdot\hat{n}_1\Delta S = 0 \text{ or } (D_1\cos\theta_1 - D_2\cos\theta_2)\,\Delta S = 0 \text{ or } (D_{1n} - D_{2n})\,\Delta S = 0$$

or

$$D_{1n} = D_{2n} \text{ or } \varepsilon_1 E_{1n} = \varepsilon_2 E_{2n} \qquad ...(6)$$

Using the same procedure, we obtain the following boundary condition, for magnetic field from Maxwell's equation (ii) :

$$B_{1n} = B_{2n} \qquad ...(7)$$

Now, let us consider a very thin Amperian rectangular loop $ABCD$ across the boundary surface of the two media [**fig. 12.2**] and apply Maxwell's equation (iii) to this loop :

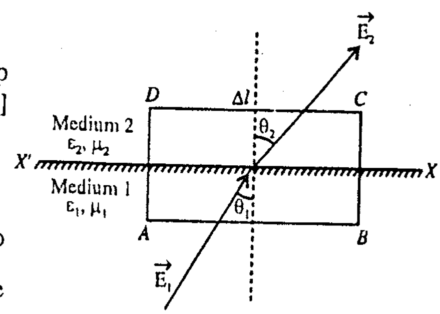

Fig. 12.2.

$$\oint_C \vec{E}\cdot d\vec{l} = -\frac{d}{dt}\int_S \vec{B}\cdot d\vec{S}$$

But in the limit as the width of the loop AD or BC goes to zero, the flux $\int_S \vec{B}\cdot d\vec{S}$ through the loop vanishes and also the contribution of the two ends AD and BC to $\oint \vec{E}\cdot d\vec{l}$ becomes zero. Hence

$$\vec{E}_1\cdot\vec{AB} + \vec{E}_2\cdot\vec{CD} = 0$$

As $\vec{AB}$ and $\vec{CD}$ have same length and are in opposite directions, hence

$$E_1\Delta l\sin\theta_1 - E_2\,\Delta l\sin\theta_2 = 0$$

or $\quad (E_1\sin\theta_1 - E_2\sin\theta_2)\,\Delta l = 0$ or $(E_{1t} - E_{2t})\,\Delta l = 0$

Thus,

$$E_{1t} = E_{2t} \qquad ...(8)$$

In the same way from Maxwell's equation (iv), we obtain

$$\frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2} \text{ or } H_{1t} = H_{2t} \qquad [\because B = \mu H] \quad ...(9)$$

12·2. Reflection and Refraction of Plane Electromagnetic Waves at a Plane Boundary of a Dielectric

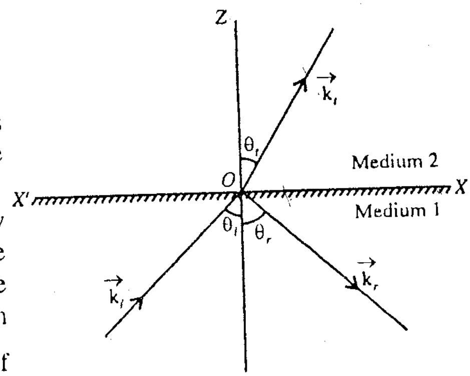

Fig. 12.3 : Reflection and refraction of an electromagnetic wave.

If a ray of light is incident on the interface of two dielectric media, it is partially reflected and the remaining part is refracted. Reflected and refracted light obey certain laws, known as laws of reflection and refraction. We shall obtain these laws on the basis of electromagnetic theory of light.

Consider a plane electromagnetic wave incident obliquely at angle (θ_i) on the interface between two dielectrics. Suppose n_1 and n_2 be the refractive indices of the two media and the plane XX' separates the medium 1 and the medium 2 as shown in fig. 12.3. Let $\vec{k}_i$, $\vec{k}_r$ and $\vec{k}_t$ be the propagation vectors of the incident, reflected and refracted rays respectively. We assume that these vectors are coplanar and lie in the X-Z plane. The electric field vector is in X-Z plane, therefore the magnetic field vector will be perpendicular to this plane. Since the field vectors ($\vec{E}_i, \vec{E}_r$ and $\vec{E}_t$) in an electromagnetic wave change periodically, hence we can write

Incident wave : $$\vec{E}_i = \vec{E}_{i0} e^{i(\vec{k}_i\cdot\vec{r}-\omega_i t)} \quad ...(1)$$

reflected wave : $$\vec{E}_r = \vec{E}_{r0} e^{i(\vec{k}_r\cdot\vec{r}-\omega_r r)} \quad ...(2)$$

and

transmitted wave : $$\vec{E}_t = \vec{E}_{t0} e^{i(\vec{k}_t\cdot\vec{r}-\omega_t t)} \quad ...(3)$$

where $\vec{E}_{i0}, \vec{E}_{r0}$ and $\vec{E}_{t0}$ are the amplitudes of the incident, reflected and refracted waves and ω_i, ω_r and ω_t are the corresponding frequencies. According to boundary conditions, the tangential component of the electric field on the interface of any two media is always continuous *i.e.*,

$$(\vec{E}_i)_{\text{tangential}} + (\vec{E}_r)_{\text{tangential}} = (\vec{E}_t)_{\text{tangential}}$$

or $$(\vec{E}_i)_x + (\vec{E}_r)_x = (\vec{E}_t)_x \quad ...(4)$$

Substituting the values from equations (1), (2) and (3)

$$(E_{i0})_x e^{i(\vec{k}_i\cdot\vec{r}-\omega_i t)} + (E_{r0})_x e^{i(\vec{k}_r\cdot\vec{r}-\omega_r t)} = (E_{t0})_x e^{i(\vec{k}_t\cdot\vec{r}-\omega_t t)} \quad ...(5)$$

In order to abey the boundary conditions, eq. (5) must be valid at all the points of the interface ($z = 0$) at all the times *i.e.*, for all the values of x, y and t. This can be satisfied if the time and space varying components of the phases are equal.

If we equate the time varying components of the phases, we get.

$$\omega_i t = \omega_r t = \omega_t t \text{ or } \omega_i = \omega_r = \omega_t = \omega \text{(say)} \quad ...(6)$$

Thus the reflected and refracted waves have the same frequency as the incident wave.

Equating the space varying components of phases, we obtain

$$(\vec{k}_i\cdot\vec{r})_{z=0} = (\vec{k}_r\cdot\vec{r})_{z=0} = (\vec{k}_t\cdot\vec{r})_{z=0} \text{ or } k_{ix}x + k_{iy}y = k_{rx}x + k_{ry}y = k_{tx}x + k_{ty}y$$

This is valid for all values of x and y, hence equating the coefficients of x and y get

$$k_{ix} = k_{rx} = k_{tx} \quad ...(7)$$

and $$k_{iy} = k_{ry} = k_{ty} \quad ...(8)$$

As the incident beam $(\vec{k}_i)$ is in the XZ plane, $k_{iy} = 0$ and hence from eq. (8)

$$k_{iy} = k_{ry} = k_{ty} = 0 \qquad \text{...(9)}$$

Thus, $\vec{k}_i$, $\vec{k}_r$ and $\vec{k}_t$ lie in the XZ plane *i.e., incident, reflected and refracted rays lie in the same plane.*

If $\hat{n}$ be the unit vector normal to the interface along Z-axis, it lies in the XZ plane. Thus, **incident, reflected, refracted rays and normal lie in the same plane.**

If θ_i, θ_r and θ_t are the angles of incidence, reflection and refraction respectively, then from eq. (7)

$$k_{ix} = k_{rx} = k_{tx}$$

or

$$k_i \sin\theta_i = k_r \sin\theta_r = k_t \sin\theta_t \qquad \text{...(10)}$$

As $k_i = \dfrac{\omega_i}{v_i}$, $k_r = \dfrac{\omega_r}{v_r}$, $\omega_i = \dfrac{\omega_r}{v_r}$ [eq. (6)] and $v_i = v_r$ (for the same medium), we find that $k_i = k_r$. Therefore from eq. (10)

$$\sin\theta_i = \sin\theta_r \quad \text{or} \quad \theta_i = \theta_r \qquad \text{...(11)}$$

Thus, **the angle of reflection is equal to the angle of incidence.** This is called **law of reflection.**

From eq. (10),
$$k_i \sin\theta_i = k_t \sin\theta_t \text{ or } \frac{\sin\theta_i}{\sin\theta_t} = \frac{k_t}{k_i} \qquad \text{...(12)}$$

But $k = \dfrac{\omega}{v}$, $\quad\therefore\quad \dfrac{\sin\theta_i}{\sin\theta_t} = \dfrac{\omega_t}{v_t} \cdot \dfrac{v_i}{\omega_i}$

From eq. (6) $\omega_t = \omega_i$, therefore

$$\frac{\sin\theta_i}{\sin\theta_t} = \frac{v_i}{v_t} = \frac{n_2}{n_1} = n \qquad \text{...(13)}$$

where n_1, n_2 are the refractive indices and v_i, v_t are the velocities of the waves in medium 1 and medium 2 respectively. Eq. (13) represents the **Snell's law of reflection.**

12·3. Polarisation by Reflection and Total Internal Reflection—Fresnel's Relations

(a) **Amplitude reflection and transmission coefficients :** The Fresnel equations describe the relations between the amplitudes of the reflected and transmitted waves with that of incident wave. These relations are expressed by **amplitude reflection** and **transmission coefficients.** *The ratio of amplitudes of reflected and incident rays, transmitted and incident rays are called amplitude (or Fresnel) reflection coefficient (r) and amplitude (or Fresnel) transmission coefficient (t) respectively.* Thus

$$\text{Amplitude (Fresnel) reflection coefficient} = \frac{\text{Amplitude of reflected ray}}{\text{Amplitude of incident ray}} \quad i.e., \; r = \frac{E_{r0}}{E_{i0}} \qquad \text{...(1)}$$

$$\text{Amplitude (Fresnel) transmission coefficient} = \frac{\text{Amplitude of transmitted ray}}{\text{Amplitude of incident ray}} \quad i.e., \; t = \frac{E_{t0}}{E_{i0}} \qquad \text{...(2)}$$

Suppose a plane polarised wave is incident on the interface of the two dielectric media and electric field vector lie in the plane of incidence [**fig. 12.4**]. In this situation, we plan to find the expressions for amplitude reflection and transmission coefficients and Fresnel's relations.

(b) **Fresnel's relations :** Let ε_1, μ_1 and ε_2, μ_2 be the permittivity and permeability of the two media respectively. The plane XX' separates the medium 1 and medium 2. Suppose the equation for the electric field of the incident wave is as follows :

$$\vec{E}_i = \vec{E}_{i0}\, e^{i(\vec{k}_i \cdot \vec{r} - \omega_i t)} \qquad \text{...(3)}$$

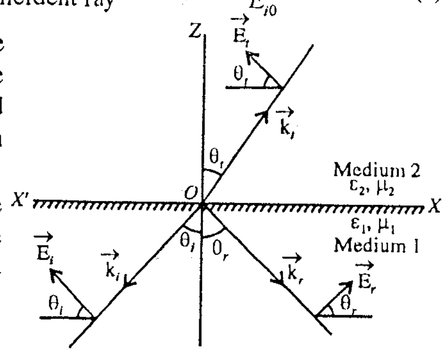

Fig. 12.4.

Therefore, the equations for reflected and refracted waves will be

$$\vec{E}_r = \vec{E}_{r0}\, e^{i(\vec{k}_r\cdot\vec{r}-\omega_r t)} \quad ...(4)$$

$$\vec{E}_t = \vec{E}_{t0}\, e^{i(\vec{k}_t\cdot\vec{r}-\omega_t t)} \quad ...(5)$$

According to the boundary condition the tangential component of the electric field $\vec{E}$ at the interface of the medium is always continuous *i.e.*,

$$(\vec{E}_i)_x + (\vec{E}_r)_x = (\vec{E}_t)_x \text{ or } -\vec{E}_i\cos\theta_i + \vec{E}_r\cos\theta_r = -\vec{E}_t\cos\theta_t$$

or

$$-E_{i0}e^{i(\vec{k}_i\cdot\vec{r}-\omega_i t)}\cos\theta_i + E_{r0}e^{i(\vec{k}_r\cdot\vec{r}-\omega_r t)}\cos\theta_r = -E_{t0}e^{i(\vec{k}_t\cdot\vec{r}-\omega_t t)}$$

But at the interface, the exponential terms are equal and above condition is satisfied at all points of the interface and time. Then, taking $\theta_i = \theta_r$,

$$-E_{i0}\cos\theta_i + E_{r0}\cos\theta_i = -E_{t0}\cos\theta_t$$

$$(E_{i0} - E_{r0})\cos\theta_i = E_{t0}\cos\theta_t$$

Dividing the above equation by E_{i0}

$$\left(1 - \frac{E_{r0}}{E_{i0}}\right)\cos\theta_i = \frac{E_{t0}}{E_{i0}}\cos\theta_t \text{ or } (1 - r)\cos\theta_i = t\cos\theta_t$$

i.e.,

$$1 - r = \frac{t\cos\theta_t}{\cos\theta_i} \quad ...(6)$$

According to another boundary condition in the absence of charges on the interface, the normal component of the electric displacement vector $\vec{D} = \varepsilon\vec{E}$ is continuous on the boundary *i.e.*,

$$\varepsilon_1(\vec{E}_i)_z + \varepsilon_1(\vec{E}_r)_z = \varepsilon_2(\vec{E}_t)_z$$

or

$$\varepsilon_1 E_{i0}\sin\theta_i + \varepsilon_1 E_{r0}\sin\theta_r = \varepsilon_2 E_{t0}\sin\theta_t$$

But $\theta_i = \theta_r$, hence

$$\varepsilon_1(E_{i0} + E_{r0})\sin\theta_i = \varepsilon_2 E_{t0}\sin\theta_t \text{ or } E_{i0} + E_{r0} = \frac{\varepsilon_2}{\varepsilon_1}E_{t0}\frac{\sin\theta_t}{\sin\theta_i}$$

Dividing the above equation by E_{i0} on both sides and putting $E_{r0}/E_{i0} = r$ and $E_{t0}/E_{i0} = t$, we obtain

$$1 + r = \frac{\varepsilon_2}{\varepsilon_1}\frac{\sin\theta_t}{\sin\theta_i}t \quad ...(7)$$

But according to Snell's law,

$$n = \frac{n_2}{n_1} = \frac{c/v_2}{c/v_1} = \frac{v_1}{v_2} = \sqrt{\frac{\mu_2\varepsilon_2}{\mu_1\varepsilon_1}} = \frac{\sin\theta_i}{\sin\theta_t}$$

For non-magnetic medium, $\mu_2 = \mu_1 = \mu$, therefore

$$n = \frac{n_2}{n_1} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} = \frac{\sin\theta_i}{\sin\theta_t} \quad ...(8)$$

Substituting eq. (8) in eq. (7), we get

$$1 + r = t\sqrt{\frac{\varepsilon_2}{\varepsilon_1}} = nt \qquad \text{...(9)}$$

Dividing eq. (6) by eq. (9)

$$\frac{1-r}{1+r} = \frac{\cos\theta_t}{n\cos\theta_i}$$

whence,

$$r = \frac{n\cos\theta_i - \cos\theta_t}{n\cos\theta_i + \cos\theta_t} \qquad \text{...(10)}$$

According to Snell's law, $n = \dfrac{\sin\theta_i}{\sin\theta_t}$

Putting it in eq. (10), we get

$$r = \frac{\dfrac{\sin\theta_i}{\sin\theta_t}\cos\theta_i - \cos\theta_t}{\dfrac{\sin\theta_i}{\sin\theta_t}\cos\theta_i + \cos\theta_t} = \frac{\sin\theta_i\cos\theta_i - \sin\theta_t\cos\theta_t}{\sin\theta_i\cos\theta_i + \sin\theta_t\cos\theta_t}$$

$$= \frac{\sin 2\theta_i - \sin 2\theta_t}{\sin 2\theta_i + \sin 2\theta_t} = \frac{2\sin(\theta_i - \theta_t)\cos(\theta_i + \theta_t)}{2\sin(\theta_i + \theta_t)\cos(\theta_i - \theta_t)}$$

i.e.,

$$r = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \qquad \text{...(11)}$$

Similary, on adding eqs. (6) and (9)

$$2 = t\left[n + \frac{\cos\theta_t}{\cos\theta_i}\right]$$

or

$$t = \frac{2\cos\theta_i}{n\cos\theta_i + \cos\theta_t} \qquad \text{...(12)}$$

Then,

$$t = \frac{2\cos\theta_i}{\dfrac{\sin\theta_i}{\sin\theta_t}\cos\theta_i + \cos\theta_t} = \frac{2\cos\theta_i\sin\theta_t}{\sin\theta_i\cos\theta_i + \sin\theta_t\cos\theta_t} \qquad \left[\because n = \frac{\sin\theta_i}{\sin\theta_t}\right]$$

$$= \frac{2\cos\theta_i\sin\theta_t}{\frac{1}{2}(\sin 2\theta_i + \sin 2\theta_t)}$$

i.e.,

$$t = \frac{2\cos\theta_i\sin\theta_t}{\sin(\theta_i + \theta_t)\cos(\theta_i - \theta_t)} \qquad \text{...(13)}$$

Equations (11) and (13) [or (10) and (12)] are **Fresnel's relations.**

Normal incidence : At the normal incidence $\theta_i = \theta_t = 0$. Therefore from eqs. (10) and (12),

$$r = \frac{n-1}{n+1} = \frac{n_2 - n_1}{n_2 + n_1} \qquad \left[\because n = \frac{n_2}{n_1}\right] \text{...(14)}$$

and

$$t = \frac{2}{n+1} = \frac{2n_1}{n_2 + n_1} \qquad \text{...(15)}$$

(c) Total internal reflection : If an electromagnetic wave is incident on the interface of the rarer medium from denser medium, then n_1 is greater than n_2. Therefore, according to Snell's law,

$$\frac{\sin\theta_i}{\sin\theta_t} = \frac{n_1}{n_2} \quad \text{or} \quad \sin\theta_t = \frac{n_1}{n_2}\sin\theta_i \qquad \text{...(16)}$$

Therefore when the value of θ_i increases, the value of θ_t increases from 0 to $\pi/2$.

If $\theta_t = \pi/2$, then $\sin\theta_t = \sin\pi/2 = 1$.

In this situation, if we write $\theta_i = \theta_C$, then

$$\sin\theta_C = \frac{n_2}{n_1} \quad ...(17)$$

This θ_C is called the **critical angle.**

Hence, critical angle is that incident angle made by the electromagnetic wave in the denser medium for which the angle of refraction is $\pi/2$ in the rarer medium.

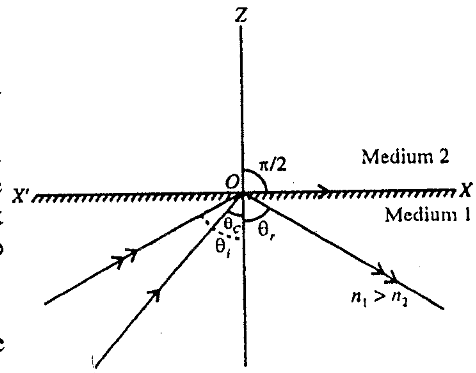

Fig. 12.5 : Total internal reflection.

As shown in **fig. 12.5,** the incident ray at the critical angle after refraction propagates along the interface of the rarer and denser media. If $\theta_i > \theta_C$, *i.e.*, the incident wave is at a greater angle than the θ_C then in this case according to equation (16),

$$\sin\theta_t > 1.$$

We know that the value of the sine of the actual angle can not be greater than 1. Therefore when $\theta_i > \theta_C$ and $n_1 > n_2$, then the wave can not enter in the other medium and it is totally reflected. This phenomenon is called **total internal reflection [fig. 12.5].** Now in case of $\theta_i > \theta_C$, the value of θ_t becomes imaginary.

From equation (10),

$$r = \frac{\frac{n_2}{n_1}\cos\theta_i - \cos\theta_t}{\frac{n_2}{n_1}\cos\theta_i + \cos\theta_t} = \frac{\cos\theta_i - \frac{n_1}{n_2}\cos\theta_t}{\cos\theta_i + \frac{n_1}{n_2}\cos\theta_t}$$

Using eq. (16), we get

$$\frac{n_1}{n_2}\cos\theta_t = \frac{n_1}{n_2}\sqrt{1-\sin^2\theta_t} = \frac{n_1}{n_2}\sqrt{1-\frac{n_1^2}{n_2^2}\sin^2\theta_i}$$

$$= i\frac{n_1}{n_2}\sqrt{\frac{n_1^2}{n_2^2}\sin^2\theta_i - 1} = i\alpha$$

where, $i = \sqrt{-1}$ and $\alpha = \frac{n_1}{n_2}\sqrt{\frac{n_1^2}{n_2^2}\sin^2\theta_i - 1}$

$$\therefore \quad r = \frac{\cos\theta_i - i\alpha}{\cos\theta_i + i\alpha} = \frac{\left(\sqrt{\cos^2\theta_i + \alpha^2}\right)e^{-i\beta}}{\left(\sqrt{\cos^2\theta_i + \alpha^2}\right)e^{i\beta}} = e^{-2i\beta} \quad ...(18)$$

where $\beta = \tan^{-1}(\alpha/\cos\theta_i)$ ...(19)

Now, $$|r| = 1, \text{ i.e., } \frac{|E_{r0}|}{|E_{i0}|} = 1 \quad ...(20)$$

Therefore, the amplitude of the reflected ray is equal to the amplitude of the incident ray *i.e.*, the incident ray is totally reflected from the interface. This phenomenon is termed as **total internal reflection.**

(d) **Polarisation by reflection and Brewster's Law :** From eq. (11), we have

$$r = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} = \frac{E_{r0}}{E_{i0}} \quad ...(21)$$

If the angle of incidence $\theta_i = \frac{\pi}{2} - \theta_t$ or $\theta_i + \theta_t = \pi/2$, then the angle between reflected and refracted waves is $\pi/2$ and $\tan(\theta_i + \theta_t) = \tan\pi/2 = \infty$. Hence from eq. (21), $r = 0$ or $E_{r0} = 0$.

Thus when reflected and refracted rays are perpendicular to each other and the electric field vector of the plane polarised wave is in the plane of incidence, the amplitude of reflected wave (E_{r0}) is zero *i.e.*,, there is no reflected wave. In such a case, the angle of incidence is called **Brewster's angle**. Let the Brewster's angle be denoted by ϕ *i.e.*,

$$\phi = \theta_i = \frac{\pi}{2} - \theta_t \qquad ...(22)$$

According to Snell's law,

$$n = \frac{n_2}{n_1} = \frac{\sin\theta_i}{\sin\theta_t} = \frac{\sin\phi}{\sin\left(\frac{\pi}{2} - \phi\right)} = \frac{\sin\phi}{\cos\phi} = \tan\phi$$

$\therefore$ Brewster's angle, $\phi = \tan^{-1}(n)$ ...(23)

Now, let an unpolarised light wave be incident at the interface. Its electric vector can be resolved into two components, one parallel to the plane of incidence and the other perpendicular to it. In **fig. 12.6**, the parallel component has been shown by double headed arrow and perpendicular component by dot. If the unpolarised wave is incident on the interface at the **Brewster's angle**, then the parallel component of the reflected light will be zero and only the perpendicular component of the wave will be reflected. Thus the reflected wave will be **linearly polarised** with vibrations perpendicular to the plane of incidence. This is why the Brewster's angle is sometimes called **polarising angle.**

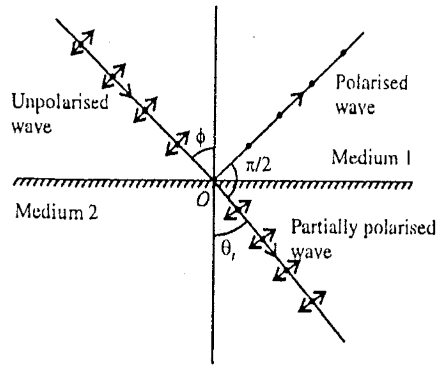

Fig. 12.6 : Polarisation by reflection-Brewster's law.

(e) Reflectance (R) and Transmittance (T) : Consider an electromagnetic wave incident at an angle of incidence θ_i [**fig. 12.7**]. If it falls on an area A of the interface of the two media, the cross sectional areas of the incident, reflected and transmitted waves are $A\cos\theta_i$, $A\cos\theta_r$ and $A\cos\theta_t$ respectively. If I represents the intensity of a wave*, then the incident, reflected and transmitted powers corresponding to the incident, reflected and transmitted waves are $I_i A\cos\theta_i$, $I_r A\cos\theta_r$ and $I_t A\cos\theta_t$ respectively. The **reflectance** R of a surface is defined as the *ratio of the reflected power to incident power i.e.*,

$$R = \frac{I_r A\cos\theta_r}{I_i A\cos\theta_i} = \frac{I_r}{I_i} \quad [\because \theta_r = \theta_i] \qquad ...(24)$$

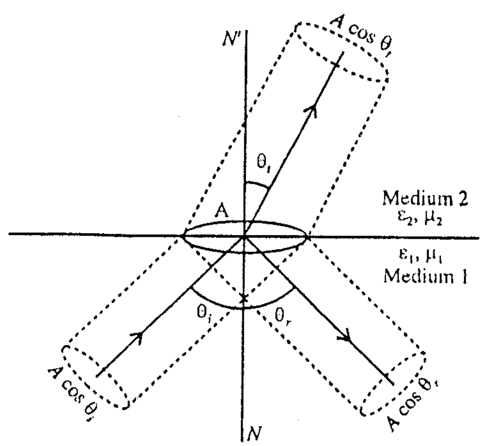

Fig. 12.7.

where $I_i = \frac{1}{2}v_i\varepsilon_i E_{i0}^2$ and $I_r = \frac{1}{2}v_r\varepsilon_r E_{r0}^2$. As the incident and the reflected waves are in the same medium, hence $\varepsilon_i = \varepsilon_r$ and $v_i = v_r$. Thus

* The intensity of a wave is the energy per unit time or power carried by the wave per unit area. Hence the power carried through an area α will be $I\alpha$. The energy carried per unit area per unit time is I = energy contained in a cylinder of cross sectional area unity and length v (speed of the wave) $\times$ energy density $= v \times \frac{1}{2}\varepsilon E^2 = \frac{1}{2}v\varepsilon E^2$.

$$R = \frac{E_{r0}^2}{E_{i0}^2} = r^2 \qquad ...(25)$$

The **transmittance** T is defined as the ratio of the transmitted power to the incident power *i.e.*,

$$T = \frac{I_t A\cos\theta_t}{I_i A\cos\theta_i} = \frac{\varepsilon_2 v_2 E_{t0}^2}{\varepsilon_1 v_1 E_{i0}^2}\frac{\cos\theta_t}{\cos\theta_i} = \frac{n_2\cos\theta_t}{n_1\cos\theta_i}\cdot\frac{E_{t0}^2}{E_{i0}^2} = \frac{n_2\cos\theta_t}{n_1\cos\theta_i}t^2 \qquad ...(26)$$

because $n = \frac{n_2}{n_1} = \frac{c/v_2}{c/v_1} = \frac{v_1}{v_2} = \sqrt{\frac{\mu_2\varepsilon_2}{\mu_1\varepsilon_1}} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$ for $\mu_1 = \mu_2$ and therfore $\varepsilon_2 v_2/\varepsilon_1 v_1 = \sqrt{\varepsilon_2/\varepsilon_1} = n_2/n_1$

For normal incidence, $$R = r^2 = \left(\frac{n-1}{n+1}\right)^2 = \left(\frac{n_2 - n_1}{n_2 + n_1}\right)^2 \qquad ...(27)$$

and $$T = nt^2 = n\left(\frac{2}{n+1}\right)^2 = \frac{4n}{(n+1)^2} = \frac{n_2}{n_1}\left(\frac{2}{n_2/n_1 + 1}\right)^2 = \frac{4n_1 n_2}{(n_2 + n_1)^2} \qquad ...(28)$$

Reflectance and transmittance are also called as the **reflection coefficient** and **transmission coefficient for intensity**.

The conservation of energy requires that the sum of reflected energy and transmitted energy must be equal to the incident energy *i.e.*, $R + T = 1$. The proof is as follows :

From eqs. (25) and (26), we have

$$T = \frac{n_2}{n_1}\frac{\cos\theta_t}{\cos\theta_i}t^2 \text{ and } R = r^2$$

But we know $$\frac{\cos\theta_t}{\cos\theta_i} = \frac{1-r}{t} \text{ and } \frac{n_2}{n_1} = \frac{1+r}{t}. \qquad \text{[from eqs. (6) and (9)]}$$

$$\therefore \quad T = \frac{1+r}{t}\cdot\frac{1-r}{t}\cdot t^2 = 1 - r^2 = 1 - R$$

Thus $$R + T = 1. \qquad ...(29)$$

12·4. Faraday Effect

In 1845 Michael Faraday discovered that when flint glass is subjected to a strong magnetic field, then it becomes optically active. When plane-polarised light passes through the glass along the direction of the applied magnetic field, then the plane of vibration is rotated. This is called **Faraday effect**. Later on, this phonomenon has been observed in isotropic solids, liquids and gases.

Fig. 12.8 shows the experimental arrangement to study the Faraday effect.

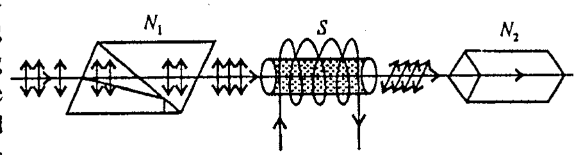

Fig. 12.8.

The tube T filled with the transparent medium is placed along the axis of the solenoid S. A strong magnetic field is produced in the solenoid by passing high electric current. In absence of the current in the solenoid, both the nicol prisms N_1 and N_2 are adjusted in the crossed position. When the magnetic field is subjected to the transparent medium by passing electric current in the solenoid, light emerges from the nicol prism N_2. Obviously the light will emerge from the nicol prism N_2 due to the rotation of plane of polarisation. In order to bring back the nicol prism N_2 in the crossed position, it is rotated by such angle so that no light energes from this nicol prism N_2. The angle by which the nicol prism has been rotated to get the new crossed position is the **angle of rotation of the plane of polarisation.** Experimentally it is found that the angle of rotation θ is proportional to the product of the magnetic field B and the distance (l) travelled by light in the transparent medium. At a particular temperature and wavelength, this angle of rotation of the plane of polarisation is given by

$$\theta = VBl$$

where the magnetic field B is in tesla and the length l of the transparent medium is in metres. The constant of proportionality V is called **Verdet constant** and is difined as the rotation per unit path (length) per unit magnetic field. For some materials, the value of the Verdet constant is given in the **table.**

Table : Values of the Verdet constant in minutes of arc per tesla per meter for λ = 5893Å

Substance	t°C	V
Water (H_2O)	20	1.31×10^4
Glass (phosphate crown)	18	1.61×10^4
Glass (light flint)	18	3.17×10^4
Phosphorus	33	13.26×10^4
Quartz (perpendicular to axis)	20	1.66×10^4
Acetone	15	1.109×10^4
Salt (NaCl)	16	3.585×10^4
CS_2	20	4.23×10^4
Ethyl alcohol	25	1.112×10^4

12·5. Electromagnetic Waves in a Conducting Medium

When we deduced the wave equation for electromagnetic fields, we considered the free charge density ρ and free current density $\vec{J}$ to be zero *i.e.*, $\rho = 0$ and $\vec{J} = 0$. Such a condition is perfectly reasonable when we are considering the wave propagation through vacuum or through dielectric media such as glass or pure water. However in case of conductors, *e.g.*, metals, we have to consider the flow of charge and hence the free current density $\vec{J}$ is not equal to zero. This obeys the Ohm's law $\vec{J} = \sigma\vec{E}$. In such a case, the Maxwell's equations are :

$$\vec{\nabla}\cdot\vec{E} = 0 \quad ...(1)$$

$$\vec{\nabla}\cdot\vec{B} = 0 \quad ...(2)$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \quad ...(3)$$

and

$$\vec{\nabla}\times\vec{B} = \mu\sigma\vec{E} + \mu\varepsilon\frac{\partial\vec{E}}{\partial t} \quad ...(4)$$

where we have taken charge density $\rho = 0$ in the steady state.

Taking cross product of $\vec{\nabla}$ operator on both sides of eq. (3), we get

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = -\frac{\partial}{\partial t}(\vec{\nabla}\times\vec{B}) = -\frac{\partial}{\partial t}\left(\mu\sigma\vec{E} + \mu\varepsilon\frac{\partial\vec{E}}{\partial t}\right) \quad \text{[using eq. (4)]}$$

or

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = -\mu\sigma\frac{\partial\vec{E}}{\partial t} - \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \quad ...(5)$$

From vector triple product, $\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}\cdot\vec{C}) - (\vec{A}\cdot\vec{B})\vec{C}$

Therefore, with the help of this equation, left hand side of eq. (5) can be rewritten as

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = \vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - (\vec{\nabla}\cdot\vec{\nabla})\vec{E} = -\nabla^2\vec{E} \;\; (\because \vec{\nabla}.\vec{E} = 0) \quad ...(6)$$

Now from eqs. (5) and (6),

$$\nabla^2\vec{E} - \sigma\mu\frac{\partial\vec{E}}{\partial t} - \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} = 0 \quad ...(7)$$

Similarly,

$$\nabla^2\vec{B} - \sigma\mu\frac{\partial\vec{B}}{\partial t} - \mu\varepsilon\frac{\partial^2\vec{B}}{\partial t^2} = 0 \quad ...(8)$$

These equations (7) and (8) have plane wave solutions for the electromagnetic waves travelling along Z-direction, given by

$$\vec{E}(z,t) = E_0 e^{i(kz-\omega t)} \quad ...(9)$$

$$\vec{B}(z,t) = B_0 e^{i(kz-\omega t)} \quad ...(10)$$

where the wave number k is complex given by

$$k^2 = \mu\varepsilon\omega^2 + i\mu\sigma\varepsilon \quad ...(11)$$

which can be easily checked by putting eqs. (9) and (10) in eqs. (7) and (8). For vacuum or dielectric, when $\sigma = 0,\ k = \omega\sqrt{\mu\varepsilon} = \omega/c.$

Taking square root of eq. (11), we find

$$k = \gamma + i\delta \quad ...(12)$$

where $\gamma = \omega\sqrt{\frac{\varepsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2}+1\right]^{\frac{1}{2}}$ and $\delta = \omega\sqrt{\frac{\varepsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2}-1\right]^{\frac{1}{2}}$ ...(13)

The imaginary part of k results in an **attenuation** (*i.e.*, decreasing amplitude with z) of the waves. In view of eq. (12), eqs. (9) and (10) become

$$\vec{E}(z,t) = E_0 e^{-\delta z} e^{i(\gamma z-\omega t)} \text{ and } \vec{B}(z,t) = B_0 e^{-\delta z} e^{i(\gamma z-\omega t)} \quad ...(14)$$

The amplitude $E_0' = E_0 e^{-\delta z}$ becomes $1/e$ in the distance $z = 1/\delta$, say d. *The distance d in which the amplitude of the electromagnetic wave reduces by a factor of $1/e$, is called* **skin depth**. Thus

$$\text{skin depth, } d = \frac{1}{\delta} \quad ...(15)$$

It is a measure of the depth to which an electromagnetic wave penetrates into a conductor.

In eq. (14) the real part of k *i.e.*, γ determines the wavelength, the speed of propagation and the index of refraction in the usual way *i.e.*,

$$\lambda = \frac{2\pi}{\gamma},\ v = \frac{\omega}{\gamma} \text{ and } n = \frac{c\gamma}{\omega} \quad ...(16)$$

Now, let us discuss the cases of poor and good conductors.

(1) **In case of a poor conductor**, $\sigma \ll \varepsilon\omega$, and hence from eq. (13)

$$\gamma = \sqrt{\varepsilon\mu} \text{ and } \delta = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}} \quad ...(17)$$

Thus *for a poor conductor, the skin depth is independent of frequency.*

(2) **In case of a good conductor**, $\sigma \gg \varepsilon\omega$, and hence from eq. (13)

$$\gamma = \delta = \sqrt{\frac{\omega\sigma\mu}{2}} \quad ...(18)$$

Thus **for a good conductor, the skin depth decreases with the increasing frequency of the electromagnetic wave.**

It is to be noted that good and poor conductors depend on the frequency. The same substance can be a good conductor at low frequencies and a poor conductor at high frequencies. It may become clear from the following example :

Example : For sea water $\mu = \mu_0 = 4\pi \times 10^{-7}$ N/A^2, $\varepsilon_r = 70$ or $\varepsilon = 70\ \varepsilon_0 = 6 \times 10^{-10}$ Coul.^{2}/N-m^2 and $\sigma = 5$ ohm^{-1} m^{-1}. Hence

$$\frac{\sigma}{\varepsilon} = \frac{5}{6 \times 10^{-10}} \simeq 10^{10}\,\text{per sec}.$$

For poor conductor, $\sigma << \varepsilon\omega$, or $\frac{\sigma}{\varepsilon} << \omega$.

Hence sea water is poor conductor, for frequencies above 10^{10} Hz (including visible region).

Sea water is a good conductor for radio frequencies (in KHz, MHz) *i.e.*, frequencies much less than 10^{10} Hz. For good conductors $\sigma >> \varepsilon\omega$, or $\frac{\sigma}{\varepsilon} >> \omega$.

12·6. Reflection and Refraction of Electromagnetic Waves by Ionosphere

The ionosphere is that region of the earth's atmosphere in which the constituent gases are ionized by radiations mainly coming from the sun. This ionized medium contains electrons and positive ions and is also called **plasma.** The region of the ionosphere extends from about 50 km above the earth to several earth radii (radius of earth = 6400 km approximately). With the altitude the ionization increases with the maximum in ionization density at about 300 km and then it decreases. It appears that the conductivity of the ionized region is almost exclusively due to the electron density N with value about 10^{11} per metre^3. In fact, ionosphere have several regions in which the ionization density either reaches to maximum value or remains roughly constant. In order of height, these regions are designated by D, E and F. Although, the height of the layers vary during 24 hrs of a day, the E and F layers have a permanent existence and that is why long distance radio communication becomes possible by reflection of radio waves by these layers of the ionosphere [**fig. 12.9**].

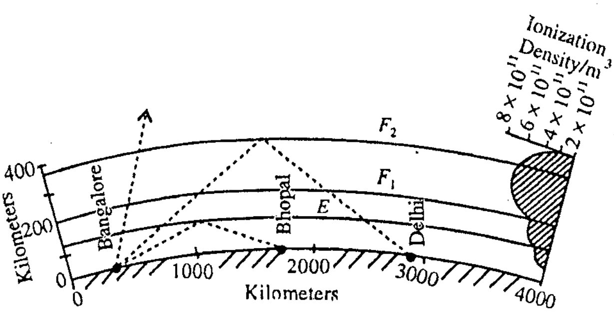

Fig. 12.9 : Ionosphere.

The radio waves, transmitted from the transmitting station and received at a distant place after reflection from the ionosphere, are called **sky waves.**

Basically, the machanism of reflection and refraction of radio waves by the ionosphere is a function of frequency. Radio waves of high frequencies in the range 3 MHz to 30 MHz can be propagated to a long distance by reflection from the ionosphere and hence are of much practical importance. This is why we will study here **the reflection and refraction of high frequency waves from the ionosphere.**

When an electromagnetic wave passes through an ionized medium, current arises due to the presence of free electrons and ions in the medium.

Consider an electomagnetic wave of frequency ω, moving through an ionised medium containing electrons and ions. Due to the electromagnetic wave, let the electric field in the ionized medium be represented by

$$E = E_0 \cos \omega t \qquad ...(1)$$

The ionised medium is containing electrons and positive ions. As the mass of an ion is much larger than the mass of an electron, hence the velocity and acceleration acquired by the ions due to the electric field is much less than those acquired by the electrons. Hence we consider the effect of the electric field on the motion of electrons only. In the present analysis, we also do not consider the effect of magnetic field of the earth.

The force on an electron due to the electric field is given by

$$F = -eE = -eE_0 \cos \omega t \quad ...(2)$$

If v_d is the drift velocity of the electron, then

$$F = m\frac{dv_d}{dt} \quad ...(3)$$

Therefore, from eqs. (2) and (3)

$$\frac{mdv_d}{dt} = -eE_0 \cos \omega t \text{ or } \frac{dv_d}{dt} = -\frac{eE_0}{m}\cos\omega t$$

Integrating it, we obtain

$$v_d = -\frac{eE_0}{m\omega}\sin\omega t \quad ...(4)$$

where the constant of integration has been taken zero by taking initial condition at $t = 0$, $v_d = 0$

Hence the current density in the ionised medium is

$$J = -Nev_d = \frac{Ne^2}{m\omega}E_0 \sin\omega t \quad ...(5)$$

where N is the electron density *i.e.*, number of electrons per metre cube in the medium.

Additionally, there is the displacement current density due to the electric field E in the charge free space *i.e.*,

$$J_d = \varepsilon_0\frac{dE}{dt} = \varepsilon_0\frac{d}{dt}(E_0 \cos\omega t) = -\varepsilon_0 E_0\omega \sin \omega t \quad ...(6)$$

We thus have the total current density (J_t) in the ionised medium

$$J_t = J + J_d = \frac{Ne^2}{m\omega}E_0 \sin\omega t - e_0E_0\omega \sin\omega t$$

i.e.,

$$J_t = -\varepsilon_0\left(1 - \frac{Ne^2}{\varepsilon_0 m\omega^2}\right)E_0\omega \sin\omega t \quad ...(7)$$

From eqs. (6) and (7), we note that the permittivity of the free space ε_0 is replaced by $\varepsilon_0(1 - Ne^2/\varepsilon_0 m\omega^2)$ for ionised medium with free electrons. Hence the permittivity of the ionised medium with electrons is given by

$$\varepsilon = \varepsilon_0\left(1 - \frac{Ne^2}{\varepsilon_0 m\omega^2}\right) \text{ or } \varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = 1 - \frac{Ne^2}{\varepsilon_0 m\omega^2} \quad ...(8)$$

Thus *the permittivity of ionised medium with electrons is less than that of free space.*

Putting the values of electronic charge, $e = 1.6 \times 10^{-19}$ C, electron mass $m = 9.1 \times 10^{-31}$ kg, $\varepsilon_0 = 8.854 \times 10^{-12}$ farad/m and $\omega = 2\pi f$ in eq. (8), we obtain

$$\varepsilon_r = 1 - \frac{(1.6 \times 10^{-19})^2 N}{8.854 \times 10^{-12} \times 9.1 \times 10^{-13} \times (2 \times 3.14)^2 \times f^2}$$

$$= 1 - \frac{80.6N}{f^2} \approx 1 - \frac{81N}{f^2} \quad ...(9)$$

The phase velocity of the electromagnetic wave in a medium is given by

$$v_p = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c\sqrt{\mu_0\varepsilon_0}}{\sqrt{\mu\varepsilon}} = \frac{c}{\sqrt{\mu_r\varepsilon_r}} \quad ...(10)$$

where c = speed of light = $1/\sqrt{\mu_0\varepsilon_0}$; μ_0 and ε_0 are the permeability and permittivity for free space and μ and ε are those for the medium respectively. $\mu_r = \mu/\mu_0$ and $\varepsilon_r = \varepsilon/\varepsilon_0$ are the relative permeability and permittivity for the medium respectively.

We assume that the permeability of the medium of the ionosphere is not changed by the presence of electrons. Hence $\mu_r = 1$ and the phase velocity from eq. (10) is

$$v_p = \frac{c}{\sqrt{\varepsilon_r}} = \frac{c}{\sqrt{1 - 81N/f^2}} \qquad \left[\because \varepsilon_r = 1 - \frac{81N}{f^2}\right] \quad ...(11)$$

The refractive index of a medium is given by

$$n = \frac{c}{v_p} = \frac{\text{Speed of light in vacuum}}{\text{Phase velocity in the medium}} = \frac{c}{c/\sqrt{1 - 81N/f^2}}$$

i.e.,

$$n = \sqrt{1 - \frac{81N}{f^2}} = \sqrt{\varepsilon_r} \quad ...(12)$$

Thus the value of refractive index is less than 1. Hence when a radio wave enters the ionized medium from an unionized medium, the ionized medium behaves as a rarer medium and the wave moves away from the normal *i.e.*, angle of refraction is more than the angle of incidence.

As we proceed up in the ionosphere from the boundary of unionized and ionized medium, the electron density first increases and hence the refractive index decreases accordingly. Thus if we consider the ionosphere to be composed of various layers of infinitesimal thickness of increasing electron density or decreasing refractive index, then as the electro-magnetic wave goes more and more inside the ionosphere, at different layers angle of refraction increases at each layer correspondingly. Ultimately a level may be reached where the electron density is such that the angle of refraction is $\pi/2$ [**fig. 12.10**]. and the wave will move horizontally (at B).

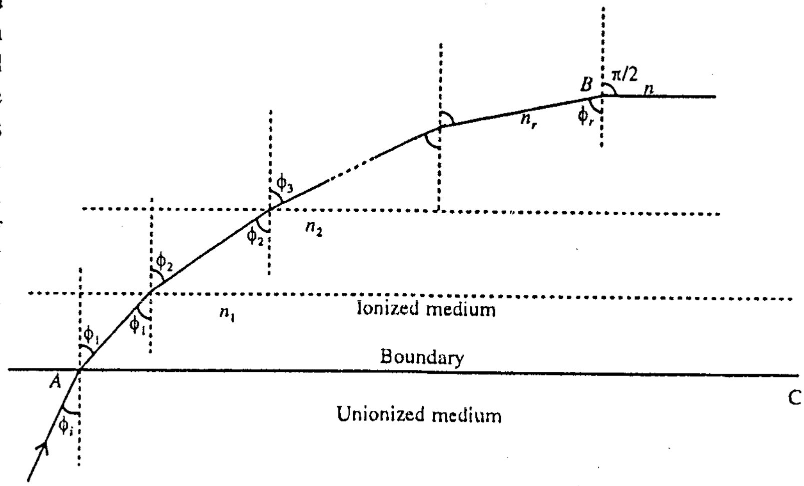

Fig. 12.10 : Refraction of electromagnetic wave at various layers of infinitesimal thickness in the ionosphere.

Now, as shown in **fig 12.10**, when radio wave crosses the boundary of the unionized medium and first layer of ionosphere, the refractive index of first layer is given by

$$n_1 = \frac{\sin\phi_i}{\sin\phi_1} \text{ or } \sin\phi_i = n_1 \sin\phi_1 \quad ...(13a)$$

When it crosses the boundary of the first and second layers, the refractive index of second layer with respect to first layer is given by

$$\frac{n_2}{n_1} = \frac{\sin\phi_1}{\sin\phi_2} \text{ or } n_2 \sin\phi_2 = n_1 \sin\phi_1 \quad ...(13b)$$

Similarly for third layer,

$$\frac{n_3}{n_2} = \frac{\sin\phi_2}{\sin\phi_3} \text{ or } n_3 \sin\phi_3 = n_2 \sin\phi_2 \quad ...(13c)$$

Ultimately as the wave crosses the r^{th} layer so that angle of refraction becomes $\pi/2$,

$$\frac{n}{n_r} = \frac{\sin\phi_r}{\sin\pi/2} \text{ or } n \sin\frac{\pi}{2} = n_r \sin\phi_r \qquad \text{...(13d)}$$

Hence from eqs. (13a), (13b), (13c),..... we have

$$\sin\phi_i = n_1 \sin\phi_1 = n_2 \sin\phi_2 = n_3 \sin\phi_3 = \ldots\ldots = n_r \sin\phi_r = n \sin\pi/2$$

where n is the refractive index at the level at B.

Thus, $$n = \frac{\sin\phi_i}{\sin\pi/2} = \sin\phi_i \qquad \text{...(14)}$$

Now the path of the electromagnetic wave becomes horizontal. Let B be the highest point reached by the wave. This point corresponds to the electron density of sufficient amount, say N', satisfying the relation [from eqs. (12) and (14)]

$$\sqrt{\frac{1-81N'}{f^2}} = \sin\phi_i \text{ or } N' = \frac{f^2\cos^2\phi_i}{81} \qquad \text{...(15)}$$

The wave is now reflected from the level of the point B (having density N') travels downwards and finally reaches on the earth [fig. 12.11].

If the maximum electron density in a layer is less than that required by the relation (15) the wave will penetrate the layer and may be reflected from the higher layer of greater electron density.

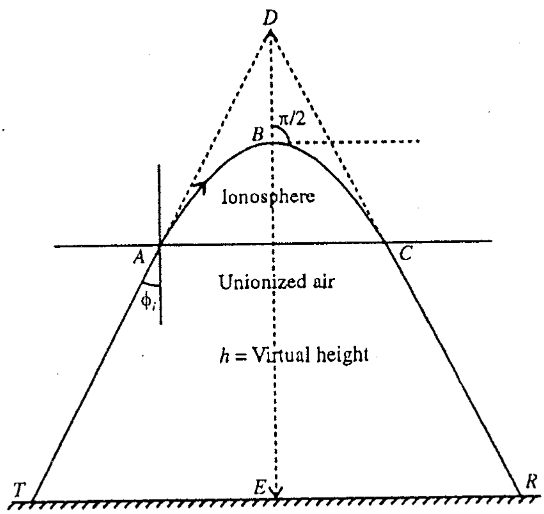

Fig. 12.11 : Reflection of radio wave in the ionosphere.

Use of light waves is inadmissible as sky waves : Let us see what happens if light (or an electromagnetic wave of very high frequency) instead of radio wave is used as sky wave. The frequency of the light wave is the order of 10^{14} Hz (for light of wavelength of 5000 Å, $f = c/\lambda = 3 \times 10^8/5000 \times 10^{-10}$). Hence for electron density (or electrons per unit volume) 10^{11} per m^3, we have from eq. (12)

$$n = \sqrt{\frac{1-81N}{f^2}} = \sqrt{1 - \frac{81\times10^{11}}{(10^{14})^2}} \approx \sqrt{1-10^{-15}} = 1$$

Thus the refractive index of the ionized medium is almost 1 and hence there is no effect of the ionized medium (ionosphere) on the propagation of very high frequency electromagnetic waves such as light waves. Thus such electromagnetic waves will penetrate the ionosphere without deviation and cannot be reflected back to the earth. Hence *light waves cannot be used for long distance communication by reflecting from the ionosphere.*

Critical frequency : *The largest electron density required for reflection occurs when the angle of incidence ϕ_i is zero, i.e., for vertical incidence.* Let this maximum electrons density be N_{max}. Hence for vertical incidence $\phi_i = 0$ and from eq. (15) $N_{max} = \frac{f^2}{81}$ or $f = \sqrt{81N_{max}} = f_C$ (say) ...(16)

For the layer having N_{max} electron density (electrons/m^3), the highest frequency that will be reflected back for vertical incidence, satisfying relation (16), is called **critical frequency** for that layer. From eq. (14), for vertical incidence, $\phi_i = 0$, the refractive index is zero and corresponds to the layer of N_{max} electron density. Now, if the frequency is increased more than the critical frequency, the wave penetrates the layer and is reflected from a higher layer of more electron density so that the relation (16) is satisfied.

Virtual height for reflection : Consider a sky wave, incident obliquely at an angle ϕ_i on the ionized medium. It travels from T to R following the path TBR **[fig. 12.11]**. The apex B of the path corresponds to an ionization density (N') sufficient to reflect back the incident wave. The paths TA and CR are travelled in air ($n = 1$) with speed of light in free space and the path ABC lies in ionized medium. The **real height** reached by the radio wave is BE, while the height DE is the **virtual height** (h) for the wave. The virtual height is always greater than the true height of the point of reflection. *The virtual height h is that height from which a radio wave sent up at an angle appears to be reflected.* It is also the height obtained by pulse measuring technique (where a radio frequency pulse of energy of short duration is sent from T and received at R), because it can be shown that the time of travel around the real curved path ABC in the ionosphere is the same as would be for the wave to travel the path ADC (if the ionosphere were replaced by a mirror like reflecting surface at D).

Maximum Usable Frequency (MUF) :

Although, the critical frequency for any layer represents the heighest frequency which is reflected from that layer at vertical incidence, it is not the highest frequency that can be reflected from the layer. The highest frequency that can be reflected depends also upon the angle of incidence, and therefore for a given layer height, upon the distance between the transmitting and receiving points of the signal. *The maximum frequency that can be reflected back for a given distance of transmission is termed as the* **maximum usable frequency** (MUF) *for that distance.* From eq. (16) the critical frequency for the layer of electron density N_{max} is given by

$$f_C = \sqrt{81 N_{max}} \qquad ...(17)$$

Now, if a wave of frequency f is sent obliquely at an angle of incidence ϕ_i, then it will be reflected from the layer of electron density N_{max} in the conditon, given by eq. (15) *i.e.*,

$$N_{max} = \frac{f^2 \cos^2 \phi_i}{81} \qquad ...(18)$$

From eqs. (17) and (18), we get

$$\frac{f_C^{\,2}}{81} = \frac{f^2 \cos^2 \phi_i}{81} \text{ or } f^2 = \frac{f_C^{\,2}}{\cos^2 \phi_i} \quad \text{or} \quad f = f_C \sec \phi_i \qquad ...(19)$$

This is the expression for **maximum usable frequency** (MUF) and is sec ϕ_i times the critical frequency for the given layer. This is called **secent law.** If the critical frequency for a layer is known, then the maximum usable frequency (MUF) for given angle of incidence or distance can be determined by the use of eq. (19).

Optimum frequency : As the maximum usable frequency may show small variations about the monthly average of upto 15%, hence a frequency somewhat lower than the predicted maximum usable frequency is used. The optimum frequency to transmit the radio waves between any two points at a distance is therefore selected as some frequency about 50 to 85% of the predicted maximum usable frequency for that distance.

Skip Distance

As the angle of incidence at the ionosphere decreases, the distance from the transmitter to the point at which the radio wave of a particular frequency returns to the earth, first decreases and at a particular angle of incidence, this distance becomes minimum which is called the **skip distance.** If the angle of incidence is further decreased, the distance from transmitter to the point of return first increases and then for more small angles of incidences the wave penetrates the layer **[fig. 12.12]**

The minimum distance from the transmitter to the point on the earth at which a sky wave of definite *fequency will be returned to the earth by reflection from the ionosohere is called* **skip distance.**

The maximum usable frequency (MUF) corresponding to angle of incidence ϕ_i, which will be reflected from a layer, having N_{max} electron density or $f_C = \sqrt{81N_{max}}$ critical frequency, is given by

$$f = f_C \sec \phi_i \quad ...(20)$$

Fig. 12.13 shows the transmission path for the reflection of radio wave from an ionized layer at P. We consider the earth to be flat. Let D be the distance between the transmitting and receiving points and h be the height of the layer from which reflection takes place. From **fig. 12.13**,

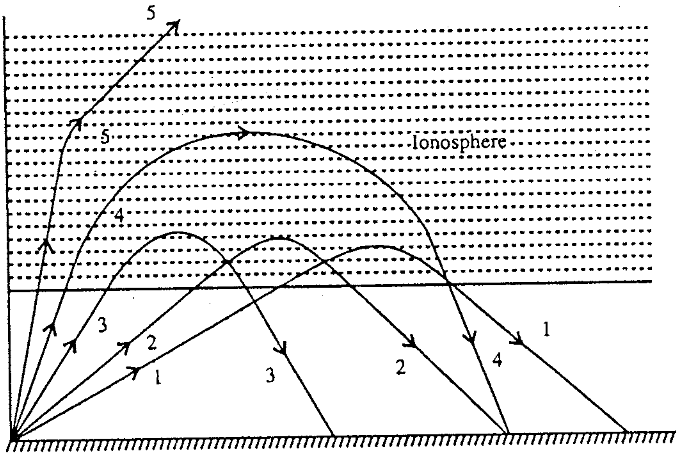

Fig. 12.12 : Illustration of skip distance.

$$\tan \phi_i = \frac{D}{2h}, \therefore\ 1 + \tan^2 \phi_i = 1 + \frac{D^2}{4h^2} \text{ or } \sec^2 \phi_i = 1 + \frac{D^2}{4h^2} \text{ or } \sec \phi_i = \sqrt{1 + \frac{D^2}{4h^2}}$$

Hence,

$$f = f_C \sqrt{1 + \frac{D^2}{4h^2}} \quad ...(21)$$

This is the **maximum usable frequency** (MUF) for the minimum distance *i.e.*, skip distance, upto which a particular frequency can be used. From relation (21), the skip distance D is given by

$$D = 2h\sqrt{\left(\frac{f^2}{f_C^2} - 1\right)} \quad ...(22)$$

As f_C and h are the characteristics of any ionospheric layer, hence each layer in the ionosphere has a particular skip distance for each maximum usable frequency f. For a particular layer, the skip distance increases with the maximum usable frequency.

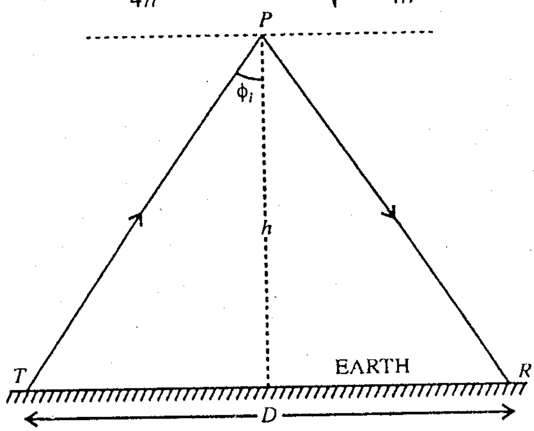

Fig. 12.13 : Determination of skip distance.

Ex. 1. *The amplitude reflection coefficient and amplitude transmission coefficient are r and t respectively for the light wave propagating from medium 1 to medium 2. If the path of the transmitted ray is reversed i.e., it becomes incident ray then above coefficients become r′ and t′ respectively. Establish the relation* $1 + rr' = tt'$.

It is known as Stoke's relation.

Sol. In the first case [**fig. 12.14 (a)**],

$$r = \frac{\sin\theta_i \cos\theta_i - \sin\theta_t \cos\theta_t}{\sin\theta_i \cos\theta_i + \sin\theta_t \cos\theta_t}$$

and

$$t = \frac{2\cos\theta_i \sin\theta_t}{\sin\theta_i \cos\theta_i + \sin\theta_t \cos\theta_t}$$

In the second case [**fig. 12.14 (b)**],

$$\theta_i' = \theta_t,\ \theta_t' = \theta_i$$

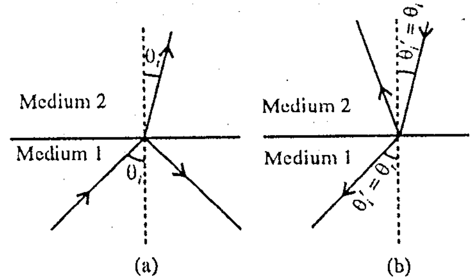

Fig. 12.14.

$$\therefore \qquad r' = \frac{\sin\theta_t \cos\theta_t - \sin\theta_i \cos\theta_i}{\sin\theta_t \cos\theta_t + \sin\theta_i \cos\theta_i} \text{ and } t' = \frac{2\cos\theta_t \sin\theta_i}{\sin\theta_t \cos\theta_t + \sin\theta_i \cos\theta_i}$$

$$\therefore \qquad 1 + rr' = 1 - \frac{(\sin\theta_i \cos\theta_i - \sin\theta_t \cos\theta_t)^2}{(\sin\theta_i \cos\theta_i + \sin\theta_t \cos\theta_t)^2}$$

$$= \frac{4 \sin\theta_i \cos\theta_i \sin\theta_t \cos\theta_t}{(\sin\theta_i \cos\theta_i + \sin\theta_t \cos\theta_t)^2} = tt' .$$

Ex. 2. *A plane polarised wave enters a dielectric from free space at an angle of incidence 30°. Determine the intensity reflection and transmission coefficients, when the electric field vector is parallel to the plane of incidence.*

Sol.

$$\sin\theta_t = \frac{\sin\theta_i}{n} = \frac{\sin 30^\circ}{1.5} = \frac{1}{2 \times 1.5} = \frac{1}{3}$$

$$\therefore \qquad \cos\theta_t = \sqrt{1 - \sin^2\theta_t} = \frac{\sqrt{8}}{3} \text{ and } \cos\theta_i = \frac{\sqrt{3}}{2}$$

Amplitude reflection coefficient,

$$r = \frac{n\cos\theta_i - \cos\theta_t}{n\cos\theta_i + \cos\theta_t}$$

Here
$$n\cos\theta_i - \cos\theta_t = 1.5 \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{8}}{3} = 1.299 - 0.943 = 0.356$$

and
$$n\cos\theta_i + \cos\theta_t = 1.299 + 0.943 = 2.242$$

$$\therefore \qquad r = \frac{0.356}{2.242} = 0.159$$

Amplitude transmission coefficient,

$$t = \frac{2\cos\theta_i}{n\cos\theta_i + \cos\theta_t} = \frac{\sqrt{3}}{2.242} = 0.733$$

$\therefore$ Intensity reflection coefficient,

$$R = |r|^2 = (0.159)^2 = \mathbf{0.025}$$

Intensity transmission coefficient, $T = 1 - R = \mathbf{0.975}$.

Ex. 3. *Determine the amplitude reflection coefficient and the transmission coefficient of an electromegnetic wave travelling in air and incident normally on a boundary between air and the dielectric having permeability μ_0 and permittivity, $\varepsilon_r = 4$.*

Sol.

$$r = \frac{n-1}{n+1} \text{ and } T = \frac{2}{n+1}$$

But
$$n = \frac{\text{Speed of light in vacuum or air}}{\text{Speed of light in medium}}$$

$$= \frac{c}{v} = \frac{1\sqrt{\mu\varepsilon}}{\sqrt{\mu_0\varepsilon_0}}\sqrt{\mu\varepsilon} = \sqrt{\mu_r \varepsilon_r}$$

Here,
$$\mu_r = \frac{\mu}{\mu_0} = \frac{\mu_0}{\mu_0} = 1 \text{ and } \varepsilon_r = 4.$$

$$\therefore \qquad n = \sqrt{4} = 2$$

Therefore,
$$r = \frac{2-1}{2+1} = \frac{1}{3} = \mathbf{0{\cdot}33} \text{ and } t = \frac{2}{2+1} = \frac{2}{3} = \mathbf{0.67}.$$

Ex. 4. *A plane electromagnetic wave normally passes from one dielectric medium to the other. If at the interface of two media, reflectance is equal to transmittance, calculate the ratio of their refractive indices.*

Sol. $$R = \left(\frac{n-1}{n+1}\right)^2 \text{ and } T = \frac{4n}{(n+1)^2}, \text{ where } n = \frac{n_2}{n_1}$$

Here, $R = T$, $\therefore$ $$\left(\frac{n-1}{n+1}\right)^2 = \frac{4n}{(n+1)^2} \text{ or } (n-1)^2 = 4n$$

or $$n^2 - 6n + 1 = 0, \ \therefore\ n = \mathbf{5{\cdot}828} \text{ or } \mathbf{0{\cdot}172}.$$

Ex. 5. *Determine the value of the relative permittivity of an ionospheric layer, having the electron density as* 5×10^5 *electrons/cc, corresponding to an electromagnetic wave of* 50 *MHz.*

Sol. Here, $N = 5 \times 10^5$ electrons/cc $= 5 \times 10^{11}$ electrons/m^3 and $f = 50$ MHz $= 5 \times 10^6$ Hz

$$\therefore \quad \varepsilon_r = 1 - \frac{81N}{f^2} = 1 - \frac{81 \times 5 \times 10^{11}}{(50 \times 10^6)^2} = 1 - 1.62 \times 10^{-2} = \mathbf{0.9838.}$$

Ex. 6. *Find the value of the frequency at which an electromagnetic wave must be propagated for a layer to have an index of refraction of* 0.5. *Electron density for the layer is* 400 *electrons/cc.*

Sol. $$n = \sqrt{1 - \frac{81N}{f^2}} \quad \therefore \quad 0.5 = \sqrt{1 - \frac{81 \times 400 \times 10^6}{f^2}} \text{ or } 0.25 = 1 - \frac{32.4 \times 10^8}{f^2}$$

or $$f^2 = \frac{324 \times 10^8}{0{\cdot}75} = 432 \times 10^8 \quad \therefore \quad f = \mathbf{2.078 \times 10^5\ Hz}$$

Ex. 7. *The height of a layer of ionosphere is* 150 *km and the critical frequency of layer is* 2.7 *MHz. If an electromagnetic wave of frequency* 4.0 *MHz is transmitted by reflection from the layer of ionosphere, calculate the skip distance.* **(Jabalpur 2003; Ujjain 2004)**

Sol. Skip distance, $D = 2h\sqrt{\dfrac{f^2}{f_C^2} - 1}$

Here, $h = 150$ km $= 150 \times 10^3$ m, $f_C = 2.7$ MHz $= 2.7 \times 10^6$ Hz, $f = 4.0$ MHz $= 4 \times 10^6$ Hz.

$$\therefore \quad D = 2 \times (150 \times 10^3)\sqrt{\left(\frac{4.0 \times 10^6}{2.7 \times 10^6}\right)^2 - 1} = 327.9 \times 10^3 \text{ m} = \mathbf{327.9\ km.}$$

Ex. 8. *The electron density of a layer of ionosphere at a height* 150 *km from earth's surface is* 9×10^{10} *per* m^3. *From this layer of ionosphere, for the transmission of electromagnetic wave by reflection upto a range of* 300 *km, find* : (i) *crtical frequency of the layer,* (ii) *the maximum usable frequency,* (iii) *angle of incidence at this layer.*

Sol. Here, $h = 150$ km $= 150 \times 10^3$m, $N_{max} = 9 \times 10^{10}$ per m^3 and $D = 250$ km $= 250 \times 10^3$m.

(i) Critical frequency $f_C = \sqrt{81 N_{max}} = \sqrt{81 \times 9 \times 10^{10}} = 2.7 \times 10^6$ Hz $=$ **2.7 MHz**

(ii) Maximum usable frequency, $$f = f_C\sqrt{1 + \frac{D^2}{4h^2}} = 2.7 \times 10^6 \times \sqrt{1 + \left(\frac{300 \times 10^3}{4 \times 150 \times 10^3}\right)^2}$$
$$= 2.7 \times 10^6 \times 1.118 = 3.02 \times 10^6 \text{ Hz} = \mathbf{3.02\ MHz.}$$

(iii) If angle of incidence at this layer is ϕ_i from the relation $f = f_C \sec \phi_i$, we have

$$\sec \phi_i = \frac{f}{f_C} = \frac{3.02 \times 10^6}{2.7 \times 10^6} = 1.1185 \ \therefore\ \cos \phi_i = 1/1.1185 \quad 0.89405, \ \therefore\ \phi_i = \mathbf{26.61°.}$$

EXERCISE 12·1

1. The light is incident from water to glass, calculate the polarizing angle on the interface of water and

flint glass. If light is incident from glass to water, then will the value of the polarizing angle obtained different ? If yes, what is this value ?

[Refractive index for the flint glass = 1.72 and refractive index for water = 4/3.]

[**Ans.** 37.8; Yes, 52.2°.]

2. A plane electromagnetic wave, moving in a homogeneous and isotropic dielectric medium, is incident obliquely on the plane surface of another dielectric medium. Calculate reflection and transmission coefficients of the electric field vector of the incident wave in the plane of incidence.

   [**Ans.** $R = \dfrac{\tan^2(\theta_i - \theta_t)}{\tan^2(\theta_i + \theta_t)}$; $T = \dfrac{\sin 2\theta_i \sin 2\theta_t}{\sin^2(\theta_i + \theta_t)\cos^2(\theta_i - \theta_t)}$.]

3. In case of normal incidence on the air-glass interface, show that only 4% light will be reflected and rest 96% will be transmitted. The refractive index of glass is 1.5.

   [**Hint** : $r = \dfrac{n-1}{n+1} = \dfrac{1.5-1}{1.5+1} = 0.2$; $\therefore\ R = r^2 = 0.04$ or 4% and $T = 1 - R = 96\%$.]

4. A light wave is normally incident on glass from water. The refractive indices of water and glass are 1.33 and 1.55 respectively. Determine the reflection and transmission coifficients for the water glass interface.

   [**Ans.** 0.0058, 0.994.]

5. Calculate the Brewster angle for a parallel polarized wave travelling from air to glass for which $\varepsilon_r = 5.0$.

   [**Ans.** 65.91°.]

6. A sky wave with a frequency of 50 MHz is incident on the *D*-layer at an angle 30°. Calculate the angle of refraction, if the electron density for the layer is 4×10^8 electrons/m^3.

   [**Ans.** 30°.]

   $$\left[\textbf{Hint: } n = \frac{\sin\phi_i}{\sin\phi_r} = \sqrt{1 - \frac{81N}{f^2}}\right].$$

7. Compute the value of relative permittivity for an ionospheric layer in case of an electromagnetic wave of frequency 50 MHz. Given that the electron density of the layer is 2.02×10^6 electrons/cc.

   [**Ans.** 0.9348.]

QUESTIONS

Long Answer Questions

1. What are boundary conditions ? Obtain boundary conditions for reflection and refraction of an electromagnetic wave from the interface of two dielectic media.
2. What are amplitude reflection and transmission coefficients ? Show that in the condition of normal incidence on the boundary of two dielectric media, the amplitude reflection and transmission coefficients are given by

   $$r = \frac{n_2 - n_1}{n_1 + n_2} \text{ and } t = \frac{2n_1}{n_1 + n_2}$$

   where n_1 and n_2 are the refractive indices of two media.
3. A plane electromagnetic wave is incident obliquely on a plane boundary between two non-conduting dielectric media. Prove the laws of reflection on the basis of electromagnetic theory.
4. Obtain Fresnel's formula when an electromagnetic plane wave is incident obliquely on the interface between two dielectrics.

5. Deduce the laws of reflection and refraction for electromagnetic waves from the boundary surface of two media of refractive indices n_1 and n_2.
6. Derive Snell's law of refraction and use it to obtain the condition for total internal reflection of an electromagnetic wave.
7. What are reflectance and transmittance ? How are they related to amplitude (Fresnel) coefficients of reflection and transmission ? If R and T represent reflectance and transmittance for the boundary surface of two media, then show that $R + T = 1$.
8. Explain polarisation of electromagnetic wave by reflection and deduce the expression for Brewster's angle.
9. What is Faraday effect ? Describe this effect experimentally. On what factors the angle of rotation depends ? What is Verdet constant ?
10. Write down the Maxwell's equations in a charge free conducting medium and explaining the propagation of electromagnetic wave obtain an expression for the skin depth. **(Jabalpur 2004; Ujjain 2004)**
11. Derive the expression for the depth of penetration of EM waves in a good conducting medium. **(Agra 2006)**
12. What is ionosphere ? Explain the role of ionosphere in reflection and refraction of electromagnetic waves. **(Indore 2003)**
13. Neglecting the effect of earth's magnetic field and assuming that the collisions with the neutral particles in ionosphere are negligible, show that the ionosphere behaves like a medium whose refractive index is given as $n = \left(1 - Ne^2/\varepsilon_0 m\omega^2\right)^{1/2} = \left(1 - 81N/f^2\right)^{1/2}$, where N is the electron density and f is the frequency of the electromagnetic wave. **(Gwalior 2003; Indore2004)**
14. Show that the relative permittivity of an ionized medium for an electromagnetic wave of frequency f is given by $\varepsilon_r = 1 - \frac{81N}{f^2}$, where N is the electron density.
15. Show that the refractive index of an ionized medium is given by $n = \sqrt{1 - \frac{81N}{f^2}}$, where the symbols have usual meaning.
16. Explain how reflection of a radio wave takes place in ionosphere.
17. Show that the skip distance for a radio wave of frequency f is given by
$$D = 2h\sqrt{\frac{f^2}{f_C^2} - 1},$$
where h is the height of the ionized layer above the earth surface from where the reflection takes place and f_C is the critical frequency of the layer.
18. How is the ionosphere helpful in long distance communication ? For what frequencies in general the ionosphere useful and why ?
19. Write short notes on the following :
(i) Polarisation by reflection.
(ii) Faraday effect.
(iii) Explanation of total internal reflection by electromagnetic theory.
(iv) Critical frequency.
(v) Virtual height.
(vi) Maximum usable frequency.

(vii) Skip distance.

(viii) Skin depth.

Short Answer Questions

1. State the boundary conditions for E.M. fields at an interface of two media. **(Agra 2006)**
2. What are Fresnel's reflection and transmission coefficients ?
3. Define reflectance and transmittance.
4. Discuss the phenomenon of total internal reflection by electromagnetic theory.
5. When an electromagnetic wave polarised parallel to the plane of incidence is incident on the interface of air and glass at Brewster's angle, discuss the reflectivity at the interface.
   [Ans. zero.]
6. Explain Faraday effect.
7. What is Verdet constant ?
8. Show that if an electromagnetic wave polarised parallel to the plane of incidence is incident on the boundary surface of two dielectric media at Brewster's angle, the amplitude of reflected wave is zero.
9. Define skin depth of a conductor. Show that it is a function of the frequency of the electromagnetic wave.
10. What is the range of frequencies used as sky waves for long distance communication ?
11. On what factors the refractive index of a layer of ionosphere for an electromagnetic wave depends ?
12. What are sky waves ?
13. What do you understand by the skip distance ? On what factors does it depend ? **(Rewa 2003)**
14. Can light waves be used for long distance communication by reflecting from the ionosphere ?
15. Show that for ionized medium, the index of refraction is given by $n = \sqrt{\varepsilon_r}$, where ε_r is the relative permittivity of the medium.
16. What is critical frequency ? Show that the critical frequency of a layer is given by $f_c = \sqrt{81 N_{max}}$, where symbols have usual meaning.
17. Discuss the conditions of total internal reflection of radio waves from the ionosphere.
18. Discuss the effect of ionosphere on the sky waves.
19. What is maximum usable frequency ? How is it related to critical frequency ?

Objective Type Questions

1. The tangential component of the electric vector ($\vec{E}$) of an electromagnetic wave on the interface of two dielectric media :
   (a) is continuous (b) is not continuous
   (c) is neither continuous nor discontinuous (d) may be continuous or discontinuous
2. Let R and T represent reflectance and transmittance at the boundary of two media. The correct relation is :
   (a) $R = T - 1$ (b) $R = T$ (c) $R = 1 - T$ (d) $R + T = 0$
3. For normal incidence, Fresnel's coefficient of reflection is :
   (a) $r = \dfrac{n}{n+1}$ (b) $r = \dfrac{n-1}{n+1}$ (c) $r = \dfrac{2}{n+1}$ (d) $r = \dfrac{n-1}{n+2}$
4. Plane of vibration of a polarised wave is rotated in :
   (a) Brewster's law (b) Faraday effect
   (c) Total internal reflection (d) Stark effect

5. The unit of Verdet constant is :
(a) Tesla
(b) Metre
(c) Tesla per metre
(d) minutes of arc per tesla

6. The angle of rotation of the plane of polarisation in the Faraday effect is given by the following relation :
(a) $\theta = VBl$
(b) $\theta = Vl$
(c) $\theta = Bl$
(d) $\theta = \alpha lc$

7. The approximate distance of the lower surface of the ionosphere is :
(a) 1000 km
(b) 200 m
(c) 50 km
(d) 300 km

8. Which of the following are reflected by the ionosphere :
(a) light waves
(b) X-rays
(c) electromagnetic waves of frequency 20 MHz
(d) electromagnetic waves of frequency 50 MHz

9. The ionized medium behaves for a radio wave as :
(a) a denser medium
(b) a rarer medium
(c) vacuum
(d) air

10. Maximum usable frequency (f) is related to the critical frequency (f_c) as :
(a) $f = f_C \cos \phi_i$
(b) $f_C = f \cos \phi_i$
(c) $f = f_C \tan \phi_i$
(d) $f = f_C \sin \phi_i$

ANSWERS

1. (a) **2.** (c) **3.** (b) **4.** (b) **5.** (d) **6.** (a) **7.** (c) **8.** (c) **9.** (b) **10.** (b) www.himpub.com

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